

# A Categorical Atlas of Parameter-Efficient Fine-Tuning

Cones, Orbits, and Gauges<sup>†</sup>

Xiaoyu Li

University of New South Wales  
xiaoyu.li2@unsw.edu.au

Zhizhou Sha

University of Texas at Austin  
zhizhousha@utexas.edu

Chiwun Yang

City University of Hong Kong  
christiannyang37@gmail.com

Dai Shi

University of Cambridge  
ds2213@cam.ac.uk

*with Claude Opus 5.5 (Anthropic), in the spirit of Grothendieck's rising sea*

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Interactive companion: <https://xiaoyulics.com/peft-atlas/>

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## ABSTRACT

Parameter-efficient fine-tuning has hundreds of methods, usually compared by parameter count and benchmark score. We compare them as maps. A pretrained checkpoint is a base point in weight space, and a weight-space method is a smooth map from a small parameter space that starts at that point. These maps form a slice category in which the frozen model is initial and full fine-tuning terminal. Three invariants of the map do most of the work. Its image near the base point governs expressivity, through the first-order image, the dimension, a min-cut bound on rank and whether that point is smooth or an apex. Its fibres, with the optimizer's metric, govern training dynamics. Base change moves the base point and covers splitting initialisations, quantisation, restarts and merging. We prove, for example, that zero-initialised LoRA, and Householder reflection adaptation at its default start under stated rank and parity conditions, begin at the apex of a cone; that no exact initialisation lets LoRA reach an update of rank above twice its own; that exactly orthogonal methods never change a weight's singular values in exact arithmetic, however often they are merged; that under Adam with no epsilon, weight decay or clipping and one shared schedule, LoRA+ from the standard zero initialisation follows exactly the trajectory of plain LoRA with its scale multiplied by the learning-rate ratio; that Adam with shared hyperparameters respects only the signed permutations inside LoRA's continuous symmetry group of full-rank factors; and that the activation function decides when a scaling adapter folds into preceding weights. We classify 382 methods from 138 papers by seven coordinates read off their formulas, with 403 simulation arrows and 485 obstructions. Every numbered result is proved in full, and results due to others are credited where we re-derive them.

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<sup>†</sup>The title pays homage to Bronstein et al. (2021), *Geometric Deep Learning: Grids, Groups, Graphs, Geodesics, and Gauges*, whose programme of reading architectures through their symmetries this atlas extends to fine-tuning.

<sup>‡</sup>This is a technical report rather than a journal or conference submission. We share it, with its proofs, catalogue and interactive companion, as an open resource for everyone working on parameter-efficient fine-tuning, and we welcome corrections and suggestions for future versions.

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# Thesis

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*The right generality turns a hard theorem into a remark,  
and the remark into a design rule.*

after Grothendieck

Parameter-efficient fine-tuning (PEFT) has hundreds of methods, usually compared by parameter count and benchmark score. Those two numbers leave open what a method can reach, how it trains, and what happens to it when it is initialised differently, quantised, restarted or merged. Each of them becomes a precise mathematical question once a method is treated as a map, and many then have exact answers. This paper compares methods that way.

Fine-tuning moves a point. A pretrained checkpoint  $\theta_0$  is a *base point* in weight space  $\Theta$ , and a weight-space PEFT method is a *relative object* over it, namely a pointed smooth map

$$\rho : (Q, q_0) \longrightarrow (\Theta, \theta_0), \quad \rho(q_0) = \theta_0,$$

from a small space  $Q$  of trainable parameters into weight space. These maps form the pointed slice category **Meth** $(\Theta, \theta_0)$  (Definition I.2). A morphism  $h : M \rightarrow N$  is a pointed smooth map with  $\rho_N \circ h = \rho_M$ , read “ $N$  simulates  $M$ ”. The frozen model is initial and full fine-tuning is terminal (Observation I.4), so every method pointed at  $\theta_0$  sits in the interval between them. Following Grothendieck’s relative point of view, the object of study is the morphism  $\rho$  itself. Almost all the work is done by three invariants of the morphism, plus one further layer for architectures. Figure 1 draws the picture.

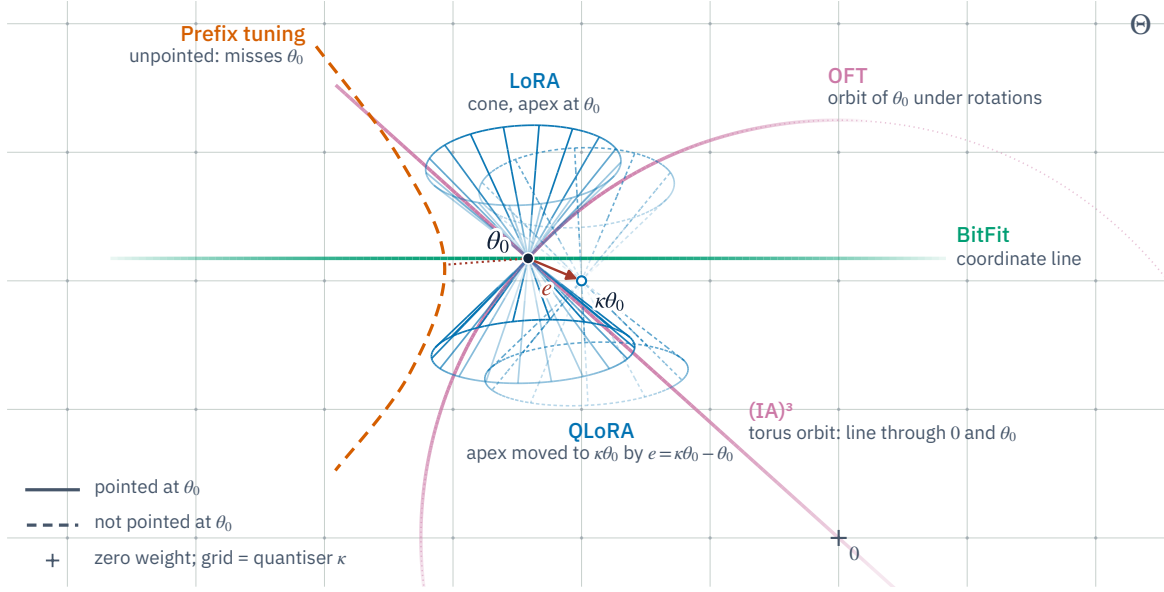
The bicategory **Para** and the lens reading of backpropagation are due to Fong et al. (2017) and Cruttwell et al. (2021). The closest earlier unified frame for PEFT is that of He et al. (2021), who read adapters, prefix tuning and LoRA as modifications of specific hidden states. Ours works on the weight side, where adapters and prefixes return as extensions of the network (Definition VI.1).

## Three invariants and one further layer

The three invariants answer three practical questions: what a method can say, how it learns, and how it travels.

**The image germ at  $\theta_0$  governs expressivity.** This is what a method can say. Near  $\theta_0$ , the weights a method reaches form a linear space, a cone, a group orbit, a torus saturation, or a meet of these. Four data are read off this germ: the first-order image  $S_1$ , the expressive dimension  $d(M)$ , the min-cut bound on rank (the cut-rank), and the type of the base point, which is either a smooth point or an apex. Zero-initialised LoRA (Hu et al., 2021) starts at the apex of the cone  $\theta_0 + \mathcal{M}_{\leq r}$ , where first steps miss directions that the method reaches later; a PiSSA-style split (Meng et al., 2024a) starts at a smooth point. Exposé II treats the germ in three parts: strata of initialisations (II.A), invariants of group-type methods (II.B), and cut and volume (II.C).

**The fibres, together with a metric, govern dynamics.** This is how a method learns. Many parameter values give the same weight, and the fibres of  $\rho$  are gauge orbits; on LoRA’s full-rank locus they are the orbits of  $\mathrm{GL}_r$  acting by  $g \cdot (B, A) = (Bg^{-1}, gA)$ . Combined with the



**Figure 1. Fine-tuning moves a point.** Weight space  $\Theta$  is drawn as a chart around the pretrained point  $\theta_0$ ; the grid is the lattice of a round-to-nearest quantiser  $\kappa$  through the zero weight 0 (marked +). Solid curves are image germs of methods pointed at  $\theta_0$  (Definition I.2), drawn darkest near  $\theta_0$  because only the germ there matters. LoRA (Hu et al., 2021) is the cone  $\theta_0 + \mathcal{M}_{\leq r}$  with its apex at  $\theta_0$  (Theorem II.2), shown as the rank- $\leq 1$  double cone  $x^2 = y^2 + z^2$  of the symmetric  $2 \times 2$  slice in perspective. OFT (Qiu et al., 2023) is the orbit of  $\theta_0$  under rotations, an arc of the circle about 0 through  $\theta_0$ , the rest of which is dotted. (IA)<sup>3</sup> (Liu et al., 2022a) is a torus orbit, whose closure is the line through 0 and  $\theta_0$  (Theorem II.5). BitFit (Ben-Zaken et al., 2021) moves along a coordinate line, parallel to the grid. The dashed curves are not pointed at  $\theta_0$ . Prefix tuning (Li and Liang, 2021) extends the network and is unpointed (Definition VI.1); drawn in weight space it is a curve that misses  $\theta_0$ , at the distance marked by the dotted segment. QLoRA (Detrmers et al., 2023) is LoRA pointed at the nearest grid node  $\kappa\theta_0$ , so its cone is LoRA’s cone translated by the defect  $e = \kappa\theta_0 - \theta_0$  (red arrow), and  $\theta_0$  is off its image unless rank  $e \leq r$  (Proposition V.2).

optimizer’s learning rates  $\Lambda \succ 0$ , the parametrisation induces a cometric on weight space,

$$K = D\rho \wedge D\rho^\top,$$

so isomorphic methods can train differently. For LoRA with per-block rates, the compact half of the gauge acts by isometries and the non-compact half gives charges conserved under gradient flow. The optimizer’s equivariance group decides what descends to the model (Exposé III). The lens structure of backpropagation sorts tangent-side from cotangent-side methods (Exposé IV). Through this lens, GaLore (Zhao et al., 2024a) with no weight decay and no full-gradient clipping is one-sided LoRA with a moving frozen factor and carried-over optimizer state (Proposition IV.2); this identity is due to Torroba-Hennigen et al. (2025). This layer rests on work on factorised and linear networks: balancedness (Arora et al., 2018; Du et al., 2018), symmetries and conservation laws (Kunin et al., 2020; Zhao et al., 2022b), and imbalance at initialisation (Min et al., 2021; Tarmoun et al., 2021).

**Base change moves  $\theta_0$ .** This is how a method travels. Initialisation by splitting, quantisation, restarts and merging all move the base point (Exposé V). QLoRA (Detrmers et al., 2023) is LoRA pointed at the quantised weights  $\kappa\theta_0$ , with defect  $e = \kappa\theta_0 - \theta_0$ , and its image misses  $\theta_0$  whenever rank  $e > r$ , which is the generic case (Proposition V.2). Merge-and-restart with a

fixed adapter shape reaches, in the limit, the orbit of the monoid generated by one cycle. For ReLoRA (Lialin et al., 2023b) this means that  $K$  cycles reach exactly the updates of rank at most  $\min(Kr, m, n)$  (Theorem V.3).

**Extensions and mergeability.** The fourth layer is the one place where a method changes the network  $f$  itself; the network otherwise enters only through its realisation map and the backpropagation lens (Remark I.3). Adapters, prefixes and ReFT first extend  $f$ . Merging is then a lifting problem, certified by a local rewriting calculus of seven rules with side conditions, which gives sufficient conditions and box-local obstructions (Exposé VI).

## Contributions

Each result below is proved in Appendix C, and the scope stated here is the scope of the proof. Where a result, or a part of one, is due to others, the item says so. We re-derive such results in our language and add only their reading for fine-tuning.

1. **A relative frame for PEFT.** Weight-space methods are the objects of the pointed slice  $\mathbf{Meth}(\Theta, \theta_0) = \mathbf{Diff}_*/(\Theta, \theta_0)$ . Every such method lies in the interval [frozen, full FT], and “ $N$  simulates  $M$ ” becomes an explicit pointed map  $h$  with  $\rho_N \circ h = \rho_M$  (Definition I.2, Observation I.4). Isomorphism invariants finer than the image, such as  $S_1$  and the singular locus, separate initialisations that the image cannot (Observation I.6, Theorem II.2). A categorical product produces a named method. Freezing either frame of LoRA gives two one-sided methods whose product is the sandwich  $R \mapsto W_0 + B_0 R A_0$ , which is LoRA-XS (Bałazy et al., 2024) in the SVD frames of  $W_0$  (Proposition I.8). Image-level meets, which are not products, organise others, such as HRA below. The bicategory **Para** is that of Fong et al. (2017) and Cruttwell et al. (2021); the slice for PEFT and its invariants are new.
2. **No initialisation buys rank above  $2r$**  (Theorem II.1). For an additive method, moving the starting point while subtracting a frozen residual keeps the method pointed at  $\theta_0$ , and the union of all such re-pointed images is  $\theta_0$  plus the difference set  $C - C$  of its update shape. For LoRA $_r$  that difference set is the matrices of rank at most  $\min(2r, m, n)$ . So no exact pointing of LoRA $_r$ , zero-init or split (PiSSA, OLoRA (Büyükyüz, 2024), CorDA (Yang et al., 2024c), LoRA-GA (Wang et al., 2024e)), makes an update of rank above  $2r$  reachable in any adapted matrix, and split initialisations attain  $2r$  when  $2r \leq \min(m, n)$ . The statement is elementary; the  $2r$  bound underlies the PiSSA-to-LoRA conversion of Meng et al. (2024a), implemented in Hugging Face (HF) PEFT.
3. **Three strata of initialisations, and two apexes** (Theorem II.2, Theorem II.7; Figure 3). For LoRA of rank  $r < \min(m, n)$  on one  $m \times n$  weight there are three strata of exact full-rank pointings:  $B_0 = 0$  with full-rank  $A_0$  (vanilla LoRA on linear layers, EVA (Paischer et al., 2024));  $A_0 = 0$  with full-rank  $B_0$  (Init[B] (Hayou et al., 2024a), HF’s default on embedding layers); and a rank- $r$  product  $B_0 A_0$  subtracted from the frozen weight (PiSSA, MiLoRA (Wang et al., 2024b), OLoRA, CorDA, LoRA-GA, Astra (Liu et al., 2026b)). Two pointed LoRA methods in these strata are isomorphic over weight space exactly when they share row  $A_0$ , col  $B_0$  or  $B_0 A_0$  respectively, so as a set the classes are  $\mathrm{Gr}(r, n) \sqcup \mathrm{Gr}(r, m) \sqcup \mathcal{M}_r$ . Zero-factor pointings sit at the apex of the cone  $W_0 + \mathcal{M}_{\leq r}$  and miss  $r(n - r)$  or  $r(m - r)$  first-order directions; split pointings sit at a smooth point, and neither image contains the other. The classification is set-level and in exact arithmetic. Rank-deficient pointings such as HF’s orthogonal init, frozen-factor sub-objects (LoRA-



FA (Zhang et al., 2023d), MiCA (Rüdiger and Raschka, 2026)) and quantised-base defects (QLoRA, LoftQ (Li et al., 2023b)) lie outside the three strata.

Let  $W_0$  have full column rank and let  $r$  be even with  $2 \leq r \leq n - 2$ . Then the image of  $r$ -reflection HRA (Yuan et al., 2024) without Gram–Schmidt, HF PEFT’s default (apply\_GS=False), is exactly  $\{W_0 R : R \in \text{SO}(n), \text{rank}(W_0 R - W_0) \leq r\}$ . This is the image-level intersection of  $\text{LoRA}_r$  with the rotation orbit  $W_0 \cdot \text{SO}(n)$ , of dimension  $rn - r(r+1)/2$ , and HRA is not the categorical product of the two. At HF’s default paired initialisation ( $u_{2i-1} = u_{2i}$ ),  $W_0$  is a singular vertex of the image and the first-order image has dimension  $kn - k(k+1)/2$  with  $k = r/2$ , so that initialisation is an apex like zero-init LoRA. Both theorems are new.

4. **Exactly orthogonal methods keep the spectrum** (Corollary II.6, Theorem V.3). If every weight a method produces, including after merge-and-restart cycles, has the form  $W_0 R$  with  $R$  exactly orthogonal on the input side, then in exact arithmetic the neuron Gram matrix  $W W^\top$  never changes. Every neuron norm, every angle between neurons (and with them the hyperspherical energy) and every singular value of  $W_0$  stay fixed, however often the method is merged, and block OFT with a fixed partition also fixes each block Gram. This covers exact-Cayley OFT (Qiu et al., 2023) and block OFT, strict GOFT (Ma et al., 2024) without its optional scaler, HRA, and ETHER (Bini et al., 2024) as defined in its paper. It does not cover variants with a per-neuron or per-input scale (HF BOFT (Liu et al., 2023b), HF OFT 0.14 to 0.15, GOFT’s default code), output-side variants, or approximately orthogonal ones such as HF PEFT’s default Cayley–Neumann OFT, qGOFT and ETHER+. With small generators each Cayley–Neumann merge is a contraction, so repeated merging can only shrink neuron norms and singular values; low-precision merges add rounding drift. The invariance itself is elementary, and preserving hyperspherical energy is the motivation of OFT (Qiu et al., 2023); new is the sorting of methods and implementations by whether they satisfy its hypothesis.
5. **Rank is a cut, dimension is a volume** (Theorem II.10, Proposition II.12; Figure 4). The rank of a map computed by a tensor network is at most the smallest product of wire dimensions across any cut separating inputs from outputs (standard). The expressive dimension is at most the budget minus the dimension of the generic gauge orbit, with equality when the gauge is locally transitive on generic fibres. At equal budget, with  $2r \leq \min(m, n)$ ,  $r^2 \leq \min(m, n)$  and  $2r^2 < m + n - 1$ ,  $\text{LoHa}_{r,r}$  (Hyeon-Woo et al., 2021) reaches at least  $(m + n - 1) - 2r^2$  fewer dimensions than  $\text{LoRA}_{2r}$ , about  $m + n$  when  $r^2 \ll (m + n)/2$ . It buys rank up to  $r^2$  in exchange for  $r \geq 3$ , and it is strictly dominated for  $r \leq 2$ . When  $2r^2 \geq m + n - 1$ , for example  $r = 64$  on  $4096 \times 4096$  or LyCORIS-style  $r = 32$  on 320- to 640-wide layers, the dimension bound no longer separates the two, and wherever it is attained, as the numerical checks find, LoHa matches or exceeds  $\text{LoRA}_{2r}$  in dimension as well as in rank. GraLoRA (Jung et al., 2025) ( $k \times k$  blocks of rank  $r/k$ , HF default hybrid\_r=0) keeps LoRA’s budget and dimension  $r(m + n - r)$  and lifts the maximum rank to  $\min(kr, m, n)$ , but caps each block at rank  $r/k$ ; for  $k \geq 2$  and  $kr \leq \min(m, n)$  neither image contains the other. The census is new.
6. **Kronecker sums cut the other way** (Theorem II.11; Figure 6). Split the dimensions as  $m = m_1 m_2$  and  $n = n_1 n_2$ . A  $k$ -term Kronecker sum  $\sum_i C_i \otimes D_i$  is exactly a matrix of rank at most  $k$  after the rearrangement of Van Loan and Pitsianis (1993), so unconstrained Kronecker methods are LoRA across a different cut of the same four-leg tensor. With equal square splits of a  $d \times d$  weight, a generic rank-one update has the largest Kronecker rank,  $d$ , while



the identity has Kronecker rank one. Hence zero-init LoRA<sub>r</sub> and unconstrained  $k$ -term Kronecker sums, for  $r, k < d$ , have incomparable images. LoKr (Yeh et al., 2023) updates  $C \otimes BA$  with inner rank  $r' \leq \min(m_2, n_2)$  have rank at most  $\min(m_1, n_1) r'$ , attained, so LoKr's image lies inside that of zero-init LoRA<sub>r</sub> exactly when  $r \geq \min(m_1, n_1) r'$ , and is incomparable with it otherwise when  $\min(m_1, m_2) \min(n_1, n_2) > 1$ . The linear algebra is standard (the rearrangement and the operator-Schmidt decomposition); the PEFT reading is new.

7. **A conserved charge** (Theorem III.5, Corollary III.6, Theorem III.9; Figure 7). With per-block rates, LoRA's charge

$$\Phi = B^\top B / \eta_B - AA^\top / \eta_A$$

is conserved by every continuous-time update  $\dot{q} = -\Lambda D\rho_q^\top \gamma(t)$  with constant  $\Lambda > 0$ , gradient or not, including minibatches, global-norm clipping and DoRA's (Liu et al., 2024b) detached norm. The compact half of the gauge gives  $\Lambda$ -isometries that leave  $K$  unchanged, so gauge-related initialisations give identical weight trajectories whenever the signal depends on  $q$  only through  $(t, \rho(q))$ . With per-rank-channel rates, off-diagonal charges and isometries exist exactly between channels with equal products  $\eta_{B,j} \eta_{A,j}$ . Discrete steps make the charge drift, with relative drift of second order in the step size, and Adam and momentum are not covered. The law is a mild extension of known results: Zhao et al. (2022b, Prop. 5.1) and Marcotte et al. (2023), after the balancedness principle of Du et al. (2018) and Arora et al. (2018) and the symmetry analysis of Kunin et al. (2020). New here is only the remark that  $\gamma$  need not be a gradient. Under an isotropic charge the weight dynamics have a closed form, with a crossover scale below which zero- $B$  LoRA behaves like LoRA-FA; this closed form is due to Tarmoun et al. (2021) and Min et al. (2021), and the PEFT reading is ours.

8. **LoRA+ is  $\alpha$  in disguise** (Theorem III.10, Corollary III.11; Figure 8). For zero-init LoRA under Adam with  $\varepsilon = 0$ , no weight decay or clipping, one learning-rate schedule for both groups, and the same data order and dropout masks, LoRA+ (Hayou et al., 2024b) with ratio  $\lambda$  follows, in exact arithmetic, the same weight trajectory as LoRA with  $\alpha$  multiplied by  $\lambda$ . Per-block  $\varepsilon$  and decoupled weight decay keep the statement once they are rescaled too, and under SGD the factor is  $\sqrt{\lambda}$ . At a fixed rank, rsLoRA (Kalajdziewski, 2023) is LoRA with a rescaled  $\alpha$  or  $B$  learning rate; how  $\alpha$  should scale with rank is a separate question. Both statements are instances of a hyperparameter torus: for a method whose update is  $s$  times a multihomogeneous function of its parameter blocks, rescaling each block's initial scale and learning rate together with the overall scale  $s$  (and, under Adam, the per-block  $\varepsilon$ , decoupled decay and clipping thresholds) leaves the weight trajectory exactly unchanged under Adam, given the same data order and dropout masks; under SGD the same holds with learning rates scaled by the square of the block factor. The LoRA/Adam case is due to Schulman and Thinking Machines Lab (2025); the general multihomogeneous torus and the SGD law are new.
9. **Adam sees a finite shadow of the gauge** (Theorem III.3, Theorem III.12, Proposition III.4; Figure 9). Gradient descent is natural exactly where cometrics agree, which for LoRA's linear gauge with block-scalar learning rates, at points where  $A$  has full row rank and  $r < n$ , means exactly along  $O(r)$ . On LoRA's full-rank locus with channel-uniform hyperparameters, the largest subgroup of  $GL_r$  under which the parameter update is equivariant is  $O(r)$  for SGD, with or without momentum, and the signed permutations  $B_r$ , of order  $2^r r!$ ,

for Adam and AdamW with any  $\beta$ 's, any  $\varepsilon \geq 0$  and decoupled decay. Positive diagonal rescalings become symmetries of Adam only jointly with per-rank-channel rescaling of the learning rates (and of  $\varepsilon$  and learning-rate-coupled decay when these are nonzero), and only without global-norm clipping. Undamped scaled GD (Riemannian preconditioning) and LoRA-RITE (Yen et al., 2024) are equivariant under all of  $\text{GL}_r$ ; damping reduces scaled GD to  $\mathcal{O}(r)$ , and wrapping it in Adam reduces it to  $B_r$ . Combination rules must be gauge-natural as well. A rule for combining LoRA factors is a well-defined function of the adapters exactly when it is constant on the factorisation fibres, which task arithmetic on the products satisfies and post-hoc product-of-sums rules such as LoraHub (Huang et al., 2023) do not (Proposition V.4). The hierarchy is new; scaled GD is due to Tong et al. (2021) and, for LoRA, Zhang and Pilanci (2024a).

10. **GaLore is one-sided LoRA with a moving frame** (Proposition IV.2). If an adapter is linear in its trainable tensor ( $W = W_k + \mathcal{L}C$  on the current base  $W_k$ , with  $C_0 = 0$ ) and the optimizer depends only on the gradients it receives (SGD or momentum, Adam without weight decay, Adafactor without parameter scaling, no global-norm clipping, the same implementation on both sides), then training  $C$  and merging gives exactly the same weights as running that optimizer on the compressed gradient  $\mathcal{L}^* \nabla_W L$  and mapping back with  $\mathcal{L}$ . In particular, GaLore with weight decay 0 is exactly one-sided LoRA whose frozen factor is GaLore's projector matrix, merged and re-initialised every period with the trainable factor's optimizer state carried over. Decoupled weight decay, parameter-scaled Adafactor, trust-ratio optimizers and full-gradient clipping break the identity, and Flora, which compresses only the momentum, is not covered. Known: Torroba-Hennigen et al. (2025) for GaLore, and Hao et al. (2024) for random projections.
11. **Base change has explicit defects and closures** (Proposition V.2, Theorem V.3; Figure 10). Quantising the frozen base moves the base point. At initialisation QLoRA misses the pretrained weight  $W$  by the generically full-rank error  $e = \kappa W - W$ , and it can reach  $W$  only if  $\text{rank } e \leq r$ . Splitting or error-fitting initialisations change this defect in an order-dependent way: split-then-quantise (QPSSA) leaves  $\kappa(W_{\text{res}}) - W_{\text{res}}$ , and one-step LoftQ leaves  $e - s \text{SVD}_r(e)$ , which is  $e$  minus its best rank- $r$  approximation only when the LoRA scale  $s = \alpha/r$  is 1, because HF's LoftQ initialisation ignores the scale. After  $K$  merge-and-restart cycles with a fixed shape, the reachable set is  $\theta_0 + C^{+K}$  for an additive shape  $C$ , or  $S^K \theta_0$  for a set  $S$  of group elements containing the identity. Methods whose shape is a fixed subspace, subgroup or submonoid gain nothing, and ReLoRA reaches exactly  $W_0 + \{\text{rank} \leq \min(Kr, m, n)\}$ . Exact-Cayley block OFT stays in  $W_0 \cdot \prod \text{SO}(b)$ , up to closure, unless its block hypergraph is connected, which BOFT's butterflies achieve only at full depth, and re-HRA reaches  $(W_0 + \mathcal{M}_{\leq Kr}) \cap W_0 \cdot \text{SO}(n)$ . New.
12. **The activation decides the merge** (Theorem VI.3, Theorem VI.5; Figure 12). A few exact rewrite rules give sufficient conditions for folding a scaling or affine PEFT edit into existing weights with no change in output. The feed-forward scale of (IA)<sup>3</sup> (Liu et al., 2022a) always folds into  $W_{\text{down}}$ . It folds into  $W_{\text{up}}$  under SwiGLU or GeGLU for every scale and under ReLU for nonnegative ones, while under GELU only entries 0 and 1 fold. Position-selective edits (ReFT (Wu et al., 2024d), Activated LoRA (Greenewald et al., 2025)) and nonlinear adapters are not box-locally mergeable, and exactly merging a generic QLoRA or LoftQ update leaves the quantised format, with QA-LoRA (Xu et al., 2023b) as the exception. For attention, a prefix (Li and Liang, 2021) acts as an input-gated parallel adapter (He et al., 2021) that rescales the content attention weights

within its layer and never reorders them (Petrov et al., 2023), while LLaMA-Adapter’s (Zhang et al., 2023h) separate softmax with a zero-initialised gate leaves those weights unchanged. The rules are elementary, and the merge into  $W_{\text{down}}$  is noted by Liu et al. (2022a); SmoothQuant (Xiao et al., 2023) and AWQ (Lin et al., 2024a) already migrate scales through activations and meet the GELU obstruction. New is only the packaging as a merge calculus for PEFT, with explicit side conditions. The calculus gives sufficient conditions and box-local obstructions; deciding mergeability of a whole network lies outside its scope.

13. **An atlas** (Exposés VII–IX, Appendix D; Figures 13 and 14). Fourteen methods are worked through the three invariants. Seven coordinates, each read off a method’s formula (kind, shape of the image germ, base point, gauge, covariance group, merge type and rebasing closure), classify 382 methods from 138 papers, 82 of them in HF PEFT. The atlas records 403 simulation arrows between methods, each exhibited by an explicit map  $h$  with  $\rho_N \circ h = \rho_M$ , and 485 obstructions, each naming the test that rules an arrow out. The 142 core methods form the periodic table and the category drawn in Figure 14. A Rosetta stone translates between the PEFT and the categorical vocabulary.
14. **Predictions and limits** (Exposés X and XI). The theory makes falsifiable predictions, each with its optimizer scope, and states plainly which of its words are analogies and which problems are open.

Some further results are classical or standard and carry their credit in their headings: the complete invariants of the basic actions (Theorem II.5), the Kempf–Ness slice of the LoRA gauge (Proposition III.7; Kempf and Ness, 1979) and the privileged basis fixed by an activation (Lemma VI.2).

**Where the category theory earns its keep.** There are seven places.

1. The pointed slice gives the interval [frozen, full FT] for weight-space methods and makes “simulation” precise (Exposé I).
2. Isomorphism invariants finer than the image ( $S_1$ , the singular locus) classify initialisations (Theorem II.2).
3. A categorical product produces a named method (LoRA-XS, Proposition I.8), and image-level meets organise others (HRA, Theorem II.7).
4. Base change separates function-preserving reparametrisations, which act on the slice, from maps that move the base point (quantisers). The resulting pointing defects are elementary, but the slice makes their order dependence explicit (Observation II.15, Proposition V.2).
5. The lens 2-functor **Para**(R) separates tangent-side from cotangent-side methods. The degree-one collapse shows that every cotangent method whose optimizer reads only its gradients (no weight decay, no parameter scaling, no full-gradient clipping) is, period by period, a forward linear method on a moving base (Proposition IV.2).
6. Naturality: gradient descent is natural exactly where cometrics agree (for LoRA’s linear gauge with block-scalar rates, along  $O(r)$  where  $A$  has full row rank and  $r < n$ ), and combination rules must be gauge-natural (Theorem III.3, Proposition V.4).
7. Symmetric monoidal structure turns rank into a min-cut (Theorem II.10).

The rest is real algebraic geometry, invariant theory, Lie generation and gradient-flow ODEs, and we say so where it happens. True but idle constructions are omitted: coends for the rank

filtration, a saturation monad, sheaf language for locality, the Grothendieck construction  $\int \mathbf{Meth} \rightarrow \mathbf{PNet}$ , and coKleisli 2-cells for mixtures.

### How the results were checked

Each of the 41 numbered results was checked by adversarial model referees (Appendix F) against its proof, against the literature, and against the methods as published and as coded, including the source of HF PEFT at commit 85be3f3 of 30 September 2026. Of these, 14 stood as first written and 27 were repaired. A second round, with one mathematical and one faithfulness referee for each result, re-examined 26 of the repaired results and narrowed many of them. Its objections mostly concerned a side clause: a missing hypothesis, a remark that went past its proof, or a description of a method that matched only one implementation. For example, HF PEFT’s LoftQ initialiser turned out to ignore the LoRA scale  $\alpha/r$  (Proposition V.2). Only the narrowed forms are stated in this paper. Some results are new, such as the HRA apex; many are known and re-derived with credit, as the contributions above record.

### How to read this paper

**Plan.** Exposé I sets up the pointed slice and its invariants. Exposé II studies the image germ: initialisations and their strata (II.A), group-type methods (II.B), and cut and volume (II.C). Exposé III treats fibres, metrics and training dynamics, Exposé IV the backpropagation lens, Exposé V base change, and Exposé VI extensions and mergeability. Exposé VII works through fourteen methods, Exposé VIII gives the classification and Exposé IX the Rosetta stone. Exposé X lists falsifiable predictions, Exposé XI separates analogy from theorem and collects open problems, Exposé XII discusses related work and Exposé XIII concludes.

**Labels.** Table 1 lists the labels. Every numbered item keeps the number it has on the companion site. Definitions, remarks and results share one counter within each exposé, so Definition I.2 is followed by Remark I.3 and Observation I.4.

**Conventions.** These carry real weight in the arguments. Table 2 collects the notation.

- *Linear box.* A linear box computes  $y = Wx$  with  $W \in \mathbb{R}^{m \times n}$ ,  $m = d_{\text{out}}$  and  $n = d_{\text{in}}$ . Row  $i$  of  $W$  is output neuron  $i$ .
- *Two Gram matrices, two sides.*  $K_{\text{out}}(W) = WW^\top$  is the neuron Gram matrix and  $K_{\text{in}}(W) = W^\top W$  the input Gram matrix. A right multiplication  $W \mapsto WR$  with  $R \in O(n)$  (the *input side*) fixes  $K_{\text{out}}$ . A left multiplication  $W \mapsto PW$  fixes  $K_{\text{in}}$ . Papers that store  $W^\top$  swap the words *input* and *output*; we translate every method into the convention above.
- *Ambient category.* Maps and dynamics live in **Diff**. Images are taken in **Set**. Every image we meet is semialgebraic, or definable in  $\mathbb{R}_{\text{exp}}$  when softmax is involved, so dimensions, smooth points and tangent cones are well defined.

**Proofs and appendices.** The main text contains no proofs. Complete proofs of all numbered results are in Appendix C, in the same order as the statements, one subsection per result. External theorems that the proofs use are quoted in Appendix B. Appendix A collects the background the paper assumes, on categories, monoidal categories and lenses, manifolds, germs and cones, and groups, symmetry and conservation; its entries keep the numbers A.1–D.34 they carry on the companion site, and readers new to category theory or Lie groups can start there. Appendix D describes the catalogue and Appendix E the numerical checks.

**Table 1.** How to read the labels. The coloured rule is the one that marks the environment in the text.

| Label                                  | What it means here                                                                                                                                                                                                                                       |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Theorem, Proposition, Lemma, Corollary | Proved in this paper, with the complete proof in Appendix C; the statement is set in italic. A heading that says <i>known</i> and names authors marks a result due to them, which we re-derive in our language, adding only its reading for fine-tuning. |
| Observation                            | True and formally easy; kept because it organises.                                                                                                                                                                                                       |
| Definition                             | A definition.                                                                                                                                                                                                                                            |
| Remark, Example                        | Commentary and instances, in upright type.                                                                                                                                                                                                               |
| Conjecture, [Conj]                     | A conjecture.                                                                                                                                                                                                                                            |
| Prediction, [Pred]                     | A falsifiable prediction, stated with its optimizer scope.                                                                                                                                                                                               |
| Analogy                                | Organising language with no theorem behind it. <i>Noether</i> , <i>Erlangen</i> , <i>gauge</i> and <i>moduli</i> are meant this way beyond what is proved (Exposé XI).                                                                                   |
| WHAT THIS EXPLAINS                     | What a result says about methods as published and as implemented.                                                                                                                                                                                        |
| [Num]                                  | Checked numerically in float64 (autograd Jacobian ranks, exact optimizer runs) by the ancillary script described in Appendix E. The companion site’s figures recompute their numbers in the browser.                                                     |
| [Sketch]                               | Only a proof sketch is given, in Appendix C; the claim is not counted as proved.                                                                                                                                                                         |
| verified, repaired                     | Review status of a numbered result, recorded in the ancillary file <code>propositions.json</code> . Repaired results were corrected or narrowed, and only that form is stated.                                                                           |

**The companion site.** This paper is the long-form version of the interactive site at <https://xiaoyulics.com/peft-atlas/>. The two share every number, so “Theorem III.10” names the same statement here, on the site and in the ancillary file `propositions.json`. The site states each result with its proof beneath it and has 20 interactive figures that recompute their numbers in the browser, among them an orbit explorer for LoRA+, a Gram lab for orthogonal methods, a gauge lab, Noether hyperbolas and a merge game. This paper gives the complete text, collects the proofs in Appendix C, and replaces the interactive figures by static ones. The catalogue, the arrows and obstructions, the numbered results with their review status, and the numerical checks accompany this paper as data files. When citing a result, please give its number; for a result marked known, please also cite the authors named in its heading, to whom the result itself is due.

Table 2. Notation.

| Symbol                                | Meaning                                                                                                                                                                                                                          |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $W$                                   | A linear box $y = Wx$ , $W \in \mathbb{R}^{m \times n}$ , with $m = d_{\text{out}}$ and $n = d_{\text{in}}$ . Row $i$ is output neuron $i$ .                                                                                     |
| $\Theta$                              | Weight space, a real vector space of adapted and frozen boxes.                                                                                                                                                                   |
| $\theta_0, W_0$                       | The pretrained point; $W_0$ for one box.                                                                                                                                                                                         |
| $\rho, (Q, q_0)$                      | The method, a pointed smooth map from trainable parameters $Q$ started at $q_0$ , with $\rho(q_0) = \theta_0$ .                                                                                                                  |
| <b>Meth</b> $(\Theta, \theta_0)$      | The pointed slice <b>Diff</b> $_{*}/(\Theta, \theta_0)$ , whose objects are methods $M = (Q, q_0, \rho)$ . A morphism $h : M \rightarrow N$ is a pointed smooth map with $\rho_N \circ h = \rho_M$ , read “ $N$ simulates $M$ ”. |
| $ M $                                 | The budget, $\dim Q$ .                                                                                                                                                                                                           |
| $d(M)$                                | The expressive dimension $\dim \text{Im } M = \max_q \text{rank } d\rho_q$ . The gap $ M  - d(M)$ is gauge waste.                                                                                                                |
| $S_1$                                 | The first-order image $d\rho_{q_0}(T_{q_0}Q)$ . The base point is regular if $\dim S_1 = d(M)$ , an apex if smaller.                                                                                                             |
| $K$                                   | The induced cometric $D\rho \wedge D\rho^\top$ , with $\Lambda \succ 0$ the learning rates. Gradient flow is $\dot{\theta} = -K\nabla_{\theta} L$ .                                                                              |
| $K_{\text{out}}, K_{\text{in}}$       | The neuron Gram matrix $WW^\top$ and the input Gram matrix $W^\top W$ .                                                                                                                                                          |
| $\mathcal{M}_{\leq r}, \mathcal{M}_r$ | Matrices of rank at most $r$ ; the smooth stratum of rank exactly $r$ , of dimension $r(m + n - r)$ . The singular locus of $\mathcal{M}_{\leq r}$ is $\mathcal{M}_{\leq r-1}$ .                                                 |
| $\text{LoRA}_r$                       | $\rho(B, A) = W_0 + sBA$ with $B \in \mathbb{R}^{m \times r}$ , $A \in \mathbb{R}^{r \times n}$ and $s = \alpha/r$ .                                                                                                             |
| $G, \text{GL}_r$                      | The weight gradient $\nabla_W L$ ; the LoRA gauge, $g \cdot (B, A) = (Bg^{-1}, gA)$ .                                                                                                                                            |
| $T_m, T_m^+$                          | The diagonal torus of invertible diagonal matrices; its identity component $\{\text{diag}(d) : d_i > 0\}$ .                                                                                                                      |
| $\text{Diag}_m$                       | The monoid of all diagonal matrices.                                                                                                                                                                                             |
| $\text{Mon}_m$                        | The monomial matrices, $T_m \rtimes S_m$ .                                                                                                                                                                                       |
| $B_r$                                 | The signed permutations, $\{\pm 1\}^r \rtimes S_r$ , of order $2^r r!$ .                                                                                                                                                         |
| $\text{CO}(n)$                        | The conformal orthogonal group $\mathbb{R}_{>0} \cdot \text{O}(n)$ ; $\text{CO}^+(2) \cong \mathbb{C}^\times$ .                                                                                                                  |



## The pointed slice

We model a weight-space PEFT method as a pointed map into the pretrained weights. The frozen model and full fine-tuning sit at the two ends of an interval, and sums and products of such maps already give methods in use. This exposé sets up the category and its first invariants. The theorems begin when we look closely at one image near one point, which is where Exposé II starts.

### Pretrained models and methods

A pretrained network is a function with a parameter slot and a chosen value in it. Its natural home is the bicategory of parametrised maps (Background B.16).

**DEFINITION I.1** (Para; pretrained model). For a cartesian category  $C$ , the bicategory  $\mathbf{Para}(C)$  is defined as follows (Cruttwell et al., 2021; Fong et al., 2017):

- its 1-cells  $X \rightarrow Y$  are pairs  $(P, f : P \times X \rightarrow Y)$ ;
- its 2-cells  $(P, f) \Rightarrow (Q, g)$  are maps  $r : Q \rightarrow P$  with  $g = f \circ (r \times X)$ .

A **pretrained model** is a pointed 1-cell  $F = (\Theta, \theta_0, f)$  of  $\mathbf{Para}(\mathbf{Diff})$ . Its **realisation** is  $\Phi_f : \Theta \rightarrow C^\infty(X, Y)$ ,  $\theta \mapsto f(\theta, -)$ . Fine-tuning is a path  $\theta(t)$  with  $\theta(0) = \theta_0$ .

**Para** and the lens reading of backpropagation are due to Fong et al. (2017) and Cruttwell et al. (2021); Gavranović et al. (2024) develop a broader programme on the same foundations.

**DEFINITION I.2** (methods and simulations). A weight-space **PEFT method** at  $\theta_0$  is a pointed smooth map  $M = (Q, q_0, \rho)$ ,  $\rho : Q \rightarrow \Theta$ , with  $\rho(q_0) = \theta_0$ . Usually  $\dim Q \ll \dim \Theta$ , and  $\theta_0$  (or a splitting of it) is frozen inside  $\rho$ . These form the pointed slice

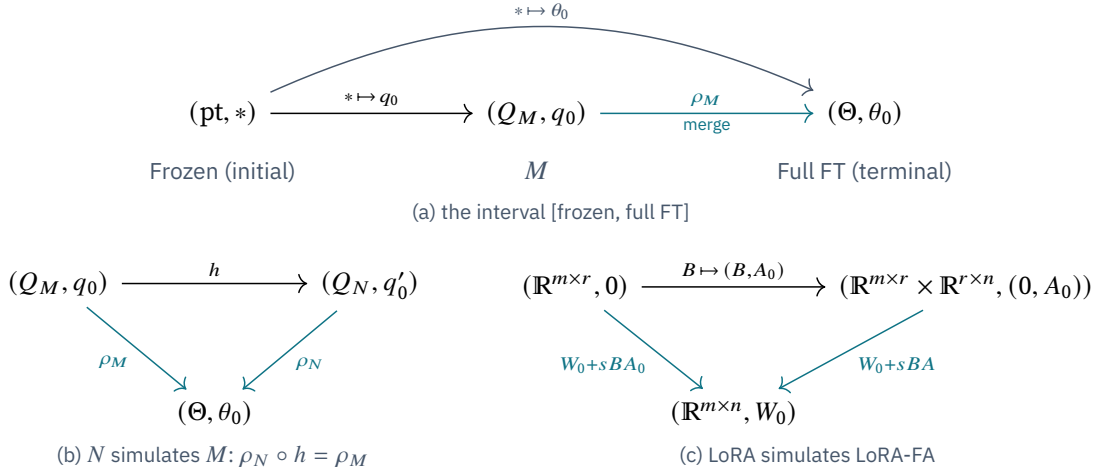
$$\mathbf{Meth}(\Theta, \theta_0) := \mathbf{Diff}_*/(\Theta, \theta_0).$$

- A **morphism**  $h : M \rightarrow N$  is a pointed smooth map with  $\rho_N \circ h = \rho_M$ . Read it as “ $N$  simulates  $M$ ”. Then  $N$  reproduces every weight that  $M$  reaches, together with its parameter path. A *weak* morphism drops the condition  $h(q_0) = q'_0$ .
- The **budget** is  $|M| = \dim Q$ .
- $M$  is **additive** if  $\rho = \theta_0 + \delta$  with  $\delta(q_0) = 0$  and  $\delta$  independent of  $\theta_0$ .
- $M$  is of **action type** if  $\rho(q) = \gamma(q) \cdot \theta_0$  for a map  $\gamma$  into a group acting on  $\Theta$ .

**EXAMPLE** (four objects). On one  $m \times n$  box, LoRA (Hu et al., 2021) is  $\rho(B, A) = W_0 + sBA$  on  $Q = \mathbb{R}^{m \times r} \times \mathbb{R}^{r \times n}$ , pointed at  $(0, A_0)$ . It is additive, with budget  $r(m + n)$ . OFT (Qiu et al., 2023) with the exact Cayley map (Background D.7) is  $\rho(Q) = W_0 C(Q)$  for skew  $Q$  (block-diagonal in OFT), pointed at  $Q = 0$ . It is of action type, with  $\gamma = C$  landing in  $\mathrm{SO}(n)$  on the input side. The frozen model is  $(\mathrm{pt}, *, \theta_0)$ , the inclusion of the single point  $\theta_0$ , and full fine-tuning is  $(\Theta, \theta_0, \mathrm{id})$ , the identity of  $\Theta$ .

**EXAMPLE** (one arrow). Take LoRA-FA (Zhang et al., 2023d),  $B \mapsto W_0 + sBA_0$  with  $A_0$  frozen, pointed at  $B = 0$ . The pointed map  $h(B) = (B, A_0)$  satisfies  $\rho_{\mathrm{LoRA}} \circ h = \rho_{\mathrm{FA}}$ , so LoRA pointed at  $(0, A_0)$  simulates LoRA-FA (Figure 2c).





**Figure 2.** The pointed slice  $\mathbf{Meth}(\Theta, \theta_0)$ . (a) The interval of weight-space methods. For a method  $M$ , the first arrow picks the initialisation  $q_0$ ; it lies over  $\Theta$  because  $M$  starts at  $\theta_0$ . The second arrow is  $\rho_M$  itself, the merge into the base weights, and the composite is the inclusion of  $\theta_0$ . The frozen model is initial and full fine-tuning terminal ([Observation I.4](#)). A pointing defect,  $\rho_M(q_0) \neq \theta_0$ , leaves the triangle commuting only up to that defect ([Exposé V](#)). Adapters and prefixes fall outside the interval because they extend the network ([Definition VI.1](#)). (b) A simulation is a pointed smooth map  $h$  that makes the triangle over  $(\Theta, \theta_0)$  commute; a weak simulation drops  $h(q_0) = q'_0$ . (c) An instance: LoRA pointed at  $(0, A_0)$  simulates LoRA-FA through  $h(B) = (B, A_0)$ .

Random-subspace training,  $\rho(z) = \theta_0 + Pz$  with a fixed random matrix  $P$ , is how [Li et al. \(2018\)](#) measured intrinsic dimension and how [Aghajanyan et al. \(2020\)](#) fine-tuned language models in small subspaces. In the slice it is a linear object, with  $d(M) = |M|$  ([Definition I.5](#)) when  $P$  has full column rank.

### What Para adds

**REMARK I.3** (orientation, and what Para adds). Each  $\rho$  is a 2-cell  $F \Rightarrow F^\rho := (Q, f \circ (\rho \times X))$  out of  $F$ . A simulation  $h: M \rightarrow N$  is a 2-cell  $F^{\rho_N} \Rightarrow F^{\rho_M}$ , since  $\rho_N \circ h = \rho_M$ . Hence  $\mathbf{Meth}(\Theta, \theta_0)$  is the **opposite of the pointed coslice of reparametrisation 2-cells under  $F$**  ([Background A.7](#)). The slice over  $F$  is a different category, and its objects are extensions ([Exposé VI](#)).

**Meth** depends only on  $(\Theta, \theta_0)$  and not on  $f$ . So for weight-space methods, **Para** is bookkeeping. The network enters in exactly three places, and we use **Para** only there:

- extensions and mergeability ([Exposé VI](#); [Definition VI.1](#), [Theorem VI.3](#));
- the realisation  $\Phi_f$ , through functional images and descent ([Observation II.15](#));
- the backpropagation lens ([Exposé IV](#); [Definition IV.1](#), [Proposition IV.2](#)).

Calling **Para** bookkeeping concerns this slice only. In the three places listed we use the framework of [Fong et al. \(2017\)](#) and [Cruttwell et al. \(2021\)](#) directly, including their lens 2-functor in [Exposé IV](#).

This is also why the next exposés work in weight space. Reach in  $\Theta$  is a property of the image, and the cometric  $K = D\rho \wedge D\rho^\top$  that shapes training is built from  $\rho$  and the optimizer's rates  $\Lambda$ . The network enters training through the gradient it hands to the method, and enters expressivity through  $\Phi_f$  once we ask about functions instead of weights. So [Exposés II](#) and [III](#) work in  $\Theta$  and flag the places where  $f$  matters.

## Frozen and full fine-tuning

**OBSERVATION I.4** (universal properties). In  $\mathbf{Meth}(\Theta, \theta_0)$ :

(a) the frozen model  $(\text{pt}, *, \theta_0)$  is initial, and full fine-tuning  $(\Theta, \theta_0, \text{id})$  is terminal. The unique morphism  $M \rightarrow \text{Full FT}$  is  $\rho_M$  itself.

(b) If the fibre product  $Q_M \times_{\Theta} Q_N$  (Background A.11) exists in  $\mathbf{Diff}$ , it is the product  $M \times N$ . It exists, for instance, when the set-level fibre product is an embedded submanifold of  $Q_M \times Q_N$ : a clean intersection (Background C.7), or linear or affine  $\rho$ 's as in Proposition I.8(a). Transversality  $\rho_M \pitchfork \rho_N$  is never the reason when  $|M| + |N| < \dim \Theta$ , since at  $(q_0, q'_0)$  it would force  $|M| + |N| \geq \dim \Theta$ . In  $\mathbf{Set}$  the fibre product always exists and  $\text{Im}(M \times N) = \text{Im } M \cap \text{Im } N$ . Whenever the product exists in  $\mathbf{Meth}$ , its underlying set is this set-level fibre product, so the image identity holds in every case. The product can fail to exist. On  $\Theta = \mathbb{R}$  ( $1 \times 1$  matrices),  $\text{LoRA}_1 \times \text{Frozen}$  would have the cross  $\{BA = 0\} \subset \mathbb{R}^2$  as underlying set. No manifold structure on the cross has the universal property, because both axes would have to lift smoothly through the point over the origin, which forces rank 2 there.

(c)  $\text{Im}$  is a functor (Background A.4) to the poset (Background A.16) of subsets of  $\Theta$  containing  $\theta_0$ . Conversely,  $\text{Im } M \subseteq \text{Im } N$  yields a morphism of the set-level slice  $\mathbf{Set}_*/(\Theta, \theta_0)$ , though not necessarily a smooth one. Take  $\rho_M(t) = t$  and  $\rho_N(u) = u^3$  on  $\mathbb{R}$ , pointed at 0. The images agree, yet no smooth  $h : \mathbb{R} \rightarrow \mathbb{R}$  satisfies  $\rho_N \circ h = \text{id}$ , that is,  $h(t)^3 = t$ .

The proof is in Appendix C.2. The obstruction in (b) covers every pointing of  $\text{LoRA}_1$ , including the standard one  $(0, a_0)$  with  $a_0 \neq 0$ , whose base point is a smooth point of the cross; the obstruction sits at the origin.

**WHAT THIS EXPLAINS** Every weight-space method pointed at  $\theta_0$  lies in the interval [frozen, full FT]. Adapters, prefixes and ReFT (Wu et al., 2024d) are extensions outside  $\mathbf{Meth}$  (Exposé VI). “Merge into the base weights” is the canonical morphism to the terminal object. In exact arithmetic on a full-precision base this is definitional, since every object of  $\mathbf{Meth}$  is mergeable by construction. Type-preserving merges into quantised bases are lossy (Proposition V.2(d)), combining several adapters is a separate question (Proposition V.4), and the remaining zero-overhead questions live on extensions (Exposé VI). What both  $M$  and  $N$  can express,  $\text{Im } M \cap \text{Im } N$ , always exists as a set. The categorical product, when it exists, is the universal method simulated by both, and its image is exactly this set. Proposition I.8(a) computes a genuine one (LoRA-XS). Proposition I.8(b) and Theorem II.7 compute meets, that is, intersections in the poset of subsets of  $\Theta$  containing  $\theta_0$ ; they are not products in  $\mathbf{Meth}$ .

**Defaults that do not start at the checkpoint.** Some defaults move the weights before training starts. In Hugging Face PEFT (Mangrulkar et al., 2022) the default FourierFT (Gao et al., 2024) draws its spectrum from a standard normal, and the default vector bank and logits of VB-LoRA (Li et al., 2024d) are random too. LoRA with `init_lora_weights=False`, LoRA-XS with its small random  $R$ , and every method on a quantised base also start away from  $\theta_0$ . Each lives over its own starting weight or carries a pointing defect, which Exposé V measures.

**Lifting obstructions beyond first order.** The gap between  $\mathbf{Set}$  and  $\mathbf{Diff}$  in (c) is a smooth lifting obstruction. Observation I.6 below detects its first-order instances. In the cube-root example,  $S_1(M) = \mathbb{R} \not\subseteq S_1(N) = 0$ . Higher-order obstructions need finer invariants. The maps  $\rho_M(t) = t^3$  and  $\rho_N(u) = u^5$  have equal images and  $S_1 = 0$  on both sides, yet the only lift,  $t^{3/5}$ , is not  $C^1$ .

### Invariants of a method

The budget  $|M|$  counts parameters. Two other numbers say more. One is the dimension of what a method reaches; the other is the dimension of what it reaches to first order at the start, which is what the first gradient steps use. For LoRA with  $r = 16$  on a  $4096 \times 4096$  matrix the budget is 131,072, the image has dimension  $r(m + n - r) = 130,816$ , and from zero initialisation the first steps use only  $mr = 65,536$  directions ([Theorem II.2](#), [Proposition II.12](#)).

**DEFINITION I.5** (invariants of an object). For  $M = (Q, q_0, \rho)$ :

- the image  $\text{Im } M = \rho(Q)$ ;
- the **expressive dimension**  $d(M) = \dim \text{Im } M = \max_q \text{rank } d\rho_q$ ;
- the **gauge waste**  $g(M) = |M| - d(M)$ ;
- the **first-order image**  $S_1(M) = d\rho_{q_0}(T_{q_0}Q)$ , and the **first-order deficit**, which is  $d(M) - \dim S_1(M)$ ;
- the **gauge group**  $\{\varphi \in \text{Diff}(Q) : \rho \circ \varphi = \rho\}$ , which is unpointed, since a gauge transformation moves  $q_0$  inside its fibre. In practice we exhibit a Lie group ([Background D.1](#)) acting on  $Q$  that preserves  $\rho$  and is locally transitive on generic fibres, and call it the **generic gauge**.

The base point  $\theta_0$  is **regular** for  $M$  if the deficit is 0, and an **apex** if the deficit is positive. We say **vertex** for the subset-level notion, the cone point of an image germ ([Background C.10](#)), and keep **apex** for the base-point notion. The two differ in general. At a crossing of two smooth branches the image is singular, yet the deficit can be 0. In every case below with a positive deficit, the image germ is (diffeomorphic to) a cone with vertex  $\theta_0$ ; the exception is DoRA ([Liu et al., 2024b](#)), which inherits LoRA’s deficit through its torus factor. A cone germ that is a  $C^1$  manifold at its vertex equals its tangent space ([Background C.2](#)) there, so it is linear. Hence the vertex of a non-linear cone is a singular point.

**REMARK** (the word “gauge”). “Gauge” here means the symmetry group of a parametrisation, as in [Definition I.5](#). There is no connection, and away from the full-rank stratum the fibres jump. The word borrows, as an analogy, the physicist’s picture of redundant coordinates that leave every observation unchanged. Only the group and its action enter the theorems.

If  $N$  simulates  $M$ , whatever  $M$  can do at its first step,  $N$  can do too.

**OBSERVATION I.6** ( $S_1$  is functorial). *If  $M \rightarrow N$  is a (pointed) morphism, then  $S_1(M) \subseteq S_1(N)$ . In particular, isomorphic methods have equal  $S_1$  and equal first-order deficit. Monotonicity holds only along pointed morphisms. Along a weak morphism, such as a gauge map moving  $q_0$  inside its fibre,  $S_1$  can change.*

The proof is in [Appendix C.4](#).

**WHAT THIS EXPLAINS**  $S_1$  is the set of weight-space directions available to the linearised (lazy, NTK) dynamics at initialisation, since the lazy-regime kernel  $J_f d\rho \wedge d\rho^\top J_f^\top$  has range  $J_f(S_1)$ . That kernel also depends on  $\Lambda$  and on the network Jacobian  $J_f$ , not only on  $S_1$ . For random- $A_0$  zero-init LoRA it approximates the full fine-tuning kernel ([Malladi et al., 2022](#)), so a large first-order deficit need not cost much on the data.  $S_1$  is an isomorphism invariant that separates initialisations the image cannot, namely zero-init  $T_A$  against  $T_B$ , and the

choice of row  $A_0$  ([Theorem II.2](#)). The image already separates split initialisations, since it determines  $P$ .

The kernel view of language-model fine-tuning, and the closeness of LoRA's kernel to full fine-tuning's, are due to [Malladi et al. \(2022\)](#). [Hayou et al. \(2024a\)](#) compared the zero initialisations  $B_0 = 0$  and  $A_0 = 0$  as training procedures; in the slice they are already different objects.

### Sums

Weight space is a vector space, so running two methods side by side and adding their updates gives a method again.

**OBSERVATION I.7** (additive monoidal structure).  $\Theta$  is a vector space, so every object of  $\mathbf{Meth}(\Theta, \theta_0)$  can be written  $\rho = \theta_0 + \delta$  with  $\delta := \rho - \theta_0$  and  $\delta(q_0) = 0$ . Put

$$M \oplus N := (Q_M \times Q_N, (q_0, q'_0), \theta_0 + \delta_M + \delta_N).$$

This is a symmetric monoidal structure ([Background B.1](#)) with the frozen model as unit. Its image is the Minkowski sum  $\theta_0 + (\text{Im } \delta_M + \text{Im } \delta_N)$  ([Background B.14](#)). It differs from the categorical product, whose image, when the product exists, is the intersection  $\text{Im } M \cap \text{Im } N$  ([Observation I.4\(b\)](#)). “Additive” in [Definition I.2](#) ( $\delta$  independent of  $\theta_0$ ) is a condition on families  $\theta_0 \mapsto M(\theta_0)$ ; it matters for base change ([Exposé V](#)) and plays no role here.

Write  $\text{LoRA}_k$  for LoRA with frozen residual  $\theta_0$ ; for split pointings the sum below has residual  $\theta_0 - P - P'$ . Concatenation  $h(B_1, A_1, B_2, A_2) = ([B_1 \ B_2], [A_1; A_2])$  is an isomorphism  $\text{LoRA}_r \oplus \text{LoRA}_s \cong \text{LoRA}_{r+s}$  of unpointed objects. It is a pointed isomorphism onto  $\text{LoRA}_{r+s}$  pointed at  $([B_0 \ B'_0], [A_0; A'_0])$ . For  $T_A$  pointings ( $B_0 = 0$ ,  $\text{rank } A_0 = r$ ), the target pointing has  $\text{rank}[A_0; A'_0] = r + s$  if and only if  $\text{row } A_0 \cap \text{row } A'_0 = 0$ ; it lies in the stratum  $T_A$  of [Theorem II.2](#) when moreover  $r + s < \min(m, n)$ .

Standard pointings ( $T_A$ , or symmetrically  $T_B$ ) exist only for  $r < \min(m, n)$  ([Theorem II.2](#)), so the monoidal question is pointwise. If  $F(r)$  and  $F(2r)$  are standard ( $2r < \min(m, n)$ ), then  $F(r) \oplus F(r) \not\cong F(2r)$  as pointed objects; for  $2r \geq \min(m, n)$  no standard  $F(2r)$  exists. Hence no assignment  $r \mapsto \text{LoRA}_r$  with  $F(1)$  and  $F(2)$  standard is **strong monoidal** ([Background B.15](#)) into pointed objects. For  $r \neq s$  and standard pointings,  $F(r) \oplus F(s) \cong F(r + s)$  as pointed objects exactly when the row space of the target's  $A_0$  is the direct sum of the two source row spaces. The assignment is strong monoidal on unpointed objects, up to weak morphisms, and on pointed objects with degenerate pointings, namely the origin  $(B_0, A_0) = (0, 0)$ , where  $S_1 = 0$ , or the stacked pointings  $F(r) = F(1)^{\oplus r}$ . Concatenation is strictly associative and unital, so discrete  $(\mathbb{N}, +)$  suffices for a plain strong monoidal functor. Symmetric strong monoidality needs the permutation groupoid, since swapping and then concatenating permutes the blocks.

With per-adapter scales  $s_r$  ( $\alpha/r$  for LoRA,  $\alpha/\sqrt{r}$  for rsLoRA) and any target scale  $s_*$ , the map  $h = ([B_1 \ B_2], [c_1 A_1; c_2 A_2])$  with  $c_i = s_{r_i}/s_*$ , where  $r_1 = r$  and  $r_2 = s$ , is an isomorphism over  $\Theta$ . It is a linear bijection and preserves row spaces, so the pointing criterion is unchanged.

The proof is in [Appendix C.5](#). The negative claim rests on  $\dim S_1$ , which any pointed isomorphism preserves ([Observation I.6](#)). It is  $mr$  for  $F(r) \oplus F(r)$  and  $2mr$  for a standard  $F(2r)$ . [Num] At  $m = 7$ ,  $\dim S_1 = 28$  for a standard  $\text{LoRA}_4$ , against 14 for  $\text{LoRA}_2 \oplus \text{LoRA}_2$  with the same  $A_0$  and 28 with independent  $A_0$ 's;  $F(1) \oplus F(2)$  and a standard  $F(3)$  both give 21 ([Appendix E](#)).

**The cat merge of Hugging Face PEFT.** Hugging Face PEFT's `add_weighted_adapter(..., combination_type="cat")` (Mangrulkar et al., 2022) evaluates the rescaled concatenation  $h$  at trained points. It takes  $s_* = 1$ . It sets the merged `lora_alpha` to the new rank  $r + s$ , multiplies each  $A$  block by  $w_i s_{r_i}$ , so that  $c_i = w_i s_{r_i}$ , and leaves the  $B$  blocks alone. It is the isomorphism  $h$  when all weights  $w_i = 1$  and the merged adapter has `use_rslora=False`. In PEFT releases up to v0.21.1 the merged adapter inherits `use_rslora` from the first source. For rsLoRA sources (Kalajdziewski, 2023) its scale is then  $(r + s)/\sqrt{r + s} = \sqrt{r + s}$ , and the merged update is  $\sqrt{r + s}$  times too large. PR #3449 on the main branch fixes this by setting `use_rslora=False` on the merged adapter.

**WHAT THIS EXPLAINS** RoSA (Nikdan et al., 2024) ( $\text{LoRA} \oplus$  a fixed-mask sparse method) and GraLoRA (Jung et al., 2025) ( $\oplus$  of  $k^2$  block-embedded LoRAs) are  $\oplus$ -words, and their expressivity is a Minkowski sum. ReLoRA (Lialin et al., 2023b) is  $\text{LoRA}^{\oplus K}$  unrolled in time, apart from its full-rank warm start. A sum of  $k$  rank- $r$  LoRAs, each with residual  $\theta_0$ , is isomorphic in **Meth** to one LoRA of rank  $kr$ , pointed at the concatenated initialisation. Under  $\alpha/r$  scaling the isomorphism rescales the  $A$  blocks, so training at equal  $\alpha$  and learning rate follows a different trajectory. The cat merge is the isomorphism  $h$  evaluated at trained points; the merged adapter, if unfrozen, is  $\text{LoRA}_{r+s}$ .

**What is and is not a sum.** LoRA-soup CAT (Prabhakar et al., 2024) freezes the trained modules and learns layer-wise weights, so it is a  $\oplus$  of  $k$  frozen line methods  $\alpha_i \mapsto \alpha_i B_i A_i$ . Its per-layer image is a span of dimension at most  $k$  (generically exactly  $k$ ), much smaller than the image of  $\text{LoRA}_{kr}$ . UniPELT (Mao et al., 2021) and MAM (He et al., 2021) are **not**  $\oplus$ -words in **Meth**( $\Theta, \theta_0$ ). Their adapter and prefix parts live on extensions (Exposé VI), and UniPELT's gates depend on the input. The obstruction in Observation I.7 is the reuse of one pointing. With a fixed pointing, stacking the same  $A_0$  twice gives a rank-deficient frame, which  $S_1$  distinguishes from a standard  $\text{LoRA}_{2r}$ .

## Products

Freeze  $A$  at  $A_0$  and train  $B$ , or freeze  $B$  at  $B_0$  and train  $A$ . The product of these two one-sided methods is LoRA-XS. As a pullback square over  $\Theta$  it reads

$$\begin{array}{ccc} \mathbb{R}^{r \times r} & \xrightarrow{R \mapsto B_0 R} & Q_{\text{FA}} = \mathbb{R}^{m \times r} \\ \downarrow R \mapsto R A_0 & \lrcorner & \downarrow B \mapsto W_0 + B A_0 \\ Q_{\text{FB}} = \mathbb{R}^{r \times n} & \xrightarrow{A \mapsto W_0 + B_0 A} & \Theta = \mathbb{R}^{m \times n} \end{array}$$

and both composites send  $R$  to  $W_0 + B_0 R A_0$ .

**PROPOSITION I.8** (a product that produces a named method; an image-level meet). (a) Let  $A_0 \in \mathbb{R}^{r \times n}$  have full row rank and  $B_0 \in \mathbb{R}^{m \times r}$  full column rank. Define the one-sided methods

$$\text{FA}(A_0) : B \mapsto W_0 + B A_0, \quad \text{FB}(B_0) : A \mapsto W_0 + B_0 A,$$

both pointed at 0. Then

$$\text{FA}(A_0) \times \text{FB}(B_0) \cong (\mathbb{R}^{r \times r}, 0, R \mapsto W_0 + B_0 R A_0).$$

The product supplies this sandwich for any frames. Assume  $\text{rank } W_0 \geq r$  ( $\sigma_r > 0$ ) and take the SVD frames  $B_0 = U_r$ ,  $A_0 = \Sigma_r V_r^\top$ . The result is **LoRA-XS pointed at**  $R = 0$ .



(b) Let  $W_0$  have full row rank (so  $m \leq n$ ), and let  $\text{LoRA}_r$  carry its standard pointing ( $B_0 = 0$ , image  $W_0 + M_{\leq r}$ ). The image-level meet of  $\text{LoRA}_r$  with the output-scaling method  $\ell \mapsto \text{diag}(\ell)W_0$  is  $\{\text{diag}(\ell)W_0 : \#\{i : \ell_i \neq 1\} \leq r\}$ , that is, “rescale at most  $r$  output neurons”.

The LoRA-XS paper (Bałazy et al., 2024) writes  $xW + xARB$  in row-vector convention with  $A = U_r \Sigma_r$  and  $B = V_r^\top$ . Transposed, the update is  $B_0 R^\top A_0$ , so the isomorphism is  $R \mapsto \alpha R^\top$ . LoRA-XS initialises  $R \sim \mathcal{N}(0, \sigma^2)$  with  $\sigma = 10^{-5}$ , a negligible pointing defect. The output-scaling method in (b) is the key/value part of (IA)<sup>3</sup> (Liu et al., 2022a). On the FFN, (IA)<sup>3</sup> scales the *input* of  $W_{\text{down}}$ ,  $W \mapsto W \text{diag}(\ell)$ ; the mirror statement then needs full column rank, or holds generically for  $r < d$ , where  $d$  is the number of rows of  $W_{\text{down}}$ .

The proof is in Appendix C.6. In (a) the fibre product is the  $r^2$ -dimensional linear subspace  $\{(B_0 R, R A_0)\}$ , and if  $\text{rank } W_0 < r$  the frame  $\Sigma_r V_r^\top$  is rank-deficient and the LoRA-XS instance breaks. [Num] The fibre product of (a) has dimension  $4 = r^2$  at  $(m, n, r) = (6, 8, 2)$  (Appendix E).

**WHAT THIS EXPLAINS** LoRA-XS is the largest method that factors through *both* frozen frames. In the frames of an initialisation it is the overlap that a split initialisation double-counts (Proposition II.3); it is LoRA-XS proper when those frames span the top- $r$  singular subspaces of  $W_0$ , as at a PiSSA initialisation (Meng et al., 2024a). The meet in (b) is only set-level. For  $n \geq 2$  the product does not exist in **Diff**, because its fibre product is singular at the origin  $(b, a, \ell) = (0, 0, \mathbf{1})$ , by the rank argument of Observation I.4(b) (Appendix C.6). Once a support is fixed, the meet is just (IA)<sup>3</sup> masked to  $r$  neurons; compare sparse activation scaling (Stoechr et al., 2024).

**The shape of the meet in (b).** The meet is the union of the  $\binom{m}{r}$  coordinate  $r$ -planes “rescale at most  $r$  output neurons”. For  $r = 1$  the fibre product is the union of the linear spaces  $\{b = 0, \ell = \mathbf{1}\}$  and  $\{a = 0, \ell = \mathbf{1}\}$  with  $m$  two-dimensional sheets  $\{b = c e_i, a = \lambda w_i, \ell_i = 1 + s c \lambda\}$ , where  $w_i$  is row  $i$  of  $W_0$ . Under the standard pointing ( $b = 0, a_0$  generic) the base point is a smooth point of  $\{b = 0, \ell = \mathbf{1}\}$ , so the singularity sits away from it, at the origin.

### The dictionary so far

Table 3 collects the translations made in this exposé. The full dictionary is the Rosetta stone of Exposé IX.



**Table 3.** The dictionary for Exposé I.

| In PEFT                             | In the pointed slice                                                                                 | Statement                                                              |
|-------------------------------------|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| pretrained checkpoint               | base point $\theta_0$ of a pointed 1-cell $(\Theta, \theta_0, f)$                                    | <a href="#">Definition I.1</a>                                         |
| a weight-space PEFT method          | an object $\rho : (Q, q_0) \rightarrow (\Theta, \theta_0)$ of <b>Meth</b> $(\Theta, \theta_0)$       | <a href="#">Definition I.2</a>                                         |
| “ $N$ can do whatever $M$ can”      | a morphism $h : M \rightarrow N$ , $\rho_N \circ h = \rho_M$ ; at the level of sets, image inclusion | <a href="#">Definition I.2</a> ,<br><a href="#">Observation I.4(c)</a> |
| frozen model, full fine-tuning      | initial and terminal objects                                                                         | <a href="#">Observation I.4(a)</a>                                     |
| merging into full-precision weights | the canonical morphism $\rho_M$ to the terminal object                                               | <a href="#">Observation I.4(a)</a>                                     |
| effective number of parameters      | expressive dimension $d(M)$ ; the rest is gauge waste                                                | <a href="#">Definition I.5</a>                                         |
| what the first steps can use        | first-order image $S_1$ , a functorial invariant                                                     | <a href="#">Observation I.6</a>                                        |
| RoSA, GraLoRA, ReLoRA stages        | $\oplus$ -words, with a Minkowski sum as image                                                       | <a href="#">Observation I.7</a>                                        |
| HF cat merge                        | the concatenation isomorphism evaluated at trained points                                            | <a href="#">Observation I.7</a>                                        |
| LoRA-XS                             | the product $\text{FA}(A_0) \times \text{FB}(B_0)$                                                   | <a href="#">Proposition I.8</a>                                        |
| adapters, prefixes, ReFT            | extensions of the network, outside <b>Meth</b>                                                       | <a href="#">Definition VI.1</a>                                        |

## The image germ: expressivity

Expressivity is read off the germ of a method’s image at the pretrained point. An initialisation decides whether that point sits at the apex of a cone, on a smooth sheet of one, or just off the image.

Exposé I turned a PEFT method into a pointed map  $\rho : (Q, q_0) \rightarrow (\Theta, \theta_0)$ . Its image  $\text{Im } M = \rho(Q)$  is everything it can reach. Fine-tuning travels only a short way from the pretrained weights, so what matters is the image near  $\theta_0$ , its *germ* at  $\theta_0$  (Background C.8).

The germ takes few shapes (Figure 1). It is a linear space when  $\rho$  is affine, as for BitFit (Ben-Zaken et al., 2021), LoRA-FA (Zhang et al., 2023d), LoRA-XS (Bałazy et al., 2024), FourierFT (Gao et al., 2024) and (IA)<sup>3</sup> (Liu et al., 2022a). It is a cone (Background C.10) when  $\rho$  is multihomogeneous and non-linear and starts where the update vanishes, as LoRA’s *BA* (Hu et al., 2021) does at zero initialisation. It is a piece of a group orbit for exactly orthogonal methods such as exact-Cayley OFT (Qiu et al., 2023). DoRA (Liu et al., 2024b) adds an output gain to LoRA’s cone (Proposition II.8), and HRA (Yuan et al., 2024) meets that cone with a rotation orbit (Theorem II.7), each under its stated hypotheses. Adapters and prefixes change the network itself, and their germs live among functions (Exposé VI).

Four invariants of the germ carry most of the story, and isomorphic methods share them (Observation I.6).

- The **first-order image**  $S_1(M) = d\rho_{q_0}(T_{q_0}Q)$ , the directions a first step can take.
- The **type of the base point**: *regular* when the first-order deficit  $d(M) - \dim S_1(M)$  vanishes, an *apex* when it is positive (Definition I.5).
- The **maximum rank** of an update, bounded by a minimum cut (Theorem II.10).
- The **expressive dimension**  $d(M) = \dim \text{Im } M$ , which falls short of the parameter count  $|M|$  by the gauge waste  $g(M) = |M| - d(M)$ .

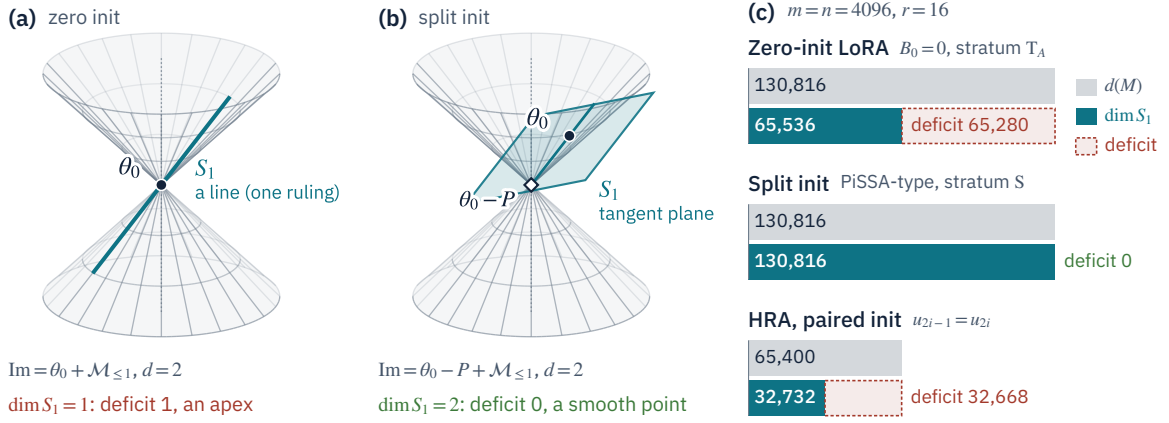
Subsection II.A asks where a method starts, Subsection II.B what a group-type method can never change (Theorem II.5, Corollary II.6), and Subsection II.C how rank differs from dimension (Proposition II.12). Expressivity here means reach, and finding a reachable point is left to Exposé III.

Which functions a LoRA-adapted network can represent is studied by Zeng and Lee (2023). The intrinsic dimension of Li et al. (2018) and Aghajanyan et al. (2020) depends on the data, while  $d(M)$  is a property of the method alone.

### II.A Initialisations: pointings and strata

Every LoRA recipe decides where training starts. LoRA (Hu et al., 2021) sets  $B_0 = 0$  and draws  $A_0$  at random. PiSSA (Meng et al., 2024a) takes both factors from the top of the SVD of  $W_0$  and subtracts their product from the frozen weight, LoRA-GA (Wang et al., 2024e) does the same with the first gradient, and CorDA (Yang et al., 2024c), OLoRA (Büyükyüz, 2024), MiLoRA (Wang et al., 2024b) and EVA (Paischer et al., 2024) choose otherwise. Each recipe is a *pointing*, a parameter  $q_0$  with the frozen residual that makes  $\rho(q_0) = \theta_0$ , and two pointings of one formula can be different objects of  $\mathbf{Meth}(\Theta, \theta_0)$ . Figure 3 draws the typical cases.

The answers occupy the rest of this subsection. For  $r < \min(m, n)$ , full-rank LoRA initialisations come in exactly three kinds up to reparametrisation, classified as sets by row  $A_0$



**Figure 3. The apex and the smooth point.** Panels (a) and (b) draw the symmetric slice  $S = \begin{pmatrix} x+y & z \\ z & x-y \end{pmatrix}$  of  $2 \times 2$  matrices, in which  $\text{rank} \leq 1$  is the grey double cone  $x^2 = y^2 + z^2$ , and the model  $\rho(c, \phi) = W_{\text{res}} + c v_\phi v_\phi^\top$ ,  $v_\phi = (\cos \phi, \sin \phi)$ , which plays LoRA ( $c$  for  $B$ ,  $v_\phi$  for  $A$ ); its image is the whole cone, so  $d = 2$ . (a) Zero initialisation,  $c_0 = 0$  and  $W_{\text{res}} = \theta_0$ : the base point  $\theta_0$  is the vertex of the cone, and the first-order image  $S_1$  (teal) is a single line, one ruling, so the deficit is 1 and  $\theta_0$  is an apex (Definition I.5, Theorem II.2(c)). (b) A split (PiSSA-type) initialisation,  $c_0 \neq 0$  and  $W_{\text{res}} = \theta_0 - P$  with  $P = c_0 v_{\phi_0} v_{\phi_0}^\top$ : the vertex moves to  $\theta_0 - P$  (open diamond),  $\theta_0$  lies on a smooth sheet of the cone, and  $S_1$  is the tangent plane there, so the deficit is 0. Both values of  $\dim S_1$  are Jacobian ranks of the model. (c) For  $m = n = 4096$  and  $r = 16$ , each pair of bars compares the expressive dimension  $d(M)$  (grey) with  $\dim S_1$  (teal); the dashed red box is the gap between them, the first-order deficit. Zero-init LoRA, stratum  $T_A$ , has  $\dim S_1 = mr = 65,536$  against  $d = r(m + n - r) = 130,816$ ; a split initialisation has  $\dim S_1 = d$  (Theorem II.2(b)); HRA's paired initialisation  $u_{2i-1} = u_{2i}$ ,  $k = r/2$ , has  $\dim S_1 = kn - \frac{k(k+1)}{2} = 32,732$  against  $d = rn - \frac{r(r+1)}{2} = 65,400$ , so it too starts at an apex (Theorem II.7(b)). The script checks each closed form against a Jacobian rank at  $(m, n, r) = (11, 9, 4)$ .

(when  $B_0 = 0$ ), by col  $B_0$  (when  $A_0 = 0$ ) or by the subtracted product  $B_0 A_0$  (PiSSA-type splits), as Theorem II.2 shows. Zero-factor starts sit at the apex of a cone with  $mr$  or  $nr$  first-order directions, and split starts at a smooth point with all  $r(m + n - r)$ . For a  $4096 \times 4096$  matrix at rank 16 that is 130,816 against 65,536 (Proposition II.3). Neither kind reaches everything the other reaches. No exact start of LoRA<sub>r</sub> reaches an update of rank above  $2r$  (Theorem II.1), and a factor behind a closed gate, such as  $A$  when  $B_0 = 0$ , does not move the weights at first order (Lemma II.4).

### Translates of one shape

An additive method adds updates from a fixed shape  $C$ , for LoRA the cone  $\mathcal{M}_{\leq r}$  of matrices of rank at most  $r$  (Background C.12). Pointing it at  $q$  slides the shape so that  $\delta(q)$  lands on  $\theta_0$ , and reachable updates become differences of two points of  $C$  (Background B.14).

**THEOREM II.1** (pointings of an additive method). *Let  $M = (Q, q_0, \theta_0 + \delta)$  be additive with shape  $C = \delta(Q)$ . Write  $-v + C$  for a translate, and  $C - C := \{c - c' : c, c' \in C\}$  for the Minkowski difference.*

(a) *For each  $q \in Q$ , the re-pointed method  $M_q := (Q, q, \theta_0 - \delta(q) + \delta(\cdot))$ , with frozen residual  $\theta_0 - \delta(q)$ , is pointed at  $\theta_0$ , and  $\text{Im } M_q = \theta_0 + (-\delta(q) + C)$ .*

(b)  $\bigcup_q \text{Im } M_q = \theta_0 + (C - C)$ .

(c) *For LoRA<sub>r</sub> on a single linear box with  $s \neq 0$ ,  $C - C = \mathcal{M}_{\leq \min(2r, m, n)}$ ; for several boxes,  $C - C$  is the product of the per-box varieties  $\mathcal{M}_{\leq \min(2r, m_i, n_i)}$ . Hence **no exact pointing of LoRA<sub>r</sub>***

(frozen residual  $\theta_0 - \delta(q)$ ), whether zero-init or split (PiSSA, OLoRA, CorDA, LoRA-GA), makes an update  $\theta - \theta_0$  of rank  $> 2r$  reachable in any adapted matrix. More robustly, from any starting point  $q_0$  of  $\text{LoRA}_r$ , pointed or not,  $\rho(q) - \rho(q_0) = \delta(q) - \delta(q_0)$  has rank  $\leq \min(2r, m, n)$ . Quantised-residual schemes (QLoRA, LoftQ, QPiSSA) are pointed only up to a defect that depends on the order of operations ( $e$  for QLoRA; Proposition V.2(b) for the others). Measured from  $\theta_0$  rather than from their starting weight, their updates contain this defect, whose rank is generically at least  $\min(m, n) - r$ . The bound  $2r$  is attained by split pointings when  $2r \leq \min(m, n)$ .

The theorem is elementary, and its rank- $2r$  consequence is known in practice. Meng et al. (2024a) convert a trained PiSSA adapter into a rank- $2r$  LoRA through

$$BA - B_0A_0 = \begin{bmatrix} B & B_0 \end{bmatrix} \begin{bmatrix} A \\ -A_0 \end{bmatrix}, \quad (1)$$

and HF PEFT ships the conversion. The theorem adds that the conversion loses nothing and that rank  $2r$  is needed in general.

**WHAT THIS EXPLAINS** An initialisation of an additive method chooses which *translate* of the shape passes through  $\theta_0$ . Splitting initialisations (PiSSA-type) buy “spectral replacement” updates  $Y - P$ , which can have rank  $2r$ . They pay by losing generic updates of every rank  $\geq 1$  (Theorem II.2(d)). Moving a weight matrix by more than rank  $2r$  needs rebasing, as in ReLoRA (Lialin et al., 2023b) (Theorem V.3), or another shape.

### Three strata

Isomorphisms of methods are pointed diffeomorphisms over  $\Theta$  (Definition I.2), such as the gauge  $(B, A) \mapsto (Bg^{-1}, gA)$  with  $g \in \text{GL}_r$  (Background D.21). They preserve the image and  $S_1$  (Observation I.6) but may change training (Exposé III). Up to isomorphism, a full-rank LoRA initialisation is determined by one subspace or one matrix.

**THEOREM II.2** (three strata of LoRA initialisations). Let  $\delta(B, A) = BA$  with  $1 \leq r < \min(m, n)$ . Consider pointed LoRA objects in three distinguished strata of full-rank pointings (pointings whose nonzero factors have full rank). Only  $S$  is open and dense in the space of pointings;  $T_A$  and  $T_B$  are nowhere dense.

- $(T_A)$ :  $B_0 = 0$  and  $\text{rank } A_0 = r$ ;
- $(T_B)$ :  $A_0 = 0$  and  $\text{rank } B_0 = r$ ;
- $(S)$ :  $\text{rank } B_0A_0 = r$ , with frozen residual  $\theta_0 - B_0A_0$ .

(a) **Classification.** Within a stratum,  $M_{q_0} \cong M_{q'_0}$  if and only if, respectively, row  $A_0 = \text{row } A'_0$ ; col  $B_0 = \text{col } B'_0$ ; or  $B_0A_0 = B'_0A'_0$ . Objects in different strata are never isomorphic. As a set, the isomorphism classes of the three strata therefore form

$$\text{Gr}(r, \mathbb{R}^n) \sqcup \text{Gr}(r, \mathbb{R}^m) \sqcup \mathcal{M}_r(m \times n). \quad (2)$$

(b) **First order.**  $\dim S_1$  is  $mr$ ,  $nr$  and  $r(m + n - r)$  respectively, while  $d = r(m + n - r)$  in every case. The deficits are  $r(n - r)$ ,  $r(m - r)$  and 0.

(c) **Geometry.** In  $T_A$  and  $T_B$ ,  $\text{Im} = \theta_0 + \mathcal{M}_{\leq r}$ , whose vertex  $\theta_0$  is an **apex**. In  $S$ ,  $\text{Im} = (\theta_0 - P) + \mathcal{M}_{\leq r}$  with  $P = B_0A_0$ , and  $\theta_0$  is a **smooth point**.

(d) **Incomparability.** The images of  $S$ -type and  $T$ -type objects are incomparable.

(e) **Scope.** The three strata do not exhaust the pointings of  $\text{LoRA}_r$ . Outside them lie:  $B_0 = 0$  with  $\text{rank } A_0 < r$ ;  $A_0 = 0$  with  $\text{rank } B_0 < r$ ;  $B_0 = A_0 = 0$ ;  $0 < \text{rank } B_0A_0 < r$ ; and pointings

with  $B_0, A_0$  both nonzero and  $B_0 A_0 = 0$ . HF PEFT ships one of the last kind: `init_lora_weights="orthogonal"` (which requires  $r$  even) lies in the stratum  $(\text{rk } B_0, \text{rk } A_0, \text{rk } B_0 A_0) = (r/2, r/2, 0)$ , an apex with  $\dim S_1 = \frac{r}{2}(m+n-\frac{r}{2})$ . It is isomorphic to no object of the three strata. Its image  $\theta_0 + \mathcal{M}_{\leq r}$  rules out  $S$ , and its  $S_1$  determines  $(\text{col } B_0, \text{row } A_0)$ , of dimensions  $(r/2, r/2)$ , against  $(0, r)$  in  $T_A$  and  $(r, 0)$  in  $T_B$ . The dimension of  $S_1$  alone does not separate it, since at  $(m, n, r) = (4, 5, 2)$  it equals  $mr$ . This placement holds in exact arithmetic; HF casts the two factors to the base dtype separately, and in bf16 the result is a generic pointing with a tiny defect. For every pointing, the image determines  $P = B_0 A_0$  (its vertex is  $\theta_0 - P$ ), and  $S_1 = \{X A_0 + B_0 Y\}$  determines  $(\text{row } A_0, \text{col } B_0)$  whenever these are proper subspaces. The classification is set-level: in the space of pointings  $T_A$  lies in the closure of  $S$ , since  $(\varepsilon B', A_0) \rightarrow (0, A_0)$ , so the quotient is not a topological disjoint union.

Numerically,  $\dim S_1 = (14, 10, 20)$  for  $(T_A, T_B, S)$  at  $(m, n, r) = (7, 5, 2)$ , and  $\dim S_1 = 11$  for HF's orthogonal init (Appendix E). Here  $\text{Gr}$  is the Grassmannian (Background C.15),  $\mathcal{M}_r$  the smooth stratum of  $\mathcal{M}_{\leq r}$  (Background C.12), and smooth and singular points are those of Background C.9.

**REMARK** (a moduli space of initialisations, as an analogy). Calling the three spaces in (2) a moduli space (Background C.20) borrows a word. The bijection is of sets only.  $T_A$  lies in the closure of  $S$ , since  $(\varepsilon B', A_0) \rightarrow (0, A_0)$ , so the quotient is not a topological disjoint union, and no universal family is built.

Most LoRA initialisation schemes in the literature are points of these three strata (Table 4). LoRA-FA and MiCA freeze one factor and are sub-objects (Background A.15), and the last three rows of the table lie outside the strata.

**WHAT THIS EXPLAINS** Zero-init LoRA starts at a *singular point*. At first order the  $A$ -direction is dead, so LoRA and LoRA-FA (which freezes  $A$  and trains  $B$ ; Zhang et al. (2023d) use AdamW), started from the same  $A_0$ , have the same  $K_{q_0}$  under plain gradient steps and the same first-order trajectory under Adam (Lemma II.4). Splitting initialisations start at a smooth point with full first-order reach. We conjecture that this is one mechanism behind the faster early progress reported for PiSSA and LoRA-GA. LoRA-GA attributes its speed-up to gradient alignment, and MiLoRA also starts at a smooth point, so the smooth start alone does not account for the reported gains. The strata classify *expressivity together with  $S_1$* . What an optimizer additionally sees is the subject of Corollary III.13.

Hayou et al. (2024a) compare the two zero-factor starts, “Init[A]” and “Init[B]”, through their training dynamics; here  $S_1$  already tells them apart. The facts about  $\mathcal{M}_{\leq r}$  used in the proof are classical (Background C.12).

**REMARK** (the first-order deficit and the data).  $S_1$  counts directions in weight space. The lazy-regime kernel  $J_f d\rho \wedge d\rho^\top J_f^\top$  also involves the network Jacobian  $J_f$  and the optimizer's metric  $\Lambda$ , and for random- $A_0$  zero-init LoRA it approximates the full fine-tuning kernel (Malladi et al., 2022). A large first-order deficit need not cost much on the data.

### Counting first-order directions

A first step of LoRA from  $(B, A)$  adds  $s(\dot{B}A + B\dot{A})$ , a matrix with rows in row  $A$  plus one with columns in col  $B$ . Counting both families and their overlap gives  $\dim S_1$ .

**Table 4.** LoRA initialisation schemes as points of the strata of [Theorem II.2](#), with the sub-objects and the exceptions. Conventions:  $y = Wx$ ,  $W \in \mathbb{R}^{m \times n}$ ; HF is Hugging Face PEFT.

| scheme                                                                                       | stratum                  | point                                                                                                                                                                                                                                                                                         |
|----------------------------------------------------------------------------------------------|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| vanilla LoRA ( <a href="#">Hu et al., 2021</a> ) (Linear, Conv)                              | $T_A$                    | a random point of $\text{Gr}(r, n)$ : Haar ( <a href="#">Background D.14</a> ) under Gaussian $A_0$ , only approximately Haar under Kaiming-uniform                                                                                                                                           |
| vanilla LoRA on nn.Embedding (loralib and HF default)                                        | $T_B$                    | $A_0 = 0$ , $B_0 \sim \mathcal{N}(0, 1)$ in the convention $y = Wx$ with $m$ the embedding dimension; deficit $r(m - r)$                                                                                                                                                                      |
| EVA ( <a href="#">Paischer et al., 2024</a> )                                                | $T_A$                    | the top- $r$ principal subspace of the input activations, at its per-layer rank                                                                                                                                                                                                               |
| LoRA-FA ( <a href="#">Zhang et al., 2023d</a> )                                              | sub-object of $T_A$      | the same point with $A$ frozen (an arrow $\text{FA}(A_0) \rightarrow \text{LoRA}_{(0, A_0)}$ )                                                                                                                                                                                                |
| “Init[B]” ( <a href="#">Hayou et al., 2024a</a> )                                            | $T_B$                    | a random point of $\text{Gr}(r, m)$                                                                                                                                                                                                                                                           |
| MiCA ( <a href="#">Rüdiger and Raschka, 2026</a> )                                           | sub-object of $T_B$      | $B_0$ the minor left-singular frame of $W_0$ , frozen: $\text{FB}(B_0) \rightarrow \text{LoRA}_{(B_0, 0)}$ , a linear image at a regular point (HF also allows $r = \min(m, n)$ , outside the theorem)                                                                                        |
| PiSSA ( <a href="#">Meng et al., 2024a</a> ) / MiLoRA ( <a href="#">Wang et al., 2024b</a> ) | $S$                      | $P$ the top / bottom rank- $r$ SVD part of $W_0$ (HF absorbs the scale: $P = sB_0A_0$ )                                                                                                                                                                                                       |
| OLoRA ( <a href="#">Büyükyüz, 2024</a> )                                                     | $S$                      | $P = sQ_r R_r$ in HF, from a truncated QR factorisation of $W_0$                                                                                                                                                                                                                              |
| CorDA ( <a href="#">Yang et al., 2024c</a> )                                                 | $S$                      | $P$ from the SVD of $W_0 C$ , where $C$ is the activation covariance                                                                                                                                                                                                                          |
| LoRA-GA ( <a href="#">Wang et al., 2024e</a> )                                               | $S$                      | $P$ from the SVD of the first gradient                                                                                                                                                                                                                                                        |
| Astra ( <a href="#">Liu et al., 2026b</a> )                                                  | $S$                      | $P = VV^\top W_0$ (times $s$ in HF and the official code, which do not absorb the scale), with $V$ eigenvectors of the output activations (the centred covariance’s tail in the paper, the uncentred second moment’s top $r$ in HF and the official code), when $\text{rank } V^\top W_0 = r$ |
| HF orthogonal                                                                                | outside: $(r/2, r/2, 0)$ | an apex with $\dim S_1 = \frac{r}{2}(m + n - \frac{r}{2})$ , for even $r$ in exact arithmetic                                                                                                                                                                                                 |
| QLoRA ( <a href="#">Dettmers et al., 2023</a> ), LoftQ ( <a href="#">Li et al., 2023b</a> )  | defect, not a point      | zero-init (QLoRA) or approximately split (LoftQ) over a quantised base ( <a href="#">Proposition V.2</a> )                                                                                                                                                                                    |
| HF init_lora_weights=False                                                                   | unpointed                | random $A$ and $B$ , with no subtraction                                                                                                                                                                                                                                                      |

**PROPOSITION II.3** (first-order rank formula: smoothness, alignment and scale). For  $\kappa(B, A) = W_{\text{base}} + sBA$  with  $s \neq 0$ ,

$$\text{rank } d\kappa_{(B, A)} = m \text{rk } A + n \text{rk } B - \text{rk } A \text{rk } B \quad \text{for all } (B, A), \quad (3)$$

and  $\{XA\} \cap \{BY\} = \{BRA : R \in \mathbb{R}^{r \times r}\}$ . For full-rank frames this overlap is the tangent space of the product  $\text{FA}(A) \times \text{FB}(B)$  of [Proposition I.8\(a\)](#), re-based at  $\kappa(B, A)$ , whose image is the meet of the two one-sided images: a generalised LoRA-XS. It is LoRA-XS proper when  $(B, A)$  span the top- $r$  singular subspaces of  $W_0$  ( $\sigma_r > 0$ ), as at a PiSSA init; bottom frames (MiLoRA) give a generalised LoRA-XS.

Under gradient flow  $\dot{B} = -\eta_B \nabla_B L$ ,  $\dot{A} = -\eta_A \nabla_A L$ , with  $G = \nabla_W L$ ,

$$\frac{d}{dt} L = -s^2 (\eta_B \|GA^\top\|_F^2 + \eta_A \|B^\top G\|_F^2) \quad (4)$$



at every time, in particular at  $(B_0, A_0)$ . To first order in the step sizes the same holds for full-batch gradient descent, and in expectation for unbiased minibatch SGD. Under Adam ( $\varepsilon \rightarrow 0$ , with bias correction) the first step changes the loss by

$$-|s|(\eta_B \|GA_0^\top\|_1 + \eta_A \|B_0^\top G\|_1) + O(\eta^2), \quad (5)$$

with entrywise  $\ell_1$  norms. This first-order term is linear in  $|s|$ , against quadratic under gradient flow.

Numerically, the rank formula (3) holds in 400 random cases, including rank-deficient factors, and the rate (4) matches finite differences (Appendix E).

**EXAMPLE** (one attention projection of a 7B model). At  $4096 \times 4096$  with  $r = 16$ , zero-factor initialisations (LoRA, QLoRA, EVA, LoRA-FA) have  $mr = 65,536$  first-order directions, and split ones have  $r(m + n - r) = 130,816$ , the dimension of LoRA’s image.

**WHAT THIS EXPLAINS** Zero-factor initialisations (LoRA, QLoRA, EVA, LoRA-FA) have  $mr$  first-order directions; split initialisations have  $r(m + n - r)$ . “Better initialisation” in the literature mixes three distinct mechanisms, and the proposition separates them.

**Smoothness.** PiSSA and LoRA-GA start off the apex, so their first step sees all  $r(m + n - r)$  directions.

**Alignment.** Under gradient flow, among  $A_0$  with  $A_0 A_0^\top = cI$ , the term  $\eta_B \|GA_0^\top\|_F^2$  of the initial rate (4), and with it the whole rate at a zero- $B$  start, is maximised by the top- $r$  right singular space of  $G$ . LoRA-GA takes exactly this for  $A_0$ , but takes  $B_0$  from the left singular vectors  $r + 1, \dots, 2r$  (its Algorithm 1; HF’s default `direction="ArB2r"`).  $G$  then has no component in the overlap  $\{B_0 R A_0\}$ , so the first gradient step approximates the rank- $2r$  truncation of  $G$ . LoRA-GA optimises gradient approximation, and gives up part of the  $B$ -term of the rate to do so. EVA uses the top principal subspace of the input activations. Its Theorem 3.2 assumes the inputs independent of the upstream errors and bounds the expected per-sample squared gradient norm with the uncentred  $\mathbb{E}[xx^\top]$ . That quantity is not the rate  $\|\mathbb{E}[G]A_0^\top\|_F^2$ , and HF’s EVA uses mean-centred PCA, so EVA’s frame is a data-only proxy. Alignment is a gradient-flow and SGD notion. Under Adam the first-step rate  $\|GA_0^\top\|_1$  of (5) depends on the basis of row  $A_0$ , not only on the subspace.

**Scale.**  $\|GA_0^\top\|_F^2$  grows with  $\|A_0\|^2$ , and initial scale trades off against learning rate (Theorem III.10). PiSSA’s  $S_r^{1/2}$ -weighted factors change it. LoRA-GA’s paper scales both factors by  $d_{\text{out}}^{1/4}/\sqrt{\gamma}$ ; HF’s default `scale="stable"` scales only  $B$  (which also carries  $S_{r+1:2r}$ ), so there  $\|GA_0^\top\|_F^2$  is unchanged. HF’s Kaiming default has  $A_0 A_0^\top \approx I/3$ , against  $A_0 A_0^\top = I$  for EVA with `whiten=False`. Comparing rates across initialisations needs  $A_0 A_0^\top$  normalised.

LoRA-GA uses all three mechanisms, with alignment on the  $A$  side only. The gradient-aligned frame is due to Wang et al. (2024e), the activation frame to Paischer et al. (2024), and the spectral split to Meng et al. (2024a); Proposition II.3 only sorts their effects into three terms.

### The zero gate

At a zero- $B$  start,  $BA$  is linear in  $B$  and  $B_0 = 0$ , so moving  $A$  alone changes nothing. The same holds whenever a trainable gate starts closed. Below,  $K = D\rho \wedge D\rho^\top$  is the cometric on weights induced by an optimizer’s metric  $\wedge$  (Background D.23, Exposé III).

**LEMMA II.4** (zero gate). Let  $Q = U \times V$  with  $U$  a vector space, let  $\rho = \theta_0 + \beta(u, v)$  with  $\beta$  smooth, and suppose  $\beta(u_0, v) = 0$  for all  $v$ ; for example  $\beta(\cdot, v)$  linear and  $u_0 = 0$ . Then  $D_v \rho_{q_0} = 0$ . Hence  $\nabla_v (L \circ \rho)(q_0) = 0$ ,  $S_1 = D_u \beta(u_0, v_0)(U)$  ( $= \beta(U, v_0)$  in the linear case), and the cometric at  $q_0$  is  $K_{q_0} = D_u \beta \Lambda_{uu} D_u \beta^\top$ , where  $\Lambda_{uu}$  is the  $uu$ -block of  $\Lambda$ ; no block-diagonality is needed, since the  $v$ -columns of  $D\rho$  vanish. The weight trajectory  $\theta(t)$  agrees with that of the method with  $v$  frozen at  $v_0$  to first order in  $t$  (gradient flow), or in the step size (one or more Adam steps). The parameter  $v$  itself may still move: under terms not of the form  $L \circ \rho$  (regularisers, decoupled weight decay), and under Adam, whose step-2 move of  $v$  is  $O(\eta)$  although its effect on  $\theta$  is  $O(\eta^2)$ .

Function-level variant. For an extension with realised function  $F(u, v, x)$ , if  $F(u_0, v, x) = F_0(x)$  for all  $v$  and  $x$ , then  $D_v F = 0$  at  $q_0$ , with the same conclusions at the level of functions. No linearity is needed.

Numerically, under Adam with  $\eta \in \{10^{-2}, 10^{-3}, 10^{-4}\}$ , the step-2 move of  $A$  in zero-init LoRA scales as  $\eta$ , while the deviation of  $\Delta W$  from the frozen- $A$  method scales as  $\eta^2$  (Appendix E).

Instances, assuming no weight decay on  $v$ :

- LoRA (Hu et al., 2021), with  $u = B$ .
- VeRA (Kopiczko et al., 2023), with  $u = b$ :  $\theta$  first moves only along  $b$ .
- AdaLoRA (Zhang et al., 2023g), whose update is  $P \text{diag}(\lambda) Q$ , with  $u = \lambda$ , the diagonal of singular values, started at 0.  $\theta$  initially moves only along the  $r$  rank-one directions  $p_i q_i^\top$ , so  $\dim S_1 = r$  for generic  $P, Q$ .  $P$  and  $Q$  do move at step one, through AdaLoRA's orthogonality penalty  $R(P, Q)$  (HF default `orth_reg_weight=0.5`), which has no first-order effect on  $\theta$ .
- LoHa (Hyeon-Woo et al., 2021) with a zero factor (HF zero-initialises `hada_w2_b`): only that factor moves.
- By the function-level variant, the gate of LLaMA-Adapter (Zhang et al., 2023h), which is the only parameter that moves at first order, and bottleneck adapters whose up-projection starts at zero, where only that factor moves (Houlsby et al. (2019) start near that point).

HRA's paired initialisation has a different form and shows the same first-order collapse (Theorem II.7(b)).

Zhu et al. (2024b) find over whole runs that a random, untrained  $A$  does nearly as well as a tuned one, and that training  $B$  alone gives better generalisation bounds. Lemma II.4 is only the first-order statement.

### Off the point: quantised bases

QLoRA (Dettmers et al., 2023) quantises the frozen weight, so it is LoRA pointed at  $\kappa\theta_0$ , with pointing defect  $e = \kappa\theta_0 - \theta_0$  (Definition V.1, Proposition V.2(a)). As  $e$  is generically of full rank,  $\theta_0$  lies outside  $\text{Im LoRA}_r(\kappa\theta_0)$  unless  $\text{rank } e \leq r$ , which is why Theorem II.1 speaks of *exact* pointings. When the initialisation also splits, the order of operations decides the defect (Proposition V.2(b)). In Table 5,  $W$  is the full-precision weight,  $e = \kappa W - W$  and  $\delta_0 = \delta(q_0)$ .

**REMARK** (LoftQ's scale in HF PEFT). HF PEFT's LoftQ initialiser does not divide by the LoRA scale  $s$  ( $\alpha/r$ , or  $\alpha/\sqrt{r}$  under rsLoRA), so for  $s \neq 1$  the deployed defect is  $e - s \text{SVD}_r(e)$ . At  $s = 2$  its Frobenius norm equals  $\|e\|$ , and the fit buys nothing over QLoRA in that norm.

**Table 5.** Pointing defects of quantised-base LoRA pipelines (Proposition V.2(b)).

| pipeline                          | examples                                                                                             | defect $\rho(q_0) - W$                                                                                |
|-----------------------------------|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| quantise, then zero-init          | QLoRA (Dettmers et al., 2023)                                                                        | $e$                                                                                                   |
| split, then quantise the residual | QPiSSA (Meng et al., 2024a); HF’s recommended PiSSA/CorDA workflow; LoftQ’s later quantisation steps | $\kappa(W_{\text{res}}) - W_{\text{res}}$                                                             |
| quantise, then fit the error      | LoftQ (Li et al., 2023b) with one iteration (HF default loftq_iter=1); HF replace_lora_weights_loftq | $e - \text{SVD}_r(e)$ , the part of $e$ left after its best rank- $r$ approximation (unit LoRA scale) |
| quantise, split, re-quantise      | HF olora_init on bitsandbytes weights                                                                | $e + \kappa(r') - r'$ , with $r' = \kappa W - \delta_0$                                               |

Quantised bases for LoRA come from Dettmers et al. (2023), fitting the adapter to the quantisation error from Li et al. (2023b), and the quantised PiSSA variant from Meng et al. (2024a).

The strata of Theorem II.2 classify what a start can express. The dynamics see a finer quotient. For a zero- $B$  start, SGD sees  $A_0^\top A_0$  and Adam sees  $A_0$  up to signed permutations of its rows (Corollary III.13). A dynamical classification of initialisations, including the degenerate pointings, is open (Exposé XI).

## II.B Erlangen: invariants of group-type methods

*A geometry is what a group leaves unchanged. Before asking how a method learns,  
ask what it cannot help keeping.  
after Klein*

A method that moves the pretrained weight by a group element can never change that group’s invariants. Orbits also place HRA as a meet, DoRA as a saturation and the scaling methods on a ladder.

### In brief.

- A method whose weights are always  $W_0 R$ , with  $R$  exactly orthogonal, can never change  $W W^\top$ , so neuron norms, angles and singular values survive any number of merges in exact arithmetic. HF PEFT’s default OFT, BOFT’s scale and GOFT’s default scaler break this.
- For injective  $W_0$  and even  $r \leq n - 2$ , HRA reaches exactly the rank- $r$  LoRA updates of the form  $W_0 R$ ,  $R \in \text{SO}(n)$ . Its HF PEFT default initialisation is an apex.
- DoRA reaches exactly the weights of LoRA plus a free gain per output neuron. From the same start, any gap between the two comes from training.
- $(\text{IA})^3$ , SSF’s scales and DoRA’s magnitude lie in the image of rank-one HiRA,  $W_0 \odot (J + BA)$ , which for  $W_0$  without zeros is LoRA seen through  $X \mapsto W_0 \odot X$ .

Klein’s Erlangen programme of 1872 (Klein, 1872) (Background D.17) defines a geometry by its group of motions and studies what the motions leave unchanged. OFT, GOFT, HRA and ETHER multiply the pretrained matrix by an orthogonal matrix,  $(\text{IA})^3$  and SSF’s scale multiply it by a diagonal one, and DoRA’s magnitude does the same to a LoRA weight. These are methods of *action type* (Definition I.2),  $\rho(q) = \gamma(q) \cdot W_0$  with  $\gamma$  landing in a group. Whatever

that group cannot change, the method cannot change, in training or after any number of merges.

Klein’s programme is the model. [Bronstein et al. \(2021\)](#) made it the organising principle of geometric deep learning, for architectures. Here it is applied to the fine-tuning map.

**ANALOGY** (what “Erlangen” means here). For a group-type method the Erlangen reading is a theorem, since its image lies in one orbit. LoRA’s cone  $W_0 + \mathcal{M}_{\leq r}$  has a singular vertex at  $W_0$  ([Theorem II.2\(c\)](#)), so it is no orbit; for cones the word names the shape of the image and its covariance group ([Definition II.14](#)).

One convention matters. A linear box computes  $y = Wx$  with  $W \in \mathbb{R}^{m \times n}$ , and row  $i$  of  $W$  is output neuron  $i$ . The two Gram matrices, over neurons and over inputs, are

$$K_{\text{out}}(W) = WW^\top, \quad K_{\text{in}}(W) = W^\top W.$$

A right multiplication  $W \mapsto WR$  acts on the input side and fixes  $K_{\text{out}}$ ; a left multiplication  $W \mapsto PW$  acts on the output side and fixes  $K_{\text{in}}$ . Papers that store  $W^\top$  swap these words, and we translate every method into this convention.

### Complete invariants

A complete invariant of a group action ([Background D.15](#)) agrees on two matrices exactly when one can be carried to the other. For the five actions in PEFT it is classical. We write  $T_m$  for the invertible diagonal matrices,  $T_m^+ = \{\text{diag}(d) : d_i > 0\}$  for its identity component ([Background D.4](#)), and  $\odot$  for the entrywise product.

**THEOREM II.5** (complete invariants of the basic actions; classical). *Let  $W, W' \in \mathbb{R}^{m \times n}$ .*

- (1) Right orthogonal (input side).  $W' = WR$  for some  $R \in \text{O}(n)$  iff  $WW^\top = W'W'^\top$ . For  $\text{SO}(n)$ , when  $\text{rank } W = n$ , also require that the then unique  $R$  have  $\det R = 1$ .
- (2) Left orthogonal.  $W' = PW$  for some  $P \in \text{O}(m)$  iff  $W^\top W = W'^\top W'$ .
- (3) Two-sided.  $W' = PWR$  for some  $P \in \text{O}(m)$  and  $R \in \text{O}(n)$  iff  $W$  and  $W'$  have the same singular values, counted with multiplicity.
- (4) Left torus. If  $W$  has no zero row,  $W' = DW$  for some  $D \in T_m^+$  iff  $w'_i \in \mathbb{R}_{>0} w_i$  for every  $i$ . For the full torus  $T_m$ , whose signs are unconstrained, the complete invariant is the set of zero rows together with the row lines.
- (5) Full Hadamard torus.  $W' = E \odot W$  for some  $E \in (\mathbb{R}^\times)^{m \times n}$  iff  $W$  and  $W'$  have the same zero pattern.

Item (1) is the orbit-separation consequence of the first fundamental theorem of invariant theory for  $\text{O}(n)$  ([Background D.16](#)); for a compact group, invariants separate orbits. Equivalently, it is Witt-type extension of isometries. The two-sided diagonal torus  $\{\text{diag}(a) \times \text{diag}(b)\} \cong (T_m \times T_n)/\mathbb{R}^\times$  that appears as LoHa’s gauge ([Proposition II.12](#)) is a different group from the full Hadamard torus in (5).

**Caveat for methods.** No PEFT method realises (4) or (5) as a group action on all of its parameters. (IA)<sup>3</sup> entries and DoRA magnitudes are unconstrained reals initialised at 1 or at the row norms, so the relevant group in (4) is  $T_m$  or its closure. HiRA’s factor  $J + BA$  can vanish entrywise. These methods can create new zeros, so for them only a one-way statement transfers: zeros of  $W_0$  remain zero. Table 6 lists the five actions with the methods that live in each.

**Table 6.** The five basic actions, their complete invariants ([Theorem II.5](#)) and the methods whose images lie in one orbit or orbit closure.

| action                       | group                              | complete invariant                | methods that live here                                                     |
|------------------------------|------------------------------------|-----------------------------------|----------------------------------------------------------------------------|
| $W \mapsto WR$ , input side  | $O(n)$                             | $K_{\text{out}} = WW^\top$        | exact-Cayley OFT and block OFT, HRA, and GOFT and ETHER as published       |
| $W \mapsto PW$ , output side | $O(m)$                             | $K_{\text{in}} = W^\top W$        | LyCORIS diag-OFT; HF PEFT OFT up to version 0.13                           |
| $W \mapsto PWR$              | $O(m) \times O(n)$                 | singular values with multiplicity | any composite of the two rows above                                        |
| $W \mapsto DW$               | $T_m^+, T_m$                       | row rays; zero rows and row lines | (IA) <sup>3</sup> on $k, v$ , SSF's scale, DoRA's magnitude (closure only) |
| $W \mapsto E \odot W$        | $(\mathbb{R}^\times)^{m \times n}$ | zero pattern                      | HiRA (one way only: $Z(W_0) \subseteq Z(W)$ )                              |

**WHAT THIS EXPLAINS** Moving a method from the input side to the output side trades  $K_{\text{out}}$  for  $K_{\text{in}}$ . Letting a scale change sign trades rays for lines, and letting it reach zero leaves only the zeros of  $W_0$ .

The interactive companion has a *Gram lab* for these invariants. It fixes an invertible  $4 \times 4$  matrix  $W_0$  with one zero entry and distinct singular values. The reader picks one of ten methods and drags, types or randomises its parameters. Each lamp is a float64 test, to  $10^{-10}$ , that an invariant of  $W = \rho(q)$  (the neuron Gram matrix, the cosines, the input Gram matrix, the singular values, the row lines, the zeros of  $W_0$ ) still equals its value at  $W_0$ , and a further lamp tests  $\text{rank } \Delta W \leq 2$ . A lamp guaranteed by a theorem is marked as such, and a red lamp names the hypothesis that was broken. Heatmaps, neuron arrows and strips show  $K_{\text{out}}$ , the cosines,  $K_{\text{in}}$ ,  $W$ ,  $\sigma(W)$ , the rank of  $\Delta W$  and  $\dim S_1$  against  $d$ . OFT's Cayley–Neumann switch (HF's default), BOFT's  $s_1 \mapsto -s_1$ , (IA)<sup>3</sup>'s  $\ell_2 \mapsto 0$  and DoRA's  $B \mapsto 0$  each break or restore a hypothesis.

### What exactly orthogonal methods can never do

If every weight a method reaches is  $W_0 R$  with  $R$  orthogonal, [Theorem II.5](#)(1) and (3) freeze the neuron Gram matrix and the singular values, and merging cannot help, since products of orthogonal matrices are orthogonal. The work lies in saying which implementations are exactly orthogonal, and on which side.

**COROLLARY II.6** (what exactly orthogonal methods can never do). *Let a method have image in  $W_0 \cdot O(n)$  (input side), and work in exact arithmetic. This covers:*

- OFT and block OFT with an exactly orthogonal parametrisation: the original paper's exact Cayley map, and HF PEFT from 0.16 with `use_cayley_neumann=False` on Linear and Conv targets;
- GOFT with strict Givens rotations and no scaler, as in the paper (`no_scaling=True` in the official `peft-givens` code);
- HRA, and ETHER as defined in its paper (not ETHER+);
- any number of merge-and-restart cycles of these.

Then:



- it preserves  $K_{\text{out}}$ , hence every neuron norm and the angle between any two nonzero neurons, and so the hyperspherical energy, which depends only on these angles;
- it preserves the whole singular spectrum of  $W_0$ ;
- block-diagonal OFT with input blocks  $J$ , with the same partition in every cycle, preserves each partial Gram matrix  $W_{:,J}W_{:,J}^\top$ .

Several methods that look orthogonal fail the hypothesis. Each item below is a separate computation, and Table 7 summarises them.

**REMARK** (scope of Corollary II.6; these statements do not follow from its hypothesis).

- *BOFT* as implemented in HF PEFT,  $W = \text{diag}(s)W_0R^\top$  with  $R$  exactly orthogonal, has  $K_{\text{out}}(W) = \text{diag}(s)K_{\text{out}}(W_0)\text{diag}(s)$ . So  $\cos'_{ij} = \text{sign}(s_i s_j) \cos_{ij}$  whenever  $s_i s_j \neq 0$ . It preserves  $|\cos|$  when every  $s_i$  is nonzero, and preserves  $\cos_{ij}$  whenever  $s_i s_j > 0$ . Hence it preserves every cosine when all  $s_i$  are nonzero and share a sign (for instance  $s \in T_m^+$ , the constraint in the re-scaled OFT of Qiu et al. (2023), Sec. 3.5); when no cosine of  $W_0$  is zero, that condition is also necessary. HF's `boft_s` is an unconstrained real initialised at 1. At the default `boft_n_butterfly_factor=1` the butterfly permutation is the identity, BOFT is block OFT times an output scale, and each input block satisfies  $W_{:,J}W_{:,J}^\top = \text{diag}(s)(W_0)_{:,J}(W_0)_{:,J}^\top \text{diag}(s)$ . HF PEFT OFT 0.14 to 0.15 (exact Cayley, input side) carried a trainable output scale `oft_s` and has the same form.
- *GOFT's official code* by default (`no_scaling=False`) adds a trainable input-side diagonal  $g$ , initialised at ones. Then  $W = W_0 \text{diag}(g)G$  and  $K_{\text{out}}(W) = W_0 \text{diag}(g)^2 W_0^\top$ , whatever the Givens part  $G$  is. This is the paper's GOFT\* ablation, and it is not covered.
- *Output-side implementations* (LyCORIS diag-OFT at its defaults; HF PEFT OFT up to 0.13, which merged to  $R^\top W_0$ ; HF OFT on embedding layers; ETHER with `flip_side=True`) have image in  $O(m) \cdot W_0$ . They preserve  $K_{\text{in}}$ , the partial input Grams  $W_{I,:}^\top W_{I,:}$  and the spectrum, and they do not preserve  $K_{\text{out}}$ . ETHER's official code with `ether_nb>1` applies a different reflection to each output-row group, which keeps neuron norms and within-group angles but neither the cross-group cosines nor the spectrum.
- *HF PEFT's default OFT* since 0.18 (`use_cayley_neumann=True`, 5 terms) builds  $R = I + 2Q + 2Q^2 + 2Q^3 + Q^4 = C(Q)(I - Q^4)$ , where  $C(Q) = (I + Q)(I - Q)^{-1}$  is the exact Cayley map (HF's exact branch evaluated at  $-Q$ ). Hence  $R^\top R = (I - Q^4)^2 \neq I$ , and the singular values of  $R$  are  $|1 - \theta_k^4|$ , where  $\pm i\theta_k$  are the eigenvalues of  $Q$ : a systematic contraction for small  $Q$ , and  $R$  is singular when some  $\theta_k = 1$ . Already the adapted weight has  $K_{\text{out}} = W_0(I - Q^4)^2 W_0^\top$ , a drift of order  $\|Q\|^4$ , and merge cycles compound it. For  $\|Q\|_{\text{op}} \leq 2^{1/4}$  every factor is a contraction, so the drift is a monotone shrinkage of neuron norms and singular values (Theorem V.3, corollary 4). In 0.16 and 0.17 the last coefficient was 2, so  $R^\top R = I + 4Q^6 + 4Q^8$ , still not orthogonal, with drift of order  $\|Q\|^6$ . (HF's docstring says the default is `False`; the dataclass default is `True`.)
- *Quasi-orthogonal variants* preserve neither  $K_{\text{out}}$  nor its normalised form exactly: qGOFT, whose  $2 \times 2$  blocks are arbitrary matrices with only a soft orthogonality penalty; ETHER+, which is two-sided and whose factor  $I - uu^\top + vv^\top$  is singular when  $u \perp v$ ; and the default HF OFT above. Once  $n > 2$ , a single non-orthogonal  $2 \times 2$  block, conformal or not, generically changes some cosine.
- *Finite precision*. The invariance is exact only in exact arithmetic; low-precision merges add a rounding drift that grows like a random walk over merge cycles.



**Table 7.** Implementations that look orthogonal, read against [Corollary II.6](#). Each row is a separate computation (see the remarks after the corollary).

| implementation                                                                                                        | map                                                                                                                                   | exactly or-<br>thogonal?            | what it keeps                                                                                                                                                                                                                                                        |
|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BOFT in HF PEFT;<br>HF PEFT OFT<br>0.14–0.15                                                                          | $\text{diag}(s) W_0 R^\top$ , $R$ exactly<br>orthogonal, $s$ free,<br>initialised at 1                                                | only $R$                            | $K_{\text{out}} \mapsto \text{diag}(s) K_{\text{out}} \text{diag}(s)$ , and block<br>Grams likewise. $ \cos_{ij} $ survives when<br>$s_i s_j \neq 0$ , $\cos_{ij}$ when $s_i s_j > 0$ . At HF’s default<br>single butterfly factor, BOFT is block OFT<br>times $s$ . |
| GOFT, official code<br>default<br>output-side OFT<br>(LyCORIS diag-OFT<br>at its defaults; HF<br>PEFT OFT up to 0.13) | $W_0 \text{diag}(g) G$ , with a<br>trainable input scale $g$<br>image in $O(m) \cdot W_0$                                             | no<br><br>yes, on the<br>other side | Nothing on this list, since<br>$K_{\text{out}} = W_0 \text{diag}(g)^2 W_0^\top$ .<br>$K_{\text{in}}$ , the partial input Grams and the<br>spectrum. Not $K_{\text{out}}$ .                                                                                           |
| HF PEFT default<br>OFT since 0.18<br>(use_cayley_<br>neumann=True)<br>qGOFT                                           | $R = I + 2Q + 2Q^2 + 2Q^3 +$<br>$Q^4 = C(Q)(I - Q^4)$ , with<br>$C$ an exact Cayley map                                               | no, $R^\top R =$<br>$(I - Q^4)^2$   | Nothing exactly. The adapted weight already<br>has $K_{\text{out}} = W_0(I - Q^4)^2 W_0^\top$ , and for small $Q$<br>each merge shrinks $K_{\text{out}}$ and the spectrum<br>further. (In 0.16–0.17, $R^\top R = I + 4Q^6 + 4Q^8$ .)                                 |
| ETHER+                                                                                                                | free $2 \times 2$ blocks with a<br>soft orthogonality<br>penalty<br>two-sided, $I - uu^\top + vv^\top$ ,<br>singular when $u \perp v$ | no                                  | Neither $K_{\text{out}}$ nor the cosines. Once $n > 2$ ,<br>one non-orthogonal block, conformal or<br>not, generically moves a cosine.<br>Neither $K_{\text{out}}$ nor the cosines.                                                                                  |

**Numerical check.** With  $s = (-0.5, 1, \dots, 1)$ , BOFT changes a cosine by 1.5 while  $|\cos|$  is fixed to  $4 \times 10^{-16}$ , and an all-negative  $s$  fixes every cosine to  $10^{-15}$ . GOFT with a scaler  $g = 1 \pm 0.05$  moves cosines and singular values. A conformal block  $\sqrt{2} \text{Rot}(0.3)$  on two of 8 input coordinates changes some cosine, by 0.17. HF’s default OFT satisfies  $R^\top R = (I - Q^4)^2$  to  $5 \times 10^{-16}$ , with  $\|R^\top R - I\| = 1.5 \times 10^{-2}$  already for block size 8 and skew entries of size about 0.07, and it is singular at  $\theta = 1$  ([Appendix E](#)).

**Credit.** [Qiu et al. \(2023\)](#) introduced orthogonal fine-tuning, motivated by preserving hyperspherical energy ([Liu et al., 2018](#)); their re-scaled OFT (Sec. 3.5) adds the positive per-neuron scale that HF BOFT carries. BOFT is due to [Liu et al. \(2023b\)](#), building on butterfly factorisations ([Dao et al., 2019](#)); GOFT and qGOFT are due to [Ma et al. \(2024\)](#), with an official peft-givens implementation; ETHER and ETHER+ to [Bini et al. \(2024\)](#); HRA to [Yuan et al. \(2024\)](#). LyCORIS is described by [Yeh et al. \(2023\)](#). The Cayley–Neumann default, BOFT’s scale and the output-side merge are read from the Hugging Face PEFT source ([Mangrulkar et al., 2022](#)) (the OFT configuration and the BOFT layer).

**WHAT THIS EXPLAINS** “Orthogonal fine-tuning preserves knowledge” corresponds to preserving a complete invariant: in exact arithmetic an exactly orthogonal method can *never* change spectra or neuron geometry, through any number of merges ([Theorem V.3](#), corollary 7), up to rounding. So a variant that must change them adds a non-orthogonal factor (the per-neuron scale of Qiu et al.’s re-scaled OFT, as in HF BOFT and HF OFT 0.14 to 0.15, or GOFT’s default input scaler). HF’s default OFT is only close to orthogonal; with block size 8 and fresh skew generators with entries near 0.05 in every cycle, its merges shrink  $K_{\text{out}}$  by about 0.2%, 1.3%, 6% and 22% after 1, 10, 50 and 200 cycles (relative Frobenius change, averaged over draws). The statement concerns reach. It does not show that preserving the invariant causes knowledge retention, and the control study of ETHER ([Bini et al.,](#)

2024) found a non-orthogonal multiplicative variant comparably stable. Earlier claims that BOFT or qGOFT preserve the full Gram matrix are corrected here.

### HRA is a meet of two images

A Householder reflection  $H_u = I - 2uu^\top / \|u\|^2$  is orthogonal and differs from  $I$  by rank one. So  $W_0 H_{u_1} \cdots H_{u_r}$  lies in the orthogonal orbit of  $W_0$  and in LoRA's cone  $W_0 + \mathcal{M}_{\leq r}$ , and for injective  $W_0$  and even  $r$  the Cartan–Dieudonné theorem ([Background D.11](#)) says that HRA fills the cone's intersection with the rotation orbit. HF PEFT repeats each reflection at initialisation, so to first order each pair can only turn its own vector against the rest of the space, and HRA starts at an apex, as zero-init LoRA does ([Theorem II.2](#)).

**THEOREM II.7** (HRA is the meet of LoRA with the rotation orbit, and it starts at an apex). *Let  $W_0 \in \mathbb{R}^{m \times n}$  be injective (rank  $W_0 = n$ ), and let  $r$  be even with  $2 \leq r \leq n - 2$ . Define  $\text{HRA}_r : (u_1, \dots, u_r) \mapsto W_0 H_{u_1} \cdots H_{u_r}$  with  $u_i \neq 0$ , where  $H_u = I - 2uu^\top / \|u\|^2$ . Write  $\text{OFT}_{\text{SO}} := W_0 \cdot \text{SO}(n)$  for the rotation orbit. (The  $\text{O}(n)$ -orbit would add the odd component:  $W_0 H_{e_1}$  lies in  $W_0 + \mathcal{M}_{\leq r}$  but not in  $\text{Im HRA}_r$ .)*

- (a) **Meet.**  $\text{Im HRA}_r = (W_0 + \mathcal{M}_{\leq r}) \cap W_0 \cdot \text{SO}(n)$ . Further,  $d(\text{HRA}_r) = rn - \frac{r(r+1)}{2}$ , so the generic fibre has dimension  $\frac{r(r+1)}{2} = rn - d$ . This gauge is non-linear. The evident continuous global symmetries are the  $r$  scalings  $u_i \mapsto c_i u_i$ ; the braid group also acts globally by Hurwitz moves  $(u_i, u_{i+1}) \mapsto (H_{u_i} u_{i+1}, u_i)$ , since  $H_a H_b = H_{H_a b} H_a$ .
- (b) **First-order collapse.** At the paired initialisation  $u_{2i-1} = u_{2i} = w_i$  ( $i \leq k = r/2$ , with the  $w_i$  independent and  $U = \text{span}\{w_i\}$ ),

$$S_1 = W_0 \cdot \{\Omega \in \mathfrak{so}(n) : P_{U^\perp} \Omega P_{U^\perp} = 0\}, \quad \dim S_1 = kn - \frac{k(k+1)}{2} < d(\text{HRA}_r).$$

- (c) **Vertex.** The germ of  $\text{Im HRA}_r$  at  $W_0$  is diffeomorphic to the germ at 0 of  $\Sigma_r = \{\Omega \in \mathfrak{so}(n) : \text{rank } \Omega \leq r\}$ , the skew-determinantal cone ([Background C.13](#)), which is not linear. So  $W_0$  is a singular point (the vertex) of the image. Together with (b), the paired initialisation is an **apex**.

- (d) **Scope.**

- Not a product. (a) is a meet in the poset of subsets of  $\Theta$  containing  $W_0$  ([Observation I.4\(c\)](#)). HRA is not the categorical product of  $\text{LoRA}_r$  and  $\text{OFT}_{\text{SO}}$ : the set-level fibre product  $\{(B, A, R) : W_0 + BA = W_0 R\}$  has local dimension  $d + r^2 = rn + \frac{r(r-1)}{2} \neq rn$  over a generic point of the meet. (It is not pure-dimensional: over  $W_0$  alone it contains  $\{(B, 0, I)\}$ , of dimension  $mr$ .)
- Exactly orthogonal OFT is smaller. The Cayley map never produces a rotation with eigenvalue  $-1$ . For exact full-block Cayley OFT (the original paper; HF with `use_cayley_neumann=False`),  $(W_0 + \mathcal{M}_{\leq r}) \cap \text{Im OFT}$  is dense in  $\text{Im HRA}_r$  but not equal to it:  $W_0 H_{e_1} H_{e_2}$ , with  $H_{e_1} H_{e_2} = \text{diag}(-1, -1, 1, \dots, 1)$ , is missing. For exact block Cayley OFT with  $b < n$  the intersection is a proper subset of  $\text{Im HRA}_r$ .
- HF's default OFT is incomparable. HF PEFT's default OFT is not orthogonal ([Corollary II.6](#); `use_cayley_neumann=True`). Since  $R^\top R = (I - Q^4)^2 = I$  forces  $Q = 0$  near  $Q = 0$ , its image meets  $\text{Im HRA}_r$  near  $W_0$  only at  $W_0$ , while its intersection with  $W_0 + \mathcal{M}_{\leq r}$  contains non-orthogonal points such as  $W_0(I + 2Q + 2Q^2 + 2Q^3 + Q^4)$  with  $\text{rank } Q = 2$ .
- Implementation. The theorem concerns HRA with  $\lambda < \infty$  (`HF apply_GS=False`). The strictly orthogonal Gram–Schmidt variant has the smaller image  $W_0(I - 2P_U)$ , of di-

mension  $r(n - r)$ . The paired initialisation is HF's default (`repeat_interleave`) and is not stated in the HRA paper. HF HRA acts on `in_features`, so injectivity requires `out_features`  $\geq$  `in_features` (and full column rank); the theorem does not cover `down_proj` or GQA key/value projections. The Gram–Schmidt variant never contains  $W_0$ , since  $I - 2P_U \neq I$ , and at HF's default paired initialisation it runs Gram–Schmidt on duplicated columns, which is numerically degenerate.

**Numerical check.** At  $n = 9$ , the paired-initialisation Jacobian has rank 8 against generic rank 15 for  $r = 2$ , and 15 against 26 for  $r = 4$  (Appendix E).

**Credit.** The sharp Cartan–Dieudonné theorem, that a rotation moving a  $k$ -dimensional space is a product of exactly  $k$  reflections, is due to Scherk (1950); see Artin (1957). Householder parametrisations in deep learning go back to Mhammedi et al. (2016), and Lie-algebra charts such as the Cayley map to Lezcano-Casado and Martínez-Rubio (2019). Yuan et al. (2024) proposed HRA as a bridge between low-rank and orthogonal adaptation, which the theorem makes precise. The paired initialisation is read from the HF PEFT HRA layer (Mangrulkar et al., 2022).

#### WHAT THIS EXPLAINS

- HRA's "bridge between low-rank and orthogonal adaptation" is, at the level of images, the meet of LoRA with the rotation orbit  $W_0 \cdot \mathrm{SO}(n)$ . It is a meet of images and no product of methods. It strictly contains  $\mathrm{LoRA} \cap \mathrm{OFT}$  for exactly orthogonal (Cayley) OFT, as in the original paper or HF with `use_cayley_neumann=False`; for HF's default Cayley–Neumann OFT the two are incomparable. Compare Proposition I.8(a), where LoRA-XS (Bałazy et al., 2024) is a genuine categorical product  $\mathrm{FA} \times \mathrm{FB}$ .
- HF recommends an even  $r$  because its paired initialisation needs pairs, and for odd  $r$  it falls back to a random Kaiming initialisation. The determinant identity  $\det R = (-1)^r$  explains why no odd- $r$  HRA can start at  $W_0$ , so that fallback is a pointing defect.
- New here (earlier analyses labelled this base point smooth): **HRA's default initialisation is an apex.** At first order, each pair can only rotate its own vector against the rest of the space. This is the multiplicative twin of LoRA's dead  $A$ -direction (Lemma II.4), so HRA belongs in the apex column beside LoRA. Whether the collapse slows early training is open (§XI).
- Rebasing commutes with the meet, so after  $K$  cycles re-HRA reaches  $(W_0 + \mathcal{M}_{\leq Kr}) \cap W_0 \cdot \mathrm{SO}(n)$  (Theorem V.3, corollary 5).

In the Gram lab, choosing HRA and resetting to  $q_0$  gives  $\dim S_1 = 3$  against  $d = 5$ , the values of (a) and (b) at  $n = 4$ ,  $r = 2$ . Moving  $u_2$  keeps the Gram, spectrum and rank lamps lit together, and the preset  $u = (e_1, e_2)$  lands on  $W_0 H_{e_1} H_{e_2}$ , which exact Cayley OFT never reaches.

#### LOFT fixes one leaf of the meet

The proof of Theorem II.7 (§C.13) already shows how the meet is built. It is the union of the orbits  $W_0 \cdot \mathrm{SO}(V)$  over the  $r$ -dimensional input subspaces  $V$ , one subgroup orbit for each point of the Grassmannian  $\mathrm{Gr}(r, n)$  (Background C.15). Call each orbit a *leaf*. Leaves overlap along lower-rank strata, so the leaves cover the meet without partitioning it; a generic point lies on exactly one leaf, fixed by its moved space  $\mathrm{im}(R - I)$ . HRA's reflection vectors move

the leaf as it trains. LOFT (Zhao et al., 2026) freezes the leaf and makes the choice of leaf the design variable, which the paper calls the adaptation support.

**REMARK** (2026: LoCO). One component of LoCO (Nguyen et al., 2026) is the Cayley rotation of a low-rank skew generator  $UV^\top - VU^\top$ , with  $U, V \in \mathbb{R}^{n \times r}$ . For injective  $W_0$  and  $2r \leq n - 2$ , its image is  $\text{Im HRA}_{2r}$  minus the points  $W_0 R$  at which  $R$  has eigenvalue  $-1$ , so like HRA it moves its leaf while it trains. Writing the generator as  $ZJZ^\top$  with  $Z = [U \ V]$  shows that its generic fibre is an orbit of the linear group  $\text{Sp}(2r, \mathbb{R})$ , acting by  $Z \mapsto ZM$ . The non-linear gauge of Theorem II.7(a) is therefore a property of the reflection parametrisation; LoCO's component reaches an open dense part of the same set with the linear gauge  $\text{Sp}(2r, \mathbb{R})$ . A sum of several LoCO components is in general only approximately orthogonal, and then leaves the meet.

**EXAMPLE** (LOFT: a leaf, and how to choose it). Let  $P_r \in \mathbb{R}^{r \times n}$  have orthonormal rows spanning  $V$ , and let  $\text{O}(V) = \{I + P_r^\top(T - I)P_r : T \in \text{O}(r)\}$  be the orthogonal maps that fix  $V^\perp$  pointwise. LOFT trains  $\rho(E) = W_0(I + P_r^\top(Q(E) - I)P_r)$  with  $E \in \mathfrak{so}(r)$ ,  $E_0 = 0$ , so  $|M| = r(r - 1)/2$ . We read it with an exactly orthogonal  $Q$  (the exponential, or the Cayley map  $C$ ); the released code defaults to a five-term Cayley–Neumann series with  $R^\top R = (I - X^6)^2$ , which is only approximately orthogonal, so like HF's default OFT (Corollary II.6) that version leaves its leaf.

- (a) *One leaf.* If  $W_0 P_r^\top$  has rank  $r$ , then  $(W_0 + W_0 P_r^\top \mathbb{R}^{r \times r} P_r) \cap W_0 \cdot \text{O}(n) = W_0 \cdot \text{O}(V)$ . For the principal support  $P_r = V_r^\top$  the first set is the image of LoRA-XS (this needs  $\sigma_r(W_0) > 0$ ), so the leaf is  $\text{LoRA-XS} \cap W_0 \cdot \text{O}(n)$ . Principal-support LOFT reaches an open dense part of its component  $W_0 \cdot \text{SO}(V)$ ; for injective  $W_0$  and a Cayley  $Q$ , its image is exactly  $\text{LoRA-XS} \cap$  the image of exact full-block Cayley OFT.
- (b) *Leaves over the Grassmannian.* If  $W_0$  is injective,  $(W_0 + \mathcal{M}_{\leq r}) \cap W_0 \cdot \text{SO}(n) = \bigcup_{V \in \text{Gr}(r, n)} W_0 \cdot \text{SO}(V)$ , which for even  $r$  is  $\text{Im HRA}_r$  (Theorem II.7(a)). LOFT's image is the whole leaf  $W_0 \cdot \text{SO}(V)$  with the exponential, and an open dense part of it with a Cayley map, which never reaches eigenvalue  $-1$ .
- (c) *A regular base point.* If  $W_0 P_r^\top$  has rank  $r$ , then  $S_1 = W_0 P_r^\top \mathfrak{so}(r) P_r$  has dimension  $r(r - 1)/2 = d(M)$ : LOFT starts at a smooth point of its leaf and has no gauge, while HRA's paired initialisation is an apex (Theorem II.7(b)). For  $r \geq 4$  there is no smooth pointed arrow  $\text{LOFT} \rightarrow \text{HRA}$  at that initialisation, although the images are nested (obstruction test O3 of §VIII).
- (d) *Choosing the leaf.* If  $Q'(0) = \text{id}$  (the exponential, or  $E \mapsto C(E/2)$ ), the gradient pulled back to  $E$  at  $\theta_0$  is  $P_r \text{skew}(W_0^\top G) P_r^\top$  with  $G = \nabla L(W_0)$ ; the Cayley map  $C$  itself doubles it, which does not change the best support. Its squared norm is at most  $2 \sum_{k \leq r/2} \mu_k^2$ , where  $\pm i\mu_k$  are the eigenvalues of  $\text{skew}(W_0^\top G)$ , with equality when  $V$  spans the invariant subspace of the top  $\lfloor r/2 \rfloor$  pairs.

Appendix C.14 proves (a)–(d).

**Numerical check.** For  $n = 8$ ,  $m = 10$  and  $r = 2, 4, 6$ , the leaf identity, the union, the arrows into exact Cayley OFT and LoRA-XS, and  $\dim S_1 = r(r - 1)/2$  all hold to about  $10^{-15}$  (Appendix E).

**Credit.** LOFT, the SkewGrad and GradSVD supports, part (d) and its bound (their Proposition 2), the reading of each support as selecting a Lie subgroup (their Remark 1) and the

suggestion of dynamic supports (their Remark 3) are from Zhao et al. (2026). Their Table 1 writes full and block OFT, GOFT, BOFT, HRA and PSOFT as choices of support, width, depth and transform in a multi-factor version of the same primitive. Parts (a)–(c) place the single-factor method inside Theorem II.7. PSOFT is due to Wu et al. (2025a); LoRA-XS to Bałazy et al. (2024).

**WHAT THIS EXPLAINS** Orthogonal methods differ in which leaves they use and how they combine them. Block OFT, GOFT and BOFT multiply rotations of fixed coordinate subspaces; HRA moves its leaf while it trains; PSOFT takes the principal leaf and adds two tori; LOFT takes one leaf, chosen before training from  $W_0$  or from the gradient at  $\theta_0$ . By part (d), Skew-Grad is the orthogonal counterpart of the alignment mechanism of Proposition II.3, and GradSVD transplants the gradient-based frames of LoRA-GA (Wang et al., 2024e). With the leaf frozen, merge-and-restart gains nothing. With finitely many supports revisited across restarts, the reach is the orbit of the connected group whose Lie algebra (Background D.8) the  $\mathfrak{so}(V_k)$  generate (Theorem V.3(b)); numerically, generic supports with  $r \geq 2$  generate all of  $\mathrm{SO}(n)$  once  $\lceil n/r \rceil$  of them are used. The principal support does not move when it is re-selected after a merge, and data-chosen supports have to be checked case by case.

### DoRA is a torus saturation of LoRA

DoRA normalises each row of LoRA’s matrix  $V = W_0 + BA$  and multiplies row  $i$  by a learned magnitude  $\mu_i$ . That is one row scaling, by  $\mu_i / \|v_i\|$ , so every DoRA weight is a diagonal matrix times a LoRA weight, and conversely up to a repair for vanishing rows. We write  $\mathrm{Diag}_m$  for the monoid of diagonal matrices (Background D.2),  $\mathrm{Mon}_m = T_m \rtimes S_m$  for the monomial matrices (Background D.3), and  $N$  for row normalisation.

**PROPOSITION II.8** (DoRA is LoRA followed by an output scale, at the level of images). Assume  $1 \leq r \leq \min(m, n)$  and  $s \neq 0$ . Let  $N$  normalise rows; DoRA’s magnitude  $\mu \in \mathbb{R}^m$  is per output neuron, as in HF PEFT and the official code (the paper’s notation says “column-wise”, in the transposed convention). Images refer to the merged (eval-mode) weight; HF’s train-mode forward with  $\text{lora\_dropout} > 0$  is not of the form  $Wx$ . Let  $W_0$  have no zero row, let  $\mathcal{U}$  be the open set of matrices with no zero row, and let  $\mathrm{Diag}_m$  be the monoid of all diagonal matrices. Then

$$\mathrm{Im DoRA}_r = \{\mathrm{diag}(\mu) N(V) : V \in (W_0 + \mathcal{M}_{\leq r}) \cap \mathcal{U}, \mu \in \mathbb{R}^m\} = \mathrm{Diag}_m \cdot (W_0 + \mathcal{M}_{\leq r}),$$

which is exactly the image of “LoRA<sub>r</sub> followed by a trainable output scale  $\ell \in \mathbb{R}^m$ ”. With  $\mu$  restricted to  $(\mathbb{R}^\times)^m$ , or equivalently after intersecting with  $\mathcal{U}$ :  $\mathrm{Im DoRA}_r \cap \mathcal{U} = T_m \cdot (W_0 + \mathcal{M}_{\leq r}) \cap \mathcal{U}$ . Moreover:

- $d(\mathrm{DoRA}_r) \leq \min(mn, r(m+n-r) + m)$ . At a parameter point with  $\mu \in (\mathbb{R}^\times)^m$  and  $X = sBA \in \mathcal{M}_r$ , with  $U = \mathrm{col} X$  and  $R = \mathrm{row} X$ , the Jacobian rank is  $r(m+n-r) + m - \dim \ker(\delta \mapsto P_{U^\perp} \mathrm{diag}(\delta) W_0 P_{R^\perp})$ ; where some  $\mu_i = 0$  it is smaller when  $r \geq 2$ , and for  $r = 1$  when some row  $w_{0,i}$  with  $\mu_i = 0$  lies outside  $R$ . For generic  $W_0$ ,  $d = \min(mn, r(m+n-r) + m)$ ; in particular  $d = r(m+n-r) + m$  iff  $(m-r)(n-r) \geq m$ , which holds for all practical shapes. For special  $W_0$  it is smaller: for  $W_0 = ac^\top$  with all  $a_i \neq 0$  and  $r < n$ ,  $d = r(m+n-r) + m - r$ , strictly below the generic value iff  $r < m$  and  $r \leq n - 2$ .
- The image covariance group (Definition II.14) contains  $\mathrm{Mon}_m \times \mathrm{GL}_n$ .
- At  $B = 0$ , in HF’s magnitude coordinates,  $\Delta W = (\mathrm{diag}(\mu_i / \|w_{0,i}\|) - I) W_0$ , which can have rank up to  $\mathrm{rank} W_0$ , so DoRA escapes LoRA’s rank bound.



**Numerical check.** The Jacobian rank is  $9 = mn$  at  $(m, n, r) = (3, 3, 2)$ , where  $r(m + n - r) + m$  gives 11; it is 36 at  $(8, 8, 2)$ , matching, and 34 there when one  $\mu_i = 0$ ; it is 24 for a rank-one  $W_0$  at  $(6, 6, 2)$  (Appendix E).

**Credit.** DoRA is due to Liu et al. (2024b). Its magnitude–direction split follows weight normalisation (Salimans and Kingma, 2016). From a different direction, Zhang et al. (2025g) argue that the magnitude of the update is the main driver of LoRA’s performance.

**WHAT THIS EXPLAINS** At the level of images, DoRA’s “magnitude–direction decoupling” is LoRA plus  $m$  free output gains. What DoRA changes beyond that lives in its training dynamics. Under gradient flow or SGD there are two different geometries. The exact gradient of the normalised parametrisation projects the direction gradient row by row, which is a change of metric. The detached backward of the DoRA paper (Sec. 4.3) and of HF PEFT (`weight_norm.detach()`) removes that projection; the resulting update is a non-gradient pseudo-gradient that no metric describes, and it need not be a descent direction. Both the paper and HF train with AdamW, under which neither variant follows a fixed metric, and the coordinates matter in their own right:  $\mu_i$  starts at  $\|w_{0,i}\|$  and  $\ell_i$  at 1, so an equal step on  $\mu_i$  and on a gain  $\ell_i$  changes the row gain by different amounts.

**Experiment D.** With one optimizer (AdamW, same learning rate, weight decay and data order) and the same  $W_0$ , compare four arms: (i) LoRA + gain  $\ell$  with  $\ell_0 = \mathbf{1}$ ; (ii) LoRA + gain in DoRA’s coordinates,  $W = \text{diag}(\mu_i / \|w_{0,i}\|)(W_0 + sBA)$  with  $\mu_0 = (\|w_{0,i}\|)_i$ , which isolates the optimizer-coordinate effect without normalisation; (iii) DoRA with the exact backward; (iv) DoRA with HF’s detached backward. Since the images are equal, any gap between arms comes from the parametrisation, the backward rule or the optimizer coordinates, and the four arms separate these. The DoRA paper’s own ablation reports (iii) and (iv) within 0.2 points. [Prediction X.2](#) uses the equal images to set up this controlled comparison.

The covariance group  $\text{Mon}_m \times \text{GL}_n$  decides where [Observation II.15](#) applies. One orthogonal rotation of the residual stream of a gain-fused pre-norm model, as in SliceGPT (Ashkboos et al., 2024a), acts on the input side of  $q, k, v$ , gate and up, where DoRA is  $\text{GL}_n$ -covariant, so DoRA on those boxes reaches the same functions after the rotation as before; on  $o$  and down the sufficient condition fails. Training outcomes are a separate question.

### The torus ladder

Write a weight as  $W_0 \odot (J + Z)$ , with  $J$  the all-ones matrix. Output scalings use  $Z = (\ell - \mathbf{1})\mathbf{1}^\top$ , of rank one; two-sided scalings use  $ab^\top - J$ , of rank at most two; HiRA allows any  $Z = BA$  of rank at most  $r$ . These choices form a ladder of images, and no rung can fill a zero of  $W_0$ .

#### PROPOSITION II.9 (the torus ladder).

- (a) **HiRA.**  $\rho(B, A) = W_0 \odot (J + BA)$ , where  $J$  is the all-ones matrix; HF PEFT has  $B_0 = 0$  and no scale factor on Linear and Conv layers (on embeddings it uses  $A_0 = 0$ ). For every  $W_0$ ,  $\text{Im HiRA}_r = W_0 + D_{W_0}(\mathcal{M}_{\leq r})$  with  $D_{W_0} : X \mapsto W_0 \odot X$ . If  $W_0$  has no zero entries,  $D_{W_0} \in \text{GL}(\Theta)$ , so inside the slice at  $\theta_0$  HiRA is LoRA transported by a diagonal automorphism: it has the same  $d = r(m + n - r)$  (for  $r \leq \min(m, n)$ ) and the same gauge. In all cases  $\text{rank } \Delta W \leq r \cdot \text{rank } W_0$ . The transport depends on  $\theta_0$  ([Proposition II.13](#)).
- (b) **Scalings.** Output-side scalings,  $\text{diag}(\ell)W_0 = W_0 \odot (J + (\ell - \mathbf{1})\mathbf{1}^\top)$ , and input-side scalings,  $W_0 \text{diag}(\ell) = W_0 \odot (J + \mathbf{1}(\ell - \mathbf{1})^\top)$ , have images in  $\text{Im HiRA}_1$ . Output-side examples:  $(IA)^3$  on keys and values, the post-linear scale of SSF, DoRA’s magnitude. Input-side examples:



$(\text{IA})^3$ 's feed-forward vector, which scales the input of  $W_{\text{down}}$ , and a LayerNorm gain  $\gamma$  folded into the next linear map. The fold is exact when the norm's output is consumed only by linear maps: the input norms of pre-norm blocks, and a final norm before an untied head (in post-norm BERT, the last block's norm feeds only the head). It does not apply to Gemma-style post-sublayer norms, which HF's default LN-tuning targets for Gemma 2 and 3, to QK-norms, or to post-norm residual paths. Where it applies,  $\gamma$  is tied across all consumers of the norm (Q, K and V, or gate and up) rather than being a free point per box, and with tied embeddings a fold into the head breaks the tie. LayerNorm biases and SSF's shift fold into the next bias, not into  $W$ . Two-sided scalings  $\text{diag}(a)W_0\text{diag}(b) = W_0 \odot (ab^\top)$  lie in  $\text{Im HiRA}_2$ . As a morphism,  $h(\ell) = (\ell - \mathbf{1}, \mathbf{1}^\top)$  lands in  $\text{HiRA}_1$  pointed at  $(0, \mathbf{1}^\top)$ , which is not HiRA's random-A default pointing.

- (c) **Orbit dimension.** The two-sided torus orbit  $\{\text{diag}(a)W_0\text{diag}(b)\}$  has dimension  $m + n - c(W_0)$ , where  $c(W_0)$  is the number of connected components of the bipartite support graph of  $W_0$ , an isolated vertex counting as its own component.
- (d) **RoAd.**  $\text{RoAd}_1$  (the paper's main variant, HF `road_1`) is  $W = RW_0$  with  $R$  block-diagonal in scaled rotations  $\alpha_i \text{Rot}(\theta_i)$  of output pairs. It is the left action of  $\text{CO}^+(2)^{m/2} \cong (\mathbb{C}^\times)^{m/2}$ : it is  $(\text{IA})^3$  over  $\mathbb{C}$ . It multiplies the input Gram matrix of each output pair by  $\alpha_i^2$ . Since  $\alpha_i$  may be 0, its image is the orbit closure, a real-linear space of dimension exactly  $m$  whenever no output-pair block of  $W_0$  is zero.  $\text{RoAd}_2$  and  $\text{RoAd}_4$  (HF `road_2`, `road_4`) let each block range over all of  $M_2(\mathbb{R})$ . They realise the block-diagonal action of the monoid  $M_2(\mathbb{R})^{m/2}$  (its group part  $\text{GL}_2^{m/2}$  is non-abelian), with linear image  $\{\text{blockdiag}(R_i)W_0\}$  of dimension  $2m$  when every pair block has rank 2. For them the Gram statement and “over  $\mathbb{C}$ ” fail, and  $\text{RoAd}_4$  carries a positive-dimensional gauge (two dimensions per row). HF pairs output  $j$  with  $j + g/2$  inside each group of size  $g$  (default 64) rather than adjacent outputs as in the paper, and also rotates the bias; the statements above do not depend on the pairing.
- (e) **Zeros and covariance.** Every  $W$  reachable by HiRA,  $(\text{IA})^3$  or SSF vanishes wherever  $W_0$  does,  $Z(W_0) \subseteq Z(W)$ , and this persists under merging. New zeros can appear ( $\ell_i = 0$ , or  $(J + BA)_{ij} = 0$ ), so the zero pattern itself is not invariant. The image covariance group of output-scaled boxes  $((\text{IA})^3 \text{ on } k, v; \text{ post-linear SSF})$  is  $\text{Mon}_m \times \text{GL}_n$ , and that of input-scaled boxes  $((\text{IA})^3 \text{ on } W_{\text{down}})$  is  $\text{GL}_m \times \text{Mon}_n$ . HiRA's is exactly  $\text{Mon}_m \times \text{Mon}_n (= \text{GL}_1 \times \text{GL}_1 \text{ when } m = n = 1)$ .

**Numerical check.** The  $\text{RoAd}_1$  orbit has dimension 8 and the  $\text{RoAd}_2$  image dimension 16 at  $m = 8$  (Appendix E).

**Credit.**  $(\text{IA})^3$  is due to Liu et al. (2022a), SSF to Lian et al. (2022), HiRA to Huang et al. (2025b), RoAd to Liao and Monz (2024), and LayerNorm tuning to Zhao et al. (2023). Feature-wise scaling as a building block goes back at least to FiLM (Perez et al., 2017).

**WHAT THIS EXPLAINS** Each scaling method here is a rung  $W_0 \odot (J + Z)$ :  $(\text{IA})^3$ , SSF's scales and the gain part of LN-tuning on norms that feed only linear maps lie in  $\text{HiRA}_1$ , on one side or the other, and HiRA is LoRA in the Hadamard frame of  $W_0$ . DoRA's image is the  $\text{Diag}_m$ -orbit of LoRA's image (image level only; Proposition II.8). DoRA and RoAd can fill zeros of  $W_0$ , so they become rungs only when  $W_0$  is zero-free. Then every image germ can be written as  $W_0 \odot (J + Z)$ , so the ladder by itself constrains nothing there and serves as bookkeeping.  $\text{RoAd}_1$  is complex  $(\text{IA})^3$  and  $\text{RoAd}_{2,4}$  are block- $M_2(\mathbb{R})$   $(\text{IA})^3$ ; for generic  $W_0$  neither lies on a low-rank rung. HiRA's “high rank” is a property of a frame: its transport  $D_{W_0}$  depends on  $\theta_0$  and does not preserve rank. Two re-based cycles reach  $W_0 \odot (J + Z_1) \odot (J + Z_2)$ , whose

relative update  $Z_1 + Z_2 + Z_1 \odot Z_2$  has rank up to  $\min(2r + r^2, m, n)$ , against  $2r$  for LoRA (Proposition II.13).

With HF defaults, (IA)<sup>3</sup> scales the output side of the attention projections it targets ( $k$  and  $v$ , or  $q$  and  $v$ ) and the input side of down, so the residual rotation above, acting on the other side, leaves its reachable functions unchanged. HiRA's covariance group stops at  $\text{Mon}_m \times \text{Mon}_n$  and fails the sufficient condition of Observation II.15. That alone does not prove that its reachable functions change; Conjecture A (stated after Observation II.15) says they do, and Prediction X.4 is the test.

Corollary II.6, Theorem II.7 and Propositions II.8 and II.9 were repaired after two rounds of adversarial review; the referee record in §XI lists what changed.

The groups of this exposé see orbits. LoRA's cone is governed instead by the rank of the update, read off a cut in a tensor network, while the size of an image is a dimension, which behaves like a volume. Rank and volume can move in opposite directions, the subject of §II.C.

## II.C Cut and volume

*Rank lives on cuts and dimension lives in volumes.  
Count the wrong one and you will compare the wrong things.  
a working maxim*

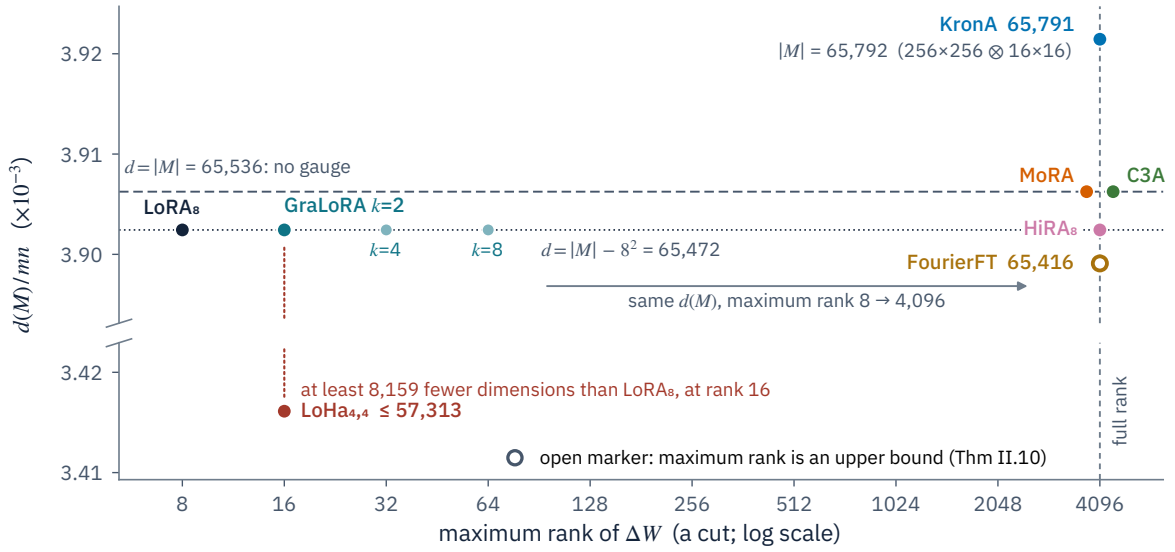
At a fixed budget, the narrowest cut of a method's diagram bounds the rank it can reach. The dimension of what it can reach is a separate count, and the two can move in opposite directions.

### In brief.

- A method's update rank is bounded by the narrowest cut of its tensor-network diagram. The dimension  $d(M)$  of what it can reach is its parameter count minus the directions that change nothing.
- The two can move in opposite directions. At the budget of LoRA<sub>8</sub> on a  $4096 \times 4096$  layer, LoHa<sub>4,4</sub> reaches rank 16 and at least 8,159 fewer dimensions; GraLoRA with  $k = 2$  reaches rank 16 at LoRA's dimension, with every block capped at rank 4.
- Kronecker sums are LoRA across another cut of the same four-leg tensor, and HiRA is LoRA after  $X \mapsto W_0 \odot X$  (for  $W_0$  without zero entries). Such changes of frame keep dimension and move rank.
- LoRA respects every change of a layer's coordinates and so reaches the same set of functions after a function-preserving rotation of the network; PiSSA, MiLoRA and CorDA splits respect only rotations and uniform scalings.

PEFT papers usually report a method's expressiveness as a rank and its cost as a parameter count. A third number, the dimension of the set of weights the method can reach, usually goes unreported.

Draw a method as a network of boxes and wires. The rank of its update is limited by the narrowest place where the network can be cut in two, so rank is a *cut* invariant. The dimension counts the independent directions in which the method can move the weights; we call it a *volume* invariant, loosely, since no measure is computed. In Figure 4, several methods raise one while lowering the other.



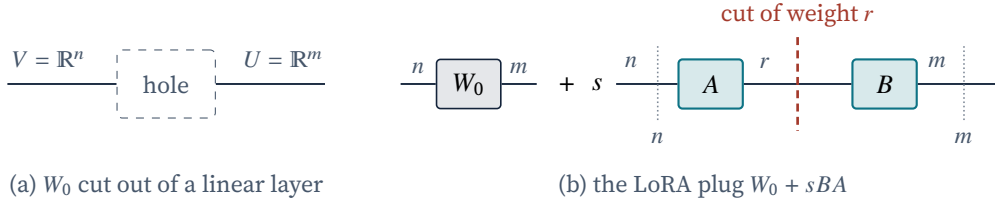
**Figure 4. Rank is not volume** (Theorem II.10, Proposition II.12). One  $4096 \times 4096$  layer at the budget of  $\text{LoRA}_8$  (Hu et al., 2021),  $|M| = 8(m + n) = 65,536$ . Each method is placed by the largest rank of  $\Delta W$  it can reach (horizontal, logarithmic scale), which is bounded by the narrowest cut of its diagram (Theorem II.10), and by its expressive dimension as a fraction of the layer,  $d(M)/(mn)$  (vertical, in units of  $10^{-3}$ , with the axis broken between 3.41 and 3.90), from the gauge census of Proposition II.12. The dashed line is  $d = |M|$ , a method without gauge, and the dotted line is  $d = |M| - 8^2 = 65,472$ , the dimension of  $\text{LoRA}_8$  with its  $\text{GL}_8$  gauge. Along the dotted line the maximum rank runs from 8 for  $\text{LoRA}_8$ , through  $\text{GraLoRA}$  (Jung et al., 2025) with  $k = 2, 4$  and  $8$  blocks (the last two light), to full rank for  $\text{HiRA}_8$  (Huang et al., 2025b).  $\text{MoRA}$  (Jiang et al., 2024) and  $\text{C3A}$  (Chen et al., 2024a) have no gauge and reach full rank; their points coincide and are drawn on either side of the full-rank line.  $\text{FourierFT}$  (Gao et al., 2024) loses one dimension for each sampled conjugate pair, and its maximum rank is known only through a bound (open marker).  $\text{KronA}$  (Edalati et al., 2022) is shown at the factor shapes closest to the budget.  $\text{LoHa}_{4,4}$  (Hyeon-Woo et al., 2021), below the break, sits at the proved upper bound of its dimension: it reaches rank 16 with at least 8,159 fewer dimensions than  $\text{LoRA}_8$  (Proposition II.12(c)). The parameters behind each point are in §E.6.

### Rank is a cut

In  $\text{LoRA}$  every signal crosses the inner wire of dimension  $r$  between  $A$  and  $B$ , so  $\Delta W = BA$  has rank at most  $r$  (Figure 5). The argument works for any method drawn as a tensor network (Background B.7), a string diagram (Background B.3) in the symmetric monoidal category  $(\mathbf{FinVect}, \otimes)$ . Split the boxes into an input side and an output side, and the network factors through the wires that cross.

**THEOREM II.10** (cut-rank bound; standard). *Let  $\Gamma$  be a tensor network, evaluated in the symmetric monoidal category  $(\mathbf{FinVect}, \otimes)$ , with input legs  $V$  and output legs  $U$ . Every wire and open leg is assumed incident to a vertex: insert an identity box on any bare strand, or equivalently count each bare input–output wire as crossing every cut. A crossing hyperedge (copy spider) contributes its dimension once. Partition the vertices into  $I \sqcup O$ . The weight  $w$  of the cut is the product of the dimensions of the internal wires between the two sides and of the open legs attached on the wrong side (input legs at  $O$ , output legs at  $I$ ). Then*

$$\text{rank}[\Gamma] \leq \min_{\text{cuts}} w,$$



**Figure 5. LoRA as a plug.** (a) Cutting  $W_0$  out of a linear layer  $y = W_0x$  leaves a hole  $V \rightarrow U$ , with  $V = \mathbb{R}^n$  and  $U = \mathbb{R}^m$ . (b) LoRA fills the hole with the frozen box  $W_0$  (grey) plus the trainable chain  $sBA$  (teal), a formal sum, over which rank is subadditive. The dashed cut puts  $A$  on the input side and  $B$  on the output side and crosses one wire of dimension  $r$ , so  $\text{rank}(sBA) \leq r$  (Theorem II.10). The dotted cuts put both boxes on one side and cross an open leg, with weights  $n$  and  $m$ ; the minimum over cuts is  $\min(n, r, m)$ .

and rank is subadditive over formal sums. The bound refers to a chosen presentation of the network.

**Numerical check.** No violation in 283 random networks (Appendix E).

**Credit.** String diagrams go back to Joyal and Street (1991), and the copy spider (Background B.6) to the Frobenius algebra of a basis (Coecke et al., 2013). Cut bounds on rank are standard tensor-network reasoning (Background B.8; Cichocki et al., 2016; Kolda and Bader, 2009). The PEFT reading is ours.

Without the convention on bare strands,  $I_d \otimes BA$  would violate the bound. The bound also depends on the drawing. A naive expansion of HRA’s product of reflections has  $2^r - 1$  terms, while the telescoping presentation

$$R - I = \sum_{k=1}^r (H_k - I) H_{k+1} \cdots H_r$$

writes  $R - I$  as  $r$  terms of rank at most one and gives the sharp bound  $r$ .

**Consequences.** For one box, Theorem II.10 gives the bounds in Table 8. Relative to  $W_0$ , splitting initialisations give  $\Delta W = BA - B_0A_0$ , of rank at most  $2r$  (Theorem II.1). The Givens bound uses the same telescoping presentation as HRA, each gate contributing a term of rank at most two, and HiRA’s bound factors  $W_0$  through a bond of dimension rank  $W_0$ .

**WHAT THIS EXPLAINS** A large min-cut is necessary for high rank and does not guarantee it. The “high-rank PEFT” literature consists of networks whose min-cut avoids an  $r$ -dimensional bottleneck. Whether the bound is attained is a separate question, and attaining it can cost volume.

### Kronecker adapters cut the other way

If  $m = m_1m_2$  and  $n = n_1n_2$ , an  $m \times n$  matrix is a tensor with output legs  $U_1, U_2$  and input legs  $V_1, V_2$  (Background B.10). LoRA’s rank is the rank across  $\{U_1U_2\} \mid \{V_1V_2\}$ ; a sum of  $k$  Kronecker products has rank at most  $k$  across  $\{U_1V_1\} \mid \{U_2V_2\}$ . The two cuts disagree sharply, since  $I \otimes I$  is one Kronecker term of full rank.

**THEOREM II.11** (Kronecker incomparability; the linear algebra is standard, the PEFT reading is new). Write  $\mathbb{R}^m = \mathbb{R}^{m_1} \otimes \mathbb{R}^{m_2}$  and  $\mathbb{R}^n = \mathbb{R}^{n_1} \otimes \mathbb{R}^{n_2}$ , and view  $\Delta \in \mathbb{R}^{m \times n}$  as a 4-leg tensor. Let  $\text{rk}$  be the rank across the cut  $\{U_1U_2\} \mid \{V_1V_2\}$  and  $\text{rk}_\otimes$  the rank across  $\{U_1V_1\} \mid \{U_2V_2\}$ .

**Table 8.** Cut bounds of [Theorem II.10](#) on rank  $\Delta W$  for one  $m \times n$  box, with the presentation each bound uses.

| method                                                                                                                                                | presentation used                                             | bound on rank $\Delta W$  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|---------------------------|
| LoRA family, BA ( <a href="#">Hu et al., 2021</a> )                                                                                                   | one bond of dimension $r$                                     | $r$                       |
| split initialisations (PiSSA type), relative to $W_0$                                                                                                 | $\Delta W = BA - B_0 A_0$ , two bonds                         | $2r$                      |
| HRA ( <a href="#">Yuan et al., 2024</a> )                                                                                                             | telescoping sum $R - I = \sum_k (H_k - I) H_{k+1} \cdots H_r$ | $r$                       |
| Givens circuit with $g$ gates ( <a href="#">Ma et al., 2024</a> )                                                                                     | the same telescoping sum                                      | $2g$                      |
| ReLoRA after $K$ phases ( <a href="#">Lialin et al., 2023b</a> )                                                                                      | formal sum of $K$ LoRAs                                       | $Kr$                      |
| GraLoRA ( <a href="#">Jung et al., 2025</a> )                                                                                                         | sum of $k^2$ block LoRAs                                      | $kr$                      |
| LoHa ( <a href="#">Hyeon-Woo et al., 2021</a> )                                                                                                       | Hadamard product through copy spiders                         | $r_1 r_2$                 |
| HiRA ( <a href="#">Huang et al., 2025b</a> )                                                                                                          | $W_0$ factored through a bond of dimension rank $W_0$         | $r \text{ rank } W_0$     |
| LoKr ( <a href="#">Yeh et al., 2023</a> )                                                                                                             | $C \otimes BA$ , inner rank $r'$                              | $\text{rank } C \cdot r'$ |
| FourierFT with $n_c$ coefficients ( <a href="#">Gao et al., 2024</a> )                                                                                | sum of real cosine patterns, each of rank $\leq 2$            | $\min(2n_c, m, n)$        |
| MoRA (reshape) ( <a href="#">Jiang et al., 2024</a> ), C3A ( <a href="#">Chen et al., 2024a</a> ), MPO adapters ( <a href="#">Liu et al., 2021a</a> ) | no narrow cut                                                 | up to full rank           |

Put

$$\kappa_1 := \min(m_1, m_2) \cdot \min(n_1, n_2), \quad \nu := \min(m_1, n_1) \cdot \min(m_2, n_2).$$

- (a)  $\text{rk}_{\otimes} \Delta \leq k$  iff  $\Delta = \sum_{i \leq k} C_i \otimes D_i$ . Every  $\Delta$  has  $\text{rk}_{\otimes} \Delta \leq \min(m_1 n_1, m_2 n_2)$ .
- (b)  $\text{rk}(C \otimes D) = \text{rk } C \cdot \text{rk } D$ , while  $\text{rk}_{\otimes}(C \otimes D) = 1$  for  $C, D \neq 0$ . So  $\nu$  is the largest rank of a single Kronecker product.
- (c) If  $u \in \mathbb{R}^m$  and  $v \in \mathbb{R}^n$  have Schmidt ranks  $\rho_u, \rho_v$  ([Background B.12](#)), then  $\text{rk}_{\otimes}(uv^\top) = \rho_u \rho_v$ . A generic rank-one update therefore has Kronecker rank  $\kappa_1$ . For aligned splits,  $(m_1 - m_2)(n_1 - n_2) \geq 0$ , which include equal square splits and every split HF PEFT's factorisation produces (it puts the smaller factor first on both sides),  $\kappa_1 = \min(m_1 n_1, m_2 n_2)$  is the maximal Kronecker rank.
- (d) Incomparability. For  $1 \leq r < \nu$  and  $1 \leq k < \kappa_1$ , the sets  $\mathcal{M}_{\leq r}$  and  $\mathcal{K}_{\leq k} := \{\Delta : \text{rk}_{\otimes} \Delta \leq k\}$  of unconstrained  $k$ -term Kronecker sums are incomparable, for every split. A generic rank-one update lies in the first and not the second, and  $C \otimes D$  with full-rank factors, of Kronecker rank 1 and rank  $\nu$ , lies in the second and not the first. For equal square splits ( $m = n = N$ , all factors  $s$ ),  $\kappa_1 = \nu = s^2 = N$ , and  $I_N$  is such a product.
- (e) Constrained Kronecker methods. Let  $\text{Im } M \subseteq W_0 + \mathcal{K}_{\leq k}$ . Then  $\text{Im LoRA}_r \not\subseteq \text{Im } M$  whenever  $k < \kappa_1$ , for every pointing of  $\text{LoRA}_r$ . Give  $\text{LoRA}_r$  its zero-init pointing, with image  $W_0 + \mathcal{M}_{\leq r}$ . Then  $\text{Im } M \not\subseteq \text{Im LoRA}_r$  iff some  $\theta \in \text{Im } M$  has  $\text{rank}(\theta - W_0) > r$ . For LoKr,  $\Delta W = C \otimes BA$  with  $C \in \mathbb{R}^{m_1 \times n_1}$ ,  $B \in \mathbb{R}^{m_2 \times r'}$  and  $A \in \mathbb{R}^{r' \times n_2}$ , put  $r'_{\text{eff}} = \min(r', m_2, n_2)$ . The maximum rank is  $\min(m_1, n_1) \cdot r'_{\text{eff}}$ , attained. So  $\text{Im LoKr}_{r'} \subseteq \text{Im LoRA}_r$  iff  $r \geq \min(m_1, n_1) r'_{\text{eff}}$ , and otherwise the two images are incomparable, provided  $\kappa_1 > 1$ . In HF PEFT the second LoKr factor is low-rank only when its rank parameter satisfies  $r' < \max(m_2, n_2)/2$ ; otherwise it is full, and LoKr is unconstrained KronA, with maximum rank  $\nu = \min(m_1, n_1) \min(m_2, n_2)$ . On a 768-wide layer ( $24 \otimes 32$ ) the maximum rank therefore jumps from 360 at  $r' = 15$  to 768 at  $r' = 16$ . With `decompose_both`,  $C$  is also capped at rank  $r'$ .



**Numerical check.** Schmidt ranks  $(1, 3) \rightarrow 3$ ,  $(2, 3) \rightarrow 6$ ,  $(3, 3) \rightarrow 9$ , each with ordinary rank 1. For the  $2 \times 4$  split of  $\mathbb{R}^8$ ,  $\text{rk}_{\otimes}(uv^{\top}) = 4$ , the maximal Kronecker rank there. LoKr with  $s = 4$ ,  $r' = 1$  never exceeds rank 4, and at  $(m_1, m_2, n_1, n_2) = (2, 8, 8, 2)$  with inner dimension 4, LoKr never exceeds rank  $4 = \min(m_1, n_1) r'_{\text{eff}}$  (Appendix E).

**Credit.** The rearrangement  $\mathcal{R}(C \otimes D) = \text{vec } C \text{ vec } D^{\top}$  (Background B.11) and the nearest-Kronecker-product problem are due to Van Loan and Pitsianis (1993). Kronecker-sum layers entered deep learning with PHM layers (Zhang et al., 2021) and Compacter (Mahabadi et al., 2021a); KronA (Edalati et al., 2022) and LoKr (Yeh et al., 2023) brought them to weight updates.

**WHAT THIS EXPLAINS** Unconstrained Kronecker sums (KronA with full factors, sums of Kronecker terms) are LoRA across a different cut of the same 4-leg tensor, so calling them higher-rank LoRAs mixes two cuts. With aligned splits, a generic rank-one update needs the maximal number of Kronecker terms, and with fewer than  $\kappa_1$  terms a Kronecker sum misses it. These methods also rest on a reshape that the network does not have. Constrained Kronecker methods behave differently. LoKr with inner rank  $r'$  lies inside a LoRA image only when the LoRA rank is at least  $\min(m_1, n_1) r'$ , about  $\sqrt{N} r'$ : 192 on 768-wide and 512 on 4096-wide layers at HF’s default  $r' = 8$ . At practical ranks, and at equal parameter budget, LoKr and LoRA<sub>r</sub> are incomparable (e).

Compacter’s PHM weights are Kronecker-structured weights inside a nonlinear bottleneck adapter and do not update  $W_0$ , and MPO adapters (Liu et al., 2021a) are a multi-core reparametrisation that is lossless at full bond dimension. The image comparison does not transfer to either without more work.

**Prediction.** On targets  $W_0 + uv^{\top}$  with generic  $u, v$ , and a loss whose only minimiser is the target (for instance regression on full-rank inputs), LoRA<sub>1</sub> reaches zero loss, while no Kronecker sum with fewer than  $\kappa_1$  terms does. HF PEFT’s balanced factorisation is non-square for many widths ( $768 \rightarrow 24 \times 32$  with  $\kappa_1 = 576$ , the maximal Kronecker rank;  $2048 \rightarrow 32 \times 64$ ;  $5120 \rightarrow 64 \times 80$ ) but square for 1024 and 4096 ( $32 \times 32$ ,  $64 \times 64$ ), where  $\kappa_1 = N$ . At an equalised budget neither family should dominate the other across tasks.

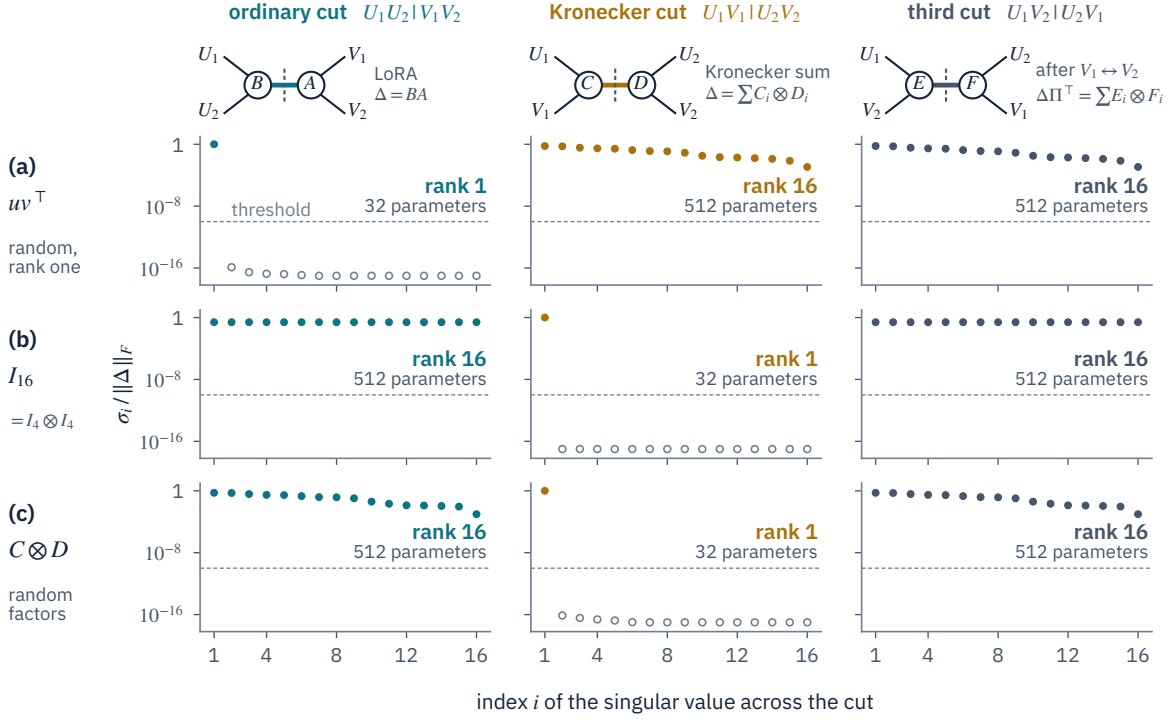
Figure 6 makes the three cuts of a  $16 \times 16$  matrix concrete. On the  $4 \otimes 4$  split, a generic  $uv^{\top}$  has rank 1 and Kronecker rank 16. LoRA<sub>1</sub> fits it with 32 parameters, and a Kronecker sum needs 16 terms and 512 parameters. For  $I_{16} = I_4 \otimes I_4$  the costs trade places.

### Dimension is a volume

Moving  $q \in Q$  along a fibre of  $\rho$  changes nothing. For LoRA,  $(Bg^{-1}, gA)$  gives the same weight as  $(B, A)$  for every  $g \in \text{GL}_r$ , so  $r^2$  of the  $r(m+n)$  parameters are spent on a symmetry and the image has dimension  $r(m+n-r)$ . This is the expressive dimension  $d(M)$  of Definition I.5, and  $|M| - d(M)$  is the gauge waste. “Gauge” means the symmetry group of a parametrisation (Background D.21). There is no connection, and away from the full-rank stratum the fibres jump in dimension.

**Credit.** The quotient  $(\mathbb{R}_*^{m \times r} \times \mathbb{R}_*^{r \times n})/\text{GL}_r$  is the classical geometry of fixed-rank matrices (Mishra et al., 2012). Putterman et al. (2024) build learners on LoRA weights that respect this  $\text{GL}_r$  symmetry.

The census below covers LoRA (Hu et al., 2021), GraLoRA (Jung et al., 2025), LoHa (Hyeon-Woo et al., 2021), HRA (Yuan et al., 2024), DoRA (Liu et al., 2024b), VeRA (Kopiczko et al., 2023), KronA (Edalati et al., 2022), tensor-train and Tucker adapters (Background B.9), and



**Figure 6. Cut the tensor (Theorem II.11).** A  $16 \times 16$  matrix  $\Delta$  is read as a four-leg tensor with output legs  $U_1, U_2$  and input legs  $V_1, V_2$ , through  $\mathbb{R}^{16} = \mathbb{R}^4 \otimes \mathbb{R}^4$  on both sides. The legs can be split into two pairs in three ways (columns): the ordinary cut  $\{U_1U_2\} | \{V_1V_2\}$ , across which LoRA factors as  $\Delta = BA$ ; the Kronecker cut  $\{U_1V_1\} | \{U_2V_2\}$ , across which a sum of Kronecker products  $\sum_i C_i \otimes D_i$  factors after the rearrangement of Van Loan and Pitsianis (1993); and the third cut  $\{U_1V_2\} | \{U_2V_1\}$ , the Kronecker cut with the two input legs swapped. The glyph above each column draws that factorisation as two boxes joined by a bond across the cut. Each panel shows the operator-Schmidt spectrum across one cut, the singular values  $\sigma_i$  of the corresponding  $16 \times 16$  reshape of  $\Delta$  divided by  $\|\Delta\|_F$ , on a logarithmic scale, for (a) a random rank-one update  $uv^\top$ , (b) the identity  $I_{16} = I_4 \otimes I_4$ , and (c)  $C \otimes D$  with random  $4 \times 4$  factors. Filled markers lie above the rank threshold (dashed line), and open markers are numerically zero. The ranks are (1, 16, 16) for  $uv^\top$  and (16, 1, 16) for  $I_{16}$  and for  $C \otimes D$ . So a generic rank-one update has the maximal Kronecker rank  $\kappa_1 = 16$ , and a Kronecker product of full-rank factors has the maximal rank  $\nu = 16$ , which makes zero-init LoRA<sub>r</sub> and  $k$ -term Kronecker sums incomparable for  $r, k < 16$  (Theorem II.11(b)–(d)). LoRA<sub>1</sub> fits  $uv^\top$  with 32 parameters where a Kronecker sum needs 16 terms and 512 parameters, and for  $I_{16}$  and  $C \otimes D$  the two costs trade places.

the linear methods LoRA-FA (Zhang et al., 2023d), LoRA-XS (Bałazy et al., 2024), MiSS (Kang and Yin, 2024), FourierFT (Gao et al., 2024), C3A (Chen et al., 2024a) and masks.

**PROPOSITION II.12** (gauge census: rank is a cut, dimension is a volume).

- (a)  $d(M) \leq |M| - \dim(\text{generic gauge orbit})$ , with equality when the gauge acts locally transitively on generic fibres.
- (b) The values in Table 9 hold for one  $m \times n$  box, with  $r \leq \min(m, n)$  throughout except for VeRA (whose published RoBERTa-base setting has  $r = 1024 > 768$ ). In the LoHa row,

$$F = r_1(m + n - r_1) + r_2(m + n - r_2) - (m + n - 1)$$

is a proved upper bound for  $d$ . The gauge column lists the exhibited Lie group. It acts locally transitively on generic fibres away from the  $mn$  cap, but not where the cap binds; for LoHa at

$(m, n, r_1, r_2) = (4, 4, 2, 2)$  the orbit has dimension 15 and the fibre 16. BitFit does not touch the  $m \times n$  box; it is linear in the biases. Hybrid GraLoRA ( $\text{hybrid}_r > 0$ ) is not covered by the table. FourierFT is injective only when its sampled frequencies contain no conjugate pair  $(u, v), (-u, -v)$ ; otherwise  $d = |M| - \#\{\text{pairs}\}$ . At LLM sizes this is close to  $|M|$ , but at  $768 \times 768$  with 1000 frequencies a pair occurs in about half of the draws (HF's default seed 777 gives exactly two pairs there).

- (c) Equal-budget trades. Assume  $2r \leq \min(m, n)$ ,  $r^2 \leq \min(m, n)$  and  $2r^2 < m + n - 1$ .  $\text{LoHa}_{r,r}$  and  $\text{LoRA}_{2r}$  both cost  $2r(m + n)$ , but LoHa reaches at least  $(m + n - 1) - 2r^2$  fewer dimensions (exactly that many when  $F$  is attained [Num]). For  $r \geq 3$  it gains rank up to  $r^2 > 2r$  in exchange. For  $r \leq 2$ ,  $r^2 \leq 2r$ , so  $\text{Im } \text{LoHa}_{r,r} \subseteq \text{Im } \text{LoRA}_{2r}$  and LoHa is strictly dominated. When  $2r^2 \geq m + n - 1$  (still with  $2r \leq \min(m, n)$ ) the bound no longer separates them, and wherever  $F$  is attained  $\text{LoHa}_{r,r}$  matches or exceeds  $\text{LoRA}_{2r}$  in dimension as well as in rank, for instance at  $(m, n, r) = (9, 9, 3)$ , at  $r = 64$  on  $4096 \times 4096$ , and at LyCORIS-style  $r = 32$  on 320- to 640-wide layers.  $\text{GraLoRA}_k$  costs the same as  $\text{LoRA}_r$ , reaches the same number of dimensions, and has maximum rank  $\min(kr, m, n)$ , but caps every block at rank  $r/k$ . A generic rank- $r$  update has rank  $r$  on every block, so for  $k \geq 2$  and  $kr \leq \min(m, n)$  the images of  $\text{LoRA}_r$  and  $\text{GraLoRA}_k$  are incomparable.

**Numerical check.** LoHa dimensions 38, 59, 93, 101 at  $(m, n, r) = (8, 7, 2), (12, 10, 2), (12, 10, 3), (20, 16, 2)$ , and  $16 = mn$  at  $(4, 4, 2)$ , where the formula gives 17 and  $d(\text{LoRA}_4) = 16$  too.  $\text{LoHa}_{3,3}$  against  $\text{LoRA}_6$ : 73 against 72 at  $(9, 9, 3)$ . GraLoRA dimensions 48, 64, 108 at  $(m, n, r, k) = (8, 8, 4, 2), (12, 8, 4, 2), (12, 12, 6, 3)$ ; GraLoRA generic rank  $8 = kr$  at  $(16, 16, 4, 2)$  and  $12 = \min(m, n)$  at  $(12, 12, 6, 3)$  (Appendix E).

**Credit.** LoHa is the low-rank Hadamard product of FedPara (Hyeon-Woo et al., 2021); GraLoRA is due to Jung et al. (2025). The Tucker gauge and dimension are due to Koch and Lubich (2010) and Uschmajew and Vandereycken (2013), and the TT case to Holtz et al. (2012), building on Oseledets (2011). The intrinsic-dimension experiments of Li et al. (2018) and Aghajanyan et al. (2020) measure how many random directions a task needs;  $d(M)$  counts the directions a method's image has, a complementary notion.

$\text{LoHa}_{r,r}$  maps into  $\text{LoRA}_{2r}$  through the face-splitting product (Background B.13), one of the arrow rules of Exposé VIII; for  $r \leq 2$  this places  $\text{LoHa}_{r,r}$  inside  $\text{LoRA}_{2r}$ .

**WHAT THIS EXPLAINS** Rank is a *cut* invariant and dimension a *volume* invariant, and the literature conflates them. When  $r \geq 3$  and  $2r^2$  is small next to  $m + n$ , as on LLM projections at the usual ranks, LoHa reaches high rank on a thin, curved subvariety, because its Hadamard product spends a whole torus on gauge. When  $2r^2 \geq m + n - 1$ , as in LyCORIS's usual Stable Diffusion settings, it matches or exceeds  $\text{LoRA}_{2r}$  in dimension too. GraLoRA keeps LoRA's dimension and multiplies the maximum rank by  $k$ , but caps every block at rank  $r/k$ . It moves the cut, and the block caps pay for the higher maximum rank.

### Transport of structure

An invertible linear change of frame fixing  $\theta_0$  carries dimension, gauge, first-order image and degree along, since they are defined from  $\rho$  and its differential (Background A.17). It carries rank along only when it respects rows and columns, which the Kronecker rearrangement, Fourier and wavelet frames and the Hadamard rescaling by  $W_0$  do not.

**Table 9.** The gauge census of [Proposition II.12\(b\)](#) for one  $m \times n$  box: budget, exhibited Lie gauge with the dimension of its generic orbit, expressive dimension and maximum rank. [Num] marks values checked as float64 Jacobian ranks ([Appendix E](#));  $F$  is defined in [Proposition II.12\(b\)](#).

| method                                                                                                          | budget $ M $        | exhibited Lie gauge (orbit dim)                                                                                                                                    | $d(M)$                                                                          | max rank                              |
|-----------------------------------------------------------------------------------------------------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------|
| LoRA <sub><math>r</math></sub>                                                                                  | $r(m+n)$            | $\mathrm{GL}_r(r^2)$                                                                                                                                               | $r(m+n-r)$                                                                      | $r$                                   |
| GraLoRA ( $k \times k$ blocks, rank $r/k$ each; HF default hybrid_r=0)                                          | $r(m+n)$            | $\prod \mathrm{GL}_{r/k}(r^2)$                                                                                                                                     | $r(m+n-r)$                                                                      | $\min(kr, m, n)$                      |
| LoHa <sub><math>r_1, r_2</math></sub>                                                                           | $(r_1 + r_2)(m+n)$  | $\mathrm{GL}_{r_1} \times \mathrm{GL}_{r_2} \times \text{diagonal torus } (r_1^2 + r_2^2 + m + n - 1)$                                                             | $\min(mn, F)$ [Num]; $F$ is a proved upper bound                                | $\min(r_1 r_2, m, n)$                 |
| HRA <sub><math>r</math></sub> (rank $W_0 = n$ , apply_GS=False; even $r$ is needed only for HF's identity init) | $rn$                | non-linear; fibre dimension $r(r+1)/2$ ; exhibited Lie gauge $(\mathbb{R}^\times)^r$                                                                               | $rn - r(r+1)/2$ (for $r \in \{n-1, n\}$ this is $\dim \mathrm{SO}(n)$ [Num])    | $r$                                   |
| DoRA <sub><math>r</math></sub>                                                                                  | $r(m+n) + m$        | $\mathrm{GL}_r$ for generic $W_0$ with $(m-r)(n-r) \geq m$ ; larger otherwise (dimension $6 > r^2$ at $(m, n, r) = (3, 3, 2)$ , and $r^2 + r$ for rank-one $W_0$ ) | $\min(mn, r(m+n-r) + m)$ for generic $W_0$ ( <a href="#">Proposition II.8</a> ) | up to full                            |
| VeRA                                                                                                            | $m+r$               | $\mathbb{R}^\times$                                                                                                                                                | $m+r-1$ [Num]                                                                   | $\min(r, m, n)$                       |
| KronA $A \otimes B$                                                                                             | $m_1 n_1 + m_2 n_2$ | $\mathbb{R}^\times$                                                                                                                                                | $m_1 n_1 + m_2 n_2 - 1$                                                         | $\min(m_1, n_1) \cdot \min(m_2, n_2)$ |
| TT / Tucker (admissible, minimal ranks)                                                                         | $\Sigma$ cores      | $\prod_e \mathrm{GL}_{r_e}$                                                                                                                                        | $ M  - \sum_e r_e^2$                                                            | up to full                            |
| linear methods (LoRA-FA, LoRA-XS, MiSS in every HF mode, FourierFT, C3A, masks)                                 | $ M $               | translations by the kernel                                                                                                                                         | rank of the linear map; $=  M $ when injective                                  | varies                                |

**PROPOSITION II.13** (transport of structure). For  $T \in \mathrm{GL}(\Theta)$  put  $\tau_T(\theta) = \theta_0 + T(\theta - \theta_0)$  and  $T_*M := (Q, q_0, \tau_T \circ \rho)$ .

- $T_*$  is a strict automorphism of  $\mathbf{Meth}(\Theta, \theta_0)$ , with inverse  $(T^{-1})_*$ . It maps  $S_1$  to  $T S_1$  and preserves  $d$ ,  $g$ , the gauge group, the first-order deficit, polynomial degree, the regular or apex type and the germ type of the image. It does not preserve rank, and, for the family extension below, it does not preserve the covariance group.
- $K^{T_*M} = T K^M T^\top$  for every  $T$ . For any update rule that queries the loss only through  $q \mapsto L(\rho(q))$ , with its own state on  $Q$ ,  $T_*M$  trained on  $L$  and  $M$  trained on  $L \circ \tau_T$  have identical  $q$ -trajectories, so their  $\theta$ -trajectories are conjugate by  $\tau_T$ . Rules that read  $\nabla_\theta L$  and solve a least-squares problem on  $\Theta$  (LoRA-Pro), and  $\theta$ -level weight decay or clipping, are covered only for orthogonal  $T$ . Orthogonality of  $T$  is exactly the condition for  $\tau_T$  to be a Frobenius isometry; for rules of the first kind it is needed only to keep the Frobenius spectrum of  $K$ , or an isometry-invariant loss class such as  $\|\theta - \theta^*\|^2$ , unchanged.

- Coordinatewise optimizers (SGD, Adam) are also equivariant under coordinate permutations of  $Q$ . The Kronecker identification below needs this, because it composes  $\mathcal{R}_*^{-1}$  with the vec reshape of  $Q$ .
- Families and rebasing. For  $T$  independent of the base point, define the family extension  $(T_*M)(\theta) := (Q, q_0, \theta + T(\rho_\theta - \theta))$ . For an additive family with fixed shape  $C$  (Definition I.2),  $T_*M$  is additive with shape  $TC$ , so transport commutes with rebasing:  $R_K(T_*M) = \tau_T(R_K(M))$  (Theorem V.3(a)). For action-type families this can fail even for  $T = 2I$ , because the transported family acts by  $I + T(g - I)$ . For OFT on  $\Theta = \mathbb{R}^{1 \times 2}$ , two cycles of  $\theta \mapsto \theta(2R - I)$  with  $R = \text{rot}(\pi)$  reach  $9\theta_0$ , outside the transported circle. Transports that depend on  $\theta_0$  may fail (HiRA below) or may not (LoRA transported by the SVD frame of  $\theta_0$ , since  $M_{\leq r}$  is  $O(m) \times O(n)$ -invariant). The covariance group (Definition II.14) is a property of a family and is compared through this extension.

Instances.

- Kronecker family. An isomorphism of slices over  $\Theta' = \mathbb{R}^{m_1 n_1 \times m_2 n_2}$  and over  $\Theta: \sum_{i \leq k} C_i \otimes D_i = \mathcal{R}_*^{-1} \text{LoRA}_k$ , with  $Q$  reshaped by vec, dynamics included for coordinatewise optimizers. KronA with unconstrained factors is  $\mathcal{R}_*^{-1} \text{LoRA}_1$ . LoKr is  $\mathcal{R}_*^{-1}$  of  $\text{LoRA}_1$  with the second factor constrained to  $\text{vec } M_{\leq r'}$ , when  $r' < \max(m_2, n_2)/2$  (HF's switch); otherwise HF's second factor is full and LoKr is KronA. LoKr's `decompose_both` option also constrains the first factor.
- Linear methods as coordinate subspaces in transported frames.
  - FourierFT: sparse in the real cosine half of the real Fourier frame. With scaling  $c$  (default 150), HF's update  $\Delta W = c \text{Re}(\text{ifft2}(S))$  is the transport of a mask, with  $g = 0$ , only when the sampled index set has no conjugate pair  $(u, v), (-u, -v)$ ; otherwise  $g$  is the number of such pairs. Its image lies in the point-symmetric subspace  $\Delta W_{j,l} = \Delta W_{-j,-l}$ , and the frame is orthogonal only up to scale (self-conjugate frequencies have twice the squared norm).
  - WaveFT: sparse in a wavelet frame, when HF's pad-IDWT-crop map is invertible (plausible for Haar on even shapes; unverified for other wavelet families).
  - C3A: block-circulant. Over  $\mathbb{R}$  the frame is given by the powers  $P^k$  of the cyclic shift (Gram matrix  $bI$ ). Over  $\mathbb{C}$ ,  $\text{circ}(w) = F^{-1} \text{diag}(Fw)F$  for first column  $w$ , and the real form has  $2 \times 2$  rotation-scaling blocks. HF adds a factor  $1/\text{in\_features}$ .
  - LoRA-XS, SVFT, TinyLoRA: masks or random subspaces in the SVD frame of  $W_0$ , a  $\theta_0$ -dependent transport.
  - SHiRA, FiSH, Trainable Tokens: masks in the standard frame.
  - MiSS/Bone in its default (balance) mode:  $\Delta W = D(\mathbf{1}_g^\top \otimes I_r)$ , which is LoRA-FA with a frozen summing projection. Its BAT mode,  $\Delta W = \text{blocks}(W_0) M + M$  blockwise, is linear in the trained block  $M$  but depends on  $\theta_0$ , so it is not additive.
  - These frame readings of FourierFT and C3A are for `init_weights=True`. HF's defaults start off  $\theta_0$ : FourierFT draws its spectrum from  $\mathcal{N}(0, 1)$  with scaling 150 (at  $768^2$  with 1000 frequencies,  $\|\Delta W_0\|_F \approx 4.3$ , about 14% of a std-0.04  $W_0$ ), and C3A is Xavier-uniform. These are objects with a pointing defect.
- HiRA =  $(D_{W_0})_* \text{LoRA}$  inside the slice at  $\theta_0$ , when  $W_0$  has no zero entries, with  $D_{W_0}: X \mapsto W_0 \odot X$ . Because  $D_{W_0}$  depends on  $\theta_0$ , re-based HiRA is not transported ReLoRA. Two cycles



**Table 10.** One  $4096 \times 4096$  box and one budget,  $|M| = 131,072$ , spent three ways (Proposition II.12).

| method              | $d(M)$         | fibre $ M  - d(M)$ | max rank | block structure                    |
|---------------------|----------------|--------------------|----------|------------------------------------|
| LoRA <sub>16</sub>  | 130,816        | 256                | 16       | none                               |
| GraLoRA, $k = 2$    | 130,816        | 256                | 32       | each of 4 blocks has rank $\leq 8$ |
| LoHa <sub>8,8</sub> | $\leq 122,753$ | $\geq 8,319$       | 64       | none                               |

reach  $W_0 \odot (J + Z_1) \odot (J + Z_2)$ , whose relative update  $Z_1 + Z_2 + Z_1 \odot Z_2$  has rank up to  $\min(2r + r^2, m, n)$ , attained, where transported ReLoRA stays at rank  $\leq 2r$ .

**Numerical check.**  $\|K^{T_*M} - TK^MT^\top\| = 2 \times 10^{-14}$  for a Gaussian  $T$ . Adam on  $T_*M$  with  $L$  and on  $M$  with  $L \circ \tau_T$  agree to  $8 \times 10^{-11}$ . Two HiRA cycles give relative-update rank 3, 8, 15 at  $r = 1, 2, 3$  ( $m = n = 30$ ) (Appendix E).

**Credit.** The transported methods are FourierFT (Gao et al., 2024), WaveFT (Bilican et al., 2025), C3A (Chen et al., 2024a), LoRA-XS (Bałazy et al., 2024), SVFT (Lingam et al., 2024), TinyLoRA (Morris et al., 2026), SHiRA (Bhardwaj et al., 2024), FISH (Sung et al., 2021), Trainable Tokens (team, 2025), MiSS (Kang and Yin, 2024) and HiRA (Huang et al., 2025b); LoRA-Pro is due to Wang et al. (2024i). Si et al. (2024c) organise PEFT around the SVD frame of  $W_0$ , a related unification.

**WHAT THIS EXPLAINS** “High rank” claims (MoRA, HiRA, LoKr, C3A, SHiRA, WaveFT) are not  $\text{GL}(\Theta)$ -invariant, although rank is  $\text{GL}_m \times \text{GL}_n$ -invariant. The transports in play ( $\mathcal{R}$ , the real Fourier (cosine) frame,  $D_{W_0}$ ) do not preserve the network function, and Theorem II.11 shows that KronA and LoRA<sub>r</sub> have incomparable images when  $r < \min(m_1, n_1) \min(m_2, n_2)$  and  $\kappa_1 > 1$ . Transport preserves every isomorphism invariant of the method ( $d$ ,  $g$ , the gauge group,  $\dim S_1$ , the deficit, the regular or apex type, the germ type and the degree) and conjugates its dynamics. Relative to the network’s  $\text{GL}_m \times \text{GL}_n$  structure it changes, in particular, rank, covariance,  $S_1$  as a subspace, and the image itself. In Figure 4 HiRA therefore sits at LoRA’s height and at full rank (Proposition II.9(a)).

**Prediction.** At matched learning-rate sweeps, the performance gap between a transported method and its source (Kronecker sums against LoRA; Fourier or wavelet methods against coordinate masks) is explained by how well each image aligns with the task’s update, measured for instance by projecting a full fine-tuning  $\Delta W$  onto each image, more than by optimizer effects.

### The plug, and a reporting rule

Cut  $W_0$  out of a linear layer and a hole  $V \rightarrow U$  remains. A weight-space method fills it with a small network of frozen and trainable boxes, a *plug* (Figure 5). Every invariant above is read off the plug: the budget (Definition I.2), the min-cut, the Jacobian rank (Definition I.5), the gauges on its wires, and the merge type, set by whether an activation sits on the path (Theorem VI.3). The plug is a weight-space cousin of the unified view of He et al. (2021), which puts adapters, prefix tuning and LoRA in one design space.

Table 10 spends one budget three ways on one  $4096 \times 4096$  box (Proposition II.12). The methods differ by at least 8,063 dimensions and by a factor of four in rank. The LoHa row uses the upper bound  $F$ , whose equality is checked numerically at small sizes and open in general.

**Prediction.** Benchmarks should report  $d(M)$  beside  $|M|$ , with the maximum rank and the cut it refers to. At budgets of a few thousand parameters, rankings at equal *effective* dimension can change.

A small case shows every readout at once. On an  $8 \times 8$  box, LoHa<sub>2,2</sub> has 64 parameters, rank at most 4, Jacobian rank 41 and a gauge orbit of dimension  $23 = 4 + 4 + 15$  that fills the fibre. LoRA<sub>4</sub> costs the same and reaches 7 more dimensions, with LoHa’s image strictly inside it (Proposition II.12(c), since  $2r^2 = 8 < 15 = m + n - 1$ ). The plug workbench of the companion site recomputes these invariants in float64 for fifteen presets and for plugs the reader edits: the budget, the degree, the min-cut bound by brute force over vertex bipartitions beside the rank attained,  $d(M)$  as a Jacobian rank, the gauge orbit with each generator checked to lie in  $\ker d\rho$ , and the merge verdict.

### Covariance and descent

A method is usually one recipe applied to every pretrained weight. Rename a layer’s input basis by  $h$  and its output basis by  $g$ , so that  $W_0$  becomes  $gW_0h^{-1}$ ; the method is covariant under  $(g, h)$  if its image moves the same way.

**DEFINITION II.14** (covariance group). For a method defined for every  $W_0$  (a natural family), the **covariance group**  $\mathcal{G}_M \subseteq \mathrm{GL}_m \times \mathrm{GL}_n$  is the largest subgroup with  $\mathrm{Im} M(gW_0h^{-1}) = g \cdot \mathrm{Im} M(W_0) \cdot h^{-1}$ . When the method depends on random frames or on data, the identity is required in distribution, with the data transformed.  $\mathcal{G}_M$  is the *geometry the method lives in*, in Klein’s sense.

Klein defined a geometry by its group of motions and studied what the group leaves unchanged (Klein, 1872) (Background D.17). For group-type methods this reading is a theorem (§II.B). For cones such as LoRA’s image it is a naming device, and the content is the shape of the image and its covariance group.

LoRA’s image  $W_0 + \mathcal{M}_{\leq r}$  is covariant under all of  $\mathrm{GL}_m \times \mathrm{GL}_n$ , since rank ignores bases, while a coordinate mask has a much smaller group. This matters because networks have symmetries of their own, such as permutations of hidden neurons, rotations of the residual stream and the query–key gauge of attention, which change the weights and keep the function.

**OBSERVATION II.15** (descent through function-preserving reparametrisations). Let  $M = \prod_b M_b$  be a box-wise method: each  $M_b$  is a member of a natural family with covariance group  $\mathcal{G}_{M_b}$ , and no trainable parameters or random frames are shared across boxes. Let  $G_{\mathrm{arch}}$  act on  $\Theta = \prod_b \Theta_b \times \Theta_{\mathrm{frozen}}$ , preserving this split and acting on each adapted box by  $W_b \mapsto g_b W_b h_b^{-1}$  with  $(g_b, h_b) \in \mathcal{G}_{M_b}$ . Assume  $f(g\theta, -) = f(\theta, -)$  on a  $G_{\mathrm{arch}}$ -stable set containing  $\mathrm{Im} M(\theta_0)$ . Then  $\mathrm{Im} M(g\theta_0) = g \cdot \mathrm{Im} M(\theta_0)$ , and the set of functions reachable by  $M$  from  $g\theta_0$  equals the set reachable from  $\theta_0$ . For methods with random frames, the equality holds in distribution, provided the frames are drawn independently per box. Boxes that share a frame need a pathwise condition: a law-preserving map  $T_{(g,h)}$  on frame space with  $\mathrm{Im} M_b(g_b W h_b^{-1}; TF) = g_b \cdot \mathrm{Im} M_b(W; F) \cdot h_b^{-1}$  for all  $W$  and  $F$ , with the same  $T$  on every box that shares  $F$  (marginal covariance in law, as in Definition II.14, is not enough). For data-dependent methods, require covariance pathwise in the data, and that the box- $b$  data at  $g\theta_0$  is the Definition II.14 transform under  $(g_b, h_b)$  of the data at  $\theta_0$ . This holds for intertwiner gauges such as hidden-unit permutations, query–key and value–output head gauges and residual rotations. The statement concerns reachable sets only;

training trajectories and outcomes can still differ, for instance under Adam or non-isotropic initialisations.

Examples of  $G_{\text{arch}}$ :

- hidden-neuron permutations;
- the query-key gauge  $(W_Q, W_K) \mapsto (gW_Q, g^{-\top}W_K)$  per head. With no positional encoding or an absolute one,  $g \in \text{GL}_{d_k}$ . Under RoPE,  $g$  must commute with every RoPE rotation, which leaves the per-frequency  $2 \times 2$  rotation-scalings  $(\mathbb{C}^\times)^{d_k/2}$ , shared across a GQA group. Under partial RoPE on  $r$  of the  $d_k$  dimensions the group is  $(\mathbb{C}^\times)^{r/2} \times \text{GL}_{d_k-r}$ . Under QK-norm,  $g$  must in addition be orthogonal when the norm's  $\varepsilon > 0$  (conformal,  $g \in \text{CO}(d_k)$ , when  $\varepsilon = 0$ ) and satisfy  $g^\top \Gamma_q R_t \Gamma_k g = \Gamma_q R_t \Gamma_k$  for every RoPE rotation  $R_t$ , where  $\Gamma_q, \Gamma_k$  are the gains. OLMo 2 normalises  $q$  and  $k$  across all heads jointly;
- a single global orthogonal rotation  $Q$  of the residual stream of a pre-norm transformer whose norm gains can all be fused into the linear layers they feed, after LayerNorm is converted to RMSNorm (SliceGPT's computational invariance; QuaRot's offline rotation). SliceGPT's per-block rotations  $Q_\ell$  need extra residual matrices and change the architecture. Sandwich and post-sublayer norms (Gemma 2 and 3, OLMo 2) are excluded, because their gains cannot be fused and  $\text{diag}(\gamma)$  does not commute with  $Q$ . The comparison is between the gain-fused model and the fused-and-rotated model: gain fusion is a diagonal input rescaling outside  $G_{\text{arch}}$  that already changes PiSSA's split and block OFT's image (Proposition II.16), and converting LayerNorm to RMSNorm changes LoRA's reachable set on the residual writers.

**Numerical check.** A tied rank-one update shared by two boxes violates the conclusion (relative distance up to 0.30), while independent per-box LoRA satisfies it. A Haar frame shared by two boxes, each covariant in law, gives a joint statistic of 0.44 from  $\theta_0$  against 0.81 from  $g\theta_0$ . Under RoPE a generic  $g \in \text{GL}_8$  changes attention scores by up to 313, a RoPE-commuting  $g$  by  $5 \times 10^{-14}$  (Appendix E).

**Credit.** RoPE is due to Su et al. (2024). Residual-stream rotations are the device of SliceGPT (Ashkboos et al., 2024a) and of QuaRot (Ashkboos et al., 2024b). The architectures with sandwich or post-sublayer norms are Gemma 2 (Gemma Team et al., 2024), Gemma 3 (Gemma Team et al., 2025) and OLMo 2 (Team OLMo et al., 2024).

**WHAT THIS EXPLAINS** Descent is decided by whether each induced pair  $(g_b, h_b)$  lies in  $\mathcal{G}_{M_b}$ . For one-sided methods ((IA)<sup>3</sup>, DoRA, OFT, BOFT) this is a question of side. For coordinate-structured methods (LoHa with  $r \geq 2$ , HiRA, masks, FourierFT, Kronecker methods) it is a question of group. They descend through hidden-unit monomial gauges, since  $P(X \odot Y)Q = (PXQ) \odot (PYQ)$  for permutation matrices  $P, Q$  and a diagonal scaling can be absorbed into one factor, and they do not descend through rotations. A global residual rotation acts on the input side of  $q, k, v$ , gate and up, and on the output side of  $o$  and down.

Table 11 sorts common methods by this criterion. Failing a sufficient condition does not prove that the reachable function sets differ.

**CONJECTURE A** (reachable sets under a residual rotation). *Against a gain-fused but unrotated control, fine-tuning the fused-and-rotated, function-identical model measurably changes the reachable function sets, and the results, of block OFT/BOFT on residual-input projections, LoHa with  $r \geq 2$ , HiRA and coordinate masks, because LLM residual streams have a privileged basis.*

**Table 11.** Methods against the sufficient condition of [Observation II.15](#). For the residual rotation the comparison is between the gain-fused model and the fused-and-rotated model.

| gauge                                            | satisfy the hypothesis (reachable functions unchanged)                                                                                                                                                                                                                                                                                                                                                  | fail the sufficient condition                                                                                                                                                                                                                          |
|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| single global residual rotation (SliceGPT style) | LoRA, full FT, BitFit, PiSSA; (IA) <sup>3</sup> with HF defaults, which scales the output side of $k$ (or $q$ , for Qwen, Gemma and Phi) and $v$ and the input side of down, while the rotation acts on the other side, where (IA) <sup>3</sup> is GL-covariant; DoRA on $q$ , $k$ , $v$ , gate and up; OFT on $o$ and down (no scale); LoHa <sub>1</sub> , whose image is $W_0 + \mathcal{M}_{\leq 1}$ | block OFT and BOFT on residual-input projections (the block normaliser contains no Hadamard or PCA rotation); BOFT on $o$ and down (its output scale $\text{boft}_s$ ); DoRA on $o$ and down; LoHa with $r \geq 2$ ; HiRA; coordinate masks; FourierFT |
| query-key and value-output head gauges           | LoRA                                                                                                                                                                                                                                                                                                                                                                                                    | (IA) <sup>3</sup> on $q$ or $k$ and on $v$ ; OFT on $o$                                                                                                                                                                                                |

QuaRot’s per-head Hadamard on  $W_v$  and  $W_{\text{out}}$  is a value-output gauge in  $G_{\text{arch}}$ . Its online Hadamards (across heads, before `down_proj`, on keys) preserve the realised function but insert new operations, so they change the architecture map and lie outside  $G_{\text{arch}}$ . The optimizer adds a further layer. Adam is only  $B_r$ -equivariant ([Theorem III.12](#)) and HF’s Kaiming-uniform  $A$  is not rotation-invariant in law, so LoRA *trained with Adam* need not descend even though its image does.

### Covariant data-aware splits

SVD-based initialisations raise the same question. PiSSA, MiLoRA and CorDA split  $W_0$  into a trainable part and a frozen residual by a truncated SVD of  $W_0$  or, for CorDA, of  $W_0$  weighted by the input second moment. Truncated SVD commutes only with rotations and uniform scalings, so these splits depend on the input coordinates, and whitening removes that dependence.

**PROPOSITION II.16** (covariant data-aware splits). *Let  $C = \mathbb{E}[xx^\top] \succ 0$  be the input second moment, so that under  $x \mapsto hx$  we have  $W \mapsto Wh^{-1}$  and  $C \mapsto hCh^\top$ . Let  $LL^\top = C$  and  $p := \min(m, n)$ . Write  $[M]_r$  for the rank- $r$  truncated SVD,  $\langle M \rangle_r := M - [M]_{p-r}$  for the bottom  $r$  singular components, and  $\text{CO}(n) = \mathbb{R}_{>0} \cdot \text{O}(n)$ . Assume  $1 \leq r < p$  and the spectral gap each split needs:  $\sigma_r > \sigma_{r+1}$  of  $W_0L$  in (a); in (b),  $\sigma_r > \sigma_{r+1}$  of  $W_0C$  or  $W_0$  for top- $r$  splits, and  $\sigma_{p-r} > \sigma_{p-r+1}$  for bottom- $r$  splits. Covariance groups are taken over the generic set where these gaps hold.*

(a) *The whitened split*

$$X^* = \arg \min_{\text{rank } X \leq r} \|(W_0 - X)L\|_F = [W_0L]_r L^{-1} = \Pi_r W_0,$$

where  $\Pi_r$  is the top- $r$  eigenprojector of the output second moment  $W_0CW_0^\top$ , is  $\text{GL}_n$ -covariant on the input side and  $\text{CO}(m)$ -covariant on the output side; its joint group is  $\text{CO}(m) \times \text{GL}_n$ . This is the classical reduced-rank-regression (SVD-LLM) solution.

(b) *The formula-level CorDA splits, IPM ( $X = [W_0C]_r C^{-1}$ , the minimiser of  $\|(W_0 - X)C\|_F$ ) and KPM (the bottom- $r$  components of  $W_0C$  mapped back by  $C^{-1}$ ,  $X = \langle W_0C \rangle_r C^{-1}$ ), with or without CorDA’s trace-relative damping  $C + \kappa(\text{tr } C/n)I$ , and the PiSSA and MiLoRA splits*

$([W_0]_r$  and the bottom- $r$  part  $\langle W_0 \rangle_r$ ), have input-side covariance group exactly  $\text{CO}(n)$ ; jointly,  $\text{CO}(m) \times \text{CO}(n)$ .

(c) Scope. An absolute damping  $C + \lambda I$  reduces the input-side group of (a) and of (b) to  $\text{O}(n)$ . The official and HF CorDA code divide each calibration sample by the absolute value of its largest entry before accumulating  $C$ , which keeps only positive scalings and coordinate permutations, so (b) describes the CorDA formulas and does not cover shipped CorDA under rotations.

**Numerical check.** Under  $h = 3Q$  with  $Q$  orthogonal, CorDA (IPM and bottom- $r$  KPM), PiSSA and bottom- $r$  MiLoRA move covariantly to  $4 \times 10^{-15}$ . Under a generic  $h$  only the whitened split does ( $2 \times 10^{-15}$ , against 0.63 and 1.1). At  $r = n$  PiSSA is  $\text{GL}_n$ -covariant. An absolute damping 0.1 breaks the whitened split under a generic  $h$  (0.14), and under  $h = 3Q$  (about  $2 \times 10^{-2}$ ), while trace-relative damping keeps it ( $3 \times 10^{-15}$ ) (Appendix E).

**Credit.** The splits are CorDA (Yang et al., 2024c), PiSSA (Meng et al., 2024a) and MiLoRA (Wang et al., 2024b). The whitened truncation is the one used by SVD-LLM (Wang et al., 2025c) for compression.

**WHAT THIS EXPLAINS** A design rule from the Erlangen programme. A sufficient condition for a data-aware Type II split to be independent of how a layer’s input coordinates are chosen is that it be a left multiple of  $W_0$  by a matrix-valued function of the output second moment,  $X = \Phi(W_0 C W_0^\top) W_0$ . The whitened (SVD-LLM-style) split and the eigenprojector split of Astra (Liu et al., 2026b) are instances; Astra’s paper form, a tail eigenprojector of the output covariance, has the same form with  $C$  replaced by the centred input covariance, which transforms in the same way. Depending on the data only through  $W_0 C W_0^\top$  is not enough, since PiSSA satisfies that vacuously and is not  $\text{GL}_n$ -covariant. Non-conformal reparametrisations, such as the diagonal rescalings from norm-gain folding or SmoothQuant (Xiao et al., 2023), change the CorDA and PiSSA splits.

Three caveats apply. An absolute damping  $C + \lambda I$  reduces the covariance to  $\text{O}(n)$ , while CorDA’s trace-relative damping keeps  $\text{CO}(n)$ . The covariance is one-sided. No rank- $r$  split that depends only on  $(W_0, C)$  can be  $\text{GL}_m \times \text{GL}_n$ -covariant, since the stabiliser of a normal form contains an orthogonal group that fixes only full-rank or zero splits, and whitening with the output second moment itself is degenerate (every singular value of  $(W_0 C W_0^\top)^{-1/2} W_0 L$  is 1). Descent through hidden-layer GL gauges would need an output-side statistic independent of  $(W_0, C)$ , such as a backpropagated-gradient second moment. Training dynamics (SGD, Adam) remain non-covariant.

EVA (Paischer et al., 2024) is not a split. It is Type I ( $B_0 = 0$ ,  $W_0$  not split) and points row  $A_0$  at the top- $r$  centred principal subspace of the inputs. At a fixed rank its image  $W_0 + \mathcal{M}_{\leq r}$  is  $\text{GL}_m \times \text{GL}_n$ -covariant under Definition II.14, and only its pointing row  $A_0$  is  $\text{CO}(n)$ -covariant. With HF’s default  $\text{rho}=2.0$  the per-layer rank is redistributed by explained-variance ratios, which are only  $\text{CO}(n)$ -invariant, so the image becomes  $\text{GL}_m \times \text{CO}(n)$ -covariant.

**CONJECTURE C** (whitened splits). *Whitened CorDA or PiSSA is more robust to function-preserving input reparametrisations, and no worse otherwise.*

Rank is a cut, dimension a volume and covariance a group. All three can be computed before training. The next exposé turns from what a method can reach to how it moves there.



## Fibres and metrics: dynamics

Expressivity lives on the image. Training lives on the fibres, through a metric. The image of a method fixes what it can reach, while its fibres and the optimizer's metric fix how it gets there. So two objects of **Meth** can be isomorphic and still train differently, and methods that look different can train identically.

Training happens upstairs, in the parameter space  $Q$ , where each fibre of  $\rho$  collects the parameters that give one weight. The optimizer's metric on  $Q$ , pushed down to weight space, is what the model feels. Every statement below names its optimizer.

The exposé has four movements. Training with rates  $\Lambda$  moves the weights by  $\dot{\theta} = -K\nabla_{\theta}L$ , with the cometric  $K = D\rho \wedge D\rho^{\top}$ , which depends on where the parameters sit in their fibre (Definition III.1 to Proposition III.4). Gradient flow conserves LoRA's charge  $\Phi = B^{\top}B/\eta_B - AA^{\top}/\eta_A$ , which sorts initialisations (Theorem III.5 to Proposition III.8); when the charge is isotropic, one scale  $\sigma^*$  separates a LoRA-FA-like regime from balanced dynamics (Theorem III.9). A 2-torus of rescalings acts on LoRA's knobs  $(\alpha, \eta_A, \eta_B, \sigma_A)$  without changing training. Under Adam with  $\varepsilon = 0$ , no decay or clipping and one shared schedule, LoRA+ with ratio  $\lambda$  is plain LoRA with  $\alpha$  multiplied by  $\lambda$  (Theorem III.10, Corollary III.11, after Schulman and Thinking Machines Lab, 2025). Finally, SGD respects the subgroup  $O(r)$  of the gauge and Adam only the signed permutations. So from a zero- $B$  start the image depends only on row  $A_0$ , the SGD trajectory on  $A_0^{\top}A_0$ , and the Adam trajectory on  $A_0$  up to signed row permutations (Theorem III.12, Corollary III.13).

### One experiment first

Figure 8 shows the cleanest consequence. Train with Adam at  $\varepsilon = 0$  from  $B_0 = 0$ , with no weight decay or clipping, one schedule for both parameter groups, and the same data order and dropout masks. Then LoRA+ with ratio  $\lambda$  ( $\eta_B = \lambda\eta_A$ ) and plain LoRA with  $\alpha$  multiplied by  $\lambda$ , at the same  $\eta_A$  and  $A_0$ , produce the same weights at every step (Theorem III.10, Corollary III.11). This LoRA/Adam statement is due to Schulman and Thinking Machines Lab (2025), for the LoRA+ of Hayou et al. (2024b); Theorem III.10 says what is added here.

### The metric a method carries

A learning rate is a metric in disguise. Gradient descent with rates  $\Lambda$  moves  $q$  by  $-\Lambda\nabla_q$ , and pushed down by  $D\rho$  this is the weight gradient multiplied by a positive semidefinite operator.

**DEFINITION III.1** (trained method; cometric; three equivalences). A *trained method* is a method together with optimizer data: a learning-rate operator  $\Lambda \succ 0$  (per-block rates), an initial scale, and an update rule. Gradient flow  $\dot{q} = -\Lambda\nabla_q(L \circ \rho)$  moves the weights by

$$\dot{\theta} = -K_q \nabla_{\theta} L, \quad K_q = D\rho_q \wedge D\rho_q^{\top}.$$

We call  $K_q$  the **induced cometric**. For a backward method with compression  $c_t$  and anchor  $a_t$  (Definition IV.1),  $K_t = a_t c_t$ .

Two methods are

- *expressively equivalent* if their images agree;
- *first-order equivalent* if their  $K_{q_0}$  agree;
- *dynamically equivalent* if they produce the same  $\theta$ -trajectory for every loss.

Cometrics compose the way derivatives do.

**OBSERVATION III.2** (chain rule for cometrics). *Let  $\sigma : Q' \rightarrow Q$  and  $\rho : Q \rightarrow X$  be smooth, with  $X = \Theta$  or a function or output space. Train  $Q'$  by gradient flow with a constant rate operator  $\Lambda' \succ 0$ , and put  $K_{q'}^\sigma := D\sigma_{q'} \Lambda' D\sigma_{q'}^\top$ . Then*

$$K_{q'}^{\rho\sigma} = D\rho_{\sigma(q')} K_{q'}^\sigma D\rho_{\sigma(q')}^\top,$$

*and no metric on  $Q$  is used.  $K$  is a degenerate cometric (positive semidefinite, of rank  $\leq \dim Q'$ ); when  $\sigma$  is not injective,  $K^\sigma$  is a field along  $\sigma$  and fails to be a cometric on  $Q$ .*

*For LoRA (Hu et al., 2021),  $\rho(B, A) = W_0 + sBA$  with  $s = \alpha/r$  ( $\alpha/\sqrt{r}$  for rsLoRA, Kala-jdziewski, 2023), block rates  $\Lambda = \text{diag}(\eta_B I, \eta_A I)$  and no LoRA dropout, under gradient flow (equivalently, to first order in the step size for GD),*

$$K_{(B,A)}(G) = s^2(\eta_B G A^\top A + \eta_A B B^\top G).$$

*Under Adam there is no state-independent linear cometric; Theorem III.10 and Theorem III.12 treat Adam.*

NUM Finite-difference Jacobians agree with the formula to  $6 \times 10^{-9}$  (Appendix E).

**WHAT THIS EXPLAINS** For LoRA, the term  $\eta_B G A^\top A$  lets the update reach all outputs while filtering its inputs through row  $A$ , and  $\eta_A B B^\top G$  does the reverse. At  $B_0 = 0$  only the first term survives, so the first-step cometric  $s^2 \eta_B G A_0^\top A_0$  is fixed by  $A_0$  (Corollary III.13).

Take  $X$  to be the prefix space (Exposé VI). HF PrefixTuning with `prefix_projection=True` (the MLP  $\sigma$  of Li and Liang, 2021) and with `prefix_projection=False` (the P-tuning v2 setting of Liu et al., 2021c, ignoring its classification head) have the same prefix image whenever `encoder_hidden_size`  $\geq$  `num_virtual_tokens`  $- 1$ . Under gradient flow their cometrics on prefix space are  $D\sigma \Lambda' D\sigma^\top$  and  $\Lambda$ , so the reparametrisation replaces the flat metric on prefix space by a point-dependent one. Prefix methods are extensions, so the target here is the prefix (or function) space rather than  $\Theta$ . This alone does not account for the instability that Li and Liang observed under AdamW.

Backpropagation as a lens, the source of this chain rule, is due to Fong et al. (2017) and Cruttwell et al. (2021).

### When a change of chart does not matter

A gauge transformation changes the parameters and keeps the weights. It leaves training unchanged exactly when the cometrics agree, which for LoRA's gauge  $\text{GL}_r$  singles out  $\mathcal{O}(r)$ .

**THEOREM III.3** (gradient descent is natural exactly where cometrics agree; for LoRA's linear gauge, exactly along  $\mathcal{O}(r)$ ). *(a) Let  $\varphi : Q \rightarrow Q'$  be a diffeomorphism with  $\rho' \circ \varphi = \rho$ , and let  $\Lambda, \Lambda'$  be the rate operators on  $Q, Q'$ . Under gradient flow, the two methods induce the same  $\theta$ -velocity at  $q$  and at  $\varphi(q)$  for every loss iff  $K_q^\rho = K_{\varphi(q)}^{\rho'}$ , that is,*

$$D\rho'_{\varphi(q)}(D\varphi_q \Lambda D\varphi_q^\top - \Lambda') D\rho_{\varphi(q)}^\top = 0.$$

*This holds whenever  $D\varphi \Lambda D\varphi^\top = \Lambda'$ , which is an isometry when  $\Lambda = \Lambda' = I$ . That condition is necessary when  $D\rho'_{\varphi(q)}$  is injective, and fails to be necessary in general. Every linear gauge of a linear parametrisation is natural; for instance, for  $\rho(x, y) = \theta_0 + x + y$  the shear  $(x, y) \mapsto$*

$(2x, y - x)$  preserves  $\rho$  and  $K$  without being orthogonal. So is every pointwise-orthogonal non-linear LoRA gauge  $\varphi(q) = \varphi_{g(q)}(q)$  with  $g(q) \in \mathcal{O}(r)$ .

(b) For LoRA's linear gauge  $\varphi_g(B, A) = (Bg^{-1}, gA)$  with block-scalar rates, at points where  $A$  has full row rank ( $B$  arbitrary, including  $B = 0$ ) and  $r < n$ , the criterion holds iff  $g \in \mathcal{O}(r)$ . The same holds symmetrically where  $B$  has full column rank and  $r < m$ .

NUM At a random point a non-orthogonal  $g$  gives a cometric mismatch of about 18, an orthogonal one about  $10^{-15}$  (Appendix E).

**WHAT THIS EXPLAINS** No learner with a fixed Euclidean metric respects all invertible reparametrisations. For LoRA's linear gauge, plain gradient descent defines a family of training dynamics indexed by the cosets  $\mathcal{O}(r) \backslash \mathrm{GL}_r$ , which  $g \mapsto g^\top g$  identifies with  $\mathrm{Sym}_r^+$ . Under gradient descent, LoRA's rate ratio  $\lambda$  corresponds to the scalar gauge  $g = \lambda^{1/4} I$  together with an overall change of rate. rsLoRA rescales  $s$ , equivalently the learning rates and the initialisation; this is a move on the hyperparameter torus and involves no choice of gauge (Theorem III.10, Corollary III.11). Riemannian preconditioning and LoRA-Pro remove the dependence on the representative (Proposition III.4). Under AdamW, as used in practice, the relevant group is the group  $B_r$  of signed permutations (Theorem III.12).

The  $\mathrm{GL}_r$  symmetry of  $W = BA$  and the quotient geometry of fixed-rank matrices are classical (Mishra et al., 2012). Putterman et al. (2024) build learners on LoRA weights that respect it.

Some published optimizers use instead a cometric that depends only on the weight, at least in an idealised form.

**PROPOSITION III.4** (cometrics that descend to the rank- $r$  manifold; known in part). Write  $U = \mathrm{col} B$  and  $V = \mathrm{row} A$ . Statements (a), (b) and the first sentence of (d) are on the full-rank locus; (c) needs only  $\mathrm{rank} A_0 = r$ , with  $B$  arbitrary, including  $B_0 = 0$ . All statements concern gradient flow, or (S)GD to first order, on the stated gradients, undamped and without Adam. Cometrics include the learning rate.

(a) Scaled GD,  $\delta B = -\eta \nabla_B (AA^\top)^{-1}$  and  $\delta A = -\eta (B^\top B)^{-1} \nabla_A$ , is  $\mathrm{GL}_r$ -equivariant with first-order cometric  $K(G) = \eta s^2 (P_U G + G P_V)$ .

(b) LoRA-Pro's equivalent gradient  $\tilde{g} = P_U G + (I - P_U) G P_V$  is the Frobenius-orthogonal projection  $\Pi_{T_{W_0 + M_r} M_r}(G)$ . The flow of its SGD variant (LoRA-Pro Alg. 1) is the Riemannian gradient flow (Background D.24) on  $W_0 + M_r$ .

(c) LoRA-FA's corrected gradient  $s^{-2} \nabla_B (A_0 A_0^\top)^{-1}$ , fed to SGD with  $A_0$  frozen, gives  $K(G) = \eta G P_V$  with  $V = \mathrm{row} A_0$ , which depends on  $A_0$  only through  $V \in \mathrm{Gr}(r, n)$ . Its original form, plain SGD on  $B$ , has  $K(G) = s^2 \eta_B G A_0^\top A_0$ .

(d) The cometrics of (a) and (b) are functions of  $W$  on the full-rank locus and define vector fields tangent to  $W_0 + M_r$ . The field of (c) is  $G \mapsto \eta G P_V$  with  $V = \mathrm{row} A_0$ . Its flow is the Frobenius gradient flow on the slice  $W_0 + \{X A_0\} = W_0 + \mathbb{R}^{m \times n} P_V$ , of dimension  $mr$ , which contains  $W_0$  and lies in  $W_0 + M_{\leq r}$ . The rank- $r$  part of the slice,  $N_V = \{\mathrm{row}(W - W_0) = V\}$ , is open and dense in it and is a submanifold of  $W_0 + M_r$ . On  $N_V$  the field is tangent to  $M_r$  and equals  $\eta G P_{\mathrm{row}(W - W_0)}$ , a function of  $W$ . So (c) descends too, as a degenerate cometric of rank  $mr$  (less than  $r(m + n - r)$  when  $r < n$ ) whose flow keeps  $\mathrm{row}(W - W_0) = \mathrm{row} A_0$  fixed. The cometric of plain GD depends on the representative; it is invariant only under  $\mathcal{O}(r)$ .

The algorithms are known. Scaled GD is due to Tong et al. (2021) and Zhang and Pilanci (2024a), and LoRA-Pro to Wang et al. (2024i). LoRA-FA in its frozen-A form is Zhang et al. (2023d); the corrected gradient of (c) comes from the 2026 revisions (v2 and v3) of that arXiv

record and from the LoRA-FA optimizer added to HF PEFT (Mangrulkar et al., 2022) in 2025. The  $\text{GL}_r$ -invariance of the scaled-GD metric and its descent to the quotient  $\mathcal{M}_r$  are classical (Mishra et al., 2012; Mishra and Sepulchre, 2016).

NUM Gauge defects are about  $10^{-15}$  for (a), against 52 for plain GD. LoRA-FA with AdamW from  $A_0$  and from  $gA_0$  differs by  $O(1)$  after a few steps, against  $10^{-15}$  with SGD (Appendix E).

**Practice.** Each published method departs from these idealisations, and the departures change the symmetry (Theorem III.12).

- The standard start  $B_0 = 0$  lies off the full-rank locus, where (a) and (b) are undefined without damping. Zhang and Pilanci (2024a) use  $(B^\top B + \delta I)^{-1}$  with an unspecified  $\delta > 0$ , which is only  $O(r)$ -equivariant. Their LLM experiments report both damped scaled GD and scaled AdamW, in which the preconditioner is applied before the Adam moments ( $B_r$ ). HF’s `create_riemannian_optimizer` damps as well.
- LoRA-Pro’s experiments use its AdamW variant (Alg. 2, with  $m \times n$  moments),  $\dot{W} = -\Pi_T(\text{Adam}(\Pi_T G))$ . It descends to  $W_0 + \mathcal{M}_r$  without being a gradient flow, and its discrete update with the Sylvester-optimal free matrix (Background D.33) is  $\text{GL}_r$ -invariant only to first order. The released code also damps both inverses ( $\text{pinv}(\cdot + 10^{-8}I)$ , so  $O(r)$ ), replaces the step at  $B_0 = 0$  by a LoRA-FA-type step, and runs with HF’s default global-norm clipping.
- LoRA-FA as published (v3, Alg. 1) and HF’s `create_lorafa_optimizer` apply AdamW after the correction (HF also damps with  $10^{-8}I$ ). The trajectory is then invariant only under signed row permutations of  $A_0$  (Theorem III.12(b)), so it depends on the basis and scale of  $A_0$  as well as on row  $A_0$ .

**REMARK** (2026: Riemannion). Riemannion (Bogachev et al., 2025) defines its optimizer on  $\mathcal{M}_r$  directly. Its variable is the point  $X \in \mathcal{M}_r$ , stored as thin factors, and apart from weight decay each step uses only  $X$ , the projected gradient and the momentum: the tangent projection of Proposition III.4(b), momentum carried along by the same projection, a Muon step (Jordan et al., 2024) that sets the nonzero singular values of the momentum to 1 and projects again, and a truncated SVD as retraction. No Gram matrix is inverted, so nothing needs damping, and with weight decay off, the  $W$ -trajectory, momentum included, does not depend on how  $X$  is factored. The published Algorithm 4 writes its weight decay through the orthonormal factors, as a term  $\gamma A_L B_R^\top$ , so this holds for  $\gamma = 0$ . Its non-linearity acts on a weight-space matrix, while Adam’s acts coordinatewise on the factors, which is what cuts the gauge down to  $B_r$  (Theorem III.12).

**WHAT THIS EXPLAINS** Under (S)GD each method is summarised by its induced weight-space cometric (Table 12). LoRA, scaled GD and LoRA-Pro share the image  $W_0 + \mathcal{M}_{\leq r}$  and differ only in cometric. LoRA-FA is the sub-object  $W_0 + \{XA_0\}$  (Corollary III.13). GaLore (with a gradient-SVD or a random projector) and MeZO are full-parameter methods, listed for comparison. Flora proper compresses only its momentum, takes full-space Adafactor steps, and has no cometric of this form (Exposé IV).

“Transformation invariance” in the sense of Definition 1 of LoRA-RITE (Yen et al., 2024) means that the  $W$ -update is invariant under the gauge ( $W$ -level descent), which is weaker than parameter-level  $\text{GL}_r$ -equivariance. LoRA-RITE’s algorithm satisfies the stronger property on the full-rank locus (Theorem III.12).

**Table 12.** Induced weight-space cometrics under gradient flow or SGD ([Observation III.2](#), [Proposition III.4](#)). The last three rows are full-parameter methods, listed for comparison;  $P$  and  $Q$  are GaLore’s left and right projectors,  $A_t$  a random projector,  $z$  the random direction of MeZO.

| Method (gradient flow or SGD)                                                                                                   | Cometric $K(G)$                                                                                |
|---------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| LoRA ( <a href="#">Hu et al., 2021</a> )                                                                                        | $s^2(\eta_B G A^\top A + \eta_A B B^\top G)$                                                   |
| LoRA-FA, original / corrected ( <a href="#">Zhang et al., 2023d</a> )                                                           | $s^2 \eta_B G A_0^\top A_0 / \eta G P_V$                                                       |
| scaled GD, undamped Riemannian preconditioning ( <a href="#">Tong et al., 2021</a> ; <a href="#">Zhang and Pilanci, 2024a</a> ) | $\eta s^2(P_U G + G P_V)$                                                                      |
| LoRA-Pro, SGD variant ( <a href="#">Wang et al., 2024i</a> )                                                                    | $\eta \Pi_{T\mathcal{M}_r} G$                                                                  |
| GaLore, one period, scale $\alpha$ ( <a href="#">Zhao et al., 2024a</a> ); <a href="#">Proposition IV.2</a>                     | $\eta \alpha P P^\top G$ , or $\eta \alpha G Q Q^\top$ when $m > n$ (default $\alpha = 0.25$ ) |
| GaLore with a random projector $A_t$ , one period (the compressed SGD of <a href="#">Hao et al., 2024</a> )                     | $\eta G A_t^\top A_t$                                                                          |
| MeZO ( <a href="#">Malladi et al., 2023</a> )                                                                                   | $\eta z z^\top G$ , equal to $\eta G$ in expectation                                           |

### Isometries and charges

Split LoRA’s gauge in two. With per-block scalar rates, the antisymmetric generators rotate the rank channels and preserve the rate metric, so the optimizer cannot feel them. Along the symmetric ones, gradient flow conserves a quadratic charge ([Background D.26](#)). For LoRA the charges form one matrix  $\Phi$ , whose vanishing is the balancedness of deep linear networks ([Background D.28](#)).

**THEOREM III.5** (Noether charges and isometries for cotangent signals; known, [Zhao et al. \(2022b\)](#)). *Let a Lie group act linearly on  $Q = \mathbb{R}^N$  with  $\rho \circ k = \rho$ , let  $\Lambda \succ 0$  be constant, and let  $\gamma(t) \in T_{\rho(q(t))}^* \Theta$  be any cotangent signal, gradient or not. Consider the continuous-time updates*

$$\dot{q} = -\Lambda D\rho_q^\top \gamma(t).$$

(i) *For every generator  $\xi$  with  $\Lambda^{-1}\xi$  symmetric, the charge  $J_\xi(q) = \frac{1}{2}\langle q, \Lambda^{-1}\xi q \rangle$  is conserved.*  
(ii) *For every generator  $\xi$  with  $\Lambda^{-1}\xi$  antisymmetric,  $k = e^{\tau\xi}$  preserves  $\langle \cdot, \Lambda^{-1}\cdot \rangle$ , so  $K_{kq} = K_q$  along the orbit; for the same cotangent  $\gamma$ , the  $\theta$ -velocities at  $q$  and at  $kq$  agree. Suppose moreover that  $\gamma$  depends on  $q$  only through  $(t, \rho(q))$ , as  $\gamma = \nabla_\theta L$  or an open-loop signal does, and that solutions are unique ( $\gamma$  measurable in  $t$  and locally Lipschitz in  $\theta$ ,  $\rho$  of class  $C_{\text{loc}}^{1,1}$ ). Then the  $\theta$ -trajectories from  $q_0$  and from  $kq_0$  coincide. If  $\gamma$  uses Euclidean norms on  $Q$ , as global-norm clipping does, the trajectory statement holds when, in addition,  $k$  is a Euclidean isometry, and can fail otherwise. So (i) and the velocity statement of (ii) hold for arbitrary  $\gamma$ , while the trajectory statement needs a gradient-like signal.*

(iii) *When  $\mathfrak{g}$  is closed under  $\xi \mapsto \Lambda\xi^\top\Lambda^{-1}$ , every generator splits into a part of type (i) and a part of type (ii), a Cartan decomposition  $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$  ([Background D.27](#)). This holds for LoRA’s  $\mathfrak{gl}_r$ , with per-block scalar rates. Without closure, there are generators of neither type. With per-rank-channel rates  $\Lambda(B, A) = (B D_B, D_A A)$ , where  $D_B = \text{diag}(\eta_{B,j})$  and  $D_A = \text{diag}(\eta_{A,j})$ , the generator  $\xi_X$  is of type (i) iff  $X D_B^{-1}$  and  $D_A^{-1} X$  are both symmetric, and of type (ii) iff both are antisymmetric. An off-diagonal pair  $(i, j)$  contributes one charge and one isometry iff  $\eta_{B,i}\eta_{A,i} = \eta_{B,j}\eta_{A,j}$ , and  $\mathfrak{g}$  is closed iff all the products  $\eta_{B,j}\eta_{A,j}$  are equal. In that case one gets a*



*D-conjugate of  $\mathfrak{so}_r \oplus \text{Sym}_r$ , whose isometries are not all Euclidean unless the rates are block-scalar, so the clipping caveat of (ii) applies. For pairwise distinct products, only the diagonal  $X$  give charges and no nonzero  $X$  gives an isometry.*

Part (i) is Zhao et al. (2022b), Prop. 5.1, including the table of symmetric and antisymmetric generators and symmetric learning rates. Laws that hold for every loss and dataset are characterised by Marcotte et al. (2023), and the principle goes back to Du et al. (2018), Arora et al. (2018) and Kunin et al. (2020). New here are only the remark that  $\gamma$  need not be a gradient, which is formally immediate (along the realised trajectory,  $\gamma(t)$  is the gradient of the time-dependent linear loss  $\langle \gamma(t), \theta \rangle$ ), and the PEFT instances below.

**NUM** With a non-gradient,  $q$ -dependent  $\gamma$  on LoRA,  $\Phi$  drifts by  $1.6 \times 10^{-12}$ . With Euclidean clipping and a non-Euclidean isometry  $k$ , the  $\theta$ -trajectories from  $q_0$  and  $kq_0$  differ by 0.06. With per-channel rates  $\eta_B = (1, 1, 1)$  and  $\eta_A = (1, 1, 2)$  at  $r = 3$ , closure fails, but  $X = E_{12} + E_{21}$  still gives a conserved charge (Appendix E).

**Instances.** All instances assume gradient flow, or SGD without momentum, with per-block rates. AdamW and heavy-ball momentum fall outside the theorem, since their buffers store  $D\rho$  at past points.

- **LoRA.** The generators  $\xi_X(B, A) = (BX, -XA)$  with  $\Lambda = \text{diag}(\eta_B, \eta_A)$  satisfy:  $\Lambda^{-1}\xi_X$  is symmetric iff  $X = X^\top$ . So the  $r(r+1)/2$  charges are the entries of

$$\Phi = \frac{B^\top B}{\eta_B} - \frac{AA^\top}{\eta_A},$$

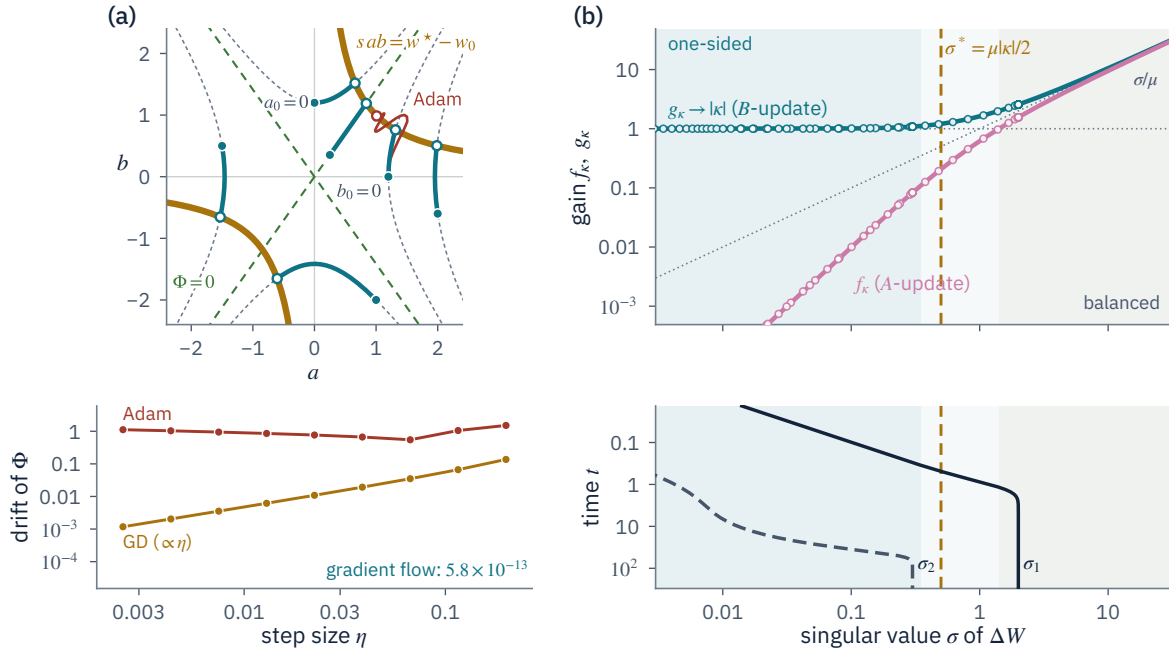
which is “balancedness”, the  $\mathfrak{p} = \text{Sym}_r$  half of  $\mathfrak{gl}_r = \mathfrak{so}_r \oplus \text{Sym}_r$ . The  $\mathfrak{so}_r$  generators act by Euclidean isometries, so (ii) holds for LoRA even under global-norm clipping.

- **DyLoRA** (Valipour et al., 2022). The gauge reduces to the torus, so only  $\text{diag } \Phi$  is conserved.
- **VeRA** (Kopiczko et al., 2023). The gauge is  $(b, d) \mapsto (cb, d/c)$ , with charge  $\|b\|^2/\eta_b - \|d\|^2/\eta_d$ . NOLA (Koochpayegani et al., 2023) is analogous.
- **AdaLoRA** (Zhang et al., 2023g) ( $P\Lambda Q$ ), without its orthogonality regulariser and between rank-budget masking events. Each torus edge gives a charge,  $\|p_k\|^2 - \lambda_k^2$  and  $\|q_k\|^2 - \lambda_k^2$ , at equal rates. The regularised objective (HF default `orth_reg_weight=0.5`) breaks these charges.
- **HRA** (Yuan et al., 2024). The rescaling  $u_i \mapsto cu_i$  gives a conserved  $\|u_i\|^2$ .
- **ReLU bottleneck adapters** (Houlsby et al., 2019). Positive per-unit rescalings give one charge per hidden unit.
- **TT adapters** (for instance Anjum et al., 2024; Yang et al., 2024d). One conserved matrix per bond.
- **DoRA’s detached-norm backward** (Liu et al., 2024b) and **global-norm gradient clipping**. Both still have the form  $\Lambda D\rho_{\text{LoRA}}^\top \gamma$ , with  $\gamma$  the detached or clipped signal, so  $\Phi$  is still conserved.

Under discrete GD with rates  $\eta_B, \eta_A$ , one step changes  $\Phi$  by exactly

$$\eta_B AG^\top GA^\top - \eta_A B^\top GG^\top B,$$

so the drift of  $\eta\Phi$ , or the drift relative to  $\Phi$ , is  $O(\eta^2)$  per step. Adam is not of the form  $\Lambda D\rho^\top \gamma$ , so the drift of  $\Phi$  under Adam measures how far Adam departs from the gauge geometry; this is a prediction (Exposé X).



**Figure 7. Noether hyperbolas and the crossover** (Theorem III.5, Corollary III.6, Theorem III.9). (a) Scalar LoRA,  $w = w_0 + sba$  with  $s = 1$ , loss  $\frac{1}{2}(w - w^*)^2$ ,  $w^* - w_0 = 1$  and rates  $\eta_A = 1$ ,  $\eta_B = 2$ . Top: six gradient-flow trajectories (teal) run from the filled dots along their own charge hyperbolas  $b^2/\eta_B - a^2/\eta_A = \Phi_0$  (dashed grey; the slice  $\Phi = 0$  dashed green) to the fibre  $sab = w^* - w_0$  (ochre). From the zero- $B$  start  $(1.2, 0)$ , Adam (red) leaves its hyperbola and lands at another point of the fibre. Bottom: the drift  $\max_{t \leq 10} |\Phi(t) - \Phi_0|$  from the same start against the step size  $\eta$  (learning rates  $\eta\eta_A$ ,  $\eta\eta_B$ ). A GD step changes the charge at order  $\eta^2$ , so over a fixed time the GD drift is proportional to  $\eta$  and vanishes in the flow limit. An Adam step changes the charge at first order in  $\eta$ , and its drift stays of order one at every step size: Adam lies outside Theorem III.5 (Corollary III.6). (b) Matrix LoRA with  $m = n = 6$ ,  $r = 2$ , loss  $\frac{1}{2}\|W - W^*\|_F^2$ ,  $B_0 = 0$  and  $A_0 A_0^\top = I$  exactly, so that  $\Phi = -\kappa I$  with  $\kappa = 1$  and the crossover lies at  $\sigma^* = \mu|\kappa|/2 = 0.5$  (dashed ochre). Top: the closed-form gains  $g_\kappa(\sigma^2/\mu^2)$  (teal, the  $B$ -update) and  $f_\kappa(\sigma^2/\mu^2)$  (purple, the  $A$ -update) of Theorem III.9, with the plateau  $|\kappa|$  and the balanced asymptote  $\sigma/\mu$  dotted. The hollow circles, read off the factors along the integrated flow for both singular directions, lie on the curves. Shading marks the one-sided regime, the crossover, and the balanced regime (grey). Bottom: the singular values  $\sigma_1(t)$  and  $\sigma_2(t)$  of  $\Delta W$ , with time running downward on a log scale.  $\sigma_1$  crosses  $\sigma^*$  and settles in the balanced regime;  $\sigma_2$  stays below  $\sigma^*$ , where the  $A$ -update supplies only 7% of its growth rate, as in LoRA-FA. Integrators and tolerances are in §E.6.

**WHAT THIS EXPLAINS** Conservation needs only the shape  $\dot{q} = -\Lambda D\rho^\top \gamma$ , so it holds for any loss and any data. With  $\eta$  the common scale of the rates, a GD step changes  $\eta\Phi$  by  $O(\eta^2)$  and an Adam step by  $O(\eta)$ , as panel (a) of Figure 7 shows.

**ANALOGY** (Noether; gauge). *Noether* names the passage from a symmetry to a conserved quantity, here proved for gradient flow without a Lagrangian. *Gauge* means the symmetry group of a parametrisation; no connection is involved.

A conserved charge is fixed at time zero, so it labels initialisations. Weight decay moves it on purpose, and its two conventions move it toward different targets.

**COROLLARY III.6** (charge at initialisation; weight decay). (a)  $\Phi_0 = B_0^\top B_0 / \eta_B - A_0 A_0^\top / \eta_A$  is the value of the conserved charge, fixed by the initialisation and the rates. Since  $O(r)$  acts on  $\Phi$  by conjugation and preserves  $\theta$ -trajectories ([Theorem III.5\(ii\)](#)), only the spectrum of  $\Phi_0$  labels a  $\theta$ -trajectory.

| Initialisation                                                                                                                                                                                                                                                                                                                                         | Charge $\Phi_0$                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| HF default LoRA on Linear and Conv layers ( $B_0 = 0$ ; Kaiming-uniform $A_0$ with entries $U(-n^{-1/2}, n^{-1/2})$ , $n$ the fan-in, which is <code>in_channels</code> times the kernel size for Conv)                                                                                                                                                | $-A_0 A_0^\top / \eta_A$ , with $\mathbb{E}[A_0 A_0^\top] = \frac{1}{3} I_r$ and fluctuations $O(n^{-1/2})$ on and off the diagonal |
| HF default LoRA on embedding layers ( $A_0 = 0$ , normal $B_0$ )                                                                                                                                                                                                                                                                                       | $+B_0^\top B_0 / \eta_B$ (Init[B]-type)                                                                                             |
| HF mica ( <a href="#">Rüdiger and Raschka, 2026</a> ) ( $A_0 = 0$ , frozen minor-singular $B_0$ )                                                                                                                                                                                                                                                      | $+B_0^\top B_0 / \eta_B$ (Init[B]-type, with $B$ frozen)                                                                            |
| EVA ( <a href="#">Paischer et al., 2024</a> ) (orthonormal rows; default <code>whiten=False</code> )                                                                                                                                                                                                                                                   | $-I / \eta_A$ exactly                                                                                                               |
| PiSSA in HF PEFT ( <a href="#">Meng et al., 2024a</a> ), and MiLoRA ( <a href="#">Wang et al., 2024b</a> ) if implemented like <code>pissa_init</code> with the bottom- $r$ singular values (MiLoRA is not in HF PEFT); $B_0^\top B_0 = A_0 A_0^\top = S_r / s$ , with $S_r$ the top- $r$ , resp. bottom- $r$ , singular values and $s$ the LoRA scale | $(S_r / s)(1 / \eta_B - 1 / \eta_A)$ , zero <b>iff</b> $\eta_A = \eta_B$ (when $S_r \neq 0$ )                                       |
| OLoRA ( <a href="#">Büyükyüz, 2024</a> ) ( $B_0 = Q_r$ , $A_0 = R_r$ )                                                                                                                                                                                                                                                                                 | $I / \eta_B - R_r R_r^\top / \eta_A$                                                                                                |
| “Init[B]” ( <a href="#">Hayou et al., 2024a</a> ) ( $A_0 = 0$ )                                                                                                                                                                                                                                                                                        | $+B_0^\top B_0 / \eta_B$                                                                                                            |

(b) Let  $G$  be any cotangent signal, with the scale  $s$  absorbed into it. Under gradient flow, if decay acts per unit time and identically on both factors,

$$\dot{B} = -\eta_B G A^\top - \lambda B, \quad \dot{A} = -\eta_A B^\top G - \lambda A,$$

then  $\dot{\Phi} = -2\lambda\Phi$ . With learning-rate-coupled decay (PyTorch SGD `weight_decay`; the decay term of PyTorch AdamW),  $\dot{B} = -\eta_B(G A^\top + \lambda B)$  and likewise for  $A$ , so that  $\dot{\Phi} = -2\lambda(B^\top B - A A^\top)$ . Then  $\Phi(t) = e^{-2\eta\lambda t}\Phi(0)$  holds when  $\eta_A = \eta_B = \eta$ . For  $\eta_A \neq \eta_B$  it fails in general, since  $-2\lambda(B^\top B - A A^\top)$  is not a function of  $\Phi$ . With an actual Adam(W) gradient term,  $\Phi$  drifts even without decay, at first order in the step size; Adam lies outside [Theorem III.5](#). So neither law describes AdamW training.

NUM Testing  $\Phi(t) = e^{-2\eta\lambda t}\Phi(0)$  at  $\eta_B / \eta_A = 4$  with coupled decay gives a relative error of 10.7; in the valid cases the error is that of the integrator, between  $10^{-8}$  and  $2 \times 10^{-4}$ . One Adam sign step moves  $\Phi$  by an amount proportional to the learning rate, while an SGD step moves  $\eta\Phi$  at second order ([Appendix E](#)).

#### WHAT THIS EXPLAINS

- Zero-product initialisations keep their charge under gradient flow, and that charge is maximally unbalanced ([Proposition III.8](#)). The relative imbalance  $\|\Phi\| / (\|B\|^2 / \eta_B + \|A\|^2 / \eta_A)$  decays as the adapter grows ([Theorem III.9](#)).
- HF PiSSA always has  $\mu_0 = B_0^\top B_0 - A_0 A_0^\top = 0$  (unweighted-balanced), and  $\Phi_0 = 0$  only at equal rates. So it starts on the slice that coupled decay targets, and off the per-time-decay slice when  $\eta_A \neq \eta_B$ .

- Both decay conventions gauge-fix at equilibrium, because each is a  $\Lambda$ -gradient flow of  $L$  plus a fibre-wise Kempf–Ness potential (Background D.30). Per-time decay is the flow of  $L + \frac{\lambda}{2}(\|B\|^2/\eta_B + \|A\|^2/\eta_A)$ ; it drives  $\Phi \rightarrow 0$  exponentially, onto the rate-weighted slice  $B^\top B/\eta_B = AA^\top/\eta_A$ . Coupled decay is the flow of  $L + \frac{\lambda}{2}(\|B\|^2 + \|A\|^2)$ ; its equilibria lie on the unweighted balanced slice  $\mu = B^\top B - AA^\top = 0$  of Proposition III.7(a), with no closed-form law for  $\Phi$  unless  $\eta_A = \eta_B$ . The two slices coincide iff  $\eta_A = \eta_B$ . In particular, coupled decay also gauge-fixes at LoRA+ rates, onto  $\mu = 0$ .

NUM At  $\eta_B/\eta_A = 4$ , coupled decay converges to  $\mu = 0$  to within  $10^{-12}$  while  $\Phi$  stays of order 1; per-time decay converges to  $\Phi = 0$  while  $\mu$  stays of order 1 (Appendix E).

The names Init[A] and Init[B], and the finding that these zero-product starts train differently, are due to Hayou et al. (2024a). Exponential decay of such charges under weight decay is the rescale-symmetry law of Kunin et al. (2020).

### The balanced slice

In Kempf–Ness theory, the least-norm vectors on a closed orbit of a reductive group form one orbit of its compact part, and they are where the moment map vanishes (Background D.29, Background D.30). For LoRA the moment map is  $\mu = B^\top B - AA^\top$ , and the least-norm representatives are the balanced ones.

**PROPOSITION III.7** (Kempf–Ness slice; Loewner floor; known, Kempf and Ness (1979); Srebro et al. (2004)). (a) On a full-rank  $\mathrm{GL}_r$ -orbit, the critical points of  $\|B\|^2 + \|A\|^2$  are exactly the balanced representatives, those with  $\mu = B^\top B - AA^\top = 0$ . They form one  $O(r)$ -orbit, they are the minima, and the minimum value is  $2\|BA\|_*$ .

(b) Assume gradient flow with block-scalar rates  $\eta_A, \eta_B$ , no weight decay, a loss that depends on  $(B, A)$  only through  $BA$ , and any cotangent signal of the form  $\Lambda D\rho^\top \gamma$  (dropout, minibatches and clipping included). From  $B_0 = 0$ ,

$$B^\top B = \frac{\eta_B}{\eta_A} (AA^\top - A_0 A_0^\top) \succeq 0,$$

so  $AA^\top \succeq A_0 A_0^\top$  for all time; the  $A$ -factor never shrinks below its initialisation in the Loewner order.  $AA^\top(t)$  need not be monotone in  $t$ ; only this floor is guaranteed. Discrete steps break the bound by  $O(\eta^2)$  per step. Under Adam it is not guaranteed and fails empirically. With per-time decay  $\lambda$ , only  $AA^\top \succeq e^{-2\lambda t} A_0 A_0^\top$  holds.

Part (a) is the theorem of Kempf and Ness (1979), in the real form of Richardson and Slodowy (1990); the nuclear-norm identity is due to Srebro et al. (2004). BaLoRA (Castin et al., 2026) turns the slice into an algorithm, moving after every step to a product-preserving representative with  $B^\top B = AA^\top$  diagonal.

NUM On 5 random orbits at  $(m, n, r) = (7, 5, 3)$  the minimum equals  $2\|BA\|_*$  to within  $10^{-14}$  (Appendix E).

**WHAT THIS EXPLAINS** Balancedness is a precise notion: it is the Kempf–Ness slice of the gauge action. An explicit penalty  $\lambda(\|B\|^2 + \|A\|^2)$  is equivalent at its minimisers to a  $(2\lambda/s)$ -nuclear-norm penalty on  $\Delta W = sBA$  restricted to rank  $\leq r$ . SGD-coupled decay with coefficient  $\lambda$  is the flow of  $L + \frac{\lambda}{2}(\|B\|^2 + \|A\|^2)$  and matches a  $(\lambda/s)$ -nuclear-norm penalty (Srebro et al., 2004, whose argument also covers rank-deficient factorisations). In both cases the stationary points lie on  $\mu = 0$ . With per-time decay at unequal rates the relevant slice is  $\Phi = 0$  (Corollary III.6). Decoupled AdamW decay is not covered.

The default start is as unbalanced as possible, and no start meets all three natural wishes.

**PROPOSITION III.8** (the balance–escape trilemma). *No LoRA initialisation is simultaneously*

- (a) *identity-at-init without a residual split* ( $B_0 A_0 = 0$ ),
- (b) *balanced* ( $B_0^\top B_0 = c A_0 A_0^\top$  for some  $c > 0$ : unweighted,  $c = 1$ , or rate-weighted,  $\Phi_0 = 0$  with  $c = \eta_B / \eta_A$ ), and
- (c) *non-stationary, meaning that  $D\rho$  at  $(B_0, A_0)$  is not the zero map; equivalently,  $\dot{W}(0) \neq 0$  for some parameter velocity, under any first-order optimizer.*

Moreover, if  $B_0 A_0 = 0$  then  $\langle B_0^\top B_0, A_0 A_0^\top \rangle_F = 0$ , so

$$\|B_0^\top B_0 - c A_0 A_0^\top\|_F^2 = \|B_0^\top B_0\|_F^2 + c^2 \|A_0 A_0^\top\|_F^2,$$

and zero-product initialisations are maximally unbalanced.

**WHAT THIS EXPLAINS** The field splits into two families of initialisation.

- **Type I** keeps  $\rho$  and picks  $q_0$  in the apex fibre  $\{BA = 0\}$ : LoRA, EVA, Init[B]. By the identity above these are maximally unbalanced.
- **Type II** changes  $\rho$  by an exact split  $\theta_0 = \theta_{\text{res}} + \delta(q_0)$ : PiSSA, MiLoRA, OLoRA, LoRA-GA (Wang et al., 2024e), CorDA (Yang et al., 2024c). LoftQ (Li et al., 2023b) and QPiSSA (Meng et al., 2024a) are approximate splits over a quantised residual, so they are not identity-at-init (Proposition V.2).

The proposition forces every balanced, non-stationary, identity-at-init LoRA initialisation to be Type II. The converse fails: OLoRA ( $B_0^\top B_0 = I$ ,  $A_0 A_0^\top = R_r R_r^\top$ ) and CorDA are Type II and generally unbalanced. A frozen residual is the price of any nonzero  $B_0 A_0$ , and so in particular of balance, as in PiSSA and MiLoRA at equal rates. By Theorem II.2, Type II objects are different objects, incomparable with LoRA.

## Two regimes and a crossover

When the charge is a multiple of the identity, the LoRA flow has a closed form. Below a scale  $\sigma^*$ , one factor carries the motion, as in LoRA-FA (Zhang et al., 2023d); above it, the factors share the work, as in balanced deep linear networks.

**THEOREM III.9** (closed-form isotropic-charge LoRA dynamics; known, Tarmoun et al. (2021) and Min et al. (2021)). *Let  $\rho(B, A) = W_{\text{res}} + sBA$  be trained by gradient flow with rates  $\eta_A, \eta_B$ , and suppose the charge is isotropic,  $\Phi = -\kappa I_r$ . Put  $\mu = s\sqrt{\eta_A \eta_B}$  (a rate, unrelated to the moment map  $\mu$  of Proposition III.7), and on the locus where  $Z = W - W_{\text{res}}$  has rank  $r$  write its compact SVD  $Z = U\Sigma V^\top$ . Then*

$$\dot{W} = -\mu^2 [G V g_\kappa(\Sigma^2 / \mu^2) V^\top + U f_\kappa(\Sigma^2 / \mu^2) U^\top G],$$

where

$$f_\kappa(x) = \sqrt{x + \kappa^2 / 4} - \kappa / 2, \quad g_\kappa = f_\kappa + \kappa.$$

The identity holds pointwise on  $\{\Phi = -\kappa I, \text{rank } Z = r\}$ ; gradient flow is used only to keep  $\Phi$  fixed. The two terms under the root are equal at the crossover scale

$$\sigma^* = \mu |\kappa| / 2.$$



- For  $\sigma \ll \sigma^*$  with  $\kappa > 0$ ,  $f \rightarrow 0$  and  $g \rightarrow \kappa$ , so  $\dot{W} \approx -\mu^2 \kappa G P_V$ . These are LoRA-FA dynamics. The contribution of  $A$  to  $\dot{W}$  is  $O(\sigma^2/(\mu^2 \kappa))$ , although  $A$  itself still drifts slowly. For  $\kappa < 0$  the roles swap,  $\dot{W} \approx -\mu^2 |\kappa| P_U G$ , and only  $A$  effectively moves ( $\text{Init}[B]$ ).
- For  $\sigma \gg \sigma^*$ ,  $f \approx g \approx \sigma/\mu$ , so

$$\dot{W} \approx -\mu [G(Z^\top Z)^{1/2} + (ZZ^\top)^{1/2}G].$$

These are the balanced dynamics of Arora, Cohen and Hazan.

- Off the rank- $r$  locus. Write  $\dot{W} = -\mu^2(GS + TG)$  with  $S = A^\top A/\eta_A$  and  $T = BB^\top/\eta_B$ , so that on the rank- $r$  locus  $S = V g_\kappa(\Sigma^2/\mu^2) V^\top$  and  $T = U f_\kappa(\Sigma^2/\mu^2) U^\top$ . For  $\kappa \geq 0$ ,  $T = f_\kappa(ZZ^\top/\mu^2)$  still holds as a full matrix function, but  $S$  gains the term  $\kappa(P_{\text{row } A} - P_V)$ ; symmetrically, for  $\kappa < 0$ ,  $T$  gains  $|\kappa|(P_{\text{col } B} - P_U)$ . At the default-LoRA start  $Z = 0$  ( $B_0 = 0$ ) this leaves  $S = \kappa P_{\text{row } A_0}$  and  $T = 0$ , so  $\dot{W} = -\mu^2 \kappa G P_{\text{row } A_0}$ , and the  $\sigma \ll \sigma^*$  regime extends continuously to  $t = 0$ .

The closed form is due to Tarmoun et al. (2021), who call this charge “ $\lambda$ -balanced”, and to Min et al. (2021); see also Dominé et al. (2025) and Kunin et al. (2024). The balanced regime is that of Arora et al. (2018). The PEFT reading is ours. Panel (b) of Figure 7 shows the closed-form singular-value dynamics and the crossover scale  $\sigma^*$ .

NUM The formula matches direct integration to  $10^{-12}$ , and the pointwise identity holds to  $10^{-15}$  for  $\kappa \in \{-2, -0.3, 0, 0.3, 2, 10\}$  and all shape cases (Appendix E).

**WHAT THIS EXPLAINS** Call this the PEFT reading; it is scoped to gradient flow and SGD.

- *Default LoRA.* With  $B_0 = 0$  and  $A_0 A_0^\top = cI$ ,  $\kappa = c/\eta_A$  and  $\sigma^* = \frac{sc}{2} \sqrt{\eta_B/\eta_A}$ ; HF’s default gives  $c \approx 1/3$ .
- *LoRA+ ( $\lambda = \eta_B/\eta_A$ )* multiplies  $\sigma^*$  by  $\sqrt{\lambda}$  at fixed  $\eta_A$ , and its balanced-regime rate is the geometric mean  $s\sqrt{\eta_A \eta_B}$ . LoRA-FA is the endpoint  $\lambda \rightarrow \infty$  of the LoRA+ family.
- *PiSSA at equal rates* has  $\kappa = 0$  and is exactly balanced forever under gradient flow; SGD drifts by  $O(\eta^2)$  per step, and Adam does not preserve it.
- *Exactness.* The theorem is exact for EVA- and LoRA-GA-type orthonormal frames. For Kaiming initialisations it holds only approximately. The eigenvalues of  $A_0 A_0^\top$  lie near  $c(1 \pm \sqrt{r/n})^2$  (Marčenko and Pastur, 1967; Auxiliary result B.35), so the crossover is a band of relative width  $O(\sqrt{r/n})$  rather than a single scale. At  $r = 256$ ,  $n = 4096$ , the eigenvalues of  $3A_0 A_0^\top$  span roughly 0.57 to 1.57.
- *Random  $A$ .* The one-sided regime fits the finding of Zhu et al. (2024b) that a random, untrained  $A$  does nearly as well as a tuned one; the theorem says that this regime lasts while  $\Delta W$  stays well below  $\sigma^*$ .

PREDICTION Under SGD, singular values of  $\Delta W$  switch from one-sided to two-sided growth as they cross  $\sigma^*$  (Prediction X.7).

### The hyperparameter torus

Rescale block  $i$  by  $c_i$  and divide  $s$  by the matching power of the  $c_i$ . The method is the same object of **Meth**, and training is the same if every step rescales alike. Adam ignores the size of a block’s gradient, so its rate must scale like  $c_i$ . An SGD step scales with the gradient, which shrinks by  $c_i$ , so its rate must scale like  $c_i^2$ .

**THEOREM III.10** (the hyperparameter torus; the LoRA/Adam case is known, [Schulman and Thinking Machines Lab \(2025\)](#)). Let  $\rho(q) = \theta_b + s \delta(q_1, \dots, q_k)$  with  $\delta$  multihomogeneous of multidegree  $(d_1, \dots, d_k)$ . Train block  $i$  from  $q_i(0) = \sigma_i u_i$  with Adam ([Kingma and Ba, 2015](#)) ( $\beta_1 \in [0, 1)$ ,  $\beta_2 \in (0, 1)$ , bias correction allowed) at learning rate  $\eta_i(t)$ , on the loss  $L \circ \rho$  with no  $q$ -dependent regulariser. Assume no gradient clipping (or per-block clipping with thresholds rescaled as below), common random numbers (the same data order and dropout masks in both runs), and exact arithmetic (the result is bit-exact in floating point when every  $c_i$  is a power of 2). Then the trajectory  $t \mapsto \rho(q(t))$  is invariant under the torus action of  $c \in \mathbb{R}_{>0}^k$ :

$$(s, \eta_i(t), \sigma_i, \varepsilon_i, \lambda_i, C_i) \mapsto \left( s \prod_i c_i^{-d_i}, c_i \eta_i(t), c_i \sigma_i, \varepsilon_i / c_i, \lambda_i / c_i, C_i / c_i \right).$$

Here  $\varepsilon_i$  is a per-block Adam  $\varepsilon$  (with the convention  $0/0 = 0$  at  $\varepsilon = 0$ ),  $\lambda_i$  a per-block decoupled (AdamW) weight decay, and  $C_i$  a per-block clipping threshold. Under SGD, with or without heavy-ball or Nesterov momentum, the same holds with  $\eta_i(t) \mapsto c_i^2 \eta_i(t)$  and coupled  $L^2$  decay  $\lambda_i \mapsto \lambda_i / c_i^2$ . In particular, any learning-rate schedule shared by all blocks, warm-up included, is preserved.

The LoRA/Adam case, with the  $\varepsilon > 0$  extension and the two effective hyperparameters, is due to [Schulman and Thinking Machines Lab \(2025\)](#) (LoRA Without Regret). New here are arbitrary multihomogeneous  $\delta$  and the SGD and heavy-ball  $c^2$  law.

NUM Trajectory differences are exactly 0.0 for power-of-two  $c$  and about  $10^{-15}$  otherwise, including a warm-up schedule and rescaled  $\varepsilon$  and decay. With  $\varepsilon$  or decay held fixed, or with global-norm clipping, they are  $10^{-3}$  to  $O(1)$  (Appendix E).

**WHAT THIS EXPLAINS** Only the torus orbit of the hyperparameters reaches the weights. LoRA is the case  $k = 2$ ,  $d = (1, 1)$ ;  $\delta$  may be any multilinear update, such as that of VeRA ([Kopiczko et al., 2023](#)) in its two scaling vectors or that of LoHa ([Hyeon-Woo et al., 2021](#)) in its four factors.

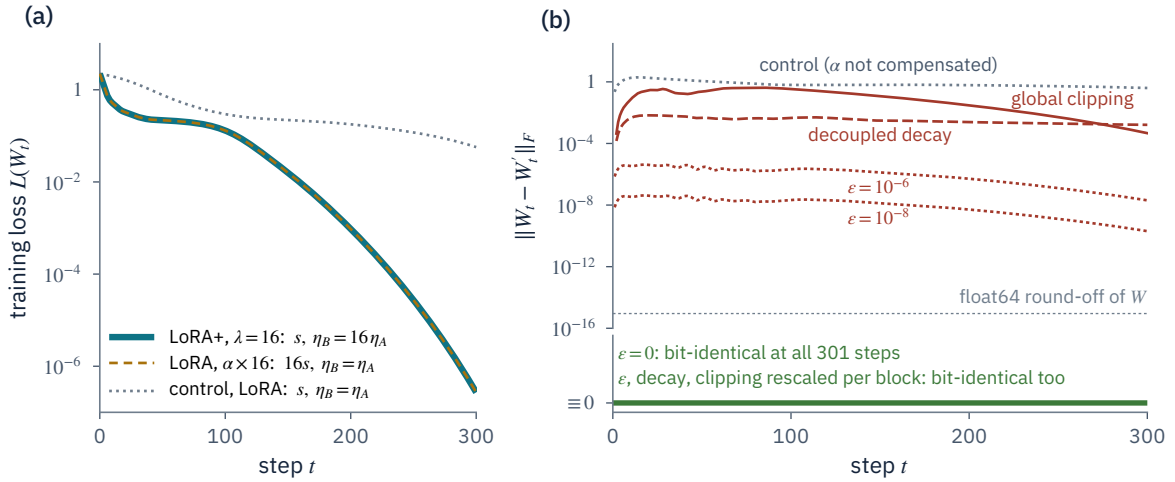
**COROLLARY III.11** (LoRA+ is  $\alpha$  in disguise; rsLoRA is LoRA; (a) and the LoRA+ reading are known, [Schulman and Thinking Machines Lab \(2025\)](#)). For linear LoRA modules ( $k = 2$ ,  $d = (1, 1)$ ,  $\sigma_B = 0$ ), under the hypotheses of [Theorem III.10](#):

(a) Effective hyperparameters. Under Adam the torus invariants of  $(s, \eta_A, \eta_B, \sigma_A)$  are  $(s\eta_A\eta_B, \sigma_A/\eta_A)$ ; the quantities  $\alpha \cdot \text{init}_A \cdot \text{LR}_B$  and  $\text{init}_A/\text{LR}_A$  of Schulman et al. are equivalent generators. Under SGD the invariants are  $(s^2\eta_A\eta_B, \sigma_A^2/\eta_A)$ .

(b) LoRA+. Under Adam with  $\varepsilon = 0$ , no weight decay, no gradient clipping and a learning-rate schedule shared by both groups, LoRA+ with ratio  $\lambda$  at scale  $s$  produces exactly the trajectory of plain LoRA at scale  $\lambda s$  (same  $\eta_A$ , same  $A_0$ ). For  $\varepsilon > 0$  the equivalence needs a per-block  $\varepsilon'_B = \lambda \varepsilon$ , and with AdamW a per-group decay  $\lambda \cdot \text{wd}$  on  $B$ . Under SGD it holds at scale  $\sqrt{\lambda} s$ , or equivalently with plain LoRA at the single rate  $\sqrt{\eta_A\eta_B}$  and  $A_0 \mapsto \lambda^{1/4} A_0$ .

(c) rsLoRA.  $\rho_{\text{rs}}(B, A) = \rho_{\text{LoRA}}(\sqrt{r} B, A)$  is an isomorphism of objects. Up to the hyperparameter torus, rsLoRA can be represented as LoRA at the same  $\alpha$  with  $\eta_B$  multiplied by  $\sqrt{r}$  under Adam (by  $r$  under SGD), or equally as LoRA with  $\alpha$  multiplied by  $\sqrt{r}$  at the same rates. The reading “same object, different metric” is a choice of representative within the orbit. Part (c) concerns a fixed rank; it does not address rsLoRA’s rank-stability prescription, how  $\alpha$  should scale with  $r$ .

LoRA+ is due to [Hayou et al. \(2024b\)](#), rsLoRA to [Kalajdziewski \(2023\)](#), and (a) with the LoRA+ reading to [Schulman and Thinking Machines Lab \(2025\)](#). On the empirical side, [Lee](#)



**Figure 8. LoRA+ is  $\alpha$  in disguise** (Theorem III.10, Corollary III.11). A seeded toy LoRA regression with  $B_0 = 0$  is trained by Adam in float64, once as LoRA+ with ratio  $\lambda = 16$  (scale  $s, \eta_B = 16\eta_A$ ) and once as plain LoRA with  $\alpha$  multiplied by 16 (scale  $16s, \eta_B = \eta_A$ ), at the same  $\eta_A$  and  $A_0$ . (a) At  $\epsilon = 0$ , without weight decay or clipping, the two loss curves coincide; the control, plain LoRA at the uncompensated scale  $s$ , ends far above them. (b) The weight gap  $\|W_t - W'_t\|_F$  between the two runs at every step, on a log scale above an exact-zero floor. At  $\epsilon = 0$  the runs are bit-identical at every step: the torus element relating them,  $c = (1, 1/16)$ , is a power of two and rescales floating-point numbers exactly. Each red curve switches on one ingredient with a single value shared by both parameter groups: Adam’s  $\epsilon$ , decoupled weight decay, or global-norm clipping. Each breaks the hyperparameter torus; the control is dotted. Rescaling the same ingredients per block, as Theorem III.10 prescribes, restores the exact zero (green floor). The values used are in §E.6.

et al. (2026) find that nine LoRA variants reach similar peak performance once the learning rate is tuned.

NUM The maximum trajectory difference is exactly 0.0 at  $\epsilon = 0$  for power-of-two  $\lambda$  and about  $10^{-14}$  otherwise, against 0.93 for the unrescaled control. It is  $5 \times 10^{-9}$  at a shared  $\epsilon = 10^{-8}$ , and 0.6 with global-norm clipping at 0.5 (Appendix E).

**WHAT THIS EXPLAINS** LoRA’s four knobs are coordinates on a 4-dimensional space acted on by a 2-torus, and only the orbit matters. LoRA+, rsLoRA and  $\alpha$ -tuning therefore collapse into one family. The slogan depends on the chart. Holding  $\eta_A$  fixed, LoRA+ is “ $\alpha$  multiplied by  $\lambda$ ”; holding  $\eta_B$  fixed, it is plain LoRA with  $\eta_A$  divided by  $\lambda$ , a slower  $A$ -timescale. In particular, Adam does not make the LoRA+ ratio non-absorbable.

**PREDICTION** Any gain of LoRA+ over LoRA with  $\alpha$  re-tuned by the factor  $\lambda$  must come from the ingredients that break the extended torus: global-norm clipping (on by default in the HF Trainer, `max_grad_norm=1.0`),  $\epsilon$  or weight decay held equal across parameter groups (as in HF PEFT’s LoRA+ optimizer), and learning-rate schedules that differ between groups. Shared schedules and warm-up preserve the torus. HF’s LoRA+ optimizer treats embedding adapters and 1-D parameters separately, so the statement covers linear modules only (Prediction X.1).

### What the optimizer can see

An optimizer is equivariant under a gauge element  $g$  if starting from  $(Bg^{-1}, gA)$  gives the transported trajectory. The Euclidean metric of SGD respects rotations of the rank channels. Adam, acting coordinatewise, respects only the signed permutations.

**REMARK** (2026: privacy noise). Privacy noise fits the same pattern. DP-SGD (Abadi et al., 2016) on the factors adds isotropic Gaussian noise  $\xi$  to the clipped factor gradients, and  $\xi g^{-1}$  has the law of  $\xi$  only for orthogonal  $g$ , so it respects  $O(r)$  and no larger subgroup. PRISM (Wang and Zhang, 2026) clips and noises the projection of each gradient onto the tangent space of  $M_r$  at  $BA$ , which depends only on  $\text{col } B$  and  $\text{row } A$ . For full-rank factors the privatised gradient then has the same law for every factorisation of  $BA$ . Its Algorithm 1 then applies damped  $r \times r$  preconditioners (its Eqs. 24–26), which the paper also calls gauge invariant. They respect only  $O(r)$ . Under  $(B, A) \mapsto (cB, A/c)$  the preconditioned factor steps do not change, so  $\delta B A$  scales by  $1/c$  and  $B \delta A$  by  $c$ . The released code returns to balanced factors,  $B^\top B = AA^\top$ , after every step, where the remaining gauge is  $O(r)$ .

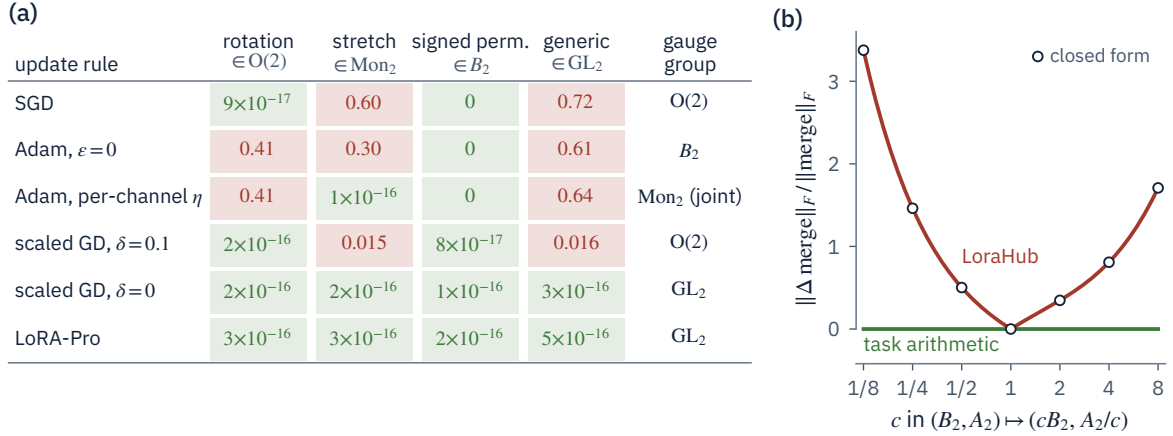
**THEOREM III.12** (the optimizer gauge hierarchy). *On LoRA’s full-rank locus, with hyperparameters shared by all rank channels, the largest subgroup of the gauge  $GL_r$  under which an update rule is equivariant, at the level of parameters, is as follows.*

- (a) SGD and GD with any block rates, with or without heavy-ball momentum:  $O(r)$ .
- (b) Adam and AdamW at fixed learning rates  $(\beta_1, \beta_2 \in [0, 1])$ , any  $\varepsilon \geq 0$ , any weight decay): the signed permutations  $B_r$ , of order  $2^r r!$ . A positive diagonal  $g = \text{diag}(d)$  is a symmetry only jointly with per-rank-channel hyperparameters, and only without global-norm clipping. At  $\varepsilon = 0$  and without weight decay it suffices to rescale the rates,  $\eta_{B,j} \mapsto \eta_{B,j}/d_j$  and  $\eta_{A,j} \mapsto d_j \eta_{A,j}$ . With  $\varepsilon > 0$  one must also give each channel its own  $\varepsilon_{B,j} = d_j \varepsilon$  and  $\varepsilon_{A,j} = \varepsilon/d_j$ . With decoupled decay multiplied by the learning rate (PyTorch AdamW) one must also set  $\lambda_{B,j} = d_j \lambda$  and  $\lambda_{A,j} = \lambda/d_j$ ; under the convention of Loshchilov and Hutter (2019), where decay is scaled only by the schedule multiplier,  $\lambda$  needs no change.
- (c) Undamped scaled GD ( $\delta = 0$ ; the Riemannian preconditioner of Tong, Ma and Chi and of Zhang and Pilanci, with or without heavy-ball momentum): all of  $GL_r$ . The  $W$ -trajectory from gauge-related initial states (with zero or correspondingly transformed momentum) is independent of the representative; a step may drop rank, so the iteration maps into  $W_0 + M_{\leq r}$ .

Scaled GD is due to Tong et al. (2021) and Zhang and Pilanci (2024a). LoRA-RITE (Yen et al., 2024) makes gauge invariance a design requirement for LoRA optimizers and reports that it pays.

**Published variants.** Damped scaled GD ( $\delta > 0$ ) is only  $O(r)$ -equivariant. Wrapped in AdamW, as in Zhang and Pilanci’s scaled AdamW (their practical variant) and in HF’s `create_riemannian_optimizer`, it is  $B_r$ -equivariant by (b). The first-order  $W$ -level dynamics of LoRA-Pro (Wang et al., 2024i) descend (Proposition III.4(b)), and so do those of its variant with free matrix  $X = 0$ ; its published parameter update with the Sylvester-optimal  $X$  is only  $O(r)$ -equivariant, and its discrete  $W$ -step depends on the representative at  $O(\eta^2)$ . LoRA-RITE states only  $W$ -level invariance (its Definition 1). Its update (unmagnified gradients, transported matrix moments, then multiplication by  $(R_B^\top)^+$ ) is  $GL_r$ -equivariant at the parameter level on the full-rank locus: under a gauge change its internal quantities change only by orthogonal matrices, and the final multiplication supplies the gauge element (Appendix C.35, which follows its Algorithm 1 line by line; the resulting step is Eq. 15 of Yen et al. (2024)). LoRA-RITE is therefore an Adam-type optimizer in class (c).

**NUM** For Adam, the defects are  $10^{-15}$  for signed permutations, 11.3 for a positive diagonal, 6.6 for a rotation, and  $5 \times 10^{-16}$  for the diagonal with jointly rescaled per-channel rates at  $\varepsilon = 0$ . With  $\varepsilon = 10^{-8}$  left unrescaled the joint torus fails by about 2%. LoRA-RITE (its Algorithm 1) from  $(B, A)$  and from  $(Bg^{-1}, gA)$  with a generic  $g$  agrees to  $3 \times 10^{-14}$  over 30



**Figure 9. Update rules that see the gauge** (Theorem III.12, Proposition III.4, Proposition V.4). (a) At a fixed seeded LoRA state  $(B, A)$  with loss gradient  $G$ , each cell is the relative gauge defect  $\|\delta W(g) - \delta W(I)\|_F / \|\delta W(I)\|_F$  of one update rule: the change in its first-order weight update  $\delta W = \delta B A + B \delta A$  when the factors are replaced by the gauge-equivalent pair  $(Bg^{-1}, gA)$ . The columns are a rotation, a stretch, a signed permutation and a generic  $g \in \text{GL}_2$ ; under each name is the smallest of the groups  $B_2 = O(2) \cap \text{Mon}_2$ ,  $O(2)$ ,  $\text{Mon}_2$  (monomial matrices) and  $\text{GL}_2$  that contains it. Green cells vanish up to float64 round-off, and they occur exactly where  $g$  lies in the rule's gauge group (right column):  $O(2)$  for SGD and for damped scaled GD,  $B_2$  for Adam,  $\text{Mon}_2$  for Adam whose per-rank-channel rates are rescaled jointly with a stretch, and all of  $\text{GL}_2$  for undamped scaled GD and for LoRA-Pro with the Sylvester-optimal free matrix, whose weight updates depend only on  $W$ . Red cells are genuine defects. (b) Three LoRA modules  $(B_i, A_i)$  with weights  $w_i$  are merged after module 2 alone is re-gauged to  $(cB_2, A_2/c)$ . Task arithmetic (Ilharco et al., 2022),  $\sum_i w_i B_i A_i$ , does not move. The LoraHub merge (Huang et al., 2023),  $(\sum_i w_i B_i)(\sum_j w_j A_j)$ , does, and the open circles, the closed form of Proposition V.4, match the recomputed change. The settings are in §E.6.

steps (referee replication). LoRA-Pro with the Sylvester  $X$  has a parameter-level defect of 1.2 under a non-orthogonal  $g$ , against  $2 \times 10^{-15}$  with  $X = 0$  or with an orthogonal  $g$  (Appendix E).

**WHAT THIS EXPLAINS** There is a hierarchy  $B_r \subset O(r) \subset \text{GL}_r$  of optimizer geometries, strict for  $r \geq 2$ .  $\text{GL}_r$ -equivariant rules, started from gauge-related states, define an optimizer on the *model* rather than on a chart. More generally any rule that maps gauge orbits to gauge orbits does (for instance a step followed by a canonical refactorisation of  $W$ ), while  $B_r$ - and  $O(r)$ -equivariant rules in general do not. Plain Adam, the default, sees a finite shadow of the gauge. LoRA+ and rsLoRA act on the hyperparameter torus (Theorem III.10) and leave the gauge of a fixed method alone; balanced or orthonormal initialisations choose a point in the fibre. The positive torus is a symmetry of the *hyperparameters*; Adam at fixed rates does not have it.

### What a zero- $B$ start chooses

Default LoRA starts at  $B_0 = 0$  with a random full-rank  $A_0$ . The image remembers only row  $A_0$  (Theorem II.2). Training remembers what the optimizer's gauge group cannot move.

**COROLLARY III.13** (what a zero- $B$  initialisation really chooses). *At  $q_0 = (0, A_0)$  with rank  $A_0 = r$ , with hyperparameters shared by the rank channels and the same data order and dropout masks, two choices of  $A_0$  give the same  $\theta$ -trajectory whenever the optimizer's gauge group relates them. The resulting invariant of  $A_0$  is:*



- row  $A_0 \in \text{Gr}(r, n)$  for an update rule that is  $\text{GL}_r$ -equivariant at every point of its trajectory, including  $B = 0$  and rank-deficient  $B$ , run without global-norm clipping. An example is  $\delta B = -\eta \nabla_B (AA^\top)^{-1}$ ,  $\delta A = -\eta (AA^\top) \nabla_A$ , which needs only  $\text{rank } A = r$ ;
- the PSD matrix  $A_0^\top A_0$  for SGD, which is exactly the initial preconditioner in

$$\dot{W}(0) = -s^2 \eta_B G A_0^\top A_0;$$

- $A_0$  up to signed permutation of its rows for Adam.

The undamped scaled-GD optimizers of [Theorem III.12\(c\)](#) are not defined at  $B = 0$ , since they need  $(B^\top B)^{-1}$ . LoRA-RITE uses pseudo-inverses and so is defined there, but its  $\text{GL}_r$ -equivariance is established only on the full-rank locus. Their damped forms are only  $\text{O}(r)$ -equivariant, so in practice their invariant is a  $\delta$ -dependent function of  $A_0^\top A_0$ , and their Adam-wrapped forms (Riemannian-preconditioned AdamW, HF's LoRA-FA optimizer) have the invariant  $A_0$  modulo  $B_r$ . Conversely, different invariants give different dynamics, for SGD and for Adam at  $\varepsilon = 0$ .

Random-A LoRA and EVA are different points of the  $B = 0$  stratum  $\{(0, A) : \text{rank } A = r\}$  of the apex fibre  $\rho^{-1}(\theta_0) = \{(B, A) : BA = 0\}$ ;  $\text{Init}[B]$  lies in another stratum of the same fibre. LoRA-FA is a different object, the sub-object  $\text{FA}(A_0) \hookrightarrow \text{LoRA}_{(0, A_0)}$  ([Proposition I.8](#)). More precisely, EVA is a point of the apex fibre of  $\text{LoRA}_{r_i}$  at its per-layer redistributed rank and adjusted  $\alpha$ , with orthonormal rows only when `whiten=False`.

Appendix [C.36](#) proves the converse (different invariants give different dynamics) by comparing first steps. [Hayou et al. \(2024a\)](#) showed that the two zero-product starts, random  $A$  with  $B = 0$  and random  $B$  with  $A = 0$ , train differently; here they lie in different strata of one apex fibre, with charges of opposite sign. EVA ([Paischer et al., 2024](#)) and random-A LoRA share a stratum.

NUM SGD from  $A_0$  and from  $QA_0$  (with  $Q$  orthogonal) agree to  $10^{-15}$ , and from  $A_0$  and  $DA_0$  (with  $D$  diagonal) differ by 0.81. Adam from  $A_0$  and from  $PA_0$  (with  $P$  a signed permutation) agree to  $10^{-16}$ , and from  $A_0$  and  $QA_0$  differ by 0.98. The example rule from  $A_0$  and from  $gA_0$  agrees to  $3 \times 10^{-15}$ . LoRA-RITE's Algorithm 1, run as written with its pseudo-inverses from  $B_0 = 0$ , gives the same weights from  $A_0$  and from  $gA_0$  for a generic  $g$  to  $10^{-13}$ , and weights that differ by  $O(1)$  for a different row space ([Appendix E](#)).

**WHAT THIS EXPLAINS** The strata of [Theorem II.2](#) are the expressive quotient. The dynamics see a finer quotient, and how fine depends on the optimizer ([Table 13](#)). Undamped scaled GD is undefined at  $B = 0$ , so scaled-GD optimizers enter in damped ( $\text{O}(r)$ ) or AdamW-wrapped ( $B_r$ ) form.

Two problems stay open. One is the apex fibre modulo  $B_r$  for Adam, stratified by  $\Phi_0$ , degenerate pointings included. The other is an Adam analogue of [Theorem III.9](#), a conserved or monotone quantity for sign-like updates.

Exposé [IV](#) reads the same cometric from the backward side of the lens, where methods that compress gradients have cometrics of their own.

**Table 13.** What an update rule reads off  $A_0$  from a zero- $B$  start (Corollary III.13).

| Update rule                                                                                                     | Equivariance group | What it reads off $A_0$ at $B_0 = 0$                    |
|-----------------------------------------------------------------------------------------------------------------|--------------------|---------------------------------------------------------|
| $GL_r$ -equivariant along the whole trajectory (the example rule; LoRA-RITE's Algorithm 1, checked numerically) | $GL_r$             | row $A_0 \in \text{Gr}(r, n)$                           |
| SGD, GD                                                                                                         | $O(r)$             | $A_0^\top A_0$                                          |
| damped scaled GD                                                                                                | $O(r)$             | $A_0^\top A_0$ (through a $\delta$ -dependent function) |
| Adam, AdamW; scaled AdamW (Zhang and Pilanci; HF's Riemannian optimizer with AdamW); HF's LoRA-FA optimizer     | $B_r$              | $A_0$ up to signed permutation of its rows              |

## The lens: tangent and cotangent methods

Backpropagation pairs every map with its reverse derivative, so a method can shape what the model expresses on the forward side or compress the gradient on the backward side. For linear methods and optimizers that read only their gradients (no weight decay, no clipping of the full gradient), the two sides give the same weights, and GaLore becomes one-sided LoRA with a moving frame and carried optimizer state.

### Two sides of one port

A layer does two jobs in training. Forward, it turns parameters and inputs into outputs. Backward, it turns a cotangent at the output into cotangents at the input and at the parameters. Backpropagation keeps the pair together, and a category theorist calls it a *lens* (Background B.17).

Every method in Exposés I to III changes the forward job, replacing the parameter port  $\Theta$  by a smaller space  $Q$  and a map  $\rho : Q \rightarrow \Theta$ . A second family edits only the backward job. GaLore (Zhao et al., 2024a) compresses the weight gradient  $G$  to  $P^\top G$ , with  $P$  from an SVD of  $G$ , runs Adam on that small matrix and maps the result back with  $P$ . Every weight trains; only the optimizer state is small. LISA (Pan et al., 2024) and MeZO (Malladi et al., 2023) follow the same pattern.

When  $\rho$  is linear the two families meet. The pullback of the forward method is then the compression of a backward method, and an optimizer that reads only its gradients cannot tell the two apart. The equivalence between GaLore and one-sided LoRA is due to Torroba-Hennigen et al. (2025), who prove a duality between linear gradient transformations and adapters (their Thm 1 and Cor. 3, with weight decay in their App. B). The frozen- $A$  case under SGD, with random projections, appears earlier in Hao et al. (2024). Our additions are the lens bookkeeping, the explicit hypotheses on the optimizer, and the table that sorts the cotangent family (Table 15).

### Backpropagation as a lens

In the language of Definition I.1, a layer is a parametrised map  $f : P \times X \rightarrow Y$ . Its reverse derivative  $R[f](x, y) = J_f(x)^\top y$  takes an output cotangent to an input cotangent, parameters included, and backpropagation is the 2-functor

$$\mathbf{Para}(R) : \mathbf{Para}(C) \longrightarrow \mathbf{Para}(\mathbf{Lens}(C)), \quad f \longmapsto (f, R[f]). \quad (6)$$

The parametrised-map construction and the reading of backpropagation as a functor into lenses are due to Fong et al. (2017) and Cruttwell et al. (2021) (Background B.16, Background B.17).

A reparametrisation  $\rho$  is a 2-cell, so the functor turns it into a lens  $(\rho, R[\rho])$ , and a reparametrisation therefore acts on both sides of the parameter port. Its tangent side is the image, which is what the method can *express* (Exposé II). Its cotangent side  $R[\rho](q, \gamma) = D\rho_q^\top \gamma$  pulls the weight gradient back to  $Q$  and, with the rates  $\Lambda$ , induces the cometric  $K = D\rho \wedge D\rho^\top$  (Definition III.1, Background D.23), which is how the method *moves*. A lens need not come from a map, though, and a backward half written down directly gives the second family. The

**Table 14.** Backward methods as compression–anchor pairs (Definition IV.1). The last row is the only reading of Flora as published in the form of Definition IV.1, the trivial lens with all compression inside opt.

| Method                                                                           | Compression $c_t$                                                                                | Anchor $a_t$                         |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|--------------------------------------|
| GaLore (Zhao et al., 2024a)                                                      | $P_t^\top$ , with $P_t$ from a gradient SVD                                                      | $P_t$                                |
| GaLore with a random projector (Hao et al., 2024; Torroba-Hennigen et al., 2025) | as GaLore, with $P_t$ random                                                                     | $P_t$                                |
| LISA (Pan et al., 2024)                                                          | layer-block coordinate projection                                                                | the same projection                  |
| MeZO (Malladi et al., 2023)                                                      | $g \mapsto \langle z_t, g \rangle$ , estimated by finite differences without a backward pass     | $\lambda \mapsto \lambda z_t$        |
| Flora as published (Hao et al., 2024)                                            | identity: only the momentum or accumulation buffer is compressed, by a random $A_t$ , inside opt | identity (full-space Adafactor step) |

two kinds of lens on the parameter port are

$$\begin{array}{ccc}
 Q & \xrightarrow{\rho} & \Theta \\
 T_q^* Q & \xleftarrow{D\rho_q^\top} & T_{\rho(q)}^* \Theta
 \end{array}
 \qquad
 \begin{array}{ccc}
 \Theta & \xrightarrow{\text{id}_\Theta} & \Theta \\
 T_\theta \Theta & \xleftarrow{a_t} & S \xleftarrow{c_t} T_\theta^* \Theta
 \end{array}$$

on the left a forward method, whose backward half is forced by its forward half, and on the right a backward method, whose forward half is the identity and whose backward half is chosen freely.

**DEFINITION IV.1** (forward and backward methods).

- A **forward** method acts on the parameter port by a lens in the image of  $R$ , namely  $(\rho, R[\rho])$ .
- A **backward**, or cotangent-side, method acts by a lens whose forward part is  $\text{id}_\Theta$  and whose backward part is a *compression*  $c_t : T_\theta^* \Theta \rightarrow S$  into a state space  $S$ , paired with an *anchor*  $a_t : S \rightarrow T_\theta \Theta$ . The update is

$$\theta \leftarrow \theta - \eta a_t(\text{opt}(c_t(\nabla L))). \quad (7)$$

Table 14 lists examples.

A backward method has its own cometric  $K_t = a_t c_t$  (Definition III.1), for instance  $K(G) = PP^\top G$  for GaLore under SGD. The update must lie in the image of the anchor. Read with their projections as anchors, Flora as published (Hao et al., 2024), APOLLO (Zhu et al., 2024a) and Fira (Chen et al., 2024e) make updates that leave  $\text{im } a$ , so they are not backward methods in the sense of Definition IV.1 beyond the trivial reading with compression and anchor the identity, which the last row of Table 14 records for Flora. Flora compresses its momentum or gradient accumulator with a random projection and hands the decompressed tensor to Adafactor (Shazeer and Stern, 2018), which acts in the full space, so its update need not lie in the image of the projection. On resampling it transports its momentum to the new frame. It fits Definition IV.1 with a nontrivial anchor only in an idealised form with an SGD or heavy-ball base.

### The degree-one collapse

If  $\rho(C) = W_k + \mathcal{L}C$ , training  $C$  and running the backward method with  $c = \mathcal{L}^*$ ,  $a = \mathcal{L}$  feed the optimizer the same gradients  $\mathcal{L}^*G$  and so make the same weight updates. A curved  $\rho$ , such as LoRA's  $BA$ , separates them after one step, and so does an optimizer that reads the parameter tensor, which is  $C$  in one procedure and  $W$  in the other.

**PROPOSITION IV.2** (the degree-one collapse; known, [Torroba-Hennigen et al. \(2025\)](#) for GaLore (their Thm 1 and Cor. 3, App. B for weight decay) and [Hao et al. \(2024\)](#) for random projections). *Let  $\rho_k(C) = W_k + \mathcal{L}C$  with  $\mathcal{L}$  linear and  $C_0 = 0$ . Let  $\text{opt}$  be a first-order optimizer, with no coupled or decoupled weight decay, whose state and output depend only on the sequence of gradients it is fed, and which does not read the parameter tensor: SGD, heavy-ball or Nesterov momentum, Adam or AMSGrad, or Adafactor with `scale_parameter=False` (fixed or relative step, with its factorisation applied to the compressed gradient in the same tensor shape as  $C$ ). Run both procedures below with the same implementation of  $\text{opt}$ , on the same minibatches, and from the same optimizer state, fresh or carried over. Then they produce the same  $W$ -trajectory over a period:*

- training  $C$  with  $\text{opt}$  and merging at the end of the period;
- the backward method with  $c = \mathcal{L}^*$  and  $a = \mathcal{L}$ , started at  $W_k$ .

AdamW with  $\lambda > 0$ , LAMB and LARS trust ratios, parameter-scaled Adafactor (`scale_parameter=True`, the default in HF and in `galore_torch`'s `GaLoreAdafactor`) and global-norm clipping of the full gradient (such as HF Trainer's default `max_grad_norm=1.0`, whenever it triggers) all break the identity. Injectivity of  $\mathcal{L}$  is not needed.

In particular, on each matrix that GaLore projects (its other parameters train in full in both procedures), GaLore with a projector  $P_k$  fixed on each period equals one-sided LoRA  $W_k + P_k C$  (frozen  $B = P_k$ , or frozen  $A$  when projecting on the right), merged and re-based every period, with  $C$ 's optimizer state (moments and step counter) carried over unchanged across re-basings, with no reset.  $P_k$  must be exactly GaLore's matrix, signs and column order included; its span alone does not suffice, because the state is reused without transport. ReLoRA-style resets give a different trajectory. The identity is exact with weight decay 0, no full-gradient clipping, and an optimizer from the admissible class; `galore_torch`'s `GaLoreAdafactor` defaults to `scale_parameter=True` and is excluded. `GaLoreAdamW` is the transformers-style AdamW, which adds  $\varepsilon$  to  $\sqrt{v}$  before the bias correction, so the LoRA side must use the same implementation. GaLore's scale  $\alpha$  multiplies the optimizer output, so it acts as a learning-rate factor; a LoRA-style scale inside  $\rho$  gives a different trajectory under Adam with  $\varepsilon > 0$ .

**REMARK** (scale and implementation). Moving GaLore's  $\alpha$  inside  $\rho$  as a LoRA scale multiplies the compressed gradient by  $\alpha$ . Under Adam this gives GaLore's trajectory only at  $\varepsilon = 0$ , or with the LoRA side's  $\varepsilon$  multiplied by  $\alpha$  as well. This is the exchange of scale for learning rate and  $\varepsilon$  on the hyperparameter torus of [Theorem III.10](#). Both sides must also run the same implementation: `galore_torch`'s AdamW adds  $\varepsilon$  to  $\sqrt{v}$  before the bias correction, as in transformers, and swapping in `torch.optim.Adam` on the LoRA side moves the trajectory measurably (see the check below).

**NUM** In a toy check under Adam the two procedures agree to  $7 \times 10^{-16}$  over one period, and to  $5 \times 10^{-15}$  over four periods with the state carried on both sides. Against the official `galore_torch` optimizers (least squares,  $W \in \mathbb{R}^{6 \times 10}$ ,  $r = 2$ ,  $T = 7$ , four periods, total movement 0.046), the two procedures agree to  $8 \times 10^{-17}$  with the state carried over. Resetting the LoRA



**Table 15.** The cotangent family by projector (rows) and schedule (columns). [Proposition IV.2](#) covers the first three rows under its optimizer hypotheses, except EVA and LoRA-GA, which initialise two-sided LoRA and sit in the frozen column only for their first step. The last row is a contrast row outside the equivalence.

| Projector / schedule                  | Frozen                                                                                                                                         | Re-based periodically                                                                                                                             | Resampled every step                                                                                  |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| random                                | LoRA-FA ( <a href="#">Zhang et al., 2023d</a> ), with $K \propto GA_0^\top A_0$ under SGD <sup>a</sup>                                         | GaLore with a random projector ( <a href="#">Hao et al., 2024</a> ; <a href="#">Torroba-Hennigen et al., 2025</a> ); idealised Flora <sup>b</sup> | MeZO ( <a href="#">Malladi et al., 2023</a> ) (rank 1, global), up to its finite-difference estimator |
| gradient or data SVD                  | EVA ( <a href="#">Paischer et al., 2024</a> ), LoRA-GA ( <a href="#">Wang et al., 2024e</a> ) (first step only)                                | GaLore ( <a href="#">Zhao et al., 2024a</a> )                                                                                                     | GaLore with $T = 1$                                                                                   |
| coordinate blocks                     | BitFit ( <a href="#">Ben-Zaken et al., 2021</a> ), FISH ( <a href="#">Sung et al., 2021</a> ), LN-tuning ( <a href="#">Zhao et al., 2023</a> ) | LISA ( <a href="#">Pan et al., 2024</a> )                                                                                                         | block coordinate descent                                                                              |
| two-sided, state-dependent (contrast) | LoRA ( <a href="#">Hu et al., 2021</a> )                                                                                                       | ReLoRA ( <a href="#">Lialin et al., 2023b</a> )                                                                                                   | none in use                                                                                           |

<sup>a</sup> For the original method under SGD. HF PEFT’s LoRA-FA optimizer preconditions by  $(A_0 A_0^\top + \delta I)^{-1}$ , which in the undamped limit gives the row-space projector  $GP_V$  of [Proposition III.4\(c\)](#); its Adam layer makes the update nonlinear in  $G$ . <sup>b</sup> Flora’s Alg. 2 with an SGD or heavy-ball base, with the momentum transported at resampling instead of carried over; Flora as published, with an Adafactor base, is outside [Definition IV.1](#) and outside [Proposition IV.2](#).

side’s state separates them by 0.029, weight decay 0.1 by 0.020, HF-default Adafactor by 0.030 (against  $6 \times 10^{-17}$  with `scale_parameter=False`), and flipping the sign of one column of  $P_k$ , with the state carried, by 0.12. Using `torch.optim.Adam` on the LoRA side moves the trajectory by about  $4 \times 10^{-5}$ , and placing  $\alpha$  inside  $\rho$  by 0.019 at  $\varepsilon = 10^{-6}$ , with exact agreement at  $\varepsilon = 0$ . The companion site reruns a check of this kind live ([Appendix E](#)).

**WHAT THIS EXPLAINS** Beyond one-sided LoRA, GaLore adds two ingredients: a frame read off the gradient’s SVD, and the carry-over of optimizer state across re-basings. The carry-over involves no change of basis. Moments written in the coordinates of  $P_{k-1}$  are read as coordinates for  $P_k$ . Projection-aware state updates, such as LDAdam ([Robert et al., 2025](#)), and Flora transport the state instead. The equivalence classifies the cotangent methods whose optimizer satisfies its hypotheses. Table 15 sorts them by where the projector comes from and how often it changes; its LoRA and ReLoRA row is a contrast row outside the equivalence.

[Proposition IV.2](#) covers the first three rows of Table 15 under its optimizer hypotheses, except EVA and LoRA-GA, which initialise two-sided LoRA and sit in the frozen column only for their first step; it covers MeZO up to the finite-difference error of its estimator. LoRA’s map  $(B, A) \mapsto W_0 + sBA$  has degree two, so its cometric moves with the state ([Observation III.2](#)) and the collapse does not apply. The other frozen entries gain nothing from merge-and-restart ([Theorem V.3](#), corollary 2); their re-based neighbours move the frame and can reach all of weight space. The LoRA-FA cometric holds for the original method of [Zhang et al. \(2023d\)](#) under SGD; the LoRA-FA optimizer of HF PEFT ([Mangrulkar et al., 2022](#)) gives the row-space projector of [Proposition III.4\(c\)](#), as noted under the table.

Flora proper is not in the table. Its Adafactor step acts on the full gradient, with parameter scaling by default, so its steps are not confined to  $\text{im } P_t$  and [Proposition IV.2](#) does not apply;

like LDAdam, it transports its optimizer state. Only the idealised form with an SGD or heavy-ball base sits in the random, re-based cell, and there its momentum is transported rather than carried.

### When a backward method is a forward one

A backward method chooses at each weight a space of directions  $D_\theta = \text{im } a(\theta)$ . It reads as a forward method only if these directions fit together into  $k$ -dimensional surfaces, the images of immersive parametrisations ([Background C.4](#)). Frobenius' theorem tests this ([Background D.25](#), [Auxiliary result B.31](#)). The bracket  $[X, Y]$ , the gap by which a small loop along  $X, Y, -X, -Y$  fails to close, must stay in  $D$ . Discrete steps and the optimizer's metric ask for more.

**PROPOSITION IV.3** (integrability; after Frobenius' theorem). *Let  $D$  be a smooth distribution of constant rank  $k$  on an open set  $U \subseteq \Theta$ , for instance the anchor distribution  $D_\theta = \text{im } a(\theta)$  of a time-independent backward method. Then  $D$  is involutive on  $U$  if and only if through every  $\theta \in U$  passes an injective immersion  $\rho_\theta : (\mathbb{R}^k, 0) \rightarrow (U, \theta)$  with  $\text{im } D\rho_\theta = D$  along its image. A single such  $\rho$  through  $\theta_0$  gives involutivity only along  $\text{im } \rho$ . For example,  $D = \text{span}\{\partial_x, \partial_y + xz\partial_z\}$  on  $\mathbb{R}^3$  has  $[\partial_x, \partial_y + xz\partial_z] = z\partial_z \notin D$  off  $z = 0$ , yet the plane  $z = 0$  is an integral manifold.*

*The dynamic content needs separate hypotheses.*

- In continuous time, if  $D$  is involutive, a backward flow with  $\dot{\theta} \in D_\theta$  stays on the leaf through  $\theta_0$ .
- In the discrete update (7) of [Definition IV.1](#), the iterates stay on a leaf for all step sizes and all optimizer outputs  $s$  (all losses) if and only if that leaf contains  $(\theta + D_\theta) \cap U$  at every iterate, that is, if and only if the leaves are affine; an anchor constant over the period gives this ([Proposition IV.2](#)).
- Under gradient flow, dynamic equivalence with a forward lens  $(\rho, R[\rho])$  forces matching frames,  $a_t c_t = D\rho \wedge D\rho^\top$  along the trajectory. Hence  $K = ac$  is symmetric positive semidefinite and, for injective  $a$ ,  $c = Ma^\top$  with  $M$  symmetric positive semidefinite on  $S$ , possibly depending on  $\theta$ . If in addition  $\rho$  is an immersion with  $\dim Q = \text{rank } D$ , then  $\rho$  is a local isometry from  $(Q, \Lambda^{-1})$  onto the leaf with metric  $(aMa^\top)^{-1}|_D$ , so that metric must be flat. Forward methods with gauge fibres escape this requirement:  $\rho(B) = BB^\top$  on symmetric matrices induces the curved Bures–Wasserstein metric, and balanced full-rank LoRA also induces a curved leaf metric. For discrete steps, exact agreement for every step size further forces  $\rho$  to be affine along each horizontal line  $q + t \wedge D\rho_q^\top g$ .

**Instances.** Every projector-type backward method of [Table 14](#) (GaLore with any period  $T$  and any projector, its random-projector variant, LISA, MeZO) has an anchor that does not depend on  $\theta$  within its period. Its distribution is constant, hence trivially involutive with affine leaves (vacuously at  $T = 1$ , where a period is a single step). Under the optimizer hypotheses of [Proposition IV.2](#) (no weight decay of either kind, no full-gradient clipping; GaLore and MeZO need weight decay 0, and LISA's AdamW with positive decay violates them), these methods are rebasing schemes of linear forward methods with period  $T$ , including  $T = 1$ , whose reach [Theorem V.3](#) describes. Each MeZO step is a step of the linear method  $\theta_t + \lambda z_t$ , up to its SPSA estimator, and GaLore with  $T = 1$  is one-sided LoRA re-based every step. What separates these methods from LoRA-type methods is the resampling schedule together with the carry-over of optimizer state; integrability plays no part in it. Flora as published (Adafactor base, with a full-rank current-gradient term in the code), APOLLO (a low-rank-estimated scaling applied to

the full gradient) and Fira (a scaled full-rank residual) are not of the form of [Definition IV.1](#) with their projections as anchors, because their updates leave  $\text{im } a$ .

Non-involutivity proper needs a  $\theta$ -dependent anchor. Full-batch GaLore with  $T = 1$ , read as the distribution  $D_W = \text{Hom}(\mathbb{R}^n, U_r(\nabla L(W)))$  (or  $\text{Hom}(V_r(\nabla L(W)), \mathbb{R}^m)$  when  $m \geq n$ , as GaLore projects) wherever a singular-value gap exists, is involutive if and only if  $U_r(\nabla L(W))$  (respectively  $V_r(\nabla L(W))$ ) is invariant along  $D$ . That fails generically for every  $r \geq 1$  once both matrix dimensions are at least 2. For every  $r$  with  $1 \leq r < \min(m, n)$  it fails for  $L = \frac{1}{2}\|W\|_F^2$ , and numerically it fails for least squares and cross-entropy with several samples. It holds, for instance, for single-sample least squares, and automatically only when the distribution is a line field ( $nr = 1$ ). Minibatch GaLore with  $T = 1$  and MeZO violate time-independence, because their anchors depend on step-wise randomness.

**NUM** For full-batch GaLore with  $T = 1$ , the relative normal part of the bracket is 0.08 to 0.94 at  $r = 1$  and  $(m, n) \in \{(4, 3), (5, 4)\}$ , for least-squares, tanh-regression and cross-entropy losses with several samples (an independent replication), and 0.21 to 0.95 at  $(m, n, r) = (4, 3, 2)$ . It vanishes only for  $n = 1$  or for single-sample least squares. At  $W = \text{diag}(2, 1)$  with  $L = \frac{1}{2}\|W\|_F^2$  and the frame  $X_i = u(W)e_i^\top$ , the bracket is  $[X_1, X_2] = -\frac{1}{3}e_2e_1^\top \notin D$ , and the general diagonal formula behind it was checked by central differences for several  $(m, n, r)$ . The cometric of  $B \mapsto BB^\top$  takes the same value at  $B$  and at  $BR$ ,  $R \in \text{O}(2)$ , to  $10^{-12}$ . The companion site recomputes the bracket live at  $W \in \mathbb{R}^{3 \times 4}$  ([Appendix E](#)).

**WHAT THIS EXPLAINS** The lens 2-functor (6) separates tangent-side from cotangent-side methods. Frobenius integrability is the test for when a  $\theta$ -dependent anchor foliates weight space by images of forward methods. For the PEFT zoo it does no discriminating work. Every cotangent method whose update lies in  $\text{im } a$ , whose optimizer does not read the parameter, and which has no decoupled decay on  $W$  and no clipping of the full gradient (GaLore with any projector and MeZO at weight decay 0, LISA without decay) is, period by period, a forward linear method on a moving base ([Exposé V](#)). At  $T = 1$  this reading is automatic.

**ANALOGY** (the lens duality). “Tangent side” and “cotangent side” are exact words here. A forward method acts on tangent vectors through  $D\rho$ , and a backward method acts on cotangent vectors through  $c$ . Calling GaLore and one-sided LoRA *dual* is exact only at degree one, and only for optimizers that do not read the parameter tensor, with no weight decay. Beyond that range the word organises the zoo and proves nothing.

### What the backward pass must remember

Fine-tuning memory consists of weights, gradients and optimizer state, and the activations stored for the backward pass. Shrinking  $Q$  shrinks the gradients and the optimizer state. As for the activations, a frozen linear box’s reverse derivative  $W^\top y$  ignores its input, and cotangents are needed only downstream of something trainable, so only a cone through the network costs activation memory.

**OBSERVATION IV.4** (backward reach and the backprop cone; an upper bound). Assume reverse mode in which each box keeps its own residual, with no checkpointing across boxes, and take boxes at the granularity where adapters are separate boxes (for LoRA-FA,  $A_0$  and  $B$  separately). A VJP residual is an input, an output, a sufficient statistic such as a ReLU mask, or a recipe to recompute one of these inside the box. Then it suffices to keep a VJP residual for

- each trainable box whose parameter-VJP depends on its input: linear and multiplicative boxes such as  $W$ ,  $A$ ,  $B$  and  $(IA)^3$  scales; additive boxes such as biases and prompts need nothing;
- each nonlinear box, bilinear boxes such as the attention products included, on a directed path from a trainable box to the loss.

Deterministic frozen linear boxes, and all boxes off this cone, need nothing; stochastic linear boxes on the cone (dropout) need their mask or random-number state. Alternatively, the  $k$  coordinates of  $P^\top \nabla L$  can be computed exactly by  $k$  forward-mode JVPs, with no reverse pass and  $O(1)$ -layer activation memory, each JVP costing a small constant multiple (about 1.5 to 3) of a forward pass. This always uses less activation memory than un-checkpointed reverse mode, but it is time-competitive with one VJP (about 3 forward passes) only for  $k$  of about 1 or 2. It also needs  $P$  before the passes, so it cannot produce GaLore’s SVD refresh, where  $P$  depends on  $\nabla L$ , more cheaply than a backward pass ( $\nabla L$  itself would take  $|\theta|$  JVPs).

**WHAT THIS EXPLAINS** The bound reads off the activation memory of the zoo.

- **LST** (Sung et al., 2022). The side network reads the backbone through trainable down-projection ladders, so the backbone’s backward pass never runs.
- **LoRA-FA** (Zhang et al., 2023d). With `lora_dropout=0` (the HF default) it stores  $A_0 x$  ( $r$  numbers per token) instead of  $x$  ( $n$  numbers); with dropout on, the  $n$ -element mask is stored too. The saving is realised only where no neighbouring nonlinear box already keeps  $x$ .
- **Top-layer adaptation** truncates the cone, under the no-checkpointing assumption. With HF’s reentrant gradient checkpointing, PEFT calls `enable_input_require_grads()`, which makes the embedding outputs require gradients, so cotangents run down to the embeddings and top-layer adaptation no longer truncates the cone.
- **LISA** (Pan et al., 2024) trains the embedding and the LM head at every step, so its cone is the whole network and it gets no truncation. On the cone, the unsampled layers’ frozen linear boxes drop their input residuals (about 20% of a block’s activations at LLaMA proportions without FlashAttention), and relative to LoRA it also drops the adapter residuals, in line with the LISA paper’s report of less activation memory than LoRA. Most of its saving against full fine-tuning is in gradients and optimizer state. A variant that froze the embedding would truncate the cone at its lowest sampled layer.
- **Prompt tuning** (Lester et al., 2021) has a  $Q$  as small as that of VeRA (Kopiczko et al., 2023), FourierFT (Gao et al., 2024) or  $(IA)^3$  (Liu et al., 2022a), and the maximal cone, the whole network, which it shares with every method that trains something in the first layer. Its prompt tokens also enlarge every stored activation.
- **MeZO** (Malladi et al., 2023) is the case  $k = 1$ , with the JVP estimated by a central difference and no reverse pass. Its estimator is unbiased for the gradient of the Gaussian-smoothed loss and, for a loss with Lipschitz Hessian, has bias  $O(\varepsilon^2)$  against  $\nabla L$ ; ReLU networks lack a Lipschitz Hessian. Exact unbiasedness,  $\mathbb{E}[(\nabla L \cdot z)z] = \nabla L$ , holds for the forward gradient computed with an exact JVP. GaLore, its random-projector variant, Flora and LISA all take the reverse route.

Ladder side-tuning is due to Sung et al. (2022), prompt tuning to Lester et al. (2021), and the zeroth-order estimator in this setting to Malladi et al. (2023). That reverse derivatives of

linear maps ignore their inputs is the observation behind many activation-saving designs; the lens of [Cruttwell et al. \(2021\)](#) makes it visible box by box.

So every cotangent method meeting the hypotheses of [Proposition IV.2](#) is, period by period, a linear forward method on a moving base. What it reaches, and what a quantiser does to the base, are the subject of Exposé [V](#).



## Base change

*Study each object over a base, so that the base can be changed.  
What survives the change is what the object really is.*

after Grothendieck

Quantising and restarting move the point a method is attached to, and combining adapters builds one update from several. Each move comes down to one computation, of the defect a quantiser leaves, the monoid one cycle generates, or the gauge a combination rule must ignore.

### In brief.

- QLoRA is LoRA anchored at the quantised weight  $\kappa\theta_0$ , so it misses  $\theta_0$  by the quantisation error  $e$ , which is generically of full rank. LoftQ and QPiSSA are built to shrink this defect, by quantising and splitting in opposite orders. HF PEFT's LoftQ initialiser ignores the LoRA scale  $s = \alpha/r$ , so with its default single iteration it beats QLoRA in Frobenius norm only for  $0 < s < 2$ .
- With a fixed shape, merge-and-restart reaches exactly the monoid one cycle generates. ReLoRA reaches full rank after  $\lceil \min(m, n)/r \rceil$  cycles, while fixed linear shapes such as BitFit, (IA)<sup>3</sup> or a fixed mask gain nothing.
- Exactly orthogonal block OFT on a fixed partition never leaves  $W_0 \cdot \prod \text{SO}(b)$ , and re-merged BOFT reaches all rotations only at full butterfly depth. HF's default Cayley–Neumann OFT is not orthogonal, and with small generators each merge can only shrink neuron norms and singular values.
- A rule combining LoRA factors is well defined only if it ignores each adapter's  $\text{GL}_r$  gauge. Task arithmetic on the products passes. LoraHub and HF's factor-wise linear merge fail through cross terms, and become exact only when the adapters share a factor.

### The point moves

Every method so far has been an arrow into one fixed point  $(\Theta, \theta_0)$ . In practice the point moves. Quantisation moves it once, ReLoRA (Lialin et al., 2023b) at every merge, and task arithmetic (Ilharco et al., 2022), LoraHub (Huang et al., 2023), TIES (Yadav et al., 2023) and DARE (Yu et al., 2023) build a new update from several adapters. A PEFT method is a formula that makes sense over any base point, so it is a family  $\theta_0 \mapsto M(\theta_0)$ , and this exposé relates its members. Four kinds of move occur (Table 16).

#### DEFINITION V.1 (base change and rebasing).

- (a) A *function-preserving* pointed reparametrisation  $\phi : (\Theta', f', \theta'_0) \rightarrow (\Theta, f, \theta_0)$ , with  $f \circ (\phi \times \text{id}_X) = f'$ , induces by postcomposition

$$\phi_! : \mathbf{Meth}(\Theta', \theta'_0) \rightarrow \mathbf{Meth}(\Theta, \theta_0), \quad (Q, q_0, \rho) \mapsto (Q, q_0, \phi \circ \rho).$$

- (b) A map  $\kappa : \Theta \rightarrow \Theta$  that does *not* preserve the function, such as a quantiser, moves the base point. Its **pointing defect** is  $e = \kappa\theta_0 - \theta_0$ .
- (c) A **rebasing scheme** (merge-and-restart) picks  $\theta_{k+1} \in \text{Im } M(\theta_k)$ . Its reachable sets are  $R_K$ , the points  $\theta_K$  it can reach in  $K$  cycles, and  $R_\infty = \bigcup_K R_K$ .

**Table 16.** Four ways the base point moves, and what each one asks us to compute.

| Move                                        | Examples                                                                                         | What to compute                                                                       | Where                                |
|---------------------------------------------|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|--------------------------------------|
| a reparametrisation that keeps the function | permuting hidden units; SliceGPT-style rotations of the residual stream (Ashkboos et al., 2024a) | nothing new; methods are carried along by composition                                 | Definition V.1(a), Observation II.15 |
| a quantiser                                 | QLoRA, LoftQ, QPiSSA, QA-LoRA                                                                    | the pointing defect, and the invariants that move with the base                       | Proposition V.2                      |
| merge and restart                           | ReLoRA, re-merged OFT and BOFT, re-HRA; GaLore period by period                                  | the monoid generated by one cycle                                                     | Theorem V.3                          |
| combining adapters                          | task arithmetic, LoraHub, TIES, DARE, KnOTS                                                      | whether the rule is a function of the adapters, and which symmetries it commutes with | Proposition V.4                      |

Only part (a) is base change in the algebraic geometer’s sense (Background A.19), and it costs nothing; whether  $\phi_i$  turns LoRA into LoRA is the covariance question of Observation II.15. Parts (b) and (c) borrow the word, and the work lies there and in the combination rules. Figure 10 shows the restart computations of Theorem V.3.

**REMARK** (what “base change” means here). In algebraic geometry a family over a base  $S$  pulls back along any map  $S' \rightarrow S$ , and the statements worth having are the ones that survive the change. Part (a) of Definition V.1 is the standard companion of pullback on slice categories, composition with a map of bases. A quantiser changes the network’s function, so there is no commuting square to transport along, and we compute the defect it leaves instead. A rebasing scheme is an iteration, and we compute its reachable set. “Gauge” keeps the sense of Exposé III, the symmetry group of a parametrisation (Background D.21); no connection is involved.

### Quantisation and splitting

For an additive method (Definition I.2) whose starting point does not depend on the base, such as zero-init LoRA (Hu et al., 2021), quantising the base only translates the image, so QLoRA (Dettmers et al., 2023) is LoRA with its vertex at  $\kappa\theta_0$ . A quantisation error is spread over all singular directions, so  $e$  is generically of full rank and the image misses  $\theta_0$ . Methods whose starting point is computed from the base, such as PiSSA (Meng et al., 2024a), CorDA (Yang et al., 2024c), OLoRA (Büyükyüz, 2024), LoftQ (Li et al., 2023b) and DoRA (Liu et al., 2024b), meet two operations that do not commute, and the order fixes the defect (tabulated in Table 5 of Exposé II.A).

**REMARK** (2026: SRR). SRR (Cho et al., 2026) joins the first two orders of Proposition V.2(b) by splitting the rank budget. It puts  $\text{SVD}_k(W)$  into the adapter, quantises  $W_{\text{res}} = W - \text{SVD}_k(W)$  and fits the quantisation error with the other  $r - k$  ranks. Its defect is therefore  $e_k$  minus its best rank- $(r - k)$  approximation, where  $e_k = \kappa(W_{\text{res}}) - W_{\text{res}}$ , and  $\text{SVD}_k$  and best approximations are taken in its activation-scaled norm. With that scaling set to the identity,  $k = r$  gives the split-first defect of QPiSSA and  $k = 0$  that of one-step LoftQ at  $s = 1$ . SRR picks  $k$  per matrix from a closed-form proxy.

**PROPOSITION V.2** (quantisation and splitting). Write  $W$  for the full-precision weight,  $\kappa$  for the quantiser,  $e = \kappa W - W$ ,  $\delta_0 = \delta(q_0)$ , and  $s$  for the LoRA scale  $\alpha/r$  ( $\alpha/\sqrt{r}$  under rsLoRA, Kalajdziewski, 2023).

(a) Additive families commute with translation,  $M(\theta + v) = \tau_v \circ M(\theta)$ . Hence QLoRA is LoRA pointed at  $\kappa\theta_0$ , with defect  $e$ , and it needs no new weight-space structure. Since  $-e$  is generically of full rank,  $\theta_0 \notin \text{Im LoRA}_r(\kappa\theta_0)$  unless  $\text{rank } e \leq r$ .

(b) For splitting families derived from an SVD or from data, the order of operations matters, and three orders occur in practice.

- Split, then quantise the residual (QPISA; HF's recommended PiSSA and CorDA workflow; the quantisation half-step of LoftQ for  $t \geq 2$ , with  $W_{\text{res}} = W - \delta(q_{t-1})$ ). The defect is  $\kappa(W_{\text{res}}) - W_{\text{res}}$ .
- Quantise, then fit the error (LoftQ with one iteration, HF's default `loftq_iter=1`; HF `replace_lora_weights_loftq`). HF sets the factors from  $\text{SVD}_r(W - \kappa W)$  without dividing by  $s$ , unlike its PiSSA and OLoRA initialisations, so  $\delta_0 = -s \text{SVD}_r(e)$  and the defect is  $e - s \text{SVD}_r(e)$ , with

$$\|e - s \text{SVD}_r(e)\|^2 = \|e - \text{SVD}_r(e)\|^2 + (1 - s)^2 \|\text{SVD}_r(e)\|^2.$$

The defect is  $e$  minus its best rank- $r$  approximation (of rank  $\text{rank } e - r$ ) iff  $s = 1$ , which is HF's bare default  $r = \alpha = 8$ . For  $s = 2$  its norm equals  $\|e\|$ , no better than QLoRA. With a callback, `replace_lora_weights_loftq` can roll a module back to the zero initialisation, whose defect is  $e$ .

- Quantise, split, re-quantise (HF `olora_init` on `bitsandbytes` weights). The defect is  $e + \kappa(r') - r'$  with  $r' = \kappa W - \delta_0$ .

Quantise-then-split has defect exactly  $e$  iff the frozen part equals  $\kappa W - \delta_0$ , for instance  $\kappa W$  kept quantised minus a frozen copy of the rank- $r$   $\delta_0$ , at a cost of about  $r(m + n)$  extra numbers. Re-quantising the residual, as HF `olora_init` does, generically adds  $\kappa(r') - r'$ .

Action families whose pointing does not depend on the base (OFT at  $R = I$ , BOFT at  $R = I$  with unit scale, HiRA at  $B = 0$ ), evaluated at  $\kappa\theta_0$ , simply carry the defect  $e$ . Methods whose pointing is computed from the base have an order ambiguity of their own. DoRA's magnitude initialised from  $\kappa\theta_0$  gives defect  $e$ , while initialised from  $\theta_0$  it gives

$$\text{diag}(\|w_{0,i}\|/(\kappa\theta_0)_i) \kappa\theta_0 - \theta_0.$$

For action families a base change also moves the absolute invariants, along with the base point. NF4's exact zero code enlarges the zero set that HiRA must preserve, and exactly orthogonal (exact-Cayley) OFT preserves the Gram matrix of  $\kappa\theta_0$  rather than that of  $\theta_0$ ; HF's default Cayley-Neumann OFT preserves it only approximately.

(c) LoftQ alternates two updates on  $\|\theta_0 - Q - LR\|$ , an exact rank- $r$  SVD step and a round-to-nearest quantisation step with blockwise scales recomputed from its input. The second is not an exact projection, so the objective need not decrease monotonically. With  $s = 1$  the output is pointed up to the attained residual. It is then an approximate section, a choice  $(Q, q_0)$  with reduced  $\|\rho(q_0) - \theta_0\|$ . In general, with  $Q_T$  and the low-rank fit  $L_T R_T$  after  $T$  iterations, HF's deployed defect is

$$(Q_T + L_T R_T - W) + (s - 1)L_T R_T,$$

because HF's internal residual also omits  $s$ . The official LoftQ checkpoints use  $r = 64$  and  $\alpha = 16$ , so  $s = 1/4$ , and then the deployed defect can grow with  $T$  while the attained residual falls.

(d) For a generic high-precision adapter  $\Delta$ , fine-tune-then-quantise and quantise-then-fine-tune do not commute. The first lands on the quantisation grid. In the second, the exactly merged weight  $\kappa\theta_0 + \Delta$  leaves the quantised type. For a uniform fixed-scale INT- $k$  grid,  $\kappa\theta_0 + \Delta$  stays on the grid iff  $\Delta$  lies on the scale lattice (within range), so there the obstruction is the off-grid update. NF4 is non-uniform and not closed under addition, and absmax grids whose scales are recomputed are not closed either, so for QLoRA's NF4 the grid itself is part of the obstruction. Type-preserving merges, such as HF PEFT's dequantise-add-requantise for bitsandbytes 4- and 8-bit layers, re-quantise with recomputed block scales, so they stay in the quantised type but are lossy. Under affine group quantisation QA-LoRA is the structured exception. Its group-pooled adapter is constant on each quantisation group and merges into the zero-points, so the merged model stays INT- $k$ .

**Numerical check.** On a  $256 \times 256$  NF4 example whose weight has a dominant low-rank part, with  $r = 8$ ,  $s = 1$  and  $\|e\| = 5.65$ , QPiSSA's defect is 1.45, one-step LoftQ's is 5.22, and the LoftQ objective rises in 27 of 60 half-steps. QPiSSA's large advantage there relies on the dominant low-rank part; on a Gaussian weight all three defects are close to  $\|e\|$ . With  $s = 2$  the one-step LoftQ defect equals  $\|e\|$  exactly, and adding an NF4-representable update to an NF4 weight leaves the NF4 grid in most entries (Appendix E).

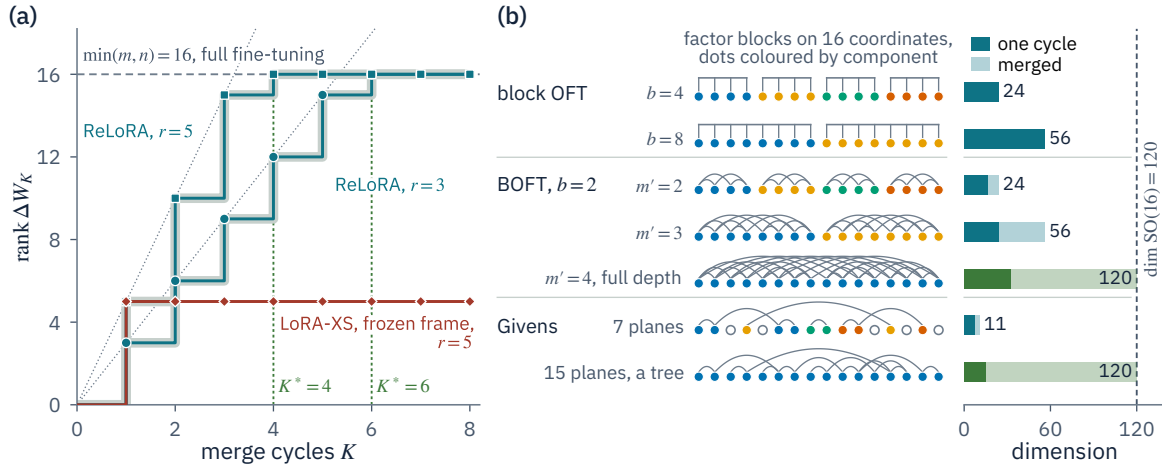
**LoftQ as deployed.** The official LoftQ checkpoints were made with  $r = 64$ ,  $\alpha = 16$  and five iterations, so  $s = 1/4$ . In a referee's simulation of that configuration on a Gaussian  $512 \times 512$  NF4 weight, the attained residual was  $0.67\|e\|$  and the deployed defect  $0.96\|e\|$ . For  $T \geq 2$ , HF's internal residual also omits  $s$ , so the objective in Proposition V.2(c) is not the defect HF starts from. With a callback, `replace_lora_weights_loftq` can roll a module back to the zero initialisation, and that module's defect is then  $e$ .

**Credit.** Training LoRA over a 4-bit NF4 base is due to QLoRA (Dettmers et al., 2023), building on the blockwise absmax quantisation of Dettmers et al. (2021). Fitting the adapter to the quantisation error is LoftQ (Li et al., 2023b), and the split-first variant is QPiSSA (Meng et al., 2024a). The data-aware and QR splits are CorDA (Yang et al., 2024c) and OLoRA (Büyükyüz, 2024). The group-pooled adapter that merges into the zero-points is QA-LoRA (Xu et al., 2023b). HiRA is due to Huang et al. (2025b), and OFT to Qiu et al. (2023). The pipelines are as implemented in HF PEFT (Mangrulkar et al., 2022).

**WHAT THIS EXPLAINS** Over a quantised base, exactly orthogonal OFT keeps the neuron geometry of  $\kappa\theta_0$  (Corollary II.6) and HiRA keeps the zeros NF4 created (Proposition II.9(e)). QA-LoRA makes merging commute with quantisation by living inside the quantiser's own degrees of freedom.

### Restarting: what merge-and-restart can reach

Merge the adapter, restart at the merged point, repeat. Each cycle adds one more copy of an additive shape or applies one more group element, so the long-run reach is whatever one cycle generates. A rank- $r$  cone generates every matrix, and a subspace, subgroup or submonoid generates only itself. Block-orthogonal factors generate the rotations of each connected piece of their block hypergraph, because the bracket of infinitesimal rotations in the planes  $(a, c)$  and  $(c, b)$  is the infinitesimal rotation in the plane  $(a, b)$ .



**Figure 10. Rebasing reach (Theorem V.3).** (a) The rank staircase for  $m \times n = 24 \times 16$ : the rank of  $\Delta W_K = \sum_{k \leq K} B_k A_k$  after  $K$  merge-and-restart cycles of ReLoRA, each adding a fresh random rank- $r$  product, for  $r = 3$  (circles) and  $r = 5$  (squares). The measured ranks (teal) lie on the bound  $\min(Kr, m, n)$  (thick grey; dotted, the line  $Kr$ ) at every  $K$ , so ReLoRA reaches full rank exactly at  $K^* = \lceil 16/r \rceil$  cycles, and not before (corollary 1, Theorem V.3(a)). LoRA-XS with its frames frozen at the injective base  $W_0$  (red,  $r = 5$ ) stays at rank 5: an idempotent method gains nothing from rebasing (corollary 2). (b) Seven families of block-orthogonal factors on  $n = 16$  coordinates: block OFT on a fixed partition with blocks of size  $b = 4$  and  $b = 8$ , BOFT with  $b = 2$  and  $m' = 2, 3, 4$  butterfly factors, and Givens rotations in 7 planes and in 15 planes forming a spanning tree. Arcs are two-element blocks and brackets larger blocks; dots are coloured by the connected component of the hypergraph the blocks generate, and hollow dots are isolated coordinates. The light bars give the dimension  $\sum_C \binom{|C|}{2}$  of the generated Lie algebra (Theorem V.3(c)), and the dark bars the dimension reached in one cycle, against  $\dim \mathfrak{so}(16) = 120$  (dashed), with green bars for the families that reach it. Only a connected hypergraph reaches 120; for BOFT that takes the full depth  $m' = 4$  (corollary 4). The bar values are in §E.6.

**THEOREM V.3** (rebasing closure). (a) Additive methods with a fixed shape  $C \ni 0$ , with frames, masks and compression operators frozen at  $\theta_0$  across cycles. Then  $R_K = \theta_0 + C^{+K}$ , the  $K$ -fold Minkowski sum, and  $R_\infty = \theta_0 + \text{Mon}(C)$ , the additive monoid generated by  $C$  (Background D.10). If  $C$  is closed under real scalings,  $R_\infty = \theta_0 + \text{span } C$ .

(b) Action methods, where one cycle applies an element of a set  $S$  with  $1 \in S \subseteq G$ ; for scaling methods,  $S \subseteq \text{End}(\Theta)$  is a subset containing the identity. Then  $R_K = S^K \theta_0$ , and  $R_\infty$  is the orbit of the monoid generated by  $S$ . Suppose  $S$  lies in the subgroup generated by connected Lie subgroups  $H_1, \dots, H_p$  (Background D.8) and contains a neighbourhood of 1 in each. Then the monoid generated by  $S$  is the connected Lie subgroup  $H$  with  $\text{Lie}(H) = \text{Lie}(\mathfrak{h}_1, \dots, \mathfrak{h}_p)$ .

(c) Block-orthogonal factors. Let the factors have coordinate blocks  $I_j \subseteq \{1, \dots, n\}$  (block OFT, BOFT factors, Givens planes). Then

$$\text{Lie}(\mathfrak{so}(I_j))_j = \bigoplus_C \mathfrak{so}(C),$$

summed over the connected components  $C$  of the hypergraph  $(\{1, \dots, n\}, \{I_j\})$ .

Corollaries.

1. **ReLoRA** (Lialin et al., 2023b).  $R_K = W_0 + \mathcal{M}_{\leq \min(Kr, m, n)}$ . It reaches full fine-tuning iff  $K \geq \lceil \min(m, n)/r \rceil$ .



2. **Idempotent methods** gain nothing from rebasing. These are the methods whose shape is exactly a subspace, subgroup or submonoid independent of the base, with frames, masks and compression operators frozen across cycles. Examples are LoRA-FA with a fixed  $A$  (Zhang et al., 2023d), LoRA-XS with frozen frames (Balaży et al., 2024), MoRA with fixed operators (Jiang et al., 2024), FourierFT with fixed indices (Gao et al., 2024), WaveFT (Bilican et al., 2025), C3A (Chen et al., 2024a), MiSS (Kang and Yin, 2024) (default balance mode; bat mode is base-dependent but still idempotent, since its blockwise maps  $W \mapsto (W + I)(I + M) - I$  form a monoid), SHiRA (Bhardwaj et al., 2024), BitFit (Ben-Zaken et al., 2021), FISH with a fixed mask (Sung et al., 2021), LN-tuning (Zhao et al., 2023), (IA)<sup>3</sup> (Liu et al., 2022a), SSF (Lian et al., 2022) and RoAd (Liao and Monz, 2024). Block OFT with a fixed partition and an exact Cayley map gains only the measure-zero set of block rotations with an eigenvalue  $-1$ , so  $R_\infty = W_0 \cdot \prod \text{SO}(b) = \overline{R_1}$ . Constrained OFT (COFT, an  $\varepsilon$ -ball) with an exact Cayley map gains up to  $\prod \text{SO}(b)$ ; HF's `coft=True` keeps the Cayley–Neumann map by default and so falls under corollary 4. Methods that recompute frames or masks from the merged weight (an SVD-recomputing LoRA-XS, ReMoRA's alternating operators, resampled indices) do gain.
3. **VeRA** (Kopiczko et al., 2023).  $\text{VeRA}^\infty = \text{LoRA-FA}(A)$  whenever the frozen  $B$  has no zero entries.
4. **OFT family**. Block OFT with an exact Cayley map is stuck in  $W_0 \cdot \prod \text{SO}(b)$ . HF's default Cayley–Neumann factor is  $R = C(Q)(I - Q^4)$ , with  $C$  the exact Cayley map (Background D.7), so  $R^\top R = (I - Q^4)^2$  and  $R$  is not orthogonal. Its singular values are  $|1 - \mu_k^4|$ , where  $\pm i\mu_k$  are the eigenvalues of  $Q$ . In the operating regime  $\|Q\|_{\text{op}} \leq 2^{1/4}$  (every  $\mu_k \leq 2^{1/4}$ ), the regime the truncated series is meant for, every factor is a contraction. Re-merging then decreases  $K_{\text{out}}$  in the Loewner order and no singular value grows, so the drift is monotone shrinkage, and the reachable set lies in  $W_0$  times the block contraction semigroup. With unconstrained  $Q$  the factors include singular ones ( $\mu_k = 1$ ), and re-merging generates  $\prod (\text{GL}^+(b) \cup \mathcal{M}_{\leq b-2})$ , or  $\prod (\text{CO}^+(2) \cup \{0\})$  when  $b = 2$ ; this clause is established at the level of a proof sketch. Iterated BOFT (exact Cayley in HF) with  $m'$  butterfly factors of block size  $b$  reaches

$$\text{diag}(\mathbb{R}^m) \cdot W_0 \cdot P \left( \prod \text{SO}(b 2^{m'-1}) \right) P^\top,$$

because the factor hypergraph has components of size  $b 2^{m'-1}$ . For injective  $W_0$  this is  $\text{diag}(\mathbb{R}^m) \cdot W_0 \cdot \text{SO}(n)$  iff  $b 2^{m'-1} = n$ , the full-depth butterfly  $m' = 1 + \log_2(n/b)$ . HF's default `boft_n_butterfly_factor=1` is block OFT times  $\text{diag}(s)$ . A single full-depth BOFT has dimension at most  $m' \frac{n}{b} \binom{b}{2}$ , which is below  $\dim \text{SO}(n)$  when  $m'(b-1) < n-1$ ; its matrices are dense, while the set they form has positive codimension.

5. **Re-HRA** (HF `apply_GS=False`), with  $W_0$  injective and  $r$  even. Then  $R_K = (W_0 + \mathcal{M}_{\leq Kr}) \cap W_0 \cdot \text{SO}(n)$ , so rebasing commutes with the meet of Theorem II.7. This multiplicative twin of ReLoRA reaches  $W_0 \cdot \text{SO}(n)$  in at most  $\lceil n/r \rceil$  cycles.
6. **GaLore (with a gradient-SVD or a random projector), LISA and MeZO**, under the optimizer hypotheses of Proposition IV.2, are rebasing schemes of linear methods whose shape moves. That is why, as a set over the sequences of shapes they can choose, they reach all of  $\Theta$ , while a fixed linear shape  $L$  reaches only  $\theta_0 + L$  (corollary 2). Low rank is the per-period obstruction only for GaLore; LISA and MeZO already make full-rank per-matrix updates and gain coverage. Flora proper takes full-space steps with a compressed momentum, so it has no per-period low-rank shape.

**Table 17.** What merge-and-restart reaches (Theorem V.3 and its corollaries).

| Method under rebasing                                                                       | One cycle                                                  | After $K$ cycles, or $R_\infty$                                    | Gain from restarting                                     |
|---------------------------------------------------------------------------------------------|------------------------------------------------------------|--------------------------------------------------------------------|----------------------------------------------------------|
| ReLoRA (Lialin et al., 2023b)                                                               | $W_0 + \mathcal{M}_{\leq r}$                               | $W_0 + \mathcal{M}_{\leq \min(Kr, m, n)}$                          | yes; full FT at $K \geq \lceil \min(m, n)/r \rceil$      |
| LoRA-FA (fixed $A$ ), LoRA-XS (frozen frames), BitFit, (IA) <sup>3</sup> , SSF, fixed masks | a subspace or submonoid                                    | the same set                                                       | none (idempotent)                                        |
| VeRA (Kopiczko et al., 2023) ( $B$ without zero entries)                                    | $W_0 + \Lambda_b B \Lambda_d A$                            | LoRA-FA( $A$ )                                                     | yes, up to LoRA-FA                                       |
| block OFT (Qiu et al., 2023), exact Cayley, fixed partition                                 | $W_0 \cdot \prod \text{Cay}(b)$                            | $W_0 \cdot \prod \text{SO}(b) = \overline{R_1}$                    | measure zero                                             |
| COFT, exact Cayley                                                                          | an $\varepsilon$ -ball in each block                       | up to $W_0 \cdot \prod \text{SO}(b)$                               | within the blocks                                        |
| block OFT, HF Cayley–Neumann default                                                        | not orthogonal                                             | a contraction monoid for small $Q$                                 | drifts; $K_{\text{out}}$ shrinks                         |
| BOFT (Liu et al., 2023b), $m'$ butterfly factors, exact Cayley                              | products of butterfly factors                              | $\text{diag} \cdot W_0 \cdot P(\prod \text{SO}(b 2^{m'-1}))P^\top$ | $\text{SO}(n)$ iff $b 2^{m'-1} = n$                      |
| re-HRA (Yuan et al., 2024) ( $W_0$ injective, $r$ even)                                     | $(W_0 + \mathcal{M}_{\leq r}) \cap W_0 \cdot \text{SO}(n)$ | $(W_0 + \mathcal{M}_{\leq Kr}) \cap W_0 \cdot \text{SO}(n)$        | yes; $W_0 \cdot \text{SO}(n)$ within $\lceil n/r \rceil$ |
| GaLore, LISA, MeZO                                                                          | a linear shape that moves                                  | all of $\Theta$ , as a set                                         | yes, because the shape moves                             |

### 7. Merging never rescues an exactly orthogonal method from its absolute invariants.

*In exact arithmetic, every exactly orthogonal input-side rebasing preserves  $K_{\text{out}}$  and the spectrum, and every output-side one preserves  $K_{\text{in}}$  and the spectrum (Corollary II.6).*

**Numerical check.** For the butterfly hypergraph at  $n = 64$ ,  $b = 4$ , the components have sizes 4, 8, 16, 32, 64 for  $m' = 1, \dots, 5$ , so it is connected only at  $m' = 5$ ; an independent check with HF’s own permutation code gives the same sizes. The Lie algebra generated by the Cayley–Neumann factors at  $b = 4$  has dimension  $16 = \dim \mathfrak{gl}(4)$ . With skew entries of size about 0.05,  $K_{\text{out}}$  moves by 0.2%, 1%, 6% and 22% (relative) after 1, 10, 50 and 200 default-OFT merges, decreasing in the Loewner order at every merge. Random skew  $Q$  with  $\|Q\|_{\text{op}} \leq 2^{1/4}$  never give  $\|R\|_{\text{op}} > 1$ , and at  $b = 2$  the generator  $Q = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$  gives  $R = 0$  (Appendix E).

**Credit.** Merge-and-restart training is ReLoRA (Lialin et al., 2023b). The orthogonal methods are OFT (Qiu et al., 2023), with its constrained variant COFT, BOFT (Liu et al., 2023b), whose butterflies come from Dao et al. (2019), and HRA (Yuan et al., 2024), whose Householder products go back to Mhammedi et al. (2016). Cayley parametrisations of orthogonal layers are studied in Lezcano-Casado and Martínez-Rubio (2019). The group theory is classical. It uses Yamabe’s theorem on arcwise connected subgroups (Yamabe, 1950) (Background D.9) and Scherk’s sharp form of the Cartan–Dieudonné theorem (Scherk, 1950) (Background D.11). That GaLore (Zhao et al., 2024a) is one-sided LoRA re-based every period is due to Torroba-Hennigen et al. (2025), and the random-projection case to Flora (Hao et al., 2024); see Proposition IV.2. The other moving-shape methods are LISA (Pan et al., 2024) and MeZO (Malladi et al., 2023).

**WHAT THIS EXPLAINS** ReLoRA is [Observation I.7](#) unrolled in time, since a sum of  $K$  rank- $r$  LoRAs is one LoRA of rank  $Kr$ , so it walks the group its cone generates. A fixed additive shape gains from restarts only when it is not closed under addition, as LoRA's cone and VeRA's shape are not. Exact-Cayley block OFT can never escape its  $n/b$  partial-Gram invariants however often it is re-merged. Butterfly permutations exist precisely to connect the block hypergraph, and they connect it fully only at full depth.

**REMARK** (2026: POET-X). POET-X ([Qiu et al., 2026](#)), a pretraining method whose  $W_0$  is a random initialisation, changes the partition between cycles, on both sides of the weight. Each round trains a block-diagonal factor conjugated by a random permutation on each side, merges both into the weight and redraws the permutations. With exact Cayley blocks every merged weight lies in  $SO(m) W_0 SO(n)$ , so it keeps the singular values of  $W_0$  ([Theorem II.5](#)), and by [Theorem V.3\(b\)](#) and (c) blocks that together connect all coordinates of a side generate all of  $SO$  on that side. Its blocks are in fact  $(I + Q)(I + Q + Q^2 + Q^3)$ , the Cayley–Neumann factor of [corollary 4](#), so while every  $\mu_k \leq 2^{1/4}$  no merge raises a singular value, and the spectrum can only drift down.

**EXAMPLE** (one projection of a 7B model). Take a  $4096 \times 4096$  projection. ReLoRA with  $r = 16$  reaches full fine-tuning after 256 cycles and not before. For injective  $W_0$ , re-HRA with  $r = 8$  reaches  $W_0 \cdot SO(4096)$  within 512 cycles. BOFT with  $b = 4$  needs  $m' = 11$  butterfly factors before re-merging reaches that orbit up to output scaling; at HF's default of one factor it stays in  $W_0 \cdot \prod SO(4)$  up to the scale. A single 11-factor BOFT has dimension at most  $11 \cdot 1024 \cdot 6 = 67,584$ , against  $\dim SO(4096) = 8,386,560$ .

**Prediction.** With an exact Cayley map and a fixed partition, re-merging adds nothing to block OFT's reach beyond closure, so re-merged exact-Cayley block OFT and COFT on a fixed partition never leave  $W_0 \cdot \prod SO(b)$ . Enlarging the reach needs a connected block hypergraph (partitions that change between cycles, or full-depth butterflies) or a COFT ball. LoRA-type cones gain from rebasing with a fixed shape, while linear additive shapes do not ([corollary 2](#)). These are statements about reach; whether extra reach improves a trained model is the experiment ([Prediction X.6](#)).

### Combining: rules that must ignore the gauge

A LoRA module ships as factors  $(B, A)$ , the model sees only  $BA$ , and  $(Bg^{-1}, gA)$  gives the same product for every  $g \in GL_r$ . A combination rule should return the same model for every choice of factors, or two people with the same adapters can merge them into different models. Task arithmetic adds the products and passes. LoraHub adds the  $B$ 's, adds the  $A$ 's and multiplies, and fails, because its cross terms  $B_i A_l$  see a rescaling  $(cB, A/c)$  of one module. The merge panel of [Figure 9](#) shows the LoraHub merge moving under that rescaling while task arithmetic stays fixed.

**PROPOSITION V.4** (combination rules: gauge and naturality). (a) A rule  $\Phi((B_i, A_i)_i)$  for combining LoRA factors descends to a function of the adapters  $(B_i A_i)_i$  iff it is constant on the fibres of  $(B_i, A_i) \mapsto (B_i A_i)$ . On the full-rank locus this is equivalent to  $\prod_i GL_r$ -invariance. It is also equivalent for  $\Phi$  continuous on the whole factor space; this clause rests on the unique closed orbit in each fibre, which lies in the closure of every orbit of the fibre ([Background D.31](#)).

- Task arithmetic  $\sum_i w_i s_i B_i A_i$ , with  $s_i$  HF's scaling ( $\alpha_i/r_i$ , or  $\alpha_i/\sqrt{r_i}$  under rsLoRA), is invariant and lands in  $\text{LoRA}_{\sum_i r_i}$ . It is HF's cat with merged scale 1 ([Observation I.7](#)).
- LoraHub's factor-wise rule  $(\sum_i w_i B_i)(\sum_j w_j A_j)$  is not invariant. Replacing  $(B_2, A_2)$  by  $(cB_2, A_2/c)$  changes it by

$$\sum_{l \neq 2} w_l w_2 [(c-1)B_2 A_l + (c^{-1}-1)B_l A_2].$$

HF's `add_weighted_adapter(..., combination_type="linear")` is a factor-wise rule of the same type. It weights  $A_i$  by  $\text{sign}(w_i)\sqrt{|w_i s_i|}$  and  $B_i$  by  $\sqrt{|w_i s_i|}$  and sums the factors separately, so it has cross terms. On adapters that share a frozen  $A$  it is still not exact up to a global factor, since the relative weights become  $\sqrt{|w_i s_i|}$ . HF's non-SVD ties, `dare_*` and `magnitude_prune` variants also act on factors; the `*_svd` variants act on full updates.

- Poly and MHR are different. Their skill inventories are trained jointly through the product-of-sums map, which is invariant under the diagonal  $\text{GL}_r$ , one common  $g$  for every skill. Numerically, for fixed generic routing with fewer skills than tasks and full-rank factors, this is the whole continuous fibre. With at least as many skills as tasks, or once the routing is counted as a parameter, there are further symmetries (skill permutations, and re-mixings  $\alpha \mapsto \alpha M$  with  $M\mathbf{1} = \mathbf{1}$ ). Their cross terms are a design choice of a jointly trained parametrisation, and raise no question of well-definedness.

(b) Well-definedness versus exactness. A joint gauge fixing makes a factor-wise rule well defined. For rank-preserving product-of-sums rules, exactness (agreement with task arithmetic for all weights  $w$ ) needs degree one in  $w$ , which for injective factors forces a shared factor; cat, which concatenates and raises the rank, is exact without one. A frozen shared factor (FFA-LoRA; LoRA-FA) gives  $(\sum_i w_i B_i)A = \sum_i w_i B_i A$  exactly. LoraHub with a shared  $A$  still returns  $(\sum_j w_j) \sum_i w_i B_i A$ , exact only up to the global factor  $\sum_j w_j$ , since its weights are unnormalised. A per-module SVD canonical form leaves a  $(\mathbb{Z}/2)^r$  of sign flips per module, plus rotations within repeated singular values, and these change the cross terms. What is needed is a joint frame that produces a shared factor, as KnOTS's joint re-factorisation of the concatenated updates through a shared  $U\Sigma$  does. Federated LoRA shows the difference. FedAvg-LoRA clients start each round from the same broadcast  $(B, A)$ , so their failure,

$$\frac{1}{2}(B_1 + B_2) \cdot \frac{1}{2}(A_1 + A_2) \neq \frac{1}{2}(B_1 A_1 + B_2 A_2),$$

is bilinearity, and it persists in a fully fixed gauge.

(c) Naturality. On updates with  $N$  coordinates, write  $T_N$ ,  $B_N$  and  $\text{Mon}_N$  for the invertible diagonal, signed permutation and monomial matrices. Task arithmetic is  $\text{GL}(\Theta)$ -equivariant on updates. DARE is coordinatewise and magnitude-free,  $D(m \odot \tau) = m \odot (D\tau)$  for every invertible diagonal  $D$  and the same mask  $m$ . So DARE is exactly equivariant under the torus  $T_N$ , sign flips included, for each mask, and under permutations in distribution; its distributional group is exactly  $\text{Mon}_N$ . TIES with global trimming (top- $k$  magnitude, sign election by total mass as in the TIES paper and HF's default "total", disjoint mean) is equivariant under  $\mathbb{R}^\times \cdot B_N$ , homotheties times signed permutations, generically (off magnitude ties at the threshold). With HF's `majority_sign_method="frequency"`, count ties are broken to +1 and the group drops to  $\mathbb{R}_{>0} \cdot S_N$ . Per-tensor trimming changes the group to  $\prod_b (\mathbb{R}^\times \cdot B_{N_b})$ , extended by permutations of equal-size blocks. DARE followed by full TIES (with trimming) is  $\mathbb{R}^\times \cdot B_N$ -equivariant in distribution; HF's and MergeKit's `dare_ties` has no magnitude trim, so with "total" election it is exactly  $T_N$ -equivariant per mask, and its group contains  $\text{Mon}_N$ . None of the sparsifying rules



(DARE, TIES, DARE-TIES) is  $O(N)$ -equivariant. Applied to factors and summed across adapters (HF's non-SVD variants), all of them are gauge-dependent, and under independent per-adapter gauges even DARE fails, through the cross terms of (a). DARE's masking commutes with  $T_r$  exactly for each mask and with  $\text{Mon}_r$  in distribution, so a single masked adapter, or a factor-wise sum under one gauge shared by all adapters, is  $T_r$ -invariant per mask and  $\text{Mon}_r$ -invariant in distribution.

**Numerical check.** DARE with a random non-unit diagonal  $D$  is equivariant to  $2 \times 10^{-15}$  per mask. TIES is homogeneous to  $9 \times 10^{-16}$  and fails by 5.6 under a generic positive diagonal. The general LoraHub change formula matches with three modules to  $2 \times 10^{-15}$ . On the factors of two adapters, a diagonal gauge on one of them moves factor-wise DARE by an amount of order one, while the same gauge shared by all adapters leaves it fixed to  $10^{-15}$ . For Poly with 3 skills,  $r = 2$  and  $6 \times 6$  weights, the fibre has dimension 32 with 2 tasks and  $4 = \dim \text{GL}_2$  with 4 tasks (Appendix E).

**HF options.** With HF's `majority_sign_method="frequency"`, sign counts are integers, so ties are common, and they are broken towards +1. Sign flips and negative homotheties therefore fail for ties and dare\_ties alike.

**Credit.** The rules studied here are task arithmetic (Ilharco et al., 2022), LoraHub (Huang et al., 2023), TIES (Yadav et al., 2023), DARE (Yu et al., 2023) and KnOTS (Stoica et al., 2024), whose joint SVD is the shared frame of (b); dare\_ties is as implemented in HF PEFT (Man-grulkar et al., 2022) and MergeKit (Goddard et al., 2024). Jointly trained skill inventories are Poly (Ponti et al., 2022) and MHR (Caccia et al., 2022). Federated averaging of LoRA factors is used in FedIT (Zhang et al., 2023b), and the frozen shared factor that removes its bilinearity is FFA-LoRA (Sun et al., 2024). That functions of LoRA weights should respect their  $\text{GL}_r$  symmetry is the starting point of the GL-equivariant networks of Putterman et al. (2024); for full models, hidden-unit permutations play that role in Git Re-Basin (Ainsworth et al., 2022). The unique closed orbit in each fibre is classical invariant theory. Kempf and Ness (1979) characterise closed orbits by norm-minimising points (Background D.30), here the balanced factorisations of Proposition III.7, and Richardson and Slodowy (1990) treat real groups. For optimizers, the same question leads to Zhang and Pilanci (2024a) and Yen et al. (2024); see Theorem III.12.

**WHAT THIS EXPLAINS** Post-hoc factor merging of independently trained modules, as in LoraHub and HF's linear, is not a function of the adapters. It depends on how each module happened to be factorised, which nobody controls after training. Federated LoRA fails through bilinearity instead, and FFA-LoRA's frozen shared factor repairs it. KnOTS's joint re-factorisation produces a shared factor. **Design rule: combine LoRAs on products, or only after a joint re-factorisation that produces a shared factor.**

**Prediction.** LoraHub behaves like update arithmetic, up to the factor  $\sum_j w_j$ , when its modules share the  $A$ -seed and  $A$  stays close to its initialisation. That regime is empirical and favoured by a large ratio  $\eta_B/\eta_A$ , as in LoRA+ (Hayou et al., 2024b). LoraHub degrades when the modules come from independent seeds (Prediction X.9).

### What base change leaves for the network

Every move here stayed inside weight space, where merging into the base is the canonical arrow to the terminal object of Observation I.4 and always exists. Adapters, prefixes and ReFT



change the network itself, and for them merging is a lifting problem whose obstructions the activation function decides. That is Exposé [VI](#).

## Extensions and mergeability

*When a lift fails, look for the place where it fails,  
and give the obstruction a name.*

after Grothendieck

Adapters, prefixes, prompts and ReFT enlarge the network before they train, so merging them back is a lifting problem. Seven local rewrite rules give sufficient conditions for a merge and name the side condition that fails when they stop.

Merging an extension back means finding a weight of the original network that computes the trained function. The rules slide an edit along the wires of a block until a weight absorbs it. They certify a merge when the edit acts alike at every position, the absorbing weights are unshared, shifts have bias slots and, under RoPE, scales are constant on rotary pairs. A stuck rewrite shows only a box-local obstruction. Three examples show what the exposé finds. The feed-forward scale of (IA)<sup>3</sup> always folds into  $W_{\text{down}}$ ; it folds into  $W_{\text{up}}$  under SwiGLU for every scale and under ReLU for nonnegative ones, and under GELU only entries 0 and 1 fold. A prefix is a gated parallel adapter whose gate depends on the input, so for generic frozen weights it cannot be folded into the head's own weights. Per hidden vector, LoReFT is LoRA on an inserted identity box, and since it acts at selected positions no merge is known.

### The weight that absorbs the edit

(IA)<sup>3</sup> (Liu et al., 2022a) rescales the hidden units of a feed-forward block by a learned vector  $\ell$ , which one would like to fold into a frozen weight after training. The weight that reads the hidden units,  $W_{\text{down}}$ , absorbs  $\ell$  under every activation. The weight that produces them,  $W_{\text{up}}$ , absorbs it under SwiGLU for every  $\ell$ , under ReLU when no entry is negative, and under GELU only for entries 0 or 1.

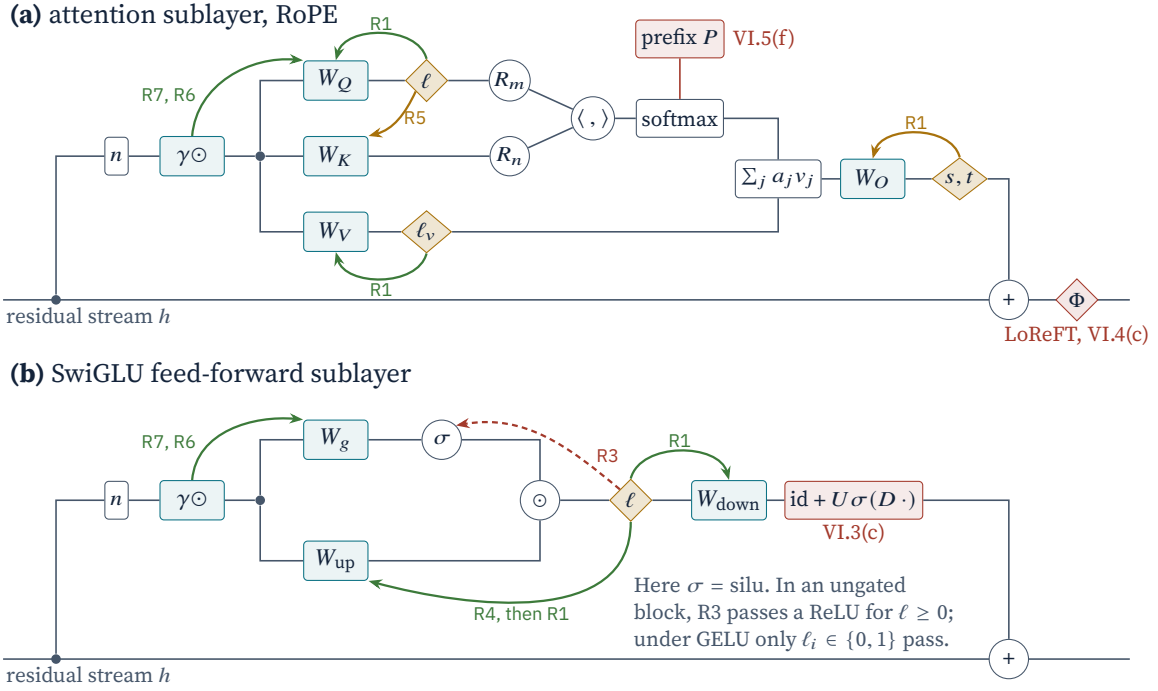
The activation belongs to the network, which a weight-space method never sees (Remark I.3). Adapters, prefixes, prompts and ReFT add boxes the pretrained network lacked, and whether they can be removed depends on their surroundings. Figure 11 runs the rules of this exposé on one block.

### Extensions, and merging as a lift

An adapter  $y \mapsto y + U\sigma(Dy)$  (Houlsby et al., 2019) adds weights  $(U, D)$ , and at  $U = 0$  the network computes the old function. A prefix (Li and Liang, 2021) adds key-value pairs to every softmax, and ReFT (Wu et al., 2024d) edits hidden states at chosen positions. Each time  $\Theta$  grows to  $\Theta \times \Theta'$  and  $f$  to a larger  $\tilde{f}$ . A prefix always takes softmax mass, and on a head with  $W_o W_v \neq 0$  no prefix leaves the output unchanged, so prefixes have no neutral setting there (Theorem VI.5(f)).

**DEFINITION VI.1** (extensions; mergeability).

- An **extension** is a function-preserving pointed embedding  $\iota : (\Theta, f, \theta_0) \rightarrow (\Theta \times \Theta', \tilde{f}, (\theta_0, \theta'_0))$ ,  $\iota(\theta) = (\theta, \theta'_0)$ , with  $\tilde{f}(\theta, \theta'_0, -) = f(\theta, -)$ . The point  $\theta'_0$  is the **neutral point** (“identity at initialisation”).
- An **unpointed extension** has no neutral point.



**Figure 11.** The merge rules of [Theorem VI.3](#) on a pre-LN transformer block, drawn as a string diagram with the residual stream along the bottom: (a) the attention sublayer under RoPE, (b) a SwiGLU feed-forward sublayer. Tide boxes are trainable boxes of the base network,  $\bullet$  is a copy node,  $\odot$  the product node of the gated block, and ochre diamonds are edits inserted by a method. Moss arrows are rewrites licensed for every parameter value, ochre arrows rewrites licensed under a side condition, and seal marks what the rules do not merge. A query scaling  $\ell$  fuses into  $W_Q$  by R1; R5 moves it onto  $W_K$  only if  $\ell$  is constant on each rotary pair and across a grouped-query sharing group. A value scaling  $\ell_v$  fuses into  $W_V$ . A norm gain is copied to every reader (R7) and absorbed (R6). SSF's scale and shift  $(s, t)$  fuse into  $W_O$  only if  $W_O$  has a bias slot. The (IA)<sup>3</sup> scale fuses into  $W_{\text{down}}$  (R1), and through the product node into  $W_{\text{up}}$  for every  $\ell$  (R4, then R1); it does not slide through the SiLU gate, and in an ungated block R3 carries it through a ReLU only for  $\ell \geq 0$ . A prefix enters the softmax with an input-dependent gate ([Theorem VI.5\(f\)](#)), a nonlinear adapter is a non-affine edit ([Theorem VI.3\(c\)](#)), and LoReFT acts at selected positions of the residual stream ([Proposition VI.4\(c\)](#)); R1–R7 merge none of the three. The companion site's merge game runs these rewrites interactively, with a float64 check of each licensed merge on seeded weights.

- A method on an extension is **mergeable** if there is  $\mu : Q \rightarrow \Theta$  with  $\Phi_{\tilde{f}} \circ \rho = \Phi_f \circ \mu$ : a lift along the realisation.
- It is **box-locally mergeable** if the modified box function stays in the box's function class: linear maps for a linear box, affine maps when the box has a bias slot. [Theorem VI.3](#) gives sufficient conditions, with explicit side conditions (position-uniformity, no weight sharing, bias slots), under which this yields a merge into the base network.

A merge is an arrow  $\mu$  that makes the square of realisations commute:

$$\begin{array}{ccc}
 Q & \xrightarrow{\mu} & \Theta \\
 \rho \downarrow & & \downarrow \Phi_f \\
 \Theta \times \Theta' & \xrightarrow{\Phi_{\tilde{f}}} & C^\infty(X, Y)
 \end{array}$$

A weight-space method merges by  $\mu = \rho$ , the canonical arrow to full fine-tuning ([Observation I.4\(a\)](#)), so the real questions live on extensions, in the slice of **Para** over the pretrained model ([Remark I.3](#)). Box-local mergeability can be read off formulas, and the classification of Exposé VIII records it as a merge type:

- M1: the edit merges into its own box;
- M1 $\rightarrow$ : the rules move it into a neighbouring box, under side conditions;
- Mq: an exact merge exists but leaves the quantised type;
- M $\infty$ : no merge; the rules are blocked box-locally, or the edit lies outside them.

**WHAT THIS EXPLAINS** “Zero inference latency” runs two questions together: whether the original architecture can compute the trained function at all, and whether a local edit finds the weight. A zero up-projection, or the zero gate of LLaMA-Adapter, makes training start at the pretrained function, and by the function-level form of [Lemma II.4](#) only  $U$ , or the gate, moves at first.

Adapters are due to [Houlsby et al. \(2019\)](#), prefix tuning to [Li and Liang \(2021\)](#), prompt tuning to [Lester et al. \(2021\)](#) and ReFT to [Wu et al. \(2024d\)](#). [He et al. \(2021\)](#) first placed adapters, prefixes and LoRA in one frame, as modifications of hidden states. The construction **Para** is from [Fong et al. \(2017\)](#) and [Cruttwell et al. \(2021\)](#) ([Background B.16](#)).

### The activation fixes the privileged basis

An edit moves through a network by sliding along wires, and a linear map slides through a box when it commutes with the box. Activations are the bottleneck. Permutations commute with every elementwise  $\sigma$ , and for ReLU so do positive rescalings, since  $\text{ReLU}(ct) = c \text{ReLU}(t)$  for  $c > 0$ . The lemma says how little else does.

**LEMMA VI.2** (the activation fixes the privileged basis; classical, [Sussmann \(1992\)](#), [Chen et al. \(1993\)](#), [Godfrey et al. \(2022\)](#)). *Let  $\sigma : \mathbb{R} \rightarrow \mathbb{R}$  be an elementwise activation that is not affine, and let  $L \in \text{GL}_d$  commute with it as maps on  $\mathbb{R}^d$ :  $L \sigma^{\times d} = \sigma^{\times d} L$ .*

- If  $\sigma \in C^2$ , then  $L = PD$  is monomial.*
- If  $\sigma$  is ReLU or leaky ReLU with slope  $\alpha \neq 1$ , then  $L = PD$  with  $D$  a positive diagonal.*
- If moreover  $\sigma$  is real-analytic at 0 ([Background C.22](#)) with  $\sigma(t) = \sum a_k t^k$ , each diagonal entry  $c$  of  $D$  satisfies  $ca_k = c^k a_k$  for all  $k$ . So  $c = 1$  for GELU and SiLU, since  $\sigma''(0) \neq 0$ , and  $c = \pm 1$  for tanh.*

The lemma is classical. [Sussmann \(1992\)](#) and [Chen et al. \(1993\)](#) found the weight-space symmetries of tanh networks, and [Godfrey et al. \(2022\)](#) computed the linear maps that commute with each common activation. Permutation symmetry is what [Ainsworth et al. \(2022\)](#) use to align whole models before merging them.

So rules R2 and R3 below exhaust the invertible linear maps that slide through  $\sigma$ : permutations for GELU and SiLU, and positive monomials for ReLU. In gated blocks such as SwiGLU every diagonal slides through the linear branch (rule R4), so their hidden-unit group is the full monomial group  $\text{Mon}_{d_{\text{ff}}}$ .

**WHAT THIS EXPLAINS** Within the lemma’s scope (global linear intertwiners of an ungated elementwise  $\sigma$ ), the symmetries a method may ignore are *derived* from  $\sigma$ . For ungated FFNs (BERT, GPT-2, ViT), GELU and SiLU give  $S_d$ , tanh gives  $B_d$ , and ReLU gives positive monomials. Gated FFNs (SwiGLU and GeGLU, as in LLaMA, Mistral, Qwen and Gemma) are not elementwise, and their hidden-unit group is the full  $\text{Mon}_{d_{\text{ff}}}$ , acting through the

linear branch (R4 of [Theorem VI.3](#)). Affine, bias-absorbing symmetries can be larger still: sigmoid satisfies  $\sigma(-z) = 1 - \sigma(z)$ . The privileged basis of LayerNorm gains comes from the diagonal gain parametrisation, and  $\sigma$  plays no part in it. A gain-free RMSNorm is  $O(d)$ -equivariant, which SliceGPT ([Ashkboos et al., 2024a](#)) and QuaRot ([Ashkboos et al., 2024b](#)) use to rotate the residual stream ([Observation II.15](#)).

**PREDICTION** Over random function-preserving permutations of hidden units, LoRA’s outcome variance equals its seed variance, since the pretrained order is exchangeable. FourierFT ([Gao et al., 2024](#)), LoKr ([Yeh et al., 2023](#)) and MoRA ([Jiang et al., 2024](#)) show larger variance, and so do BOFT ([Liu et al., 2023b](#)) and OFT ([Qiu et al., 2023](#)) on modules whose input side is the hidden layer (down\_proj). On up\_proj and gate\_proj a hidden-unit permutation only reorders BOFT’s output rows, so BOFT is exactly equivariant there. [Prediction X.3](#) turns this into a permutation test.

### Seven local rules

Draw a transformer block as a string diagram ([Background B.3](#)), with a copy node where a wire feeds several boxes and a product node in a gated block. An edit is a small box on a wire. To merge, slide it by equalities of maps until a trainable box absorbs it; where it gets stuck names the failing side condition. A positional rotation (RoPE, [Su et al., 2024](#)), a shared weight, a missing bias slot or a second reader of a wire can each block a slide, and all four occur in production models.

**ANALOGY** (what “string diagram” and “spider” mean here). The diagrams are string diagrams of a symmetric monoidal category, read literally. Copy is the cartesian diagonal  $x \mapsto (x, x)$  ([Background B.4](#)), which commutes with every map, and the product node is  $\odot$ . “Spider” is only a name for both; they fail the Frobenius law ([Background B.6](#)), which is never used.

String diagrams for monoidal categories go back to [Joyal and Street \(1991\)](#); spiders are from [Coecke et al. \(2013\)](#).

**THEOREM VI.3** (the merge calculus; known in part). *Work with activation-side diagrams in a symmetric monoidal category of continuous, piecewise-smooth maps with copy and discard nodes. (We call copy and the Hadamard product “spiders” by analogy only; in this cartesian setting they do not satisfy the Frobenius law.) A trainable box is a node whose weight ranges over a stated class: all linear maps, or all affine maps when the box has a bias slot. R2 and R4–R7 are identities of maps; R1 is a closure property and R3 a characterisation.*



| Rule                         | Statement                                                                                       | When it holds                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R1 (fusion)                  | composites and parallel sums of linear (affine) boxes are linear (affine)                       | the operators involved are position-independent                                                                                                                                                                                                                                                                                                                                                                                                |
| R2 (permutation slide)       | $P\sigma^{\times d} = \sigma^{\times d}P$                                                       | every elementwise $\sigma$                                                                                                                                                                                                                                                                                                                                                                                                                     |
| R3 (positive-diagonal slide) | $\text{diag}(\ell)\sigma^{\times d} = \sigma^{\times d}\text{diag}(\ell)$ for all $\ell \geq 0$ | iff $\sigma$ is positively homogeneous (ReLU, leaky ReLU; not GELU, SiLU, tanh); such $\sigma$ are piecewise linear                                                                                                                                                                                                                                                                                                                            |
| R4 (spider slide)            | $\text{diag}(\ell)(a \odot b) = (\text{diag}(\ell)a) \odot b$                                   | any $\ell$                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| R5 (cup slide)               | $\langle Mq, k \rangle = \langle q, M^\top k \rangle$                                           | always. Under RoPE, $\langle R_m Mq, R_n k \rangle = \langle q, M^\top R_{n-m} k \rangle$ holds for every $M$ , but the slide onto the key side, $\langle R_m Mq, R_n k \rangle = \langle R_m q, R_n M^\top k \rangle$ for all $m, n$ , holds iff $M^\top$ commutes with every $R_{n-m}$ that occurs; for diagonal $M = \text{diag}(\ell)$ and distinct frequencies, iff $\ell$ is constant on each rotary pair (in either pairing convention) |
| R6 (norm absorption)         | $W(\gamma \odot n(x) + \beta) = (W \text{diag } \gamma) n(x) + W\beta$                          | always; the shift $W\beta$ needs a bias slot, and an RMSNorm gain has no $\beta$                                                                                                                                                                                                                                                                                                                                                               |
| R7 (copy slide)              | applying $D$ and then copying equals copying and then applying $D$ to every copy                | always                                                                                                                                                                                                                                                                                                                                                                                                                                         |

Consequences:

(a) **Affine edits.** Let a method insert a position-uniform affine edit  $E_q$  on a wire  $w$  of the base network  $f$ , for every  $q \in Q$ . Suppose every reader of  $w$  is a trainable, unshared box (no weight tying) whose class is closed under precomposition with  $E_q$ ; a shift needs a bias slot. Then the method merges, by  $W_c \mapsto W_c E_q$  for all readers  $c$  at once. Dually, it merges into an untied trainable box that writes  $w$  whenever  $E_q$  composed with that box stays in its class. Absorption into a box that the extension itself inserts does not count. Edits that violate these hypotheses fall outside (a): on the reader side, R1–R7 do not merge position-selective edits or edits on a wire that is also read by a skip connection, a norm or the network output. Whole-network non-mergeability is open (Exposé XI). Position-selective edits (ReFT/LoReFT, and position-gated LoRA variants such as Activated LoRA) are not box-locally mergeable. Residual-stream edits need case analysis: an edit merges into the box that produces the wire when that box can absorb it (an untied embedding table; for a fixed-position edit at layer 0, a learned absolute positional table), a shift folds into a writer’s bias when one exists, a diagonal edit after a post-LN folds into that LN’s affine parameters, and an orthogonal edit in an RMSNorm model merges globally, SliceGPT/QuaRot-style. No merge is known for LoReFT’s non-orthogonal low-rank edit.

(b) **Diagonal methods.** A diagonal method merges into a box reached by R1–R7 if all of the following hold: the rules move every copy of  $\text{diag}(\ell)$  into trainable boxes for all  $\ell$  in the method’s parameter domain (with  $\ell \geq 0$  when R3 is used); no position-dependent operator lies on the path (for RoPE,  $\ell$  is constant on each rotary pair); the target box is unshared, or  $\ell$  is constant across the sharing group; and every shift has a bias slot. Conversely, the RoPE condition is necessary for sliding a query scaling onto  $W_K$  when  $W_Q$  and  $W_K$  have full rank and at least two occurring relative offsets have rotation angles that are not multiples of  $\pi$ . Without such genericity hypotheses the conditions are not necessary: a dead rotary pair, or zero entries of  $\ell$  under ReLU, can still merge.

(c) **Non-affine edits.** For a linear or affine box, an edit whose modified box function is non-affine in the box input, for some parameter value, is not box-locally mergeable. For linear and affine boxes this is the definition; for non-linear boxes such as attention, see [Theorem VI.5\(f\)](#).

(d) **Type relativity.** A generic high-precision update is off the quantisation grid, so exact merging of a QLoRA or LoftQ update leaves the quantised type, a type-preserving merge is lossy, and QA-LoRA’s group-constant adapters are the exception ([Proposition V.2\(d\)](#)).

The rules are elementary, and the individual merges are known: (IA)<sup>3</sup> by [Liu et al. \(2022a\)](#) (Sec. 3.3), SSF by re-parameterisation in [Lian et al. \(2022\)](#), and LoRA by linearity in [Hu et al. \(2021\)](#). Moving scales through ReLU, SwiGLU’s linear branch and norm gains, and the GELU obstruction, are in SmoothQuant ([Xiao et al., 2023](#)) and AWQ ([Lin et al., 2024a](#)). New here is only the packaging as one merge calculus for PEFT, with explicit side conditions. QLoRA is [Dettmers et al. \(2023\)](#), LoftQ [Li et al. \(2023b\)](#), QA-LoRA [Xu et al. \(2023b\)](#) and Activated LoRA [Greenewald et al. \(2025\)](#). [Theorem VI.3](#), [Proposition VI.4](#) and [Theorem VI.5](#) were repaired after two rounds of adversarial review; the referee record (§XI) lists what changed.

NUM Under RoPE,  $\langle R_m M q, R_n k \rangle = \langle q, M^\top R_{n-m} k \rangle$  holds for a generic  $M$  to  $10^{-15}$  while the slide onto the key side fails. A query scaling slid onto  $W_K$  leaves a 7.6% least-squares residual unless  $\ell$  is pair-constant. Folding post-LN gains into  $W_1$  changes the output. Legal rewrites agree to  $10^{-15}$  (Appendix E).

Under SwiGLU, R4 moves  $\ell$  onto the linear branch and R1 fuses it into  $W_{\text{up}}$ , for  $\ell$  of either sign:

$$\text{diag}(\ell)(\text{silu}(W_g x) \odot W_u x) = \text{silu}(W_g x) \odot (\text{diag}(\ell) W_u x) = \text{silu}(W_g x) \odot ((\text{diag}(\ell) W_u) x). \quad (8)$$

**Corollaries: scaling and reparametrisation methods.** All weight-space reparametrisation methods are mergeable, because their nonlinearities sit on the parameter side (Cayley in OFT ([Qiu et al., 2023](#)), the norm in DoRA ([Liu et al., 2024b](#)),  $u/\|u\|$  in HRA ([Yuan et al., 2024](#))). In HF PEFT practice ([Mangrulkar et al., 2022](#)), an adapter on one side of a tied embedding and LM head is not merged faithfully (ensure\_weight\_tying=False by default). The scaling methods merge as follows.

- (IA)<sup>3</sup> on keys and values is an output scaling of  $W_K, W_V$  and fuses into them. On the FFN, HF applies it to the input of  $W_{\text{down}}$ , and it fuses there. Both hold for every  $\sigma$ .
- Alternatively, the FFN scaling fuses into  $W_{\text{up}}$  under ReLU when  $\ell \geq 0$ , a restriction of the domain (IA)<sup>3</sup> trains over, and under SwiGLU/GeGLU for every  $\ell$ , by R4 applied to  $\text{silu}(W_g x) \odot W_u x$ , as in (8). Under GELU it does not fuse into  $W_{\text{up}}$ , except for entries with  $\ell_i \in \{0, 1\}$ , when the pre-activations range over  $\mathbb{R}$  (the argument is in Appendix C.47). The  $W_{\text{up}}$  statement is practically idle, since  $W_{\text{down}}$  always works, and AWQ’s code already migrates scales this way (into up\_proj under SwiGLU, into fc1 under ReLU, and through an inserted scaled activation under GELU).
- A query scaling fuses into  $W_Q$  (R1). It moves onto  $W_K$  (R5) only without RoPE or with  $\ell$  constant on rotary pairs, and only without grouped-query (GQA) sharing or with  $\ell$  constant across the group.
- SSF fuses into the preceding linear box when that box has a bias slot; in bias-free LLaMA-type layers its shift needs a new bias. LN-tuning edits existing LN parameters, so it is trivially mergeable. Folding a LayerNorm’s  $\gamma, \beta$  into the following box works in pre-LN blocks, where the norm feeds only the next boxes, when that box has a bias slot for  $W\beta$ ;

an RMSNorm gain has no  $\beta$  and folds without one (the SliceGPT/QuaRot fold). Neither folds into a tied embedding or LM head without breaking the tie.

Not box-locally mergeable: nonlinear Houlsby, Pfeiffer and parallel adapters; routed mixtures (MoLE, LoRAMoE, X-LoRA), unless the gate is constant; AdapterFusion, even with a constant gate, since it combines nonlinear adapters; LST; prefix-type methods ([Theorem VI.5](#)).

The nonlinear adapters are [Houlsby et al. \(2019\)](#), [Pfeiffer et al. \(2020a\)](#) (with AdapterFusion) and the parallel adapter of [He et al. \(2021\)](#). The routed mixtures are MoLE ([Wu et al., 2024c](#)), LoRAMoE ([Dou et al., 2023](#)) and X-LoRA ([Buehler and Buehler, 2024](#)). LST is [Sung et al. \(2022\)](#) and LN-tuning [Zhao et al. \(2023\)](#). Masking (IA)<sup>3</sup> to a few neurons connects to the sparse activation scaling of [Stoehr et al. \(2024\)](#).

**WHAT THIS EXPLAINS** Exact merging into existing weights is certified by local rewriting, a sufficient test together with box-local obstructions, under explicit side conditions (position-uniformity, no sharing, bias slots), and the rewrite also tells you *which* weight absorbs the update. It is not a decision procedure for whole-network mergeability; a stuck rewrite certifies only a box-local obstruction. RoPE pins a query scaling to  $W_Q$  unless it is pair-constant. Bias-free layers leave a shift no slot, and a new bias, though nearly free at inference, changes the architecture. Post-LN feeds the normalised wire to the residual too, so R7 cannot carry its gains forward. Zero inference latency is a weaker goal (a new bias costs almost nothing), and for quantised bases the type caveat (d) applies.

**REMARK** (residual-stream edits that do merge). Some edits outside (a) merge anyway, all position-uniform. A diagonal edit  $h \mapsto s \odot h + t$  after a post-LN folds into that norm's gain and shift,  $\gamma' = s \odot \gamma$  and  $\beta' = s \odot \beta + t$ . A shift  $h \mapsto h + t$  after a pre-LN block with biases folds into its FFN output bias. An orthogonal edit  $h \mapsto Qh$  after block  $l$  of an untied pre-RMSNorm model merges globally, SliceGPT-style: rotate the embedding table, fold the gains of blocks  $1, \dots, l$  into their input weights composed with  $Q^\top$ , compose their output weights with  $Q$ , and leave later blocks unchanged. No merge is known for LoReFT's edit at interior positions.

### Linear insertions, and ReFT

A linear inserted module fuses at once by R1, and the question is which weight-space method it becomes. ReFT, read one hidden vector at a time, becomes LoRA on a box the network lacks, the identity, written in homogeneous coordinates  $(h, 1)$  as  $[I_d \mid 0]$  so that its last column holds the bias.

**PROPOSITION VI.4** (linear insertions, and ReFT). (a) A linear adapter in parallel with a single linear box,  $W_0x + sUDx$ , is LoRA: it has the same  $\rho$  ([He et al., 2021](#), Table 1). A linear adapter parallel to a whole sublayer (He et al.'s parallel adapter around attention or the FFN) edits the weightless skip connection rather than a base box, and is not box-locally mergeable when the sublayer is non-affine.

(b) Let  $W_0 \in \mathbb{R}^{m \times n}$  be bias-free (as LLaMA-type  $W_o$  and  $W_{\text{down}}$  are), and  $r \geq 1$ . A serial linear adapter after  $W_0$ ,  $(I + UD)W_0$  with  $U \in \mathbb{R}^{m \times r}$  and  $D \in \mathbb{R}^{r \times m}$  (scale absorbed), has image

$$W_0 + \{Y \in \mathcal{M}_{\leq r} : \text{row } Y \subseteq \text{row } W_0\}.$$

It is expressively equivalent to LoRA iff  $\text{rank } W_0 = n$ , and weakly (unpointed) isomorphic to LoRA iff  $W_0$  is invertible. The isomorphism  $(U, D) \mapsto (U, DW_0)$  sends a base point  $(U_0, D_0)$  to  $(U_0, D_0W_0)$ . For a fixed  $T_A$  pointing  $(0, A_0)$  with  $\text{rank } A_0 = r$  and  $1 \leq r < m = n$ , a

pointed isomorphism exists iff  $W_0$  is invertible and  $\text{row } A_0 = \text{row}(D_0 W_0)$ , and then  $U_0 = 0$  automatically. With a bias  $b_0$ , the serial adapter also moves the bias by  $U D b_0$ , so on the affine box its image differs from weight-only LoRA's. Houlsby's near-identity initialisation (weights from a zero-mean Gaussian with standard deviation  $10^{-2}$ , truncated at two standard deviations) makes practical serial adapters slightly unpointed; the pointed statement applies to zero-up initialisations.

(c) LoReFT (Wu et al., 2024d) is  $\Phi(h) = h + R^\top (Wh + b - Rh)$ , with  $R \in \mathbb{R}^{r \times d}$  having orthonormal rows, applied at selected positions. On one hidden vector it is the extension by an **identity box** followed by a rank- $\leq r$  affine edit,

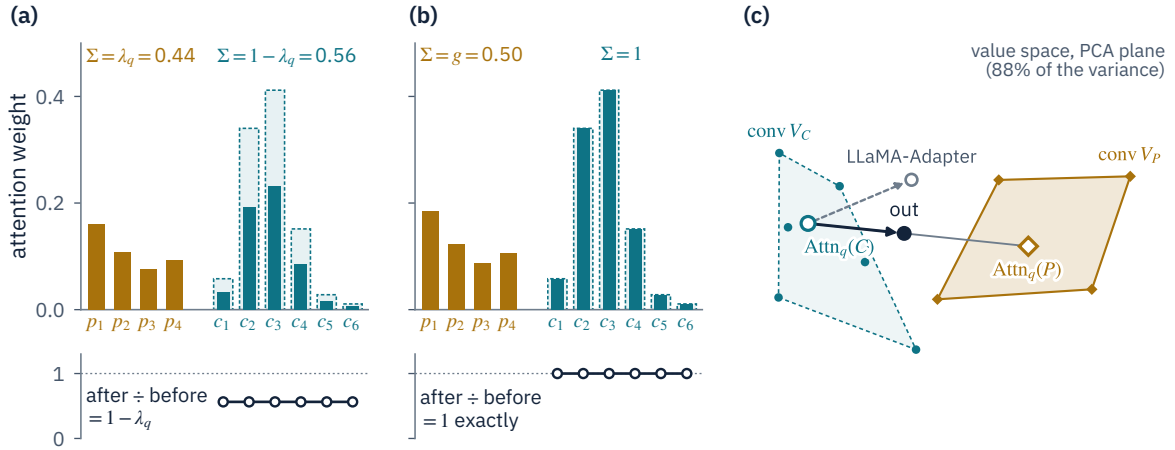
$$\Phi(h) = (I + R^\top (W - R))h + R^\top b,$$

which is  $\text{LoRA}_r$  on the affine identity box  $[I_d \mid 0]$  in homogeneous coordinates, with  $B = R^\top$  constrained to the Stiefel manifold and  $A = [W - R \mid b]$ . Its image is  $(I, 0) + \mathcal{M}_{\leq r}(d \times (d+1))$ , of dimension  $r(2d+1-r)$ . The bias is confined to  $\text{col } B$ , so for  $r < d$  this is not  $\text{LoRA}_r \oplus \text{BitFit}$ , whose dimension is larger by  $d - r$ . It is simulated, through  $h(R, W, b) = (R^\top, [W - R \mid b])$ , by  $\text{LoRA}_r$  on that affine box pointed at  $(R_0^\top, 0)$ , a  $T_B$  pointing with  $\text{col } B_0 = \text{row } R_0$ ; this homogeneous-coordinate LoRA has a bias column in  $A$ , unlike HF LoRA. It is not simulated by  $T_A$ -pointed  $\text{LoRA}_r$ , since its  $\dim S_1 = r(d+1)$  exceeds  $rd$ . It is not isomorphic to  $\text{LoRA}_r$ : its parameter count is smaller by  $r(r+1)/2$ , and its residual symmetry is  $O(r)$ , acting by  $(R, W, b) \mapsto (OR, OW, Ob)$ . It is pointed iff  $W = R$  and  $b = 0$ . The reference implementation (pyreft) initialises  $W$  and  $b$  by the default `nn.Linear init`, independently of  $R$ , so practical LoReFT has a pointing defect  $R^\top((W_0 - R_0)h + b_0)$ : at initialisation it replaces the  $R$ -component of  $h$  by a random affine function, and the simulation above is then only a weak morphism. pyreft also applies dropout to the whole output,  $h$  included, so the training-time map is not  $\Phi$ .

**NUM** The Jacobian rank is  $22 = r(2d+1-r)$  at  $d = 6, r = 2$ , against 26 for  $\text{LoRA}_r \oplus \text{bias}$ . The serial adapter's image has dimension 7 at  $(m, n, r) = (5, 3, 1)$  and at  $(4, 8, 1)$ , against 11 for LoRA at  $(4, 8, 1)$  (Appendix E).

**WHAT THIS EXPLAINS** Per hidden vector, LoReFT (and DiReFT) adds no new geometric type. Its image is that of  $\text{LoRA}_r$  on an affine identity box the architecture did not have. Wu et al. (2024d) already describe DiReFT as LoRA applied to hidden representations at certain positions, and name position as what separates ReFT from PEFT; the new content here is the image dimension, the bias confinement, the  $O(r)$  gauge and the initialisation defect. ReFT as a framework also allows nonlinear interventions (pyreft's `act_fn`), to which this does not apply. The main obstruction to merging is position-selectivity: LoReFT acts only at selected prompt positions, outside Theorem VI.3(a), so no position-independent weight can absorb it. Even applied at every position, at interior residual-stream sites the stream is read by the skip connection and by every later block, and no linear box of the original architecture sits there (at the embedding output an untied embedding table could absorb a position-uniform edit, Theorem VI.3(a)). By Definition VI.1 it is box-locally mergeable into its own inserted box, which does not help.

LoReFT, DiReFT and the reference implementation pyreft are due to Wu et al. (2024d). Serial adapters are due to Houlsby et al. (2019). The parallel form, and the reading of LoRA as a linear adapter parallel to one box, are in He et al. (2021). LoRA is Hu et al. (2021).



**Figure 12. A prefix is a gate (Theorem VI.5).** One query  $q$  attends to six content tokens  $C$  and four prefix tokens  $P$ , all seeded synthetic data. (a) Prefix tuning (Li and Liang, 2021), one softmax over  $P \uplus C$ . The prefix (ochre) takes the mass  $\lambda_q = Z_q(P)/(Z_q(P) + Z_q(C))$ , and each content weight (teal) is its value without the prefix (dashed outline) times  $1 - \lambda_q$ . The strip below plots the ratio after/before for every content token; it is the same for all six, so the content weights keep their ratios and their order (part (c), due to Petrov et al. (2023)). (b) LLaMA-Adapter (Zhang et al., 2023h) normalises  $P$  and  $C$  separately and adds the prefix branch with a gate  $g$ : the prefix weights sum to  $g$ , and the content weights equal the no-prefix ones exactly (part (d)). (c) The value vectors projected on their two leading principal axes, with the hulls of the content values (dashed) and of the prefix values (ochre). The prefix-tuning output (black dot) lies on the segment from  $\text{Attn}_q(C)$  to  $\text{Attn}_q(P) \in \text{conv } V_P$ , the fraction  $\lambda_q$  of the way (parts (b) and (c)); the identity (b) is due to He et al. (2021)), and a larger  $\lambda_q$  moves it further towards the prefix hull. The LLaMA-Adapter output  $\text{Attn}_q(C) + g \text{Attn}_q(P)$  (open grey) is an additive step that leaves the segment. The rigidity holds within one layer, relative to the same layer without the prefix: prefixes at lower layers change the queries, keys and values of higher ones. The values used are in §E.6.

### Attention is a homomorphism followed by a perspective map

Fix a query  $q$ . Attention over a context  $K$  of key-value pairs computes

$$\text{Attn}_q(K) = \frac{\sum_{(k,v) \in K} e^{\langle q,k \rangle / \sqrt{d}} v}{\sum_{(k,v) \in K} e^{\langle q,k \rangle / \sqrt{d}}}. \quad (9)$$

The pair (total weight, weighted sum of values) is additive in the context, and only the division  $(Z, S) \mapsto S/Z$ , the perspective map of convex analysis (Background D.34), is nonlinear. With a prefix  $P$  before the content  $C$ , the output averages what  $P$  alone and  $C$  alone would give, weighted by their shares of the mass, so the prefix is a parallel branch gated by its share.

**THEOREM VI.5** (attention is a homomorphism followed by a perspective map; known in part). *Fix a query  $q$ . Multisets of key-value pairs form a commutative monoid under  $\uplus$ . Define*

$$\Sigma_q(K) = \left( \sum e^{\langle q,k \rangle / \sqrt{d}}, \sum e^{\langle q,k \rangle / \sqrt{d}} v \right).$$

(a)  $\Sigma_q$  is a monoid homomorphism into  $(\mathbb{R}_{\geq 0} \times \mathbb{R}^{d_v}, +)$ , landing in  $\{Z > 0\} \cup \{(0, 0)\}$ . On nonempty multisets,  $\text{Attn}_q = \pi \circ \Sigma_q$  with the partial map  $\pi(Z, S) = S/Z$ , defined for  $Z > 0$ .

(b) **Prefix is a gated parallel adapter.** For nonempty  $P$  and  $C$ ,

$$\text{Attn}_q(P \uplus C) = (1 - \lambda_q) \text{Attn}_q(C) + \lambda_q \text{Attn}_q(P), \quad \lambda_q = \frac{Z_q(P)}{Z_q(P) + Z_q(C)}.$$



(c) **Per-layer pattern rigidity.** Within a single layer, for fixed  $q$  and a fixed content multiset  $C$ : the content attention weights are multiplied by  $1 - \lambda_q$ , their ratios are unchanged, and the output lies on the segment from  $\text{Attn}_q(C)$  to  $\text{Attn}_q(P) \in \text{conv } V_P$ , where  $V_P$  is the set of prefix values. This rigidity is relative to the same layer's computation without the prefix; it is relative to the pretrained pattern only at the lowest prefixed layer. Lower-layer prefixes change the queries and the content keys and values of higher layers, so attention patterns at the level of the network are not rigid (Petrov et al., 2023, §6; the same authors show that prefix-tuning a pretrained transformer can be a universal approximator, Petrov et al., 2024).

(d) **LLaMA-Adapter** normalises  $P$  and  $C$  separately and adds a zero-initialised gate. It computes  $\text{Attn}_q(C) + \tanh(g_l) \text{Attn}_q(P)$ , with one gate per head in the paper; HF PEFT uses one raw zero-initialised scalar per layer, without  $\tanh$ . It uses two perspective maps, leaves the content weights exactly unchanged, and is pointed at  $g = 0$ . Its prompt keys and values are the frozen  $W_k, W_v$  applied to one learned prompt per adapted layer, and it acts in the top layers only.

(e) **Prompt  $\rightarrow$  deep prefix in causal decoders.** Assume that the soft prompt comes first, that attention is causal, and that content positions are offset by the prompt length  $m$  in both methods (as HF PEFT does via `cache_position`). Then at every layer the states at prompt positions depend only on  $p$ , and  $h : p \mapsto (K_j(p), V_j(p))_j$  satisfies

$$\Phi \circ \rho_{\text{prefix}} \circ h = \Phi \circ \rho_{\text{prompt}}$$

as functions of the content tokens, read at content positions. The two methods live on different extensions, so this is an arrow of the unpointed slice over the realisation  $C^\infty(X_{\text{content}}, Y_{\text{content}})$ , not of **Meth**. Under RoPE without the offset, a different  $h$  works: rotate each prefix key by  $-m$  (positions  $-m, \dots, -1$ ). For additive relative biases (ALiBi, T5) the answer depends on how the implementation places the prefix keys. The construction does not apply in bidirectional encoders of depth  $\geq 2$ .

(f) **Obstructions.** Call a head standard if it is vanilla softmax attention without torch-style `add_bias_kv` (which would absorb a one-token prefix), `add_zero_attn`, learned attention sinks or softmax-plus-one. On length-one contexts, every weight setting of a standard head is affine in  $x$ . Some prefix makes the prefixed head non-affine on length-one contexts iff  $W_o \neq 0$  and ( $W_q \neq 0$  or  $W_k^\top b_q \neq 0$ ); for bias-free heads, iff  $W_o \neq 0$  and  $W_q \neq 0$ . For such frozen weights and generic prefixes the prefixed head is not affine, because the gate  $\lambda(x)$  is not constant. So prefix tuning and P-tuning v2 are not box-locally mergeable into a standard head. When  $W_o W_v \neq 0$ , no prefix is neutral, so prompt and prefix tuning are unpointed.

The identity is standard. Part (b) is due to He et al. (2021); see also Chen et al. (2022b). Part (c) and the soft-prompt-to-prefix map in (e) are due to Petrov et al. (2023, §2.2 and §6), informally already Li and Liang (2021, §7.2). LLaMA-Adapter is Zhang et al. (2023h), P-tuning v2 is Liu et al. (2021c), and prompt tuning is Lester et al. (2021); on what prompts can express, see Wang et al. (2023c). Ours are the monoid organisation, the explicit position hypotheses and the slice-category typing of (e).

NUM On a 3-layer, 2-head causal decoder, prompt-to-prefix agreement is  $9 \times 10^{-16}$  with RoPE and the offset, against 1.3 without it and  $2 \times 10^{-15}$  with keys rotated by  $-m$ . A head with  $W_o = 0$  stays affine although  $W_q^\top W_k \neq 0$ , and one with  $W_q \neq 0$ ,  $W_q^\top W_k = 0$  does not (referee replication). The identities (a)–(d) and the non-affinity in (f) are checked to rounding error (Appendix E); Figure 12 draws them for one query.

**WHAT THIS EXPLAINS** At the lowest prefixed layer the content attention equals the pretrained one, rescaled by  $1 - \lambda_q$ . At higher layers it changes only through the modified inputs, to first order additively in each position (Petrov et al., 2023, §6), so new query-key interactions cannot be imposed directly at a layer. No network-level impossibility follows. LLaMA-Adapter’s zero-init attention is a structural change; categorically, its contribution is *pointedness*. Prompt tuning maps into P-tuning v2 in causal decoders. Petrov et al. (2023) and Li and Liang (2021) assert that the inclusion is strict. It is strict in a one-layer, one-token example, where a prompt fixes the prefix value as a function of its key, and we do not prove the generic statement. Encoders remain open.

### Where each method lands

Table 18 collects the verdicts of this exposé.

The verdicts are box-local, and the attention results are per layer. Two questions remain open (Exposé XI): whether prefix tuning is non-mergeable for the whole network, and whether prompt tuning is strictly weaker than P-tuning v2 in general causal decoders, together with how the two relate in encoders.

**Table 18.** Extensions and their merge types. The verdicts are box-local (Definition VI.1), and the attention results are per layer.

| Method                                         | What it adds                                                          | Neutral point                            | Merge                | Reason                                                                                             |
|------------------------------------------------|-----------------------------------------------------------------------|------------------------------------------|----------------------|----------------------------------------------------------------------------------------------------|
| LoRA, DoRA, OFT, HRA                           | nothing; weight-space reparametrisations                              | their pointing in <b>Meth</b>            | M1                   | nonlinearity on the parameter side; $\mu = \rho$                                                   |
| (IA) <sup>3</sup> on $k, v$                    | output scaling of $W_K, W_V$                                          | $\ell = 1$                               | M1                   | R1                                                                                                 |
| (IA) <sup>3</sup> on the FFN                   | input scaling of $W_{\text{down}}$                                    | $\ell = 1$                               | M1, M1 $\rightarrow$ | R1 into $W_{\text{down}}$ ; R4 into $W_{\text{up}}$ under SwiGLU, R3 under ReLU with $\ell \geq 0$ |
| SSF                                            | scale and shift after a box                                           | $\gamma = 1, \beta = 0$                  | M1 $\rightarrow$     | R1 into the preceding box if it has a bias slot                                                    |
| LN-tuning                                      | new values of existing $\gamma, \beta$                                | the pretrained values                    | M1                   | in place; folding forward needs pre-LN, and $\beta$ a bias slot                                    |
| linear adapter parallel to one box             | $sUDx$ beside $W_0x$                                                  | $U = 0$                                  | M1                   | Proposition VI.4(a): it is LoRA                                                                    |
| serial linear adapter                          | $(I + UD)$ after $W_0$                                                | $U_0 D_0 W_0 = 0$                        | M1                   | Proposition VI.4(b)                                                                                |
| Houlsby, Pfeiffer, nonlinear parallel adapters | $U\sigma(D \cdot)$                                                    | $U = 0$ ; Houlsby's init starts near it  | M $\infty$           | non-affine edit, Theorem VI.3(c)                                                                   |
| routed mixtures (MoLE, LoRAMoE, X-LoRA)        | input-dependent gates over LoRA modules                               | varies                                   | M $\infty$           | Theorem VI.3(c); mergeable if the gate is constant                                                 |
| AdapterFusion, LST                             | a gated mix of nonlinear adapters; a side network                     | none                                     | M $\infty$           | non-affine edit, Theorem VI.3(c)                                                                   |
| LoReFT                                         | identity box with a rank- $\leq r$ affine edit, at selected positions | $W = R, b = 0$ ; pyreft's init misses it | M $\infty$           | position-selective, outside Theorem VI.3(a); no merge known                                        |
| prefix tuning, P-tuning v2                     | key-value pairs in every layer                                        | none, generically                        | M $\infty$           | box-locally, Theorem VI.5(f)                                                                       |
| prompt tuning                                  | soft tokens at the input                                              | none, generically                        | M $\infty$           | a deep prefix in causal decoders (Theorem VI.5(e)), so Theorem VI.5(f)                             |
| LLaMA-Adapter                                  | a separately normalised, gated prompt in the top layers               | $g = 0$ , Theorem VI.5(d)                | M $\infty$           | as in Theorem VI.5(f), once the prompt has two or more tokens                                      |
| QLoRA, LoftQ                                   | nothing; LoRA on a quantised base                                     | pointed off $\theta_0$ , with a defect   | Mq                   | Theorem VI.3(d), Proposition V.2(d); QA-LoRA stays on the grid                                     |

## Worked examples

This exposé reads fourteen methods through the three invariants and the merge layer. For each method it says what the method reaches, how training moves it, what a moved base point does and how it merges, and it names the numbered result behind every claim.

A method is a pointed map  $\rho : (Q, q_0) \rightarrow (\Theta, \theta_0)$ . Its image germ says what it can express, its fibres together with the optimizer's metric say how it moves, base change says what a moved  $\theta_0$  does, and extensions cover adapters and prompts. The fourteen cards below apply that toolkit; Exposé VIII turns the answers into coordinates and arrows.

Each card gives the object  $(Q, q_0, \rho)$  for one linear layer  $y = Wx$  with  $W \in \mathbb{R}^{m \times n}$ , and then reads it through the invariants. The **IMAGE** line is the image germ at  $\theta_0$  (Exposé II); the **FIBRE** line is the fibre, the gauge and the induced metric (Exposés III and IV); the **BASE** line is base change, meaning pointings, quantised bases and merge-and-restart (Exposé V); and the **MERGE** line is the extension and the merge type (Exposé VI). The prefix card adds an **ARROWS** line. Lines that an invariant leaves empty or trivial are omitted. The merge types are those of Exposé VIII, where M1 merges into its own box, M1 $\rightarrow$  into a neighbouring box under side conditions, Mq leaves the quantised type when merged exactly, and M $\infty$  is not box-locally mergeable. The header of each card names the source paper and the modification kind used in the catalogue, and its last line lists the results its claims rest on. Their complete proofs are in Appendix C; claims marked [Num] are numerical checks (Appendix E).

**Credits.** Balancedness is due to Du et al. (2018) and Arora et al. (2018), conserved charges to Zhao et al. (2022b), and the balanced slice to Kempf and Ness (1979). The crossover is the closed form of Tarmoun et al. (2021) and Min et al. (2021). The comparison of zero- $B$  and zero- $A$  starts is due to Hayou et al. (2024a), and scaled GD to Tong et al. (2021) and Zhang and Pilanci (2024a). GaLore as one-sided LoRA with carried state is due to Torroba-Hennigen et al. (2025). Prefixes as gated adapters are due to He et al. (2021), and the rigid content ratios and the prompt-to-prefix map to Petrov et al. (2023). The merge of (IA)<sup>3</sup> into  $W_{\text{down}}$  is in Liu et al. (2022a), and scale migration through SwiGLU is used by AWQ (Lin et al., 2024a). LoKr's transport is the rearrangement of Van Loan and Pitsianis (1993). The  $2r$  bound underlies the PiSSA-to-LoRA conversion of Meng et al. (2024a).

**EXAMPLE** (LoRA). Hu et al. (2021) · additive

$$\rho(B, A) = W_0 + sBA, \quad q_0 = (0, A_0), \quad B \in \mathbb{R}^{m \times r}, \quad A \in \mathbb{R}^{r \times n}.$$

**IMAGE** The cone  $W_0 + \mathcal{M}_{\leq r}$ , of dimension  $r(m + n - r)$  for  $r < \min(m, n)$ . With rank  $A_0 = r$  the first-order image is  $S_1 = \{XA_0\}$ , of dimension  $mr$ , so  $\theta_0$  is an apex with deficit  $r(n - r)$  (Figure 3a).

**FIBRE** Gauge  $\text{GL}_r$ , acting by  $(B, A) \mapsto (Bg^{-1}, gA)$ . Under gradient flow with block learning rates  $\eta_B, \eta_A$  and no dropout, the induced cometric is  $K(G) = s^2(\eta_B GA^\top A + \eta_A BB^\top G)$ . The Lie algebra splits as  $\mathfrak{gl}_r = \mathfrak{so}_r \oplus \text{Sym}_r$ . The isometric half  $\mathfrak{so}_r$  leaves  $K$  unchanged, and the  $\text{Sym}_r$  half carries the charge  $\Phi = B^\top B / \eta_B - AA^\top / \eta_A$ , conserved under gradient flow and equal to  $-A_0 A_0^\top / \eta_A$  from the zero- $B$  start. For isotropic  $\Phi = -\kappa I$  the flow is solved in closed form (Min et al., 2021; Tarmoun et al., 2021). With  $\mu = s\sqrt{\eta_A \eta_B}$  and the crossover scale  $\sigma^* = \mu|\kappa|/2$ , zero- $B$  LoRA is one-sided well below  $\sigma^*$ , where effectively only  $B$  moves as in

LoRA-FA, and well above it follows the balanced dynamics of [Arora et al. \(2018\)](#) (Figure 7). On the full-rank locus, with channel-uniform hyperparameters, SGD keeps  $O(r)$  of the gauge, Adam and AdamW keep the signed permutations  $B_r$ , and undamped scaled GD keeps all of  $GL_r$  (Figure 9). Under Adam with  $\varepsilon = 0$ , no weight decay, no gradient clipping and a learning-rate schedule shared by both blocks, LoRA+ with learning-rate ratio  $\lambda$  follows exactly the trajectory of LoRA with  $\alpha$  multiplied by  $\lambda$  ([Schulman and Thinking Machines Lab, 2025](#)) (Figure 8).

**BASE**  $K$  merge-and-restart cycles reach exactly  $W_0 + \mathcal{M}_{\leq \min(Kr, m, n)}$ , so all of  $\Theta$  at  $K^* = \lceil \min(m, n)/r \rceil$ . Exact full-rank pointings form three strata (zero  $B$ , zero  $A$ , and a split that subtracts  $B_0 A_0$  from the frozen weight). Their isomorphism classes are fixed by row  $A_0$ , col  $B_0$  and  $B_0 A_0$  respectively, so as a set they are  $\text{Gr}(r, n) \sqcup \text{Gr}(r, m) \sqcup \mathcal{M}_r$ .

**MERGE** M1, since the trained update  $sBA$  adds to  $W_0$ .

**Results.** [Theorem II.2](#) · [Proposition II.3](#) · [Observation III.2](#) · [Theorem III.5](#) · [Corollary III.6](#) · [Theorem III.9](#) · [Corollary III.11](#) · [Theorem III.12](#) · [Theorem V.3](#)

**EXAMPLE** (DoRA). [Liu et al. \(2024b\)](#) · hybrid

$$\rho(\mu, B, A) = \text{diag}(\mu) N(W_0 + sBA), \quad q_0 = (\mu_0, 0, A_0), \quad \mu_{0,i} = \|w_{0,i}\|.$$

Here  $N$  normalises each row to unit length and  $w_{0,i}$  is the  $i$ -th row of  $W_0$ .

**IMAGE** For  $W_0$  without zero rows and  $r \leq \min(m, n)$  the image is  $\text{Diag}_m \cdot (W_0 + \mathcal{M}_{\leq r}) = \{\text{diag}(\ell)(W_0 + X) : \text{rank } X \leq r\}$ , exactly LoRA followed by a free per-output gain. Its dimension is  $\min(mn, r(m + n - r) + m)$  for generic  $W_0$  [Num], and smaller for special  $W_0$ , for instance  $r(m + n - r) + m - r$  when  $W_0$  has rank one. Its updates can exceed rank  $r$ . From  $B_0 = 0$  the base point is an apex, as for LoRA.

**FIBRE** The gauge contains  $GL_r$ , acting on  $(B, A)$  as for LoRA. For generic  $W_0$  with  $(m - r)(n - r) \geq m$  the budget  $r(m + n) + m$  exceeds the dimension by exactly  $r^2$ , so  $GL_r$  is the whole continuous gauge. Gradient flow conserves the LoRA charge  $\Phi$ , also through the detached norm. The two backward rules give different geometries. The exact gradient of the row normalisation projects the direction gradient row by row, which is a change of metric. The detached backward of the DoRA paper (its Sec. 4.3) and of HF PEFT (`weight_norm.detach()`) removes that projection and gives a non-gradient update, which need not be a descent direction. Under AdamW, which both the paper and HF use, neither variant follows a fixed metric, and the magnitude coordinates matter in their own right. Since  $\mu_i$  has size about  $\|w_{0,i}\|$  and an output gain  $\ell_i$  has size 1, an equal step on each changes the row gain by different amounts.

**MERGE** M1. Since the images agree, any gap between DoRA and LoRA plus an output gain, started at the same weights, comes from training dynamics. Experiment D ([Proposition II.8](#), [Prediction X.2](#)) separates its three possible sources, the parametrisation, the backward rule and the optimizer coordinates.

**Results.** [Proposition II.8](#) · [Proposition II.12](#) · [Theorem III.5](#) · [Prediction X.2](#)



**EXAMPLE** (OFT / BOFT). Qiu et al. (2023); Liu et al. (2023b) · multiplicative

$$W = W_0 R^\top \text{ (OFT)}, \quad W = \text{diag}(s) W_0 R^\top \text{ (BOFT)}, \quad R_0 = I, \quad s_0 = \mathbf{1}.$$

Here  $R$  is block-diagonal with Cayley blocks  $C(Q) = (I + Q)(I - Q)^{-1}$ ,  $Q$  skew-symmetric of size  $b$  (OFT), or a product of  $m'$  butterfly factors (BOFT).

**IMAGE** With exact Cayley maps the image is an orbit in the level set of the neuron Gram matrix  $K_{\text{out}} = WW^\top$ , so every neuron norm, every angle between neurons and every singular value of  $W_0$  stays fixed, and block OFT also fixes the  $n/b$  partial Grams  $W_{:,J} W_{:,J}^\top$ . BOFT has  $K_{\text{out}}(W) = \text{diag}(s) K_{\text{out}}(W_0) \text{diag}(s)$ . It keeps  $|\cos|$  between neurons when every  $s_i \neq 0$ , and the cosines themselves when all  $s_i$  share a sign; with one butterfly factor it sends each block Gram to its  $\text{diag}(s)$ -rescaling. HF's default Cayley–Neumann OFT,  $R = C(Q)(I - Q^4)$ , is only approximately orthogonal, since  $R^\top R = (I - Q^4)^2$ . The base point is regular. For BOFT this holds when the factors' plane sets are disjoint (for instance  $b = 2$ ), so that the factor Lie algebras form a direct sum.

**FIBRE** For exact-Cayley block OFT the gauge is the stabiliser of  $W_0$  in the block group, which is trivial when  $W_0$  is injective.

**BASE** Exact-Cayley block OFT on a fixed partition is idempotent up to closure, and stays in  $W_0 \cdot \prod \text{SO}(b)$  however often it is merged. Iterated BOFT with  $m'$  factors reaches  $\text{diag} \cdot W_0 \cdot P(\prod \text{SO}(b 2^{m'-1})) P^\top$  for a fixed permutation  $P$ , and, for injective  $W_0$ ,  $\text{diag} \cdot W_0 \cdot \text{SO}(n)$  only at full butterfly depth. At  $n = 64$  and  $b = 4$  the block hypergraph has components of sizes 4, 8, 16, 32, 64 for  $m' = 1, \dots, 5$ , so it is connected only at  $m' = 5$  [Num] (Figure 10b). With small generators ( $\|Q\|_{\text{op}} \leq 2^{1/4}$ ) each merge of HF's default is a contraction, so re-merging can only shrink neuron norms and singular values.

**MERGE** M1.

**Results.** Theorem II.5 · Corollary II.6 · Theorem V.3

**EXAMPLE** ((IA)<sup>3</sup>). Liu et al. (2022a) · multiplicative

$$W = \text{diag}(\ell) W_0 \text{ } (W_K, W_V), \quad W = W_0 \text{diag}(\ell) \text{ } (W_{\text{down}}), \quad \ell_0 = \mathbf{1}.$$

**IMAGE** Linear, the closure of the orbit of  $W_0$  under a torus of one-sided diagonal scalings. Zeros of  $W_0$  stay zero, and row lines (column lines on  $W_{\text{down}}$ ) stay fixed while all  $\ell_i \neq 0$ . The image lies inside that of rank-1 HiRA, HiRA<sub>1</sub>. Its covariance group is  $\text{Mon} \times \text{GL}$  on the output-scaled boxes and  $\text{GL} \times \text{Mon}$  on  $W_{\text{down}}$ .

**BASE** Idempotent.

**MERGE** M1 $\rightarrow$ . It fuses into  $W_K$ ,  $W_V$  and  $W_{\text{down}}$  for every activation. The FFN scale fuses alternatively into  $W_{\text{up}}$  under SwiGLU or GeGLU for every  $\ell$ , through the linear branch, a scale migration that AWQ already uses; under ReLU for  $\ell \geq 0$ ; and under GELU only for entries  $\ell_i \in \{0, 1\}$ .

**Results.** Theorem II.5 · Proposition II.9 · Theorem VI.3

**EXAMPLE** (VeRA). [Kopiczko et al. \(2023\)](#) · additive

$$\rho(b, d) = W_0 + \Lambda_b B \Lambda_d A, \quad q_0 = (0, d_0).$$

Here  $\Lambda_b = \text{diag}(b) \in \mathbb{R}^{m \times m}$  and  $\Lambda_d = \text{diag}(d) \in \mathbb{R}^{r \times r}$  are trained, while  $B \in \mathbb{R}^{m \times r}$  and  $A \in \mathbb{R}^{r \times n}$  are frozen, random and shared across layers.

**IMAGE** A toric cone of dimension  $m + r - 1$  [Num], with apex at  $\theta_0$ . At  $b_0 = 0$  only  $b$  receives a gradient, so the first-order image has dimension at most  $m$ , below  $m + r - 1$  when  $r \geq 2$ .

**FIBRE** The gauge is  $\mathbb{R}^\times$ , with  $c$  acting by  $(b, d) \mapsto (cb, d/c)$ , and gradient flow conserves the charge  $\|b\|^2/\eta_b - \|d\|^2/\eta_d$ .

**BASE** If  $B$  has no zero entry, rebasing reaches exactly  $\text{LoRA-FA}(A)$ , the method  $W_0 + \{XA\}$ , which is the target of the arrow  $h(b, d) = \Lambda_b B \Lambda_d A$  from VeRA to  $\text{LoRA-FA}(A)$ .

**MERGE** M1.

**Results.** [Lemma II.4](#) · [Proposition II.12](#) · [Theorem III.5](#) · [Theorem V.3](#)

**EXAMPLE** (LoHa / LoKr). [Hyeon-Woo et al. \(2021\)](#); [Yeh et al. \(2023\)](#) · additive

$$W_0 + (B_1 A_1) \odot (B_2 A_2) \text{ (LoHa)}, \quad W_0 + C \otimes BA \text{ (LoKr)}.$$

**IMAGE**  $\text{LoHa}_{r,r}$  has updates of rank at most  $\min(r^2, m, n)$ , by the cut-rank bound. Its dimension is at most  $\min(mn, 2r(m+n) - 2r^2 - (m+n-1))$ , with equality below  $mn$  [Num]. The gauge is  $\text{GL}_r \times \text{GL}_r$  times the two-sided diagonal torus  $(B_1, A_1, B_2, A_2) \mapsto (D_1 B_1, A_1 D_2, D_1^{-1} B_2, A_2 D_2^{-1})$ , and their joint generic orbits have dimension  $2r^2 + m + n - 1$ . HF zero-initialises one factor, so the base point is an apex at which only that factor moves to first order. At the common budget  $2r(m+n)$  of  $\text{LoHa}_{r,r}$  and  $\text{LoRA}_{2r}$  three regimes occur. For  $r \leq 2$ ,  $r^2 \leq 2r$ , so the image of  $\text{LoHa}_{r,r}$  lies inside that of  $\text{LoRA}_{2r}$ , strictly when  $2r^2 < m + n - 1$ . For  $r \geq 3$  and  $2r^2 < m + n - 1$ , LoHa trades at least  $(m + n - 1) - 2r^2$  dimensions for rank up to  $r^2$ . When  $2r^2 \geq m + n - 1$ , for instance at  $r = 64$  on  $4096 \times 4096$  or at LyCORIS-style  $r = 32$  on 320- to 640-wide layers, it matches or exceeds  $\text{LoRA}_{2r}$  in dimension as well as in rank. At 131,072 parameters on a  $4096 \times 4096$  matrix,  $\text{LoRA}_{16}$  reaches 130,816 dimensions and  $\text{LoHa}_{8,8}$  at most 122,753 (Figure 4).

**IMAGE** A  $k$ -term Kronecker sum  $\sum_i C_i \otimes D_i$  is a matrix of rank at most  $k$  after the Van Loan–Pitsianis rearrangement  $\mathcal{R}$ , which sends  $C \otimes D$  to  $\text{vec}(C) \text{vec}(D)^\top$ . So LoKr is a constrained  $\text{LoRA}_1$  transported by  $\mathcal{R}^{-1}$ , which is LoRA across a different cut of the same four-leg tensor. Write  $m = m_1 m_2$  and  $n = n_1 n_2$ . For  $C \in \mathbb{R}^{m_1 \times n_1}$  and inner rank  $r' \leq \min(m_2, n_2)$ , LoKr’s maximum rank is  $\min(m_1, n_1) r'$ , attained. It lies inside zero-init  $\text{LoRA}_r$  iff  $r \geq \min(m_1, n_1) r'$ , and is otherwise incomparable with it when  $\min(m_1, m_2) \min(n_1, n_2) > 1$  (Figure 6). In HF PEFT the second factor is low-rank only when  $r < \max(m_2, n_2)/2$ ; above that LoKr is an unconstrained Kronecker product.

**MERGE** M1.

**Results.** [Theorem II.10](#) · [Theorem II.11](#) · [Proposition II.12](#) · [Proposition II.13](#)

**EXAMPLE** (FourierFT). [Gao et al. \(2024\)](#) · additive

$$W_0 + \alpha \operatorname{Re} F^{-1}(c), \quad c \text{ supported on } n_f \text{ fixed frequencies.}$$

Here  $F$  is the two-dimensional discrete Fourier transform and  $\alpha$  a fixed scaling (HF's default 150).

**IMAGE** A coordinate mask transported into the real cosine frame. Its image lies in the point-symmetric subspace  $\Delta W_{j,l} = \Delta W_{-j,-l}$ . Started at  $\theta_0$  ( $c_0 = 0$ , HF's `init_weights=True`) it is linear and regular. Its dimension is  $n_f$  minus the number of sampled conjugate pairs  $(u, v), (-u, -v)$ , each of which adds one gauge dimension. At LLM sizes this is close to  $n_f$ , but at  $768 \times 768$  with 1000 frequencies a pair occurs in about half the draws, and HF's default seed 777 gives exactly two. HF's default random spectrum ( $\mathcal{N}(0, 1)$ ) starts elsewhere, with a defect about 14% the size of a  $768 \times 768$   $W_0$  of entry standard deviation 0.04 at 1000 frequencies. Its covariance group  $\Pi_{\text{cyc}}$  excludes most neuron permutations, so permuting hidden units can change what it reaches.

**BASE** Idempotent with fixed indices.

**MERGE** M1.

**Results.** [Proposition II.12](#) · [Proposition II.13](#) · [Observation II.15](#) · [Theorem V.3](#)

**EXAMPLE** (BitFit). [Ben-Zaken et al. \(2021\)](#) · selective

$$b \mapsto b_0 + \beta \text{ on every bias,} \quad W = W_0.$$

**IMAGE** A translation of the bias block, linear and regular. It has no gauge, so its expressive dimension equals its parameter count.

**BASE** Idempotent.

**MERGE** Nothing to merge, since BitFit trains the model's own biases. Llama and Mistral have no biases, so it trains nothing there.

**Results.** [Proposition II.12](#) · [Theorem V.3](#)

**EXAMPLE** (Adapters). [Houlsby et al. \(2019\)](#) · architectural

$$x \mapsto x + U\sigma(Dx), \quad D \in \mathbb{R}^{r \times d}, \quad U \in \mathbb{R}^{d \times r}.$$

**IMAGE** Function level. The adapter is an extension of the network, with neutral point  $U = 0$  (zero-up initialisation). Houlsby et al.'s own near-identity initialisation, with weights drawn from a zero-mean Gaussian of standard deviation  $10^{-2}$  truncated at two standard deviations, starts near the neutral point and leaves a small defect.

**FIBRE** Under Houlsby's GELU the gauge is only the permutations of hidden units. A ReLU, as in the Pfeiffer and parallel adapters, adds positive per-unit rescalings, giving  $S_r \propto \mathbb{R}_{>0}^r$ . At  $U = 0$  only  $U$  moves to first order (the zero gate, function-level variant).

**MERGE**  $M_\infty$ , since it is not box-locally mergeable for non-linear  $\sigma$ . With  $\sigma = \text{id}$ , an adapter parallel to a single linear box is LoRA, with the same  $\rho$ ; one parallel to a whole sublayer edits the weightless skip connection and stays  $M_\infty$  when the sublayer is non-affine. A serial linear adapter  $(I + UD)W_0$  after a bias-free  $W_0$  reaches exactly the rank- $\leq r$  updates whose rows lie in row  $W_0$ , which is LoRA's image iff  $\text{rank } W_0 = n$ .

**Results.** [Lemma II.4](#) · [Definition VI.1](#) · [Lemma VI.2](#) · [Theorem VI.3](#) · [Proposition VI.4](#)

**EXAMPLE** (Prefix / prompt tuning). [Li and Liang \(2021\)](#); [Lester et al. \(2021\)](#) · augmentation

$$\text{Attn}_q(P \uplus C) = (1 - \lambda_q) \text{Attn}_q(C) + \lambda_q \text{Attn}_q(P), \quad \lambda_q = \frac{Z_q(P)}{Z_q(P) + Z_q(C)}.$$

Here  $P$  and  $C$  are the prefix and content multisets of key–value pairs, and  $Z_q(K) = \sum_{k \in K} e^{\langle q, k \rangle / \sqrt{d}}$ .

**IMAGE** Prefix and prompt tuning are unpointed extensions on the sequence wire, in the sense of [Theorem VI.5\(f\)](#): on a head with  $W_o W_v \neq 0$  no prefix is neutral. A prefix acts as an input-gated parallel adapter on attention. Within its layer it multiplies the content attention weights by  $1 - \lambda_q$  and keeps their ratios, and the output moves on the segment towards  $\text{Attn}_q(P) \in \text{conv } V_P$  (Figure 12). This rigidity is relative to the same layer without the prefix, and relative to the pretrained pattern only at the lowest prefixed layer.

**ARROWS** In causal decoders whose content positions are offset by the prompt length, as in HF PEFT, every soft prompt is realised exactly by the deep prefix made of its own per-layer keys and values. This is an arrow of the unpointed slice, from prompt tuning into deep prefix tuning (P-tuning v2). [Petrov et al. \(2023\)](#) and [Li and Liang \(2021\)](#) assert that the inclusion is strict. It is strict in a one-layer, one-token example, where a prompt fixes the prefix value as a function of its key; the generic statement is not proved here, and encoders remain open. LLaMA-Adapter ([Zhang et al., 2023h](#)) normalises the prefix separately and adds a zero-initialised gate, so it leaves the content attention weights unchanged and supplies the missing neutral point.

**MERGE**  $M_\infty$ . For a standard softmax head with  $W_o \neq 0$  and  $W_q \neq 0$ , a generic prefix makes the head non-affine even on single-token inputs, so prefix tuning cannot be folded into the head’s own weights.

**Results.** [Definition VI.1](#) · [Theorem VI.5](#)

**EXAMPLE** (PiSSA / LoftQ / QLoRA). [Meng et al. \(2024a\)](#); [Li et al. \(2023b\)](#); [Dettmers et al. \(2023\)](#) · additive

$$\begin{aligned} \rho(B, A) &= W_{\text{res}} + sBA, & sB_0A_0 &= [W_0]_r, & W_{\text{res}} &= W_0 - [W_0]_r & \text{(PiSSA),} \\ \rho(B, A) &= \kappa\theta_0 + sBA, & B_0 &= 0 & & \text{(QLoRA).} \end{aligned}$$

Here  $[X]_r$  is the best rank- $r$  approximation of  $X$ ,  $W_{\text{res}}$  is frozen, and  $\kappa$  quantises and dequantises. [Meng et al. \(2024a\)](#) take  $s = 1$ .

**IMAGE** PiSSA is a split (Type II) start. Its image is the cone with vertex  $W_0 - [W_0]_r$ , and  $\theta_0$  is a smooth point of it (Figure 3b). Its image and zero-init LoRA’s are incomparable. No exact pointing of  $\text{LoRA}_r$ , zero-init or split, makes an update of rank above  $2r$  reachable, and split pointings attain  $2r$  when  $2r \leq \min(m, n)$ .

**FIBRE** Gauge  $\text{GL}_r$ . HF’s PiSSA factors give  $\Phi_0 = (S_r/s)(1/\eta_B - 1/\eta_A)$ , so the start is balanced,  $\Phi_0 = 0$ , at equal rates. By the balance–escape trilemma a balanced, non-stationary, identity-at-init start needs the frozen residual.

**BASE** QLoRA is LoRA pointed at  $\kappa\theta_0$ , with generically full-rank defect  $e = \kappa\theta_0 - \theta_0$ , so its image misses  $\theta_0$  unless  $\text{rank } e \leq r$  (Figure 1). Splitting and error fitting change the defect, and the order of operations matters. QPiSSA splits, then quantises the residual, and leaves  $\kappa(W_{\text{res}}) - W_{\text{res}}$ . One-step LoftQ quantises, then fits the error. HF’s LoftQ initialisation

ignores the LoRA scale  $s = \alpha/r$ , so the defect is  $e - s[e]_r$ . It is  $e$  minus its best rank- $r$  approximation only when  $s = 1$ , and for  $s = 2$ , as with  $\alpha = 2r$ , its norm equals  $\|e\|$ .

**MERGE** PiSSA merges into its box (M1). On a quantised base the type is Mq. Exact merges of generic high-precision adapters leave the quantised type, and type-preserving merges are lossy; QA-LoRA's group-constant adapters under affine group quantisation are the exception (Xu et al., 2023b).

**Results.** [Theorem II.1](#) · [Theorem II.2](#) · [Corollary III.6](#) · [Proposition III.8](#) · [Proposition V.2](#)

**EXAMPLE** (ReLoRA). [Lialin et al. \(2023b\)](#) · optimizer

$$\theta_{k+1} = \theta_k + sB_k A_k, \quad \text{fresh } (B_k, A_k) \text{ at each stage.}$$

**BASE** Rebased LoRA. After  $K$  stages it reaches exactly  $W_0 + \mathcal{M}_{\leq \min(Kr, m, n)}$ , so all of  $\Theta$  at  $K^* = \lceil \min(m, n)/r \rceil$  (Figure 10a). Over weight space it is  $\text{LoRA}_r^{\oplus K}$  unrolled in time.

**FIBRE** Each stage is an apex with gauge  $\text{GL}_r$ .

**MERGE** Each stage merges (M1).

**Results.** [Observation I.7](#) · [Theorem V.3](#)

**EXAMPLE** (GaLore). [Zhao et al. \(2024a\)](#) · optimizer

$$W_{t+1} = W_t - \eta P_t \text{opt}(P_t^\top G_t), \quad P_t \in \mathbb{R}^{m \times r}.$$

Here  $G_t$  is the weight gradient, and  $P_t$  is refreshed from its SVD every  $T$  steps.

**FIBRE** A backward lens with compression  $P_t^\top$  and anchor  $P_t$ . Under an optimizer that reads only its gradients, such as Adam with weight decay 0 and no full-gradient clipping, each period is exactly one-sided LoRA  $W_k + P_k C$  with  $C_0 = 0$ , merged and re-initialised at the end of the period, with its optimizer state carried over rather than reset or transported. Decoupled weight decay and full-gradient clipping break this identity.

**BASE** The anchor is constant on each period, so it is integrable trivially, including at  $T = 1$ , and integrability does not separate GaLore from LoRA. What separates them is the resampling schedule together with the carried state. As a set over its possible sequences of projectors, the moving shape reaches all of  $\Theta$ .

**MERGE** Not applicable, since GaLore writes  $W$  itself.

**Results.** [Definition IV.1](#) · [Proposition IV.2](#) · [Proposition IV.3](#) · [Theorem V.3](#)

**EXAMPLE** (Merging). [Ilharco et al. \(2022\)](#); [Huang et al. \(2023\)](#) · composition

$$\theta_0 + \sum_i \lambda_i \tau_i \text{ (task arithmetic),} \quad \left( \sum_i w_i B_i \right) \left( \sum_j w_j A_j \right) \text{ (LoraHub).}$$

**FIBRE** Task arithmetic is  $\text{GL}(\Theta)$ -natural. TIES-Merging (Yadav et al., 2023), with global trimming and sign election by total mass, is natural under homotheties times signed permutations. DARE (Yu et al., 2023) is natural under every invertible coordinate rescaling for each mask and under permutations in law, so its distributional group is the monomial group. None of these sparsifying rules is natural under rotations.

**FIBRE** Post-hoc factor-wise rules (LoraHub, and HF's linear merge with its non-SVD ties and dare variants) depend on each adapter's gauge, since replacing  $(B_2, A_2)$  by  $(cB_2, A_2/c)$  changes LoraHub's result by  $\sum_{l \neq 2} w_l w_2 [(c-1)B_2 A_l + (c^{-1}-1)B_l A_2]$  (Figure 9). A shared



factor, such as a common frozen  $A$ , or a joint frame that produces one, such as the joint  $U\Sigma$  of KnOTS (Stoica et al., 2024), removes the gauge dependence. With a shared  $A$ , LoraHub returns  $(\sum_j w_j) \sum_i w_i B_i A$ , exact up to the global factor  $\sum_j w_j$ .

**MERGE** Merging an adapter into a full-precision base is the canonical arrow to the terminal object, full fine-tuning.

**Results.** [Observation I.4](#) · [Proposition V.4](#)

Exposé VIII turns the answers on these cards into seven coordinates, each read off a method's formula without training, and into the arrows between methods.

## The classification

A method is a pointed map  $\rho : (Q, q_0) \rightarrow (\Theta, \theta_0)$ . Its image germ says what it can express (Exposé II), its fibres and the optimizer’s metric how it moves (Exposé III), base change what a moved  $\theta_0$  does (Exposé V), and extensions cover adapters and prompts (Exposé VI). Exposé VII applied this toolkit to fourteen methods. This exposé turns the answers into coordinates and arrows.

Seven coordinates, each read off the formula, place all 382 catalogued methods in a periodic table (Table 19; §VIII.A and §VIII.B). Some combinations of coordinates are empty by theorem; for instance, no linear method starts at an apex. A method  $N$  simulates a method  $M$  when a pointed smooth map  $h$  gives  $\rho_N \circ h = \rho_M$ . Five rules produce every arrow, and the atlas records 403 of them, each with its map  $h$ ; three tests rule arrows out, and PiSSA and LoRA, for instance, have no arrow either way (§VIII.C). Parameters are not reach. At 131,072 parameters on a  $4096 \times 4096$  matrix, LoRA<sub>16</sub> reaches 130,816 directions and LoHa<sub>8,8</sub> at most 122,753 (§VIII.B).

### VIII.A Seven axes, each decidable from the formula

He et al. (2021) gave adapters four design axes (functional form, insertion form, modified representation and composition function), and Chen et al. (2023a) searched a similar design space. The seven axes below extend the idea to weight-space, backward and post-hoc methods, and each axis is tied to a result.

**DEFINITION** (coordinates of a method). The *coordinates* of a method  $M = (Q, q_0, \rho)$  are its answers to the seven questions below, each read off  $\rho$ ,  $q_0$  and the box that  $\rho$  acts on, with no training.

The axes and their values are as follows. Table 21 states the question each axis asks and defines every value, Table 20 gives the coordinates of LoRA, (IA)<sup>3</sup> and prefix tuning, and the ancillary file catalogue.json repeats the definitions with examples for every value.

1. **Kind.** R (reparametrisation) · E (pointed extension) · E° (unpointed extension) · B (backward lens or schedule) · P (post-hoc operation) · F<sub>T</sub> (family over tasks, inputs or tenants).
2. **Shape** of the image germ. Lin ( $\rho$  affine) · Cone (multihomogeneous, non-linear) · Cone\* (translated cone, split pointing) · Orb (group orbit) · Sat (group  $\times$  cone) · Meet (orbit  $\cap$  cone) · Fun (function level).
3. **Base point**, read from  $S_1$ . regular · apex (deficit  $> 0$ ) · neutral · defect ( $\rho(q_0) \neq \theta_0$ ) · unpointed.
4. **Gauge**, from wire insertions and spider slides, checked by Jacobian rank. none · scalar · torus · GL · GL  $\times$  torus · non-linear.
5. **Covariance group** (the Klein geometry of the image). GL  $\times$  GL · CO  $\times$  CO · GL  $\times$  CO · Mon  $\times$  GL · Mon  $\times$  Mon ·  $\Pi_X$  (imposed structure) · frame (random or data, in law).
6. **Merge type**. M1 (into its box) · M1 $\rightarrow$  (into a neighbour via R1–R7, under side conditions) · Mq (an exact merge leaves the quantised type) · M $\infty$  (not box-locally mergeable).
7. **Rebasing closure**. idempotent ·  $\rightarrow$ span ·  $\rightarrow$ group · schedule.

**Table 19.** The periodic table of the 142 core methods. Rows are kinds (axis 1 of §VIII.A), with reparametrisations split by how they couple to  $W_0$ : additively, by a group action, by a Hadamard product or by selecting coordinates. Columns are the shapes of the image germ (axis 2). A dot marks an empty cell. All seven coordinates of the methods named here are in Table 22, and those of every catalogued method in Table 36.

| Kind                       | Lin                                                                                                                                          | Cone                                                                                                                                                                                                                                                   | Cone*                                                              | Orb                                            | Sat               | Meet      | Fun                                                                                                                         |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|------------------------------------------------|-------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------|
| R, additive                | C3A, FFA-LoRA, FourierFT, FRoD, InfLoRA, LoRA-FA, LoRA-SB, LoRA-XS, MiCA, MiSS (formerly Bone), MoRA, OSF, SVFT, TinyLoRA, WaveFT            | AdaLoRA, AdaMSS, BD-LoRA, DEFT, DeLoRA, DyLoRA, EVA, FacT, GraLoRA, KaSA, KronA, LoHa, LoKr, LongLoRA, LoRA, LoRA+, LoRA-Pro, LoRA-RITE, NOLA, O-LoRA, PEANuT, PVerA, QA-LoRA, QLoRA, RandLoRA, Riemannian LoRA, RoSA, rsLoRA, Uni-LoRA, VB-LoRA, VeRA | Astra, CorDA, LoftQ, LoRA-GA, LoRETTA, MiLoRA, OLoRA, PiSSA, PRISM | -                                              | -                 | -         | -                                                                                                                           |
| R, action                  | (IA) <sup>3</sup> , PEQA, RED, SSF                                                                                                           | -                                                                                                                                                                                                                                                      | -                                                                  | GOFT, LOFT, OFT, OFTv2, Piggyback, qGOFT, RoAd | BOFT, DoRA, PSOFT | HRA, LoCO | -                                                                                                                           |
| R, Hadamard                | -                                                                                                                                            | GLoRA, HiRA                                                                                                                                                                                                                                            | -                                                                  | -                                              | -                 | -         | -                                                                                                                           |
| R, selective               | BEFT, BitFit, Child-Tuning, FISH Mask, Linear probing, LN Tuning, LT-SFT, SHiRA, Super-Tuning (Super / Supra), Surgical FT, Trainable Tokens | Diff Pruning                                                                                                                                                                                                                                           | -                                                                  | -                                              | -                 | -         | -                                                                                                                           |
| E (pointed ext.)           | Custom Diffusion, Textual Inversion                                                                                                          | -                                                                                                                                                                                                                                                      | -                                                                  | -                                              | -                 | -         | Adapter (Houlsby), Compacter, ControlNet, LLaMA-Adapter, LoReFT, Parallel Adapter, Pfeiffer Adapter, ShadowPEFT             |
| E <sup>0</sup> (unpointed) | -                                                                                                                                            | -                                                                                                                                                                                                                                                      | -                                                                  | -                                              | -                 | -         | AutoPrompt, Cartridges, CoOp, CPT, LST, MAM Adapter, MPT, P-Tuning, P-Tuning v2, Prefix-Tuning, Prompt Tuning, UniPELT, VPT |
| B (backward)               | Flora, GaLore, LISA, LP-FT, MeZO                                                                                                             | ReLoRA                                                                                                                                                                                                                                                 | -                                                                  | -                                              | -                 | -         | -                                                                                                                           |
| P (post hoc)               | DARE, Intruder-dimension reduction, KnOTS, LCM-LoRA, LoRA Soups (CAT), Task Arithmetic, TIES-Merging, WiSE-FT                                | LoraHub                                                                                                                                                                                                                                                | -                                                                  | -                                              | -                 | -         | Proxy-Tuning                                                                                                                |
| F <sub>T</sub> (family)    | -                                                                                                                                            | HyperDreamBooth, Laplace-LoRA, MHR, Polytropon (Poly), Punica, S-LoRA, Text-to-LoRA (T2L)                                                                                                                                                              | -                                                                  | -                                              | -                 | -         | AdapterFusion, aLoRA, Arrow, HydraLoRA, HyperFormer, Lily, LoRAMoE, MoLE, PHATGOOSE, X-LoRA                                 |

**Table 20.** The seven questions, answered for three methods. Every answer is read off the formula with no training; the values of each axis are defined in Table 21.

| Axis         | Question                                                                                 | LoRA                                                                               | (IA) <sup>3</sup>                                                          | Prefix tuning                      |
|--------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|----------------------------------------------------------------------------|------------------------------------|
| 1 Kind       | Where does it act?                                                                       | R                                                                                  | R                                                                          | E°                                 |
| 2 Shape      | What is the image germ?                                                                  | Cone,<br>$d = r(m + n - r)$                                                        | Lin, a torus-orbit<br>closure                                              | Fun                                |
| 3 Base point | Is $\rho(q_0) = \theta_0$ ? Is<br>$\dim S_1 = d(M)$ ?                                    | apex, deficit $r(n - r)$                                                           | regular                                                                    | unpointed                          |
| 4 Gauge      | Which reparametrisations of<br>$Q$ leave $\rho$ unchanged?                               | $GL_r$                                                                             | none                                                                       | non-linear<br>(Li-Liang's<br>MLP)  |
| 5 Covariance | For which $(g, h)$ is<br>$\text{Im } M(gW_0h^{-1}) =$<br>$g \text{ Im } M(W_0) h^{-1}$ ? | $GL \times GL$                                                                     | $\text{Mon} \times GL$ ;<br>$GL \times \text{Mon}$ on<br>$W_{\text{down}}$ | $GL \times GL$ ,<br>function level |
| 6 Merge      | Does the update fuse into its<br>box, or by R1–R7 into a<br>neighbour?                   | M1                                                                                 | $M1 \rightarrow$                                                           | $M_\infty$                         |
| 7 Rebasing   | What does merge-and-restart<br>reach?                                                    | $\rightarrow \text{span}$ , all of $\Theta$ at<br>$K = \lceil \min(m, n)/r \rceil$ | idempotent                                                                 | –                                  |

### Deciding each axis

Each axis has a procedure that uses only  $\rho$ ,  $q_0$  and the box.

- *Kind*. Check whether the trained model is  $f(\rho(q), -)$  with the original architecture (R), whether new boxes are inserted and some parameter value makes them the identity (E) or none does (E°), whether only the gradient path or the schedule changes (B), whether the inputs are trained checkpoints (P), and whether the adapter depends on a task, an input or a tenant (F<sub>T</sub>). See [Definition I.2](#), [Definition IV.1](#) and [Definition VI.1](#).
- *Shape*. Read the degree of  $\rho - \theta_0$  in  $q$ . An affine map gives Lin and a multihomogeneous non-linear one gives Cone. A group acting on  $\theta_0$  gives Orb, a torus acting on another method's image gives Sat ([Proposition II.8](#)), and an intersection gives Meet ([Theorem II.7](#)).
- *Base point*. Evaluate  $\rho(q_0)$  and compute  $S_1 = d\rho_{q_0}(T_{q_0}Q)$ , then compare  $\dim S_1$  with the generic rank  $d(M)$  ([Definition I.5](#)). For LoRA this is [Theorem II.2\(b\)](#).
- *Gauge*. Insert  $gg^{-1}$  on each internal wire between trainable boxes and slide diagonals through copy spiders, then confirm  $|M| = d(M)$  by a Jacobian rank at a random point ([Proposition II.12\(a\)](#)).
- *Covariance*. Test the identity of [Definition II.14](#) on generators of  $GL_m \times GL_n$ , in distribution when frames are random or come from data.
- *Merge*. Run the rewrite rules R1–R7 of [Theorem VI.3](#) with their side conditions, adding the type condition of [Proposition V.2\(d\)](#) for a quantised base. The rules are sufficient conditions. They certify M1 and  $M1 \rightarrow$ , and  $M_\infty$  records a box-local obstruction. Whether a whole network admits some other merge is left open.
- *Rebasing*. Apply [Theorem V.3\(a\)](#) to additive shapes and [Theorem V.3\(b\)](#) and (c) to actions.

### Conventions

- *Null coordinates*. “–” (null in the classified data) means that the axis does not apply, and no axis has a separate “not applicable” value. The merge axis keeps a legacy key na, used for

backward lenses and schedules, which write the full weight directly and so have nothing to merge; it means the same as “–”.

- *When an axis applies.* Shape and gauge need a parameter space, so they are “–” for closed-form post-hoc operations with no tunable coefficient (Fisher merging without mixing weights, RegMean, uniform model soups), which are single points. Base point is “–” only for methods with no base point of their own, the serving families S-LoRA and Punica. Rebasing needs a weight-space merge, so it is “–” for every  $M_\infty$  method; function-level closures belong in the notes. Covariance applies to every method with a formula.
- *Function-level covariance.* For extensions and families (shape Fun), covariance is the function-level analogue of [Definition II.14](#): the inserted or generated family on each adapted wire is invariant under conjugation by  $(g, h)$  on that wire. Routed mixtures are included, since router and experts transform together.
- *Post-hoc operations.* The parameter space of a P operation is its tunable coefficients ( $\lambda$  in task arithmetic, TIES, DARE, LoRA soups and CAT, LoraHub), and its covariance is the equivariance of the combination rule ([Proposition V.4](#)).
- *Nearest keys.* Entries without an exact key are coded to the nearest one:  $\text{CO} \times \text{Borel} \rightarrow \Pi$  (OLoRA);  $\text{O}(r) \rightarrow \text{GL}$  (LoReFT, PoLAR); stabiliser  $\rightarrow \text{none}$  (OFT);  $\text{CO}(m) \times \text{GL}(n) \rightarrow \text{CO} \times \text{CO}$  (Astra, whitened splits);  $\text{O}(m) \times \text{O}(n) \rightarrow \text{CO} \times \text{CO}$  (LoRA-GA);  $\mathbb{R}^\times B_m \times \mathbb{R}^\times B_n \rightarrow \text{Mon} \times \text{Mon}$  (LoRA-SB);  $\mathbb{R}^\times B_N$  and the torus  $T_N \rightarrow \text{Mon} \times \text{Mon}$  (TIES, DARE). One-sided keys ( $\text{Mon} \times \text{GL}$ ,  $\text{GL} \times \text{CO}$ ) name the group up to exchanging the two sides; the tables of [§VIII.B](#) write (output side)  $\times$  (input side).
- *Instances.* Arrows and obstructions relate instances of methods: a rank, a pointing, a hyperparameter regime. The instance is recorded in the note that accompanies each arrow and obstruction in the ancillary files ([§VIII.B](#)).

**WHAT THIS EXPLAINS** Methods with different shapes or base-point types are never isomorphic, however alike their formulas ([Observation I.4\(c\)](#), [Observation I.6](#)).

**Table 21.** The seven axes. For each axis, the question it answers and the definition of every value, with the test that decides it where one is needed. “–” (a null coordinate) means that the axis does not apply.

| Value                                                                                                                                                                                                                                                                                                           | Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1 Kind</b> (where the method acts). <i>Does the method reparametrise the existing weights, extend the architecture (with or without a neutral point), act only on the backward pass or the schedule, operate post hoc on several trained points, or form a family over an auxiliary base <math>T</math>?</i> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| R<br>reparametrisation                                                                                                                                                                                                                                                                                          | A pointed smooth map $\rho : (Q, q_0) \rightarrow (\Theta, \theta_0)$ into the existing weight space, an object of $\mathbf{Meth}(\Theta, \theta_0) = \mathbf{Diff}_*/(\Theta, \theta_0)$ . Test: the trained model is $f(\rho(q), -)$ with the original architecture $f$ .                                                                                                                                                                                            |
| E<br>pointed extension                                                                                                                                                                                                                                                                                          | A function-preserving embedding $\iota(\theta) = (\theta, \theta'_0)$ into a larger architecture $\tilde{f}$ with a neutral point $\theta'_0$ , so that $\tilde{f}(\theta, \theta'_0, -) = f(\theta, -)$ , followed by a reparametrisation of $\Theta \times \Theta'$ . Test: new boxes are inserted, and some parameter value makes them the identity.                                                                                                                |
| E°<br>unpointed extension                                                                                                                                                                                                                                                                                       | An extension with no neutral point: no parameter value reproduces the frozen function (for prompts and prefixes, <a href="#">Theorem VI.5(f)</a> proves this on a head with $W_o W_v \neq 0$ ). Test: try to solve $\tilde{f}(\theta_0, p, -) = f(\theta_0, -)$ for $p$ .                                                                                                                                                                                              |
| B<br>backward lens or schedule                                                                                                                                                                                                                                                                                  | The forward map is $\text{id}_\Theta$ and the update is $\theta \leftarrow \theta - \eta a_t(\text{opt}(c_t(\nabla L)))$ with a compression $c_t$ and an anchor $a_t$ ; or the method is a sequence of merge-and-restart stages (a rebasing scheme); or it is an optimizer that compresses only its own state (Flora). Test: the method changes how gradients or optimizer state are compressed, or when the base is re-pointed, and leaves the parametrisation alone. |



| Value                           | Definition                                                                                                                                                                                                                                                                      |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P<br>post-hoc operation         | An operation $\Theta^N \rightarrow \Theta$ (or on tuples of factors) applied to several trained points. Test: the inputs are already-trained adapters or checkpoints.                                                                                                           |
| $F_T$<br>family over a base $T$ | A $T$ -point $a : T \rightarrow Q$ of a method, giving $\rho \circ a : T \rightarrow \Theta$ , where $T$ is a set of tasks (hypernetworks), the inputs $X$ (routing), a simplex (mixing) or a set of tenants $U$ (serving). Test: the adapter depends on an auxiliary variable. |

**2 Shape** of the image germ (expressivity). *What is the germ at  $\theta_0$  of the reachable set  $\text{Im } \rho$ ? Read it from the formula: is  $\rho$  affine, multihomogeneous, a group action, a group acting on a cone, an intersection, or defined only at function level?*

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lin<br>linear (flat)                  | $\rho$ is affine in $q$ , so $\text{Im } \rho$ is an affine subspace, and $d(M)$ is the rank of the linear map, which equals $ M $ when the map is injective (FourierFT loses one dimension per sampled conjugate frequency pair). Every point is regular, and rebasing is idempotent when frames, masks and operators are frozen across cycles.                                                              |
| Cone<br>cone with vertex at the start | $\rho - \rho(q_0)$ is multihomogeneous (or a union of linear spaces), and the image is a non-linear cone with vertex at the start $\rho(q_0)$ , which is then a singular point (an apex). The start is $\theta_0$ for pointed methods, and the quantised or truncated base $\theta_0 + e$ for QLoRA-type defects (Proposition V.2). Determinantal, Segre, Hadamard and tensor-network varieties are included. |
| Cone*<br>translated cone (split)      | $\rho = \theta_{\text{res}} + \delta(q)$ with $\delta$ multihomogeneous and $q_0$ chosen so that $\delta(q_0) = \theta_0 - \theta_{\text{res}}$ has full rank. The image is a cone whose apex $\theta_{\text{res}}$ is not $\theta_0$ , and $\theta_0$ is a smooth point (Type II objects).                                                                                                                   |
| Orb<br>group orbit                    | $\rho(q) = \gamma(q) \cdot \theta_0$ with $\gamma(q)$ in a group (or monoid) acting linearly on $\Theta$ . Exactly orthogonal actions preserve the Gram matrix and the spectrum. Torus and Hadamard actions keep only the zeros of $W_0$ (and its row lines while the scales are nonzero), since unconstrained scales can create new zeros.                                                                   |
| Sat<br>torus saturation               | A group or monoid acting on a cone or an orbit: $\text{Im } M = T \cdot \text{Im } N$ for a torus or diagonal monoid $T$ and a method $N$ .                                                                                                                                                                                                                                                                   |
| Meet<br>meet (pullback)               | The image is an intersection of a cone and an orbit (an image-level meet), or the object is a fibre product of two methods (a categorical product, which exists only when the set-level fibre product is a manifold).                                                                                                                                                                                         |
| Fun<br>function level                 | There is no weight-space image: the method lives on an extension and is compared through the realisation $\Phi_f$ .                                                                                                                                                                                                                                                                                           |

**3 Base point** (the position of  $\theta_0$ ). *Is  $\theta_0 = \rho(q_0)$ , and if so, is the first-order image  $S_1 = d\rho_{q_0}(T_{q_0}Q)$  as large as the expressive dimension  $d(M)$ ?*

|                               |                                                                                                                                                                                                                                                                         |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| regular                       | $\dim S_1 = d(M)$ , a first-order deficit of 0: linear methods, orbit methods with an injective orbit map, and split (Type II) initialisations.                                                                                                                         |
| apex<br>first-order deficient | $\dim S_1 < d(M)$ . For cone-shaped germs $\theta_0$ is the vertex. LoRA's deficit is $r(n - r)$ ; HRA's paired initialisation has $\dim S_1 = kn - k(k + 1)/2$ against $d = rn - r(r + 1)/2$ , with $k = r/2$ . DoRA inherits LoRA's deficit through its torus factor. |
| neutral<br>pointed extension  | An extension evaluated at its neutral point: zero up-projection, zero gate, identity box.                                                                                                                                                                               |
| defect<br>pointing defect     | $\rho(q_0) \neq \theta_0$ , by design or by base change; the defect $e = \rho(q_0) - \theta_0$ is part of the data.                                                                                                                                                     |
| unpointed                     | No parameter value reproduces the frozen function.                                                                                                                                                                                                                      |

**4 Gauge** (the generic fibre of  $\rho$ ). *Which transformations of the trainable parameters leave  $\rho$  unchanged, and what is the dimension  $|M| - d(M)$  of the generic fibre? Read it off by inserting  $gg^{-1}$  on internal wires between trainable boxes and sliding diagonals through spiders, and confirm it by the generic Jacobian rank.*

|                            |                                                                                                                       |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------|
| none<br>trivial            | $\rho$ is injective near generic points. There are no charges, and factor-wise operations are well defined.           |
| scalar<br>one scalar slide | A one-parameter group $(b, d) \mapsto (cb, d/c)$ through frozen boxes, with one conserved charge under gradient flow. |
| torus                      | A diagonal group (per rank channel or per hidden unit), whose charges are diagonal entries.                           |

| Value                                                                                                                                                                                                                                                                                                                                                                   | Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GL<br>general linear on<br>internal wires                                                                                                                                                                                                                                                                                                                               | $GL_r$ , or a product of general linear groups on bonds, acting by $(Bg^{-1}, gA)$ . The compact half $\mathfrak{so}(r)$ gives isometries, and the non-compact half $\text{Sym}_r$ gives $r(r+1)/2$ charges.                                                                                                                                                                                                                                                                                                                                                                                                             |
| GL $\times$ torus<br>with a Hadamard<br>torus                                                                                                                                                                                                                                                                                                                           | GL on each factor plus the torus $(D_1XD_2, D_1^{-1}YD_2^{-1})$ sliding through the copy spider, of dimension $r_1^2 + r_2^2 + m + n - 1$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| non-linear                                                                                                                                                                                                                                                                                                                                                              | The generic fibre is not an orbit of a linear group action.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>5 Covariance group</b> (the Erlangen geometry of the image). <i>For which <math>(g, h) \in GL_m \times GL_n</math> does <math>\text{Im } M(gW_0h^{-1}) = g \text{ Im } M(W_0) h^{-1}</math> hold for every <math>W_0</math> (in distribution if random frames or data are involved)?</i>                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| GL $\times$ GL<br>basis-free                                                                                                                                                                                                                                                                                                                                            | The image transforms with every change of basis on both sides. It satisfies the sufficient condition of <a href="#">Observation II.15</a> for every box-wise linear function-preserving gauge.                                                                                                                                                                                                                                                                                                                                                                                                                           |
| CO $\times$ CO<br>spectral frame                                                                                                                                                                                                                                                                                                                                        | The method uses an SVD or another orthogonally equivariant frame of $W_0$ , of gradients or of data. Truncated SVDs are also equivariant under positive scalings, so the group is $\text{CO}(m) \times \text{CO}(n) = (\mathbb{R}_{>0}\text{O}(m)) \times (\mathbb{R}_{>0}\text{O}(n))$ ( <a href="#">Proposition II.16</a> ). Methods that use bare singular frames without singular values (LoRA-GA) are scale-invariant rather than covariant and get only $\text{O}(m) \times \text{O}(n)$ ; they are coded with this nearest key.                                                                                   |
| GL $\times$ CO( $n$ )<br>input-side<br>orthogonal                                                                                                                                                                                                                                                                                                                       | A right action of $\text{O}(n)$ or a subgroup of it. The covariance on the input side is the normaliser of the structure group: $\text{CO}(n)$ for full OFT, the block normaliser for block OFT.                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Mon $\times$ GL<br>one-sided torus                                                                                                                                                                                                                                                                                                                                      | A diagonal torus acts on one side, so only monomial changes of basis are allowed there.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Mon $\times$ Mon<br>coordinate<br>geometry                                                                                                                                                                                                                                                                                                                              | Hadamard or support structure in the standard basis on both sides. It satisfies the sufficient condition of <a href="#">Observation II.15</a> for neuron permutations but not for rotations.                                                                                                                                                                                                                                                                                                                                                                                                                             |
| $\Pi_X$<br>imposed structure                                                                                                                                                                                                                                                                                                                                            | The method imposes a reshape (a tensor factorisation), a block partition, a cyclic order or a pairing that the network does not have, and so breaks even neuron permutations (the test of <a href="#">Lemma VI.2</a> ).                                                                                                                                                                                                                                                                                                                                                                                                  |
| frame<br>random or data<br>frame                                                                                                                                                                                                                                                                                                                                        | Covariance holds only in law, through the distribution of frozen random matrices or of data statistics.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>6 Merge type</b> (lifting along the realisation). <i>After training, does the modified box function stay in the box's function class, and if not, can the rewrite rules R1–R7 of <a href="#">Theorem VI.3</a> (fusion, permutation slide, positive-diagonal slide, spider slide, cup slide, norm absorption, copy slide) move it into a neighbouring linear box?</i> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| M1<br>fuses into its own<br>box                                                                                                                                                                                                                                                                                                                                         | All non-linearity is on the parameter side, and the merge is the canonical map $\rho$ to the terminal object.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| M1 $\rightarrow$<br>fuses into a<br>neighbour                                                                                                                                                                                                                                                                                                                           | A diagonal method merges by R1–R7 into a neighbouring linear box under side conditions: a position-independent path (under RoPE, scales constant on rotary pairs), an unshared target box (no tying, and no GQA sharing unless the scale is constant across the group), and a bias slot for shifts. $(\text{IA})^3$ always fuses into $W_K$ , $W_V$ and $W_{\text{down}}$ ; $W_{\text{up}}$ is an alternative only under SwiGLU or GeGLU (every scale) or ReLU (nonnegative scales), and under GELU only for scales in $\{0, 1\}$ . The rules give sufficient conditions; they do not decide whole-network mergeability. |
| Mq<br>leaves the<br>quantised type                                                                                                                                                                                                                                                                                                                                      | An exact merge of a generic high-precision update leaves the quantised type (NF4 and recomputed-scale absmax grids are not even closed under addition). Type-preserving merges (dequantise, add and requantise, as HF PEFT does for bitsandbytes layers) stay on the grid but are lossy. QA-LoRA's group-constant adapters are the exception.                                                                                                                                                                                                                                                                            |

| Value                                                                                                                                                                                                                                                                  | Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $M_\infty$<br>obstructed                                                                                                                                                                                                                                               | The activation side is non-affine in the input for some parameter value (non-linear adapters, softmax-gated prefixes, input-dependent routing), or the edit acts only at selected positions (ReFT, position-gated LoRA variants). This is box-local non-mergeability under the rewrite calculus of <a href="#">Theorem VI.3</a> . Whole-network non-mergeability is open, and some residual-stream edits do merge: into an untied embedding table, a writer’s bias or a post-LN affine, or globally for orthogonal edits in RMSNorm models. |
| <b>7 Rebasing closure</b> (the reach of merge-and-restart). <i>What does the reachable set <math>R_K = \theta_0 + C^{+K}</math> (additive) or <math>R_K = S^K \theta_0</math> (action) converge to as the number <math>K</math> of merge-and-restart cycles grows?</i> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| idempotent                                                                                                                                                                                                                                                             | The shape is exactly a subspace, subgroup or submonoid independent of the base, with frames, masks and operators frozen across cycles, so cycles gain nothing (up to closure). Recomputing frames or masks from the merged weight does gain.                                                                                                                                                                                                                                                                                                |
| $\rightarrow \text{span}$<br>converges to the<br>span                                                                                                                                                                                                                  | For a cone closed under real scaling, $R_\infty = \theta_0 + \text{span } C$ . For LoRA, $R_K = W_0 + \mathcal{M}_{\leq \min(Kr, m, n)}$ , and full fine-tuning is reached exactly at $K = \lceil \min(m, n)/r \rceil$ .                                                                                                                                                                                                                                                                                                                    |
| $\rightarrow \text{group}$<br>converges to a<br>generated group                                                                                                                                                                                                        | For action methods, $R_\infty$ is the orbit of the monoid generated by one cycle’s image. It is the connected Lie subgroup generated when that image contains a neighbourhood of $e$ in each generating subgroup ( <a href="#">Theorem V.3(b)</a> ). For block-orthogonal factors it is $\prod_C \text{SO}(C)$ over the connected components $C$ of the block hypergraph. HF’s Cayley–Neumann OFT does not satisfy the hypothesis: for small generators it generates only a contraction semigroup.                                          |
| schedule<br>a rebasing scheme<br>itself                                                                                                                                                                                                                                | The method is defined as a sequence of base changes with a moving shape, and its reach is the sum of the shapes visited.                                                                                                                                                                                                                                                                                                                                                                                                                    |

## VIII.B The periodic table

Table 19, at the head of this exposé, places the 142 core methods by kind and by the shape of the image germ, and splits reparametrisations by how they touch  $W_0$ : additively, by a group action, by a Hadamard product or by selecting coordinates. The split refines the additive, selective, reparametrisation-based and hybrid classes of [Lialin et al. \(2023a\)](#); the survey of [Han et al. \(2024b\)](#) uses a similar split.

Some combinations of coordinates are empty by theorem. An affine  $\rho$  has the same differential everywhere, so no linear method is an apex, and in the catalogue none is. Corollary 2 of [Theorem V.3](#) ties idempotent rebasing to shapes that are fixed subspaces, subgroups or submonoids, and of the catalogue’s 92 idempotent methods, 84 have linear shapes and 7 are orbits. The other, WASI ([Nguyen et al., 2025](#)), trains on the rank- $K$  variety  $\mathcal{M}_{\leq K}$  itself, a set that does not move with the base.

### The table by method

Table 22 gives all seven coordinates of the methods the theory works with, together with the expressive dimension and the maximum rank. In it,  $d$  is the expressive dimension,  $\rho_{\max}$  the maximum rank, and  $k$  (in GraLoRA) the number of blocks per side. “ $\Pi$ ” means imposed structure and “fr” a random or data frame. Covariance is written (output side)  $\times$  (input side).  $F_X$  and  $F_U$  are families over inputs and over tenants, coded  $F_T$  in the catalogue, and  $\text{LoRA}_{(B_0, A_0)}$  is LoRA pointed at  $(B_0, A_0)$ .

**Table 22.** The periodic table by method: kind, shape (with  $d$  and  $\rho_{\max}$ ), base point, gauge, covariance, merge type and rebasing closure. The notation is explained before the table, and the marks  $\dagger$ ,  $\approx$  and (fn.) and the numbered notes after it.

| Method                                   | Kind                    | Shape ( $d; \rho_{\max}$ )                                                      | Base pt.                               | Gauge                                       | Covariance                                                                  | Merge | Rebasing                                                 |
|------------------------------------------|-------------------------|---------------------------------------------------------------------------------|----------------------------------------|---------------------------------------------|-----------------------------------------------------------------------------|-------|----------------------------------------------------------|
| Frozen / Full FT                         | R (initial / terminal)  | $\{\theta_0\} / \Theta$                                                         | – / regular                            | –                                           | all / $\text{GL} \times \text{GL}$                                          | –     | id                                                       |
| LoRA                                     | R                       | Cone $r(m+n-r); r$                                                              | apex                                   | $\text{GL}_r$                               | $\text{GL} \times \text{GL}$                                                | M1    | $\rightarrow \Theta$ at $K = \lceil \min(m, n)/r \rceil$ |
| rsLoRA, LoRA+                            | R <sup>1</sup>          | = LoRA                                                                          | apex                                   | $\text{GL}_r$                               | $\text{GL} \times \text{GL}$                                                | M1    | $\rightarrow \Theta$                                     |
| LoRA-Pro, Riem. precondition., LoRA-RITE | R + metric <sup>2</sup> | = LoRA                                                                          | apex                                   | $\text{GL}_r$ <sup>2</sup>                  | $\text{GL} \times \text{GL}$                                                | M1    | $\rightarrow \Theta$                                     |
| DeLoRA                                   | R                       | Cone $\subseteq \mathcal{M}_{\leq r}$ , $\ \Delta W\ _F \leq \lambda \ W_0\ $   | apex (HF) / split (paper)              | non-linear, fibre $r^2+1$ <sup>3</sup>      | $\text{GL} \times \text{GL}$                                                | M1    | $\rightarrow \Theta$                                     |
| GraLoRA                                  | R                       | $\prod$ block cones, $r(m+n-r); \min(kr, m, n)$                                 | apex                                   | $\prod \text{GL}_{r/k}$                     | $\Pi$ (blocks)                                                              | M1    | $\rightarrow \Theta$                                     |
| DyLoRA                                   | R                       | filtration of cones                                                             | apex                                   | $T_r$                                       | $\text{GL} \times \text{GL}$                                                | M1    | $\rightarrow \Theta$                                     |
| AdaLoRA                                  | R                       | budgeted union of cones ( $P \wedge Q$ )                                        | apex ( $\dim S_1 = r$ )                | non-linear, fibre $r(r+1)$ <sup>4</sup>     | $\text{GL} \times \text{GL}$                                                | M1    | $\rightarrow \Theta$                                     |
| LoRA-FA (v1/v2)                          | R                       | Lin $\{XA_0\}, mr$                                                              | regular                                | none                                        | $\text{GL} \times \text{fr}$                                                | M1    | id                                                       |
| MISS / Bone                              | R                       | Lin = LoRA-FA( $\mathbf{1}^\top \otimes I_r$ ) (balance mode)                   | regular                                | none                                        | $\text{GL} \times \Pi$                                                      | M1    | id                                                       |
| MiCA                                     | R                       | Lin $\{U_{\min} Y\}$                                                            | regular                                | none                                        | $\text{CO} \times \text{CO}$                                                | M1    | id                                                       |
| LoRA-XS / SVFT / TinyLoRA                | R                       | Lin in SVD frame ( $r^2 /  \Omega  / u$ )                                       | regular (LoRA-XS, TinyLoRA $\approx$ ) | none                                        | $\text{CO} \times \text{CO}$                                                | M1    | id (frames frozen)                                       |
| LoRA-SB                                  | R                       | Lin $BRA$ in gradient frame, $r^2$                                              | defect (by design)                     | none                                        | $\text{Mon} \times \text{Mon}$ <sup>5</sup>                                 | M1    | id                                                       |
| VeRA                                     | R                       | toric cone, $m+r-1; r$                                                          | apex                                   | $\mathbf{R}^\times$                         | $\text{Mon} \times \text{fr}$                                               | M1    | $\rightarrow \text{LoRA-FA}(A)$                          |
| NOLA                                     | R                       | bilinear cone, $k+l-1$                                                          | apex                                   | $\mathbf{R}^\times$                         | fr                                                                          | M1    | $\rightarrow \text{span}\{B_i A_j\}$                     |
| RandLoRA                                 | R                       | sum of VeRA-type cones; full rank                                               | apex                                   | torus                                       | fr                                                                          | M1    | $\rightarrow \Theta$                                     |
| VB-LoRA, Uni-LoRA                        | R                       | bank / random-tied LoRA images                                                  | defect ( $\approx$ )                   | discrete                                    | $\Pi / \text{fr}$ <sup>6</sup>                                              | M1    | $\rightarrow \Theta$                                     |
| DoRA                                     | R                       | Sat $\text{Diag}_m \cdot \text{Cone}$ , $\min(mn, r(m+n-r) + m)$ ; full         | apex                                   | $\text{GL}_r$                               | $\text{Mon} \times \text{GL}$                                               | M1    | $\rightarrow \Theta$                                     |
| PiSSA / MiLoRA                           | R (Type II)             | translated cone                                                                 | regular (smooth)                       | $\text{GL}_r$                               | $\text{CO} \times \text{CO}$                                                | M1    | $\rightarrow \Theta$                                     |
| OLoRA                                    | R (Type II)             | translated cone (QR)                                                            | regular (smooth)                       | $\text{GL}_r$                               | $\text{CO} \times \text{Borel}$ <sup>7</sup>                                | M1    | $\rightarrow \Theta$                                     |
| LoRA-GA / CorDA                          | R (Type II)             | translated cone (gradient / covariance)                                         | regular (smooth)                       | $\text{GL}_r$                               | $\text{O} \times \text{O} / \text{CO} \times \text{CO}$ (data) <sup>8</sup> | M1    | $\rightarrow \Theta$                                     |
| Astra                                    | R (Type II)             | translated cone (output-activation second moment)                               | regular (smooth)                       | $\text{GL}_r$                               | $\text{CO} \times \text{GL}$ (data) <sup>9</sup>                            | M1    | $\rightarrow \Theta$                                     |
| EVA                                      | R (Type I)              | = LoRA <sup>10</sup>                                                            | apex                                   | $\text{GL}_r$                               | $\text{GL} \times \text{GL}$ <sup>10</sup>                                  | M1    | $\rightarrow \Theta$                                     |
| QLoRA                                    | R at $\kappa \theta_0$  | cone at $\kappa \theta_0$                                                       | defect $e$                             | $\text{GL}_r$                               | $\text{Mon}$ (grid)                                                         | Mq    | $\rightarrow \Theta$                                     |
| LoftQ                                    | R (approx. Type II)     | translated cone over a quantised residual                                       | defect <sup>11</sup>                   | $\text{GL}_r$                               | $\text{Mon}$                                                                | Mq    | $\rightarrow \Theta$                                     |
| LoHa                                     | R                       | Hadamard variety, $\leq \min(mn, 2r(m+n) - 2r^2 - (m+n-1))$ ; $\min(r^2, m, n)$ | apex                                   | $\text{GL}_r^2 \times \text{torus}$         | $\text{Mon} \times \text{Mon}$                                              | M1    | $\rightarrow \Theta$                                     |
| LoKr / KronA                             | R                       | $\mathcal{R}^{-1}$ -transported cone; rank $C \cdot r'$                         | apex                                   | $\mathbf{R}^\times (\times \text{GL}_{r'})$ | $\Pi_\otimes$                                                               | M1    | $\rightarrow \Theta$                                     |

| Method                                  | Kind            | Shape ( $d; \rho_{\max}$ )                                                                                  | Base pt.                             | Gauge                                                    | Covariance                                                                             | Merge                    | Rebasing                                                                                                                   |
|-----------------------------------------|-----------------|-------------------------------------------------------------------------------------------------------------|--------------------------------------|----------------------------------------------------------|----------------------------------------------------------------------------------------|--------------------------|----------------------------------------------------------------------------------------------------------------------------|
| TT / Tucker<br>(Fact, LoTR;<br>LoRETTA) | R               | meet of transported<br>cones <sup>12</sup>                                                                  | apex /<br>regular <sup>12</sup>      | $\prod \mathrm{GL}_{r_e}$                                | $\Pi_{\otimes}$                                                                        | M1                       | $\rightarrow \Theta$                                                                                                       |
| MoRA                                    | R               | Lin, $\hat{r}^2$ ; high rank                                                                                | regular                              | none                                                     | $\Pi$ (blocks)                                                                         | M1                       | id (fixed<br>operators)                                                                                                    |
| HiRA                                    | R               | $(DW_0)_* \text{Cone}$ ; $r$ rank $W_0$                                                                     | apex                                 | $\mathrm{GL}_r$                                          | $\mathrm{Mon} \times \mathrm{Mon}$                                                     | M1                       | $\rightarrow \Theta$ (not<br>transported<br>ReLoRA)                                                                        |
| FourierFT /<br>WaveFT                   | R               | Lin (sparse in the real<br>Fourier / wavelet frame)                                                         | regular <sup>13</sup>                | none <sup>13</sup>                                       | $\Pi_{\mathrm{cyc}} / \Pi$                                                             | M1                       | id (fixed<br>indices)                                                                                                      |
| C3A                                     | R               | Lin (block-circulant),<br>$mn/b$ ; full                                                                     | regular                              | none                                                     | $\Pi_{\mathrm{cyc}}$                                                                   | M1                       | id                                                                                                                         |
| SHiRA / FISH                            | R               | Lin (fixed coordinate<br>mask)                                                                              | regular                              | none                                                     | $\mathrm{Mon} \times \mathrm{Mon}$                                                     | M1                       | id                                                                                                                         |
| Trainable<br>Tokens                     | R               | Lin (fixed row mask on<br>the embedding and a tied<br>head)                                                 | regular                              | none                                                     | $\mathrm{GL} \times \mathrm{Mon}$ <sup>14</sup>                                        | M1                       | id                                                                                                                         |
| Diff pruning                            | R               | $\cup k$ -sparse subspaces                                                                                  | apex                                 | none                                                     | $\mathrm{Mon} \times \mathrm{Mon}$                                                     | M1                       | $\rightarrow \Theta$                                                                                                       |
| BitFit /<br>LN-tuning                   | R               | Lin (bias / LN parameters)                                                                                  | regular                              | none                                                     | $\mathrm{GL} \times \mathrm{GL} /$<br>$\mathrm{Mon}$                                   | M1 /<br>M1 $\rightarrow$ | id                                                                                                                         |
| OFT (block)                             | R               | Orb of $\prod \mathrm{SO}(b)$ (exact<br>Cayley); quasi-orthogonal<br>(HF) <sup>15</sup>                     | regular                              | stabiliser <sup>15</sup>                                 | $\mathrm{GL} \times N_{\mathrm{blk}}$                                                  | M1                       | id up to closure<br>(exact); $K_{\mathrm{out}}$<br>shrinks (HF) <sup>15</sup>                                              |
| BOFT (HF)                               | R               | $\mathrm{diag}(\mathbb{R}^m) \times$ butterfly<br>product                                                   | regular $\dagger$                    | inter-factor                                             | $\mathrm{Mon} \times \Pi$                                                              | M1                       | $\rightarrow \mathrm{diag} \cdot W_0 \cdot$<br>$\prod \mathrm{SO}(b 2^{m'-1})$ ;<br>$\mathrm{SO}(n)$ only at<br>full depth |
| GOFT (strict)<br>/ qGOFT                | R               | Givens product / free<br>$2 \times 2$ blocks <sup>16</sup>                                                  | regular                              | none /<br>torus <sup>16</sup>                            | $\mathrm{GL} \times \Pi$                                                               | M1                       | $\rightarrow W_0 \mathrm{SO}(n) /$<br>$W_0 \mathbb{R}^{n \times n}$ <sup>16</sup>                                          |
| HRA                                     | R               | Meet<br>$(W_0 + \mathcal{M}_{\leq r}) \cap W_0 \mathrm{SO}(n)$<br>(image level), $rn - \frac{r(r+1)}{2}$    | <b>apex</b>                          | non-linear,<br>fibre $\frac{r(r+1)}{2}$                  | $\mathrm{GL} \times \mathrm{CO}$                                                       | M1                       | $\rightarrow W_0 \mathrm{SO}(n)$ in<br>$[n/r]$                                                                             |
| (IA) <sup>3</sup> / SSF /<br>RED        | R               | Lin = closure of a torus<br>orbit                                                                           | regular                              | none                                                     | $\mathrm{Mon} \times \mathrm{GL} /$<br>$\mathrm{GL} \times \mathrm{Mon}$ <sup>17</sup> | M1 $\rightarrow$         | id                                                                                                                         |
| RoAd                                    | R               | RoAd <sub>1</sub> : orbit closure of<br>$(\mathbb{C}^\times)^{m/2}, m$ ; RoAd <sub>2,4</sub> : Lin,<br>$2m$ | regular                              | none (RoAd <sub>4</sub> :<br>per block)                  | $\mathrm{Mon}_{\mathrm{pair}} \times \mathrm{GL}$                                      | M1                       | id                                                                                                                         |
| Houlsby /<br>Pfeiffer                   | E               | Fun (serial, non-linear)                                                                                    | neutral <sup>18</sup>                | $S_r /$<br>$S_r \ltimes \mathbb{R}_{>0}^r$ <sup>18</sup> | $\mathrm{GL} \times \mathrm{GL}$ (fn.)                                                 | $M_\infty$               | –                                                                                                                          |
| Parallel<br>adapter                     | E               | Fun <sup>19</sup>                                                                                           | neutral                              | $S_r \ltimes \mathbb{R}_{>0}^r$ <sup>19</sup>            | $\mathrm{GL} \times \mathrm{GL}$ (fn.)                                                 | $M_\infty$ <sup>19</sup> | –                                                                                                                          |
| Compacter                               | E               | Fun with PHM weights <sup>20</sup>                                                                          | neutral                              | torus <sup>20</sup>                                      | $\Pi_{\otimes}$                                                                        | $M_\infty$               | –                                                                                                                          |
| LST                                     | E $^\circ$      | Fun (side network) <sup>21</sup>                                                                            | unpointed <sup>21</sup>              | $\mathrm{GL}$ (side<br>network)                          | $\mathrm{GL} \times \mathrm{GL}$ (fn.)                                                 | $M_\infty$               | –                                                                                                                          |
| LLaMA-<br>Adapter                       | E               | Fun (separately<br>normalised prefix)                                                                       | neutral<br>( $g = 0$ )               | none                                                     | $\mathrm{GL} \times \mathrm{GL}$ (fn.)                                                 | $M_\infty$               | –                                                                                                                          |
| LoReFT                                  | E <sup>22</sup> | LoRA <sub>r</sub> on $\mathrm{Aff}(I_d)$ ,<br>$r(2d+1-r)$                                                   | defect <sup>22</sup>                 | $O(r)$ <sup>22</sup>                                     | $\mathrm{GL} \times \mathrm{GL}$ (fn.)                                                 | $M_\infty$               | –                                                                                                                          |
| Prompt<br>tuning /<br>P-tuning v1       | E $^\circ$      | Fun (input state) <sup>23</sup>                                                                             | unpointed                            | none /<br>non-linear <sup>23</sup>                       | $\mathrm{GL} \times \mathrm{GL}$ (fn.)                                                 | $M_\infty$               | –                                                                                                                          |
| Prefix /<br>P-tuning v2                 | E $^\circ$      | Fun (gated parallel<br>adapter, per layer)                                                                  | unpointed                            | non-linear /<br>none <sup>24</sup>                       | $\mathrm{GL} \times \mathrm{GL}$ (fn.)                                                 | $M_\infty$               | –                                                                                                                          |
| MPT / CPT                               | E $^\circ$      | $P^* \odot uv^\top$ / ball <sup>25</sup>                                                                    | unpointed /<br>pointed <sup>25</sup> | torus /<br>none <sup>25</sup>                            | $\mathrm{GL} \times \mathrm{GL} /$<br>$\mathrm{CO} \times \mathrm{CO}$ (fn.)           | $M_\infty$               | –                                                                                                                          |
| GaLore                                  | B               | moving Lin $\{P_t X\}$                                                                                      | regular                              | none                                                     | $\mathrm{CO} \times \mathrm{CO}$<br>(gradient)                                         | –                        | $\rightarrow \Theta$                                                                                                       |
| GaLore with<br>a random<br>projector    | B               | moving Lin $\{XA_t\}$                                                                                       | regular                              | none                                                     | fr                                                                                     | –                        | $\rightarrow \Theta$                                                                                                       |
| Flora                                   | B <sup>26</sup> | Lin = $\Theta$ <sup>26</sup>                                                                                | regular                              | none                                                     | fr (in law) <sup>26</sup>                                                              | –                        | –                                                                                                                          |
| LISA                                    | B               | moving layer blocks                                                                                         | regular                              | none                                                     | $\mathrm{GL} \times \mathrm{GL}$ <sup>27</sup>                                         | –                        | $\rightarrow \Theta$                                                                                                       |



| Method                     | Kind      | Shape ( $d; \rho_{\max}$ )                         | Base pt.                          | Gauge                                          | Covariance                                        | Merge                       | Rebasing             |
|----------------------------|-----------|----------------------------------------------------|-----------------------------------|------------------------------------------------|---------------------------------------------------|-----------------------------|----------------------|
| MeZO                       | B         | moving random line                                 | regular                           | none                                           | $O(\Theta)$ in law                                | –                           | $\rightarrow \Theta$ |
| ReLoRA                     | B         | cone per stage $\rightarrow \mathcal{M}_{\leq Kr}$ | apex per stage                    | $GL_r$ per stage                               | $GL \times GL$                                    | M1                          | schedule             |
| Task arithmetic            | P         | $\text{Lin } \theta_0 + \sum \lambda_i \tau_i$     | regular ( $\lambda = 0$ )         | none                                           | $GL(\Theta)$                                      | M1                          | id                   |
| TIES / DARE                | P         | Lin in $\lambda$ (coordinatewise rule)             | regular                           | none                                           | $\mathbb{R}^\times \cdot B_N / \text{Mon}_N^{28}$ | M1                          | id                   |
| LoRA soups / CAT           | P         | $\oplus$ of frozen line methods <sup>29</sup>      | regular                           | none                                           | $GL \times GL$                                    | M1                          | id                   |
| LoraHub / Poly / MHR       | P / $F_T$ | factor-wise product of sums                        | apex                              | $\prod GL_r$ , broken by LoraHub <sup>30</sup> | $GL \times GL / \Pi^{30}$                         | M1 per task                 | –                    |
| MoLE / LoRAMoE / X-LoRA    | $F_X$     | input-dependent mixture                            | neutral / unpointed <sup>31</sup> | per expert / scalar / none <sup>31</sup>       | $GL \times GL$ (fn.) <sup>31</sup>                | $M_\infty^{31}$             | –                    |
| S-LoRA / Punica            | $F_U$     | LoRA cone per tenant over a shared base            | –                                 | $\prod_u GL_{r_u}$                             | $GL \times GL$                                    | M1 per tenant <sup>32</sup> | –                    |
| Text-to-LoRA / HyperFormer | $F_T$     | a $T$ -point of LoRA / of non-linear adapters      | apex / defect <sup>33</sup>       | $GL_r / GL^{33}$                               | $GL \times GL$                                    | M1 / $M_\infty^{33}$        | –                    |
| UniPELT / MAM              | $E^{34}$  | Fun, not a $\oplus$ -word <sup>34</sup>            | unpointed <sup>34</sup>           | $GL^{34}$                                      | $GL \times GL$ (fn.)                              | $M_\infty^{34}$             | –                    |

**Notes on the table.** † BOFT’s identity is regular when the factors’ plane sets are disjoint (for example  $b = 2$ ), because then the factor Lie algebras form a direct sum. “ $\approx$ ” marks implementations whose default initialisation is near, but not exactly at,  $\theta_0$ : the FourierFT and C3A defaults in HF PEFT, VB-LoRA, Uni-LoRA, TinyLoRA, and LoRA-XS ( $R_0 \sim \mathcal{N}(0, 10^{-10})$ ). They are objects of the unpointed slice with a pointing defect, small for LoRA-XS and TinyLoRA but not for HF’s FourierFT ( $\mathcal{N}(0, 1)$  spectrum with scaling 150, about 14% of a std-0.04  $W_0$  at  $768^2$  with 1000 frequencies), and the table classifies their exactly pointed variants. “ $CO \times CO$ ”: frames built from truncated SVDs are equivariant under positive scalings as well as orthogonal maps, so the spectral-frame geometry is  $CO(m) \times CO(n)$  (Proposition II.16); bare singular frames used without their singular values (LoRA-GA) are scale-invariant rather than covariant, which leaves  $O(m) \times O(n)$ . “(fn.)” marks the function-level analogue of Definition II.14: the inserted or generated family on each adapted wire is invariant under conjugation by  $(g, h)$  on that wire. “–” means that the axis does not apply (null in the classified data). Where the classified data refines an entry (Kind, Gauge, Covariance and Merge of LST, UniPELT and MAM, MPT, Compacter, Houlsby, DeLoRA, AdaLoRA, LoRETTA, LoRA-SB, Trainable Tokens, VB-LoRA, LISA, HyperFormer, S-LoRA and Punica), the table follows the data.

Dimensions hold under the hypotheses of the results that compute them. HRA’s is for  $W_0$  of full column rank and even  $r \leq n - 2$  (Theorem II.7); DoRA’s is for generic  $W_0$  with no zero rows, and it drops for special  $W_0$  (Proposition II.8); GraLoRA’s is for HF’s default `hybrid_r=0` (Proposition II.12); and LoHa’s entry is an upper bound, with equality checked numerically below the  $mn$  cap (Proposition II.12).

**Notes by row.** The superscript numbers in Table 22 refer to these notes.

- 1 rsLoRA and LoRA+ are LoRA up to the hyperparameter torus.
- 2 LoRA-Pro, Riemannian preconditioning and LoRA-RITE are reparametrisations with a descending metric (their SGD forms). The  $GL_r$  gauge is respected only undamped; LoRA-Pro’s parameter step keeps  $O(r)$ , and LoRA-RITE keeps  $GL_r$  on the full-rank locus (Theorem III.12).
- 3 DeLoRA: its per-term scalings form a  $2r$ -torus inside the fibre; for  $r = 1$  the fibre is that torus.
- 4 AdaLoRA: the fibre consists of two torus edges and the absorption of  $\Lambda$ .

- 5 LoRA-SB: the covariance is  $\mathbb{R}^{\times} B_m \times \mathbb{R}^{\times} B_n$ , because the frame is the SVD of a sign-gradient.
- 6 VB-LoRA:  $\Pi$ , the chunk partition tied across boxes by the shared bank. Uni-LoRA: a random frame.
- 7 OLoRA: the QR frame makes the input side Borel. The catalogue key is  $\Pi$ , since the order of the inputs matters.
- 8 LoRA-GA:  $O \times O$ , from data; its bare singular frames are scale-invariant and depend on the sign convention. CorDA:  $CO \times CO$ , from data, at the level of the formula.
- 9 Astra: the split is  $\Phi(W_0 C W_0^T) W_0$  ([Proposition II.16](#)).
- 10 EVA is LoRA with row  $A_0$  spanned by principal components of the activations. Its covariance is  $GL \times GL$  at a fixed rank (frame CO), and  $GL \times CO$  with HF's default  $\text{rho}=2$ .
- 11 LoftQ: the defect is the attained residual when  $s = 1$ ; HF ignores the scale  $s = \alpha/r$ .
- 12 TT and Tucker methods sit at the apex for zero-core initialisations. LoRETTA's compensated split initialisation is regular, and its shape is a meet of translated cones ( $\text{Cone}^*$ ).
- 13 FourierFT and WaveFT are regular with `init_weights=True`; HF's FourierFT default is a defect ( $\approx$ ). The gauge gains one dimension for each sampled conjugate pair.
- 14 Trainable Tokens: GL on the embedding dimensions, Mon on the vocabulary.
- 15 OFT: with the exact Cayley map the shape is the orbit of  $\Pi \text{SO}(b)$  in the level set of  $K_{\text{out}}$ , and rebasing is the identity up to closure. HF's default Cayley–Neumann map is only quasi-orthogonal, and its merges shrink  $K_{\text{out}}$  monotonically when  $\|Q\|_{\text{op}} \leq 2^{1/4}$ . The gauge is the stabiliser of  $W_0$ , trivial for injective  $W_0$ ; the catalogue key is “none”.
- 16 GOFT and qGOFT: qGOFT replaces each Givens rotation by an arbitrary  $2 \times 2$  block, and its gauge is a torus of dimension  $d - 2$ . GOFT's official code adds an input scaler by default ([Corollary II.6](#)), and the row is for strict GOFT. When the planes connect all coordinates, as in GOFT's layout, re-merging reaches  $W_0 \text{SO}(n)$  for GOFT. For qGOFT it reaches  $W_0 \mathbb{R}^{n \times n}$ , the matrices whose columns lie in  $\text{col } W_0$ : its blocks include singular ones and ones of negative determinant, and on connected planes they generate the monoid of all  $n \times n$  matrices ([Theorem V.3\(b\)](#)).
- 17  $(\text{IA})^3$ , SSF and RED:  $\text{Mon} \times GL$  for output scalings,  $GL \times \text{Mon}$  for scaling  $W_{\text{down}}$ .
- 18 Houlsby and Pfeiffer adapters are neutral with a zero up-projection; Houlsby's own near-identity initialisation leaves a small defect. Houlsby's gauge is only  $S_r$  (GELU in the reference code, swish in AdapterHub); Pfeiffer's is  $S_r \ltimes \mathbb{R}_{>0}^r$  (ReLU).
- 19 A parallel adapter with  $\sigma = \text{id}$ , parallel to one linear box, is LoRA and merges as M1. The gauge  $S_r \ltimes \mathbb{R}_{>0}^r$  is for ReLU, as in the code of He et al.
- 20 Compacter: the PHM weights are Kronecker-structured and shared. The torus consists of per-term slides; a GL mixing of the PHM index leaves the rank- $\rho$  class.
- 21 LST: the side network is fed through trainable down-projection ladders. The up-projection has rank at most  $d/r$ , so no parameter value reproduces  $f$ .
- 22 LoReFT is an extension at an affine identity box over positions. The pyreft initialisation leaves a defect; LoReFT is neutral iff  $W = R$  and  $b = 0$ . The catalogue key for its gauge is GL.
- 23 Prompt tuning has no gauge. P-tuning v1 adds an LSTM or MLP 2-cell, and its gauge is non-linear, through that encoder.
- 24 Prefix tuning has a non-linear gauge through the MLP of Li and Liang; P-tuning v2 trains the prefix directly and has none.
- 25 MPT's prompt is  $P^* \odot uv^T$ , and MPT is unpointed. CPT's prompt lies in a ball around the demonstrations and is pointed at the in-context-learning context. MPT's gauge is a torus of dimension  $\ell + d$  with  $P^*$  trainable ( $\mathbb{R}^{\times}$  with  $P^*$  frozen); CPT has none.
- 26 Flora compresses only its momentum and takes full-space Adafactor steps ([Proposition IV.2](#)); its random frame is the momentum projection, in law.
- 27 LISA: each stage frees whole boxes.
- 28 TIES elects signs by total mass, and its covariance is  $\mathbb{R}^{\times} \cdot B_N$ . DARE's is  $\text{Mon}_N$ : the torus exactly for each mask, the permutations in law.
- 29 CAT is a  $\oplus$  of frozen line methods,  $\text{span}\{B_i A_i\}$  per layer, inside  $\text{LoRA}_{kr}$ .

- 30 LoraHub, which combines adapters post hoc, is gauge-dependent and breaks  $\Pi GL_r$ ; Poly and MHR, trained jointly, keep a diagonal  $GL_r$ . The covariance is  $GL \times GL$  on LoraHub's products and  $\Pi$  (blocks) for MHR.
- 31 LoRAMoE is neutral, MoLE and X-LoRA are unpointed. The gauge is  $GL_r$  per expert for LoRAMoE, a scalar for MoLE (only the gate trains) and none for X-LoRA (only the head trains). Router and experts transform together. Merging is M1 when the gate is constant.
- 32 S-LoRA and Punica compute relative to a shared base, and each tenant's fibre is a LoRA cone. Each tenant is left unmerged by design, to share the base batch.
- 33 Text-to-LoRA is at the apex; HyperFormer's random hypernetworks leave a defect. The gauge is  $GL_r$  on the generated rank wire for Text-to-LoRA and  $GL$  on the task-embedding wire for HyperFormer, and factor targets are gauge-sensitive. Merging is M1 per task for Text-to-LoRA and  $M_\infty$  for HyperFormer.
- 34 UniPELT's gates also make it an  $F_X$ . Its adapter and prefix parts sit on extensions, so it is not a  $\oplus$ -word. It is unpointed: the prefix has no neutral value, and UniPELT's gate scales the prefix keys rather than their softmax mass. The  $GL$  gauge comes from UniPELT's LoRA summand and MAM's prefix MLP. Merging is  $M_\infty$ , also for UniPELT's input-gated LoRA part.

### Checks on the table

Several entries rest on numerical checks reported in Appendix E: the GraLoRA and LoHa dimensions and the LoHa orbit, the HRA apex (by the Jacobian rank at the paired initialisation), the RoAd orbit dimension, the VeRA rebasing span, the DoRA dimension caps, the FourierFT conjugate pairs, the LoraHub and Poly fibres, and the Cayley–Neumann contraction [Num].

The classified data pass the following structural checks. Every arrow and every obstruction joins two catalogued methods and names a valid rule or test. A check at the level of method identifiers flags 26 pairs that carry both an arrow and an obstruction, and 8 arrows from a method to itself. Each flagged pair concerns two instances of the target: a different rank, pointing, hyperparameter regime or variant. For example,  $PiSSA_r$  maps into  $LoRA_{2r}$  at the degenerate pointing with product 0 (Theorem II.1), while the O3 obstruction concerns  $LoRA_r$  at its apex, whose image is incomparable with  $PiSSA$ 's (Theorem II.2(d)). Each self-arrow is an inclusion between instances of one family, such as  $LoRA_r \hookrightarrow LoRA_{r+1}$ ,  $ReLoRA^K \hookrightarrow ReLoRA^{K+1}$  or BOFT with  $m'$  butterfly factors inside BOFT with  $m' + 1$ .

The data deviate from the conventions of §VIII.A in eight recorded places, which are kept as published in the ancillary files.

- (1) Sixteen function-level families and post-hoc compositions carry no covariance (LoRAMoE, MixLoRA, HydraLoRA, ICedit, MoV, Lily, MoLE, X-LoRA, PHATGOOSE, Arrow, LoraRetriever, LoRA-Flow, SMEAR, GOAT, AdapterSoup and LoRA Switch). Under the conventions they carry their expert family's function-level value:  $GL \times GL$  for dense LoRA experts and routers, and  $Mon \times GL$  for MoV.
- (2) Initial-state tuning and state-offset tuning are  $M_\infty$  with rebasing “idempotent”; under the conventions their rebasing is “–”.
- (3) Flora is coded as a per-period linear method with rebasing “schedule”. Flora takes full-space steps and compresses only its momentum (Proposition IV.2), so its per-step shape is all of  $\Theta$  and its rebasing is “–”, as in Table 22.
- (4) EVA is coded  $GL \times GL$ , which holds at a fixed rank; HF's default  $\rho=2$  gives  $GL \times CO(n)$  (Proposition II.16).

**Table 23.** The classification of all 382 catalogued methods: the number of methods with each coordinate value, over all methods / over the 142 core methods. “–” counts null coordinates, including the legacy merge key na. Per-method values are in Table 36.

| Axis       | Value: all / core                                                                                                 |
|------------|-------------------------------------------------------------------------------------------------------------------|
| Kind       | R 208/85 · E 33/10 · E° 27/13 · B 29/6 · P 31/11 · F <sub>T</sub> 54/17                                           |
| Shape      | Lin 106/45 · Cone 134/43 · Cone* 20/9 · Orb 13/7 · Sat 5/3 · Meet 2/2 · Fun 93/32 · – 9/1                         |
| Base point | regular 113/56 · apex 111/40 · neutral 40/13 · defect 59/13 · unpointed 57/18 · – 2/2                             |
| Gauge      | none 139/62 · scalar 9/6 · torus 40/9 · GL 152/51 · GL × torus 3/1 · non-linear 23/10 · – 16/3                    |
| Covariance | GL × GL 166/53 · CO × CO 47/20 · GL × CO 12/5 · Mon × GL 21/10 · Mon × Mon 52/18 · Π 49/19 · frame 19/10 · – 16/7 |
| Merge      | M1 242/98 · M1→ 11/5 · Mq 11/2 · M∞ 97/32 · – 21/5                                                                |
| Rebasing   | idempotent 92/43 · →span 130/45 · →group 12/7 · schedule 29/9 · – 119/38                                          |

- (5) OFTv2’s rebasing note says that its Cayley–Neumann factors generate  $\prod \mathrm{GL}^+(b)$  and that  $K_{\mathrm{out}}$  drifts. [Theorem V.3](#) gives a contraction semigroup, with monotone shrinkage of  $K_{\mathrm{out}}$ , for small generators.
- (6) Fisher merging’s optional mixing weights would make it a coefficient family under the post-hoc convention.
- (7) The entries MoV / MoLoRA and Child-Tuning (its D and F variants) each cover two variants.
- (8) The “core” flag is not a coordinate. It marks every method named in Table 22 plus a curated set.

### The whole catalogue

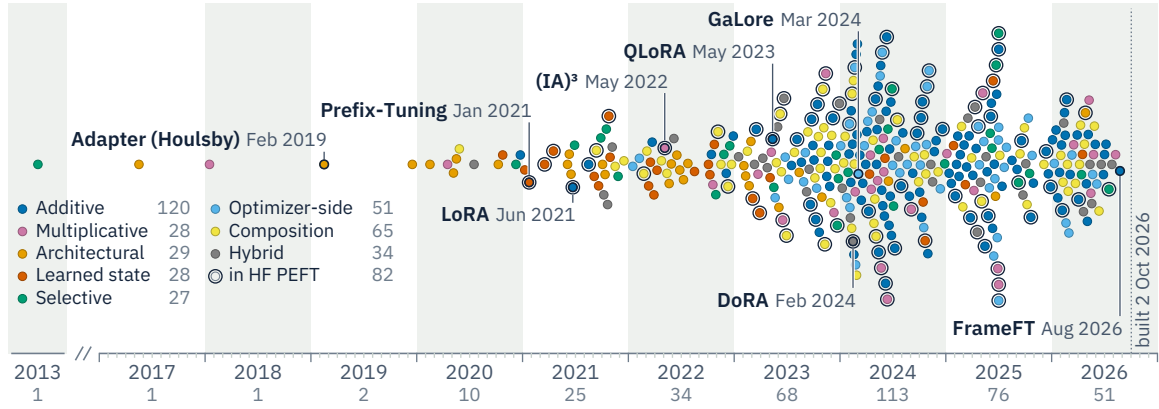
Everything above rests on a catalogue of 382 methods and 138 papers, checked against arXiv and, for library claims, against Hugging Face PEFT ([Mangrulkar et al., 2022](#)) as of 30 September 2026; 82 of the methods ship in that library. Each method entry gives the formula, budget, locus, kind, merge verdict, initialisation, relations, sources, coordinates and a categorical reading, most with arrows and obstructions. Appendix D lists every method with its seven coordinates and its reference (Table 36) and every arrow with its map  $h$  (Table 37); the ancillary files `catalogue.csv`, `arrows.csv` and `obstructions.csv` carry the full entries. Table 23 counts the methods per coordinate value. Cones and linear shapes account for 240 of the 382 methods, and apex and regular base points are about equally common.

Earlier maps made this one possible: the surveys of [Lialin et al. \(2023a\)](#), [Xu et al. \(2023a\)](#), [Han et al. \(2024b\)](#) and [Mao et al. \(2024b\)](#), the delta-tuning study of [Ding et al. \(2022\)](#), and the libraries AdapterHub ([Pfeiffer et al., 2020b](#)), Adapters ([Poth et al., 2023](#)) and Hugging Face PEFT ([Mangrulkar et al., 2022](#)), whose source settled many details.

Figure 13 dates each method by its first arXiv submission. Landmarks come in rough order, with bottleneck adapters in 2019 ([Houlsby et al., 2019](#)), prefixes ([Li and Liang, 2021](#)) and LoRA’s cone ([Hu et al., 2021](#)) in 2021, the torus of  $(\mathrm{IA})^3$  in 2022 ([Liu et al., 2022a](#)), base change with QLoRA in 2023 ([Dettmers et al., 2023](#)), and GaLore’s moving frame in 2024 ([Zhao et al., 2024a](#)), the year of 113 of the 382 methods.

### What a budget buys

[Proposition II.12](#) separates the budget  $|M|$ , what a method trains, from the expressive dimension  $d(M)$ , the number of directions in which it can move the weights. The gap is gauge waste



**Figure 13. The catalogue in time.** Each of the 382 catalogued methods is one dot, placed at the month of its paper’s first arXiv submission, read from the YYMM prefix of its identifier; within a month the dots are spread evenly in the order of their arXiv numbers, and the six entries without an identifier sit at mid-year. The dots form a centred beeswarm, coloured by modification kind (legend, with the number of entries of each kind) and ringed when the method ships in Hugging Face PEFT (82 entries). The one entry before 2017, linear probing on DeCAF features (Donahue et al., 2013), sits left of the axis break. The number under each year counts its entries, from 10 in 2020 to 113 in 2024; the busiest month is February 2024, with 20. The 2026 count ends at the catalogue’s build date, 2 October 2026 (dotted line). Labels mark the bottleneck adapter (Houlsby et al., 2019), prefix tuning (Li and Liang, 2021), LoRA (Hu et al., 2021), (IA)<sup>3</sup> (Liu et al., 2022a), QLoRA (Dettmers et al., 2023), DoRA (Liu et al., 2024b), GaLore (Zhao et al., 2024a) and the newest entry, FrameFT (Adepu et al., 2026). Additive weight edits are the largest kind (120 entries), followed by composition and routing (65) and optimizer-side methods (51).

(Definition I.5): per matrix,  $r^2$  for LoRA, one for VeRA (numerically), zero for injective linear methods, and at least  $2r^2 + m + n - 1$  for  $\text{LoHa}_{r,r}$  (Proposition II.12). At 131,072 parameters on a  $4096 \times 4096$  matrix,  $\text{LoRA}_{16}$  reaches 130,816 directions at rank up to 16, and  $\text{LoHa}_{8,8}$  at most 122,753 at rank up to 64. Rankings at matched budget and at matched  $d(M)$  can differ (Prediction X.10). Figure 4 compares maximum rank and  $d(M)$  at equal budget for eight methods on such a layer, and the companion site’s budget calculator computes both counts for the adapted matrices of common base models.

### VIII.C Arrow rules: when $N$ simulates $M$

“Generalises” and “is a special case of” get one reading here.  $N$  simulates  $M$  when a pointed smooth map  $h : Q_M \rightarrow Q_N$  satisfies  $\rho_N \circ h = \rho_M$  (Definition I.2), so  $N$  reaches every weight that  $M$  reaches. A morphism  $M \rightarrow N$  is exhibited by an explicit pointed  $h$ . Five rules generate every arrow in the atlas, and three tests rule arrows out (Table 24).

#### The five rules

Write  $\text{LoRA}_{(B_0, A_0)}$  for LoRA pointed at  $(B_0, A_0)$ , and let  $\text{FA}(A_0) : B \mapsto W_0 + BA_0$  and  $\text{FB}(B_0) : A \mapsto W_0 + B_0A$  be the two one-sided LoRAs of Proposition I.8.

- **A1, interval.** Frozen  $\rightarrow M \rightarrow$  Full FT, with  $h = \text{const}_{q_0}$  and  $h = \rho_M$  respectively.
- **A2, freezing gives sub-objects.** Freezing a block of coordinates  $q''$  at its base value  $q_0''$  gives a monomorphism  $h(q') = (q', q_0'')$ . Examples:  $\text{FA}(A_0) \hookrightarrow \text{LoRA}_{(0, A_0)}$ ,  $\text{FB}(B_0) \hookrightarrow \text{LoRA}$ ,  $\text{MiCA} \hookrightarrow \text{LoRA}_{(B_0, 0)}$ , and  $\text{DyLoRA}_k \hookrightarrow \text{DyLoRA}_{k+1}$  by zero padding.



**Table 24.** The arrow rules A1–A5, transport T and the obstruction tests O1–O3. An arrow  $M \rightarrow N$  is a pointed smooth map  $h : \mathcal{Q}_M \rightarrow \mathcal{Q}_N$  with  $\rho_N \circ h = \rho_M$ .

| Rule                       | Statement                                                                                                               | The map $h$ , or the test                                                                                                                                                                                                                                                        | Examples                                                                                                                                                                           |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>A1</b><br>interval      | Frozen $\rightarrow M \rightarrow$ Full FT for every weight-space $M$ .                                                 | $h = \text{const}_{q_0}; h = \rho_M$                                                                                                                                                                                                                                             | Merging is the arrow to Full FT.                                                                                                                                                   |
| <b>A2</b><br>freezing      | Freezing coordinates at their base values gives a sub-object.                                                           | $h(q') = (q', q_0'')$                                                                                                                                                                                                                                                            | FA( $A_0$ ) $\hookrightarrow$ LoRA pointed at $(0, A_0)$ ; MiCA $\hookrightarrow$ LoRA pointed at $(B_0, 0)$ .                                                                     |
| <b>A3</b><br>factorisation | $\rho_M = \rho_N \circ h$ for an explicit pointed map $h$ , usually polynomial.                                         | VeRA $\rightarrow$ FA( $A$ ):<br>$h(b, d) = \Lambda_b B \Lambda_d$ .<br>LoHa $_{r_1, r_2} \rightarrow$ LoRA $_{r_1 r_2}$ :<br>face-splitting products.<br>(IA) $^3 \rightarrow$ HiRA $_1$ pointed at $(0, \mathbf{1}^\top)$ : $h(\ell) = (\ell - \mathbf{1}, \mathbf{1}^\top)$ . | Isomorphisms: rsLoRA $\cong$ LoRA by $h = (\sqrt{r} B, A)$ ; MiSS (balance mode) $\cong$ FA( $\mathbf{1}^\top \otimes I_r$ ); a linear adapter parallel to one layer $\cong$ LoRA. |
| <b>A4</b><br>products      | A product comes with its two projections.                                                                               | $R \mapsto B_0 R, R \mapsto R A_0$                                                                                                                                                                                                                                               | LoRA-XS $\rightarrow$ FA( $A_0$ ), FB( $B_0$ ). HRA's image lies in those of OFT $_{S_0}$ and LoRA $_r$ , an image-level inclusion only.                                           |
| <b>A5</b><br>sums          | $M \rightarrow M \oplus N$ , and $K$ stages embed in $K + 1$ .                                                          | $q \mapsto (q, q_0^N)$                                                                                                                                                                                                                                                           | LoRA $_r \oplus$ LoRA $_s \cong$ LoRA $_{r+s}$ (unpointed);<br>ReLoRA $^K \rightarrow$ ReLoRA $^{K+1}$ .                                                                           |
| <b>T</b><br>transport      | $T_*$ for $T \in \text{GL}(\Theta)$ is an automorphism of the slice, giving one object in two frames.                   | no arrow; record the frame $T$                                                                                                                                                                                                                                                   | Sums of Kronecker products are $\mathcal{R}_*^{-1}$ LoRA; HiRA is $(D_{W_0})_*$ LoRA at $\theta_0$ when $W_0$ has no zero entry.                                                   |
| <b>O1</b><br>image         | No arrow if $\text{Im } M \not\subseteq \text{Im } N$ , as when $N$ keeps an invariant that $M$ breaks.                 | compare images                                                                                                                                                                                                                                                                   | LoRA $\not\rightarrow$ exact-Cayley OFT, which keeps $K_{\text{out}}$ .                                                                                                            |
| <b>O2</b><br>size          | No arrow if $d(M) > d(N)$ or $\rho_{\max}(M) > \rho_{\max}(N)$ .                                                        | compare dimension and maximum rank                                                                                                                                                                                                                                               | LoRA $_r \not\rightarrow$ LoRA-FA, since $r(m + n - r) > mr$ .                                                                                                                     |
| <b>O3</b><br>first order   | No arrow if $S_1(M) \not\subseteq S_1(N)$ , or if $M$ and $N$ lie in different strata of <a href="#">Theorem II.2</a> . | compare $S_1$ and strata                                                                                                                                                                                                                                                         | T $_A$ $\not\rightarrow$ T $_B$ -pointed LoRA; PiSSA and LoRA, either way;<br>LoRA $\not\rightarrow$ AdaLoRA, though on one matrix their images agree.                             |

• **A3, factorisation through a polynomial map.** If  $\rho_M = \rho_N \circ h$  for an explicit smooth pointed  $h$ , then  $M \rightarrow N$ ; one checks  $h(q_0^M) = q_0^N$ . Examples:

- VeRA  $\rightarrow$  FA( $A$ ),  $h = (\Lambda_b B \Lambda_d)$ ;
- NOLA  $\rightarrow$  LoRA $_r$ ,  $h = (\sum_i \alpha_i B_i, \sum_j \beta_j A_j)$ ;
- LoHa $_{r_1, r_2} \rightarrow$  LoRA $_{r_1 r_2}$ , by the face-splitting product  $h = (B_1 \bullet B_2, (A_1^\top \bullet A_2^\top)^\top)$  ([Background B.13](#));
- LoRA $_r \hookrightarrow$  LoRA $_{r+1}$ ,  $h = ([B \ 0], [A; a_0])$ ;
- DyLoRA $_k \hookrightarrow$  DyLoRA $_{k+1}$ ;
- LoRA-XS  $\rightarrow$  PiSSA,  $h(R) = (U_r(S_r + R)S_r^{-1/2}, S_r^{1/2}V_r^\top)$ ;
- HRA $_r \rightarrow$  LoRA $_r$  pointed at  $h(q_0)$ , via  $R - I = \sum_i H_1 \cdots H_{i-1}(H_i - I)$ ;
- (IA) $^3 \rightarrow$  HiRA $_1$  pointed at  $(0, \mathbf{1}^\top)$ ,  $h(\ell) = (\ell - \mathbf{1}, \mathbf{1}^\top)$ ;
- isomorphisms: MiSS  $\cong$  FA( $\mathbf{1}^\top \otimes I_r$ ); rsLoRA  $\cong$  LoRA,  $h = (\sqrt{r} B, A)$ ; a linear adapter parallel to one linear box  $\cong$  LoRA ([Proposition VI.4](#)).

- **A4, products and meets.** A product comes with two projections, and it is universal among the methods that both factors simulate. Examples:  $\text{LoRA-XS} \rightarrow \text{FA}(A_0)$  and  $\text{LoRA-XS} \rightarrow \text{FB}(B_0)$ , at the level of objects;  $\text{HRA} \rightarrow \text{OFT}_{S_0}$  and  $\text{HRA} \rightarrow \text{LoRA}_r$ , at the level of images only. An intersection of images gives only inclusions of images, not arrows of **Meth**; for example  $\text{Im LoRA}_r \cap \text{Im (IA)}^3$  exists only as a set, and there is no product in **Diff**.
- **A5, sums.**  $M \rightarrow M \oplus N$ ,  $q \mapsto (q, q_0^N)$ , for additive  $N$  pointed at  $\theta_0$ ; for example  $\text{LoRA}_r \rightarrow \text{LoRA}_r \oplus \text{sparse (RoSA, Nikdan et al., 2024)}$ . Also  $\text{LoRA}_r \oplus \text{LoRA}_s \cong \text{LoRA}_{r+s}$ , unpointed (**Observation I.7**), by concatenation with the  $A$ -blocks rescaled under  $\alpha/r$  scales; it is pointed exactly at the concatenated initialisation, which is a standard zero- $B$  pointing iff the stacked  $A_0$  has full row rank and  $r + s < \min(m, n)$ . Rebasings stages give  $M^{\oplus K} \rightarrow M^{\oplus(K+1)}$ .

**Transport is not an arrow.**  $T_*$  for  $T \in \text{GL}(\Theta)$  is an automorphism of the slice (**Proposition II.13**). Methods related by transport, such as the Kronecker family and LoRA (across  $\Theta' \cong \Theta$ ), are “the same object in another frame”. Sums of  $k$  Kronecker terms are  $\mathcal{R}_*^{-1} \text{LoRA}_k$  after the Van Loan–Pitsianis rearrangement (**Theorem II.11**); HiRA is  $(D_{W_0})_* \text{LoRA}$  inside the slice at  $\theta_0$  when  $W_0$  has no zero entries (**Proposition II.9**); FourierFT is a mask in the real cosine Fourier frame, a transport only without sampled conjugate pairs and only for `init_weights=True`. Transports that depend on  $\theta_0$  (HiRA, SVD frames) agree with their source only at the base point. For additive families and a base-independent  $T$ , transport also commutes with merge-and-restart; for action-type or  $\theta_0$ -dependent transports it need not.

### Three tests

There is no arrow  $M \rightarrow N$  if any of the following holds.

- **O1, image.**  $\text{Im } M \not\subseteq \text{Im } N$ , for instance because  $N$  preserves an absolute invariant (Gram matrix, spectrum, row lines, zero pattern) that  $M$  breaks. Examples:  $\text{LoRA} \not\rightarrow \text{OFT}$ ,  $\text{LoRA} \not\rightarrow (\text{IA})^3$ ,  $\text{DoRA} \not\rightarrow \text{LoRA}_r$ .
- **O2, dimension and cut-rank.**  $d(M) > d(N)$ , or  $\rho_{\max}(M) > \rho_{\max}(N)$ . Examples:  $\text{LoHa}_{r,r}$  does not map into  $\text{LoRA}_{r'}$  for  $r' < r^2$ , since its generic rank is  $r^2$ ; and  $\text{LoRA}_r$  does not map into  $k$ -term Kronecker sums for  $k < \min(m_1, m_2) \min(n_1, n_2)$ , the Kronecker rank of a generic rank-one update (**Theorem II.11**).
- **O3, first order and base point.**  $S_1(M) \not\subseteq S_1(N)$ , or  $M$  and  $N$  lie in different strata of **Theorem II.2** (Type I apex against Type II smooth point); the  $S_1$  test applies even when the images agree. Examples:  $\text{T}_A\text{-stratum LoRA} \not\rightarrow \text{T}_B\text{-stratum LoRA}$ ;  $\text{PiSSA} \not\rightarrow \text{LoRA}$  and  $\text{LoRA} \not\rightarrow \text{PiSSA}$ .

**Why the rules give arrows and the tests rule them out.** Each rule and test is an instance of a result stated earlier. In each rule  $h$  is smooth and pointed, and  $\rho_N \circ h = \rho_M$  is checked by multiplying out. A1 is **Observation I.4(a)**, since  $\rho_M \circ \text{const}_{q_0} = \theta_0$  and  $\text{id} \circ \rho_M = \rho_M$ . A2 holds because freezing  $q''$  at  $q_0''$  leaves the rest of the formula unchanged. For A3, take VerA, where  $W_0 + \Lambda_b B \Lambda_d A = W_0 + (\Lambda_b B \Lambda_d) A$  and  $h$  sends  $b_0 = 0$  to the base point 0 of LoRA-FA. A4 is the universal property of the fibre product (**Observation I.4(b)**, **Proposition I.8(a)**); for a meet such as HRA only the inclusion of images holds (**Theorem II.7**). A5 is the identity

$$\rho_{M \oplus N}(q, q_0^N) = \theta_0 + \delta_M(q) + \delta_N(q_0^N) = \rho_M(q), \quad (10)$$

from **Observation I.7**.

**Table 25.** The 485 recorded obstructions  $M \not\rightarrow N$ , by the test that rules the arrow out. The full list, with the failing invariant of each, is the ancillary file `obstructions.csv`.

| Test | Count | What fails                                                                             |
|------|-------|----------------------------------------------------------------------------------------|
| O1   | 288   | an image invariant (Gram matrix, spectrum, row lines, zero pattern) or image inclusion |
| O2   | 129   | dimension or cut-rank                                                                  |
| O3   | 68    | first-order image or base-point stratum                                                |

An arrow gives  $\text{Im } M = \rho_N(h(Q_M)) \subseteq \text{Im } N$ , which is O1. Images are semialgebraic, or definable when softmax enters ([Background C.17](#)), so dimension is monotone under inclusion; and every update in  $\text{Im } M - \theta_0$  also lies in  $\text{Im } N - \theta_0$ , so its rank is bounded by  $\rho_{\max}(N)$ . Together these give O2. The chain rule at  $q_0$  gives  $S_1(M) \subseteq S_1(N)$  ([Observation I.6](#)), the first half of O3. For the strata, the images of T-type and S-type objects are incomparable ([Theorem II.2\(d\)](#)). Inside the T strata,  $S_1(T_A) = \{XA_0\}$  contains matrices whose column space leaves  $\text{col } B'_0$ , so it does not lie in  $S_1(T_B) = \{B'_0Y\}$ , and symmetrically.

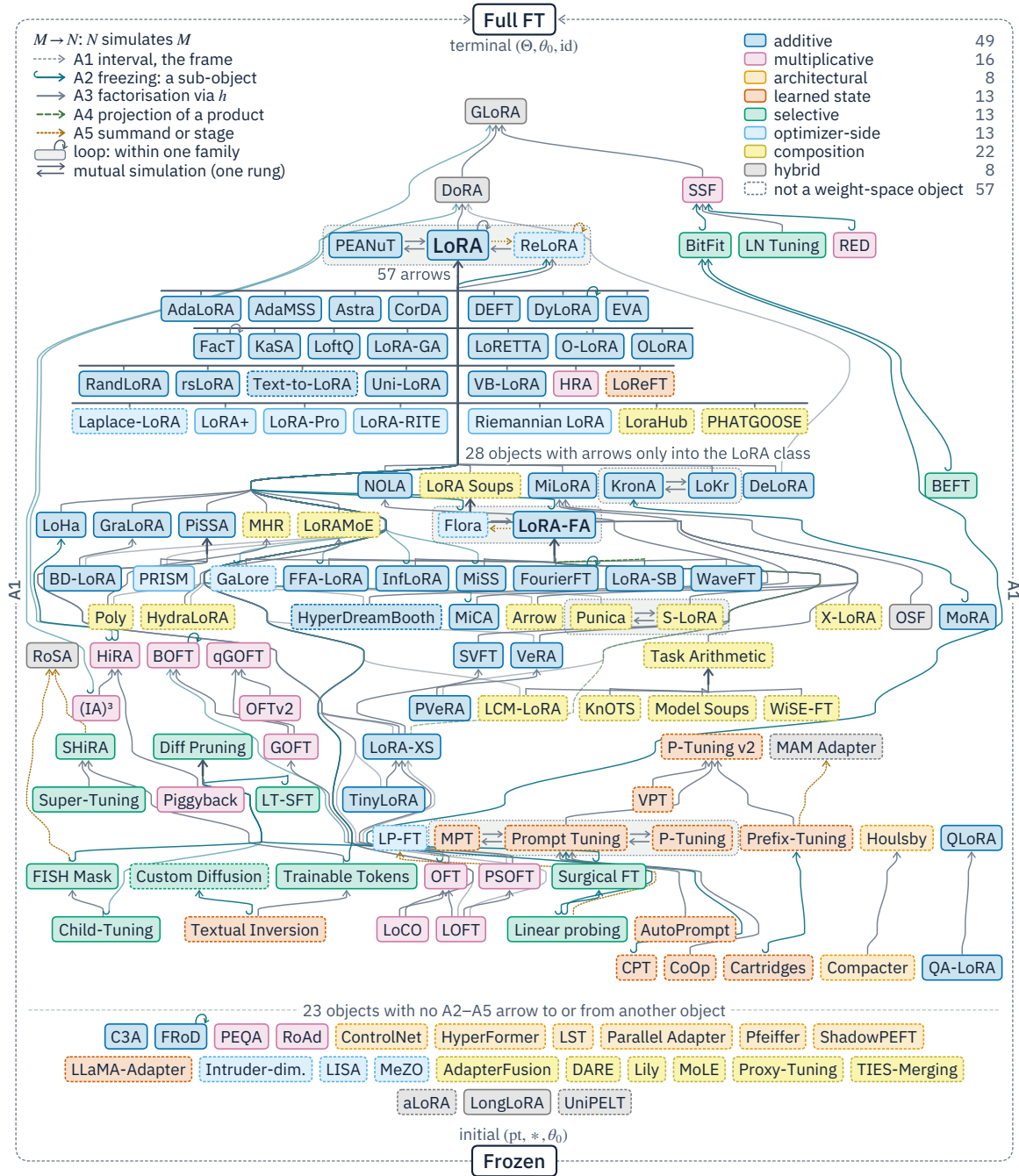
Transport  $T_*$  is an automorphism of the slice with inverse  $(T^{-1})_*$  ([Proposition II.13](#)). It identifies  $M$  and  $T_*M$  as one object in two frames and does not by itself give an arrow between them.

**Which published relations survive.** Some become exact. VeRA is LoRA-FA( $A$ ) with a constrained factor, and rsLoRA and a linear parallel adapter are LoRA. Others are weaker than they sound. HRA's arrow to LoRA <sub>$r$</sub>  lands on a degenerate pointing ( $B_0A_0 = 0$ ) outside the strata of [Theorem II.2](#), and LoRA is a special case of AdaLoRA only unpointed, since their first-order images have dimensions  $mr$  and  $r$  ([Lemma II.4](#)).

### The recorded arrows and obstructions

The atlas records 403 arrows, each with its map  $h$  ([Table 37](#) in [Appendix D](#)): 329 by factorisation (A3), 61 by freezing (A2), 11 by sums (A5) and 2 by products (A4). The interval arrows of A1 hold for every weight-space method and are not listed. The atlas also records 485 obstructions, each naming the test that rules an arrow out ([Table 25](#)). Of the recorded arrows, 161 join two core methods, and [Figure 14](#) draws the category of core methods as a layered Hasse diagram ([Background A.16](#)).

The atlas is meant to grow. A new method enters by answering the seven questions and exhibiting its arrows, each with its map  $h$ ; a disputed comparison is settled by an explicit  $h$  or by one of the three tests.



**Figure 14. The category of core methods.** A layered Hasse diagram (Background A.16), from Frozen (initial, bottom) to Full FT (terminal, top); the dashed outer frame stands for the A1 arrows Frozen  $\rightarrow M \rightarrow$  Full FT of every weight-space method  $M$  (Observation I.4(a)). Chips are the 142 core methods, coloured by modification kind and dashed for the 57 that are not objects of the weight-space slice. An arrow  $M \rightarrow N$  points up and means that  $N$  simulates  $M$  through a pointed smooth map  $h$  with  $\rho_N \circ h = \rho_M$  (Definition I.2). All 161 recorded arrows between two core methods are drawn, coloured by rule (Table 24): 29 by freezing (A2, hooked tail), 122 by factorisation (A3), 2 by products (A4) and 8 by sums or stages (A5). The 7 loops land on another member of a family, as  $\text{LoRA}_r \rightarrow \text{LoRA}_{r+1}$  does, and the 5 classes of mutually simulating families share a rung in a grey box. Arrows into a method with four or more incoming arrows from outside its class end in one trunk (57 into LoRA), and the 28 methods whose arrows all land in the LoRA class form the comb beneath it. The 26 lighter arrows are composites, also reached by a longer path of arrows.

## Rosetta stone

*A half-seen analogy between two theories is the most fertile thing a mathematician can hold.  
One day the hunch turns into certainty, and the twin theories show the source they share.*  
after Weil's Rosetta-stone letter of 1940

This exposé is a dictionary from fine-tuning phrases to the objects of the relative theory, with the result behind each pairing that needs one.

### In brief.

- LoRA+ is  $\alpha$  in disguise. Under Adam with  $\varepsilon = 0$ , no weight decay or clipping and one schedule, LoRA+ with ratio  $\lambda$  trains exactly like LoRA with  $\alpha$  multiplied by  $\lambda$ , given the same data order and dropout masks (Schulman and Thinking Machines Lab, 2025).
- Under Adam without weight decay or full-gradient clipping, GaLore is one-sided LoRA, restarted every period with its optimizer state kept (Torroba-Hennigen et al., 2025). QLoRA is LoRA attached at the quantised weight, off  $\theta_0$  by a generically full-rank error.
- Read from the mathematical side, some constructions are known methods. In the SVD frames of  $W_0$ , when  $\text{rank } W_0 \geq r$ , the product of the two one-sided LoRAs is LoRA-XS, and for injective  $W_0$  and even  $r \leq n - 2$  LoRA's image meets the rotation orbit of  $W_0$  in the image of HRA (without Gram-Schmidt, the HF default).

A fine-tuning paper says that LoRA+ gives  $B$  a larger learning rate than  $A$ . The relative theory says that, under the hypotheses of Theorem III.10, LoRA+ lies on the torus orbit of plain LoRA with a larger  $\alpha$ . Passing between such sentences lets each side borrow the other's theorems. Table 28, at the end of this exposé, is a dictionary for the passage. It lists all 36 entries, each a PEFT phrase, its categorical or geometric counterpart, and a gloss citing the results behind it; the hypotheses live in the cited statements. The sections below read six entries closely, one from each of Exposés I to VI, and then read the dictionary from the right.

### How to read an entry

An entry claims that a phrase names a definite object, on which claims about the phrase can be checked. There are three strengths (Table 26).

**Credit.** The language of parametrised maps and lenses on our right-hand side comes from Fong et al. (2017) and Cruttwell et al. (2021). Inside PEFT, He et al. (2021) wrote an earlier dictionary of this kind, reading adapters, prefix tuning and LoRA as modifications of one hidden state.

The box below fixes each borrowed word's sense. Exposé XI says more (Table 30).

#### ANALOGY (the borrowed words).

- *Noether*: conserved charges from symmetries, proved for continuous-time flows with constant rates (Theorem III.5); Adam and momentum break them.
- *Gauge*: the symmetry group of a parametrisation, with no connection.
- *Erlangen*: a theorem for group-type methods (Theorem II.5), and a name for the shape of cones such as LoRA's image.



**Table 26.** The three strengths of a dictionary entry.

| Strength   | What the entry asserts                                                     | Example                                                                                                                                                                                           |
|------------|----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition | The phrase gets a precise, computable meaning.                             | “Effective number of parameters” is $d(M)$ , the generic rank of $d\rho$ (Definition I.5).                                                                                                        |
| Theorem    | The phrase hides a claim, proved under stated hypotheses.                  | “No initialisation buys rank above $2r$ ” means that on one $m \times n$ box the exact pointings of $\text{LoRA}_r$ together reach $\theta_0 + \mathcal{M}_{\leq \min(2r, m, n)}$ (Theorem II.1). |
| Analogy    | A borrowed word names a pattern and adds nothing to the theorem behind it. | “Noether charge”, “gauge”, “Erlangen”, “moduli space”.                                                                                                                                            |

- *Moduli space*: LoRA initialisations in three full-rank strata, classified only as a set (Theorem II.2).
- *Lens duality*: exact only at degree one, for optimizers that read neither the parameters nor the full gradient (Proposition IV.2).

Most of the right-hand column concerns images, which measure expressivity. Learnability is a separate property, and entries with dynamical claims state their optimizer.

### Six entries, one from each exposé

#### I. “Method $N$ can do whatever $M$ can”.

**EXAMPLE** (entries 3 and 4, Exposé I). PEFT “method  $N$  can do whatever  $M$  can”; the frozen model and full fine-tuning.

CATEGORY THEORY · GEOMETRY a morphism  $h : M \rightarrow N$  of the pointed slice, a pointed smooth map with  $\rho_N \circ h = \rho_M$ ; the initial and terminal objects.

The weak reading asks that  $N$  reach every weight  $M$  reaches. The useful one asks that  $N$  also follow  $M$ ’s parameter paths, through a map  $h$  with  $\rho_N \circ h = \rho_M$ . Methods at  $\theta_0$  with such arrows form the pointed slice **Meth**( $\Theta, \theta_0$ ) (Definition I.2). The frozen model is initial, full fine-tuning is terminal, and the arrow from  $M$  to full fine-tuning is  $\rho_M$  itself (Observation I.4(a)). Image inclusion gives only a map of sets over  $\Theta$ , which need not be smooth (Observation I.4(c)).

**WHAT THIS EXPLAINS** Every weight-space method pointed at  $\theta_0$  lies between the frozen model and full fine-tuning, and in exact arithmetic on a full-precision base its merge is that canonical arrow. Merging questions remain for extensions such as adapters, prefixes and ReFT (reading VI below), for quantised bases (Proposition V.2) and for combinations of adapters (Proposition V.4).

**A first-order test, and defaults that start elsewhere.** The first-order image  $S_1$  detects part of the gap between sets and smooth maps. Take  $\rho_M(t) = t$  and  $\rho_N(u) = u^3$  on  $\mathbb{R}$ , both pointed at 0. The images agree, but  $S_1(M) = \mathbb{R}$  is not inside  $S_1(N) = 0$ , so no smooth simulation  $M \rightarrow N$  exists (Observation I.4(c), Observation I.6). Several initialisations start away from  $\theta_0$  and carry a pointing defect (Exposé V): HF’s default FourierFT spectrum and VB-LoRA bank, `init_lora_weights=False`, LoRA-XS’s small random  $R$ , and quantised bases.

## II. “The effective number of parameters”.

**EXAMPLE** (entries 6 and 11, Exposé II). PEFT expressivity, the effective number of parameters; redundant parameters.

CATEGORY THEORY · GEOMETRY the image germ of  $\rho$  at  $\theta_0$  and its dimension  $d(M)$ ; the gauge group (automorphisms of  $Q$  over  $\Theta$ ) and the generic fibre.

A parameter count measures  $Q$ , while expressivity lives in the image. The difference is spent on the fibres of  $\rho$ , along which the gauge group moves. LoRA spends  $r^2$  parameters there, since  $(Bg^{-1}, gA)$  and  $(B, A)$  give the same weights for every  $g \in \text{GL}_r$ .

With  $|M| = \dim Q$  and  $d(M) = \dim \text{Im } M$  (Definition I.5),

$$d(M) \leq |M| - \dim(\text{generic gauge orbit}), \quad (11)$$

with equality when the gauge is locally transitive on generic fibres (Proposition II.12(a)). For  $\text{LoRA}_r$  on an  $m \times n$  box with  $r \leq \min(m, n)$ ,  $d = r(m + n - r)$ . Compare  $\text{LoHa}_{r,r}$  with  $\text{LoRA}_{2r}$  at their shared budget  $2r(m + n)$ , with  $2r \leq \min(m, n)$ . If  $2r^2 < m + n - 1$ , LoHa reaches at least  $(m + n - 1) - 2r^2$  fewer dimensions; if moreover  $r^2 \leq \min(m, n)$ , it trades them for rank up to  $r^2$  when  $r \geq 3$ , and it lies strictly inside  $\text{LoRA}_{2r}$  when  $r \leq 2$  (Proposition II.12(c)). If  $2r^2 \geq m + n - 1$ , as for  $r = 64$  on  $4096 \times 4096$  or for  $r = 32$  on layers 320 to 640 wide, LoHa matches or beats  $\text{LoRA}_{2r}$  in rank, and in dimension by its numerically checked dimension formula. Rank, in turn, is bounded by a minimum cut of the method’s tensor network (Theorem II.10).

**WHAT THIS EXPLAINS** Report  $d(M)$  beside  $|M|$  in every comparison (Prediction X.10).

**Credit.** LoHa is the Hadamard-product parametrisation of FedPara (Hyeon-Woo et al., 2021). Its dimension bound is proved, and equality is checked numerically (Exposé XI, Appendix E).

## III. “The LoRA+ ratio, $\alpha$ , the init scale, rsLoRA”.

**EXAMPLE** (entry 23, Exposé III). PEFT the LoRA+ ratio,  $\alpha$ , the initial scale, rsLoRA.

CATEGORY THEORY · GEOMETRY coordinates on the orbit space of a hyperparameter torus.

Multiplying  $B$  by  $c > 0$  and dividing the scale  $s$  by  $c$  leaves the weights unchanged, and Adam with  $\varepsilon = 0$  ignores a positive rescaling of a block’s gradient history. Rescaling the block’s learning rate too then reproduces the run step for step, and the per-block factors form a torus acting on the hyperparameters.

Theorem III.10 makes this exact for any multihomogeneous method, assuming no parameter-dependent regulariser, no global clipping, the same data order and dropout masks, and exact arithmetic. For LoRA (Corollary III.11(b)), under Adam with  $\varepsilon = 0$ , no weight decay or clipping and one schedule for both groups, LoRA+ with ratio  $\lambda$  at scale  $s$  produces exactly the trajectory of plain LoRA at scale  $\lambda s$ , with the same  $\eta_A$  and  $A_0$ . For  $\varepsilon > 0$  this needs  $\varepsilon'_B = \lambda \varepsilon$ , and with AdamW a decay  $\lambda \cdot \text{wd}$  on  $B$ . Under SGD the scale is  $\sqrt{\lambda} s$ .

**WHAT THIS EXPLAINS** Only the torus orbit of  $(s, \eta_A, \eta_B, \sigma_A)$  matters. Under Adam it is fixed by  $s\eta_A\eta_B$  and  $\sigma_A/\eta_A$  (Corollary III.11(a)), so at a fixed rank LoRA+, rsLoRA and  $\alpha$ -tuning are one family. A gain of LoRA+ over LoRA with  $\alpha$  multiplied by  $\lambda$  needs something that breaks the torus, such as global-norm clipping (a Hugging Face Trainer default), an  $\varepsilon$  or decay shared across groups, or different schedules (Prediction X.1). Figure 8 runs both trainings.

**Credit.** The LoRA and Adam case, with its  $\varepsilon > 0$  extension and the two effective hyperparameters, is due to Schulman and Thinking Machines Lab (2025), for the method of Hayou

et al. (2024b). The torus for any multihomogeneous method and the  $c^2$  law for SGD are added here.

**The same fibres under gradient flow.** Entry 22 reads the same fibres under gradient flow. With block-scalar rates the gauge yields the conserved charge

$$\Phi = \frac{B^\top B}{\eta_B} - \frac{AA^\top}{\eta_A} \quad (12)$$

(Theorem III.5), after Du et al. (2018), Arora et al. (2018), Zhao et al. (2022b) and Marcotte et al. (2023). Balanced points  $B^\top B = AA^\top$  form the Kempf–Ness slice (Proposition III.7; Kempf and Ness, 1979; Richardson and Slodowy, 1990).

#### IV. “GaLore versus LoRA”.

**EXAMPLE** (entry 25, Exposé IV). PEFT GaLore versus LoRA.

CATEGORY THEORY · GEOMETRY cotangent-side and tangent-side lens 2-cells, which coincide for linear charts (the degree-one collapse).

Backpropagation makes each map a lens, running forward from parameters to outputs and backward from output gradients to parameter gradients. LoRA restricts the forward side. GaLore compresses the gradient before the optimizer and expands its output after.

Let  $\rho_k(C) = W_k + \mathcal{L}C$  with  $\mathcal{L}$  linear and  $C_0 = 0$ . Let the optimizer’s state and output depend only on the gradients fed to it, with one implementation on both sides (SGD, heavy-ball or Nesterov momentum, Adam without weight decay, Adafactor without parameter scaling). Then training  $C$  and merging at the end of a period gives the same  $W$ -trajectory as the backward method with compression  $\mathcal{L}^*$  and anchor  $\mathcal{L}$ , started at  $W_k$  from the same optimizer state (Proposition IV.2). Decoupled weight decay, LAMB and LARS trust ratios, parameter-scaled Adafactor and full-gradient clipping break the identity.

**WHAT THIS EXPLAINS** Under these hypotheses, GaLore with a projector  $P_k$  fixed on each period is one-sided LoRA  $W_k + P_k C$ , merged and restarted every period with  $C$ ’s optimizer state carried over unchanged. It adds two things to one-sided LoRA, a frame from the gradient’s SVD and that carry-over, which reads moments written for  $P_{k-1}$  as if written for  $P_k$ . LDAdam (Robert et al., 2025) transports them instead.

**Credit.** The identity for GaLore (Zhao et al., 2024a) is due to Torroba-Hennigen et al. (2025), and its SGD case with a frozen random factor to Hao et al. (2024).

**The rest of the cotangent family.** Under the same hypotheses the identity (Proposition IV.2, Proposition IV.3) covers LoRA-FA and GaLore with random projectors, MeZO up to its finite-difference estimate, GaLore with gradient SVDs, and BitFit and LISA (without weight decay) with coordinate blocks. Flora as published, APOLLO and Fira update outside their projected subspace and are not covered.

#### V. “QLoRA and LoftQ”.

**EXAMPLE** (entry 28, Exposé V). PEFT QLoRA, LoftQ.

CATEGORY THEORY · GEOMETRY base change along a map that does not preserve the function; a pointing defect; an approximate section.

A quantiser  $\kappa$  moves the pretrained point to  $\kappa\theta_0$  and changes the function. Additive methods commute with translation,  $M(\theta + \nu) = \tau_\nu \circ M(\theta)$ , so QLoRA (Detrmers et al., 2023) is LoRA attached at  $\kappa\theta_0$ , with pointing defect  $e = \kappa\theta_0 - \theta_0$  (Definition V.1), and  $\theta_0$  lies outside

**Table 27.** Three orders of quantising and splitting, and the pointing defect  $\rho(q_0) - W$  each leaves (Proposition V.2(b)).

| Order                             | Used by                                                          | Pointing defect                                                                                                          |
|-----------------------------------|------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| split, then quantise the residual | QPiSSA; HF's PiSSA and CorDA workflow                            | $\kappa(W_{\text{res}}) - W_{\text{res}}$                                                                                |
| quantise, then fit the error      | LoftQ with one iteration (HF default <code>loftq_iter=1</code> ) | $e - s \text{SVD}_r(e)$ , as HF does not divide the fit by $s$ ; at $s = 1$ , $e$ minus its best rank- $r$ approximation |
| quantise, split, re-quantise      | HF <code>olora_init</code> on <code>bitsandbytes</code> weights  | $e + \kappa(r') - r'$ , with $r' = \kappa W - \delta_0$                                                                  |

its image unless  $\text{rank } e \leq r$  (Proposition V.2(a)). Splitting initialisations come in three orders with three different defects (Proposition V.2(b); Table 27). Write  $W$  for the full-precision weight,  $\delta_0$  for the initial adapter,  $W_{\text{res}} = W - \delta_0$  and  $s = \alpha/r$ .

**WHAT THIS EXPLAINS** Measured from its own start, QLoRA obeys the rank bound of any LoRA (Theorem II.1(c)); measured from  $\theta_0$ , its update also contains the generically full-rank defect. Since  $e - \text{SVD}_r(e)$  is orthogonal to  $\text{SVD}_r(e)$ , one-step LoftQ in HF beats QLoRA in Frobenius norm only for  $0 < s < 2$ . Further LoftQ iterations need not decrease their objective, and at  $s = 1$  the output is an approximate section, pointed up to its attained residual (Proposition V.2(c)).

**Credit.** QLoRA is due to Dettmers et al. (2023), LoftQ to Li et al. (2023b), and PiSSA and QPiSSA to Meng et al. (2024a).

## VI. “Merging an adapter into the weights”.

**EXAMPLE** (entry 5, Exposé VI). PEFT merging an adapter into the weights; zero inference latency.

CATEGORY THEORY · GEOMETRY for weight-space methods, the canonical arrow to the terminal object; for extensions, a lift along the realisation  $\Phi_f$ , certified box by box by rewriting.

Reading I settled merging for weight-space methods on a full-precision base. Adapters, prefixes and (IA)<sup>3</sup> scalings insert new boxes and live on an extension of  $f$ , so merging one is a lifting problem along the realisation  $\theta \mapsto f(\theta, -)$ .

A method  $\rho$  on an extension  $\tilde{f}$  is mergeable if some  $\mu : Q \rightarrow \Theta$  has  $\Phi_{\tilde{f}} \circ \rho = \Phi_f \circ \mu$  (Definition VI.1). Theorem VI.3 gives sufficient conditions box by box, through seven rewrite rules with side conditions such as position-uniform edits, unshared absorbing weights and bias slots for shifts. For instance, HF applies the feed-forward vector  $\ell$  of (IA)<sup>3</sup> (Liu et al., 2022a) to the input of  $W_{\text{down}}$ , where it always fuses. It fuses into  $W_{\text{up}}$  instead under SwiGLU or GeGLU for every  $\ell$ , and under ReLU when  $\ell \geq 0$ . Under GELU, an entry  $\ell_i$  on a nonzero row of  $W_{\text{up}}$  fuses there only if it is 0 or 1.

**WHAT THIS EXPLAINS** “Zero inference latency” is certified by local rewriting, a sufficient test that also names the absorbing weight; the  $W_{\text{up}}$  route is idle in practice, as  $W_{\text{down}}$  always works. Nonlinear adapters, input-routed mixtures and prefixes do not merge box-locally. At its layer a prefix is a gated parallel adapter (Theorem VI.5(b)), and it is not box-locally mergeable for generic frozen weights and prefixes (Theorem VI.5(f)); whole-network merging of prefixes is open.

**Credit.** The  $(\text{IA})^3$  merge is noted by Liu et al. (2022a), and the gated-adapter reading of prefix tuning is due to He et al. (2021). Scale migration through SwiGLU, ReLU and norm layers, with the GELU obstruction, already appears in SmoothQuant (Xiao et al., 2023) and AWQ (Lin et al., 2024a). The activation symmetries behind slides R2 and R3 (Lemma VI.2) are classical (Chen et al., 1993; Godfrey et al., 2022; Sussmann, 1992).

### Reading from the right

Some mathematical constructions are methods already in use.

- The product  $\text{FA}(A_0) \times \text{FB}(B_0)$  of the two one-sided LoRAs is the sandwich  $R \mapsto W_0 + B_0 R A_0$ . In the SVD frames of  $W_0$ , with  $\sigma_r > 0$ , it is LoRA-XS pointed at  $R = 0$  (Proposition I.8(a); Bałazy et al., 2024).
- For injective  $W_0$  and even  $r$  with  $2 \leq r \leq n - 2$ , LoRA's image meets the rotation orbit  $W_0 \cdot \text{SO}(n)$  exactly in the image of HRA without Gram–Schmidt, the HF default (Theorem II.7; Yuan et al., 2024). This is a meet of sets, and HRA is not the categorical product.
- Transport along a fixed  $T \in \text{GL}(\Theta)$  is an automorphism of the slice. It carries coordinate masks to FourierFT (Gao et al., 2024) (with no conjugate frequency pair sampled), WaveFT (Bilican et al., 2025) (when its padded inverse wavelet transform is invertible) and C3A (Chen et al., 2024a), each started at zero, and to LoRA-XS and SVFT (Lingam et al., 2024) in the SVD frame of  $W_0$ , which depends on  $\theta_0$ . The inverse of the rearrangement of Van Loan and Pitsianis (1993) carries  $\text{LoRA}_k$  to unconstrained sums of  $k$  Kronecker products (Proposition II.13).
- The monoidal sum  $\oplus$  adds updates, so its image is a Minkowski sum. RoSA (Nikdan et al., 2024), GraLoRA (Jung et al., 2025) and the stages of ReLoRA (Lialin et al., 2023b), unrolled in time, are words in  $\oplus$  (Observation I.7).

**WHAT THIS EXPLAINS** Products and meets can be tried on other pairs too. Meets of images always exist as sets of weights, while products can fail to exist (Observation I.4(b)). Which meets admit a polynomial chart, as LoRA-XS and HRA's image do, is an open problem (§XI, problem 4).

### The full dictionary

Table 28 gives every entry. Result numbers in the glosses point to the statements whose hypotheses the entry inherits; entries without a number are definitions or bookkeeping.

**Table 28.** Rosetta stone: the full dictionary from PEFT phrases to their categorical or geometric counterparts. The gloss names the results behind each pairing; their hypotheses are part of the entry.

| # | PEFT                  | Categorical / geometric                                                                                                                                                                      | Gloss                                                                                                                                                                                         |
|---|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | pretrained checkpoint | base point $\theta_0 : 1 \rightarrow \Theta$ ; a pointed 1-cell $(\Theta, \theta_0, f)$ of <b>Para</b> ( <b>Diff</b> )                                                                       | Fine-tuning moves a point; everything is measured relative to it (Def. I.1).                                                                                                                  |
| 2 | PEFT method           | object of the pointed slice $\mathbf{Meth}(\Theta, \theta_0) = \mathbf{Diff}_*/(\Theta, \theta_0)$ , equivalently the opposite of the pointed coslice of reparametrisation 2-cells under $F$ | A method is a pointed map $\rho : (Q, q_0) \rightarrow (\Theta, \theta_0)$ . The slice depends only on $(\Theta, \theta_0)$ ; <b>Para</b> is bookkeeping for weight-space methods (Rem. I.3). |

*continued on the next page*



Table 28, continued

| #  | PEFT                                          | Categorical / geometric                                                                                                                           | Gloss                                                                                                                                                                                                                                                                                                                                                                                       |
|----|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3  | “method $N$ can do whatever $M$ can”          | morphism (simulation)<br>$h : M \rightarrow N$ with $\rho_N \circ h = \rho_M$                                                                     | Smooth simulations also carry parameter paths; at the level of sets they reduce to image inclusion (Obs. I.4).                                                                                                                                                                                                                                                                              |
| 4  | frozen model / full fine-tuning               | initial / terminal object                                                                                                                         | Every weight-space method lies in the interval [frozen, full FT]; adapters, prefixes and ReFT are extensions, not objects of <b>Meth</b> (Obs. I.4).                                                                                                                                                                                                                                        |
| 5  | merging an adapter into the weights           | the canonical morphism to the terminal object (it is $\rho$ itself); for extensions, a lift along the realisation $\Phi_f$                        | For weight-space methods merging is definitional; for extensions zero inference overhead is a lifting problem, certified box-locally by rewriting (sufficient conditions and box-local obstructions) under side conditions such as position-uniformity, no weight sharing and bias slots (Thm VI.3).                                                                                        |
| 6  | expressivity / effective number of parameters | image germ of $\rho$ at $\theta_0$ ; its dimension $d(M)$ , the generic rank of $d\rho$                                                           | Rank is a cut invariant; dimension is a volume invariant (Thm II.10, Prop. II.12).                                                                                                                                                                                                                                                                                                          |
| 7  | lazy / NTK-regime capability                  | first-order image<br>$S_1 = d\rho_{q_0}(T_{q_0}Q)$ , a functorial isomorphism invariant                                                           | The lazy kernel has range $J_f(S_1)$ ; $S_1$ separates zero-init pointings ( $T_A$ vs $T_B$ , the row space of $A_0$ ) that the image cannot, while the image already separates split inits (Obs. I.6).                                                                                                                                                                                     |
| 8  | zero-init vs PiSSA-type init                  | apex (singular vertex) vs smooth point of the image; Type I (new $q_0$ in the apex fibre) vs Type II (new $\rho$ by splitting)                    | Zero- $B$ init loses $r(n-r)$ first-order directions (zero- $A$ init $r(m-r)$ ); split inits are different objects, incomparable with LoRA (Thm II.2, Prop. III.8).                                                                                                                                                                                                                         |
| 9  | the zoo of LoRA initialisations               | points of three distinguished strata of full-rank pointings, classified as a set by $\text{Gr}(r, n) \sqcup \text{Gr}(r, m) \sqcup \mathcal{M}_r$ | Vanilla LoRA on linear layers, EVA, Init[B], HF’s default LoRA on embedding layers, PiSSA, MiLoRA, OLoRA, CorDA, LoRA-GA and Astra are points; LoRA-FA and MiCA are frozen-factor sub-objects of the zero- $B$ and zero- $A$ strata; HF’s orthogonal init lies outside the three strata, and LoftQ/QLoRA are defect pointings. The optimizer sees a finer quotient (Thm II.2, Cor. III.13). |
| 10 | no init buys rank above $2r$                  | union over pointings,<br>$\bigcup_q \text{Im } M_q = \theta_0 + (C - C) = \theta_0 + \mathcal{M}_{\leq \min(2r, m, n)}$                           | An exact initialisation picks a translate of the shape; from any starting point the update has rank at most $2r$ . This is the known rank- $2r$ PiSSA-to-LoRA conversion (Thm II.1).                                                                                                                                                                                                        |
| 11 | redundant parameters                          | gauge group, the automorphisms of $Q$ over $\Theta$ ; generic fibre                                                                               | LoRA wastes $r^2$ parameters; LoHa wastes at least $r_1^2 + r_2^2 + m + n - 1$ below the $mn$ cap (exactly that many, numerically), and more where the cap binds (Prop. II.12).                                                                                                                                                                                                             |
| 12 | high-rank PEFT (LoHa, LoKr, HiRA, MoRA, C3A)  | tensor networks whose min-cut avoids the bond; transport of structure along $\text{GL}(\Theta)$                                                   | High rank is not $\text{GL}(\Theta)$ -invariant (though rank is $\text{GL}_m \times \text{GL}_n$ -invariant) and need not mean large volume; a large min-cut is necessary for high rank and is not sufficient (Thms II.10–II.11, Prop. II.13).                                                                                                                                              |

continued on the next page

Table 28, continued

| #  | PEFT                                                           | Categorical / geometric                                                                                                                                                                                                                                                     | Gloss                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----|----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 13 | Kronecker adapters (unconstrained Kronecker sums, KronA, LoKr) | LoRA across a different cut of the same 4-leg tensor (Van Loan–Pitsianis rearrangement, a coordinate permutation)                                                                                                                                                           | For aligned splits (equal square splits, and every split HF PEFT’s factorisation produces), a generic rank-one update has maximal Kronecker rank, so unconstrained Kronecker sums with fewer terms and zero-init LoRA, below the rank of a full-rank Kronecker product are incomparable. LoKr with inner rank $r' \leq \min(m_2, n_2)$ sits inside zero-init LoRA, exactly when $r \geq \min(m_1, n_1) r'$ ; Compacter’s PHM weights live in a nonlinear adapter (Thm II.11). |
| 14 | “new method = old method”                                      | transport $T_*$ along $T \in \mathrm{GL}(\Theta)$ : an automorphism of the category, not an arrow inside it                                                                                                                                                                 | FourierFT (when no conjugate frequency pair is sampled), WaveFT, C3A, LoRA-XS, SVFT and SHiRA are coordinate masks in transported frames, and HiRA (for $W_0$ without zero entries) is LoRA in a transported frame; for LoRA-XS, SVFT and HiRA the frame depends on $\theta_0$ (Prop. II.13).                                                                                                                                                                                 |
| 15 | orthogonal fine-tuning preserves knowledge                     | the image lies in a fibre of the complete invariant $W \mapsto WW^\top$ (first fundamental theorem for $O(n)$ )                                                                                                                                                             | In exact arithmetic, exactly orthogonal input-side methods (OFT with an exact Cayley map, strict GOFT without its optional scaler, HRA) can never change the spectrum or neuron angles, however often they are merged; HF’s default Cayley–Neumann OFT is only approximately orthogonal (Cor. II.6).                                                                                                                                                                          |
| 16 | HRA bridges LoRA and OFT                                       | image-level meet with the rotation orbit, $(W_0 + \mathcal{M}_{\leq r}) \cap W_0 \mathrm{SO}(n)$ , for injective $W_0$ and even $r \leq n - 2$ ; not a product of methods                                                                                                   | LoRA meets exact-Cayley OFT in a strictly smaller set, and HF’s default Cayley–Neumann OFT, which is not orthogonal, is incomparable with HRA. At HRA’s paired identity init the image germ is the skew-determinantal cone, so that init is an apex (Thm II.7).                                                                                                                                                                                                               |
| 17 | LoRA-XS                                                        | categorical product $\mathrm{FA}(A_0) \times \mathrm{FB}(B_0)$ of the two one-sided LoRAs, in the SVD frames of $W_0$ (with rank $W_0 \geq r$ )                                                                                                                             | The overlap that a split init double-counts (Prop. I.8, Prop. II.3).                                                                                                                                                                                                                                                                                                                                                                                                          |
| 18 | DoRA’s magnitude/direction decoupling                          | diagonal saturation $\mathrm{Diag}_m \cdot (W_0 + \mathcal{M}_{\leq r})$ , the image of LoRA followed by a trainable output scale                                                                                                                                           | Equal images, for $W_0$ without zero rows. Under gradient flow the exact gradient changes the metric and the detached backward is a non-gradient update; under AdamW the magnitude coordinates matter too (Prop. II.8).                                                                                                                                                                                                                                                       |
| 19 | (IA) <sup>3</sup> , SSF, LN tuning, RoAd, HiRA                 | the scalings and HiRA are rungs of the Hadamard-torus ladder $W_0 \odot (J + Z)$ ; RoAd <sub>1</sub> is the complex torus $(\mathbb{C}^\times)^{m/2}$ , and RoAd <sub>2</sub> and RoAd <sub>4</sub> apply arbitrary $2 \times 2$ blocks (group part block-GL <sub>2</sub> ) | On every rung zeros of $W_0$ stay zero. RoAd can fill them, so it is a rung only for zero-free $W_0$ , and for generic $W_0$ it lies on no low-rank rung. Covariance $\mathrm{Mon}_m \times \mathrm{GL}_n$ on output-scaled boxes, $\mathrm{GL}_m \times \mathrm{Mon}_n$ on down_proj, exactly $\mathrm{Mon}_m \times \mathrm{Mon}_n$ for HiRA, never $\mathrm{GL}_m \times \mathrm{GL}_n$ (Prop. II.9).                                                                      |

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Table 28, continued

| #  | PEFT                                                                                           | Categorical / geometric                                                                                                                     | Gloss                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 20 | “the geometry a method lives in”                                                               | covariance group $\mathcal{G}_M$ (Klein): the largest group of pairs $(g, h)$ with $\text{Im } M(gW_0h^{-1}) = g \text{ Im } M(W_0) h^{-1}$ | Descent depends on whether each induced pair $(g, h)$ lies in $\mathcal{G}_M$ , a question of side for one-sided methods and of group for coordinate-structured ones. LoRA and (IA) <sup>3</sup> (HF defaults) provably keep their reachable function sets under a global residual rotation of a gain-fused pre-norm model; block OFT, LoHa ( $r \geq 2$ ), HiRA and masks fail the sufficient condition ( <a href="#">Obs. II.15</a> , <a href="#">Conj. A</a> ). |
| 21 | how a method trains                                                                            | induced cometric $K = D\rho \wedge D\rho^\top$ (pulled-back gradient of the lens)                                                           | Two isomorphic objects can train differently; gradient descent is natural exactly where cometrics agree, which for LoRA’s linear gauge means $\mathcal{O}(r)$ ( <a href="#">Thm III.3</a> ).                                                                                                                                                                                                                                                                       |
| 22 | balancedness<br>$B^\top B = AA^\top$                                                           | Noether charge of the non-compact half $\text{Sym}_r$ of $\mathfrak{gl}_r$ ; the Kempf–Ness slice of the gauge action                       | Conserved under gradient flow with constant rates for any cotangent signal of the form $\wedge D\rho^\top \gamma$ (DoRA’s detached norm, clipping), after <a href="#">Zhao et al. (2022b)</a> ( <a href="#">Thm III.5</a> , <a href="#">Prop. III.7</a> ).                                                                                                                                                                                                         |
| 23 | LoRA+ ratio, $\alpha$ , init scale, rsLoRA                                                     | coordinates on the orbit space of a hyperparameter torus                                                                                    | Under Adam with $\varepsilon = 0$ , no weight decay, no clipping and a shared schedule, LoRA+ with ratio $\lambda$ is exactly LoRA with $\alpha$ times $\lambda$ (the LoRA case is <a href="#">Schulman and Thinking Machines Lab, 2025</a> ); at a fixed rank, rsLoRA is isomorphic to LoRA ( <a href="#">Thm III.10</a> , <a href="#">Cor. III.11</a> ).                                                                                                         |
| 24 | transformation-invariant optimizer (Riemannian preconditioning, ScaledGD, LoRA-Pro, LoRA-RITE) | update rule equivariant under the whole gauge $\text{GL}_r$ , so it descends to the quotient $\mathcal{M}_r$                                | On the full-rank locus, undamped scaled GD and LoRA-RITE are $\text{GL}_r$ -equivariant in the parameters; damped variants and LoRA-Pro’s step are only $\mathcal{O}(r)$ -equivariant, and Adam only $B_r$ -equivariant ( <a href="#">Prop. III.4</a> , <a href="#">Thm III.12</a> ).                                                                                                                                                                              |
| 25 | GaLore vs LoRA                                                                                 | cotangent-side vs tangent-side lens 2-cells, which coincide for linear charts (the degree-one collapse)                                     | Without weight decay or full-gradient clipping, GaLore is one-sided LoRA with a moving frame, with optimizer state carried across re-basings rather than transported ( <a href="#">Prop. IV.2</a> , credited to <a href="#">Torroba-Hennigen et al., 2025</a> ).                                                                                                                                                                                                   |
| 26 | when a gradient-projection method is really a reparametrisation                                | Frobenius integrability of the anchor distribution                                                                                          | GaLore, LISA and MeZO keep a $\theta$ -independent anchor within each period, hence affine leaves, and under the hypotheses of <a href="#">Prop. IV.2</a> are rebasing schemes of linear forward methods; Frobenius integrability matters only for $\theta$ -dependent anchors such as full-batch GaLore with $T = 1$ ( <a href="#">Prop. IV.3</a> ).                                                                                                              |

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Table 28, continued

| #  | PEFT                                                | Categorical / geometric                                                                                                                                                                              | Gloss                                                                                                                                                                                                                                                                                                                                    |
|----|-----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 27 | activation memory                                   | backward reach of the composite lens (the backprop cone)                                                                                                                                             | An upper bound: LST avoids the backbone's backward pass; LoRA-FA stores $A_0x$ ; LISA's cone is the whole network because its embedding is always trained, though its frozen layers need not store their linear-input residuals; MeZO has no reverse pass (Obs. IV.4).                                                                   |
| 28 | QLoRA / LoftQ                                       | base change along a map that does not preserve the function; pointing defect; approximate section                                                                                                    | Split-then-quantise, quantise-then-fit (LoftQ's default, whose HF init ignores the LoRA scale) and quantise-split-requantise leave three different defects (Prop. V.2).                                                                                                                                                                  |
| 29 | ReLoRA, re-OFT, iterated BOFT/HRA                   | the monoid (or Lie subgroup) generated by the reachable displacements                                                                                                                                | ReLoRA reaches exactly $W_0 + \mathcal{M}_{\leq \min(Kr, m, n)}$ ; exact-Cayley block OFT is stuck; iterated BOFT reaches the orbit $W_0 \text{SO}(n)$ , up to output scaling, only at full butterfly depth; re-HRA reaches $(W_0 + \mathcal{M}_{\leq Kr}) \cap W_0 \text{SO}(n)$ (Thm V.3).                                             |
| 30 | task arithmetic vs TIES/DARE                        | $\text{GL}(\Theta)$ -natural vs coordinatewise operations (TIES with sign election by total mass: homotheties times signed permutations; DARE: the monomial group, its permutations in distribution) | Coordinate merges depend on the basis; factor-space merges depend on the gauge (Prop. V.4).                                                                                                                                                                                                                                              |
| 31 | LoraHub / Poly factor mixing                        | a rule on factors that is not invariant under $\prod \text{GL}_r$ , hence not a function of the adapters                                                                                             | Post-hoc LoraHub (and HF's linear combination) needs a joint gauge fixing that produces a shared factor (KnOTS) or a shared frozen factor (FFA-LoRA); Poly/MHR are trained jointly and, for fixed generic routing with fewer skills than tasks, have only the diagonal $\text{GL}_r$ symmetry, which is checked numerically (Prop. V.4). |
| 32 | adapters (Houlsby, parallel), LLaMA-Adapter, LoReFT | pointed extensions: function-preserving embeddings with a neutral point                                                                                                                              | LoReFT is LoRA on an inserted affine identity box (bias confined to col $B$ ), applied at selected positions; pyreft's default init is not neutral (Prop. VI.4).                                                                                                                                                                         |
| 33 | prompt / prefix tuning                              | unpointed extensions on the sequence wire                                                                                                                                                            | Prefix = gated parallel adapter (attention = monoid homomorphism followed by a perspective map); content-attention ratios are rigid only within the layer where the prefix acts; prompt tuning maps into deep prefix tuning in causal decoders with matching position offsets (Thm VI.5).                                                |
| 34 | privileged neuron basis                             | automorphism group of the elementwise activation (the $\sigma$ -spider)                                                                                                                              | For ungated FFNs: permutations for GELU/SiLU, signed permutations for tanh, positive monomials for ReLU; gated FFNs (SwiGLU, GeGLU) have the full monomial group (Lemma VI.2).                                                                                                                                                           |

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Table 28, continued

| #  | PEFT                                                    | Categorical / geometric                                                                                                  | Gloss                                                                                                                                                                                                                 |
|----|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 35 | RoSA, GraLoRA, HF<br>cat merges,<br>ReLoRA stages       | words in the convolution<br>monoidal product $\oplus$ of additive<br>methods (Minkowski sum of<br>images)                | $\oplus$ is not the categorical product, whose<br>image is the intersection when it exists.<br>LoRA-soup CAT is a $\oplus$ of frozen line methods<br>(a span), and UniPELT/MAM are not $\oplus$ -words<br>(Obs. I.7). |
| 36 | target_modules /<br>applying a method<br>to every layer | assigning a 2-cell to each<br>generator box and composing<br>horizontally; $T$ -points for<br>routing, tasks and tenants | Non-local methods (VB-LoRA, Compacter,<br>FacT, Uni-LoRA) share parameters across<br>boxes and are not generator-wise.                                                                                                |



## Falsifiable predictions

*Say what you proved, say what you only named,  
and say what would show you wrong.*

a working maxim

This exposé states eleven experiments that could show the framework wrong, each for a named optimizer.

### In brief.

- A fair LoRA+ baseline multiplies  $\alpha$  by the ratio  $\lambda$ . Under Adam, with  $\varepsilon$ , decay and per-block clipping thresholds rescaled and one shared schedule, the two runs give identical weights (after [Schulman and Thinking Machines Lab, 2025](#)).
- LoRA's charge  $\Phi = B^\top B / \eta_B - A A^\top / \eta_A$  is constant under gradient flow and, when a step size  $\eta$  multiplies fixed block rates, moves by  $O(\eta^2)$  per SGD step, a cheap diagnostic of what the optimizer adds.
- Some claims need no training run. Generic rank-one targets separate LoRA from Kronecker adapters, exact block OFT re-merged on a fixed partition never leaves  $W_0 \cdot \Pi \text{SO}(b)$ , and LoraHub's merge depends on gauge choices unless the modules share a factor.

### Why the theory should risk something

A framework that only re-describes known methods cannot be wrong, and so it earns little trust. The claims below say what a training run will do. Each prediction carries the optimizer scope of the result it rests on; a claim about images needs none, because no run can leave the image. Table 29 lists them. Each keeps the number it carries on the companion site, and the subsections group them by subject (optimizers, images, symmetry, base change and merging), so the numbers do not run in order.

Read the kind column as a warning label. *Theorem* means the sharp half cannot fail inside its hypotheses. *Theorem + data* pairs a proved mechanism with an empirical claim about its size. *Conjecture* means we expect it and have not proved it. The toys check mechanisms in float64 on small matrices, and the experiments on real models are still to be run. Each prediction below lists its scope, the results that support it, a test, the outcome that would refute it, and the toy or figure that shows its mechanism.

### Predictions about optimizers

LoRA has four knobs: the scale  $s = \alpha/r$ , the rates  $\eta_A, \eta_B$  and the initial size  $\sigma_A$  of  $A$ . A two-dimensional torus of rescalings moves all four without changing any weight. LoRA+ turns one knob, the ratio  $\lambda = \eta_B/\eta_A$ , and under Adam that turn lands on the orbit of plain LoRA with a larger  $\alpha$ .

**PREDICTION X.1** (LoRA+ against a re-tuned  $\alpha$ ). Any gain of LoRA+ with ratio  $\lambda$  over plain LoRA with  $\alpha$  multiplied by  $\lambda$  (same  $\eta_A$ , same  $A_0$ ) comes from the ingredients that break the extended torus: global-norm clipping, an  $\varepsilon$  or weight decay shared by the two parameter groups, and schedules that differ between the groups. With all of them off, or rescaled per block, the two runs produce the same weights at every step in exact arithmetic.

**Table 29.** The eleven predictions, the optimizer scope each inherits, the results it rests on, and its kind.

| No.  | Prediction                                   | Optimizer scope                                    | Rests on                | Kind           |
|------|----------------------------------------------|----------------------------------------------------|-------------------------|----------------|
| X.1  | LoRA+ against $\alpha$ re-tuned by $\lambda$ | Adam, AdamW                                        | Thm III.10, Cor. III.11 | theorem        |
| X.2  | DoRA against LoRA + output gain              | AdamW, shared by all arms                          | Prop. II.8, Conj. D     | conjecture     |
| X.3  | Hidden-unit permutations                     | coordinatewise (SGD, Adam)                         | Lemma VI.2, Obs. II.15  | theorem + data |
| X.4  | Residual-stream rotations                    | reach: any; outcomes: depend on optimizer and init | Obs. II.15, Conj. A     | conjecture     |
| X.5  | Rank-one targets                             | any (image claim)                                  | Thm II.11               | theorem        |
| X.6  | Re-based OFT                                 | any (reach); exact Cayley map                      | Thm V.3                 | theorem        |
| X.7  | The crossover scale $\sigma^*$               | gradient flow, SGD                                 | Thm III.9               | theorem + data |
| X.8  | Drift of the charge $\Phi$                   | gradient flow, SGD, Adam                           | Thm III.5               | theorem + data |
| X.9  | LoraHub with shared or independent seeds     | post hoc; regime set by training                   | Prop. V.4               | theorem + data |
| X.10 | Rankings at equal effective dimension        | any (bookkeeping)                                  | Prop. II.12             | theorem + data |
| X.11 | Whitened data-aware splits                   | split: any; training is not covariant              | Prop. II.16, Conj. C    | conjecture     |

**SCOPE** Adam and AdamW ( $\beta_1 \in [0, 1)$ ,  $\beta_2 \in (0, 1)$ , bias correction allowed); linear LoRA modules from  $B_0 = 0$ ; no regulariser on the factors; common data order and dropout masks. A shared schedule, warm-up included, preserves the torus. HF PEFT’s LoRA+ optimizer treats embeddings and 1-D parameters separately, so only linear modules are covered.

**SUPPORT** Theorem III.10 and Corollary III.11(b), both verified. For  $\varepsilon > 0$  the LoRA run needs  $\varepsilon_B = \lambda\varepsilon$  on  $B$ , and with AdamW a decay  $\lambda \cdot \text{wd}$  on  $B$ . Under SGD the matching scale is  $\sqrt{\lambda} s$ .

**TEST** Train LoRA+ with  $\lambda = 16$  against LoRA with  $\alpha$  multiplied by 16,  $\varepsilon$  on  $A$  and  $16\varepsilon$  on  $B$ , no decay and no clipping; the weights agree bit for bit, since 16 is a power of two. Then switch on one torus breaker at a time (shared  $\varepsilon$ , shared AdamW decay, `max_grad_norm=1.0`, group-specific schedules) and record the gap each opens.

**WOULD FAIL IF** a gap beyond rounding appeared with every hypothesis met. A standard HF run clips by default and its LoRA+ optimizer shares  $\varepsilon$  and decay, so there the experiment measures how much of the gain each breaker carries.

**TOY** Figure 8 runs LoRA+ with ratio  $\lambda$  against LoRA with  $\alpha \cdot \lambda$  under Adam with  $\varepsilon = 0$ . The loss curves coincide and the weight gap is exactly zero, and the gap jumps when a shared  $\varepsilon > 0$ , decoupled weight decay or global-norm clipping is switched on.

**REMARK** (why the test avoids  $\varepsilon = 0$ ). Literal  $\varepsilon = 0$  needs the convention  $0/0 = 0$ . At  $B_0 = 0$  the gradient of  $A$  vanishes, and a stock Adam divides zero by zero there.

**Credit.** The LoRA and Adam case, its extension to  $\varepsilon > 0$  and the two effective hyperparameters are due to Schulman and Thinking Machines Lab (2025), for the method of Hayou et al.

(2024b). The torus for any multihomogeneous method, and the  $c^2$  law for SGD, are added here.

Under gradient flow LoRA conserves the charge  $\Phi = B^\top B / \eta_B - A A^\top / \eta_A$ , and when  $\Phi$  is isotropic the dynamics have a closed form. Small singular values of the update grow through  $B$  alone, as in LoRA-FA; large ones grow through both factors in balance.

**PREDICTION X.7** (the crossover scale). Under SGD, the singular values of  $\Delta W$  switch from one-sided to two-sided growth as they cross  $\sigma^* = \mu|\kappa|/2$ , where  $\mu = s\sqrt{\eta_A\eta_B}$  and  $\Phi = -\kappa I_r$ . For default LoRA with  $B_0 = 0$  and  $A_0 A_0^\top = cI$ ,  $\kappa = c/\eta_A$  and

$$\sigma^* = \frac{sc}{2} \sqrt{\frac{\eta_B}{\eta_A}}. \quad (13)$$

LoRA+ with ratio  $\lambda$  multiplies  $\sigma^*$  by  $\sqrt{\lambda}$ .

**SCOPE** Gradient flow exactly; SGD without momentum up to the  $O(\eta^2)$  drift of the charge per step. For Kaiming initialisations  $A_0 A_0^\top$  is only close to  $cI$  (HF’s default has  $c \approx 1/3$ ), and the crossover becomes a band of relative width  $O(\sqrt{r/n})$ . Adam is not covered.

**SUPPORT** [Theorem III.9](#), verified; known from the linear-network literature, with the PEFT reading ours.

**TEST** Train with SGD from orthonormal rows of  $A_0$  scaled by  $\sqrt{c}$ , so that the charge is exactly isotropic. Take the SVD  $\Delta W = U\Sigma V^\top$  at intervals and read off  $f_i = u_i^\top (B B^\top / \eta_B) u_i$  and  $g_i = v_i^\top (A^\top A / \eta_A) v_i$ . Below  $\sigma^*$ ,  $f_i \approx 0$  and  $g_i \approx \kappa$ ; above it,  $f_i \approx g_i \approx \sigma_i / \mu$ . Repeat at ratio  $\lambda$ .

**WOULD FAIL IF** the gains departed from the closed-form values  $f_i = f_\kappa(\sigma_i^2/\mu^2)$  and  $g_i = g_\kappa(\sigma_i^2/\mu^2)$  of [Theorem III.9](#), with  $f_\kappa(x) = \sqrt{x + \kappa^2/4} - \kappa/2$  and  $g_\kappa = f_\kappa + \kappa$ , while  $\Phi$  stayed isotropic. If  $\Phi$  has drifted, the test says nothing.

**TOY** Figure 7(b) plots the closed-form isotropic-charge singular-value dynamics and the crossover  $\sigma^*$ .

**Credit.** The closed form is due to [Tarmoun et al. \(2021\)](#), who call the charge “ $\lambda$ -balanced”, and to [Min et al. \(2021\)](#); see also [Dominé et al. \(2025\)](#) and [Kunin et al. \(2024\)](#). The balanced regime above  $\sigma^*$  is the dynamics of [Arora et al. \(2018\)](#).

**WHAT THIS EXPLAINS** LoRA+ moves  $\sigma^*$  up by  $\sqrt{\lambda}$ , so more of the update spends longer in the LoRA-FA regime. PiSSA at equal rates has  $\kappa = 0$ , and under gradient flow it is balanced from the first step.

Part of LoRA’s gauge is a symmetry of  $\rho$ , and gradient flow conserves its charge whatever signal drives it. A discrete step conserves it up to second order in the step size, and nothing forces Adam’s sign-like step to conserve it.

**PREDICTION X.8** (drift of the charge).  $\Phi$  is constant under gradient flow, moves by  $O(\eta^2)$  per SGD step at fixed block rates  $\eta_A, \eta_B$ , and drifts under Adam. We expect its drift under Adam to track other measures of how far Adam breaks LoRA’s gauge.

**SCOPE** Gradient flow driven by any signal  $\Delta D \rho^\top \gamma$ , which covers minibatches, LoRA dropout, global-norm clipping and DoRA’s detached norm; SGD without momentum or weight decay. Momentum, Adam and AdamW fall outside [Theorem III.5](#).

**SUPPORT** [Theorem III.5](#) (repaired; known after [Zhao et al., 2022b](#)); weight decay moves the charge on purpose ([Corollary III.6](#)). With a step size  $\eta$  multiplying the block rates, a step  $B \leftarrow B - \eta \eta_B \nabla_B$ ,  $A \leftarrow A - \eta \eta_A \nabla_A$  changes the charge by exactly

$$\Delta \Phi = \eta^2 (\eta_B \nabla_B^\top \nabla_B - \eta_A \nabla_A \nabla_A^\top). \quad (14)$$

The identity uses only that both factor gradients come from one weight-space gradient  $G$ , so it covers minibatches, LoRA dropout on the input of  $A$ , and global clipping when  $\nabla_A, \nabla_B$  denote the gradients actually applied.

**TEST** Log  $\Phi$  during training. Under SGD at step sizes  $\eta, \eta/2, \eta/4$ , the per-step change should match (14) to rounding and shrink about fourfold each time. Under Adam, compare  $\|\Phi_t - \Phi_0\|$  (our suggestion) with the gap between runs started at  $A_0$  and  $gA_0, g \notin B_r$ , which by [Corollary III.13](#) can differ.

**WOULD FAIL IF** the SGD drift departed from (14), so that the update was not a plain gradient step, or  $\Phi$ -drift under Adam showed no relation to any other measure of gauge breaking.

**TOY** Figure 7(a): scalar LoRA rides its hyperbola  $b^2/\eta_B - a^2/\eta_A = \Phi$  under gradient flow, and the charge drifts under discrete GD and under Adam.

**Credit.** The conservation law, with its symmetric and antisymmetric halves, is Prop. 5.1 of [Zhao et al. \(2022b\)](#); laws independent of loss and data are in [Marcotte et al. \(2023\)](#). The principle goes back to [Du et al. \(2018\)](#), [Arora et al. \(2018\)](#) and [Kunin et al. \(2020\)](#).

### Predictions about images

DoRA splits each output neuron into a length and a direction. At the level of images that adds exactly one free gain per output neuron to LoRA, so anything else that separates the two happens in training, where Adam notices the coordinates a parameter is written in.

**PREDICTION X.2** (DoRA against LoRA + output gain). **CONJECTURE D** (DoRA against a gain in DoRA's coordinates). Started from the same weights and trained with AdamW at the same settings, LoRA followed by a trainable output gain matches DoRA ([Liu et al., 2024b](#)) once the gain is trained in DoRA's coordinates ( $\mu_i$  starting at  $\|w_{0,i}\|$  in place of  $\ell_i$  starting at 1).

**SCOPE** AdamW, as in the DoRA paper and HF PEFT, with one learning rate, decay and data order for every arm. Images refer to the merged weight, with  $r \leq \min(m, n)$  and  $W_0$  free of zero rows. Every arm trains  $r(m + n) + m$  parameters.

**SUPPORT** [Proposition II.8](#) (repaired):  $\text{Im DoRA}_r = \text{Diag}_m \cdot (W_0 + M_{\leq r})$ , the image of  $\text{LoRA}_r$  followed by an output gain. The first form of Conjecture D, that any gap comes from the parametrisation, followed from this equality and could not fail; the form above can.

**TEST** Start four arms at  $W_0$  with the same  $A_0$ , as in Experiment D of Exposé II.B: (i) LoRA with an output gain,  $\ell_0 = \mathbf{1}$ ; (ii) the gain in DoRA's coordinates,  $W = \text{diag}(\mu_i / \|w_{0,i}\|)(W_0 + sBA)$ ,  $\mu_{0,i} = \|w_{0,i}\|$ ; (iii) DoRA with the exact gradient of its normalisation; (iv) DoRA with the detached norm of the DoRA paper (Sec. 4.3) and HF PEFT (`weight_norm.detach()`). Since the images are equal, any gap between arms comes from the parametrisation, the backward rule or the optimizer coordinates, and the four arms separate these. This design is ours.

**WOULD FAIL IF** arm (ii) stayed clearly behind arm (iv). Comparing (iii) with (iv) then shows how much of the gap the detached backward carries.

**TOY** None for training. The companion site's Gram lab shows the image-level half: at  $B = 0$  DoRA acts as an output torus, and its update escapes rank  $r$ .

A rank-one update writes one new feature into many neurons, and LoRA does this with a single rank. A Kronecker sum measures rank across a different cut of the same four-leg tensor, where, for the splits HF PEFT uses, a generic rank-one update is as complex as any matrix can be.

**PREDICTION X.5** (rank-one targets). Split  $m = m_1 m_2$  and  $n = n_1 n_2$ . On targets  $W_0 + uv^\top$  with generic  $u, v$ , and a loss whose only minimiser is the target, LoRA<sub>1</sub> can reach zero loss while no sum of fewer than

$$\kappa_1 = \min(m_1, m_2) \min(n_1, n_2) \quad (15)$$

Kronecker products can. For HF PEFT's splits  $\kappa_1$  is the largest possible Kronecker rank:  $d$  for its square splits of  $d = 1024$  or  $4096$  (for instance the  $64 \times 64$  split of  $4096$ ), and  $576$  for its  $24 \times 32$  split of  $768$ . For LoRA ranks below  $\min(m_1, n_1) \min(m_2, n_2)$  and fewer than  $\kappa_1$  terms, neither image contains the other.

**SCOPE** The negative half concerns images and holds for every optimizer; the positive half asks the optimizer to find a point that exists.

**SUPPORT** [Theorem II.11](#) (repaired), for unconstrained Kronecker sums and for LoKr, since  $C \otimes BA$  is one Kronecker term. With Schmidt decompositions  $u = \sum_i \sigma_i x_i \otimes y_i$  and  $v = \sum_j \tau_j z_j \otimes w_j$ , the best  $k$ -term sum leaves the squared error  $\sum \sigma_i^2 \tau_j^2$  over all but the  $k$  largest products  $\sigma_i \tau_j$ .

**TEST** On one linear layer with whitened inputs and more samples than parameters (or with the loss  $\frac{1}{2} \|W - W_0 - uv^\top\|_F^2$ ), train LoRA<sub>1</sub> and sums of  $k = 1, \dots, \kappa_1$  unconstrained Kronecker products, each with its own learning-rate sweep, and plot the final loss against  $k$  beside the floor.

**WOULD FAIL IF** a sum of fewer than  $\kappa_1$  terms went below the floor. LoRA<sub>1</sub> stalling above zero would be an optimization failure, against which the framework claims nothing.

**TOY** [Figure 6](#): for a random rank-one update the operator-Schmidt spectrum has one nonzero value across the ordinary cut and  $\kappa_1$  across the Kronecker cut.

**Credit.** The rearrangement and the nearest-Kronecker-product problem are due to [Van Loan and Pitsianis \(1993\)](#) ([Background B.11](#)). The PEFT reading, that Kronecker sums are LoRA across another cut, is ours. Kronecker adapters in practice are KronA ([Edalati et al., 2022](#)) and LoKr ([Yeh et al., 2023](#)).

Parameter count measures what a method pays, expressive dimension what it can reach. The difference is gauge, parameters that move without moving the weights. It ranges from one parameter for VeRA through  $r^2$  for LoRA <sub>$r$</sub>  to at least  $2r^2 + m + n - 1$  for LoHa <sub>$r, r$</sub> , which has the budget of LoRA <sub>$2r$</sub> .

**PREDICTION X.10** (effective dimension). Benchmarks should report the expressive dimension  $d(M)$  beside the budget  $|M|$ . At budgets of a few thousand parameters, rankings at equal effective dimension can differ from rankings at equal parameter count.

**SCOPE** Optimizer-free bookkeeping; the change of ranking is the empirical claim.

**SUPPORT** [Proposition II.12](#) (repaired). LoRA <sub>$r$</sub>  wastes  $r^2$  parameters, VeRA ([Kopiczko et al., 2023](#)) one (numerically, [Appendix E](#); for  $r \leq n$  and generic frozen factors also by [Background C.14](#)) and FourierFT one per sampled conjugate pair. At the budget of LoRA <sub>$2r$</sub> , with  $2r \leq \min(m, n)$ ,  $r^2 \leq \min(m, n)$  and  $2r^2 < m + n - 1$ , LoHa <sub>$r, r$</sub>  reaches at least  $(m + n - 1) - 2r^2$  fewer dimensions, and is strictly dominated for  $r \leq 2$ . When  $2r^2 \geq m + n - 1$ , as for  $r = 64$  on  $4096 \times 4096$  or  $r = 32$  on  $320$ - to  $640$ -wide layers, LoHa matches or beats LoRA <sub>$2r$</sub>  in dimension too, by its numerically checked dimension formula.

**TEST** On a small-budget benchmark, compare methods at matched  $|M|$  and at matched  $d(M)$ , computing  $d$  by a float64 Jacobian rank where no formula is proved. Report the maximum rank too, and name its cut.

**WOULD FAIL IF** the two rankings agreed at every budget tested, so that gauge is too small a share of the budget to matter there.



**TOY** Figure 4 places LoRA, LoHa, GraLoRA, HiRA, KronA, MoRA, FourierFT and C3A at one budget on a  $4096 \times 4096$  layer by maximum rank and by  $d(M)/(mn)$ . The companion site’s budget calculator sets  $|M|$  beside  $d(M)$  on real base models.

### Predictions about symmetry

The hidden units of a feed-forward block carry no order. Permute them with the matching rows and columns, and the network computes the same function. A method that never looks at unit order should not notice. A method built on a frame, a reshape or a block partition of the hidden coordinates should.

**PREDICTION X.3** (the permutation test). Over random function-preserving permutations of hidden units, LoRA’s outcome variance equals its seed variance. FourierFT, LoKr and MoRA show larger variance, and so do BOFT and OFT on modules whose input side is the hidden layer (`down_proj`). On `up_proj` and `gate_proj` the permutation only reorders BOFT’s output rows, so BOFT is exactly equivariant there.

**SCOPE** The LoRA half holds for any coordinatewise optimizer (SGD, Adam, AdamW), which commutes with coordinate permutations of  $Q$  (the third item of [Proposition II.13](#)), and an initialisation of  $A$  with i.i.d. entries. The rest is empirical.

**SUPPORT** [Lemma VI.2](#) (classical): the linear maps that commute with GELU or SiLU are the permutations, and with ReLU the positive monomials. Gated SwiGLU or GeGLU blocks admit the full monomial group through their linear branch (rule R4 of [Theorem VI.3](#)). [Observation II.15](#) turns covariance under those maps into equal reachable function sets.

**TEST** Apply  $K$  random permutations to the rows of `up_proj` and `gate_proj` and the columns of `down_proj`, and confirm that outputs agree to rounding. Fine-tune each method once per permuted copy with a fresh seed, and  $K$  times on the original with  $K$  seeds; compare the two variances, method by method.

**WOULD FAIL IF** LoRA’s permutation variance exceeded its seed variance, pointing to an implementation tied to unit order, or the listed methods showed no excess.

**TOY** No variance test. The companion site’s merge game runs the rule behind the lemma: a positive diagonal slides through ReLU and stops at GELU.

**Credit.** Lemma VI.2 is classical ([Chen et al., 1993](#); [Godfrey et al., 2022](#); [Sussmann, 1992](#)). The methods under test are FourierFT ([Gao et al., 2024](#)), LoKr ([Yeh et al., 2023](#)), MoRA ([Jiang et al., 2024](#)), BOFT ([Liu et al., 2023b](#)) and OFT ([Qiu et al., 2023](#)).

**WHAT THIS EXPLAINS** If the prediction holds, seed variance understates the variance of methods tied to unit order across equally good checkpoints.

SliceGPT rotates the residual stream of a model without changing what it computes. A method whose image rotates with the weights reaches the same functions afterwards; a method built on the stream’s coordinates may lose or gain reach.

**PREDICTION X.4** (the rotation test). Take a pre-norm transformer with its norm gains fused into the linear maps they feed and LayerNorm converted to RMSNorm, and rotate its residual stream by one global orthogonal matrix. This provably leaves unchanged the reachable function sets of LoRA, of  $(IA)^3$  with HF defaults, and of DoRA on the input-rotated boxes ( $q, k, v, \text{gate}, \text{up}$ ). [Conjecture A](#), restated: against the gain-fused, unrotated model, the rotation measurably changes the fine-tuning results of block OFT/BOFT on residual-input projections, LoHa ( $r \geq 2$ ), HiRA and coordinate masks, because LLM residual streams have a privileged basis.

**SCOPE** The proved half concerns reachable function sets, for every optimizer and for box-wise methods with no shared parameters or random frames. Training outcomes are exactly unchanged only for (IA)<sup>3</sup>, and in law for LoRA and DoRA under SGD with an isotropic initialisation. HF’s Kaiming-uniform  $A$  is not rotation-invariant in law, and Adam is only  $B_r$ -equivariant ([Theorem III.12](#)), so both break this. Sandwich and post-norm models (Gemma-2/3, OLMo-2) and SliceGPT’s per-block rotations are excluded.

**SUPPORT** [Observation II.15](#) (repaired), a sufficient condition. The conjectured methods fail it, which by itself proves nothing.

**TEST** Keep the gain-fused model as control, rotate a copy, and confirm that outputs agree to rounding. Fine-tune every method on both over several seeds. LoRA’s reach provably does not change, so its change in results measures the share of the optimizer and the initialisation.

**WOULD FAIL IF** the listed methods changed no more than LoRA does.

**Credit.** The rotation is that of SliceGPT ([Ashkboos et al., 2024a](#)). In QuaRot ([Ashkboos et al., 2024b](#)), the per-head Hadamard fused offline into  $W_v$  and  $W_{out}$  is a value–output head gauge, which [Observation II.15](#) covers; the online Hadamards preserve the function but insert new operations, outside its hypothesis.

PiSSA and CorDA choose their split from a truncated SVD, which follows the coordinates under rotations and uniform scalings and changes when they are stretched unevenly. Whitening the input first removes that dependence from the split.

**PREDICTION X.11** (whitened splits). [Conjecture C](#), restated: whitened CorDA or PiSSA is more robust to function-preserving input reparametrisations, and no worse otherwise.

**SCOPE** The covariance of the split is optimizer-free; the robustness of trained results is the empirical part. Absolute damping  $C + \lambda I$  cuts the covariance to  $O(n)$ , while CorDA’s trace-relative damping keeps  $CO(n)$ . The released CorDA code divides each calibration sample by its largest entry, which breaks even rotations, so only the published formulas are covered.

**SUPPORT** [Proposition II.16](#) (repaired), for  $C = \mathbb{E}[xx^\top] \succ 0$ ,  $1 \leq r < \min(m, n)$  and spectral gaps. The whitened split  $\Pi_r W_0$ , with  $\Pi_r$  the top- $r$  eigenprojector of  $W_0 C W_0^\top$ , is  $GL_n$ -covariant on the input side. The PiSSA and MiLoRA splits and the paper-form CorDA splits have input-side group exactly  $CO(n)$ , so norm-gain folding or SmoothQuant rescalings change them.

**TEST** Prepare function-preserving versions of one checkpoint that differ by non-conformal input maps (folded RMSNorm gains, or a per-channel rescaling with its inverse folded into the producer). Initialise PiSSA, paper-form CorDA and the whitened split on each, fine-tune with one optimizer, and compare the spread across versions; compare all three on the unmodified checkpoint for the “no worse” half.

**WOULD FAIL IF** PiSSA and CorDA varied no more than the whitened split across versions, or the whitened split did worse on the unmodified checkpoint.

**Credit.** The splits are PiSSA ([Meng et al., 2024a](#)), MiLoRA ([Wang et al., 2024b](#)) and CorDA ([Yang et al., 2024c](#)); the whitened form is the truncation-aware whitening of SVD-LLM ([Wang et al., 2025c](#)). EVA ([Paischer et al., 2024](#)) starts at  $B_0 = 0$  and splits nothing.

### Predictions about base change and merging

Merging and restarting lets a method compose its own moves. For an additive cone such as ReLoRA’s, composition keeps buying rank. For block OFT with a fixed partition, a product of block rotations is again a block rotation.

**PREDICTION X.6** (re-based OFT). With an exact Cayley map and a fixed partition, merge-and-restart OFT reaches nothing beyond the closure of one cycle,  $W_0 \cdot \prod \text{SO}(b)$ , so fixed blocks gain nothing. Partitions whose hypergraph is connected generate  $\text{SO}(n)$ , so re-based exact OFT gains reach only if its partition changes between cycles or its butterflies reach full depth. LoRA-type cones gain from rebasing with a fixed shape; fixed linear shapes (masks, BitFit, (IA)<sup>3</sup>) gain nothing.

SCOPE Reachable sets, hence optimizer-free, and silent on trained performance. HF PEFT’s default Cayley–Neumann OFT is not orthogonal; with small generators ( $\|Q\|_{\text{op}} \leq 2^{1/4}$ ) each merge is a contraction, so repeated merging can only shrink neuron norms and singular values. Iterated exact BOFT with  $m'$  butterfly factors reaches  $W_0 \cdot \text{SO}(n)$ , up to its output scale, only at full depth  $b \cdot 2^{m'-1} = n$ .

SUPPORT [Theorem V.3\(c\)](#) (repaired) and its corollaries 2 and 4.

TEST Run  $K$  cycles of merge-and-restart OFT with `use_cayley_neumann=False` and a fixed partition, logging the partial Grams  $W_{:,J} W_{:,J}^T$  and the loss. Repeat with a fresh random partition each cycle, and with ReLoRA at the same per-cycle budget. Only the fixed-partition run keeps its partial Grams fixed to rounding.

WOULD FAIL IF a partial Gram moved beyond rounding under the fixed partition, so that the map was not exactly orthogonal. Gains in loss from restarts alone are outside the claim.

TOY Figure 10(b) shows the components of the hypergraph generated by block-OFT, butterfly and Givens factors on 16 coordinates and the dimension of the Lie algebra they generate against  $\dim \mathfrak{so}(16) = 120$ ; a fixed block partition never connects the hypergraph.

**Credit.** OFT ([Qiu et al., 2023](#)) and BOFT ([Liu et al., 2023b](#)) introduced the block and butterfly parametrisations; ReLoRA ([Lialin et al., 2023b](#)) introduced merge-and-restart for LoRA.

LoraHub mixes adapters factor by factor. A factor belongs to no adapter in particular, since  $(B, A)$  and  $(cB, A/c)$  give the same update, so the mix depends on choices the adapters do not record unless the modules share a factor.

**PREDICTION X.9** (LoraHub seeds). LoraHub’s factor-wise merge  $(\sum_i w_i B_i)(\sum_j w_j A_j)$  behaves like update arithmetic, up to the factor  $\sum_j w_j$ , when its modules share the  $A$ -seed and  $A$  stays close to its initialisation. It degrades when the modules come from independent seeds.

SCOPE The algebra is post hoc and optimizer-free. Whether  $A$  stays near  $A_0$  is empirical; a large  $\eta_B/\eta_A$ , as in LoRA+, favours it.

SUPPORT [Proposition V.4](#) (repaired). Replacing  $(B_2, A_2)$  by  $(cB_2, A_2/c)$  changes the merge by

$$\sum_{l \neq 2} w_l w_2 [(c-1)B_2 A_l + (c^{-1}-1)B_l A_2], \quad (16)$$

so it is not a function of the adapters. With one shared frozen  $A$  it returns  $(\sum_j w_j) \sum_i w_i B_i A$  exactly. HF’s `add_weighted_adapter` with `linear`, and its non-SVD `ties` and `dare_*` variants, are factor-wise rules of the same kind.

TEST Train  $N$  modules on  $N$  tasks from a shared  $A$ -seed and from independent seeds. Merge each set by LoraHub’s rule at fixed weights and compare with  $(\sum_j w_j) \sum_i w_i B_i A_i$ , as matrices and on the downstream metric, logging  $\|A_i - A_0\|/\|A_0\|$ .

WOULD FAIL IF independent seeds merged as well as shared ones, so that the cross terms  $B_i A_j$ ,  $i \neq j$ , cost nothing in practice.

TOY Figure 9, merge panel: a gauge of one module moves the LoraHub merge and leaves task arithmetic unchanged.

**Credit.** LoraHub is due to [Huang et al. \(2023\)](#). FFA-LoRA ([Sun et al., 2024](#)) freezes a shared factor, and KnOTS ([Stoica et al., 2024](#)) re-factorises the modules jointly through a shared  $U\Sigma$ . Task arithmetic ([Ilharco et al., 2022](#)) combines updates and has no gauge to break.

**WHAT THIS EXPLAINS** A safe rule combines LoRAs on the products  $B_i A_i$ , or on factors only when the modules share one.

### One more expectation, and how to read a failure

One expectation is not on the list. By [Proposition II.13](#), unconstrained Kronecker sums are LoRA after a rearrangement of entries, and FourierFT (with no sampled conjugate pair), WaveFT (when HF’s pad-IDWT-crop map is invertible) and C3A are coordinate masks in a fixed frame. An optimizer that sees the loss only through the trainable parameters runs each exactly as its source on a transported loss. At matched learning-rate sweeps we therefore expect the gap between a transported method and its source to follow how well each image aligns with the task’s update, measured by projecting a full fine-tuning  $\Delta W$  onto each image.

A failure inside the hypotheses of [X.1](#), [X.5](#) or [X.6](#), or of a proved half, would mean an error in a proof, and we would mark that result refuted; a failure outside them shows which hypothesis matters. A failure of [X.2](#), [X.4](#) or [X.11](#) would refute [Conjecture D](#), [Conjecture A](#) or [Conjecture C](#), respectively.

## What is analogy, and what is open

This exposé gives a plain account of which borrowed words carry theorems, what is open, whose results are re-derived here, and what the referees changed. Referees repaired 27 of the 41 results, and a second review narrowed 19 of those again.

### What the borrowed words mean

Words borrowed from geometry and physics compress a great deal, and each also carries connotations no theorem here supports. Table 30 gives, for seven such words, the sense each has in this paper, the results behind it, and what it leaves out.

**Table 30.** Analogy versus theorem: seven borrowed words, their meaning here, what is proved under each, and what is not claimed.

| Word                | Meaning here                                                                                                                                                                                                                                        | Proved                                                                                                                                                                                                                                                                                                                                                                 | Not claimed                                                                                                                                                                                                                                      |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Grothendieck</b> | The relative point of view: a method is a map $\rho : (Q, q_0) \rightarrow (\Theta, \theta_0)$ , methods form the pointed slice <b>Meth</b> $(\Theta, \theta_0)$ , and changing $\theta_0$ is base change.                                          | Frozen model initial and full fine-tuning terminal; products are fibre products in <b>Diff</b> and can fail to exist ( <a href="#">Obs. I.4</a> ); a symmetric monoidal sum ( <a href="#">Obs. I.7</a> ); automorphisms by transport ( <a href="#">Prop. II.13</a> ); pushforward along function-preserving maps ( <a href="#">Def. V.1(a)</a> ).                      | No schemes, sheaves, descent theory or stacks. Quantisation and rebasing borrow only the word “base change”, and idle constructions such as $\int \mathbf{Meth} \rightarrow \mathbf{PNet}$ are left out. “Exposé” and the rising sea are homage. |
| <b>Erlangen</b>     | After <a href="#">Klein (1872)</a> , a method’s geometry is its covariance group $\mathcal{G}_M$ ( <a href="#">Def. II.14</a> , <a href="#">Background D.17</a> ); what a group-type method can never change is a complete invariant of the action. | Complete invariants of the orthogonal, torus and Hadamard actions ( <a href="#">Thm II.5</a> ); exactly orthogonal input-side methods keep $K_{\text{out}}$ and the spectrum through any number of merges ( <a href="#">Cor. II.6</a> , <a href="#">Thm V.3</a> ); covariance groups of named methods ( <a href="#">Prop. II.9(e)</a> , <a href="#">Prop. II.16</a> ). | For cones such as LoRA’s image the word only names the shape and its covariance. Keeping an invariant concerns reach; that it causes knowledge retention is not shown.                                                                           |
| <b>Noether</b>      | The pattern that turns a continuous symmetry of $\rho$ into a conserved quantity.                                                                                                                                                                   | <a href="#">Thm III.5</a> , for gradient flow driven by any signal $\wedge D\rho^\top \gamma$ , a mild extension of <a href="#">Zhao et al. (2022b)</a> ; for LoRA the charges are the entries of $\Phi$ .                                                                                                                                                             | No Lagrangian. Discrete steps drift by $O(\eta^2)$ relative to the charge, and Adam and momentum break the law.                                                                                                                                  |
| <b>Gauge</b>        | The symmetry group of a parametrisation ( <a href="#">Def. I.5</a> , <a href="#">Background D.21</a> ); for LoRA, $\text{GL}_r$ acting by $(Bg^{-1}, gA)$ .                                                                                         | Generic gauges and fibre dimensions ( <a href="#">Prop. II.12</a> ); the part each optimizer respects, $B_r \subset O(r) \subset \text{GL}_r$ ( <a href="#">Thm III.12</a> ); the gauge dependence of factor-wise merging ( <a href="#">Prop. V.4</a> ).                                                                                                               | No connection or curvature. Fibres are gauge orbits only on the full-rank stratum; away from it the fibres jump.                                                                                                                                 |

*continued on the next page*



Table 30, continued

| Word          | Meaning here                                                                                                                                    | Proved                                                                                                                                                            | Not claimed                                                                                                                                                                                              |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Moduli</b> | A set of isomorphism classes ( <a href="#">Background C.20</a> ).                                                                               | Three strata of full-rank LoRA pointings are classified, as a set, by $\text{Gr}(r, n) \sqcup \text{Gr}(r, m) \sqcup \mathcal{M}_r$ ( <a href="#">Thm II.2</a> ). | No topology: the natural quotient topology is not a disjoint union ( $T_A$ lies in the closure of $S$ ). No stack is built, and the strata miss some pointings (HF's orthogonal init, defect pointings). |
| <b>Fibre</b>  | The preimage $\rho^{-1}(\theta)$ , the parameters that give the same weights, as in the title.                                                  | Fibre dimensions ( <a href="#">Prop. II.12(a)</a> ); which points of the apex fibre $\{BA = 0\}$ an optimizer can tell apart ( <a href="#">Cor. III.13</a> ).     | $\rho$ is no fibre bundle; the fibre dimension jumps between strata.                                                                                                                                     |
| <b>Lens</b>   | Backpropagation as the 2-functor <b>Para</b> ( $R$ ), which sorts tangent-side from cotangent-side methods ( <a href="#">Background B.17</a> ). | The degree-one collapse: GaLore with weight decay 0 is one-sided LoRA with a moving frame and carried optimizer state ( <a href="#">Prop. IV.2</a> ).             | The duality is exact only at degree one, for optimizers that do not read the parameter tensor, with no weight decay; decoupled decay, default Adafactor and full-gradient clipping break it.             |

**Credit.** **Para** and the lens reading of backpropagation are due to [Fong et al. \(2017\)](#) and [Cruttwell et al. \(2021\)](#). The ETHER control study that tempers the knowledge-retention reading of orthogonal methods is [Bini et al. \(2024\)](#).

Three scopes travel with every claim. Expressivity is a different property from learnability. Only Exposé [III](#) is dynamical, and each of its results names its optimizer. Mergeability is proved box by box under explicit side conditions (position-uniformity, no weight sharing, bias slots); the rewrite rules give sufficient conditions only, and whole-network statements are open.

Most of the mathematics is real algebraic geometry, invariant theory, Lie generation and gradient-flow ODEs. The category theory does its own work in seven places: the interval [frozen, full FT] and simulation; invariants finer than the image; a product that yields LoRA-XS, and meets that organise others; base change along maps that change the function; the lens; naturality of gradient descent and of combination rules; and rank as a min-cut.

Some statements rest on less than a full proof. The equalities in the dimensions of LoHa ([Proposition II.12](#)) and VeRA are checked numerically, and the bounds are proved; for VeRA with  $r \leq n$  and generic frozen factors the equality also follows from the toric-cone description of [Background C.14](#), and the generic equality for DoRA ([Proposition II.8](#)) is proved. The case of unconstrained generators in corollary 4 of [Theorem V.3](#) is recorded as a proof sketch, which [Appendix C.43](#) completes. The continuous case of [Proposition V.4\(a\)](#) and the converse in [Corollary III.13](#) are proved in [Appendix C](#), and [Proposition IV.3](#) is argued around Frobenius' theorem ([Background D.25](#)). The ancillary script `framework_checks.py` re-runs the main numbers in about a minute ([Appendix E](#)).

## Open problems

Each problem grew out of a proof that stopped short, and each is stated so that a solution would be recognisable.

1. **LoHa's dimension.** [Proposition II.12](#) proves

$$d(\text{LoHa}_{r_1, r_2}) \leq \min(mn, r_1(m + n - r_1) + r_2(m + n - r_2) - (m + n - 1)), \quad (17)$$

with equality checked only numerically. Prove it in general, and decide whether the Hadamard variety is closed.

2. **Initialisations, dynamically.** [Theorem II.2](#) classifies three strata by expressivity, and [Corollary III.13](#) shows that Adam sees  $A_0$  only up to signed row permutations. Classify the apex fibre modulo  $B_r$  for Adam, stratified by the charge  $\Phi_0$  ([Corollary III.6](#)), including the degenerate pointings outside the three strata of [Theorem II.2](#).
3. **HRA's apex in practice.** HRA's paired initialisation, HF PEFT's default, starts at an apex ([Theorem II.7\(b\)](#), for injective  $W_0$  and even  $r \leq n - 2$ ), with  $\dim S_1 = kn - \frac{k(k+1)}{2}$ ,  $k = r/2$ , against  $d = rn - \frac{r(r+1)}{2}$ . Decide whether this apex is a practical first-order bottleneck that slows early training, by comparison with a regular initialisation at matched budget.
4. **Charts for meets.** LoRA-XS ([Proposition I.8\(a\)](#)) and HRA's image ([Theorem II.7\(a\)](#)) are meets with a polynomial chart; the meet of LoRA with output scaling ([Proposition I.8\(b\)](#)) is a union of coordinate planes. Characterise the pairs of named methods whose meet is parametrisable by a polynomial chart.
5. **The frontier.** Incomparable pairs are common: split against zero-init LoRA ([Theorem II.2\(d\)](#)), Kronecker sums or GraLoRA against LoRA ([Theorem II.11](#), [Proposition II.12\(c\)](#)). Find the size- $k$  frontier of **Meth**, whose maximal elements are pairwise incomparable.
6. **An Adam analogue of the crossover.** [Theorem III.5](#) and [Theorem III.9](#) hold for gradient flow. Find a conserved or monotone quantity for sign-like updates, whose symmetry is only  $B_r$  ([Theorem III.12](#)).
7. **Prefixes and prompts.** For a vanilla softmax head with  $W_o \neq 0$  and  $W_q \neq 0$ , a generic prefix is not box-locally mergeable ([Theorem VI.5\(f\)](#)); prove or refute its non-mergeability for the whole network. In causal decoders every soft prompt is a deep prefix ([Theorem VI.5\(e\)](#)). [Petrov et al. \(2023\)](#) and [Li and Liang \(2021\)](#) assert that prompt tuning is strictly weaker than deep prefix tuning (P-tuning v2, [Liu et al., 2021c](#)), and the inclusion is strict in a one-layer, one-token example. Prove the generic strictness, and settle encoders.

Conjectures [A](#), [C](#) and [D](#) are open as well. They appear in Exposé [X](#) as Predictions [X.4](#), [X.11](#) and [X.2](#).

## Debts: known results we re-derive

Much of what this paper proves was known in another language or setting. A re-derivation earns its place only if the new language adds something, so [Table 31](#) states what we add next to each debt.

**Table 31.** Known results we re-derive, whom they are due to, and what is added here.

| Here                          | Due to                                                                                                                                                                                              | Added here                                                                                                                                                                      |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Def. I.1,<br>Exposé IV        | Fong et al. (2017); Cruttwell et al. (2021)                                                                                                                                                         | The pointed slice of methods; tangent and cotangent sides.                                                                                                                      |
| Thm II.1                      | The PiSSA-to-LoRA conversion of Meng et al. (2024a), in HF PEFT (Mangrulkar et al., 2022)                                                                                                           | The rank- $2r$ bound for every exact pointing, and from any start.                                                                                                              |
| Thm II.5                      | Classical: the first fundamental theorem for $O(n)$ ; Witt’s extension theorem (Background D.16)                                                                                                    | Which invariant each group-type method keeps.                                                                                                                                   |
| Thm II.10                     | The standard cut-rank bound for tensor networks (Background B.8)                                                                                                                                    | A rank census of PEFT plugs.                                                                                                                                                    |
| Thm II.11                     | Van Loan and Pitsianis (1993); the operator-Schmidt decomposition (Background B.12)                                                                                                                 | Kronecker sums as LoRA across another cut; the LoKr inclusion criterion.                                                                                                        |
| Prop. II.12,<br>TT/Tucker row | Koch and Lubich (2010); Uschmajew and Vandereycken (2013); Holtz et al. (2012)                                                                                                                      | The same count, at minimal ranks, beside other PEFT families.                                                                                                                   |
| Prop. III.4,<br>Thm III.12(c) | Tong et al. (2021); Zhang and Pilanci (2024a); LoRA-Pro (Wang et al., 2024i); LoRA-FA (Zhang et al., 2023d); descent to the quotient: Mishra et al. (2012), Mishra and Sepulchre (2016)             | One cometric table for (S)GD; the hierarchy $B_r \subset O(r) \subset GL_r$ .                                                                                                   |
| Thm III.5                     | Zhao et al. (2022b); Marcotte et al. (2023); the principle: Du et al. (2018), Arora et al. (2018), Kunin et al. (2020)                                                                              | The formally immediate remark that the signal need not be a gradient; PEFT instances (DoRA’s detached norm, clipping, VeRA, AdaLoRA without its regulariser, HRA).              |
| Prop. III.7                   | Kempf and Ness (1979), in the real form of Richardson and Slodowy (1990); Srebro et al. (2004)                                                                                                      | Balance as the Kempf–Ness slice of LoRA’s gauge; two decay conventions, two slices (Cor. III.6).                                                                                |
| Thm III.9                     | Tarmoun et al. (2021); Min et al. (2021); Dominé et al. (2025); Kunin et al. (2024)                                                                                                                 | The PEFT reading of $\sigma^*$ , its shift under LoRA+, PiSSA’s balance, the Kaiming band.                                                                                      |
| Thm III.10,<br>Cor. III.11    | Schulman and Thinking Machines Lab (2025), for LoRA+ (Hayou et al., 2024b)                                                                                                                          | Any multihomogeneous method; the SGD $c^2$ law; per-block $\varepsilon$ , decay and clipping.                                                                                   |
| Prop. IV.2                    | Torroba-Hennigen et al. (2025); for random projections, Flora (Hao et al., 2024)                                                                                                                    | The projector-by-schedule table of cotangent methods; optimizer state carried across re-basings.                                                                                |
| Prop. IV.3                    | Frobenius’ theorem (Background D.25)                                                                                                                                                                | When a $\theta$ -dependent anchor foliates weight space by images of forward methods; GaLore, LISA and MeZO keep a fixed anchor per period, so the test does not separate them. |
| Lemma VI.2                    | Sussmann (1992); Chen et al. (1993); Godfrey et al. (2022)                                                                                                                                          | Which symmetries a method may ignore; the monomial group of gated FFNs.                                                                                                         |
| Thm VI.3                      | (IA) <sup>3</sup> merging; Liu et al. (2022a), Sec. 3.3; SSF’s re-parameterisation (Lian et al., 2022); LoRA’s linearity (Hu et al., 2021); scale migration through SwiGLU: AWQ (Lin et al., 2024a) | One calculus of sufficient conditions, with explicit side conditions.                                                                                                           |

*continued on the next page*

Table 31, continued

| Here     | Due to                                                                                                                                                                                                                                                                                   | Added here                                                                                                |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Thm VI.5 | The identity is standard; (b) <a href="#">He et al. (2021)</a> ; (c) and the prompt-to-prefix map (e) <a href="#">Petrov et al. (2023)</a> , Sec. 2.2, who also show that prefix-tuning a pretrained transformer can be a universal approximator ( <a href="#">Petrov et al., 2024</a> ) | The homomorphism-then-perspective organisation; explicit position hypotheses and the slice typing of (e). |

**Credit.** We also build on work without re-deriving it. [Malladi et al. \(2022\)](#) found the lazy kernel of random- $A_0$  LoRA close to that of full fine-tuning, which [Observation I.6](#) reads through  $S_1$ . The Init[B] scheme in the table of [Theorem II.2](#) is from [Hayou et al. \(2024a\)](#). With a fixed support, the meet of [Proposition I.8\(b\)](#) is  $(IA)^3$  masked to  $r$  neurons; compare the sparse activation scaling of [Stoehr et al. \(2024\)](#).

### The referee record

The framework was synthesised from four proposals and three referee reports. Every numbered result then went to two independent referees, who checked it against its proof, the literature and the Hugging Face PEFT source, that is, for its proof and for its fit to the methods as published and as coded. Of the 41 results, 14 stood as first written, with wording fixes, and 27 were repaired by an added hypothesis, a narrowed claim, a replaced proof or a withdrawn reading. A second review re-examined 26 of the repaired statements in 48 reports. It raised a major objection to 18 of them, mostly to a side clause: a missing hypothesis, a remark that went past its proof, or a description of a method that matched only one implementation. For example, HF PEFT’s LoftQ initialiser turned out to ignore the LoRA scale  $\alpha/r$  ([Proposition V.2](#)). In all it found a false or overreaching clause in 19 statements, and those were narrowed again. Only the final statements appear in this paper; the referee notes of both rounds are in the ancillary file `propositions.json`. Table 32 records what changed, result by result.

**Table 32.** The referee record: the status of each numbered result after both rounds of review, and what the referees changed.

| Result                                                          | Status         | What the referees changed                                                                                                                                      |
|-----------------------------------------------------------------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Obs. I.4</a> , universal properties                 | repaired       | Products need a fibre product in <b>Diff</b> and can fail to exist ( $\text{LoRA}_1 \times \text{Frozen}$ on $1 \times 1$ weights).                            |
| <a href="#">Obs. I.6</a> , $S_1$ is functorial                  | verified       | Monotonicity stated only along pointed morphisms.                                                                                                              |
| <a href="#">Obs. I.7</a> , additive monoidal structure          | repaired twice | $r \mapsto \text{LoRA}_r$ is strong monoidal only unpointed or at degenerate pointings ( $\dim S_1$ is $mr$ against $2mr$ ); HF’s cat merge described exactly. |
| <a href="#">Prop. I.8</a> , a product and a meet                | verified       | LoRA-XS needs rank $W_0 \geq r$ ; the set-level meet in (b) is no new method.                                                                                  |
| <a href="#">Thm II.1</a> , pointings of an additive method      | verified       | Stated for exact pointings, with a robust form; credited to <a href="#">Meng et al. (2024a)</a> .                                                              |
| <a href="#">Thm II.2</a> , three strata of LoRA initialisations | repaired       | Full-rank pointings only, and only as a set; HF’s orthogonal init and QLoRA/LoftQ defects lie outside.                                                         |
| <a href="#">Prop. II.3</a> , first-order rank formula           | repaired       | Scale and gradient flow made explicit; Adam’s first step has an $\ell_1$ rate; LoRA-GA’s $A_0$ , not its $B_0$ , maximises the rate.                           |

continued on the next page

Table 32, continued

| Result                                      | Status         | What the referees changed                                                                                                                                         |
|---------------------------------------------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lemma II.4, zero gate                       | verified       | Hypotheses stated; a function-level variant for LLaMA-Adapter; AdaLoRA instance corrected.                                                                        |
| Thm II.5, complete invariants               | verified       | Methods that can create zeros ((IA) <sup>3</sup> , HiRA) keep only “zeros of $W_0$ stay zero”.                                                                    |
| Cor. II.6, what orthogonal methods never do | repaired twice | Exactly orthogonal input-side maps only; BOFT keeps cosines up to sign; HF’s Cayley–Neumann OFT and GOFT’s scaler excluded.                                       |
| Thm II.7, HRA as a meet; its apex           | repaired twice | An image-level meet with $W_0 \cdot \text{SO}(n)$ , no product; strictly above $\text{LoRA} \cap \text{OFT}$ only for exactly orthogonal OFT.                     |
| Prop. II.8, DoRA’s image                    | repaired twice | $\text{Diag}_m \cdot (W_0 + M_{\leq r})$ for $W_0$ without zero rows; the rank formula needs nonzero magnitudes; Conjecture D made falsifiable.                   |
| Prop. II.9, the torus ladder                | repaired       | $\text{RoAd}_1$ is $(\text{IA})^3$ over $\mathbb{C}$ ; only zeros of $W_0$ persist; $(\text{IA})^3$ ’s FFN vector acts on the input of $W_{\text{down}}$ .        |
| Thm II.10, cut-rank bound                   | verified       | A large min-cut is necessary for high rank and is not sufficient.                                                                                                 |
| Thm II.11, Kronecker incomparability        | repaired twice | $\kappa_1$ is the maximal Kronecker rank for aligned splits; $\text{LoKr}$ with $r' \leq \min(m_2, n_2)$ lies in $\text{LoRA}_r$ iff $r \geq \min(m_1, n_1) r'$ . |
| Prop. II.12, gauge census                   | repaired twice | GraLoRA’s maximum rank is $\min(kr, m, n)$ ; LoHa loses dimensions to $\text{LoRA}_{2r}$ only when $2r^2 < m + n - 1$ .                                           |
| Prop. II.13, transport of structure         | repaired twice | Any $T \in \text{GL}(\Theta)$ ; commutes with rebasing for additive families and fixed $T$ ; FourierFT needs no conjugate pair.                                   |
| Obs. II.15, descent                         | repaired twice | Reachable sets only, for box-wise methods without shared frames; residual rotations need a gain-fused pre-norm model.                                             |
| Prop. II.16, covariant splits               | repaired twice | MiLoRA and CorDA-KPM are bottom- $r$ splits; trace-relative damping keeps $\text{CO}(n)$ ; shipped CorDA code is not covered.                                     |
| Obs. III.2, chain rule for cometrics        | verified       | Scope stated: gradient flow, no LoRA dropout, constant rates.                                                                                                     |
| Thm III.3, naturality of gradient descent   | verified       | The title “natural only along isometries” was false in general and was replaced.                                                                                  |
| Prop. III.4, cometrics that descend         | repaired       | Undamped (S)GD only; the published methods use AdamW or damping.                                                                                                  |
| Thm III.5, Noether charges                  | repaired twice | Isometries need the signal to factor through $(t, \rho(q))$ ; relabelled known; the per-channel-rate remark needs distinct rate products.                         |
| Cor. III.6, charge at initialisation; decay | repaired       | The decay commentary was reversed: coupled decay tends to $\mu = 0$ , per-time decay to $\Phi = 0$ .                                                              |
| Prop. III.7, Kempf–Ness slice               | verified       | Uniqueness up to $\text{O}(r)$ proved; the Loewner floor renamed.                                                                                                 |
| Prop. III.8, balance–escape trilemma        | verified       | Balance generalised to $B_0^\top B_0 = c A_0 A_0^\top$ .                                                                                                          |
| Thm III.9, isotropic-charge dynamics        | verified       | Pointwise on the rank- $r$ locus; the Kaiming crossover is a band.                                                                                                |
| Thm III.10, the hyperparameter torus        | verified       | Hypotheses explicit; extended to schedules, per-block $\varepsilon$ , decay and clipping.                                                                         |
| Cor. III.11, LoRA+ and rsLoRA               | verified       | A shared warm-up keeps the torus; linear modules only.                                                                                                            |
| Thm III.12, optimizer gauge hierarchy       | repaired twice | Adam respects $B_r$ , positive diagonals only with per-channel rates, $\varepsilon$ and decay; damping cuts $\text{GL}_r$ to $\text{O}(r)$ .                      |

continued on the next page

Table 32, continued

| Result                                                 | Status         | What the referees changed                                                                                              |
|--------------------------------------------------------|----------------|------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Cor. III.13</a> , what zero- $B$ chooses   | repaired       | The apex fibre is $\{BA = 0\}$ ; the $GL_r$ case needs equivariance along the whole trajectory.                        |
| <a href="#">Prop. IV.2</a> , degree-one collapse       | repaired       | Decoupled decay, default Adafactor and full clipping break it; GaLore carries optimizer state.                         |
| <a href="#">Prop. IV.3</a> , integrability             | repaired twice | The “only if” was false; the top- $r$ gradient field is non-involutive for every $r \geq 1$ .                          |
| <a href="#">Obs. IV.4</a> , the backprop cone          | repaired twice | An upper bound; forward mode pays off only for $k = O(1)$ .                                                            |
| <a href="#">Prop. V.2</a> , quantisation and splitting | repaired twice | Three orders, three defects; HF’s LoftQ init ignores the LoRA scale; QA-LoRA merges on-grid.                           |
| <a href="#">Thm V.3</a> , rebasing closure             | repaired twice | BOFT reaches $SO(n)$ only at full depth; HF’s Cayley–Neumann merges contract.                                          |
| <a href="#">Prop. V.4</a> , combination rules          | repaired twice | The “iff” needs full rank or continuity; factor-wise DARE is gauge-dependent too.                                      |
| <a href="#">Lemma VI.2</a> , activations fix the basis | verified       | Gated FFNs have the full monomial group.                                                                               |
| <a href="#">Thm VI.3</a> , the merge calculus          | repaired twice | Sufficient conditions only; RoPE needs scales constant on rotary pairs.                                                |
| <a href="#">Prop. VI.4</a> , linear insertions, ReFT   | repaired twice | LoReFT has dimension $r(2d + 1 - r)$ and a pyreft pointing defect; the serial-adaptor criterion needs rank $A_0 = r$ . |
| <a href="#">Thm VI.5</a> , attention as a homomorphism | repaired twice | Rigidity is per layer; the prompt-to-prefix map credited to <a href="#">Petrov et al. (2023)</a> .                     |

A repair is a record of an error that was caught. We keep the record public so that a reader can weigh each result by its history, and so that the next error, when someone finds it, has an obvious place to go.



## Related work

This paper draws on three bodies of work: the PEFT literature, with its surveys and unified views; the theory of factorised models, their symmetries and their optimisation; and applied category theory. We take them in turn and say, for each, what we use and where our questions differ. Results that we re-derive from earlier work carry that credit in their own headings and in the contributions of Exposé 0; here we place the paper as a whole. The catalogue of Appendix D cites all 382 methods and 138 papers it classifies, and this section names the representatives of each family.

**Surveys, unified views and design spaces.** The closest earlier unified frame is that of He et al. (2021). They read adapters (Houlsby et al., 2019), prefix tuning (Li and Liang, 2021) and LoRA (Hu et al., 2021) as modifications of specific hidden states that differ along four axes: functional form, insertion form, modified representation and composition function. Their observation that prefix tuning is a softmax-gated parallel adapter is part (b) of Theorem VI.5, and their MAM adapter, like UniPELT (Mao et al., 2021), is a hybrid in our classification. The delta-tuning study of Ding et al. (2022) sorts methods into addition-, specification- and reparametrisation-based families and adds optimisation and optimal-control perspectives. The guide of Lialin et al. (2023a), the reviews of Xu et al. (2023a) and Han et al. (2024b), the LoRA survey of Mao et al. (2024b), the vision survey of Xin et al. (2024), the prompting survey of Liu et al. (2021b) and the modular view of Pfeiffer et al. (2023) drew the family boundaries that our catalogue starts from.

Other work searches over configurations. Chen et al. (2023a) parametrise a design space by layer grouping, parameter allocation, tunable groups and strategy assignment, and AutoPEFT (Zhou et al., 2023b) and NOAH (Zhang et al., 2022b) search it automatically. Si et al. (2024c) organise PEFT as subspace tuning relative to the SVD frame of  $W_0$ , which is close in spirit to our transport of structure (Proposition II.13), and Fu et al. (2022) read many methods as sparse fine-tuning of an equivalent model. Recent comparisons under careful learning-rate tuning (He et al., 2026a; Lee et al., 2026) find that vanilla LoRA matches most of its variants; Corollary III.11 gives one exact mechanism behind such coincidences. Libraries made the field comparable in practice: AdapterHub (Pfeiffer et al., 2020b), Adapters (Poth et al., 2023), LLM-Adapters (Hu et al., 2023) and Hugging Face PEFT (Mangrulkar et al., 2022), whose source settled many of the conventions used here.

These works classify methods by where a module is inserted and how it is composed, and compare them by benchmark. We classify by invariants of the map  $\rho$  into weight space. Two methods with different code can then be shown to have the same image, as DoRA and LoRA followed by an output scale do when  $W_0$  has no zero row (Proposition II.8). Two methods can be shown incomparable, as LoRA and GraLoRA are at equal budget (Proposition II.12) and zero-init LoRA and Kronecker sums with few terms are (Theorem II.11). Each of the seven coordinates of Exposé VIII is read off a formula and tied to a result.

**Low-rank adaptation and its variants.** LoRA (Hu et al., 2021) trains  $W_0 + sBA$  from  $B_0 = 0$ . Its variants change the scale (rsLoRA, Kalajdzievski, 2023), the per-factor learning rates (LoRA+, Hayou et al., 2024b), which factor is trained (LoRA-FA, Zhang et al., 2023d; the asymmetry study of Zhu et al., 2024b), or the split into magnitude and direction (DoRA, Liu et al., 2024b, after weight normalisation, Salimans and Kingma, 2016; BiDoRA, Qin et al.,

2024b; DeLoRA, Bini et al., 2025). Others change the shape of the factors: shared or random bases (VeRA, Kopiczko et al., 2023; NOLA, Koohpayegani et al., 2023; VB-LoRA, Li et al., 2024d; Tied-LoRA, Renduchintala et al., 2023; PRoLoRA, Wang et al., 2024g; RandLoRA, Albert et al., 2025a; Uni-LoRA, Li et al., 2025a), a small core between frozen or orthonormal frames (LoRA-XS, Bałazy et al., 2024; SVFT, Lingam et al., 2024; LoRA-SB, Ponkshe et al., 2024; PoLAR, Lion et al., 2025; TinyLoRA, Morris et al., 2026), a single factor and its transpose (SingLoRA, Bensaïd et al., 2025), or one matrix shared across input shards (MiSS, Kang and Yin, 2024).

A further group changes the training procedure. ReLoRA (Lialin et al., 2023b), COLA (Xia et al., 2024), PeriodicLoRA (Meng et al., 2024b) and ScaLoRA (Zhang et al., 2025c) merge and restart; Delta-LoRA (Zi et al., 2023) also updates  $W_0$ ; LoRA-Pro (Wang et al., 2024i), AltLoRA (Yu et al., 2025) and LoFT (Tastan et al., 2025) match the full-model gradient or optimizer state; Flat-LoRA (Li et al., 2024c) seeks flat minima. Schulman and Thinking Machines Lab (2025) map a low-regret regime in which LoRA matches full fine-tuning, and prove the LoRA/Adam case of [Theorem III.10](#). Our results place several of these variants exactly: LoRA-XS is the categorical product of the two one-sided methods obtained by freezing LoRA’s frames at the SVD frames of  $W_0$  ([Proposition I.8](#)), LoRA+ and rsLoRA are points of a hyperparameter torus under the stated optimizer conditions ([Corollary III.11](#)), ReLoRA reaches exactly the updates of rank at most  $\min(Kr, m, n)$  after  $K$  cycles ([Theorem V.3](#)), and on the full-rank locus the SGD variant of LoRA-Pro follows, under gradient flow, the Riemannian gradient flow on  $W_0 + \mathcal{M}_r$  ([Proposition III.4](#)).

**Multiplicative and orthogonal methods.** Orthogonal fine-tuning (OFT, Qiu et al., 2023) multiplies  $W_0$  by a learned orthogonal matrix to preserve hyperspherical energy (Liu et al., 2018), following orthogonal over-parametrised training (Liu et al., 2020). BOFT (Liu et al., 2023b) builds the rotation from butterfly factors (Dao et al., 2019); OFTv2 (Qiu et al., 2025) rotates inputs and replaces the Cayley inverse by a Neumann series; GOFT and qGOFT (Ma et al., 2024) use Givens rotations, ETHER (Bini et al., 2024) hyperplane reflections and HRA (Yuan et al., 2024) products of Householder reflections, a parametrisation that goes back to Mhammedi et al. (2016). PSOFT (Wu et al., 2025a) confines the rotation to a principal subspace, LoCO (Nguyen et al., 2026) uses Cayley rotations of low-rank skew generators, and LOFT (Zhao et al., 2026) chooses the support of the rotation from the task. Spectral methods rotate or rescale the singular frame of  $W_0$  (SVDiff, Han et al., 2023; Spectral Adapter, Zhang and Pilanci, 2024b; SVF, Sun et al., 2025), and POET-X (Qiu et al., 2026) trains weights as orthogonal equivalence transforms of a fixed matrix. Lie-algebra charts such as the exponential and Cayley maps are studied by Lezcano-Casado and Martínez-Rubio (2019), and LieRA (Si et al., 2025) lifts multiplicative updates to tensors through the exponential map.

Diagonal and affine scalings form the other half of the multiplicative family: (IA)<sup>3</sup> (Liu et al., 2022a), SSF (Lian et al., 2022), RED (Wu et al., 2024a) and LayerNorm tuning (Zhao et al., 2023), preceded by FiLM (Perez et al., 2017). RoAd (Liao and Monz, 2024) applies  $2 \times 2$  rotations, and HiRA (Huang et al., 2025b) and ABBA (Singhal et al., 2025) use Hadamard products with  $W_0$  or with a second learned factor. These papers motivate orthogonality and scaling by what they preserve. We make the preserved quantities exact. The complete invariants of the basic actions ([Theorem II.5](#)) say what an exactly orthogonal method can never change in exact arithmetic ([Corollary II.6](#)), also after any number of merges ([Theorem V.3](#)), and they separate exact methods from approximately orthogonal implementations such as the Cayley-

Neumann default of HF PEFT. HRA’s bridge between low-rank and orthogonal adaptation becomes, for  $W_0$  of full column rank and even  $r \leq n - 2$ , an image-level meet with an apex at the default initialisation ([Theorem II.7](#)), and the torus ladder ([Proposition II.9](#)) places one- and two-sided rescalings inside the images of rank-one and rank-two HiRA.

**Structured shapes.** A second line changes the shape of the update. Kronecker products enter with PHM layers ([Zhang et al., 2021](#)) and Compacter ([Mahabadi et al., 2021a](#)), and KronA ([Edalati et al., 2022](#)), KAdaptation ([He et al., 2022](#)), LoKr ([Yeh et al., 2023](#)) and KRAdapter ([Albert et al., 2025b](#)) apply them to weight updates. Hadamard products of low-rank factors come from FedPara ([Hyeon-Woo et al., 2021](#)), known as LoHa in LyCORIS. Tensor-network updates include LoRETTA ([Yang et al., 2024d](#)), TT-LoRA ([Anjum et al., 2024](#)), LoTR ([Bershatsky et al., 2024](#)), FacT ([Jie and Deng, 2022b](#)), the Tucker-core FLoRA ([Si et al., 2024b](#)), LoRTA ([Hounie et al., 2024](#)), MetaTT ([Lopez-Piqueres et al., 2025](#)) and QuanTA ([Chen et al., 2024f](#)), resting on the decompositions surveyed by [Kolda and Bader \(2009\)](#), the tensor train of [Oseledets \(2011\)](#) and the tensor networks of [Cichocki et al. \(2016\)](#) and [Wang et al. \(2023a\)](#). Square and block shapes appear in MoRA ([Jiang et al., 2024](#)), MELoRA ([Ren et al., 2024](#)), GraLoRA ([Jung et al., 2025](#)), MoRe ([Tan et al., 2024a](#)), built on Monarch matrices ([Dao et al., 2022](#); [Qiu et al., 2024](#)), and DiaBlo ([Gurses et al., 2025](#)). Fixed-frame coordinate methods include FourierFT ([Gao et al., 2024](#)), WaveFT ([Bilican et al., 2025](#)), LoCA ([Du et al., 2025](#)), SSH ([Shen et al., 2025](#)) and the circulant C3A ([Chen et al., 2024a](#)). Sparse supports appear in SHiRA ([Bhardwaj et al., 2024](#)), RoSA ([Nikdan et al., 2024](#)), SLTrain ([Han et al., 2024a](#)) and DSEE ([Chen et al., 2021](#)), and in the selective methods BitFit ([Ben-Zaken et al., 2021](#)), diff pruning ([Guo et al., 2020](#)) and FISH ([Sung et al., 2021](#)); Sine-LoRA ([Ji et al., 2024](#)) raises rank with an elementwise nonlinearity.

Many of these papers aim for high rank at a small budget, and we separate the two quantities in that aim. Rank is bounded by a min-cut of a tensor network ([Theorem II.10](#)), and dimension by the budget minus the generic gauge orbit ([Proposition II.12](#)), so a method can gain rank at the same dimension, as GraLoRA does, or gain rank while it loses dimension, as LoHa does in the regime  $2r^2 < m + n - 1$ . The rearrangement of [Van Loan and Pitsianis \(1993\)](#) makes unconstrained Kronecker sums LoRA across a different cut of the same tensor ([Theorem II.11](#)), and the gauges and dimensions of Tucker and tensor-train shapes are those of [Koch and Lubich \(2010\)](#), [Uschmajew and Vandereycken \(2013\)](#) and [Holtz et al. \(2012\)](#).

**Initialisation and rank allocation.** Initialisation methods choose the frames that zero-initialised LoRA leaves free, or split  $W_0$ . PiSSA ([Meng et al., 2024a](#)) and MiLoRA ([Wang et al., 2024b](#)) move the principal or minor singular components of  $W_0$  into the adapter, OLoRA ([Büyükkakyüz, 2024](#)) uses a QR split, and CorDA ([Yang et al., 2024c](#)) and Astra ([Liu et al., 2026b](#)) use activation statistics. LoRA-GA ([Wang et al., 2024e](#)), LoRA-One ([Zhang et al., 2025d](#)) and GoRA ([He et al., 2025a](#)) align the adapter with the first gradient, EVA ([Paischer et al., 2024](#)) with activations, KaSA ([Wang et al., 2024a](#)) and MiCA ([Rüdiger and Raschka, 2026](#)) with minor singular directions, LoRA-Null ([Tang et al., 2025](#)) with a null space of activations, and KeepLoRA ([Luo et al., 2026](#)) with a residual subspace; [Zhang et al. \(2025g\)](#) argue that the magnitude of the update matters most. [Hayou et al. \(2024a\)](#) showed that the two zero-product starts, Init[A] and Init[B], train differently, and [Li et al. \(2024a\)](#) show that a Nyström initialisation changes the convergence rate of scaled gradient descent. Rank allocation is a separate axis, with AdaLoRA ([Zhang et al., 2023g](#)), DyLoRA ([Valipour et al., 2022](#)), SoRA ([Ding et al., 2023](#)), IncreLoRA ([Zhang et al., 2023a](#)), ALoRA ([Liu et al., 2024d](#)), AutoLoRA ([Zhang et al., 2024b](#)), ElaLoRA ([Chang et al., 2025](#)), LoRA-drop ([Zhou et al., 2024](#)), the dynamic-rank

DoRA of Mao et al. (2024a), FlexLoRA (Liu et al., 2026c) and MatryoshkaLoRA (Modoranu et al., 2026); PLoP (Hayou et al., 2025) chooses where adapters go.

These works compare initialisations by training outcomes. We first compare them as objects. The three strata of Theorem II.2 are told apart by  $S_1$  before any step is taken, zero-factor starts are apexes of a cone, and no exact pointing reaches rank above  $2r$  (Theorem II.1). The  $2r$  bound underlies the PiSSA-to-LoRA conversion of Meng et al. (2024a), and split initialisations attain it when  $2r \leq \min(m, n)$ . The optimizer then decides how much of the chosen frame training actually sees (Corollary III.13).

**Quantised bases and function-preserving reparametrisations.** QLoRA (Dettmers et al., 2023) trains LoRA over a 4-bit base, building on blockwise quantisation (Dettmers et al., 2021). LoftQ (Li et al., 2023b) fits the adapter to the quantisation error, and LQ-LoRA (Guo et al., 2023), IR-QLoRA (Qin et al., 2024a), ApiQ (Liao et al., 2024), LoQT (Loeschcke et al., 2024) and SRR (Cho et al., 2026) refine how the error is split; QA-LoRA (Xu et al., 2023b), PEQA (Kim et al., 2023b) and L4Q (Jeon et al., 2024) keep the merged model quantised. Outside PEFT, SmoothQuant (Xiao et al., 2023) and AWQ (Lin et al., 2024a) migrate scales through activations, SliceGPT (Ashkboos et al., 2024a) and QuaRot (Ashkboos et al., 2024b) rotate the residual stream without changing the network function, and SVD-LLM (Wang et al., 2025c) uses the whitened truncation that reappears in Proposition II.16. Base change (Exposé V) separates the two kinds of map. Function-preserving reparametrisations act on the slice and leave reachable function sets unchanged under the hypotheses of Observation II.15, while quantisers move the base point and leave explicit, order-dependent defects (Proposition V.2).

**Theory of LoRA and of fine-tuning.** *Intrinsic dimension.* Li et al. (2018) measured how many random directions a task needs, Aghajanyan et al. (2020) found that fine-tuning a language model needs few, Qin et al. (2021) found a shared low-dimensional subspace for soft prompts, and Gur-Ari et al. (2018) observed that gradients concentrate in a small subspace early in training. These notions depend on the data, while the expressive dimension  $d(M)$  is a property of the method alone. *Kernels.* Malladi et al. (2022) found that prompt-based fine-tuning often follows kernel dynamics, in which LoRA approximately preserves the kernel; the background is the neural tangent kernel (Jacot et al., 2018) and lazy training (Chizat et al., 2019). Jang et al. (2024) show that LoRA in the NTK regime has no spurious local minima, and Li et al. (2026b) extend the kernel view to zeroth-order fine-tuning. *Expressivity.* Zeng and Lee (2023) bound the rank LoRA needs to represent a smaller target network.

*Dynamics.* Hayou et al. (2024b) derive LoRA+ from an infinite-width analysis in the spirit of Yang and Hu (2021), and Chen et al. (2026a) extend it to joint limits in width and rank. Xu et al. (2025) analyse gradient flow for LoRA on matrix factorisation, Kang et al. (2026) spectral flows of Muon type, Zhang et al. (2026g) the role of the scale factor and Mu and Klabjan (2025) convergence rates of LoRA gradient descent. *Landscapes.* Kim et al. (2025) show that LoRA training converges either to a low-rank, small-magnitude global minimiser or to a qualitatively different solution, in the tradition of benign landscapes for factorised problems (Burer and Monteiro, 2003; Ge et al., 2017) and of implicit low-rank bias in matrix factorisation (Arora et al., 2019; Gunasekar et al., 2017).

Empirically, LoRA and full fine-tuning differ in what they learn and forget (Biderman et al., 2024; Shuttleworth et al., 2024), fine-tuning has its own scaling laws (Zhang et al., 2024a), and reinforcement-learning fine-tuning changes small and structured parts of the network (Mukherjee et al., 2025; Shenfeld et al., 2025; Zhu et al., 2025). Our dynamical results



are exact statements under stated optimizer hypotheses (Exposé III). They complement asymptotic and statistical analyses, and they say nothing about generalisation.

**Weight-space symmetry and conserved quantities.** The weight-space symmetries of tanh networks were found by Sussmann (1992) and Chen et al. (1993), and Godfrey et al. (2022) computed the linear maps that commute with each common activation; Lemma VI.2 restates these facts. Rescaling symmetries affect sharpness (Dinh et al., 2017) and motivate optimizers that respect them, such as Path-SGD (Neyshabur et al., 2015). Under gradient flow, symmetries give conservation laws: balancedness of linear layers (Arora et al., 2018; Du et al., 2018), the Noether-type laws of Kunin et al. (2020) and Tanaka and Kunin (2021), the conserved quantities of Zhao et al. (2022b), whose Prop. 5.1 is the core of Theorem III.5, the complete count of independent laws by Marcotte et al. (2023), and its extension beyond Euclidean flows, momentum included, by Marcotte et al. (2024). Symmetry teleportation (Zhao et al., 2022a) moves along symmetry orbits to speed up optimisation. Imbalance at initialisation governs the closed-form dynamics of Tarmoun et al. (2021) and Min et al. (2021), which Theorem III.9 re-derives for LoRA, after the exact solutions of Saxe et al. (2014) for deep linear networks; the lazy and rich regimes are analysed by Dominé et al. (2025) and Kunin et al. (2024).

Kempf–Ness theory (Kempf and Ness, 1979; Richardson and Slodowy, 1990) characterises closed orbits by norm-minimising points, which for LoRA are the balanced factorisations (Proposition III.7), with the nuclear-norm identity of Srebro et al. (2004); Balanced LoRA (Castin et al., 2026) turns this slice into an algorithm. Permutation symmetry is used to align and merge whole models (Ainsworth et al., 2022; Entezari et al., 2022; Singh and Jaggi, 2020) and to build networks that process weights (Navon et al., 2023; Zhou et al., 2023a), and Putterman et al. (2024) build learners on LoRA weights that respect the  $GL_r$  symmetry. Our additions are the PEFT instances. Per-block rates split LoRA’s gauge into isometries and conserved charges (Theorem III.5), the optimizer decides which part of the gauge descends to the weights (Theorem III.12), and combination rules must respect the gauge (Proposition V.4). FedRot-LoRA (Zhang et al., 2026c) and FLoRG (Meng et al., 2026) meet the same gauge in federated averaging, and PRISM (Wang and Zhang, 2026) in differentially private training (Abadi et al., 2016).

**Riemannian and preconditioned optimisation on low-rank manifolds.** Optimisation over fixed-rank matrices has a mature Riemannian theory (Absil et al., 2008; Vandereycken, 2013), including the quotient geometry of factorisations modulo  $GL_r$  (Mishra et al., 2012; Mishra and Sepulchre, 2016) and dynamical low-rank approximation (Koch and Lubich, 2010); natural gradients (Amari, 1998) and Riemannian gradient flows of deep linear networks (Bah et al., 2022) belong to the same circle of ideas. Scaled gradient descent (Tong et al., 2021) preconditions each factor by the Gram matrix of the other, and Zhang and Pilanci (2024a) bring it to LoRA. LoRA-RITE (Yen et al., 2024) makes invariance under reparametrisation of the factors a design requirement, LoRA-S (Zheng and Wu, 2026) lifts matrix-space optimizers through the quotient, Riemannion (Bogachev et al., 2025) takes a Muon step (Jordan et al., 2024) on  $\mathcal{M}_r$  directly, and PoLAR (Lion et al., 2025) trains orthonormal frames by Riemannian landing; Aliev and Rakhuba (2026) study the Riemannian structure of structured orthogonal classes. We sort these update rules, together with SGD, Adam (Kingma and Ba, 2015) and AdamW (Loshchilov and Hutter, 2019), by the largest subgroup of the gauge they respect (Theorem III.12). On LoRA’s full-rank locus with channel-uniform hyperparameters, this is  $O(r)$  for SGD, the signed permutations for Adam and AdamW, and all of  $GL_r$  for undamped

scaled GD and LoRA-RITE. [Proposition III.4](#) identifies which cometrics descend to the rank- $r$  manifold.

**Gradient projection and memory.** GaLore ([Zhao et al., 2024a](#)) projects gradients onto a low-rank subspace that is refreshed periodically. Its relatives include GaLore 2 ([Su et al., 2025](#)), Q-GaLore ([Zhang et al., 2024e](#)), Fira ([Chen et al., 2024e](#)), APOLLO ([Zhu et al., 2024a](#)), Online Subspace Descent ([Liang et al., 2024](#)), LDAdam ([Robert et al., 2025](#)) and LoRA-Pre ([Wang et al., 2026c](#)), which compresses the momentum; Flora ([Hao et al., 2024](#)) uses random projections, and [He et al. \(2024b\)](#) analyse when such methods converge. Memory is also saved by compressing optimizer state ([Dettmers et al., 2021](#); [Shazeer and Stern, 2018](#); [Zhang et al., 2024d](#)), by zeroth-order training (MeZO, [Malladi et al., 2023](#); see the benchmark of [Zhang et al., 2024c](#)), by fused updates (LOMO, [Lv et al., 2023b](#); AdaLomo, [Lv et al., 2023a](#)), by sampling layers (LISA, [Pan et al., 2024](#); BAdam, [Luo et al., 2024](#)), by side networks (LST, [Sung et al., 2022](#)) and by compressing activations (VeLoRA, [Miles et al., 2024](#)). [Torroba-Hennigen et al. \(2025\)](#) proved the duality between linear gradient transformations and adapters that [Proposition IV.2](#) restates in lens form with explicit optimizer hypotheses. The integrability condition ([Proposition IV.3](#)) and the backward cone ([Observation IV.4](#)) sort the rest of this family. In continual learning, gradient projection memory ([Saha et al., 2021](#)) and orthogonal gradient descent ([Farajtabar et al., 2019](#)) project updates away from the subspaces that earlier tasks use, an idea that O-LoRA ([Wang et al., 2023b](#)) and InfLoRA ([Liang and Li, 2024](#)) carry over to adapters.

**Prompts and prefixes, and their limits.** Soft prompts ([Lester et al., 2021](#); [Liu et al., 2021d](#)), deep prefixes ([Li and Liang, 2021](#); [Liu et al., 2021c](#)) and their variants ([Asai et al., 2022](#); [Razdaibiedina et al., 2023](#); [Shi and Lipani, 2023](#); [Vu et al., 2021](#); [Wang et al., 2023d](#); in vision, [Jia et al., 2022](#); [Khattak et al., 2022](#); [Zhou et al., 2021](#)) extend the network and leave its weights in place. [Petrov et al. \(2023\)](#) show that a prefix cannot change the relative attention pattern over the content and can only bias the output of an attention layer, while [Petrov et al. \(2024\)](#) show that prefix-tuning a pretrained transformer can still be a universal approximator. [Wang et al. \(2023c\)](#) prove universality and limitations of prompt tuning, [Oymak et al. \(2023\)](#) analyse prompt-attention, [Le et al. \(2024\)](#) the statistical benefit of the prefix reparametrisation, and [Galim et al. \(2024\)](#) show that for state-space models prefix tuning is bounded by initial-state tuning. [Chen et al. \(2022b\)](#) connect prefixes to adapters through kernel estimators, [Khashabi et al. \(2021\)](#) and [Su et al. \(2021\)](#) study what soft prompts encode and how they transfer, and [Mondal and Shihab \(2026\)](#) give exact resource-capped tests for when a prefix can reproduce a LoRA update at one head.

[Theorem VI.5](#) organises the attention side of this work as a monoid homomorphism followed by a perspective map. It recovers the gated-adapter reading of [He et al. \(2021\)](#) and the per-layer rigidity of [Petrov et al. \(2023\)](#), adds that LLaMA-Adapter’s ([Zhang et al., 2023h](#)) separate softmax leaves the content attention weights unchanged, and shows that for a vanilla softmax head with  $W_o \neq 0$  and  $W_q \neq 0$  a generic prefix cannot be folded into the head’s own weights. Representation edits (ReFT, [Wu et al., 2024d](#); RED, [Wu et al., 2024a](#); LoFiT, [Yin et al., 2024b](#)) and steering directions ([Li et al., 2023a](#); [Zou et al., 2023](#)) also act on activations, and [Adila et al. \(2026\)](#) relate weight updates to activation shifts directly; [Proposition VI.4](#) treats linear insertions and ReFT.

**Model merging and composition.** Weight averaging ([Wortsman et al., 2021, 2022](#)), Fisher-weighted merging ([Matena and Raffel, 2021](#)), RegMean ([Jin et al., 2022](#)), task arithmetic



(Ilharco et al., 2022) and its tangent-space analysis (Ortiz-Jimenez et al., 2023), TIES (Yadav et al., 2023), DARE (Yu et al., 2023), AdaMerging (Yang et al., 2023b) and TSV-Merge (Gargiulo et al., 2024) combine full fine-tuning updates, and MergeKit (Goddard et al., 2024) packages many of these rules. Linear mode connectivity (Frankle et al., 2019) and permutation alignment (Ainsworth et al., 2022; Entezari et al., 2022) help explain when averaging works, and surveys and large-scale studies map the field (Wang et al., 2025e; Yadav et al., 2024; Yang et al., 2024a). For adapters, LoraHub (Huang et al., 2023), LoRA Soups (Prabhakar et al., 2024), AdapterSoup (Chronopoulou et al., 2023), PEM arithmetic (Zhang et al., 2023c), ZipLoRA (Shah et al., 2023), Mix-of-Show (Gu et al., 2023) and orthogonal adaptation (Po et al., 2023) combine modules. KnOTS (Stoica et al., 2024), Core Space (Panariello et al., 2025) and Compress-then-Merge (He et al., 2026c) first express the modules in a shared frame, and OrthoMerge (Yang et al., 2026a) averages OFT rotations in the Lie algebra. Poly (Ponti et al., 2022), MHR (Caccia et al., 2022) and routed mixtures (Buehler and Buehler, 2024; Dou et al., 2023; Wu et al., 2024c) train combinations jointly, and serving systems such as S-LoRA (Sheng et al., 2023) and Punica (Chen et al., 2023b) keep many adapters separate at inference. Liu et al. (2026a) find that adaptive merging of recycled LoRAs gives limited benefit over training a new adapter.

We contribute two exact constraints to this literature. A rule for combining LoRA factors is a function of the adapters only when it is constant on factorisation fibres; task arithmetic on the products satisfies this, and post-hoc product-of-sums rules such as LoraHub’s do not (Proposition V.4). Folding a trained edit into existing weights is certified by a local rewrite calculus whose side conditions depend on the activation (Theorem VI.3); it gives sufficient conditions and box-local obstructions, and SmoothQuant and AWQ already use the same scale migration.

**Category theory and geometry for machine learning.** Fong et al. (2017) defined learners as the morphisms of a symmetric monoidal category, Fong and Johnson (2019) related learners to lenses, Cockett et al. (2019) axiomatised reverse derivative categories, and Cruttwell et al. (2021) gave a semantics of gradient-based learning in  $\mathbf{Para}(\mathbf{Lens}(C))$ , which we use directly for the lens of Exposé IV. Parametrised optics (Capucci et al., 2021), learners’ languages (Spivak, 2022), reverse derivative ascent (Wilson and Zanasi, 2021) and the categorical account of automatic differentiation by Elliott (2018) belong to the same programme, which Shiebler et al. (2021) survey. Gavranović et al. (2024) propose monads valued in a 2-category of parametric maps as an algebraic theory of architectures, developed at length in Gavranović (2024). Geometric deep learning (Bronstein et al., 2021), after group-equivariant (Cohen and Welling, 2016) and gauge-equivariant (Cohen et al., 2019) networks, organises architectures by their symmetries in the spirit of Klein’s programme (Klein, 1872). The calculus of string diagrams we use was put on a rigorous footing by Joyal and Street (1991), and the spiders we use for copying come from Coecke et al. (2013).

These works give categorical semantics to architectures and learning algorithms. We study a narrower object, the fine-tuning map relative to one fixed checkpoint, as an object of the slice  $\mathbf{Meth}(\Theta, \theta_0)$ . The category theory is used in the seven places listed in Exposé 0, and the rest of the work is real algebraic geometry, invariant theory, Lie theory and gradient-flow analysis. Our Erlangen language applies Klein’s idea to the fine-tuning map, where geometric deep learning applies it to architectures, and our gauge is the symmetry group of a parametrisation, with no connection, unlike the gauge of gauge-equivariant networks (Exposé XI).

## Conclusion

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Fine-tuning moves a point. Treating a PEFT method as a pointed map  $\rho : (Q, q_0) \rightarrow (\Theta, \theta_0)$  into weight space, and studying the map more than the module that implements it, turns many informal comparisons between methods into questions with exact answers. Three invariants of the map do most of the work. The image germ at  $\theta_0$  governs what a method can reach, through its first-order image  $S_1$ , its expressive dimension  $d(M)$ , its cut-rank and the type of its base point. The fibres, together with the optimizer’s metric, govern how it trains. Base change governs how it travels when it is initialised by splitting, quantised, restarted or merged. Extensions such as adapters and prefixes add one further layer, where the network itself enters and merging becomes a lifting problem.

Several results follow, each with the scope of its proof. Zero-initialised LoRA starts at the apex of a cone, and so does HRA at the paired initialisation of HF PEFT when  $W_0$  has full column rank and  $r \leq n - 2$  is even ([Theorem II.2](#), [Theorem II.7](#)). No exact initialisation lets LoRA reach an update of rank above  $2r$  ([Theorem II.1](#)). Exactly orthogonal methods never change the singular values of a weight in exact arithmetic, however often they are merged ([Corollary II.6](#), [Theorem V.3](#)). Rank is a cut and dimension is a volume, so at equal budget a structured method can gain rank at the same dimension, as GraLoRA does, or trade dimension for rank, as LoHa does when  $2r^2 < m + n - 1$  and  $r \geq 3$  ([Proposition II.12](#)). Under Adam with no  $\varepsilon$ , weight decay or clipping and one shared schedule, LoRA+ follows exactly the trajectory of LoRA with  $\alpha$  multiplied by the learning-rate ratio ([Corollary III.11](#); the LoRA/Adam case is due to [Schulman and Thinking Machines Lab, 2025](#)), and on LoRA’s full-rank locus Adam with channel-uniform hyperparameters respects only the signed permutations inside the  $GL_r$  gauge ([Theorem III.12](#)). Whether a scaling adapter folds into the weights before it depends on the activation, and a local rewrite calculus certifies when it does ([Theorem VI.3](#)). The atlas reads the same coordinates off 382 methods from 138 papers and records 403 simulation arrows and 485 obstructions, each backed by an explicit map or a named test.

Several of these results are short once the setting is in place, which is what the rising sea on the title page refers to ([Appendix F](#)). In the slice, re-pointing an additive method sweeps out the difference set  $C - C$  of its update shape, and for LoRA, that set is the matrices of rank at most  $2r$ . The bound on every exact initialisation is then a few lines ([Theorem II.1](#)). Once LoRA’s factors are seen to carry a torus of rescalings that fixes the weight trajectory, LoRA+ is a point on that torus ([Corollary III.11](#)). In the lens, GaLore without weight decay becomes one-sided LoRA on a moving base by a one-line induction ([Proposition IV.2](#)). Read off a tensor diagram, the maximum rank of a structured adapter is a min-cut ([Theorem II.10](#)). Three of these four facts were known before this paper, from the PiSSA-to-LoRA conversion ([Meng et al., 2024a](#)), [Schulman and Thinking Machines Lab \(2025\)](#) and [Torroba-Hennigen et al. \(2025\)](#). What the setting adds is a short proof, an exact scope and the same proof for every method of the same shape. Other results did not dissolve. The classification of initialisations ([Theorem II.2](#)) and rebasing closure ([Theorem V.3](#)) required real algebraic geometry and Lie theory, and their proofs say where.

For practitioners these statements become checks that can be made before training. Compare methods at equal expressive dimension and cut-rank as well as at equal budget.

Ask whether an initialisation is an apex, and which part of its frame the optimizer will see ([Corollary III.13](#)). Before treating a new hyperparameter as a new method, check whether it lies on a hyperparameter torus. Before merging adapters factor by factor, check that the rule is constant on factorisation fibres ([Proposition V.4](#)). Before relying on an orthogonal method to keep a spectrum, check that its implementation is exactly orthogonal. Implementations matter throughout. Several results distinguish a method as published from its HF PEFT implementation, such as OFT’s Cayley–Neumann default, HRA’s paired initialisation and LoftQ’s treatment of the scale  $\alpha/r$  ([Proposition V.2](#)), and the catalogue records which form each statement covers.

The limits are part of the statements. Expressivity is not learnability, and only [Exposé III](#) is dynamical. Its conservation laws hold under gradient flow, drift under discrete steps by an amount of second order in the step size relative to the charge, and do not cover Adam or momentum. The classification of initialisations is set-level and in exact arithmetic. Mergeability is certified box by box under explicit side conditions, and the rewrite rules give sufficient conditions and local obstructions; they are not a decision procedure for whole networks. Words such as Noether, Erlangen, gauge and moduli are analogies beyond what is proved, and [Exposé XI](#) says where each one stops.

The falsifiable predictions of [Exposé X](#) say where the theory can be tested, each with its optimizer scope. The open problems of [Exposé XI](#) say where it should grow next: a dynamical classification of initialisations modulo the optimizer’s shadow of the gauge, a conserved or monotone quantity for sign-like updates such as Adam’s, the frontier of pairwise incomparable methods at a fixed budget, and whole-network statements about prefix tuning. We hope the frame serves as a habit of asking three questions of any new method: what its map can reach, how its fibres meet the optimizer, and what happens to it when the base point moves.

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## Technical appendix

The appendices fall into three groups. Each can be read on its own, and every entry, proof and table points back to the place in the main text that uses it.

**Preliminaries.** Appendix A collects the mathematical background the exposés assume: categories, slices and base change; monoidal categories, string diagrams and lenses; manifolds, germs and cones; and groups, symmetry and conservation laws. Its 93 entries keep the numbers A.1–D.34 they carry on the companion site, and each states a notion with its hypotheses, shows it on a fine-tuning example and lists the results that rely on it. Appendix B quotes the external theorems that the proofs use, each in the exact form in which it is applied and with its source.

**Theory.** Appendix C gives complete proofs of the numbered results, in the order of the main text and with one subsection per result. Each subsection restates its result, in the form recorded after two rounds of review, before the proof.

**Data and reproducibility.** Appendix D describes the catalogue of 382 methods from 138 papers. It lists every method with its seven coordinates and its reference, and every one of the 403 recorded simulation arrows with its map  $h$ . Appendix E documents the numerical checks marked [Num] in the main text and explains how to rerun them from the data files that accompany this paper.

**Process.** Appendix F records how this paper and its companion site were made: who did what, how the literature was searched and checked, how the theory was generalised and refereed, and what the work cost in agent runs and tokens.

### APPENDIX A

## Mathematical background

This appendix is for readers who know linear algebra, multivariable calculus, basic probability and the PEFT literature, and who have not studied category theory, Lie groups or real algebraic geometry. It skips what ML readers already use, such as the SVD, rank, orthogonal matrices, eigenvalues, Jacobians, gradients, Adam, LoRA and the transformer block. Each of the 93 entries gives a short definition or statement with its hypotheses, a small example from fine-tuning, the places in the paper that use it, and usually a textbook for further reading. Definitions carry an ochre rule and standard facts quoted from the literature a teal one; the one remark, [Background A.20](#), is set like a definition.

Part A covers categories, slices and base change, Part B monoidal categories, string diagrams and lenses, Part C manifolds, cones and real algebraic geometry, and Part D groups, symmetry and conservation laws. The parts can be read in any order, since each entry refers to the entries it depends on at their first use. The exposés cite an entry at the first use of its term, as in [Background A.8](#), and each entry ends with the results that rely on it.

A few symbols from the exposés recur in the examples. A method  $M = (Q, q_0, \rho)$  is a smooth map  $\rho : Q \rightarrow \Theta$  from trainable parameters to weights with  $\rho(q_0) = \theta_0$  ([Back-](#)



ground A.8), and  $|M| = \dim Q$  is its parameter budget. The expressive dimension  $d(M)$ , the first-order image  $S_1$  and the first-order deficit are defined in Background C.3; the base point is *regular* for  $M$  when the deficit is 0 and an *apex* when it is positive. LoRA (Hu et al., 2021) is  $\rho(B, A) = W_0 + sBA$  with  $B \in \mathbb{R}^{m \times r}$  and  $A \in \mathbb{R}^{r \times n}$ , and its full-rank pointings fall into the strata  $T_A$ ,  $T_B$  and  $S$  of Background C.19. *Block-scalar rates* give each factor one learning rate,  $\eta_B$  for  $B$  and  $\eta_A$  for  $A$  (Background D.23).

Table 33 lists every entry with its area and the first places it is used.

**Table 33.** Index of the background entries, with the area of each and the first places in the paper that use it.

| No.  | Entry                                                          | Area            | First places used                                                                     |
|------|----------------------------------------------------------------|-----------------|---------------------------------------------------------------------------------------|
| A.1  | category                                                       | Categories      | Table 2, Def. I.2                                                                     |
| A.2  | commutative diagram                                            | Categories      | Figure 2, Def. I.2                                                                    |
| A.3  | isomorphism                                                    | Categories      | Obs. I.6, Obs. I.7                                                                    |
| A.4  | functor; functorial invariant                                  | Categories      | Obs. I.4(c), Obs. I.6                                                                 |
| A.5  | natural transformation; naturality                             | Categories      | Thm III.3, Prop. V.4                                                                  |
| A.6  | slice category                                                 | Categories      | Table 2, Def. I.2                                                                     |
| A.7  | coslice; pointed objects; opposite category                    | Categories      | Def. I.2, Rem. I.3                                                                    |
| A.8  | the pointed slice <b>Meth</b> ( $\Theta, \theta_0$ )           | Categories      | Table 2, Def. I.2                                                                     |
| A.9  | initial and terminal objects; universal property               | Limits          | Obs. I.4(a), Figure 2                                                                 |
| A.10 | product                                                        | Limits          | Obs. I.4(b), Prop. I.8                                                                |
| A.11 | pullback; fibre product                                        | Limits          | Obs. I.4(b), Prop. I.8                                                                |
| A.12 | limit; creation of limits                                      | Limits          | Obs. I.4(b)                                                                           |
| A.13 | adjunction                                                     | Limits          | Obs. I.4(b)                                                                           |
| A.14 | two standard facts about limits                                | Limits          | Obs. I.4(b), §I                                                                       |
| A.15 | monomorphism; sub-object                                       | Limits          | Def. I.2, Cor. III.13                                                                 |
| A.16 | posets; meets; the expressivity order                          | Limits          | Obs. I.4(c), Prop. I.8(b)                                                             |
| A.17 | automorphism of a category; transport of structure             | Base change     | Prop. II.13, Exposé II.C                                                              |
| A.18 | lift; lifting problem; obstruction                             | Base change     | Obs. I.4(c), Obs. I.6                                                                 |
| A.19 | base change; pushforward $\phi_*$ ; the relative point of view | Base change     | Exposés 0–I, Def. V.1                                                                 |
| A.20 | constructions the paper names and leaves unused                | Base change     | Exposé 0 (where the category theory earns its keep), Exposé XI (seven borrowed words) |
| B.1  | monoidal and symmetric monoidal categories                     | Monoidal        | Obs. I.7, Thm II.10                                                                   |
| B.2  | direct sum and tensor product in <b>FinVect</b>                | Monoidal        | Thm II.10, Exposé II.C                                                                |
| B.3  | string diagrams                                                | Monoidal        | Exposé II.C, Thm II.10                                                                |
| B.4  | cartesian categories; copy and discard                         | Monoidal        | Def. I.1, Thm VI.3                                                                    |
| B.5  | compact closed categories; cups, caps and wire-bending         | Monoidal        | Thm II.11, Thm VI.3                                                                   |
| B.6  | Frobenius algebras and spiders                                 | Monoidal        | Exposé II.C, Thm II.10                                                                |
| B.7  | tensor networks, contraction, bond dimension                   | Tensor networks | Thm II.10, Exposé II.C                                                                |
| B.8  | the min-cut bound on rank                                      | Tensor networks | Thm II.10, Exposé II.C                                                                |

*continued on the next page*

| No.  | Entry                                                                       | Area            | First places used                                                                                             |
|------|-----------------------------------------------------------------------------|-----------------|---------------------------------------------------------------------------------------------------------------|
| B.9  | tensor trains, Tucker and MPO;<br>gauge on bonds                            | Tensor networks | <a href="#">Prop. II.12</a> , <a href="#">Exposé II.C</a>                                                     |
| B.10 | Kronecker product, vec, tensor<br>product of subspaces                      | Tensor networks | <a href="#">Thm II.11</a> , <a href="#">Exposé II.C</a>                                                       |
| B.11 | Van Loan–Pitsianis rearrangement;<br>Kronecker rank                         | Tensor networks | <a href="#">Thm II.11</a> , <a href="#">Prop. II.13</a>                                                       |
| B.12 | Schmidt and operator-Schmidt<br>decompositions                              | Tensor networks | <a href="#">Thm II.11</a> , <a href="#">Figure 6</a>                                                          |
| B.13 | rank of a Hadamard product;<br>face-splitting product                       | Tensor networks | <a href="#">Prop. II.12</a> , <a href="#">Thm II.10</a>                                                       |
| B.14 | sums of methods; Minkowski sums,<br>differences and translates              | Para, lenses    | <a href="#">Obs. I.7</a> , <a href="#">Thm II.1</a>                                                           |
| B.15 | strong monoidal functors; $(\mathbb{N}, +)$<br>and the permutation groupoid | Para, lenses    | <a href="#">Obs. I.7</a> , <a href="#">Exposé XI</a> (the referee record)                                     |
| B.16 | bicategories and the Para<br>construction                                   | Para, lenses    | <a href="#">Def. I.1</a> , <a href="#">Rem. I.3</a>                                                           |
| B.17 | lenses, reverse derivatives and<br><b>Para</b> (R)                          | Para, lenses    | <a href="#">Exposé IV</a> , <a href="#">Def. IV.1</a>                                                         |
| C.1  | smooth manifolds and smooth<br>maps                                         | Manifolds       | <a href="#">Def. I.2</a> , <a href="#">Def. I.5</a>                                                           |
| C.2  | tangent spaces, differentials and<br>covectors                              | Manifolds       | <a href="#">Table 2</a> , <a href="#">Def. I.5</a>                                                            |
| C.3  | rank, expressive dimension and<br>first-order image                         | Manifolds       | <a href="#">Table 2</a> , <a href="#">Def. I.5</a>                                                            |
| C.4  | immersions, submersions and<br>embedded submanifolds                        | Manifolds       | <a href="#">Obs. I.4</a> , <a href="#">Prop. IV.3</a>                                                         |
| C.5  | constant-rank theorem                                                       | Manifolds       | <a href="#">Def. I.5</a> , <a href="#">Prop. II.12</a>                                                        |
| C.6  | fibres and fibre dimension                                                  | Manifolds       | <a href="#">Exposé III</a> , <a href="#">Def. I.5</a>                                                         |
| C.7  | fibre products, transversality and<br>clean intersection                    | Manifolds       | <a href="#">Obs. I.4</a> , <a href="#">Prop. I.8</a>                                                          |
| C.8  | germs                                                                       | Singularities   | <a href="#">Thm II.7</a> , <a href="#">Exposé II</a> (the image germ)                                         |
| C.9  | smooth and singular points                                                  | Singularities   | <a href="#">Def. I.5</a> , <a href="#">Thm II.2</a>                                                           |
| C.10 | cones and vertices                                                          | Singularities   | <a href="#">Def. I.5</a> , <a href="#">Thm II.1</a>                                                           |
| C.11 | first-order image versus image                                              | Singularities   | <a href="#">Def. I.5</a> , <a href="#">Obs. I.6</a>                                                           |
| C.12 | the determinantal variety                                                   | Varieties       | <a href="#">Table 2</a> , <a href="#">Thm II.1</a>                                                            |
| C.13 | the skew-determinantal cone                                                 | Varieties       | <a href="#">Thm II.7</a> , <a href="#">§II.B</a>                                                              |
| C.14 | Segre and toric cones                                                       | Varieties       | <a href="#">Prop. II.8</a> , <a href="#">Prop. II.12</a>                                                      |
| C.15 | the Grassmannian                                                            | Varieties       | <a href="#">Thm II.2</a> , <a href="#">Thm II.7</a>                                                           |
| C.16 | images of smooth maps need not<br>be manifolds                              | Tame geometry   | <a href="#">Obs. I.4</a> , <a href="#">Thm II.2</a>                                                           |
| C.17 | semialgebraic and definable sets                                            | Tame geometry   | <a href="#">Exposé 0</a> (where the category theory earns<br>its keep), <a href="#">§VIII.C</a> (arrow rules) |
| C.18 | Tarski–Seidenberg and definable<br>dimension                                | Tame geometry   | <a href="#">Def. I.5</a> , <a href="#">§VIII.C</a> (arrow rules)                                              |
| C.19 | stratification and local dimension                                          | Tame geometry   | <a href="#">Thm II.2</a> , <a href="#">Thm II.7</a>                                                           |
| C.20 | moduli sets and moduli spaces                                               | Tame geometry   | <a href="#">Thm II.2</a> , <a href="#">Exposé IX</a>                                                          |
| C.21 | generic properties                                                          | Tame geometry   | <a href="#">Thm II.2</a> , <a href="#">Prop. II.8</a>                                                         |
| C.22 | real-analytic functions and their<br>zero sets                              | Tame geometry   | <a href="#">Lem. VI.2</a> , <a href="#">Thm VI.5</a>                                                          |
| D.1  | Lie groups; the matrix groups of<br>this paper                              | Lie theory      | <a href="#">Table 2</a> , <a href="#">Def. I.5</a>                                                            |

*continued on the next page*

| No.  | Entry                                                                                     | Area           | First places used                                                       |
|------|-------------------------------------------------------------------------------------------|----------------|-------------------------------------------------------------------------|
| D.2  | tori, diagonal monoids and saturation                                                     | Lie theory     | <a href="#">Thm II.5</a> , <a href="#">Prop. II.8</a>                   |
| D.3  | permutations, monomial matrices, semidirect products, normalisers                         | Lie theory     | <a href="#">Prop. II.8</a> , <a href="#">Prop. II.9</a>                 |
| D.4  | connectedness and the identity component                                                  | Lie theory     | <a href="#">Thm V.3</a> , <a href="#">§II.B</a>                         |
| D.5  | Lie algebras; the commutator; $\mathfrak{gl}$ , $\mathfrak{so}$ , $\text{Sym}$ , $\wedge$ | Lie theory     | <a href="#">Thm II.7</a> , <a href="#">Thm III.5</a>                    |
| D.6  | the exponential map; one-parameter subgroups                                              | Lie theory     | <a href="#">Thm III.5</a> , <a href="#">Prop. III.7</a>                 |
| D.7  | the Cayley map and its Neumann truncation                                                 | Lie theory     | <a href="#">Cor. II.6</a> , <a href="#">Thm II.7</a>                    |
| D.8  | Lie subgroups and subalgebras; the generated Lie algebra                                  | Lie theory     | <a href="#">Thm V.3</a> , <a href="#">Figure 10</a>                     |
| D.9  | Yamabe’s theorem                                                                          | Lie theory     | <a href="#">Thm V.3</a> , <a href="#">§V</a>                            |
| D.10 | monoids, units, monoid orbits, homomorphisms                                              | Lie theory     | <a href="#">Thm V.3</a> , <a href="#">Prop. II.9</a>                    |
| D.11 | Cartan–Dieudonné theorem; Scherk’s sharp form                                             | Lie theory     | <a href="#">Thm II.7</a> , <a href="#">§II.B</a>                        |
| D.12 | group actions                                                                             | Invariants     | <a href="#">Def. I.2</a> , <a href="#">Table 2</a>                      |
| D.13 | orbits, stabilisers, free actions                                                         | Invariants     | <a href="#">Thm II.5</a> , <a href="#">Prop. II.9</a>                   |
| D.14 | Haar measure; invariant measures on homogeneous spaces                                    | Invariants     | <a href="#">Thm II.2</a> , <a href="#">§II.A</a>                        |
| D.15 | invariants and complete invariants                                                        | Invariants     | <a href="#">Thm II.5</a> , <a href="#">§II.B</a>                        |
| D.16 | first fundamental theorem for $O(n)$ ; Witt’s extension theorem                           | Invariants     | <a href="#">Thm II.5</a> , <a href="#">Exposé XII</a>                   |
| D.17 | Klein’s Erlangen programme                                                                | Invariants     | <a href="#">Def. II.14</a> , <a href="#">Exposé II.B</a>                |
| D.18 | equivariance and invariance of maps, update rules and merge rules                         | Invariants     | <a href="#">Def. II.14</a> , <a href="#">Obs. II.15</a>                 |
| D.19 | multihomogeneous maps; multidegree                                                        | Invariants     | <a href="#">Thm III.10</a> , <a href="#">Exposé II</a> (the image germ) |
| D.20 | orbit space of a positive torus                                                           | Invariants     | <a href="#">Thm III.10</a> , <a href="#">Cor. III.11</a>                |
| D.21 | gauge symmetry of a parametrisation                                                       | Gauge, metrics | <a href="#">Def. I.5</a> , <a href="#">Table 2</a>                      |
| D.22 | quotients and descent; the quotient manifold theorem                                      | Gauge, metrics | <a href="#">Prop. V.4</a> , <a href="#">Prop. III.4</a>                 |
| D.23 | metrics, cometrics and the pulled-back cometric                                           | Gauge, metrics | <a href="#">Def. III.1</a> , <a href="#">Obs. III.2</a>                 |
| D.24 | Riemannian gradient flow; tangent projection; retraction                                  | Gauge, metrics | <a href="#">Prop. III.4</a> , <a href="#">Thm III.12</a>                |
| D.25 | distributions, Lie brackets and Frobenius’ theorem                                        | Gauge, metrics | <a href="#">Prop. IV.3</a> , <a href="#">§IV</a>                        |
| D.26 | Noether’s theorem for gradient flows                                                      | Conservation   | <a href="#">Thm III.5</a> , <a href="#">Figure 7</a>                    |
| D.27 | Cartan decomposition; polar decomposition                                                 | Conservation   | <a href="#">Thm III.5</a> , <a href="#">Prop. III.7</a>                 |
| D.28 | balancedness; the conserved $B^\top B - AA^\top$                                          | Conservation   | <a href="#">Prop. III.7</a> , <a href="#">Prop. III.8</a>               |
| D.29 | the moment map                                                                            | Conservation   | <a href="#">Prop. III.7</a> , <a href="#">Cor. III.6</a>                |
| D.30 | Kempf–Ness theorem, real form                                                             | Conservation   | <a href="#">Prop. III.7</a> , <a href="#">Cor. III.6</a>                |
| D.31 | the closed orbit in each fibre of $(B, A) \mapsto BA$                                     | Conservation   | <a href="#">Prop. V.4</a> , <a href="#">§V</a>                          |

continued on the next page

| No.  | Entry                                    | Area         | First places used                    |
|------|------------------------------------------|--------------|--------------------------------------|
| D.32 | Ky Fan’s maximum principle               | Matrix tools | <a href="#">Prop. II.3, §II.B</a>    |
| D.33 | Sylvester equations                      | Matrix tools | <a href="#">Thm III.12, Figure 9</a> |
| D.34 | the perspective map; convex combinations | Matrix tools | <a href="#">Thm VI.5, Figure 12</a>  |

## Part A. Categories without tears

Part A collects the category theory the exposés use, from the definition of a category up to base change, and illustrates every notion in one running example, the pointed slice  $\mathbf{Meth}(\Theta, \theta_0)$  of weight-space methods. Entries [Background A.1](#) to [Background A.8](#) set up categories, functors and slices, [Background A.9](#) to [Background A.16](#) cover universal properties, limits and orders, and [Background A.17](#) to [Background A.19](#) cover transport, lifting and base change, with [Background A.20](#) listing the constructions the paper names and leaves unused.

### Categories and slices

**BACKGROUND A.1** (category). A **category**  $C$  has objects and, for any two objects  $X, Y$ , a set  $C(X, Y)$  of **morphisms**, or arrows,  $f : X \rightarrow Y$ . Arrows  $f : X \rightarrow Y$  and  $g : Y \rightarrow Z$  have a composite  $g \circ f : X \rightarrow Z$ . Composition is associative, and every object has an identity arrow  $\text{id}_X$  with  $f \circ \text{id}_X = f = \text{id}_Y \circ f$ .

The paper uses **Set** (sets and functions), **Diff** (smooth manifolds and smooth maps), **FinVect** (finite-dimensional real vector spaces and linear maps, where tensor networks are evaluated) and the pointed categories  $\mathbf{Set}_*$  and  $\mathbf{Diff}_*$  of [Background A.7](#). In  $\mathbf{Meth}(\Theta, \theta_0)$  the objects are methods  $M = (Q, q_0, \rho)$ , and an arrow  $h : M \rightarrow N$  is a pointed smooth map  $h : Q_M \rightarrow Q_N$  with  $\rho_N \circ h = \rho_M$ , read “ $N$  simulates  $M$ ”. Simulations compose, since  $\rho_P \circ k \circ h = \rho_N \circ h = \rho_M$  for  $h : M \rightarrow N$  and  $k : N \rightarrow P$ . For instance,  $h(B) = (B, A_0)$  is an arrow from  $\text{LoRA-FA}(A_0)$  ([Zhang et al., 2023d](#)) to  $\text{LoRA}$  pointed at  $(0, A_0)$ . The simulation arrows of the atlas are arrows of this category, each recorded with its map  $h$ .

**Used in.** Table 2, [Definition I.2](#), Exposés 0–I, [§VIII.C](#), [Figure 14](#). Reading: [Leinster \(2014, Ch. 1\)](#).

**BACKGROUND A.2** (commutative diagram). A **diagram** is a family of objects joined by arrows, drawn as a directed graph. It **commutes** if any two directed paths with the same start and the same end compose to the same arrow. A triangle commutes when one side equals the composite of the other two, and a square commutes when its two routes between opposite corners agree.

A simulation is a commuting triangle over  $\Theta$ , whose two paths from  $Q_M$  to  $\Theta$  are

$$Q_M \xrightarrow{h} Q_N \xrightarrow{\rho_N} \Theta \quad \text{and} \quad Q_M \xrightarrow{\rho_M} \Theta.$$

For  $M = \text{FA}(A_0)$ ,  $N$   $\text{LoRA}$  pointed at  $(0, A_0)$  and  $h(B) = (B, A_0)$ , both paths send  $B$  to the weight  $W_0 + BA_0$  (scale absorbed into  $B$ ), so the triangle commutes. [Figure 2\(a\)](#) draws, for a general method, the triangle of [Figure 15](#), which commutes only up to a pointing defect when the method starts away from  $\theta_0$  ([Background A.19](#)). Commuting squares, used for merging on extensions, return in [Background A.18](#).

**Used in.** [Figure 2](#), [Definition I.2](#), [Definition VI.1](#), [Definition V.1](#), [Exposé V](#). Reading: [Leinster \(2014, §1.1\)](#).

**BACKGROUND A.3** (isomorphism). An arrow  $f : X \rightarrow Y$  is an **isomorphism**, written  $X \cong Y$ , if some  $g : Y \rightarrow X$  satisfies  $g \circ f = \text{id}_X$  and  $f \circ g = \text{id}_Y$ . The inverse  $g$  is then unique. In **Diff** the isomorphisms are the diffeomorphisms, so a smooth bijection need not be one, as  $t \mapsto t^3$  shows. In **Meth** an isomorphism  $M \cong N$  is a pointed diffeomorphism  $h : Q_M \rightarrow Q_N$  with  $\rho_N \circ h = \rho_M$ . Its inverse lies over  $\Theta$  automatically, since  $\rho_M \circ h^{-1} = \rho_N$ .

The LoRA gauge ([Background D.21](#))  $(B, A) \mapsto (Bg^{-1}, gA)$ ,  $g \in \text{GL}_r$ , is an isomorphism from LoRA pointed at  $(B_0, A_0)$  to LoRA pointed at  $(B_0g^{-1}, gA_0)$ . Other examples are rsLoRA ([Kalajdziewski, 2023](#))  $\cong$  LoRA by  $(B, A) \mapsto (\sqrt{r}B, A)$ , and  $\text{LoRA}_r \oplus \text{LoRA}_s \cong \text{LoRA}_{r+s}$  by concatenating factors ([Background B.14](#)). Isomorphic methods have the same image, the same  $S_1$  and the same first-order deficit ([Background C.3](#), [Observation I.6](#)). They can still train differently, because the optimizer acts on the coordinates of  $Q$ . For LoRA's gauge with block-scalar rates (one rate per factor), gradient flow is unchanged exactly when  $g \in \text{O}(r)$ , at points where  $A$  has full row rank and  $r < n$  ([Theorem III.3](#); [Background D.23](#)).

**Used in.** [Observation I.6](#), [Observation I.7](#), [§II.A](#), [Theorem II.2](#), [Proposition VI.4](#), [§VIII.C](#).

**BACKGROUND A.4** (functor; functorial invariant). A **functor**  $F : \mathcal{C} \rightarrow \mathcal{D}$  sends each object  $X$  to an object  $FX$  and each arrow  $f : X \rightarrow Y$  to an arrow  $Ff : FX \rightarrow FY$ , with  $F(g \circ f) = Fg \circ Ff$  and  $F(\text{id}_X) = \text{id}_{FX}$ . A functor sends isomorphisms to isomorphisms, so  $FX$  is an isomorphism invariant of  $X$ . A **forgetful** functor discards structure, as  $\mathbf{Diff}_* \rightarrow \mathbf{Diff}$  discards the base point.

The image is a functor  $\text{Im}$  from  $\mathbf{Meth}(\Theta, \theta_0)$  to the subsets of  $\Theta$  containing  $\theta_0$ , ordered by inclusion (a poset, [Background A.16](#)), because an arrow  $h : M \rightarrow N$  gives  $\text{Im } M = \rho_N(h(Q_M)) \subseteq \text{Im } N$ . In the same way the chain rule at  $q_0$  makes  $S_1$  a functor to the linear subspaces of  $\Theta$ , with  $S_1(M) \subseteq S_1(N)$  along every pointed arrow. Read backwards, functoriality gives the atlas test O1 and the first half of O3 ([§VIII.C](#)). If  $\text{Im } M \not\subseteq \text{Im } N$  or  $S_1(M) \not\subseteq S_1(N)$ , then  $N$  does not simulate  $M$ . Base change  $\phi_!$  ([Background A.19](#)) and transport  $T_*$  ([Background A.17](#)) are functors between slices.

**Used in.** [Observation I.4\(c\)](#), [Observation I.6](#), [§VIII.C](#), [Proposition II.13](#), [Definition V.1](#), [§IV](#). Reading: [Leinster \(2014, §1.2\)](#).

**BACKGROUND A.5** (natural transformation; naturality). For functors  $F, G : \mathcal{C} \rightarrow \mathcal{D}$ , a **natural transformation**  $\alpha : F \Rightarrow G$  is a family of arrows  $\alpha_X : FX \rightarrow GX$  with  $Gf \circ \alpha_X = \alpha_Y \circ Ff$  for every  $f : X \rightarrow Y$ . This naturality square says that applying  $\alpha$  and then moving along  $f$  agrees with moving first and applying  $\alpha$  after. A group  $K$  can be viewed as a category with one object whose arrows are the elements of  $K$ . A functor from it to **Set** is then a set with a  $K$ -action, and a natural transformation between two such functors is exactly a  $K$ -equivariant map.

The copy map  $\Delta_X(x) = (x, x)$  is natural,  $\Delta_Y \circ f = (f \times f) \circ \Delta_X$ , which is rule R7 of the merge calculus ([Background B.4](#)). Elsewhere in the paper, natural for a group of changes of chart means equivariant in this sense ([Background D.18](#)). Task arithmetic ([Ilharco et al., 2022](#)) is natural for all of  $\text{GL}(\Theta)$ , and DARE ([Yu et al., 2023](#)) for invertible diagonal matrices at a fixed mask ([Proposition V.4\(c\)](#)). Gradient flow on LoRA commutes with the gauge  $g$  exactly



when the induced cometrics (Background D.23) agree, which for block-scalar rates singles out  $O(r)$  (Theorem III.3).

**Used in.** Exposé 0 (where the category theory earns its keep), Theorem III.3, §III, Proposition V.4, §VI, Theorem VI.3. Reading: Leinster (2014, §1.3).

**BACKGROUND A.6** (slice category). For an object  $X$  of  $C$ , the **slice**  $C/X$  has as objects the arrows  $a : A \rightarrow X$ , called objects over  $X$ , and as morphisms  $(A, a) \rightarrow (B, b)$  the arrows  $h : A \rightarrow B$  of  $C$  with  $b \circ h = a$ . Composition and identities are those of  $C$ . The object  $X$  is the **base**, and  $a$  is the **structure map**.

$\mathbf{Diff}/\Theta$  consists of smooth maps  $\rho : Q \rightarrow \Theta$  into weight space without base points, and its morphisms are the weak morphisms of Definition I.2, which need not send  $q_0$  to  $q'_0$ . A slice always has a terminal object, the identity  $(X, \text{id}_X)$ , which here is full fine-tuning. Working in a slice is the relative point of view of Background A.19, in which an object is studied together with its map to a base.

**Used in.** Table 2, Definition I.2, Exposé 0 (where the category theory earns its keep), Rosetta entries 3 and 4 (§IX), the seven borrowed words of Exposé XI. Reading: Mac Lane (1998, §II.6).

**BACKGROUND A.7** (coslice; pointed objects; opposite category). The **coslice**  $X/C$  has as objects the arrows  $x : X \rightarrow A$  out of  $X$ , and as morphisms  $(A, x) \rightarrow (B, x')$  the arrows  $h : A \rightarrow B$  with  $h \circ x = x'$ . For the one-point space  $X = \text{pt}$ , an arrow  $\text{pt} \rightarrow Q$  is a point  $q_0$ , so  $\mathbf{Diff}_* := \text{pt}/\mathbf{Diff}$  is the category of **pointed manifolds**  $(Q, q_0)$  and **pointed maps**, those with  $h(q_0) = q'_0$ . Likewise  $\mathbf{Set}_* := \text{pt}/\mathbf{Set}$ . The **opposite category**  $C^{\text{op}}$  has the same objects and every arrow reversed, and  $(C/X)^{\text{op}} \cong X/C^{\text{op}}$ .

A method sends its initialisation to the checkpoint,  $\rho(q_0) = \theta_0$ , so it is an arrow of  $\mathbf{Diff}_*$ , and a weak morphism forgets the base points and lives in  $\mathbf{Diff}$ . Opposites enter in Remark I.3. A simulation  $h : M \rightarrow N$  runs from  $Q_M$  to  $Q_N$ . The network with  $Q_M$ -parameters is obtained from the one with  $Q_N$ -parameters by precomposing with  $h$ ,

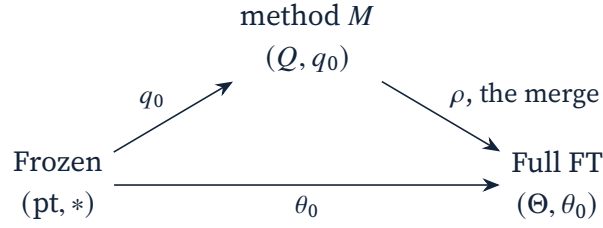
$$f \circ (\rho_M \times \text{id}_X) = f \circ (\rho_N \times \text{id}_X) \circ (h \times \text{id}_X),$$

with  $X$  the input space. Read as a change of parameters for networks,  $h$  therefore points the other way, from the  $Q_N$ -network to the  $Q_M$ -network, and Remark I.3 records this reversal with an opposite category (Background B.16).

**Used in.** Definition I.2, Remark I.3, Observation I.4, §I.

**BACKGROUND A.8** (the pointed slice  $\mathbf{Meth}(\Theta, \theta_0)$ ). Combining Background A.6 and Background A.7,  $\mathbf{Meth}(\Theta, \theta_0) := \mathbf{Diff}_*/(\Theta, \theta_0)$ . An object is a pointed smooth map  $\rho : (Q, q_0) \rightarrow (\Theta, \theta_0)$ , and a morphism  $h : M \rightarrow N$  is a pointed smooth map with  $\rho_N \circ h = \rho_M$ . Equivalently, an object is a factorisation  $\text{pt} \xrightarrow{q_0} Q \xrightarrow{\rho} \Theta$  of the point  $\theta_0 : \text{pt} \rightarrow \Theta$ , and a morphism is a map between the middle terms that is compatible with both ends.

On one box, LoRA is  $(\mathbb{R}^{m \times r} \times \mathbb{R}^{r \times n}, (0, A_0), (B, A) \mapsto W_0 + sBA)$ , and OFT (Qiu et al., 2023) is  $Q \mapsto W_0 C(Q)$  on skew matrices (block-diagonal in OFT), pointed at 0, with  $C$  the Cayley map (Background D.7). Full fine-tuning is  $(\Theta, \theta_0, \text{id})$  and the frozen model is  $(\text{pt}, *, \theta_0)$ . The network  $f$  appears nowhere in the definition, so the category depends only on  $(\Theta, \theta_0)$



**Figure 15.** The interval of  $\mathbf{Meth}(\Theta, \theta_0)$ . The left arrow is the only arrow out of the initial object, the right arrow  $\rho$  is the only arrow into the terminal object, and the triangle commutes because  $\rho(q_0) = \theta_0$ . Figure 2(a) draws the same triangle in the main text, where a pointing defect  $\rho_M(q_0) \neq \theta_0$  leaves it commuting only up to that defect.

(**Remark I.3**). The factorisation picture already shows why the frozen model and full fine-tuning sit at the two ends (**Background A.9**).

**Used in.** Table 2, **Definition I.2**, §I, **Remark I.3**, Rosetta entries 3 and 4 (§IX).

### Universal properties

**BACKGROUND A.9** (initial and terminal objects; universal property). An object  $I$  is **initial** if every object  $X$  receives exactly one arrow  $I \rightarrow X$ , and  $T$  is **terminal** if every  $X$  has exactly one arrow  $X \rightarrow T$ . Each is unique up to a unique isomorphism when it exists. Characterising an object by the arrows into or out of it in this way is called a **universal property**. In **Set** the empty set is initial and a one-point set is terminal, and in **Set**<sub>\*</sub> and **Diff**<sub>\*</sub> the one-point space is both.

In  $\mathbf{Meth}(\Theta, \theta_0)$  the frozen model  $(\text{pt}, *, \theta_0)$  is initial. The only pointed map  $\text{pt} \rightarrow Q$  is  $* \mapsto q_0$ , and it lies over  $\Theta$  because  $\rho(q_0) = \theta_0$ . Full fine-tuning  $(\Theta, \theta_0, \text{id})$  is terminal, since an arrow  $h$  from  $M$  must satisfy  $\text{id} \circ h = \rho_M$ . So the unique arrow from  $M$  to Full FT is  $\rho_M$  itself, the merge, and every weight-space method that starts at  $\theta_0$  sits in the commuting triangle of Figure 15.

**Used in.** **Observation I.4(a)**, §I, Figure 2, Rosetta entries 3 and 4 (§IX), rule A1 in §VIII.C, §V.

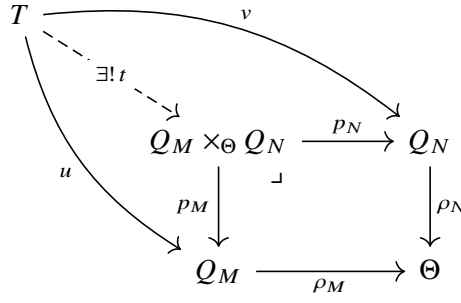
**BACKGROUND A.10** (product). A **product** of  $X$  and  $Y$  is an object  $X \times Y$  with projections  $p_X : X \times Y \rightarrow X$  and  $p_Y : X \times Y \rightarrow Y$  such that for every object  $T$  and arrows  $u : T \rightarrow X$ ,  $v : T \rightarrow Y$  there is exactly one arrow  $\langle u, v \rangle : T \rightarrow X \times Y$  with  $p_X \circ \langle u, v \rangle = u$  and  $p_Y \circ \langle u, v \rangle = v$ . It is unique up to unique isomorphism, and it need not exist. In a slice  $C/Z$ , the product of  $(X, a)$  and  $(Y, b)$  is the pullback of  $a$  and  $b$  in  $C$  (**Background A.11**).

In  $\mathbf{Meth}$  a product  $M \times N$  has arrows to  $M$  and to  $N$ , so both factors simulate it, and every method  $T$  with arrows  $u : T \rightarrow M$  and  $v : T \rightarrow N$  maps to it by exactly one arrow compatible with  $u$  and  $v$ . When it exists, its image is  $\text{Im } M \cap \text{Im } N$ . For the one-sided LoRAs with frozen frames  $B_0$  of full column rank and  $A_0$  of full row rank,

$$\text{FA}(A_0) \times \text{FB}(B_0) \cong (\mathbb{R}^{r \times r}, 0, R \mapsto W_0 + B_0 R A_0),$$

with projections  $R \mapsto B_0 R$  and  $R \mapsto R A_0$ . In the SVD frames of  $W_0$ , when  $\text{rank } W_0 \geq r$ , this is LoRA-XS (Bałazy et al., 2024) (**Proposition I.8(a)**). The sum  $\oplus$  of **Observation I.7** is a different operation, whose image is a Minkowski sum (**Background B.14**).

**Used in.** **Observation I.4(b)**, **Proposition I.8**, §I, **Proposition II.3**, rule A4 in §VIII.C, §IX. Reading: Leinster (2014, §5.1).



**Figure 16.** A pullback square. It commutes,  $\rho_M \circ p_M = \rho_N \circ p_N$ , and any test object  $T$  whose maps  $u, v$  agree over  $\Theta$  factors through the corner by exactly one  $t$ . The small corner mark is the usual sign of a pullback.

**BACKGROUND A.11** (pullback; fibre product). Given  $f : X \rightarrow Z$  and  $g : Y \rightarrow Z$ , a **pullback** is an object  $P$  with arrows  $p : P \rightarrow X$  and  $q : P \rightarrow Y$  such that  $f \circ p = g \circ q$ , and such that whenever  $u : T \rightarrow X$  and  $v : T \rightarrow Y$  satisfy  $f \circ u = g \circ v$ , exactly one  $t : T \rightarrow P$  has  $p \circ t = u$  and  $q \circ t = v$  (Figure 16). In **Set** the pullback is the **fibre product**  $X \times_Z Y = \{(x, y) : f(x) = g(y)\}$ . In **Diff**, if this set is an embedded submanifold of  $X \times Y$ , it is the pullback, and transversality (Background C.7) is the standard sufficient condition for that. A pullback in **Diff** can fail to exist.

Products in **Meth** are fibre products  $Q_M \times_{\Theta} Q_N = \{(q, q') : \rho_M(q) = \rho_N(q')\}$ . For  $\text{FA}(A_0)$  and  $\text{FB}(B_0)$ , with the ranks of Background A.10, this set is the linear space  $\{(B_0 R, R A_0)\}$ , so the product exists. For  $\text{LoRA}_1$  and the frozen model on  $1 \times 1$  weights the set is the cross  $\{ba = 0\} \subset \mathbb{R}^2$ , no manifold structure on it has the universal property, and the product does not exist (Background A.13). Pulling back along the point  $\theta_0 : \text{pt} \rightarrow \Theta$  gives the fibre  $\rho^{-1}(\theta_0)$ , the underlying set of  $M \times \text{Frozen}$  whenever that product exists.

**HRA is a meet of images, not a product.** Take  $W_0$  injective and  $r$  even with  $2 \leq r \leq n - 2$ . The fibre product of  $\text{LoRA}_r$  with the rotation orbit  $W_0 \cdot \text{SO}(n)$  has local dimension  $d(\text{HRA}_r) + r^2 = rn + r(r - 1)/2$  over generic points of the meet (Background C.19). This exceeds  $rn = |\text{HRA}_r|$  because  $r \geq 2$ , so HRA (Yuan et al., 2024) is not the product of the two methods, and Theorem II.7 keeps only the meet of their images (Background A.16).

**Used in.** Observation I.4(b), Proposition I.8, Proposition II.3, Theorem II.7(d), Corollary III.13.

**BACKGROUND A.12** (limit; creation of limits). A **cone** over a diagram  $D$  in  $C$  is an object  $L$  with an arrow to every object of  $D$ , such that each triangle formed with an arrow of  $D$  commutes. A **limit** of  $D$  is a cone through which every other cone factors by exactly one arrow. This categorical cone is unrelated to the geometric cones of Background C.10. Terminal objects, products and pullbacks are the limits of the empty diagram, of two objects, and of two arrows with a common target. A functor  $U : C \rightarrow \mathcal{D}$  **creates limits** if, for every diagram  $D$  in  $C$  and every limit cone of  $U \circ D$ , exactly one cone over  $D$  is sent by  $U$  to that limit cone, and this cone is a limit of  $D$ .

The forgetful functor  $\mathbf{Diff}_* \rightarrow \mathbf{Diff}$  creates limits (Background A.14). Concretely, when the underlying manifolds have a pullback, the pullback of pointed manifolds is that pullback, pointed at the pair of base points. This is how Observation I.4(b) computes products of methods as fibre products in **Diff**.

**Used in.** Observation I.4(b). Reading: Mac Lane (1998, Ch. V); Riehl (2016, Ch. 3).

**BACKGROUND A.13** (adjunction). A functor  $F : C \rightarrow \mathcal{D}$  is **left adjoint** to  $G : \mathcal{D} \rightarrow C$ , and  $G$  is **right adjoint** to  $F$ , written  $F \dashv G$ , if there are bijections  $\mathcal{D}(FX, Y) \cong C(X, GY)$  natural in  $X$  and  $Y$ . Many forgetful functors are right adjoints, whose left adjoints add the forgotten structure freely.

The forgetful functor  $\mathbf{Diff}_* \rightarrow \mathbf{Diff}$  is right adjoint to  $X \mapsto (X \sqcup \text{pt}, \text{pt})$ , which adds a disjoint base point, since a pointed map out of  $X \sqcup \text{pt}$  is the same as an arbitrary map out of  $X$  (a manifold may have components of different dimensions). Right adjoints preserve limits (Background A.14), so a product of methods, if it exists, is also a pullback in  $\mathbf{Diff}$  and can be tested with unpointed maps. Observation I.4(b) uses this to rule out  $\text{LoRA}_1 \times \text{Frozen}$  on  $1 \times 1$  weights. The axis curves  $t \mapsto (t, 0)$  and  $t \mapsto (0, t)$  in the cross would lift smoothly through the point over the origin, so the differential of the projection from the product to the parameter plane  $\mathbb{R}^2$  of  $\text{LoRA}_1$  would have rank 2 there. That projection would then be a submersion near the point (Background C.4), and its image would contain an open set, which the cross does not.

**Used in.** Observation I.4(b). Reading: Leinster (2014, Ch. 2 and Ch. 6).

**FACT A.14** (two standard facts about limits). *Let  $C$  be a category.*

- (a) *For every object  $X$  of  $C$ , the forgetful functor  $X/C \rightarrow C$  creates limits. In particular  $\mathbf{Diff}_* \rightarrow \mathbf{Diff}$  does.*
- (b) *A right adjoint preserves every limit that exists in its domain.*

**The idea behind (a) and (b).** (a) The structure arrows  $X \rightarrow D_j$  of a diagram in  $X/C$  form a cone in  $C$ , which factors through the limit by exactly one arrow, and that arrow is the structure map of the limit. (b) For  $F \dashv G$ ,

$$C(X, G \lim D) \cong \mathcal{D}(FX, \lim D) \cong \lim \mathcal{D}(FX, D_j) \cong \lim C(X, G D_j),$$

naturally in  $X$ .

Observation I.4(b) combines them. Products in  $\mathbf{Meth}$  are pullbacks in  $\mathbf{Diff}_*$ , which by (a) exist whenever the underlying pullback exists in  $\mathbf{Diff}$ , and by (b) and Background A.13 only then. The transversality hypothesis of Background C.7 fails at the base point whenever  $|M| + |N| < \dim \Theta$ , with  $|M| = \dim \mathcal{Q}_M$  (Background C.3), so the products that exist, such as  $\text{LoRA-XS}$ , come from other structure (Background A.11).

**Used in.** Observation I.4(b), §I. Reading: Mac Lane (1998, Ch. V); Riehl (2016, Ch. 3 and Ch. 4).

**BACKGROUND A.15** (monomorphism; sub-object). An arrow  $m : A \rightarrow B$  is a **monomorphism** if  $m \circ u = m \circ v$  implies  $u = v$  for all arrows  $u, v : T \rightarrow A$ . A **sub-object** of  $B$  is a monomorphism into  $B$ , two of them being identified when they differ by an isomorphism over  $B$ . Testing with maps out of one- and two-point manifolds shows that the monomorphisms of  $\mathbf{Diff}$ , and those of  $\mathbf{Meth}$ , are exactly the arrows whose underlying map is injective. Such a map need not be an embedding, as  $t \mapsto t^3$  shows.

Freezing coordinates at their base values gives an injective affine map, hence a sub-object, written  $\hookrightarrow$  in the atlas (rule A2). Examples are  $\text{FA}(A_0) \hookrightarrow \text{LoRA}$  pointed at  $(0, A_0)$ , by  $B \mapsto (B, A_0)$ , and  $\text{MiCA}$  (Rüdiger and Raschka, 2026)  $\hookrightarrow \text{LoRA}$  pointed at  $(B_0, 0)$ , by  $A \mapsto (B_0, A)$ . These maps are embeddings, a stronger property than injectivity. The paper writes sub-object and never monomorphism. In Corollary III.13,  $\text{LoRA-FA}$  is a sub-object of

LoRA and so a different object, while random-A LoRA and EVA (Paischer et al., 2024) are the same map  $\rho$  with two different base points in its fibre over  $\theta_0$ .

**Used in.** Definition I.2, rule A2 in §VIII.C, Corollary III.13.

**BACKGROUND A.16** (posets; meets; the expressivity order). A **preorder** is a category with at most one arrow between any two objects; write  $x \leq y$  when there is one. It is a **poset** if  $x \leq y \leq x$  forces  $x = y$ . In a poset the product of  $x$  and  $y$  is their **meet**  $x \wedge y$ , the greatest lower bound, and for subsets ordered by inclusion the meet is the intersection. Two elements are **incomparable** if neither  $x \leq y$  nor  $y \leq x$ , and  $x$  is **maximal** if  $x \leq y$  implies  $x = y$ . Every category has a **poset reflection**, in which  $X \leq Y$  when some arrow  $X \rightarrow Y$  exists and  $X, Y$  are identified when arrows run both ways. A **Hasse diagram** draws a finite poset with  $y$  above  $x$  when  $x < y$ , joining them by an edge when nothing lies strictly between.

The poset reflection of **Meth**, which we call the expressivity order, orders methods by simulation. Figure 14 draws it for the core methods as a layered Hasse diagram, from the frozen model at the bottom to full fine-tuning at the top. A meet of images always exists and need not be the image of a product. For injective  $W_0$  and even  $r \leq n - 2$ , the image of HRA without Gram–Schmidt is the meet of LoRA’s cone with the rotation orbit  $W_0 \cdot \text{SO}(n)$  (Background D.11), and HRA is not their product (Theorem II.7). Split and zero-init LoRA have incomparable images (Theorem II.2(d)), so neither simulates the other. The frontier problem asks for the maximal methods of budget at most  $k$ , which are automatically pairwise incomparable.

**Used in.** Observation I.4(c), Proposition I.8(b), Theorem II.2, Theorem II.7, §II.B, Figure 14, the open problems of Exposé XI. Reading: Davey and Priestley (2002, Ch. 1 and Ch. 2).

### Base change and lifting

**BACKGROUND A.17** (automorphism of a category; transport of structure). An **automorphism** of a category  $C$  is a functor  $F : C \rightarrow C$  with an inverse functor, so that both composites are identity functors. It carries initial objects, products, isomorphisms and every other notion defined from arrows alone to notions of the same kind. **Transport of structure** builds such a functor from an invertible map of the underlying data.

For  $T \in \text{GL}(\Theta)$  put  $\tau_T(\theta) = \theta_0 + T(\theta - \theta_0)$  and  $T_*M := (Q, q_0, \tau_T \circ \rho)$ . An arrow  $h : M \rightarrow N$  is also an arrow  $T_*M \rightarrow T_*N$ , since  $\tau_T \circ \rho_N \circ h = \tau_T \circ \rho_M$ , so  $T_*$  is a functor, with inverse  $(T^{-1})_*$ . It keeps  $d(M)$  and the first-order deficit (Background C.3) and the gauge group (Background D.21), and moves  $S_1$  to  $TS_1$ . It does not keep rank, which depends on the matrix shape of  $\Theta$  that  $T$  need not respect. Sums of  $k$  Kronecker products are  $(\mathcal{R}^{-1})_*\text{LoRA}_k$ , LoRA on the matrix rearranged by Van Loan and Pitsianis and carried back (Background B.11), and HiRA (Huang et al., 2025b) is  $(D_{W_0})_*\text{LoRA}$  with  $D_{W_0}(X) = W_0 \odot X$ , when  $W_0$  has no zero entries. The atlas records  $M$  and  $T_*M$  as one object in two frames, with no arrow between them.

**Used in.** Proposition II.13, Exposé II.C, rule T in §VIII.C, §IX.



**BACKGROUND A.18** (lift; lifting problem; obstruction). Given arrows  $p : E \rightarrow B$  and  $f : A \rightarrow B$ , a **lift** of  $f$  along  $p$  is an arrow  $g : A \rightarrow E$  with  $p \circ g = f$ . A **lifting problem** asks whether a lift exists, and an **obstruction** is an invariant whose value shows that none does. A morphism  $h : M \rightarrow N$  of **Meth** is exactly a pointed smooth lift of  $\rho_M$  along  $\rho_N$ .

In **Set** a lift exists once  $f(A) \subseteq p(E)$ , by choosing preimages, so smoothness is the real constraint. For  $\rho_M(t) = t$  and  $\rho_N(u) = u^3$  on  $\mathbb{R}$  the images agree, and the only lift is the cube root, which is not differentiable at 0. A pointed smooth lift forces  $S_1(M) \subseteq S_1(N)$  by the chain rule, so  $S_1$  is a first-order obstruction, and here  $S_1(M) = \mathbb{R} \not\subseteq S_1(N) = 0$ . For  $t^3$  against  $u^5$  both  $S_1$  vanish, yet the only lift  $t^{3/5}$  is not  $C^1$ , so that obstruction is invisible to  $S_1$ . Merging a method on an extension is again a lifting problem, to find  $\mu$  with  $\Phi_f \circ \mu = \Phi_{\tilde{f}} \circ \rho$ , a lift along the realisation  $\Phi_f$  (**Background B.16**, **Definition VI.1**).

**Used in.** **Observation I.4(c)**, §I, **Observation I.6**, **Definition VI.1**, §VI, Rosetta entry 5 (§IX).

**BACKGROUND A.19** (base change; pushforward  $\phi_!$ ; the relative point of view). A map of bases  $\phi : S' \rightarrow S$  induces the **pushforward**  $\phi_! : C/S' \rightarrow C/S$ ,  $(X, x) \mapsto (X, \phi \circ x)$ , by postcomposition. When every object over  $S$  has a pullback along  $\phi$ , it also induces the **pullback**  $\phi^* : C/S \rightarrow C/S'$ ,  $(Y, y) \mapsto (Y \times_S S', \text{projection})$ , and then  $\phi_! \dashv \phi^*$ . In algebraic geometry, base change usually means  $\phi^*$ .

**Definition V.1(a)** uses  $\phi_!$  on pointed slices. A function-preserving pointed reparametrisation  $\phi : (\Theta', \theta'_0) \rightarrow (\Theta, \theta_0)$ , with  $f \circ (\phi \times X) = f'$ , gives  $\phi_! : \mathbf{Meth}(\Theta', \theta'_0) \rightarrow \mathbf{Meth}(\Theta, \theta_0)$ ,  $\rho \mapsto \phi \circ \rho$ , for instance for a permutation of hidden units. Whether  $\phi_!$  carries LoRA to LoRA is the covariance question of **Observation II.15**. A quantiser changes the function and induces no such functor, so QLoRA (**Dettmers et al., 2023**) and ReLoRA (**Lialin et al., 2023b**) only borrow the name, and **Exposé V** computes pointing defects and reachable sets for them.

Grothendieck's relative point of view replaces a space  $X$  by a morphism  $X \rightarrow S$ , an object over a base. Properties then belong to the morphism, and the useful ones are those that survive pulling back along every change of base  $S' \rightarrow S$ . The fibre over a point  $s$  is itself the pullback along  $s : \text{pt} \rightarrow S$ . The paper takes this stance in a modest form. A method is a morphism over the base point  $\theta_0$ , and quantisation, restarts and merging are read as moves of that base point.

**Used in.** **Exposés 0–I**, **Exposé 0** (where the category theory earns its keep), **Definition V.1**, **Exposé V**, Rosetta entry 28 (§IX), the seven borrowed words of **Exposé XI**. Reading: **Vakil (2025)**, on fibred products and base change.

**BACKGROUND A.20** (constructions the paper names and leaves unused). Coends, monads, sheaves, stacks, schemes, Grothendieck's descent theory (descent data, effective descent morphisms), the Grothendieck construction  $\int \mathbf{Meth} \rightarrow \mathbf{PNet}$  and coKleisli 2-cells appear in the paper only in lists of constructions it leaves out (**Exposé 0**, where the category theory earns its keep, and **Exposé XI**, on what the borrowed words mean). No result depends on them. The elementary descent of **Background D.22**, factoring a map that is constant on orbits through the orbit space, is a different and much simpler notion, and several results use it.

For orientation only, the Grothendieck construction would assemble the slices of all pretrained models,  $\mathbf{Meth}(\Theta, \theta_0)$  for each  $(\Theta, \theta_0)$ , into one category of pairs (network, method), with a projection to the category **PNet** whose objects are pointed networks and whose arrows

are function-preserving reparametrisations. A monad is an endofunctor  $T$  with natural transformations  $\text{id} \Rightarrow T$  and  $TT \Rightarrow T$  satisfying unit and associativity laws, and a sheaf assigns data to open sets so that compatible local data glue to unique global data.

**Used in.** Exposé 0 (where the category theory earns its keep), the seven borrowed words and what the borrowed words mean in Exposé XI.

## Part B. Monoidal categories, string diagrams and lenses

Part A compared methods through the arrows between them. Part B covers the algebra behind drawing a method as boxes and wires, reading a rank bound off a cut of the drawing, and treating backpropagation as a pair of maps. Entries [Background B.1](#) to [Background B.6](#) set up monoidal categories and their diagrams, [Background B.7](#) to [Background B.13](#) turn them into rank and dimension counts for matrices built from small factors, and [Background B.14](#) to [Background B.17](#) cover sums of methods, parametrised maps and lenses.

### Monoidal categories

**BACKGROUND B.1** (monoidal and symmetric monoidal categories). A **monoidal category** is a category  $C$  with a functor  $\otimes : C \times C \rightarrow C$ , a unit object  $I$ , and natural isomorphisms  $(A \otimes B) \otimes C \cong A \otimes (B \otimes C)$  and  $I \otimes A \cong A \cong A \otimes I$ , the **coherence isomorphisms**, which satisfy the pentagon and triangle equations. Mac Lane’s coherence theorem then makes every diagram built from them commute, so brackets can be dropped. The category is **symmetric** if it also has natural isomorphisms  $\sigma_{A,B} : A \otimes B \rightarrow B \otimes A$  with  $\sigma_{B,A}\sigma_{A,B} = \text{id}$  that are compatible with the associator (the hexagon equation). Examples are (**FinVect**,  $\otimes$ ,  $\mathbb{R}$ ), (**FinVect**,  $\oplus$ , 0) and (**Diff**,  $\times$ , pt).

On methods,  $M \oplus N$  runs the two parameter spaces side by side and adds their updates ([Background B.14](#)). This makes **Meth**( $\Theta$ ,  $\theta_0$ ) symmetric monoidal with the frozen model as unit. Its coherence isomorphisms are those of  $\times$  on pointed manifolds, and they lie over  $\Theta$  because addition is associative and commutative.

**Used in.** [Observation I.7](#), §I, Exposé 0 (where the category theory earns its keep), [Theorem II.10](#), [Theorem VI.3](#). Reading: [Mac Lane \(1998, Ch. VII and Ch. XI\)](#).

**BACKGROUND B.2** (direct sum and tensor product in **FinVect**). The category **FinVect** has finite-dimensional real vector spaces as objects and linear maps as morphisms, and two monoidal products. The **direct sum**  $V \oplus W$  is a **biproduct**, both the categorical product and the coproduct, with  $\dim(V \oplus W) = \dim V + \dim W$ ; on matrices  $f \oplus g$  is block-diagonal. The **tensor product**  $V \otimes W$  has  $\dim(V \otimes W) = \dim V \cdot \dim W$  and is not a categorical product; on matrices  $f \otimes g$  is the Kronecker product ([Background B.10](#)). Each bilinear map  $V \times W \rightarrow U$  factors through a unique linear map  $V \otimes W \rightarrow U$ .

Two boxes drawn side by side therefore mean a block-diagonal map in the activation diagrams of Exposé VI, whose monoidal product is  $\times = \oplus$ , and a Kronecker product in a tensor network. Dimensions multiply along  $\otimes$ , which is why the weight of a cut in [Theorem II.10](#) is a product of wire dimensions. LoRA’s residual branch  $W_0 + BA$  uses  $\oplus$ , since it sends one input to two branches and adds their outputs ([Figure 17](#)).

**Used in.** [Theorem II.10](#), Exposé II.C, [Theorem VI.3](#), [Observation I.7](#).

**BACKGROUND B.3** (string diagrams). A **string diagram** draws a morphism  $f : A \rightarrow B$  of a monoidal category as a box with an input wire  $A$  and an output wire  $B$ . Composition  $g \circ f$  joins boxes in series,  $f \otimes g$  places them side by side, an identity is a bare wire, the unit  $I$  is drawn as no wire, and in a symmetric category  $\sigma$  is a crossing. Joyal and Street (1991) proved that the pictures are sound and complete. In a monoidal category, diagrams related by a planar deformation denote the same morphism; in a symmetric one, diagrams with the same boxes joined in the same way do. Laws such as  $(g \otimes g') \circ (f \otimes f') = (g \circ f) \otimes (g' \circ f')$  can then be read off the picture.

The plug of Exposé II.C (§II.C) is a string diagram. LoRA is a box  $A : \mathbb{R}^n \rightarrow \mathbb{R}^r$  followed by  $B : \mathbb{R}^r \rightarrow \mathbb{R}^m$ , drawn beside  $W_0$  (Figure 17, read left to right). The merge calculus moves an edit along wires by equalities of maps, so a successful slide proves that the merged network computes the same function.

**Used in.** Exposé II.C, Theorem II.10, §II.C, §VI and the analogy on string diagrams there, Theorem VI.3. Reading: Selinger (2011).

**BACKGROUND B.4** (cartesian categories; copy and discard). A monoidal category is **cartesian** when  $\otimes$  is the categorical product  $\times$  and  $I$  is a terminal object  $1$ . Every object then has a **copy** map  $\Delta_X : X \rightarrow X \times X$ ,  $x \mapsto (x, x)$ , and a **discard** map  $!_X : X \rightarrow 1$ , and copy is natural,  $(f \times f) \circ \Delta_X = \Delta_Y \circ f$  for every  $f : X \rightarrow Y$ . Conversely, by a theorem of Fox (1976), a symmetric monoidal category is cartesian exactly when every object carries a commutative comonoid, a copy and a discard map obeying coassociativity, counit and commutativity laws (Background B.6), compatible with  $\otimes$  and natural in all morphisms. **Set**, **Diff** and **(FinVect,  $\oplus$ )** are cartesian. **(FinVect,  $\otimes$ )** is not, since  $x \mapsto x \otimes x$  is not linear.

Naturality of copy is rule R7 of the merge calculus. A linear edit  $E$  on a wire read by  $W_Q$ ,  $W_K$  and  $W_V$  equals  $E$  applied on each of the three branches, so it is absorbed as  $W_Q E$ ,  $W_K E$ ,  $W_V E$ . The paper builds **Para(C)** (Background B.16) over a cartesian  $C$ . The linear copy of Background B.6 copies basis vectors only and is a different map.

**Used in.** Definition I.1, §VI and the analogy on string diagrams there, Theorem VI.3.

**BACKGROUND B.5** (compact closed categories; cups, caps and wire-bending). A symmetric monoidal category is **compact closed** if every object  $A$  has a dual  $A^*$  with a **cup**  $\eta_A : I \rightarrow A^* \otimes A$  and a **cap**  $\varepsilon_A : A \otimes A^* \rightarrow I$  satisfying the snake equations

$$(\varepsilon_A \otimes \text{id}_A)(\text{id}_A \otimes \eta_A) = \text{id}_A, \quad (\text{id}_{A^*} \otimes \varepsilon_A)(\eta_A \otimes \text{id}_{A^*}) = \text{id}_{A^*},$$

with the unitors  $I \otimes A \cong A \cong A \otimes I$  of Background B.1 suppressed. Drawn as bent wires, cups and caps turn an input wire into an output wire and back. In **FinVect** the cap is evaluation, the cup is  $1 \mapsto \sum_i e_i^* \otimes e_i$ , and bending gives  $\text{Hom}(V, U) \cong U \otimes V^*$ . With a fixed basis, a box slid around a cup or cap comes out as its transpose.

Bending the input wire of an  $m \times n$  matrix gives its vectorisation in  $\mathbb{R}^m \otimes \mathbb{R}^n$ , and the Van Loan–Pitsianis rearrangement (Background B.11) is a composite of bends and crossings. Rule R5 of the merge calculus,  $\langle Mq, k \rangle = \langle q, M^\top k \rangle$ , slides  $M$  around the cap that computes an attention score. The cut-rank bound uses only  $\otimes$  and crossings.

**Used in.** Theorem II.11, Theorem VI.3, Exposé II.C. Reading: Coecke and Kissinger (2017).

**BACKGROUND B.6** (Frobenius algebras and spiders). A **Frobenius algebra** on an object  $V$  is a monoid  $\mu : V \otimes V \rightarrow V$ ,  $\eta : I \rightarrow V$  and a comonoid  $\delta : V \rightarrow V \otimes V$ ,  $\varepsilon : V \rightarrow I$  obeying the **Frobenius law**

$$(\mu \otimes \text{id})(\text{id} \otimes \delta) = \delta\mu = (\text{id} \otimes \mu)(\delta \otimes \text{id}).$$

A basis  $e_1, \dots, e_n$  of  $\mathbb{R}^n$  gives one, with *copy*  $\delta(e_i) = e_i \otimes e_i$ , *merge* (the multiplication; unrelated to merging an adapter into the weights)  $\mu(e_i \otimes e_j) = e_i$  if  $i = j$  and 0 otherwise,  $\eta = \sum_i e_i$  and  $\varepsilon(e_i) = 1$ . It is commutative and special ( $\mu\delta = \text{id}$ ). For such algebras the **spider theorem** says that every connected diagram of  $\mu, \eta, \delta, \varepsilon$  with  $k \geq 1$  inputs and  $l$  outputs equals one **spider**, which sends  $e_i^{\otimes k}$  to  $e_i^{\otimes l}$  and every other basis tensor to 0.

Merge multiplies coordinates,  $\mu(x \otimes y) = x \odot y$ . So for matrices  $X, Y : \mathbb{R}^n \rightarrow \mathbb{R}^m$  the Hadamard product is the spider composite

$$X \odot Y = \mu_m \circ (X \otimes Y) \circ \delta_n.$$

LoHa (Hyeon-Woo et al., 2021) is a copy spider on the input feeding  $A_1$  and  $A_2$ , then  $B_1$  and  $B_2$ , then a merge spider on the output, and in Theorem II.10 a spider split by the cut counts its dimension once. The copy and  $\odot$  nodes of Exposé VI live in a cartesian category (Background B.4), where copy is  $x \mapsto (x, x)$ ; together they fail the Frobenius law, so there “spider” is only a name.

**Used in.** Exposé II.C, Theorem II.10, Proposition II.12, the analogy on string diagrams in §VI. Reading: Coecke et al. (2013).

## Tensor networks and rank

**BACKGROUND B.7** (tensor networks, contraction, bond dimension). A **tensor network**  $\Gamma$  is a string diagram in  $(\mathbf{FinVect}, \otimes)$ , possibly with spiders. Each box is a tensor and each wire a finite-dimensional space. A wire joining two boxes means **contraction**, a sum over the index the wire carries. The open wires (legs) are the inputs  $V$  and outputs  $U$ , and contracting every internal wire gives the value  $\llbracket \Gamma \rrbracket : V \rightarrow U$ . The dimension of an internal wire is its **bond dimension**. One value has many presentations, and quantities read off a network, such as its cuts, depend on the presentation.

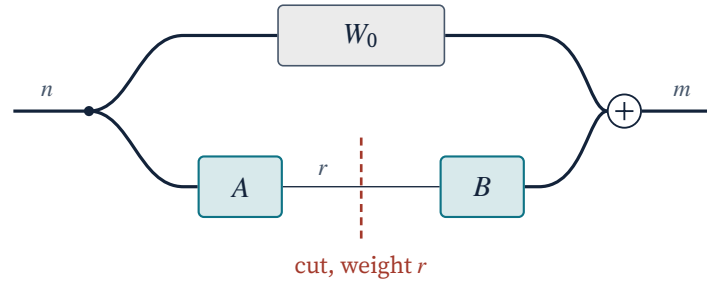
LoRA’s update is the two-box network  $B \circ A$  with one bond of dimension  $r$ , and  $\Delta W = BA$  contracts it,  $(BA)_{ij} = \sum_{k \leq r} B_{ik} A_{kj}$ . HiRA factors  $W_0$  through a bond of dimension  $\text{rk } W_0$ , while MoRA (Jiang et al., 2024), C3A (Chen et al., 2024a) and MPO adapters have no narrow cut between inputs and outputs (Table 8, after Theorem II.10).

**Used in.** Theorem II.10, Exposé II.C, §II.C, Exposé 0 (where the category theory earns its keep), the known results credited in Exposé XII. Reading: Orús (2014).

**FACT B.8** (the min-cut bound on rank). Let  $\Gamma$  be a tensor network with input legs  $V$  and output legs  $U$ , with every wire attached to a box (put an identity box on any bare strand). Split the boxes into an input side  $I$  and an output side  $O$ . The **weight**  $w$  of the cut is the product of the dimensions of the wires joining the two sides and of the legs attached on the wrong side (input legs at  $O$ , output legs at  $I$ ); a spider with legs on both sides contributes its dimension once. Then

$$\text{rank} \llbracket \Gamma \rrbracket \leq \min_{\text{cuts}} w.$$

For sums of networks,  $\text{rank}(X + Y) \leq \text{rank } X + \text{rank } Y$ .



**Figure 17.** LoRA as a string diagram, read left to right, with the scale  $s$  absorbed into  $B$ . The fork sends the input to both branches and the  $+$  node adds their outputs. These are the copy and sum maps of the biproduct  $\oplus$  (Background B.2), so the diagram says  $W = W_0 + BA$ . The frozen  $W_0$  is grey, and the trainable  $A \in \mathbb{R}^{r \times n}$  and  $B \in \mathbb{R}^{m \times r}$  are tinted. The dashed cut separates  $A$  from  $B$  and crosses one wire of dimension  $r$ , so  $\text{rank}(W - W_0) = \text{rank } BA \leq r$ . For  $W$  itself, subadditivity over the two branches gives only  $\text{rank } W \leq \text{rank } W_0 + r$ .

**The idea of the proof.** Contract each side separately. The value becomes a map into the tensor product of the crossing wires and wrong-side legs followed by a map out of it, and that space has dimension  $w$ .

For LoRA the cut between  $A$  and  $B$  crosses one wire, so  $\text{rank } BA \leq r$  (Figure 17). For LoHa the cut with  $A_1, A_2$  on the input side crosses two bonds and gives  $r_1 r_2$ . The bound depends on the presentation. HRA's product of  $r$  reflections expands into  $2^r - 1$  terms of rank at most one, while the telescoping sum

$$R - I = \sum_{k=1}^r (H_k - I) H_{k+1} \cdots H_r$$

gives  $r$ .

**Used in.** Theorem II.10, Exposé II.C, §II.C, Proposition II.12, the known results credited in Exposé XII. Reading: Kolda and Bader (2009).

**BACKGROUND B.9** (tensor trains, Tucker and MPO; gauge on bonds). A **tensor train** (TT) writes a tensor with legs of dimensions  $n_1, \dots, n_d$  as a chain of cores  $G_k \in \mathbb{R}^{r_{k-1} \times n_k \times r_k}$ ,  $r_0 = r_d = 1$ , joined by bonds of dimensions  $r_1, \dots, r_{d-1}$ . An **MPO** is the same chain with one input and one output leg on each core. **Tucker** joins a core  $C \in \mathbb{R}^{r_1 \times \cdots \times r_d}$  to factor matrices  $U_k \in \mathbb{R}^{n_k \times r_k}$ . Inserting  $g^{-1}g = \text{id}$ ,  $g \in \text{GL}_{r_e}$ , on a bond  $e$  and absorbing each factor into its neighbouring box leaves the value unchanged, so the gauge (Background D.21) contains  $\prod_e \text{GL}_{r_e}$ . When every  $r_e$  is the rank of the matching unfolding (minimal ranks), the tensors of these ranks form a manifold of dimension  $\sum(\text{core sizes}) - \sum_e r_e^2$ .

LoRA is the two-core case,  $B$  and  $A$  joined by one bond, with gauge  $(B, A) \mapsto (Bg^{-1}, gA)$  and, for  $r \leq \min(m, n)$ , image dimension  $r(m + n) - r^2$ . TT and Tucker adapters have  $d(M) = |M| - \sum_e r_e^2$  at minimal ranks (Proposition II.12), and under gradient flow each bond carries a conserved matrix (§III).

**Used in.** Proposition II.12, Exposé II.C, §III. Reading: Oseledets (2011); Holtz et al. (2012); Koch and Lubich (2010).



**BACKGROUND B.10** (Kronecker product, vec, tensor product of subspaces). For  $C \in \mathbb{R}^{m_1 \times n_1}$  and  $D \in \mathbb{R}^{m_2 \times n_2}$ , the **Kronecker product**  $C \otimes D \in \mathbb{R}^{m_1 m_2 \times n_1 n_2}$  is the block matrix  $[c_{ij} D]$ , the matrix of  $C \otimes D$  in the product basis. It obeys the mixed-product rule  $(C \otimes D)(E \otimes F) = CE \otimes DF$ , and  $\text{rank}(C \otimes D) = \text{rank } C \cdot \text{rank } D$ . The map  $u \otimes v \mapsto uv^\top$  identifies  $\mathbb{R}^m \otimes \mathbb{R}^n$  with  $\mathbb{R}^{m \times n}$ . For subspaces  $U \subseteq \mathbb{R}^m$  and  $V \subseteq \mathbb{R}^n$ ,  $U \otimes V$  is then the space of matrices with column space in  $U$  and row space in  $V$ , of dimension  $\dim U \cdot \dim V$ .

LoKr’s (Yeh et al., 2023) update is  $C \otimes BA$  with  $C$  ranging over all  $m_1 \times n_1$  matrices,  $B \in \mathbb{R}^{m_2 \times r'}$  and  $A \in \mathbb{R}^{r' \times n_2}$ . For inner rank  $r' \leq \min(m_2, n_2)$  its rank  $\text{rk } C \cdot \text{rk}(BA)$  is at most  $\min(m_1, n_1) r'$ , and this is attained. So Im LoKr lies inside the image  $W_0 + \mathcal{M}_{\leq r'}$  of zero-init LoRA $_r$  exactly when  $r \geq \min(m_1, n_1) r'$  (Theorem II.11(e)). At a LoRA point  $(B, A)$ , the directions reached by moving one factor alone overlap in

$$\{XA\} \cap \{BY\} = \text{col } B \otimes \text{row } A,$$

of dimension  $\text{rk } A \cdot \text{rk } B$  (Proposition II.3).

**Used in.** Theorem II.11, Exposé II.C, Proposition II.3, Theorem II.2, the LoHa/LoKr example of Exposé VII. Reading: Horn and Johnson (1991, Ch. 4).

**FACT B.11** (Van Loan–Pitsianis rearrangement; Kronecker rank). Let  $m = m_1 m_2$  and  $n = n_1 n_2$ , and cut  $\Delta \in \mathbb{R}^{m \times n}$  into an  $m_1 \times n_1$  grid of  $m_2 \times n_2$  blocks  $\Delta_{ij}$ . The **rearrangement**  $\mathcal{R}(\Delta) \in \mathbb{R}^{m_1 n_1 \times m_2 n_2}$  has the rows  $\text{vec}(\Delta_{ij})^\top$ , ordered with  $i$  running fastest. Then

$$\mathcal{R}(C \otimes D) = \text{vec}(C) \text{vec}(D)^\top,$$

and  $\mathcal{R}$  is a linear bijection that permutes entries, hence a Frobenius isometry. So the **Kronecker rank**  $\text{rk}_\otimes \Delta$ , the least  $k$  with  $\Delta = \sum_{i \leq k} C_i \otimes D_i$ , equals  $\text{rank } \mathcal{R}(\Delta)$ , and the nearest sum of  $k$  Kronecker products in Frobenius norm comes from the rank- $k$  truncated SVD of  $\mathcal{R}(\Delta)$  (Van Loan and Pitsianis 1993).

**The idea of the proof.** Block  $(i, j)$  of  $C \otimes D$  is  $c_{ij} D$ , so its row in  $\mathcal{R}$  is  $c_{ij} \text{vec}(D)^\top$ . Linearity extends this to sums, and Eckart–Young applies to  $\mathcal{R}(\Delta)$  because  $\mathcal{R}$  preserves the Frobenius norm.

In string-diagram terms  $\mathcal{R}$  bends and crosses wires, regrouping the four legs of  $\Delta$  from  $U_1 U_2 \mid V_1 V_2$  to  $U_1 V_1 \mid U_2 V_2$ . A sum of  $k$  Kronecker products is therefore LoRA $_k$  transported by  $\mathcal{R}^{-1}$  (Proposition II.13), with rank taken across another cut. For  $N = s^2$ ,  $I_N = I_s \otimes I_s$  has Kronecker rank 1 and rank  $N$ . Figure 6 shows the singular values across the ordinary cut, this Kronecker cut and a third one, for a random rank-one update, the identity and a Kronecker product  $C \otimes D$ . In this paper a *Kronecker sum* means a sum of Kronecker products; matrix analysis uses the same name for  $C \otimes I + I \otimes D$ .

**Used in.** Theorem II.11, Proposition II.13, Figure 6, Prediction X.5, the known results and key references of Exposé XII. Reading: Van Loan (2000).

**BACKGROUND B.12** (Schmidt and operator-Schmidt decompositions). Every  $u \in \mathbb{R}^{m_1} \otimes \mathbb{R}^{m_2}$  has a **Schmidt decomposition**  $u = \sum_{i \leq \rho} \sigma_i x_i \otimes y_i$  with  $\sigma_i > 0$  and orthonormal families  $(x_i), (y_i)$ . It is the SVD of the  $m_1 \times m_2$  reshape of  $u$  ( $x \otimes y \mapsto xy^\top$ ), and the **Schmidt rank**  $\rho_u$  is the rank of that reshape, at most  $\min(m_1, m_2)$ . The SVD of  $\mathcal{R}(\Delta)$  likewise gives the **operator-Schmidt decomposition**  $\Delta = \sum_{k \leq K} s_k C_k \otimes D_k$ , with  $s_k > 0$ , Frobenius-orthonormal families  $(C_k)$  and  $(D_k)$ , and  $K = \text{rk}_\otimes \Delta$ .

If  $u = \sum_i \sigma_i x_i \otimes y_i$  and  $v = \sum_j \tau_j z_j \otimes w_j$  have Schmidt ranks  $\rho_u$  and  $\rho_v$ , the mixed-product rule gives

$$uv^\top = \sum_{i,j} \sigma_i \tau_j (x_i z_j^\top) \otimes (y_i w_j^\top),$$

an operator-Schmidt decomposition with  $\rho_u \rho_v$  terms, so  $\text{rk}_\otimes(uv^\top) = \rho_u \rho_v$ . A generic rank-one update, which LoRA<sub>1</sub> reaches, thus needs  $\kappa_1 = \min(m_1, m_2) \min(n_1, n_2)$  Kronecker terms (Theorem II.11(c)).

**Used in.** Theorem II.11, Figure 6, Prediction X.5, the known results credited in Exposé XII. Reading: Nielsen and Chuang (2000, §2.5).

**FACT B.13** (rank of a Hadamard product; face-splitting product). *For  $X, Y \in \mathbb{R}^{m \times n}$ ,  $\text{rank}(X \odot Y) \leq \text{rank } X \cdot \text{rank } Y$ . Explicitly, if  $X = B_1 A_1$  and  $Y = B_2 A_2$  with inner dimensions  $r_1$  and  $r_2$ , then*

$$X \odot Y = (B_1 \bullet B_2) (A_1^\top \bullet A_2^\top)^\top,$$

where the **face-splitting product**  $B_1 \bullet B_2 \in \mathbb{R}^{m \times r_1 r_2}$  has as row  $i$  the Kronecker product of row  $i$  of  $B_1$  and row  $i$  of  $B_2$ . In particular

$$W_0 \odot (BA) = \sum_{k \leq r} \text{diag}(b_k) W_0 \text{diag}(a_k),$$

summed over the columns  $b_k$  of  $B$  and the rows  $a_k$  of  $A$ , has rank at most  $r \text{rank } W_0$  (Horn and Johnson, 1991, Ch. 5).

**The idea of the proof.**  $X \odot Y$  is the submatrix of  $X \otimes Y$  on the rows  $(i, i)$  and the columns  $(j, j)$ . In pictures it is the spider composite  $\mu \circ (X \otimes Y) \circ \delta$  of Background B.6, which factors through the two inner wires.

LoHa <sub>$r_1, r_2$</sub>  therefore has rank at most  $r_1 r_2$ , and  $h = (B_1 \bullet B_2, (A_1^\top \bullet A_2^\top)^\top)$  is the atlas arrow LoHa <sub>$r_1, r_2$</sub>   $\rightarrow$  LoRA <sub>$r_1 r_2$</sub> . HiRA's update  $W_0 \odot BA$  has rank at most  $r \text{rank } W_0$ , a bound that allows full rank when  $W_0$  has full rank (Proposition II.9(a)).

**Used in.** Proposition II.12, Theorem II.10, Proposition II.9, §VIII.C, Rosetta entries 6 and 11 (§IX).

## Sums, Para and lenses

**BACKGROUND B.14** (sums of methods; Minkowski sums, differences and translates). For subsets  $S, T$  of a vector space, the **Minkowski sum** is  $S + T = \{s + t : s \in S, t \in T\}$ , the **Minkowski difference** is  $S - T = \{s - t : s \in S, t \in T\}$ , a **translate** of  $S$  is  $v + S = \{v + s : s \in S\}$ , and  $C^{+K}$  is the sum of  $K$  copies of  $C$ . A sum is the image of  $S \times T$  under addition, so for semialgebraic sets  $\dim(S + T) \leq \dim S + \dim T$  (Background C.18), and likewise for differences. If  $0 \in C$ , the additive monoid (Background D.10) generated by  $C$  is  $\text{Mon}(C) = \bigcup_K C^{+K}$ , and if moreover  $\mathbb{R}C = C$  it is  $\text{span } C$ . Writing  $\rho_M = \theta_0 + \delta_M$ , the sum of methods

$$M \oplus N = (Q_M \times Q_N, (q_0, q'_0), \theta_0 + \delta_M + \delta_N)$$

has image  $\theta_0 + (\text{Im } \delta_M + \text{Im } \delta_N)$ , and  $\oplus$  is a symmetric monoidal structure with the frozen model as unit (Background B.1).

Concatenation  $([B_1 \ B_2], [A_1; A_2])$  is an isomorphism  $\text{LoRA}_r \oplus \text{LoRA}_s \cong \text{LoRA}_{r+s}$  of unpointed objects, with the  $A$ -blocks rescaled to the target scale when the scales differ, as under  $\alpha/r$  scaling (Observation I.7). It is also a pointed isomorphism onto  $\text{LoRA}_{r+s}$  pointed at the

concatenated initialisation. RoSA (Nikdan et al., 2024) and GraLoRA (Jung et al., 2025) are  $\oplus$ -words. The categorical product of [Observation I.4](#), when it exists, has image  $\text{Im } M \cap \text{Im } N$  ([Background A.10](#)). Merging and restarting  $K$  times with a fixed shape  $C$  reaches  $\theta_0 + C^{+K}$ . Since  $\mathcal{M}_{\leq r}^{+K} = \mathcal{M}_{\leq \min(Kr, m, n)}$ , ReLoRA reaches every matrix after  $\lceil \min(m, n)/r \rceil$  cycles ([Theorem V.3](#)).

Re-pointing an additive method  $\rho = \theta_b + \delta$  with shape  $C = \text{Im } \delta$  at a parameter  $q$ , so that  $\theta_b = \theta_0 - \delta(q)$ , gives the translate  $\theta_0 + (C - \delta(q))$  as image, and all exact pointings together reach  $\theta_0 + (C - C)$ . For LoRA,  $\mathcal{M}_{\leq r}$  is closed under negation and  $\mathcal{M}_{\leq r} + \mathcal{M}_{\leq r} = \mathcal{M}_{\leq \min(2r, m, n)}$  by splitting an SVD ([Background C.12](#)), so  $C - C = \mathcal{M}_{\leq \min(2r, m, n)}$ , which is why no exact pointing reaches an update of rank above  $2r$  ([Theorem II.1](#)).

**Used in.** [Observation I.7](#), §I, [Theorem II.1](#), [Proposition II.13](#), [Theorem V.3](#), §V, §IX.

**BACKGROUND B.15** (strong monoidal functors;  $(\mathbb{N}, +)$  and the permutation groupoid). A **strong monoidal functor**  $F : (C, \otimes, I) \rightarrow (D, \otimes, J)$  is a functor with isomorphisms  $\phi_{A,B} : FA \otimes FB \rightarrow F(A \otimes B)$ , natural in  $A$  and  $B$ , and  $\phi_0 : J \rightarrow FI$ , compatible with associators and unitors. It is **symmetric** if also  $F(\sigma_{A,B}) \circ \phi_{A,B} = \phi_{B,A} \circ \sigma_{FA,FB}$ . The **discrete monoidal category**  $(\mathbb{N}, +)$  has the natural numbers as objects, only identity morphisms,  $r \otimes s = r + s$  and identity symmetry. The **permutation groupoid**  $\mathbb{P}$  has the same objects, the symmetric group  $S_r$  as the automorphisms of  $r$  and no other morphisms, with  $\otimes$  given by  $+$  and block sums of permutations, and the block transposition in  $S_{r+s}$  as symmetry.

Concatenation ([Background B.14](#)) makes  $r \mapsto \text{LoRA}_r$  strong monoidal from  $(\mathbb{N}, +)$  into unpointed methods. It is not symmetric there, because  $\phi_{s,r} \circ \sigma$  lists the blocks in the other order. On  $\mathbb{P}$  the block transposition acts on  $\text{LoRA}_{r+s}$  by permuting inner channels,  $(B, A) \mapsto (BP^{-1}, PA)$ , and the symmetry square commutes. Pointed objects behave differently. If  $F(r)$  and  $F(2r)$  are standard  $T_A$  pointings ( $B_0 = 0$  and  $A_0$  of full row rank, [Background C.19](#)), which needs  $2r < \min(m, n)$ , then  $F(r) \oplus F(r) \not\equiv F(2r)$  as pointed objects, since  $\dim S_1$  is  $mr$  against  $2mr$  ([Observation I.7](#)).

**Used in.** [Observation I.7](#), the referee record in Exposé XI. Reading: Mac Lane (1998, Ch. XI).

**BACKGROUND B.16** (bicategories and the Para construction). A **bicategory** has objects, **1-cells** between objects and **2-cells** between parallel 1-cells. 2-cells compose vertically and horizontally, and composition of 1-cells is associative and unital up to coherent invertible 2-cells. For a symmetric monoidal category  $C$ , here always cartesian, the bicategory **Para**( $C$ ) has the objects of  $C$ . A 1-cell  $X \rightarrow Y$  is a pair  $(P, f : P \times X \rightarrow Y)$ , and  $(P', g) : Y \rightarrow Z$  composes with it to  $(P' \times P, (p', p, x) \mapsto g(p', f(p, x)))$ . A 2-cell  $(P, f) \Rightarrow (Q, g)$  is a **reparametrisation**  $r : Q \rightarrow P$  with  $g = f \circ (r \times \text{id}_X)$ . A **2-functor** (pseudofunctor) between bicategories sends objects, 1-cells and 2-cells to objects, 1-cells and 2-cells, and preserves both compositions up to coherent invertible 2-cells.

A linear layer is the 1-cell  $(\mathbb{R}^{m \times n}, (W, x) \mapsto Wx) : \mathbb{R}^n \rightarrow \mathbb{R}^m$ , and stacking a second layer  $\mathbb{R}^m \rightarrow \mathbb{R}^k$  composes 1-cells, with parameter space  $\mathbb{R}^{k \times m} \times \mathbb{R}^{m \times n}$ . A pretrained model is a pointed 1-cell  $F = (\Theta, \theta_0, f)$  of **Para**(**Diff**), and its realisation  $\Phi_f : \Theta \rightarrow C^\infty(X, Y), \theta \mapsto f(\theta, -)$ , records the function at each weight. LoRA is a 2-cell  $F \Rightarrow (Q, f \circ (\rho \times \text{id}_X))$ , the network with  $(B, A)$  in its parameter slot, and a simulation is a 2-cell between two such cells ([Remark I.3](#)). An adapter extension is a 2-cell into  $F$ , and merging it is a lift through  $\Phi_f$  ([Definition VI.1](#)).

Composition is associative only up to  $(P'' \times P') \times P \cong P'' \times (P' \times P)$ , hence a bicategory; over a strict monoidal category it is a 2-category.

**Used in.** Definition I.1, §I, Remark I.3, Definition VI.1, §IV, the key references of Exposé XII. Reading: Leinster (1998).

**BACKGROUND B.17** (lenses, reverse derivatives and  $\mathbf{Para}(\mathbf{R})$ ). In a cartesian category, a **lens**  $(A, A') \rightarrow (B, B')$  is a pair of maps, a forward part  $f : A \rightarrow B$  and a backward part  $f^\sharp : A \times B' \rightarrow A'$ . Lenses compose by

$$(g, g^\sharp) \circ (f, f^\sharp) = (g \circ f, (a, c') \mapsto f^\sharp(a, g^\sharp(f(a), c')))$$

and form the category  $\mathbf{Lens}(C)$ . For smooth maps between open subsets of Euclidean spaces, the **reverse derivative** is

$$\mathbf{R}[f](x, y) = J_f(x)^\top y.$$

The chain rule  $J_{g \circ f}(x)^\top = J_f(x)^\top J_g(f(x))^\top$  says that  $f \mapsto (f, \mathbf{R}[f])$  is a functor into  $\mathbf{Lens}$ . Applied to parametrised maps it gives the 2-functor  $\mathbf{Para}(\mathbf{R}) : \mathbf{Para}(C) \rightarrow \mathbf{Para}(\mathbf{Lens}(C))$  of Cruttwell et al. (2021), which is backpropagation.

LoRA's forward part is  $\rho(B, A) = W_0 + sBA$ , and its backward part pulls the weight gradient  $G$  back to

$$\mathbf{R}[\rho]((B, A), G) = (sGA^\top, sB^\top G).$$

GaLore (Zhao et al., 2024a) keeps the forward part  $\text{id}_\Theta$  and uses the compression  $G \mapsto P^\top G$  with anchor  $P$ , a lens whose backward part is not the reverse derivative of its forward part (Definition IV.1). For linear  $\rho(C) = W_k + \mathcal{L}C$ ,  $\mathbf{R}[\rho](C, G) = \mathcal{L}^*G$  does not depend on  $C$ , which is why the two readings agree at degree one (Proposition IV.2).

**Used in.** Exposé IV, §IV, Definition IV.1, Proposition IV.2, Rosetta entry 25 (§IX), Exposé 0 (where the category theory earns its keep), the seven borrowed words of Exposé XI. Reading: Cruttwell et al. (2021).

## Part C. Manifolds, germs and cones

Parts A and B used only the arrows of a method and its wiring. Part C turns to its geometry. A PEFT method in this paper is a smooth map  $\rho : (Q, q_0) \rightarrow (\Theta, \theta_0)$ , and its expressivity is read off the set  $\text{Im } M = \rho(Q)$  near  $\theta_0$ . This part collects the geometry behind that reading, from manifolds, differentials and rank for the map to germs, cones, singular points and semialgebraic dimension for the image, which is often not a manifold.

### Manifolds and smooth maps

**BACKGROUND C.1** (smooth manifolds and smooth maps). A **smooth manifold** of dimension  $k$  is a space covered by **charts**, bijections from open pieces of it onto open subsets of  $\mathbb{R}^k$ , whose changes of chart are smooth. Every manifold in this paper is an open subset of a vector space or an embedded submanifold of one (Background C.4), or is diffeomorphic to one, as quotients (Background D.22) and the Grassmannian (Background C.15) are. A **diffeomorphism** is a smooth bijection with smooth inverse, and  $\text{Diff}(Q)$  is the group of diffeomorphisms of  $Q$  under composition. A **local diffeomorphism** maps an open

neighbourhood of each point diffeomorphically onto an open set; by the inverse function theorem,  $\rho$  does this near  $q$  whenever  $d\rho_q$  (Background C.2) is invertible.

LoRA's (Hu et al., 2021) parameter space  $\mathbb{R}^{m \times r} \times \mathbb{R}^{r \times n}$  is a vector space, and HRA's (Yuan et al., 2024)  $\{(u_1, \dots, u_r) : u_i \neq 0\}$  is open in  $(\mathbb{R}^n)^r$ . The rotation group  $\mathrm{SO}(n)$  is a compact manifold of dimension  $n(n-1)/2$ , and the Cayley map (Background D.7) is a diffeomorphism from  $\mathfrak{so}(n)$  onto an open dense subset of it. A method is a pointed smooth map, and its gauge group  $\{\varphi \in \mathrm{Diff}(Q) : \rho \circ \varphi = \rho\}$  (Background D.21) contains LoRA's  $(B, A) \mapsto (Bg^{-1}, gA)$  for every  $g \in \mathrm{GL}_r$ .

**Used in.** Definition I.2, Definition I.5, Theorem II.7, Proposition IV.3. Reading: Lee (2013, Ch. 1 and 2).

**BACKGROUND C.2** (tangent spaces, differentials and covectors). For a manifold  $Q \subseteq \mathbb{R}^N$ , the **tangent space**  $T_q Q$  is the set of velocities  $c'(0)$  of smooth curves  $c$  in  $Q$  with  $c(0) = q$ . It is a linear space of dimension  $\dim Q$ , all of  $\mathbb{R}^N$  when  $Q$  is open. The **differential** of a smooth  $\rho$  at  $q$  is the linear map  $d\rho_q : T_q Q \rightarrow T_{\rho(q)} \Theta$ ,  $c'(0) \mapsto (\rho \circ c)'(0)$ . In coordinates it is the Jacobian, and the chain rule reads  $d(\rho \circ h)_q = d\rho_{h(q)} \circ dh_q$ . The **cotangent space**  $T_\theta^* \Theta$  is the dual of  $T_\theta \Theta$ . The differential  $dL_\theta$  of a loss is a **covector**, and it becomes a gradient vector  $\nabla L(\theta)$  only once an inner product is chosen.

For LoRA,  $d\rho_{(B,A)}(\dot{B}, \dot{A}) = s(\dot{B}A + B\dot{A})$ . For a rotation orbit,  $T_l \mathrm{SO}(n) = \mathfrak{so}(n)$  and  $T_{W_0 R}(W_0 \cdot \mathrm{SO}(n)) = W_0 R \mathfrak{so}(n)$ . **Observation I.6** is the chain rule at  $q_0$ . **Theorem III.5** drives the parameters by any cotangent signal  $\gamma(t) \in T_{\rho(q)}^* \Theta$ , pulled back to  $Q$  as  $\gamma \circ d\rho_q$ , which is  $D\rho^\top \gamma$  in coordinates.

**Used in.** the notation (Table 2), Definition I.5, Observation I.6, Theorem II.2, Theorem II.7, Theorem III.5, Definition IV.1. Reading: Lee (2013, Ch. 3 and 11).

**BACKGROUND C.3** (rank, expressive dimension and first-order image). The **rank** of  $\rho$  at  $q$  is  $\mathrm{rank} d\rho_q$ . It is lower semicontinuous, since a nonzero  $k \times k$  minor of the Jacobian stays nonzero nearby, so  $\{q : \mathrm{rank} d\rho_q \geq k\}$  is open. The **generic rank** is  $\max_q \mathrm{rank} d\rho_q$ . If  $\rho$  is real-analytic and  $Q$  is connected, the generic rank is attained on an open dense set whose complement has measure zero (Background C.22). The paper writes  $|M| = \dim Q$  for the parameter budget,  $d(M)$  for the generic rank,  $S_1 = d\rho_{q_0}(T_{q_0} Q)$  for the **first-order image** and  $d(M) - \dim S_1$  for the **first-order deficit**. For definable  $\rho$  on an open definable  $Q$ , as for every method in this paper, Background C.18(d) shows that  $d(M) = \dim \mathrm{Im} M$ , the **expressive dimension**. The deficit vanishes exactly when  $q_0$  is a point of maximal rank; the base point  $\theta_0$  is then called **regular** for  $M$ , and an **apex** when the deficit is positive (Definition I.5).

**Proposition II.3** gives LoRA's rank at every point,  $m \mathrm{rk} A + n \mathrm{rk} B - \mathrm{rk} A \mathrm{rk} B$ . At a random pair it is  $r(m + n - r) = d$ ; at the zero-init point  $(0, A_0)$  it is  $mr$ , a deficit of  $r(n - r)$ . Since the points of lower rank have measure zero, a float64 Jacobian rank at one random point computes  $d(M)$  with probability one, up to the tolerance that decides numerical rank. The plug workbench of the companion site (§II.C) and the check script (Appendix E) do exactly this.

**Used in.** the notation (Table 2), Definition I.5, Proposition II.3, Proposition II.12, the plug workbench (§II.C), reproducing the numbers (Appendix E).



**BACKGROUND C.4** (immersions, submersions and embedded submanifolds). A smooth  $\rho : Q \rightarrow \Theta$  is an **immersion** if every  $d\rho_q$  is injective ( $\text{rank dim } Q$ ), and a **submersion** if every  $d\rho_q$  is surjective ( $\text{rank dim } \Theta$ ). A subset  $X \subseteq \mathbb{R}^N$  is an **embedded submanifold** of dimension  $k$  if near each of its points it is the zero set of  $N - k$  smooth functions with linearly independent gradients. A submersion is an open map, so its image contains an open set. An **embedding**, an injective immersion that is a homeomorphism onto its image, has an embedded submanifold as image; an injective immersion such as the figure eight need not.

The one-sided method  $\text{FA}(A_0) : B \mapsto W_0 + BA_0$  is an injective immersion with affine image  $W_0 + \{BA_0\}$  when  $A_0$  has full row rank. On pairs of full rank,  $(B, A) \mapsto BA$  is a submersion onto  $\mathcal{M}_r$  ([Background C.12](#)). [Observation I.4\(b\)](#) uses openness, in the argument recalled in [Background A.13](#). [Proposition IV.3](#) asks for injective immersions  $(\mathbb{R}^k, 0) \rightarrow (U, \theta)$  tangent to a distribution.

**Used in.** [Observation I.4](#), [Proposition IV.3](#), integrability (§IV). Reading: [Lee \(2013, Ch. 4 and 5\)](#).

**FACT C.5** (constant-rank theorem). *Let  $\rho : Q \rightarrow \Theta$  be smooth with  $\text{rank } d\rho_q = k$  for all  $q$  near  $p$ . Then there are charts centred at  $p$  and at  $\rho(p)$  in which  $\rho$  reads  $(x_1, \dots, x_{\dim Q}) \mapsto (x_1, \dots, x_k, 0, \dots, 0)$ . Hence some neighbourhood  $V$  of  $p$  has  $\rho(V)$  an embedded  $k$ -dimensional submanifold, and  $\rho^{-1}(\rho(p)) \cap V$  an embedded submanifold of dimension  $\dim Q - k$ . By [Background C.3](#) the hypothesis holds wherever the rank is maximal.*

**The idea of the proof.** Choose coordinates in which the top-left  $k \times k$  block of the Jacobian is invertible. The inverse function theorem straightens the first  $k$  output coordinates. Constant rank then forces the other outputs to depend only on those  $k$ , and a second change of coordinates in the target removes them.

The paper uses this theorem without naming it. Near a point of maximal rank,  $\rho$  is locally a projection onto a sheet of dimension  $d(M)$ , with fibres of dimension  $|M| - d(M)$ , the gauge waste. For LoRA at a full-rank pair the sheet is a piece of  $W_0 + \mathcal{M}_r$ , and the fibre a piece of a  $\text{GL}_r$ -orbit. [Proposition IV.3](#) makes the analogous constant-rank assumption for distributions, where Frobenius' theorem ([Background D.25](#)) takes the place of this one.

**Used in.** [Definition I.5](#), [Proposition II.12](#), [Background C.18](#). Reading: [Lee \(2013, Ch. 4, the rank theorem\)](#).

**BACKGROUND C.6** (fibres and fibre dimension). The **fibre** of  $\rho$  over  $\theta$  is  $\rho^{-1}(\theta)$ , the set of parameters that give the weight  $\theta$ . If  $\text{rank } d\rho_q = d(M)$ , [Background C.5](#) makes the fibre through  $q$  a submanifold of dimension  $|M| - d(M)$  near  $q$ . For real-analytic  $\rho$  on connected  $Q$  this holds for all  $q$  outside a closed set of measure zero, and these are the **generic fibres**. Fibres through points of lower rank can have a different dimension. The fibres of a fibre bundle all look alike locally, so a map whose fibre dimension jumps is no fibre bundle.

For LoRA on the full-rank stratum, the fibre through  $(B, A)$  is the orbit  $\{(Bg^{-1}, gA) : g \in \text{GL}_r\}$ , of dimension  $r^2 = r(m+n) - r(m+n-r)$  ([Background D.21](#)). For zero-init LoRA the fibre over  $\theta_0$  is  $\{(B, A) : BA = 0\}$ . It contains  $\{B = 0\}$ , of dimension  $rn > r^2$ , and is itself a union of strata, such as  $\{(0, A) : \text{rank } A = r\}$  ([Corollary III.13](#)).

**Used in.** [Exposé III](#), [Definition I.5](#), [Proposition II.12](#), [Corollary III.13](#), Rosetta entries 6 and 11 ([Exposé IX](#)), the seven borrowed words ([Exposé XI](#)).

**BACKGROUND C.7** (fibre products, transversality and clean intersection). For  $\rho_M : Q_M \rightarrow \Theta$  and  $\rho_N : Q_N \rightarrow \Theta$ , the set-level **fibre product** is  $P = \{(q, q') : \rho_M(q) = \rho_N(q')\} \subseteq Q_M \times Q_N$ . The maps are **transverse**,  $\rho_M \pitchfork \rho_N$ , if  $d\rho_M(T_q Q_M) + d\rho_N(T_{q'} Q_N) = T_{\theta} \Theta$  at every  $(q, q') \in P$ ; then  $P$  is an embedded submanifold of dimension  $|M| + |N| - \dim \Theta$ , and it is the pullback of  $\rho_M$  and  $\rho_N$  in **Diff** ([Background A.11](#)). The intersection is **clean** if  $P$  is an embedded submanifold with  $T_{(q, q')} P = \{(v, w) : d\rho_M v = d\rho_N w\}$  at every point. Transverse maps meet cleanly, and so do affine maps.

Transversality at  $(q_0, q'_0)$  needs  $|M| + |N| \geq \dim \Theta$ , which fails for any two small methods, so the products the paper builds come from clean cases. The affine methods  $\text{FA}(A_0)$  and  $\text{FB}(B_0)$ , with  $B_0$  of full column rank and  $A_0$  of full row rank, have the linear fibre product  $\{(B_0 R, R A_0)\}$  of dimension  $r^2$ , which is LoRA-XS ([Bałazy et al., 2024](#)) ([Proposition I.8](#)). On  $\Theta = \mathbb{R}$ , LoRA<sub>1</sub> and the frozen model have the cross  $\{BA = 0\}$  as fibre product. It is no manifold at the origin, and no product exists ([Background A.13](#)).

**Used in.** [Observation I.4](#), [Proposition I.8](#), [Theorem II.7\(d\)](#). Reading: [Guillemin and Pollack \(1974, Ch. 1\)](#); [Lee \(2013, Ch. 6\)](#).

### Germes, cones and singular points

**BACKGROUND C.8** (germs). Two subsets  $X, Y \subseteq \Theta$  have the same **germ** at  $\theta_0$  if  $X \cap U = Y \cap U$  for some neighbourhood  $U$  of  $\theta_0$ ; two maps have the same germ at  $q_0$  if they agree near  $q_0$ . The germs  $(X, \theta_0)$  and  $(Y, 0)$  are diffeomorphic if a diffeomorphism  $\varphi$  between neighbourhoods  $U \ni \theta_0$  and  $V \ni 0$  has  $\varphi(\theta_0) = 0$  and  $\varphi(X \cap U) = Y \cap V$ . Local properties, such as the dimension at  $\theta_0$ , smoothness at  $\theta_0$  and the tangent cone ([Background C.11](#)), depend only on the germ.

The image germ of a method is the germ of  $\text{Im } M = \rho(Q)$  at  $\theta_0$ . It uses all of  $Q$ , so it can be larger than the image of a small neighbourhood of  $q_0$ . For zero-init LoRA,  $sBA$  with  $B$  small and  $A$  arbitrary is a small update with any row space, so the germ is that of the whole cone  $\theta_0 + \mathcal{M}_{\leq r}$ ; parameters near  $(0, A_0)$  only give updates with row space inside row  $A$  for some  $A$  near  $A_0$ . For injective  $W_0$  and even  $r$  with  $2 \leq r \leq n - 2$ , HRA's image germ at  $W_0$  is the image of the germ of  $\Sigma_r$  at 0 under the embedding  $\Omega \mapsto W_0 C(\Omega)$  of  $\mathfrak{so}(n)$  into  $\mathbb{R}^{m \times n}$  ([Background C.13](#), [Background D.7](#)).

**Used in.** the image germ ([Exposé II](#)), the three invariants ([Exposé 0](#)), [Theorem II.7](#), the seven axes ([§VIII.A](#)).

**BACKGROUND C.9** (smooth and singular points). A point  $x$  of a set  $X \subseteq \mathbb{R}^N$  is a **smooth point** if  $X \cap U$  is a  $C^1$  embedded submanifold for some neighbourhood  $U$  of  $x$ , and a **singular point** otherwise. The singular points form the **singular locus**  $\text{Sing } X$ . Algebraic geometry defines  $\text{Sing}$  through the Jacobian of defining polynomials, and for  $\mathcal{M}_{\leq r}$  the two notions agree. Smoothness is a property of a set. The paper's regular base point and apex, defined by a zero or positive first-order deficit ([Background C.3](#), [Definition I.5](#)), are properties of a method at  $q_0$ .

The cross  $\{BA = 0\} \subseteq \mathbb{R}^2$  is singular at the origin and smooth elsewhere. For  $1 \leq r < \min(m, n)$ ,  $\text{Sing } \mathcal{M}_{\leq r} = \mathcal{M}_{\leq r-1}$ . A split initialisation has image  $(\theta_0 - P) + \mathcal{M}_{\leq r}$  with  $\text{rank } P = r$ , so  $\theta_0$  is a smooth point; zero-init LoRA starts at the singular vertex. A figure-eight curve

passing through its crossing with nonzero velocity has deficit 0 there, though its image is singular, so smooth and regular are different notions.

**Used in.** Definition I.5, Theorem II.2, Figure 3, Theorem II.7, Erlangen (§II.B). Reading: Harris (1992, Lecture 14).

**BACKGROUND C.10** (cones and vertices). A set  $X$  in a vector space is a **cone** with **vertex**  $v$  if  $v \in X$  and  $v + t(x - v) \in X$  for all  $x \in X$  and  $t \geq 0$ , and **linear** if it is an affine subspace. If  $\rho = \theta_0 + \delta$  with  $\delta(tu, w) = t\delta(u, w)$  for  $t \geq 0$  on a block  $u$  of parameters, and  $Q$  is closed under  $u \mapsto tu$ , then  $\text{Im } M$  is a cone with vertex  $\theta_0$ . A cone that is a  $C^1$  embedded submanifold near its vertex  $v$  equals  $v + T_v X$ , so the vertex of a non-linear cone is a singular point.

**Why a smooth cone is linear.** Each ray from  $v$  lies in  $X$ , so  $X \subseteq v + T_v X$ . Near  $v$ ,  $X$  is a submanifold of full dimension in this affine space, so it contains a neighbourhood of  $v$  there, and scaling spreads it over all of  $v + T_v X$ .

$\mathcal{M}_{\leq r}$  is a closed cone with vertex 0, and it is not linear for  $1 \leq r < \min(m, n)$ , since a rank- $r$  and a rank-one matrix can sum to rank  $r + 1$ . Zero-init LoRA has  $\delta = sBA$ , linear in  $B$ , so its image  $W_0 + \mathcal{M}_{\leq r}$  has singular vertex  $W_0$ . Its deficit  $r(n - r)$  makes  $\theta_0$  an apex too; the vertex is a property of the set, the apex one of the method. Multihomogeneous updates such as LoHa's (Hyeon-Woo et al., 2021) give cones in the same way (Background D.19).

**Used in.** Definition I.5, the image germ (Exposé II), Theorem II.1, Theorem II.2, Figure 3, Theorem II.7.

**FACT C.11** (first-order image versus image). *The **tangent cone** of a set  $X$  at  $x$  is the set of limits of  $t_k(x_k - x)$  with  $x_k \in X$ ,  $x_k \rightarrow x$  and  $t_k > 0$ ; for a closed cone with vertex  $v$  it is  $X - v$ . For a method  $M$ : (a)  $S_1$  lies in the tangent cone of  $\text{Im } M$  at  $\theta_0$ ; (b)  $\dim S_1 \leq d(M)$ ; (c) if  $\text{Im } M$  is a  $C^1$  manifold of dimension  $d(M)$  near  $\theta_0$ , then  $S_1 \subseteq T_{\theta_0} \text{Im } M$ , with equality exactly when the deficit is 0.*

For (a),  $d\rho_{q_0}(\xi)$  is the velocity of the curve  $\rho(q_0 + t\xi)$  inside the image; for (c),  $\rho$  maps a neighbourhood of  $q_0$  into the manifold. For  $T_A$ -pointed LoRA ( $B_0 = 0$ , Background C.19) the tangent cone at  $\theta_0$  is  $\mathcal{M}_{\leq r}$  itself, and inside it  $S_1 = \{XA_0\}$ , the matrices with row space in row  $A_0$ , is a linear space of dimension  $mr < r(m + n - r)$ . Figure 3(a) draws  $S_1$  as one ruling of the cone, a line through the apex. At a split pointing case (c) applies, and  $S_1 = T_P \mathcal{M}_r$ , the tangent plane of Figure 3(b).

**Used in.** Definition I.5, Observation I.6, Theorem II.2, Figure 3. Reading: Harris (1992, Lecture 20).

## Matrix varieties

**BACKGROUND C.12** (the determinantal variety).  $\mathcal{M}_{\leq r} = \{X \in \mathbb{R}^{m \times n} : \text{rank } X \leq r\}$ , the **determinantal variety**, is the zero set of all  $(r + 1) \times (r + 1)$  minors, a closed cone with vertex 0. Let  $1 \leq r < \min(m, n)$ . The matrices of rank exactly  $r$  form an open dense subset  $\mathcal{M}_r$  of  $\mathcal{M}_{\leq r}$ , an embedded submanifold of dimension  $r(m + n - r)$ , and  $\text{Sing } \mathcal{M}_{\leq r} = \mathcal{M}_{\leq r-1}$ . If  $X \in \mathcal{M}_r$  has column space  $U$  and row space  $V$ , and  $X = BA$  with  $B \in \mathbb{R}^{m \times r}$  and  $A \in \mathbb{R}^{r \times n}$ , then

$$T_X \mathcal{M}_r = \{\dot{B}A + B\dot{A}\} = \{D : P_{U^\perp} D P_{V^\perp} = 0\}.$$

Also  $\mathcal{M}_{\leq r} - \mathcal{M}_{\leq r} = \mathcal{M}_{\leq \min(2r, m, n)}$ .

Zero-init LoRA has image  $W_0 + \mathcal{M}_{\leq r}$ , of dimension 130,816 for a  $4096 \times 4096$  matrix at  $r = 16$ . On full-rank pairs,  $(B, A) \mapsto BA$  is a submersion onto  $\mathcal{M}_r$ , whose fibres are  $\mathrm{GL}_r$ -orbits, which identifies  $\mathcal{M}_r$  with the quotient  $(\mathbb{R}_*^{m \times r} \times \mathbb{R}_*^{r \times n})/\mathrm{GL}_r$  (Background D.22). The difference formula is why no pointing of  $\mathrm{LoRA}_r$  reaches an update of rank above  $2r$  (Background B.14, Theorem II.1).

**Used in.** the notation (Table 2), translates of one shape (§II.A), Theorem II.1, Theorem II.2, Proposition II.8, Exposé II.C (§II.C), Proposition III.4. Reading: Harris (1992, Lecture 9).

**BACKGROUND C.13** (the skew-determinantal cone). Every skew-symmetric matrix has even rank. For even  $r$  with  $2 \leq r \leq n - 2$ , the **skew-determinantal cone**  $\Sigma_r = \{\Omega \in \mathfrak{so}(n) : \mathrm{rank} \Omega \leq r\}$  is a closed cone with vertex 0 and dimension  $r(n-r) + r(r-1)/2 = rn - r(r+1)/2$ , counting a range in  $\mathrm{Gr}(r, n)$  and a nondegenerate skew form on it. Its smooth points are the matrices of rank exactly  $r$ , and  $\mathrm{Sing} \Sigma_r = \Sigma_{r-2}$ . It is not linear, since skew matrices of rank  $r$  and 2 can sum to rank  $r + 2 \leq n$ , so its vertex is singular.

For injective  $W_0$  and even  $r$ , HRA's image near  $W_0$  consists of the  $W_0 R$  with  $R \in \mathrm{SO}(n)$  near  $I$  and  $\mathrm{rank}(R - I) \leq r$ . The Cayley map satisfies  $\mathrm{rank}(C(\Omega) - I) = \mathrm{rank} \Omega$  (Background D.7), so the image germ at  $W_0$  is the image of the germ of  $\Sigma_r$  at 0 under the embedding  $\Omega \mapsto W_0 C(\Omega)$ ,  $W_0$  is a singular vertex, and  $d(\mathrm{HRA}_r) = \dim \Sigma_r$ . At the paired initialisation  $u_{2i-1} = u_{2i} = w_i$  ( $i \leq k = r/2$ , with the  $w_i$  independent), put  $U = \mathrm{span}\{w_1, \dots, w_k\}$ , so  $\dim U = k$ . Then  $S_1 = W_0 \cdot S$  with

$$S = \{\Omega \in \mathfrak{so}(n) : P_{U^\perp} \Omega P_{U^\perp} = 0\} = U \wedge \mathbb{R}^n$$

(Background D.5), a linear space of skew matrices of rank at most  $2k = r$  and of dimension  $kn - k(k+1)/2 < d(\mathrm{HRA}_r)$ . It lies inside  $\Sigma_r$ , so  $S_1$  lies inside the tangent cone  $W_0 \cdot \Sigma_r$  (Theorem II.7(b)).

**Used in.** Theorem II.7, HRA is a meet (§II.B).

**BACKGROUND C.14** (Segre and toric cones). The **Segre cone** in  $\mathbb{R}^p \otimes \mathbb{R}^q \cong \mathbb{R}^{p \times q}$  is the set  $\{a \otimes b\} = \mathcal{M}_{\leq 1}$  of rank-one tensors, a closed cone of dimension  $p + q - 1$  that spans the whole space and is non-linear when  $p, q \geq 2$ . It is the simplest **toric cone**, the closure of the image of a torus  $(\mathbb{R}^\times)^k$  under a monomial map, here  $(a, b) \mapsto (a_i b_j)_{ij}$ . Since rank-one matrices span  $\mathbb{R}^{p \times q}$ ,  $k$  generic rank-one matrices span  $\min(k, pq)$  dimensions.

Under the Van Loan–Pitsianis rearrangement (Background B.11), KronA's (Edalati et al., 2022) update  $A \otimes B$  becomes  $\mathrm{vec}(A) \mathrm{vec}(B)^\top$ , so its image is a linear copy of a Segre cone (Proposition II.12). VeRA's (Kopiczko et al., 2023) update  $\Lambda_b B \Lambda_d A$  has entries  $\sum_k (b_i d_k) B_{ik} A_{kj}$ , a fixed linear function of the rank-one matrix  $bd^\top$ . This function is injective when  $B$  has no zero entry and  $A$  has full row rank, so the image is a toric cone of dimension  $m + r - 1$  with vertex  $\theta_0$ , as VeRA's entry in Exposé VII records. Proposition II.8 uses the spanning fact for DoRA (Liu et al., 2024b), with  $m$  points of the Segre cone in  $\mathbb{R}^{m-r} \otimes \mathbb{R}^{n-r}$ .

**Used in.** Proposition II.8, Proposition II.12, VeRA (Exposé VII). Reading: Harris (1992, Lecture 2); Cox et al. (2011, Ch. 1).

**BACKGROUND C.15** (the Grassmannian). The **Grassmannian**  $\mathrm{Gr}(r, n)$  is the set of  $r$ -dimensional linear subspaces of  $\mathbb{R}^n$ , a compact manifold of dimension  $r(n - r)$ . One model is the full-row-rank  $r \times n$  matrices up to  $A \mapsto gA$ ,  $g \in \mathrm{GL}_r$ , through  $A \mapsto \mathrm{row} A$ ; another is the orthogonal projectors  $P = P^\top = P^2$  with  $\mathrm{tr} P = r$ . Near  $V$ , the graphs of

linear maps  $V \rightarrow V^\perp$  give a chart onto  $\text{Hom}(V, V^\perp) \cong \mathbb{R}^{r(n-r)}$ .  $\text{Gr}(r, n)$  carries a unique  $O(n)$ -invariant probability measure, the uniform or Haar measure, and the row space of a Gaussian  $A$  has this law ([Background D.14](#)).

$T_A$ -pointed LoRA is classified up to isomorphism by row  $A_0 \in \text{Gr}(r, n)$  ([Theorem II.2](#)), and its deficit  $r(n-r)$  equals  $\dim \text{Gr}(r, n)$ . First steps keep the row space of the update inside row  $A_0$ , and the missing directions are those that tilt it. For injective  $W_0$  and even  $r$  with  $2 \leq r \leq n-2$ , HRA's image is the union of the orbits  $W_0 \cdot \text{SO}(V)$  over  $V \in \text{Gr}(r, n)$ , and its dimension  $r(n-r) + r(r-1)/2$  counts a subspace and a rotation inside it. LoRA-FA ([Zhang et al., 2023d](#)), which trains  $B$  with  $A_0$  frozen, sees  $A_0$  only through  $V = \text{row } A_0$  once its gradient is corrected as in [Proposition III.4\(c\)](#).

**Used in.** [Theorem II.2](#), the checks ([Appendix E](#)), [Theorem II.7](#), LOFT (§II.B) and the LOFT example there, [Proposition III.4](#), [Corollary III.13](#). Reading: Lee (2013, Ch. 1); Harris (1992, Lecture 6).

### Tame sets and genericity

**FACT C.16** (images of smooth maps need not be manifolds). *Even a polynomial map can have an image that is no manifold. (a) Cusps.  $t \mapsto (t^2, t^3)$  is injective with image  $\{y^2 = x^3\}$ , which is not  $C^1$  at 0. (b) Crossings and cones. The cross  $\{BA = 0\} \subseteq \mathbb{R}^2$  and the cone  $\mathcal{M}_{\leq r}$  are singular at the origin. (c) Missing limits.  $(x, y) \mapsto (x, xy)$  has image  $\{u \neq 0\} \cup \{(0, 0)\}$  in the  $(u, v)$ -plane, neither open nor closed. (d) Dense images. The line of irrational slope is an injective immersion into the torus with dense image.*

PEFT images show (b) routinely, as in the LoRA cone and HRA's skew cone, so the paper measures them with tools that need no manifold structure, such as germs ([Background C.8](#)), tangent cones ([Background C.11](#)) and the dimension theory of semialgebraic and definable sets ([Background C.17](#), [Background C.18](#)). Whether LoHa's Hadamard image is closed, the question raised by (c), is open.

**Used in.** [Observation I.4](#), [Theorem II.2](#), [Theorem II.7](#), the open problems ([Exposé XI](#)).

**BACKGROUND C.17** (semialgebraic and definable sets). A **semialgebraic set** is a finite union of sets  $\{x \in \mathbb{R}^N : p(x) = 0, q_1(x) > 0, \dots, q_k(x) > 0\}$  with polynomials  $p, q_i$ , and a map is semialgebraic if its graph is. A set is **definable** in  $\mathbb{R}_{\text{exp}}$  if a first-order formula defines it from polynomials with real coefficients, exp, equality, order, Boolean operations and quantifiers over  $\mathbb{R}$ . Wilkie (1996) proved that  $\mathbb{R}_{\text{exp}}$  is **o-minimal**, meaning every definable subset of  $\mathbb{R}$  is a finite union of points and intervals. Semialgebraic sets are the definable sets that need no exp.

LoRA, DoRA (whose row norms involve square roots), HRA and exact-Cayley OFT ([Qiu et al., 2023](#)) are built from polynomials, quotients and square roots, so their images are semialgebraic by [Background C.18\(a\)](#). A softmax brings in exp, and maps that contain one are definable in  $\mathbb{R}_{\text{exp}}$ .

**Used in.** where the category theory works ([Exposé 0](#)), the arrow rules (§VIII.C). Reading: Bochnak et al. (1998, Ch. 2); Wilkie (1996).



**FACT C.18** (Tarski–Seidenberg and definable dimension). (a) The image of a semialgebraic set under a semialgebraic map is semialgebraic (Tarski–Seidenberg); images of definable sets under definable maps are definable. (b) Every definable set  $X$  is a finite disjoint union of  $C^1$  embedded submanifolds, and  $\dim X$ , the largest of their dimensions, does not depend on the choice. (c)  $X \subseteq Y$  implies  $\dim X \leq \dim Y$ ,  $\dim f(X) \leq \dim X$  for definable  $f$ , and  $\dim(\overline{X} \setminus X) < \dim X$  for nonempty  $X$ . (d) For a definable  $C^1$  map  $\rho$  on an open definable  $Q \subseteq \mathbb{R}^k$ ,  $\dim \rho(Q) = \max_Q \operatorname{rank} d\rho_Q$ .

**The idea of (d).** Near a point of maximal rank, [Background C.5](#) gives a sheet of that dimension inside  $\rho(Q)$ . Conversely, a definable section of  $\rho$ ,  $C^1$  on a top-dimensional piece of  $\rho(Q)$ , composes with  $\rho$  to the identity there, so the rank is at least that dimension.

Part (d) is the identity  $d(M) = \dim \operatorname{Im} M$  of [Definition I.5](#). Part (c) is the monotonicity behind test O2 ([§VIII.C](#)), since an arrow  $M \rightarrow N$  gives  $\operatorname{Im} M \subseteq \operatorname{Im} N$  and hence  $d(M) \leq d(N)$ ; for instance  $\operatorname{LoRA}_r \not\rightarrow \operatorname{LoRA}\text{-FA}$  because  $r(m+n-r) > mr$ .

**Used in.** [Definition I.5](#), the arrow rules ([§VIII.C](#)), where the category theory works ([Exposé 0](#)). Reading: [van den Dries \(1998, Ch. 3 and 4\)](#); [Bochnak et al. \(1998, Ch. 2\)](#).

**BACKGROUND C.19** (stratification and local dimension). A **stratification** of a set  $X$  is a partition into finitely many embedded submanifolds, the **strata**, such that the closure of each stratum is a union of strata. The **local dimension**  $\dim_x X$  is the dimension of the germ of  $X$  at  $x$ , the largest dimension of a stratum whose closure contains  $x$ .  $X$  is **pure-dimensional** if  $\dim_x X$  is the same at every point. Definable sets admit stratifications, so these notions make sense for every image in this paper.

The cross  $\{ba = 0\} \subseteq \mathbb{R}^2$  is stratified as the origin and the four open half-axes, and it is pure of dimension 1. Likewise  $M_{\leq r} = M_0 \sqcup M_1 \sqcup \cdots \sqcup M_r$  with  $\overline{M_k} = M_{\leq k}$ , and it is pure of dimension  $r(m+n-r)$ . For  $1 \leq r < \min(m, n)$ , [Theorem II.2](#) singles out three strata of full-rank LoRA pointings,  $T_A (B_0 = 0, \operatorname{rank} A_0 = r)$ ,  $T_B (A_0 = 0, \operatorname{rank} B_0 = r)$  and the open dense  $S$  ( $\operatorname{rank} B_0 A_0 = r$ ), and the fibre  $\{BA = 0\}$  over  $\theta_0$  is itself stratified ([Corollary III.13](#)). The HRA fibre product of [Theorem II.7\(d\)](#) has local dimension  $d(\operatorname{HRA}_r) + r^2 = rn + r(r-1)/2$  over generic points of the meet and contains  $\{(B, 0, I)\}$ , of dimension  $mr$ , over  $W_0$ , so it is not pure-dimensional when  $mr > rn + r(r-1)/2$ .

**Used in.** [Theorem II.2](#), the three strata ([§II.A](#)), [Theorem II.7](#), [Corollary III.13](#), the open problems ([Exposé XI](#)), the seven borrowed words ([Exposé XI](#)). Reading: [van den Dries \(1998, Ch. 4\)](#); [Bochnak et al. \(1998, Ch. 9\)](#).

**BACKGROUND C.20** (moduli sets and moduli spaces). Classifying objects up to isomorphism gives a set of classes. A **moduli space** also carries a topology or geometry for which continuous or smooth families of objects correspond to continuous or smooth maps into it, ideally with a universal family over it from which every family is pulled back. A disjoint union of spaces  $X \sqcup Y$  has the topology in which  $X$  and  $Y$  are both open and closed.

[Theorem II.2](#) classifies the three strata of LoRA pointings, as a set, by  $\operatorname{Gr}(r, n) \sqcup \operatorname{Gr}(r, m) \sqcup M_r$ . The quotient topology does not split this way, since  $(\varepsilon B', A_0) \in S$  tends to  $(0, A_0) \in T_A$  as  $\varepsilon \rightarrow 0$ , so  $T_A$  lies in the closure of  $S$ . No universal family is built, and the paper uses “moduli” as an analogy.

**Used in.** [Theorem II.2](#), the Rosetta stone ([Exposé IX](#)), the seven borrowed words ([Exposé XI](#)). Reading: [Harris \(1992, Lecture 4\)](#).

**BACKGROUND C.21** (generic properties). A property of points of  $\mathbb{R}^N$  holds **generically** if the set where it fails lies in  $\{f = 0\}$  for some nonzero polynomial  $f$ . The set  $\{f = 0\}$  is closed, nowhere dense and of Lebesgue measure zero, so a generic property holds with probability one at a random point whose law has a density, such as a Gaussian or uniform initialisation. For points of a parametrised set, such as a Segre cone, “generic” refers to generic parameters. To prove genericity it suffices to write failure as the vanishing of polynomials and exhibit one point where one of them is nonzero. A weaker notion, **almost every** point, asks only that the failure set have Lebesgue measure zero, as a countable union of proper algebraic subsets does.

[Theorem II.2\(d\)](#) uses a generic rank-one update, [Proposition II.8](#) gives DoRA’s dimension  $\min(mn, r(m + n - r) + m)$  for generic  $W_0$ , and [Proposition V.2](#) notes that QLoRA’s (Dettmers et al., 2023) defect  $e = \kappa W - W$  has full rank for almost every  $W$ , in the weaker sense. The quantiser  $\kappa$  is defined cell by cell rather than by one polynomial, so the argument runs cell by cell, and the failure set is a countable union of proper algebraic subsets of quantisation cells, which has measure zero. A pretrained  $W_0$  is not random, and structured weights can lie in the exceptional set. For a rank-one  $W_0 = ac^\top$  with all  $a_i \neq 0$  and  $r < n$ , [Proposition II.8](#) finds DoRA’s dimension  $r(m + n - r) + m - r$ , strictly below the generic value exactly when  $r < m$  and  $r \leq n - 2$ .

**Used in.** [Theorem II.2](#), [Proposition II.8](#), [Theorem II.11](#), [Proposition II.12](#), [Proposition V.2](#).

**BACKGROUND C.22** (real-analytic functions and their zero sets). A function is **real-analytic** on an open set if near every point it equals a convergent power series. Polynomials, exp, softmax, tanh, GELU and SiLU are real-analytic, and so are sums, products, compositions and quotients with nonvanishing denominator. On a connected open set, a real-analytic function that vanishes on a nonempty open subset vanishes identically, and the zero set of one that is not identically zero is closed, nowhere dense and of measure zero.

[Lemma VI.2](#) compares Taylor coefficients at 0. A nonzero scale  $c$  with  $\sigma(ct) = c \sigma(t)$  must satisfy  $ca_k = c^k a_k$  for every coefficient  $a_k$  of  $\sigma$ , so  $c = 1$  for GELU and SiLU, whose  $a_2 \neq 0$ . [Theorem VI.5\(f\)](#) uses the zero-set fact. A prefixed head is analytic in the prefix, and one prefix with a nonzero second derivative makes the prefixes with affine output a nowhere-dense set. The same fact makes the maximal-rank set of an analytic  $\rho$  open and dense ([Background C.3](#)).

**Used in.** [Lemma VI.2](#), [Theorem VI.5](#), Rosetta entry 5 (Exposé IX). Reading: Krantz and Parks (2002).

## Part D. Groups, symmetry and conservation

Part D turns to symmetry. Most PEFT parametrisations have symmetries, since different trainable parameters can give the same weights and some transformations of weight space carry a method’s image to itself. This part collects the group theory the paper uses to describe those symmetries, the invariants they leave untouched, and the quantities they conserve under gradient flow.

## Groups and their Lie algebras

**BACKGROUND D.1** (Lie groups; the matrix groups of this paper). A **Lie group** is a group that is also a smooth manifold, with smooth multiplication and inversion. Every closed subgroup of  $\mathrm{GL}_n(\mathbb{R})$  is a Lie group (Cartan’s closed-subgroup theorem), and all groups below are of this kind. Its dimension is its dimension as a manifold.

| group                           | dimension      | elements                                                                                                                                          |
|---------------------------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| $\mathrm{GL}_r$                 | $r^2$          | invertible $r \times r$ matrices                                                                                                                  |
| $\mathrm{O}(n), \mathrm{SO}(n)$ | $n(n-1)/2$     | $R^\top R = I$ ; for $\mathrm{SO}(n)$ also $\det R = 1$                                                                                           |
| $\mathrm{CO}(n)$                | $n(n-1)/2 + 1$ | $cR$ with $c > 0, R \in \mathrm{O}(n)$ , so $\mathrm{CO}(n) = \mathbb{R}_{>0} \cdot \mathrm{O}(n)$ ; equivalently $g^\top g \in \mathbb{R}_{>0}I$ |
| $\mathrm{Sp}(2r, \mathbb{R})$   | $r(2r+1)$      | $M^\top J M = J$ , with $J = \begin{pmatrix} 0 & I_r \\ -I_r & 0 \end{pmatrix}$                                                                   |
| $T_m$                           | $m$            | invertible diagonal $m \times m$ matrices ( <a href="#">Background D.2</a> )                                                                      |

LoRA’s gauge is  $\mathrm{GL}_r$ , acting on factors by  $g \cdot (B, A) = (Bg^{-1}, gA)$ . Exactly orthogonal OFT keeps the weight inside the orbit  $W_0 \cdot \mathrm{O}(n)$ , and HRA with even  $r$  keeps it inside  $W_0 \cdot \mathrm{SO}(n)$ .  $\mathrm{CO}(n)$  is exactly the input-side covariance group ([Background D.17](#)) of the PiSSA ([Meng et al., 2024a](#)), MiLoRA ([Wang et al., 2024b](#)) and paper-form CorDA ([Yang et al., 2024c](#)) splits. LoCO’s ([Nguyen et al., 2026](#)) skew generator satisfies  $UV^\top - VU^\top = ZJZ^\top$  with  $Z = [U \ V]$ , so it is unchanged under  $Z \mapsto ZM$  for every  $M \in \mathrm{Sp}(2r, \mathbb{R})$ .

**Used in.** the notation (Table 2), [Definition I.5](#), [Theorem II.5](#), [Proposition II.16](#), LOFT (§II.B), [Theorem III.5](#), [Theorem V.3](#). Reading: Hall (2015, Ch. 1).

**BACKGROUND D.2** (tori, diagonal monoids and saturation). The **diagonal torus**  $T_m = \{\mathrm{diag}(d) : d_i \neq 0\} \cong (\mathbb{R}^\times)^m$  is a commutative Lie group with  $2^m$  connected components, and  $T_m^+ = \{\mathrm{diag}(d) : d_i > 0\} \cong \mathbb{R}_{>0}^m$  is the component of the identity.  $\mathrm{Diag}_m$ , the set of all diagonal matrices including singular ones, is a monoid under multiplication ([Background D.10](#)) whose invertible elements form  $T_m$ . Identifying  $\mathbb{R}^2$  with  $\mathbb{C}$ , the scaled rotations  $a \mathrm{Rot}(\theta)$  with  $a > 0$  form  $\mathrm{CO}^+(2) \cong \mathbb{C}^\times$ , through  $a \mathrm{Rot}(\theta) \mapsto ae^{i\theta}$ . If a group or monoid  $G$  acts on a space containing a set  $X$ , the **saturation**  $G \cdot X = \{g \cdot x : g \in G, x \in X\}$  is the smallest  $G$ -stable set containing  $X$ .

(IA)<sup>3</sup> ([Liu et al., 2022a](#)) on keys and values maps  $\ell \mapsto \mathrm{diag}(\ell)W_0$ , so its image  $\mathrm{Diag}_m \cdot W_0$  is the closure of the torus orbit  $T_m \cdot W_0$ . DoRA’s image is the saturation  $\mathrm{Diag}_m \cdot (W_0 + \mathcal{M}_{\leq r})$  of LoRA’s cone. RoAd<sub>1</sub> ([Liao and Monz, 2024](#)) is the left action of  $\mathrm{CO}^+(2)^{m/2} \cong (\mathbb{C}^\times)^{m/2}$  on pairs of output rows, which makes it (IA)<sup>3</sup> over  $\mathbb{C}$ . LoHa’s gauge contains a two-sided diagonal torus.

**Used in.** [Theorem II.5](#), DoRA (§II.B), [Proposition II.8](#), [Proposition II.9](#), [Proposition II.12](#), the hyperparameter torus (§III).

**BACKGROUND D.3** (permutations, monomial matrices, semidirect products, normalisers).  $S_d$  is the group of permutations of  $d$  objects, realised by permutation matrices. A matrix is **monomial** if each row and each column has exactly one nonzero entry. Every monomial matrix factors uniquely as  $DP$  with  $D \in T_m$  and  $P$  a permutation matrix, and  $T_m$  is normal in the group  $\mathrm{Mon}_m$  of monomial matrices, so  $\mathrm{Mon}_m = T_m \rtimes S_m$ . Here a **semidirect product**  $N \rtimes H$  is a group with a normal subgroup  $N$  and a subgroup  $H$  such that  $N \cap H = \{e\}$

and  $NH$  is the whole group. The **signed permutations**  $B_r = \{\pm 1\}^r \rtimes S_r = \text{Mon}_r \cap \text{O}(r)$  are the monomial matrices with entries  $\pm 1$ , and  $|B_r| = 2^r r!$ . The **normaliser** of a subset  $H \subseteq G$  is  $N_G(H) = \{g \in G : gHg^{-1} = H\}$ .

A monomial  $g$  satisfies  $g \text{Diag}_m g^{-1} = \text{Diag}_m$ , and in  $\text{GL}_m$  the normaliser of  $T_m$  is exactly  $\text{Mon}_m$ . This puts  $\text{Mon}_m \times \text{GL}_n$  inside the covariance group (Background D.17) of output scalings such as (IA)<sup>3</sup> and DoRA’s magnitude. Adam (Kingma and Ba, 2015) acts coordinatewise, so on LoRA’s full-rank locus with channel-uniform hyperparameters (one learning rate,  $\varepsilon$  and decay per factor) it is equivariant, of LoRA’s gauge, only under  $B_r \subset \text{O}(r) \subset \text{GL}_r$  (Theorem III.12). By Lemma VI.2, an invertible linear map that commutes with a non-affine  $C^2$  activation is monomial.

**Used in.** DoRA (§II.B), Proposition II.8, Proposition II.9, Theorem III.12, Corollary III.13, Proposition V.4, Lemma VI.2, the privileged basis (§VI).

**BACKGROUND D.4** (connectedness and the identity component). The **identity component**  $G^\circ$  of a Lie group  $G$  is the connected component containing  $e$ . It is an open normal subgroup.  $\text{GL}_n$  has two components, told apart by the sign of the determinant, and  $\text{GL}^+(n) = \{\det > 0\}$  is the identity component; likewise  $\text{SO}(n) \subset \text{O}(n)$  and  $T_m^+ \subset T_m$ . A subset  $U$  is **symmetric** if  $U^{-1} = U$ . Every neighbourhood of  $e$  contains a symmetric one, and in a connected group a symmetric neighbourhood  $U$  of  $e$  generates the group as a monoid, so every element is a finite product  $u_1 \cdots u_k$  with  $u_i \in U$ .

In merge-and-restart of a method of action type (Background D.12), each cycle applies one element of a set  $S$ . If  $S$  contains a neighbourhood of the identity in a connected group  $H$ , repeated merging reaches all of  $H$  (Theorem V.3(b)). HF’s (Mangrulkar et al., 2022) Cayley–Neumann OFT factors have  $\det R \geq 0$ , so the invertible ones lie in  $\text{GL}^+(b)$ ; they generate  $\prod \text{GL}^+(b)$  as a group, or  $\prod \text{CO}^+(2)$  when  $b = 2$ .

**Used in.** complete invariants (§II.B), Theorem V.3, the corollaries of Theorem V.3 (§V).

**BACKGROUND D.5** (Lie algebras; the commutator;  $\mathfrak{gl}$ ,  $\mathfrak{so}$ ,  $\text{Sym}$ ,  $\wedge$ ). The **Lie algebra** of a matrix group  $G \subseteq \text{GL}_n$  is  $\mathfrak{g} = \{X : e^{tX} \in G \text{ for all } t \in \mathbb{R}\}$ , the tangent space of  $G$  at  $I$ . It is a vector space closed under the **commutator**  $[X, Y] = XY - YX$ . The paper uses  $\mathfrak{gl}_r$  (all  $r \times r$  matrices, for  $\text{GL}_r$ ),  $\mathfrak{so}(n)$  (skew matrices  $\Omega^\top = -\Omega$ , for  $\text{O}(n)$  and  $\text{SO}(n)$ ), the diagonal matrices (for  $T_m$ ),  $\mathbb{R}I \oplus \mathfrak{so}(n)$  (for  $\text{CO}(n)$ ) and  $\mathfrak{sp}(2r) = \{X : X^\top J + JX = 0\}$ . For a subspace  $V \subseteq \mathbb{R}^n$ ,  $\mathfrak{so}(V)$  consists of the skew maps of  $V$  extended by 0 on  $V^\perp$ . The symmetric matrices  $\text{Sym}_r$  do not form a Lie algebra, since  $[S, S']$  is skew. Write  $v \wedge w = vw^\top - wv^\top \in \mathfrak{so}(n)$ ,  $U \wedge V$  for the span of the  $u \wedge v$  with  $u \in U$ ,  $v \in V$ , and  $L_{ab} = e_a \wedge e_b = E_{ab} - E_{ba}$ . Then  $[L_{ac}, L_{cb}] = L_{ab}$  for distinct  $a, b, c$ .

For injective  $W_0$ , at HRA’s paired initialisation  $u_{2i-1} = u_{2i} = w_i$  ( $i \leq k = r/2$ , with independent  $w_i$  and  $U = \text{span}\{w_i\}$  of dimension  $k$ ), the first-order image is  $S_1 = W_0 \cdot \{\Omega \in \mathfrak{so}(n) : P_{U^\perp} \Omega P_{U^\perp} = 0\} = W_0 \cdot (U \wedge \mathbb{R}^n)$ , a subspace of  $W_0 \cdot \mathfrak{so}(n)$  of dimension  $kn - k(k+1)/2$  (Background C.13). LOFT (Zhao et al., 2026) trains a skew  $E \in \mathfrak{so}(r)$  inside a fixed  $r$ -dimensional input subspace. The bracket identity lets rotations of overlapping coordinate blocks generate rotations of their union (Background D.8).

**Used in.** Theorem II.7, the LOFT example and LOFT fixes one leaf (§II.B), Theorem III.5, Theorem V.3. Reading: Hall (2015, Ch. 3).

**BACKGROUND D.6** (the exponential map; one-parameter subgroups). For  $X \in \mathfrak{g}$ , the curve  $t \mapsto e^{tX}$  of matrix exponentials is a **one-parameter subgroup**, a smooth homomorphism  $(\mathbb{R}, +) \rightarrow G$ , and every continuous one-parameter subgroup has this form for a unique  $X$ , its **generator**. The exponential map  $\mathfrak{g} \rightarrow G$  has derivative  $\text{id}$  at 0, so it maps a neighbourhood of 0 diffeomorphically onto a neighbourhood of  $I$ . It is onto for  $\text{SO}(n)$  and not onto for  $\text{GL}^+(n)$ , since  $\text{diag}(-1, -2)$  is  $e^X$  for no real  $X$ .

The one-parameter subgroup  $t \mapsto (Be^{-tX}, e^{tX}A)$  of LoRA's gauge moves along a fibre without changing  $BA$ . [Theorem III.5](#) builds its isometries as  $k = e^{\tau\xi}$  from generators  $\xi$  with  $\Lambda^{-1}\xi$  antisymmetric. LOFT may use the exponential as its map  $\mathfrak{so}(r) \rightarrow \text{SO}(r)$ , and the Kempf–Ness argument ([Background D.30](#)) differentiates the norm along  $e^{t\xi}$ .

**Used in.** [Theorem III.5](#), [Proposition III.7](#), LOFT (§II.B), [Theorem V.3](#). Reading: [Hall \(2015, Ch. 2\)](#).

**BACKGROUND D.7** (the Cayley map and its Neumann truncation). For skew  $\Omega$  the eigenvalues are imaginary, so  $I - \Omega$  is invertible, and the **Cayley map** is

$$C(\Omega) = (I + \Omega)(I - \Omega)^{-1}.$$

It lands in  $\text{SO}(n)$ , and  $C(\Omega) - I = 2\Omega(I - \Omega)^{-1}$ , so  $\text{rank}(C(\Omega) - I) = \text{rank } \Omega$ . Its expansion  $C(\Omega) = I + 2\Omega + 2\Omega^2 + \dots$  shows that its derivative at 0 is  $2 \text{id}$ . It is a diffeomorphism from  $\mathfrak{so}(n)$  onto the open dense set of rotations with no eigenvalue  $-1$ , with inverse  $R \mapsto (R - I)(R + I)^{-1}$ , so it misses, for instance,  $\text{diag}(-1, -1, 1, \dots, 1)$ .

Exact-Cayley OFT and block OFT are orthogonal, so they preserve  $K_{\text{out}} = WW^\top$  ([Background D.15](#)) and the spectrum. HF PEFT's default (since version 0.18) truncates  $(I - Q)^{-1}$  to  $I + Q + Q^2 + Q^3$ , which gives  $R = C(Q)(I - Q^4)$ . As  $Q^4$  is symmetric and commutes with  $C(Q)$ ,  $R^\top R = (I - Q^4)^2 \neq I$ , and for small  $Q$  each merge shrinks  $K_{\text{out}}$ . The rank identity is how [Theorem II.7\(c\)](#) identifies HRA's image germ with the cone of skew matrices of rank at most  $r$ .

**Used in.** the thesis (Exposé 0), [Corollary II.6](#), [Theorem II.7](#), the Gram lab of the companion site (§II.B), the corollaries of [Theorem V.3](#) (§V).

**BACKGROUND D.8** (Lie subgroups and subalgebras; the generated Lie algebra). A **Lie subgroup** of  $G$  is a subgroup with a Lie group structure for which the inclusion is an injective immersion; it need not be closed. Connected Lie subgroups  $H \subseteq G$  correspond one to one to Lie subalgebras  $\mathfrak{h} \subseteq \mathfrak{g}$ , with  $H$  the subgroup generated by  $e^{\mathfrak{h}}$ . The Lie algebra **generated** by subalgebras  $\mathfrak{h}_1, \dots, \mathfrak{h}_p$ , written  $\text{Lie}\langle \mathfrak{h}_1, \dots, \mathfrak{h}_p \rangle$ , is the smallest subalgebra containing them, the span of their elements and all iterated brackets of them.

Block OFT, BOFT ([Liu et al., 2023b](#)) factors and Givens rotations ([Ma et al., 2024](#)) use  $\mathfrak{so}(I_j)$  for coordinate blocks  $I_j \subseteq \{1, \dots, n\}$ . Since  $[L_{ac}, L_{cb}] = L_{ab}$ , two overlapping blocks generate  $\mathfrak{so}$  of their union, and  $\text{Lie}\langle \mathfrak{so}(I_j) \rangle_j = \bigoplus_C \mathfrak{so}(C)$ , summed over the connected components  $C$  of the hypergraph with vertices  $\{1, \dots, n\}$  and hyperedges  $I_j$ . Two vertices share a component when a chain of hyperedges, each meeting the next, joins them. Butterfly factors connect this hypergraph only at full depth.

**Used in.** [Theorem V.3](#), [Figure 10\(b\)](#), the corollaries of [Theorem V.3](#) (§V), LOFT fixes one leaf (§II.B). Reading: [Lee \(2013, Ch. 19\)](#); [Hall \(2015, Ch. 5\)](#).



**FACT D.9** (Yamabe’s theorem). *Every path-connected subgroup of a Lie group is a Lie subgroup. Consequently, if  $H_1, \dots, H_p$  are connected Lie subgroups of  $G$ , the subgroup they generate is the connected Lie subgroup  $H$  with  $\text{Lie}(H) = \text{Lie}(\mathfrak{h}_1, \dots, \mathfrak{h}_p)$  (Yamabe, 1950); a short proof is in Goto (1969).*

The subgroup generated by the  $H_i$  is path-connected, because products of paths are paths, so the theorem makes it a Lie subgroup and the correspondence of Background D.8 names it. With Background D.4, this is why a re-merged action method whose cycles contain a neighbourhood of the identity in each  $H_i$  reaches exactly the connected subgroup with Lie algebra  $\text{Lie}(\mathfrak{h}_1, \dots, \mathfrak{h}_p)$ , and why exact-Cayley block-orthogonal factors reach  $\prod_C \text{SO}(C)$ .

**Used in.** Theorem V.3, the corollaries of Theorem V.3 (§V).

**BACKGROUND D.10** (monoids, units, monoid orbits, homomorphisms). A **monoid** is a set with an associative product and an identity. Its invertible elements form its **group of units**. The submonoid **generated** by a subset  $S$  consists of all finite products  $s_1 \cdots s_k$  of elements of  $S$ , the empty product being the identity. A monoid acts on a set when  $1 \cdot x = x$  and  $s \cdot (t \cdot x) = (st) \cdot x$ , and the **monoid orbit** of  $x$  is  $\{s \cdot x\}$ . A **homomorphism** of monoids preserves products and the identity. Examples are  $\text{Diag}_m$ ,  $M_2(\mathbb{R})$  and  $\text{End}(\Theta)$  under multiplication, with units  $T_m$ ,  $\text{GL}_2(\mathbb{R})$  and  $\text{GL}(\Theta)$ , and finite multisets under disjoint union  $\uplus$ . Additive monoids  $\text{Mon}(C)$  are in Background B.14.

Merge-and-restart reaches the orbit of the monoid that one cycle generates, which is  $\theta_0 + \text{Mon}(C)$  for an additive method  $\rho = \theta_0 + \delta$  with shape  $C = \text{Im } \delta$  (Background B.14) and  $\langle S \rangle \cdot \theta_0$  for a method of action type (Background D.12) whose cycle applies elements of  $S$ . RoAd<sub>2</sub>’s blocks range over  $M_2(\mathbb{R})$ . For a fixed query, attention’s pair (total weight, weighted sum of values) is a homomorphism from multisets of key–value pairs to  $(\mathbb{R}_{\geq 0} \times \mathbb{R}^{d_v}, +)$  (Background D.34).

**Used in.** Theorem V.3, restarting (§V), DoRA (§II.B), Proposition II.9, Theorem VI.5, attention (§VI).

**FACT D.11** (Cartan–Dieudonné theorem; Scherk’s sharp form). *For nonzero  $u \in \mathbb{R}^n$ , the reflection  $H_u = I - 2uu^\top / \|u\|^2$  fixes the hyperplane  $u^\perp$  and has  $\det H_u = -1$ . Every  $R \in \text{O}(n)$  is a product of at most  $n$  reflections (Cartan–Dieudonné). In Scherk’s sharp form, if  $k = \text{rank}(R - I)$ , then  $R$  is a product of exactly  $k$  reflections and of no fewer. Hence  $\det R = (-1)^k$ , and every  $R \in \text{SO}(n)$  is a product of  $\text{rank}(R - I)$  reflections, an even number. See Scherk (1950) and Artin (1957, Ch. III).*

**The idea of the proof.** A product of  $j$  reflections fixes an intersection of  $j$  hyperplanes, so  $\text{rank}(R - I) \leq j$ . Conversely, pick  $x$  with  $Rx \neq x$  and put  $u = Rx - x$ . Then  $H_u$  swaps  $Rx$  and  $x$  and fixes every fixed vector of  $R$ , so  $H_u R$  fixes one more dimension, and induction on  $k$  finishes.

HRA multiplies  $r$  reflections, so  $\text{rank}(R - I) \leq r$ , and  $\det R = 1$  for even  $r$ . Scherk’s form gives the converse. For injective  $W_0$  and even  $r$ , HRA reaches all of  $(W_0 + \mathcal{M}_{\leq r}) \cap W_0 \cdot \text{SO}(n)$ , padding with pairs  $H_u H_u = I$ , and re-merged HRA reaches  $W_0 \cdot \text{SO}(n)$  within  $\lceil n/r \rceil$  cycles.

**Used in.** Theorem II.7, HRA is a meet (§II.B), the corollaries of Theorem V.3 (§V).

## Actions, orbits and invariants

**BACKGROUND D.12** (group actions). A (left) **action** of a group  $G$  on a set  $X$  is a map  $(g, x) \mapsto g \cdot x$  with  $e \cdot x = x$  and  $g \cdot (h \cdot x) = (gh) \cdot x$ . A **right** action  $x \cdot g$  satisfies  $x \cdot (gh) = (x \cdot g) \cdot h$ , and  $g \cdot x := x \cdot g^{-1}$  turns it into a left one. The action is **smooth** if  $G \times X \rightarrow X$  is smooth, and **linear** if  $X$  is a vector space and each map  $x \mapsto g \cdot x$  is linear.

On weights,  $W \mapsto WR$  is a right action of  $O(n)$  on the input side and  $W \mapsto PW$  a left action of  $O(m)$  on the output side, and  $(g, h) \cdot W = gWh^{-1}$  is a left action of  $GL_m \times GL_n$ . On LoRA's factors,  $g \cdot (B, A) = (Bg^{-1}, gA)$  is a linear left action of  $GL_r$ . A method is of **action type** when  $\rho(q) = \gamma(q) \cdot \theta_0$  for a map  $\gamma$  into a group acting on  $\Theta$ , as for OFT and HRA. (This  $\gamma$  follows the exposés and is unrelated to the cotangent signal  $\gamma(t)$  of [Background C.2](#) and [Background D.26](#).) (IA)<sup>3</sup> has the same form with the monoid  $\text{Diag}_m$  in place of a group, and DoRA's magnitude applies that monoid to LoRA's weight.

**Used in.** [Definition I.2](#), the notation ([Table 2](#)), [Erlangen \(§II.B\)](#), [Theorem II.5](#), [Theorem III.5](#).

**BACKGROUND D.13** (orbits, stabilisers, free actions). The **orbit** of  $x$  is  $G \cdot x = \{g \cdot x : g \in G\}$ , its closure is the **orbit closure**, and its **stabiliser** is the closed subgroup  $G_x = \{g : g \cdot x = x\}$ . For a smooth action of a Lie group,  $g \mapsto g \cdot x$  induces an injective immersion  $G/G_x \rightarrow X$  with image  $G \cdot x$ , so the orbit is an immersed submanifold of dimension  $\dim G - \dim G_x$ . The **infinitesimal stabiliser**  $\{\xi \in \mathfrak{g} : \xi \cdot x = 0\}$  is the Lie algebra of  $G_x$ . The action is **free** if every stabiliser is trivial, and the **generic stabiliser** is the stabiliser at a generic point, one in a suitable dense open set. Each map  $x \mapsto g \cdot x$  carries an orbit onto itself, so all its points look alike; orbits are **homogeneous**.

$GL_r$  acts freely on pairs with rank  $A = r$ , since  $gA = A$  forces  $g = I$ , so LoRA's generic fibre, a single orbit, has dimension  $r^2$ . LoHa's joint gauge  $GL_{r_1} \times GL_{r_2} \times T_m \times T_n$  has a one-dimensional generic stabiliser, so its orbits have dimension  $r_1^2 + r_2^2 + m + n - 1$ . By homogeneity, the two-sided torus orbit  $\{\text{diag}(a)W_0 \text{diag}(b)\}$  has dimension  $m + n - c(W_0)$  at each of its points, with  $c(W_0)$  the number of connected components of the support graph, the bipartite graph on the  $m$  rows and  $n$  columns with an edge  $\{i, j\}$  where  $(W_0)_{ij} \neq 0$ . The images of (IA)<sup>3</sup> and RoAd<sub>1</sub> are orbit closures.

**Used in.** [Erlangen \(§II.B\)](#), [Theorem II.5](#), [Proposition II.9](#), the image germ ([Exposé II](#)), [Proposition II.12](#), [Rosetta entries 6 and 11 \(Exposé IX\)](#). Reading: [Lee \(2013, Ch. 7 and 21\)](#).

**BACKGROUND D.14** (Haar measure; invariant measures on homogeneous spaces). A compact Lie group  $K$  carries a unique invariant probability measure, its **Haar measure**. If  $K$  acts transitively on  $X$ , so that  $X \cong K/K_x$  is a **homogeneous space**, then  $X$  carries a unique  $K$ -invariant probability measure, the image of Haar measure under  $k \mapsto k \cdot x$ . The Grassmannian  $\text{Gr}(r, n)$  of  $r$ -dimensional subspaces of  $\mathbb{R}^n$  ([Background C.15](#)) is a homogeneous space of  $O(n)$ .

If  $A_0 \in \mathbb{R}^{r \times n}$  has i.i.d. Gaussian entries, its law is unchanged by  $A_0 \mapsto A_0 R$  for  $R \in O(n)$ , and  $A_0$  has rank  $r$  almost surely. So row  $A_0$  is an  $O(n)$ -invariant random point of  $\text{Gr}(r, n)$ , and by uniqueness it is Haar-distributed. Vanilla LoRA with a Gaussian  $A_0$  therefore picks a Haar-random point of  $\text{Gr}(r, n)$ , the space that classifies its stratum  $T_A$ . Kaiming-uniform entries are not rotation invariant, so under HF's default the point is not exactly Haar-distributed.

**Used in.** the three strata ([§II.A](#)), [Theorem II.2](#). Reading: [Mattila \(1995, Ch. 3\)](#).

**BACKGROUND D.15** (invariants and complete invariants). A map  $f : X \rightarrow Y$  is **invariant** under an action if  $f(g \cdot x) = f(x)$  for all  $g$  and  $x$ , so it is constant on orbits. It is a **complete invariant** if moreover  $f(x) = f(x')$  implies  $x' \in G \cdot x$ ; it then identifies the orbit space  $X/G$  with the image of  $f$ .

| action on $W \in \mathbb{R}^{m \times n}$                     | complete invariant                         |
|---------------------------------------------------------------|--------------------------------------------|
| $W \mapsto WR, R \in O(n)$ (input side)                       | $K_{\text{out}} = WW^\top$                 |
| $W \mapsto PW, P \in O(m)$ (output side)                      | $K_{\text{in}} = W^\top W$                 |
| $W \mapsto PWR, P \in O(m), R \in O(n)$                       | singular values with multiplicity          |
| $W \mapsto DW, D \in T_m^+, W$ without zero rows              | the rays $\mathbb{R}_{>0} w_i$ of the rows |
| $W \mapsto E \odot W, E \in (\mathbb{R}^\times)^{m \times n}$ | the zero pattern                           |

If a method's image lies in one orbit, every invariant of that action stays fixed through training and merging. This is why exactly orthogonal methods never change  $K_{\text{out}}$  or the spectrum ([Corollary II.6](#)). Isomorphism invariants of methods, such as  $S_1$ ,  $d(M)$  and the first-order deficit, play the same role one level up, since methods with different values cannot be isomorphic.

**Used in.** [Theorem II.5](#), complete invariants and the Gram lab (§II.B), [Corollary II.6](#), [Observation I.6](#), the seven borrowed words ([Exposé XI](#)).

**FACT D.16** (first fundamental theorem for  $O(n)$ ; Witt's extension theorem). (*First fundamental theorem.*) Every polynomial in the rows  $w_1, \dots, w_m \in \mathbb{R}^n$  of  $W$  that is invariant under  $W \mapsto WR, R \in O(n)$ , is a polynomial in the inner products  $\langle w_i, w_j \rangle$ , the entries of  $WW^\top$ . For a compact group acting linearly, invariant polynomials separate orbits, so  $WW^\top$  is a complete invariant. (Witt's extension theorem, Euclidean case.) A linear isometry between two subspaces of  $\mathbb{R}^n$  extends to an element of  $O(n)$ . See [Weyl \(1939, Ch. II\)](#) and [Goodman and Wallach \(2009, Ch. 5\)](#).

**Witt gives the orbit separation directly.** If  $WW^\top = W'W'^\top$ , the map  $\sum_i c_i w_i \mapsto \sum_i c_i w'_i$  is well defined and isometric on the row spaces, because  $\|\sum_i c_i w_i\|^2 = c^\top WW^\top c$ . Extend it by an isometry between the orthogonal complements to  $R^\top \in O(n)$ ; then  $W' = WR$ .

This is item (1) of [Theorem II.5](#), the reason exactly orthogonal input-side methods are pinned to  $K_{\text{out}}(W_0)$ . Applied to  $W^\top$ , it gives  $K_{\text{in}}$  for the output side.

**Used in.** [Theorem II.5](#), the known results ([Exposé XI](#)).

**BACKGROUND D.17** (Klein's Erlangen programme). [Klein \(1872\)](#) proposed to define a geometry by a space together with a group acting on it, and to study the properties the group preserves. Euclidean geometry keeps lengths and angles, the invariants of rigid motions. Affine geometry keeps parallelism and ratios along parallel lines but not lengths or angles, because its group is larger. A larger group leaves fewer invariants.

For a group-type method the reading is a theorem, because the image lies in one orbit and so the method can change nothing the group preserves. For a cone such as LoRA's image the word only names its **covariance group**, the largest  $\mathcal{G}_M \subseteq \text{GL}_m \times \text{GL}_n$  with  $\text{Im } M(gW_0h^{-1}) = g \text{ Im } M(W_0) h^{-1}$  ([Background D.18](#)). For LoRA this is all of  $\text{GL}_m \times \text{GL}_n$ , for (IA)<sup>3</sup> on keys and values it is  $\text{Mon}_m \times \text{GL}_n$ , and for the PiSSA split it is  $\text{CO}(m) \times \text{CO}(n)$ .

**Used in.** Erlangen and what "Erlangen" means here (§II.B), [Definition II.14](#), the borrowed words ([Exposé IX](#)), the seven borrowed words ([Exposé XI](#)). Reading: [Sharpe \(1997, Ch. 4\)](#).

**BACKGROUND D.18** (equivariance and invariance of maps, update rules and merge rules). A map  $F : X \rightarrow Y$  between spaces with  $G$ -actions is **equivariant** if  $F(g \cdot x) = g \cdot F(x)$ , and **invariant** if  $F(g \cdot x) = F(x)$ . An update rule on parameters is equivariant under a subgroup  $H$  of the gauge if, started from  $h \cdot q$  with correspondingly transformed optimizer state, it produces  $h$  applied to the iterates started from  $q$ . A rule that uses randomness is **equivariant in distribution** if  $F(g \cdot x)$  and  $g \cdot F(x)$  have the same law for every  $x$ .

On LoRA's full-rank locus, with channel-uniform hyperparameters, the largest such subgroup of  $GL_r$  is  $O(r)$  for SGD,  $B_r$  for Adam and all of  $GL_r$  for undamped scaled GD (Background D.23, Theorem III.12). An equivariant rule started from gauge-related points traces the same weights, since  $\rho(h \cdot q) = \rho(q)$ . Merge rules should be invariant under each adapter's gauge. Task arithmetic (Ilharco et al., 2022)  $\sum_i w_i s_i B_i A_i$  is, while LoraHub's (Huang et al., 2023)  $(\sum_i w_i B_i)(\sum_j w_j A_j)$  is not. On a weight space  $\mathbb{R}^N$ , DARE's (Yu et al., 2023) random drop-and-rescale is equivariant under the invertible diagonal matrices  $T_N$  for each fixed mask, and under the monomial matrices  $\text{Mon}_N$  (Background D.3) in distribution.

**Used in.** Definition II.14, Observation II.15, Theorem III.12, Corollary III.13, Figure 9, Proposition V.4.

**BACKGROUND D.19** (multihomogeneous maps; multidegree). Split the parameters into blocks  $q = (q_1, \dots, q_k)$ . A map  $\delta$  is **multihomogeneous** of **multidegree**  $(d_1, \dots, d_k)$  if  $\delta(c_1 q_1, \dots, c_k q_k) = c_1^{d_1} \dots c_k^{d_k} \delta(q)$  for all real  $c_i$ . LoRA's  $BA$  has multidegree  $(1, 1)$ , and LoHa's  $(B_1 A_1) \odot (B_2 A_2)$  has multidegree  $(1, 1, 1, 1)$ .

If some  $d_i \geq 1$ , the image of  $\delta$  contains 0 and is closed under multiplication by every nonnegative scalar, so it is a cone with vertex 0 (Background C.10). For a method  $\rho = \theta_b + s \delta$  started where  $\delta(q_0) = 0$ , so that  $\theta_b = \theta_0$ , the image is the cone  $\theta_0 + s \text{Im } \delta$  with vertex  $\theta_0$ . For  $c \in \mathbb{R}_{>0}^k$ , the change  $q_i \mapsto c_i q_i$  together with  $s \mapsto s \prod_i c_i^{-d_i}$  leaves  $\rho$  unchanged. That rescaling is an isomorphism of methods and the source of the hyperparameter torus (Background D.20).

**Used in.** the image germ (Exposé II), Theorem III.10, the seven axes (§VIII.A).

**BACKGROUND D.20** (orbit space of a positive torus). Let  $\mathbb{R}_{>0}^k$  act on positive variables  $x \in \mathbb{R}_{>0}^N$  by  $x_j \mapsto x_j \prod_i c_i^{a_{ji}}$ , with a real weight matrix  $a = (a_{ji})$ . In logarithms this is translation by the column space of  $a$ , so the orbits are parallel affine subspaces and the orbit space is a vector space of dimension  $N - \text{rank } a$ . The monomials  $\prod_j x_j^{b_j}$  with  $b^\top a = 0$  are invariant, and a basis of such vectors  $b$  gives a complete set of invariants.

Under the hypotheses of Theorem III.10, taken with Adam at  $\varepsilon = 0$ , no weight decay or clipping and a shared schedule (otherwise the per-block  $\varepsilon$ , decay and clipping thresholds rescale as well), the torus acts on LoRA's hyperparameters by  $s \mapsto s/(c_A c_B)$ ,  $\eta_A \mapsto c_A \eta_A$ ,  $\eta_B \mapsto c_B \eta_B$ ,  $\sigma_A \mapsto c_A \sigma_A$ , with  $\sigma_A$  the scale of  $A$ 's initialisation, without changing the weight trajectory. The invariants are  $s \eta_A \eta_B$  and  $\sigma_A / \eta_A$ , so only two of the four knobs matter. LoRA+ (Hayou et al., 2024b) with ratio  $\lambda$  replaces  $\eta_B$  by  $\lambda \eta_B$ , which changes the invariants exactly as replacing  $\alpha$  by  $\lambda \alpha$  does. Under SGD the rates scale by  $c_i^2$ , and the invariants become  $s^2 \eta_A \eta_B$  and  $\sigma_A^2 / \eta_A$ .

**Used in.** Theorem III.10, Corollary III.11, Figure 8, Rosetta entry 23 (Exposé IX), Prediction X.1.

## Gauge, quotients and metrics

**BACKGROUND D.21** (gauge symmetry of a parametrisation). The **gauge group** of a method  $\rho : Q \rightarrow \Theta$  is the group of diffeomorphisms  $\varphi$  of  $Q$  with  $\rho \circ \varphi = \rho$ . If a Lie group  $G$  acts on  $Q$  with  $\rho(g \cdot q) = \rho(q)$ , each orbit lies inside a fibre  $\rho^{-1}(\theta)$  (Background C.6), so  $d(M) \leq |M| - \dim(\text{generic orbit})$ . The action is **locally transitive** on a fibre when its orbits are open in the fibre, that is, when orbits and fibre have the same dimension there; on generic fibres this gives  $d(M) = |M| - \dim(\text{generic orbit})$ , and we then call  $G$  the generic gauge. Here the word means only this symmetry group; no connection is involved.

LoRA's  $\text{GL}_r$  acts freely and locally transitively on full-rank fibres, so  $d = r(m+n) - r^2 = r(m+n-r)$ , and  $r^2$  parameters are gauge waste. HRA's generic fibres have dimension  $r(r+1)/2$ , while its  $r$  scalings  $u_i \mapsto c_i u_i$  sweep out only  $r$  of those dimensions, so the scalings do not fill the fibres; Theorem II.7(a) describes the rest of the gauge as non-linear. For  $r = 2$  the fibre through  $(u_1, u_2)$  has dimension 3, made up of the two scalings and of rotating  $u_1$  and  $u_2$  by a common angle inside their plane, which keeps the angle between them and hence  $H_{u_1} H_{u_2}$ , the rotation of that plane by twice the angle. The plane depends on the point, so this move is not a linear action on parameters. The **Hurwitz moves**  $(u_i, u_{i+1}) \mapsto (H_{u_i} u_{i+1}, u_i)$  preserve the product of reflections, since  $H_a H_b = H_{H_a b} H_a$ , and they satisfy the braid relations, so the braid group on  $r$  strands, unrelated to the signed permutations  $B_r$ , acts on HRA's parameters. These moves are discrete and add no dimension to the fibre.

**Used in.** Definition I.5, invariants of a method (§I), the notation (Table 2), rank and dimension (§II.C), Proposition II.12, Theorem II.7, the analogy “Noether; gauge” (§III), the seven borrowed words (Exposé XI).

**FACT D.22** (quotients and descent; the quotient manifold theorem). *For an action of  $G$  on  $X$ , the **orbit space**  $X/G$  is the set of orbits, with the map  $\pi : x \mapsto G \cdot x$ . A map  $f : X \rightarrow Y$  that is constant on orbits factors uniquely as  $f = \tilde{f} \circ \pi$ ; this is **descent**. (Quotient manifold theorem.) If a Lie group acts smoothly, freely and properly on a manifold  $X$ , then  $X/G$  is a manifold of dimension  $\dim X - \dim G$ ,  $\pi$  is a submersion, and  $\tilde{f}$  is smooth whenever  $f$  is. Properness means that  $(g, x) \mapsto (g \cdot x, x)$  is a proper map, one under which preimages of compact sets are compact (Lee, 2013, Thm 21.10).*

$\text{GL}_r$  acts freely and properly on pairs  $(B, A)$  of full rank, and  $(B, A) \mapsto BA$  identifies the quotient with  $\mathcal{M}_r$ , of dimension  $r(m+n) - r^2$ . A rule for combining LoRA factors is a function of the adapters exactly when it is constant on the fibres of  $(B_i, A_i) \mapsto (B_i A_i)$ , and a cometric on factors that depends only on  $W$  descends to  $\mathcal{M}_r$ .

**Used in.** Proposition V.4, Proposition III.4, Observation II.15, when a change of chart does not matter (§III).

**BACKGROUND D.23** (metrics, cometrics and the pulled-back cometric). A **metric** on a vector space  $V$  is a positive-definite inner product, equivalently a symmetric positive-definite map  $g : V \rightarrow V^*$ , that is,  $\langle gv, v \rangle > 0$  for  $v \neq 0$ . A **cometric** is a symmetric positive semidefinite map  $K : V^* \rightarrow V$ . It turns a covector, such as a gradient  $\nabla_\theta L$ , into a velocity  $-K \nabla_\theta L$ . A metric gives the cometric  $K = g^{-1}$ , while a cometric may be degenerate. Metrics pull back along immersions, and cometrics push forward along any smooth map. For instance, gradient flow  $\dot{q} = -\Lambda \nabla_q (L \circ \rho)$  moves the weights by  $\dot{\theta} = -K_q \nabla_\theta L$ , with

$$K_q = D\rho_q \wedge D\rho_q^\top.$$



For LoRA with **block-scalar rates**, one rate per factor,  $\Lambda = \text{diag}(\eta_B I, \eta_A I)$  on  $(B, A)$  and  $K(G) = s^2(\eta_B G A^\top A + \eta_A B B^\top G)$ . Two points of one fibre give the same weight velocity for every loss exactly when their cometrics agree. At points where  $A$  has full row rank and  $r < n$ , this holds for LoRA's linear gauge and plain gradient descent exactly along  $\mathcal{O}(r)$ . A rule descends to a cometric on weights when it gives the same weight velocity at all points of each fibre; with a point-dependent preconditioner in place of  $\Lambda$ , undamped scaled GD (Tong et al., 2021; Zhang and Pilanci, 2024a) and the SGD variant of LoRA-Pro (Wang et al., 2024i) do, on the full-rank locus. Scaled GD multiplies the  $B$ -gradient by  $(A A^\top)^{-1}$  and the  $A$ -gradient by  $(B^\top B)^{-1}$ , and undamped means that no ridge term  $\delta I$  is added before inverting (Proposition III.4).

**Used in.** Definition III.1, Observation III.2, Theorem III.3, Proposition III.4, the metric a method carries (§III), the three invariants (Exposé 0).

**BACKGROUND D.24** (Riemannian gradient flow; tangent projection; retraction). On a submanifold  $N \subseteq \Theta$  with the metric induced by the Frobenius inner product, the Riemannian gradient of  $L|_N$  at  $\theta$  is the orthogonal projection  $\Pi_{T_\theta N} \nabla L(\theta)$  of the ambient gradient onto the tangent space, and the **Riemannian gradient flow** is  $\dot{\theta} = -\Pi_{T_\theta N} \nabla L(\theta)$ . A discrete step along the tangent space leaves  $N$ , so it is followed by a **retraction**, a smooth map  $R_\theta$  from a neighbourhood of 0 in  $T_\theta N$  to  $N$  with  $R_\theta(0) = \theta$  and  $DR_\theta(0) = \text{id}$ .

At  $W = W_0 + X$  with  $X \in \mathcal{M}_r$ ,  $U = \text{col } X$  and  $V = \text{row } X$ , the tangent space is  $\{D : P_{U^\perp} D P_{V^\perp} = 0\}$  (Background C.12) and the projection is  $P_U G + G P_V - P_U G P_V$ . This is LoRA-Pro's equivalent gradient, so the SGD variant of LoRA-Pro follows the Riemannian gradient flow on  $W_0 + \mathcal{M}_r$ . Truncating the SVD of  $X + \xi$  to rank  $r$  is a retraction onto  $\mathcal{M}_r$ , the one Riemannion (Bogachev et al., 2025) uses.

**Used in.** Proposition III.4, when a change of chart does not matter (§III), Theorem III.12. Reading: Absil et al. (2008, Ch. 3 and 4).

**FACT D.25** (distributions, Lie brackets and Frobenius' theorem). A **distribution**  $D$  of rank  $k$  on an open set  $U$  assigns to each point  $\theta$  a  $k$ -dimensional subspace  $D_\theta$  of directions, smoothly in  $\theta$ ; the word is unrelated to probability distributions. The **Lie bracket** of vector fields is  $[X, Y] = DY \cdot X - DX \cdot Y$ ; flowing along  $X, Y, -X, -Y$  for time  $t$  misses the starting point by  $t^2[X, Y] + \mathcal{O}(t^3)$ .  $D$  is **involutive** if  $[X, Y]$  lies in  $D$  whenever  $X$  and  $Y$  do. (Frobenius.)  $D$  is involutive iff every point lies on a  $k$ -dimensional **integral manifold**, a submanifold whose tangent space at each of its points  $\theta$  is  $D_\theta$ . A maximal connected integral manifold is a **leaf** (Lee, 2013, Ch. 19). The Frobenius algebras of Background B.6 and the Frobenius norm share only the name.

On  $\mathbb{R}^3$ ,  $D = \text{span}\{\partial_x, \partial_y + x \partial_z\}$  is not involutive, since  $[\partial_x, \partial_y + x \partial_z] = \partial_z \notin D$ , so no surface is tangent to  $D$ . A backward method with anchor  $a(\theta)$  chooses  $D_\theta = \text{im } a(\theta)$ , and it reads as a forward method only along integral manifolds. GaLore (Zhao et al., 2024a) with a projector fixed on each period, LISA (Pan et al., 2024) and MeZO (Malladi et al., 2023) have constant distributions, which are involutive with affine leaves. Full-batch GaLore with  $T = 1$ , read as  $D_W = \text{Hom}(\mathbb{R}^n, U_r(\nabla L(W)))$  wherever a singular-value gap exists, already fails to be involutive for  $L = \frac{1}{2}\|W\|_F^2$ .

**Used in.** when a backward method is a forward one (§IV), Proposition IV.3.

## Conservation and balance

**FACT D.26** (Noether’s theorem for gradient flows). *Let a Lie group act linearly on  $Q = \mathbb{R}^N$  with  $\rho(k \cdot q) = \rho(q)$ , let  $\Lambda \succ 0$  be a constant rate matrix, and let  $\gamma(t)$  be any time-dependent covector on  $\Theta$ , a gradient or not. Along every solution of  $\dot{q} = -\Lambda D\rho_q^\top \gamma(t)$ , and for every generator  $\xi$  of the action with  $\Lambda^{-1}\xi$  symmetric, the **charge***

$$J_\xi(q) = \frac{1}{2} \langle q, \Lambda^{-1} \xi q \rangle$$

*is constant (Kunin et al., 2020; Marcotte et al., 2023; Zhao et al., 2022b).*

**The one-line proof.** Invariance of  $\rho$  gives  $D\rho_q(\xi q) = 0$ . Hence  $\frac{d}{dt} J_\xi = \langle \Lambda^{-1} \xi q, \dot{q} \rangle = -\langle \xi q, D\rho_q^\top \gamma \rangle = -\langle D\rho_q \xi q, \gamma \rangle = 0$ .

For scalar LoRA  $w = w_0 + ba$  with rates  $\eta_B, \eta_A$ , the charge  $b^2/\eta_B - a^2/\eta_A$  is conserved, so every trajectory runs along a hyperbola, the Noether hyperbolas of Figure 7(a). Noether (1918) derived conservation laws from symmetries of a Lagrangian; this version needs no Lagrangian and holds for any loss and data. For LoRA with rates  $\eta_B, \eta_A$ , the generators  $\xi_X(B, A) = (BX, -XA)$  pass the test exactly for symmetric  $X$ , and the  $r(r+1)/2$  charges are the entries of  $\Phi = B^\top B/\eta_B - AA^\top/\eta_A$ . Minibatches, LoRA dropout, global-norm clipping and DoRA’s detached norm keep the required form. Momentum and Adam do not, and under discrete SGD the charge drifts, by an amount relative to the charge of order  $\eta^2$  per step.

**Used in.** Theorem III.5, isometries and charges and the analogy “Noether; gauge” (§III), Figure 7, the things one can check (Appendix E), Prediction X.8, the seven borrowed words (Exposé XI).

**BACKGROUND D.27** (Cartan decomposition; polar decomposition). The map  $\vartheta(\xi) = -\xi^\top$  is an automorphism of  $\mathfrak{gl}_r$  of order two, a **Cartan involution**. Its  $+1$  and  $-1$  eigenspaces give the **Cartan decomposition**  $\mathfrak{gl}_r = \mathfrak{so}(r) \oplus \text{Sym}_r$ ,  $\xi = \frac{1}{2}(\xi - \xi^\top) + \frac{1}{2}(\xi + \xi^\top)$ , with  $[\mathfrak{so}, \mathfrak{so}] \subseteq \mathfrak{so}$ ,  $[\mathfrak{so}, \text{Sym}] \subseteq \text{Sym}$  and  $[\text{Sym}, \text{Sym}] \subseteq \mathfrak{so}$ . Its group version is the **polar decomposition**, which says that  $(k, S) \mapsto k e^S$  is a diffeomorphism  $O(r) \times \text{Sym}_r \rightarrow \text{GL}_r$ . Hence  $g \mapsto g^\top g$  identifies the cosets  $O(r)g$  with the positive-definite matrices,  $O(r) \backslash \text{GL}_r \cong \text{Sym}_r^+$ .

With block-scalar rates (Background D.23), the antisymmetric generators  $X \in \mathfrak{so}(r)$  of LoRA’s gauge act by rotations  $(BO^\top, OA)$  that preserve the rate metric, so the optimizer cannot feel them; they are **isometries**. The symmetric generators  $X \in \text{Sym}_r$  give the conserved charges of Background D.26. For general rates the split needs  $\mathfrak{g}$  to be closed under  $\xi \mapsto \Lambda \xi^\top \Lambda^{-1}$ . Gradient descent on LoRA is thus a family of dynamics indexed by  $O(r) \backslash \text{GL}_r$ , and the Kempf–Ness argument needs only curves  $e^{t\xi}$  with  $\xi$  symmetric.

**Used in.** Theorem III.5, isometries and charges (§III), the three invariants and the thesis (Exposé 0), Proposition III.7, when a change of chart does not matter (§III). Reading: Knapp (2002, §VI.2); Horn and Johnson (2013, §7.3).

**BACKGROUND D.28** (balancedness; the conserved  $B^\top B - AA^\top$ ). A factorisation  $W = BA$  is **balanced** if  $B^\top B = AA^\top$ , and rate-weighted balanced if  $\Phi = B^\top B/\eta_B - AA^\top/\eta_A = 0$ , that is,  $B^\top B = cAA^\top$  with  $c = \eta_B/\eta_A$ . For a deep linear network  $W = W_L \cdots W_1$  the same word means  $W_{j+1}^\top W_{j+1} = W_j W_j^\top$  for every  $j$ .

Under gradient flow with block-scalar rates,  $\Phi$  is conserved (Background D.26), so rate-weighted balancedness, once it holds, holds forever; at equal rates the two notions agree. Zero-product starts  $B_0 A_0 = 0$  have  $\langle B_0^\top B_0, A_0 A_0^\top \rangle_F = 0$ , which makes them maximally unbalanced

for given norms of the two Gram matrices, since the inner product of two positive semidefinite matrices is never negative. From  $B_0 = 0$ , conservation gives  $B^\top B = \frac{\eta_B}{\eta_A}(AA^\top - A_0A_0^\top)$ , so  $AA^\top$  never drops below  $A_0A_0^\top$  in the Loewner order. HF's PiSSA starts with  $B_0^\top B_0 = A_0A_0^\top$ . When  $\Phi$  is isotropic, singular values of  $\Delta W$  well above the crossover scale  $\sigma^*$  of [Theorem III.9](#), set by the charge, the scale and the rates, follow the balanced dynamics of [Arora et al. \(2018\)](#).

**Used in.** isometries and charges (§III), [Proposition III.7](#), [Proposition III.8](#), [Theorem III.9](#), the bibliography. Reading: [Arora et al. \(2018\)](#); [Du et al. \(2018\)](#).

**BACKGROUND D.29** (the moment map). Let a group  $G \subseteq \text{GL}(V)$  that is closed under transpose act on a real inner-product space  $V$ , and split  $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$  into its antisymmetric and symmetric elements. The (real) **moment map**  $\mu : V \rightarrow \mathfrak{p}$  is defined by

$$\langle \mu(v), \xi \rangle = \frac{1}{2} \frac{d}{dt} \Big|_{t=0} \|e^{t\xi} v\|^2 = \langle \xi v, v \rangle \quad (\xi \in \mathfrak{p}).$$

It records how the norm changes along the symmetric directions of the orbit; the anti-symmetric directions preserve the norm.

For LoRA's gauge,  $\frac{d}{dt} \Big|_0 (\|Be^{-tX}\|^2 + \|e^{tX}A\|^2) = -2 \text{tr}(X\mu)$  for symmetric  $X$ , with  $\mu = B^\top B - AA^\top$ , so this  $\mu$  is the moment map up to sign. Its zero set is the balanced slice, and at equal rates  $\eta_A = \eta_B = \eta$  it equals  $\eta\Phi$ , a multiple of the Noether charge.

**Used in.** the balanced slice (§III), [Proposition III.7](#), [Corollary III.6](#), [Figure 7](#).

**FACT D.30** (Kempf–Ness theorem, real form). *Let  $G \subseteq \text{GL}(V)$  be a real algebraic group, cut out by polynomial equations, that is closed under transpose (such a group is **reductive**), for instance  $\text{GL}_r$ , acting on LoRA factors, and let  $K = G \cap \text{O}(V)$ . For  $v \in V$ , (i)  $g = e$  is a critical point of  $g \mapsto \|g \cdot v\|^2$  iff  $\mu(v) = 0$ , and then  $v$  has least norm on  $G \cdot v$ ; (ii)  $G \cdot v$  is closed iff it contains a point with  $\mu = 0$ ; (iii) on a closed orbit, the points with  $\mu = 0$  form a single  $K$ -orbit; (iv) the closure of every orbit contains exactly one closed orbit ([Birkes, 1971](#); [Kempf and Ness, 1979](#); [Richardson and Slodowy, 1990](#)).*

**The idea of the proof.** For symmetric  $\xi$ ,  $\|e^{t\xi} v\|^2 = \sum_i e^{2t\lambda_i} \|v_i\|^2$  in an eigenbasis of  $\xi$ , which is convex in  $t$ . By the polar decomposition every orbit point is  $k e^\xi v$  with  $k \in K$  norm-preserving, so critical points are minima and two minima differ by an element of  $K$ .

For  $m = n = r = 1$ ,  $t \in \mathbb{R}^\times$  acts by  $t \cdot (b, a) = (b/t, ta)$  and  $\mu = b^2 - a^2$ . The orbits are the hyperbolas  $ba = c \neq 0$ , which are closed and meet  $\mu = 0$  in the two points  $\pm(b, a)$  with  $|b| = |a|$ , one  $\text{O}(1)$ -orbit; the two punctured axes, which are not closed and avoid  $\mu = 0$ ; and the origin. For LoRA in general,  $K = \text{O}(r)$ , and a full-rank orbit is closed, being a whole fibre of  $(B, A) \mapsto BA$ . The least-norm factorisations of a fixed  $BA$  are the balanced ones, they form one  $\text{O}(r)$ -orbit, and there  $\|B\|_F^2 + \|A\|_F^2 = 2\|BA\|_*$ . Weight decay adds a norm to the loss ([Corollary III.6](#)). With learning-rate-coupled decay, which multiplies the decay of each factor by its rate,  $\dot{\Phi} = -2\lambda\mu$ , so the equilibria lie on  $\mu = 0$ , and  $\mu$  decays exponentially at equal rates. With per-time decay, the same  $\lambda$  for both factors,  $\dot{\Phi} = -2\lambda\Phi$ , so  $\Phi \rightarrow 0$ .

**Used in.** the balanced slice (§III), [Proposition III.7](#), [Corollary III.6](#), combining (§V), the known results (Exposé XI), the classification (Exposé VIII).

**FACT D.31** (the closed orbit in each fibre of  $(B, A) \mapsto BA$ ). For  $\text{GL}_r$  acting on factor pairs, fix  $W$  of rank  $k \leq r$ . Among the pairs with  $BA = W$ , those with  $\text{rank } B = \text{rank } A = k$  form a single orbit. It is closed, and it lies in the closure of every orbit in the fibre  $\{BA = W\}$ . Consequently a continuous  $\text{GL}_r$ -invariant function is constant on each fibre of  $(B, A) \mapsto BA$ . For a complex reductive group acting on a vector space, every fibre of the quotient map likewise contains exactly one closed orbit (Mumford et al., 1994, Ch. 1).

For  $m = n = r = 1$  (Background D.30), the fibre  $\{ba = 0\}$  consists of the origin and the two punctured axes, and the origin, its only closed orbit, lies in the closure of both axes. This is the step behind Proposition V.4(a). A continuous rule for combining LoRA factors that is invariant under each adapter’s  $\text{GL}_r$  is automatically a function of the adapters  $B_i A_i$ , also off the full-rank locus. Continuity matters. The projector onto row  $A$ , defined where  $\text{rank } A = r$ , is invariant, yet it differs at  $(0, A)$  and  $(0, A')$ , which have the same product 0.

**Used in.** Proposition V.4, combining (§V).

### Matrix and convexity tools

**FACT D.32** (Ky Fan’s maximum principle). Let  $S \in \mathbb{R}^{n \times n}$  be symmetric with eigenvalues  $\lambda_1 \geq \dots \geq \lambda_n$ , and let  $1 \leq k \leq n$ . Then

$$\max\{\text{tr}(PSP^\top) : P \in \mathbb{R}^{k \times n}, PP^\top = I_k\} = \lambda_1 + \dots + \lambda_k,$$

attained when the rows of  $P$  span a top- $k$  eigenspace. For any matrix  $X$  with singular values  $\sigma_1 \geq \sigma_2 \geq \dots$ , applying this to  $S = X^\top X$  gives  $\max_P \|XP^\top\|_F^2 = \sigma_1^2 + \dots + \sigma_k^2$ , and for square  $X$  also  $\|PXP^\top\|_F^2 \leq \|XP^\top\|_F^2 \leq \sigma_1^2 + \dots + \sigma_k^2$ . See Fan (1949) and Bhatia (1997, Ch. III).

Among  $A_0$  with  $A_0 A_0^\top = cI$ , LoRA’s initial rate  $\|GA_0^\top\|_F^2$  under gradient flow is largest when row  $A_0$  is the top- $r$  right singular space of  $G$ , which is LoRA-GA’s (Wang et al., 2024e) choice for  $A_0$ . For LOFT, the squared norm of the pulled-back gradient  $P_r \text{skew}(W_0^\top G) P_r^\top$  is at most  $2 \sum_{k \leq \lfloor r/2 \rfloor} \mu_k^2$ , where  $\pm i\mu_k$  are the eigenvalues of the skew matrix  $\text{skew}(W_0^\top G)$ , whose singular values are  $\mu_1, \mu_1, \mu_2, \mu_2, \dots$ . The floor appears because a skew matrix compressed to an odd-dimensional subspace has a zero eigenvalue, and Cauchy interlacing bounds the rest.

**Used in.** LOFT (§II.B), Proposition II.3.

**FACT D.33** (Sylvester equations). For  $S \in \mathbb{R}^{p \times p}$ ,  $T \in \mathbb{R}^{q \times q}$  and  $C \in \mathbb{R}^{p \times q}$ , the **Sylvester equation**  $SX + XT = C$  has a unique solution  $X$  for every  $C$  iff  $S$  and  $-T$  have no eigenvalue in common, since the eigenvalues of  $X \mapsto SX + XT$  are the sums  $\lambda_i(S) + \lambda_j(T)$ . If  $S$  and  $T$  are symmetric positive definite, this holds and  $X = \int_0^\infty e^{-tS} C e^{-tT} dt$ . The Bartels–Stewart algorithm (Bartels and Stewart, 1972) solves the equation through Schur forms of  $S$  and  $T$  (Horn and Johnson, 1991, Ch. 4).

LoRA-Pro’s equivalent gradient leaves a free  $r \times r$  matrix  $X$ . The published method picks it by solving  $B^\top B X + X A A^\top = C$ , with  $C$  built from the gradient. On the full-rank locus both Gram matrices are positive definite, so  $X$  exists and is unique. The first-order weight velocity does not depend on  $X$ , while the resulting parameter update is only  $O(r)$ -equivariant.

**Used in.** Theorem III.12, when a change of chart does not matter (§III), Figure 9.

**BACKGROUND D.34** (the perspective map; convex combinations). The **perspective map** is  $\pi(Z, S) = S/Z$  on  $\{(Z, S) \in \mathbb{R} \times \mathbb{R}^d : Z > 0\}$ . It is constant on rays,  $\pi(tZ, tS) = \pi(Z, S)$  for  $t > 0$ , and it turns sums into convex combinations,

$$\pi(Z_1 + Z_2, S_1 + S_2) = \lambda \pi(Z_1, S_1) + (1 - \lambda) \pi(Z_2, S_2), \quad \lambda = \frac{Z_1}{Z_1 + Z_2}.$$

For a fixed query  $q$ , attention is  $\text{Attn}_q(K) = \pi(\Sigma_q(K))$ , where

$$\Sigma_q(K) = \left( \sum e^{\langle q, k \rangle / \sqrt{d}}, \sum e^{\langle q, k \rangle / \sqrt{d}} v \right)$$

is a monoid homomorphism on multisets of key–value pairs ([Background D.10](#)). With a prefix  $P$  before the content  $C$ , the identity above gives

$$\text{Attn}_q(P \uplus C) = (1 - \lambda_q) \text{Attn}_q(C) + \lambda_q \text{Attn}_q(P),$$

with  $\text{Attn}_q(P)$  in the convex hull  $\text{conv } V_P$  of the prefix values, so a prefix acts as a parallel branch gated by its share of the mass.

**Used in.** [Theorem VI.5](#), attention (§VI), [Figure 12](#). Reading: [Boyd and Vandenberghe \(2004, §2.3.3 and §3.2.6\)](#).



## Auxiliary results

This appendix quotes the results from outside the paper that the proofs in Appendix C use. Each is stated in the form in which it is applied, with a precise source, and is followed by the places that use it. Several are textbook facts that a specialist applies without comment. They are listed so that every step of Appendix C can be traced to a statement with hypotheses. For a few elementary facts we give the one-line reason instead of a reference. Background definitions (manifolds, germs, group actions, lenses) are in Appendix A, and where an entry has a companion there we name it.

Three results of the main text are external results restated in the paper's conventions and proved there: [Theorem II.10](#) (the min-cut bound, here [Auxiliary result B.14](#)), [Theorem III.5](#) (Noether charges, [Auxiliary result B.29](#)) and [Proposition III.7\(a\)](#) (the Kempf–Ness slice, [Auxiliary result B.26](#)). The entries below give their source forms.

The entries are grouped by subject: smooth maps and limits (B.1–B.4), real algebraic and tame geometry (B.5–B.10), matrix analysis (B.11–B.21), Lie groups and invariant theory (B.22–B.27), dynamics, optimisation and differentiation (B.28–B.35), three elementary facts (B.36), and spectral projectors, Riemannian submanifolds and floating-point scaling (B.37–B.39). Notation follows the exposés:  $\mathcal{M}_{\leq r}$  is the set of  $m \times n$  matrices of rank at most  $r$ ,  $\mathcal{M}_r$  those of rank exactly  $r$ ,  $P_U$  the orthogonal projector onto a subspace  $U$ , and  $\|\cdot\|$  the Frobenius norm unless marked otherwise. Entries of Appendix A are cited as “Background A.1” to “Background D.34”; the numbers of its Part B are unrelated to the numbers B.1–B.39 used here.

### Smooth maps, fibre products and limits

**AUXILIARY RESULT B.1** (rank, the rank theorem and submersions). ([Lee, 2013, Prop. 4.1, Thm 4.12, Prop. 4.28, Thm 5.12](#)) Let  $F : Q \rightarrow \Theta$  be a smooth map between manifolds of dimensions  $N$  and  $D$ .

- (a) For every  $k$ , the set  $\{q : \text{rank } dF_q \geq k\}$  is open. In particular the set where  $\text{rank } dF_q$  takes its maximal value is open and nonempty, and if  $\text{rank } dF_p = D$  then  $F$  is a submersion on a neighbourhood of  $p$ .
- (b) (Rank theorem.) If  $\text{rank } dF_q = k$  for all  $q$  in a neighbourhood of  $p$ , there are charts centred at  $p$  and at  $F(p)$  in which  $F$  reads  $(x_1, \dots, x_N) \mapsto (x_1, \dots, x_k, 0, \dots, 0)$ . Hence  $p$  has a neighbourhood  $V$  such that  $F(V)$  is an embedded submanifold of dimension  $k$  and  $F^{-1}(F(p)) \cap V$  is an embedded submanifold of dimension  $N - k$ .
- (c) A submersion is an open map. In particular, if  $\text{rank } dF = D$  on a neighbourhood  $U$  of  $p$ , then  $F(U)$  contains an open neighbourhood of  $F(p)$ .

*Reason for (a).* In coordinates,  $\text{rank } dF_q \geq k$  says that some  $k \times k$  minor of the Jacobian is nonzero at  $q$ , an open condition; Lee's Prop. 4.1 is the case  $k = D$ . Companion: [Background C.5](#).

**Used in.** [Observation I.4\(b\)](#), where the lifts of both axes force  $\text{rank } d\iota_{p^*} = 2$ , the rank stays 2 near  $p^*$ , and the image of a neighbourhood of  $p^*$  contains an open subset of  $\mathbb{R}^2$ ; [Definition I.5](#) (with [Auxiliary result B.9](#)); [Proposition II.12\(a\)](#) (fibre dimension  $|M| - d(M)$  near points of maximal rank); [Proposition III.4\(d\)](#) (rank  $r$  is an open condition on the slice). The proofs of [Proposition I.8](#), [Theorem II.7](#) and [Proposition IV.3](#) in Appendix C also cite it.

**AUXILIARY RESULT B.2** (transversality and fibre products). (*Lee, 2013*, Cor. 5.30 and Thm 6.30); (*Guillemin and Pollack, 1974*, §1.5) Let  $\rho_M : Q_M \rightarrow \Theta$  and  $\rho_N : Q_N \rightarrow \Theta$  be smooth, with  $\dim Q_M = |M|$  and  $\dim Q_N = |N|$ , and let  $P = \{(q, q') : \rho_M(q) = \rho_N(q')\} \subseteq Q_M \times Q_N$ .

- (a) If  $d\rho_M(T_q Q_M) + d\rho_N(T_{q'} Q_N) = T_\theta \Theta$  at every  $(q, q') \in P$ , then  $P$  is an embedded submanifold of  $Q_M \times Q_N$  of dimension  $|M| + |N| - \dim \Theta$ .
- (b) Transversality at a single point of  $P$  forces  $|M| + |N| \geq \dim \Theta$ .
- (c) If  $S \subseteq X$  is an embedded submanifold, every smooth map  $T \rightarrow X$  with image in  $S$  is smooth as a map into  $S$ . Consequently, whenever  $P$  is an embedded submanifold of  $Q_M \times Q_N$  (under (a), or when  $\rho_M$  and  $\rho_N$  are affine, so that  $P$  is an affine subspace),  $P$  with its two projections is the pullback of  $\rho_M$  and  $\rho_N$  in **Diff**.

*Reason.*  $P = (\rho_M \times \rho_N)^{-1}(\Delta_\Theta)$ , and  $\rho_M \times \rho_N$  is transverse to the diagonal  $\Delta_\Theta$ , of codimension  $\dim \Theta$ , exactly when the condition in (a) holds; Lee's Thm 6.30(a) gives (a). Part (b) is a dimension count, since the left side of the condition has dimension at most  $|M| + |N|$ . Companion: [Background C.7](#).

**Used in.** [Observation I.4\(b\)](#), [Proposition I.8\(a\)](#), [Theorem II.7\(d\)](#). The proof of [Proposition III.4](#) in [Appendix C](#) also cites it.

**AUXILIARY RESULT B.3** (limits in slices and coslices; right adjoints). (*Mac Lane, 1998*, §III.4, §V.1 Exercise 1, §V.5 Thm 1) Let  $C$  be a category.

- (a) In the slice  $C/X$ , a product of  $a : A \rightarrow X$  and  $b : B \rightarrow X$  is a pullback  $A \times_X B$  of  $a$  and  $b$  in  $C$ , with its map to  $X$ ; one exists exactly when the other does.
- (b) For an object  $X$ , the projection from the coslice  $X/C$  (the comma category  $(X \downarrow C)$ ) to  $C$  creates limits: every limit cone in  $C$  of the underlying diagram lifts to exactly one cone in  $X/C$  over the given diagram, and that cone is a limit.
- (c) A functor that has a left adjoint preserves all limits that exist in its domain. The forgetful functors  $\mathbf{Set}_* \rightarrow \mathbf{Set}$  and  $\mathbf{Diff}_* \rightarrow \mathbf{Diff}$  have the left adjoint  $X \mapsto X \sqcup \text{pt}$ .

*Form used.* A product in the pointed slice  $\mathbf{Meth}(\Theta, \theta_0) = \mathbf{Diff}_*/(\Theta, \theta_0)$  is a pullback over  $(\Theta, \theta_0)$  in  $\mathbf{Diff}_*$  by (a). By (b) a pullback in  $\mathbf{Diff}$  of the underlying maps lifts to  $\mathbf{Diff}_*$  with base point  $(q_0, q'_0)$ , and by (c) any pullback in  $\mathbf{Diff}_*$  is a pullback in  $\mathbf{Diff}$ , so it may be tested with unpointed curves. Companions: [Background A.11](#), [Background A.12](#), [Background A.13](#).

**Used in.** [Observation I.4\(b\)](#) (products in  $\mathbf{Meth}$  as pullbacks, tested with unpointed curves). The proof of [Proposition I.8](#) in [Appendix C](#) also cites it.

**AUXILIARY RESULT B.4** (smooth descent along submersions; quotient manifold theorem). (*Lee, 2013*, Thms 4.29, 4.30, 21.10)

- (a) Let  $\pi : X \rightarrow Y$  be a surjective smooth submersion. A map  $\tilde{f} : Y \rightarrow Z$  is smooth iff  $\tilde{f} \circ \pi$  is smooth, and every smooth  $f : X \rightarrow Z$  that is constant on the fibres of  $\pi$  factors uniquely as  $f = \tilde{f} \circ \pi$  with  $\tilde{f}$  smooth.
- (b) If a Lie group  $G$  acts smoothly, freely and properly on a manifold  $X$ , then  $X/G$  has a unique smooth structure for which  $\pi : X \rightarrow X/G$  is a submersion, and  $\dim X/G = \dim X - \dim G$ .

*Instance.* On pairs  $(B, A)$  of full rank,  $(B, A) \mapsto BA$  is a surjective submersion onto  $\mathcal{M}_r$  whose fibres are the  $\text{GL}_r$ -orbits  $\{(Bg^{-1}, gA)\}$  ([Auxiliary result B.5\(b\)](#)), so a function of the factors

that is constant on these orbits is a smooth function of  $W$  on the full-rank locus. Companion: [Background D.22](#).

**Used in.** [Proposition III.4\(d\)](#) (cometrics that are functions of  $W$ ), [Proposition V.4\(a\)](#) (descent of combination rules on the full-rank locus).

### Real algebraic and tame geometry

**AUXILIARY RESULT B.5** (determinantal varieties). (*Harris, 1992, Lectures 9, 12 and 14*); (*Schneider and Uschmajew, 2015, Thm 3.2*) Let  $1 \leq r < \min(m, n)$ .

- (a)  $\mathcal{M}_{\leq r} \subseteq \mathbb{R}^{m \times n}$  is the common zero set of the  $(r+1) \times (r+1)$  minors. It is closed and closed under all real scalings, so it is a closed cone with vertex 0, of dimension  $r(m+n-r)$ .
- (b)  $\mathcal{M}_r$  is open and dense in  $\mathcal{M}_{\leq r}$  and is an embedded submanifold of dimension  $r(m+n-r)$ . If  $X \in \mathcal{M}_r$  has column space  $U$  and row space  $V$ , and  $X = BA$  with  $B \in \mathbb{R}^{m \times r}$ ,  $A \in \mathbb{R}^{r \times n}$ , then

$$T_X \mathcal{M}_r = \{\dot{B}A + B\dot{A}\} = \{D : P_{U^\perp} D P_{V^\perp} = 0\},$$

and  $(B, A) \mapsto BA$  is a submersion from the pairs of full rank onto  $\mathcal{M}_r$  whose fibres are the orbits  $\{(Bg^{-1}, gA) : g \in \text{GL}_r\}$ .

- (c) At a point  $X$  of rank  $s < r$ , with column space  $U$  and row space  $V$ , the tangent cone of  $\mathcal{M}_{\leq r}$  is  $\{D : \text{rank}(P_{U^\perp} D P_{V^\perp}) \leq r-s\}$ , which is not a linear space. Hence  $\mathcal{M}_{\leq r}$  is not a  $C^1$  submanifold near  $X$ , and  $\text{Sing } \mathcal{M}_{\leq r} = \mathcal{M}_{\leq r-1}$ , algebraically and in the  $C^1$  sense.

*Reason for the last claim of (c).*  $P_{U^\perp} D P_{V^\perp}$  ranges over all  $(m-s) \times (n-s)$  matrices, and since  $r-s < \min(m-s, n-s)$  two of rank  $r-s$  can sum to a matrix of larger rank. The tangent cone of a  $C^1$  submanifold is its tangent space, a linear space. Companion: [Background C.12](#).

**Used in.** [Theorem II.1\(c\)](#), [Theorem II.2\(a\)](#), (b), (c), (e) (the image determines its vertex), [Proposition II.3](#), [Proposition II.8](#) (tangent space at  $(I, X)$ ), [Proposition II.12](#), [Proposition II.16](#) (the translation stabiliser of  $\mathcal{M}_{\leq r}$  is  $\{0\}$ ), [Proposition III.4\(b\)](#), (d), [Proposition V.4\(a\)](#). The proofs of [Theorem II.7](#) and [Proposition II.9](#) in Appendix C also cite it.

**AUXILIARY RESULT B.6** (cones that are  $C^1$  at their vertex). *elementary* Let  $C \subseteq \mathbb{R}^N$  be a closed set with  $0 \in C$  and  $tC \subseteq C$  for every  $t > 0$ . If  $C$  is a  $C^1$  embedded submanifold near 0, then  $C$  equals the linear subspace  $T_0 C$ . Consequently, the vertex of a closed cone that is not a linear subspace is a singular point.

*Reason.* The tangent cone of  $C$  at 0, the set of limits of  $t_k x_k$  with  $x_k \in C$ ,  $x_k \rightarrow 0$  and  $t_k > 0$ , equals  $C$  because  $C$  is a closed cone, and the tangent cone of a  $C^1$  submanifold at a point is its tangent space. Companions: [Background C.9](#), [Background C.10](#).

**Used in.** [Definition I.5](#), [Theorem II.2\(a\)](#), (c), [Theorem II.7\(c\)](#), [Theorem II.11\(e\)](#) (the closed cone  $\mathcal{K}_{\leq k}$ ).

**AUXILIARY RESULT B.7** (real normal forms; the skew-determinantal cone). (*Horn and Johnson, 2013, Ch. 2*)

- (a) Every  $\Omega \in \mathfrak{so}(n)$  is orthogonally similar to a block-diagonal matrix with  $2 \times 2$  blocks  $\mu_k \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ ,  $\mu_1 \geq \mu_2 \geq \dots > 0$ , and zeros. Hence  $\text{rank } \Omega$  is even, the nonzero eigenvalues are  $\pm i\mu_k$ , the singular values are  $\mu_1, \mu_1, \mu_2, \mu_2, \dots$ , and  $\Omega^4$  is symmetric positive semidefinite with eigenvalues  $\mu_k^4$ . A skew matrix of odd size has 0 as an eigenvalue.

- (b) Every  $R \in \mathrm{SO}(n)$  is orthogonally similar to a block-diagonal matrix with  $2 \times 2$  rotation blocks of angles  $\varphi_k \in (0, \pi]$  and ones. Then  $\mathrm{rank}(R - I)$  is twice the number of blocks,  $\mathrm{im}(R - I)$  is the sum of the rotation planes, and  $R = e^X$  for a skew  $X$  whose eigenvalues are  $\pm i\varphi_k$  and 0.
- (c) For even  $r$  with  $2 \leq r \leq n - 2$ , the set  $\Sigma_r = \{\Omega \in \mathfrak{so}(n) : \mathrm{rank} \Omega \leq r\}$  is a closed cone of dimension  $r(n - r) + r(r - 1)/2 = rn - r(r + 1)/2$ . Its points of rank exactly  $r$  form an open dense submanifold of that dimension, and  $\Sigma_r$  is not a linear space.

Reason for (c). A skew matrix of rank  $r$  is a range  $V \in \mathrm{Gr}(r, n)$  together with a nondegenerate skew form on  $V$ , an open subset of  $\mathfrak{so}(r)$ ; and skew matrices of ranks  $r$  and 2 can sum to rank  $r + 2 \leq n$ . Companion: [Background C.13](#).

**Used in.** [Corollary II.6](#) (the Cayley–Neumann factor), [Theorem II.7\(a\)](#), (c), (d) (the dimension count and the density argument), [Theorem V.3](#), corollary 4, the LOFT example of Exposé II.B.

**AUXILIARY RESULT B.8** (Segre cones). ([Harris, 1992](#), Lecture 2) The Segre cone  $\mathcal{M}_{\leq 1}(p \times q) = \{ab^\top : a \in \mathbb{R}^p, b \in \mathbb{R}^q\}$  is a closed cone of dimension  $p + q - 1$ , it is not a linear space when  $p, q \geq 2$ , and it spans  $\mathbb{R}^{p \times q}$ . Hence  $k$  generic rank-one matrices span a space of dimension  $\min(k, pq)$ : the parameters  $(a_1, b_1, \dots, a_k, b_k)$  for which  $a_1 b_1^\top, \dots, a_k b_k^\top$  span less form a proper algebraic subset of  $(\mathbb{R}^p \times \mathbb{R}^q)^k$ .

Reason for the last claim. Spanning dimension at least  $d$  is the nonvanishing of some  $d \times d$  minor of the coefficient matrix, a polynomial in the parameters, and distinct matrix units  $E_{ij}$  show that it is not identically zero for  $d = \min(k, pq)$ . Companion: [Background C.14](#).

**Used in.** [Proposition II.8](#) (the  $m$  points  $(E^\top e_i) \otimes (F^\top w_{0,i})$  give the generic value of  $d(\mathrm{DoRA}_r)$ ), [Proposition II.12](#) (the KronA image), the VeRA card of Exposé VII.

**AUXILIARY RESULT B.9** (semialgebraic and definable sets; dimension). ([Bochnak et al., 1998](#), Ch. 2 and 9); ([van den Dries, 1998](#), Ch. 4, 6 and 7); ([Wilkie, 1996](#))

- (a) (Tarski–Seidenberg.) The image of a semialgebraic set under a semialgebraic map is semialgebraic. (Wilkie.) The expansion  $\mathbb{R}_{\exp}$  of the real field by the exponential function is o-minimal, so sets and maps defined by first-order formulas from polynomials and  $\exp$  (softmax included) are definable, and images of definable sets under definable maps are definable.
- (b) Every definable set  $X$  is a finite disjoint union of  $C^1$  embedded submanifolds (cells), and  $\dim X$ , the largest of their dimensions, does not depend on the decomposition. If  $X \subseteq Y$  then  $\dim X \leq \dim Y$ , and  $\dim f(X) \leq \dim X$  for every definable map  $f$ .
- (c) (Definable choice and generic smoothness.) A definable surjection has a definable section, and a definable map is  $C^1$  outside a definable subset of its domain of smaller dimension.
- (d) Consequently, for a definable  $C^1$  map  $\rho$  on an open definable set  $Q \subseteq \mathbb{R}^k$ ,

$$\dim \rho(Q) = \max_{q \in Q} \mathrm{rank} d\rho_q.$$

Reason for (d). At a point of maximal rank, the rank theorem ([Auxiliary result B.1\(b\)](#)) puts a sheet of that dimension inside  $\rho(Q)$ . Conversely, by (c) a definable section  $s$  of  $\rho$  is  $C^1$  on an open subset  $S$  of a top-dimensional cell of  $\rho(Q)$ , and  $\rho \circ s = \mathrm{id}_S$  gives  $\mathrm{rank} d\rho_{s(y)} \geq \dim S = \dim \rho(Q)$ . Companions: [Background C.17](#), [Background C.18](#).

**Used in.** [Definition I.5](#) ( $d(M) = \dim \mathrm{Im} M = \max_q \mathrm{rank} d\rho_q$ ), the obstruction  $d(M) > d(N)$  of Exposé VIII.C, [Proposition II.12\(a\)](#). The proofs of [Theorem II.7](#), [Proposition II.8](#), [Proposition II.9](#), [Proposition II.13](#) and [Theorem V.3](#) in Appendix C also cite it.

**AUXILIARY RESULT B.10** (genericity; zero sets; the identity theorem). (*Mityagin, 2020*); (*Krantz and Parks, 2002, Ch. 1 and 2*)

- (a) Let  $U \subseteq \mathbb{R}^N$  be open and connected and  $f : U \rightarrow \mathbb{R}$  real-analytic, for instance a polynomial. If  $f$  vanishes on a nonempty open subset of  $U$ , then  $f \equiv 0$ . If  $f \not\equiv 0$ , its zero set is closed, nowhere dense and of Lebesgue measure zero.
- (b) Hence a property that fails only on the common zero set of finitely many real-analytic functions on a connected domain, not all identically zero, holds on an open dense set of full measure, and with probability one under every law with a density. To prove such genericity it suffices to write failure as the vanishing of these functions and to exhibit one point where one of them does not vanish. A countable union of such exceptional sets still has measure zero.
- (c) A function equal to a convergent power series  $\sum_k a_k t^k$  near 0 determines its coefficients. In particular, if  $\sigma(ct) = c \sigma(t)$  near 0 for some  $c \neq 0$ , then  $ca_k = c^k a_k$  for every  $k$ .

Companions: [Background C.21](#), [Background C.22](#).

**Used in.** [Theorem II.2\(d\)](#) (a generic rank-one update), [Proposition II.8](#), [Theorem II.11\(c\)](#) (generic Schmidt ranks), [Proposition II.12](#) (the generic rank of GraLoRA), [Proposition V.2\(a\)](#) ( $e$  has full rank for almost every  $W$ , cell by cell of the quantiser), [Lemma VI.2\(c\)](#), [Theorem VI.3](#) (the GELU obstruction), [Theorem VI.5\(f\)](#) (prefixes with affine output form a nowhere-dense analytic subset). The proofs of [Proposition I.8](#), [Theorem II.1](#), [Corollary II.6](#), [Theorem II.7](#), [Proposition II.13](#), [Proposition IV.3](#) and [Theorem V.3](#) in Appendix C also cite it.

## Matrix analysis

**AUXILIARY RESULT B.11** (singular value decomposition; Eckart–Young–Mirsky). (*Eckart and Young, 1936*; *Mirsky, 1960*); (*Horn and Johnson, 2013, Ch. 2 and 7*)

- (a) Every  $X \in \mathbb{R}^{m \times n}$  of rank  $k$  has a singular value decomposition  $X = \sum_{i \leq k} \sigma_i u_i v_i^\top$  with  $\sigma_1 \geq \dots \geq \sigma_k > 0$  and orthonormal families  $(u_i)$ ,  $(v_i)$ ; two matrices lie in one orbit of  $(P, R) \mapsto PXR$ ,  $P \in O(m)$ ,  $R \in O(n)$ , iff they have the same singular values. Applied to the  $m_1 \times m_2$  reshape of  $u \in \mathbb{R}^{m_1} \otimes \mathbb{R}^{m_2}$  ( $x \otimes y \mapsto xy^\top$ ), the decomposition is the Schmidt decomposition  $u = \sum_i \sigma_i x_i \otimes y_i$ , whose number of terms, the Schmidt rank, is the rank of the reshape. Every matrix of rank at most  $r$  factors as  $QV$  with  $Q \in \mathbb{R}^{m \times r}$  having orthonormal columns.
- (b) (Eckart–Young–Mirsky.) For  $0 \leq s < k$  the truncation  $[X]_s = \sum_{i \leq s} \sigma_i u_i v_i^\top$  minimises  $\|X - Y\|$  over  $\text{rank } Y \leq s$ , with minimum  $(\sum_{i > s} \sigma_i^2)^{1/2}$ , and the minimiser is unique iff  $\sigma_s > \sigma_{s+1}$ . The matrices  $[X]_s$  and  $X - [X]_s$  are Frobenius-orthogonal, so for every real  $c$

$$\|X - c[X]_s\|^2 = \|X - [X]_s\|^2 + (1 - c)^2 \|[X]_s\|^2.$$

- (c) (Reduced-rank regression.) For  $L \in \text{GL}_n$ ,  $\arg \min_{\text{rank } Y \leq s} \|(X - Y)L\| = [XL]_s L^{-1}$ , unique iff  $\sigma_s(XL) > \sigma_{s+1}(XL)$ ; and  $[XL]_s = \Pi_s XL$ , where  $\Pi_s$  is the orthogonal projector onto the top- $s$  eigenspace of  $XLL^\top X^\top$ .

Reason for (c). Substitute  $Z = YL$ , which has the same rank as  $Y$ , and apply (b) to  $XL$ ; the left singular vectors of  $XL$  are the eigenvectors of  $XLL^\top X^\top$ .

**Used in.** [Proposition I.8\(a\)](#) (the SVD frames of LoRA-XS), [Theorem II.1\(c\)](#) (splitting an SVD), [Theorem II.5\(3\)](#), [Theorem II.11\(c\)](#) (Schmidt decompositions), [Proposition II.16\(a\)](#) and the lemma in (b), [Proposition III.7\(a\)](#) (the balanced representative), [Theorem III.9](#) (compact SVD), [Proposition V.2\(b\)](#),



(c) (LoftQ’s rank- $r$  step and its defect), [Proposition VI.4\(c\)](#) (the factorisation  $R^\top V$ ), [Prediction X.5](#) (the distance to  $k$ -term Kronecker sums). The proofs of [Proposition II.8](#) and [Theorem V.3](#) in [Appendix C](#) also cite it.

**AUXILIARY RESULT B.12** (rank inequalities; the Hadamard rank bound). (*Horn and Johnson, 1991, Ch. 4 and 5*) For matrices of compatible sizes:

- (a)  $\text{rank}(X + Y) \leq \text{rank } X + \text{rank } Y$  and  $\text{rank}(XY) \leq \min(\text{rank } X, \text{rank } Y)$ ;
- (b)  $\text{rank}(C \otimes D) = \text{rank } C \cdot \text{rank } D$ ;
- (c) (Hadamard rank bound)  $\text{rank}(X \odot Y) \leq \text{rank } X \cdot \text{rank } Y$ . Explicitly, if  $X = B_1 A_1$  and  $Y = B_2 A_2$  with inner dimensions  $r_1$  and  $r_2$ , then

$$X \odot Y = (B_1 \bullet B_2) (A_1^\top \bullet A_2^\top)^\top,$$

where row  $i$  of the face-splitting product  $B_1 \bullet B_2 \in \mathbb{R}^{m \times r_1 r_2}$  is the Kronecker product of the rows  $i$  of  $B_1$  and  $B_2$ . In particular  $W_0 \odot (BA) = \sum_{k \leq r} \text{diag}(b_k) W_0 \text{diag}(a_k)$ , summed over the columns  $b_k$  of  $B$  and the rows  $a_k$  of  $A$ , has rank at most  $r \text{rank } W_0$ .

Reason for (c).  $X \odot Y$  is the submatrix of  $X \otimes Y$  on the rows  $(i, i)$  and the columns  $(j, j)$ , so (b) applies. Companion: [Background B.13](#).

**Used in.** [Theorem II.1\(c\)](#), [Proposition II.9\(a\)](#) (HiRA), [Theorem II.10](#) (subadditivity), [Theorem II.11\(b\)](#), [Proposition II.12](#) (LoHa, GraLoRA), [Proposition II.13](#) (two HiRA cycles), [Theorem V.3](#), corollaries 1 and 5, the face-splitting arrow of rule A3 ([Exposé VIII.C](#)).

**AUXILIARY RESULT B.13** (the Van Loan–Pitsianis rearrangement; operator-Schmidt decomposition). (*Van Loan and Pitsianis, 1993*); (*Van Loan, 2000*) Let  $m = m_1 m_2$  and  $n = n_1 n_2$ , cut  $\Delta \in \mathbb{R}^{m \times n}$  into an  $m_1 \times n_1$  grid of  $m_2 \times n_2$  blocks  $\Delta_{ij}$ , and let  $\mathcal{R}(\Delta) \in \mathbb{R}^{m_1 n_1 \times m_2 n_2}$  have the rows  $\text{vec}(\Delta_{ij})^\top$ , ordered with  $i$  running fastest.

- (a)  $\mathcal{R}$  is a linear bijection that permutes entries, hence a Frobenius isometry, and  $\mathcal{R}(C \otimes D) = \text{vec } C \text{ vec } D^\top$  for  $C \in \mathbb{R}^{m_1 \times n_1}$  and  $D \in \mathbb{R}^{m_2 \times n_2}$ .
- (b) Hence  $\Delta = \sum_{i \leq k} C_i \otimes D_i$  for some  $C_i, D_i$  iff  $\text{rank } \mathcal{R}(\Delta) \leq k$ . The Kronecker rank  $\text{rank}_\otimes \Delta$  equals  $\text{rank } \mathcal{R}(\Delta) \leq \min(m_1 n_1, m_2 n_2)$ , and the nearest sum of  $k$  Kronecker products in Frobenius norm is  $\mathcal{R}^{-1}([\mathcal{R}(\Delta)]_k)$  ([Auxiliary result B.11\(b\)](#)).
- (c) The SVD of  $\mathcal{R}(\Delta)$  gives the operator-Schmidt decomposition  $\Delta = \sum_{k \leq K} s_k C_k \otimes D_k$  with  $s_k > 0$ , Frobenius-orthonormal families  $(C_k)$ ,  $(D_k)$  and  $K = \text{rank}_\otimes \Delta$ . Conversely, any expansion of this form with positive coefficients and Frobenius-orthonormal families is one, so its number of terms is  $\text{rank}_\otimes \Delta$ .

Companions: [Background B.11](#), [Background B.12](#).

**Used in.** [Theorem II.11\(a\)–\(e\)](#), [Proposition II.13](#) (the Kronecker family as  $\mathcal{R}_*^{-1} \text{LoRA}_k$ ), [Prediction X.5](#). The proof of [Proposition II.12](#) in [Appendix C](#) also cites it.

**AUXILIARY RESULT B.14** (the min-cut bound for tensor networks). (*Cui et al., 2016*) Let  $\Gamma$  be a tensor network evaluated in  $(\mathbf{FinVect}, \otimes)$ , every wire attached to a box, and let  $I \sqcup O$  be a partition of its boxes. Contracting each side separately factors the value  $\llbracket \Gamma \rrbracket$  through the tensor product of the wires that cross the partition and of the open legs attached on the wrong side, so  $\text{rank} \llbracket \Gamma \rrbracket$  is at most the product  $w$  of their dimensions, and at most  $\min w$  over all partitions. For fixed bond dimensions the largest rank over all choices of tensors can be strictly smaller than

the minimal cut weight; equality holds, for instance, when every bond dimension is a power of one integer.

*Form used.* [Theorem II.10](#) states the bound with the paper’s conventions (bare strands, copy spiders counted once, formal sums) and proves it; only the upper bound is used. Companion: [Background B.8](#).

*Used in.* [Theorem II.10](#) and its consequences, [Proposition II.12](#) (maximum ranks), the “high-rank PEFT” remark of [Exposé II.C](#).

**AUXILIARY RESULT B.15** (manifolds of fixed Tucker and tensor-train rank). ([Koch and Lubich, 2010](#)) (Tucker); ([Holtz et al., 2012](#)) (TT); ([Uschmajew and Vandereycken, 2013](#)) (hierarchical Tucker)

- (a) (Tucker.) For an admissible multilinear rank  $(r_1, \dots, r_d)$  of tensors in  $\mathbb{R}^{n_1 \times \dots \times n_d}$ , the tensors of exactly that rank form an embedded submanifold of dimension  $\prod_k r_k + \sum_k r_k (n_k - r_k)$ . In the parametrisation by a core in  $\mathbb{R}^{r_1 \times \dots \times r_d}$  and factor matrices of full column rank, with  $\prod_k r_k + \sum_k n_k r_k$  parameters, the fibres are the orbits of  $\prod_k \text{GL}_{r_k}$ , so the dimension is the parameter count minus  $\sum_k r_k^2$ .
- (b) (Tensor train.) For an admissible TT rank  $(r_1, \dots, r_{d-1})$ , with  $r_0 = r_d = 1$ , the tensors of exactly that rank form an embedded submanifold of dimension  $\sum_{k=1}^d r_{k-1} n_k r_k - \sum_{k=1}^{d-1} r_k^2$ , the number of core entries minus the dimension of the bond gauge  $\prod_{k=1}^{d-1} \text{GL}_{r_k}$ .

Here admissible means realised by some tensor; at minimal ranks the bond gauge acts freely. Companion: [Background B.9](#).

*Used in.* [Proposition II.12](#)(b), row “TT / Tucker”.

**AUXILIARY RESULT B.16** (circulant matrices and the discrete Fourier transform). ([Gray, 2006, Thm 3.1](#))

- (a) A circulant matrix with first column  $w \in \mathbb{R}^b$  satisfies  $\text{circ}(w) = F^{-1} \text{diag}(Fw) F$ , with  $F$  the discrete Fourier matrix. The powers  $P^0, \dots, P^{b-1}$  of the cyclic shift are a basis of the circulants, with Gram matrix  $bI$  in the Frobenius inner product.
- (b) Over  $\mathbb{R}$ , since the eigenvalues of a real circulant at conjugate frequencies are conjugate,  $\text{circ}(w)$  is orthogonally similar to a block-diagonal matrix of  $2 \times 2$  rotation–scaling blocks and  $1 \times 1$  blocks at the self-conjugate frequencies.
- (c) For a real coefficient  $c$  and a frequency  $(u, v)$ , the real part of the inverse two-dimensional transform of  $c e_{(u,v)}$  is the cosine pattern

$$\text{Re ifft2}(c e_{(u,v)})_{j,l} = \frac{c}{mn} \cos(2\pi(uj/m + vl/n)).$$

It is the same for  $(u, v)$  and  $(-u, -v)$ , and it is even under  $(j, l) \mapsto (-j, -l)$ , indices taken modulo  $m$  and  $n$ .

*Used in.* [Proposition II.12](#) (the FourierFT gauge from conjugate pairs), [Proposition II.13](#) (FourierFT and C3A as masks in transported frames). The proof of [Theorem II.10](#) in [Appendix C](#) also cites it.

**AUXILIARY RESULT B.17** (Ky Fan’s maximum principle; interlacing). (*Fan, 1949*); (*Bhatia, 1997, Ch. III*); (*Horn and Johnson, 1991, Ch. 3*)

(a) (Ky Fan.) Let  $S \in \mathbb{R}^{n \times n}$  be symmetric with eigenvalues  $\lambda_1 \geq \dots \geq \lambda_n$  and let  $1 \leq k \leq n$ . Then

$$\max\{\text{tr}(PSP^\top) : P \in \mathbb{R}^{k \times n}, PP^\top = I_k\} = \lambda_1 + \dots + \lambda_k,$$

attained when the rows of  $P$  span a top- $k$  eigenspace. For  $S = X^\top X$  this reads

$$\max_P \|XP^\top\|^2 = \sigma_1(X)^2 + \dots + \sigma_k(X)^2.$$

(b) (Interlacing for compressions.) If  $P \in \mathbb{R}^{k \times n}$  and  $Q \in \mathbb{R}^{l \times m}$  have orthonormal rows, then  $\sigma_j(QXP^\top) \leq \sigma_j(X)$  for every  $j$  and every  $X \in \mathbb{R}^{m \times n}$ ; for symmetric  $S$ ,  $\lambda_j(PSP^\top) \leq \lambda_j(S)$ .

(c) Consequently, if  $S \in \mathfrak{so}(n)$  has eigenvalues  $\pm i\mu_1, \pm i\mu_2, \dots$  with  $\mu_1 \geq \mu_2 \geq \dots$ , then for every  $P \in \mathbb{R}^{k \times n}$  with orthonormal rows

$$\|PSP^\top\|^2 \leq 2 \sum_{j \leq \lfloor k/2 \rfloor} \mu_j^2,$$

with equality when the row space of  $P$  contains the invariant subspace of the top  $\lfloor k/2 \rfloor$  rotation pairs of  $S$ .

Reason for (c).  $PSP^\top$  is skew of size  $k$ , so by [Auxiliary result B.7\(a\)](#) its singular values are  $\nu_1, \nu_1, \nu_2, \nu_2, \dots$ , with an extra 0 when  $k$  is odd. By (b),  $\nu_j = \sigma_{2j-1}(PSP^\top) \leq \sigma_{2j-1}(S) = \mu_j$ , and  $\|PSP^\top\|^2 = 2 \sum_{j \leq \lfloor k/2 \rfloor} \nu_j^2$ . If the row space contains an invariant subspace  $W$  of the top pairs, then  $W^\perp$  is invariant too, and the compression is  $S|_W$  plus zeros. Companion: [Background D.32](#).

**Used in.** [Proposition II.3](#) (alignment: among  $A_0$  with  $A_0 A_0^\top = cI$  the rate  $\|GA_0^\top\|^2$  is largest on the top- $r$  right singular space of  $G$ ), the LOFT example of [Exposé II.B](#), part (d).

**AUXILIARY RESULT B.18** (singular values of products; the Loewner order). (*Horn and Johnson, 1991, Ch. 3*); (*Bhatia, 1997, Ch. III*); (*Horn and Johnson, 2013, Ch. 7*)

(a) (Horn.)  $\sigma_{i+j-1}(XY) \leq \sigma_i(X) \sigma_j(Y)$ . In particular

$$\sigma_i(XY) \leq \sigma_i(X) \|Y\|_{\text{op}}, \quad \sigma_i(XY) \leq \|X\|_{\text{op}} \sigma_i(Y),$$

and a product of contractions (matrices with  $\|\cdot\|_{\text{op}} \leq 1$ ) is a contraction.

(b) If  $\|Y\|_{\text{op}} \leq 1$ , then  $Y^\top Y \preceq I$ , and  $WY^\top YW^\top \preceq WW^\top$  for every  $W$ .

(c) (Weyl monotonicity.) If  $S \preceq T$  are symmetric, then  $\lambda_i(S) \leq \lambda_i(T)$  for every  $i$ . With (b),  $\sigma_i(WY^\top) \leq \sigma_i(W)$  whenever  $\|Y\|_{\text{op}} \leq 1$ .

**Used in.** [Corollary II.6](#) (the Cayley–Neumann factor contracts for small  $Q$ ), [Theorem V.3](#), corollary 4 (each default-OFT merge decreases  $K_{\text{out}}$  in the Loewner order and lowers every singular value).

**AUXILIARY RESULT B.19** (polar, Cartan and  $KAK$  decompositions). (*Horn and Johnson, 2013, Ch. 2 and 7*)

(a) (Polar.) Every  $g \in \text{GL}_r$  is uniquely  $g = k e^S$  with  $k \in \text{O}(r)$  and  $S \in \text{Sym}_r$ , and  $(k, S) \mapsto k e^S$  is a diffeomorphism  $\text{O}(r) \times \text{Sym}_r \rightarrow \text{GL}_r$ . Hence  $g \mapsto g^\top g = e^{2S}$  identifies the cosets  $\text{O}(r)g$  with the positive definite matrices,  $\text{O}(r) \backslash \text{GL}_r \cong \text{Sym}_r^+$ .

- (b) (Cartan.)  $\mathfrak{gl}_r = \mathfrak{so}(r) \oplus \text{Sym}_r$  through  $\xi = \frac{1}{2}(\xi - \xi^\top) + \frac{1}{2}(\xi + \xi^\top)$ , with  $[\mathfrak{so}, \mathfrak{so}] \subseteq \mathfrak{so}$ ,  $[\mathfrak{so}, \text{Sym}] \subseteq \text{Sym}$  and  $[\text{Sym}, \text{Sym}] \subseteq \mathfrak{so}$ .
- (c) (KAK.) Every  $g \in \text{GL}^+(b)$  is  $k_1 \text{diag}(d) k_2$  with  $k_1, k_2 \in \text{SO}(b)$  and  $d \in \mathbb{R}_{>0}^b$ .

*Reason.* (a) combines the polar form  $g = kP$ ,  $P$  positive definite and unique, with the fact that  $\exp : \text{Sym}_r \rightarrow \text{Sym}_r^+$  is a diffeomorphism (spectral theorem). (c) is the SVD with signs absorbed: if both singular frames have determinant  $-1$ , flip the first column of each. Companion: [Background D.27](#).

**Used in.** [Theorem III.3](#) (gradient descent on LoRA as a family indexed by  $\text{O}(r) \backslash \text{GL}_r$ ), [Theorem III.5\(iii\)](#), [Proposition III.7\(a\)](#) (reduction to  $h(t) = f(e^{t\xi} \cdot x)$  with  $\xi$  symmetric), [Theorem V.3](#), corollary 4 (the monoid generated by Cayley–Neumann factors).

**AUXILIARY RESULT B.20** (Sylvester equations). (*Bhatia and Rosenthal, 1997*); (*Horn and Johnson, 1991, Ch. 4*) For  $S \in \mathbb{R}^{p \times p}$  and  $T \in \mathbb{R}^{q \times q}$ , the map  $X \mapsto SX + XT$  on  $\mathbb{R}^{p \times q}$  has the eigenvalues  $\lambda_i(S) + \lambda_j(T)$ . So  $SX + XT = C$  has a unique solution for every  $C$  iff  $S$  and  $-T$  have no common eigenvalue. If  $S$  and  $T$  are symmetric positive definite, the solution is  $X = \int_0^\infty e^{-tS} C e^{-tT} dt$ .

Companion: [Background D.33](#).

**Used in.** [Proposition III.4](#) (practice: LoRA-Pro’s Sylvester-optimal free matrix exists and is unique on the full-rank locus), [Theorem III.12](#) (published variants). The proof of [Proposition IV.3](#) in [Appendix C](#) also cites it.

**AUXILIARY RESULT B.21** (the nuclear norm as a factorisation norm). (*Srebro et al., 2004, Lemma 1*); see also *Recht et al. (2010)* For  $X \in \mathbb{R}^{m \times n}$  and every  $r \geq \text{rank } X$ ,

$$\|X\|_* = \min \left\{ \frac{1}{2}(\|B\|^2 + \|A\|^2) : B \in \mathbb{R}^{m \times r}, A \in \mathbb{R}^{r \times n}, BA = X \right\} = \min_{BA=X} \|B\| \|A\|,$$

attained at  $B = U\Sigma^{1/2}O$ ,  $A = O^\top \Sigma^{1/2}V^\top$  for an SVD  $X = U\Sigma V^\top$  padded to inner size  $r$  and any  $O \in \text{O}(r)$ .

**Used in.** [Proposition III.7\(a\)](#) (the minimum value  $2\|BA\|_*$  on a full-rank orbit) and its commentary (weight decay on the factors as a nuclear-norm penalty).

## Lie groups and invariant theory

**AUXILIARY RESULT B.22** (the Cartan–Dieudonné theorem in Scherk’s sharp form). (*Scherk, 1950*); (*Artin, 1957, Ch. III*) For  $u \neq 0$  let  $H_u = I - 2uu^\top / \|u\|^2$ , the reflection that fixes  $u^\perp$ ;  $\det H_u = -1$ . A product of  $j$  reflections fixes an intersection of  $j$  hyperplanes, so it satisfies  $\text{rank}(R - I) \leq j$ . Every  $R \in \text{O}(n)$  is a product of at most  $n$  reflections (Cartan–Dieudonné). For the Euclidean form the bound is sharp (Scherk): if  $k = \text{rank}(R - I)$ , then  $R$  is a product of exactly  $k$  reflections and of no fewer. Hence  $\det R = (-1)^{\text{rank}(R-I)}$ , and every  $R \in \text{SO}(n)$  is a product of  $\text{rank}(R - I)$  reflections, an even number.

*Form used.* Scherk’s theorem for a nondegenerate quadratic space adds two reflections when the moved space  $\text{im}(R - I)$  is totally isotropic; a definite form has no nonzero isotropic vectors, so this case does not occur. Companion: [Background D.11](#).

**Used in.** [Theorem II.7\(a\)](#) ( $\supseteq$ , and the parity of  $r$ ), [Theorem V.3](#), corollary 5 (re-HRA).

**AUXILIARY RESULT B.23** (the Cayley map). (*Cayley, 1846*); (*Lezcano-Casado and Martínez-Rubio, 2019*) Let  $\Omega \in \mathfrak{so}(n)$ .

- (a)  $I - \Omega$  is invertible, since the eigenvalues of  $\Omega$  are imaginary, and  $C(\Omega) = (I + \Omega)(I - \Omega)^{-1}$  lies in  $\mathrm{SO}(n)$  and commutes with every polynomial in  $\Omega$ .
- (b)  $C(\Omega) - I = 2\Omega(I - \Omega)^{-1}$ , so  $\mathrm{rank}(C(\Omega) - I) = \mathrm{rank} \Omega$ ; and  $C(\Omega) = I + 2\Omega + 2\Omega^2 + \dots$ , so  $dC_0 = 2 \mathrm{id}$ .
- (c)  $C$  is a diffeomorphism from  $\mathfrak{so}(n)$  onto the open dense subset  $\{R \in \mathrm{SO}(n) : -1 \notin \mathrm{spec} R\}$ , with inverse  $R \mapsto (R - I)(R + I)^{-1}$ . In particular it misses  $\mathrm{diag}(-1, -1, 1, \dots, 1)$ .
- (d) Every element of  $\mathrm{SO}(n)$  is a product of two elements of the image of  $C$ .
- (e) (Neumann truncation.)  $(I - Q)(I + Q + Q^2 + Q^3) = I - Q^4$ , so for skew  $Q$

$$R := I + 2Q + 2Q^2 + 2Q^3 + Q^4 = C(Q)(I - Q^4), \quad R^\top R = (I - Q^4)^2,$$

because  $Q^4$  is symmetric and commutes with the orthogonal matrix  $C(Q)$ ; the singular values of  $R$  are  $|1 - \mu_k^4|$ , where  $\pm i\mu_k$  are the eigenvalues of  $Q$ .

Reason for (d). By [Auxiliary result B.7\(b\)](#),  $R = e^X$  with  $X$  skew and eigenvalues  $\pm i\varphi$ ,  $\varphi \in [0, \pi]$ . Then  $e^{X/2}$  has eigenvalues  $e^{\pm i\varphi/2} \neq -1$ , and  $R = (e^{X/2})^2$ . Companion: [Background D.7](#).

**Used in.** [Corollary II.6](#) (HF PEFT's default OFT), [Theorem II.7\(c\)](#) (the germ of the image at  $W_0$ ) and (d) (exact Cayley OFT never produces eigenvalue  $-1$ ), [Theorem V.3](#), corollaries 2 and 4, the LOFT example of [Exposé II.B](#). The proof of [Proposition V.2](#) in [Appendix C](#) also cites it.

**AUXILIARY RESULT B.24** (Lie subgroups, generation, and Yamabe's theorem). (*Lee, 2013, Prop. 7.14, Thms 19.25, 19.26*); (*Yamabe, 1950*); (*Goto, 1969*)

- (a) In a connected Lie group every neighbourhood of the identity generates the group. A symmetric neighbourhood  $U = U^{-1}$  therefore generates it as a monoid.
- (b) For every Lie subalgebra  $\mathfrak{h}$  of the Lie algebra of  $G$  there is a unique connected Lie subgroup of  $G$  with Lie algebra  $\mathfrak{h}$ , namely the subgroup generated by  $\exp \mathfrak{h}$ . Lie subgroups are weakly embedded, so an inclusion  $H \subseteq H'$  of Lie subgroups is smooth and gives  $\mathrm{Lie}(H) \subseteq \mathrm{Lie}(H')$ .
- (c) (Yamabe.) Every arcwise connected subgroup of a Lie group is a Lie subgroup.
- (d) Consequently, the subgroup generated by connected Lie subgroups  $H_1, \dots, H_p$  of  $G$  is the connected Lie subgroup  $H$  with  $\mathrm{Lie}(H) = \mathrm{Lie}\langle \mathfrak{h}_1, \dots, \mathfrak{h}_p \rangle$ , the smallest Lie subalgebra containing every  $\mathfrak{h}_i$ .

Reason for (d). The generated subgroup is arcwise connected, so it is a connected Lie subgroup  $H$  by (c), and  $\mathrm{Lie}(H) \supseteq \mathfrak{h}_i$  for every  $i$  by (b). The connected Lie subgroup  $H'$  with Lie algebra  $\mathrm{Lie}\langle \mathfrak{h}_i \rangle$  contains every  $H_i$ , so  $H \subseteq H'$ , hence  $\mathrm{Lie}(H) = \mathrm{Lie}(H')$  and  $H = H'$  by uniqueness. Companions: [Background D.4](#), [Background D.8](#), [Background D.9](#).

**Used in.** [Theorem V.3\(b\)](#), (c) and corollary 4; the LOFT example of [Exposé II.B](#) (supports revisited across restarts).

**AUXILIARY RESULT B.25** (first fundamental theorem for  $\mathrm{O}(n)$ ; Witt's extension theorem). (*Weyl, 1939, Ch. II*); (*Goodman and Wallach, 2009, Ch. 5*); (*Artin, 1957, Ch. III*)

- (a) Every polynomial in the rows  $w_1, \dots, w_m \in \mathbb{R}^n$  of  $W$  that is invariant under  $W \mapsto WR$ ,  $R \in \mathrm{O}(n)$ , is a polynomial in the inner products  $\langle w_i, w_j \rangle$ . Since the invariant polynomials



of a compact group acting linearly separate its orbits,  $WW^\top$  is a complete invariant of the right  $O(n)$ -action.

- (b) (Witt, Euclidean case.) A linear isometry between two subspaces of  $\mathbb{R}^n$  extends to an element of  $O(n)$ .

*Form used.* Part (b) gives the orbit statement of (a) directly: if  $WW^\top = W'W'^\top$ , the map  $\sum_i c_i w_i \mapsto \sum_i c_i w'_i$  is a well-defined isometry of the row spaces, and its extension is  $R^\top$  with  $W' = WR$ . Companion: [Background D.16](#).

**Used in.** [Theorem II.5\(1\), \(2\)](#), [Corollary II.6](#), [Corollary III.13](#) (left  $O(r)$ -orbits of full-rank  $A_0$  are classified by  $A_0^\top A_0$ ).

**AUXILIARY RESULT B.26** (the Kempf–Ness theorem in the real form used). ([Kempf and Ness, 1979](#)); ([Richardson and Slodowy, 1990](#)); ([Birkes, 1971](#)) Let  $V$  be a real inner-product space,  $G \subseteq GL(V)$  a real algebraic group that is closed under transpose,  $K = G \cap O(V)$ , and  $\mathfrak{p}$  the symmetric elements of its Lie algebra. Define the moment map  $\mu : V \rightarrow \mathfrak{p}$  by  $\langle \mu(v), \xi \rangle = \frac{1}{2} \frac{d}{dt} \Big|_{t=0} \|e^{t\xi} v\|^2 = \langle \xi v, v \rangle$ . Then:

- (i)  $g = e$  is a critical point of  $g \mapsto \|g \cdot v\|^2$  iff  $\mu(v) = 0$ , and then  $\|v\| \leq \|g \cdot v\|$  for every  $g \in G$ ;
- (ii)  $G \cdot v$  is closed iff it meets  $\mu^{-1}(0)$ ;
- (iii) if  $G \cdot v$  is closed,  $G \cdot v \cap \mu^{-1}(0)$  is a single  $K$ -orbit;
- (iv) the closure of every orbit contains exactly one closed orbit.

*Form used.* For  $GL_r$  acting on  $V = \mathbb{R}^{m \times r} \times \mathbb{R}^{r \times n}$  by  $(Bg^{-1}, gA)$  with the Frobenius inner product, the image is closed under transpose (the adjoint of the action of  $g$  is the action of  $g^\top$ ),  $K = O(r)$ , and  $\frac{1}{2} \frac{d}{dt} \Big|_0 (\|Be^{-t\xi}\|^2 + \|e^{t\xi} A\|^2) = \langle \xi, AA^\top - B^\top B \rangle$ . So the moment map is  $-\mu$  with  $\mu = B^\top B - AA^\top$  as in [Proposition III.7](#), its zero set is the balanced slice, a full-rank orbit is closed (it is a whole fibre of  $(B, A) \mapsto BA$ ), and by [Auxiliary result B.21](#) the least value of  $\|B\|^2 + \|A\|^2$  on it is  $2\|BA\|_*$ . Companions: [Background D.29](#), [Background D.30](#).

**Used in.** [Proposition III.7\(a\)](#), [Corollary III.6](#) (decay as a Kempf–Ness potential), [Proposition V.4\(a\)](#) (part (iv)).

**AUXILIARY RESULT B.27** (closed orbits in the fibres of  $(B, A) \mapsto BA$ ). ([Goodman and Wallach, 2009, Ch. 5](#)); ([Mumford et al., 1994, Ch. 1](#)); part (c) elementary

- (a) (First fundamental theorem for  $GL_r$ .) The polynomial invariants of  $GL_r(\mathbb{C})$  acting on  $\mathbb{C}^{m \times r} \times \mathbb{C}^{r \times n}$  by  $(Bg^{-1}, gA)$  are generated by the entries of  $BA$ , the contractions of the rows of  $B$  with the columns of  $A$ .
- (b) (Reductive quotients.) For a complex reductive group acting on an affine variety, every fibre of the quotient map contains exactly one closed orbit, and it lies in the closure of every orbit in that fibre.
- (c) (Real form used.) Fix  $W \in \mathbb{R}^{m \times n}$  of rank  $k \leq r$ . Among the real pairs with  $BA = W$ , those with  $\text{rank } B = \text{rank } A = k$  form a single  $GL_r$ -orbit  $O_W$ . It is closed and lies in the closure of every orbit in the fibre  $\{BA = W\}$ . Hence a continuous  $GL_r$ -invariant function of  $(B, A)$  is constant on each fibre of  $(B, A) \mapsto BA$ .

*Reason for (c).* If  $BA = W$  with  $\text{rank } B = \text{rank } A = k$ , then  $\text{col } B = \text{col } W$  and  $\text{row } A = \text{row } W$ ; a gauge brings  $B$  to  $[U_0 \ 0]$  for a fixed basis  $U_0$  of  $\text{col } W$ , which fixes the first  $k$  rows of  $A$ , and the stabiliser of  $[U_0 \ 0]$  clears the remaining rows, which lie in  $\text{row } W$ . The set  $O_W = \{BA = W, \text{rank } B \leq k, \text{rank } A \leq k\}$  is closed. For a general pair in the fibre, split  $\mathbb{R}^r = C_1 \oplus C_2 \oplus C_3 \oplus C_4$

with  $\text{im } A = C_1 \oplus C_2$ ,  $\ker B = C_2 \oplus C_3$  and  $\dim C_1 = k$ , and let  $g_t$  act by  $1, t, 1, t^{-1}$  on the four summands. As  $t \rightarrow 0$ ,  $(Bg_t^{-1}, g_t A)$  converges to a pair in the fibre whose factors have rank at most  $k$ , so to a point of  $O_W$ . Companion: [Background D.31](#).

**Used in.** [Proposition V.4\(a\)](#) (a continuous combination rule that is invariant under each adapter's  $\text{GL}_r$  is a function of the adapters, also off the full-rank locus).

## Dynamics, optimisation and differentiation

**AUXILIARY RESULT B.28** (existence and uniqueness under Carathéodory conditions). ([Hale, 1980](#), Ch. I, Thms 5.1 and 5.3) Let  $D \subseteq \mathbb{R}^{N+1}$  be open and let  $F : D \rightarrow \mathbb{R}^N$  be measurable in  $t$  for each fixed  $x$  and continuous in  $x$  for each fixed  $t$ , with  $|F(t, x)| \leq m_U(t)$  on each compact  $U \subseteq D$  for an integrable  $m_U$ . Then through every  $(t_0, x_0) \in D$  there is an absolutely continuous solution of  $\dot{x} = F(t, x)$ , holding for almost every  $t$ . If moreover  $|F(t, x) - F(t, y)| \leq k_U(t)|x - y|$  on each compact  $U$  for an integrable  $k_U$ , the solution through each point is unique.

*Form used.* If  $\gamma$  is measurable in  $t$  and locally Lipschitz in  $\theta$ , with locally integrable bounds and Lipschitz constants, and  $\rho$  is of class  $C_{\text{loc}}^{1,1}$ , then  $F(t, q) = -\Lambda D\rho_q^\top \gamma(t, \rho(q))$  satisfies these conditions, so the trajectory from  $kq_0$  is unique.

**Used in.** [Theorem III.5\(ii\)](#) (the trajectory statement).

**AUXILIARY RESULT B.29** (Noether's theorem for gradient flows). ([Kunin et al., 2020](#)); ([Zhao et al., 2022b](#), Prop. 5.1); ([Marcotte et al., 2023](#)); ([Du et al., 2018](#)) Let a Lie group act linearly on  $Q = \mathbb{R}^N$  with  $\rho(k \cdot q) = \rho(q)$ , and let  $\Lambda > 0$  be constant. Along every solution of  $\dot{q} = -\Lambda D\rho_q^\top \gamma(t)$ , where  $\gamma(t)$  is any cotangent signal, gradient or not, and for every generator  $\xi$  of the action with  $\Lambda^{-1}\xi$  symmetric, the charge

$$J_\xi(q) = \frac{1}{2} \langle q, \Lambda^{-1} \xi q \rangle$$

is constant. For LoRA with block rates  $\eta_B, \eta_A$  and generators  $\xi_X(B, A) = (BX, -XA)$ ,  $X$  symmetric, the charges are the entries of  $\Phi = B^\top B / \eta_B - AA^\top / \eta_A$ .

*Form used.* The cited sources treat gradient flows, [Zhao et al. \(2022b\)](#) with  $\Lambda = I$ ; their proof uses only  $D\rho_q(\xi q) = 0$ , so it covers arbitrary signals  $\gamma$  and constant  $\Lambda$ , which is how [Theorem III.5\(i\)](#) states and proves it. Companions: [Background D.26](#), [Background D.28](#).

**Used in.** [Theorem III.5\(i\)](#), [Corollary III.6](#), [Proposition III.7\(b\)](#), [Theorem III.9](#) (the charge stays at  $-\kappa I$ ).

**AUXILIARY RESULT B.30** (Riemannian gradients on embedded submanifolds). ([Absil et al., 2008](#), §3.6) Let  $N \subseteq \mathbb{R}^D$  be an embedded submanifold with the metric induced by the Euclidean (Frobenius) inner product, and  $f$  smooth near  $N$ . The Riemannian gradient of  $f|_N$  at  $x$  is  $P_{T_x N} \nabla f(x)$ , the orthogonal projection of the Euclidean gradient onto the tangent space, and the Riemannian gradient flow is  $\dot{x} = -P_{T_x N} \nabla f(x)$ .

*Form used.* For  $N = W_0 + \mathcal{M}_r$  at  $W = W_0 + X$ , with  $U = \text{col } X$  and  $V = \text{row } X$ , [Auxiliary result B.5\(b\)](#) gives the projection  $G \mapsto P_U G + G P_V - P_U G P_V$ . Companion: [Background D.24](#).

**Used in.** [Proposition III.4\(b\)](#) (LoRA-Pro's equivalent gradient).

**AUXILIARY RESULT B.31** (Frobenius’ theorem). ([Lee, 2013](#), Prop. 19.3, Thms 19.12 and 19.21) Let  $D$  be a smooth distribution of constant rank  $k$  on an open set  $U$ .

- (a) If  $N$  is an integral manifold of  $D$  ( $T_\theta N = D_\theta$  for  $\theta \in N$ ) and  $X, Y$  are sections of  $D$ , then  $[X, Y]_\theta \in D_\theta$  for every  $\theta \in N$ . Hence an integrable distribution is involutive.
- (b) (Frobenius.) Every involutive distribution is completely integrable: each point has a chart in which  $D$  is spanned by the first  $k$  coordinate fields. In such a chart a curve with  $\dot{\theta}(t) \in D_{\theta(t)}$  keeps its last coordinates constant, so it stays in one integral manifold.
- (c) (Global Frobenius.) The maximal connected integral manifolds of an involutive distribution form a foliation of  $U$ .

Companion: [Background D.25](#).

Used in. [Proposition IV.3](#).

**AUXILIARY RESULT B.32** (flatness and the Bures–Wasserstein metric). ([Lee, 2018](#), Ch. 7); ([Takatsu, 2011](#)); ([Bhatia et al., 2019](#))

- (a) Riemannian curvature is preserved by local isometries, and a Riemannian manifold is flat iff it is locally isometric to an open subset of a Euclidean space.
- (b) On the positive definite  $n \times n$  matrices, the Bures–Wasserstein metric is the quotient of the Euclidean (Frobenius) metric under  $B \mapsto BB^\top$ , whose fibres are the orbits  $B \mapsto BR$ ,  $R \in O(n)$ ; up to a constant factor fixed by normalisation,  $B \mapsto BB^\top$  is a Riemannian submersion onto it. Its sectional curvature is nonnegative, and for  $n \geq 2$  it is positive on some planes, so the metric is not flat.

*Form used.* By (a), an immersion  $\rho$  with  $\dim Q = \text{rank } D$  that is a local isometry from the flat  $(Q, \Lambda^{-1})$  onto a leaf forces the leaf metric to be flat. The map  $\rho(B) = BB^\top$  has positive-dimensional fibres, so it is no immersion; its cometric  $D\rho D\rho^\top(G) = 2(G\theta + \theta G)$  at  $\theta = BB^\top$  does not depend on the point of the fibre, and by (b) it is the cometric of the curved Bures–Wasserstein metric.

Used in. [Proposition IV.3](#) (dynamic equivalence with a forward lens forces a flat leaf metric when  $\dim Q = \text{rank } D$ ).

**AUXILIARY RESULT B.33** (the cost of forward and reverse automatic differentiation). ([Griewank and Walther, 2008](#), Ch. 3 and 4) Let  $L : \mathbb{R}^N \rightarrow \mathbb{R}$  be evaluated by a straight-line program.

- (a) Forward mode evaluates  $L(\theta)$  together with one directional derivative,  $DL(\theta)[p]$ , which equals  $\langle \nabla L(\theta), p \rangle$ , at a cost bounded by a small constant multiple of the cost of evaluating  $L$ . It carries each tangent alongside its value and stores no trace.
- (b) Reverse mode evaluates  $\nabla L(\theta)$  at a cost bounded by a small constant multiple of the cost of evaluating  $L$ , independent of  $N$ , but needs the intermediate values of the evaluation, stored or recomputed, during the reverse sweep.

*Form used.* The  $k$  coordinates  $\langle p_i, \nabla L(\theta) \rangle$  of  $P^\top \nabla L$  cost  $k$  forward passes and no reverse pass;  $\nabla L$  itself by forward mode would cost  $N$  of them.

Used in. [Observation IV.4](#).

**AUXILIARY RESULT B.34** (central differences and Gaussian smoothing). (*Nesterov and Spokoiny, 2017*); part (a) by Taylor’s theorem Let  $L : \mathbb{R}^N \rightarrow \mathbb{R}$  be twice differentiable with an  $\ell_2$ -Lipschitz Hessian (constant  $L_2$ ),  $\varepsilon > 0$  and  $z \in \mathbb{R}^N$ .

- (a)  $\left| \frac{L(\theta + \varepsilon z) - L(\theta - \varepsilon z)}{2\varepsilon} - \langle \nabla L(\theta), z \rangle \right| \leq \frac{L_2}{6} \varepsilon^2 \|z\|^3.$
- (b) For  $z \sim \mathcal{N}(0, I_N)$ ,  $\mathbb{E}[\langle \nabla L(\theta), z \rangle z] = \nabla L(\theta)$ , and for Lipschitz  $L$ ,  $\mathbb{E}\left[\frac{1}{2\varepsilon}(L(\theta + \varepsilon z) - L(\theta - \varepsilon z)) z\right] = \nabla L_\varepsilon(\theta)$ , the gradient of the Gaussian smoothing  $L_\varepsilon(\theta) = \mathbb{E}_u L(\theta + \varepsilon u)$ .
- (c) Hence the central two-point estimator is unbiased for  $\nabla L_\varepsilon$ , and by (a) its bias relative to  $\nabla L$  is at most  $\frac{L_2}{6} \varepsilon^2 \mathbb{E}\|z\|^4 = \frac{L_2}{6} \varepsilon^2 N(N+2).$

**Used in.** Observation IV.4 (MeZO’s estimator), Proposition IV.3 (instances: each MeZO step is a step of a linear method up to its estimator). The proof of Definition IV.1 in Appendix C also cites it.

**AUXILIARY RESULT B.35** (extreme eigenvalues of Wishart-type matrices). (*Marčenko and Pastur, 1967*); (*Yin et al., 1988*); (*Bai and Yin, 1993*) Let  $A \in \mathbb{R}^{r \times n}$  have independent, identically distributed entries with mean 0, variance  $c/n$  and finite fourth moment. As  $r, n \rightarrow \infty$  with  $r/n \rightarrow y \in (0, 1]$ , the empirical eigenvalue distribution of  $AA^\top$  converges to the Marčenko–Pastur law supported on  $[c(1 - \sqrt{y})^2, c(1 + \sqrt{y})^2]$ , and the largest and smallest eigenvalues converge almost surely to  $c(1 + \sqrt{y})^2$  and  $c(1 - \sqrt{y})^2$ .

*Form used.* HF PEFT’s Kaiming-uniform  $A_0$  has entries  $U(-n^{-1/2}, n^{-1/2})$ , so  $c = 1/3$ , and the eigenvalues of  $A_0 A_0^\top$  fill a band of relative width  $O(\sqrt{r/n})$  around  $c$ . No proof depends on this entry.

**Used in.** the commentary on Theorem III.9 (exactness for Kaiming initialisations), Prediction X.7.

## Elementary facts

**AUXILIARY RESULT B.36** (three elementary facts). *elementary*; part (c) in *Horn and Johnson, 2013, Ch. 1*

- (a) If  $\sigma : \mathbb{R} \rightarrow \mathbb{R}$  is differentiable at 0 and  $\sigma(\ell t) = \ell \sigma(t)$  for all  $\ell \geq 0$  and  $t \in \mathbb{R}$ , then  $\sigma$  is linear.
- (b) A real vector space is not a finite union of proper subspaces. In particular, a hyperplane of  $\mathbb{R}^d$  that is contained in a finite union of coordinate hyperplanes is one of them.
- (c) For  $X \in \mathbb{R}^{p \times q}$  and  $Y \in \mathbb{R}^{q \times p}$ ,  $XY$  and  $YX$  have the same nonzero eigenvalues, with multiplicities.

*Reasons.* (a) Taking  $\ell = 0$  gives  $\sigma(0) = 0$ , and  $\sigma(t) = t \sigma(1)$  for  $t > 0$  and  $\sigma(t) = -t \sigma(-1)$  for  $t < 0$ ; the one-sided derivatives at 0 are  $\sigma(1)$  and  $-\sigma(-1)$ , so differentiability forces  $\sigma(t) = t \sigma(1)$ . ReLU and leaky ReLU satisfy the identity without being linear. (b) If  $V = V_1 \cup \dots \cup V_k$  with  $k$  minimal, pick  $x \in V_1$  outside the other  $V_i$  and  $y \notin V_1$ . No point  $y + tx$  lies in  $V_1$ , so two of them,  $t \neq t'$ , lie in one  $V_i$  with  $i \geq 2$ , and then  $x \in V_i$ , a contradiction. The second claim applies the first to  $H = \bigcup_j (H \cap \{x_j = 0\})$ .

**Used in.** Theorem VI.3 (rule R3: positively homogeneous activations), Lemma VI.2(b) (the kink argument for ReLU), Theorem III.9 (the nonzero eigenvalues of  $T = \beta \beta^\top$  are those of  $\beta^\top \beta$ ). The proof of Corollary III.13 in Appendix C also cites it.

### Spectral projectors, Riemannian submanifolds and floating point

**AUXILIARY RESULT B.37** (smooth spectral projectors). (*Kato, 1995, Ch. I, §5 and Ch. II, §1*); (*Bhatia, 1997, Ch. III*) Let  $S : V \rightarrow \text{Sym}(m)$  be smooth on an open set  $V$  of a real vector space, with eigenvalues  $\lambda_1(W) \geq \dots \geq \lambda_m(W)$ , and let  $\lambda_r(W_*) > \lambda_{r+1}(W_*)$  at some  $W_* \in V$ . Then the gap  $\lambda_r > \lambda_{r+1}$  persists on a neighbourhood  $V_*$  of  $W_*$ , and on  $V_*$  the orthogonal projector  $\Pi(W)$  onto the sum of the eigenspaces of  $\lambda_1(W), \dots, \lambda_r(W)$  is the Riesz projector

$$\Pi(W) = \frac{1}{2\pi i} \oint_{\Gamma} (z - S(W))^{-1} dz$$

for a fixed contour  $\Gamma$  in  $\mathbb{C}$  that encloses  $\lambda_1(W_*), \dots, \lambda_r(W_*)$  and no other eigenvalue of  $S(W_*)$ . In particular  $\Pi$  is smooth on  $V_*$ .

*Reason.* By Weyl's perturbation inequality the eigenvalues are 1-Lipschitz in the operator norm, so near  $W_*$  the contour still separates the top  $r$  eigenvalues and  $z - S(W)$  is invertible on  $\Gamma$ . Matrix inversion is smooth, so the integral is smooth in  $W$ , and at each  $W$  it is the orthogonal projector onto the enclosed eigenspaces.

**Used in.** Proposition IV.3 (step 9: the GaLore field  $\Pi(W)$  is smooth where a singular-value gap exists).

**AUXILIARY RESULT B.38** (isometries, totally geodesic submanifolds and the Gauss equation). (*Lee, 2018, Ch. 4, 5 and 8*); (*Kobayashi, 1958*) Let  $(M, g)$  be a Riemannian manifold.

- (a) A geodesic is determined by its initial point and initial velocity, and an isometry of  $(M, g)$  maps geodesics to geodesics.
- (b) Let  $F \subseteq M$  be an embedded submanifold with the induced metric. If every geodesic of  $M$  with initial velocity tangent to  $F$  stays in  $F$  for small times ( $F$  is totally geodesic), the second fundamental form of  $F$  vanishes, and by the Gauss equation the sectional curvature of  $F$  on every 2-plane  $\Pi \subseteq T_x F$  equals the sectional curvature of  $M$  on  $\Pi$ .

*Form used.* Let  $\tau$  be an isometry and  $F$  an embedded submanifold, open in its fixed-point set, with  $d\tau = \text{id}$  on  $TF$ . By (a) a geodesic  $\gamma$  tangent to  $F$  and  $\tau \circ \gamma$  have the same initial data, so  $\tau \circ \gamma = \gamma$  and  $\gamma$  stays in  $F$  for small times. So  $F$  is totally geodesic, and (b) applies.

**Used in.** Proposition IV.3 (step 6: balanced full-rank LoRA induces a curved leaf metric).

**AUXILIARY RESULT B.39** (exact scaling by powers of two in floating point). (*Higham, 2002, Ch. 2*) In binary floating-point arithmetic with correctly rounded operations (IEEE 754, rounding to nearest), let  $c = 2^k$  with  $k \in \mathbb{Z}$ . If all operands, exact results and rounded results below stay in the normal range (no overflow and no underflow), then  $\text{fl}(cx) = cx$ , and for floating-point numbers  $x, y$ ,

$$\text{fl}(cx \pm cy) = c \text{fl}(x \pm y), \quad \text{fl}(cx \cdot y) = c \text{fl}(xy), \quad \text{fl}(cx/y) = c \text{fl}(x/y), \quad \text{fl}(\sqrt{c^2 x}) = c \text{fl}(\sqrt{x}).$$

*Reason.* In the normal range, multiplication by  $2^k$  only shifts the exponent, so it maps the set of floating-point numbers onto itself and scales all distances between them by  $c$ . Rounding to the nearest floating-point number, with its tie-breaking rule, therefore commutes with it, and each identity follows from  $\text{fl}(cz) = c \text{fl}(z)$  applied to the exact result  $z$ .

**Used in.** Theorem III.10 (bit-exact trajectories when every block factor is a power of two).



## Deferred proofs

This appendix proves the numbered results of the main text in the order in which they appear there, one subsection per result. Each subsection first restates the result in its final form, the form recorded after the two rounds of review, and then proves it. External theorems enter through Appendix B, cited as Auxiliary results B.1–B.39, and background notions through Appendix A. Exposé 0 states its headline results without numbers; each of them is proved below or in the later parts of this appendix under its number. Definitions I.1 and I.2 need no proof, since  $\mathbf{Para}(C)$  is a bicategory (Cruttwell et al., 2021; Fong et al., 2017) and  $\mathbf{Meth}(\Theta, \theta_0)$  is a slice of a coslice; Remark I.3 and the two definitions that contain a claim, Definitions I.5 and II.14, receive short proofs. Unless we say otherwise, parameter spaces are open definable subsets of a vector space and the maps  $\rho$  are semialgebraic or definable in  $\mathbb{R}_{\text{exp}}$  (Background C.17), dimensions are those of Auxiliary result B.9, “generic” has the meaning of Auxiliary result B.10, and LoRA on one  $m \times n$  box is  $\rho(B, A) = W_{\text{res}} + sBA$  with  $s \neq 0$ ,  $B \in \mathbb{R}^{m \times r}$  and  $A \in \mathbb{R}^{r \times n}$ . For subspaces  $V \subseteq \mathbb{R}^n$  and  $U \subseteq \mathbb{R}^m$  we write  $\mathcal{R}(V) = \{Z \in \mathbb{R}^{m \times n} : \text{row } Z \subseteq V\}$  and  $\mathcal{C}(U) = \{Z : \text{col } Z \subseteq U\}$ , and  $x \wedge y = xy^\top - yx^\top$ . Values marked [Num] in a statement are checked numerically and not proved; Appendix E documents them.

### C.1 Remark I.3

**REMARK I.3** (orientation, and what Para adds). *Each  $\rho$  is a 2-cell  $F \Rightarrow F^\rho := (Q, f \circ (\rho \times X))$  out of  $F$ . A simulation  $h : M \rightarrow N$  is a 2-cell  $F^{\rho_N} \Rightarrow F^{\rho_M}$ , since  $\rho_N \circ h = \rho_M$ . Hence  $\mathbf{Meth}(\Theta, \theta_0)$  is the opposite of the pointed coslice of reparametrisation 2-cells under  $F$ . In particular  $\mathbf{Meth}(\Theta, \theta_0)$  depends only on  $(\Theta, \theta_0)$  and not on  $f$ .*

*Proof.* Let  $\mathcal{P}$  be the category whose objects are the 1-cells  $X \rightarrow Y$  of  $\mathbf{Para}(\mathbf{Diff})$  and whose arrows are the 2-cells, composed vertically. By Definition I.1 a 2-cell  $(P, f) \Rightarrow (Q, g)$  is a smooth map  $r : Q \rightarrow P$  with  $g = f \circ (r \times X)$ . The vertical composite of  $r : (P, f) \Rightarrow (Q, g)$  and  $r' : (Q, g) \Rightarrow (Q', g')$  is  $r \circ r' : Q' \rightarrow P$ , a 2-cell because  $g' = g \circ (r' \times X) = f \circ ((r \circ r') \times X)$ .

An object of the coslice  $F/\mathcal{P}$  is a 2-cell out of  $F = (\Theta, f)$ . A 2-cell from  $F$  to  $(Q, g)$  is a smooth  $\rho : Q \rightarrow \Theta$  with  $g = f \circ (\rho \times X)$ , so its target is forced to be  $F^\rho$  and the object is determined by the pair  $(Q, \rho)$ . An arrow from  $\rho_M : F \Rightarrow F^{\rho_M}$  to  $\rho_N : F \Rightarrow F^{\rho_N}$  is a 2-cell  $k : F^{\rho_M} \Rightarrow F^{\rho_N}$  whose vertical composite with  $\rho_M$  is  $\rho_N$ . Concretely, it is a smooth map  $k : Q_N \rightarrow Q_M$  with

$$f \circ (\rho_N \times X) = f \circ (\rho_M \times X) \circ (k \times X) \quad \text{and} \quad \rho_M \circ k = \rho_N.$$

The second condition implies the first. So the arrows  $(Q_M, \rho_M) \rightarrow (Q_N, \rho_N)$  of  $F/\mathcal{P}$  are exactly the smooth maps  $k : Q_N \rightarrow Q_M$  over  $\Theta$ , which are the arrows  $N \rightarrow M$  of  $\mathbf{Diff}/\Theta$ , and composition is reversed. Restricting to pointed parameter spaces with  $\rho(q_0) = \theta_0$  and to pointed  $k$  gives the pointed coslice, which is therefore  $\mathbf{Meth}(\Theta, \theta_0)^{\text{op}}$ . The network  $f$  entered only through the first condition, which is implied by the second for every  $f$ , so  $\mathbf{Meth}(\Theta, \theta_0)$  depends only on  $(\Theta, \theta_0)$ .  $\square$

## C.2 Observation I.4

**OBSERVATION I.4** (universal properties). In **Meth**( $\Theta, \theta_0$ ):

(a) the frozen model  $(\text{pt}, *, \theta_0)$  is initial, and full fine-tuning  $(\Theta, \theta_0, \text{id})$  is terminal. The unique morphism  $M \rightarrow \text{Full FT}$  is  $\rho_M$  itself.

(b) If the fibre product  $Q_M \times_{\Theta} Q_N$  exists in **Diff**, it is the product  $M \times N$ . It exists, for instance, when the set-level fibre product is an embedded submanifold of  $Q_M \times Q_N$ : a clean intersection, or linear or affine  $\rho$ 's as in [Proposition I.8\(a\)](#). Transversality  $\rho_M \pitchfork \rho_N$  is never the reason when  $|M| + |N| < \dim \Theta$ , since at  $(q_0, q'_0)$  it would force  $|M| + |N| \geq \dim \Theta$ . In **Set** the fibre product always exists and  $\text{Im}(M \times N) = \text{Im } M \cap \text{Im } N$ . Whenever the product exists in **Meth**, its underlying set is this set-level fibre product, so the image identity holds in every case. The product can fail to exist. On  $\Theta = \mathbb{R}$  ( $1 \times 1$  matrices),  $\text{LoRA}_1 \times \text{Frozen}$  would have the cross  $\{BA = 0\} \subset \mathbb{R}^2$  as underlying set. No manifold structure on the cross has the universal property, because both axes would have to lift smoothly through the point over the origin, which forces rank 2 there.

(c)  $\text{Im}$  is a functor to the poset of subsets of  $\Theta$  containing  $\theta_0$ . Conversely,  $\text{Im } M \subseteq \text{Im } N$  yields a morphism of the set-level slice  $\mathbf{Set}_*/(\Theta, \theta_0)$ , though not necessarily a smooth one. Take  $\rho_M(t) = t$  and  $\rho_N(u) = u^3$  on  $\mathbb{R}$ , pointed at 0. The images agree, yet no smooth  $h : \mathbb{R} \rightarrow \mathbb{R}$  satisfies  $\rho_N \circ h = \text{id}$ , that is,  $h(t)^3 = t$ .

*Proof.* Recall that **Meth**( $\Theta, \theta_0$ ) = **Diff** $_*/(\Theta, \theta_0)$  ([Definition I.2](#)).

(a) Let  $M = (Q, q_0, \rho)$ . An arrow from the frozen model to  $M$  is a pointed smooth map  $h : \text{pt} \rightarrow Q$  with  $\rho \circ h = \theta_0$ . Pointedness forces  $h(*) = q_0$ , and the condition over  $\Theta$  then holds because  $\rho(q_0) = \theta_0$ . So there is exactly one such arrow. An arrow  $h : M \rightarrow (\Theta, \theta_0, \text{id})$  is a pointed smooth map  $h : Q \rightarrow \Theta$  with  $\text{id} \circ h = \rho$ . The only candidate is  $h = \rho$ , which is smooth and pointed.

(b) By [Auxiliary result B.3\(a\)](#), a product of  $M$  and  $N$  in the slice **Diff** $_*/(\Theta, \theta_0)$  is the same as a pullback of  $\rho_M$  and  $\rho_N$  in **Diff** $_*$ . Suppose the pullback  $P$  of the underlying maps exists in **Diff**, with projections  $\pi_M, \pi_N$ . The constant maps  $\text{pt} \rightarrow Q_M$  and  $\text{pt} \rightarrow Q_N$  at  $q_0$  and  $q'_0$  agree over  $\Theta$ , both composing to  $\theta_0$ , so they factor through a unique point  $p_0 \in P$ . With base point  $p_0$ ,  $P$  is the pullback in **Diff** $_*$ . For a pointed test object  $T$  and pointed maps  $h_M, h_N$  that agree over  $\Theta$ , the induced map  $T \rightarrow P$  sends the base point to a point whose projections are  $q_0$  and  $q'_0$ , and that point is  $p_0$  by uniqueness. This is the creation of limits by **Diff** $_* \rightarrow \mathbf{Diff}$  ([Auxiliary result B.3\(b\)](#)).

If the set-level fibre product  $S = \{(q, q') : \rho_M(q) = \rho_N(q')\}$  is an embedded submanifold of  $Q_M \times Q_N$ , then  $S$  with its two projections is the pullback in **Diff** ([Auxiliary result B.2\(c\)](#)). Indeed, two smooth maps  $T \rightarrow Q_M$  and  $T \rightarrow Q_N$  that agree over  $\Theta$  give a smooth map  $T \rightarrow Q_M \times Q_N$  with image in  $S$ , which is smooth into  $S$ , and the factorisation is unique because  $S \rightarrow Q_M \times Q_N$  is injective. A clean intersection has, by definition, an embedded submanifold  $S$  ([Background C.7](#)), and for affine  $\rho_M$  and  $\rho_N$  on vector spaces  $S$  is an affine subspace.

Transversality at  $(q_0, q'_0)$  asks  $d\rho_M(T_{q_0}Q_M) + d\rho_N(T_{q'_0}Q_N) = T_{\theta_0}\Theta$ . The left side has dimension at most  $|M| + |N|$ , so transversality fails at the base point whenever  $|M| + |N| < \dim \Theta$  ([Auxiliary result B.2\(b\)](#)).

In **Set** the fibre product is  $S$ , and its image in  $\Theta$  is  $\text{Im } M \cap \text{Im } N$ : a pair  $(q, q') \in S$  lies over  $\rho_M(q) = \rho_N(q')$ , which belongs to both images, and a point  $\theta$  of both images has preimages  $q$  and  $q'$  with  $(q, q') \in S$ . Now suppose that a product  $P$  exists in **Meth**, and test it with the pointed 0-manifold  $T = \{*, t\}$ , pointed at  $*$ . A pointed map  $T \rightarrow P$  is the choice of the

image of  $t$ , that is, a point of  $P$ . A pair of pointed maps  $T \rightarrow Q_M$  and  $T \rightarrow Q_N$  that agree over  $\Theta$  is a point  $(q, q')$  of  $S$ . The universal property makes these correspond bijectively through  $p \mapsto (\pi_M(p), \pi_N(p))$ . So the canonical map  $P \rightarrow Q_M \times Q_N$  is a bijection onto  $S$ , and  $\text{Im}(M \times N) = \rho_M(\pi_M(P)) = \text{Im } M \cap \text{Im } N$ .

Take  $\Theta = \mathbb{R}$  and  $\text{LoRA}_1 : (B, A) \mapsto \theta_0 + sBA$  on  $\mathbb{R}^2$ , with  $s \neq 0$  and frozen residual  $\theta_0$ , so that its pointings are the  $(B_0, A_0)$  with  $B_0 A_0 = 0$ . Its set-level fibre product with the frozen model is the cross  $X = \{(B, A) : BA = 0\}$ . Suppose a product  $P$  exists, and let  $\iota : P \rightarrow \mathbb{R}^2$  be its projection to the parameters of  $\text{LoRA}_1$ . The forgetful functor  $\mathbf{Diff}_* \rightarrow \mathbf{Diff}$  has a left adjoint, so it preserves this pullback (Auxiliary result B.3(c)), and  $P$  is also the pullback in  $\mathbf{Diff}$  of  $\rho_{\text{LoRA}_1}$  and the inclusion of  $\theta_0$ . Hence every smooth curve  $c : \mathbb{R} \rightarrow \mathbb{R}^2$  with  $\rho \circ c \equiv \theta_0$  has a unique smooth lift  $\tilde{c} : \mathbb{R} \rightarrow P$  with  $\iota \circ \tilde{c} = c$ , and by the previous paragraph  $\iota$  is a bijection onto  $X$ . Let  $p^*$  be the point of  $P$  over  $(0, 0)$ . The axis curves  $c_1(t) = (t, 0)$  and  $c_2(t) = (0, t)$  lie in  $X$ , so they lift to smooth curves through  $p^*$ , and  $d_{\iota_{p^*}}$  maps their velocities to  $c'_1(0) = (1, 0)$  and  $c'_2(0) = (0, 1)$ . Hence  $\text{rank } d_{\iota_{p^*}} = 2$ . The rank cannot exceed 2 and cannot drop near  $p^*$  (Auxiliary result B.1(a)), so  $\iota$  is a submersion on a neighbourhood  $V$  of  $p^*$ , and  $\iota(V)$  contains a nonempty open subset of  $\mathbb{R}^2$  (Auxiliary result B.1(c)). But  $\iota(V) \subseteq X$ , which contains no nonempty open set. So no product exists. The argument used only the point over the origin, so it covers every pointing, including the standard one  $(0, a_0)$  with  $a_0 \neq 0$ , whose base point is a smooth point of  $X$ . (A split pointing, with  $B_0 A_0 \neq 0$  and residual  $\theta_0 - sB_0 A_0$ , has a hyperbola over  $\theta_0$ , an embedded submanifold, and there the product exists.)

(c) If  $h : M \rightarrow N$  is an arrow, then  $\rho_M = \rho_N \circ h$  gives  $\text{Im } M = \rho_N(h(Q_M)) \subseteq \text{Im } N$ , and both images contain  $\theta_0$ ; functoriality is automatic because the target is a poset. Conversely, let  $\text{Im } M \subseteq \text{Im } N$ . Choose, by the axiom of choice, a preimage  $s(\theta) \in \rho_N^{-1}(\theta)$  for each  $\theta \in \text{Im } M$ , with  $s(\theta_0) = q'_0$ , and put  $h = s \circ \rho_M$ . Then  $\rho_N \circ h = \rho_M$  and  $h(q_0) = q'_0$ , so  $h$  is an arrow of  $\mathbf{Set}_*/(\Theta, \theta_0)$ . For  $\rho_M(t) = t$  and  $\rho_N(u) = u^3$  both images are  $\mathbb{R}$ . A map with  $\rho_N \circ h = \text{id}$  satisfies  $h(t)^3 = t$ , so  $h(t) = t^{1/3}$ , the unique real cube root, whose difference quotient  $t^{-2/3}$  at 0 is unbounded. So no  $C^1$  arrow, and in particular no smooth one, exists.  $\square$

### C.3 Definition I.5

**DEFINITION I.5** (invariants of an object). For  $M = (Q, q_0, \rho)$ :

- the image  $\text{Im } M = \rho(Q)$ ;
- the **expressive dimension**  $d(M) = \dim \text{Im } M = \max_q \text{rank } d\rho_q$ ;
- the **gauge waste**  $g(M) = |M| - d(M)$ ;
- the **first-order image**  $S_1(M) = d\rho_{q_0}(T_{q_0}Q)$ , and the **first-order deficit**  $d(M) - \dim S_1(M)$ ;
- the **gauge group**  $\{\varphi \in \text{Diff}(Q) : \rho \circ \varphi = \rho\}$ , which is unpointed, since a gauge transformation moves  $q_0$  inside its fibre. In practice we exhibit a Lie group acting on  $Q$  that preserves  $\rho$  and is locally transitive on generic fibres, and call it the **generic gauge**.

The base point  $\theta_0$  is **regular** for  $M$  if the deficit is 0, and an **apex** if the deficit is positive. We say **vertex** for the subset-level notion, the cone point of an image germ, and keep **apex** for the base-point notion. The two differ in general. At a crossing of two smooth branches the image is singular, yet the deficit can be 0. In every case below with a positive deficit, the image germ is (diffeomorphic to) a cone with vertex  $\theta_0$ ; the exception is DoRA, which inherits LoRA's deficit through its torus factor. A cone germ that is a  $C^1$  manifold at its vertex equals its tangent space there, so it is linear. Hence the vertex of a non-linear cone is a singular point.

The definition contains three claims: the two descriptions of  $d(M)$  agree, the deficit is never negative (so that regular and apex exhaust the cases), and a cone germ that is  $C^1$  at its vertex is linear.

*Proof.*  $d(M)$  is well defined. Let  $Q$  be an open definable subset of  $\mathbb{R}^N$  and  $\rho$  definable and smooth, and let  $k = \max_q \text{rank } d\rho_q$ . The set where the rank equals  $k$  is open and nonempty, because a nonzero  $k \times k$  minor of the Jacobian stays nonzero nearby (Auxiliary result B.1(a)). At a point  $p$  of it the rank is constant nearby, so by the rank theorem (Auxiliary result B.1(b)) a small open ball  $V \ni p$ , a definable set, has image  $\rho(V)$  an embedded  $k$ -dimensional submanifold. It is definable, of dimension  $k$ , and contained in  $\text{Im } M$ , so  $\dim \text{Im } M \geq k$  (Auxiliary result B.9(b)). Conversely, let  $e = \dim \text{Im } M$ . By Auxiliary result B.9(b),  $\text{Im } M$  contains a definable  $C^1$  cell  $Y$  of dimension  $e$ . By definable choice  $\rho$  has a definable section  $\sigma : Y \rightarrow Q$ , and by generic smoothness  $\sigma$  is  $C^1$  on a nonempty open subset  $Y'$  of  $Y$  (Auxiliary result B.9(c)). Differentiating  $\rho \circ \sigma = \text{id}_{Y'}$  at  $y \in Y'$  gives  $d\rho_{\sigma(y)} \circ d\sigma_y = \text{id}_{T_y Y}$ , so  $\text{rank } d\rho_{\sigma(y)} \geq e$ , and  $k \geq e$ . Hence  $\dim \text{Im } M = \max_q \text{rank } d\rho_q$ , which is Auxiliary result B.9(d). The parameter spaces of the paper are open definable sets, for instance  $\{(u_1, \dots, u_r) : u_i \neq 0\}$  for HRA, and the maps are built from polynomials, quotients, square roots and, for softmax, the exponential (Background C.17).

The deficit is nonnegative.  $\dim S_1(M) = \text{rank } d\rho_{q_0} \leq \max_q \text{rank } d\rho_q = d(M)$ . Likewise  $d(M) \leq \dim Q = |M|$ , so  $g(M) \geq 0$ .

Cones. Let  $X \subseteq \mathbb{R}^D$  agree near  $v$  with a cone  $K$  with vertex  $v$ , meaning  $v \in K$  and  $v + t(x - v) \in K$  for all  $x \in K$  and  $t \geq 0$ , and suppose  $X$  is a  $C^1$  embedded submanifold near  $v$ , with tangent space  $T$  at  $v$ . Translate so that  $v = 0$ . Near 0,  $X$  is the graph of a  $C^1$  map  $\varphi$  from a neighbourhood of 0 in  $T$  to  $T^\perp$  with  $\varphi(0) = 0$  and  $d\varphi_0 = 0$ . For  $x \in K$  and small  $t > 0$  the point  $tx$  lies in  $K$  near 0, hence in  $X$ , so  $tP_{T^\perp}x = \varphi(tP_Tx) = o(t)$ , and therefore  $P_{T^\perp}x = 0$ . Thus  $K \subseteq T$ . Near 0,  $X$  is then a graph over  $T$  that lies in  $T$ , so  $\varphi = 0$  and  $X$  contains a neighbourhood  $W$  of 0 in  $T$ . For  $y \in T$  and small  $\varepsilon > 0$  we have  $\varepsilon y \in W \subseteq K$ , so  $y = \varepsilon^{-1}(\varepsilon y) \in K$ . Hence  $K = T$ , and the germ of  $X$  at  $v$  is the germ of the affine space  $v + T$ . If  $K$  is not an affine subspace,  $X$  cannot be a  $C^1$  manifold at  $v$ , so  $v$  is a singular point (compare Auxiliary result B.6). Finally, a figure-eight curve through its crossing with nonzero velocity has deficit 0 there and a singular image, so vertex and apex are different notions.  $\square$

#### C.4 Observation I.6

**OBSERVATION I.6** ( $S_1$  is functorial). *If  $M \rightarrow N$  is a (pointed) morphism, then  $S_1(M) \subseteq S_1(N)$ . In particular, isomorphic methods have equal  $S_1$  and equal first-order deficit. Monotonicity holds only along pointed morphisms. Along a weak morphism, such as a gauge map moving  $q_0$  inside its fibre,  $S_1$  can change.*

*Proof.* Let  $h : M \rightarrow N$  be a pointed arrow, so  $\rho_N \circ h = \rho_M$  and  $h(q_0) = q'_0$ . The chain rule at  $q_0$  gives  $d(\rho_N)_{q'_0} \circ dh_{q_0} = d(\rho_M)_{q_0}$ , so  $S_1(M) = d(\rho_N)_{q'_0}(dh_{q_0}(T_{q_0}Q_M)) \subseteq d(\rho_N)_{q'_0}(T_{q'_0}Q_N) = S_1(N)$ . If  $h$  is an isomorphism,  $h^{-1}$  is a pointed arrow  $N \rightarrow M$ , so  $S_1(N) \subseteq S_1(M)$  as well. Moreover  $\rho_M = \rho_N \circ h$  with  $h$  bijective gives  $\text{Im } M = \text{Im } N$ , so  $d(M) = d(N)$  (equivalently,  $\text{rank } d(\rho_M)_q = \text{rank } d(\rho_N)_{h(q)}$  since  $dh_q$  is invertible), and the deficits agree.

Without  $h(q_0) = q'_0$  the chain rule compares  $S_1(M)$  with the first-order image of  $N$  at  $h(q_0)$ , which need not be the base point of  $N$ . For instance, take LoRA with frozen residual  $\theta_0$  and  $1 \leq r < \min(m, n)$ , and let  $M$  and  $N$  be its pointings at  $(0, A_0)$  and at  $(B_0, 0)$ , with  $A_0$  and  $B_0$  of full rank; both points lie in the fibre over  $\theta_0$ . The identity of  $Q$  is a weak morphism

$M \rightarrow N$ , yet  $S_1(M) = \{XA_0\} = \mathcal{R}(\text{row } A_0)$  and  $S_1(N) = \{B_0Y\} = \mathcal{C}(\text{col } B_0)$  are not nested:  $\mathcal{R}(\text{row } A_0)$  contains  $e_i v^\top$  for every  $i$  and every  $v \in \text{row } A_0$ , which is not in  $\mathcal{C}(\text{col } B_0)$  when  $e_i \notin \text{col } B_0$ , and symmetrically.  $\square$

### C.5 Observation I.7

**OBSERVATION I.7** (additive monoidal structure).  $\Theta$  is a vector space, so every object of  $\mathbf{Meth}(\Theta, \theta_0)$  can be written  $\rho = \theta_0 + \delta$  with  $\delta := \rho - \theta_0$  and  $\delta(q_0) = 0$ . Put

$$M \oplus N := (Q_M \times Q_N, (q_0, q'_0), \theta_0 + \delta_M + \delta_N).$$

This is a symmetric monoidal structure with the frozen model as unit. Its image is the Minkowski sum  $\theta_0 + (\text{Im } \delta_M + \text{Im } \delta_N)$ . It differs from the categorical product, whose image, when the product exists, is the intersection  $\text{Im } M \cap \text{Im } N$ . “Additive” in [Definition I.2](#) ( $\delta$  independent of  $\theta_0$ ) is a condition on families  $\theta_0 \mapsto M(\theta_0)$ ; it matters for base change and plays no role here.

Write  $\text{LoRA}_k$  for LoRA with frozen residual  $\theta_0$ ; for split pointings the sum below has residual  $\theta_0 - P - P'$ . Concatenation  $h(B_1, A_1, B_2, A_2) = ([B_1 \ B_2], [A_1; A_2])$  is an isomorphism  $\text{LoRA}_r \oplus \text{LoRA}_s \cong \text{LoRA}_{r+s}$  of unpointed objects. It is a pointed isomorphism onto  $\text{LoRA}_{r+s}$  pointed at  $([B_0 \ B'_0], [A_0; A'_0])$ . For  $T_A$  pointings ( $B_0 = 0$ ,  $\text{rank } A_0 = r$ ), the target pointing has  $\text{rank}[A_0; A'_0] = r + s$  if and only if  $\text{row } A_0 \cap \text{row } A'_0 = 0$ ; it lies in the stratum  $T_A$  of [Theorem II.2](#) when moreover  $r + s < \min(m, n)$ .

Standard pointings ( $T_A$ , or symmetrically  $T_B$ ) exist only for  $r < \min(m, n)$ , so the monoidal question is pointwise. If  $F(r)$  and  $F(2r)$  are standard ( $2r < \min(m, n)$ ), then  $F(r) \oplus F(r) \not\cong F(2r)$  as pointed objects; for  $2r \geq \min(m, n)$  no standard  $F(2r)$  exists. Hence no assignment  $r \mapsto \text{LoRA}_r$  with  $F(1)$  and  $F(2)$  standard is **strong monoidal** into pointed objects. For  $r \neq s$  and standard pointings,  $F(r) \oplus F(s) \cong F(r + s)$  as pointed objects exactly when the row space of the target’s  $A_0$  is the direct sum of the two source row spaces. The assignment is strong monoidal on unpointed objects, up to weak morphisms, and on pointed objects with degenerate pointings, namely the origin  $(B_0, A_0) = (0, 0)$ , where  $S_1 = 0$ , or the stacked pointings  $F(r) = F(1)^{\oplus r}$ . Concatenation is strictly associative and unital, so discrete  $(\mathbb{N}, +)$  suffices for a plain strong monoidal functor. Symmetric strong monoidality needs the permutation groupoid, since swapping and then concatenating permutes the blocks.

With per-adapter scales  $s_r$  ( $\alpha/r$  for LoRA,  $\alpha/\sqrt{r}$  for rsLoRA) and any target scale  $s_*$ , the map  $h = ([B_1 \ B_2], [c_1 A_1; c_2 A_2])$  with  $c_i = s_{r_i}/s_*$ , where  $r_1 = r$  and  $r_2 = s$ , is an isomorphism over  $\Theta$ . It is a linear bijection and preserves row spaces, so the pointing criterion is unchanged.

*Proof.* The monoidal structure.  $M \oplus N$  is an object:  $Q_M \times Q_N$  is a manifold,  $\theta_0 + \delta_M + \delta_N$  is smooth and takes the value  $\theta_0$  at  $(q_0, q'_0)$ . For arrows  $h : M \rightarrow M'$  and  $k : N \rightarrow N'$ , the pointed map  $h \times k$  lies over  $\Theta$  because  $\delta_{M'} \circ h = \rho_{M'} \circ h - \theta_0 = \delta_M$  and likewise for  $k$ ; this makes  $\oplus$  a functor. The associator  $(Q_1 \times Q_2) \times Q_3 \cong Q_1 \times (Q_2 \times Q_3)$ , the unitors  $\text{pt} \times Q \cong Q \cong Q \times \text{pt}$  and the symmetry  $Q_M \times Q_N \cong Q_N \times Q_M$  are the structure maps of the cartesian monoidal structure of  $\mathbf{Diff}_*$ . They lie over  $\Theta$  because addition is associative and commutative and the frozen model has  $\delta = 0$ . Their coherence (pentagon, triangle, hexagon and the symmetry being an involution) holds in  $\mathbf{Diff}_*$ , and the forgetful functor  $\mathbf{Meth}(\Theta, \theta_0) \rightarrow \mathbf{Diff}_*$  is faithful, so it holds in  $\mathbf{Meth}(\Theta, \theta_0)$ .

*Image.*  $\{\theta_0 + \delta_M(q) + \delta_N(q')\} = \theta_0 + (\text{Im } \delta_M + \text{Im } \delta_N)$ . The categorical product has image  $\text{Im } M \cap \text{Im } N$  when it exists ([Observation I.4\(b\)](#)), and the two differ in general: for zero-init



LoRA<sub>1</sub> on a  $2 \times 2$  box,  $M \oplus M$  has image  $\theta_0 + \mathcal{M}_{\leq 2}$ , while any product of  $M$  with itself has image  $\theta_0 + \mathcal{M}_{\leq 1}$ .

*Concatenation.* The map  $h$  only regroups coordinates, so it is a linear bijection between the parameter spaces, and  $\theta_0 + s[B_1 \ B_2][A_1; A_2] = \theta_0 + sB_1A_1 + sB_2A_2$ , so it lies over  $\Theta$ . It is a pointed isomorphism onto LoRA <sub>$r+s$</sub>  pointed at the image of  $(q_0, q'_0)$ , the concatenated point. For split pointings with  $P = sB_0A_0$  and  $P' = sB'_0A'_0$ , the sum is  $(\theta_0 - P - P') + s[B_1 \ B_2][A_1; A_2]$ , which is LoRA <sub>$r+s$</sub>  with residual  $\theta_0 - P - P'$ . For  $T_A$  pointings, the concatenated pointing has  $B = 0$  and  $\text{rank}[A_0; A'_0] = \dim(\text{row } A_0 + \text{row } A'_0) = r + s - \dim(\text{row } A_0 \cap \text{row } A'_0)$ , which is  $r + s$  exactly when the intersection is 0. The stratum  $T_A$  of rank  $r + s$  is defined for  $r + s < \min(m, n)$ , which gives the last condition.

*Row and column types.* We use two elementary facts about the spaces  $\mathcal{R}(V)$  and  $\mathcal{C}(U)$  of the introduction, which return in [Theorem II.2](#). First,  $\mathcal{R}(V) = \{XA : X \in \mathbb{R}^{m \times k}\}$  for any  $A$  of full row rank with  $\text{row } A = V$ , since  $Z = ZA^+A$  when  $\text{row } Z \subseteq V$ ; so  $\dim \mathcal{R}(V) = m \dim V$ , and  $\mathcal{R}(V)$  determines  $V$  as the sum of the row spaces of its elements. Likewise  $\mathcal{C}(U) = \{BY\}$  for  $\text{col } B = U$ , of dimension  $n \dim U$ , determines  $U$ . Second, if  $V \neq 0$  and  $\mathcal{R}(V) = \mathcal{C}(U)$ , then  $U = \mathbb{R}^m$ , because  $\mathcal{R}(V)$  contains  $e_i v^\top$  for every  $i$  and every nonzero  $v \in V$ , and  $e_i v^\top \in \mathcal{C}(U)$  forces  $e_i \in U$ .

*The negative claim.* Standard pointings are the strata  $T_A$  and  $T_B$  of [Theorem II.2](#), which are defined only for ranks below  $\min(m, n)$ . Let  $F(r)$  and  $F(2r)$  be standard, so  $2r < \min(m, n)$ . If  $F(r)$  is pointed at  $(0, A_0)$ , then  $F(r) \oplus F(r)$  is pointed at  $((0, A_0), (0, A_0))$  and its first-order image is  $\{X_1 A_0 + X_2 A_0\} = \mathcal{R}(\text{row } A_0)$ , of dimension  $mr$ . If  $F(r)$  is in  $T_B$ , it is  $\mathcal{C}(\text{col } B_0)$ , of dimension  $nr$ . A standard  $F(2r)$  has first-order image  $\mathcal{R}(V'')$  with  $\dim V'' = 2r$  or  $\mathcal{C}(U'')$  with  $\dim U'' = 2r$ . A pointed isomorphism would make the two first-order images equal ([Observation I.6](#)). Two row-type spaces determine their subspaces, of dimensions  $r \neq 2r$ ; two column-type spaces likewise; and a row-type space with  $V \neq 0$  equals a column-type space only if the column subspace is all of  $\mathbb{R}^m$ , impossible for  $U''$  of dimension  $2r < m$  (and symmetrically). So  $F(r) \oplus F(r) \not\cong F(2r)$ . A strong monoidal functor  $F$  would supply a pointed isomorphism  $F(1) \oplus F(1) \cong F(2)$ , so none exists when  $F(1)$  and  $F(2)$  are standard.

*The case  $r \neq s$ .* Let  $F(r)$ ,  $F(s)$  and  $F(r + s)$  be standard of type  $T_A$ , pointed at  $(0, A_0)$ ,  $(0, A'_0)$  and  $(0, A''_0)$ ; the type  $T_B$  is the transpose. Then  $S_1(F(r) \oplus F(s)) = \{XA_0 + X'A'_0\} = \mathcal{R}(\text{row } A_0 + \text{row } A'_0)$  and  $S_1(F(r + s)) = \mathcal{R}(\text{row } A''_0)$ , with  $\dim \text{row } A''_0 = r + s$ . A pointed isomorphism forces  $\text{row } A''_0 = \text{row } A_0 + \text{row } A'_0$ , and the dimension count makes the sum direct. Conversely, if  $\text{row } A''_0 = \text{row } A_0 \oplus \text{row } A'_0$ , then  $[A_0; A'_0]$  has full row rank and the row space of  $A''_0$ , so  $A''_0 = g[A_0; A'_0]$  for a unique  $g \in \text{GL}_{r+s}$ . Concatenation followed by the gauge  $(B, A) \mapsto (Bg^{-1}, gA)$ , which preserves  $BA$ , is a pointed isomorphism  $F(r) \oplus F(s) \rightarrow F(r + s)$ .

*Strong monoidality.* The concatenation maps  $\phi_{r,s} = h$  satisfy  $\phi_{r+s,t} \circ (\phi_{r,s} \times \text{id}) = \phi_{r,s+t} \circ (\text{id} \times \phi_{s,t}) \circ a$ , both sides sending  $(B_1, A_1, B_2, A_2, B_3, A_3)$  to  $([B_1 \ B_2 \ B_3], [A_1; A_2; A_3])$ , and  $\phi_{0,r}$ ,  $\phi_{r,0}$  are the unitors, with LoRA<sub>0</sub> the frozen model. Discrete  $(\mathbb{N}, +)$  has only identity arrows, so naturality is vacuous, and  $r \mapsto \text{LoRA}_r$  is strong monoidal on unpointed objects. As maps between chosen pointings, the  $\phi_{r,s}$  are weak morphisms. They are pointed isomorphisms when every  $F(r)$  is pointed at the origin, since  $h$  maps origin to origin, and when  $F(r)$  is pointed at the concatenation of  $r$  copies of the base point of  $F(1)$ , since  $h$  maps concatenations to concatenations. The symmetry of discrete  $(\mathbb{N}, +)$  is the identity, so symmetry would require  $\phi_{r,s} = \phi_{s,r} \circ \sigma$ , where  $\sigma$  swaps the factors. The left side sends  $(B_1, A_1, B_2, A_2)$  to  $([B_1 \ B_2], [A_1; A_2])$  and the right side to  $([B_2 \ B_1], [A_2; A_1])$ , which differ for  $r, s \geq 1$ . On the permutation groupoid, a permutation  $\pi \in S_r$  acts on LoRA <sub>$r$</sub>  by the gauge  $(B, A) \mapsto (BP_\pi^{-1}, P_\pi A)$ ,

which is functorial and lies over  $\Theta$ , the maps  $\phi_{r,s}$  are natural for block sums of permutations, and with the block transposition  $\tau_{r,s} \in S_{r+s}$  as symmetry,  $F(\tau_{r,s}) \circ \phi_{r,s} = \phi_{s,r} \circ \sigma$ , because permuting the channel blocks of  $([B_1 \ B_2], [A_1; A_2])$  gives  $([B_2 \ B_1], [A_2; A_1])$ . The origin and the stacked pointings are fixed by channel permutations, so the same holds for them.

*Rescaled concatenation.*  $s_*[B_1 \ B_2][c_1 A_1; c_2 A_2] = s_* c_1 B_1 A_1 + s_* c_2 B_2 A_2 = s_r B_1 A_1 + s_s B_2 A_2$ . The scales are nonzero, so  $h$  is a linear bijection, and  $\text{row}(c_i A_i) = \text{row } A_i$ , so the criterion for the concatenated pointing is the same as before.  $\square$

## C.6 Proposition I.8

**PROPOSITION I.8** (a product that produces a named method; an image-level meet). (a) Let  $A_0 \in \mathbb{R}^{r \times n}$  have full row rank and  $B_0 \in \mathbb{R}^{m \times r}$  full column rank. Define the one-sided methods

$$\text{FA}(A_0) : B \mapsto W_0 + B A_0, \quad \text{FB}(B_0) : A \mapsto W_0 + B_0 A,$$

both pointed at 0. Then

$$\text{FA}(A_0) \times \text{FB}(B_0) \cong (\mathbb{R}^{r \times r}, 0, R \mapsto W_0 + B_0 R A_0).$$

The product supplies this sandwich for any frames. Assume  $\text{rank } W_0 \geq r$  ( $\sigma_r > 0$ ) and take the SVD frames  $B_0 = U_r$ ,  $A_0 = \Sigma_r V_r^\top$ . The result is **LoRA-XS pointed at  $R = 0$** .

(b) Let  $W_0$  have full row rank (so  $m \leq n$ ), and let  $\text{LoRA}_r$  carry its standard pointing ( $B_0 = 0$ , image  $W_0 + M_{\leq r}$ ). The image-level meet of  $\text{LoRA}_r$  with the output-scaling method  $\ell \mapsto \text{diag}(\ell)W_0$  is  $\{\text{diag}(\ell)W_0 : \#\{i : \ell_i \neq 1\} \leq r\}$ , that is, “rescale at most  $r$  output neurons”.

The LoRA-XS paper writes  $xW + xARB$  in row-vector convention with  $A = U_r \Sigma_r$  and  $B = V_r^\top$ . Transposed, the update is  $B_0 R^\top A_0$ , so the isomorphism is  $R \mapsto \alpha R^\top$ . LoRA-XS initialises  $R \sim \mathcal{N}(0, \sigma^2)$  with  $\sigma = 10^{-5}$ , a negligible pointing defect. The output-scaling method in (b) is the key/value part of (IA)<sup>3</sup>. On the FFN, (IA)<sup>3</sup> scales the *input* of  $W_{\text{down}}$ ,  $W \mapsto W \text{diag}(\ell)$ ; the mirror statement then needs full column rank, or holds generically for  $r < d$ , where  $d$  is the number of rows of  $W_{\text{down}}$ .

*Proof.* (a) Write  $B_0^+ = (B_0^\top B_0)^{-1} B_0^\top$  and  $A_0^+ = A_0^\top (A_0 A_0^\top)^{-1}$ , so that  $B_0^+ B_0 = I_r = A_0 A_0^+$ . A point of the set-level fibre product is a pair  $(B, A)$  with  $B A_0 = B_0 A =: D$ . Then  $\text{row } D \subseteq \text{row } A_0$  and  $\text{col } D \subseteq \text{col } B_0$ , so  $D = B_0 B_0^+ D A_0^+ A_0 = B_0 R A_0$  with  $R = B_0^+ D A_0^+$ , and  $R$  is unique because  $B_0$  is injective and  $A_0$  surjective. From  $B A_0 = B_0 R A_0$  and  $A_0 A_0^+ = I$  we get  $B = B_0 R$ , and from  $B_0 A = B_0 R A_0$  we get  $A = R A_0$ . So the fibre product is the  $r^2$ -dimensional linear subspace  $\{(B_0 R, R A_0)\}$  of  $\mathbb{R}^{m \times r} \times \mathbb{R}^{r \times n}$ , and  $j : R \mapsto (B_0 R, R A_0)$  is a linear isomorphism onto it, with inverse  $(B, A) \mapsto B_0^+ B$ . Both composites  $R \mapsto W_0 + B_0 R A_0$  agree, and  $j(0) = (0, 0)$  is the pair of base points.

The universal property: let  $T$  be a pointed manifold and  $h_1 : T \rightarrow \mathbb{R}^{m \times r}$ ,  $h_2 : T \rightarrow \mathbb{R}^{r \times n}$  pointed smooth maps with  $W_0 + h_1(t) A_0 = W_0 + B_0 h_2(t)$ . By the previous paragraph  $(h_1(t), h_2(t)) = j(R(t))$  with  $R(t) = B_0^+ h_1(t)$ , a pointed smooth map, and it is the only map with this property because  $j$  is injective. So  $(\mathbb{R}^{r \times r}, 0, R \mapsto W_0 + B_0 R A_0)$  with the projections  $R \mapsto B_0 R$  and  $R \mapsto R A_0$  is the product in **Meth**( $\Theta, W_0$ ) ([Auxiliary result B.2\(c\)](#), [Auxiliary result B.3](#)). This holds for any frames of full rank. With  $B_0 = U_r$  and  $A_0 = \Sigma_r V_r^\top$  from an SVD of  $W_0$  ([Auxiliary result B.11\(a\)](#)), the frames have full rank exactly when  $\sigma_r > 0$ ; if  $\text{rank } W_0 < r$ ,  $\Sigma_r V_r^\top$  is rank-deficient and the hypothesis fails.

The row-vector convention: LoRA-XS acts on row vectors by  $x \mapsto x(W_{\text{row}} + \alpha ARB)$ , where  $W_{\text{row}} = W_0^\top$  has SVD  $U'\Sigma V'^\top$ ,  $A = U'_r \Sigma_r$  and  $B = V'_r{}^\top$ . In our convention the update is the transpose,  $\alpha V'_r R^\top \Sigma_r U'_r{}^\top$ . Since  $W_0 = V'\Sigma U'^\top$ , the left and right singular frames of  $W_0$  are  $U_r = V'_r$  and  $V_r = U'_r$ , so the update is  $U_r(\alpha R^\top) \Sigma_r V_r^\top = B_0(\alpha R^\top) A_0$ . Hence LoRA-XS is the product composed with the linear isomorphism  $R \mapsto \alpha R^\top$ , which fixes 0. A random initial  $R$  makes  $\rho(R_0) - W_0 = \alpha U_r R_0^\top \Sigma_r V_r^\top$  nonzero, of the size of  $\sigma$ , a pointing defect.

(b) The image of  $\text{LoRA}_r$  with its standard pointing is  $W_0 + \mathcal{M}_{\leq r}$  (Theorem II.2(c)), and the output-scaling method has image  $\{\text{diag}(\ell)W_0\}$ . A point  $\text{diag}(\ell)W_0$  lies in  $W_0 + \mathcal{M}_{\leq r}$  iff  $\text{rank}(\text{diag}(\ell - \mathbf{1})W_0) \leq r$ . The rows of  $\text{diag}(\ell - \mathbf{1})W_0$  are  $(\ell_i - 1)w_i$ , and the nonzero ones are linearly independent because the rows of  $W_0$  are, so the rank is  $\#\{i : \ell_i \neq 1\}$ . The meet in the poset of subsets is the intersection, which is the stated set.

For the input side,  $W_0 \text{diag}(\ell) - W_0 = W_0 \text{diag}(\ell - \mathbf{1})$  consists of the columns  $(\ell_j - 1)c_j$  of  $W_0$ , so its rank is the rank of the columns of  $W_0$  indexed by  $J = \{j : \ell_j \neq 1\}$ . If  $W_0$  has full column rank this is  $|J|$ , and the meet is  $\{W_0 \text{diag}(\ell) : |J| \leq r\}$ . For  $W_{\text{down}} \in \mathbb{R}^{d \times d_{\text{ff}}}$  with  $d < d_{\text{ff}}$ , full column rank is impossible. For generic  $W_0$  every set of at most  $d$  columns is independent (the failure is the vanishing of all maximal minors of the chosen columns), so the rank is  $\min(|J|, d)$ , and when  $r < d$  the condition  $\min(|J|, d) \leq r$  is again  $|J| \leq r$ .

No product in (b). The commentary after the proposition says that, for  $n \geq 2$ ,  $\text{LoRA}_r$  and the output-scaling method have no product; the argument is that of Observation I.4(b). Their set-level fibre product is  $\mathcal{F} = \{(B, A, \ell) : W_0 + sBA = \text{diag}(\ell)W_0\}$ , whatever the pointing of  $\text{LoRA}_r$  (with residual  $W_0$ ). Suppose a product  $P$  exists. By the proof of Observation I.4(b) its canonical map  $\iota : P \rightarrow \mathbb{R}^{m \times r} \times \mathbb{R}^{r \times n} \times \mathbb{R}^m$  is a bijection onto  $\mathcal{F}$ ,  $P$  is also the pullback in **Diff**, and smooth curves in  $\mathcal{F}$  lift. Let  $p^*$  be the point over  $(0, 0, \mathbf{1})$ . For every  $B'$  and  $A'$  the curves  $t \mapsto (tB', 0, \mathbf{1})$  and  $t \mapsto (0, tA', \mathbf{1})$  lie in  $\mathcal{F}$  and lift through  $p^*$ , so with  $\text{pr}$  the projection to the LoRA parameters,  $d(\text{pr} \circ \iota)_{p^*}$  is onto  $\mathbb{R}^{m \times r} \times \mathbb{R}^{r \times n}$ . By Auxiliary result B.1(a),(c),  $\text{pr} \circ \iota$  is a submersion near  $p^*$  and its image contains a nonempty open set of pairs  $(B, A)$ . But every pair in  $\text{pr}(\mathcal{F})$  has  $BA = \text{diag}(\ell - \mathbf{1})W_0/s$ , so the first row of  $BA$  is parallel to the first row  $w_{0,1}$  of  $W_0$ : the  $2 \times 2$  minors of the  $2 \times n$  matrix with rows  $e_1^\top BA$  and  $w_{0,1}^\top$  vanish. For  $n \geq 2$  one of these minors is a polynomial in  $(B, A)$  that is not identically zero, so its zero set has empty interior (Auxiliary result B.10(a)), a contradiction. (For  $m = n = 1$  the fibre product is the graph of  $\ell = 1 + sBA/W_0$ , and the product exists.)  $\square$

## C.7 Theorem II.1

**THEOREM II.1** (pointings of an additive method). *Let  $M = (Q, q_0, \theta_0 + \delta)$  be additive with shape  $C = \delta(Q)$ . Write  $-v + C$  for a translate, and  $C - C := \{c - c' : c, c' \in C\}$  for the Minkowski difference.*

(a) *For each  $q \in Q$ , the re-pointed method  $M_q := (Q, q, \theta_0 - \delta(q) + \delta(\cdot))$ , with frozen residual  $\theta_0 - \delta(q)$ , is pointed at  $\theta_0$ , and  $\text{Im } M_q = \theta_0 + (-\delta(q) + C)$ .*

(b)  $\cup_q \text{Im } M_q = \theta_0 + (C - C)$ .

(c) *For  $\text{LoRA}_r$  on a single linear box with  $s \neq 0$ ,  $C - C = \mathcal{M}_{\leq \min(2r, m, n)}$ ; for several boxes,  $C - C$  is the product of the per-box varieties  $\mathcal{M}_{\leq \min(2r, m_i, n_i)}$ . Hence **no exact pointing of  $\text{LoRA}_r$  (frozen residual  $\theta_0 - \delta(q)$ ), whether zero-init or split (PiSSA, OLoRA, CorDA, LoRA-GA), makes an update  $\theta - \theta_0$  of rank  $> 2r$  reachable in any adapted matrix.** More robustly, from any starting point  $q_0$  of  $\text{LoRA}_r$ , pointed or not,  $\rho(q) - \rho(q_0) = \delta(q) - \delta(q_0)$  has rank  $\leq \min(2r, m, n)$ . Quantised-residual schemes (QLoRA, LoftQ, QPiSSA) are pointed only up to a defect that depends on the order of operations ( $e$  for QLoRA; Proposition V.2(b) for the others).*

Measured from  $\theta_0$  rather than from their starting weight, their updates contain this defect, whose rank is generically at least  $\min(m, n) - r$ . The bound  $2r$  is attained by split pointings when  $2r \leq \min(m, n)$ .

*Proof.* (a)  $M_q$  has parameter space  $Q$  and the smooth map  $\rho_q = \theta_0 - \delta(q) + \delta(\cdot)$ , with  $\rho_q(q) = \theta_0$ , so it is pointed at  $\theta_0$ , and  $\text{Im } M_q = \theta_0 - \delta(Q) + \delta(Q) = \theta_0 + (-\delta(Q) + C)$ .

(b) As  $q$  ranges over  $Q$ ,  $\delta(q)$  ranges over  $C$ , so  $\bigcup_q \text{Im } M_q = \theta_0 + \bigcup_{c' \in C} (C - c') = \theta_0 + (C - C)$ .

(c) On one box  $C = \{sBA\}$ . Since  $s \neq 0$  and every matrix of rank at most  $\min(r, m, n)$  factors as  $BA$  with inner dimension  $r$  (Auxiliary result B.11(a)),  $C = \mathcal{M}_{\leq \min(r, m, n)}$ , which we write  $\mathcal{M}_{\leq r}$ . For  $X, Y \in C$ ,  $\text{rank}(X - Y) \leq \text{rank } X + \text{rank } Y \leq 2r$  and  $\text{rank}(X - Y) \leq \min(m, n)$  (Auxiliary result B.12(a)). Conversely, let  $Z$  have rank  $k \leq \min(2r, m, n)$  and SVD  $Z = \sum_{i \leq k} \sigma_i u_i v_i^\top$  (Auxiliary result B.11(a)). Put  $Z_1 = \sum_{i \leq \min(k, r)} \sigma_i u_i v_i^\top$  and  $Z_2 = Z - Z_1$ , both of rank at most  $r$ . The cone  $C$  is closed under negation, since  $-BA = (-B)A$ , so  $Z = Z_1 - (-Z_2) \in C - C$ . On several boxes, each with its own factors,  $\delta$  is the product of the per-box maps, so  $C$  is the product of the per-box shapes and  $C - C$  the product of the per-box differences.

An exact pointing of  $\text{LoRA}_r$  is a re-pointed method  $M_q$  of (a), so its image lies in  $\theta_0 + (C - C)$ , and in every box the update  $\theta - \theta_0$  has rank at most  $\min(2r, m_b, n_b)$ . For an arbitrary starting point  $q_0$  and an arbitrary residual  $W_{\text{res}}$ ,  $\rho(q) - \rho(q_0) = \delta(q) - \delta(q_0) \in C - C$  with no condition at all.

For a pipeline with pointing defect  $e_{\text{pt}} = \rho(q_0) - \theta_0$ , the update measured from  $\theta_0$  is

$$\theta - \theta_0 = (\rho(q) - \rho(q_0)) + e_{\text{pt}},$$

a matrix of rank at most  $2r$  plus the defect. The defects are those of Proposition V.2:  $e = \kappa\theta_0 - \theta_0$  for QLoRA,  $e - s \text{SVD}_r(e)$  for one-step LoftQ, and  $\kappa(W_{\text{res}}) - W_{\text{res}}$  for QPiSSA. Each is a quantisation error, or a quantisation error minus a matrix of rank at most  $r$ , so it has rank at least  $\min(m, n) - r$  whenever the quantisation error involved has full rank (Auxiliary result B.12(a)). Full rank is the generic case recorded in Proposition V.2(a). For instance, a round-to-nearest quantiser with fixed scales is constant on each open cell of constant codes, so there  $e(W) = c - W$ , which is rank-deficient only on a translate of  $\mathcal{M}_{\leq \min(m, n) - 1}$ , a proper algebraic subset of the cell; the cells cover all weights outside a null set (Auxiliary result B.10(b)).

Attainment: let  $2r \leq \min(m, n)$ , and take a split pointing with  $P = sB_0A_0$  of rank  $r$ . Choose  $Y = sBA$  of rank  $r$  with  $\text{col } Y \cap \text{col } P = 0$  and  $\text{row } Y \cap \text{row } P = 0$ , possible since  $2r \leq \min(m, n)$ . Writing  $Y = U_Y V_Y^\top$  and  $P = U_P V_P^\top$  with factors of full column rank  $r$ , we have  $Y - P = [U_Y \ U_P] [V_Y \ -V_P]^\top$ , and both  $m \times 2r$  and  $n \times 2r$  factors have independent columns, so  $\text{rank}(Y - P) = 2r$ . The weight  $\theta_0 - P + Y$  lies in the image of the split pointing, with update  $Y - P$  of rank  $2r$ .  $\square$

## C.8 Theorem II.2

**THEOREM II.2** (three strata of LoRA initialisations). *Let  $\delta(B, A) = BA$  with  $1 \leq r < \min(m, n)$ . Consider pointed LoRA objects in three distinguished strata of full-rank pointings (pointings whose nonzero factors have full rank). Only  $S$  is open and dense in the space of pointings;  $T_A$  and  $T_B$  are nowhere dense.*

- $(T_A)$ :  $B_0 = 0$  and  $\text{rank } A_0 = r$ ;
- $(T_B)$ :  $A_0 = 0$  and  $\text{rank } B_0 = r$ ;
- $(S)$ :  $\text{rank } B_0 A_0 = r$ , with frozen residual  $\theta_0 - B_0 A_0$ .



(a) **Classification.** Within a stratum,  $M_{q_0} \cong M_{q'_0}$  if and only if, respectively,  $\text{row } A_0 = \text{row } A'_0$ ;  $\text{col } B_0 = \text{col } B'_0$ ; or  $B_0 A_0 = B'_0 A'_0$ . Objects in different strata are never isomorphic. As a set, the isomorphism classes of the three strata therefore form  $\text{Gr}(r, \mathbb{R}^n) \sqcup \text{Gr}(r, \mathbb{R}^m) \sqcup \mathcal{M}_r(m \times n)$ .

(b) **First order.**  $\dim S_1$  is  $mr$ ,  $nr$  and  $r(m + n - r)$  respectively, while  $d = r(m + n - r)$  in every case. The deficits are  $r(n - r)$ ,  $r(m - r)$  and 0.

(c) **Geometry.** In  $T_A$  and  $T_B$ ,  $\text{Im} = \theta_0 + \mathcal{M}_{\leq r}$ , whose vertex  $\theta_0$  is an **apex**. In  $S$ ,  $\text{Im} = (\theta_0 - P) + \mathcal{M}_{\leq r}$  with  $P = B_0 A_0$ , and  $\theta_0$  is a **smooth point**.

(d) **Incomparability.** The images of  $S$ -type and  $T$ -type objects are incomparable.

(e) **Scope.** The three strata do not exhaust the pointings of  $\text{LoRA}_r$ . Outside them lie:  $B_0 = 0$  with  $\text{rank } A_0 < r$ ;  $A_0 = 0$  with  $\text{rank } B_0 < r$ ;  $B_0 = A_0 = 0$ ;  $0 < \text{rank } B_0 A_0 < r$ ; and pointings with  $B_0, A_0$  both nonzero and  $B_0 A_0 = 0$ . HF PEFT ships one of the last kind: `init_lora_weights="orthogonal"` (which requires  $r$  even) lies in the stratum  $(\text{rk } B_0, \text{rk } A_0, \text{rk } B_0 A_0) = (r/2, r/2, 0)$ , an apex with  $\dim S_1 = \frac{r}{2}(m + n - \frac{r}{2})$ . It is isomorphic to no object of the three strata. Its image  $\theta_0 + \mathcal{M}_{\leq r}$  rules out  $S$ , and its  $S_1$  determines  $(\text{col } B_0, \text{row } A_0)$ , of dimensions  $(r/2, r/2)$ , against  $(0, r)$  in  $T_A$  and  $(r, 0)$  in  $T_B$ . The dimension of  $S_1$  alone does not separate it, since at  $(m, n, r) = (4, 5, 2)$  it equals  $mr$ . This placement holds in exact arithmetic; HF casts the two factors to the base dtype separately, and in bf16 the result is a generic pointing with a tiny defect. For every pointing, the image determines  $P = B_0 A_0$  (its vertex is  $\theta_0 - P$ ), and  $S_1 = \{X A_0 + B_0 Y\}$  determines  $(\text{row } A_0, \text{col } B_0)$  whenever these are proper subspaces. The classification is set-level: in the space of pointings  $T_A$  lies in the closure of  $S$ , since  $(\varepsilon B', A_0) \rightarrow (0, A_0)$ , so the quotient is not a topological disjoint union.

*Proof.* Absorb the scale  $s$  into  $B$ ; this changes neither the strata nor the images. The space of pointings is  $\mathbb{R}^{m \times r} \times \mathbb{R}^{r \times n}$ , and the residual is determined by the pointing.

*Openness.*  $\text{rank } B_0 A_0 \leq \min(\text{rank } B_0, \text{rank } A_0)$ , and if both factors have rank  $r$  then  $B_0$  is injective and  $A_0$  surjective, so  $\text{rank } B_0 A_0 = r$ . Hence  $S = \{\text{rank } B_0 = \text{rank } A_0 = r\}$  is the set where some  $r \times r$  minor of  $B_0 A_0$  is nonzero: it is open, and dense because its complement is a proper algebraic subset (Auxiliary result B.10(a)). The stratum  $T_A$  lies in the proper linear subspace  $\{B_0 = 0\}$ , which is closed with empty interior, so  $T_A$  is nowhere dense, and likewise  $T_B$ .

*A translation argument.* We use the following fact here and in Theorem II.11 and Proposition II.16. Let  $1 \leq r < \min(m, n)$  and  $D \neq 0$ , of rank  $k$ . If  $k \geq r + 1$  put  $X = 0$ . Otherwise choose  $X$  of rank  $r + 1 - k \leq r$  with  $\text{col } X \cap \text{col } D = 0$  and  $\text{row } X \cap \text{row } D = 0$ , possible since  $r + 1 \leq \min(m, n)$ . Factoring  $D = U_D V_D^\top$  and  $X = U_X V_X^\top$  with factors of full column rank,  $D + X = [U_D \ U_X][V_D \ V_X]^\top$  with both factors of full column rank, so  $\text{rank}(D + X) = r + 1$ . In either case  $X \in \mathcal{M}_{\leq r}$  and  $D + X \notin \mathcal{M}_{\leq r}$ . Consequently,

$$c + \mathcal{M}_{\leq r} = c' + \mathcal{M}_{\leq r} \implies c = c', \quad (18)$$

since otherwise, with  $D = c - c'$ , the point  $c + X = c' + (D + X)$  lies in the left side and not in the right side. The translation stabiliser of  $\mathcal{M}_{\leq r}$  is  $\{0\}$ .

(a), “if”. In  $T_A$ , let  $\text{row } A_0 = \text{row } A'_0$  with both of rank  $r$ . Then  $A'_0 = A'_0 A_0^+ A_0 = g A_0$  with  $g = A'_0 A_0^+ \in \mathbb{R}^{r \times r}$ , invertible because  $\text{rank } A'_0 = r$ . The map  $\varphi_g(B, A) = (B g^{-1}, g A)$  is a diffeomorphism of the parameter space with  $\rho \circ \varphi_g = \rho$  (both residuals are  $\theta_0$ ) and  $\varphi_g(0, A_0) = (0, A'_0)$ , a pointed isomorphism over  $\Theta$ . In  $T_B$ ,  $\text{col } B_0 = \text{col } B'_0$  gives  $B'_0 = B_0 k$  with  $k \in \text{GL}_r$ , and  $\varphi_{k^{-1}}(B_0, 0) = (B'_0, 0)$ . In  $S$ , let  $B_0 A_0 = B'_0 A'_0 = P$  of rank  $r$ . Then  $\text{col } B_0 = \text{col } P = \text{col } B'_0$ , so  $B'_0 = B_0 k$  for a unique  $k \in \text{GL}_r$ , and  $B_0 k A'_0 = B_0 A_0$  gives  $k A'_0 = A_0$  because  $B_0$  is injective. So  $(B'_0, A'_0) = \varphi_{k^{-1}}(B_0, A_0)$ , and the residuals  $\theta_0 - P$  agree.



(a), “only if”. In  $T_A$ ,  $d\rho_{(0,A_0)}(\dot{B}, \dot{A}) = \dot{B}A_0$ , so  $S_1 = \{XA_0\} = \mathcal{R}(\text{row } A_0)$ , which determines row  $A_0$  (proof of [Observation I.7](#)); pointed isomorphisms preserve  $S_1$  ([Observation I.6](#)). In  $T_B$ ,  $S_1 = C(\text{col } B_0)$  determines col  $B_0$ . In  $S$  the image is  $(\theta_0 - P) + \mathcal{M}_{\leq r}$  by (c), it is an isomorphism invariant, and by (18) it determines  $P$ .

*Different strata.*  $S_1(T_A) = \mathcal{R}(V)$  with  $\dim V = r \geq 1$  and  $S_1(T_B) = C(U)$  with  $\dim U = r < m$ . They differ by the second fact in the proof of [Observation I.7](#). (When  $m \neq n$  their dimensions  $mr$  and  $nr$  already differ.)  $S$  differs from both because  $\dim S_1(S) = r(m + n - r)$  by (b), which exceeds  $mr$  by  $r(n - r) > 0$  and  $nr$  by  $r(m - r) > 0$ . Isomorphism classes in  $T_A$  correspond to row  $A_0$ , and every  $r$ -dimensional subspace occurs; in  $T_B$  to col  $B_0$ ; in  $S$  to  $P$ , and every rank- $r$  matrix has a full-rank factorisation. This gives the stated disjoint union of sets.

(b) At  $(B_0, A_0)$ ,  $S_1 = \{XA_0 + B_0Y\}$ , of dimension  $m \text{ rank } A_0 + n \text{ rank } B_0 - \text{rank } A_0 \text{ rank } B_0$  by [Proposition II.3](#):  $mr$  in  $T_A$ ,  $nr$  in  $T_B$  and  $r(m + n - r)$  in  $S$ . In every case the image is a translate of  $\mathcal{M}_{\leq r}$  by (c), of dimension  $r(m + n - r)$  ([Auxiliary result B.5\(a\)](#)), so  $d = r(m + n - r)$ , and the deficits are  $r(n - r)$ ,  $r(m - r)$  and 0.

(c) In  $T_A$  and  $T_B$  the residual is  $\theta_0$  and the image is  $\theta_0 + \{BA\} = \theta_0 + \mathcal{M}_{\leq r}$ . This is a cone with vertex  $\theta_0$ , and it is not an affine subspace, because  $\sum_{i \leq r} E_{ii}$  and  $E_{r+1, r+1}$  lie in  $\mathcal{M}_{\leq r}$  while their sum has rank  $r + 1 \leq \min(m, n)$ . By the cone part of [Definition I.5](#) ([Auxiliary result B.6](#)),  $\theta_0$  is a singular point, and by (b) the deficit is positive, so  $\theta_0$  is an apex. In  $S$  the image is  $(\theta_0 - P) + \mathcal{M}_{\leq r}$ , and  $\theta_0 = (\theta_0 - P) + P$  with  $P \in \mathcal{M}_r$ . The stratum  $\mathcal{M}_r$  is open in  $\mathcal{M}_{\leq r}$  and an embedded submanifold of dimension  $r(m + n - r)$  ([Auxiliary result B.5\(b\)](#)), so the image is a  $C^1$  submanifold near  $\theta_0$ , which is a smooth point; its deficit is 0.

(d) The  $T$ -image is  $\theta_0 + \mathcal{M}_{\leq r}$  and the  $S$ -image  $(\theta_0 - P) + \mathcal{M}_{\leq r}$  with  $P \neq 0$ . The translation argument with  $D = P$  gives  $X \in \mathcal{M}_{\leq r}$  with  $\theta_0 + X$  outside the  $S$ -image, and with  $D = -P$  it gives  $X' \in \mathcal{M}_{\leq r}$  with  $\theta_0 - P + X'$  outside the  $T$ -image. (For a generic rank-one  $X$ ,  $\text{rank}(P + X) = \text{rank}(X - P) = r + 1$  already.)

(e) Every pointing has a rank triple  $(\text{rank } B_0, \text{rank } A_0, \text{rank } B_0A_0)$ . The strata are  $(0, r, 0)$ ,  $(r, 0, 0)$  and  $(r, r, r)$ . If  $B_0 = 0$  and  $A_0 \neq 0$  is not of rank  $r$ , or  $A_0 = 0$  and  $B_0 \neq 0$  is not of rank  $r$ , or  $B_0 = A_0 = 0$ , the pointing is in the first three listed families. Otherwise both factors are nonzero, and either  $\text{rank } B_0A_0 = r$ , which forces both ranks to be  $r$  and gives  $S$ , or  $0 < \text{rank } B_0A_0 < r$ , or  $B_0A_0 = 0$ . So the list is exhaustive.

For every pointing,  $S_1 = \{XA_0 + B_0Y\} = \{Z : Z(\text{row } A_0)^\perp \subseteq \text{col } B_0\}$ . Indeed,  $(XA_0 + B_0Y)w = B_0Yw$  for  $w \in \ker A_0 = (\text{row } A_0)^\perp$ ; conversely, if  $Z \ker A_0 \subseteq \text{col } B_0$ , then  $Z = ZA_0^+A_0 + B_0B_0^+Z(I - A_0^+A_0)$ . Its annihilator in the Frobenius inner product is  $\{W : WA_0^\top = 0, B_0^\top W = 0\}$ , the matrices with column space in  $(\text{col } B_0)^\perp$  and row space in  $(\text{row } A_0)^\perp$ , that is  $(\text{col } B_0)^\perp \otimes (\text{row } A_0)^\perp$ . When both subspaces are proper, both complements are nonzero, and the column spaces of the elements of the annihilator span  $(\text{col } B_0)^\perp$  while their row spaces span  $(\text{row } A_0)^\perp$ . So  $S_1$  determines  $(\text{col } B_0, \text{row } A_0)$  and their dimensions, and so does every pointed isomorphism class ([Observation I.6](#)). The image is  $(\theta_0 - P) + \mathcal{M}_{\leq r}$  with  $P = B_0A_0$  (residual  $\theta_0 - P$ ), and it determines  $P$  by (18).

HF's orthogonal initialisation draws one  $Q \in O(r)$  and sets  $A_0 = (G_A Q_{[0::2]})^\top / 10$  and  $B_0 = G_B Q_{[1::2]} / 10$ , with  $G_A \in \mathbb{R}^{n \times r/2}$  and  $G_B \in \mathbb{R}^{m \times r/2}$  random and  $Q_{[0::2]}, Q_{[1::2]}$  the even- and odd-indexed rows of  $Q$ . Distinct rows of  $Q$  are orthogonal, so  $B_0A_0 = G_B Q_{[1::2]} Q_{[0::2]}^\top G_A^\top / 100 = 0$ , and both factors have rank  $r/2$  for generic  $G_A, G_B$ . The residual is  $\theta_0$  and the image is  $\theta_0 + \mathcal{M}_{\leq r}$ , whose vertex is singular. By [Proposition II.3](#),  $\dim S_1 = m \frac{r}{2} + n \frac{r}{2} - \frac{r^2}{4} = \frac{r}{2}(m + n - \frac{r}{2})$ , and the deficit  $r(m + n - r) - \frac{r}{2}(m + n - \frac{r}{2}) = \frac{r}{2}(m + n - \frac{3r}{2})$  is positive, so the base point is an apex. Its image differs from every  $S$ -image by (18), and its pair of subspace dimensions

$(r/2, r/2)$ , determined by  $S_1$  since both subspaces are proper, differs from  $(0, r)$  and  $(r, 0)$ ; so it is isomorphic to no object of the three strata. At  $(m, n, r) = (4, 5, 2)$ ,  $\frac{r}{2}(m + n - \frac{r}{2}) = 8 = mr$ .

Finally, for  $B'$  of rank  $r$  and  $\varepsilon \neq 0$ ,  $(\varepsilon B', A_0) \in S$ , and  $(\varepsilon B', A_0) \rightarrow (0, A_0) \in T_A$  as  $\varepsilon \rightarrow 0$ , so  $T_A$  lies in the closure of  $S$ .  $\square$

### C.9 Proposition II.3

**PROPOSITION II.3** (first-order rank formula: smoothness, alignment and scale). *For  $\kappa(B, A) = W_{\text{base}} + sBA$  with  $s \neq 0$ ,*

$$\text{rank } d\kappa_{(B,A)} = m \text{rk } A + n \text{rk } B - \text{rk } A \text{rk } B \quad \text{for all } (B, A),$$

*and  $\{XA\} \cap \{BY\} = \{BRA : R \in \mathbb{R}^{r \times r}\}$ . For full-rank frames this overlap is the tangent space of the product  $\text{FA}(A) \times \text{FB}(B)$  of [Proposition I.8\(a\)](#), re-based at  $\kappa(B, A)$ , whose image is the meet of the two one-sided images: a generalised LoRA-XS. It is LoRA-XS proper when  $(B, A)$  span the top- $r$  singular subspaces of  $W_0$  ( $\sigma_r > 0$ ), as at a PiSSA init; bottom frames (MiLoRA) give a generalised LoRA-XS.*

*Under gradient flow  $\dot{B} = -\eta_B \nabla_B L$ ,  $\dot{A} = -\eta_A \nabla_A L$ , with  $G = \nabla_W L$ ,*

$$\frac{d}{dt} L = -s^2 (\eta_B \|GA^\top\|_F^2 + \eta_A \|B^\top G\|_F^2)$$

*at every time, in particular at  $(B_0, A_0)$ . To first order in the step sizes the same holds for full-batch gradient descent, and in expectation for unbiased minibatch SGD. Under Adam ( $\varepsilon \rightarrow 0$ , with bias correction) the first step changes the loss by*

$$-|s|(\eta_B \|GA_0^\top\|_1 + \eta_A \|B_0^\top G\|_1) + O(\eta^2),$$

*with entrywise  $\ell_1$  norms. This first-order term is linear in  $|s|$ , against quadratic under gradient flow.*

*Proof. Rank.*  $d\kappa_{(B,A)}(\dot{B}, \dot{A}) = s(\dot{B}A + B\dot{A})$ , and  $s \neq 0$ , so the image of the differential is  $\{XA\} + \{BY\} = \mathcal{R}(\text{row } A) + \mathcal{C}(\text{col } B)$ . With  $a = \text{rank } A$  and  $b = \text{rank } B$ , the two summands have dimensions  $ma$  and  $nb$  (proof of [Observation I.7](#)). A matrix  $D$  lies in both iff  $\text{row } D \subseteq \text{row } A$  and  $\text{col } D \subseteq \text{col } B$ . The space of such  $D$  is  $\text{col } B \otimes \text{row } A$ , of dimension  $ab$ , and every such  $D$  satisfies  $D = BB^+DA^+A = B(B^+DA^+)A$ , since  $BB^+$  and  $A^+A$  are the projectors onto  $\text{col } B$  and  $\text{row } A$ . Conversely  $BRA$  lies in both summands. So  $\{XA\} \cap \{BY\} = \{BRA\}$  and  $\text{rank } d\kappa = ma + nb - ab$ , for every  $(B, A)$ .

*The overlap.* Let  $B$  have full column rank and  $A$  full row rank. Re-based at  $W = \kappa(B, A)$ , the one-sided methods  $X \mapsto W + XA$  and  $Y \mapsto W + sBY$  are those of [Proposition I.8\(a\)](#) with frames  $A$  and  $sB$ , so their product is  $R \mapsto W + sBRA$ , a linear method whose image  $W + \{BRA\}$  is its own tangent space at  $W$ . That image is the meet  $(W + \{XA\}) \cap (W + \{BY\})$  by the identity just proved. If  $\text{col } B = \text{col } U_r$  and  $\text{row } A = \text{row } V_r^\top$  for the top- $r$  singular frames of  $W_0$  with  $\sigma_r > 0$ , then  $B = U_r k_1$  and  $A = k_2 V_r^\top$  with  $k_1, k_2 \in \text{GL}_r$ , and  $\{BRA\} = \{U_r R' \Sigma_r V_r^\top\}$  with  $R' = k_1 R k_2 \Sigma_r^{-1}$  ranging over  $\mathbb{R}^{r \times r}$ , which is LoRA-XS's update space ([Proposition I.8\(a\)](#)). For other frames, such as the bottom singular frames of MiLoRA, the overlap is the same kind of sandwich in those frames.

*Gradient flow.* With  $W = \kappa(B, A)$ , the chain rule gives  $\nabla_B(L \circ \kappa) = sGA^\top$  and  $\nabla_A(L \circ \kappa) = sB^\top G$ . Along the flow,

$$\frac{d}{dt}L = \langle G, s(\dot{B}A + B\dot{A}) \rangle = s\langle GA^\top, \dot{B} \rangle + s\langle B^\top G, \dot{A} \rangle = -s^2(\eta_B \|GA^\top\|_F^2 + \eta_A \|B^\top G\|_F^2).$$

One step of full-batch gradient descent moves  $W$  by  $s(\Delta B A + B \Delta A) + s \Delta B \Delta A$  with  $\Delta B = -\eta_B sGA^\top$  and  $\Delta A = -\eta_A sB^\top G$ . For  $L$  of class  $C^2$  near  $W$ , the change of loss is  $\langle G, s(\Delta B A + B \Delta A) \rangle$  plus terms of second order in the step sizes, and the first term is the expression above with  $\eta$  in place of  $\eta dt$ . With an unbiased minibatch gradient  $\hat{G}$ ,  $\mathbb{E}\hat{G} = G$ , the first-order change is  $-s^2(\eta_B \langle GA^\top, \hat{G}A^\top \rangle + \eta_A \langle B^\top G, B^\top \hat{G} \rangle)$ , whose expectation is the same expression.

*Adam.* With bias correction, Adam's first moment and second moment after one step are  $\hat{m}_1 = g$  and  $\hat{v}_1 = g \odot g$ , where  $g$  is the first gradient, so the first update is  $-\eta g / (|g| + \varepsilon)$  entrywise, which tends to  $-\eta \text{sign}(g)$  as  $\varepsilon \rightarrow 0$  (entries with  $g = 0$  do not move). For  $B$ ,  $g = sGA_0^\top$ , so  $\delta B = -\eta_B \text{sign}(s) \text{sign}(GA_0^\top)$ , and its first-order effect on the loss is  $\langle G, s \delta B A_0 \rangle = s\langle GA_0^\top, \delta B \rangle = -\eta_B |s| \|GA_0^\top\|_1$ . The  $A$ -term is the same with  $B_0^\top G$ , and the cross term  $s \delta B \delta A$  and the Taylor remainder are  $O(\eta^2)$ .

*The alignment remark after the proposition.* Among  $A_0$  with  $A_0 A_0^\top = cI$ , write  $A_0 = \sqrt{c} P$  with  $P$  of orthonormal rows. The  $B$ -term of the gradient-flow rate is  $s^2 \eta_B \|GA_0^\top\|_F^2 = s^2 \eta_B c \|GP^\top\|_F^2$ , and by Ky Fan's maximum principle (Auxiliary result B.17(a)) its largest value over  $P$  is  $s^2 \eta_B c \sum_{i \leq r} \sigma_i(G)^2$ , attained when row  $A_0$  is a top- $r$  right singular subspace of  $G$ . Under Adam the first-step term  $\|GA_0^\top\|_1$  is not invariant under  $A_0 \mapsto OA_0$  with  $O \in O(r)$ , so it depends on the basis of row  $A_0$  and not only on the subspace.  $\square$

## C.10 Lemma II.4

**LEMMA II.4** (zero gate). Let  $Q = U \times V$  with  $U$  a vector space, let  $\rho = \theta_0 + \beta(u, v)$  with  $\beta$  smooth, and suppose  $\beta(u_0, v) = 0$  for all  $v$ ; for example  $\beta(\cdot, v)$  linear and  $u_0 = 0$ . Then  $D_v \rho_{q_0} = 0$ . Hence  $\nabla_v(L \circ \rho)(q_0) = 0$ ,  $S_1 = D_u \beta(u_0, v_0)(U)$  ( $= \beta(U, v_0)$  in the linear case), and the cometric at  $q_0$  is  $K_{q_0} = D_u \beta \Lambda_{uu} D_u \beta^\top$ , where  $\Lambda_{uu}$  is the  $uu$ -block of  $\Lambda$ ; no block-diagonality is needed, since the  $v$ -columns of  $D\rho$  vanish. The weight trajectory  $\theta(t)$  agrees with that of the method with  $v$  frozen at  $v_0$  to first order in  $t$  (gradient flow), or in the step size (one or more Adam steps). The parameter  $v$  itself may still move: under terms not of the form  $L \circ \rho$  (regularisers, decoupled weight decay), and under Adam, whose step-2 move of  $v$  is  $O(\eta)$  although its effect on  $\theta$  is  $O(\eta^2)$ .

*Function-level variant.* For an extension with realised function  $F(u, v, x)$ , if  $F(u_0, v, x) = F_0(x)$  for all  $v$  and  $x$ , then  $D_v F = 0$  at  $q_0$ , with the same conclusions at the level of functions. No linearity is needed.

*Proof.* Since  $\beta(u_0, v) = 0$  for every  $v$ , differentiating in  $v$  gives  $D_v \beta(u_0, v) = 0$  for every  $v$ , in particular at  $v_0$ . So  $D\rho_{q_0} = [D_u \beta(u_0, v_0) \ 0]$ . Then  $\nabla_v(L \circ \rho)(q_0) = D_v \rho_{q_0}^\top \nabla L(\theta_0) = 0$ ,  $S_1 = D_u \beta(u_0, v_0)(U)$ , and in the linear case  $D_u \beta(0, v_0)u = \beta(u, v_0)$ . For any symmetric  $\Lambda$  written in blocks,

$$K_{q_0} = \begin{bmatrix} D_u \beta & 0 \end{bmatrix} \begin{bmatrix} \Lambda_{uu} & \Lambda_{uv} \\ \Lambda_{vu} & \Lambda_{vv} \end{bmatrix} \begin{bmatrix} D_u \beta^\top \\ 0 \end{bmatrix} = D_u \beta \Lambda_{uu} D_u \beta^\top.$$

*Gradient flow.* For  $\dot{q} = -\Lambda \nabla(L \circ \rho)(q)$ , the weights  $\theta(t) = \rho(q(t))$  satisfy  $\dot{\theta}(0) = -K_{q_0} \nabla L(\theta_0)$ . The method with  $v$  frozen,  $u \mapsto \rho(u, v_0)$  with the metric  $\Lambda_{uu}$ , has the same velocity at  $t = 0$ , so the two weight trajectories agree to first order in  $t$ .

*Adam.* Let  $\eta$  be the step size and  $\varepsilon > 0$  fixed. At step one the  $v$ -gradient vanishes, so Adam's first and second moments in  $v$  vanish and  $v$  does not move; the  $u$ -update is computed from the same gradient as for the frozen method. So after one step the two methods have the same parameters and the same weights. At step two, the gradient in  $v$  is  $D_v\beta(u_1, v_0)^\top \nabla L$ . Since  $\beta$  is smooth and  $D_v\beta(u_0, \cdot) \equiv 0$ , we have  $\|D_v\beta(u, v)\| = O(\|u - u_0\|)$  uniformly for  $v$  near  $v_0$ , and  $\|u_1 - u_0\| = O(\eta)$ , so this gradient is  $O(\eta)$ , and the move of  $v$  is at most of order  $\eta$ ; it is of exact order  $\eta$  when  $\varepsilon$  is negligible against the gradient, since Adam normalises it. Its effect on  $\theta$  is

$$\rho(u, v) - \rho(u, v_0) = \int_0^1 D_v\beta(u, v_0 + \tau(v - v_0))(v - v_0) d\tau = O(\|u - u_0\| \|v - v_0\|) = O(\eta^2).$$

For any fixed number  $k$  of steps the same estimate propagates: the Adam update is a smooth function of the gradient history when  $\varepsilon > 0$ , and by induction on the step,  $v_t - v_0 = O(\eta)$ , the  $u$ -iterates of the two methods differ by  $O(\eta^2)$  (their gradients differ by  $O(\eta)$  and are multiplied by  $\eta$ ), and the weights differ by  $O(\eta^2)$  by the display. For  $\varepsilon \rightarrow 0$  the same argument holds as long as no entry of the frozen method's gradient vanishes in the first  $k$  steps, so that the normalisation stays smooth. Terms that are not of the form  $L \circ \rho$ , such as a regulariser  $R(v)$  or decoupled weight decay  $v \mapsto (1 - \eta\lambda)v$ , move  $v$  already at step one, which is why the statement concerns the weight trajectory.

*Function-level variant.* If  $F(u_0, v, x) = F_0(x)$  for all  $v$  and  $x$ , then  $D_v F(u_0, v_0, x) = 0$  for every  $x$ , and the argument above applies verbatim with the realised function in place of  $\theta$ .  $\square$

### C.11 Theorem II.5

**THEOREM II.5** (complete invariants of the basic actions; classical). *Let  $W, W' \in \mathbb{R}^{m \times n}$ .*

- (1) Right orthogonal (input side).  $W' = WR$  for some  $R \in O(n)$  iff  $WW^\top = W'W'^\top$ . For  $SO(n)$ , when  $\text{rank } W = n$ , also require that the then unique  $R$  have  $\det R = 1$ .
- (2) Left orthogonal.  $W' = PW$  for some  $P \in O(m)$  iff  $W^\top W = W'^\top W'$ .
- (3) Two-sided.  $W' = PWR$  for some  $P \in O(m)$  and  $R \in O(n)$  iff  $W$  and  $W'$  have the same singular values, counted with multiplicity.
- (4) Left torus. If  $W$  has no zero row,  $W' = DW$  for some  $D \in T_m^+$  iff  $w'_i \in \mathbb{R}_{>0} w_i$  for every  $i$ . For the full torus  $T_m$ , whose signs are unconstrained, the complete invariant is the set of zero rows together with the row lines.
- (5) Full Hadamard torus.  $W' = E \odot W$  for some  $E \in (\mathbb{R}^\times)^{m \times n}$  iff  $W$  and  $W'$  have the same zero pattern.

*Proof.* Write  $w_i, w'_i \in \mathbb{R}^n$  for the rows of  $W$  and  $W'$ .

(1) If  $W' = WR$  with  $R$  orthogonal, then  $W'W'^\top = WRR^\top W^\top = WW^\top$ . Conversely, let  $WW^\top = W'W'^\top$ . For  $c \in \mathbb{R}^m$ ,

$$\left\| \sum_i c_i w_i \right\|^2 = c^\top WW^\top c = c^\top W'W'^\top c = \left\| \sum_i c_i w'_i \right\|^2.$$

Hence  $\varphi : \sum_i c_i w_i \mapsto \sum_i c_i w'_i$  is well defined on row  $W$  (if  $\sum_i c_i w_i = \sum_i c'_i w_i$ , the vector  $\sum_i (c_i - c'_i) w_i$  has norm 0), linear, isometric and onto row  $W'$ . In particular both row spaces have dimension  $\text{rank } WW^\top$ . By Witt's extension theorem (Auxiliary result B.25(b)),  $\varphi$  extends

to some  $S \in O(n)$ , for instance by an isometry between the orthogonal complements, which have equal dimension. Then  $Sw_i = w'_i$  for every  $i$ , that is  $W' = WS^\top$ , and  $R = S^\top \in O(n)$ .

For  $SO(n)$ : if  $\text{rank } W < n$ , the complement  $(\text{row } W)^\perp$  is nonzero, and composing the extension with a reflection of that complement changes  $\det S$  without changing  $Sw_i$ ; so some  $R \in SO(n)$  exists as soon as some  $R \in O(n)$  does. If  $\text{rank } W = n$ , then  $W$  has full column rank and  $WR = W'$  determines  $R = W^+W'$  uniquely, so  $W' \in W \cdot SO(n)$  iff this  $R$  has determinant 1.

(2)  $W' = PW$  iff  $W'^\top = W^\top P^\top$  with  $P^\top \in O(m)$ , and by (1) applied to  $W^\top$  and  $W'^\top$  this happens iff  $W^\top W = W'^\top W'$ .

(3) If  $W' = PWR$ , then  $W'^\top W' = R^\top (W^\top W)R$  has the eigenvalues of  $W^\top W$ , so the singular values agree with multiplicity. Conversely, if  $W = U\Sigma V^\top$  and  $W' = U'\Sigma V'^\top$  are singular value decompositions with the same  $\Sigma$  ([Auxiliary result B.11\(a\)](#)), then  $W' = (U'U^\top)W(VV'^\top)$  with both factors orthogonal.

(4) For  $D = \text{diag}(d)$  the rows of  $DW$  are  $d_i w_i$ . If  $d \in \mathbb{R}_{>0}^m$ , then  $w'_i = d_i w_i \in \mathbb{R}_{>0} w_i$ . Conversely, if  $w'_i = d_i w_i$  with  $d_i > 0$  for every  $i$  ( $d_i$  is unique because  $w_i \neq 0$ ), then  $W' = \text{diag}(d)W$  with  $\text{diag}(d) \in T_m^+$ . For  $T_m$ , the rows  $d_i w_i$  with  $d_i \neq 0$  vanish exactly where  $w_i$  does and otherwise span the line  $\mathbb{R}w_i$ . Conversely, if  $W$  and  $W'$  have the same zero rows and  $\mathbb{R}w'_i = \mathbb{R}w_i$  for every nonzero row, then  $w'_i = d_i w_i$  with  $d_i \neq 0$ , and with  $d_i = 1$  on the zero rows  $W' = \text{diag}(d)W$ ,  $\text{diag}(d) \in T_m$ .

(5)  $(E \odot W)_{ij} = E_{ij}W_{ij}$  vanishes iff  $W_{ij}$  does, because  $E_{ij} \neq 0$ . Conversely, if  $W$  and  $W'$  have the same zero pattern, put  $E_{ij} = W'_{ij}/W_{ij}$  on the support and  $E_{ij} = 1$  off it; then  $E$  has no zero entry and  $W' = E \odot W$ .  $\square$

Item (1) is also the orbit-separation statement of the first fundamental theorem for  $O(n)$  ([Auxiliary result B.25\(a\)](#)); the proof above uses only the extension of isometries.

## C.12 Corollary II.6

**COROLLARY II.6** (what exactly orthogonal methods can never do). *Let a method have image in  $W_0 \cdot O(n)$  (input side), and work in exact arithmetic. This covers:*

- *OFT and block OFT with an exactly orthogonal parametrisation: the original paper's exact Cayley map, and HF PEFT from 0.16 with `use_cayley_neumann=False` on Linear and Conv targets;*
- *GOFT with strict Givens rotations and no scaler, as in the paper (`no_scaling=True` in the official `peft-givens` code);*
- *HRA, and ETHER as defined in its paper (not ETHER+);*
- *any number of merge-and-restart cycles of these.*

*Then:*

- *it preserves  $K_{\text{out}}$ , hence every neuron norm and the angle between any two nonzero neurons, and so the hyperspherical energy, which depends only on these angles;*
- *it preserves the whole singular spectrum of  $W_0$ ;*
- *block-diagonal OFT with input blocks  $J$ , with the same partition in every cycle, preserves each partial Gram matrix  $W_{:,J}W_{:,J}^\top$ .*

*Proof.* Coverage. Each listed parametrisation produces  $W = W_0 R$  with  $R \in O(n)$ , read in the convention  $y = Wx$ , where an input-side rotation acts on the columns of  $W_0$ . The exact Cayley map  $C(\Omega) = (I + \Omega)(I - \Omega)^{-1}$  of a skew matrix is orthogonal ([Auxiliary result B.23\(a\)](#)); HF



PEFT's input-side OFT merges to  $W_0 R^\top$ , and  $R^\top$  is orthogonal too. A block-diagonal matrix with orthogonal blocks is orthogonal. A Givens rotation is orthogonal, and so is a product of them. HRA multiplies Householder reflections  $H_u = I - 2uu^\top/\|u\|^2$ , which are orthogonal, and ETHER as defined in its paper multiplies  $W_0$  on the input side by one reflection  $I - 2\hat{u}\hat{u}^\top$  with  $\|\hat{u}\| = 1$ . For merge and restart, if every weight reachable in cycle  $k$  is  $W_{k-1}R_k$  with  $R_k \in O(n)$ , then by induction  $W_k = W_0 R_1 \cdots R_k$ , and  $R_1 \cdots R_k \in O(n)$ .

*Gram matrix and spectrum.* For  $W = W_0 R$  with  $R \in O(n)$ ,  $K_{\text{out}}(W) = W_0 R R^\top W_0^\top = K_{\text{out}}(W_0)$  (Theorem II.5(1)). The neuron norms are  $\|w_i\|^2 = K_{\text{out}}(W)_{ii}$ , and the cosine between two nonzero neurons is  $K_{\text{out}}(W)_{ij}/(K_{\text{out}}(W)_{ii}K_{\text{out}}(W)_{jj})^{1/2}$ , so both are preserved. The hyperspherical energy of Liu et al. (2018) is a function of the distances between the normalised neurons,  $\|\hat{w}_i - \hat{w}_j\|^2 = 2 - 2\cos_{ij}$ , so it is preserved as well. The singular values of  $W$  are the square roots of the eigenvalues of  $K_{\text{out}}(W) = K_{\text{out}}(W_0)$  (equivalently, Theorem II.5(3) with  $P = I$ ).

*Partial Grams.* Let  $R = \text{blockdiag}(R_J)_J$  be block-diagonal for a partition of the input coordinates into blocks  $J$ , with orthogonal blocks. The columns of  $W_0 R$  indexed by  $J$  are  $(W_0 R)_{:,J} = (W_0)_{:,J} R_J$ , so  $W_{:,J} W_{:,J}^\top = (W_0)_{:,J} R_J R_J^\top (W_0)_{:,J}^\top = (W_0)_{:,J} (W_0)_{:,J}^\top$ . With the same partition in every cycle,  $R_1 \cdots R_k$  is again block-diagonal with orthogonal blocks, so the identity survives merging.  $\square$

The remarks after the corollary in §II.B make several claims that do not follow from its hypothesis. We prove them in the same order. Throughout,  $K_{\text{out}}(W) = W W^\top$ , and for an input-side factor  $G$  we have  $K_{\text{out}}(W_0 G) = W_0 G G^\top W_0^\top$ .

*BOFT and HF OFT 0.14–0.15.* For  $W = \text{diag}(s) W_0 R^\top$  with  $R$  orthogonal,

$$K_{\text{out}}(W) = \text{diag}(s) W_0 R^\top R W_0^\top \text{diag}(s) = \text{diag}(s) K_{\text{out}}(W_0) \text{diag}(s),$$

with entries  $s_i s_j K_{\text{out}}(W_0)_{ij}$ . If  $s_i s_j \neq 0$  and the neurons  $i, j$  of  $W_0$  are nonzero,

$$\cos'_{ij} = \frac{s_i s_j K_{\text{out}}(W_0)_{ij}}{|s_i| |s_j| (K_{\text{out}}(W_0)_{ii} K_{\text{out}}(W_0)_{jj})^{1/2}} = \text{sign}(s_i s_j) \cos_{ij}.$$

So  $|\cos_{ij}|$  is preserved whenever  $s_i s_j \neq 0$ , and  $\cos_{ij}$  itself whenever  $s_i s_j > 0$ . If every  $s_i$  is nonzero and all share a sign, every cosine is preserved. Conversely, if every cosine of  $W_0$  is nonzero and every cosine is preserved, then every  $s_i$  is nonzero (otherwise neuron  $i$  of  $W$  vanishes and its cosines are undefined) and  $\text{sign}(s_i s_j) = 1$  for all  $i \neq j$ , so all  $s_i$  share a sign. With the butterfly permutation equal to the identity,  $R$  is block-diagonal with orthogonal blocks  $R_J$  over a partition of the input coordinates, and  $W_{:,J} = \text{diag}(s) (W_0)_{:,J} R_J^\top$ , so  $W_{:,J} W_{:,J}^\top = \text{diag}(s) (W_0)_{:,J} (W_0)_{:,J}^\top \text{diag}(s)$ . HF PEFT OFT 0.14–0.15 has the same form with its output scale `oft_s` in place of  $s$ , which gives the same conclusions.

*GOFT with its scaler.* For  $W = W_0 \text{diag}(g) G$  with  $G$  orthogonal, whatever  $G$  is,

$$K_{\text{out}}(W) = W_0 \text{diag}(g) G G^\top \text{diag}(g) W_0^\top = W_0 \text{diag}(g)^2 W_0^\top.$$

This differs from  $K_{\text{out}}(W_0)$  in general (for  $W_0 = I_n$  it is  $\text{diag}(g)^2$ ), so the image is not contained in  $W_0 \cdot O(n)$  and the corollary does not apply.

*Output-side implementations.* If  $W = P W_0$  with  $P \in O(m)$ , then  $K_{\text{in}}(W) = W_0^\top P^\top P W_0 = K_{\text{in}}(W_0)$ , and the singular values are preserved by Theorem II.5(3). If  $P$  is block-diagonal with orthogonal blocks  $P_I$  over a partition of the output coordinates, the rows indexed by  $I$  satisfy  $W_{I,:} = P_I (W_0)_{I,:}$ , so  $W_{I,:}^\top W_{I,:} = (W_0)_{I,:}^\top (W_0)_{I,:}$ . In general  $K_{\text{out}}(P W_0) = P K_{\text{out}}(W_0) P^\top$

differs from  $K_{\text{out}}(W_0)$ , for instance for  $W_0 = E_{11}$  and  $P$  a rotation moving  $e_1$ . ETHER's code with `ether_nb > 1` multiplies the rows of output group  $g$  on the right by a reflection  $H_g$ ,  $W_{I_g, \cdot} = (W_0)_{I_g, \cdot} H_g$ . Within a group,  $W_{I_g, \cdot} W_{I_g, \cdot}^\top = (W_0)_{I_g, \cdot} (W_0)_{I_g, \cdot}^\top$ , which keeps neuron norms and within-group angles. Across groups the inner products become  $w_{0,i}^\top H_g H_{g'} w_{0,j}$ . For two groups of one row each, with  $w_{0,1} = w_{0,2} = e_1$  in  $\mathbb{R}^2$ ,  $H_1 = I$  and  $H_2$  the reflection that swaps  $e_1$  and  $e_2$ , the two neurons become  $e_1$  and  $e_2$ : the cosine drops from 1 to 0, and the singular values change from  $(\sqrt{2}, 0)$  to  $(1, 1)$ .

*HF PEFT's default OFT.* Let  $Q$  be skew. Then  $(I - Q)(I + Q + Q^2 + Q^3) = I - Q^4$  and  $I - Q$  is invertible ([Auxiliary result B.23\(a\),\(e\)](#)), so

$$R := I + 2Q + 2Q^2 + 2Q^3 + Q^4 = (I + Q)(I + Q + Q^2 + Q^3) = (I + Q)(I - Q)^{-1}(I - Q^4) = C(Q)(I - Q^4),$$

where all factors are polynomials in  $Q$  or their inverses and commute. The matrix  $Q^4$  is symmetric, since  $(Q^4)^\top = (-Q)^4$ , and  $C(Q)$  is orthogonal, so  $R^\top R = (I - Q^4)C(Q)^\top C(Q)(I - Q^4) = (I - Q^4)^2$ . If  $\pm i\theta_k$  are the eigenvalues of  $Q$ , then  $Q^4$  has eigenvalues  $\theta_k^4$  (and 0), so the singular values of  $R$  are  $|1 - \theta_k^4|$  (and 1). Thus  $R^\top R = I$  only if every  $\theta_k$  is 0 or  $2^{1/4}$ ; in particular  $R^\top R \neq I$  for every  $Q \neq 0$  with  $\|Q\|_{\text{op}} < 2^{1/4}$ , and  $R$  is singular when some  $\theta_k = 1$ . The adapted weight is  $W = W_0 R^\top$ , with  $K_{\text{out}}(W) = W_0 R^\top R W_0^\top = W_0 (I - Q^4)^2 W_0^\top$ , and  $(I - Q^4)^2 - I = Q^8 - 2Q^4$  is of order  $\|Q\|^4$ . When  $\|Q\|_{\text{op}} \leq 2^{1/4}$ , every  $|1 - \theta_k^4| \leq 1$ , so  $R$  is a contraction, and each merge  $W \mapsto W R^\top$  gives  $W R^\top R W^\top \preceq W W^\top$  and lowers every singular value ([Auxiliary result B.18\(b\),\(c\)](#)); [Theorem V.3](#), corollary 4, records the iteration. In PEFT 0.16 and 0.17 the last coefficient was 2,  $R = p(Q)$  with  $p(x) = 1 + 2x + 2x^2 + 2x^3 + 2x^4$ . Then  $R^\top = p(-Q)$ , and writing  $p = E + O$  with even part  $E = 1 + 2x^2 + 2x^4$  and odd part  $O = 2x + 2x^3$ ,

$$p(x)p(-x) = E^2 - O^2 = (1 + 4x^2 + 8x^4 + 8x^6 + 4x^8) - (4x^2 + 8x^4 + 4x^6) = 1 + 4x^6 + 4x^8,$$

so  $R^\top R = I + 4Q^6 + 4Q^8$ . Its eigenvalues are  $1 + 4\theta_k^6(\theta_k^2 - 1)$ , so it differs from  $I$  unless every  $\theta_k$  is 0 or 1, and the deviation is of order  $\|Q\|^6$ .

*One non-orthogonal plane changes a cosine.* We prove the following, which covers qGOFT, ETHER+ and HF's default OFT. Let  $n > 2$  and let  $G \in \mathbb{R}^{n \times n}$  be an input-side factor such that  $S = G G^\top$  acts as the identity on the orthogonal complement of a plane  $\Pi$ , maps  $\Pi$  into itself, and  $S|_\Pi \neq I_\Pi$  (that is,  $G$  is not orthogonal on  $\Pi$ ). Then for  $W_0$  outside a nowhere-dense null set some cosine of  $W_0 G$  differs from the corresponding cosine of  $W_0$ . Indeed, for two rows  $w_1, w_2$  of  $W_0$  put

$$\phi(W_0) = (w_1^\top S w_2)^2 \|w_1\|^2 \|w_2\|^2 - (w_1^\top w_2)^2 (w_1^\top S w_1)(w_2^\top S w_2),$$

a polynomial in  $W_0$ . Where both cosines are defined,  $\cos'_{12} = \cos_{12}$  forces  $\phi(W_0) = 0$ . Choose a unit vector  $c \perp \Pi$  (possible since  $n > 2$ ) and  $x \in \Pi$  with  $x^\top S x \neq \|x\|^2$  (possible since  $S|_\Pi - I_\Pi$  is a nonzero symmetric map), and take  $w_1 = c$ ,  $w_2 = c + x$ . Then  $w_1^\top S w_2 = 1$ ,  $w_1^\top S w_1 = 1$ ,  $w_2^\top S w_2 = 1 + x^\top S x$ ,  $\|w_2\|^2 = 1 + \|x\|^2$  and  $w_1^\top w_2 = 1$ , so  $\phi = \|x\|^2 - x^\top S x \neq 0$ . By [Auxiliary result B.10\(b\)](#),  $\phi \neq 0$  on an open dense set of full measure, which proves the claim. A conformal block  $\sqrt{c}$  Rot with  $c \neq 1$  on two input coordinates has  $S|_\Pi = cI$ , so it is covered. For qGOFT, take one block non-orthogonal and the others equal to  $I$ . For ETHER+, take the output factor equal to  $I$  (its two vectors equal) and the input factor  $I - uu^\top + vv^\top$  with unit  $u \neq \pm v$ : it is symmetric, equal to  $I$  on  $\{u, v\}^\perp$ , preserves  $\Pi = \text{span}\{u, v\}$ , and on  $\Pi$  has eigenvalues  $1 \pm (u^\top v)^2$ , so it is not orthogonal there; for  $u \perp v$  it kills  $u$  and is singular. For HF's default OFT, take  $Q = \theta(e_1 e_2^\top - e_2 e_1^\top)$  with  $\theta \notin \{0, 2^{1/4}\}$ ; then  $R^\top R = (I - Q^4)^2$  is  $(1 - \theta^4)^2 I$  on  $\Pi = \text{span}\{e_1, e_2\}$  and  $I$  elsewhere.

*Finite precision.* The last remark concerns floating-point arithmetic, and Appendix E reports the measured drift; in exact arithmetic the corollary holds as proved.

### C.13 Theorem II.7

**THEOREM II.7** (HRA is the meet of LoRA with the rotation orbit, and it starts at an apex). Let  $W_0 \in \mathbb{R}^{m \times n}$  be injective ( $\text{rank } W_0 = n$ ), and let  $r$  be even with  $2 \leq r \leq n - 2$ . Define  $\text{HRA}_r : (u_1, \dots, u_r) \mapsto W_0 H_{u_1} \cdots H_{u_r}$  with  $u_i \neq 0$ , where  $H_u = I - 2uu^\top / \|u\|^2$ . Write  $\text{OFT}_{\text{SO}} := W_0 \cdot \text{SO}(n)$  for the rotation orbit. (The  $\text{O}(n)$ -orbit would add the odd component:  $W_0 H_{e_1}$  lies in  $W_0 + \mathcal{M}_{\leq r}$  but not in  $\text{Im HRA}_r$ .)

- (a) **Meet.**  $\text{Im HRA}_r = (W_0 + \mathcal{M}_{\leq r}) \cap W_0 \cdot \text{SO}(n)$ . Further,  $d(\text{HRA}_r) = rn - \frac{r(r+1)}{2}$ , so the generic fibre has dimension  $\frac{r(r+1)}{2} = rn - d$ . This gauge is non-linear. The evident continuous global symmetries are the  $r$  scalings  $u_i \mapsto c_i u_i$ ; the braid group also acts globally by Hurwitz moves  $(u_i, u_{i+1}) \mapsto (H_{u_i} u_{i+1}, u_i)$ , since  $H_a H_b = H_{H_a b} H_a$ .
- (b) **First-order collapse.** At the paired initialisation  $u_{2i-1} = u_{2i} = w_i$  ( $i \leq k = r/2$ , with the  $w_i$  independent and  $U = \text{span}\{w_i\}$ ),

$$S_1 = W_0 \cdot \{\Omega \in \mathfrak{so}(n) : P_{U^\perp} \Omega P_{U^\perp} = 0\}, \quad \dim S_1 = kn - \frac{k(k+1)}{2} < d(\text{HRA}_r).$$

- (c) **Vertex.** The germ of  $\text{Im HRA}_r$  at  $W_0$  is diffeomorphic to the germ at 0 of the cone  $\Sigma_r = \{\Omega \in \mathfrak{so}(n) : \text{rank } \Omega \leq r\}$  (the skew-determinantal cone), which is not linear. So  $W_0$  is a singular point (the vertex) of the image. Together with (b), the paired initialisation is an **apex**.

- (d) **Scope.**

- Not a product. (a) is a meet in the poset of subsets of  $\Theta$  containing  $W_0$  (Observation I.4(c)). HRA is not the categorical product of  $\text{LoRA}_r$  and  $\text{OFT}_{\text{SO}}$ : the set-level fibre product  $\{(B, A, R) : W_0 + BA = W_0 R\}$  has local dimension  $d + r^2 = rn + \frac{r(r-1)}{2} \neq rn$  over a generic point of the meet. (It is not pure-dimensional: over  $W_0$  alone it contains  $\{(B, 0, I)\}$ , of dimension  $mr$ .)
- Exactly orthogonal OFT is smaller. The Cayley map never produces a rotation with eigenvalue  $-1$ . For exact full-block Cayley OFT (the original paper; HF with `use_cayley_neumann=False`),  $(W_0 + \mathcal{M}_{\leq r}) \cap \text{Im OFT}$  is dense in  $\text{Im HRA}_r$  but not equal to it:  $W_0 H_{e_1} H_{e_2}$ , with  $H_{e_1} H_{e_2} = \text{diag}(-1, -1, 1, \dots, 1)$ , is missing. For exact block Cayley OFT with  $b < n$  the intersection is a proper subset of  $\text{Im HRA}_r$ .
- HF's default OFT is incomparable. HF PEFT's default OFT is not orthogonal (Corollary II.6); its flag is `use_cayley_neumann=True`. Since  $R^\top R = (I - Q^4)^2 = I$  forces  $Q = 0$  near  $Q = 0$ , its image meets  $\text{Im HRA}_r$  near  $W_0$  only at  $W_0$ , while its intersection with  $W_0 + \mathcal{M}_{\leq r}$  contains non-orthogonal points such as  $W_0(I + 2Q + 2Q^2 + 2Q^3 + Q^4)$  with  $\text{rank } Q = 2$ .
- Implementation. The theorem concerns HRA with  $\lambda < \infty$  (HF `apply_gs=False`). The strictly orthogonal Gram–Schmidt variant has the smaller image  $W_0(I - 2P_U)$ , of dimension  $r(n - r)$ . The paired initialisation is HF's default (`repeat_interleave`) and is not stated in the HRA paper. HF HRA acts on `in_features`, so injectivity requires `out_features`  $\geq$  `in_features` (and full column rank); the theorem does not cover `down_proj` or GQA key/value projections. The Gram–Schmidt variant never contains  $W_0$ , since  $I - 2P_U \neq I$ , and at HF's default paired initialisation it runs Gram–Schmidt on duplicated columns, which is numerically degenerate.

*Proof.* For  $u \neq 0$  write  $\hat{u} = u/\|u\|$ . The reflection  $H_u = I - 2\hat{u}\hat{u}^\top$  is symmetric and orthogonal, has determinant  $-1$ , fixes  $u^\perp$  pointwise, and  $H_u - I$  has rank one. For  $g \in O(n)$ ,  $gH_u g^{-1} = H_{gu}$ . For  $x, y \in \mathbb{R}^n$  put  $x \wedge y = xy^\top - yx^\top \in \mathfrak{so}(n)$ , and for a subspace  $A \subseteq \mathbb{R}^n$  let  $\mathbb{R}^n \wedge A = \text{span}\{x \wedge a : x \in \mathbb{R}^n, a \in A\}$ . Because  $W_0$  is injective,  $X \mapsto W_0 X$  is an injective linear map  $\mathbb{R}^{n \times n} \rightarrow \mathbb{R}^{m \times n}$ ; in particular  $W_0 R = W_0 R'$  implies  $R = R'$ .

*Two lemmas.* (i) For every subspace  $A$  of dimension  $a$ ,

$$\mathbb{R}^n \wedge A = \{\Omega \in \mathfrak{so}(n) : P_{A^\perp} \Omega P_{A^\perp} = 0\}, \quad \dim(\mathbb{R}^n \wedge A) = an - \frac{a(a+1)}{2}.$$

For  $x, y \in A^\perp$  and  $z \in \mathbb{R}^n$ ,  $a' \in A$ ,  $y^\top(z \wedge a')x = (y^\top z)(a'^\top x) - (y^\top a')(z^\top x) = 0$ , which gives “ $\subseteq$ ”. Take an orthonormal basis  $e_1, \dots, e_a$  of  $A$  and  $e_{a+1}, \dots, e_n$  of  $A^\perp$ . The matrices  $e_p \wedge e_q$  ( $p < q$ ) form a basis of  $\mathfrak{so}(n)$ , and  $P_{A^\perp} \Omega P_{A^\perp} = 0$  says exactly that the coefficients with  $p, q > a$  vanish, so the right side is spanned by the  $e_p \wedge e_q$  with  $p \leq a$ , which lie in  $\mathbb{R}^n \wedge A$ . Its dimension is  $\dim \mathfrak{so}(n) - \dim \mathfrak{so}(n-a) = an - a(a+1)/2$ .

(ii) Let  $R(u) = H_{u_1} \cdots H_{u_r}$ , put  $g_k = H_{u_1} \cdots H_{u_{k-1}}$  (so  $g_1 = I$ ) and  $a_k = g_k \hat{u}_k$ , and let  $A(u) = \text{span}\{a_1, \dots, a_r\}$ . Then the right-translated differential satisfies

$$\{dR_u(\dot{u}) R(u)^{-1} : \dot{u} \in (\mathbb{R}^n)^r\} = \mathbb{R}^n \wedge A(u).$$

Indeed, varying only  $u_k$  in the direction  $\dot{u}_k$  moves  $\hat{u}_k$  by  $\delta = (I - \hat{u}_k \hat{u}_k^\top) \dot{u}_k / \|\dot{u}_k\|$ , which ranges over  $\hat{u}_k^\perp$ , and  $H_{u_k}$  by  $\dot{H} = -2(\delta \hat{u}_k^\top + \hat{u}_k \delta^\top)$ . Since  $H_{u_k} \hat{u}_k = -\hat{u}_k$  and  $H_{u_k} \delta = \delta$ , we get  $\dot{H} H_{u_k} = 2(\delta \hat{u}_k^\top - \hat{u}_k \delta^\top) = 2 \delta \wedge \hat{u}_k$ . Hence

$$\dot{R} R^{-1} = g_k \dot{H} H_{u_{k+1}} \cdots H_{u_r} R^{-1} = g_k (\dot{H} H_{u_k}) g_k^{-1} = 2 (g_k \delta) \wedge a_k,$$

using  $H_{u_{k+1}} \cdots H_{u_r} R^{-1} = (g_k H_{u_k})^{-1}$  and  $g(x \wedge y) g^\top = gx \wedge gy$  for orthogonal  $g$ . As  $\delta$  ranges over  $\hat{u}_k^\perp$ ,  $g_k \delta$  ranges over  $a_k^\perp$ , and  $a_k^\perp \wedge a_k = \mathbb{R}^n \wedge a_k$  because  $a_k \wedge a_k = 0$ . Summing over  $k$  gives the claim.

(a), “ $\subseteq$ ”.  $R = H_{u_1} \cdots H_{u_r}$  is orthogonal with  $\det R = (-1)^r = 1$ , and it fixes  $\cap_i u_i^\perp$ , of dimension at least  $n - r$ , pointwise. So  $\text{rank}(R - I) \leq r$  and  $W_0 R - W_0 = W_0(R - I) \in \mathcal{M}_{\leq r}$ .

(a), “ $\supseteq$ ”. Let  $W_0 R = W_0 + X$  with  $R \in \text{SO}(n)$  and  $\text{rank } X \leq r$ . Since  $W_0$  is injective,  $k' := \text{rank}(R - I) = \text{rank } X \leq r$ . By the Cartan–Dieudonné theorem in Scherk’s sharp form (Auxiliary result B.22),  $R$  is a product of exactly  $k'$  reflections and  $\det R = (-1)^{k'}$ , so  $k'$  is even. Append  $(r - k')/2$  pairs  $H_v H_v = I$ ; then  $R$  is a product of exactly  $r$  reflections and  $W_0 R \in \text{Im HRA}_r$ .

*Leaves.* For  $V \in \text{Gr}(r, n)$  let  $\text{SO}(V) = \{S \in \text{SO}(n) : Sx = x \text{ for } x \in V^\perp\}$ . Then

$$(W_0 + \mathcal{M}_{\leq r}) \cap W_0 \cdot \text{SO}(n) = \bigcup_{V \in \text{Gr}(r, n)} W_0 \cdot \text{SO}(V).$$

If  $S \in \text{SO}(V)$ , then  $S - I$  vanishes on  $V^\perp$ , so  $\text{rank}(S - I) \leq r$ . Conversely, if  $\text{rank}(R - I) \leq r$ , the fixed space  $F = \ker(R - I)$  has dimension at least  $n - r$ , and any  $r$ -dimensional  $V \supseteq F^\perp$  has  $V^\perp \subseteq F$ , so  $R \in \text{SO}(V)$ . A point with  $\text{rank}(R - I) = r$  lies on exactly one leaf,  $V = \text{im}(R - I)$ .

*The  $O(n)$ -orbit.*  $W_0 H_{e_1} - W_0 = W_0(H_{e_1} - I)$  has rank one, so  $W_0 H_{e_1} \in W_0 + \mathcal{M}_{\leq r}$ . If  $W_0 H_{e_1} = W_0 R$  with  $R = H_{u_1} \cdots H_{u_r}$ , injectivity gives  $R = H_{e_1}$ , whose determinant is  $-1 \neq (-1)^r$ .

*Dimension and fibre.*  $\rho = \text{HRA}_r$  is a rational map on the connected open set  $Q = (\mathbb{R}^n \setminus \{0\})^r$ , and  $d\rho_u(\dot{u}) = W_0 dR_u(\dot{u})$  has the rank of  $dR_u$ , which by lemma (ii) equals  $\dim(\mathbb{R}^n \wedge A(u))$ . By lemma (i) this is  $an - a(a+1)/2$  with  $a = \dim A(u) \leq r$ , a non-decreasing function of  $a$  for  $a \leq n$ . For  $u_k = e_k$  we get  $g_k = H_{e_1} \cdots H_{e_{k-1}}$  and  $a_k = e_k$ , so  $a = r$  is attained. By Auxiliary result B.9(d),

$$d(\text{HRA}_r) = \max_u \text{rank } d\rho_u = rn - \frac{r(r+1)}{2}.$$

The computation uses neither the parity of  $r$  nor  $r \leq n - 2$ , so the same value holds for every  $1 \leq r \leq n$ ; for  $r \in \{n - 1, n\}$  it equals  $\dim \mathrm{SO}(n)$ . At a point of maximal rank the fibre is locally a submanifold of dimension  $rn - d = r(r + 1)/2$  (Auxiliary result B.1(b)), and the points of maximal rank, those with  $a_1, \dots, a_r$  independent, are open and dense in  $Q$  (Auxiliary result B.10). As a check, a generic point of the meet is fixed by its moved space  $\mathrm{im}(R - I) \in \mathrm{Gr}(r, n)$ , of dimension  $r(n - r)$ , and by a fixed-point-free element of  $\mathrm{SO}(r)$  on it, an open subset of dimension  $r(r - 1)/2$ ; the sum is  $d$ .

*Symmetries.*  $H_{cu} = H_u$  for  $c \neq 0$ , so the scalings  $(\mathbb{R}^\times)^r$  preserve  $\rho$ . From  $gH_bg^{-1} = H_gb$  with  $g = H_a$  we get  $H_aH_b = H_{H_ab}H_a$ , so the Hurwitz move  $\sigma_i : (u_i, u_{i+1}) \mapsto (H_{u_i}u_{i+1}, u_i)$  preserves the product, and  $H_{u_i}u_{i+1} \neq 0$ . It is invertible, with inverse  $(u_i, u_{i+1}) \mapsto (u_{i+1}, H_{u_{i+1}}u_i)$ , the moves  $\sigma_i, \sigma_j$  with  $|i - j| \geq 2$  commute, and  $\sigma_1\sigma_2\sigma_1 = \sigma_2\sigma_1\sigma_2$ : on a triple both composites give  $(a, b, c) \mapsto (H_aH_bc, H_ab, a)$ , using  $H_{H_ab}H_ac = H_aH_bc$ . So the braid group acts. The scalings have  $r$ -dimensional orbits, while the generic fibre has dimension  $r(r + 1)/2 > r$  for  $r \geq 2$ , and the braid moves are discrete. So the exhibited symmetries are not locally transitive on generic fibres; the remaining fibre directions come from refactorising  $R$  inside its moved space, and this is what we mean by a non-linear gauge.

(b) At the paired initialisation,  $g_{2i-1} = (H_{w_1}H_{w_1}) \cdots (H_{w_{i-1}}H_{w_{i-1}}) = I$ , so  $a_{2i-1} = \hat{w}_i$  and  $a_{2i} = H_{w_i}\hat{w}_i = -\hat{w}_i$ . Hence  $A(u) = U$ ,  $R(u) = I$ , and by lemmas (ii) and (i)

$$S_1 = W_0 \cdot (\mathbb{R}^n \wedge U) = W_0 \cdot \{\Omega \in \mathfrak{so}(n) : P_{U^\perp}\Omega P_{U^\perp} = 0\}, \quad \dim S_1 = kn - \frac{k(k+1)}{2},$$

since  $W_0$  is injective. Equivalently, a direct computation for one pair gives

$$\frac{d}{d\varepsilon}(H_{w+\varepsilon a}H_{w+\varepsilon b})\Big|_{\varepsilon=0} = \frac{2}{\|w\|}(v\hat{w}^\top - \hat{w}v^\top), \quad v = (I - \hat{w}\hat{w}^\top)(a - b),$$

and because every pair equals  $I$  at the base point the differential of the product is the sum of the pair differentials. Finally, with  $r = 2k$ ,

$$d - \dim S_1 = 2kn - k(2k + 1) - kn + \frac{k(k+1)}{2} = \frac{k(2n-3k-1)}{2},$$

which is positive because  $n \geq 2k + 2$  gives  $2n - 3k - 1 \geq k + 3$ .

(c) By Auxiliary result B.23(b),(c), the Cayley map  $C(\Omega) = (I + \Omega)(I - \Omega)^{-1}$  is a diffeomorphism from  $\mathfrak{so}(n)$  onto the open set  $\mathcal{O} = \{R \in \mathrm{SO}(n) : -1 \notin \mathrm{spec} R\}$ , which contains  $I$ , and  $\mathrm{rank}(C(\Omega) - I) = \mathrm{rank} \Omega$ . By (a),  $\mathrm{Im} \mathrm{HRA}_r \cap W_0 \cdot \mathcal{O} = W_0 \cdot C(\Sigma_r)$ . The map  $\Omega \mapsto W_0C(\Omega)$  is a diffeomorphism of  $\mathfrak{so}(n)$  onto an open subset of the embedded submanifold  $W_0 \cdot \mathrm{SO}(n)$  (a linear injection of  $\mathrm{SO}(n)$ ), and it carries  $\Sigma_r$  onto  $\mathrm{Im} \mathrm{HRA}_r$  near  $W_0$ . So the germs correspond.  $\Sigma_r$  is a closed cone and is not linear (Auxiliary result B.7(c)): with  $r = 2k$ , the matrices  $\sum_{j \leq k} e_{2j-1} \wedge e_{2j}$  and  $e_{r+1} \wedge e_{r+2}$  lie in  $\Sigma_r$  (here  $r + 2 \leq n$ ), and their sum has rank  $r + 2$ . By Auxiliary result B.6,  $\Sigma_r$  is not a  $C^1$  submanifold near 0. If  $\mathrm{Im} \mathrm{HRA}_r$  were a  $C^1$  submanifold near  $W_0$ , it would be one of  $W_0 \cdot \mathrm{SO}(n)$ , and its preimage  $\Sigma_r$  under the diffeomorphism would be  $C^1$  near 0. So  $W_0$  is a singular point, the vertex of the germ. A singular image alone does not give a positive deficit; the apex statement is (b).

(d), *not a product.* Let  $\mathcal{F} = \{(B, A, R) \in \mathbb{R}^{m \times r} \times \mathbb{R}^{r \times n} \times \mathrm{SO}(n) : W_0 + BA = W_0R\}$  be the set-level fibre product. The points of the meet of the form  $W_0C(\Omega)$  with  $\mathrm{rank} \Omega = r$  form a  $d$ -dimensional submanifold  $\mathcal{G}$ , the image under the diffeomorphism of (c) of the rank- $r$  part of  $\Sigma_r$ , which is open and dense in  $\Sigma_r$  and of dimension  $d$  (Auxiliary result B.7(c)).  $\mathcal{G}$  is open in the meet, since there  $\mathrm{rank}(R - I) \geq r$  and  $-1 \notin \mathrm{spec} R$  are open conditions, and it is dense: an  $R$  in the meet with eigenvalue  $-1$  is a limit of rotations with the same moved



planes and the angles  $\pi$  replaced by  $\pi - \varepsilon$  (Auxiliary result B.7(b)). Over a point  $W_0R \in \mathcal{G}$ , the update  $W_0(R - I)$  has rank exactly  $r$ , so near such a point  $\mathcal{F}$  consists of the pairs  $(B, A)$  of full rank with  $BA \in W_0(R - I)$  for  $W_0R \in \mathcal{G}$ , with  $R$  recovered from  $BA$  by injectivity. The map  $(B, A) \mapsto BA$  is a submersion from pairs of full rank onto  $\mathcal{M}_r$  with fibres the  $\mathrm{GL}_r$ -orbits, of dimension  $r^2$  (Auxiliary result B.5(b)). A submersion is transverse to every submanifold, so by Auxiliary result B.2(a), applied with target  $\mathcal{M}_r$ ,  $\mathcal{F}$  is there a submanifold of dimension  $d + r^2 = rn + r(r - 1)/2$ . If HRA, or any method with an  $rn$ -dimensional parameter space, were the product, its underlying set would be in bijection with  $\mathcal{F}$  through its two smooth projections (Observation I.4(b)), so  $\mathcal{F}$  would be the image of an  $rn$ -manifold  $P$  under a smooth map  $\iota$  into the ambient space  $E = \mathbb{R}^{m \times r} \times \mathbb{R}^{r \times n} \times \mathbb{R}^{n \times n}$ . This is impossible, because  $\mathcal{F}$  contains an embedded submanifold  $\mathcal{N}$  of dimension  $D = rn + r(r - 1)/2 > rn$  (the part over  $\mathcal{G}$ ). Fix  $p \in \mathcal{N}$  and let  $\pi : E \rightarrow T_p\mathcal{N} \cong \mathbb{R}^D$  be the orthogonal projection, which is 1-Lipschitz. Its restriction to  $\mathcal{N}$  has invertible differential at  $p$ , so by the inverse function theorem  $\pi(\mathcal{N}')$  contains a closed ball  $\tilde{B}$  of positive volume for a compact neighbourhood  $\mathcal{N}'$  of  $p$  in  $\mathcal{N}$ . Cover  $P$  by countably many compact sets  $K_j = \varphi_j([0, 1]^{rn})$ , images of cubes under charts, on which  $\iota \circ \varphi_j$  is Lipschitz with a constant  $L_j$ . Cutting the cube into  $k^{rn}$  subcubes of side  $1/k$  covers  $\pi(\iota(K_j))$  by  $k^{rn}$  sets of diameter at most  $L_j\sqrt{rn}/k$ , so its outer Lebesgue measure is at most  $k^{rn} \omega_D (L_j\sqrt{rn}/k)^D$ , which tends to 0 as  $k \rightarrow \infty$  because  $D > rn$ . The countably many null sets  $\pi(\iota(K_j))$  would then cover  $\pi(\mathcal{N}') \supseteq \tilde{B}$ , which is impossible. For the parenthetical remark: over  $W_0$  itself  $\mathcal{F}$  contains  $\{(B, 0, I)\}$ , of dimension  $mr$ , while over points of  $\mathcal{G}$  its fibre has dimension  $r^2$ . So the fibres of  $\mathcal{F}$  over the meet jump, and whenever  $mr > d + r^2$  (for instance when  $m \geq n + r$ , as at  $(m, n, r) = (12, 9, 4)$ , where  $mr = 48 > 42$ ) the local dimension of  $\mathcal{F}$  at these points exceeds  $d + r^2$  and  $\mathcal{F}$  is not pure-dimensional.

(d), *exactly orthogonal OFT*. If  $C(\Omega)x = -x$ , then  $y = (I - \Omega)^{-1}x$  satisfies  $(I + \Omega)y = -(I - \Omega)y$ , so  $2y = 0$  and  $x = 0$ ; this is the statement that  $C$  misses eigenvalue  $-1$ . Exact full-block Cayley OFT has image  $W_0 \cdot C(\mathfrak{so}(n)) = W_0 \cdot \mathcal{O}$  (input-side forms  $W_0C(\Omega)^\top = W_0C(-\Omega)$  give the same set). Its intersection with  $W_0 + \mathcal{M}_{\leq r}$  is  $W_0 \cdot C(\Sigma_r) \subseteq \mathrm{Im} \mathrm{HRA}_r$ , which contains  $\mathcal{G}$  and is therefore dense in  $\mathrm{Im} \mathrm{HRA}_r$ . The point  $W_0H_{e_1}H_{e_2}$  lies in  $\mathrm{Im} \mathrm{HRA}_r$  (pad with pairs), and by injectivity it equals  $W_0R$  only for  $R = \mathrm{diag}(-1, -1, 1, \dots, 1)$ , which has eigenvalue  $-1$ ; so it is missing. For block Cayley OFT with block size  $b < n$  the image consists of  $W_0R$  with  $R$  block-diagonal for a partition with at least two blocks. Choose coordinates  $p, q$  in different blocks and  $v = (e_p + e_q)/\sqrt{2}$ . Then  $R = H_{e_p}H_v \in \mathrm{SO}(n)$  has  $\mathrm{rank}(R - I) \leq 2 \leq r$  and  $Re_p = H_{e_p}e_q = e_q$ , so  $R$  is not block-diagonal. Hence  $W_0R \in \mathrm{Im} \mathrm{HRA}_r$  lies outside the image of block OFT, by injectivity.

(d), *HF's default OFT*. Its image consists of the  $W_0R(Q)$ ,  $R(Q) = I + 2Q + 2Q^2 + 2Q^3 + Q^4$ , with  $Q$  skew and block-diagonal (the input-side form uses  $R(Q)^\top = R(-Q)$ , of the same kind). By the proof of Corollary II.6,  $R(Q)^\top R(Q) = (I - Q^4)^2$  has eigenvalues  $(1 - \theta_k^4)^2$ , so  $R(Q)$  is orthogonal only if every  $\theta_k \in \{0, 2^{1/4}\}$ . If  $\|Q\|_{\mathrm{op}} < 2^{1/4}$  this forces every  $\theta_k = 0$ , hence  $Q = 0$  since a skew matrix is normal. As  $\mathrm{Im} \mathrm{HRA}_r \subseteq W_0 \cdot \mathcal{O}(n)$  and  $W_0$  is injective, the parameters near  $Q = 0$  reach  $\mathrm{Im} \mathrm{HRA}_r$  only at  $W_0$ . For  $Q = \theta(e_1e_2^\top - e_2e_1^\top)$  with  $\theta \notin \{0, 2^{1/4}\}$  (one plane, inside one block),  $R(Q) - I = Q(2I + 2Q + 2Q^2 + Q^3)$  has rank at most  $2 \leq r$  while  $R(Q)$  is not orthogonal, so  $W_0R(Q) \in W_0 + \mathcal{M}_{\leq r}$  lies outside  $W_0 \cdot \mathcal{O}(n) \supseteq \mathrm{Im} \mathrm{HRA}_r$ . Conversely, an orthogonal  $R(Q)$  acts on each plane of  $Q$  with  $\theta_k = 2^{1/4}$  as  $-C(2^{1/4}J)$ , a rotation by  $\pi + 2 \arctan 2^{1/4}$ , and as the identity on  $\ker Q$ ; so a rotation by a small angle  $\varphi \neq 0$  in one plane, which lies in  $\mathrm{Im} \mathrm{HRA}_r$  by (a), is not of the form  $R(Q)$ . The two images are therefore incomparable.

(d), *implementation*. For orthonormal  $\hat{u}_1, \dots, \hat{u}_r$ ,  $(I - 2\hat{u}_1\hat{u}_1^\top)(I - 2\hat{u}_2\hat{u}_2^\top) = I - 2\hat{u}_1\hat{u}_1^\top - 2\hat{u}_2\hat{u}_2^\top$  because  $\hat{u}_1^\top\hat{u}_2 = 0$ , and by induction  $\prod_i (I - 2\hat{u}_i\hat{u}_i^\top) = I - 2P_U$  with  $U = \text{span}\{\hat{u}_i\}$ . So the Gram–Schmidt variant has image  $\{W_0(I - 2P_U) : U \in \text{Gr}(r, n)\}$ . The map  $U \mapsto W_0(I - 2P_U)$  is injective ( $P_U$  determines  $U$ , and  $W_0$  is injective) and is an immersion of the compact manifold  $\text{Gr}(r, n)$ , so the image has dimension  $r(n - r)$ . It never contains  $W_0$ , since  $I - 2P_U \neq I$  for  $U \neq 0$ . With duplicated columns, Gram–Schmidt meets a column that lies in the span of the previous ones, whose residual vanishes, and normalising it divides by zero up to rounding. Injectivity of  $W_0 \in \mathbb{R}^{\text{out} \times \text{in}}$  needs  $\text{out} \geq \text{in}$  and full column rank, which fails for `down_proj` and for GQA key and value projections.  $\square$

### C.14 The LOFT example of Exposé II.B

**EXAMPLE** (LOFT: a leaf, and how to choose it; §II.B). Let  $P_r \in \mathbb{R}^{r \times n}$  have orthonormal rows spanning  $V$ , and let  $\text{O}(V) = \{I + P_r^\top(T - I)P_r : T \in \text{O}(r)\}$  be the orthogonal maps that fix  $V^\perp$  pointwise. LOFT trains  $\rho(E) = W_0(I + P_r^\top(Q(E) - I)P_r)$  with  $E \in \mathfrak{so}(r)$ ,  $E_0 = 0$ , so  $|M| = r(r - 1)/2$ . We read it with an exactly orthogonal  $Q$  (the exponential, or the Cayley map  $C$ ); the released code defaults to a five-term Cayley–Neumann series with  $R^\top R = (I - X^6)^2$ , which is only approximately orthogonal, so like HF’s default OFT (Corollary II.6) that version leaves its leaf.

- (a) One leaf. If  $W_0 P_r^\top$  has rank  $r$ , then  $(W_0 + W_0 P_r^\top \mathbb{R}^{r \times r} P_r) \cap W_0 \cdot \text{O}(n) = W_0 \cdot \text{O}(V)$ . For the principal support  $P_r = V_r^\top$  the first set is the image of LoRA-XS (this needs  $\sigma_r(W_0) > 0$ ), so the leaf is  $\text{LoRA-XS} \cap W_0 \cdot \text{O}(n)$ . Principal-support LOFT reaches an open dense part of its component  $W_0 \cdot \text{SO}(V)$ ; for injective  $W_0$  and a Cayley  $Q$ , its image is exactly  $\text{LoRA-XS} \cap$  the image of exact full-block Cayley OFT.
- (b) Leaves over the Grassmannian. If  $W_0$  is injective,  $(W_0 + \mathcal{M}_{\leq r}) \cap W_0 \cdot \text{SO}(n) = \bigcup_{V \in \text{Gr}(r, n)} W_0 \cdot \text{SO}(V)$ , which for even  $r$  is  $\text{Im HRA}_r$  (Theorem II.7(a)). LOFT’s image is the whole leaf  $W_0 \cdot \text{SO}(V)$  with the exponential, and an open dense part of it with a Cayley map, which never reaches eigenvalue  $-1$ .
- (c) A regular base point. If  $W_0 P_r^\top$  has rank  $r$ , then  $S_1 = W_0 P_r^\top \mathfrak{so}(r) P_r$  has dimension  $r(r - 1)/2 = d(M)$ : LOFT starts at a smooth point of its leaf and has no gauge, while HRA’s paired initialisation is an apex (Theorem II.7(b)). For  $r \geq 4$  there is no smooth pointed arrow  $\text{LOFT} \rightarrow \text{HRA}$  at that initialisation, although the images are nested (obstruction test O3 of §VIII).
- (d) Choosing the leaf. If  $Q'(0) = \text{id}$  (the exponential, or  $E \mapsto C(E/2)$ ), the gradient pulled back to  $E$  at  $\theta_0$  is  $P_r \text{skew}(W_0^\top G) P_r^\top$  with  $G = \nabla L(W_0)$ ; the Cayley map  $C$  itself doubles it, which does not change the best support. Its squared norm is at most  $2 \sum_{k \leq r/2} \mu_k^2$ , where  $\pm i\mu_k$  are the eigenvalues of  $\text{skew}(W_0^\top G)$ , with equality when  $V$  spans the invariant subspace of the top  $\lfloor r/2 \rfloor$  pairs.

*Proof.* (a) A rotation  $T \in \text{O}(r)$  gives  $W_0(I + P_r^\top(T - I)P_r) = W_0 + W_0 P_r^\top(T - I)P_r$ , so “ $\supseteq$ ” holds. Conversely let  $W = W_0 + BXP_r$  with  $B = W_0 P_r^\top$ , and suppose  $W = W_0 S$  with  $S \in \text{O}(n)$ . Then  $WW^\top = W_0 W_0^\top$ , and expanding with  $P_r P_r^\top = I_r$  gives  $B(X + X^\top + XX^\top)B^\top = 0$ . Since  $B$  has full column rank,  $X + X^\top + XX^\top = 0$ , that is  $(I + X)(I + X)^\top = I$ , so  $T = I + X \in \text{O}(r)$  and  $W = W_0(I + P_r^\top(T - I)P_r) \in W_0 \cdot \text{O}(V)$ . For  $P_r = V_r^\top$  we have  $B = U_r \Sigma_r$ , and  $\{U_r \Sigma_r X V_r^\top\} = \{U_r R \Sigma_r V_r^\top\}$  through  $R = \Sigma_r X \Sigma_r^{-1}$ , which is LoRA-XS’s image. With the exponential,  $T$  ranges over  $\text{SO}(r)$  and LOFT reaches the whole component  $W_0 \cdot \text{SO}(V)$ ; with a Cayley map it reaches the  $S(T) = I + P_r^\top(T - I)P_r$  with  $T \in \text{SO}(r)$  and no eigenvalue  $-1$ , an open dense part of it. For

injective  $W_0$ ,  $W_0 S$  determines  $S$ , so meeting with exact full-block Cayley OFT keeps exactly the  $S(T)$  with  $T \in \text{SO}(r)$  and no eigenvalue  $-1$  (the eigenvalues of  $S(T)$  are those of  $T$  together with 1), which is the Cayley image.

(b) Every  $S \in \text{SO}(V)$  has  $\text{rank}(S - I) \leq r$ , which gives “ $\supseteq$ ”. Conversely, if  $W_0 S - W_0$  has rank at most  $r$  and  $W_0$  is injective, then  $\text{rank}(S - I) \leq r$ . The fixed space  $F = \ker(S - I)$  has dimension at least  $n - r$ , and  $F^\perp$  is invariant. Any  $r$ -dimensional  $V \supseteq F^\perp$  works:  $S$  is the identity on  $V^\perp \subseteq F$  and has determinant 1 on  $V$ . The statements about LOFT’s image are those of (a).

(c) The differential at 0 is  $D\rho_0(E) = W_0 P_r^\top Q'(0)(E) P_r$ , with  $Q'(0)$  invertible. If this vanishes, then  $W_0 P_r^\top Q'(0)(E) = 0$  because  $P_r$  has full row rank, and  $E = 0$  because  $W_0 P_r^\top$  has full column rank. So  $\dim S_1 = |M| \geq d(M) \geq \dim S_1$ . A smooth pointed arrow would give  $S_1(\text{LOFT}) \subseteq S_1(\text{HRA})$  ([Observation I.6](#)), and by [Theorem II.7\(b\)](#) and injectivity this asks every  $\Omega \in \mathfrak{so}(V)$  to satisfy  $P_{U^\perp} \Omega P_{U^\perp} = 0$ . Since  $\dim(V \cap U^\perp) \geq r - r/2 \geq 2$ , any two orthonormal  $a, b \in V \cap U^\perp$  give  $a \wedge b = ab^\top - ba^\top$  violating it.

(d) With  $Q'(0) = \text{id}$ ,

$$\frac{d}{dt} L(\rho(tE)) \Big|_{t=0} = \langle G, W_0 P_r^\top E P_r \rangle = \langle P_r W_0^\top G P_r^\top, E \rangle = \langle P_r \text{skew}(W_0^\top G) P_r^\top, E \rangle,$$

since  $E$  is skew. The squared norm of this gradient is  $\|P_r \text{skew}(W_0^\top G) P_r^\top\|^2$ , the compression of the skew matrix  $\text{skew}(W_0^\top G)$  to  $V$ , so the bound and its equality case are [Auxiliary result B.17\(c\)](#) with  $k = r$ .  $\square$

### C.15 Proposition II.8

**PROPOSITION II.8** (DoRA is LoRA followed by an output scale, at the level of images). Assume  $1 \leq r \leq \min(m, n)$  and  $s \neq 0$ . Let  $N$  normalise rows; DoRA’s magnitude  $\mu \in \mathbb{R}^m$  is per output neuron, as in HF PEFT and the official code (the paper’s notation says “column-wise”, in the transposed convention). Images refer to the merged (eval-mode) weight; HF’s train-mode forward with `lora_dropout > 0` is not of the form  $Wx$ . Let  $W_0$  have no zero row, let  $\mathcal{U}$  be the open set of matrices with no zero row, and let  $\text{Diag}_m$  be the monoid of all diagonal matrices. Then

$$\text{Im DoRA}_r = \{\text{diag}(\mu) N(V) : V \in (W_0 + \mathcal{M}_{\leq r}) \cap \mathcal{U}, \mu \in \mathbb{R}^m\} = \text{Diag}_m \cdot (W_0 + \mathcal{M}_{\leq r}),$$

which is exactly the image of “LoRA<sub>r</sub> followed by a trainable output scale  $\ell \in \mathbb{R}^m$ ”. With  $\mu$  restricted to  $(\mathbb{R}^\times)^m$ , or equivalently after intersecting with  $\mathcal{U}$ :  $\text{Im DoRA}_r \cap \mathcal{U} = T_m \cdot (W_0 + \mathcal{M}_{\leq r}) \cap \mathcal{U}$ . Moreover:

- $d(\text{DoRA}_r) \leq \min(mn, r(m + n - r) + m)$ . At a parameter point with  $\mu \in (\mathbb{R}^\times)^m$  and  $X = sBA \in \mathcal{M}_r$ , with  $U = \text{col } X$  and  $R = \text{row } X$ , the Jacobian rank is  $r(m + n - r) + m - \dim \ker(\delta \mapsto P_{U^\perp} \text{diag}(\delta) W_0 P_{R^\perp})$ ; where some  $\mu_i = 0$  it is smaller when  $r \geq 2$ , and for  $r = 1$  when some row  $w_{0,i}$  with  $\mu_i = 0$  lies outside  $R$ . For generic  $W_0$ ,  $d = \min(mn, r(m + n - r) + m)$ ; in particular  $d = r(m + n - r) + m$  iff  $(m - r)(n - r) \geq m$ , which holds for all practical shapes. For special  $W_0$  it is smaller: for  $W_0 = ac^\top$  with all  $a_i \neq 0$  and  $r < n$ ,  $d = r(m + n - r) + m - r$ , strictly below the generic value iff  $r < m$  and  $r \leq n - 2$ .
- The image covariance group ([Definition II.14](#)) contains  $\text{Mon}_m \times \text{GL}_n$ .
- At  $B = 0$ , in HF’s magnitude coordinates,  $\Delta W = (\text{diag}(\mu_i / \|w_{0,i}\|) - I) W_0$ , which can have rank up to  $\text{rank } W_0$ , so DoRA escapes LoRA’s rank bound.

*Proof.* DoRA is the map  $\rho(\mu, B, A) = \text{diag}(\mu)N(V)$  with  $V = W_0 + sBA$ , defined on the open set  $\mathcal{D} = \{(\mu, B, A) : V \in \mathcal{U}\}$ , where  $N(V)$  divides row  $i$  by  $\|v_i\|$ . Since  $s \neq 0$  and every matrix of rank at most  $r$  factors with inner dimension  $r$  (Auxiliary result B.11(a)),  $\{sBA\} = \mathcal{M}_{\leq r}$ .

*The image.*  $\text{diag}(\mu)N(V) = \text{diag}(\mu_i/\|v_i\|)V \in \text{Diag}_m \cdot (W_0 + \mathcal{M}_{\leq r})$ , which gives the first inclusion; the first equality is the definition of  $\rho$ . Conversely, take  $\text{diag}(\ell)V$  with  $V = W_0 + X$ ,  $\text{rank } X \leq r$ , and let  $Z = \{i : v_i = 0\}$ . For  $0 < \varepsilon \leq 1$  put

$$V' = W_0 + \left(I - \varepsilon \sum_{i \in Z} E_{ii}\right)X.$$

Its update still has rank at most  $r$ . Its rows outside  $Z$  are those of  $V$ , and for  $i \in Z$  we have  $x_i = -w_{0,i}$ , so row  $i$  of  $V'$  is  $w_{0,i} - (1 - \varepsilon)w_{0,i} = \varepsilon w_{0,i} \neq 0$ . Hence  $V' \in \mathcal{U}$ . With  $\ell'_i = 0$  on  $Z$  and  $\ell'_i = \ell_i$  elsewhere,  $\text{diag}(\ell')V' = \text{diag}(\ell)V$ , since both vanish on the rows in  $Z$ . Taking  $\mu_i = \ell'_i\|v'_i\|$  gives  $\text{diag}(\mu)N(V') = \text{diag}(\ell)V$ . The set  $\text{Diag}_m \cdot (W_0 + \mathcal{M}_{\leq r})$  is by definition the image of  $(\ell, B, A) \mapsto \text{diag}(\ell)(W_0 + sBA)$ .

*The torus form.* If  $\mu \in (\mathbb{R}^\times)^m$ , then  $\text{diag}(\mu_i/\|v_i\|) \in T_m$  and the result has no zero row. Conversely, if  $DV \in \mathcal{U}$  with  $D \in T_m$  and  $V \in W_0 + \mathcal{M}_{\leq r}$ , then  $V \in \mathcal{U}$  and  $\mu_i = D_{ii}\|v_i\| \neq 0$  realises  $DV$ . Finally,  $\text{diag}(\ell)V \in \mathcal{U}$  forces every  $\ell_i \neq 0$  and  $V \in \mathcal{U}$ , so  $\text{Im DoRA}_r \cap \mathcal{U} = T_m \cdot (W_0 + \mathcal{M}_{\leq r}) \cap \mathcal{U}$ , and these are exactly the values at  $\mu \in (\mathbb{R}^\times)^m$ .

*A change of coordinates.* Let  $\rho'(\ell, B, A) = \text{diag}(\ell)(W_0 + sBA)$  be LoRA followed by an output scale. The map  $\varphi(\mu, B, A) = (\mu_i/\|v_i\|, B, A)$  is a diffeomorphism of  $\mathcal{D}$  onto  $\{(\ell, B, A) : W_0 + sBA \in \mathcal{U}\}$ , with inverse  $(\ell, B, A) \mapsto (\ell_i\|v_i\|, B, A)$ , and  $\rho = \rho' \circ \varphi$ . So the Jacobian rank of  $\rho$  at  $p$  equals that of  $\rho'$  at  $\varphi(p)$ , and  $\mu_i = 0$  iff  $\ell_i = 0$ .

*Upper bound.*  $\rho'$  factors through  $(\ell, B, A) \mapsto (\ell, W_0 + sBA) \in \mathbb{R}^m \times (W_0 + \mathcal{M}_{\leq r})$ , whose image has dimension  $m + r(m + n - r)$  (Auxiliary result B.5(a); for  $r = \min(m, n)$ ,  $\mathcal{M}_{\leq r} = \mathbb{R}^{m \times n}$  has dimension  $mn = r(m + n - r)$ ). A polynomial map does not raise dimension (Auxiliary result B.9(b)), and trivially  $d \leq mn$ .

*The Jacobian rank.* The differential of  $\rho'$  at  $(\ell, B, A)$  is  $(\delta, \dot{B}, \dot{A}) \mapsto \text{diag}(\delta)V + \text{diag}(\ell)s(\dot{B}A + B\dot{A})$ . Let  $\ell$  be invertible and  $X = sBA \in \mathcal{M}_r$ , so that  $B$  and  $A$  have full rank and  $\{s(\dot{B}A + B\dot{A})\} = T_X\mathcal{M}_r = \{D : P_{U^\perp}DP_{R^\perp} = 0\}$ , of dimension  $r(m + n - r)$  (Auxiliary result B.5(b)). Substituting  $\delta = \text{diag}(\ell)\delta'$ , the image of the differential is  $\text{diag}(\ell)[\{\text{diag}(\delta')V\} + T_X\mathcal{M}_r]$ , whose dimension is that of the bracket. The first summand has dimension  $m$ , because  $V$  has no zero row. A matrix  $\text{diag}(\delta')V$  lies in  $T_X\mathcal{M}_r$  iff  $P_{U^\perp}\text{diag}(\delta')VP_{R^\perp} = 0$ , and  $VP_{R^\perp} = W_0P_{R^\perp}$  because  $XP_{R^\perp} = 0$ . So the intersection has the dimension of the kernel  $K$  of  $\delta \mapsto P_{U^\perp}\text{diag}(\delta)W_0P_{R^\perp}$ , and the rank is  $r(m + n - r) + m - \dim K$ .

*Zero magnitudes.* Let  $Z_0 = \{i : \ell_i = 0\} \neq \emptyset$ , keep  $X \in \mathcal{M}_r$ , and write  $S = \{\text{diag}(\delta)V\}$  and  $T = T_X\mathcal{M}_r$ . Let  $\Pi_0$  keep the rows in  $Z_0$  and  $\Pi_1$  the others. In the image of the differential, the rows in  $Z_0$  are  $\delta_i v_i$  and involve only  $\delta_{Z_0}$ , while the other rows do not involve  $\delta_{Z_0}$ ; so the image has dimension  $|Z_0| + \dim \Pi_1(S + T)$ , after the same substitution as above on the rows outside  $Z_0$ . On the other hand, the value  $F = \dim(S + T)$  of the formula satisfies  $F = \dim \Pi_1(S + T) + \dim((S + T) \cap \ker \Pi_1)$ , and  $(S + T) \cap \ker \Pi_1$  contains every  $\sum_{i \in Z_0} e_i(c_i v_i + y_i)^\top$  with  $y_i \in R$ , since  $e_i v_i^\top \in S$  and  $e_i y^\top \in T$  for  $y \in R$ . As  $v_i - w_{0,i} \in R$ , the rank at the point is at most

$$F - \sum_{i \in Z_0} (\dim \text{span}(w_{0,i}, R) - 1) \leq F - |Z_0|(r - 1),$$

which is smaller than  $F$  whenever  $r \geq 2$  or some  $w_{0,i}$  with  $i \in Z_0$  lies outside  $R$ , in particular at generic points with some  $\mu_i = 0$ .



*The generic value.* If  $r = \min(m, n)$ , the image is  $\text{Diag}_m \cdot \mathbb{R}^{m \times n} = \mathbb{R}^{m \times n}$  and  $d = mn$ , which is the claimed value. Let  $r < \min(m, n)$ , and let  $E \in \mathbb{R}^{m \times (m-r)}$  and  $F \in \mathbb{R}^{n \times (n-r)}$  have orthonormal columns spanning  $U^\perp$  and  $R^\perp$ . Then  $P_{U^\perp} D P_{R^\perp} = 0$  iff  $E^\top D F = 0$ , and

$$E^\top \text{diag}(\delta) W_0 F = \sum_i \delta_i (E^\top e_i) (F^\top w_{0,i})^\top,$$

so  $m - \dim K$  is the dimension  $k$  of the span of the  $m$  rank-one matrices  $t_i = (E^\top e_i) (F^\top w_{0,i})^\top$  in  $\mathbb{R}^{(m-r) \times (n-r)}$ , and the rank is  $r(m+n-r) + k$  with  $k \leq \min(m, (m-r)(n-r))$ . Since  $r(m+n-r) + (m-r)(n-r) = mn$ , it suffices to reach  $k = \min(m, (m-r)(n-r))$ . A witness: choose  $E$  whose rows  $p_i = E^\top e_i$  are in general position (every  $m-r$  of them independent; orthonormalising the columns of a generic matrix keeps this), let  $U = (\text{col } E)^\perp$ ,  $R = \text{span}\{e_1, \dots, e_r\}$  and  $F = [e_{r+1} \dots e_n]$ . Assign to each  $i$  a column index  $c(i) \in \{1, \dots, n-r\}$ : if  $m \leq (m-r)(n-r)$ , use each index at most  $m-r$  times; otherwise give each index to exactly  $m-r$  of the first  $(m-r)(n-r)$  values of  $i$ , and arbitrary indices to the rest. Put  $w_{0,i} = e_{r+c(i)}$ , so  $F^\top w_{0,i} = e_{c(i)}$  and  $t_i = p_i e_{c(i)}^\top$  has  $p_i$  in column  $c(i)$ . Matrices supported in different columns are independent, and in one column at most  $m-r$  vectors in general position are independent, while exactly  $m-r$  of them form a basis; so the  $t_i$  span  $\min(m, (m-r)(n-r))$  dimensions. This  $W_0$  has no zero row, and with  $B, A$  of full rank spanning  $U$  and  $R$  and  $\mu_i = \|v_i\|$  the point lies in  $\mathcal{D}$ , since  $v_i = w_{0,i} + x_i$  with  $x_i \in R \perp w_{0,i}$ . Now write  $P_{U^\perp} = I - B(B^\top B)^{-1} B^\top$  and  $P_{R^\perp} = I - A^\top (A A^\top)^{-1} A$ . After clearing the denominators  $\det(B^\top B)$  and  $\det(A A^\top)$ , the  $k \times k$  minors of the matrix of  $\delta \mapsto P_{U^\perp} \text{diag}(\delta) W_0 P_{R^\perp}$  are polynomials in  $(W_0, B, A)$ , and one of them is nonzero at the witness. Viewed as a polynomial in  $(B, A)$ , its coefficients are polynomials in  $W_0$ , not all identically zero; for  $W_0$  outside their common zero set it is nonzero on an open dense set of  $(B, A)$  ([Auxiliary result B.10](#)), which meets the open dense set where  $B, A$  have full rank and  $W_0 + sBA \in \mathcal{U}$ . At such a point with  $\ell = \mathbf{1}$  the rank is  $r(m+n-r) + \min(m, (m-r)(n-r))$ . With the upper bound,  $d = \min(mn, r(m+n-r) + m)$  for  $W_0$  outside a proper algebraic subset, and  $d = r(m+n-r) + m$  iff  $(m-r)(n-r) \geq m$ .

*Rank-one  $W_0$ .* Let  $W_0 = ac^\top$  with all  $a_i \neq 0$  and  $r < n$ . At a point with  $\ell$  invertible and  $X \in \mathcal{M}_r$ ,  $P_{U^\perp} \text{diag}(\delta) ac^\top P_{R^\perp} = (P_{U^\perp} (\delta \odot a)) (P_{R^\perp} c)^\top$ . If  $c \notin R$ , this vanishes iff  $\delta \odot a \in U$ , so  $K = \text{diag}(a)^{-1} U$  has dimension  $r$  and the rank is  $r(m+n-r) + m - r$ . If  $c \in R$ , then  $K = \mathbb{R}^m$  and the rank is  $r(m+n-r)$ . The points with  $\ell$  invertible,  $X \in \mathcal{M}_r$  and  $c \notin R$  form an open dense subset of the domain (since  $r < n$ ), and the points of maximal rank form an open nonempty subset, so the two meet and  $d = r(m+n-r) + m - r$ . This is at most  $mn$ , since  $(m-r)(n-r-1) \geq 0$ . It is strictly below the generic value  $r(m+n-r) + \min(m, (m-r)(n-r))$  iff  $m-r < (m-r)(n-r)$ , that is iff  $r < m$  and  $r \leq n-2$ .

*Covariance.* Let  $g = P_\pi D$  be monomial, with  $P_\pi$  a permutation matrix and  $D \in T_m$ , and let  $h \in \text{GL}_n$ . Then  $g W_0 h^{-1}$  has no zero row, and  $g \text{diag}(x) g^{-1} = \text{diag}(P_\pi x)$ , so  $g \text{Diag}_m g^{-1} = \text{Diag}_m$ ; also  $g \mathcal{M}_{\leq r} h^{-1} = \mathcal{M}_{\leq r}$ . Hence

$$g (\text{Diag}_m \cdot (W_0 + \mathcal{M}_{\leq r})) h^{-1} = \text{Diag}_m \cdot (g W_0 h^{-1} + \mathcal{M}_{\leq r}) = \text{Im DoRA}_r(g W_0 h^{-1}).$$

$\text{Mon}_m \times \text{GL}_n$  is a subgroup with this property, so it lies in the covariance group (proof of [Definition II.14](#)).

*The update at  $B = 0$ .* HF initialises  $\mu_i = \|w_{0,i}\|$  and then trains  $\mu$ ; at  $B = 0$  the weight is  $\text{diag}(\mu_i / \|w_{0,i}\|) W_0$ , so  $\Delta W = (\text{diag}(\mu_i / \|w_{0,i}\|) - I) W_0$ . With  $\mu_i = 2\|w_{0,i}\|$  for every  $i$ ,  $\Delta W = W_0$ , of rank  $\text{rank } W_0$ , which exceeds  $2r$  whenever  $\text{rank } W_0 > 2r$ , beyond the bound of [Theorem II.1\(c\)](#).  $\square$



## C.16 Proposition II.9

**PROPOSITION II.9** (the torus ladder).

- (a) **HiRA.**  $\rho(B, A) = W_0 \odot (J + BA)$ , where  $J$  is the all-ones matrix; HF PEFT has  $B_0 = 0$  and no scale factor on Linear and Conv layers (on embeddings it uses  $A_0 = 0$ ). For every  $W_0$ ,  $\text{Im HiRA}_r = W_0 + D_{W_0}(\mathcal{M}_{\leq r})$  with  $D_{W_0} : X \mapsto W_0 \odot X$ . If  $W_0$  has no zero entries,  $D_{W_0} \in \text{GL}(\Theta)$ , so inside the slice at  $\theta_0$  HiRA is LoRA transported by a diagonal automorphism: it has the same  $d = r(m + n - r)$  (for  $r \leq \min(m, n)$ ) and the same gauge. In all cases  $\text{rank } \Delta W \leq r \cdot \text{rank } W_0$ . The transport depends on  $\theta_0$  (Proposition II.13).
- (b) **Scalings.** Output-side scalings,  $\text{diag}(\ell)W_0 = W_0 \odot (J + (\ell - \mathbf{1})\mathbf{1}^\top)$ , and input-side scalings,  $W_0 \text{diag}(\ell) = W_0 \odot (J + \mathbf{1}(\ell - \mathbf{1})^\top)$ , have images in  $\text{Im HiRA}_1$ . Output-side examples:  $(\text{IA})^3$  on keys and values, the post-linear scale of SSF, DoRA’s magnitude. Input-side examples:  $(\text{IA})^3$ ’s feed-forward vector, which scales the input of  $W_{\text{down}}$ , and a LayerNorm gain  $\gamma$  folded into the next linear map. The fold is exact when the norm’s output is consumed only by linear maps: the input norms of pre-norm blocks, and a final norm before an untied head (in post-norm BERT, the last block’s norm feeds only the head). It does not apply to Gemma-style post-sublayer norms, which HF’s default LN-tuning targets for Gemma 2 and 3, to QK-norms, or to post-norm residual paths. Where it applies,  $\gamma$  is tied across all consumers of the norm (Q, K and V, or gate and up) rather than being a free point per box, and with tied embeddings a fold into the head breaks the tie. LayerNorm biases and SSF’s shift fold into the next bias, not into  $W$ . Two-sided scalings  $\text{diag}(a)W_0 \text{diag}(b) = W_0 \odot (ab^\top)$  lie in  $\text{Im HiRA}_2$ . As a morphism,  $h(\ell) = (\ell - \mathbf{1}, \mathbf{1}^\top)$  lands in  $\text{HiRA}_1$  pointed at  $(0, \mathbf{1}^\top)$ , which is not HiRA’s random-A default pointing.
- (c) **Orbit dimension.** The two-sided torus orbit  $\{\text{diag}(a)W_0 \text{diag}(b)\}$  has dimension  $m + n - c(W_0)$ , where  $c(W_0)$  is the number of connected components of the bipartite support graph of  $W_0$ , an isolated vertex counting as its own component.
- (d) **RoAd.**  $\text{RoAd}_1$  (the paper’s main variant, HF road\_1) is  $W = RW_0$  with  $R$  block-diagonal in scaled rotations  $\alpha_i \text{Rot}(\theta_i)$  of output pairs. It is the left action of  $\text{CO}^+(2)^{m/2} \cong (\mathbb{C}^\times)^{m/2}$ : it is  $(\text{IA})^3$  over  $\mathbb{C}$ . It multiplies the input Gram matrix of each output pair by  $\alpha_i^2$ . Since  $\alpha_i$  may be 0, its image is the orbit closure, a real-linear space of dimension exactly  $m$  whenever no output-pair block of  $W_0$  is zero.  $\text{RoAd}_2$  and  $\text{RoAd}_4$  (HF road\_2, road\_4) let each block range over all of  $M_2(\mathbb{R})$ . They realise the block-diagonal action of the monoid  $M_2(\mathbb{R})^{m/2}$  (its group part  $\text{GL}_2^{m/2}$  is non-abelian), with linear image  $\{\text{blockdiag}(R_i)W_0\}$  of dimension  $2m$  when every pair block has rank 2. For them the Gram statement and “over  $\mathbb{C}$ ” fail, and  $\text{RoAd}_4$  carries a positive-dimensional gauge (two dimensions per row). HF pairs output  $j$  with  $j + g/2$  inside each group of size  $g$  (default 64) rather than adjacent outputs as in the paper, and also rotates the bias; the statements above do not depend on the pairing.
- (e) **Zeros and covariance.** Every  $W$  reachable by HiRA,  $(\text{IA})^3$  or SSF vanishes wherever  $W_0$  does,  $Z(W_0) \subseteq Z(W)$ , and this persists under merging. New zeros can appear ( $\ell_i = 0$ , or  $(J + BA)_{ij} = 0$ ), so the zero pattern itself is not invariant. The image covariance group of output-scaled boxes  $((\text{IA})^3$  on  $k, v$ ; post-linear SSF) is  $\text{Mon}_m \times \text{GL}_n$ , and that of input-scaled boxes  $((\text{IA})^3$  on  $W_{\text{down}}$ ) is  $\text{GL}_m \times \text{Mon}_n$ . HiRA’s is exactly  $\text{Mon}_m \times \text{Mon}_n (= \text{GL}_1 \times \text{GL}_1$  when  $m = n = 1$ ).

*Proof.* (a)  $W_0 \odot (J + BA) = W_0 + W_0 \odot (BA)$ , and  $\{BA\} = \mathcal{M}_{\leq r}$ , so  $\text{Im HiRA}_r = W_0 + D_{W_0}(\mathcal{M}_{\leq r})$ . In the standard basis  $E_{ij}$  of  $\Theta = \mathbb{R}^{m \times n}$ ,  $D_{W_0}$  is diagonal with entries  $(W_0)_{ij}$ , so it is invertible

iff no entry of  $W_0$  vanishes. Then, with  $\tau(\theta) = W_0 + D_{W_0}(\theta - W_0)$ , HiRA is  $\tau \circ \rho_{\text{LoRA}}$  for LoRA with residual  $W_0$  and scale 1. Since  $\tau$  is an affine bijection,  $D(\tau \circ \rho) = D_{W_0} \circ D\rho$  has the same rank as  $D\rho$  at every point, so  $d$  is unchanged (Auxiliary result B.9(d)), and  $\tau \circ \rho \circ \varphi = \tau \circ \rho$  iff  $\rho \circ \varphi = \rho$ , so the gauge group is unchanged; this is the transport  $(D_{W_0})_*$  of Proposition II.13. For  $r \leq \min(m, n)$ ,  $d(\text{LoRA}_r) = r(m + n - r)$  (Auxiliary result B.5). For the rank, with columns  $b_k$  of  $B$  and rows  $a_k$  of  $A$ ,  $W_0 \odot (BA) = \sum_{k \leq r} \text{diag}(b_k) W_0 \text{diag}(a_k)$  (Auxiliary result B.12(c)), a sum of  $r$  matrices of rank at most  $W_0$ . The map  $D_{W_0}$  is built from  $W_0 = \theta_0$ , which is the dependence on the base point.

(b) Entrywise,  $(\text{diag}(\ell)W_0)_{ij} = \ell_i(W_0)_{ij} = (W_0)_{ij}(1 + (\ell_i - 1))$ , so  $\text{diag}(\ell)W_0 = W_0 \odot (J + (\ell - \mathbf{1})\mathbf{1}^\top)$ , and  $(\ell - \mathbf{1})\mathbf{1}^\top = BA$  with  $B = \ell - \mathbf{1}$  and  $A = \mathbf{1}^\top$ , a point of  $\text{HiRA}_1$ . The input side is the transpose computation. For two-sided scalings,  $\text{diag}(a)W_0 \text{diag}(b) = W_0 \odot (ab^\top)$  and  $ab^\top - J = [a, -\mathbf{1}] [b, \mathbf{1}]^\top$  has rank at most 2. The map  $h(\ell) = (\ell - \mathbf{1}, \mathbf{1}^\top)$  is smooth, sends  $\mathbf{1}$  to  $(0, \mathbf{1}^\top)$  and satisfies  $\rho_{\text{HiRA}_1} \circ h(\ell) = \text{diag}(\ell)W_0$ , so it is a pointed morphism into  $\text{HiRA}_1$  pointed there. For a norm whose output  $\gamma \odot \hat{x}$  (with  $\hat{x}$  the normalised input) feeds only linear maps  $W$ ,  $W(\gamma \odot \hat{x}) = (W \text{diag}(\gamma))\hat{x}$ , so the gain is an input-side scaling of each consumer, the same  $\gamma$  for all of them; a bias  $\beta$  gives  $W\beta$ , a change of the next bias. When the norm's output is added to the residual stream (post-sublayer norms, post-norm residual paths) or is used inside attention scores (QK-norms), no linear map of the adapted box absorbs  $\gamma$ . With tied embeddings the head is  $E^\top$ , and replacing it by  $E^\top \text{diag}(\gamma)$  unties it.

(c) The orbit map  $\Psi(a, b) = \text{diag}(a)W_0 \text{diag}(b)$  of  $T_m \times T_n$  has differential at  $(\mathbf{1}, \mathbf{1})$

$$(\alpha, \beta) \mapsto \text{diag}(\alpha)W_0 + W_0 \text{diag}(\beta) = ((\alpha_i + \beta_j)(W_0)_{ij})_{ij}.$$

Its kernel is cut out by  $\alpha_i + \beta_j = 0$  along every edge  $(i, j)$  of the bipartite support graph. On a connected component the value at one vertex determines all the others (with  $\beta = -\alpha$  across each edge), and an isolated vertex carries one free coordinate; so the kernel has dimension  $c(W_0)$  and the rank is  $m + n - c(W_0)$ . Because  $\Psi(gg') = g \cdot \Psi(g')$  for the commutative group  $T_m \times T_n$ , the differential has this rank at every point of the group, so the orbit, an immersed submanifold of dimension  $\dim(T_m \times T_n)$  minus that of the stabiliser, has dimension  $m + n - c(W_0)$ .

(d) Identify an output pair, a  $2 \times n$  block  $X$  with rows  $x_1, x_2$ , with the complex row  $z = x_1 + ix_2 \in \mathbb{C}^n$ . For  $\text{Rot}(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$  the rows of  $\alpha \text{Rot}(\theta)X$  are  $\alpha(\cos \theta x_1 - \sin \theta x_2)$  and  $\alpha(\sin \theta x_1 + \cos \theta x_2)$ , which correspond to  $\alpha e^{i\theta} z$ . So  $\text{RoAd}_1$  multiplies each pair by a complex scalar, the left action of  $\text{CO}^+(2)^{m/2} \cong (\mathbb{C}^\times)^{m/2}$ , the complex analogue of  $(\text{IA})^3$ 's real output scaling. The input Gram matrix of the pair becomes  $(\alpha \text{Rot} X)^\top (\alpha \text{Rot} X) = \alpha^2 X^\top X$ . As  $(\alpha, \theta)$  ranges over  $\mathbb{R} \times \mathbb{R}$ ,  $\alpha e^{i\theta}$  ranges over all of  $\mathbb{C}$ , so the image is  $\prod_i \mathbb{C} \cdot z_i$ , the closure of the orbit, a real-linear space of dimension  $2 \# \{i : z_i \neq 0\}$ , equal to  $m$  when no pair block vanishes. In  $\text{RoAd}_2$  each row of a block has its own polar pair, so each row of the  $2 \times 2$  block is an arbitrary vector of  $\mathbb{R}^2$  and the block is an arbitrary element of  $M_2(\mathbb{R})$ ;  $\text{RoAd}_4$  also reaches every element of  $M_2(\mathbb{R})$ , with eight parameters for four entries, so its fibres have dimension 4 per block, two per row. The image  $\{R_i X_i\}$  of one block is the set of  $2 \times n$  matrices whose rows lie in row  $X_i$ , of dimension  $2 \text{rank } X_i$ , so the total is  $2m$  when every block has rank 2. For  $R = \text{diag}(1, 2)$  and  $X = I_2$ ,  $(RX)^\top (RX) = \text{diag}(1, 4)$  is not a multiple of  $X^\top X$ , and  $R$  does not commute with  $\text{Rot}(\pi/2)$ , so it is not multiplication by a complex number. The pairing of outputs and the rotated bias do not enter these computations.

(e) Zeros.  $W_{ij} = (W_0)_{ij}(1 + (BA)_{ij})$  for HiRA,  $\ell_i(W_0)_{ij}$  or  $(W_0)_{ij}\ell_j$  for  $(\text{IA})^3$ , and  $\gamma_i(W_0)_{ij}$  for SSF's post-linear scale (its shift changes the bias), so  $(W_0)_{ij} = 0$  implies  $W_{ij} = 0$ . After a

merge the new base  $W_1$  has  $Z(W_0) \subseteq Z(W_1)$ , and induction gives the claim for any number of cycles. New zeros:  $\ell_i = 0$  kills row  $i$ , and  $B = -e_i$ ,  $A = e_j^\top$  give  $(J + BA)_{ij} = 0$ .

(e) *Output and input scalings.* Here  $\text{Im } M(W_0) = \text{Diag}_m \cdot W_0$ . For  $g$  monomial and  $h \in \text{GL}_n$ ,  $g \text{ diag}(\ell) W_0 h^{-1} = \text{diag}(\ell') g W_0 h^{-1}$  with  $\text{diag}(\ell') = g \text{ diag}(\ell) g^{-1}$ , and  $\ell'$  ranges over  $\mathbb{R}^m$  with  $\ell$ , so  $\text{Mon}_m \times \text{GL}_n$  is contained in the covariance group. Conversely, let  $(g, h)$  be in it and take  $W_0 = e_j c^\top$  with  $c \neq 0$ . Then  $\text{Im } M(e_j c^\top) = \mathbb{R} e_j c^\top$ , and  $g \text{ Im } M(W_0) h^{-1} = \mathbb{R} (g e_j)(h^{-\top} c)^\top$  is a line, while  $\text{Im } M(g W_0 h^{-1}) = \{y(h^{-\top} c)^\top : \text{supp } y \subseteq \text{supp}(g e_j)\}$  has dimension  $|\text{supp}(g e_j)|$ . Equality for every  $j$  forces every column of  $g$  to have one nonzero entry, so  $g$  is monomial. The input side is the transpose.

(e) *HiRA.* For permutation matrices  $P, P'$  and diagonal  $D, D'$  we have  $P(X \odot Y)P' = (PXP') \odot (PYP')$  and  $D(X \odot Y)D' = (DXD') \odot Y$ . For  $g = PD$  and  $h^{-1} = D'P'$  this gives

$$g(W_0 \odot (J + Z))h^{-1} = (gW_0 h^{-1}) \odot (J + PZP'),$$

and  $PM_{\leq r}P' = M_{\leq r}$ , so  $\text{Mon}_m \times \text{Mon}_n$  is contained in the covariance group. Conversely, let  $(g, h)$  be in it and take  $W_0 = E_{ij}$ , whose image  $E_{ij} \odot (J + M_{\leq r}) = \mathbb{R} E_{ij}$  is a line. With  $u = g e_i$  and  $v = h^{-\top} e_j$ , covariance requires  $\text{Im } M(uv^\top) = \mathbb{R} uv^\top$ . Suppose  $uv^\top$  has two nonzero entries  $(k, l) \neq (k', l')$ . Taking  $Z = E_{kl} \in M_{\leq 1}$  gives  $Y = uv^\top + u_k v_l E_{kl} \in \text{Im } M(uv^\top)$ . If  $Y = \lambda uv^\top$ , the entry  $(k', l')$  forces  $\lambda = 1$  and the entry  $(k, l)$  forces  $u_k v_l = 0$ , a contradiction. So  $|\text{supp } u| = |\text{supp } v| = 1$  for all  $i, j$ , which makes  $g$  and  $h^{-\top}$ , hence  $h$ , monomial. For  $m = n = 1$ ,  $\text{Mon}_1 = \text{GL}_1$ .  $\square$

### C.17 Theorem II.10

**THEOREM II.10** (cut-rank bound; standard). *Let  $\Gamma$  be a tensor network, evaluated in the symmetric monoidal category  $(\mathbf{FinVect}, \otimes)$ , with input legs  $V$  and output legs  $U$ . Every wire and open leg is assumed incident to a vertex: insert an identity box on any bare strand, or equivalently count each bare input–output wire as crossing every cut. A crossing hyperedge (copy spider) contributes its dimension once. Partition the vertices into  $I \sqcup O$ . The weight  $w$  of the cut is the product of the dimensions of the internal wires between the two sides and of the open legs attached on the wrong side (input legs at  $O$ , output legs at  $I$ ). Then*

$$\text{rank}[\Gamma] \leq \min_{\text{cuts}} w,$$

*and rank is subadditive over formal sums. The bound refers to a chosen presentation of the network.*

*Proof.* Fix a cut  $I \sqcup O$ . Write  $V_I, V_O$  for the tensor products of the input legs attached to  $I$  and to  $O$ ,  $U_I, U_O$  likewise for the output legs, and  $C$  for the tensor product of the spaces carried by the crossing wires, each crossing hyperedge counted once. Contracting every wire internal to  $I$  produces a tensor whose free indices are those of  $V_I, U_I$  and  $C$ ; read it as a linear map  $T_I : V_I \rightarrow U_I \otimes C$ . Likewise the  $O$  side gives  $T_O : C \otimes V_O \rightarrow U_O$ . A crossing hyperedge is a single summation index shared by tensors on both sides, so it is summed once, over a basis of its space, when the two sides are joined. Contracting the crossing indices joins the two halves, and up to the symmetry isomorphisms that reorder tensor factors,

$$[\Gamma] = (\text{id}_{U_I} \otimes T_O) \circ (T_I \otimes \text{id}_{V_O}) : V_I \otimes V_O \rightarrow U_I \otimes C \otimes V_O \rightarrow U_I \otimes U_O.$$

A composite factors through its middle space, so its rank is at most  $\dim U_I \cdot \dim C \cdot \dim V_O = w$ . Only the symmetric monoidal structure was used: no cups or caps. A bare strand from an

input to an output, carried by an identity box, has one of its two legs on the wrong side of every cut, which is the convention of the statement; without it,  $I_d \otimes BA$  would violate the bound. Minimising over cuts gives the bound, and subadditivity over formal sums is  $\text{rank}(X + Y) \leq \text{rank } X + \text{rank } Y$  (Auxiliary result B.12(a)). The weight depends on the drawing, which is why the bound refers to a presentation; compare Auxiliary result B.14.  $\square$

The bounds of Table 8. Each follows from a cut in the stated presentation or from subadditivity. The LoRA family  $sBA$  factors through the bond of dimension  $r$ . A split initialisation gives  $\Delta W = sBA - sB_0A_0$ , a sum of two such terms (Theorem II.1(c)). For HRA, the telescoping identity

$$R - I = \sum_{k=1}^r (H_k \cdots H_r - H_{k+1} \cdots H_r) = \sum_{k=1}^r (H_k - I) H_{k+1} \cdots H_r$$

writes  $R - I$  as  $r$  terms of rank at most  $\text{rank}(H_k - I) = 1$ , so  $\text{rank}(W_0 R - W_0) \leq r$ ; a Givens rotation differs from  $I$  in a  $2 \times 2$  block, so the same identity gives  $2g$  for  $g$  gates. ReLoRA after  $K$  phases is a sum of  $K$  rank- $r$  terms, and GraLoRA factors through  $\mathbb{R}^{kr}$  (proof of Proposition II.12). LoHa's Hadamard product, drawn with copy spiders on the two legs, factors through the face-splitting bond of dimension  $r_1 r_2$  (Auxiliary result B.12(c)). HiRA's update  $W_0 \odot BA = \sum_{k \leq r} \text{diag}(b_k) W_0 \text{diag}(a_k)$  factors through  $r$  copies of a bond of dimension  $\text{rank } W_0$ . LoKr gives  $\text{rank}(C \otimes BA) = \text{rank } C \cdot \text{rank}(BA) \leq \text{rank } C \cdot r'$  (Auxiliary result B.12(b)). FourierFT's update is a sum of  $n_c$  real cosine patterns  $\cos(2\pi(uj/m + vl/n)) = \text{Re}(x_u y_v^\top)$  with  $(x_u)_j = e^{2\pi i u j / m}$  and  $(y_v)_l = e^{2\pi i v l / n}$  (Auxiliary result B.16(c)), and  $\text{Re}(xy^\top) = \text{Re } x \text{Re } y^\top - \text{Im } x \text{Im } y^\top$  has rank at most 2. MoRA's reshape, C3A's circulant blocks and MPO adapters have no narrow cut in their presentation, so the theorem bounds them only by  $\min(m, n)$ .

### C.18 Theorem II.11

**THEOREM II.11** (Kronecker incomparability; the linear algebra is standard, the PEFT reading is new). Write  $\mathbb{R}^m = \mathbb{R}^{m_1} \otimes \mathbb{R}^{m_2}$  and  $\mathbb{R}^n = \mathbb{R}^{n_1} \otimes \mathbb{R}^{n_2}$ , and view  $\Delta \in \mathbb{R}^{m \times n}$  as a 4-leg tensor. Let  $\text{rk}$  be the rank across the cut  $\{U_1 U_2\} \mid \{V_1 V_2\}$  and  $\text{rk}_\otimes$  the rank across  $\{U_1 V_1\} \mid \{U_2 V_2\}$ . Put

$$\kappa_1 := \min(m_1, m_2) \cdot \min(n_1, n_2), \quad \nu := \min(m_1, n_1) \cdot \min(m_2, n_2).$$

- (a)  $\text{rk}_\otimes \Delta \leq k$  iff  $\Delta = \sum_{i \leq k} C_i \otimes D_i$ . Every  $\Delta$  has  $\text{rk}_\otimes \Delta \leq \min(m_1 n_1, m_2 n_2)$ .
- (b)  $\text{rk}(C \otimes D) = \text{rk } C \cdot \text{rk } D$ , while  $\text{rk}_\otimes(C \otimes D) = 1$  for  $C, D \neq 0$ . So  $\nu$  is the largest rank of a single Kronecker product.
- (c) If  $u \in \mathbb{R}^m$  and  $v \in \mathbb{R}^n$  have Schmidt ranks  $\rho_u, \rho_v$ , then  $\text{rk}_\otimes(uv^\top) = \rho_u \rho_v$ . A generic rank-one update therefore has Kronecker rank  $\kappa_1$ . For aligned splits,  $(m_1 - m_2)(n_1 - n_2) \geq 0$ , which include equal square splits and every split HF PEFT's factorisation produces (it puts the smaller factor first on both sides),  $\kappa_1 = \min(m_1 n_1, m_2 n_2)$  is the maximal Kronecker rank.
- (d) Incomparability. For  $1 \leq r < \nu$  and  $1 \leq k < \kappa_1$ , the sets  $\mathcal{M}_{\leq r}$  and  $\mathcal{K}_{\leq k} := \{\Delta : \text{rk}_\otimes \Delta \leq k\}$  of unconstrained  $k$ -term Kronecker sums are incomparable, for every split. A generic rank-one update lies in the first and not the second, and  $C \otimes D$  with full-rank factors, of Kronecker rank 1 and rank  $\nu$ , lies in the second and not the first. For equal square splits ( $m = n = N$ , all factors  $s$ ),  $\kappa_1 = \nu = s^2 = N$ , and  $I_N$  is such a product.
- (e) Constrained Kronecker methods. Let  $\text{Im } M \subseteq W_0 + \mathcal{K}_{\leq k}$ . Then  $\text{Im LoRA}_r \not\subseteq \text{Im } M$  whenever  $k < \kappa_1$ , for every pointing of  $\text{LoRA}_r$ . Give  $\text{LoRA}_r$  its zero-init pointing, with image  $W_0 + \mathcal{M}_{\leq r}$ . Then  $\text{Im } M \not\subseteq \text{Im LoRA}_r$  iff some  $\theta \in \text{Im } M$  has  $\text{rank}(\theta - W_0) > r$ . For LoKr,  $\Delta W = C \otimes BA$  with  $C \in \mathbb{R}^{m_1 \times n_1}$ ,  $B \in \mathbb{R}^{m_2 \times r'}$  and  $A \in \mathbb{R}^{r' \times n_2}$ , put  $r'_{\text{eff}} = \min(r', m_2, n_2)$ . The



maximum rank is  $\min(m_1, n_1) \cdot r'_{\text{eff}}$ , attained. So  $\text{Im LoKr}_{r'} \subseteq \text{Im LoRA}_r$  iff  $r \geq \min(m_1, n_1) r'_{\text{eff}}$ , and otherwise the two images are incomparable, provided  $\kappa_1 > 1$ . In HF PEFT the second LoKr factor is low-rank only when its rank parameter satisfies  $r' < \max(m_2, n_2)/2$ ; otherwise it is full, and LoKr is unconstrained KronA, with maximum rank  $\nu = \min(m_1, n_1) \min(m_2, n_2)$ . On a 768-wide layer ( $24 \otimes 32$ ) the maximum rank therefore jumps from 360 at  $r' = 15$  to 768 at  $r' = 16$ . With `decompose_both`,  $C$  is also capped at rank  $r'$ .

*Proof.* Index  $\mathbb{R}^m$  by pairs  $(i_1, i_2)$  and  $\mathbb{R}^n$  by  $(j_1, j_2)$ . The rank across  $\{U_1 U_2\} \mid \{V_1 V_2\}$  is the ordinary rank of  $\Delta$ . The rank across  $\{U_1 V_1\} \mid \{U_2 V_2\}$  is the rank of the rearrangement  $\mathcal{R}(\Delta) \in \mathbb{R}^{m_1 n_1 \times m_2 n_2}$  with entries  $\mathcal{R}(\Delta)_{(i_1 j_1), (i_2 j_2)} = \Delta_{(i_1 i_2), (j_1 j_2)}$ , a linear bijection that permutes entries, with  $\mathcal{R}(C \otimes D) = \text{vec } C \text{ vec } D^\top$  (Auxiliary result B.13(a)). (In this proof and the next,  $\mathcal{R}$  denotes this rearrangement.)

(a)  $\text{rank } \mathcal{R}(\Delta) \leq k$  iff  $\mathcal{R}(\Delta) = \sum_{i \leq k} x_i y_i^\top$ , iff  $\Delta = \sum_{i \leq k} \mathcal{R}^{-1}(x_i y_i^\top) = \sum_{i \leq k} C_i \otimes D_i$  with  $\text{vec } C_i = x_i$  and  $\text{vec } D_i = y_i$ . The matrix  $\mathcal{R}(\Delta)$  has size  $m_1 n_1 \times m_2 n_2$ , which gives the second claim (Auxiliary result B.13(b)).

(b)  $\text{rank}(C \otimes D) = \text{rank } C \cdot \text{rank } D$  (Auxiliary result B.12(b)), and  $\mathcal{R}(C \otimes D) = \text{vec } C \text{ vec } D^\top$  has rank 1 for  $C, D \neq 0$ . The largest value of  $\text{rank } C \cdot \text{rank } D$  is  $\min(m_1, n_1) \min(m_2, n_2) = \nu$ , attained by factors of full rank.

(c) Take Schmidt decompositions  $u = \sum_{i \leq \rho_u} \sigma_i x_i \otimes y_i$  and  $v = \sum_{j \leq \rho_v} \tau_j z_j \otimes w_j$  with positive coefficients and orthonormal families (Auxiliary result B.11(a)). By the mixed-product rule,

$$uv^\top = \sum_{i,j} \sigma_i \tau_j (x_i z_j^\top) \otimes (y_i w_j^\top).$$

The two families  $(x_i z_j^\top)_{ij}$  and  $(y_i w_j^\top)_{ij}$  are Frobenius-orthonormal, because  $\langle x_i z_j^\top, x_{i'} z_{j'}^\top \rangle = (x_i^\top x_{i'}) (z_j^\top z_{j'})$  and similarly for the second family, and the  $\rho_u \rho_v$  coefficients are positive. By Auxiliary result B.13(c) this is an operator-Schmidt decomposition, so  $\text{rank}_\otimes(uv^\top) = \rho_u \rho_v$ . The Schmidt rank of  $u$  is the rank of its  $m_1 \times m_2$  reshape, which equals  $\min(m_1, m_2)$  outside the zero set of the maximal minors, a proper algebraic subset (Auxiliary result B.10(b)); likewise for  $v$ . So for generic  $u, v$ ,  $\text{rank}_\otimes(uv^\top) = \min(m_1, m_2) \min(n_1, n_2) = \kappa_1$ . If  $(m_1 - m_2)(n_1 - n_2) \geq 0$ , then either  $m_1 \leq m_2$  and  $n_1 \leq n_2$ , so  $\kappa_1 = m_1 n_1 = \min(m_1 n_1, m_2 n_2)$ , or  $m_1 \geq m_2$  and  $n_1 \geq n_2$ , so  $\kappa_1 = m_2 n_2 = \min(m_1 n_1, m_2 n_2)$ ; in both cases  $\kappa_1$  is the bound of (a). Equal square splits are aligned, and so is any split with the smaller factor first on both sides.

(d) A generic rank-one  $uv^\top$  lies in  $\mathcal{M}_{\leq 1} \subseteq \mathcal{M}_{\leq r}$  and has  $\text{rank}_\otimes = \kappa_1 > k$  by (c), so it is not in  $\mathcal{K}_{\leq k}$ . A product  $C \otimes D$  with factors of full rank has  $\text{rank}_\otimes = 1 \leq k$  and  $\text{rank } \nu > r$  by (b). Neither argument uses alignment. For equal square splits  $\kappa_1 = \nu = s \cdot s = N$  and  $I_N = I_s \otimes I_s$ .

(e) Any pointing of  $\text{LoRA}_r$  has image  $W_{\text{res}} + \mathcal{M}_{\leq r}$  for its frozen residual  $W_{\text{res}}$ . Suppose  $W_{\text{res}} + \mathcal{M}_{\leq r} \subseteq W_0 + \mathcal{K}_{\leq k}$ , and put  $c = W_{\text{res}} - W_0$ , so  $c + \mathcal{M}_{\leq r} \subseteq \mathcal{K}_{\leq k}$ . The set  $\mathcal{K}_{\leq k}$  is closed (the zero set of the  $(k+1) \times (k+1)$  minors of  $\mathcal{R}(\cdot)$ ) and closed under scaling. For  $X \in \mathcal{M}_{\leq r}$  and  $t > 0$ ,  $c + tX \in \mathcal{K}_{\leq k}$ , hence  $c/t + X \in \mathcal{K}_{\leq k}$ , and letting  $t \rightarrow \infty$  gives  $X \in \mathcal{K}_{\leq k}$ . So  $\mathcal{M}_{\leq r} \subseteq \mathcal{K}_{\leq k}$ , which contradicts (d) when  $k < \kappa_1$ . The second clause restates  $\text{Im LoRA}_r = W_0 + \mathcal{M}_{\leq r}$  for the zero-init pointing.

For LoKr,  $\text{rank}(C \otimes BA) = \text{rank } C \cdot \text{rank}(BA) \leq \min(m_1, n_1) \cdot \min(r', m_2, n_2)$  (Auxiliary result B.12(a),(b)), with equality for  $C$  of full rank and  $BA$  of rank  $r'_{\text{eff}}$ , which exists by Auxiliary result B.11(a). If  $r \geq \min(m_1, n_1) r'_{\text{eff}}$ , every update has rank at most  $r$  and  $\text{Im LoKr}_{r'} \subseteq W_0 + \mathcal{M}_{\leq r}$ . Otherwise the maximising update has rank above  $r$ , so  $\text{Im LoKr}_{r'} \not\subseteq \text{Im LoRA}_r$ ; and since  $\text{Im LoKr}_{r'} \subseteq W_0 + \mathcal{K}_{\leq 1}$ , the first clause with  $k = 1 < \kappa_1$  gives  $\text{Im LoRA}_r \not\subseteq \text{Im LoKr}_{r'}$ . When HF's second factor is full, LoKr is  $C \otimes D$  with both factors free, whose maximum rank



is  $\nu$  by (b). On a 768-wide layer HF factors  $768 = 24 \cdot 32$  on both sides, so  $m_1 = n_1 = 24$  and  $m_2 = n_2 = 32$ : for  $r' = 15 < 16$  the maximum rank is  $24 \cdot 15 = 360$ , and for  $r' = 16$  the second factor is full and the maximum is  $24 \cdot 32 = 768$ . The `decompose_both` option factors  $C$  through rank  $r'$ , which caps rank  $C$  at  $r'$ . The thresholds and factorisations are read from the HF PEFT source (Mangrulkar et al., 2022).  $\square$

### C.19 Proposition II.12

**PROPOSITION II.12** (gauge census: rank is a cut, dimension is a volume).

- (a)  $d(M) \leq |M| - \dim(\text{generic gauge orbit})$ , with equality when the gauge acts locally transitively on generic fibres.
- (b) The values in Table 9 hold for one  $m \times n$  box, with  $r \leq \min(m, n)$  throughout except for VeRA (whose published RoBERTa-base setting has  $r = 1024 > 768$ ). In the LoHa row,

$$F = r_1(m + n - r_1) + r_2(m + n - r_2) - (m + n - 1)$$

is a proved upper bound for  $d$ . The gauge column lists the exhibited Lie group. It acts locally transitively on generic fibres away from the  $mn$  cap, but not where the cap binds; for LoHa at  $(m, n, r_1, r_2) = (4, 4, 2, 2)$  the orbit has dimension 15 and the fibre 16. BitFit does not touch the  $m \times n$  box; it is linear in the biases. Hybrid GraLoRA (`hybrid_r` > 0) is not covered by the table. FourierFT is injective only when its sampled frequencies contain no conjugate pair  $(u, v)$ ,  $(-u, -v)$ ; otherwise  $d = |M| - \#\{\text{pairs}\}$ . At LLM sizes this is close to  $|M|$ , but at  $768 \times 768$  with 1000 frequencies a pair occurs in about half of the draws (HF's default seed 777 gives exactly two pairs there).

- (c) Equal-budget trades. Assume  $2r \leq \min(m, n)$ ,  $r^2 \leq \min(m, n)$  and  $2r^2 < m + n - 1$ . LoHa $_{r,r}$  and LoRA $_{2r}$  both cost  $2r(m + n)$ , but LoHa reaches at least  $(m + n - 1) - 2r^2$  fewer dimensions (exactly that many when  $F$  is attained [Num]). For  $r \geq 3$  it gains rank up to  $r^2 > 2r$  in exchange. For  $r \leq 2$ ,  $r^2 \leq 2r$ , so  $\text{Im LoHa}_{r,r} \subseteq \text{Im LoRA}_{2r}$  and LoHa is strictly dominated. When  $2r^2 \geq m + n - 1$  (still with  $2r \leq \min(m, n)$ ) the bound no longer separates them, and wherever  $F$  is attained LoHa $_{r,r}$  matches or exceeds LoRA $_{2r}$  in dimension as well as in rank, for instance at  $(m, n, r) = (9, 9, 3)$ , at  $r = 64$  on  $4096 \times 4096$ , and at LyCORIS-style  $r = 32$  on 320- to 640-wide layers. GraLoRA $_k$  costs the same as LoRA $_r$ , reaches the same number of dimensions, and has maximum rank  $\min(kr, m, n)$ , but caps every block at rank  $r/k$ . A generic rank- $r$  update has rank  $r$  on every block, so for  $k \geq 2$  and  $kr \leq \min(m, n)$  the images of LoRA $_r$  and GraLoRA $_k$  are incomparable.

*Proof.* (a) By Auxiliary result B.9(d),  $d(M)$  is the maximal rank of  $d\rho$ , attained on an open set, which is dense when  $\rho$  is real-analytic on a connected domain (Auxiliary result B.10). At a point  $q$  of maximal rank the fibre through  $q$  is, near  $q$ , a submanifold of dimension  $|M| - d(M)$  (Auxiliary result B.1(b)), and it contains the gauge orbit of  $q$ , an immersed submanifold. At generic  $q$  this gives  $\dim(\text{orbit}) \leq |M| - d(M)$ . If the gauge acts locally transitively on generic fibres, the orbit is open in the fibre and the two dimensions agree.

(b), LoRA.  $\text{GL}_r$  acts on pairs of full rank by  $(Bg^{-1}, gA)$ , freely, and its orbits are the fibres of  $(B, A) \mapsto BA$  over  $\mathcal{M}_r$  (Auxiliary result B.5(b)), so  $d = r(m + n) - r^2 = r(m + n - r)$ . The maximum rank is  $r$ .

GraLoRA. With  $k \mid m, n, r$ , the update is a  $k \times k$  grid of blocks of size  $p \times q = \frac{m}{k} \times \frac{n}{k}$ , each an independent LoRA of rank  $t = r/k$ , so the image is  $W_0$  plus the product of  $k^2$  copies of

$\mathcal{M}_{\leq t}(p \times q)$ , of dimension  $k^2 t(p + q - t) = r(m + n - r)$  since  $t \leq \min(p, q)$ . The gauge  $\prod \mathrm{GL}_t$  acts freely on generic parameters with orbits of total dimension  $k^2 t^2 = r^2$ , equal to the fibre dimension. For the rank, write block  $(i, j)$  as  $U_{ij} V_{ij}^\top$ ; then  $\Delta W = \text{blockdiag}(L_1, \dots, L_k) S$  with  $L_i = [U_{i1} \cdots U_{ik}] \in \mathbb{R}^{p \times r}$  and  $S \in \mathbb{R}^{kr \times n}$  stacking the  $V_{ij}^\top$  placed in column block  $j$ , so  $\text{rank } \Delta W \leq \min(kr, m, n)$ . (Placing rank- $t$  blocks only on the diagonal would give rank  $r$ , not  $kr$ .) To see that  $\min(kr, m, n)$  is attained, assume  $p \leq q$  (otherwise transpose) and put  $a = \min(kt, p)$ , so that  $\min(kr, m, n) = ka$ . Write  $a = k\alpha + \beta$  with  $0 \leq \beta < k$ , and let  $N_{ij} = \alpha + [(j - i) \bmod k < \beta]$ . Then every row and column sum of  $N$  is  $a \leq p \leq q$ , and  $N_{ij} \leq t$ , because  $\alpha \leq t$  with  $\alpha = t$  only when  $\beta = 0$ . In block row  $i$  pick  $N_{ij}$  distinct rows for each  $j$  (in all  $a \leq p$  rows), in block column  $j$  pick  $N_{ij}$  distinct columns for each  $i$  (in all  $a \leq q$  columns), and put a 1 at the matched pairs. The resulting 0/1 matrix has at most one 1 in each row and column, so its rank is  $\sum_{ij} N_{ij} = ka$ , and its block  $(i, j)$  has rank  $N_{ij} \leq t$ , so it lies in the image. Rank at least  $ka$  is the nonvanishing of a minor, a polynomial in the block factors, so it holds for generic parameters ([Auxiliary result B.10](#)).

**LoHa.** The update is  $(B_1 A_1) \odot (B_2 A_2)$  with budget  $(r_1 + r_2)(m + n)$ . The torus acts by  $(B_1, A_1, B_2, A_2) \mapsto (D_1 B_1, A_1 D_2, D_1^{-1} B_2, A_2 D_2^{-1})$  and preserves the update, because  $(D_1 X D_2) \odot (D_1^{-1} Y D_2^{-1}) = X \odot Y$  for diagonal  $D_1, D_2$ ; together with  $\mathrm{GL}_{r_1} \times \mathrm{GL}_{r_2}$  acting on the two LoRA factors this gives an action of  $G = \mathrm{GL}_{r_1} \times \mathrm{GL}_{r_2} \times T_m \times T_n$ . We compute the stabiliser at generic parameters. If  $m = n = r_1$ , then  $r_1(m + n - r_1) = mn$  and  $F \geq mn$  (since  $r_2(m + n - r_2) - (m + n - 1) = (r_2 - 1)(m + n - r_2 - 1) \geq 0$ ), so the bound  $d \leq \min(mn, F)$  is trivial; otherwise, by symmetry under transposition, assume  $m \geq r_1 + 1$ . Let  $(g_1, g_2, D_1, D_2)$  fix the point. From  $D_1 B_1 g_1^{-1} = B_1$ , every row of  $B_1$  is a left eigenvector of  $g_1$ ; for generic  $B_1$  any  $r_1$  of its rows form a basis and the remaining rows have all coordinates nonzero in that basis, which forces a single eigenvalue, so  $g_1 = cI$  and  $D_1 = cI$ . From  $g_1 A_1 D_2 = A_1$  and nonzero columns of  $A_1$ ,  $D_2 = c^{-1}I$ . From  $D_1^{-1} B_2 g_2^{-1} = B_2$  and  $B_2$  of full column rank ( $m \geq r_2$ ),  $g_2 = c^{-1}I$ , and then the last equation holds. So the stabiliser is  $\{(cI, c^{-1}I, cI, c^{-1}I)\}$ , of dimension 1, and the generic orbit has dimension  $r_1^2 + r_2^2 + m + n - 1$ . By (a),  $d \leq (r_1 + r_2)(m + n) - r_1^2 - r_2^2 - (m + n - 1) = F$ , and trivially  $d \leq mn$ . Where  $d = F$  the orbit has the dimension of the fibre and the action is locally transitive; at  $(4, 4, 2, 2)$  the orbit has dimension  $4 + 4 + 7 = 15$  while the [Num] value  $d = 16$  gives a fibre of dimension  $32 - 16 = 16$ . For the rank,  $\text{rank}(X \odot Y) \leq r_1 r_2$  ([Auxiliary result B.12\(c\)](#)) and  $\leq \min(m, n)$ . To attain  $k = \min(r_1 r_2, m, n)$ , choose distinct pairs  $(a_i, b_i) \in [r_1] \times [r_2]$  for  $i \leq k$ , let the first  $k$  rows of  $B_1$  be  $e_{a_i}^\top$  and the first  $k$  columns of  $A_1$  be  $e_{a_i}$ , likewise  $B_2, A_2$  with the  $b_i$ , and set the rest to zero. Then  $(B_1 A_1)_{ij} = [a_i = a_j]$  and  $(B_2 A_2)_{ij} = [b_i = b_j]$  for  $i, j \leq k$ , so the Hadamard product is  $I_k$  in the top-left corner and zero elsewhere, of rank  $k$ ; by [Auxiliary result B.10](#) this rank is generic.

**HRA.** The proof of [Theorem II.7\(a\)](#) computes  $d = rn - r(r + 1)/2$  and the fibre dimension  $r(r + 1)/2$  for every  $1 \leq r \leq n$ , with no use of parity or of  $r \leq n - 2$ ; for  $r \in \{n - 1, n\}$  the value is  $\dim \mathrm{SO}(n)$ . The scalings  $(\mathbb{R}^\times)^r$  are the exhibited Lie gauge, and  $W_0 H_{u_1} \cdots H_{u_r} - W_0$  has rank at most  $r$ . Evenness of  $r$  is used only by HF's paired initialisation, which needs pairs to start at  $W_0$ .

**DoRA.** The value of  $d$  is [Proposition II.8](#).  $\mathrm{GL}_r$  acts on  $(B, A)$  by  $(B g^{-1}, g A)$ , fixing  $\mu$ , freely on pairs of full rank and preserving  $\rho$ . For generic  $W_0$  with  $(m - r)(n - r) \geq m$  the fibre has dimension  $r(m + n) + m - r(m + n - r) - m = r^2$ , so  $\mathrm{GL}_r$  is locally transitive on generic fibres. Otherwise  $d = mn$  and the fibre has dimension  $r(m + n) + m - mn = r^2 + m - (m - r)(n - r) > r^2$ , which is 6 at  $(3, 3, 2)$ ; for rank-one  $W_0$ ,  $d = r(m + n - r) + m - r$  and the fibre has dimension  $r^2 + r$ . The update can have rank up to  $\min(m, n)$  ([Proposition II.8](#), last item).

**VeRA.** The update is  $\text{diag}(b) B \text{diag}(d) A$  with frozen random  $B \in \mathbb{R}^{m \times r}$ ,  $A \in \mathbb{R}^{r \times n}$  and trainable  $b \in \mathbb{R}^m$ ,  $d \in \mathbb{R}^r$ , so  $|M| = m + r$ , and  $(b, d) \mapsto (cb, c^{-1}d)$  preserves it. The value  $d(M) = m + r - 1$  is checked numerically [Num]. The rank is at most  $\text{rank}(B \text{diag}(d) A) \leq \min(r, m, n)$ , and for invertible  $\text{diag}(b)$ ,  $\text{diag}(d)$  it equals  $\text{rank}(BA')$  with  $A' = \text{diag}(d)A$ , which is  $\min(r, m, n)$  for generic frozen matrices (Auxiliary result B.10). None of this uses  $r \leq \min(m, n)$ .

**KronA.** The update  $A \otimes B$  satisfies  $\mathcal{R}(A \otimes B) = \text{vec } A \text{vec } B^\top$ , so the image is  $W_0$  plus the preimage under the linear bijection  $\mathcal{R}$  (Auxiliary result B.13(a)) of the Segre cone  $\mathcal{M}_{\leq 1}(m_1 n_1 \times m_2 n_2)$ , of dimension  $m_1 n_1 + m_2 n_2 - 1$  (Auxiliary result B.8). If  $A \otimes B = A' \otimes B' \neq 0$ , then  $\text{vec } A \text{vec } B^\top = \text{vec } A' \text{vec } B'^\top$ , so  $A' = cA$  and  $B' = c^{-1}B$ ; the fibre over a nonzero update is the orbit of  $\mathbb{R}^\times$ . The maximum rank is  $\nu$  by Theorem II.11(b).

**Tensor-train and Tucker adapters.** The update is a fixed reshape, a linear bijection, of a tensor in TT or Tucker format, so  $d$  is the dimension of the set of tensors of the given admissible minimal ranks, which is the parameter count minus  $\sum_e r_e^2$ , with the bond gauge  $\prod_e \text{GL}_{r_e}$  acting on the cores (Auxiliary result B.15).

**Linear methods.** For  $\rho(q) = \theta_0 + Lq$  with  $L$  linear,  $d = \text{rank } L$ , the fibres are the cosets  $q + \ker L$ , and translation by  $\ker L$  acts transitively on each. LoRA-FA, LoRA-XS, FourierFT, C3A and masks are of this form; so is MiSS in each HF mode, including the BAT mode, whose update blocks  $(W_0 M + M)$  is linear in the trained block  $M$  (Proposition II.13).

**FourierFT.** HF's update is  $c \text{Re}(\text{ifft2}(S))$  with  $S$  supported on a sampled set  $\Omega$  of  $|M|$  frequencies. By Auxiliary result B.16(c), frequency  $p = (u, v)$  contributes  $s_p \kappa_p$  with  $\kappa_p(j, l) = \frac{1}{mn} \cos(2\pi(uj/m + vl/n))$ , and  $\kappa_p = \kappa_{-p}$ . Write  $\chi_p(j, l) = e^{2\pi i(uj/m + vl/n)}$ , so that  $\langle \chi_p, \chi_{p'} \rangle = mn [p = p']$  and  $\cos(\dots) = \frac{1}{2}(\chi_p + \chi_{-p})$ . Hence  $\langle \kappa_p, \kappa_{p'} \rangle = 0$  unless  $p' = \pm p$ , and every  $\kappa_p \neq 0$ . So the patterns of distinct classes  $\{p, -p\}$  are linearly independent, and the rank of  $s \mapsto \sum_p s_p \kappa_p$  is the number of classes met by  $\Omega$ , which is  $|\Omega|$  minus the number of conjugate pairs inside  $\Omega$ . Each pair  $\{p, -p\} \subseteq \Omega$  contributes the kernel direction  $s_p = 1, s_{-p} = -1$ . The frequencies of pairs in random draws and for HF's default seed are numerical observations.

(c) Assume  $2r \leq \min(m, n)$ ,  $r^2 \leq \min(m, n)$  and  $2r^2 < m + n - 1$ . Both methods cost  $2r(m + n)$ , and  $d(\text{LoRA}_{2r}) = 2r(m + n - 2r)$ . By (b),  $d(\text{LoHa}_{r,r}) \leq F = 2r(m + n - r) - (m + n - 1)$ , and

$$d(\text{LoRA}_{2r}) - F = (m + n - 1) - 2r^2 > 0.$$

So LoHa reaches at least  $(m + n - 1) - 2r^2$  fewer dimensions; since  $F < d(\text{LoRA}_{2r}) \leq mn$ , the cap does not bind, and the gap is exactly  $(m + n - 1) - 2r^2$  when  $d(\text{LoHa}_{r,r}) = F$ . Its maximum rank is  $\min(r^2, m, n) = r^2$  by (b), against  $2r$  for  $\text{LoRA}_{2r}$ , and  $r^2 > 2r$  iff  $r \geq 3$ . For  $r \leq 2$ , every LoHa update has rank at most  $r^2 \leq 2r$ , so  $\text{Im } \text{LoHa}_{r,r} \subseteq W_0 + \mathcal{M}_{\leq 2r} = \text{Im } \text{LoRA}_{2r}$  (both pointed at  $W_0$ , as HF zero-initialises one LoHa factor), and the inclusion is strict because the dimensions differ. When  $2r^2 \geq m + n - 1$ ,  $F \geq d(\text{LoRA}_{2r})$  and the bound does not separate them; wherever  $d(\text{LoHa}_{r,r}) = F$ , LoHa reaches at least as many dimensions, and its maximum rank  $\min(r^2, m, n)$  is at least  $\min(2r, m, n) = 2r$ , because  $r \geq 2$  here (for  $r = 1$ ,  $2 \geq m + n - 1$  contradicts  $m, n \geq 2$ ). The examples satisfy  $2r^2 \geq m + n - 1$ :  $18 \geq 17$  at  $(9, 9, 3)$ ,  $8192 \geq 8191$  at  $r = 64$  on  $4096 \times 4096$ , and  $2048 \geq m + n - 1$  for  $r = 32$  and  $m, n \leq 640$ .

**GraLoRA<sub>k</sub>** has budget  $k^2 \cdot t(p + q) = r(m + n)$ , dimension  $r(m + n - r)$  and maximum rank  $\min(kr, m, n)$  by (b). Let  $k \geq 2$  and  $kr \leq \min(m, n)$ . A rank- $r$  update  $BA$  has blocks  $B_i A_j$ , with  $B_i$  the rows of  $B$  in block row  $i$  and  $A_j$  the columns of  $A$  in block column  $j$ ; for generic  $B, A$  every  $B_i$  ( $p \times r$ ,  $r \leq p$ ) and  $A_j$  ( $r \times q$ ,  $r \leq q$ ) has rank  $r$ , so every block has rank  $r > r/k$ ,

and the update is not in GraLoRA's image. Conversely GraLoRA reaches rank  $kr > r$ , outside  $W_0 + \mathcal{M}_{\leq r}$ . Both images contain  $W_0$ , so they are incomparable.  $\square$

## C.20 Proposition II.13

**PROPOSITION II.13** (transport of structure). *For  $T \in \text{GL}(\Theta)$  put  $\tau_T(\theta) = \theta_0 + T(\theta - \theta_0)$  and  $T_*M := (Q, q_0, \tau_T \circ \rho)$ .*

- $T_*$  is a strict automorphism of  $\mathbf{Meth}(\Theta, \theta_0)$ , with inverse  $(T^{-1})_*$ . It maps  $S_1$  to  $T S_1$  and preserves  $d$ ,  $g$ , the gauge group, the first-order deficit, polynomial degree, the regular or apex type and the germ type of the image. It does not preserve rank, and, for the family extension below, it does not preserve the covariance group.
- $K^{T_*M} = T K^M T^\top$  for every  $T$ . For any update rule that queries the loss only through  $q \mapsto L(\rho(q))$ , with its own state on  $Q$ ,  $T_*M$  trained on  $L$  and  $M$  trained on  $L \circ \tau_T$  have identical  $q$ -trajectories, so their  $\theta$ -trajectories are conjugate by  $\tau_T$ . Rules that read  $\nabla_\theta L$  and solve a least-squares problem on  $\Theta$  (LoRA-Pro), and  $\theta$ -level weight decay or clipping, are covered only for orthogonal  $T$ . Orthogonality of  $T$  is exactly the condition for  $\tau_T$  to be a Frobenius isometry; for rules of the first kind it is needed only to keep the Frobenius spectrum of  $K$ , or an isometry-invariant loss class such as  $\|\theta - \theta^*\|^2$ , unchanged.
- Coordinatewise optimizers (SGD, Adam) are also equivariant under coordinate permutations of  $Q$ . The Kronecker identification below needs this, because it composes  $\mathcal{R}_*^{-1}$  with the vec reshape of  $Q$ .
- Families and rebasing. For  $T$  independent of the base point, define the family extension  $(T_*M)(\theta) := (Q, q_0, \theta + T(\rho_\theta - \theta))$ . For an additive family with fixed shape  $C$  (Definition I.2),  $T_*M$  is additive with shape  $TC$ , so transport commutes with rebasing:  $R_K(T_*M) = \tau_T(R_K(M))$  (Theorem V.3(a)). For action-type families this can fail even for  $T = 2I$ , because the transported family acts by  $I + T(g - I)$ . For OFT on  $\Theta = \mathbb{R}^{1 \times 2}$ , two cycles of  $\theta \mapsto \theta(2R - I)$  with  $R = \text{rot}(\pi)$  reach  $9\theta_0$ , outside the transported circle. Transports that depend on  $\theta_0$  may fail (HiRA below) or may not (LoRA transported by the SVD frame of  $\theta_0$ , since  $\mathcal{M}_{\leq r}$  is  $O(m) \times O(n)$ -invariant). The covariance group (Definition II.14) is a property of a family and is compared through this extension.

Instances.

- Kronecker family. An isomorphism of slices over  $\Theta' = \mathbb{R}^{m_1 n_1 \times m_2 n_2}$  and over  $\Theta: \sum_{i \leq k} C_i \otimes D_i = \mathcal{R}_*^{-1} \text{LoRA}_k$ , with  $Q$  reshaped by vec, dynamics included for coordinatewise optimizers. KronA with unconstrained factors is  $\mathcal{R}_*^{-1} \text{LoRA}_1$ . LoKr is  $\mathcal{R}_*^{-1}$  of  $\text{LoRA}_1$  with the second factor constrained to vec  $\mathcal{M}_{\leq r'}$  when  $r' < \max(m_2, n_2)/2$  (HF's switch); otherwise HF's second factor is full and LoKr is KronA. LoKr's `decompose_both` option also constrains the first factor.
- Linear methods as coordinate subspaces in transported frames. FourierFT: sparse in the real cosine half of the real Fourier frame. With scaling  $c$  (default 150), HF's update  $\Delta W = c \text{Re}(\text{ifft2}(S))$  is the transport of a mask, with  $g = 0$ , only when the sampled index set has no conjugate pair  $(u, v), (-u, -v)$ ; otherwise  $g$  is the number of such pairs. Its image lies in the point-symmetric subspace  $\Delta W_{j,l} = \Delta W_{-j,-l}$ , and the frame is orthogonal only up to scale (self-conjugate frequencies have twice the squared norm). WaveFT: sparse in a wavelet frame, when HF's `pad-IDWT-crop` map is invertible (plausible for Haar on even shapes; unverified for other wavelet families). C3A: block-circulant; over  $\mathbb{R}$  the frame is given



by the powers  $P^k$  of the cyclic shift (Gram matrix  $bI$ ); over  $\mathbb{C}$ ,  $\text{circ}(w) = F^{-1} \text{diag}(Fw)F$  for first column  $w$ , and the real form has  $2 \times 2$  rotation-scaling blocks; HF adds a factor  $1/\text{in\_features}$ . LoRA-XS, SVFT, TinyLoRA: masks or random subspaces in the SVD frame of  $W_0$ , a  $\theta_0$ -dependent transport. SHiRA, FISH, Trainable Tokens: masks in the standard frame. MiSS/Bone in its default (balance) mode:  $\Delta W = D(\mathbf{1}_g^\top \otimes I_r)$ , which is LoRA-FA with a frozen summing projection; its BAT mode,  $\Delta W = \text{blocks}(W_0) M + M$  blockwise, is linear in the trained block  $M$  but depends on  $\theta_0$ , so it is not additive. These frame readings of FourierFT and C3A are for  $\text{init\_weights}=\text{True}$ ; HF's defaults start off  $\theta_0$  (FourierFT draws its spectrum from  $\mathcal{N}(0, 1)$  with scaling 150, so at  $768^2$  with 1000 frequencies  $\|\Delta W_0\|_F \approx 4.3$ , about 14% of a std-0.04  $W_0$ ; C3A is Xavier-uniform), and these are objects with a pointing defect.

- HiRA =  $(D_{W_0})_* \text{LoRA}$  inside the slice at  $\theta_0$ , when  $W_0$  has no zero entries, with  $D_{W_0} : X \mapsto W_0 \odot X$ . Because  $D_{W_0}$  depends on  $\theta_0$ , re-based HiRA is not transported ReLoRA. Two cycles reach  $W_0 \odot (J + Z_1) \odot (J + Z_2)$ , whose relative update  $Z_1 + Z_2 + Z_1 \odot Z_2$  has rank up to  $\min(2r + r^2, m, n)$ , attained, where transported ReLoRA stays at rank  $\leq 2r$ .

*Proof. The automorphism.*  $\tau_T$  is an affine bijection fixing  $\theta_0$  with linear part  $T$ , so  $T_*M$  is pointed. If  $h : M \rightarrow N$  satisfies  $\rho_N \circ h = \rho_M$ , then  $\tau_T \circ \rho_N \circ h = \tau_T \circ \rho_M$ , so  $h$  is also an arrow  $T_*M \rightarrow T_*N$ ; let  $T_*$  act as the identity on arrows. This is a functor, and  $(T^{-1})_*T_* = \text{id}$  on the nose because  $\tau_{T^{-1}} \circ \tau_T = \text{id}$ . So  $T_*$  is a strict automorphism.

*Invariants.*  $D(\tau_T \circ \rho)_q = T D\rho_q$  for every  $q$ . So  $S_1(T_*M) = T S_1(M)$ , and the rank of the differential is unchanged at every point; hence  $d$  (Auxiliary result B.9(d)),  $\dim S_1$ , the deficit,  $g = |M| - d$ , and the regular or apex type are unchanged. Since  $\tau_T$  is injective,  $\tau_T \circ \rho \circ \varphi = \tau_T \circ \rho$  iff  $\rho \circ \varphi = \rho$ , so the gauge groups coincide. Composition with an affine map does not raise polynomial degree, and  $\tau_{T^{-1}}$  undoes it, so the degree is unchanged. Finally  $\tau_T$  is a diffeomorphism of  $\Theta$  fixing  $\theta_0$  that carries  $\text{Im } M$  onto  $\text{Im } T_*M$ , so the germs at  $\theta_0$  are diffeomorphic. Rank is not preserved: any  $T$  with  $T(E_{11}) = I_n$  ( $n \geq 2, m = n$ ) sends a rank-one update to a full-rank one.

*Covariance is not preserved.* Take the family extension of LoRA<sub>1</sub> by  $T = D_V : X \mapsto V \odot X$ , with  $V$  fixed, without zero entries and of rank at least 2; its image at  $W$  is  $W + V \odot M_{\leq 1}$ . If  $(g, h)$  were in its covariance group for every pair, comparing the images at  $gWh^{-1}$  and  $W$  would give  $V \odot M_{\leq 1} = g(V \odot M_{\leq 1})h^{-1}$ , so  $V \odot M_{\leq 1}$  would be a union of  $\text{GL}_m \times \text{GL}_n$ -orbits, that is of rank strata  $\mathcal{M}_k$ . It has dimension  $m + n - 1$ , since  $D_V$  is invertible, while  $\dim \mathcal{M}_k = k(m + n - k) > m + n - 1$  for  $2 \leq k \leq \min(m, n)$ . So it would lie in  $\mathcal{M}_{\leq 1}$ , and in particular  $V \odot J = V$  would have rank at most one, a contradiction. Hence the covariance group of the transported family is a proper subgroup of  $\text{GL}_m \times \text{GL}_n$ , the covariance group of LoRA.

*Cometric and dynamics.*  $K^{T_*M} = D(\tau_T \rho) \Lambda D(\tau_T \rho)^\top = T D\rho \Lambda D\rho^\top T^\top = T K^M T^\top$ . The method  $T_*M$  trained on  $L$  minimises  $L \circ \tau_T \circ \rho$  on  $Q$ , and  $M$  trained on  $L \circ \tau_T$  minimises  $(L \circ \tau_T) \circ \rho$ , the same function. A rule that sees the loss only through this function, from the same initial state (and, for stochastic rules, the same random stream), therefore produces the same  $q$ -trajectory, and the weights are  $\tau_T(\rho(q(t)))$  against  $\rho(q(t))$ . A rule that reads  $\nabla_\theta L$  and takes the Frobenius least-squares step on  $\Theta$  moves  $q$  by  $-(D\rho^\top T^\top T D\rho)^+ D\rho^\top T^\top \nabla L$  under  $T_*M$ , and by  $-(D\rho^\top D\rho)^+ D\rho^\top T^\top \nabla L$  under  $M$  with loss  $L \circ \tau_T$ , since  $\nabla_\theta(L \circ \tau_T) = T^\top \nabla L$ . They agree when  $T^\top T = I$  and differ in general: for  $\Theta = \mathbb{R}^2$ ,  $\rho(q) = qe_1$  and  $T = \text{diag}(2, 1)$  the steps are  $-\frac{1}{2}\partial_1 L$  and  $-2\partial_1 L$ . Weight decay and clipping on  $\theta$  use the Frobenius norm, which



$\tau_T$  preserves iff  $T$  is orthogonal. For orthogonal  $T$ ,  $TKT^\top$  is similar to  $K$ , so the Frobenius spectrum of the cometric is kept, and  $\|\tau_T(\theta) - \tau_T(\theta^*)\| = \|\theta - \theta^*\|$ .

*Permutations of  $Q$ .* A coordinatewise rule updates each coordinate from that coordinate's own gradient history with shared hyperparameters. If  $q(t)$  is its trajectory for an objective  $f$  from  $q_0$ , then  $\pi q(t)$  is its trajectory for  $f \circ \pi^{-1}$  from  $\pi q_0$ , for any coordinate permutation  $\pi$ .

*Families and rebasing.* For an additive family,  $\rho_\theta = \theta + \delta$  with  $\delta$  independent of  $\theta$  and shape  $C = \delta(Q)$ , the extension is  $\theta + T\delta$ , additive with shape  $TC$ . After  $K$  merge-and-restart cycles an additive family reaches  $R_K = \theta_0 + C + \dots + C$  ( $K$  summands), and since  $T$  is linear,  $\theta_0 + (TC) + \dots + (TC) = \theta_0 + T(C + \dots + C) = \tau_T(R_K)$ . For OFT on  $\mathbb{R}^{1 \times 2}$  the family is  $\{\theta R : R \in \text{SO}(2)\}$  and its extension by  $T = 2I$  is  $\theta \mapsto \theta + 2(\theta R - \theta) = \theta(2R - I)$ . With  $R = \text{rot}(\pi) = -I$ , two cycles give  $\theta_0(-3I)(-3I) = 9\theta_0$ . Every rebasing of OFT stays in  $\theta_0 \cdot \text{SO}(2)$ , whose transport  $\{\theta_0(2S - I) : S \in \text{SO}(2)\}$  consists of vectors of norm at most  $3\|\theta_0\|$ , so  $9\theta_0$  lies outside it when  $\theta_0 \neq 0$ . For LoRA transported by  $X \mapsto U(\theta)XV(\theta)^\top$  with orthogonal  $U(\theta), V(\theta)$ , the transported shape is  $UM_{\leq r}V^\top = M_{\leq r}$ , so the transported family equals LoRA's, and both sides of the rebasing identity are  $\theta_0 + M_{\leq \min(Kr, m, n)}$  (by the splitting argument of [Theorem II.1\(c\)](#)).

*Kronecker family.* By [Auxiliary result B.13\(a\)](#),  $\mathcal{R} : \Theta \rightarrow \Theta'$  is a linear bijection with

$$\mathcal{R}\left(\sum_{i \leq k} C_i \otimes D_i\right) = \sum_i \text{vec } C_i \text{vec } D_i^\top = BA,$$

where  $B$  has columns  $\text{vec } C_1, \dots, \text{vec } C_k$  and  $A$  has rows  $\text{vec } D_1^\top, \dots, \text{vec } D_k^\top$ . So the  $k$ -term Kronecker method is  $\mathcal{R}^{-1} \circ \rho_{\text{LoRA}_k}$  up to the  $\text{vec}$  relabelling of  $Q$ , and the argument for  $T_*$  applies verbatim to an isomorphism between two weight spaces. Its dynamics agree with LoRA $_k$ 's on the pulled-back loss for rules that query the loss through  $\rho$ , and the  $\text{vec}$  relabelling is a coordinate permutation, harmless for coordinatewise rules. KronA is  $k = 1$ . For LoKr,  $\mathcal{R}(C \otimes BA) = \text{vec } C \text{vec } (BA)^\top$ , LoRA $_1$  with second factor in  $\text{vec } M_{\leq r'}$ , when HF factors the second factor; otherwise it is KronA. With `decompose_both` the first factor is factored too.

*FourierFT.* By the computation in the proof of [Proposition II.12](#), the patterns  $\kappa_p$  of distinct classes  $\{p, -p\}$  are orthogonal; with the sine patterns  $\sin(2\pi(uj/m + vl/n))$  of the classes with  $p \neq -p$  they form an orthogonal basis of  $\mathbb{R}^{m \times n}$ . Let  $T$  send the standard basis to this basis (cosines on the positions of the coefficients, sines elsewhere). If  $\Omega$  has no conjugate pair, FourierFT is  $T_*$  of the mask on  $\Omega$ , with  $g = 0$ ; otherwise the two coefficients of a pair hit the same pattern and  $g$  is the number of pairs. Each  $\kappa_p$  is even under  $(j, l) \mapsto (-j, -l)$  (indices modulo  $m, n$ ), so the image lies in the point-symmetric subspace. A self-conjugate frequency has  $\kappa_p = \frac{1}{mn}\chi_p$  with  $\chi_p$  real, of squared norm  $\frac{1}{mn}$ , while for  $p \neq -p$  the squared norm is  $\frac{1}{2mn}$ . HF's default draws 1000 coefficients from  $\mathcal{N}(0, 1)$  and multiplies by 150; for generic frequencies  $\mathbb{E}\|\Delta W_0\|_F^2 = 150^2 \cdot 1000 / (2 \cdot 768^2) \approx 19.1$ , a root-mean-square norm of 4.37. A single draw deviates from it by about 2%, since the sum of 1000 squared standard normals has relative standard deviation  $\sqrt{2/1000}$ , which gives the value  $\|\Delta W_0\|_F \approx 4.3$  quoted. Against  $\|W_0\|_F \approx 0.04 \cdot 768 \approx 30.7$  this is about 14%.

*Other frames.* If HF's pad-IDWT-crop map  $P$  is invertible, WaveFT is  $P_*$  of a mask by definition. For C3A, the powers  $P^0, \dots, P^{b-1}$  of the cyclic shift span the  $b \times b$  circulants and have Gram matrix  $bI$ , and  $\text{circ}(w) = F^{-1} \text{diag}(Fw)F$ ; the real form follows from conjugate eigenvalues ([Auxiliary result B.16\(a\), \(b\)](#)). So a block-circulant update is a mask in the frame  $\{E_{ij}^{(\text{blk})} \otimes P^k\}$ . LoRA-XS's update  $U_r R \Sigma_r V_r^\top = \sum_{i,j \leq r} R_{ij} \sigma_j u_i v_j^\top$  is a scaled mask in the orthonormal frame  $(u_i v_j^\top)$  built from the SVD of  $W_0$ ; SVFT's update  $UMV^\top$  with sparse  $M$  is a mask there too, and TinyLoRA's is a random subspace of it. SHiRA, FISH and Trainable Tokens

update a fixed set of entries. In MiSS's balance mode HF computes the update on an input  $x$  as  $D$  times the sum of the  $g$  chunks of  $x$  of size  $r$ , that is  $D(\mathbf{1}_g^\top \otimes I_r)x$ ; this is LoRA-FA with the frozen projection  $\mathbf{1}_g^\top \otimes I_r$ . Its BAT mode is linear in  $M$  with coefficients from  $W_0$ , so it is linear but not additive.

**HiRA.** By [Proposition II.9\(a\)](#), HiRA is  $(D_{W_0})_*\text{LoRA}$  when  $W_0$  has no zero entries. A merge gives  $W_1 = W_0 \odot (J + Z_1)$ , and the next cycle  $W_1 \odot (J + Z_2) = W_0 \odot (J + Z_1) \odot (J + Z_2) = W_0 \odot (J + Z_1 + Z_2 + Z_1 \odot Z_2)$ . With  $Z_k = B_k A_k$ ,

$$Z_1 + Z_2 + Z_1 \odot Z_2 = [B_1, B_2, B_1 \bullet B_2] [A_1; A_2; (A_1^\top \bullet A_2^\top)^\top]$$

([Auxiliary result B.12\(c\)](#)), which factors through  $\mathbb{R}^p$  with  $p = 2r + r^2$ . Call the two factors  $L \in \mathbb{R}^{m \times p}$  and  $L' \in \mathbb{R}^{p \times n}$ . The rows of  $L$  and the columns of  $L'$  are points  $(x, y, x \otimes y)$  of a variety  $\mathcal{V}$  that spans  $\mathbb{R}^r \oplus \mathbb{R}^r \oplus \mathbb{R}^{r^2}$ : a linear form vanishing on  $\mathcal{V}$  vanishes on  $(x, 0, 0)$ , on  $(0, y, 0)$  and then on every  $(0, 0, x \otimes y)$ . Choosing points of  $\mathcal{V}$  one at a time outside the span of the previous ones,  $m$  rows can be chosen to span  $\min(m, p)$  dimensions, and likewise  $n$  columns. If  $p \leq \min(m, n)$ , then  $L$  has full column rank and  $L'$  full row rank, so  $\text{rank } LL' = p$ . If  $p > \min(m, n)$ , say  $m \leq n$ , choose  $L$  of rank  $m$ , that is  $L : \mathbb{R}^p \rightarrow \mathbb{R}^m$  onto; then  $L(\mathcal{V})$  spans  $\mathbb{R}^m$ , so  $n \geq m$  columns of  $L'$  can be chosen with images spanning  $\mathbb{R}^m$ , and  $\text{rank } LL' = m$ ; the case  $n < m$  is the transpose. In every case  $\text{rank } \min(2r + r^2, m, n)$  is attained, hence generic ([Auxiliary result B.10](#)). Transported ReLoRA instead reaches  $D_{W_0}(Z_1 + Z_2)$ , whose preimage  $Z_1 + Z_2$  has rank at most  $2r$ .  $\square$

## C.21 Definition II.14

**DEFINITION II.14** (covariance group). *For a method defined for every  $W_0$  (a natural family), the **covariance group**  $\mathcal{G}_M \subseteq \text{GL}_m \times \text{GL}_n$  is the largest subgroup with  $\text{Im } M(gW_0h^{-1}) = g \text{ Im } M(W_0) h^{-1}$ . When the method depends on random frames or on data, the identity is required in distribution, with the data transformed.  $\mathcal{G}_M$  is the geometry the method lives in, in Klein's sense.*

The definition contains one claim: a largest such subgroup exists.

*Proof.* Let  $\mathcal{S}$  be the set of  $(g, h) \in \text{GL}_m \times \text{GL}_n$  with  $\text{Im } M(gWh^{-1}) = g \text{ Im } M(W) h^{-1}$  for every  $W$  in the domain of the family (which we assume stable under  $\text{GL}_m \times \text{GL}_n$ , or at least under the pairs considered). The pair  $(I, I)$  lies in  $\mathcal{S}$ . If  $(g_1, h_1), (g_2, h_2) \in \mathcal{S}$ , then for every  $W$

$$\text{Im } M(g_1 g_2 W h_2^{-1} h_1^{-1}) = g_1 \text{ Im } M(g_2 W h_2^{-1}) h_1^{-1} = g_1 g_2 \text{ Im } M(W) h_2^{-1} h_1^{-1}.$$

If  $(g, h) \in \mathcal{S}$ , apply its identity at  $g^{-1}Wh$ :  $\text{Im } M(W) = g \text{ Im } M(g^{-1}Wh) h^{-1}$ , so  $\text{Im } M(g^{-1}Wh) = g^{-1} \text{ Im } M(W) h$ . So  $\mathcal{S}$  is a subgroup, every subgroup with the property is contained in it, and  $\mathcal{G}_M = \mathcal{S}$ . For random frames or data, read each identity as equality of the laws of the random sets, the pushforwards of the frame and data laws under  $F \mapsto \text{Im } M(W; F)$ , with the data transformed by the action of  $(g, h)$  (for instance  $x \mapsto hx$ , so  $C \mapsto hCh^\top$ ). The same computation applies, with each equality an equality of laws and with  $X \mapsto gXh^{-1}$  acting on laws by pushforward; it uses only that the data transformations compose as a group action.  $\square$

## C.22 Observation II.15

**OBSERVATION II.15** (descent through function-preserving reparametrisations). Let  $M = \prod_b M_b$  be a box-wise method: each  $M_b$  is a member of a natural family with covariance group  $\mathcal{G}_{M_b}$ , and no trainable parameters or random frames are shared across boxes. Let  $G_{\text{arch}}$  act on  $\Theta = \prod_b \Theta_b \times \Theta_{\text{frozen}}$ , preserving this split and acting on each adapted box by  $W_b \mapsto g_b W_b h_b^{-1}$  with  $(g_b, h_b) \in \mathcal{G}_{M_b}$ . Assume  $f(g\theta, -) = f(\theta, -)$  on a  $G_{\text{arch}}$ -stable set containing  $\text{Im } M(\theta_0)$ . Then  $\text{Im } M(g\theta_0) = g \cdot \text{Im } M(\theta_0)$ , and the set of functions reachable by  $M$  from  $g\theta_0$  equals the set reachable from  $\theta_0$ . For methods with random frames, the equality holds in distribution, provided the frames are drawn independently per box. Boxes that share a frame need a pathwise condition: a law-preserving map  $T_{(g,h)}$  on frame space with  $\text{Im } M_b(g_b W h_b^{-1}; TF) = g_b \text{Im } M_b(W; F) h_b^{-1}$  for all  $W$  and  $F$ , with the same  $T$  on every box that shares  $F$  (marginal covariance in law, as in [Definition II.14](#), is not enough). For data-dependent methods, require covariance pathwise in the data, and that the box- $b$  data at  $g\theta_0$  is the [Definition II.14](#) transform under  $(g_b, h_b)$  of the data at  $\theta_0$ . This holds for intertwiner gauges such as hidden-unit permutations, query-key and value-output head gauges and residual rotations. The statement concerns reachable sets only; training trajectories and outcomes can still differ, for instance under Adam or non-isotropic initialisations.

Examples of  $G_{\text{arch}}$ :

- hidden-neuron permutations;
- the query-key gauge  $(W_Q, W_K) \mapsto (gW_Q, g^{-\top}W_K)$  per head. With no positional encoding or an absolute one,  $g \in \text{GL}_{d_k}$ . Under RoPE,  $g$  must commute with every RoPE rotation, which leaves the per-frequency  $2 \times 2$  rotation-scalings  $(\mathbb{C}^\times)^{d_k/2}$ , shared across a GQA group. Under partial RoPE on  $r$  of the  $d_k$  dimensions the group is  $(\mathbb{C}^\times)^{r/2} \times \text{GL}_{d_k-r}$ . Under QK-norm,  $g$  must in addition be orthogonal when the norm's  $\varepsilon > 0$  (conformal,  $g \in \text{CO}(d_k)$ , when  $\varepsilon = 0$ ) and satisfy  $g^\top \Gamma_q R_t \Gamma_k g = \Gamma_q R_t \Gamma_k$  for every RoPE rotation  $R_t$ , where  $\Gamma_q, \Gamma_k$  are the gains. OLMo 2 normalises  $q$  and  $k$  across all heads jointly;
- a single global orthogonal rotation  $Q$  of the residual stream of a pre-norm transformer whose norm gains can all be fused into the linear layers they feed, after LayerNorm is converted to RMSNorm (SliceGPT's computational invariance; QuaRot's offline rotation). SliceGPT's per-block rotations  $Q_\ell$  need extra residual matrices and change the architecture. Sandwich and post-sublayer norms (Gemma 2 and 3, OLMo 2) are excluded, because their gains cannot be fused and  $\text{diag}(\gamma)$  does not commute with  $Q$ . The comparison is between the gain-fused model and the fused-and-rotated model: gain fusion is a diagonal input rescaling outside  $G_{\text{arch}}$  that already changes PiSSA's split and block OFT's image ([Proposition II.16](#)), and converting LayerNorm to RMSNorm changes LoRA's reachable set on the residual writers.

*Proof.* The deterministic statement. No parameters are shared, so  $\text{Im } M(\theta) = \prod_b \text{Im } M_b(W_b) \times \{\theta_{\text{frozen}}\}$ . For  $g \in G_{\text{arch}}$ , which preserves the split,

$$\begin{aligned} \text{Im } M(g\theta_0) &= \prod_b \text{Im } M_b(g_b W_{0,b} h_b^{-1}) \times \{g\theta_{0,\text{frozen}}\} \\ &= \prod_b g_b \text{Im } M_b(W_{0,b}) h_b^{-1} \times \{g\theta_{0,\text{frozen}}\} = g \cdot \text{Im } M(\theta_0), \end{aligned}$$

using  $(g_b, h_b) \in \mathcal{G}_{M_b}$  in each factor. The reachable functions are then

$$\{\Phi_f(\theta) : \theta \in \text{Im } M(g\theta_0)\} = \{\Phi_f(g\theta) : \theta \in \text{Im } M(\theta_0)\} = \{\Phi_f(\theta) : \theta \in \text{Im } M(\theta_0)\},$$

since  $f(g\theta, -) = f(\theta, -)$  for  $\theta \in \text{Im } M(\theta_0)$ . (Stability of the set under  $G_{\text{arch}}$  is not needed.)

*Independent frames.* With frames  $F_b$  drawn independently per box, the law of the tuple of sets  $\text{Im } M_b(g_b W_{0,b} h_b^{-1}; F_b)$  is the product of the marginal laws, and by covariance in law each marginal equals the law of  $g_b \text{Im } M_b(W_{0,b}; F_b) h_b^{-1}$ . So the tuple has the law of the tuple of sets  $g_b \text{Im } M_b(W_{0,b}; F_b) h_b^{-1}$ . The reachable function set is a fixed measurable function of the image, and the function-level identity above holds pathwise, so the laws of the reachable function sets agree.

*Shared frames.* Under the pathwise condition,  $\text{Im } M(g\theta_0; TF) = g \cdot \text{Im } M(\theta_0; F)$  for every  $F$ , with the same  $T$  in every box that uses  $F$ , and  $TF$  has the law of  $F$ . Hence  $\text{Im } M(g\theta_0; F)$  and  $g \cdot \text{Im } M(\theta_0; F)$  have the same law. Marginal covariance does not suffice. Take two boxes in  $\mathbb{R}^{2 \times 1}$  sharing a Haar frame  $F \in \text{O}(2)$ , with the natural family  $\text{Im } M_b(W; F) = W + \mathbb{R} F a(W)$ , where  $a(W) = (w_1^2, w_2^2) / \|(w_1^2, w_2^2)\|$  for  $W \neq 0$ . For  $g \in \text{O}(2)$ , both  $Fa(gW)$  and  $gFa(W)$  are uniform on the circle, so each box is covariant in law under  $(g, 1)$ . At  $\theta_0 = (e_1, e_2)$  the two lines have directions  $Fe_1$  and  $Fe_2$ , which are orthogonal, and so are those of  $g \cdot \text{Im } M(\theta_0; F)$ . For  $g$  the rotation by  $\pi/4$ ,  $g\theta_0 = ((e_1 + e_2)/\sqrt{2}, (e_2 - e_1)/\sqrt{2})$  has  $a(ge_1) = a(ge_2) = (e_1 + e_2)/\sqrt{2}$ , so the two lines of  $\text{Im } M(g\theta_0; F)$  are parallel. The joint laws differ, and the conclusion fails; the numerical check reports a function-level instance.

*Data.* Suppose each  $M_b$  is covariant pathwise in its data,

$$\text{Im } M_b(g_b W h_b^{-1}; (g_b, h_b) \cdot D) = g_b \text{Im } M_b(W; D) h_b^{-1},$$

and the box- $b$  data at  $g\theta_0$  is  $(g_b, h_b) \cdot D_b$ . Then the deterministic argument applies with the data fixed. For an intertwiner gauge the input of box  $b$  in the network with weights  $g\theta$  is  $h_b$  times its input with weights  $\theta$ , for every network input, so calibration inputs transform by  $x \mapsto h_b x$  and second moments by  $C \mapsto h_b C h_b^\top$ , as [Definition II.14](#) requires. The examples below are of this kind: a hidden-unit permutation permutes the inputs of the next box; the query-key and value-output gauges leave the inputs of  $W_Q, W_K, W_V$  unchanged ( $h = I$ ) and transform the input of  $W_O$ ; a residual rotation  $Q$  rotates every normalised residual input by  $Q$ . Training is not covered: [Theorem III.12](#) shows that Adam is not equivariant under general rotations of LoRA's gauge, and a Kaiming-uniform  $A_0$  is not rotation-invariant in law.

*Hidden-neuron permutations.* For a permutation matrix  $P$ ,  $\sigma(Pz) = P\sigma(z)$  for an entry-wise activation and  $P(u \odot v) = Pu \odot Pv$ , so  $(W_{\text{up}}, W_{\text{gate}}, W_{\text{down}}) \mapsto (PW_{\text{up}}, PW_{\text{gate}}, W_{\text{down}} P^\top)$  preserves the block.

*The query-key gauge.* Without positional rotation the score is  $x^\top W_Q^\top W_K y$ , and since  $(gW_Q)^\top g^{-\top} W_K = W_Q^\top W_K$ , every  $g \in \text{GL}_{d_k}$  works. Under RoPE the score between positions  $s$  and  $t$  is  $x^\top W_Q^\top R_{t-s} W_K y$  with commuting rotations  $R_\tau$  acting by angle  $\tau\omega_j$  on the  $j$ -th coordinate plane. After the gauge it is  $x^\top W_Q^\top g^\top R_{t-s} g^{-\top} W_K y$ . For  $W_Q, W_K$  of full row rank, invariance for all inputs and positions is equivalent to  $g^\top R_\tau g^{-\top} = R_\tau$  for every  $\tau$ , that is,  $g^\top$  commutes with every  $R_\tau$ . RoPE's frequencies  $\omega_j = \text{base}^{-2j/d_k}$  are distinct and lie in  $(0, 1] \subset (0, \pi)$ , so  $R_1$  has the distinct eigenvalues  $e^{\pm i\omega_j}$ ; a matrix commuting with  $R_1$  preserves each coordinate plane and commutes with a nontrivial rotation on it, so it is a rotation-scaling  $aI_2 + bJ$  there. The commutant is  $(\mathbb{C}^\times)^{d_k/2}$  in the invertible case, and it is closed under transposition, so the condition on  $g^\top$  is the same condition on  $g$ . In a GQA group one  $W_K$  is paired with several  $W_Q$ , so all of them use the same  $g$ . Under partial RoPE,  $R_1$  is the identity on the  $d_k - r$  unrotated coordinates and has no eigenvalue 1 on the others, so commuting matrices preserve this splitting, and the group is  $(\mathbb{C}^\times)^{r/2} \times \text{GL}_{d_k-r}$ .

*QK-norm.* The score becomes  $(\Gamma_q N(q))^\top R_t (\Gamma_k N(k))$  with  $q = W_Q x$ ,  $k = W_K y$  and  $N(v) = v (\|v\|^2/d_k + \varepsilon)^{-1/2}$ . If  $\varepsilon > 0$  and  $g$  is orthogonal, then  $g^{-\top} = g$  and  $N(gv) = gN(v)$ , so the

score becomes  $N(q)^\top g^\top \Gamma_q R_t \Gamma_k g N(k)$ , which is invariant under the stated relation. If  $\varepsilon = 0$ , then  $N$  removes scale, and  $g = cO$  with  $c > 0$ ,  $O$  orthogonal, gives  $N(gq) = ON(q)$  and  $N(g^{-\top}k) = ON(k)$ ; the relation is then required of  $O = g/c$ . Conversely, let  $W_Q, W_K$  have full row rank, so that  $q$  and  $k$  range over  $\mathbb{R}^{d_k}$ . Invariance at  $t = 0$  gives  $\langle \Gamma_q N(gq), \Gamma_k N(g^{-\top}k) \rangle = \langle \Gamma_q N(q), \Gamma_k N(k) \rangle$  for all  $q, k$ . The values of  $N$  span  $\mathbb{R}^{d_k}$  (they fill a ball when  $\varepsilon > 0$  and a sphere when  $\varepsilon = 0$ ), and the gains are invertible, so we may choose  $k_1, \dots, k_{d_k}$  for which the vectors  $\Gamma_k N(g^{-\top}k_i)$  form a basis. Solving the resulting linear equations gives  $\Gamma_q N(gq) = A' \Gamma_q N(q)$  for a fixed matrix  $A'$ , so  $N(gq) = A N(q)$  with  $A = \Gamma_q^{-1} A' \Gamma_q$ , and  $A$  is invertible because the values  $N(gq)$  span  $\mathbb{R}^{d_k}$ . If  $\varepsilon > 0$ , letting  $q \rightarrow 0$  (where  $N(v) = v/\sqrt{\varepsilon} + O(\|v\|^3)$ ) gives  $A = g$ , and then  $\|gq\| = \|q\|$  for all  $q$ , so  $g$  is orthogonal. If  $\varepsilon = 0$ ,  $\|N(v)\| = \sqrt{d_k}$  for all  $v \neq 0$ , so  $A$  is orthogonal, and  $gq$  is a positive multiple of  $Aq$  for every  $q$ , so  $A^{-1}g$  has every vector as an eigenvector and  $g = cA$  with  $c > 0$ . With  $g$  orthogonal (or  $O = g/c$ ), invariance for all  $t$  and all  $N(q), N(k)$  forces the stated relation.

*Residual rotations.* In a pre-norm transformer with gains fused and RMSNorm  $N(h) = h(\|h\|^2/d + \varepsilon)^{-1/2}$ , replace the residual stream  $h$  by  $Qh$ : embeddings  $E \mapsto QE$ , the writers  $W_O, W_{\text{down}} \mapsto QW_O, QW_{\text{down}}$ , and every reader (the  $q, k, v$ , gate and up projections and the head)  $W \mapsto WQ^\top$ . Since  $\|Qh\| = \|h\|$ ,  $N(Qh) = QN(h)$ , so each reader sees  $WQ^\top QN(h) = WN(h)$ , and the function is unchanged. A gain that is not fused would sit between  $N$  and the reader as  $\text{diag}(\gamma)$ , and  $\text{diag}(\gamma)Q \neq Q\text{diag}(\gamma)$  unless  $\gamma$  is constant. A post-sublayer norm writes  $h + \text{diag}(\gamma)N(W_O z)$ , and  $\text{diag}(\gamma)N(QW_O z) = \text{diag}(\gamma)QN(W_O z) \neq Q\text{diag}(\gamma)N(W_O z)$ , so the rotation does not preserve the function. LayerNorm subtracts the mean, the projection  $I - \mathbf{1}\mathbf{1}^\top/d$ , which commutes with  $Q$  only if  $Q\mathbf{1} = \pm\mathbf{1}$ ; SliceGPT therefore folds the projection into the writers and uses RMSNorm. In the original model a LayerNorm ignores the component of the residual stream along  $\mathbf{1}$ , so LoRA's updates of a writer reach functions only through  $(I - \mathbf{1}\mathbf{1}^\top/d)\Delta W$ ; in the converted model the writer is  $(I - \mathbf{1}\mathbf{1}^\top/d)W_O$  and an update  $\Delta W$  with a component along  $\mathbf{1}$  is seen by every later RMSNorm, so the reachable function sets differ. Gain fusion replaces a reader  $W$  by  $W\text{diag}(\gamma)$ , a non-conformal input rescaling when  $\gamma$  is not constant; by Proposition II.16 it changes PiSSA's split, and it changes block OFT's image  $\{W\text{diag}(\gamma)R\}$  against  $\{WR\text{diag}(\gamma)\}$  unless  $\text{diag}(\gamma)$  commutes with the block-orthogonal group, that is unless  $\gamma$  is constant on each block. Consecutive SliceGPT rotations  $Q_{\ell-1} \neq Q_\ell$  require the matrix  $Q_{\ell-1}^\top Q_\ell$  on the residual path, a new operation.  $\square$

### C.23 Proposition II.16

**PROPOSITION II.16** (covariant data-aware splits). *Let  $C = \mathbb{E}[xx^\top] \succ 0$  be the input second moment, so that under  $x \mapsto hx$  we have  $W \mapsto Wh^{-1}$  and  $C \mapsto hCh^\top$ . Let  $LL^\top = C$  and  $p := \min(m, n)$ . Write  $[M]_r$  for the rank- $r$  truncated SVD,  $\langle M \rangle_r := M - [M]_{p-r}$  for the bottom  $r$  singular components, and  $\text{CO}(n) = \mathbb{R}_{>0} \cdot \text{O}(n)$ . Assume  $1 \leq r < p$  and the spectral gap each split needs:  $\sigma_r > \sigma_{r+1}$  of  $W_0 L$  in (a); in (b),  $\sigma_r > \sigma_{r+1}$  of  $W_0 C$  or  $W_0$  for top- $r$  splits, and  $\sigma_{p-r} > \sigma_{p-r+1}$  for bottom- $r$  splits. Covariance groups are taken over the generic set where these gaps hold.*

(a) *The whitened split*

$$X^* = \arg \min_{\text{rank } X \leq r} \|(W_0 - X)L\|_F = [W_0 L]_r L^{-1} = \Pi_r W_0,$$

where  $\Pi_r$  is the top- $r$  eigenprojector of the output second moment  $W_0 C W_0^\top$ , is  $\text{GL}_n$ -covariant on the input side and  $\text{CO}(m)$ -covariant on the output side; its joint group is  $\text{CO}(m) \times \text{GL}_n$ . This is the classical reduced-rank-regression (SVD-LLM) solution.



(b) The formula-level CorDA splits, IPM ( $X = [W_0C]_r C^{-1}$ , the minimiser of  $\|(W_0 - X)C\|_F$ ) and KPM (the bottom- $r$  components of  $W_0C$  mapped back by  $C^{-1}$ ,  $X = \langle W_0C \rangle_r C^{-1}$ ), with or without CorDA's trace-relative damping  $C + \kappa(\text{tr } C/n)I$ , and the PiSSA and MiLoRA splits ( $[W_0]_r$  and the bottom- $r$  part  $\langle W_0 \rangle_r$ ), have input-side covariance group exactly  $\text{CO}(n)$ ; jointly,  $\text{CO}(m) \times \text{CO}(n)$ .

(c) Scope. An absolute damping  $C + \lambda I$  reduces the input-side group of (a) and of (b) to  $\text{O}(n)$ . The official and HF CorDA code divide each calibration sample by the absolute value of its largest entry before accumulating  $C$ , which keeps only positive scalings and coordinate permutations, so (b) describes the CorDA formulas and does not cover shipped CorDA under rotations.

*Proof.* A split  $X = X(W_0, C)$  of rank at most  $r$  yields the method with frozen residual  $W_0 - X$  and image  $(W_0 - X) + M_{\leq r}$ . Under  $(g, h)$ , with  $C' = hCh^\top$ , covariance of images means  $(gW_0h^{-1} - X(gW_0h^{-1}, C')) + M_{\leq r} = g(W_0 - X(W_0, C))h^{-1} + M_{\leq r}$ , using  $gM_{\leq r}h^{-1} = M_{\leq r}$ . Since  $r < p$ , the translation stabiliser of  $M_{\leq r}$  is  $\{0\}$  (equation (18)), so this is equivalent to

$$X(gW_0h^{-1}, hCh^\top) = gX(W_0, C)h^{-1}.$$

*Lemma.* Let  $1 \leq s < p$ ,  $g \in \text{GL}_m$  and  $k \in \text{GL}_n$ . Then  $[gMk]_s = g[M]_s k$  for every  $M \in \mathbb{R}^{m \times n}$  for which both truncations are unique iff  $g \in \text{CO}(m)$  and  $k \in \text{CO}(n)$ . “If”: for  $g = aP$  and  $k = bQ$  with  $a, b > 0$  and  $P, Q$  orthogonal, an SVD  $M = \sum_i \sigma_i u_i v_i^\top$  gives the SVD  $gMk = \sum_i (ab\sigma_i)(Pu_i)(Q^\top v_i)^\top$  with the same ordering, so  $[gMk]_s = g[M]_s k$ . “Only if”: let  $U_s \subseteq \mathbb{R}^m$  be any  $s$ -dimensional subspace and  $y \perp U_s$  a unit vector, and likewise  $V_s \subseteq \mathbb{R}^n$  and  $x \perp V_s$  (possible because  $s < p$ ). Let  $M_s$  have rank  $s$  with column space  $U_s$  and row space  $V_s$ , and for small  $t > 0$  put  $M = M_s + t yx^\top$ . Then  $[M]_s = M_s$ , and  $gMk = gM_s k + t(gy)(k^\top x)^\top$ , whose truncation is unique for small  $t$  because  $gM_s k$  has rank  $s$ . The identity gives  $[gMk]_s = gM_s k$ , so the residual  $t(gy)(k^\top x)^\top$  of the truncation must have its column space orthogonal to that of  $[gMk]_s$  and its row space orthogonal to that of  $[gMk]_s$  (Auxiliary result B.11(a):  $[X]_s$  and  $X - [X]_s$  are built from disjoint sets of singular vectors). So  $gy \perp gU_s$  and  $k^\top x \perp k^\top V_s$ . Since every pair of orthogonal unit vectors  $u, y$  occurs with  $u \in U_s$  and  $y \perp U_s$ ,  $g$  maps orthogonal vectors to orthogonal vectors. Then for orthonormal eigenvectors  $e, f$  of  $g^\top g$  with eigenvalues  $\alpha, \beta$ ,  $0 = \langle g(e+f), g(e-f) \rangle = \alpha - \beta$ , so  $g^\top g$  is scalar and  $g \in \text{CO}(m)$ ; likewise  $k^\top \in \text{CO}(n)$ .

(a) By reduced-rank regression (Auxiliary result B.11(c)),  $X^* = [W_0L]_r L^{-1}$ , unique under the gap, and  $[W_0L]_r = \Pi_r W_0 L$  with  $\Pi_r$  the top- $r$  eigenprojector of  $W_0 L L^\top W_0^\top = W_0 C W_0^\top$ ; so  $X^* = \Pi_r W_0$ , which does not depend on the choice of  $L$ . Under  $(g, h)$  the output second moment becomes  $gW_0h^{-1} hCh^\top h^{-\top} W_0^\top g^\top = gW_0 C W_0^\top g^\top$ , and with the square root  $hL$  of  $hCh^\top$ ,

$$X^*(gW_0h^{-1}, hCh^\top) = [gW_0L]_r L^{-1} h^{-1}.$$

This equals  $gX^*h^{-1} = g[W_0L]_r L^{-1} h^{-1}$  for all  $W_0$  (with gaps) iff  $[gM]_r = g[M]_r$  for all  $M = W_0L$ , which range over all matrices with a gap. By the lemma with  $k = I$  this holds iff  $g \in \text{CO}(m)$ , with no condition on  $h$ . So the joint group is  $\text{CO}(m) \times \text{GL}_n$ , and its input and output parts are  $\text{GL}_n$  and  $\text{CO}(m)$ .

(b), undamped. For IPM,  $X(W, C) = [WC]_r C^{-1}$  minimises  $\|(W - X)C\|_F$  (Auxiliary result B.11(c) with  $L = C$ ). Under  $(g, h)$ , with  $M = W_0C$ ,

$$X(gW_0h^{-1}, hCh^\top) = [gW_0Ch^\top]_r h^{-\top} C^{-1} h^{-1} = [gMh^\top]_r h^{-\top} C^{-1} h^{-1},$$

and covariance asks this to equal  $g[M]_r C^{-1} h^{-1}$ , that is  $[gMh^\top]_r = g[M]_r h^\top$ . As  $W_0$  varies,  $M = W_0C$  ranges over all matrices, so by the lemma covariance holds iff  $g \in \text{CO}(m)$  and

$h^\top \in \text{CO}(n)$ , that is  $h \in \text{CO}(n)$ . KPM:  $X = \langle W_0 C \rangle_r C^{-1} = W_0 - [W_0 C]_{p-r} C^{-1}$ , and the same computation with  $s = p - r \in [1, p - 1]$  gives the condition  $[g M h^\top]_{p-r} = g [M]_{p-r} h^\top$ , with the gap  $\sigma_{p-r} > \sigma_{p-r+1}$ . PiSSA:  $[g W_0 h^{-1}]_r = g [W_0]_r h^{-1}$  for all  $W_0$  iff  $g \in \text{CO}(m)$  and  $h^{-1} \in \text{CO}(n)$ ; MiLoRA:  $\langle W \rangle_r = W - [W]_{p-r}$  reduces to the lemma at  $s = p - r$ . In each case the input-side group (with  $g = I$ ) is  $\text{CO}(n)$  and the joint group is  $\text{CO}(m) \times \text{CO}(n)$ .

(b), *trace-relative damping*. Write  $C_\kappa = C + \kappa(\text{tr } C/n)I$  with  $\kappa > 0$  and consider  $X_\kappa(W, C) = X(W, C_\kappa)$  for IPM (KPM is identical with  $s = p - r$ ). For  $h = cQ \in \text{CO}(n)$ ,  $h C_\kappa h^\top = c^2 Q C Q^\top + \kappa(\text{tr } C/n) c^2 I = (h C h^\top)_\kappa$ , since  $\text{tr}(c^2 Q C Q^\top) = c^2 \text{tr } C$ ; so damping commutes with  $\text{CO}(n)$  and the damped splits inherit the covariance of the undamped ones. Conversely, let  $(g, h)$  be in the joint group, put  $S_1 = (h C h^\top)_\kappa$ ,  $S_2 = C_\kappa$  and  $N = W_0 h^{-1}$ . Covariance reads  $[g N S_1]_r S_1^{-1} = g [N h S_2]_r S_2^{-1} h^{-1}$ . With  $M = N S_1$ , which ranges over all matrices, and  $G = S_1^{-1} h S_2$ , this becomes  $[g M]_r G = g [M G]_r$ , or  $[g M' G^{-1}]_r = g [M']_r G^{-1}$  with  $M' = M G$ . By the lemma,  $g \in \text{CO}(m)$  and  $G \in \text{CO}(n)$  for every  $C \succ 0$ , that is,

$$h S_2^2 h^\top = c(C)^2 S_1^2 \quad \text{for every } C \succ 0.$$

Both sides are continuous in  $C$ , and  $c(C)^2 = \text{tr}(h S_2^2 h^\top) / \text{tr}(S_1^2)$  is too, so the identity extends to  $C = \nu \nu^\top$  with  $\nu \neq 0$ . There, with  $w = h \nu$ ,  $a = \kappa \|\nu\|^2 / n > 0$  and  $b = \kappa \|w\|^2 / n$ ,  $S_2^2 = (\|\nu\|^2 + 2a) \nu \nu^\top + a^2 I$  and  $S_1^2 = (\|w\|^2 + 2b) w w^\top + b^2 I$ , so

$$h h^\top = \alpha_w I + \beta_w w w^\top \quad (\alpha_w = c^2 b^2 / a^2),$$

for every  $w \in \mathbb{R}^n$ , since  $w = h \nu$  ranges over all vectors. So  $h h^\top - \alpha_w I$  has rank at most one for every  $w$ . If  $h h^\top$  were not scalar, it would have an eigenvalue  $\alpha$  of multiplicity  $n - 1$  with  $h h^\top = \alpha I + \gamma e e^\top$ ,  $\gamma \neq 0$ ; taking  $w$  neither parallel nor orthogonal to  $e$ , the matrix  $(\alpha - \alpha_w)I + \gamma e e^\top$  would have rank at most one with range along  $w$ , which is impossible (for  $n \geq 3$  it has rank at least  $n - 1 \geq 2$  unless  $\alpha_w = \alpha$ , and then its range is  $\mathbb{R}e \neq \mathbb{R}w$ ; for  $n = 2$  its range is  $\mathbb{R}e$  or  $e^\perp$ ). So  $h h^\top$  is scalar and  $h \in \text{CO}(n)$ . The input-side group of the damped splits is therefore exactly  $\text{CO}(n)$ , and the joint group  $\text{CO}(m) \times \text{CO}(n)$ .

(c), *absolute damping*.  $Q(C + \lambda I)Q^\top = Q C Q^\top + \lambda I$  for  $Q \in \text{O}(n)$ , so the damped splits of (a) and (b) are  $\text{O}(n)$ -covariant. We show that no other  $h$  is in their input-side groups.

For (b): the undamped splits are invariant under  $C \mapsto tC$ , so the damped split at  $(W, tC)$  equals the undamped split at  $(W, C + (\lambda/t)I)$ , which tends to the undamped split at  $(W, C)$  as  $t \rightarrow \infty$  wherever the gap holds, because a truncation is continuous where it is unique (it is the unique minimiser of a continuous function over the closed set  $\mathcal{M}_{\leq s}$ ). The same holds at the transformed data  $(W h^{-1}, t h C h^\top)$ . So every  $h$  in the damped group lies in the undamped group  $\text{CO}(n)$ .

For (a): with  $S = (h^\top h)^{-1}$ , the transformed damped moment is  $W_0 h^{-1} (h C h^\top + \lambda I) h^{-\top} W_0^\top = W_0 (C + \lambda S) W_0^\top$ , so covariance reads  $\Pi_r(W_0 (C + \lambda S) W_0^\top) W_0 = \Pi_r(W_0 (C + \lambda I) W_0^\top) W_0$ . Both projectors have range in  $\text{col } W_0$ , onto which  $W_0$  maps, so this is equality of the projectors. Replacing  $C$  by  $tC$  and letting  $t \rightarrow 0$  gives  $\Pi_r(W_0 S W_0^\top) = \Pi_r(W_0 W_0^\top)$  for every  $W_0$  for which both have a gap at  $r$ . Suppose  $S$  is not scalar, with orthonormal eigenvectors  $e, f$  for eigenvalues  $\alpha > \beta$ , and complete them to an orthonormal eigenbasis  $e, f, g_3, \dots, g_n$  of  $S$ . Choose orthonormal  $z_1, \dots, z_p$  with  $z_r = \cos \phi e + \sin \phi f$ ,  $z_{r+1} = -\sin \phi e + \cos \phi f$  for some  $\phi \notin \frac{\pi}{2}\mathbb{Z}$ , and the other  $z_i$  among the  $g_j$ ; put  $W_0 = \sum_{i \leq p} d_i e_i z_i^\top$  with  $d_1 > \dots > d_p > 0$ . Then  $W_0 W_0^\top = \text{diag}(d_1^2, \dots, d_p^2)$  (padded), with top- $r$  eigenspace  $\text{span}\{e_1, \dots, e_r\}$ , while  $W_0 S W_0^\top$  is diagonal except for the entry  $d_r d_{r+1}(\beta - \alpha) \sin \phi \cos \phi \neq 0$  in positions  $(r, r + 1)$  and  $(r + 1, r)$ . Taking  $d_i$  for  $i < r$  much larger and  $d_i$  for  $i > r + 1$  much smaller than  $d_r, d_{r+1}$ , the top- $r$  eigenspace of  $W_0 S W_0^\top$  is

$\text{span}\{e_1, \dots, e_{r-1}\}$  plus the top eigenvector of the  $2 \times 2$  block, which is not  $e_r$ ; both matrices have a gap at  $r$ , and the projectors differ. Hence  $S$  is scalar and  $h \in \text{CO}(n)$ .

It remains to exclude  $h = cQ$  with  $c \neq 1$ ; since  $Q$  is in the group, it suffices to exclude  $h = cI$ . For (a),  $h = cI$  turns the condition into  $\Pi_r(W_0(C + \lambda c^{-2}I)W_0^\top) = \Pi_r(W_0(C + \lambda I)W_0^\top)$ . Replacing  $C$  by  $tC$ , the function  $P(\mu) = \Pi_r(W_0CW_0^\top + \mu W_0W_0^\top)$  would satisfy  $P(\mu) = P(c^{-2}\mu)$  for all  $\mu > 0$ , hence  $P(\mu) = P(c^{-2j}\mu)$  for all  $j \in \mathbb{Z}$ , and letting  $j \rightarrow \pm\infty$  gives  $P(0) = P(\infty)$ , the top- $r$  eigenprojectors of  $W_0CW_0^\top$  and of  $W_0W_0^\top$  (both limits exist under gaps). Take  $W_0$  rectangular-diagonal with entries  $d_1 > \dots > d_p > 0$  and  $C = \text{diag}(c_i)$  with  $c_i = i/d_i^2$  for  $i \leq p$  and  $c_i = 1$  beyond. Then  $W_0W_0^\top$  has top- $r$  eigenspace  $\text{span}\{e_1, \dots, e_r\}$  and  $W_0CW_0^\top = \text{diag}(1, 2, \dots, p)$  (padded) has top- $r$  eigenspace  $\text{span}\{e_{p-r+1}, \dots, e_p\}$ , which differ since  $r < p$ . For (b),  $h = cI$  gives, by scale invariance and linearity in  $W$ ,  $X(W_0, C + \lambda c^{-2}I) = X(W_0, C + \lambda I)$ , and the same iteration applied to  $G(\mu) = X(W_0, C + \mu I)$  gives  $G(0) = G(\infty)$  with  $G(0) = X(W_0, C)$  and  $G(\infty) = X(W_0, I)$ . With the same  $W_0$  and  $C$ ,  $W_0C$  is rectangular-diagonal with the increasing entries  $i/d_i$ , so for IPM  $G(0) = \sum_{i>p-r} d_i E_{ii}$  while  $G(\infty) = [W_0]_r = \sum_{i \leq r} d_i E_{ii}$ ; for KPM the roles of top and bottom are exchanged, with the same conclusion. So  $cI$  is not in the group, and the damped input-side groups are exactly  $\text{O}(n)$ .

(c), *per-sample normalisation*. The shipped code divides a sample  $x$  by  $v(x) = |\max_j x_j|$ . If  $h = cP$  with  $c > 0$  and  $P$  a permutation matrix,  $v(hx) = c v(x)$ , so the accumulated  $C$  transforms as  $hCh^\top$  up to the factor  $c^{-2}$ , to which the splits are insensitive, and  $h \in \text{CO}(n)$ . Conversely, suppose  $v(hx) = c v(x)$  for all  $x$  and some  $c > 0$  ( $n \geq 2$ ). For  $x = -e_j$ ,  $v(x) = 0$ , so  $\max_i (-h_{ij}) = 0$  and  $h \geq 0$  entrywise; for  $x = e_j$ ,  $\max_i h_{ij} = c$ ; for  $x = \mathbf{1}$ , every row sum of  $h$  is at most  $c$ . So a row  $i_j$  with  $h_{ij} = c$  has no other nonzero entry, the rows  $i_j$  are distinct, and  $h = cP$ . Under a rotation the normalisation does not commute with  $h$ , and the shipped split is not covariant. For instance, with  $n = m = 2$ ,  $r = 1$ , samples  $x_1 = e_1$  and  $x_2 = e_1 + e_2$  (so  $v = 1$  for both and  $C = x_1x_1^\top + x_2x_2^\top = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ , up to a factor), and  $h = Q$  the rotation by  $\pi/4$ : the rotated samples are  $(1, 1)/\sqrt{2}$  and  $(0, \sqrt{2})$ , normalised to  $(1, 1)$  and  $(0, 1)$ , so the shipped moment is  $C' = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ , while  $QCQ^\top = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 5 \end{pmatrix}$ . With  $W_0 = Q$ , the shipped IPM split at the rotated data is  $X(I, C') = [C']_1 C'^{-1}$ , the top eigenprojector of  $C'$ , while covariance would require  $X(Q, C)Q^\top = Q[C]_1 C^{-1}Q^\top$ , the top eigenprojector of  $QCQ^\top$ . The two symmetric matrices do not commute, so their top eigenvectors differ, and covariance fails.  $\square$

## C.24 Definition III.1

**DEFINITION III.1** (trained method; cometric; three equivalences). A trained method is a method together with optimizer data: a learning-rate operator  $\Lambda \succ 0$  (per-block rates), an initial scale, and an update rule. Gradient flow  $\dot{q} = -\Lambda \nabla_q (L \circ \rho)$  moves the weights by

$$\dot{\theta} = -K_q \nabla_\theta L, \quad K_q = D\rho_q \Lambda D\rho_q^\top.$$

We call  $K_q$  the **induced cometric**. For a backward method with compression  $c_t$  and anchor  $a_t$  (Definition IV.1),  $K_t = a_t c_t$ . Two methods are expressively equivalent if their images agree, first-order equivalent if their  $K_{q_0}$  agree, and dynamically equivalent if they produce the same  $\theta$ -trajectory for every loss.

A definition needs no proof. This subsection fixes the conventions used in all proofs of Exposé III and checks the two identities that the definition records.

**Conventions.** Weight space  $\Theta = \mathbb{R}^{m \times n}$  carries the Frobenius inner product  $\langle X, Y \rangle = \text{tr}(X^\top Y)$ , and every parameter space carries its Euclidean (Frobenius) inner product. Gradients are identified with covectors through these inner products, and the transpose of a linear map is its adjoint. A covector on  $\Theta$  is written as a matrix, usually  $G$ . When  $G = \nabla_\theta L$  is a weight gradient we say so; otherwise  $G$  is an arbitrary cotangent signal.

LoRA is  $\rho(B, A) = W_0 + sBA$  with  $B \in \mathbb{R}^{m \times r}$ ,  $A \in \mathbb{R}^{r \times n}$  and  $s > 0$ . Its differential and the adjoint of the differential are

$$D\rho_{(B,A)}(\dot{B}, \dot{A}) = s(\dot{B}A + B\dot{A}), \quad D\rho_{(B,A)}^\top(G) = (sGA^\top, sB^\top G), \quad (19)$$

since  $\langle G, s(\dot{B}A + B\dot{A}) \rangle = \langle sGA^\top, \dot{B} \rangle + \langle sB^\top G, \dot{A} \rangle$ . With  $G = \nabla_\theta L(\rho(B, A))$  this gives  $\nabla_B(L \circ \rho) = sGA^\top$  and  $\nabla_A(L \circ \rho) = sB^\top G$ , written  $\nabla_B$  and  $\nabla_A$ . Block-scalar rates are  $\Lambda(B, A) = (\eta_B B, \eta_A A)$  with  $\eta_A, \eta_B > 0$ .

The gauge group  $\text{GL}_r$  acts by  $g \cdot (B, A) = (Bg^{-1}, gA)$ , and  $\rho(g \cdot q) = \rho(q)$  (Background D.21). At  $q' = g \cdot q$  the weight is unchanged, hence so is the weight gradient  $G$ , and (19) gives

$$\nabla_{B'} = sGA^\top g^\top = \nabla_B g^\top, \quad \nabla_{A'} = sg^{-\top} B^\top G = g^{-\top} \nabla_A. \quad (20)$$

The full-rank locus is  $\{\text{rank } B = \text{rank } A = r\}$ . There  $BA$  has rank  $r$ ,  $U := \text{col } B = \text{col}(BA)$  and  $V := \text{row } A = \text{row}(BA)$ , and the orthogonal projectors onto them are  $P_U = B(B^\top B)^{-1}B^\top$  and  $P_V = A^\top(AA^\top)^{-1}A$ .

An update rule is *equivariant* under  $g$  (Background D.18) if, for every sequence of losses (the same data order and dropout masks in both runs), the run started from  $g \cdot q_0$  with correspondingly transformed optimizer state produces  $g \cdot q_t$  at every step  $t$ . Equivariance gives identical weight trajectories, since  $\rho(g \cdot q_t) = \rho(q_t)$ . It is checked one step at a time. If  $q'_t = g \cdot q_t$ , the two runs see the same weight and the same loss, hence the same  $G$ , and their gradients are related by (20). It then suffices that the steps are transported,

$$\delta B' = \delta B g^{-1}, \quad \delta A' = g \delta A, \quad (21)$$

together with the optimizer state, and induction over  $t$  concludes.

*Verification of the identities in Definition III.1.* By the chain rule  $\nabla_q(L \circ \rho) = D\rho_q^\top \nabla_\theta L$ , so gradient flow moves the weights by  $\dot{\theta} = D\rho_q \dot{q} = -D\rho_q \Lambda D\rho_q^\top \nabla_\theta L = -K_q \nabla_\theta L$ . The operator  $K_q$  is symmetric, and  $\langle \gamma, K_q \gamma \rangle = \langle D\rho_q^\top \gamma, \Lambda D\rho_q^\top \gamma \rangle \geq 0$ . Since  $\Lambda \succ 0$ ,  $\ker K_q = \ker D\rho_q^\top$ , so  $\text{rank } K_q = \text{rank } D\rho_q \leq \dim Q$ . Thus  $K_q$  is a cometric in the sense of Background D.23, possibly degenerate.

A backward method updates  $\theta \leftarrow \theta - \eta a_t(\text{opt}(c_t(\nabla L)))$  (Definition IV.1). When  $\text{opt}$  is the identity, that is, plain (S)GD on the compressed gradient, and  $\eta$  is absorbed into  $a_t$ , the continuous-time update is  $\dot{\theta} = -a_t c_t \nabla_\theta L$ , which is the statement  $K_t = a_t c_t$ . This operator is symmetric positive semidefinite when  $a_t = c_t^\top$ , as for GaLore ( $P_t P_t^\top$ ), LISA (a coordinate projection) and MeZO ( $z_t z_t^\top$ ).

Finally, for two methods trained by gradient flow, dynamical equivalence implies first-order equivalence. For a linear loss  $L(\theta) = \langle \gamma, \theta \rangle$  the initial weight velocities are  $-K_{q_0} \gamma$  and  $-K'_{q_0} \gamma$ , and  $\gamma$  is arbitrary.  $\square$

### C.25 Observation III.2

**OBSERVATION III.2** (chain rule for cometrics). Let  $\sigma : Q' \rightarrow Q$  and  $\rho : Q \rightarrow X$  be smooth, with  $X = \Theta$  or a function or output space. Train  $Q'$  by gradient flow with a constant rate operator  $\Lambda' \succ 0$ , and put  $K_{q'}^\sigma := D\sigma_{q'} \Lambda' D\sigma_{q'}^\top$ . Then

$$K_{q'}^{\rho\sigma} = D\rho_{\sigma(q')} K_{q'}^\sigma D\rho_{\sigma(q')}^\top,$$

and no metric on  $Q$  is used.  $K$  is a degenerate cometric (positive semidefinite, of rank  $\leq \dim Q'$ ); when  $\sigma$  is not injective,  $K^\sigma$  is a field along  $\sigma$  and fails to be a cometric on  $Q$ .

For LoRA,  $\rho(B, A) = W_0 + sBA$  with  $s = \alpha/r$  (or  $\alpha/\sqrt{r}$  for rsLoRA), with block rates  $\Lambda = \text{diag}(\eta_B I, \eta_A I)$  and no LoRA dropout, under gradient flow (equivalently, to first order in the step size for GD),

$$K_{(B,A)}(G) = s^2(\eta_B G A^\top A + \eta_A B B^\top G).$$

Under Adam there is no state-independent linear cometric; [Theorem III.10](#) and [Theorem III.12](#) treat Adam.

*Proof. Chain rule.* By definition  $K_{q'}^{\rho\sigma} = D(\rho\sigma)_{q'} \Lambda' D(\rho\sigma)_{q'}^\top$ , and  $D(\rho\sigma)_{q'} = D\rho_{\sigma(q')} D\sigma_{q'}$ . Hence

$$K_{q'}^{\rho\sigma} = D\rho_{\sigma(q')} (D\sigma_{q'} \Lambda' D\sigma_{q'}^\top) D\rho_{\sigma(q')}^\top = D\rho_{\sigma(q')} K_{q'}^\sigma D\rho_{\sigma(q')}^\top.$$

Equivalently, the flow  $\dot{q}' = -\Lambda' \nabla_{q'} (L \circ \rho \circ \sigma) = -\Lambda' D\sigma^\top D\rho^\top dL$  moves  $x = \rho(\sigma(q'))$  by  $\dot{x} = -D\rho K^\sigma D\rho^\top dL$ . In this formula  $D\rho_{\sigma(q')}^\top : T_x^* X \rightarrow T_{\sigma(q')}^* Q$  and  $D\sigma_{q'}^\top : T_{\sigma(q')}^* Q \rightarrow T_{q'}^* Q'$  are dual maps, and  $K_{q'}^\sigma : T_{\sigma(q')}^* Q \rightarrow T_{\sigma(q')}^* Q$  is built from  $\Lambda'$  alone. No inner product on  $Q$  enters.

*Degeneracy.* As in §C.24,  $K_{q'}^{\rho\sigma}$  is symmetric and positive semidefinite, and its rank is at most  $\text{rank } D\sigma_{q'} \leq \dim Q'$ . The operator  $K_{q'}^\sigma$  depends on  $q'$  and not only on  $\sigma(q')$ . For example, with  $\sigma(a, b) = ab$  from  $\mathbb{R}^2$  to  $\mathbb{R}$  and  $\Lambda' = I$ ,  $K_{(a,b)}^\sigma = a^2 + b^2$  takes the value 2 at  $(1, 1)$  and  $17/4$  at  $(2, 1/2)$ , two points over the same value  $\sigma = 1$ . So for non-injective  $\sigma$ ,  $K^\sigma$  is a field along  $\sigma$  (a section of  $\sigma^*(TQ \otimes TQ)$ ) and need not descend to  $Q$ .

*LoRA.* By (19),  $\Lambda D\rho^\top(G) = (\eta_B s G A^\top, \eta_A s B^\top G)$ , and applying  $D\rho$  gives

$$K_{(B,A)}(G) = s((\eta_B s G A^\top) A + B(\eta_A s B^\top G)) = s^2(\eta_B G A^\top A + \eta_A B B^\top G).$$

A discrete GD step  $\delta B = -\eta_B s G A^\top$ ,  $\delta A = -\eta_A s B^\top G$  changes the weight by

$$\rho(B + \delta B, A + \delta A) - \rho(B, A) = s(\delta B A + B \delta A) + s \delta B \delta A = -K_{(B,A)}(G) + s^3 \eta_A \eta_B G A^\top B^\top G,$$

and the last term is of order  $\eta^2$ . With LoRA dropout the factors are trained on masked inputs. The same computation then gives  $K_{(B,A)}$  applied to the masked signal  $\tilde{G}$  (for one example with output gradient  $\delta$ , input  $x$  and mask  $m$ ,  $\tilde{G} = \delta(m \odot x)^\top$ ), which differs from  $\nabla_\theta L$ . This is why the formula with  $G = \nabla_\theta L$  assumes no dropout.

*Adam.* Adam's step depends on its moment buffers, so it is not a function of  $q$  and  $G$  alone, and even the first step from zero state has no linear cometric. Take  $B = 0$  and  $A$  of full row rank. Then  $\nabla_A = 0$ , so  $A$  does not move, and the bias-corrected first step on  $B$  is  $\delta B = -\eta_B \phi(s G A^\top)$ , with  $\phi(x) = x/(|x| + \varepsilon)$  entrywise and  $0/0 = 0$ . (Without bias correction the step is a positive multiple of the same expression with a rescaled  $\varepsilon$ .) The weight moves by  $\delta W = s \delta B A = -s \eta_B \phi(s G A^\top) A$ , which is nonzero whenever  $G A^\top \neq 0$ , because  $A$  has full row rank. Suppose  $\delta W = -K(G)$  for a linear map  $K$ . At  $\varepsilon = 0$ ,  $\phi(cx) = \phi(x)$  for  $c > 0$  gives  $K(G) = K(cG) = cK(G)$ , so  $K = 0$ . For  $\varepsilon > 0$ ,  $|\phi| < 1$  makes  $G \mapsto \delta W$  bounded, and a bounded linear map is 0. Both conclusions contradict  $\delta W \neq 0$ .  $\square$



### C.26 Theorem III.3

**THEOREM III.3** (gradient descent is natural exactly where cometrics agree; for LoRA's linear gauge, exactly along  $\mathcal{O}(r)$ ). (a) Let  $\varphi : Q \rightarrow Q'$  be a diffeomorphism with  $\rho' \circ \varphi = \rho$ , and let  $\Lambda, \Lambda'$  be the rate operators on  $Q, Q'$ . Under gradient flow, the two methods induce the same  $\theta$ -velocity at  $q$  and at  $\varphi(q)$  for every loss iff  $K_q^\rho = K_{\varphi(q)}^{\rho'}$ , that is,

$$D\rho'_{\varphi(q)}(D\varphi_q \Lambda D\varphi_q^\top - \Lambda') D\rho_{\varphi(q)}^\top = 0.$$

This holds whenever  $D\varphi \Lambda D\varphi^\top = \Lambda'$ , which is an isometry when  $\Lambda = \Lambda' = I$ . That condition is necessary when  $D\rho'_{\varphi(q)}$  is injective, and fails to be necessary in general. Every linear gauge of a linear parametrisation is natural; for instance, for  $\rho(x, y) = \theta_0 + x + y$  the shear  $(x, y) \mapsto (2x, y - x)$  preserves  $\rho$  and  $K$  without being orthogonal. So is every pointwise-orthogonal non-linear LoRA gauge  $\varphi(q) = \varphi_{g(q)}(q)$  with  $g(q) \in \mathcal{O}(r)$ .

(b) For LoRA's linear gauge  $\varphi_g(B, A) = (Bg^{-1}, gA)$  with block-scalar rates, at points where  $A$  has full row rank ( $B$  arbitrary, including  $B = 0$ ) and  $r < n$ , the criterion holds iff  $g \in \mathcal{O}(r)$ . The same holds symmetrically where  $B$  has full column rank and  $r < m$ .

*Proof.* (a) Fix  $q \in Q$  and put  $\theta = \rho(q) = \rho'(\varphi(q))$ . Gradient flow with rates  $\Lambda$  on  $Q$  moves the weight at  $q$  by  $-K_q^\rho \nabla_\theta L$  (§C.24); gradient flow with rates  $\Lambda'$  on  $Q'$  moves it at  $\varphi(q)$  by  $-K_{\varphi(q)}^{\rho'} \nabla_\theta L$ . As  $L$  ranges over the linear losses  $\langle \gamma, \cdot \rangle$ , the covector  $\nabla_\theta L(\theta) = \gamma$  ranges over all of  $T_\theta^* \Theta$ , so the two velocities agree for every loss iff the linear maps  $K_q^\rho$  and  $K_{\varphi(q)}^{\rho'}$  agree. Differentiating  $\rho = \rho' \circ \varphi$  gives  $D\rho_q = D\rho'_{\varphi(q)} D\varphi_q$ , so

$$K_q^\rho - K_{\varphi(q)}^{\rho'} = D\rho'_{\varphi(q)}(D\varphi_q \Lambda D\varphi_q^\top - \Lambda') D\rho_{\varphi(q)}^\top,$$

which is the displayed criterion.

If  $D\varphi_q \Lambda D\varphi_q^\top = \Lambda'$ , the bracket vanishes. For  $\Lambda = \Lambda' = I$  the condition says that  $D\varphi_q$  is orthogonal; in general it says that  $\varphi$  is an isometry from  $(Q, \langle \cdot, \cdot \rangle, \Lambda^{-1} \cdot)$  to  $(Q', \langle \cdot, \cdot \rangle, \Lambda'^{-1} \cdot)$ . If  $D\rho'_{\varphi(q)}$  is injective, it has a left inverse  $E$ , and multiplying  $D\rho' M D\rho'^\top = 0$  by  $E$  on the left and by  $E^\top$  on the right gives  $M = 0$ ; so the condition is then necessary.

It is not necessary in general. Let  $\rho(x) = \theta_0 + \mathcal{L}x$  be affine with linear part  $\mathcal{L}$ , and let  $P$  be a linear gauge,  $\mathcal{L}P = \mathcal{L}$ , with  $\Lambda' = \Lambda$ . Then  $D\rho' = \mathcal{L}$  and  $D\varphi = P$ , and  $\mathcal{L}P\Lambda P^\top \mathcal{L}^\top = \mathcal{L}\Lambda \mathcal{L}^\top$ , so the criterion holds for every  $\Lambda$ . For  $\rho(x, y) = \theta_0 + x + y$  and  $\Lambda = I$ , the shear  $P(x, y) = (2x, y - x)$  satisfies  $\mathcal{L}P = \mathcal{L}$ , while in block form  $PP^\top = \begin{pmatrix} 4 & -2 \\ -2 & 2 \end{pmatrix} \neq I$ .

For a pointwise gauge  $\varphi(q) = \varphi_{g(q)}(q)$  of LoRA, with  $g : Q \rightarrow \text{GL}_r$  smooth and  $\varphi$  a diffeomorphism, the product rule gives  $D\varphi_q = \varphi_{g(q)} + V_q$ . Here  $\varphi_{g(q)}$  is the linear map  $(\dot{B}, \dot{A}) \mapsto (\dot{B}g(q)^{-1}, g(q)\dot{A})$ , and  $V_q(v) = \frac{d}{d\epsilon}|_{\epsilon=0} \varphi_{g(q+\epsilon v)}(q)$  is tangent to the  $\text{GL}_r$ -orbit of  $\varphi(q)$ . That orbit lies in a fibre of  $\rho$ , so  $D\rho_{\varphi(q)} V_q = 0$  and  $D\rho_{\varphi(q)} D\varphi_q = D\rho_{\varphi(q)} \varphi_{g(q)}$ . The criterion depends on  $D\varphi_q$  only through  $D\rho_{\varphi(q)} D\varphi_q$ , so it reduces to the criterion for the linear gauge  $\varphi_{g(q)}$  at  $q$ , which holds for  $g(q) \in \mathcal{O}(r)$  by (b).

(b) Here  $Q' = Q$ ,  $\rho' = \rho$ ,  $\Lambda' = \Lambda$  is block-scalar and  $\varphi_g(q) = g \cdot q$ . By [Observation III.2](#),

$$K_{g \cdot q}(G) = s^2 (\eta_B G A^\top g^\top g A + \eta_A B g^{-1} g^{-\top} B^\top G),$$

so  $K_q = K_{g \cdot q}$  iff, for every  $G \in \mathbb{R}^{m \times n}$ ,

$$GX = YG, \quad X = \eta_B A^\top (g^\top g - I) A \in \mathbb{R}^{n \times n}, \quad Y = \eta_A B (I - g^{-1} g^{-\top}) B^\top \in \mathbb{R}^{m \times m}.$$

*Matrix-unit lemma:* if  $GX = YG$  for every  $G \in \mathbb{R}^{m \times n}$ , then  $X = cI_n$  and  $Y = cI_m$  for some  $c \in \mathbb{R}$ . Take  $G = E_{ij} = e_i e_j^\top$ . Comparing the  $(k, l)$  entries of  $E_{ij}X$  and  $YE_{ij}$  gives  $\delta_{ik}X_{jl} = Y_{ki}\delta_{jl}$ . With  $k = i$  and  $l \neq j$  this gives  $X_{jl} = 0$ ; with  $l = j$  and  $k \neq i$  it gives  $Y_{ki} = 0$ ; with  $k = i$  and  $l = j$  it gives  $X_{jj} = Y_{ii}$  for all  $i, j$ .

Now  $\text{rank } X \leq r < n$ , so  $X = cI_n$  forces  $c = 0$ , that is,  $A^\top(g^\top g - I)A = 0$ . Since  $A$  has full row rank,  $AA^\top$  is invertible, and multiplying by  $(AA^\top)^{-1}A$  on the left and by  $A^\top(AA^\top)^{-1}$  on the right gives  $g^\top g = I$ . Conversely,  $g \in O(r)$  makes  $X$  and  $Y$  vanish. Only the full row rank of  $A$  and  $r < n$  were used, and  $B$  is arbitrary, including  $B = 0$ . If instead  $B$  has full column rank and  $r < m$ , then  $\text{rank } Y \leq r < m$  forces  $c = 0$ , so  $B(I - g^{-1}g^{-\top})B^\top = 0$ ; multiplying by  $(B^\top B)^{-1}B^\top$  on the left and by  $B(B^\top B)^{-1}$  on the right gives  $g^{-1}g^{-\top} = I$ , and again  $g \in O(r)$ .  $\square$

**REMARK** (the family of plain-GD dynamics; LoRA+ under GD). The formula for  $K_{g \cdot q}$  depends on  $g$  only through  $g^\top g$ , since  $g^{-1}g^{-\top} = (g^\top g)^{-1}$ , and  $g^\top g = h^\top h$  iff  $hg^{-1} \in O(r)$ . So along a gauge orbit the cometrics of plain gradient descent are indexed by  $O(r) \backslash \text{GL}_r$ , which  $g \mapsto g^\top g$  identifies with  $\text{Sym}_r^+$  (Background D.27, Auxiliary result B.19(a)). For  $g = cI$  with  $c > 0$ , gradient flow with a single rate  $\eta$  started from  $g \cdot q$ , pulled back by  $g^{-1}$ , satisfies  $\dot{B} = c\dot{B}' = -\eta c^2 \nabla_B$  and  $\dot{A} = c^{-1}\dot{A}' = -\eta c^{-2} \nabla_A$  by (20). This is LoRA with rates  $(\eta_A, \eta_B) = (\eta c^{-2}, \eta c^2)$ , and the same holds step by step for GD, whose steps are linear in the gradients. Choosing  $c = \lambda^{1/4}$  and  $\eta = \sqrt{\lambda} \eta_A$  gives LoRA+ with ratio  $\lambda = \eta_B/\eta_A$ . Under gradient descent, LoRA+ is therefore the scalar gauge  $\lambda^{1/4}I$  together with an overall change of rate.

## C.27 Proposition III.4

**PROPOSITION III.4** (cometrics that descend to the rank- $r$  manifold; known in part). Write  $U = \text{col } B$  and  $V = \text{row } A$ . Statements (a), (b) and the first sentence of (d) are on the full-rank locus; (c) needs only  $\text{rank } A_0 = r$ , with  $B$  arbitrary, including  $B_0 = 0$ . All statements concern gradient flow, or (S)GD to first order, on the stated gradients, undamped and without Adam. Cometrics include the learning rate.

(a) Scaled GD,  $\delta B = -\eta \nabla_B (AA^\top)^{-1}$  and  $\delta A = -\eta (B^\top B)^{-1} \nabla_A$ , is  $\text{GL}_r$ -equivariant with first-order cometric  $K(G) = \eta s^2 (P_U G + G P_V)$ .

(b) LoRA-Pro's equivalent gradient  $\tilde{g} = P_U G + (I - P_U) G P_V$  is the Frobenius-orthogonal projection  $\Pi_{T_{W-W_0} \mathcal{M}_r}(G)$ . The flow of its SGD variant (LoRA-Pro Alg. 1) is the Riemannian gradient flow on  $W_0 + \mathcal{M}_r$ .

(c) LoRA-FA's corrected gradient  $s^{-2} \nabla_B (A_0 A_0^\top)^{-1}$ , fed to SGD with  $A_0$  frozen, gives  $K(G) = \eta G P_V$  with  $V = \text{row } A_0$ , which depends on  $A_0$  only through  $V \in \text{Gr}(r, n)$ . Its original form, plain SGD on  $B$ , has  $K(G) = s^2 \eta_B G A_0^\top A_0$ .

(d) The cometrics of (a) and (b) are functions of  $W$  on the full-rank locus and define vector fields tangent to  $W_0 + \mathcal{M}_r$ . The field of (c) is  $G \mapsto \eta G P_V$  with  $V = \text{row } A_0$ . Its flow is the Frobenius gradient flow on the slice  $W_0 + \{X A_0\} = W_0 + \mathbb{R}^{m \times n} P_V$ , of dimension  $mr$ , which contains  $W_0$  and lies in  $W_0 + \mathcal{M}_{\leq r}$ . The rank- $r$  part of the slice,  $N_V = \{\text{row}(W - W_0) = V\}$ , is open and dense in it and is a submanifold of  $W_0 + \mathcal{M}_r$ . On  $N_V$  the field is tangent to  $\mathcal{M}_r$  and equals  $\eta G P_{\text{row}(W - W_0)}$ , a function of  $W$ . So (c) descends too, as a degenerate cometric of rank  $mr$  (less than  $r(m + n - r)$  when  $r < n$ ) whose flow keeps  $\text{row}(W - W_0) = \text{row } A_0$  fixed. The cometric of plain GD depends on the representative; it is invariant only under  $O(r)$ .

*Proof.* In (a), (b) and the first sentence of (d),  $(B, A)$  lies on the full-rank locus and the conventions of §C.24 apply. Write  $\mathcal{T} := T_{W-W_0} \mathcal{M}_r$ , where  $W - W_0 = sBA$  has rank  $r$ , column space  $U$  and row space  $V$ .

(a) The steps are  $\delta B = -\eta \nabla_B (AA^\top)^{-1} = -\eta s GA^\top (AA^\top)^{-1}$  and  $\delta A = -\eta (B^\top B)^{-1} \nabla_A = -\eta s (B^\top B)^{-1} B^\top G$ . Hence

$$\begin{aligned} \delta W &= s(\delta BA + B\delta A) + s\delta B\delta A = -\eta s^2(GA^\top (AA^\top)^{-1}A + B(B^\top B)^{-1}B^\top G) + O(\eta^2) \\ &= -\eta s^2(GP_V + P_U G) + O(\eta^2), \end{aligned}$$

and the flow version gives  $\dot{W} = -\eta s^2(P_U G + GP_V)$  exactly. Left multiplication by  $P_U$  and right multiplication by  $P_V$  are orthogonal projections of  $(\mathbb{R}^{m \times n}, \langle \cdot, \cdot \rangle)$ , so  $K(G) = \eta s^2(P_U G + GP_V)$  is symmetric positive semidefinite. For equivariance, let  $q' = g \cdot q$ . By (20) and  $(A'A'^\top)^{-1} = (gAA^\top g^\top)^{-1} = g^{-\top}(AA^\top)^{-1}g^{-1}$ ,

$$\delta B' = -\eta \nabla_{B'} g^\top g^{-\top} (AA^\top)^{-1} g^{-1} = \delta B g^{-1},$$

and  $(B'^\top B')^{-1} = (g^{-\top} B^\top B g^{-1})^{-1} = g(B^\top B)^{-1} g^\top$  gives  $\delta A' = -\eta g(B^\top B)^{-1} g^\top g^{-\top} \nabla_A = g \delta A$ . So the discrete step satisfies (21) for every  $g \in \text{GL}_r$ .

(b) *The tangent space.* We show that

$$\{XA + BY : X \in \mathbb{R}^{m \times r}, Y \in \mathbb{R}^{r \times n}\} = \{D : P_{U^\perp} D P_{V^\perp} = 0\} = \mathcal{T}.$$

The first inclusion  $\subseteq$  holds because  $AP_{V^\perp} = 0$  and  $P_{U^\perp} B = 0$ . Conversely, if  $P_{U^\perp} D P_{V^\perp} = 0$ , then  $D = P_U D + P_{U^\perp} D P_V = B[(B^\top B)^{-1} B^\top D] + [P_{U^\perp} D A^\top (AA^\top)^{-1}]A$ . The common space has dimension  $mn - (m-r)(n-r) = r(m+n-r) = \dim \mathcal{M}_r$  (Background C.12). The curves  $t \mapsto s(B + t\dot{B})(A + t\dot{A})$  lie in  $\mathcal{M}_r$  for small  $t$ , so  $\{\dot{B}A + B\dot{A}\} \subseteq \mathcal{T}$ , and equality of dimensions gives equality.

*The projection.* Let  $\mathcal{N} = \{P_{U^\perp} E P_{V^\perp} : E \in \mathbb{R}^{m \times n}\}$ . For  $D \in \mathcal{T}$  we have  $\langle D, P_{U^\perp} E P_{V^\perp} \rangle = \langle P_{U^\perp} D P_{V^\perp}, E \rangle = 0$ , so  $\mathcal{N} \perp \mathcal{T}$ . Moreover  $G = (G - P_{U^\perp} G P_{V^\perp}) + P_{U^\perp} G P_{V^\perp}$ , with the first term in  $\mathcal{T}$  and the second in  $\mathcal{N}$ . Hence

$$\Pi_{\mathcal{T}}(G) = G - (I - P_U)G(I - P_V) = P_U G + GP_V - P_U G P_V = P_U G + (I - P_U)GP_V = \tilde{g}.$$

*LoRA-Pro's SGD variant.* Algorithm 1 of Wang et al. (2024i) takes the steps  $\delta B = -\eta g^B$  and  $\delta A = -\eta g^A$  along a pair that minimises  $\|s(g^B A + B g^A) - G\|_F$  (their Theorem 2.1). As  $(g^B, g^A)$  ranges over all pairs,  $s(g^B A + B g^A)$  ranges over  $\mathcal{T}$ , and the closest point of  $\mathcal{T}$  to  $G$  is  $\Pi_{\mathcal{T}} G$ . So every minimiser satisfies  $s(g^B A + B g^A) = \Pi_{\mathcal{T}} G$ ; explicitly, the minimisers are

$$g^A = s^{-1}(B^\top B)^{-1} B^\top G + XA, \quad g^B = s^{-1}(I - P_U)GA^\top (AA^\top)^{-1} - BX, \quad X \in \mathbb{R}^{r \times r},$$

for which  $sBg^A = P_U G + sBXA$  and  $sg^B A = (I - P_U)GP_V - sBXA$ . For every choice of the free matrix  $X$ ,

$$\delta W = s(\delta BA + B\delta A) + s\delta B\delta A = -\eta \Pi_{\mathcal{T}} G + O(\eta^2).$$

The flow version is  $\dot{W} = -\eta \Pi_{T_{W-W_0} \mathcal{M}_r} \nabla_{\theta} L(W)$ . Since  $\Pi_{\mathcal{T}} \nabla_{\theta} L$  is the Riemannian gradient of  $L$  restricted to  $W_0 + \mathcal{M}_r$  for the metric induced by the Frobenius inner product (Background D.24, Auxiliary result B.30; Absil et al., 2008, Ch. 3), this is the Riemannian gradient flow on  $W_0 + \mathcal{M}_r$ , run at rate  $\eta$ .

(c) Here  $A = A_0$  is frozen with rank  $A_0 = r$ , and  $B$  is arbitrary. The corrected step is  $\delta B = -\eta s^{-2} \nabla_B (A_0 A_0^\top)^{-1} = -\eta s^{-1} G A_0^\top (A_0 A_0^\top)^{-1}$ , so

$$\Delta W = s\delta B A_0 = -\eta G A_0^\top (A_0 A_0^\top)^{-1} A_0 = -\eta GP_V,$$

exactly, since  $A_0$  does not move.  $P_V$  depends only on  $V = \text{row } A_0$ . The original LoRA-FA step  $\delta B = -\eta_B \nabla_B = -\eta_B s G A_0^\top$  gives  $\Delta W = s \delta B A_0 = -s^2 \eta_B G A_0^\top A_0$ .

(d) On the full-rank locus  $P_U$  and  $P_V$  are the projectors onto  $\text{col}(W - W_0)$  and  $\text{row}(W - W_0)$ , so the cometrics of (a) and (b) are functions of  $W$ . Their values lie in  $\mathcal{T}$ : this is clear for  $\Pi_{\mathcal{T}}$ , and  $P_{U^\perp}(P_U G + G P_V)P_{V^\perp} = 0$ .

For (c), the trajectory satisfies  $W - W_0 = s B A_0 \in \{X A_0 : X \in \mathbb{R}^{m \times r}\}$ , and  $\{X A_0\} = \{Z : \text{row } Z \subseteq V\} = \mathbb{R}^{m \times n} P_V$ , a linear space of dimension  $mr$ . Right multiplication by  $P_V$  is the Frobenius-orthogonal projection onto it, so  $\dot{W} = -\eta \nabla_\theta L(W) P_V$  is the gradient flow of  $L$  restricted to the affine slice  $W_0 + \mathbb{R}^{m \times n} P_V$  with its flat Frobenius metric. The slice contains  $W_0$  ( $X = 0$ ) and lies in  $W_0 + \mathcal{M}_{\leq r}$ , since  $\text{rank } X A_0 \leq r$ . Because  $X \mapsto X A_0$  is injective,  $\text{rank } X A_0 = \text{rank } X$ , and  $\text{row}(X A_0) = V$  iff  $\text{rank } X = r$ . So  $N_V = W_0 + \{X A_0 : \text{rank } X = r\}$ , which is open in the slice because rank  $r$  is an open condition (Auxiliary result B.1(a)), and dense because every  $X \in \mathbb{R}^{m \times r}$  ( $m \geq r$ ) is a limit of rank- $r$  matrices  $X + tE$  as  $t \rightarrow 0$ : if  $E$  has  $I_r$  in its first  $r$  rows and zeros elsewhere, the top  $r \times r$  minor of  $X + tE$  is a monic polynomial of degree  $r$  in  $t$ , so it vanishes for only finitely many  $t$ .  $N_V$  is an open subset of an affine subspace, hence an embedded submanifold of  $\mathbb{R}^{m \times n}$ , and it lies in the embedded submanifold  $W_0 + \mathcal{M}_r$ . The inclusion  $N_V \hookrightarrow W_0 + \mathcal{M}_r$  is smooth by Auxiliary result B.2(c), an immersion because its composite with the inclusion  $W_0 + \mathcal{M}_r \hookrightarrow \mathbb{R}^{m \times n}$  is one, and a homeomorphism onto its image because both sets carry the subspace topology of  $\mathbb{R}^{m \times n}$ . So it is an embedding, and  $N_V$  is an embedded submanifold of  $W_0 + \mathcal{M}_r$  (Background C.4). At  $W \in N_V$ ,  $\text{row}(W - W_0) = V$ , so  $G P_V = G P_{\text{row}(W - W_0)}$  is a function of  $W$ , and, with  $U_W = \text{col}(W - W_0)$ ,  $P_{U_W^\perp}(G P_V)P_{V^\perp} = 0$  shows that it lies in  $T_{W - W_0} \mathcal{M}_r$  (it lies in  $\mathbb{R}^{m \times n} P_V = T N_V$  as well). The operator  $G \mapsto \eta G P_V$  has rank  $\dim \mathbb{R}^{m \times n} P_V = mr$ , and  $mr < r(m + n - r)$  when  $r < n$ . Along the flow  $W(t)$  stays in the slice, so  $\text{row}(W(t) - W_0) \subseteq V$  for all  $t$ , with equality whenever  $W(t) \in N_V$ .

The last sentence of (d) is Theorem III.3(b).  $\square$

**REMARK** (the departures listed under Practice). The symmetry statements in the Practice paragraph after Proposition III.4 are proved with Theorem III.12: damping  $(B^\top B + \delta I)^{-1}$ ,  $(A A^\top + \delta I)^{-1}$  with  $\delta > 0$  leaves exactly  $O(r)$ ; an AdamW step applied after the preconditioner, as in scaled AdamW and in HF's LoRA-FA optimizer, leaves exactly the signed permutations  $B_r$ ; and LoRA-Pro's Sylvester-optimal free matrix leaves exactly  $O(r)$  at the parameter level. For LoRA-FA with the undamped correction and AdamW at  $\varepsilon = 0$ , the first step from  $B_0 = 0$  is  $\Delta W = -s \eta_B \text{sign}(G A_0^\top (A_0 A_0^\top)^{-1}) A_0$ . Replacing  $A_0$  by  $2A_0$ , which has the same row space, doubles this step, so the trajectory depends on  $A_0$  beyond its row space (see also §C.36).

## C.28 Theorem III.5

**THEOREM III.5** (Noether charges and isometries for cotangent signals; known, Zhao et al. (2022b)). Let a Lie group act linearly on  $Q = \mathbb{R}^N$  with  $\rho \circ k = \rho$ , let  $\Lambda \succ 0$  be constant, and let  $\gamma(t) \in T_{\rho(q(t))}^* \Theta$  be any cotangent signal, gradient or not. Consider the continuous-time updates

$$\dot{q} = -\Lambda D\rho_q^\top \gamma(t).$$

- (i) For every generator  $\xi$  with  $\Lambda^{-1} \xi$  symmetric, the charge  $J_\xi(q) = \frac{1}{2} \langle q, \Lambda^{-1} \xi q \rangle$  is conserved.
- (ii) For every generator  $\xi$  with  $\Lambda^{-1} \xi$  antisymmetric,  $k = e^{\tau \xi}$  preserves  $\langle \cdot, \Lambda^{-1} \cdot \rangle$ , so  $K_{kq} = K_q$  along the orbit; for the same cotangent  $\gamma$ , the  $\theta$ -velocities at  $q$  and at  $kq$  agree. Suppose moreover that  $\gamma$  depends on  $q$  only through  $(t, \rho(q))$ , as  $\gamma = \nabla_\theta L$  or an open-loop signal does, and that

solutions are unique ( $\gamma$  measurable in  $t$  and locally Lipschitz in  $\theta$ ,  $\rho$  of class  $C_{\text{loc}}^{1,1}$ ). Then the  $\theta$ -trajectories from  $q_0$  and from  $kq_0$  coincide. If  $\gamma$  uses Euclidean norms on  $Q$ , as global-norm clipping does, the trajectory statement holds when, in addition,  $k$  is a Euclidean isometry, and can fail otherwise. So (i) and the velocity statement of (ii) hold for arbitrary  $\gamma$ , while the trajectory statement needs a gradient-like signal.

(iii) When  $\mathfrak{g}$  is closed under  $\xi \mapsto \Lambda \xi^\top \Lambda^{-1}$ , every generator splits into a part of type (i) and a part of type (ii), a Cartan decomposition  $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$ . This holds for LoRA's  $\mathfrak{gl}_r$ , with per-block scalar rates. Without closure, there are generators of neither type. With per-rank-channel rates  $\Lambda(B, A) = (BD_B, D_AA)$ , where  $D_B = \text{diag}(\eta_{B,j})$  and  $D_A = \text{diag}(\eta_{A,j})$ , the generator  $\xi_X$  is of type (i) iff  $XD_B^{-1}$  and  $D_A^{-1}X$  are both symmetric, and of type (ii) iff both are antisymmetric. An off-diagonal pair  $(i, j)$  contributes one charge and one isometry iff  $\eta_{B,i}\eta_{A,i} = \eta_{B,j}\eta_{A,j}$ , and  $\mathfrak{g}$  is closed iff all the products  $\eta_{B,j}\eta_{A,j}$  are equal. In that case one gets a  $D$ -conjugate of  $\mathfrak{so}_r \oplus \text{Sym}_r$ , whose isometries are not all Euclidean unless the rates are block-scalar, so the clipping caveat of (ii) applies. For pairwise distinct products, only the diagonal  $X$  give charges and no nonzero  $X$  gives an isometry.

*Proof.* Part (i) is Noether's theorem in the form of [Auxiliary result B.29](#), which allows an arbitrary cotangent signal; we give the short argument in full. Solutions are absolutely continuous and the identities below hold for almost every  $t$ . For  $\xi$  in the Lie algebra  $\mathfrak{g} \subseteq \mathfrak{gl}(N)$  of the group,  $\rho(e^{\tau\xi}q) = \rho(q)$  for all  $\tau$ ; differentiating at  $\tau = 0$  gives the infinitesimal invariance

$$D\rho_q(\xi q) = 0 \quad (q \in Q, \xi \in \mathfrak{g}). \quad (22)$$

(i) If  $\Lambda^{-1}\xi$  is symmetric, the gradient of  $J_\xi$  is  $\Lambda^{-1}\xi q$ . Along a solution, using that  $\Lambda$  is symmetric and (22),

$$\frac{d}{dt}J_\xi(q) = \langle \Lambda^{-1}\xi q, \dot{q} \rangle = -\langle \Lambda^{-1}\xi q, \Lambda D\rho_q^\top \gamma \rangle = -\langle \xi q, D\rho_q^\top \gamma \rangle = -\langle D\rho_q \xi q, \gamma \rangle = 0.$$

Nothing about  $\gamma$  was used, so it may depend on  $q$  in any way.

(ii) *Isometry and cometric.* If  $\Lambda^{-1}\xi$  is antisymmetric, then  $\xi^\top \Lambda^{-1} = -\Lambda^{-1}\xi$ , so

$$\frac{d}{d\tau}(e^{\tau\xi^\top} \Lambda^{-1} e^{\tau\xi}) = e^{\tau\xi^\top} (\xi^\top \Lambda^{-1} + \Lambda^{-1}\xi) e^{\tau\xi} = 0,$$

and  $k = e^{\tau\xi}$  satisfies  $k^\top \Lambda^{-1} k = \Lambda^{-1}$ , equivalently  $k \Lambda k^\top = \Lambda$ . From  $\rho \circ k = \rho$  we get  $D\rho_q = D\rho_{kq} k$ , so

$$K_q = D\rho_{kq} k \Lambda k^\top D\rho_{kq}^\top = D\rho_{kq} \Lambda D\rho_{kq}^\top = K_{kq}.$$

Since  $\rho(kq) = \rho(q)$ , the same covector  $\gamma$  acts at both points, and the velocities  $-K_q \gamma$  and  $-K_{kq} \gamma$  agree. The rest of the argument uses only the two properties  $k \Lambda k^\top = \Lambda$  and  $\rho \circ k = \rho$ , so it applies to every such  $k$ , not only to those of the form  $e^{\tau\xi}$ .

*Trajectories.* Let  $\gamma = \gamma(t, \rho(q))$  and let  $q(t)$  solve  $\dot{q} = F(t, q) := -\Lambda D\rho_q^\top \gamma(t, \rho(q))$  from  $q_0$ . Put  $p(t) = kq(t)$ . Then, using  $D\rho_q^\top = k^\top D\rho_{kq}^\top$  and  $\rho(p) = \rho(q)$ ,

$$\dot{p} = k\dot{q} = -k \Lambda k^\top D\rho_p^\top \gamma(t, \rho(q)) = -\Lambda D\rho_p^\top \gamma(t, \rho(p)) = F(t, p),$$

so  $p$  solves the same equation from  $kq_0$ . Under the stated regularity, read locally uniformly in  $t$ ,  $F$  is measurable in  $t$  and locally bounded and locally Lipschitz in  $p$ , because  $D\rho$  and  $\rho$  are locally Lipschitz and  $\gamma$  is locally Lipschitz in  $\theta$ . By Gronwall's inequality two solutions with the same initial value coincide ([Auxiliary result B.28](#)), so  $p$  is the solution from  $kq_0$ , and  $\rho(p(t)) = \rho(q(t))$  for all  $t$ .



*Clipping.* Suppose  $\gamma$  depends on  $q$  through  $(t, \rho(q))$  and through Euclidean norms  $\|D\rho_q^\top v\|$  of pulled-back covectors  $v$  that are themselves functions of  $(t, \rho(q))$ . Global-norm clipping is of this kind:

$$\gamma = \min(1, C/\|D\rho_q^\top \nabla_\theta L\|) \nabla_\theta L.$$

If  $k$  is also a Euclidean isometry, then  $\|D\rho_{kq}^\top v\| = \|k^{-\top} D\rho_q^\top v\| = \|D\rho_q^\top v\|$ , so  $\gamma$  takes the same value at  $q$  and at  $kq$ , and the argument above goes through. Otherwise it can fail. Take  $\mathcal{Q} = \mathbb{R}^2$ ,  $\Theta = \mathbb{R}$ ,  $\Lambda = \text{diag}(1, 4)$ ,  $\rho(q) = \frac{1}{2}q^\top \Lambda^{-1}q$  and  $L(\theta) = \theta^2/2$ . The group  $\{k : k^\top \Lambda^{-1}k = \Lambda^{-1}\}$  acts linearly and preserves  $\rho$ ; its generator  $\xi = \Lambda^{1/2}J\Lambda^{-1/2}$ , with  $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ , has  $\Lambda^{-1}\xi = \Lambda^{-1/2}J\Lambda^{-1/2}$  antisymmetric, and  $k = e^{(\pi/2)\xi} = \Lambda^{1/2}e^{(\pi/2)J}\Lambda^{-1/2}$  maps  $q_0 = (1, 0)$  to  $kq_0 = (0, 2)$ , both with  $\theta = 1/2$ . Here  $D\rho_q^\top \gamma = \gamma \Lambda^{-1}q$  and  $\dot{\theta} = -\gamma q^\top \Lambda^{-1}q = -2\theta\gamma$ . The parameter gradients  $L'(\theta)\Lambda^{-1}q$  have norms  $1/2$  at  $q_0$  and  $1/4$  at  $kq_0$ , so with  $C = 1/10$  the clipping factors are  $1/5$  and  $2/5$ , and the initial velocities are  $\dot{\theta} = -1/10$  and  $\dot{\theta} = -1/5$ . The  $\theta$ -trajectories differ, while without clipping they coincide by the argument above.

(iii) *Splitting.* Put  $\vartheta(\xi) = -\Lambda\xi^\top \Lambda^{-1}$ . It is an involution of  $\mathfrak{gl}(N)$  ( $\vartheta^2(\xi) = \Lambda\Lambda^{-1}\xi\Lambda\Lambda^{-1} = \xi$ ) and a Lie algebra automorphism:  $\vartheta([\xi, \zeta]) = -\Lambda(\xi^\top \zeta^\top - \zeta^\top \xi^\top)\Lambda^{-1} = [\vartheta\xi, \vartheta\zeta]$ . A generator is of type (ii) iff  $\Lambda^{-1}\xi$  is antisymmetric, iff  $\Lambda\xi^\top \Lambda^{-1} = -\xi$ , iff  $\vartheta\xi = \xi$ ; it is of type (i) iff  $\vartheta\xi = -\xi$ . If  $\mathfrak{g}$  is closed under  $\xi \mapsto \Lambda\xi^\top \Lambda^{-1}$ , it is  $\vartheta$ -stable, and

$$\xi = \frac{1}{2}(\xi + \Lambda\xi^\top \Lambda^{-1}) + \frac{1}{2}(\xi - \Lambda\xi^\top \Lambda^{-1})$$

writes every  $\xi \in \mathfrak{g}$  as a type (i) part (in the  $-1$  eigenspace  $\mathfrak{p}$  of  $\vartheta$ ) plus a type (ii) part (in the  $+1$  eigenspace  $\mathfrak{k}$ ). Since  $\vartheta$  is an automorphism,  $[\mathfrak{k}, \mathfrak{k}] \subseteq \mathfrak{k}$ ,  $[\mathfrak{k}, \mathfrak{p}] \subseteq \mathfrak{p}$  and  $[\mathfrak{p}, \mathfrak{p}] \subseteq \mathfrak{k}$ , which is the Cartan decomposition (Background D.27, Auxiliary result B.19(b)). If some  $\xi \in \mathfrak{g}$  has  $\Lambda\xi^\top \Lambda^{-1} \notin \mathfrak{g}$ , then  $\vartheta\xi \neq \pm\xi$ , so  $\xi$  is of neither type.

*LoRA.* The generators of the gauge are  $\xi_X(B, A) = (BX, -XA)$  for  $X \in \mathfrak{gl}_r$  (the derivative of  $\tau \mapsto e^{-\tau X} \cdot (B, A)$ ), and  $X \mapsto \xi_X$  is injective. For per-rank-channel rates  $\Lambda(B, A) = (BD_B, D_A A)$ ,

$$\Lambda^{-1}\xi_X(B, A) = (BXD_B^{-1}, -D_A^{-1}XA), \quad (\Lambda^{-1}\xi_X)^\top(B, A) = (B(XD_B^{-1})^\top, -(D_A^{-1}X)^\top A),$$

because  $\langle B', BM \rangle = \langle B'M^\top, B \rangle$  and  $\langle A', NA \rangle = \langle N^\top A', A \rangle$ . Hence  $\Lambda^{-1}\xi_X$  is symmetric iff  $XD_B^{-1}$  and  $D_A^{-1}X$  are symmetric, and antisymmetric iff both are antisymmetric. Entrywise,  $(XD_B^{-1})_{ij} = X_{ij}/\eta_{B,j}$  and  $(D_A^{-1}X)_{ij} = X_{ij}/\eta_{A,i}$ . For  $i \neq j$ , the two symmetry conditions read  $X_{ji} = X_{ij}\eta_{B,i}/\eta_{B,j}$  and  $X_{ji} = X_{ij}\eta_{A,i}/\eta_{A,j}$ ; the antisymmetry conditions are the same with a minus sign. Both admit  $X_{ij} \neq 0$  iff  $\eta_{B,i}/\eta_{B,j} = \eta_{A,i}/\eta_{A,j}$ , that is, iff  $\eta_{B,i}\eta_{A,i} = \eta_{B,j}\eta_{A,j}$ , and then the pair  $(i, j)$  carries a one-dimensional space of each type. On the diagonal the symmetry conditions are empty and the antisymmetry conditions force  $X_{ii} = 0$ . So for pairwise distinct products the type (i) generators are exactly the diagonal  $X$ , and the only type (ii) generator is  $X = 0$ .

For closure,  $\xi_X^\top(B, A) = (BX^\top, -X^\top A)$ , so  $\Lambda\xi_X^\top \Lambda^{-1}(B, A) = (BD_B^{-1}X^\top D_B, -D_A X^\top D_A^{-1}A)$ . An operator  $(B, A) \mapsto (BM, -NA)$  lies in  $\mathfrak{g} = \{\xi_Y\}$  iff  $M = N$ , so closure holds iff  $D_B^{-1}X^\top D_B = D_A X^\top D_A^{-1}$  for all  $X$ . The  $(i, j)$  entries are  $X_{ji}\eta_{B,j}/\eta_{B,i}$  and  $X_{ji}\eta_{A,i}/\eta_{A,j}$ , which agree for all  $X$  iff all products  $\eta_{B,j}\eta_{A,j}$  are equal. With block-scalar rates ( $D_B = \eta_B I$ ,  $D_A = \eta_A I$ ) this holds, the type (i) generators are the symmetric  $X$  and the type (ii) ones the antisymmetric  $X$ , and  $\mathfrak{gl}_r = \mathfrak{so}_r \oplus \text{Sym}_r$ .

When all products equal  $p$ ,  $D_B = pD_A^{-1}$ , and the type (i) generators are  $X = SD_B$  with  $S$  symmetric (then  $D_A^{-1}X = pD_A^{-1}SD_A^{-1}$  is symmetric too); the type (ii) generators are  $X = KD_B$

with  $K$  antisymmetric. With  $D = D_B^{1/2}$ , conjugation  $X \mapsto DXD^{-1}$  maps  $SD_B$  to  $DSD$  and  $KD_B$  to  $DKD$ , so  $\mathfrak{k} \oplus \mathfrak{p} = D^{-1}(\mathfrak{so}_r \oplus \text{Sym}_r)D$ . The isometries are the group elements  $D^{-1}OD$  with  $O \in \text{SO}(r)$ , and  $D^{-1}OD$  is orthogonal iff  $O$  commutes with  $D^2 = D_B$ . Unless  $D_B$  is a multiple of the identity (the block-scalar case), some of these isometries are not orthogonal, so they act on  $Q$  by non-Euclidean maps and the clipping caveat of (ii) applies.  $\square$

**REMARK** (instances and the discrete step). Each instance listed after [Theorem III.5](#) applies (i) to the generator named there; we record the checks. All assume gradient flow, or SGD without momentum, with per-block rates.

- **LoRA**. For symmetric  $X$ ,  $J_{\xi_X}(B, A) = \frac{1}{2}(\langle B, BX \rangle / \eta_B - \langle A, XA \rangle / \eta_A) = \frac{1}{2} \text{tr}(X\Phi)$  with  $\Phi = B^\top B / \eta_B - AA^\top / \eta_A$ , so the  $r(r+1)/2$  charges are the entries of  $\Phi$ . The  $\mathfrak{so}_r$  generators act by  $(BO^\top, OA)$  with  $O \in \text{SO}(r)$ , which are Euclidean isometries commuting with  $\Lambda$ , so (ii) holds under global-norm clipping.
- **DyLoRA**. At each step the loss uses  $\rho_b(B, A) = W_0 + s_b B E_b A$  for a sampled  $b$ , with  $E_b = \text{diag}(1, \dots, 1, 0, \dots, 0)$  of rank  $b$ . The update is  $-\Lambda D \rho_{b(t)}^\top \gamma$ , and the proof of (i) needs only  $D \rho_{b(t)}(\xi q) = 0$ . Now  $\rho_b(g \cdot q) = \rho_b(q)$  for all  $(B, A)$  iff  $g^{-1} E_b g = E_b$ , that is, iff  $g$  preserves  $\text{span}(e_1, \dots, e_b)$  and  $\text{span}(e_{b+1}, \dots, e_r)$ . When every  $b \in \{1, \dots, r\}$  can be sampled,  $g$  then preserves each line  $\text{span}(e_b) = \text{span}(e_1, \dots, e_b) \cap \text{span}(e_b, \dots, e_r)$ , so it is diagonal. Diagonal  $X$  are symmetric, which gives the charges  $\text{diag } \Phi$ .
- **VeRA**.  $\rho(b, d) = W_0 + \text{diag}(b) B_f \text{diag}(d) A_f$  with frozen  $B_f, A_f$  is invariant under  $(b, d) \mapsto (cb, d/c)$ , with generator  $(b, -d)$  and charge  $\frac{1}{2}(\|b\|^2 / \eta_b - \|d\|^2 / \eta_d)$ . NOLA's two coefficient vectors behave in the same way.
- **AdaLoRA**.  $\rho(P, \lambda, Q) = W_0 + P \text{diag}(\lambda) Q$  is invariant under  $(p_k, \lambda_k) \mapsto (cp_k, \lambda_k/c)$  and under  $(\lambda_k, q_k) \mapsto (\lambda_k/c, cq_k)$ ; at equal rates the charges are proportional to  $\|p_k\|^2 - \lambda_k^2$  and  $\|q_k\|^2 - \lambda_k^2$ . The orthogonality regulariser  $\|P^\top P - I\|^2 + \|QQ^\top - I\|^2$  is not invariant under these scalings, so the regularised loss does not factor through  $\rho$ , and rank-budget masking changes  $\rho$ .
- **HRA**.  $\rho(u_1, \dots, u_r) = W_0 \prod_i (I - 2u_i u_i^\top / \|u_i\|^2)$  is invariant under  $u_i \mapsto cu_i$ , with charge  $\frac{1}{2} \|u_i\|^2 / \eta$ .
- **ReLU bottleneck adapters**  $h \mapsto h + U \text{relu}(Dh + \beta)$ . Multiplying row  $j$  of  $D$  and  $\beta_j$  by  $c > 0$  and column  $j$  of  $U$  by  $1/c$  preserves the function, since  $\text{relu}(cz) = c \text{relu}(z)$ . The target is a function space, which [Observation III.2](#) allows, and the derivative of  $\rho$  along the scaling direction vanishes by homogeneity even at the kinks. Each hidden unit gives the charge  $\frac{1}{2}((\|D_{j,:}\|^2 + \beta_j^2) / \eta_D - \|U_{:,j}\|^2 / \eta_U)$ , with a common rate for  $D$  and  $\beta$ .
- **TT adapters**. Neighbouring cores share a bond of dimension  $r_k$ , on which  $\text{GL}(r_k)$  acts by  $h^{-1}$  on one side and  $h$  on the other. The symmetric generators give one conserved symmetric  $r_k \times r_k$  matrix per bond, the Gram matrix of the left unfolding of one core divided by its rate minus that of the right unfolding of the next core divided by its rate.
- **DoRA's detached-norm backward, global-norm clipping, LoRA dropout**. DoRA writes  $W = V \text{diag}(g/\|V\|_c)$  with  $V = W_0 + sBA$ , column norms  $\|V\|_c$  and magnitudes  $g$ . Treating  $\|V\|_c$  as a constant in the backward pass gives  $\nabla_B = s\gamma A^\top$  and  $\nabla_A = sB^\top \gamma$  with  $\gamma = \nabla_W L \text{diag}(g/\|V\|_c)$ . Clipping multiplies the factor gradients by one scalar, and dropout replaces  $G$  by the masked signal of [§C.25](#). In each case the update of  $(B, A)$  is  $-\Lambda D \rho_{\text{LoRA}}^\top \gamma$  for some  $\gamma$ , so  $\Phi$  is conserved.

- *Discrete GD.* With  $s$  absorbed into  $G$ , one step  $B^+ = B - \eta_B G A^\top$ ,  $A^+ = A - \eta_A B^\top G$  gives

$$\begin{aligned}\frac{1}{\eta_B} B^{+\top} B^+ &= \frac{1}{\eta_B} B^\top B - (A G^\top B + B^\top G A^\top) + \eta_B A G^\top G A^\top, \\ \frac{1}{\eta_A} A^+ A^{+\top} &= \frac{1}{\eta_A} A A^\top - (B^\top G A^\top + A G^\top B) + \eta_A B^\top G G^\top B,\end{aligned}$$

so  $\Phi^+ - \Phi = \eta_B A G^\top G A^\top - \eta_A B^\top G G^\top B$  exactly. This is of order  $\eta$  while  $\Phi$  is of order  $1/\eta$ , so  $\eta\Phi$ , or  $\Phi$  relative to its size, drifts by  $O(\eta^2)$  per step.

- *Momentum and Adam.* A heavy-ball step adds a buffer of gradients taken at past points,  $D\rho_{q_s}^\top G_s$  with  $s < t$ , and Adam rescales gradients coordinatewise. Neither step has the form  $-\Lambda D\rho_q^\top \gamma$  at the current  $q$  with a constant  $\Lambda$ , so (i) does not apply.

### C.29 Corollary III.6

**COROLLARY III.6** (charge at initialisation; weight decay). (a)  $\Phi_0 = B_0^\top B_0 / \eta_B - A_0 A_0^\top / \eta_A$  is the value of the conserved charge, fixed by the initialisation and the rates. Since  $O(r)$  acts on  $\Phi$  by conjugation and preserves  $\theta$ -trajectories ([Theorem III.5\(ii\)](#)), only the spectrum of  $\Phi_0$  labels a  $\theta$ -trajectory.

| Initialisation                                                                                                                                                                                                                                                                                                                                         | Charge $\Phi_0$                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| HF default LoRA on Linear and Conv layers ( $B_0 = 0$ ; Kaiming-uniform $A_0$ with entries $U(-n^{-1/2}, n^{-1/2})$ , $n$ the fan-in, which is <code>in_channels</code> times the kernel size for Conv)                                                                                                                                                | $-A_0 A_0^\top / \eta_A$ , with $\mathbb{E}[A_0 A_0^\top] = \frac{1}{3} I_r$ and fluctuations $O(n^{-1/2})$ on and off the diagonal |
| HF default LoRA on embedding layers ( $A_0 = 0$ , normal $B_0$ )                                                                                                                                                                                                                                                                                       | $+B_0^\top B_0 / \eta_B$ (Init[B]-type)                                                                                             |
| HF mica ( <a href="#">Rüdiger and Raschka, 2026</a> ) ( $A_0 = 0$ , frozen minor-singular $B_0$ )                                                                                                                                                                                                                                                      | $+B_0^\top B_0 / \eta_B$ (Init[B]-type, with $B$ frozen)                                                                            |
| EVA ( <a href="#">Paischer et al., 2024</a> ) (orthonormal rows; default <code>whiten=False</code> )                                                                                                                                                                                                                                                   | $-I / \eta_A$ exactly                                                                                                               |
| PiSSA in HF PEFT ( <a href="#">Meng et al., 2024a</a> ), and MiLoRA ( <a href="#">Wang et al., 2024b</a> ) if implemented like <code>pissa_init</code> with the bottom- $r$ singular values (MiLoRA is not in HF PEFT); $B_0^\top B_0 = A_0 A_0^\top = S_r / s$ , with $S_r$ the top- $r$ , resp. bottom- $r$ , singular values and $s$ the LoRA scale | $(S_r / s)(1 / \eta_B - 1 / \eta_A)$ , zero <b>iff</b> $\eta_A = \eta_B$ (when $S_r \neq 0$ )                                       |
| OLoRA ( <a href="#">Büyükkakyüz, 2024</a> ) ( $B_0 = Q_r$ , $A_0 = R_r$ )                                                                                                                                                                                                                                                                              | $I / \eta_B - R_r R_r^\top / \eta_A$                                                                                                |
| “Init[B]” ( <a href="#">Hayou et al., 2024a</a> ) ( $A_0 = 0$ )                                                                                                                                                                                                                                                                                        | $+B_0^\top B_0 / \eta_B$                                                                                                            |

(b) Let  $G$  be any cotangent signal, with the scale  $s$  absorbed into it. Under gradient flow, if decay acts per unit time and identically on both factors,

$$\dot{B} = -\eta_B G A^\top - \lambda B, \quad \dot{A} = -\eta_A B^\top G - \lambda A,$$

then  $\dot{\Phi} = -2\lambda\Phi$ . With learning-rate-coupled decay (PyTorch SGD `weight_decay`; the decay term of PyTorch AdamW),  $\dot{B} = -\eta_B (G A^\top + \lambda B)$  and likewise for  $A$ , so that  $\dot{\Phi} = -2\lambda(B^\top B - A A^\top)$ . Then  $\Phi(t) = e^{-2\eta\lambda t} \Phi(0)$  holds when  $\eta_A = \eta_B = \eta$ . For  $\eta_A \neq \eta_B$  it fails in general, since  $-2\lambda(B^\top B - A A^\top)$  is not a function of  $\Phi$ . With an actual Adam(W) gradient term,  $\Phi$  drifts even without decay, at first order in the step size; Adam lies outside [Theorem III.5](#). So neither law describes AdamW training.

*Proof.* (a) *Conservation and conjugation.* With block-scalar rates the entries of  $\Phi$  are the charges of [Theorem III.5\(i\)](#) (§C.28, remark on instances), so under gradient flow  $\Phi(t) = \Phi_0$ , the value fixed by  $(B_0, A_0)$  and  $(\eta_A, \eta_B)$ . For  $O \in O(r)$ , the map  $k : (B, A) \mapsto (BO, O^\top A)$  is the gauge element  $g = O^\top$ , and

$$\Phi(BO, O^\top A) = O^\top \frac{B^\top B}{\eta_B} O - O^\top \frac{AA^\top}{\eta_A} O = O^\top \Phi(B, A) O.$$

The linear map  $k$  is orthogonal on  $Q$  and commutes with block-scalar  $\Lambda$ , so  $k\Lambda k^\top = \Lambda$ , and  $\rho \circ k = \rho$ . The trajectory argument of [Theorem III.5\(ii\)](#) uses only these two properties (§C.28), so the  $\theta$ -trajectories from  $q_0$  and from  $kq_0$  coincide for every signal that depends on  $q$  only through  $(t, \rho(q))$ , and also under global-norm clipping, since  $k$  is a Euclidean isometry. This covers all of  $O(r)$ , reflections included. So initialisations whose charges are conjugate under  $O(r)$  produce the same  $\theta$ -trajectory, and a function of  $\Phi_0$  that labels  $\theta$ -trajectories must be invariant under  $\Phi_0 \mapsto O^\top \Phi_0 O$ . By the spectral theorem the  $O(r)$ -conjugacy class of the symmetric matrix  $\Phi_0$  is determined by its eigenvalues, so such a function is a function of the spectrum of  $\Phi_0$ .

*The table.* Each entry follows by substituting the initialisation into  $\Phi_0$ .

- *HF default, Linear and Conv.*  $B_0 = 0$  gives  $\Phi_0 = -A_0 A_0^\top / \eta_A$ . HF initialises  $A_0$  with `kaiming_uniform_` at  $a = \sqrt{5}$ , whose bound is  $\sqrt{2/(1+a^2)}\sqrt{3/n} = n^{-1/2}$ , with  $n$  the fan-in (`in_channels` times the product of the kernel sizes for Conv). An entry  $U(-n^{-1/2}, n^{-1/2})$  has variance  $1/(3n)$  and fourth moment  $1/(5n^2)$ . A diagonal entry of  $A_0 A_0^\top$  is a sum of  $n$  independent squares, with mean  $1/3$  and variance  $n(1/(5n^2) - 1/(9n^2)) = 4/(45n)$ ; an off-diagonal entry is a sum of  $n$  independent products, with mean 0 and variance  $n \cdot 1/(9n^2) = 1/(9n)$ . So  $\mathbb{E}[A_0 A_0^\top] = \frac{1}{3}I_r$ , with fluctuations of standard deviation  $2/(3\sqrt{5n})$  on the diagonal and  $1/(3\sqrt{n})$  off it.
- *Embeddings, Init[B] and mica.*  $A_0 = 0$  gives  $\Phi_0 = B_0^\top B_0 / \eta_B$ . For mica,  $B$  is frozen, so its rate is 0 and [Theorem III.5](#) does not apply to it; the entry records the formula and its sign type, and the training dynamics are those of the one-sided method  $A \mapsto W_0 + sB_0 A$ , along which  $B_0^\top B_0$  is constant and  $AA^\top$  moves.
- *EVA.*  $B_0 = 0$  and orthonormal rows,  $A_0 A_0^\top = I$ , give  $\Phi_0 = -I/\eta_A$ .
- *PiSSA in HF PEFT.* `pissa_init` takes the top- $r$  singular triples  $(U_r, S_r, V_r)$  of  $W_0$ , divides  $S_r$  by the scale  $s$ , and sets  $A_0 = (S_r/s)^{1/2} V_r^\top$  and  $B_0 = U_r (S_r/s)^{1/2}$ . With  $U_r^\top U_r = V_r^\top V_r = I$ ,  $B_0^\top B_0 = A_0 A_0^\top = S_r/s$ , so  $\Phi_0 = (S_r/s)(1/\eta_B - 1/\eta_A)$ , which vanishes iff  $\eta_A = \eta_B$  when  $S_r \neq 0$ . The same computation with the bottom- $r$  triples gives the MiLoRA entry.
- *OLoRA.*  $B_0 = Q_r$  has orthonormal columns, so  $\Phi_0 = I/\eta_B - R_r R_r^\top / \eta_A$ .

(b) Write both decay conventions as  $\dot{B} = -\eta_B G A^\top - c_B B$  and  $\dot{A} = -\eta_A B^\top G - c_A A$ , with  $c_B = c_A = \lambda$  per unit time and  $c_B = \eta_B \lambda$ ,  $c_A = \eta_A \lambda$  when coupled. Then

$$\frac{d}{dt} \frac{B^\top B}{\eta_B} = -(AG^\top B + B^\top G A^\top) - 2c_B \frac{B^\top B}{\eta_B}, \quad \frac{d}{dt} \frac{AA^\top}{\eta_A} = -(B^\top G A^\top + AG^\top B) - 2c_A \frac{AA^\top}{\eta_A},$$

and subtracting, the  $G$ -terms cancel:

$$\dot{\Phi} = -2c_B \frac{B^\top B}{\eta_B} + 2c_A \frac{AA^\top}{\eta_A}.$$

Per unit time this is  $\dot{\Phi} = -2\lambda\Phi$ , so  $\Phi(t) = e^{-2\lambda t}\Phi(0)$ . Coupled, it is  $\dot{\Phi} = -2\lambda(B^\top B - AA^\top)$ ; when  $\eta_A = \eta_B = \eta$  the right-hand side is  $-2\lambda\eta\Phi$ , so  $\Phi(t) = e^{-2\eta\lambda t}\Phi(0)$ . For  $\eta_A \neq \eta_B$  there are states with the same  $\Phi$  and different  $B^\top B - AA^\top$ : replacing  $(B^\top B, AA^\top)$  by  $(B^\top B + \eta_B C, AA^\top + \eta_A C)$

with  $C \geq 0$  (realised by suitable factors, since  $m, n \geq r$ ) leaves  $\Phi$  unchanged and changes  $B^\top B - AA^\top$  by  $(\eta_B - \eta_A)C$ . Since  $\dot{\Phi}$  does not involve  $G$ , two such initial states have equal  $\Phi(0)$  and different  $\dot{\Phi}(0)$ , so no law  $\dot{\Phi} = F(\Phi)$ , and in particular no exponential law, holds for all trajectories. Some trajectories do satisfy one: with  $B = 0$  and  $G = 0$ ,  $\Phi(t) = e^{-2\eta_A \lambda t} \Phi(0)$ ; so the law fails in general and not always.

*Adam.* One Adam step can move  $\eta\Phi$  at first order. Take  $m = n = r = 1$ ,  $s$  absorbed into  $G$ , rates  $\eta_B = \eta_A = \eta$ ,  $(b, a) = (2, 1)$ ,  $G = 1$  and zero optimizer state. The bias-corrected first step at  $\varepsilon = 0$  is  $\delta b = -\eta \text{sign}(Ga) = -\eta$  and  $\delta a = -\eta \text{sign}(bG) = -\eta$ , so

$$\eta\Phi^+ - \eta\Phi = (2 - \eta)^2 - (1 - \eta)^2 - (4 - 1) = -2\eta,$$

of first order in  $\eta$ ; the GD step of the previous remark moves  $\eta\Phi$  by  $O(\eta^2)$ . Adam's step is not of the form  $-\Lambda D\rho^\top \gamma$ , so it lies outside [Theorem III.5](#), and neither decay law describes AdamW training.  $\square$

**REMARK** (the two decay conventions as gauge fixings). Per-time decay is the  $\Lambda$ -gradient flow of  $L + \frac{\lambda}{2}(\|B\|^2/\eta_B + \|A\|^2/\eta_A)$ , since  $-\eta_B \nabla_B (\frac{\lambda}{2}\|B\|^2/\eta_B) = -\lambda B$ . Coupled decay is the  $\Lambda$ -gradient flow of  $L + \frac{\lambda}{2}(\|B\|^2 + \|A\|^2)$ . At an equilibrium of the coupled flow,  $\nabla_B = -\lambda B$  and  $\nabla_A = -\lambda A$ ; since  $B^\top \nabla_B = sB^\top G A^\top = \nabla_A A^\top$ , this gives  $\lambda(B^\top B - AA^\top) = 0$ , so equilibria lie on the unweighted balanced slice  $\mu = B^\top B - AA^\top = 0$ . At an equilibrium of the per-time flow,  $\eta_B \nabla_B = -\lambda B$  and  $\eta_A \nabla_A = -\lambda A$ , and the same identity gives  $\Phi = 0$ ; per-time decay also drives  $\Phi \rightarrow 0$  exponentially by (b). The two slices  $\{B^\top B = AA^\top\}$  and  $\{B^\top B = (\eta_B/\eta_A)AA^\top\}$  coincide iff  $\eta_A = \eta_B$ . HF's PiSSA has  $B_0^\top B_0 = A_0 A_0^\top$  at every rate, so it starts on the coupled-decay slice  $\mu = 0$ , and off the per-time slice  $\Phi = 0$  when  $\eta_A \neq \eta_B$ .

### C.30 Proposition III.7

**PROPOSITION III.7** (Kempf–Ness slice; Loewner floor; known, [Kempf and Ness \(1979\)](#); [Srebro et al. \(2004\)](#)). (a) On a full-rank  $\text{GL}_r$ -orbit, the critical points of  $\|B\|^2 + \|A\|^2$  are exactly the balanced representatives, those with  $\mu = B^\top B - AA^\top = 0$ . They form one  $O(r)$ -orbit, they are the minima, and the minimum value is  $2\|BA\|_*$ .

(b) Assume gradient flow with block-scalar rates  $\eta_A, \eta_B$ , no weight decay, a loss that depends on  $(B, A)$  only through  $BA$ , and any cotangent signal of the form  $\Lambda D\rho^\top \gamma$  (dropout, minibatches and clipping included). From  $B_0 = 0$ ,

$$B^\top B = \frac{\eta_B}{\eta_A}(AA^\top - A_0 A_0^\top) \succeq 0,$$

so  $AA^\top \succeq A_0 A_0^\top$  for all time; the  $A$ -factor never shrinks below its initialisation in the Loewner order.  $AA^\top(t)$  need not be monotone in  $t$ ; only this floor is guaranteed. Discrete steps break the bound by  $O(\eta^2)$  per step. Under Adam it is not guaranteed and fails empirically. With per-time decay  $\lambda$ , only  $AA^\top \succeq e^{-2\lambda t} A_0 A_0^\top$  holds.

*Proof.* (a) Let  $x = (B, A)$  have  $\text{rank } B = \text{rank } A = r$ , put  $f(B, A) = \|B\|^2 + \|A\|^2$ , and let  $O = \text{GL}_r \cdot x$ . Every factorisation  $B'A'$  of  $BA$  with inner dimension  $r$  lies in  $O$ : from  $\text{col } B' = \text{col}(BA) = \text{col } B$  we get  $B' = Bg^{-1}$  for a unique  $g \in \text{GL}_r$ , and then  $Bg^{-1}A' = BA$  gives  $A' = gA$  because  $B$  is injective. As in Kempf–Ness theory ([Background D.30](#), [Auxiliary result B.26](#)),  $x$  is a critical point of  $f$  on  $O$  when  $e \in \text{GL}_r$  is a critical point of  $g \mapsto f(g \cdot x)$ . With  $e^{t\xi} \cdot x = (Be^{-t\xi}, e^{t\xi}A)$ ,

$$\left. \frac{d}{dt} \right|_{t=0} f(e^{t\xi} \cdot x) = -2\langle B, B\xi \rangle + 2\langle A, \xi A \rangle = -2\text{tr}(\xi(B^\top B - AA^\top)) = -2\text{tr}(\xi_s \mu),$$



where  $\xi_s$  is the symmetric part of  $\xi$  and the antisymmetric part drops out because  $\mu$  is symmetric. This vanishes for all  $\xi$  iff  $\mu = 0$  (take  $\xi = \mu$ ). So the critical points are exactly the balanced points of  $O$ .

*Convexity along symmetric directions.* The function  $f$  is  $O(r)$ -invariant,  $f(BO^\top, OA) = f(B, A)$ , and  $\mu$  is  $O(r)$ -equivariant,  $\mu(O \cdot x) = O\mu(x)O^\top$ . By the polar decomposition every  $g \in \text{GL}_r$  is  $g = Oe^\xi$  with  $O \in O(r)$  and  $\xi$  symmetric ([Background D.27](#), [Auxiliary result B.19\(a\)](#); [Horn and Johnson, 2013](#), §7.3), so every point of  $O$  is  $O \cdot (e^\xi \cdot x)$  and  $f(g \cdot x) = f(e^\xi \cdot x)$ . For symmetric  $\xi = R \text{diag}(d)R^\top$  with  $R$  orthogonal, put  $h(t) = f(e^{t\xi} \cdot x)$ . Orthogonal invariance of the Frobenius norm gives

$$h(t) = \|BRE^{-t \text{diag}(d)}\|^2 + \|e^{t \text{diag}(d)}R^\top A\|^2 = \sum_i \left( \|(BR)_{:,i}\|^2 e^{-2td_i} + \|(R^\top A)_{i,:}\|^2 e^{2td_i} \right),$$

$$h''(t) = 4 \sum_i d_i^2 \left( \|(BR)_{:,i}\|^2 e^{-2td_i} + \|(R^\top A)_{i,:}\|^2 e^{2td_i} \right) \geq 0.$$

So  $h$  is convex.

*Balanced points are the minima.* Let  $x$  be balanced and  $y = O \cdot (e^\xi \cdot x) \in O$ . Then  $h'(0) = -2 \text{tr}(\xi \mu(x)) = 0$ , and convexity gives  $f(y) = h(1) \geq h(0) = f(x)$ . So every balanced point is a global minimum on  $O$ ; conversely every minimum is a critical point, hence balanced.

*One  $O(r)$ -orbit.* Let  $x$  and  $y = O \cdot (e^\xi \cdot x)$  both be balanced. Then  $e^\xi \cdot x = O^{-1} \cdot y$  is balanced by equivariance of  $\mu$ , so  $h'(0) = 0$  and  $h'(1) = -2 \text{tr}(\xi \mu(e^\xi \cdot x)) = 0$ . Since  $h'$  is non-decreasing,  $h' \equiv 0$  on  $[0, 1]$ , so  $h''(0) = 0$ . Every column  $(BR)_{:,i}$  is nonzero because  $BR$  has full column rank, so all  $d_i = 0$ ,  $\xi = 0$  and  $y = O \cdot x$ . Conversely the  $O(r)$ -orbit of a balanced point consists of balanced points.

*Existence and value.* Let  $BA = U\Sigma V^\top$  be a compact SVD ([Auxiliary result B.11\(a\)](#)) with  $\Sigma \in \mathbb{R}^{r \times r}$  positive diagonal. The factorisation  $(U\Sigma^{1/2}, \Sigma^{1/2}V^\top)$  lies in  $O$  by the first paragraph, is balanced since both Gram matrices equal  $\Sigma$ , and has  $f = 2 \text{tr} \Sigma = 2\|BA\|_*$ . This is the minimum value.

(b) Under these hypotheses the update has the form of [Theorem III.5](#), so  $\Phi$  is conserved. With  $B_0 = 0$ ,  $\Phi_0 = -A_0 A_0^\top / \eta_A$ , and  $B^\top B / \eta_B - AA^\top / \eta_A = -A_0 A_0^\top / \eta_A$  rearranges to  $B^\top B = \frac{\eta_B}{\eta_A} (AA^\top - A_0 A_0^\top)$ . Since  $B^\top B \succeq 0$ ,  $AA^\top \succeq A_0 A_0^\top$ .

*No monotonicity.* Take  $m = n = r = 1$ ,  $a(0) = a_0 > 0$ ,  $b(0) = 0$  and the signal  $\gamma = -1$  on  $[0, 1]$  and  $\gamma = +1$  on  $[1, 2]$ , the gradients of the minibatch losses  $-w$  and  $w$ . The flow is  $\dot{b} = -\eta_B s \gamma a$ ,  $\dot{a} = -\eta_A s \gamma b$ . On  $[0, 1]$ ,  $a(t) = a_0 \cosh(s\sqrt{\eta_A \eta_B} t)$  increases. On  $[1, 2]$  the system is the time reversal of the first one, so it retraces its path and returns to  $(a_0, 0)$  at  $t = 2$ , and  $a$  decreases. The floor  $a^2 \geq a_0^2$  holds throughout, but  $a^2$  is not monotone.

*Discrete steps.* By the discrete formula of [§C.28](#), after  $k$  GD steps (with  $s$  absorbed into  $G_j$ )

$$\begin{aligned} A_k A_k^\top &= \eta_A \left( \frac{1}{\eta_B} B_k^\top B_k - \Phi_k \right) \succeq -\eta_A \Phi_k = A_0 A_0^\top - \sum_{j < k} (\eta_A \eta_B A_j G_j^\top G_j A_j^\top - \eta_A^2 B_j^\top G_j G_j^\top B_j) \\ &\succeq A_0 A_0^\top - \eta_A \eta_B \sum_{j < k} A_j G_j^\top G_j A_j^\top, \end{aligned}$$

so the floor holds up to an error of order  $\eta^2$  per step. The identity itself breaks at that order already at the first step, where  $B_1 = -\eta_B G_0 A_0^\top$  and  $A_1 = A_0$  give  $B_1^\top B_1 = \eta_B^2 A_0 G_0^\top G_0 A_0^\top$  against  $\frac{\eta_B}{\eta_A} (A_1 A_1^\top - A_0 A_0^\top) = 0$ .

*Adam.* Take  $m = n = r = 1$ ,  $s = 1$ ,  $(b_0, a_0) = (0, 1)$ , minibatch gradients  $G_1 = 1$  and  $G_2 = -1$ , any  $\varepsilon \geq 0$  and bias correction. At step 1,  $\nabla_a = b_0 G_1 = 0$ , so  $a_1 = 1$ , and  $b_1 = -\eta_B \phi(G_1 a_0)$  has

sign  $-1$ . At step 2,  $\nabla_a = b_1 G_2 > 0$  is the first nonzero gradient of  $a$ , so Adam's step on  $a$  is  $-\eta_A \hat{m}/(\sqrt{\hat{v}} + \varepsilon)$  with  $\hat{m} = \nabla_a/(1 + \beta_1) > 0$ ; at  $\varepsilon = 0$  it equals  $-\eta_A \sqrt{1 + \beta_2}/(1 + \beta_1)$ . So  $a_2 < 1 = a_0$  for every  $\varepsilon \geq 0$ , and at  $\varepsilon = 0$  the violation is of first order in  $\eta_A$ .

*Per-time decay.* By [Corollary III.6\(b\)](#),  $\Phi(t) = e^{-2\lambda t} \Phi(0) = -e^{-2\lambda t} A_0 A_0^\top / \eta_A$ , so  $AA^\top = e^{-2\lambda t} A_0 A_0^\top + \frac{\eta_A}{\eta_B} B^\top B \succeq e^{-2\lambda t} A_0 A_0^\top$ .  $\square$

**REMARK** (nuclear-norm penalties). By (a),  $\min\{\lambda(\|B\|^2 + \|A\|^2) : sBA = \Delta W\} = 2\lambda\|\Delta W\|_*/s$  for  $\Delta W$  of rank  $r$ ; [Srebro et al. \(2004\)](#) show ([Auxiliary result B.21](#)) that  $\min\{\frac{1}{2}(\|B\|^2 + \|A\|^2) : BA = Z\} = \|Z\|_*$  for every inner dimension at least rank  $Z$ , which covers rank-deficient  $\Delta W$ . So the explicit penalty  $\lambda(\|B\|^2 + \|A\|^2)$  matches a  $(2\lambda/s)$ -nuclear-norm penalty at its minimisers, and coupled decay with coefficient  $\lambda$ , the flow of  $L + \frac{\lambda}{2}(\|B\|^2 + \|A\|^2)$ , matches a  $(\lambda/s)$ -nuclear-norm penalty.

### C.31 Proposition III.8

**PROPOSITION III.8** (the balance–escape trilemma). *No LoRA initialisation is simultaneously*

- (a) *identity-at-init without a residual split* ( $B_0 A_0 = 0$ ),
- (b) *balanced* ( $B_0^\top B_0 = c A_0 A_0^\top$  for some  $c > 0$ : *unweighted*,  $c = 1$ , or *rate-weighted*,  $\Phi_0 = 0$  with  $c = \eta_B/\eta_A$ ), and
- (c) *non-stationary*, meaning that  $D\rho$  at  $(B_0, A_0)$  is not the zero map; equivalently,  $\dot{W}(0) \neq 0$  for some parameter velocity, under any first-order optimizer.

Moreover, if  $B_0 A_0 = 0$  then  $\langle B_0^\top B_0, A_0 A_0^\top \rangle_F = 0$ , so

$$\|B_0^\top B_0 - c A_0 A_0^\top\|_F^2 = \|B_0^\top B_0\|_F^2 + c^2 \|A_0 A_0^\top\|_F^2,$$

and zero-product initialisations are maximally unbalanced.

*Proof.* Assume (a) and (b). Then  $0 = B_0^\top(B_0 A_0) = (B_0^\top B_0)A_0 = c A_0 A_0^\top A_0$ , and multiplying by  $A_0^\top$  on the left gives  $c(A_0^\top A_0)^2 = 0$ . The matrix  $A_0^\top A_0$  is symmetric positive semidefinite with vanishing square, so its eigenvalues vanish and  $A_0^\top A_0 = 0$ , that is,  $A_0 = 0$ . Then  $B_0^\top B_0 = c A_0 A_0^\top = 0$ , so  $B_0 = 0$ . At  $(0, 0)$ ,  $D\rho(\dot{B}, \dot{A}) = s(\dot{B} \cdot 0 + 0 \cdot \dot{A}) = 0$ , which contradicts (c). The equivalence in (c) is the definition of the zero map: some parameter velocity has  $\dot{W} = D\rho(\dot{B}, \dot{A}) \neq 0$  iff  $D\rho \neq 0$ . At  $(0, 0)$  also  $\nabla_B = sGA^\top = 0$  and  $\nabla_A = sB^\top G = 0$ , so every update rule built from these gradients and the current parameters, including Adam, momentum and weight decay started from zero state, stays at  $(0, 0)$ .

For the identity,  $\langle B_0^\top B_0, A_0 A_0^\top \rangle_F = \text{tr}(B_0^\top B_0 A_0 A_0^\top) = \text{tr}(B_0^\top (B_0 A_0) A_0^\top) = 0$ , and expanding the square gives the displayed formula. For positive semidefinite  $P$  and  $N$ ,  $\langle P, N \rangle_F = \text{tr}(P^{1/2} N P^{1/2}) \geq 0$ , so  $\|P - cN\|_F^2 \leq \|P\|_F^2 + c^2 \|N\|_F^2$ , with equality iff  $\langle P, N \rangle_F = 0$ . Zero-product initialisations attain this maximum for the given norms of the two Gram matrices.  $\square$

## C.32 Theorem III.9

**THEOREM III.9** (closed-form isotropic-charge LoRA dynamics; known, [Tarmoun et al. \(2021\)](#) and [Min et al. \(2021\)](#)). Let  $\rho(B, A) = W_{\text{res}} + sBA$  be trained by gradient flow with rates  $\eta_A, \eta_B$ , and suppose the charge is isotropic,  $\Phi = -\kappa I_r$ . Put  $\mu = s\sqrt{\eta_A\eta_B}$  (a rate, unrelated to the moment map  $\mu$  of [Proposition III.7](#)), and on the locus where  $Z = W - W_{\text{res}}$  has rank  $r$  write its compact SVD  $Z = U\Sigma V^\top$ . Then

$$\dot{W} = -\mu^2 [G V g_\kappa(\Sigma^2/\mu^2) V^\top + U f_\kappa(\Sigma^2/\mu^2) U^\top G], \quad f_\kappa(x) = \sqrt{x + \kappa^2/4} - \kappa/2, \quad g_\kappa = f_\kappa + \kappa.$$

The identity holds pointwise on  $\{\Phi = -\kappa I, \text{rank } Z = r\}$ ; gradient flow is used only to keep  $\Phi$  fixed. The two terms under the root are equal at the crossover scale  $\sigma^* = \mu|\kappa|/2$ .

- For  $\sigma \ll \sigma^*$  with  $\kappa > 0$ ,  $f \rightarrow 0$  and  $g \rightarrow \kappa$ , so  $\dot{W} \approx -\mu^2 \kappa G P_V$ . These are LoRA-FA dynamics. The contribution of  $A$  to  $\dot{W}$  is  $O(\sigma^2/(\mu^2\kappa))$ , although  $A$  itself still drifts slowly. For  $\kappa < 0$  the roles swap,  $\dot{W} \approx -\mu^2 |\kappa| P_U G$ , and only  $A$  effectively moves ( $\text{Init}[B]$ ).
- For  $\sigma \gg \sigma^*$ ,  $f \approx g \approx \sigma/\mu$ , so  $\dot{W} \approx -\mu [G(Z^\top Z)^{1/2} + (ZZ^\top)^{1/2} G]$ . These are the balanced dynamics of Arora, Cohen and Hazan.
- Off the rank- $r$  locus. Write  $\dot{W} = -\mu^2(GS + TG)$  with  $S = A^\top A/\eta_A$  and  $T = BB^\top/\eta_B$ , so that on the rank- $r$  locus  $S = V g_\kappa(\Sigma^2/\mu^2) V^\top$  and  $T = U f_\kappa(\Sigma^2/\mu^2) U^\top$ . For  $\kappa \geq 0$ ,  $T = f_\kappa(ZZ^\top/\mu^2)$  still holds as a full matrix function, but  $S$  gains the term  $\kappa(P_{\text{row } A} - P_V)$ ; symmetrically, for  $\kappa < 0$ ,  $T$  gains  $|\kappa|(P_{\text{col } B} - P_U)$ . At the default-LoRA start  $Z = 0$  ( $B_0 = 0$ ) this leaves  $S = \kappa P_{\text{row } A_0}$  and  $T = 0$ , so  $\dot{W} = -\mu^2 \kappa G P_{\text{row } A_0}$ , and the  $\sigma \ll \sigma^*$  regime extends continuously to  $t = 0$ .

*Proof.* Let  $G$  be the signal driving the flow,  $\dot{B} = -\eta_B s G A^\top$  and  $\dot{A} = -\eta_A s B^\top G$ ; under gradient flow  $G = \nabla_\theta L$ , and [Theorem III.5\(i\)](#) keeps  $\Phi = -\kappa I$  for all time if it holds at  $t = 0$ . Everything below uses only the value of  $\Phi$  at the point considered.

*Normalisation.* Put  $\beta = B/\sqrt{\eta_B}$  and  $\alpha = A/\sqrt{\eta_A}$ . Then  $\beta^\top \beta - \alpha \alpha^\top = \Phi = -\kappa I$  and  $\beta \alpha = BA/\sqrt{\eta_A\eta_B} = Z/\mu$ . Let  $T = \beta \beta^\top = BB^\top/\eta_B$  and  $S = \alpha^\top \alpha = A^\top A/\eta_A$ . Then

$$\dot{W} = s(\dot{B}A + B\dot{A}) = -s^2(\eta_B G A^\top A + \eta_A B B^\top G) = -s^2 \eta_A \eta_B (GS + TG) = -\mu^2 (GS + TG).$$

*Two quadratic identities.* Using  $\alpha \alpha^\top = \beta^\top \beta + \kappa I$  and  $\beta^\top \beta = \alpha \alpha^\top - \kappa I$ ,

$$\frac{ZZ^\top}{\mu^2} = \beta \alpha \alpha^\top \beta^\top = \beta(\beta^\top \beta + \kappa I)\beta^\top = T^2 + \kappa T, \quad \frac{Z^\top Z}{\mu^2} = \alpha^\top \beta^\top \beta \alpha = \alpha^\top (\alpha \alpha^\top - \kappa I) \alpha = S^2 - \kappa S.$$

These hold at every point with  $\Phi = -\kappa I$ .

*Spectra on the rank- $r$  locus.* If  $\text{rank } Z = r$ , then  $\beta$  and  $\alpha^\top$  have full column rank, so  $\beta^\top \beta \succ 0$  and  $\alpha \alpha^\top \succ 0$ . The nonzero eigenvalues of  $T = \beta \beta^\top$  are the eigenvalues of  $\beta^\top \beta = \alpha \alpha^\top - \kappa I$  ([Auxiliary result B.36\(c\)](#)), which are positive and exceed  $-\kappa$ ; so they exceed  $\max(0, -\kappa)$ . Likewise the nonzero eigenvalues of  $S$  are those of  $\alpha \alpha^\top = \beta^\top \beta + \kappa I$ , which exceed  $\max(0, \kappa)$ . The range of  $T$  is  $\text{col } \beta = \text{col } Z = U$ , and that of  $S$  is  $\text{row } \alpha = \text{row } Z = V$ .

*Inversion.* The polynomial  $p(t) = t^2 + \kappa t$  is strictly increasing on  $[-\kappa/2, \infty)$ , which contains the nonzero eigenvalues of  $T$  because  $\max(0, -\kappa) \geq -\kappa/2$ , and  $p(t) > 0$  there while  $p(0) = 0$ . So  $p$  is injective on the spectrum of  $T$ , the eigenspaces of  $T$  are those of  $p(T) = ZZ^\top/\mu^2 = U(\Sigma^2/\mu^2)U^\top$  (by the compact SVD of  $Z$ , [Auxiliary result B.11\(a\)](#)), and on the eigenvector  $u_i$  with  $ZZ^\top u_i = \sigma_i^2 u_i$  the eigenvalue of  $T$  is the unique root  $t > \max(0, -\kappa)$  of  $t^2 + \kappa t = \sigma_i^2/\mu^2$ , namely  $f_\kappa(\sigma_i^2/\mu^2)$ . On  $U^\perp$  both  $T$  and  $ZZ^\top$  vanish. Hence  $T = U f_\kappa(\Sigma^2/\mu^2) U^\top$ . In the same way  $q(s) = s^2 - \kappa s$  is strictly increasing on  $[\kappa/2, \infty)$ , the root above  $\max(0, \kappa)$  of  $s^2 - \kappa s = x$

is  $\kappa/2 + \sqrt{x + \kappa^2/4} = g_\kappa(x)$ , and  $S = Vg_\kappa(\Sigma^2/\mu^2)V^\top$ . The compact SVD matters:  $f_\kappa(0) = |\kappa|$  for  $\kappa < 0$  and  $g_\kappa(0) = \kappa$  for  $\kappa > 0$ , so the full matrix functions  $f_\kappa(ZZ^\top/\mu^2)$  and  $g_\kappa(Z^\top Z/\mu^2)$  would put nonzero values on  $\ker Z^\top$  and  $\ker Z$ , where  $T$  and  $S$  vanish. Substituting into  $\dot{W} = -\mu^2(GS + TG)$  gives the formula. The crossover:  $x = \sigma^2/\mu^2$  equals  $\kappa^2/4$  iff  $\sigma = \mu|\kappa|/2$ .

*The regimes.* For  $\kappa > 0$ ,  $f_\kappa(x) = x/(\sqrt{x + \kappa^2/4} + \kappa/2) \in [0, x/\kappa]$  and  $g_\kappa(x) - \kappa = f_\kappa(x)$ . So  $\dot{W} = -\mu^2\kappa GP_V + E$  with  $E = -\mu^2(GVf_\kappa(\Sigma^2/\mu^2)V^\top + TG)$  and  $\|E\| \leq 2(\sigma_1^2/\kappa)\|G\|$ , where  $\sigma_1$  is the largest singular value of  $Z$ . The contribution  $-\mu^2TG = sB\dot{A}$  of  $A$  has gain at most  $f_\kappa(\sigma_1^2/\mu^2) \leq \sigma_1^2/(\mu^2\kappa)$ , against the gain  $g_\kappa \geq \kappa$  of the  $B$ -term. The factor  $A$  still moves, at speed  $\|\dot{A}\| \leq \eta_A s \|B\|_2 \|G\|$  with  $\|B\|_2^2 = \eta_B \|T\|_2 \leq \eta_B \sigma_1^2/(\mu^2\kappa)$ . For  $\kappa < 0$  the same bounds hold with the roles exchanged:  $g_\kappa(x) \in [0, x/|\kappa|]$  and  $f_\kappa = g_\kappa + |\kappa|$ , so  $\dot{W} = -\mu^2|\kappa|P_UG + O(\sigma_1^2/|\kappa|)\|G\|$ . For large  $x$ ,  $0 \leq \sqrt{x + \kappa^2/4} - \sqrt{x} \leq |\kappa|/2$  gives  $|f_\kappa(x) - \sqrt{x}| \leq |\kappa|$  and  $|g_\kappa(x) - \sqrt{x}| \leq |\kappa|$ . At  $x = \sigma_i^2/\mu^2$  the relative error is at most  $|\kappa|\mu/\sigma_i = 2\sigma^*/\sigma_i$ , so for  $\sigma_i \gg \sigma^*$ ,  $f \approx g \approx \sigma_i/\mu$  and

$$\dot{W} \approx -\mu^2[GV(\Sigma/\mu)V^\top + U(\Sigma/\mu)U^\top G] = -\mu[G(Z^\top Z)^{1/2} + (ZZ^\top)^{1/2}G].$$

This is the end-to-end dynamics of a balanced depth-2 linear network with rate  $\mu$  (Arora et al., 2018, Thm. 1 with  $N = 2$ ); at  $\kappa = 0$  it is exact, since  $f_0(x) = g_0(x) = \sqrt{x}$ .

*Off the rank- $r$  locus.* Let  $\kappa > 0$  and  $\text{rank } Z < r$ . The nonzero eigenvalues of  $T$  are those of  $\beta^\top \beta \succeq 0$ , and on them  $p(t) = t^2 + \kappa t > 0$ ; if  $ZZ^\top v = 0$  then  $T(T + \kappa I)v = 0$  with  $T + \kappa I \succ 0$ , so  $Tv = 0$ . Hence  $\ker T = \ker ZZ^\top$  and the inversion above gives  $T = f_\kappa(ZZ^\top/\mu^2)$  as a full matrix function, using  $f_\kappa(0) = 0$  for  $\kappa \geq 0$ . For  $S$ , decompose  $\text{row } A = \text{row } Z \oplus (\text{row } A \cap \ker Z)$  orthogonally, which is possible since  $\text{row } Z \subseteq \text{row } A$  and  $(\text{row } Z)^\perp = \ker Z$ . On  $(\text{row } A)^\perp = \ker \alpha$ ,  $S = 0$ . If  $v = \alpha^\top w \in \text{row } A$  has  $Zv = 0$ , then  $0 = \beta\alpha^\top w = \beta(\beta^\top \beta + \kappa I)w$ , so  $\beta^\top \beta(\beta^\top \beta + \kappa I)w = 0$ ; as  $\beta^\top \beta + \kappa I \succ 0$  commutes with  $\beta^\top \beta \succeq 0$ , this forces  $\beta^\top \beta w = 0$ , hence  $\beta w = 0$ , and  $Sv = \alpha^\top(\alpha\alpha^\top)w = \alpha^\top(\beta^\top \beta + \kappa I)w = \kappa v$ . On  $\text{row } Z$  the eigenvalues of  $S$  exceed  $\kappa$  and the inversion gives  $Vg_\kappa(\Sigma^2/\mu^2)V^\top$ . Together,  $S = Vg_\kappa(\Sigma^2/\mu^2)V^\top + \kappa(P_{\text{row } A} - P_V)$ . The case  $\kappa < 0$  is symmetric under  $(B, A, Z, \eta_B, \eta_A, \kappa) \mapsto (A^\top, B^\top, Z^\top, \eta_A, \eta_B, -\kappa)$ , which exchanges  $S$  and  $T$ . At  $\kappa = 0$  the correction terms vanish. At  $Z = 0$  with  $B_0 = 0$  and  $A_0 A_0^\top = cI$ ,  $\kappa = c/\eta_A > 0$ ,  $T = 0$  and  $S = \kappa P_{\text{row } A_0}$ , so  $\dot{W} = -\mu^2\kappa GP_{\text{row } A_0} = -s^2\eta_B c GP_{\text{row } A_0}$ , which agrees with the direct value  $-s^2\eta_B G A_0^\top A_0$  because  $A_0^\top A_0 = cP_{\text{row } A_0}$ . As  $Z \rightarrow 0$  along the flow with  $V \rightarrow \text{row } A_0$ , the rank- $r$  formula tends to the same value, since  $g_\kappa(0) = \kappa$  and  $f_\kappa(0) = 0$ .  $\square$

### C.33 Theorem III.10

**THEOREM III.10** (the hyperparameter torus; the LoRA/Adam case is known, Schulman and Thinking Machines Lab (2025)). Let  $\rho(q) = \theta_b + s\delta(q_1, \dots, q_k)$  with  $\delta$  multihomogeneous of multidegree  $(d_1, \dots, d_k)$ . Train block  $i$  from  $q_i(0) = \sigma_i u_i$  with Adam ( $\beta_1 \in [0, 1)$ ,  $\beta_2 \in (0, 1)$ , bias correction allowed) at learning rate  $\eta_i(t)$ , on the loss  $L \circ \rho$  with no  $q$ -dependent regulariser. Assume no gradient clipping (or per-block clipping with thresholds rescaled as below), common random numbers (the same data order and dropout masks in both runs), and exact arithmetic (the result is bit-exact in floating point when every  $c_i$  is a power of 2). Then the trajectory  $t \mapsto \rho(q(t))$  is invariant under the torus action of  $c \in \mathbb{R}_{>0}^k$ :

$$(s, \eta_i(t), \sigma_i, \varepsilon_i, \lambda_i, C_i) \mapsto \left(s \prod_i c_i^{-d_i}, c_i \eta_i(t), c_i \sigma_i, \varepsilon_i/c_i, \lambda_i/c_i, C_i/c_i\right).$$

Here  $\varepsilon_i$  is a per-block Adam  $\varepsilon$  (with the convention  $0/0 = 0$  at  $\varepsilon = 0$ ),  $\lambda_i$  a per-block decoupled (AdamW) weight decay, and  $C_i$  a per-block clipping threshold. Under SGD, with or without heavy-ball or Nesterov momentum, the same holds with  $\eta_i(t) \mapsto c_i^2 \eta_i(t)$  and coupled  $L^2$  decay

$\lambda_i \mapsto \lambda_i/c_i^2$ . In particular, any learning-rate schedule shared by all blocks, warm-up included, is preserved.

*Proof.* The rescaling is an isomorphism. Write  $c \odot q = (c_1 q_1, \dots, c_k q_k)$  and  $\rho' = \theta_b + s' \delta$  with  $s' = s \prod_i c_i^{-d_i}$ . Multihomogeneity (Background D.19) gives  $\rho'(c \odot q) = \theta_b + s \prod_i c_i^{-d_i} \prod_i c_i^{d_i} \delta(q) = \rho(q)$  for all  $q$ , so  $q \mapsto c \odot q$  is a diffeomorphism over  $\Theta$  that maps the base point  $q(0)$  to  $q'(0) = c \odot q(0)$ ; it is an isomorphism in **Meth**. Let  $\ell_t = L_t \circ \rho$  and  $\ell'_t = L_t \circ \rho'$  be the step- $t$  losses, with the same minibatch and dropout masks in both runs and no  $q$ -dependent regulariser. Differentiating  $\ell'_t(c \odot q) = \ell_t(q)$  gives

$$\nabla_{q_i} \ell'_t(c \odot q) = c_i^{-1} \nabla_{q_i} \ell_t(q). \quad (23)$$

*The optimizer.* For block  $i$  at step  $t \geq 1$ , Adam(W) computes the gradient  $g_t$  of  $\ell_t$  at  $q_{t-1}$ , clips it per block,  $g_t \leftarrow g_t \min(1, C_i/\|g_t\|)$  (with factor 1 when  $g_t = 0$ , and no clipping step when there is no clipping), and sets

$$\begin{aligned} m_t &= \beta_1 m_{t-1} + (1 - \beta_1) g_t, & v_t &= \beta_2 v_{t-1} + (1 - \beta_2) g_t \odot g_t, \\ q_t &= q_{t-1} - \eta_i(t) \left( \frac{m_t/b_{1,t}}{\sqrt{v_t/b_{2,t} + \varepsilon_i}} + \lambda_i q_{t-1} \right), \end{aligned}$$

with  $m_0 = v_0 = 0$ , entrywise square root and division,  $b_{j,t} = 1 - \beta_j^t$  with bias correction and  $b_{j,t} = 1$  without, and  $0/0 = 0$ . PyTorch's AdamW first multiplies  $q_{t-1}$  by  $1 - \eta_i(t) \lambda_i$  and then takes the Adam step, which is the same update.

*Induction.* Run the original hyperparameters from  $q(0)$  and the transformed ones from  $q'(0) = c \odot q(0)$ , which is what  $\sigma_i \mapsto c_i \sigma_i$  means. Suppose that after step  $t - 1$ , in every block,

$$q'_{t-1} = c_i q_{t-1}, \quad m'_{t-1} = c_i^{-1} m_{t-1}, \quad v'_{t-1} = c_i^{-2} v_{t-1},$$

which holds at  $t = 1$ . By (23) the raw gradients satisfy  $g'_t = c_i^{-1} g_t$ . With  $C'_i = C_i/c_i$  the clipping factors agree,  $\min(1, C'_i/\|g'_t\|) = \min(1, C_i/\|g_t\|)$ , so the clipped gradients also satisfy  $g'_t = c_i^{-1} g_t$ . The moment recursions are linear in  $g_t$  and  $g_t \odot g_t$ , so  $m'_t = c_i^{-1} m_t$  and  $v'_t = c_i^{-2} v_t$ . With  $\varepsilon'_i = \varepsilon_i/c_i$  the normalised step is unchanged,

$$\frac{m'_t/b_{1,t}}{\sqrt{v'_t/b_{2,t} + \varepsilon'_i}} = \frac{c_i^{-1} m_t/b_{1,t}}{c_i^{-1} (\sqrt{v_t/b_{2,t} + \varepsilon_i})} = \frac{m_t/b_{1,t}}{\sqrt{v_t/b_{2,t} + \varepsilon_i}},$$

including the entries where  $v_t = 0$ , at which  $m_t = 0$  as well and both sides are 0. The decay term transforms as  $\lambda'_i q'_{t-1} = (\lambda_i/c_i) c_i q_{t-1} = \lambda_i q_{t-1}$ . With  $\eta'_i(t) = c_i \eta_i(t)$  the whole step is multiplied by  $c_i$ , so  $q'_t = c_i q_t$ , which closes the induction. Then  $\rho'(q'_t) = \rho(q_t)$  for all  $t$ . The value of  $\eta_i(t)$  at each step was arbitrary, so any schedule works; a schedule shared by all blocks,  $\eta_i(t) = \eta_i \varphi(t)$ , becomes  $c_i \eta_i \varphi(t)$ , the same schedule.

*SGD.* With coupled  $L^2$  decay the step- $t$  direction is  $g_t + \lambda_i q_{t-1}$ , the heavy-ball buffer is  $b_t = \mu_m b_{t-1} + (1 - \tau)(g_t + \lambda_i q_{t-1})$  (with momentum  $\mu_m$  and dampening  $\tau$ ), the direction is  $b_t$ , or  $g_t + \lambda_i q_{t-1} + \mu_m b_t$  for Nesterov, and  $q_t = q_{t-1} - \eta_i(t) \cdot (\text{direction})$ . Suppose  $q'_{t-1} = c_i q_{t-1}$  and  $b'_{t-1} = c_i^{-1} b_{t-1}$ . Then  $g'_t = c_i^{-1} g_t$  and, with  $\lambda'_i = \lambda_i/c_i^2$ ,  $\lambda'_i q'_{t-1} = c_i^{-1} \lambda_i q_{t-1}$ ; every quantity entering the direction is multiplied by  $c_i^{-1}$ , so  $b'_t = c_i^{-1} b_t$  and the direction is multiplied by  $c_i^{-1}$ . With  $\eta'_i(t) = c_i^2 \eta_i(t)$  the step is multiplied by  $c_i$ , and  $q'_t = c_i q_t$ . Per-block clipping with  $C_i/c_i$  is handled as for Adam.

*Floating point.* If every  $c_i$  is a power of 2, multiplication by  $c_i^{\pm 1}$  and by  $\prod_i c_i^{\mp d_i}$  is exact, and it commutes with every correctly rounded addition, multiplication, division and square root,



barring overflow and underflow ([Auxiliary result B.39](#)). Then the forward pass computes bit-identical weights and outputs in both runs, the backward pass computes gradients that are exactly  $c_i^{-1}$  times the original ones, and every quantity in the induction above holds exactly in floating point, so the two trajectories agree bit for bit.  $\square$

### C.34 Corollary III.11

**COROLLARY III.11** (LoRA+ is  $\alpha$  in disguise; rsLoRA is LoRA; (a) and the LoRA+ reading are known, [Schulman and Thinking Machines Lab \(2025\)](#)). For linear LoRA modules ( $k = 2$ ,  $d = (1, 1)$ ,  $\sigma_B = 0$ ), under the hypotheses of [Theorem III.10](#):

(a) Effective hyperparameters. Under Adam the torus invariants of  $(s, \eta_A, \eta_B, \sigma_A)$  are  $(s\eta_A\eta_B, \sigma_A/\eta_A)$ ; the quantities  $\alpha \cdot \text{init}_A \cdot \text{LR}_B$  and  $\text{init}_A/\text{LR}_A$  of Schulman et al. are equivalent generators. Under SGD the invariants are  $(s^2\eta_A\eta_B, \sigma_A^2/\eta_A)$ .

(b) LoRA+. Under Adam with  $\varepsilon = 0$ , no weight decay, no gradient clipping and a learning-rate schedule shared by both groups, LoRA+ with ratio  $\lambda$  at scale  $s$  produces exactly the trajectory of plain LoRA at scale  $\lambda s$  (same  $\eta_A$ , same  $A_0$ ). For  $\varepsilon > 0$  the equivalence needs a per-block  $\varepsilon'_B = \lambda\varepsilon$ , and with AdamW a per-group decay  $\lambda \cdot \text{wd}$  on  $B$ . Under SGD it holds at scale  $\sqrt{\lambda} s$ , or equivalently with plain LoRA at the single rate  $\sqrt{\eta_A\eta_B}$  and  $A_0 \mapsto \lambda^{1/4} A_0$ .

(c) rsLoRA.  $\rho_{\text{rs}}(B, A) = \rho_{\text{LoRA}}(\sqrt{r} B, A)$  is an isomorphism of objects. Up to the hyperparameter torus, rsLoRA can be represented as LoRA at the same  $\alpha$  with  $\eta_B$  multiplied by  $\sqrt{r}$  under Adam (by  $r$  under SGD), or equally as LoRA with  $\alpha$  multiplied by  $\sqrt{r}$  at the same rates. The reading “same object, different metric” is a choice of representative within the orbit. Part (c) concerns a fixed rank; it does not address rsLoRA’s rank-stability prescription, how  $\alpha$  should scale with  $r$ .

*Proof.* Here  $\delta(B, A) = BA$  has multidegree  $(1, 1)$ , and  $\sigma_B = 0$  means  $B_0 = 0$ , which every  $c_B$  leaves unchanged.

(a) Under Adam,  $(c_A, c_B)$  acts by  $s \mapsto s/(c_A c_B)$ ,  $\eta_A \mapsto c_A \eta_A$ ,  $\eta_B \mapsto c_B \eta_B$  and  $\sigma_A \mapsto c_A \sigma_A$ . In logarithms this is translation by the column space of

$$a = \begin{pmatrix} -1 & -1 \\ 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{pmatrix} \quad (\text{rows } \log s, \log \eta_A, \log \eta_B, \log \sigma_A),$$

which has rank 2, so the orbit space has dimension 2 and a basis of the vectors  $b$  with  $b^\top a = 0$  gives a complete set of invariant monomials ([Background D.20](#)). The vectors  $(1, 1, 1, 0)$  and  $(0, -1, 0, 1)$  satisfy  $b^\top a = 0$  and are independent; they give  $s\eta_A\eta_B$  and  $\sigma_A/\eta_A$ . Since  $\alpha \propto s$  at fixed  $r$ , Schulman et al.’s quantities are  $s\sigma_A\eta_B = (s\eta_A\eta_B)(\sigma_A/\eta_A)$  and  $\sigma_A/\eta_A$ , with exponent vectors  $(1, 0, 1, 1)$  and  $(0, -1, 0, 1)$ , another basis of the same lattice of invariant monomials. Under SGD the rates scale by  $c_A^2$  and  $c_B^2$ ; the weight matrix has rows  $(-1, -1)$ ,  $(2, 0)$ ,  $(0, 2)$ ,  $(1, 0)$ , again of rank 2, and  $(2, 1, 1, 0)$  and  $(0, -1, 0, 2)$  give the invariants  $s^2\eta_A\eta_B$  and  $\sigma_A^2/\eta_A$ .

(b) LoRA+ has  $\eta_B = \lambda\eta_A$ . Under Adam apply the torus element  $c_A = 1$ ,  $c_B = 1/\lambda$ : it maps  $(s, \eta_A, \lambda\eta_A, \sigma_A)$  to  $(\lambda s, \eta_A, \eta_A, \sigma_A)$ , plain LoRA at scale  $\lambda s$  with the same  $\eta_A$  and the same  $A_0$ ; a shared schedule  $\varphi(t)$  stays shared. [Theorem III.10](#) gives  $\varepsilon'_B = \varepsilon/c_B = \lambda\varepsilon$  and  $\text{wd}'_B = \text{wd}/c_B = \lambda \cdot \text{wd}$ , which are the stated per-block values, and both vanish when  $\varepsilon = \text{wd} = 0$ . Under SGD,  $c_A = 1$  and  $c_B = \lambda^{-1/2}$  give  $\eta'_B = c_B^2 \lambda \eta_A = \eta_A$  and  $s' = \sqrt{\lambda} s$ . The single-rate form uses  $c_A = \lambda^{1/4}$  and  $c_B = \lambda^{-1/4}$ : then  $\eta'_A = \sqrt{\lambda} \eta_A$ ,  $\eta'_B = \lambda^{-1/2} \lambda \eta_A = \sqrt{\lambda} \eta_A$ , both equal to  $\sqrt{\eta_A\eta_B}$ ,  $s' = s$ , and  $\sigma'_A = \lambda^{1/4} \sigma_A$ , that is,  $A_0 \mapsto \lambda^{1/4} A_0$ .

(c)  $\rho_{\text{rs}}(B, A) = W_0 + (\alpha/\sqrt{r})BA = W_0 + (\alpha/r)(\sqrt{r}B)A = \rho_{\text{LoRA}}(\sqrt{r}B, A)$ , and  $(B, A) \mapsto (\sqrt{r}B, A)$  is a pointed diffeomorphism ( $0 \mapsto 0$  in the  $B$ -factor), hence an isomorphism of objects. The rsLoRA scale  $s_{\text{rs}} = \alpha/\sqrt{r}$  is the LoRA scale for  $\alpha' = \sqrt{r}\alpha$ , which is the second representation. Applying the torus element  $c_A = 1, c_B = \sqrt{r}$  to rsLoRA gives  $s' = s_{\text{rs}}/\sqrt{r} = \alpha/r$ , the LoRA scale at the same  $\alpha$ , with  $\eta'_B = \sqrt{r}\eta_B$  under Adam and  $\eta'_B = r\eta_B$  under SGD, and with  $\varepsilon'_B = \varepsilon/\sqrt{r}$  and  $\text{wd}'_B = \text{wd}/\sqrt{r}$  when these are nonzero. Both representations lie on the torus orbit of rsLoRA's hyperparameters, which is what “up to the hyperparameter torus” means. The rank  $r$  is fixed throughout.  $\square$

### C.35 Theorem III.12

**THEOREM III.12** (the optimizer gauge hierarchy). *On LoRA's full-rank locus, with hyperparameters shared by all rank channels, the largest subgroup of the gauge  $\text{GL}_r$  under which an update rule is equivariant, at the level of parameters, is as follows.*

(a) SGD and GD with any block rates, with or without heavy-ball momentum:  $\text{O}(r)$ .

(b) Adam and AdamW at fixed learning rates  $(\beta_1, \beta_2 \in [0, 1])$ , any  $\varepsilon \geq 0$ , any weight decay): the signed permutations  $B_r$ , of order  $2^r r!$ . A positive diagonal  $g = \text{diag}(d)$  is a symmetry only jointly with per-rank-channel hyperparameters, and only without global-norm clipping. At  $\varepsilon = 0$  and without weight decay it suffices to rescale the rates,  $\eta_{B,j} \mapsto \eta_{B,j}/d_j$  and  $\eta_{A,j} \mapsto d_j \eta_{A,j}$ . With  $\varepsilon > 0$  one must also give each channel its own  $\varepsilon_{B,j} = d_j \varepsilon$  and  $\varepsilon_{A,j} = \varepsilon/d_j$ . With decoupled decay multiplied by the learning rate (PyTorch AdamW) one must also set  $\lambda_{B,j} = d_j \lambda$  and  $\lambda_{A,j} = \lambda/d_j$ ; under the convention of [Loshchilov and Hutter \(2019\)](#), where decay is scaled only by the schedule multiplier,  $\lambda$  needs no change.

(c) Undamped scaled GD ( $\delta = 0$ ; the Riemannian preconditioner of Tong, Ma and Chi and of Zhang and Pilanci, with or without heavy-ball momentum): all of  $\text{GL}_r$ . The  $W$ -trajectory from gauge-related initial states (with zero or correspondingly transformed momentum) is independent of the representative; a step may drop rank, so the iteration maps into  $W_0 + \mathcal{M}_{\leq r}$ .

*Proof.* We use the conventions of §C.24: at  $g \cdot q$  the gradients are (20), and equivariance under  $g$  means (21) for all gradients, together with transported optimizer state. The set of such  $g$  is a subgroup. On the full-rank locus  $\nabla_B = sGA^\top$  ranges over all of  $\mathbb{R}^{m \times r}$  as  $G$  ranges over  $\mathbb{R}^{m \times n}$ , because  $A$  has full row rank; this is the only use of the locus in the inclusions  $\subseteq$  below.

(a) An SGD step is  $\delta B = -\eta_B \nabla_B$ ,  $\delta A = -\eta_A \nabla_A$ . At  $g \cdot q$ ,  $\delta B' = -\eta_B \nabla_{Bg}^\top$ , and (21) requires  $\nabla_B(g^\top - g^{-1}) = 0$  for all  $\nabla_B \in \mathbb{R}^{m \times r}$ , that is,  $g^\top = g^{-1}$ . Conversely, for  $g \in \text{O}(r)$ ,  $\delta B' = \delta B g^\top = \delta B g^{-1}$  and  $\delta A' = -\eta_A g^{-\top} \nabla_A = g \delta A$ . A heavy-ball or Nesterov buffer is a linear combination of past gradients, all taken at points related by the same  $g$ , so for orthogonal  $g$  it transforms like the gradients,  $b'_B = b_B g^{-1}$  and  $b'_A = g b_A$ , and the step is transported. The first step from a zero buffer is a positive multiple of an SGD step (the multiple is 1 for heavy ball and  $1 + \mu_m$  for Nesterov with momentum coefficient  $\mu_m$ ), so the inclusion  $\subseteq \text{O}(r)$  also holds with momentum.

(b) The inclusion  $\supseteq$ . Let  $g$  be a signed permutation, so  $g^{-1} = g^\top$  and  $\nabla_{B'} = \nabla_B g^{-1}$  permutes the columns of  $\nabla_B$  and flips some signs; likewise  $\nabla_{A'} = g \nabla_A$  permutes and flips rows. Adam is coordinatewise: the first moment  $m$  is permuted with the same sign flips, the second moment  $v$  is permuted without sign changes, so the normalised step is transported, and so is a decoupled or coupled decay term, which is linear in the parameters. Hence the step satisfies (21). Since  $g$  is orthogonal it preserves  $\|\nabla_B\|$  and  $\|\nabla_A\|$ , so global-norm clipping does not affect this inclusion.

*The inclusion  $\subseteq$ .* Consider the first step from zero state at a full-rank point. With bias correction it is  $\delta B = -\eta(\phi(\nabla_B) + \lambda B)$  with  $\phi(x) = x/(|x| + \varepsilon)$  entrywise and  $0/0 = 0$ ; without bias correction  $\phi$  is replaced by a positive multiple of  $x/(|x| + \varepsilon')$  with a rescaled  $\varepsilon'$ , which changes nothing below. At  $g \cdot q$  the decay term  $\lambda B g^{-1}$  is already transported, so equivariance requires  $\phi(\nabla_{B g^\top}) = \phi(\nabla_B) g^{-1}$ , row by row:

$$\phi(x g^\top) = \phi(x) h \quad \text{for every row vector } x \in \mathbb{R}^{1 \times r}, \quad h := g^{-1}. \quad (24)$$

(With coupled decay the argument of  $\phi$  is  $\nabla_B + \lambda B$ ; letting  $\nabla_B$  grow along a fixed direction leads to the limit (25) below in the same way.)

Case  $\varepsilon = 0$ . Then  $\phi = \text{sign}$  and (24) reads

$$\text{sign}(x g^\top) = \text{sign}(x) h \quad \text{for every } x. \quad (25)$$

Fix  $s \in \{\pm 1\}^r$ . The open orthant  $\{x : \text{sign}(x) = s\}$  is not contained in the finite union of the hyperplanes  $\{x : (x g^\top)_j = 0\}$ , which are proper because  $g$  is invertible (a finite union of hyperplanes has empty interior), so it contains an  $x$  with  $x g^\top$  free of zeros. For that  $x$ , (25) gives  $sh \in \{\pm 1\}^r$ . Thus  $sh \in \{\pm 1\}^r$  for every  $s$ . Flipping  $s_i$  changes  $(sh)_j$  by  $2s_i h_{ij}$ , a difference of two elements of  $\{\pm 1\}$ , so  $h_{ij} \in \{0, \pm 1\}$ . If column  $j$  of  $h$  had no nonzero entry,  $(sh)_j = 0$ ; if it had  $t \geq 2$  nonzero entries, the choice  $s_i = h_{ij}$  on its support would give  $(sh)_j = t \geq 2$ . So every column of  $h$  has exactly one entry, equal to  $\pm 1$ , and these entries lie in distinct rows because  $h$  is invertible. Hence  $h$ , and with it  $g = h^{-1}$ , is a signed permutation matrix.

Case  $\varepsilon > 0$ . For a fixed  $x$  and  $\tau \rightarrow \infty$ ,  $\phi(\tau y) \rightarrow \text{sign}(y)$  entrywise for every  $y$ , so applying (24) to  $\tau x$  and letting  $\tau \rightarrow \infty$  gives (25), and the case  $\varepsilon = 0$  shows  $g \in B_r$ . (For  $\tau \rightarrow 0$ ,  $\phi(\tau y) = \tau y / \varepsilon + O(\tau^2)$ , and the same comparison gives  $g^\top = h$ , that is,  $g \in O(r)$ , which is consistent.) The order of  $B_r$  is  $2^r r!$ :  $r!$  permutations times  $2^r$  sign patterns.

*The joint torus.* Let  $g = \text{diag}(d)$  with  $d_j > 0$ , and give column  $j$  of  $B$  the hyperparameters  $(\eta'_{B,j}, \varepsilon'_{B,j}, \lambda'_{B,j})$  and row  $j$  of  $A$  the hyperparameters  $(\eta'_{A,j}, \varepsilon'_{A,j}, \lambda'_{A,j})$ . By (20), column  $j$  of  $\nabla_{B'}$  is  $d_j$  times column  $j$  of  $\nabla_B$ , while (21) asks for column  $j$  of  $\delta B'$  to be  $1/d_j$  times that of  $\delta B$ . Suppose that before a step  $B'_{:,j} = B_{:,j}/d_j$ ,  $m'_{:,j} = d_j m_{:,j}$  and  $v'_{:,j} = d_j^2 v_{:,j}$  (true for zero initial state). After the moment updates these relations persist, and the normalised step of column  $j$  is

$$\frac{d_j \hat{m}}{d_j \sqrt{\hat{v}} + \varepsilon'_{B,j}} = \frac{\hat{m}}{\sqrt{\hat{v}} + \varepsilon'_{B,j}/d_j},$$

which equals the original normalised step iff  $\varepsilon'_{B,j} = d_j \varepsilon$  (at  $\varepsilon = 0$  both are 0). The update of column  $j$  is then  $-\eta'_{B,j}(\text{step} + \lambda'_{B,j} B_{:,j}/d_j)$ , which equals  $-\frac{1}{d_j} \eta(\text{step} + \lambda B_{:,j})$  iff  $\eta'_{B,j} = \eta/d_j$  and  $\eta'_{B,j} \lambda'_{B,j}/d_j = \eta \lambda/d_j$ , that is,  $\lambda'_{B,j} = d_j \lambda$ . Under the convention of Loshchilov and Hutter (2019) the decay term is  $-\zeta_t \lambda'_{B,j} B'_{:,j}$  with the schedule multiplier  $\zeta_t$  alone, and it is transported iff  $\lambda'_{B,j} = \lambda$ . Row  $j$  of  $\nabla_{A'}$  is  $1/d_j$  times that of  $\nabla_A$  and the step must be multiplied by  $d_j$ , which gives  $\varepsilon'_{A,j} = \varepsilon/d_j$ ,  $\eta'_{A,j} = d_j \eta$  and  $\lambda'_{A,j} = \lambda/d_j$  (no change under Loshchilov–Hutter). The relations persist by induction. With channel-uniform hyperparameters a positive diagonal  $g \neq I$  is not in  $B_r$ , so by the inclusion  $\subseteq$  it is a symmetry only jointly with these per-channel values. The induction compares the two runs at equal clipping factors. Global-norm clipping multiplies all gradients by  $\min(1, C/(\|\nabla_B, \nabla_A\|))$ , and  $\|(\nabla_{B'}, \nabla_{A'})\|^2 = \sum_j d_j^2 \|\nabla_{B, :, j}\|^2 + \sum_j d_j^{-2} \|\nabla_{A, :, j}\|^2$  differs from the original norm for  $d \neq \mathbf{1}$  in general; the two runs then weight their gradient histories differently, and the relations above fail. This is why the joint symmetry is stated without global-norm clipping.

(c) By [Proposition III.4\(a\)](#) the undamped scaled-GD step satisfies (21) for every  $g \in \text{GL}_r$ . With heavy-ball momentum on the preconditioned gradients,  $p_B = \nabla_B(AA^\top)^{-1}$  transforms as  $p_{B'} = p_B g^{-1}$ , so a buffer of such terms transforms in the same way and so does the step. With momentum on the raw gradients followed by preconditioning, the buffer transforms as  $b_{B'} = b_B g^\top$  because all past points are related by the same  $g$ , and  $b_{B'}(A'A'^\top)^{-1} = b_B g^\top g^{-\top}(AA^\top)^{-1}g^{-1}$  is again transported. The  $A$ -factor is symmetric. So from gauge-related initial states, with zero or correspondingly transformed momentum,  $q'_t = g \cdot q_t$  for all  $t$  and the  $W$ -trajectories coincide. Each iterate lies in  $W_0 + \mathcal{M}_{\leq r}$ , and the iteration is defined as long as both factors keep full rank; a step can lower the rank of  $B + \delta B$  or  $A + \delta A$ , so the map goes into  $W_0 + \mathcal{M}_{\leq r}$  rather than  $W_0 + \mathcal{M}_r$ .

The hierarchy  $B_r \subset \text{O}(r) \subset \text{GL}_r$  is strict for  $r \geq 2$ , since  $B_r$  is finite,  $\text{O}(r)$  is infinite and compact, and  $\text{GL}_r$  is not compact.  $\square$

*Proof of the statements about published variants. Damped scaled GD.* With a damping constant  $\delta > 0$  the two inverses become  $(AA^\top + \delta I)^{-1}$  and  $(B^\top B + \delta I)^{-1}$ . Equivariance of the  $B$ -step requires, for all  $\nabla_B$ ,

$$\nabla_B g^\top (gAA^\top g^\top + \delta I)^{-1} = \nabla_B (AA^\top + \delta I)^{-1} g^{-1}, \quad \text{that is,} \quad g^\top (gAA^\top g^\top + \delta I)^{-1} = (AA^\top + \delta I)^{-1} g^{-1}.$$

Inverting both sides gives  $(gAA^\top g^\top + \delta I)g^{-\top} = g(AA^\top + \delta I)$ , that is,  $gAA^\top + \delta g^{-\top} = gAA^\top + \delta g$ , so  $g^{-\top} = g$  and  $g \in \text{O}(r)$ . Conversely, for orthogonal  $g$  we have  $(gAA^\top g^\top + \delta I)^{-1} = g(AA^\top + \delta I)^{-1}g^\top$ , and both steps are transported. The computation does not use full rank, so it holds at every point, including  $B = 0$ .

*Scaled AdamW* applies Adam to the preconditioned gradient  $p_B = \nabla_B(AA^\top + \delta I)^{-1}$  (and similarly for  $A$ ), with  $\delta \geq 0$ . For a signed permutation  $g$ ,  $p_{B'} = p_B g^{-1}$  by the previous paragraph, and Adam commutes with signed permutations, so the rule is  $B_r$ -equivariant. Conversely, the first step requires  $\phi(\nabla_B M') = \phi(\nabla_B M)g^{-1}$  for all  $\nabla_B$ , with  $M = (AA^\top + \delta I)^{-1}$  and  $M' = g^\top (gAA^\top g^\top + \delta I)^{-1}$  invertible. Writing  $y = xM$  and letting  $y$  grow, as in the case  $\varepsilon > 0$  above, gives  $\text{sign}(yM^{-1}M') = \text{sign}(y)g^{-1}$  for all  $y$ , and the orthant argument, which used only invertibility of the matrix on the left, gives  $g^{-1} \in B_r$ .

*LoRA-Pro with the Sylvester-optimal  $X$ .* Write  $\mathcal{P} = \mathbb{R}^{m \times r} \times \mathbb{R}^{r \times n}$  with its Frobenius inner product,  $u = (u_B, u_A) \in \mathcal{P}$  for a parameter velocity, and  $\mathcal{K}_q = \ker D\rho_q$ . On the full-rank locus  $\mathcal{K}_q = \{(-BX, XA) : X \in \mathbb{R}^{r \times r} : \text{if } \dot{B}A = -B\dot{A}, \text{ then } \dot{B} = -BX \text{ with } X = \dot{A}A^\top(AA^\top)^{-1}, \text{ and } BXA = B\dot{A} \text{ gives } \dot{A} = XA\}$ . By [§C.27\(b\)](#), LoRA-Pro's directions  $(g^B, g^A)$  form the affine space  $\mathcal{M}_q(G) = \{u : D\rho_q u = \Pi_{\mathcal{T}} G\}$ , and LoRA-Pro chooses  $X$  by minimising the Frobenius distance from  $(g^B, g^A)$  to LoRA's own gradients  $D\rho_q^\top G = (sGA^\top, sB^\top G)$ ; the minimiser solves a Sylvester equation ([Background D.33](#), [Auxiliary result B.20](#)). Since  $D\rho_q^\top G \perp \mathcal{K}_q$ , for  $u \in \mathcal{M}_q(G)$  the inner product  $\langle u, D\rho_q^\top G \rangle$  is constant, and minimising  $\|u - D\rho_q^\top G\|^2$  over  $\mathcal{M}_q(G)$  is the same as minimising  $\|u\|^2$ . So LoRA-Pro's direction is the minimum-norm solution  $u_q(G)$  of  $D\rho_q u = \Pi_{\mathcal{T}} G$ , that is,  $u_q(G) = D\rho_q^+ G$ .

We show that  $u_{g \cdot q}(G) = \varphi_g(u_q(G))$  for all  $G$  iff  $g \in \text{O}(r)$ , where  $\varphi_g(u) = (u_B g^{-1}, g u_A)$  is the linear action. Since  $D\rho_{g \cdot q} \varphi_g = D\rho_q$  and  $\Pi_{\mathcal{T}}$  depends only on  $W$ , substituting  $u = \varphi_g(w)$  shows that  $u_{g \cdot q}(G) = \varphi_g(w^*)$ , where  $w^*$  minimises  $\langle w, \mathcal{N}w \rangle$  over  $\mathcal{M}_q(G)$  with  $\mathcal{N} = \varphi_g^\top \varphi_g$ . The adjoint of  $\varphi_g$  is  $\varphi_g^\top(v_B, v_A) = (v_B g^{-\top}, g^\top v_A)$ , since  $\langle w_B g^{-1}, v_B \rangle = \langle w_B, v_B g^{-\top} \rangle$  and  $\langle g w_A, v_A \rangle = \langle w_A, g^\top v_A \rangle$ , so  $\mathcal{N}(w_B, w_A) = (w_B g^{-1} g^{-\top}, g^\top g w_A) = (w_B (g^\top g)^{-1}, (g^\top g) w_A)$ . So equivariance holds iff, on every coset of  $\mathcal{K}_q$  (as  $G$  varies,  $\mathcal{M}_q(G)$  runs through all cosets, since  $D\rho_q$  maps onto  $\mathcal{T}$ ), the  $\mathcal{N}$ -minimum-norm point equals the Euclidean one. The two minimisers are the intersections of the coset with  $\mathcal{K}_q^{\perp \mathcal{N}} = \mathcal{N}^{-1}(\mathcal{K}_q^\perp)$  and with  $\mathcal{K}_q^\perp$ ; they agree on every coset

iff these two complements are equal (each point of either complement is the minimiser on its own coset), iff  $\mathcal{N}\mathcal{K}_q = \mathcal{K}_q$  ( $\mathcal{N}$  is symmetric and invertible). Put  $H = g^\top g \succ 0$ . Then  $\mathcal{N}(-BX, XA) = (-BXH^{-1}, HXA)$  lies in  $\mathcal{K}_q$  iff some  $Y$  has  $BY = BXH^{-1}$  and  $YA = HXA$ , that is (as  $B$  is injective and  $A$  has full row rank),  $XH^{-1} = HX$ . Requiring this for every  $X \in \mathbb{R}^{r \times r}$ , the matrix-unit lemma of §C.26 gives  $H^{-1} = cI = H$ , so  $c^2 = 1$ , and  $c = 1$  because  $H$  is positive definite. Hence  $g^\top g = I$  and  $g \in O(r)$ . Conversely, orthogonal  $g$  give  $\mathcal{N} = I$ . So the parameter update with the Sylvester-optimal  $X$  is equivariant exactly under  $O(r)$ .

The variant with  $X = 0$  is  $g^A = s^{-1}(B^\top B)^{-1}B^\top G$ ,  $g^B = s^{-1}(I - P_U)GA^\top(AA^\top)^{-1}$ . At  $g \cdot q$ ,  $P_U$  is unchanged and

$$g^{A'} = s^{-1}g(B^\top B)^{-1}g^\top g^{-\top}B^\top G = g g^A, \quad g^{B'} = s^{-1}(I - P_U)GA^\top g^\top g^{-\top}(AA^\top)^{-1}g^{-1} = g^B g^{-1},$$

so it is  $GL_r$ -equivariant and its discrete  $W$ -step descends exactly. The first-order  $W$ -step of either variant is  $-\eta\Pi_{\mathcal{T}}G$  (Proposition III.4(b)). For the Sylvester variant,  $\varphi_g^{-1}u_{g \cdot q}(G) - u_q(G) \in \mathcal{K}_q$  equals  $(-BY, YA)$  for some  $Y$ , and the second-order term  $s \delta B \delta A = \eta^2 s g^B g^A$  becomes  $\eta^2 s (g^B - BY)(g^A + YA)$  at  $g \cdot q$ . The difference  $\eta^2 s (g^B YA - BY g^A - BY^2 A)$  is exactly  $\eta^2$  times a matrix that does not depend on  $\eta$ , and it is nonzero in general. Numerically, at random full-rank points with  $(m, n, r) = (7, 6, 3)$ ,  $s = 0.8$  and  $g$  a Gaussian matrix plus  $2I$ , the largest entry of the difference of the two  $W$ -steps at  $\eta = 0.1$  ranges from  $8 \times 10^{-4}$  to  $1.2 \times 10^{-2}$  over seven random draws, and it falls by a factor of exactly 100 each time  $\eta$  falls by 10. It is of order  $10^{-15}$  (rounding) for an orthogonal  $g$  and for the variant with  $X = 0$ .

*LoRA-RITE*. In the notation of Yen et al. (2024) the update is  $AB^\top$  with  $A \in \mathbb{R}^{m \times r}$  and  $B \in \mathbb{R}^{n \times r}$ , and the gauge is  $A_2 = A_1 R$ ,  $B_2 = B_1 R^{-\top}$  with  $R \in GL_r$ . We follow their Algorithm 1 line by line on the full-rank locus, where its pseudo-inverses are inverses. The rule is symmetric in  $A$  and  $B$ , so we treat the step of  $A$ . Algorithm 1 writes  $B = U_B R_B$  by polar decomposition, with  $U_B$  having orthonormal columns, and forms, at step  $t$ :

- the unmagnified gradient  $\tilde{\nabla}_{A,t} = \nabla_{A,t} R_{B_t}^{-1}$  and the basis change  $P_{A,t} = U_{B_t}^\top U_{B_{t-1}}$ ;
- a second moment  $\tilde{V}_{A,t}$ , a combination with fixed weights of  $P_{A,t} \tilde{V}_{A,t-1} P_{A,t}^\top$  and  $\tilde{\nabla}_{A,t}^\top \tilde{\nabla}_{A,t}$ ;
- the escaped mass

$$\rho_{A,t} = \rho_{A,t-1} + d_\lambda(\tilde{V}_{A,t-1}, P_{A,t} \tilde{V}_{A,t-1} P_{A,t}^\top), \quad d_\lambda(E_1, E_2) = \max_i |\lambda_i(E_1) - \lambda_i(E_2)|,$$

which depends only on the spectra of its two arguments;

- the preconditioned step, the first moment and the step

$$\tilde{S}_{A,t} = \tilde{\nabla}_{A,t}(\tilde{V}_{A,t} + \rho_{A,t} I)^{-1/2}, \quad \tilde{M}_{A,t} = \beta_1 \tilde{M}_{A,t-1} P_{A,t}^\top + (1 - \beta_1) \tilde{S}_{A,t}, \quad \delta A_t = -\eta \tilde{M}_{A,t} R_{B_t}^{-\top}.$$

Run the algorithm from  $(A_1, B_1)$  and from  $(A_2, B_2) = (A_1 R, B_1 R^{-\top})$ , with zero initial moments and escaped mass. Suppose that at step  $t$  the parameters of the two runs are related by  $R$ . Then  $B_{2,t}$  and  $B_{1,t}$  have the same column space, so  $U_{B_{2,t}} = U_{B_{1,t}} O_t$  for some  $O_t \in O(r)$ , and  $R_{B_{2,t}} = U_{B_{2,t}}^\top B_{2,t} = O_t^\top R_{B_{1,t}} R^{-\top}$ . With  $G$  the gradient in the weight,  $\nabla_{A_2} = G B_2 = \nabla_{A_1} R^{-\top}$ , so

$$\tilde{\nabla}_{A_{2,t}} = \nabla_{A_{1,t}} R^{-\top} R^\top R_{B_{1,t}}^{-1} O_t = \tilde{\nabla}_{A_{1,t}} O_t.$$

Suppose also that the internal states are related by  $\tilde{V}_{t-1}^{(2)} = O_{t-1}^\top \tilde{V}_{t-1}^{(1)} O_{t-1}$ ,  $\tilde{M}_{t-1}^{(2)} = \tilde{M}_{t-1}^{(1)} O_{t-1}$  and  $\rho_{t-1}^{(2)} = \rho_{t-1}^{(1)}$ ; at  $t = 1$  this holds because all three start at zero. Then  $P_{A,t}^{(2)} = O_t^\top P_{A,t}^{(1)} O_{t-1}$ , so  $P_{A,t}^{(2)} \tilde{V}_{t-1}^{(2)} P_{A,t}^{(2)\top} = O_t^\top P_{A,t}^{(1)} \tilde{V}_{t-1}^{(1)} P_{A,t}^{(1)\top} O_t$  and  $\tilde{V}_t^{(2)} = O_t^\top \tilde{V}_t^{(1)} O_t$ . Conjugation by an orthogonal



matrix keeps eigenvalues, so  $\rho_t^{(2)} = \rho_t^{(1)}$ , and it commutes with  $(\cdot + \rho I)^{-1/2}$ , so  $\tilde{S}_t^{(2)} = \tilde{S}_t^{(1)} O_t$ . Since  $\tilde{M}_{t-1}^{(2)} P_{A,t}^{(2)\top} = \tilde{M}_{t-1}^{(1)} P_{A,t}^{(1)\top} O_t$ , also  $\tilde{M}_t^{(2)} = \tilde{M}_t^{(1)} O_t$ . Finally  $R_{B_{2,t}}^{-\top} = O_t^\top R_{B_{1,t}}^{-\top} R$ , so

$$\delta A_{2,t} = -\eta \tilde{M}_t^{(1)} O_t O_t^\top R_{B_{1,t}}^{-\top} R = \delta A_{1,t} R,$$

the transported step. The step of  $B$  is the same computation with  $A$  and  $B$  exchanged and  $R$  replaced by  $R^{-\top}$ , so  $\delta B_{2,t} = \delta B_{1,t} R^{-\top}$ . The new parameters are again related by  $R$ , and the new internal states by the orthogonal matrices  $O_t$ , which completes the induction. The resulting step agrees with Eq. 15 of [Yen et al. \(2024\)](#),  $\delta A_2 = \delta A_1 B_1^\top (B_2^\top)^+$ , since  $(B_2^\top)^+ = B_1 (B_1^\top B_1)^{-1} R$  when  $B_1$  has full column rank, and with the numerical replication reported after [Theorem III.12](#), which agrees to  $3 \times 10^{-14}$ . LoRA-RITE is therefore an Adam-type optimizer in class (c).  $\square$

### C.36 Corollary III.13

**COROLLARY III.13** (what a zero- $B$  initialisation really chooses). *At  $q_0 = (0, A_0)$  with rank  $A_0 = r$ , with hyperparameters shared by the rank channels and the same data order and dropout masks, two choices of  $A_0$  give the same  $\theta$ -trajectory whenever the optimizer's gauge group relates them. The resulting invariant of  $A_0$  is:*

- row  $A_0 \in \text{Gr}(r, n)$  for an update rule that is  $\text{GL}_r$ -equivariant at every point of its trajectory, including  $B = 0$  and rank-deficient  $B$ , run without global-norm clipping. An example is  $\delta B = -\eta \nabla_B (A A^\top)^{-1}$ ,  $\delta A = -\eta (A A^\top) \nabla_A$ , which needs only rank  $A = r$ ;
- the PSD matrix  $A_0^\top A_0$  for SGD, which is exactly the initial preconditioner in  $\dot{W}(0) = -s^2 \eta_B G A_0^\top A_0$ ;
- $A_0$  up to signed permutation of its rows for Adam.

The undamped scaled-GD optimizers of [Theorem III.12\(c\)](#) are not defined at  $B = 0$ , since they need  $(B^\top B)^{-1}$ . LoRA-RITE uses pseudo-inverses and so is defined there, but its  $\text{GL}_r$ -equivariance is established only on the full-rank locus. Their damped forms are only  $\text{O}(r)$ -equivariant, so in practice their invariant is a  $\delta$ -dependent function of  $A_0^\top A_0$ , and their Adam-wrapped forms (Riemannian-preconditioned AdamW, HF's LoRA-FA optimizer) have the invariant  $A_0$  modulo  $B_r$ . Conversely, different invariants give different dynamics, for SGD and for Adam at  $\varepsilon = 0$ .

Random- $A$  LoRA and EVA are different points of the  $B = 0$  stratum  $\{(0, A) : \text{rank } A = r\}$  of the apex fibre  $\rho^{-1}(\theta_0) = \{(B, A) : BA = 0\}$ ;  $\text{Init}[B]$  lies in another stratum of the same fibre. LoRA-FA is a different object, the sub-object  $\text{FA}(A_0) \hookrightarrow \text{LoRA}_{(0, A_0)}$  ([Proposition I.8](#)). More precisely, EVA is a point of the apex fibre of  $\text{LoRA}_{r_l}$  at its per-layer redistributed rank and adjusted  $\alpha$ , with orthonormal rows only when `whiten=False`.

*Proof. Equivariance at  $B = 0$ .* Only the inclusions  $\supseteq$  of [Theorem III.12](#) are used. For SGD and  $g \in \text{O}(r)$  the identities  $\delta B' = -\eta_B \nabla_B g^\top = \delta B g^{-1}$  and  $\delta A' = g \delta A$  hold at every point, with no rank condition, and similarly for momentum buffers. For Adam and  $g \in B_r$  the coordinatewise argument also holds at every point, and the initial state  $(m, v) = (0, 0)$  is fixed by  $B_r$ . Orthogonal  $g$  preserve  $\|\nabla_B\|$  and  $\|\nabla_A\|$  by (20), so global-norm clipping factors agree in the two runs. The start transforms as  $g \cdot (0, A_0) = (0, g A_0)$ , since  $0 \cdot g^{-1} = 0$ . By the induction of §C.24,  $A_0$  and  $g A_0$  give the same  $\theta$ -trajectory whenever  $g$  lies in the optimizer's group. In the first bullet the hypothesis is equivariance under  $\text{GL}_r$  at every point of the trajectory, and the same induction applies.

*The example rule.* At  $g \cdot q$ , by (20) and  $(A'A'^\top)^{-1} = g^{-\top}(AA^\top)^{-1}g^{-1}$ ,

$$\delta B' = -\eta \nabla_B g^\top g^{-\top} (AA^\top)^{-1} g^{-1} = \delta B g^{-1}, \quad \delta A' = -\eta g A A^\top g^\top g^{-\top} \nabla_A = g \delta A.$$

Only  $\text{rank } A = r$  was used, so  $B$  may be 0 or rank deficient. Since  $\delta A = -\eta A A^\top s B^\top G = A N$  with  $N = -\eta s A^\top B^\top G$ , the step maps  $A$  to  $A(I + N)$ , which keeps  $\text{rank } r$  whenever  $I + N$  is invertible, in particular for small  $\eta$ ; in continuous time  $\dot{A} = AN(t)$  gives  $A(t) = A_0 \Psi(t)$  with  $\Psi$  an invertible fundamental matrix, so  $\text{rank } A(t) = r$  for all  $t$ .

*The invariants.* The invariant of  $A_0$  is its orbit under the optimizer's group acting by  $A_0 \mapsto g A_0$ . For  $O(r)$ ,  $A'_0 = O A_0$  for some  $O \in O(r)$  iff  $A_0^\top A'_0 = A_0^\top A_0$  (Theorem II.5(2)). For  $B_r$  the orbit is the set of signed row permutations of  $A_0$ . For  $\text{GL}_r$ , two full-row-rank matrices satisfy  $A'_0 = g A_0$  for some (unique)  $g$  iff they have the same row space (proof of Theorem II.2), so the invariant is  $\text{row } A_0 \in \text{Gr}(r, n)$  (Background C.15). At  $B_0 = 0$  the LoRA cometric of Observation III.2 is  $K_{(0, A_0)}(G) = s^2 \eta_B G A_0^\top A_0$ , so  $\dot{W}(0) = -s^2 \eta_B G A_0^\top A_0$ .

*Practical forms.* Undamped scaled GD needs  $(B^\top B)^{-1}$ , which does not exist at  $B = 0$ . The damped forms are  $O(r)$ -equivariant at every point (§C.35, published variants), so their trajectories from  $(0, A_0)$  depend on  $A_0$  only through  $A_0^\top A_0$ , in a way that depends on  $\delta$ . A preconditioned step wrapped in Adam(W) is  $B_r$ -equivariant at every point (same place), so the invariant is  $A_0$  modulo  $B_r$ .

*Converse for SGD.* The first SGD step from  $(0, A_0)$  is  $\delta B = -\eta_B s G A_0^\top$ ,  $\delta A = -\eta_A s B_0^\top G = 0$ , so  $\delta W_1 = s \delta B A_0 = -s^2 \eta_B G A_0^\top A_0$  exactly. If  $A_0$  and  $A'_0$  give the same trajectory for every loss, the linear losses  $\langle G, \cdot \rangle$  give  $G A_0^\top A_0 = G A_0'^\top A'_0$  for every  $G$ , so  $A_0^\top A_0 = A_0'^\top A'_0$ . Hence different invariants give different first steps for some loss.

*Converse for Adam at  $\varepsilon = 0$ .* The first step from  $(0, A_0)$  with zero state has  $\nabla_A = 0$ , so  $\delta A = 0$ , and  $\delta B = -\eta_B \text{sign}(s G A_0^\top)$ , so  $\delta W_1 = -s \eta_B \text{sign}(G A_0^\top) A_0$  (without bias correction, a positive multiple of this). If  $A_0$  and  $A'_0$  give the same trajectory for every loss, then row by row

$$F(x) := \text{sign}(x A_0^\top) A_0 = \sum_{j=1}^r \text{sign}(x \cdot a_j) a_j$$

equals  $F'(x) = \sum_j \text{sign}(x \cdot a'_j) a'_j$  for every  $x \in \mathbb{R}^{1 \times n}$ , where  $a_j$  and  $a'_j$  are the rows of  $A_0$  and  $A'_0$ . The rows  $a_j$  are linearly independent, so the hyperplanes  $H_j = a_j^\perp$  are pairwise distinct, and each  $H_j$  contains points  $x$  lying on no other  $H_k$  (a hyperplane is not covered by finitely many proper subspaces of it, Auxiliary result B.36(b)). At such a point, crossing  $H_j$  in the direction of increasing  $x \cdot a_j$  changes  $F$  by  $2a_j \neq 0$ , while  $F$  is constant on the connected components of the complement of  $\bigcup_j H_j$ . So the closure of the set of discontinuities of  $F$  is  $\bigcup_j H_j$ , and likewise for  $F'$ . From  $F = F'$  we get  $\bigcup_j H_j = \bigcup_k H'_k$ , and since a hyperplane contained in a finite union of hyperplanes equals one of them (Auxiliary result B.36(b) again), there is a permutation  $\pi$  with  $H_j = H'_{\pi(j)}$ , that is,  $a'_{\pi(j)} = c_j a_j$  with  $c_j \neq 0$ . Crossing  $H_j$  as above changes  $F'$  by  $2 \text{sign}(c_j) a'_{\pi(j)} = 2|c_j| a_j$ , and comparing with  $2a_j$  gives  $|c_j| = 1$ . So  $A'_0$  is a signed row permutation of  $A_0$ . For the example  $\text{GL}_r$ -rule the same argument is immediate: its first step is  $\delta W_1 = -\eta s^2 G P_{\text{row } A_0}$ , which determines  $\text{row } A_0$ .

*Strata and the sub-object.* Since  $s > 0$ ,  $\rho^{-1}(\theta_0) = \{(B, A) : W_0 + sBA = W_0\} = \{BA = 0\}$ . Random-A LoRA and EVA start at  $(0, A_0)$  with  $\text{rank } A_0 = r$ , in the stratum  $\{(0, A) : \text{rank } A = r\}$ , at points that differ for a generic random draw; Init[B] starts at  $(B_0, 0)$  with  $\text{rank } B_0 = r$ , in the disjoint stratum  $\{(B, 0) : \text{rank } B = r\}$ . The map  $B \mapsto (B, A_0)$  is injective, pointed ( $0 \mapsto (0, A_0)$ ) and commutes with the maps to  $\Theta$ , since  $\rho(B, A_0) = W_0 + sBA_0$ ; after the rescaling  $B \mapsto sB$  its source is  $\text{FA}(A_0)$  of Proposition I.8. An injective morphism is a monomorphism, so  $\text{FA}(A_0)$

is a sub-object of  $\text{LoRA}_{(0,A_0)}$  (Background A.15). Its image  $W_0 + \{XA_0\}$  has dimension  $mr$ , smaller than the dimension  $r(m+n-r)$  of the image  $W_0 + \mathcal{M}_{\leq r}$  of  $\text{LoRA}_{(0,A_0)}$  when  $r < n$ , and images are isomorphism invariants, so the two objects are not isomorphic. The last sentence describes EVA's construction and needs no proof.  $\square$

### C.37 Definition IV.1

**DEFINITION IV.1** (forward and backward methods).

- A **forward** method acts on the parameter port by a lens in the image of  $\mathbb{R}$ , namely  $(\rho, \mathbb{R}[\rho])$ .
- A **backward**, or cotangent-side, method acts by a lens whose forward part is  $\text{id}_\Theta$  and whose backward part is a compression  $c_t : T_\Theta^* \Theta \rightarrow S$  into a state space  $S$ , paired with an anchor  $a_t : S \rightarrow T_\Theta \Theta$ . The update is

$$\theta \leftarrow \theta - \eta a_t(\text{opt}(c_t(\nabla L))).$$

*Examples:* GaLore, with  $c_t = P_t^\top$  and  $a_t = P_t$  for  $P_t$  from a gradient SVD; GaLore with a random projector, the same with  $P_t$  random; LISA, a layer-block coordinate projection paired with the same projection; MeZO, with  $c_t(g) = \langle z_t, g \rangle$  estimated by finite differences without a backward pass, and  $a_t(\lambda) = \lambda z_t$ . Flora as published fits only as the trivial lens, with compression and anchor the identity and the random compression of its momentum inside opt.

A definition needs no proof. This subsection fixes the conventions used in the proofs of Exposé IV and checks the identities that the definition and its table record.

**Conventions.** As in §C.24, weight space  $\Theta = \mathbb{R}^{m \times n}$ , or a product of such boxes, carries the Frobenius inner product, covectors are written as matrices, and the transpose  $\mathcal{L}^*$  of a linear map  $\mathcal{L}$  is its adjoint. We model a first-order optimizer as a machine with a state. At step  $t$  it holds a state  $\varsigma_t$ , receives a tensor  $g_t$  (the gradient it is fed) and the tensor  $x_t$  it updates, returns an output  $U_t = F_t(\varsigma_t, g_t, x_t)$ , moves to the state  $\varsigma_{t+1} = H_t(\varsigma_t, g_t, x_t)$ , and the update is  $x_{t+1} = x_t - \eta_t U_t$ . The maps may depend on the step counter  $t$ , for instance through Adam's bias corrections or Adafactor's relative step size. The optimizer *does not read the parameter tensor* when  $F_t$  and  $H_t$  do not depend on  $x_t$ . SGD, heavy-ball and Nesterov momentum, Adam and AMSGrad, and Adafactor with `scale_parameter=False` are of this kind; Adafactor's factored second moments and its clipping of the update by  $\text{RMS}(U_t)$  read only  $g_t$  and the state. The following ingredients read  $x_t$ , or a gradient other than  $g_t$ :

- coupled weight decay, which feeds  $g_t + \lambda x_t$  in place of  $g_t$ ;
- decoupled weight decay (AdamW with  $\lambda > 0$ ), which adds  $-\eta_t \lambda x_t$  to the update;
- LAMB and LARS, which multiply the update by a trust ratio  $\phi(\|x_t\|)/\|U_t\|$ ;
- Adafactor with parameter scaling, which multiplies the step by  $\max(\epsilon_2, \text{RMS}(x_t))$  (the option `scale_parameter=True`);
- clipping by the global norm of the full gradient, which rescales  $g_t$  by  $\min(1, \tau/\|G_t\|)$ , where  $G_t$  is the gradient of every trainable tensor of the procedure.

*Verification of the identities in Definition IV.1. The lens of a reparametrisation.* A smooth  $\rho : Q \rightarrow \Theta$  has reverse derivative  $\mathbb{R}[\rho](q, \gamma) = J_\rho(q)^\top \gamma = D\rho_q^\top \gamma$  (Background B.17). For a composite  $\rho \circ \sigma$  the chain rule gives

$$\mathbb{R}[\rho \circ \sigma](q', \gamma) = D\sigma_{q'}^\top D\rho_{\sigma(q')}^\top \gamma = \mathbb{R}[\sigma](q', \mathbb{R}[\rho](\sigma(q'), \gamma)),$$

which is composition of lenses, so  $\rho \mapsto (\rho, R[\rho])$  respects composition of reparametrisations. The gradient that a forward method feeds its optimizer is the pullback  $\nabla_q(L \circ \rho) = D\rho_q^\top \nabla_\theta L = R[\rho](q, \nabla_\theta L)$ .

*A forward step to first order.* One step of a forward method sets  $q \leftarrow q - \eta U$  with  $U = \text{opt}(D\rho_q^\top \nabla L)$ , so the weight moves to

$$\rho(q - \eta U) = \theta - \eta D\rho_q U + O(\eta^2).$$

To first order in  $\eta$  this is the backward update with  $c = D\rho_q^\top$  and  $a = D\rho_q$ . The two agree exactly when  $\rho$  is affine along the step, which is the situation of [Proposition IV.2](#).

*The examples.* For GaLore with left projection,  $P_t \in \mathbb{R}^{m \times r}$  has orthonormal columns,  $S = \mathbb{R}^{r \times n}$ ,  $c_t(G) = P_t^\top G$  and  $a_t(U) = P_t U$ , so the update  $W \leftarrow W - \eta \alpha P_t \text{opt}(P_t^\top G)$  is the displayed one with  $\alpha$  absorbed into  $\eta$ ; with right projection,  $c_t(G) = G P_t$  and  $a_t(U) = U P_t^\top$ . In each case  $a_t = c_t^*$ , since  $\langle P_t U, G \rangle = \langle U, P_t^\top G \rangle$ . LISA's  $S$  is the space of the sampled layers (with the embedding and the head),  $c_t$  restricts a gradient to it and  $a_t$  includes it back, so again  $a_t = c_t^*$ . For MeZO,  $S = \mathbb{R}$ ,  $a_t(\lambda) = \lambda z_t$  and  $c_t(g) = \langle z_t, g \rangle = a_t^*(g)$ . MeZO does not compute  $c_t(\nabla L)$ ; it estimates it by the central difference  $(L(\theta + \varepsilon z_t) - L(\theta - \varepsilon z_t)) / (2\varepsilon)$ , whose error is  $O(\varepsilon^2)$  for a loss with Lipschitz Hessian ([Auxiliary result B.34\(a\)](#)), so it is the displayed update up to its estimator. Flora as published (its reference implementation) returns  $(1 - \beta)\hat{g}_t + \beta \text{up}_t(M_t)$ , where  $\hat{g}_t$  is the Adafactor-normalised full gradient and  $\text{up}_t(M_t)$  the decompressed momentum ([Hao et al., 2024](#)). The first term is generically of full rank, so the update does not lie in  $\text{im } P_t$ , and the only reading in the form of the definition is  $c_t = a_t = \text{id}$  with the random compression inside  $\text{opt}$ .

*The cometric of a backward method.* With plain (S)GD on the compressed gradient,  $\text{opt} = \text{id}$ , the update is  $\theta \leftarrow \theta - \eta a_t c_t \nabla L$ , so the induced cometric is  $K_t = a_t c_t$  ([Definition III.1](#)). For the first three examples  $a_t = c_t^*$ , so  $K_t = a_t a_t^*$  is symmetric positive semidefinite; for GaLore with left projection,  $K_t(G) = P_t P_t^\top G$ .  $\square$

### C.38 Proposition IV.2

**PROPOSITION IV.2** (the degree-one collapse; known, [Torroba-Hennigen et al. \(2025\)](#) for GaLore (their Thm 1 and Cor. 3, App. B for weight decay) and [Hao et al. \(2024\)](#) for random projections). *Let  $\rho_k(C) = W_k + \mathcal{L}C$  with  $\mathcal{L}$  linear and  $C_0 = 0$ . Let  $\text{opt}$  be a first-order optimizer, with no coupled or decoupled weight decay, whose state and output depend only on the sequence of gradients it is fed, and which does not read the parameter tensor: SGD, heavy-ball or Nesterov momentum, Adam or AMSGrad, or Adafactor with `scale_parameter=False` (fixed or relative step, with its factorisation applied to the compressed gradient in the same tensor shape as  $C$ ). Run both procedures below with the same implementation of  $\text{opt}$ , on the same minibatches, and from the same optimizer state, fresh or carried over. Then they produce the same  $W$ -trajectory over a period:*

- training  $C$  with  $\text{opt}$  and merging at the end of the period;
- the backward method with  $c = \mathcal{L}^*$  and  $a = \mathcal{L}$ , started at  $W_k$ .

*AdamW with  $\lambda > 0$ , LAMB and LARS trust ratios, parameter-scaled Adafactor (`scale_parameter=True`, the default in HF and in `galore_torch`'s `GaLoreAdafactor`) and global-norm clipping of the full gradient (such as HF Trainer's default `max_grad_norm=1.0`, whenever it triggers) all break the identity. Injectivity of  $\mathcal{L}$  is not needed.*

In particular, on each matrix that GaLore projects (its other parameters train in full in both procedures), GaLore with a projector  $P_k$  fixed on each period equals one-sided LoRA  $W_k + P_k C$  (frozen  $B = P_k$ , or frozen  $A$  when projecting on the right), merged and re-based every period, with  $C$ 's optimizer state (moments and step counter) carried over unchanged across re-basings, with no reset.  $P_k$  must be exactly GaLore's matrix, signs and column order included; its span alone does not suffice, because the state is reused without transport. ReLoRA-style resets give a different trajectory. The identity is exact with weight decay 0, no full-gradient clipping, and an optimizer from the admissible class; `galore_torch`'s `GaLoreAdafactor` defaults to `scale_parameter=True` and is excluded. `GaLoreAdamW` is the transformers-style AdamW, which adds  $\varepsilon$  to  $\sqrt{v}$  before the bias correction, so the LoRA side must use the same implementation. GaLore's scale  $\alpha$  multiplies the optimizer output, so it acts as a learning-rate factor; a LoRA-style scale inside  $\rho$  gives a different trajectory under Adam with  $\varepsilon > 0$ .

*Proof.* We use the optimizer model of §C.37. Fix a period of  $T$  steps, write  $L_t$  for the loss of the minibatch used at step  $t$  (the same in both procedures) and  $\eta_t$  for the step size.

*The two procedures.* In the first, the trainable tensor is  $C$  with  $C_0 = 0$  and the weight is  $W_t^{(1)} := \rho_k(C_t) = W_k + \mathcal{L}C_t$ . By the chain rule the gradient fed to opt is

$$g_t^{(1)} = \nabla_C(L_t \circ \rho_k)(C_t) = \mathcal{L}^* \nabla L_t(W_t^{(1)}),$$

opt returns  $U_t^{(1)}$ , and  $C_{t+1} = C_t - \eta_t U_t^{(1)}$ . In the second,  $W_0^{(2)} = W_k$ , the gradient fed to opt is  $g_t^{(2)} = c(\nabla L_t(W_t^{(2)})) = \mathcal{L}^* \nabla L_t(W_t^{(2)})$ , opt returns  $U_t^{(2)}$ , and  $W_{t+1}^{(2)} = W_t^{(2)} - \eta_t \mathcal{L}U_t^{(2)}$ .

*Induction.* We show that for  $0 \leq t \leq T$  the weights  $W_t^{(1)}$  and  $W_t^{(2)}$  agree and the two optimizer states agree. At  $t = 0$ ,  $W_0^{(1)} = W_k + \mathcal{L}0 = W_k = W_0^{(2)}$ , and the states agree by hypothesis. Suppose both claims hold at  $t$ . The two procedures evaluate the same minibatch loss at the same weight, so they compute the same weight gradient and feed opt the same tensor  $g_t := g_t^{(1)} = g_t^{(2)}$ . Since opt is the same implementation and does not read the parameter tensor, its output and its next state are the same function of the state, of  $g_t$  and of  $t$  in both procedures. Hence  $U_t^{(1)} = U_t^{(2)} =: U_t$  and the next states agree. Then

$$W_{t+1}^{(1)} = W_k + \mathcal{L}(C_t - \eta_t U_t) = W_t^{(1)} - \eta_t \mathcal{L}U_t = W_{t+1}^{(2)}.$$

At the end of the period merging sets the weight to  $\rho_k(C_T) = W_T^{(1)} = W_T^{(2)}$ . The argument never inverts  $\mathcal{L}$ , so injectivity is not needed. Every operation that opt applies to  $g_t$  and its state, such as Adafactor's factorisation in the shape of  $C$  (which is the shape of  $g_t$  in both procedures) and its update clipping, is covered, since it reads nothing else.

*The excluded ingredients.* Each of them reads a tensor that differs between the procedures, and in each case the first step already separates the trajectories for suitable data; it suffices to exhibit this. Take  $\mathcal{L}$  injective,  $W_k \neq 0$  and a minibatch gradient  $G_0$  with  $\mathcal{L}^*G_0 \neq 0$ .

- *Decoupled decay.* The first procedure sets  $C_1 = C_0 - \eta_0 U_0 - \eta_0 \lambda C_0 = -\eta_0 U_0$ , so  $W_1^{(1)} = W_k - \eta_0 \mathcal{L}U_0$ ; the second sets  $W_1^{(2)} = W_k - \eta_0 \mathcal{L}U_0 - \eta_0 \lambda W_k$ . They differ by  $\eta_0 \lambda W_k \neq 0$ . In general decay pulls  $W$  towards  $W_k$  in the first procedure and towards 0 in the second.
- *Coupled decay.* The first procedure feeds  $\mathcal{L}^*G_0 + \lambda C_0 = \mathcal{L}^*G_0$ , the second  $\mathcal{L}^*(G_0 + \lambda W_k)$ . These differ whenever  $\mathcal{L}^*W_k \neq 0$ , and for SGD the first steps then differ, since  $\mathcal{L}$  is injective.
- *Trust ratios and parameter scaling.* The first procedure reads  $\|C_0\| = \text{RMS}(C_0) = 0$ , the second  $\|W_k\| > 0$  and  $\text{RMS}(W_k) > 0$ . The resulting step sizes differ for generic  $W_k$ , because the first does not depend on  $W_k$  and the second does.



- *Global-norm clipping of the full gradient.* The first procedure clips by the norm of its trainable gradient  $\mathcal{L}^*G_0$ , the second (GaLore in HF Trainer) by the norm of the full weight gradient  $G_0$ . Under SGD with  $\|\mathcal{L}^*G_0\| \leq \tau < \|G_0\|$  the first step is  $-\eta_0 \mathcal{L} \mathcal{L}^*G_0$  in the first procedure and  $-\eta_0(\tau/\|G_0\|) \mathcal{L} \mathcal{L}^*G_0$  in the second; they differ because  $\mathcal{L} \mathcal{L}^*G_0 \neq 0$ .

*GaLore.* On a projected matrix with left projection, GaLore’s update in period  $k$  is  $W \leftarrow W - \eta \alpha P_k \text{opt}(P_k^\top G)$ . This is the backward method with  $c = P_k^\top$  and  $a = P_k$ , and for  $\mathcal{L}C = P_k C$  we have  $\mathcal{L}^* = P_k^\top$ ; the factor  $\alpha$  multiplies the output of  $\text{opt}$ , so it is part of the step size. With right projection,  $\mathcal{L}C = CP_k^\top$  and  $\mathcal{L}^*G = GP_k$ . Now induct over periods. Suppose that at the start of period  $k$  the whole model, every projected and unprojected tensor, is the same in GaLore and in the re-based one-sided LoRA, and that so are all optimizer states. The unprojected parameters train in full with the same optimizer in both, so they follow the same trajectory once the projected matrices do. GaLore computes  $P_k$  from the gradient at the start of the period, and the one-sided LoRA uses that same matrix as its frozen factor. The first part of the proposition, applied with  $\mathcal{L}C = P_k C$ , gives the same trajectory through the period, and merging  $P_k C_T$  into  $W_k$  gives the same  $W_{k+1}$ . GaLore keeps the state of  $\text{opt}$ , which lives on tensors of the shape of  $C$ , and the one-sided LoRA carries  $C$ ’s state over while resetting  $C$  to 0; so the states at the start of period  $k + 1$  agree too. This proves the corollary.

*The exact matrix.* Replacing  $P_k$  by  $P_k O$  with  $O \in \text{O}(r)$  keeps the span and changes the trajectory when state is carried. Take heavy-ball momentum with coefficient  $\beta > 0$  and SGD steps,  $U_t = m_{t+1} = \beta m_t + g_t$ , and a carried momentum  $m$  whose first row is nonzero, and let  $O = \text{diag}(-1, 1, \dots, 1)$ . With  $P_k$  the first step of the period moves  $W$  by  $-\eta P_k(\beta m + P_k^\top G)$ . With  $P_k O$  the compressed gradient is  $OP_k^\top G$ , the momentum is reused without transport, and  $W$  moves by  $-\eta P_k O(\beta m + OP_k^\top G) = -\eta P_k(\beta Om + P_k^\top G)$ . The difference is  $-\eta \beta P_k(Om - m) \neq 0$ , since  $P_k$  is injective and  $Om - m$  is minus twice the first row of  $m$ . The same computation applies to Adam’s first moment.

*Resets.* ReLoRA resets the optimizer state at every re-basing. For heavy-ball momentum the first step of period  $k + 1$  is then  $-\eta P_{k+1}g$  instead of  $-\eta P_{k+1}(\beta m + g)$ , which differs whenever the carried momentum  $m$  is nonzero.

*Implementation and scale.* With bias corrections  $b_1 = 1 - \beta_1^t$  and  $b_2 = 1 - \beta_2^t$ , transformers-style AdamW and `torch.optim.Adam` take the steps

$$\eta \frac{\sqrt{b_2}}{b_1} \frac{m_t}{\sqrt{v_t} + \varepsilon} \quad \text{and} \quad \eta \frac{m_t/b_1}{\sqrt{v_t/b_2} + \varepsilon} = \eta \frac{\sqrt{b_2}}{b_1} \frac{m_t}{\sqrt{v_t} + \varepsilon \sqrt{b_2}}.$$

For  $\varepsilon > 0$  the two differ, because they use different effective  $\varepsilon$ ; hence the hypothesis of one implementation. Finally, put GaLore’s  $\alpha$  inside  $\rho$ ,  $\rho(C) = W_k + \alpha P_k C$ . The compressed gradient becomes  $\alpha P_k^\top G$ , Adam’s moments become  $\alpha m_t$  and  $\alpha^2 v_t$ , and the weight moves by

$$-\eta \alpha P_k \frac{\alpha \hat{m}_t}{\alpha \sqrt{\hat{v}_t} + \varepsilon} = -\eta \alpha P_k \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \varepsilon/\alpha},$$

against  $-\eta \alpha P_k \hat{m}_t/(\sqrt{\hat{v}_t} + \varepsilon)$  for GaLore. These agree for  $\varepsilon = 0$ , or when the LoRA side’s  $\varepsilon$  is multiplied by  $\alpha$ , and differ for  $\varepsilon > 0$  and  $\alpha \neq 1$  whenever  $\hat{m}_t \neq 0$ .  $\square$

**REMARK** (numerical checks). The values reported after [Proposition IV.2](#) ( $7 \times 10^{-16}$  over one period and  $5 \times 10^{-15}$  over four under Adam with the state carried;  $8 \times 10^{-17}$  against the official `galore_torch` optimizers; the separations 0.029, 0.020, 0.030 and 0.12 for a reset,

weight decay 0.1, HF-default Adafactor and a flipped column) instantiate the identity and the four breaking mechanisms above (Appendix E).

### C.39 Proposition IV.3

**PROPOSITION IV.3** (integrability; after Frobenius’ theorem). *Let  $D$  be a smooth distribution of constant rank  $k$  on an open set  $U \subseteq \Theta$ , for instance the anchor distribution  $D_\theta = \text{im } a(\theta)$  of a time-independent backward method. Then  $D$  is involutive on  $U$  if and only if through every  $\theta \in U$  passes an injective immersion  $\rho_\theta : (\mathbb{R}^k, 0) \rightarrow (U, \theta)$  with  $\text{im } D\rho_\theta = D$  along its image. A single such  $\rho$  through  $\theta_0$  gives involutivity only along  $\text{im } \rho$ . For example,  $D = \text{span}\{\partial_x, \partial_y + xz\partial_z\}$  on  $\mathbb{R}^3$  has  $[\partial_x, \partial_y + xz\partial_z] = z\partial_z \notin D$  off  $z = 0$ , yet the plane  $z = 0$  is an integral manifold.*

*The dynamic content needs separate hypotheses.*

- In continuous time, if  $D$  is involutive, a backward flow with  $\dot{\theta} \in D_\theta$  stays on the leaf through  $\theta_0$ .
- In the discrete update of [Definition IV.1](#), the iterates stay on a leaf for all step sizes and all optimizer outputs  $s$  (all losses) if and only if that leaf contains  $(\theta + D_\theta) \cap U$  at every iterate, that is, if and only if the leaves are affine; an anchor constant over the period gives this ([Proposition IV.2](#)).
- Under gradient flow, dynamic equivalence with a forward lens  $(\rho, R[\rho])$  forces matching frames,  $a_t c_t = D\rho \wedge D\rho^\top$  along the trajectory. Hence  $K = ac$  is symmetric positive semidefinite and, for injective  $a$ ,  $c = Ma^\top$  with  $M$  symmetric positive semidefinite on  $S$ , possibly depending on  $\theta$ . If in addition  $\rho$  is an immersion with  $\dim Q = \text{rank } D$ , then  $\rho$  is a local isometry from  $(Q, \Lambda^{-1})$  onto the leaf with metric  $(aMa^\top)^{-1}|_D$ , so that metric must be flat. Forward methods with gauge fibres escape this requirement:  $\rho(B) = BB^\top$  on symmetric matrices induces the curved Bures–Wasserstein metric, and balanced full-rank LoRA also induces a curved leaf metric. For discrete steps, exact agreement for every step size further forces  $\rho$  to be affine along each horizontal line  $q + t \wedge D\rho_q^\top g$ .

**Instances.** Every projector-type backward method (GaLore with any period  $T$  and any projector, its random-projector variant, LISA, MeZO) has an anchor that does not depend on  $\theta$  within its period. Its distribution is constant, hence trivially involutive with affine leaves (vacuously at  $T = 1$ , where a period is a single step). Under the optimizer hypotheses of [Proposition IV.2](#) (no weight decay of either kind, no full-gradient clipping; GaLore and MeZO need weight decay 0, and LISA’s AdamW with positive decay violates them), these methods are rebasing schemes of linear forward methods with period  $T$ , including  $T = 1$ , whose reach [Theorem V.3](#) describes. Each MeZO step is a step of the linear method  $\theta_t + \lambda z_t$ , up to its SPSA estimator, and GaLore with  $T = 1$  is one-sided LoRA re-based every step. What separates these methods from LoRA-type methods is the resampling schedule together with the carry-over of optimizer state; integrability plays no part in it. Flora as published (Adafactor base, with a full-rank current-gradient term in the code), APOLLO (a low-rank-estimated scaling applied to the full gradient) and Fira (a scaled full-rank residual) are not of the form of [Definition IV.1](#) with their projections as anchors, because their updates leave  $\text{im } a$ .

Non-involutivity proper needs a  $\theta$ -dependent anchor. Full-batch GaLore with  $T = 1$ , read as the distribution  $D_W = \text{Hom}(\mathbb{R}^n, U_r(\nabla L(W)))$  (or  $\text{Hom}(V_r(\nabla L(W)), \mathbb{R}^m)$  when  $m \geq n$ , as GaLore projects) wherever a singular-value gap exists, is involutive if and only if  $U_r(\nabla L(W))$  (respectively  $V_r(\nabla L(W))$ ) is invariant along  $D$ . That fails generically for every  $r \geq 1$  once both matrix dimensions are at least 2. For every  $r$  with  $1 \leq r < \min(m, n)$  it fails for  $L = \frac{1}{2}\|W\|_F^2$ ,

and numerically it fails for least squares and cross-entropy with several samples. It holds, for instance, for single-sample least squares, and automatically only when the distribution is a line field ( $nr = 1$ ). Minibatch GaLore with  $T = 1$  and MeZO violate time-independence, because their anchors depend on step-wise randomness.

*Proof.* Throughout, a leaf of an involutive  $D$  is a maximal connected integral manifold, and the leaves form a foliation of  $U$  (Auxiliary result B.31(c)).

(1) *The Frobenius equivalence.* Suppose  $D$  is involutive and let  $\theta \in U$ . By Auxiliary result B.31(b) there is a chart  $\varphi : V \rightarrow (-\delta, \delta)^{\dim \Theta}$  centred at  $\theta$  in which  $D$  is spanned by the first  $k$  coordinate fields. Let  $\psi : \mathbb{R}^k \rightarrow (-\delta, \delta)^k$  be the diffeomorphism  $\psi_i(y) = \frac{2\delta}{\pi} \arctan y_i$  and put  $\rho_\theta(y) = \varphi^{-1}(\psi(y), 0)$ . Then  $\rho_\theta(0) = \theta$ ,  $\rho_\theta$  is an injective immersion into  $U$ , and  $\text{im } D\rho_\theta(y)$  is spanned by the first  $k$  coordinate fields at  $\rho_\theta(y)$ , which is  $D_{\rho_\theta(y)}$ . Conversely, suppose such immersions exist through every point, and let  $X, Y$  be sections of  $D$  near  $\theta$ . An immersion is locally an embedding (Auxiliary result B.1(b)), so  $\rho_\theta$  maps a neighbourhood of 0 onto an embedded integral manifold through  $\theta$ , and Auxiliary result B.31(a) gives  $[X, Y]_\theta \in D_\theta$ . Since  $\theta$  was arbitrary,  $D$  is involutive.

The same application of Auxiliary result B.31(a) to the local pieces of  $\text{im } \rho$  shows that a single integral immersion  $\rho$  forces  $[X, Y] \in D$  at the points of  $\text{im } \rho$ ; off  $\text{im } \rho$  it says nothing, as the example shows. In the example,  $X_1 = \partial_x$  and  $X_2 = \partial_y + xz \partial_z$  are linearly independent everywhere, so  $D$  has constant rank 2, and  $[X_1, X_2] = \partial_x(xz) \partial_z = z \partial_z$ . A vector of  $D$  at  $(x, y, z)$  has the form  $(a, b, bxz)$ , which equals  $(0, 0, z)$  only if  $a = b = 0$  and  $z = 0$ ; so  $[X_1, X_2] \notin D$  wherever  $z \neq 0$ , and  $D$  is involutive on no open set meeting  $\{z \neq 0\}$ . On the plane  $z = 0$ ,  $D = \text{span}\{\partial_x, \partial_y\}$  is its tangent plane, so  $\rho(u, v) = (u, v, 0)$  is an injective immersion with  $\text{im } D\rho = D$  along its image, and there  $[X_1, X_2] = 0 \in D$ .

(2) *Continuous time.* Let  $D$  be involutive, let  $\theta(t)$ ,  $t \in J$  an interval containing 0, be an absolutely continuous curve in  $U$  with  $\theta(0) = \theta_0$  and  $\dot{\theta}(t) \in D_{\theta(t)}$  for almost every  $t$ , and let  $N$  be the leaf through  $\theta_0$ . Put  $J_N = \{t \in J : \theta(t) \in N\}$ , which contains 0. Fix  $t_* \in J$  and a flat chart around  $\theta(t_*)$  as in Auxiliary result B.31(b). For  $t$  near  $t_*$  the curve stays in the chart and keeps its last  $\dim \Theta - k$  coordinates constant, so it stays in the slice  $\Sigma$  through  $\theta(t_*)$ . A slice is a connected integral manifold, and a connected integral manifold that meets a leaf lies in it (Auxiliary result B.31(c)), so if  $\Sigma$  meets  $N$  it lies in  $N$ . If  $t_* \in J_N$ , the slice  $\Sigma$  meets  $N$  at  $\theta(t_*)$ , so a neighbourhood of  $t_*$  lies in  $J_N$ :  $J_N$  is open. If  $t_n \rightarrow t_*$  with  $t_n \in J_N$ , then for large  $n$  the point  $\theta(t_n)$  lies in  $\Sigma \cap N$ , so  $\Sigma \subseteq N$  and  $t_* \in J_N$ :  $J_N$  is closed. Hence  $J_N = J$ .

(3) *Discrete steps.* The update is  $\theta' = \theta - \eta a(\theta)s$  with  $\eta > 0$  and  $s \in S$  arbitrary. As  $\eta$  and  $s$  range,  $-\eta a(\theta)s$  ranges over the linear space  $\text{im } a(\theta) = D_\theta$ , so the possible next iterates in  $U$  form  $(\theta + D_\theta) \cap U$ . They all lie on the leaf  $N$  through  $\theta$  iff  $(\theta + D_\theta) \cap U \subseteq N$ ; this is the first equivalence. Call the leaves *affine* when every leaf  $N$  equals  $(\theta + D_\theta) \cap U$  for each  $\theta \in N$ . If the containment holds at every point of every leaf, then for each  $\theta \in N$  a small  $k$ -disc of the plane  $\theta + D_\theta$  around  $\theta$  lies in  $N$ . Take a flat chart around  $\theta$  (Auxiliary result B.31(b)) and shrink the disc into it. The leaves form a foliation (Auxiliary result B.31(c)), so  $N$  meets the chart in countably many slices, on each of which the last  $\dim \Theta - k$  coordinates are constant. On the connected disc these coordinates are continuous and take countably many values, so they are constant, and the disc lies in the slice through  $\theta$ , a  $k$ -dimensional integral manifold, in which it is open by invariance of domain. Hence its tangent planes, all equal to  $D_\theta$ , are the planes  $D_x$  at its points, so  $D$  is locally constant along  $N$ , so constant on the connected  $N$ ; every path in  $N$  then has velocity in  $D_\theta$ , so  $N \subseteq \theta + D_\theta$ , and with the containment  $N = (\theta + D_\theta) \cap U$ . The converse is clear. For an anchor that is constant over the period,  $D_\theta = V$  is defined on  $U = \Theta$ ,

sections of  $D$  are  $V$ -valued functions whose derivatives and brackets are again  $V$ -valued, and the leaves are the planes  $\theta + V$ ; the condition holds, as [Proposition IV.2](#) uses.

(4) *Frames.* A backward method under gradient flow ( $\text{opt} = \text{id}$ ) moves the weight by  $\dot{\theta} = -a_t c_t \nabla L$ ; the forward method moves it by  $\dot{\theta} = -D\rho_q \Lambda D\rho_q^\top \nabla L$  ([Definition III.1](#)). We read “for every loss” as including time-dependent losses, as minibatch training produces. Let both methods follow the same trajectory up to time  $t_1$  for a loss  $L$ , and switch at  $t_1$  to  $L + \langle \gamma, \cdot \rangle$ . The trajectories still agree up to  $t_1$ , and their right derivatives at  $t_1$  agree, so  $a_{t_1} c_{t_1} (\nabla L + \gamma) = K_{q(t_1)} (\nabla L + \gamma)$  for every covector  $\gamma$ , that is,  $a_{t_1} c_{t_1} = D\rho \Lambda D\rho^\top$  at  $q(t_1)$ . This operator is symmetric positive semidefinite, with range in  $\text{im } a = D$ . If  $a$  is injective, let  $a^+ = (a^\top a)^{-1} a^\top$ , so  $P = aa^+$  is the orthogonal projector onto  $\text{im } a$ , and put  $M = a^+(ac)(a^+)^\top$ , which is symmetric positive semidefinite. Since  $ac$  is symmetric with range in  $\text{im } a$ ,  $ac = P(ac) = (ac)P$ , so  $ac = P(ac)P = aMa^\top$ , and  $a(c - Ma^\top) = 0$  gives  $c = Ma^\top$ .

Now let  $\rho$  be an immersion with  $\dim Q = \text{rank } D = k$ . Then  $D\rho_q \Lambda D\rho_q^\top$  has rank  $k$  and range in  $D$ , so  $\text{im } D\rho_q = D_{\rho(q)}$ , and  $\rho$  maps small open sets of  $Q$  diffeomorphically onto open pieces of leaves. On  $D$  the cometric  $K = aMa^\top$  is invertible as a map from  $D^*$  to  $D$ , and its inverse is a metric  $g_D$  on  $D$ :  $g_D(v, v) = \langle \xi, v \rangle$  whenever  $K\xi = v$ . For  $u \in T_q Q$  put  $v = D\rho_q u$ . Since  $D\rho_q$  is injective,  $D\rho_q^\top$  is onto, so there is  $\xi$  with  $D\rho_q^\top \xi = \Lambda^{-1}u$ , and then  $K\xi = D\rho_q \Lambda \Lambda^{-1}u = v$ . Hence

$$g_D(D\rho_q u, D\rho_q u) = \langle \xi, D\rho_q u \rangle = \langle D\rho_q^\top \xi, u \rangle = \langle \Lambda^{-1}u, u \rangle.$$

So  $\rho$  is a local isometry from  $Q$  with the constant metric  $\Lambda^{-1}$ , which is flat, onto the leaf with metric  $g_D = (aMa^\top)^{-1}|_D$ , and by [Auxiliary result B.32\(a\)](#) the leaf metric is flat.

(5) *Gauge fibres:  $BB^\top$ .* Let  $\rho(B) = BB^\top$  on  $Q = \mathbb{R}^{n \times n}$  with  $\Lambda = I$ , landing in  $\text{Sym}(n)$ , and pair covectors with  $\text{Sym}(n)$  by the trace. Then  $D\rho_B(\dot{B}) = \dot{B}B^\top + B\dot{B}^\top$ , and for symmetric  $G$ ,  $\langle G, \dot{B}B^\top + B\dot{B}^\top \rangle = \langle 2GB, \dot{B} \rangle$ , so  $D\rho_B^\top G = 2GB$  and

$$K(G) = D\rho_B(2GB) = 2(G\theta + \theta G), \quad \theta = BB^\top.$$

This depends only on  $\theta$ , not on the point of the fibre  $\{BR : R \in O(n)\}$ . On the positive definite matrices it is invertible ([Auxiliary result B.20](#)), so  $D = \text{Sym}(n)$  there. Put  $a = K^{1/2}$  and  $c = a^\top$ . The backward flow  $\dot{\theta} = -K(\theta)\nabla L$  and the forward flow pushed to  $\theta$  solve the same ordinary differential equation with smooth right-hand side, so they agree for every loss. The leaf metric  $K^{-1}$  is a constant multiple of the Bures–Wasserstein metric ([Auxiliary result B.32\(b\)](#)), which is not flat for  $n \geq 2$ . Here  $\dim Q = n^2 > \dim \text{Sym}(n)$ , so  $\rho$  is no immersion, in line with (4).

(6) *Gauge fibres: balanced full-rank LoRA.* Take  $m = n = r \geq 2$ ,  $s = 1$  and equal block rates  $\eta_A = \eta_B = 1$ , and restrict to balanced factors,  $B^\top B = AA^\top$ , with  $W = BA$  invertible; gradient flow preserves balancedness by the Noether charge of [Theorem III.5](#). From  $B^\top B = AA^\top$ ,  $W^\top W = A^\top (B^\top B) A = (A^\top A)^2$  and  $WW^\top = B(AA^\top)B^\top = (BB^\top)^2$ , so  $A^\top A = |W| := (W^\top W)^{1/2}$  and  $BB^\top = |W^\top| := (WW^\top)^{1/2}$ , the unique positive definite square roots. By [Observation III.2](#),

$$K_W(G) = G|W| + |W^\top|G,$$

a function of  $W$  alone, invertible by [Auxiliary result B.20](#); as in (5) a backward method with  $a = K^{1/2}$ ,  $c = a^\top$  reproduces the flow, and the leaf is an open subset of  $\text{GL}_n$  with metric  $g_W(X, X) = \langle K_W^{-1}X, X \rangle$ . The transpose  $\tau(W) = W^\top$  is an isometry of  $g$ :  $K_{W^\top}(G^\top) = (K_W G)^\top$ , so  $K_{W^\top}^{-1}(X^\top) = (K_W^{-1}X)^\top$  and  $g_{W^\top}(X^\top, X^\top) = g_W(X, X)$ . Its fixed points in  $\text{GL}_n$  are the invertible



symmetric matrices, and the positive definite ones form an open subset  $F$  of them, an embedded submanifold on which  $d\tau = \text{id}$ . A  $g$ -geodesic  $\gamma$  with  $\gamma(0) \in F$  and  $\dot{\gamma}(0)$  tangent to  $F$  is carried by  $\tau$  to a geodesic with the same initial data (Auxiliary result B.38(a)), so  $\tau \circ \gamma = \gamma$  and  $\gamma$  stays in  $F$  for small times. Every geodesic tangent to  $F$  stays in  $F$ , so  $F$  is totally geodesic, and by the Gauss equation its intrinsic sectional curvatures equal those of  $g$  on its tangent planes (Auxiliary result B.38(b)). On  $F$ ,  $|W| = |W^\top| = W$ , and the induced metric is  $\langle L_W^{-1}X, X \rangle$  with  $L_W(G) = GW + WG$  on  $\text{Sym}(n)$ , half the cometric of (5); this is a constant multiple of the Bures–Wasserstein metric, which has a plane of positive sectional curvature for  $n \geq 2$  (Auxiliary result B.32(b)). So  $g$  has a plane of nonzero sectional curvature and is not flat.

(7) *Discrete steps and horizontal lines.* A plain GD step of the forward method from  $q$ , with  $g = \nabla L(\rho(q))$ , lands at  $\rho(q - \eta v)$  with  $v = \Lambda D\rho_q^\top g$ ; the backward step lands at  $\rho(q) - \eta ac$ . Agreement for small  $\eta$  forces  $ac = D\rho_q \Lambda D\rho_q^\top$  as in (4), so the backward step is  $\rho(q) - \eta D\rho_q v$ . Agreement for every  $\eta > 0$ , and for  $-g$  as well as  $g$  (another loss), says  $\rho(q + tv) = \rho(q) + t D\rho_q v$  for all real  $t$  in the domain:  $\rho$  is affine along the line  $q + t \Lambda D\rho_q^\top g$ .

(8) *Instances.* Within a period, GaLore’s anchor is a fixed matrix  $P_k$ , LISA’s a fixed coordinate inclusion, and MeZO’s the fixed vector  $z_t$  for its single step; none depends on  $\theta$ , so (3) applies on  $U = \Theta$  (vacuously when a period is one step). Each pair has  $c = a^*$  (§C.37), so with  $\mathcal{L} = a$  it is the backward method of Proposition IV.2, and under that proposition’s optimizer hypotheses each period is a period of the linear forward method  $\theta_k + \mathcal{L}C$ , merged at its end. For MeZO,  $\mathcal{L}(\lambda) = \lambda z_t$ , and the identification holds up to the error of the central difference (Auxiliary result B.34). Integrability holds for all of them, so it cannot tell them apart, and by Proposition IV.2 what distinguishes them from LoRA is how the projector changes and which state is carried. Flora’s update contains the full-rank term  $(1 - \beta)\hat{g}_t$ , APOLLO’s is  $G \text{diag}(s)$  with  $G$  the full gradient (Zhu et al., 2024a), and Fira’s adds a scaled multiple of  $(I - P_t P_t^\top)G$  (Chen et al., 2024e); for gradients with  $(I - P_t P_t^\top)G \neq 0$  each update leaves  $\text{im } P_t$ , so none is of the displayed form with  $a = P_t$ .

(9) *The GaLore field.* Let  $L$  be smooth. On the open set where  $\sigma_r(\nabla L(W)) > \sigma_{r+1}(\nabla L(W))$ , let  $\Pi(W)$  be the orthogonal projector onto  $U_r(\nabla L(W))$ , the top- $r$  eigenspace of  $S(W) = \nabla L(W) \nabla L(W)^\top$ . It is a smooth function of  $W$  there, being a Riesz projector of  $S(W)$  whose spectral gap persists (Auxiliary result B.37). The fields  $X_E(W) = \Pi(W)E$ ,  $E \in \mathbb{R}^{m \times n}$ , span  $D_W = \{Z : \text{col } Z \subseteq U_r\}$ . Write  $\Pi'(Z) = D\Pi(W)[Z]$ . Differentiating  $\Pi^2 = \Pi$  gives  $\Pi'\Pi + \Pi\Pi' = \Pi'$ , hence  $\Pi\Pi'\Pi = 0$  and  $(I - \Pi)\Pi'(I - \Pi) = 0$ . With  $N(Z) := (I - \Pi)\Pi'(Z)\Pi$  and the symmetry of  $\Pi'(Z)$ ,

$$\Pi'(Z) = N(Z) + N(Z)^\top, \quad (26)$$

so  $\Pi'(Z) = 0$  iff  $N(Z) = 0$ . The bracket is  $[X_E, X_F] = \Pi'(\Pi E)F - \Pi'(\Pi F)E$ , and its component normal to  $D$  is

$$(I - \Pi)[X_E, X_F] = N(\Pi E)\Pi F - N(\Pi F)\Pi E.$$

*Criterion.* If  $U_r$  is invariant along  $D$ , that is  $\Pi'(Z) = 0$  for  $Z \in D$ , the normal component vanishes and  $D$  is involutive. Conversely, let  $D$  be involutive and  $n \geq 2$ . For  $u, u' \in U_r$  and  $x, y \in \mathbb{R}^n$  linearly independent, take  $\Pi E = ux^\top$  and  $\Pi F = u'y^\top$ : involutivity gives  $N(ux^\top)u'y^\top = N(u'y^\top)ux^\top$ . The left side has its rows in  $\text{span}\{y\}$  and the right side in  $\text{span}\{x\}$ , so both vanish, and  $N(ux^\top)u' = 0$ . As  $u'$  ranges over  $U_r$  and  $N(Z)$  vanishes on  $U_r^\perp$ ,  $N(ux^\top) = 0$  for all  $u \in U_r$  and  $x \in \mathbb{R}^n$  (given  $x$ , pick  $y$  independent of it), so  $N = 0$  on  $D$  by linearity, and  $U_r$  is invariant along  $D$ . For right projection apply this to the transpose. The failure proved next



covers every  $1 \leq r < \min(m, n)$  with  $m, n \geq 2$ ; the remaining case with a gap is  $n = r = 1$ , where  $D_W = U_1$  is a line field, and a line field is always involutive, since  $[fX, gX] = (fXg - gXf)X$ .

*Failure for  $L = \frac{1}{2}\|W\|_F^2$ .* Here  $\nabla L(W) = W$ . Let  $p = \min(m, n) \geq 2$ ,  $1 \leq r < p$ , and  $W = \sum_{i \leq p} \sigma_i e_i e_i^\top$  with  $\sigma_1 > \dots > \sigma_p > 0$ , so  $U_r = \text{span}\{e_1, \dots, e_r\}$  and  $S = WW^\top$  is diagonal with entries  $\lambda_i = \sigma_i^2$  ( $i \leq p$ ) and 0. For a smooth symmetric family  $S(t)$  with  $S(0) = S$ , differentiating  $[S(t), \Pi(t)] = 0$  gives  $[\dot{S}, \Pi] + [S, \dot{\Pi}] = 0$ ; the  $(i, j)$  entry with  $i \leq r < j$  reads  $-\dot{S}_{ij} + (\lambda_i - \lambda_j)\dot{\Pi}_{ij} = 0$ , and  $\dot{\Pi}$  vanishes on the diagonal blocks by the identities above. So  $\dot{\Pi}_{ij} = \dot{S}_{ij}/(\lambda_i - \lambda_j)$  for  $i \leq r < j$ . Take  $Z = e_1 x^\top \in D$ , so  $\dot{S} = ZW^\top + WZ^\top$ . Then  $\dot{S}_{1j} = x^\top W^\top e_j = \sigma_j x_j$  for  $r < j \leq p$ ,  $\dot{S}_{1j} = 0$  for  $j > p$ , and  $\dot{S}_{ij} = 0$  for  $2 \leq i \leq r < j$ . Hence

$$N(e_1 x^\top) = \sum_{j=r+1}^p \frac{\sigma_j x_j}{\sigma_1^2 - \sigma_j^2} e_j e_1^\top.$$

With  $E = e_1 e_{r+1}^\top$  and  $F = e_1 e_1^\top$  we get  $N(\Pi F) = 0$  and

$$(I - \Pi)[X_E, X_F] = N(e_1 e_{r+1}^\top) e_1 e_1^\top = \frac{\sigma_{r+1}}{\sigma_1^2 - \sigma_{r+1}^2} e_{r+1} e_1^\top \neq 0,$$

so  $D$  is not involutive at  $W$ . At  $m = n = 2$ ,  $r = 1$  and  $W = \text{diag}(2, 1)$ , the frame  $X_i = X_{e_1 e_i^\top}$  gives the full bracket  $[X_1, X_2] = \Pi'(e_1 e_1^\top) e_1 e_2^\top - \Pi'(e_1 e_2^\top) e_1 e_1^\top = 0 - \frac{1}{3}(e_1 e_2^\top + e_2 e_1^\top) e_1 e_1^\top = -\frac{1}{3} e_2 e_1^\top$ ; the frame  $X_i = u(W) e_i^\top$ , with  $u$  the top left singular vector, gives the same value, since  $Du[e_1 e_1^\top] = 0$  and  $Du[e_1 e_2^\top] = \frac{1}{3} e_2$  at this  $W$ .

*Genericity.* For least squares  $L(W) = \frac{1}{2}\|WX - Y\|_F^2$  the normal component of the bracket of two fixed frame fields is a real-analytic function of  $(W, X, Y)$  on the open set where the gap exists. It does not vanish identically on the connected component containing  $(W, I_n, 0)$  with  $W$  as above, where it is the value just computed, so it is nonzero on an open dense subset of that component ([Auxiliary result B.10](#)). The values for cross-entropy and for several samples are the numerical checks reported with the proposition.

*Single-sample least squares.* With one sample,  $\nabla L(W) = (Wx - y)x^\top$  has rank one and, where  $Wx \neq y$ ,  $U_1 = \text{span}\{Wx - y\}$ . Moving  $W$  along  $Z = uv^\top$  with  $u \in U_1$  changes  $Wx - y$  by  $u(v^\top x) \in U_1$ , so  $U_1$  is constant along  $D$ , and  $D$  is involutive by the criterion.

*Time-dependence.* Minibatch GaLore with  $T = 1$  draws a new minibatch, and MeZO a new  $z_t$ , at every step, so their anchors are functions of  $(t, \theta)$  and not of  $\theta$  alone.  $\square$

#### C.40 Observation IV.4

**OBSERVATION IV.4** (backward reach and the backprop cone; an upper bound). Assume reverse mode in which each box keeps its own residual, with no checkpointing across boxes, and take boxes at the granularity where adapters are separate boxes (for LoRA-FA,  $A_0$  and  $B$  separately). A VJP residual is an input, an output, a sufficient statistic such as a ReLU mask, or a recipe to recompute one of these inside the box. Then it suffices to keep a VJP residual for

- each trainable box whose parameter-VJP depends on its input: linear and multiplicative boxes such as  $W$ ,  $A$ ,  $B$  and  $(IA)^3$  scales; additive boxes such as biases and prompts need nothing;
- each nonlinear box, bilinear boxes such as the attention products included, on a directed path from a trainable box to the loss.

Deterministic frozen linear boxes, and all boxes off this cone, need nothing; stochastic linear boxes on the cone (dropout) need their mask or random-number state. Alternatively, the  $k$

coordinates of  $P^\top \nabla L$  can be computed exactly by  $k$  forward-mode JVPs, with no reverse pass and  $O(1)$ -layer activation memory, each JVP costing a small constant multiple (about 1.5 to 3) of a forward pass. This always uses less activation memory than un-checkpointed reverse mode, but it is time-competitive with one VJP (about 3 forward passes) only for  $k$  of about 1 or 2. It also needs  $P$  before the passes, so it cannot produce GaLore’s SVD refresh, where  $P$  depends on  $\nabla L$ , more cheaply than a backward pass ( $\nabla L$  itself would take  $|\theta|$  JVPs).

*Proof. The computation.* The network is a directed acyclic graph of boxes. A box receives inputs  $x$  (activations), may carry parameters  $p$ , and produces an output  $y = f(x, p)$ ; the loss is the output of the last box. Reverse mode propagates cotangents backwards. For a box with output cotangent  $\bar{y}$  it computes the input cotangents  $\bar{x} = J_x f(x, p)^\top \bar{y}$  and, for a trainable box, the parameter cotangent  $\bar{p} = J_p f(x, p)^\top \bar{y}$  (Background B.17). Call the *cone* the set of boxes on a directed path from a trainable box to the loss.

*Pruning.* The parameter gradients are sums over paths from the parameters to the loss of products of Jacobians, so only cotangents of wires on such paths contribute. A wire whose value does not depend on any trainable parameter carries a cotangent that is never multiplied into a parameter gradient, so its cotangent need not be formed. Hence it suffices to compute  $\bar{y}$  for the outputs of boxes on the cone,  $\bar{x}$  for those inputs of cone boxes that are produced on the cone, and  $\bar{p}$  for the trainable boxes. Boxes off the cone take no part in the backward sweep and need no residual.

*What each box on the cone needs.* We list, for each kind of box, what the backward sweep reads besides  $\bar{y}$  and the weights, which are stored anyway.

- A deterministic frozen linear or affine box,  $y = Wx + b$ :  $\bar{x} = W^\top \bar{y}$  does not involve  $x$ , so no residual.
- A dropout box,  $y = m \odot x / (1 - p)$ :  $\bar{x} = m \odot \bar{y} / (1 - p)$ , so it needs its mask, or the random state that regenerates it.
- A trainable linear box,  $y = Wx$ :  $\bar{W} = \bar{y} x^\top$ , summed over tokens, so it needs its input  $x$ ;  $\bar{x} = W^\top \bar{y}$  needs nothing more. A trainable multiplicative box,  $y = \ell \odot x$ :  $\bar{\ell} = \sum \bar{y} \odot x$ , so it needs  $x$ , and  $\bar{x} = \bar{\ell} \odot \bar{y}$ .
- A trainable additive box,  $y = x + b$ , or a prompt,  $y = [P; x]$ :  $\bar{b} = \sum \bar{y}$  and  $\bar{P}$  is the slice of  $\bar{y}$  at the prompt positions, and  $\bar{x}$  is  $\bar{y}$  or its slice; no residual.
- A nonlinear box,  $y = f(x)$ , possibly with frozen parameters:  $\bar{x} = J_f(x)^\top \bar{y}$  depends on  $x$  through  $J_f(x)$ , so it needs something from which  $J_f(x)^\top \bar{y}$  can be formed: the input, the output (for tanh, sigmoid or softmax), a mask (for ReLU), or a recipe to recompute one of them. A bilinear box is nonlinear in its joint input; for  $y = a \odot b$ ,  $\bar{a} = \bar{y} \odot b$  and  $\bar{b} = \bar{y} \odot a$ , and for the attention products  $QK^\top$  and  $PV$  the same holds with matrix products.

A trainable nonlinear box lies on the cone, being on a path from itself to the loss, so it is covered by the second bullet, and also by the first if its parameter-VJP reads its input. Processing the cone in reverse topological order, each box has  $\bar{y}$  available (it is a sum of input cotangents of its readers on the cone) together with its residual, so the sweep computes every parameter gradient exactly. This proves that the residuals listed suffice. For LoRA-FA,  $A_0$  is a frozen linear box and needs nothing, while  $B$  is a trainable linear box whose input is  $A_0 x$ , so  $B$  keeps  $A_0 x$ .

*The forward-mode route.* The  $i$ -th coordinate of  $P^\top \nabla L$  is  $\langle p_i, \nabla L(\theta) \rangle = DL(\theta)[p_i]$ , the directional derivative along the  $i$ -th column of  $P$ . By Auxiliary result B.33(a), forward mode computes it alongside one evaluation of  $L$  at a small constant multiple of that cost, carrying

each tangent with its value and discarding both once consumed. In a layered network this keeps the values and tangents of  $O(1)$  layers alive at a time, whereas un-checkpointed reverse mode keeps the residuals of every box on the cone until the backward sweep reaches it. The  $k$  coordinates cost  $k$  such passes. One reverse pass costs about three forward passes, so with a per-JVP cost of 1.5 to 3 forward passes the forward route is no slower only when  $1.5k$  to  $3k$  is at most about 3, that is, for  $k$  of about 1 or 2. The directions  $p_i$  must be fixed before the passes. GaLore’s refresh computes  $P$  from the SVD of the full gradient  $\nabla L$  of each projected matrix, and  $\nabla L$  by forward mode costs one pass per coordinate,  $|\theta|$  passes, against one reverse pass ([Auxiliary result B.33\(b\)](#)).

**MeZO.** The central difference estimates  $DL(\theta)[z]$  with error  $O(\varepsilon^2)$  when  $L$  has a Lipschitz Hessian, and is unbiased for the gradient of the Gaussian smoothing of  $L$  ([Auxiliary result B.34](#)).  $\square$

#### C.41 Definition V.1

**DEFINITION V.1** (base change and rebasing).

- (a) A function-preserving *pointed reparametrisation*  $\phi : (\Theta', f', \theta'_0) \rightarrow (\Theta, f, \theta_0)$ , with  $f \circ (\phi \times \text{id}_X) = f'$ , induces by postcomposition

$$\phi_! : \mathbf{Meth}(\Theta', \theta'_0) \rightarrow \mathbf{Meth}(\Theta, \theta_0), \quad (Q, q_0, \rho) \mapsto (Q, q_0, \phi \circ \rho).$$

- (b) A map  $\kappa : \Theta \rightarrow \Theta$  that does not preserve the function, such as a quantiser, moves the base point. Its **pointing defect** is  $e = \kappa\theta_0 - \theta_0$ .
- (c) A **rebasing scheme** (merge-and-restart) picks  $\theta_{k+1} \in \text{Im } M(\theta_k)$ . Its reachable sets are  $R_K$ , the points  $\theta_K$  it can reach in  $K$  cycles, and  $R_\infty = \bigcup_K R_K$ .

A definition needs no proof. We check that  $\phi_!$  is a functor that preserves the realised functions, and record the recursion for the reachable sets that the proofs of [Theorem V.3](#) use.

*Verification of the claims in Definition V.1.* (a) Since  $\phi(\theta'_0) = \theta_0$ , the map  $\phi \circ \rho$  sends  $q_0$  to  $\theta_0$ , so  $\phi_!M$  is an object of  $\mathbf{Meth}(\Theta, \theta_0)$ . A morphism  $h : M \rightarrow N$  is a pointed smooth map with  $\rho_N \circ h = \rho_M$  ([Definition I.2](#)); then  $(\phi \circ \rho_N) \circ h = \phi \circ \rho_M$ , so the same  $h$  is a morphism  $\phi_!M \rightarrow \phi_!N$ . Setting  $\phi_!h = h$  preserves identities and composition, so  $\phi_!$  is a functor, the postcomposition functor of slice categories ([Background A.19](#)). It changes neither the parameter space nor the realised functions: for  $q \in Q$ ,

$$f(\phi(\rho(q)), x) = f'(\rho(q), x) \quad \text{for all } x \in X,$$

so  $\phi_!M$  computes exactly the functions  $M$  computes, and  $\text{Im } \phi_!M = \phi(\text{Im } M)$ .

(b) is a definition. (c) Put  $R_0 = \{\theta_0\}$ . Each cycle trains the method  $M(\theta_k)$  over the current base and merges, so the points reachable in  $K + 1$  cycles are

$$R_{K+1} = \bigcup_{\theta \in R_K} \text{Im } M(\theta).$$

When every  $M(\theta)$  is pointed at  $\theta$ , that is  $\rho_\theta(q_0) = \theta$ , we have  $\theta \in \text{Im } M(\theta)$ , so  $R_K \subseteq R_{K+1}$  and  $R_\infty$  is an increasing union.  $\square$

## C.42 Proposition V.2

**PROPOSITION V.2** (quantisation and splitting). Write  $W$  for the full-precision weight,  $\kappa$  for the quantiser,  $e = \kappa W - W$ ,  $\delta_0 = \delta(q_0)$ , and  $s$  for the LoRA scale  $\alpha/r$  ( $\alpha/\sqrt{r}$  under rsLoRA).

(a) Additive families commute with translation,  $M(\theta + v) = \tau_v \circ M(\theta)$ . Hence QLoRA is LoRA pointed at  $\kappa\theta_0$ , with defect  $e$ , and it needs no new weight-space structure. Since  $-e$  is generically of full rank,  $\theta_0 \notin \text{Im LoRA}_r(\kappa\theta_0)$  unless  $\text{rank } e \leq r$ .

(b) For splitting families derived from an SVD or from data, the order of operations matters, and three orders occur in practice.

- Split, then quantise the residual (QPissa; HF's recommended PiSSA and CorDA workflow; the quantisation half-step of LoftQ for  $t \geq 2$ , with  $W_{\text{res}} = W - \delta(q_{t-1})$ ). The defect is  $\kappa(W_{\text{res}}) - W_{\text{res}}$ .
- Quantise, then fit the error (LoftQ with one iteration, HF's default `loftq_iter=1`; HF `replace_lora_weights_loftq`). HF sets the factors from  $\text{SVD}_r(W - \kappa W)$  without dividing by  $s$ , unlike its PiSSA and OLoRA initialisations, so  $\delta_0 = -s \text{SVD}_r(e)$  and the defect is  $e - s \text{SVD}_r(e)$ , with

$$\|e - s \text{SVD}_r(e)\|^2 = \|e - \text{SVD}_r(e)\|^2 + (1 - s)^2 \|\text{SVD}_r(e)\|^2.$$

The defect is  $e$  minus its best rank- $r$  approximation (of rank  $\text{rank } e - r$ ) iff  $s = 1$ , which is HF's bare default  $r = \alpha = 8$ . For  $s = 2$  its norm equals  $\|e\|$ , no better than QLoRA. With a callback, `replace_lora_weights_loftq` can roll a module back to the zero initialisation, whose defect is  $e$ .

- Quantise, split, re-quantise (HF `olora_init` on `bitsandbytes` weights). The defect is  $e + \kappa(r') - r'$  with  $r' = \kappa W - \delta_0$ .

Quantise-then-split has defect exactly  $e$  iff the frozen part equals  $\kappa W - \delta_0$ , for instance  $\kappa W$  kept quantised minus a frozen copy of the rank- $r$   $\delta_0$ , at a cost of about  $r(m + n)$  extra numbers. Re-quantising the residual, as HF `olora_init` does, generically adds  $\kappa(r') - r'$ .

Action families whose pointing does not depend on the base (OFT at  $R = I$ , BOFT at  $R = I$  with unit scale, HiRA at  $B = 0$ ), evaluated at  $\kappa\theta_0$ , simply carry the defect  $e$ . Methods whose pointing is computed from the base have an order ambiguity of their own. DoRA's magnitude initialised from  $\kappa\theta_0$  gives defect  $e$ , while initialised from  $\theta_0$  it gives  $\text{diag}(\|w_{0,i}\|/\|(\kappa\theta_0)_i\|) \kappa\theta_0 - \theta_0$ . For action families a base change also moves the absolute invariants, along with the base point. NF4's exact zero code enlarges the zero set that HiRA must preserve, and exactly orthogonal (exact-Cayley) OFT preserves the Gram matrix of  $\kappa\theta_0$  rather than that of  $\theta_0$ ; HF's default Cayley–Neumann OFT preserves it only approximately.

(c) LoftQ alternates two updates on  $\|\theta_0 - Q - LR\|$ , an exact rank- $r$  SVD step and a round-to-nearest quantisation step with blockwise scales recomputed from its input. The second is not an exact projection, so the objective need not decrease monotonically. With  $s = 1$  the output is pointed up to the attained residual. It is then an approximate section, a choice  $(Q, q_0)$  with reduced  $\|\rho(q_0) - \theta_0\|$ . In general, with  $Q_T$  and the low-rank fit  $L_T R_T$  after  $T$  iterations, HF's deployed defect is  $(Q_T + L_T R_T - W) + (s - 1)L_T R_T$ , because HF's internal residual also omits  $s$ . The official LoftQ checkpoints use  $r = 64$  and  $\alpha = 16$ , so  $s = 1/4$ , and then the deployed defect can grow with  $T$  while the attained residual falls.

(d) For a generic high-precision adapter  $\Delta$ , fine-tune-then-quantise and quantise-then-fine-tune do not commute. The first lands on the quantisation grid. In the second, the exactly merged weight  $\kappa\theta_0 + \Delta$  leaves the quantised type. For a uniform fixed-scale INT- $k$  grid,  $\kappa\theta_0 + \Delta$  stays

on the grid iff  $\Delta$  lies on the scale lattice (within range), so there the obstruction is the off-grid update. NF4 is non-uniform and not closed under addition, and absmax grids whose scales are recomputed are not closed either, so for QLoRA's NF4 the grid itself is part of the obstruction. Type-preserving merges, such as HF PEFT's dequantise–add–requantise for bitsandbytes 4- and 8-bit layers, re-quantise with recomputed block scales, so they stay in the quantised type but are lossy. Under affine group quantisation QA-LoRA is the structured exception. Its group-pooled adapter is constant on each quantisation group and merges into the zero-points, so the merged model stays INT- $k$ .

*Proof. Quantisers.* We use two properties of the blockwise quantisers in practice (NF4 and INT- $k$ , with a fixed scale or the absmax scale of each block, Dettmers et al., 2021, 2023). First,  $\kappa$  rounds each entry, divided by its block's scale, to the nearest point of a finite codebook and multiplies back, so its values, the *representable* matrices, form a Lebesgue-null set: a finite set for fixed scales, and for absmax scales a finite union of sets with one free scale per block of  $\beta \geq 2$  entries. Second,  $\kappa$  is idempotent,  $\kappa(\kappa X) = \kappa X$ , and it fixes exactly the representable matrices.

(a) An additive family has  $\rho_\theta(q) = \theta + \delta(q)$  with  $\delta$  independent of the base and  $\delta(q_0) = 0$  (Definition I.2). Then  $\rho_{\theta+v}(q) = v + \rho_\theta(q) = \tau_v(\rho_\theta(q))$ , which is the first claim. QLoRA trains zero-initialised LoRA over the frozen base  $\kappa\theta_0$ , so it is  $\text{LoRA}_r(\kappa\theta_0)$ , with  $\rho(q_0) = \kappa\theta_0$  and defect  $\rho(q_0) - \theta_0 = e$ . Since every matrix of rank at most  $r$  has the form  $sBA$  ( $s \neq 0$ ),  $\theta_0 = \kappa\theta_0 + sBA$  for some  $(B, A)$  iff  $\text{rank}(-e) \leq r$ .

The word “generically”. For a fixed-scale quantiser,  $\kappa$  is constant on each open cell of a decomposition of  $\Theta$  into countably many cells and a null set of boundaries, so on a cell  $e = K - W$  with  $K$  constant. The rank-deficient matrices are the common zeros of the maximal minors, a proper algebraic subset, and  $W \mapsto K - W$  is a bijection, so  $e$  has full rank for almost every  $W$  in each cell, hence for almost every  $W$  (Auxiliary result B.10). For an absmax quantiser with blocks of  $\beta \geq 2$  consecutive entries of a row, take the cells on which every code and the position of every block maximum are constant. With the scale stored exactly, the block maximum is represented exactly, so  $e$  vanishes at those positions, and each other entry is  $e_{ij} = a c_{ij} - W_{ij}$  with  $a$  the scale of its block and  $c_{ij}$  its code; varying  $W_{ij}$  alone moves  $e_{ij}$  alone. So on each cell  $e$  fills an open subset of the space  $Z$  of matrices vanishing at the block maxima, and its rank is, for almost every  $W$  in the cell, the largest rank in  $Z$ . A row of length  $n$  has at most  $\lceil n/\beta \rceil + 1$  forced zeros, so  $Z$  contains a matrix of rank  $r + 1$  whenever  $r + 1 \leq n - \lceil n/\beta \rceil - 1$  and  $r < m$  (place  $r + 1$  nonzero entries in distinct rows and columns, avoiding the forced zeros, greedily). Hence  $\text{rank } e > r$ , and  $\theta_0 \notin \text{Im } \text{LoRA}_r(\kappa\theta_0)$ , for almost every  $W$  in this range. Full rank holds for almost every  $W$  in each cell whose zero pattern admits a matrix of full rank; the other cells, in which the block maxima of many rows fall in a few common columns, have positive but tiny measure (for instance, a column of zeros needs the maximum of its block in every row, probability  $\beta^{-m}$  per column for weights with independent continuous entries).

(b) In each pipeline the initial weight is  $\rho(q_0) = F + \delta_0$ , where  $F$  is the frozen part and  $\delta_0$  the initial adapter, and the defect is  $F + \delta_0 - W$ .

- *Split first.*  $F = \kappa(W_{\text{res}})$  with  $W_{\text{res}} = W - \delta_0$ , so the defect is  $\kappa(W_{\text{res}}) + \delta_0 - W = \kappa(W_{\text{res}}) - W_{\text{res}}$ . In LoftQ's quantisation half-step for  $t \geq 2$  the same computation applies with  $\delta_0$  replaced by the current fit  $\delta(q_{t-1}) = L_{t-1}R_{t-1}$ .



- *Quantise, then fit.*  $F = \kappa W$  and HF sets  $LR = \text{SVD}_r(W - \kappa W) = -\text{SVD}_r(e)$  (the best rank- $r$  approximation is odd,  $\text{SVD}_r(-X) = -\text{SVD}_r(X)$ ), and the forward pass multiplies by  $s$ , so  $\delta_0 = -s \text{SVD}_r(e)$  and the defect is  $e - s \text{SVD}_r(e)$ . Write  $P = \text{SVD}_r(e)$ . By [Auxiliary result B.11\(b\)](#),  $P$  and  $e - P$  are Frobenius-orthogonal, so  $\|e - sP\|^2 = \|(e - P) + (1 - s)P\|^2 = \|e - P\|^2 + (1 - s)^2\|P\|^2$ . For  $e \neq 0$ ,  $P \neq 0$ , so the defect equals  $e - P$  iff  $s = 1$ ; then it has rank  $\text{rank } e - r$  when  $\text{rank } e \geq r$ , being the sum of the singular terms beyond the  $r$ -th. HF's defaults  $r = \alpha = 8$  give  $s = \alpha/r = 1$ . For  $s = 2$ ,  $\|e - 2P\|^2 = \|e - P\|^2 + \|P\|^2 = \|e\|^2$ , again by orthogonality; in general  $\|e - sP\| < \|e\|$  iff  $(1 - s)^2 < 1$ , that is  $0 < s < 2$ . A module rolled back to the zero initialisation has  $\delta_0 = 0$  and defect  $\kappa W - W = e$ .
- *Quantise, split, re-quantise.* The split is computed from  $\kappa W$ , the residual  $r' = \kappa W - \delta_0$  is re-quantised, and  $F = \kappa(r')$ . The defect is  $\kappa(r') + \delta_0 - W = (\kappa(r') - r') + (\kappa W - W) = e + \kappa(r') - r'$ .

For quantise-then-split with frozen part  $F$ , the defect  $F + \delta_0 - W$  equals  $e = \kappa W - W$  iff  $F = \kappa W - \delta_0$ . Storing  $\kappa W$  quantised together with the frozen factors of  $\delta_0 = sL_0R_0$ , which hold  $r(m + n)$  numbers, realises this  $F$  exactly. Re-quantising instead adds  $\kappa(r') - r'$ , which vanishes iff  $r'$  is representable, that is, iff  $r'$  lies in the null set of representable matrices.

*Action families.* OFT at  $R = I$  gives  $\rho(q_0) = \kappa\theta_0 I = \kappa\theta_0$ , BOFT at  $R = I$  with unit scale gives  $\text{diag}(\mathbf{1}) \kappa\theta_0 I = \kappa\theta_0$ , and HiRA at  $B = 0$  gives  $\kappa\theta_0 + \kappa\theta_0 \odot (0 \cdot A) = \kappa\theta_0$ ; each has defect  $e$ . DoRA at  $B = 0$  computes  $\rho(q_0) = \text{diag}(\mu_{0,i}/\|(\kappa\theta_0)_i\|)\kappa\theta_0$ , normalising the rows of its frozen base  $\kappa\theta_0$  (assumed nonzero) and rescaling them by the magnitude vector  $\mu_0$ . If  $\mu_0$  holds the row norms of  $\kappa\theta_0$ , then  $\rho(q_0) = \kappa\theta_0$  and the defect is  $e$ ; if it holds those of  $\theta_0$ , the defect is the displayed matrix.

*Moved invariants.* HiRA's update  $W \mapsto W \odot (\mathbf{1} + BA)$  preserves every zero entry of its base ([Proposition II.9\(e\)](#)). NF4 contains the code 0, so every entry of  $\theta_0$  whose ratio to its block's absmax rounds to 0 becomes an exact zero of  $\kappa\theta_0$ , while zero entries of  $\theta_0$  stay zero; the zero set of  $\kappa\theta_0$  therefore contains that of  $\theta_0$ , and generically more. Exact-Cayley OFT has  $\rho(Q)\rho(Q)^\top = \kappa\theta_0 C(Q)C(Q)^\top \kappa\theta_0^\top = \kappa\theta_0 \kappa\theta_0^\top$  ([Corollary II.6](#)). HF's default factor is  $R = C(Q)(I - Q^4)$  ([Auxiliary result B.23\(e\)](#)), so  $\rho\rho^\top = \kappa\theta_0(I - Q^4)^2 \kappa\theta_0^\top = \kappa\theta_0 \kappa\theta_0^\top + O(\|Q\|^4)$ .

(c) LoftQ starts from  $L_0R_0 = 0$  and, for  $t \geq 1$ , sets  $Q_t = \kappa(\theta_0 - L_{t-1}R_{t-1})$  and  $L_tR_t = \text{SVD}_r(\theta_0 - Q_t)$ . By [Auxiliary result B.11\(b\)](#) the SVD step minimises  $\|\theta_0 - Q_t - LR\|$  over rank- $r$  products, so it never increases the objective. The quantisation step returns  $\kappa(y)$  for  $y = \theta_0 - L_{t-1}R_{t-1}$ , which need not be the representable matrix nearest to  $y$ , and the objective can increase. An example with blocks equal to the rows of a  $2 \times 3$  matrix, absmax scales, the codebook  $\{0, \pm\frac{1}{2}, \pm 1\}$  and  $r = 1$ :

$$\theta_0 = \begin{pmatrix} 1 & 1 & -\frac{5}{4} \\ \frac{5}{4} & 1 & -1 \end{pmatrix}, \quad Q_1 = \kappa(\theta_0) = \frac{5}{4} \begin{pmatrix} 1 & 1 & -1 \\ 1 & 1 & -1 \end{pmatrix}, \quad \theta_0 - Q_1 = \frac{1}{4} \begin{pmatrix} -1 & -1 & 0 \\ 0 & -1 & 1 \end{pmatrix}.$$

Here  $(\theta_0 - Q_1)(\theta_0 - Q_1)^\top = \frac{1}{16} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$  has eigenvalues  $\frac{3}{16}$  and  $\frac{1}{16}$ , with top eigenvector  $(1, 1)/\sqrt{2}$ , so  $L_1R_1 = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} (\theta_0 - Q_1) = \frac{1}{8} \begin{pmatrix} -1 & -2 & 1 \\ -1 & -2 & 1 \end{pmatrix}$  and the objective after the SVD step is  $\|\theta_0 - Q_1 - L_1R_1\|^2 = \frac{1}{16}$ . The next quantisation step acts on  $y = \theta_0 - L_1R_1 = \frac{1}{8} \begin{pmatrix} 9 & 10 & -11 \\ 11 & 10 & -9 \end{pmatrix}$ . Each row has absmax  $\frac{11}{8}$ , and the ratios  $\frac{9}{11}$  and  $\frac{10}{11}$  round to 1, so  $Q_2 = \frac{11}{8} \begin{pmatrix} 1 & 1 & -1 \\ 1 & 1 & -1 \end{pmatrix}$  and

$$\|\theta_0 - Q_2 - L_1R_1\|^2 = \|y - Q_2\|^2 = 2\left(\frac{1}{16} + \frac{1}{64}\right) = \frac{5}{32} > \frac{1}{16}.$$

The objective rose, and  $Q_1$ , which is representable, is closer to  $y$  than  $\kappa(y)$ . The deployed weight after  $T$  iterations is  $Q_T + sL_TR_T$ , because HF stores the fitted factors and the forward pass multiplies by  $s$ ; its defect is

$$Q_T + sL_TR_T - W = (Q_T + L_TR_T - W) + (s - 1)L_TR_T.$$

For  $s = 1$  this is minus the attained residual, so the output is pointed up to that residual. For  $s \neq 1$  the second term  $(s - 1)L_T R_T$  is proportional to the fit itself and does not shrink with the residual; in the referee simulation of the official configuration ( $r = 64$ ,  $\alpha = 16$ ,  $s = 1/4$ , five iterations, a Gaussian  $512 \times 512$  NF4 weight) the attained residual falls to  $0.67\|e\|$  while the deployed defect rises from  $0.91\|e\|$  to  $0.96\|e\|$  (Appendix E).

(d) Fine-tuning and then quantising gives  $\kappa(\theta_0 + \Delta)$ , a value of  $\kappa$ . Exact merging after quantising gives  $\kappa\theta_0 + \Delta$ , which is representable iff  $\Delta$  lies in the translate of the representable set by  $-\kappa\theta_0$ , a null set; so for generic  $\Delta$  it leaves the quantised type and the two orders differ. For a uniform grid with a fixed scale  $a$  per block, the representable matrices are  $a$  times integer codes in range, so  $\kappa\theta_0 + \Delta$  is representable iff  $\Delta = a(c' - c_0)$  with  $c'$  in range, that is, iff  $\Delta$  lies on the scale lattice. NF4 is not closed under addition. Its smallest positive code is  $q_1 \approx 0.0796$ , and  $2q_1 \approx 0.1592$  is not a code (the next codes are  $0.0796$  and  $0.1609$ ). The blocks  $(1, q_1, 0, \dots)$  and  $(0, q_1, 0, \dots)$  are representable, with absmax scales  $1$  and  $q_1$ , but their sum  $(1, 2q_1, 0, \dots)$  has absmax  $1$  and needs the code  $2q_1$ . Absmax grids with uniform codes of step  $h$  are not closed either:  $(1, h, 0, \dots)$  and  $(h, 0, 0, \dots)$  are representable, while their sum  $(1 + h, h, 0, \dots)$  would need the code  $h/(1 + h)$ , which is not a multiple of  $h$  for  $0 < h < 1$ . A type-preserving merge returns  $\kappa(\kappa\theta_0 + \Delta)$ , which is representable and differs from  $\kappa\theta_0 + \Delta$  whenever the latter is not, so it is lossy for generic  $\Delta$ . Finally, affine group quantisation stores each group  $g$  as  $a_g c_g + \beta_g \mathbf{1}$ , with integer codes  $c_g$  and a high-precision scale  $a_g$  and zero-point  $\beta_g$ . QA-LoRA pools its input over the quantisation groups, so its update is constant on each group,  $\Delta_g = \delta_g \mathbf{1}$  (Xu et al., 2023b), and  $a_g c_g + \beta_g \mathbf{1} + \delta_g \mathbf{1} = a_g c_g + (\beta_g + \delta_g) \mathbf{1}$  keeps the codes and moves only the zero-point.  $\square$

### C.43 Theorem V.3

**THEOREM V.3** (rebasings closure). (a) Additive methods with a fixed shape  $C \ni 0$ , with frames, masks and compression operators frozen at  $\theta_0$  across cycles. Then  $R_K = \theta_0 + C^{+K}$ , the  $K$ -fold Minkowski sum, and  $R_\infty = \theta_0 + \text{Mon}(C)$ , the additive monoid generated by  $C$ . If  $C$  is closed under real scalings,  $R_\infty = \theta_0 + \text{span } C$ .

(b) Action methods, where one cycle applies an element of a set  $S$  with  $1 \in S \subseteq G$ ; for scaling methods,  $S \subseteq \text{End}(\Theta)$  is a subset containing the identity. Then  $R_K = S^K \theta_0$ , and  $R_\infty$  is the orbit of the monoid generated by  $S$ . Suppose  $S$  lies in the subgroup generated by connected Lie subgroups  $H_1, \dots, H_p$  and contains a neighbourhood of  $1$  in each. Then the monoid generated by  $S$  is the connected Lie subgroup  $H$  with  $\text{Lie}(H) = \text{Lie}(\mathfrak{h}_1, \dots, \mathfrak{h}_p)$ .

(c) Block-orthogonal factors. Let the factors have coordinate blocks  $I_j \subseteq \{1, \dots, n\}$  (block OFT, BOFT factors, Givens planes). Then  $\text{Lie}(\mathfrak{so}(I_j))_j = \bigoplus_C \mathfrak{so}(C)$ , summed over the connected components  $C$  of the hypergraph  $(\{1, \dots, n\}, \{I_j\})$ .

Corollaries.

1. **ReLoRA.**  $R_K = W_0 + \mathcal{M}_{\leq \min(Kr, m, n)}$ . It reaches full fine-tuning iff  $K \geq \lceil \min(m, n)/r \rceil$ .
2. **Idempotent methods** gain nothing from rebasing. These are the methods whose shape is exactly a subspace, subgroup or submonoid independent of the base, with frames, masks and compression operators frozen across cycles: LoRA-FA with a fixed  $A$ , LoRA-XS with frozen frames, MoRA with fixed operators, FourierFT with fixed indices, WaveFT, C3A, MiSS (default balance mode; bat mode is base-dependent but still idempotent, since its blockwise maps  $W \mapsto (W + I)(I + M) - I$  form a monoid), SHiRA, BitFit, FISH with a fixed mask, LN-tuning, (IA)<sup>3</sup>, SSF and RoAd. Block OFT with a fixed partition and an exact

Cayley map gains only the measure-zero set of block rotations with an eigenvalue  $-1$ , so  $R_\infty = W_0 \cdot \prod \text{SO}(b) = \overline{R_1}$ . Constrained OFT (COFT, an  $\varepsilon$ -ball) with an exact Cayley map gains up to  $\prod \text{SO}(b)$ ; HF's `coft=True` keeps the Cayley–Neumann map by default and so falls under corollary 4. Methods that recompute frames or masks from the merged weight (an SVD-recomputing LoRA-XS, ReMoRA's alternating operators, resampled indices) do gain.

3. **VeRA.**  $\text{VeRA}^\infty = \text{LoRA-FA}(A)$  whenever the frozen  $B$  has no zero entries.
4. **OFT family.** Block OFT with an exact Cayley map is stuck in  $W_0 \cdot \prod \text{SO}(b)$ . HF's default Cayley–Neumann factor is  $R = C(Q)(I - Q^4)$ , with  $C$  the exact Cayley map, so  $R^\top R = (I - Q^4)^2$  and  $R$  is not orthogonal. Its singular values are  $|1 - \mu_k^4|$ , where  $\pm i\mu_k$  are the eigenvalues of  $Q$ . In the operating regime  $\|Q\|_{\text{op}} \leq 2^{1/4}$ , every factor is a contraction. Re-merging then decreases  $K_{\text{out}}$  in the Loewner order and no singular value grows, so the drift is monotone shrinkage, and the reachable set lies in  $W_0$  times the block contraction semigroup. With unconstrained  $Q$  the factors include singular ones ( $\mu_k = 1$ ), and re-merging generates  $\prod (\text{GL}^+(b) \cup \mathcal{M}_{\leq b-2})$ , or  $\prod (\text{CO}^+(2) \cup \{0\})$  when  $b = 2$ ; this clause is established at the level of a proof sketch. Iterated BOFT (exact Cayley in HF) with  $m'$  butterfly factors of block size  $b$  reaches  $\text{diag}(\mathbb{R}^m) \cdot W_0 \cdot P(\prod \text{SO}(b 2^{m'-1}))P^\top$ , because the factor hypergraph has components of size  $b 2^{m'-1}$ . For injective  $W_0$  this is  $\text{diag}(\mathbb{R}^m) \cdot W_0 \cdot \text{SO}(n)$  iff  $b 2^{m'-1} = n$ , the full-depth butterfly  $m' = 1 + \log_2(n/b)$ . HF's default `boft_n_butterfly_factor=1` is block OFT times  $\text{diag}(s)$ . A single full-depth BOFT has dimension at most  $m' \frac{n}{b} \binom{b}{2}$ , which is below  $\dim \text{SO}(n)$  when  $m'(b-1) < n-1$ ; its matrices are dense, while the set they form has positive codimension.
5. **Re-HRA** (HF `apply_GS=False`), with  $W_0$  injective and  $r$  even. Then  $R_K = (W_0 + \mathcal{M}_{\leq Kr}) \cap W_0 \cdot \text{SO}(n)$ , so rebasing commutes with the meet of [Theorem II.7](#). This multiplicative twin of ReLoRA reaches  $W_0 \cdot \text{SO}(n)$  in at most  $\lceil n/r \rceil$  cycles.
6. **GaLore (with a gradient-SVD or a random projector), LISA and MeZO**, under the optimizer hypotheses of [Proposition IV.2](#), are rebasing schemes of linear methods whose shape moves. That is why, as a set over the sequences of shapes they can choose, they reach all of  $\Theta$ , while a fixed linear shape  $L$  reaches only  $\theta_0 + L$  (corollary 2). Low rank is the per-period obstruction only for GaLore; LISA and MeZO already make full-rank per-matrix updates and gain coverage. Flora proper takes full-space steps with a compressed momentum, so it has no per-period low-rank shape.
7. **Merging never rescues an exactly orthogonal method from its absolute invariants.** In exact arithmetic, every exactly orthogonal input-side rebasing preserves  $K_{\text{out}}$  and the spectrum, and every output-side one preserves  $K_{\text{in}}$  and the spectrum.

*Proof.* (a) With frames, masks and compression operators frozen, every cycle trains the same additive shape over the current base:  $\text{Im } M(\theta) = \theta + C$  for every  $\theta$ . By the recursion of [§C.41](#),  $R_1 = \theta_0 + C$  and  $R_{K+1} = \bigcup_{\theta \in R_K} (\theta + C) = R_K + C$ , so  $R_K = \theta_0 + C^{+K}$  by induction. Since  $0 \in C$ , the sets  $C^{+K}$  increase, and their union is the additive monoid  $\text{Mon}(C)$  generated by  $C$  ([Background D.10](#)); hence  $R_\infty = \theta_0 + \text{Mon}(C)$ . If  $\lambda C \subseteq C$  for every real  $\lambda$ , then  $-C = C$ , so  $\text{Mon}(C)$  is closed under negation and is an additive subgroup, and  $\lambda(x_1 + \dots + x_K) = \lambda x_1 + \dots + \lambda x_K$  shows that it is closed under scaling. So it is a linear subspace containing  $C$  and contained in  $\text{span } C$ , and equals  $\text{span } C$ .

(b) Each cycle applies one element of  $S$  to the current point, so  $\text{Im } M(\theta) = S\theta$ , and the recursion gives  $R_K = S^K \theta_0$ . Since  $1 \in S$ , the sets  $S^K$  increase and their union is the monoid  $\text{Mon}(S)$  generated by  $S$ , so  $R_\infty = \text{Mon}(S) \theta_0$ , the orbit of  $\theta_0$  under that monoid. For the

second statement, let  $U_i \subseteq S$  be a neighbourhood of 1 in  $H_i$ . Then  $U_i \cap U_i^{-1}$  is a symmetric neighbourhood of 1 in the connected Lie group  $H_i$ , so it generates  $H_i$  as a monoid (Auxiliary result B.24(a)), and  $H_i \subseteq \text{Mon}(S)$ . Each  $H_i$  is a group, so the monoid generated by  $H_1 \cup \dots \cup H_p$  is closed under inverses and equals the subgroup they generate, which is the connected Lie subgroup  $H$  with  $\text{Lie}(H) = \text{Lie}\langle \mathfrak{h}_1, \dots, \mathfrak{h}_p \rangle$  (Auxiliary result B.24(d)). So  $H \subseteq \text{Mon}(S)$ . Conversely  $S \subseteq H$  by hypothesis, and  $H$  is closed under products, so  $\text{Mon}(S) \subseteq H$ .

(c) Write  $L_{ab} = E_{ab} - E_{ba}$ , so  $\mathfrak{so}(I) = \text{span}\{L_{ab} : a, b \in I\}$ . For distinct  $a, b, c$ ,

$$L_{ac}L_{cb} = E_{ab}, \quad L_{cb}L_{ac} = E_{ba}, \quad \text{so} \quad [L_{ac}, L_{cb}] = L_{ab},$$

while  $L_{ab}L_{cd} = L_{cd}L_{ab} = 0$  when  $\{a, b\} \cap \{c, d\} = \emptyset$ . Let  $\mathfrak{g} = \text{Lie}\langle \mathfrak{so}(I_j) \rangle_j$ . If  $a \neq b$  lie in one component  $C$ , join them by a shortest chain  $a = c_0, c_1, \dots, c_t = b$  of distinct coordinates with consecutive ones in a common block. By induction on  $t$ ,  $L_{ab} \in \mathfrak{g}$ : for  $t = 1$ ,  $L_{ab} \in \mathfrak{so}(I_j)$  for a block containing both; for  $t > 1$ ,  $L_{ac_{t-1}} \in \mathfrak{g}$  by induction and  $L_{c_{t-1}b} \in \mathfrak{g}$ , and their bracket is  $L_{ab}$ . So  $\bigoplus_C \mathfrak{so}(C) \subseteq \mathfrak{g}$ . Conversely,  $\bigoplus_C \mathfrak{so}(C)$  is a Lie subalgebra, since each  $\mathfrak{so}(C)$  is one and elements of different components have disjoint supports and commute, and it contains every  $\mathfrak{so}(I_j)$ , because each block lies in one component. Hence  $\mathfrak{g} = \bigoplus_C \mathfrak{so}(C)$ , of dimension  $\sum_C \binom{|C|}{2}$ .

*Corollary 1.* LoRA's shape  $\{sBA\}$  is  $\mathcal{M}_{\leq r}$ , which contains 0 and is closed under real scalings, so (a) applies. By Auxiliary result B.12(a), a sum of  $K$  matrices of rank at most  $r$  has rank at most  $\min(Kr, m, n)$ . Conversely, a matrix of rank  $k \leq \min(Kr, m, n)$  is a sum of  $k$  rank-one terms of its SVD (Auxiliary result B.11(a)), which split into  $K$  groups of at most  $r$  terms, each in  $\mathcal{M}_{\leq r}$ . So  $C^{+K} = \mathcal{M}_{\leq \min(Kr, m, n)}$ . This is all of  $\mathbb{R}^{m \times n}$  iff  $Kr \geq \min(m, n)$ , that is, iff  $K \geq \lceil \min(m, n)/r \rceil$ .

*Corollary 2.* If the shape is an additive submonoid  $C$  independent of the base (in particular a linear subspace), then  $C^{+K} = C$  and  $R_K = R_1$  for all  $K$  by (a); if it is a submonoid  $S$  of a group or of  $\text{End}(\Theta)$ , then  $S^K = S$  and  $R_K = R_1$  by (b). The listed methods are of this kind once their frames, masks and operators are frozen: the updates of LoRA-FA ( $\{XA\}$ ), LoRA-XS ( $\{UMV^T\}$ ), MoRA (linear in its square matrix), FourierFT and WaveFT (fixed coefficient positions), C3A (block-circulant matrices), MiSS in balance mode (linear in its shared block), SHiRA, BitFit, FISH and LN-tuning (fixed coordinates) form linear subspaces; the scalings of  $(IA)^3$ ,  $W \mapsto \text{diag}(\ell)W$ , form a monoid in  $\text{End}(\Theta)$ ; SSF's maps  $y \mapsto s \odot y + t$  form a monoid under composition,  $s' \odot (s \odot y + t) + t' = (s's) \odot y + (s't + t')$ ; RoAd's  $2 \times 2$  blocks range over the scaled rotations or over all  $2 \times 2$  matrices, both monoids; and MiSS's bat-mode maps satisfy  $(W + I)(I + M)(I + M') - I$  with  $(I + M)(I + M') = I + M''$ ,  $M'' = M + M' + MM'$  again in the class, so they form a monoid.

For block OFT with an exact Cayley map and a fixed partition,  $S = \prod \text{Cay}(b)$  with  $\text{Cay}(b) = C(\mathfrak{so}(b)) = \{R \in \text{SO}(b) : -1 \notin \text{spec } R\}$ , open and dense in  $\text{SO}(b)$  (Auxiliary result B.23(c)). Every element of  $\text{SO}(b)$  is a product of two elements of  $\text{Cay}(b)$  (Auxiliary result B.23(d)), so  $S^2 = \prod \text{SO}(b)$ , and  $R_K = W_0 \cdot \prod \text{SO}(b)$  for  $K \geq 2$ . By continuity of  $g \mapsto W_0 g$  and compactness of  $\prod \text{SO}(b)$ ,  $\overline{R_1} = W_0 \cdot \overline{\prod \text{Cay}(b)} = W_0 \cdot \prod \text{SO}(b)$ , and  $R_\infty \setminus R_1$  lies in the image of the Haar-null set  $\{\prod_j \det(I + R_j) = 0\}$ . For COFT with an exact Cayley map,  $S$  is the image under  $C$  of a ball about 0, which contains a neighbourhood of 1 in  $\prod \text{SO}(b)$  because  $C$  is a local diffeomorphism at 0; by (b) with the single connected group  $\prod \text{SO}(b)$ ,  $\text{Mon}(S) = \prod \text{SO}(b)$ .

Methods whose shape changes with the base escape (a). For linear shapes  $L_1 \neq L_2$  used in two cycles (resampled indices, alternating operators),  $R_2 \supseteq \theta_0 + L_1 + L_2 \supseteq \theta_0 + L_1 = R_1$  whenever  $L_2 \not\subseteq L_1$ . For LoRA-XS with frames recomputed from the merged weight, take  $r = 1$



and  $W_0 = \text{diag}(2, 1)$ ;  $R_1 = \{\text{diag}(2+t, 1)\}$ ; the point  $\text{diag}(-\frac{1}{2}, 1) \in R_1$  has top singular direction  $e_2$ , so the second cycle adds  $\{te_2e_2^\top\}$  and reaches  $\text{diag}(-\frac{1}{2}, 1+t) \notin R_1$ .

*Corollary 3.* VeRA's shape is  $C = \{\Lambda_b B \Lambda_d A\}$  with  $B, A$  frozen and  $b \in \mathbb{R}^m, d \in \mathbb{R}^r$  trainable. It contains 0, is closed under real scalings ( $b \mapsto \lambda b$ ), and lies in  $\{XA : X \in \mathbb{R}^{m \times r}\}$ . The entries of  $\Lambda_b B \Lambda_d$  are  $b_i B_{ij} d_j$ , so  $b = e_i, d = e_j$  gives  $B_{ij} E_{ij}$ ; if every  $B_{ij} \neq 0$ , the span of the matrices  $\Lambda_b B \Lambda_d$  is  $\mathbb{R}^{m \times r}$ , and  $\text{span } C = \{XA\}$ . By (a),  $R_\infty = W_0 + \{XA\}$ , the image of LoRA-FA with the same  $A$ .

*Corollary 4. Exact Cayley.* This is corollary 2.

*The Cayley–Neumann factor.* By [Auxiliary result B.23\(e\)](#),  $R = C(Q)(I - Q^4)$ ,  $R^\top R = (I - Q^4)^2$ , and the singular values of  $R$  are  $|1 - \mu_k^4|$ , each twice, and 1 for a zero eigenvalue of  $Q$ . So  $R$  is orthogonal only when every  $\mu_k^4 \in \{0, 2\}$ . If  $\|Q\|_{\text{op}} \leq 2^{1/4}$ , then  $0 \leq \mu_k^4 \leq 2$  and  $|1 - \mu_k^4| \leq 1$ , so  $R$  is a contraction. For an input-side merge  $W \mapsto WR$  with block-diagonal  $R$ ,

$$K_{\text{out}}(WR) = WRR^\top W^\top = W(I - Q^4)^2 W^\top \preceq WW^\top = K_{\text{out}}(W),$$

because  $C(Q)$  commutes with  $Q^4$  and  $(I - Q^4)^2 \preceq I$  ([Auxiliary result B.18\(b\)](#)); by [Auxiliary result B.18\(c\)](#) no singular value of  $W$  grows. Products of block-diagonal contractions are block-diagonal contractions ([Auxiliary result B.18\(a\)](#)), so the reachable set lies in  $W_0$  times that semigroup.

*Unconstrained  $Q$ .* Fix one block of size  $b$  and let  $S_b = \{R(Q) : Q \in \mathfrak{so}(b)\}$ ; since the blocks are independent and  $R(0) = I$ , the monoid generated by the block-diagonal factors is  $\prod \text{Mon}(S_b)$ . Write  $Q = O(\oplus_k \mu_k J \oplus 0)O^\top$  with  $O \in \text{O}(b)$ ,  $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$  and  $\mu_k \geq 0$ ; then  $I - Q^4 = O(\oplus_k (1 - \mu_k^4) I_2 \oplus I)O^\top$ .

$\subseteq$ .  $\det R = \det C(Q) \det(I - Q^4) = \prod_k (1 - \mu_k^4)^2 \geq 0$ , and  $\text{rank } R = \text{rank}(I - Q^4) = b - 2\#\{k : \mu_k = 1\}$ . So an invertible factor lies in  $\text{GL}^+(b)$  and a singular one has rank at most  $b - 2$ . Products of elements of  $\text{GL}^+(b)$  stay in it, and any product with a factor of rank at most  $b - 2$  has rank at most  $b - 2$ . Hence  $\text{Mon}(S_b) \subseteq \text{GL}^+(b) \cup \mathcal{M}_{\leq b-2}$ .

*Positive scalings.*  $C(-Q) = C(Q)^{-1}$  commutes with  $Q^4$ , so  $R(Q)R(-Q) = (I - Q^4)^2 = O(\oplus_k (1 - \mu_k^4)^2 I_2 \oplus I)O^\top$ . As  $\mu_k$  ranges over  $[0, \infty)$ ,  $(1 - \mu_k^4)^2$  takes every positive value ( $\mu_k^4 = 1 + \sqrt{\lambda}$  gives  $\lambda$ ). Taking  $O$  a permutation matrix,  $\text{Mon}(S_b)$  contains every positive diagonal matrix that is constant on the pairs of a matching of the coordinates and 1 off it. For  $b \geq 3$  these generate all positive diagonal matrices: their logarithms include every multiple of  $e_i + e_j, i \neq j$ , and  $e_i = \frac{1}{2}((e_i + e_j) + (e_i + e_k) - (e_j + e_k))$ , so products of commuting such matrices realise every  $\text{diag}(e^x)$ . For  $b = 2$  they are the positive scalars.

*Rotations.* For  $\|Q\|_{\text{op}} < 1$ ,  $I - Q^4$  is positive definite with eigenvalues  $1 - \mu_k^4$  on the planes of  $Q$ . Choose  $Q' = O(\oplus_k \mu'_k J \oplus 0)O^\top$  with the same  $O$  and  $(1 - \mu'_k)^2 = (1 - \mu_k^4)^{-1}$ ; then  $R(Q')R(-Q') = (I - Q^4)^{-1}$ , and  $C(Q) = R(Q)R(Q')R(-Q') \in \text{Mon}(S_b)$ . The set  $\{C(Q) : \|Q\|_{\text{op}} < 1\}$  is a neighbourhood of  $I$  in  $\text{SO}(b)$  ([Auxiliary result B.23\(c\)](#)), symmetric because  $C(-Q) = C(Q)^{-1}$ , so it generates  $\text{SO}(b)$  as a monoid ([Auxiliary result B.24\(a\)](#)). With the positive diagonals and the decomposition  $\text{GL}^+(b) = \text{SO}(b) D^+ \text{SO}(b)$  ([Auxiliary result B.19\(c\)](#)),  $\text{GL}^+(b) \subseteq \text{Mon}(S_b)$  for  $b \geq 3$ ; for  $b = 2$ ,  $\text{SO}(2) \cdot \mathbb{R}_{>0} = \text{CO}^+(2)$ .

*Singular blocks.* Let  $V \subseteq \mathbb{R}^b$  have dimension  $b-2$  and choose  $O$  mapping the last coordinate plane onto  $V^\perp$ , and  $Q_V = O(0 \oplus J)O^\top$ . Then  $Q_V^4$  is the projector onto  $V^\perp$ ,  $C(Q_V)$  acts as  $J$  on  $V^\perp$  (since  $(I+J)(I-J)^{-1} = J$ ) and as the identity on  $V$ , and  $R(Q_V) = C(Q_V)(I - Q_V^4) = P_V$ , the orthogonal projector onto  $V$ . If  $P_U$  with  $\dim U = k, 1 \leq k \leq b-2$ , is in the monoid, choose  $U' \subseteq U$  of dimension  $k-1$  and  $W' \subseteq U^\perp$  of dimension  $b-1-k \leq \dim U^\perp$ , and put  $V = U' \oplus W'$ , of dimension  $b-2$ . Since  $W' \perp U$ ,  $P_U P_V = P_U(P_{U'} + P_{W'}) = P_{U'}$ . By downward induction



every orthogonal projector of rank at most  $b - 2$  lies in the monoid. A matrix  $X$  of rank  $k \leq b - 2$  is  $X = U \operatorname{diag}(\sigma_1, \dots, \sigma_k, 1, \dots, 1) P_k V^\top$  by its SVD, with  $P_k$  the projector onto the first  $k$  coordinates. Changing the sign of the last column of  $U$ , or of the last row of  $V^\top$ , leaves this product unchanged, because  $P_k$  kills the last coordinate; so both outer factors may be taken in  $\operatorname{GL}^+(b)$ , and  $X \in \operatorname{Mon}(S_b)$ . For  $b = 2$ ,  $R(\mu J) = (1 - \mu^4) C(\mu J)$  is a real multiple of a rotation, so  $\operatorname{Mon}(S_2) \subseteq \operatorname{CO}^+(2) \cup \{0\}$ , and  $R(J) = 0$ . Altogether  $\operatorname{Mon}(S_b) = \operatorname{GL}^+(b) \cup \mathcal{M}_{\leq b-2}$  for  $b \geq 3$  and  $\operatorname{CO}^+(2) \cup \{0\}$  for  $b = 2$ , which is the clause; the steps above complete the sketch recorded in the main text.

**BOFT.** Group the coordinates into runs of  $b/2$  consecutive coordinates, indexed by  $p = 0, 1, \dots, 2n/b - 1$ . Up to one fixed relabelling of coordinates, the  $l$ -th butterfly factor of BOFT ( $l = 1, \dots, m'$ ) is block-diagonal with one orthogonal  $b \times b$  block, an exact Cayley image, on each pair of runs whose indices differ exactly in binary digit  $l - 1$  (Dao et al., 2019; Liu et al., 2023b); for  $l = 1$  these are  $n/b$  consecutive blocks of size  $b$ , block OFT. Two runs lie in one component of the factor hypergraph iff their indices agree in all digits from  $m'$  on, so the components consist of  $2^{m'}$  runs,  $b 2^{m'-1}$  coordinates. HF's BOFT multiplies the output by a trainable scale,  $W \mapsto \operatorname{diag}(s) W R$ , and left and right multiplications commute, so after  $K$  cycles the weight is  $\operatorname{diag}(s_K \cdots s_1) W_0 R_1 \cdots R_K$ ; the products of scale vectors already fill  $\mathbb{R}^m$  at  $K = 1$ . The orthogonal part is generated by the set  $S$  of products of the  $m'$  butterfly factors. Let  $H_l$  be the group of all block rotations of level  $l$ , connected;  $S$  lies in the group they generate, and setting all but one factor to the identity shows that  $S$  contains a neighbourhood of 1 in each  $H_l$ , the exact Cayley image being open. By (b) and (c),  $\operatorname{Mon}(S)$  is the connected Lie subgroup with Lie algebra  $\bigoplus_{\mathbb{C}} \mathfrak{so}(C)$ , which is the group of block rotations on the components,  $P(\prod \operatorname{SO}(b 2^{m'-1})) P^\top$  for the permutation  $P$  that makes the components consecutive. This proves the reach. If  $b 2^{m'-1} = n$  there is one component and the reach is  $\operatorname{diag}(\mathbb{R}^m) \cdot W_0 \cdot \operatorname{SO}(n)$ . If  $c := b 2^{m'-1} < n$ , the group  $G = P(\prod \operatorname{SO}(c)) P^\top$  has dimension  $n(c - 1)/2 < n(n - 1)/2$ ; and for  $W_0$  of full column rank the reachable sets differ (for  $W_0 = 0$  both are  $\{0\}$ ). Indeed, if  $\operatorname{diag}(\mathbb{R}^m) W_0 G = \operatorname{diag}(\mathbb{R}^m) W_0 \operatorname{SO}(n)$ , then each  $R \in \operatorname{SO}(n)$  satisfies  $W_0 R = \operatorname{diag}(s) W_0 g$  for some  $s$  and  $g \in G$ , so  $M = R g^{-1}$  satisfies  $W_0 M = \operatorname{diag}(s) W_0$ : every row  $w_i$  of  $W_0$  is an eigenvector of  $M^\top$ . These rows span  $\mathbb{R}^n$ , so  $M^\top$  is diagonalisable with real eigenvalues; being orthogonal,  $M$  is an orthogonal involution  $I - 2P_E$  whose  $-1$ -eigenspace  $E$  is spanned by the rows of  $W_0$  it contains. So  $M$  ranges over a finite set  $F_0$ , and  $\operatorname{SO}(n) \subseteq F_0 G$ , a finite union of submanifolds of lower dimension, which is impossible. With  $m' = 1$  the single level is block OFT, so HF's default `boft_n_butterfly_factor=1` is block OFT times  $\operatorname{diag}(s)$ .

*A single BOFT.* Its parameters number  $m' \frac{n}{b} \binom{b}{2}$ , so its image has at most this dimension (Auxiliary result B.9(b)), and  $m' \frac{n}{b} \frac{b(b-1)}{2} < \frac{n(n-1)}{2}$  iff  $m'(b - 1) < n - 1$ ; the image then has positive codimension in  $\operatorname{SO}(n)$  and is Haar-null. At full depth,  $2^{m'} = 2n/b$  runs form one component, and for every pair of coordinates  $(i, j)$  there is a chain  $j = c_0, c_1, \dots, c_{m'} = i$  in which  $c_{l-1}$  and  $c_l$  lie in a common block of the  $l$ -th factor applied: each factor lets the run index flip one binary digit, a different one for each factor, and lets the position inside the run be chosen freely. So the entry  $(i, j)$  of the product, a polynomial in the block entries, is nonzero at a product of signed permutation matrices of determinant 1 (one per block) that routes  $j$  to  $i$  along the chain. It is therefore not identically zero on the connected group  $\prod_l H_l$ , and since the Cayley parametrisation is a diffeomorphism onto an open subset of that group, the entry is nonzero for parameters outside a nowhere-dense analytic subset

([Auxiliary result B.10](#)). All  $n^2$  entries are nonzero for generic parameters, so the matrices are dense.

*Corollary 5.* With `apply_GS=False`, one HRA cycle applies  $H_{u_1} \cdots H_{u_r}$  on the input side. Let  $S_k = \{R \in \text{SO}(n) : \text{rank}(R - I) \leq k\}$  for even  $k$ . A product of  $r$  reflections lies in  $\text{SO}(n)$  ( $r$  even) and has  $\text{rank}(R - I) \leq r$  ([Auxiliary result B.22](#)); conversely, if  $R \in S_r$  then  $j = \text{rank}(R - I)$  is even and  $R$  is a product of exactly  $j$  reflections ([Auxiliary result B.22](#)), which pairs  $H_u H_u = I$  pad to  $r$ . So one cycle applies exactly the elements of  $S_r$ , and by (b)  $R_K = W_0 \cdot S_r^K$ . Next,  $S_r^K = S_{Kr}$ . For  $\subseteq$ ,  $R_1 R_2 - I = R_1(R_2 - I) + (R_1 - I)$  gives  $\text{rank}(R_1 R_2 - I) \leq \text{rank}(R_2 - I) + \text{rank}(R_1 - I)$  ([Auxiliary result B.12\(a\)](#)). For  $\supseteq$ , write  $R \in S_{Kr}$  as a product of  $j = \text{rank}(R - I) \leq Kr$  reflections,  $j$  even, split them into  $K$  consecutive groups of even size at most  $r$ , and pad each group with pairs to size  $r$ . Finally, since  $W_0$  is injective,  $\text{rank}(W_0 R - W_0) = \text{rank}(R - I)$ , so  $W_0 \cdot S_{Kr} = (W_0 + \mathcal{M}_{\leq Kr}) \cap W_0 \cdot \text{SO}(n)$ . When  $Kr \geq n$ ,  $S_{Kr} = \text{SO}(n)$ , which gives the last sentence.

*Corollary 6.* By [Proposition IV.2](#) and [§C.39\(8\)](#), each period of these methods is a period of the linear method  $\theta_k + \mathcal{L}_k C$  with shape  $L_k = \text{im } \mathcal{L}_k$ , so  $R_K = \theta_0 + L_1 + \cdots + L_K$  for the chosen shapes. For GaLore with left projection,  $L_k = \{Z : \text{col } Z \subseteq \text{col } P_k\}$ , and column spaces that together span  $\mathbb{R}^m$  give all of  $\mathbb{R}^{m \times n}$  after  $\lceil m/r \rceil$  periods; for LISA, layer blocks covering every layer do the same; for MeZO,  $L_k = \mathbb{R} z_k$ , and  $z_k$  running through a basis give  $\Theta$ . A fixed linear shape  $L$  is its own additive monoid, so it reaches only  $\theta_0 + L$  ([corollary 2](#)). A GaLore period moves a projected matrix by an update of rank at most  $r$ ; a LISA period moves each sampled matrix by an unconstrained update, and a MeZO step by a multiple of the Gaussian  $z_k$ , whose restriction to each matrix has full rank almost surely. Flora’s steps contain a full-rank term ([§C.37](#)).

*Corollary 7.* For orthogonal  $R$ ,  $(WR)(WR)^\top = WW^\top$  and  $(RW)^\top(RW) = W^\top W$ ;  $WR$  and  $RW$  have the singular values of  $W$ . By induction over cycles, every input-side rebasing preserves  $K_{\text{out}}$  and every output-side one  $K_{\text{in}}$ , and both preserve the spectrum ([Corollary II.6](#)).  $\square$

**REMARK** (numerical checks). The component sizes 4, 8, 16, 32, 64 for  $n = 64$ ,  $b = 4$  and  $m' = 1, \dots, 5$  reported after [Theorem V.3](#) are the  $b 2^{m'-1}$  of the BOFT paragraph; the Loewner-monotone decrease of  $K_{\text{out}}$  over 200 default-OFT merges, the contraction bound for random skew  $Q$  with  $\|Q\|_{\text{op}} \leq 2^{1/4}$ , and  $R = 0$  at  $b = 2$ ,  $Q = J$ , instantiate the Cayley-Neumann paragraph ([Appendix E](#)).

## C.44 Proposition V.4

**PROPOSITION V.4** (combination rules: gauge and naturality). (a) A rule  $\Phi((B_i, A_i)_i)$  for combining LoRA factors descends to a function of the adapters  $(B_i A_i)_i$  iff it is constant on the fibres of  $(B_i, A_i) \mapsto (B_i A_i)$ . On the full-rank locus this is equivalent to  $\prod_i \text{GL}_r$ -invariance. It is also equivalent for  $\Phi$  continuous on the whole factor space; this clause rests on the unique closed orbit in each fibre, which lies in the closure of every orbit of the fibre.

- Task arithmetic  $\sum_i w_i s_i B_i A_i$ , with  $s_i$  HF’s scaling ( $\alpha_i/r_i$ , or  $\alpha_i/\sqrt{r_i}$  under rsLoRA), is invariant and lands in  $\text{LoRA}_{\sum_i r_i}$ . It is HF’s cat with merged scale 1.
- LoraHub’s factor-wise rule  $(\sum_i w_i B_i)(\sum_j w_j A_j)$  is not invariant. Replacing  $(B_2, A_2)$  by  $(cB_2, A_2/c)$  changes it by

$$\sum_{l \neq 2} w_l w_2 [(c-1)B_2 A_l + (c^{-1}-1)B_l A_2].$$

HF's `add_weighted_adapter(..., combination_type="linear")` is a factor-wise rule of the same type. It weights  $A_i$  by  $\text{sign}(w_i)\sqrt{|w_i s_i|}$  and  $B_i$  by  $\sqrt{|w_i s_i|}$  and sums the factors separately, so it has cross terms. On adapters that share a frozen  $A$  it is still not exact up to a global factor, since the relative weights become  $\sqrt{|w_i s_i|}$ . HF's non-SVD ties, `dare_*` and `magnitude_prune` variants also act on factors; the `*_svd` variants act on full updates.

- Poly and MHR are different. Their skill inventories are trained jointly through the product-of-sums map, which is invariant under the diagonal  $\text{GL}_r$ , one common  $g$  for every skill. Numerically, for fixed generic routing with fewer skills than tasks and full-rank factors, this is the whole continuous fibre. With at least as many skills as tasks, or once the routing is counted as a parameter, there are further symmetries (skill permutations, and re-mixings  $\alpha \mapsto \alpha M$  with  $M\mathbf{1} = \mathbf{1}$ ). Their cross terms are a design choice of a jointly trained parametrisation, and raise no question of well-definedness.

(b) Well-definedness versus exactness. A joint gauge fixing makes a factor-wise rule well defined. For rank-preserving product-of-sums rules, exactness (agreement with task arithmetic for all weights  $w$ ) needs degree one in  $w$ , which for injective factors forces a shared factor; `cat`, which concatenates and raises the rank, is exact without one. A frozen shared factor (FFA-LoRA; LoRA-FA) gives  $(\sum_i w_i B_i)A = \sum_i w_i B_i A$  exactly. LoraHub with a shared  $A$  still returns  $(\sum_j w_j) \sum_i w_i B_i A$ , exact only up to the global factor  $\sum_j w_j$ , since its weights are unnormalised. A per-module SVD canonical form leaves a  $(\mathbb{Z}/2)^r$  of sign flips per module, plus rotations within repeated singular values, and these change the cross terms. What is needed is a joint frame that produces a shared factor, as KnOTS's joint re-factorisation of the concatenated updates through a shared  $U\Sigma$  does. Federated LoRA shows the difference. FedAvg-LoRA clients start each round from the same broadcast  $(B, A)$ , so their failure,  $\frac{1}{2}(B_1 + B_2) \cdot \frac{1}{2}(A_1 + A_2) \neq \frac{1}{2}(B_1 A_1 + B_2 A_2)$ , is bilinearity, and it persists in a fully fixed gauge.

(c) Naturality. On updates with  $N$  coordinates, write  $T_N$ ,  $B_N$  and  $\text{Mon}_N$  for the invertible diagonal, signed permutation and monomial matrices. Task arithmetic is  $\text{GL}(\Theta)$ -equivariant on updates. DARE is coordinatewise and magnitude-free,  $D(m \odot \tau) = m \odot (D\tau)$  for every invertible diagonal  $D$  and the same mask  $m$ . So DARE is exactly equivariant under the torus  $T_N$ , sign flips included, for each mask, and under permutations in distribution; its distributional group is exactly  $\text{Mon}_N$ . TIES with global trimming (top- $k$  magnitude, sign election by total mass as in the TIES paper and HF's default "total", disjoint mean) is equivariant under  $\mathbb{R}^\times \cdot B_N$ , homotheties times signed permutations, generically (off magnitude ties at the threshold). With HF's `majority_sign_method="frequency"`, count ties are broken to +1 and the group drops to  $\mathbb{R}_{>0} \cdot S_N$ . Per-tensor trimming changes the group to  $\prod_b (\mathbb{R}^\times \cdot B_{N_b})$ , extended by permutations of equal-size blocks. DARE followed by full TIES (with trimming) is  $\mathbb{R}^\times \cdot B_N$ -equivariant in distribution; HF's and MergeKit's `dare_ties` has no magnitude trim, so with "total" election it is exactly  $T_N$ -equivariant per mask, and its group contains  $\text{Mon}_N$ . None of the sparsifying rules (DARE, TIES, DARE-TIES) is  $O(N)$ -equivariant. Applied to factors and summed across adapters (HF's non-SVD variants), all of them are gauge-dependent, and under independent per-adapter gauges even DARE fails, through the cross terms of (a). DARE's masking commutes with  $T_r$  exactly for each mask and with  $\text{Mon}_r$  in distribution, so a single masked adapter, or a factor-wise sum under one gauge shared by all adapters, is  $T_r$ -invariant per mask and  $\text{Mon}_r$ -invariant in distribution.

*Proof.* Write  $\pi((B_i, A_i)_i) = (B_i A_i)_i$ , and let adapter  $i$  have rank  $r_i$  (the statement's  $r$  when all ranks agree); its gauge group  $\text{GL}_{r_i}$  acts by  $(B_i g_i^{-1}, g_i A_i)$ , and  $\pi$  is invariant under  $\prod_i \text{GL}_{r_i}$ .

(a) *Descent.* If  $\Phi = \varphi \circ \pi$ , then  $\Phi$  is constant on the fibres of  $\pi$ . Conversely, if  $\Phi$  is constant on fibres,  $\varphi(\pi(x)) := \Phi(x)$  is a well-defined function on the image of  $\pi$ .

*Full-rank locus.* Let  $\text{rank } B_i = \text{rank } A_i = r_i$  and  $B'_i A'_i = B_i A_i$ . The product has rank  $r_i$ , so  $\text{rank } B'_i = \text{rank } A'_i = r_i$ , and the full-rank locus is a union of whole fibres. Moreover  $\text{col } B'_i = \text{col}(B_i A_i) = \text{col } B_i$ , so  $B'_i = B_i g_i^{-1}$  for a unique  $g_i \in \text{GL}_{r_i}$ ; then  $B_i A_i = B_i g_i^{-1} A'_i$ , and injectivity of  $B_i$  gives  $A'_i = g_i A_i$ . So each fibre in the full-rank locus is a single  $\prod_i \text{GL}_{r_i}$ -orbit (as in [Theorem II.2](#)), and there constancy on fibres is the same as invariance.

*Continuous rules.* Let  $\Phi$  be continuous on the whole factor space and invariant, and let  $x, x'$  lie in the fibre over  $(W_i)_i$ ,  $\text{rank } W_i = k_i \leq r_i$ . By [Auxiliary result B.27\(c\)](#), in each factor the pairs with product  $W_i$  and both factors of rank  $k_i$  form one  $\text{GL}_{r_i}$ -orbit  $O_{W_i}$ , which lies in the closure of every orbit of the fibre. Applying this factor by factor, there are gauges  $g(t), g'(t) \in \prod_i \text{GL}_{r_i}$  with  $g(t) \cdot x \rightarrow o$  and  $g'(t) \cdot x' \rightarrow o'$  as  $t \rightarrow 0$ , where  $o, o' \in \prod_i O_{W_i}$ , and  $o' = h \cdot o$  for some gauge  $h$ . By invariance and continuity,

$$\Phi(x) = \lim_{t \rightarrow 0} \Phi(g(t) \cdot x) = \Phi(o) = \Phi(h \cdot o) = \Phi(o') = \lim_{t \rightarrow 0} \Phi(g'(t) \cdot x') = \Phi(x').$$

So  $\Phi$  is constant on fibres; the converse holds because orbits lie in fibres. The review record graded this clause as a sketch; with the degeneration of [Auxiliary result B.27\(c\)](#), whose reason is given in [Appendix B](#), the steps above complete it. Invariance alone does not suffice off the full-rank locus: on the invariant open set  $\{\text{rank } A = r\}$  the projector  $P_{\text{row } A}$  is  $\text{GL}_r$ -invariant, since  $\text{row}(gA) = \text{row } A$ , yet  $(0, A)$  and  $(0, A')$  with  $\text{row } A \neq \text{row } A'$  lie in the fibre over 0 and receive different values. This  $\Phi$  has no continuous extension to the whole factor space.

*Task arithmetic.*  $\sum_i w_i s_i B_i A_i$  depends only on the products, so it is invariant, and

$$\sum_i w_i s_i B_i A_i = \left[ w_1 s_1 B_1 \mid \cdots \mid w_N s_N B_N \right] \begin{bmatrix} A_1 \\ \vdots \\ A_N \end{bmatrix}$$

is a LoRA update of rank at most  $\sum_i r_i$  with scale 1, the concatenation  $\text{cat}$  of [Observation I.7](#).

*LoraHub.* Expanding,  $(\sum_i w_i B_i)(\sum_j w_j A_j) = \sum_{i,j} w_i w_j B_i A_j$ . Under the gauge  $(B_2, A_2) \mapsto (cB_2, A_2/c)$  the terms with  $i, j \neq 2$  and the term  $w_2^2 B_2 A_2$  are unchanged, each  $w_l w_2 B_l A_2$  ( $l \neq 2$ ) gains the factor  $c$ , and each  $w_l w_2 B_l A_2$  gains  $c^{-1}$ , which gives the displayed change. For  $c \neq 1$ , nonzero weights and factors for which the matrices  $B_2 A_l, B_l A_2$  ( $l \neq 2$ ) are linearly independent (true for generic factors), the change is nonzero. HF's linear combination is  $(\sum_i \beta_i B_i)(\sum_j \alpha_j A_j)$  with  $\beta_i = \sqrt{|w_i s_i|}$  and  $\alpha_j = \text{sign}(w_j) \sqrt{|w_j s_j|}$  ([Mangrulkar et al., 2022](#)), a product of sums with the same cross terms and the same failure. If all adapters share  $A$ , it equals  $(\sum_j \alpha_j) \sum_i \beta_i B_i A$ , which is a fixed multiple of  $\sum_i w_i s_i B_i A$  for all  $B_i$  only if the  $\beta_i$  are proportional to the  $w_i s_i$ , that is, only if all  $|w_i s_i|$  with  $w_i s_i \neq 0$  are equal (and the signs agree); for instance  $w_1 s_1 = 1$  and  $w_2 s_2 = 4$  give relative weights 1 : 2 instead of 1 : 4.

*Poly and MHR.* For task  $t$  with routing weights  $\alpha_{tj}$  over skills  $j$  the update is  $W_t = (\sum_j \alpha_{tj} B_j)(\sum_j \alpha_{tj} A_j)$ . A common  $g$  gives  $(\sum_j \alpha_{tj} B_j g^{-1})(\sum_j \alpha_{tj} g A_j) = W_t$ , so the diagonal  $\text{GL}_r$  preserves the map. Per-skill gauges  $(g_j)$  that preserve it for all routings in an open set are diagonal on any pair of mixed skills with complementary column spaces:  $W_t = \sum_{i,j} \alpha_{ti} \alpha_{tj} B_i A_j$  is a polynomial in  $\alpha$ , and comparing the coefficients of  $\alpha_{ti} \alpha_{tj}$ ,  $i \neq j$ , gives

$$B_i(g_i^{-1} g_j - I)A_j + B_j(g_j^{-1} g_i - I)A_i = 0.$$

If  $\text{col } B_i \cap \text{col } B_j = 0$ , the two terms vanish separately, and for injective  $B_i$  and surjective  $A_j$  this forces  $g_i = g_j$ . That for one fixed generic routing with fewer skills than tasks the



continuous fibre is exactly the diagonal orbit is the numerical check reported below. With at least as many skills as tasks the fibre is larger. If there are more skills than tasks, a vector  $v \neq 0$  with  $\sum_j v_j \alpha_{tj} = 0$  for every  $t$  exists, and  $B_j \mapsto B_j + v_j X$  leaves every  $\sum_j \alpha_{tj} B_j$  unchanged for every  $X$ . If there are equally many and the routing matrix is invertible, the task-wise sums  $(C_t, D_t) = (\sum_j \alpha_{tj} B_j, \sum_j \alpha_{tj} A_j)$  are free coordinates, and each task has its own  $\text{GL}_r$  gauge. Counting the routing as a parameter, relabelling the skills together with the columns of  $\alpha$  preserves the map, and so does  $(\alpha, B) \mapsto (\alpha M, M^{-1}B)$  for invertible  $M$  with  $M\mathbf{1} = \mathbf{1}$ , where  $B$  stacks the skills (and likewise for  $A$ ), since  $(\alpha M)(M^{-1}B) = \alpha B$  and  $M\mathbf{1} = \mathbf{1}$  keeps the rows of  $\alpha$  summing to 1. Poly and MHR train skills and routing jointly through this map, so it is the method's own  $\rho$ ; the question of (a) concerns combining modules that were factorised independently.

(b) *Gauge fixing*. If a section  $\sigma$  of  $\pi$  chooses the factors from the products, then  $\Phi \circ \sigma$  is a function of the products. *Exactness*. Task arithmetic  $T(w) = \sum_i w_i B_i A_i$  (scales absorbed) is linear in  $w$ . Consider rank-preserving products of sums in which each adapter's factor carries its own weight  $w_i$ , in one sum or in both. If both sums are weighted, as in LoraHub, the rule  $P(w)$  is homogeneous of degree two in  $w$ , and  $P(w) = T(w)$  for all  $w$  would give, replacing  $w$  by  $\lambda w$ ,  $\lambda^2 P(w) = \lambda T(w)$  for all  $\lambda$ , hence  $T \equiv 0$ . So an exact rule of this kind has degree one in  $w$ : one of its factors does not depend on  $w$ , say  $(\sum_i w_i B_i)X$ . Then exactness for all  $w$  means  $B_i X = B_i A_i$  for every  $i$ , and for injective  $B_i$  this gives  $A_i = X$ : the  $A$ -factor is shared. The symmetric case  $X(\sum_j w_j A_j)$  with surjective  $A_j$  forces a shared  $B$ . cat realises  $T(w)$  exactly, as shown in (a), with inner dimension  $\sum_i r_i$ . With a frozen shared  $A$ ,  $(\sum_i w_i B_i)A = \sum_i w_i B_i A$ , and LoraHub's rule gives  $(\sum_i w_i B_i)(\sum_j w_j)A$ .

*SVD canonical forms*. If  $B_i A_i = U_i \Sigma_i V_i^\top$  and the canonical factors are  $B_i = U_i \Sigma_i^{1/2}$ ,  $A_i = \Sigma_i^{1/2} V_i^\top$  (or any convention built from  $U_i, \Sigma_i, V_i$ ), then changing the signs of the  $j$ -th column of  $U_i$  and of  $V_i$  gives another SVD and another canonical pair, multiplying column  $j$  of  $B_i$  and row  $j$  of  $A_i$  by  $-1$ . The product  $B_i A_i$  is unchanged, while a cross term  $B_i A_l$  ( $l \neq i$ ) changes by  $-2 b_{ij} a_{lj}^\top$ , where  $b_{ij}$  is column  $j$  of  $B_i$  and  $a_{lj}^\top$  row  $j$  of  $A_l$ , which is nonzero generically. When  $\sigma_j = \sigma_{j+1}$ , rotating the corresponding columns of  $U_i$  and  $V_i$  by the same  $O \in \text{O}(2)$  also preserves the product and changes the cross terms. *KnOTS*. Write the horizontal concatenation  $[\Delta_1 \mid \cdots \mid \Delta_N] = U\Sigma[V_1^\top \mid \cdots \mid V_N^\top]$  by an SVD (Stoica et al., 2024); then  $\Delta_i = (U\Sigma)V_i^\top$  share the left factor  $U\Sigma$ , and  $(U\Sigma)(\sum_i w_i V_i^\top) = \sum_i w_i \Delta_i$  exactly. *FedAvg-LoRA*. Expanding,

$$\frac{1}{2}(B_1 + B_2) \cdot \frac{1}{2}(A_1 + A_2) - \frac{1}{2}(B_1 A_1 + B_2 A_2) = -\frac{1}{4}(B_1 - B_2)(A_1 - A_2),$$

which is nonzero whenever  $(B_1 - B_2)(A_1 - A_2) \neq 0$ , whatever gauge the broadcast pair is in.

(c) Write a merge rule as  $\Psi(\tau_1, \dots, \tau_K)$  for task vectors  $\tau_k \in \mathbb{R}^N$  (or the matrices with  $N$  entries, flattened). It is  $g$ -equivariant if  $\Psi(g\tau_1, \dots, g\tau_K) = g\Psi(\tau_1, \dots, \tau_K)$ , read for random rules either for each realisation of the randomness (per mask) or as equality of laws (in distribution).

*Task arithmetic* is linear, hence  $\text{GL}(\Theta)$ -equivariant.

*DARE*.  $\text{DARE}(\tau) = m \odot \tau / (1 - p)$ , with a mask  $m$  of independent Bernoulli( $1 - p$ ) entries,  $0 < p < 1$ , drawn independently for each task, followed by a weighted sum. For invertible diagonal  $D$ ,  $D(m \odot \tau) = m \odot (D\tau)$ , so  $\text{DARE}(D\tau) = D \text{DARE}(\tau)$  for each mask, and the weighted sum preserves this. For a permutation matrix  $P$ ,  $P(m \odot \tau) = (Pm) \odot (P\tau)$  and  $Pm$  has the law of  $m$ , so  $\text{DARE}(P\tau)$  has the law of  $P \text{DARE}(\tau)$ , jointly over the tasks. Together,  $\text{Mon}_N = S_N \ltimes T_N$  acts equivariantly in distribution. Conversely, suppose  $\text{DARE}(g\tau)$  and  $g \text{DARE}(\tau)$  have the



same law for every  $\tau$ , and take  $\tau = e_k$ . The law of  $g \text{ DARE}(e_k)$  is supported on the two points 0 and  $ge_k/(1-p)$ , while  $\text{DARE}(ge_k)$  takes the  $2^q$  distinct values  $m \odot ge_k/(1-p)$ , each with positive probability, where  $q$  is the number of nonzero entries of  $ge_k$ . So  $q = 1$ : every column of  $g$  has a single nonzero entry, and  $g \in \text{Mon}_N$ .

*TIES with global trimming* (Yadav et al., 2023). Each  $\tau_k$  is trimmed to its  $k_0$  entries of largest magnitude, giving  $\hat{\tau}_k$ ; the sign  $\gamma_i = \text{sign}(\sum_k \hat{\tau}_{k,i})$  is elected by total mass; and the output is  $\lambda$  times the disjoint mean,  $\frac{1}{|A_i|} \sum_{k \in A_i} \hat{\tau}_{k,i}$  over  $A_i = \{k : \text{sign } \hat{\tau}_{k,i} = \gamma_i\}$ . Let  $g = cPS$  with  $c \neq 0$ ,  $P$  a permutation matrix and  $S = \text{diag}(s_1, \dots, s_N)$  a diagonal sign matrix, and let  $j$  be the coordinate that  $P$  moves to  $i$ . The magnitudes of  $g\tau_k$  are  $|c|$  times the permuted magnitudes of  $\tau_k$ , so, off ties at the threshold, trimming keeps the corresponding entries and  $\widehat{g\tau_k} = g\hat{\tau}_k$ . The elected sign at  $i$  becomes  $\text{sign}(c)s_j\gamma_j$ , the set  $A_i$  for the transformed input is  $A_j$ , and the mean at  $i$  is  $cs_j$  times the mean at  $j$ . So the output is  $g$  times the output, and TIES is  $\mathbb{R}^\times \cdot B_N$ -equivariant off ties. A non-scalar positive diagonal fails: with  $N = 2$ , one task,  $k_0 = 1$ ,  $\tau = (2, 1)$  and  $D = \text{diag}(1, 3)$ , TIES returns  $\lambda(2, 0)$  on  $\tau$  and  $\lambda(0, 3) \neq D\lambda(2, 0)$  on  $D\tau$ .

*Frequency election*. Here  $\gamma_i = \text{sign}(\sum_k \text{sign } \hat{\tau}_{k,i})$ , with 0 replaced by +1. Positive homotheties and permutations leave the counts unchanged or permute them, so the argument above gives  $\mathbb{R}_{>0} \cdot S_N$ -equivariance. Sign changes fail on ties, which have positive probability because the counts are integers. With two tasks and  $\hat{\tau}_{1,i} = 1$ ,  $\hat{\tau}_{2,i} = -1$ , the tie elects +1 and the output at  $i$  is  $\lambda$ ; after flipping coordinate  $i$  the entries are  $-1, +1$ , the tie again elects +1, and the output at  $i$  is  $\lambda$ , not  $-\lambda$ . The same example defeats  $-I$ .

*Per-tensor trimming*. If coordinates are grouped into tensors  $b$  of sizes  $N_b$  and each tensor is trimmed to its own top entries, an independent homothety  $c_b$  and signed permutation inside each tensor preserve the trimmed sets, and permuting tensors of equal size (and equal  $k_0$ ) preserves them too; the election and the mean are coordinatewise, so the argument above applies blockwise.

*DARE followed by TIES*. For  $g \in \mathbb{R}^\times \cdot B_N \subseteq \text{Mon}_N$ , the masked inputs  $(\text{DARE}(g\tau_k))_k$  have the law of  $(g \text{ DARE}(\tau_k))_k$ , and TIES commutes with  $g$  off magnitude ties among nonzero entries (ties among entries zeroed by the mask do not matter, since a kept zero is zero), so the composite is equivariant in distribution. HF's and MergeKit's `dare_ties` (Goddard et al., 2024; Mangrulkar et al., 2022) omit the trim: for invertible diagonal  $D$  and fixed masks,  $\text{DARE}(D\tau_k) = D \text{ DARE}(\tau_k)$ , the elected sign at  $i$  is multiplied by  $\text{sign } d_i$ , the sets  $A_i$  are unchanged and the mean is multiplied by  $d_i$ ; so it is exactly  $T_N$ -equivariant per mask, and with permutations in distribution its group contains  $\text{Mon}_N$ .

*No rotations*. Let  $R$  rotate the plane of  $e_1, e_2$  by  $30^\circ$ , so  $Re_1 = (\cos 30^\circ, \sin 30^\circ, 0, \dots)$ . For DARE,  $R \text{ DARE}(e_1)$  takes the values 0 and  $Re_1/(1-p)$ , each with two nonzero entries or none, while  $\text{DARE}(Re_1)$  has exactly one nonzero entry with probability  $2p(1-p) > 0$ ; the laws differ. For TIES with one task and  $k_0 = 1$ ,  $\text{TIES}(e_1) = \lambda e_1$ , so  $R \text{ TIES}(e_1)$  has two nonzero entries, while  $\text{TIES}(Re_1) = \lambda \cos 30^\circ e_1$  has one. With one task, `dare_ties` reduces to DARE, and DARE followed by trimmed TIES returns, with positive probability, the trimmed vector  $\lambda \cos 30^\circ e_1/(1-p)$  on  $Re_1$ , which no value of  $R \text{ TIES}(\text{DARE}(e_1)) \in \{0, \lambda Re_1/(1-p)\}$  matches.

*Factor-level rules*. HF's non-SVD variants apply the sparsifier to each  $B_i$  and each  $A_j$  with independent masks and then form  $(\sum_i v_i \text{DARE}(B_i))(\sum_j v_j \text{DARE}(A_j))$  (and the analogues for TIES). The  $B$ -masks and the  $A$ -masks are independent and DARE is unbiased,  $\mathbb{E} \text{DARE}(X) = X$ , so the expected output is the LoraHub-type rule  $(\sum_i v_i B_i)(\sum_j v_j A_j)$ . A gauge  $(cB_2, A_2/c)$  on one adapter changes this expectation by the formula of (a), which is nonzero for generic factors, so the output's law changes: even DARE is not invariant under

independent per-adapter gauges, not even in distribution. On the other hand, let  $g \in T_r$  act on all adapters at once,  $B_i \mapsto B_i g^{-1}$  and  $A_j \mapsto g A_j$ . Right multiplication by a diagonal matrix scales columns, which commutes with an entrywise mask, so  $M \odot (B_i g^{-1}) = (M \odot B_i) g^{-1}$  and  $M' \odot (g A_j) = g(M' \odot A_j)$ , and for every choice of masks

$$\left( \sum_i v_i M_i \odot B_i g^{-1} \right) \left( \sum_j v_j M'_j \odot g A_j \right) = \left( \sum_i v_i M_i \odot B_i \right) g^{-1} g \left( \sum_j v_j M'_j \odot A_j \right).$$

For a permutation  $g = P$ ,  $M \odot (B_i P^\top) = ((MP) \odot B_i) P^\top$ , and  $MP$  has the law of  $M$ ; with all masks permuted by the same  $P$  their joint law is unchanged, so the output's law is invariant. A single masked adapter is the case of one summand. This gives  $T_r$ -invariance per mask and  $\text{Mon}_r$ -invariance in distribution.  $\square$

**REMARK** (numerical checks). The checks reported after [Proposition V.4](#), namely DARE's per-mask equivariance under a random diagonal to  $2 \times 10^{-15}$ , TIES's homogeneity and its failure by 5.6 under a positive diagonal, the LoraHub formula with three modules, the order-one change of factor-wise DARE under a one-adapter gauge against  $10^{-15}$  under a shared one, and the Poly fibre dimensions 32 (two tasks) and  $4 = \dim \text{GL}_2$  (four tasks) for three skills,  $r = 2$  and  $6 \times 6$  weights, instantiate the statements proved above; the last one is the numerical part of the Poly bullet ([Appendix E](#)).

### C.45 Definition VI.1

**DEFINITION VI.1** (extensions; mergeability).

- An **extension** is a function-preserving pointed embedding

$$\iota : (\Theta, f, \theta_0) \rightarrow (\Theta \times \Theta', \tilde{f}, (\theta_0, \theta'_0)), \quad \iota(\theta) = (\theta, \theta'_0),$$

with  $\tilde{f}(\theta, \theta'_0, -) = f(\theta, -)$ . The point  $\theta'_0$  is the **neutral point** (“identity at initialisation”).

- An **unpointed extension** has no neutral point.
- A method on an extension is **mergeable** if there is  $\mu : Q \rightarrow \Theta$  with  $\Phi_{\tilde{f}} \circ \rho = \Phi_f \circ \mu$ : a lift along the realisation.
- It is **box-locally mergeable** if the modified box function stays in the box's function class: linear maps for a linear box, affine maps when the box has a bias slot. [Theorem VI.3](#) gives sufficient conditions, with explicit side conditions (position-uniformity, no weight sharing, bias slots), under which this yields a merge into the base network.

A definition needs no proof. We check the facts that the definition and the square after it use. Here  $\Phi_f : \Theta \rightarrow C^\infty(X, Y)$ ,  $\theta \mapsto f(\theta, -)$ , is the realisation, and a box's function class consists of the maps it can compute with some weight, applied identically at every position.

*Verification of the claims in Definition VI.1. Extensions.* The map  $\iota(\theta) = (\theta, \theta'_0)$  is smooth, injective and immersive, and a homeomorphism onto the closed set  $\Theta \times \{\theta'_0\}$ , so it is an embedding. It is pointed,  $\iota(\theta_0) = (\theta_0, \theta'_0)$ , and function-preserving,  $\Phi_{\tilde{f}} \circ \iota = \Phi_f$ , which is the displayed condition.

*Weight-space methods merge.* A method  $(Q, q_0, \rho)$  on  $\Theta$  itself, read on the trivial extension  $(\Theta'$  a point), satisfies the merge condition with  $\mu = \rho$ , the arrow to full fine-tuning of [Observation I.4\(a\)](#). More generally, a method on an extension whose  $\rho$  factors as  $\iota \circ \mu$  merges by  $\mu$ .

Merges are lifts, and need not be unique. The realisation  $\Phi_f$  need not be injective. Every function-preserving reparametrisation  $\phi$  of the base gives another lift  $\phi \circ \mu$ ; permutations of hidden units are examples (Lemma VI.2). Let a pointed method merge, with  $\rho(q_0)$  the neutral point  $(\theta_0, \theta'_0)$ . Then

$$\Phi_f(\mu(q_0)) = \Phi_{\tilde{f}}(\theta_0, \theta'_0) = \Phi_f(\theta_0),$$

so  $\mu(q_0)$  realises the pretrained function, without having to equal  $\theta_0$ .

*Box-local mergeability.* If the edit of a method modifies a single linear (affine) box of the base network, and for every  $q$  the modified box function is linear (affine), position-independent and computed by an unshared weight  $W(q)$  of that box, then replacing the box's weight by  $W(q)$  and keeping the others gives a  $\mu$  with  $\Phi_f \circ \mu = \Phi_{\tilde{f}} \circ \rho$ . The side conditions are exactly what makes this replacement legitimate, and Theorem VI.3(a) spells them out.  $\square$

#### C.46 Lemma VI.2

**LEMMA VI.2** (the activation fixes the privileged basis; classical, Sussmann (1992), Chen et al. (1993), Godfrey et al. (2022)). Let  $\sigma : \mathbb{R} \rightarrow \mathbb{R}$  be an elementwise activation that is not affine, and let  $L \in \text{GL}_d$  commute with it as maps on  $\mathbb{R}^d$ :  $L \sigma^{\times d} = \sigma^{\times d} L$ .

- (a) If  $\sigma \in C^2$ , then  $L = PD$  is monomial.
- (b) If  $\sigma$  is ReLU or leaky ReLU with slope  $\alpha \neq 1$ , then  $L = PD$  with  $D$  a positive diagonal.
- (c) If moreover  $\sigma$  is real-analytic at 0 with  $\sigma(t) = \sum a_k t^k$ , each diagonal entry  $c$  of  $D$  satisfies  $ca_k = c^k a_k$  for all  $k$ . So  $c = 1$  for GELU and SiLU, since  $\sigma''(0) \neq 0$ , and  $c = \pm 1$  for tanh.

*Proof.* Write  $\ell_i$  for row  $i$  of  $L$ ; it is nonzero because  $L$  is invertible. Row  $i$  of the identity  $L \sigma^{\times d}(x) = \sigma^{\times d}(Lx)$  reads

$$\sum_j L_{ij} \sigma(x_j) = \sigma(\ell_i \cdot x) \quad \text{for all } x \in \mathbb{R}^d. \quad (27)$$

Once each row has a single nonzero entry, the positions of these entries are distinct, since otherwise a column of  $L$  would vanish; so  $L = PD$  with  $P$  a permutation matrix and  $D$  an invertible diagonal matrix.

(a) For  $j \neq k$ , the mixed partial  $\partial_j \partial_k$  of the left side of (27) vanishes, since each term depends on one coordinate, and the mixed partial of the right side is  $L_{ij} L_{ik} \sigma''(\ell_i \cdot x)$ . Since  $\sigma$  is  $C^2$  and not affine,  $\sigma''(t_0) \neq 0$  for some  $t_0$ , and since  $\ell_i \neq 0$  there is  $x$  with  $\ell_i \cdot x = t_0$ . Hence  $L_{ij} L_{ik} = 0$  for all  $j \neq k$ : row  $i$  has a single nonzero entry.

(b) Let  $\sigma(t) = t$  for  $t \geq 0$  and  $\sigma(t) = \alpha t$  for  $t < 0$ , with  $\alpha \neq 1$  ( $\alpha = 0$  is ReLU). Then  $\sigma$  is smooth off 0, and not differentiable at 0, where its one-sided derivatives 1 and  $\alpha$  differ. The left side of (27) is a sum of functions of separate coordinates, so it is differentiable at  $x$  iff each term with  $L_{ij} \neq 0$  is differentiable in  $x_j$ ; its non-differentiability set is  $\bigcup_{j: L_{ij} \neq 0} \{x_j = 0\}$ . The right side is smooth where  $\ell_i \cdot x \neq 0$ , and at a point with  $\ell_i \cdot x = 0$  its one-sided directional derivatives along  $\pm \ell_i$  are  $\|\ell_i\|^2$  and  $-\alpha \|\ell_i\|^2$ , which are not negatives of each other, so it is not differentiable there; its non-differentiability set is the hyperplane  $\{\ell_i \cdot x = 0\}$ . The two sets are equal. By Auxiliary result B.36(b) the hyperplane is one of the coordinate hyperplanes  $\{x_{j_0} = 0\}$ , and each  $\{x_j = 0\}$  with  $L_{ij} \neq 0$  is contained in it and so equal to it; hence  $L_{ij} \neq 0$  only for  $j = j_0$ . Then (27) reads  $L_{ij_0} \sigma(t) = \sigma(L_{ij_0} t)$  for all  $t$ . At  $t = 1$  this is  $L_{ij_0} = \sigma(L_{ij_0})$ ; if  $L_{ij_0} < 0$  the right side is  $\alpha L_{ij_0}$ , which would force  $\alpha = 1$ . So  $L_{ij_0} > 0$ , and  $D$  is positive.

(c) A permutation matrix commutes with every elementwise map (rule R2 of Theorem VI.3), so  $D \sigma^{\times d} = P^{-1} L \sigma^{\times d} = P^{-1} \sigma^{\times d} L = \sigma^{\times d} P^{-1} L = \sigma^{\times d} D$ , that is,  $c \sigma(t) = \sigma(ct)$  for

each diagonal entry  $c \neq 0$  and all  $t$ . Near 0 both sides are power series,  $\sum_k c a_k t^k = \sum_k a_k c^k t^k$ , so  $c a_k = c^k a_k$  for every  $k$  (Auxiliary result B.10(c)). For GELU,  $\sigma(t) = t \Phi_{\mathcal{N}}(t)$  with the standard normal distribution function  $\Phi_{\mathcal{N}}(t) = \frac{1}{2} + t/\sqrt{2\pi} + O(t^3)$ , so  $a_2 = 1/\sqrt{2\pi} \neq 0$ ; for SiLU,  $\sigma(t) = t/(1 + e^{-t}) = \frac{t}{2} + \frac{t^2}{4} + O(t^4)$ , so  $a_2 = \frac{1}{4} \neq 0$ . In both cases  $c = c^2$ , and  $c \neq 0$  gives  $c = 1$ . For tanh,  $a_3 = -\frac{1}{3} \neq 0$ , so  $c = c^3$  and  $c = \pm 1$ ; both occur, tanh being odd.  $\square$

### C.47 Theorem VI.3

**THEOREM VI.3** (the merge calculus; known in part). *Work with activation-side diagrams in a symmetric monoidal category of continuous, piecewise-smooth maps with copy and discard nodes. (We call copy and the Hadamard product “spiders” by analogy only; in this cartesian setting they do not satisfy the Frobenius law.) A trainable box is a node whose weight ranges over a stated class: all linear maps, or all affine maps when the box has a bias slot. R2 and R4–R7 are identities of maps; R1 is a closure property and R3 a characterisation.*

| Rule                         | Statement                                                                                       | When it holds                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R1 (fusion)                  | composites and parallel sums of linear (affine) boxes are linear (affine)                       | the operators involved are position-independent                                                                                                                                                                                                                                                                                                                                                                                                |
| R2 (permutation slide)       | $P\sigma^{\times d} = \sigma^{\times d}P$                                                       | every elementwise $\sigma$                                                                                                                                                                                                                                                                                                                                                                                                                     |
| R3 (positive-diagonal slide) | $\text{diag}(\ell)\sigma^{\times d} = \sigma^{\times d}\text{diag}(\ell)$ for all $\ell \geq 0$ | iff $\sigma$ is positively homogeneous (ReLU, leaky ReLU; not GELU, SiLU, tanh); such $\sigma$ are piecewise linear                                                                                                                                                                                                                                                                                                                            |
| R4 (spider slide)            | $\text{diag}(\ell)(a \odot b) = (\text{diag}(\ell)a) \odot b$                                   | any $\ell$                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| R5 (cup slide)               | $\langle Mq, k \rangle = \langle q, M^\top k \rangle$                                           | always. Under RoPE, $\langle R_m Mq, R_n k \rangle = \langle q, M^\top R_{n-m} k \rangle$ holds for every $M$ , but the slide onto the key side, $\langle R_m Mq, R_n k \rangle = \langle R_m q, R_n M^\top k \rangle$ for all $m, n$ , holds iff $M^\top$ commutes with every $R_{n-m}$ that occurs; for diagonal $M = \text{diag}(\ell)$ and distinct frequencies, iff $\ell$ is constant on each rotary pair (in either pairing convention) |
| R6 (norm absorption)         | $W(\gamma \odot n(x) + \beta) = (W \text{diag } \gamma) n(x) + W\beta$                          | always; the shift $W\beta$ needs a bias slot, and an RMSNorm gain has no $\beta$                                                                                                                                                                                                                                                                                                                                                               |
| R7 (copy slide)              | applying $D$ and then copying equals copying and then applying $D$ to every copy                | always                                                                                                                                                                                                                                                                                                                                                                                                                                         |

Consequences:

(a) **Affine edits.** Let a method insert a position-uniform affine edit  $E_q$  on a wire  $w$  of the base network  $f$ , for every  $q \in Q$ . Suppose every reader of  $w$  is a trainable, unshared box (no weight tying) whose class is closed under precomposition with  $E_q$ ; a shift needs a bias slot. Then the method merges, by  $W_c \mapsto W_c E_q$  for all readers  $c$  at once. Dually, it merges into an untied trainable box that writes  $w$  whenever  $E_q$  composed with that box stays in its class. Absorption into a box that the extension itself inserts does not count. Edits that violate these hypotheses fall outside (a): on the reader side, R1–R7 do not merge position-selective edits or edits on a wire that is also read by a skip connection, a norm or the network output. Whole-network non-mergeability is open. Position-selective edits (ReFT/LoReFT, and position-gated LoRA variants such as Activated LoRA) are not box-locally mergeable. Residual-stream edits need case analysis: an edit merges

into the box that produces the wire when that box can absorb it (an untied embedding table; for a fixed-position edit at layer 0, a learned absolute positional table), a shift folds into a writer's bias when one exists, a diagonal edit after a post-LN folds into that LN's affine parameters, and an orthogonal edit in an RMSNorm model merges globally, SliceGPT/QuaRot-style. No merge is known for LoReFT's non-orthogonal low-rank edit.

(b) **Diagonal methods.** A diagonal method merges into a box reached by R1–R7 if all of the following hold: the rules move every copy of  $\text{diag}(\ell)$  into trainable boxes for all  $\ell$  in the method's parameter domain (with  $\ell \geq 0$  when R3 is used); no position-dependent operator lies on the path (for RoPE,  $\ell$  is constant on each rotary pair); the target box is unshared, or  $\ell$  is constant across the sharing group; and every shift has a bias slot. Conversely, the RoPE condition is necessary for sliding a query scaling onto  $W_K$  when  $W_Q$  and  $W_K$  have full rank and at least two occurring relative offsets have rotation angles that are not multiples of  $\pi$ . Without such genericity hypotheses the conditions are not necessary: a dead rotary pair, or zero entries of  $\ell$  under ReLU, can still merge.

(c) **Non-affine edits.** For a linear or affine box, an edit whose modified box function is non-affine in the box input, for some parameter value, is not box-locally mergeable. For linear and affine boxes this is the definition; for non-linear boxes such as attention, see [Theorem VI.5\(f\)](#).

(d) **Type relativity.** A generic high-precision update is off the quantisation grid, so exact merging of a QLoRA or LoftQ update leaves the quantised type, a type-preserving merge is lossy, and QA-LoRA's group-constant adapters are the exception ([Proposition V.2\(d\)](#)).

*Proof.* A diagram denotes a map on activations, and two diagrams related by an identity of maps compute the same network function; a rewrite that ends with every edit inside trainable boxes of the base network therefore exhibits a merge  $\mu$ , the new weights of those boxes. “Position-independent” means that the same map acts at every token position.

R1. Composites and sums of linear (affine) maps are linear (affine). If each acts identically at every position, so does the result, which is then computed by one weight (and bias) at all positions.

R2. For a permutation  $\pi$  with matrix  $P$ ,  $(P\sigma^{\times d}(x))_i = \sigma(x_{\pi^{-1}(i)}) = \sigma((Px)_i)$ .

R3. Coordinatewise the identity says  $\ell_i \sigma(t) = \sigma(\ell_i t)$  for all  $\ell_i \geq 0$  and  $t$ ; with  $d = 1$  this is positive homogeneity of degree one, and conversely positive homogeneity gives the identity in every coordinate. A positively homogeneous  $\sigma$  satisfies  $\sigma(t) = t \sigma(1)$  for  $t \geq 0$  and  $\sigma(t) = |t| \sigma(-1)$  for  $t < 0$ , so it is linear on each half-line. ReLU and leaky ReLU are of this form. GELU, SiLU and tanh are differentiable at 0 and not linear, so by [Auxiliary result B.36\(a\)](#) they are not positively homogeneous.

R4 holds entrywise,  $\ell_i a_i b_i = (\ell_i a_i) b_i$ .

R5.  $\langle Mq, k \rangle = q^\top M^\top k = \langle q, M^\top k \rangle$ . Under RoPE,  $R_m$  is block-diagonal with the planar rotations  $\rho(m\omega_j)$  on the rotary pairs, in either pairing convention, so  $R_m^\top = R_{-m}$  and  $R_m^\top R_n = R_{n-m}$ . Hence  $\langle R_m Mq, R_n k \rangle = q^\top M^\top R_{n-m} k = \langle q, M^\top R_{n-m} k \rangle$  for every  $M$ , while  $\langle R_m q, R_n M^\top k \rangle = \langle q, R_{n-m} M^\top k \rangle$ . These agree for all  $q, k$  iff  $M^\top R_{n-m} = R_{n-m} M^\top$ , and for all occurring  $(m, n)$  iff  $M^\top$  commutes with every occurring  $R_\Delta$ . For  $M = \text{diag}(\ell)$  the commutation holds pair by pair. On a pair with entries  $(\ell_a, \ell_b)$  and rotation angle  $\varphi$ ,

$$\begin{pmatrix} \ell_a & 0 \\ 0 & \ell_b \end{pmatrix} \begin{pmatrix} \cos \varphi & -\sin \varphi \\ \sin \varphi & \cos \varphi \end{pmatrix} - \begin{pmatrix} \cos \varphi & -\sin \varphi \\ \sin \varphi & \cos \varphi \end{pmatrix} \begin{pmatrix} \ell_a & 0 \\ 0 & \ell_b \end{pmatrix} = (\ell_b - \ell_a) \sin \varphi \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

which vanishes iff  $\ell_a = \ell_b$  or  $\varphi \in \pi\mathbb{Z}$ . So the slide holds iff  $\ell$  is constant on every pair whose angle  $\Delta\omega_j$  is not a multiple of  $\pi$  for some occurring  $\Delta$ . For RoPE,  $0 < \omega_j \leq 1$ , so every pair



qualifies as soon as the offset  $\Delta = 1$  occurs; distinct frequencies are not needed for diagonal  $M$ .

R6. By linearity,  $W(\gamma \odot n(x) + \beta) = W \text{diag}(\gamma) n(x) + W\beta$ . The first term is a linear box applied to the normalised wire, and the constant  $W\beta$  is absorbed by a bias slot,  $b \mapsto b + W\beta$ ; an RMSNorm gain has  $\beta = 0$ .

R7. With  $\Delta(x) = (x, x)$  the copy map,  $\Delta \circ D = (D \times D) \circ \Delta$  for every map  $D$ : the naturality of the cartesian diagonal ([Background B.4](#)).

(a) Let  $E_q(x) = A_q x + c_q$  act at every position on  $w$ . By R7 applying  $E_q$  before the copy node that feeds the readers equals applying it on each reader's input. For a reader  $c$  with weight  $W_c$  and bias  $b_c$ ,  $W_c(A_q x + c_q) + b_c = (W_c A_q)x + (W_c c_q + b_c)$ , which lies in the class of  $c$  by the closure hypothesis (for a linear class this requires  $c_q = 0$ , and for a shift a bias slot). The readers are unshared, so their weights can be changed independently; and  $E_q$  is the same at every position, so one weight serves all positions (R1). Setting  $\mu(q)$  to the base weights with  $W_c \mapsto W_c A_q$  and  $b_c \mapsto b_c + W_c c_q$  for every reader gives  $\Phi_f \circ \mu = \Phi_{\tilde{f}} \circ \rho$ . Dually, if  $w$  is the output of an untied box  $p(x) = W_p x + b_p$  and  $E_q \circ p$  lies in its class, replace  $p$  by  $E_q \circ p$ , that is,  $W_p \mapsto A_q W_p$  and  $b_p \mapsto A_q b_p + c_q$  (R1). A merge must land in the base network  $\Theta$ , so absorption into a box that only the extension contains is not a merge. On the reader side the rules have nothing to absorb an edit read by a skip connection, a norm or the network output, since those readers carry no trainable weight, and R1 does not fuse a position-selective edit into a weight that acts at every position. A position-selective edit modifies the box function at some positions only; for parameter values at which it is not the identity, the modified computation is not one map applied identically at every position, so it is not in the box's class, and the edit is not box-locally mergeable.

*Residual-stream cases.* (i) If  $w$  is the output of an untied embedding table  $\mathcal{E}$ , the dual case applies: replacing each row  $e(v)$  by  $A_q e(v) + c_q$  absorbs a position-uniform affine edit. For an additive edit  $h \mapsto h + c$  at one fixed position  $p_0$  of layer 0 in a model with a learned absolute positional table, replacing row  $p_0$  of that table by itself plus  $c$  absorbs it. (ii) A shift  $h \mapsto h + t$  of the stream after a block whose last box has a bias  $b$  is absorbed by  $b \mapsto b + t$ , since the block adds its output to the stream. (iii) In a post-LN model the normalised wire is the stream,  $h = \gamma \odot n(x) + \beta$ , and a diagonal edit gives  $s \odot h + t = (s \odot \gamma) \odot n(x) + (s \odot \beta + t)$ , absorbed by  $\gamma \mapsto s \odot \gamma$ ,  $\beta \mapsto s \odot \beta + t$ . (iv) In an untied pre-RMSNorm model, let block  $j$  map the stream by  $h \mapsto h + F_j(h)$  with  $F_j(h) = W_{\text{out}}^{(j)} G_j(W_{\text{in}}^{(j)} \text{diag}(\gamma_j) n(h))$ , where  $n(h) = h/\text{rms}(h)$ ,  $G_j$  is the internal computation (attention with its positional encoding applied after the input projections, or the gated MLP), and  $W_{\text{in}}^{(j)}$  collects the input projections. Let  $Q$  be orthogonal and the edit be  $h \mapsto Qh$  after block  $l$ . Replace the embedding rows  $e(v)$  by  $Qe(v)$ , and for  $j \leq l$  replace  $W_{\text{in}}^{(j)}$  by  $W_{\text{in}}^{(j)} \text{diag}(\gamma_j) Q^\top$ ,  $\gamma_j$  by  $\mathbf{1}$  and  $W_{\text{out}}^{(j)}$  by  $QW_{\text{out}}^{(j)}$  (and any bias  $b$  of an output projection by  $Qb$ ). Since  $\text{rms}(Qh) = \text{rms}(h)$ ,  $n(Qh) = Qn(h)$ , and by induction the new streams are  $Qh_j$  for  $j \leq l$ :  $Qh_{j-1} + QW_{\text{out}}^{(j)} G_j(W_{\text{in}}^{(j)} \text{diag}(\gamma_j) Q^\top Qn(h_{j-1})) = Q(h_{j-1} + F_j(h_{j-1}))$ . After block  $l$  the new network carries  $Qh_l$ , as the edited one does, and the later blocks are unchanged, so the outputs agree. This is the construction of [Ashkboos et al. \(2024a,b\)](#).

(b) The rules used along the path are identities of maps under their side conditions: R3 needs  $\ell \geq 0$ , the slide of R5 under RoPE needs  $\ell$  constant on the rotary pairs, and R6 needs a bias slot for the shift. Applying them moves every copy of  $\text{diag}(\ell)$  next to a trainable box, and R1 fuses it; this is (a) applied at the box reached. An unshared target can absorb its copy independently, and a target shared across a group absorbs a single matrix, which is possible for every member exactly when  $\ell$  is constant across the group. For the converse under

RoPE, suppose the query scaling  $D = \text{diag}(\ell)$  merges into some  $W'_K$  with  $W_Q$  unchanged. Equality of all logits for all inputs gives  $x^\top W_Q^\top D R_\Delta W_K x' = x^\top W_Q^\top R_\Delta W'_K x'$  for every occurring offset  $\Delta = n - m$ , so  $W_Q^\top (R_\Delta W'_K - D R_\Delta W_K) = 0$ . If  $W_Q$  has full row rank,  $W_Q^\top$  is injective and  $W'_K = R_{-\Delta} D R_\Delta W_K$  for every occurring  $\Delta$ ; if  $W_K$  has full row rank too, it has a right inverse, and  $R_{-\Delta} D R_\Delta$  is one matrix for all occurring  $\Delta$ . On a rotary pair  $j$ , let  $\Delta$  and  $\Delta'$  be occurring offsets with  $(\Delta - \Delta')\omega_j \notin \pi\mathbb{Z}$ ; then  $D$  commutes with  $R_{\Delta-\Delta'}$  on that pair, and by the computation in R5,  $\ell$  is constant on it. In causal attention  $\Delta' = 0$  occurs (a query attends to its own position), so it suffices that one occurring offset has an angle on pair  $j$  that is not a multiple of  $\pi$ , which the hypothesis provides pair by pair. The genericity hypotheses are needed. If  $W_Q$  has a dead pair (both rows zero) and  $\ell$  is constant on every other pair, then  $W_Q^\top D R_\Delta = W_Q^\top R_\Delta D$ , since both sides vanish on the dead pair and  $D$  commutes with  $R_\Delta$  on the others, so  $W'_K = D W_K$  works with  $\ell$  arbitrary on the dead pair. Under ReLU,  $\text{diag}(\ell)\text{ReLU}(Wx) = \text{ReLU}(\text{diag}(\ell)Wx)$  holds for every  $\ell \geq 0$ , zero entries included.

(c) For a linear or affine box the class is the linear or affine maps, so a non-affine modified box function is outside it, which is the definition of failing box-local mergeability.

(d) is [Proposition V.2\(d\)](#).  $\square$

**The corollaries after Theorem VI.3.** *Weight-space methods* merge by  $\mu = \rho$  (§C.45); their nonlinearities act on the parameters. If an embedding and an LM head share one tensor, a merge that changes one side must change the other or break the tie, which is why an adapter on one side is not merged faithfully when the tie is not maintained. (IA)<sup>3</sup> on keys and values is  $\text{diag}(\ell)W_K x = (\text{diag}(\ell)W_K)x$  and likewise for  $W_V$  (R1). On the FFN, applied to the input of  $W_{\text{down}}$ , it is  $W_{\text{down}} \text{diag}(\ell)h = (W_{\text{down}} \text{diag}(\ell))h$  (R1); neither step involves  $\sigma$ . Into  $W_{\text{up}}$ : under ReLU, R3 gives  $\text{diag}(\ell)\text{ReLU}(W_{\text{up}}x) = \text{ReLU}(\text{diag}(\ell)W_{\text{up}}x)$  for  $\ell \geq 0$ , while an entry  $\ell_i < 0$  cannot fuse, since  $\ell_i \text{ReLU}(w_i \cdot x) \leq 0$  and is negative somewhere, whereas  $\text{ReLU}(w'_i \cdot x) \geq 0$ . Under SwiGLU or GeGLU, R4 and R1 give (8) for every  $\ell$ . Under GELU, suppose  $\ell_i \text{GELU}(w_i \cdot x) = \text{GELU}(w'_i \cdot x)$  for all  $x$ , with  $w_i \neq 0$  so that the pre-activation ranges over  $\mathbb{R}$ . If  $\ell_i = 0$ ,  $w'_i = 0$  works. If  $\ell_i \neq 0$ , the right side vanishes on the hyperplane  $w_i \cdot x = 0$ , and  $\text{GELU}(t) = t \Phi_{\mathcal{N}}(t)$  vanishes only at  $t = 0$ , so  $w'_i \cdot x = 0$  there and  $w'_i = c w_i$ . Then  $\ell_i \text{GELU}(t) = \text{GELU}(ct)$  for all  $t$ , and comparing the coefficients  $a_1 = \frac{1}{2}$  and  $a_2 = 1/\sqrt{2\pi}$  ([Auxiliary result B.10\(c\)](#)) gives  $c = \ell_i$  and  $\ell_i = \ell_i^2$ , so  $\ell_i = 1$ . Hence a fused  $W'_{\text{up}}$  exists iff every  $\ell_i \in \{0, 1\}$ . A query scaling fuses into  $W_Q$  by R1; onto  $W_K$  it needs R5's slide, hence  $\ell$  constant on the rotary pairs under RoPE (necessary under the genericity hypotheses of (b)), and with grouped-query attention one  $W_K$  serves several query heads  $h$ , so the same argument with full-rank  $W_Q^{(h)}$  forces  $W'_K = D_h W_K$  for every  $h$  in the group, hence  $D_h$  constant across the group when  $W_K$  has full row rank. SSF after a box  $Wx + b$  gives  $s \odot (Wx + b) + t = (\text{diag}(s)W)x + (s \odot b + t)$ , which needs a bias slot when  $t \neq 0$ . LN-tuning changes existing parameters, so  $\mu$  is the new parameter vector. Folding a LayerNorm into the following boxes in a pre-LN block, where the norm feeds only those boxes, is R7 followed by R6, which needs a bias slot for  $W\beta$ ; an RMSNorm gain has no  $\beta$ . Folding a gain into a tied embedding or head changes the other side of the tie. Not box-locally mergeable: for  $\sigma$  non-affine and nonzero  $u, d$ , the map  $y \mapsto y + u \sigma(d^\top y)$  is non-affine, so a serial Houlsby or Pfeiffer adapter after a box  $W$  with  $d^\top W \neq 0$ , and a parallel adapter  $x \mapsto Wx + u \sigma(d^\top x)$ , modify the box function non-affinely for some parameter value; a routed mixture  $x \mapsto Wx + \sum_e g_e(x) B_e A_e x$  with a softmax gate is real-analytic in its input and parameters and non-affine at a witness (two experts with  $B_1 A_1 = I$ ,  $B_2 A_2 = 0$  and gate logits differing by  $\gamma^\top x$ ,  $\gamma \neq 0$ , give  $Wx + g(\gamma^\top x)x$  with  $g$  the logistic function, non-affine along any line on which  $\gamma^\top x$  varies), hence non-affine

for generic parameters ([Auxiliary result B.10](#)), and linear, so mergeable, when the gate is constant; AdapterFusion with a constant gate is a fixed combination of nonlinear adapters, still non-affine; LST adds the output of a nonlinear side network; and prefix-type methods are treated in [Theorem VI.5\(f\)](#). In each case (c) applies. For LLaMA-Adapter with a two-token prompt  $p_1, p_2$  and gate  $g \neq 0$ , the prompt pairs are  $k_i = W_k p_i + b_k$  and  $v_i = W_v p_i + b_v$ , and a head on a length-one context computes  $W_o(W_v x + b_v) + g W_o \text{Attn}_q(x)(P) + b_o$  with  $\text{Attn}_q(P) = v_2 + \logit^{-1}(\langle q, W_k \delta \rangle / \sqrt{d}) W_v \delta$ ,  $\delta = p_1 - p_2$ . If  $W_q^\top W_k \neq 0$  and  $W_o W_v \neq 0$ , choose  $u$  and  $\delta$  with  $\langle W_q u, W_k \delta \rangle \neq 0$  and  $W_o W_v \delta \neq 0$  (each condition fails only on a proper subspace or a nowhere-dense set). Along  $x = tu$  the logistic factor is a bounded, non-constant real-analytic function of  $t$ , hence not affine, so the head is not affine. By [Auxiliary result B.10](#) the same holds for generic prompts with two or more tokens, which is the entry for LLaMA-Adapter in the summary table of [Exposé VI](#); with one prompt token,  $\text{Attn}_q(P)$  is constant and the head stays affine.

#### C.48 Proposition VI.4

**PROPOSITION VI.4** (linear insertions, and ReFT). (a) A linear adapter in parallel with a single linear box,  $W_0 x + sUDx$ , is LoRA: it has the same  $\rho$ . A linear adapter parallel to a whole sublayer (He et al.’s parallel adapter around attention or the FFN) edits the weightless skip connection rather than a base box, and is not box-locally mergeable when the sublayer is non-affine.

(b) Let  $W_0 \in \mathbb{R}^{m \times n}$  be bias-free (as LLaMA-type  $W_o$  and  $W_{\text{down}}$  are), and  $r \geq 1$ . A serial linear adapter after  $W_0$ ,  $(I + UD)W_0$  with  $U \in \mathbb{R}^{m \times r}$  and  $D \in \mathbb{R}^{r \times m}$  (scale absorbed), has image  $W_0 + \{Y \in \mathcal{M}_{\leq r} : \text{row } Y \subseteq \text{row } W_0\}$ . It is expressively equivalent to LoRA iff  $\text{rank } W_0 = n$ , and weakly (unpointed) isomorphic to LoRA iff  $W_0$  is invertible. The isomorphism  $(U, D) \mapsto (U, DW_0)$  sends a base point  $(U_0, D_0)$  to  $(U_0, D_0 W_0)$ . For a fixed  $T_A$  pointing  $(0, A_0)$  with  $\text{rank } A_0 = r$  and  $1 \leq r < m = n$ , a pointed isomorphism exists iff  $W_0$  is invertible and  $\text{row } A_0 = \text{row}(D_0 W_0)$ , and then  $U_0 = 0$  automatically. With a bias  $b_0$ , the serial adapter also moves the bias by  $UDb_0$ , so on the affine box its image differs from weight-only LoRA’s. Houlsby’s near-identity initialisation (weights from a zero-mean Gaussian with standard deviation  $10^{-2}$ , truncated at two standard deviations) makes practical serial adapters slightly unpointed; the pointed statement applies to zero-up initialisations.

(c) LoReFT is  $\Phi(h) = h + R^\top(W h + b - R h)$ , with  $R \in \mathbb{R}^{r \times d}$  having orthonormal rows, applied at selected positions. On one hidden vector it is the extension by an **identity box** followed by a rank- $\leq r$  affine edit,

$$\Phi(h) = (I + R^\top(W - R))h + R^\top b,$$

which is  $\text{LoRA}_r$  on the affine identity box  $[I_d \mid 0]$  in homogeneous coordinates, with  $B = R^\top$  constrained to the Stiefel manifold and  $A = [W - R \mid b]$ . Its image is  $(I, 0) + \mathcal{M}_{\leq r}(d \times (d+1))$ , of dimension  $r(2d+1-r)$ . The bias is confined to  $\text{col } B$ , so for  $r < d$  this is not  $\text{LoRA}_r \oplus \text{BitFit}$ , whose dimension is larger by  $d - r$ . It is simulated, through  $h(R, W, b) = (R^\top, [W - R \mid b])$ , by  $\text{LoRA}_r$  on that affine box pointed at  $(R_0^\top, 0)$ , a  $T_B$  pointing with  $\text{col } B_0 = \text{row } R_0$ ; this homogeneous-coordinate LoRA has a bias column in  $A$ , unlike HF LoRA. It is not simulated by  $T_A$ -pointed  $\text{LoRA}_r$ , since its  $\dim S_1 = r(d+1)$  exceeds  $rd$ . It is not isomorphic to  $\text{LoRA}_r$ : its parameter count is smaller by  $r(r+1)/2$ , and its residual symmetry is  $O(r)$ , acting by  $(R, W, b) \mapsto (OR, OW, Ob)$ . It is pointed iff  $W = R$  and  $b = 0$ . The reference implementation (pyreft) initialises  $W$  and  $b$  by the default `nn.Linear init`, independently of  $R$ , so practical LoReFT has a pointing defect

$R^\top((W_0 - R_0)h + b_0)$ : at initialisation it replaces the  $R$ -component of  $h$  by a random affine function, and the simulation above is then only a weak morphism. *pyreft* also applies dropout to the whole output,  $h$  included, so the training-time map is not  $\Phi$ .

*Proof.* (a) The adapted box computes  $x \mapsto (W_0 + sUD)x$ , so  $\rho(U, D) = W_0 + sUD$  on  $\mathbb{R}^{m \times r} \times \mathbb{R}^{r \times n}$ , which is LoRA's  $\rho(B, A) = W_0 + sBA$  under  $(U, D) = (B, A)$ , with the zero-up pointing  $U = 0$  (He et al., 2021, Table 1). A linear adapter around a whole sublayer  $F$  gives  $x \mapsto x + F(x) + sUDx$ ; the term  $sUDx$  is added to the weightless skip connection, which no box of the base network computes, so there is no base box whose function the edit modifies. For a non-affine sublayer such as  $F(x) = W_2\sigma(W_1x)$  with  $\sigma$  non-affine, no base box both reads  $x$  and writes linearly into the block's output:  $W_1$ 's output passes through  $\sigma$ , and  $W_2$  reads  $\sigma(W_1x)$  rather than  $x$ . So no box of  $F$  can absorb  $sUDx$  within its class. If  $F(x) = Mx + c$  were affine,  $x + (M + sUD)x + c$  would fuse into  $M$ .

(b) *Image.*  $\rho(U, D) = (I + UD)W_0 = W_0 + UDW_0$ . Each  $UDW_0$  has rank at most  $r$  and its rows are combinations of the rows of  $W_0$ . Conversely, if  $\text{rank } Y \leq r$  and  $\text{row } Y \subseteq \text{row } W_0$ , then  $Y = YW_0^+W_0$ , since  $W_0^+W_0$  is the orthogonal projector onto  $\text{row } W_0$ ;  $X = YW_0^+$  has rank at most  $r$ , so  $X = UD$  with  $U \in \mathbb{R}^{m \times r}$ ,  $D \in \mathbb{R}^{r \times m}$ , and  $Y = UDW_0$ . *Expressive equivalence.* LoRA's image is  $W_0 + \mathcal{M}_{\leq r}$ . If  $\text{row } W_0 = \mathbb{R}^n$  the two images agree. If not, pick  $v \notin \text{row } W_0$ ; then  $e_1v^\top \in \mathcal{M}_{\leq 1} \subseteq \mathcal{M}_{\leq r}$  (here  $r \geq 1$ ) has a row outside  $\text{row } W_0$ , so it lies in LoRA's image minus  $W_0$  but not in the adapter's.

*Weak isomorphism.* If  $W_0$  is invertible, then  $m = n$  and  $h(U, D) = (U, DW_0)$  is a linear bijection  $\mathbb{R}^{m \times r} \times \mathbb{R}^{r \times m} \rightarrow \mathbb{R}^{m \times r} \times \mathbb{R}^{r \times n}$  with  $W_0 + U(DW_0) = \rho(U, D)$  (LoRA with  $s = 1$ , or  $U$  rescaled), so it is an isomorphism over  $\Theta$ ; it sends  $(U_0, D_0)$  to  $(U_0, D_0W_0)$ . Conversely, a weak isomorphism is a diffeomorphism  $h$  with  $\rho_{\text{LoRA}} \circ h = \rho$ , so the images agree, which forces  $\text{rank } W_0 = n$  (Observation I.4(c)), and the dimensions agree,  $2mr = r(m + n)$ , which forces  $m = n$ ; so  $W_0$  is square of full rank.

*Pointed isomorphism.* Let  $1 \leq r < m = n$ ,  $\text{rank } A_0 = r$ , and let the adapter be pointed at  $(U_0, D_0)$ , that is,  $U_0D_0W_0 = 0$ . “If”:  $W_0$  invertible gives  $U_0D_0 = 0$ . Since  $\text{row}(D_0W_0) = \text{row } A_0$  has dimension  $r$ ,  $D_0W_0$  and hence  $D_0$  have rank  $r$ , so  $D_0$  has a right inverse and  $U_0 = 0$ . Two  $r \times n$  matrices of rank  $r$  with the same row space differ by an invertible left factor, so  $A_0 = gD_0W_0$  for some  $g \in \text{GL}_r$ . The map  $(U, D) \mapsto (Ug^{-1}, gDW_0)$  is a linear bijection over  $\Theta$  that sends  $(U_0, D_0) = (0, D_0)$  to  $(0, A_0)$ . “Only if”: a pointed isomorphism is a weak one, so  $W_0$  is invertible, and it preserves  $S_1$  (Observation I.6). Differentiating,

$$\begin{aligned} S_1(\text{adapter at } (U_0, D_0)) &= \{(XD_0 + U_0Y)W_0\} = \{XD_0W_0 + U_0Y'\}, \\ S_1(\text{LoRA at } (0, A_0)) &= \{XA_0\}, \end{aligned}$$

with  $X \in \mathbb{R}^{m \times r}$  and  $Y' = YW_0$  ranging over  $\mathbb{R}^{r \times n}$  because  $W_0$  is invertible; and  $\{XA_0\} = \{Z : \text{row } Z \subseteq \text{row } A_0\}$ . If  $(U_0)_{ij} \neq 0$ , choose  $v \notin \text{row } A_0$ , possible since  $r < n$ , and  $Y' = e_jv^\top$ : row  $i$  of  $U_0Y'$  is  $(U_0)_{ij}v^\top$ , outside  $\text{row } A_0$ , a contradiction. So  $U_0 = 0$ , and  $\{XD_0W_0\} = \{XA_0\}$  gives  $\text{row}(D_0W_0) = \text{row } A_0$ .

*Bias.*  $(I + UD)(W_0x + b_0) = (W_0 + UDW_0)x + (b_0 + UDb_0)$ . On the affine box the adapter's image contains maps with bias  $b_0 + UDb_0 \neq b_0$  whenever  $b_0 \neq 0$  (choose  $UD$  with  $UDb_0 \neq 0$ ), while weight-only LoRA keeps the bias  $b_0$ . *Houlsby initialisation.* With both factors drawn at random,  $U_0D_0W_0 \neq 0$  almost surely, so the initial weight misses  $W_0$ .

(c) *Algebra.* Expanding,  $\Phi(h) = h + R^\top(W - R)h + R^\top b = (I + R^\top(W - R))h + R^\top b$ . In homogeneous coordinates  $(h, 1)$  an affine map  $h \mapsto Mh + c$  is the matrix  $[M \mid c] \in \mathbb{R}^{d \times (d+1)}$ ,



the identity box is  $[I_d \mid 0]$ , and

$$[I + R^\top(W - R) \mid R^\top b] = [I_d \mid 0] + R^\top[W - R \mid b] = [I_d \mid 0] + BA, \quad B = R^\top, A = [W - R \mid b].$$

*Image.* As  $(W, b)$  ranges freely,  $V = [W - R \mid b]$  ranges over all of  $\mathbb{R}^{r \times (d+1)}$ , so the image is  $[I_d \mid 0] + \{R^\top V\}$  over Stiefel  $R^\top$ . Every matrix of rank at most  $r$  factors as  $R^\top V$  with  $R^\top$  having orthonormal columns ([Auxiliary result B.11\(a\)](#)), so the image is  $[I_d \mid 0] + \mathcal{M}_{\leq r}(d \times (d+1))$ , of dimension  $r(d + (d+1) - r) = r(2d + 1 - r)$  ([Background C.12](#)). The bias  $R^\top b$  lies in  $\text{col } R^\top = \text{col } B$ . LoRA <sub>$r$</sub>   $\oplus$  BitFit has image  $\{[I_d + Y \mid c] : \text{rank } Y \leq r, c \in \mathbb{R}^d\}$ , of dimension  $r(2d - r) + d$ , which exceeds  $r(2d + 1 - r)$  by  $d - r$ ; for  $r < d$  the images differ. *Simulation.* The map  $h(R, W, b) = (R^\top, [W - R \mid b])$  is smooth, and LoRA <sub>$r$</sub>  on the affine box evaluated at  $h(R, W, b)$  is  $[I_d \mid 0] + R^\top[W - R \mid b] = \Phi$ , so  $h$  is a morphism to that LoRA <sub>$r$</sub> . It sends the neutral point  $(R_0, R_0, 0)$  to  $(R_0^\top, 0)$ , a pointing in the stratum  $T_B$  ( $A_0 = 0$ ,  $B_0$  of rank  $r$ ) with  $\text{col } B_0 = \text{row } R_0$ .  $S_1$ . At the neutral point  $[W - R \mid b] = 0$ , so the derivative of  $R^\top[W - R \mid b]$  is  $R_0^\top[\dot{W} - \dot{R} \mid \dot{b}]$ , and as  $(\dot{W}, \dot{b})$  range freely  $S_1 = \{R_0^\top V\}$ , of dimension  $r(d+1)$  because  $R_0^\top$  is injective. LoRA <sub>$r$</sub>  pointed at  $(0, A_0)$  with rank  $A_0 = r$  has  $S_1 = \{XA_0\}$ , of dimension  $dr$ . A pointed morphism would give an inclusion of  $S_1$ 's ([Observation I.6](#)), impossible since  $r(d+1) > rd$ .

*Not isomorphic.* The parameter space is the Stiefel manifold of orthonormal  $r$ -frames in  $\mathbb{R}^d$ , of dimension  $rd - r(r+1)/2$ , times  $\mathbb{R}^{r \times d} \times \mathbb{R}^r$ , of total dimension  $2rd + r - r(r+1)/2$ ; LoRA <sub>$r$</sub>  on the affine box has dimension  $r(2d+1)$ . The difference  $r(r+1)/2$  is positive, so no diffeomorphism, hence no isomorphism, exists. *Symmetry.* For  $O \in O(r)$ ,  $OR$  has orthonormal rows and  $(OR)^\top(OW - OR) = R^\top(W - R)$ ,  $(OR)^\top Ob = R^\top b$ , so  $\Phi$  is unchanged. Conversely, if  $R^\top V = R'^\top V'$  with rank  $V = r$ , then the column space of the product is  $\text{col } R^\top = \text{col } R'^\top$ , so  $R' = OR$  for some  $O \in O(r)$ , and injectivity of  $R^\top$  gives  $V' = OV$ , that is,  $W' = OW$  and  $b' = Ob$ : the generic fibres are the  $O(r)$ -orbits, of dimension  $r(r-1)/2$ , against  $r^2$  for LoRA. *Pointedness.*  $\Phi = \text{id}$  iff  $R^\top[W - R \mid b] = 0$  iff  $W = R$  and  $b = 0$ , because  $R^\top$  is injective.

*The pyreft initialisation.* At  $(R_0, W_0, b_0)$  with  $W_0, b_0$  drawn independently of  $R_0$ ,  $\Phi_0(h) - h = R_0^\top((W_0 - R_0)h + b_0)$ , which is nonzero almost surely. Since  $R_0 R_0^\top = I_r$ ,  $R_0 \Phi_0(h) = R_0 h + (W_0 h + b_0 - R_0 h) = W_0 h + b_0$ , while  $(I - R_0^\top R_0) \Phi_0(h) = (I - R_0^\top R_0)h$ : the  $R_0$ -component of  $h$  is replaced by the random affine function  $W_0 h + b_0$  and the rest is kept. The map  $h$  sends this initial point to  $(R_0^\top, [W_0 - R_0 \mid b_0])$ , not to the base point  $(R_0^\top, 0)$ , so it is a weak morphism only. The remark on dropout describes the implementation ([Wu et al., 2024d](#)).  $\square$

## C.49 Theorem VI.5

**THEOREM VI.5** (attention is a homomorphism followed by a perspective map; known in part). Fix a query  $q$ . Multisets of key-value pairs form a commutative monoid under  $\uplus$ . Define

$$\Sigma_q(K) = \left( \sum e^{\langle q, k \rangle / \sqrt{d}}, \sum e^{\langle q, k \rangle / \sqrt{d}} v \right).$$

(a)  $\Sigma_q$  is a monoid homomorphism into  $(\mathbb{R}_{\geq 0} \times \mathbb{R}^{d_v}, +)$ , landing in  $\{Z > 0\} \cup \{(0, 0)\}$ . On nonempty multisets,  $\text{Attn}_q = \pi \circ \Sigma_q$  with the partial map  $\pi(Z, S) = S/Z$ , defined for  $Z > 0$ .

(b) **Prefix is a gated parallel adapter.** For nonempty  $P$  and  $C$ ,

$$\text{Attn}_q(P \uplus C) = (1 - \lambda_q) \text{Attn}_q(C) + \lambda_q \text{Attn}_q(P), \quad \lambda_q = \frac{Z_q(P)}{Z_q(P) + Z_q(C)}.$$

(c) **Per-layer pattern rigidity.** Within a single layer, for fixed  $q$  and a fixed content multiset  $C$ : the content attention weights are multiplied by  $1 - \lambda_q$ , their ratios are unchanged, and the



output lies on the segment from  $\text{Attn}_q(C)$  to  $\text{Attn}_q(P) \in \text{conv } V_P$ , where  $V_P$  is the set of prefix values. This rigidity is relative to the same layer's computation without the prefix; it is relative to the pretrained pattern only at the lowest prefixed layer. Lower-layer prefixes change the queries and the content keys and values of higher layers, so attention patterns at the level of the network are not rigid.

(d) **LLaMA-Adapter** normalises  $P$  and  $C$  separately and adds a zero-initialised gate. It computes  $\text{Attn}_q(C) + \tanh(g_l) \text{Attn}_q(P)$ , with one gate per head in the paper; HF PEFT uses one raw zero-initialised scalar per layer, without  $\tanh$ . It uses two perspective maps, leaves the content weights exactly unchanged, and is pointed at  $g = 0$ . Its prompt keys and values are the frozen  $W_k, W_v$  applied to one learned prompt per adapted layer, and it acts in the top layers only.

(e) **Prompt  $\rightarrow$  deep prefix in causal decoders.** Assume that the soft prompt comes first, that attention is causal, and that content positions are offset by the prompt length  $m$  in both methods (as HF PEFT does via `cache_position`). Then at every layer the states at prompt positions depend only on  $p$ , and  $h : p \mapsto (K_j(p), V_j(p))_j$  satisfies

$$\Phi \circ \rho_{\text{prefix}} \circ h = \Phi \circ \rho_{\text{prompt}}$$

as functions of the content tokens, read at content positions. The two methods live on different extensions, so this is an arrow of the unpointed slice over the realisation  $C^\infty(X_{\text{content}}, Y_{\text{content}})$ , not of **Meth**. Under RoPE without the offset, a different  $h$  works: rotate each prefix key by  $-m$  (positions  $-m, \dots, -1$ ). For additive relative biases (ALiBi, T5) the answer depends on how the implementation places the prefix keys. The construction does not apply in bidirectional encoders of depth  $\geq 2$ .

(f) **Obstructions.** Call a head standard if it is vanilla softmax attention without torch-style `add_bias_kv` (which would absorb a one-token prefix), `add_zero_attn`, learned attention sinks or softmax-plus-one. On length-one contexts, every weight setting of a standard head is affine in  $x$ . Some prefix makes the prefixed head non-affine on length-one contexts iff  $W_o \neq 0$  and  $(W_q \neq 0$  or  $W_k^\top b_q \neq 0)$ ; for bias-free heads, iff  $W_o \neq 0$  and  $W_q \neq 0$ . For such frozen weights and generic prefixes the prefixed head is not affine, because the gate  $\lambda(x)$  is not constant. So prefix tuning and P-tuning v2 are not box-locally mergeable into a standard head. When  $W_o W_v \neq 0$ , no prefix is neutral, so prompt and prefix tuning are unpointed.

*Proof.* Write  $\epsilon_q(k) = e^{\langle q, k \rangle / \sqrt{d}} > 0$ , and for a multiset  $K$  put  $Z_q(K) = \sum_{(k,v) \in K} \epsilon_q(k)$  and  $S_q(K) = \sum_{(k,v) \in K} \epsilon_q(k)v$ , so  $\Sigma_q(K) = (Z_q(K), S_q(K))$ .

(a) Sums over a multiset union split,  $\Sigma_q(K \uplus K') = \Sigma_q(K) + \Sigma_q(K')$ , and  $\Sigma_q(\emptyset) = (0, 0)$ , so  $\Sigma_q$  is a monoid homomorphism. Each term of  $Z_q$  is positive, so  $Z_q(K) > 0$  for nonempty  $K$  and  $Z_q(\emptyset) = 0$ . By definition,  $\text{Attn}_q(K) = S_q(K)/Z_q(K) = \pi(\Sigma_q(K))$  for nonempty  $K$ ; this  $\pi$  is the perspective map of [Background D.34](#).

(b) With  $\lambda = Z_P/(Z_P + Z_C)$  and  $Z_P, Z_C > 0$ ,

$$\pi(Z_P + Z_C, S_P + S_C) = \frac{S_P + S_C}{Z_P + Z_C} = \frac{Z_P}{Z_P + Z_C} \frac{S_P}{Z_P} + \frac{Z_C}{Z_P + Z_C} \frac{S_C}{Z_C} = \lambda \text{Attn}_q(P) + (1 - \lambda) \text{Attn}_q(C).$$

(c) In  $P \uplus C$  a content pair  $(k, v) \in C$  has weight  $\epsilon_q(k)/(Z_P + Z_C) = (1 - \lambda) \epsilon_q(k)/Z_C$ , which is  $1 - \lambda$  times its weight in  $C$  alone; so all content weights scale by one factor and their ratios are unchanged. By (b) the output lies on the segment from  $\text{Attn}_q(C)$  to  $\text{Attn}_q(P)$ , with  $\lambda \in (0, 1)$ , and  $\text{Attn}_q(P) = \sum_{(k,v) \in P} (\epsilon_q(k)/Z_P) v$  is a convex combination of the prefix values. At the lowest prefixed layer the query and the content pairs are computed from inputs that no prefix has touched, so they are those of the pretrained model, and the content weights

are the pretrained ones times  $1 - \lambda_q$ . At a higher layer the inputs come from lower layers whose outputs (b) has changed, so  $q$  and  $C$  themselves differ from the pretrained ones, and the statement compares only with the same layer's computation on those inputs. The failure of network-level rigidity is shown in [Petrov et al. \(2023, §6\)](#), and universality of prefix-tuning in [Petrov et al. \(2024\)](#).

(d) LLaMA-Adapter ([Zhang et al., 2023h](#)) computes the two softmaxes separately, so its output is  $\pi(\Sigma_q(C)) + \tanh(g_l) \pi(\Sigma_q(P))$ , with two perspective maps. A content weight is  $\epsilon_q(k)/Z_q(C)$ , the same as without the adapter. At  $g = 0$ ,  $\tanh(0) = 0$  (and HF's raw gate is 0), so the output is  $\text{Attn}_q(C)$ , the pretrained one, and the method is pointed there. The remaining clauses describe how the method is built.

(e) Number the positions of the prompt-tuned model  $0, \dots, m-1$  for the soft prompt  $p = (p_0, \dots, p_{m-1})$  and  $m, \dots, m+n-1$  for the content, and give the content the same positions  $m, \dots, m+n-1$  in the prefix-tuned model. Let  $K_j(p), V_j(p)$  be the keys and values, after the positional encoding, that the prompt-tuned model forms at the prompt positions in layer  $j$ , as they are cached. We show by induction over layers  $j$  that (i) the hidden states of the prompt-tuned model at prompt positions depend only on  $p$ , and (ii) the hidden states at content positions agree in the prompt-tuned model and in the prefix-tuned model with prefix  $h(p) = (K_j(p), V_j(p))_j$ . At the input, (i) holds trivially and (ii) holds because both models embed the same content tokens at the same positions. Suppose (i) and (ii) hold at the input of layer  $j$ . Under causal masking a prompt position attends only to prompt positions, so its output depends only on prompt states, which gives (i) at the next layer. A content position  $t$  attends to the prompt positions and to the content positions  $u \leq t$ . Its query, and the content keys and values, are computed from equal content states with equal positional encodings, by (ii). Its prompt keys and values are  $K_j(p), V_j(p)$  in the prompt-tuned model by definition, and they are supplied verbatim by the prefix in the other. With rotary encodings, a score is  $\langle R_t q, R_s k_s \rangle$  in both models, the cached key  $R_s k_s$  being the same object. So the query and the multiset of key-value pairs are the same, and the attention outputs agree; the position-wise sublayers then give (ii) at the next layer. Reading the content positions at the top gives  $\Phi \circ \rho_{\text{prefix}} \circ h = \Phi \circ \rho_{\text{prompt}}$ . The prompt-tuned model and the prefix-tuned model are methods on different extensions of the base network (input tokens in one, key-value slots in the other), so  $h$  is a map over the common realisation as functions of the content, an arrow of the slice over  $C^\infty(X_{\text{content}}, Y_{\text{content}})$ .

Without the offset, the prefix-tuned model places the content at positions  $0, \dots, n-1$ . For RoPE put  $h'(p) = (R_{-m} K_j(p), V_j(p))_j$ , rotating the cached key from prompt position  $s$  to  $s-m$ . A content query at position  $t-m$  then scores  $\langle R_{t-m} q, R_{s-m} k_s \rangle = \langle q, R_{s-t} k_s \rangle = \langle R_t q, R_s k_s \rangle$ , content-content scores depend only on differences of positions and are unchanged, and values carry no rotation, so the induction goes through with  $h'$ . For additive relative biases the score of a content query against a prefix key depends on the distance the implementation assigns to that key, and the induction goes through exactly when these distances equal  $t-s$ . In a bidirectional encoder of depth at least 2, prompt positions attend to content positions, so already the second layer's prompt states depend on the content, and step (i) fails.

(f) *Affine without prefix*. On a context of one token  $x$  the softmax has one term, equal to 1, so a standard head computes  $W_o(W_v x + b_v) + b_o$ , which is affine in  $x$  for every weight setting. (With `add_zero_attn`, a sink or softmax-plus-one there is a second term, and this fails.)

*With a prefix*. Let the prefix be  $P = \{(k_i, v_i)\}$  with  $N_P \geq 1$  fixed pairs, and write  $q(x) = W_q x + b_q$ ,  $k(x) = W_k x + b_k$ ,  $v(x) = W_v x + b_v$ , the content score  $\varsigma(x) = \langle q(x), k(x) \rangle / \sqrt{d}$  and

$Z_P(x) = \sum_i \epsilon_{q(x)}(k_i)$ . By (b),

$$\text{head}_P(x) = W_o \left[ (1 - \lambda(x)) v(x) + \lambda(x) \text{Attn}_{q(x)}(P) \right] + b_o, \quad \lambda(x) = \frac{Z_P(x)}{Z_P(x) + e^{\varsigma(x)}} \in (0, 1). \quad (28)$$

*Only if.* If  $W_o = 0$  the output is the constant  $b_o$ . If  $W_q = 0$  and  $W_k^\top b_q = 0$ , then  $q = b_q$  is constant, every prefix score  $\langle b_q, k_i \rangle / \sqrt{d}$  is constant, and the content score  $\langle b_q, W_k x + b_k \rangle / \sqrt{d} = ((W_k^\top b_q)^\top x + \langle b_q, b_k \rangle) / \sqrt{d}$  is constant; so  $\lambda$  and  $\text{Attn}_q(P)$  are constant and the output is affine.

*If.* Suppose  $W_o \neq 0$  and  $(W_q \neq 0 \text{ or } W_k^\top b_q \neq 0)$ . Take all  $N_P$  prefix pairs equal to  $(k_1, v_1)$ ; then  $\text{Attn}_{q(x)}(P) = v_1$  and  $\lambda(x) = \text{logit}^{-1}(\delta(x))$  with  $\delta(x) = \langle q(x), k_1 \rangle / \sqrt{d} + \log N_P - \varsigma(x)$ . Restrict to a line  $x = tu$ . If  $W_q \neq 0$ , choose  $u$  with  $W_q u \neq 0$ ;  $\varsigma(tu)$  is a fixed polynomial of degree at most two in  $t$ , and  $\langle q(tu), k_1 \rangle$  has the slope  $\langle W_q u, k_1 \rangle$ , which  $k_1$  can make any real number, so  $k_1$  can be chosen with  $\delta(tu)$  non-constant. If  $W_q = 0$  and  $W_k^\top b_q \neq 0$ , choose  $u$  with  $\langle W_k^\top b_q, u \rangle \neq 0$ ; then  $\varsigma(tu)$  has nonzero slope while the prefix score is constant, so  $\delta(tu)$  is non-constant for every  $k_1$ . In both cases  $\lambda(t) := \lambda(tu)$  is real-analytic, takes values in  $(0, 1)$  and is non-constant, so  $\lambda''(t_0) \neq 0$  for some  $t_0$ ; otherwise  $\lambda$  would be affine and bounded, hence constant. Choose  $v_*$  with  $W_o v_* \neq 0$  and put  $v_1 = M v_*$ . Since  $v(tu)$  is affine in  $t$ ,

$$\frac{d^2}{dt^2} \text{head}_P(tu) = \lambda''(t) (M W_o v_* - W_o v(tu)) - 2\lambda'(t) W_o W_v u,$$

and at  $t_0$  the term  $M\lambda''(t_0)W_o v_*$  dominates for large  $M$ , so the second derivative is nonzero and the head is not affine. For bias-free heads  $W_k^\top b_q = 0$ , which gives the second form of the criterion.

*Generic prefixes.* The prefixed output is real-analytic in  $(x, P)$  on the connected space of inputs and prefixes with  $N_P$  pairs. It is affine in  $x$  for a given  $P$  iff all second partial derivatives in  $x$  vanish at every  $x$ ; this is the common zero set, over  $x$  and pairs of coordinates, of real-analytic functions of  $P$ . The witness above is a prefix at which one of these functions is nonzero, so that function is not identically zero, and the prefixes with affine output lie in its zero set, which is closed, nowhere dense and null ([Auxiliary result B.10](#)). The same argument applied to  $\lambda$  shows that the gate is non-constant for generic prefixes. Box-locally, the class of a standard head is the set of functions it computes with some weights; on length-one contexts these are all affine, while for generic prefixes the prefixed head is not, so prefix tuning and P-tuning v2 are not box-locally mergeable into the head.

*No neutral prefix.* A neutral prefix would leave the head unchanged on every input, in particular on length-one contexts, where by (28) this means  $\lambda(x)(W_o \text{Attn}_{q(x)}(P) - W_o v(x)) = 0$ , and since  $\lambda(x) > 0$ ,  $W_o \text{Attn}_{q(x)}(P) = W_o(W_v x + b_v)$  for all  $x$ . The left side lies in the compact set  $W_o \text{conv } V_P$ ; if  $W_o W_v \neq 0$ , choose  $x_0$  with  $W_o W_v x_0 \neq 0$ , and along  $x = tx_0$  the right side is unbounded. So no prefix is neutral. For a base network consisting of one such head, the extension by prefixes therefore has no neutral point, and by (e), at the first layer of a causal decoder a soft prompt acts through its key-value pairs exactly as a one-layer prefix, so the same base network admits no neutral prompt: prompt and prefix tuning are unpointed.  $\square$

**REMARK** (numerical checks). The values reported after [Theorem VI.5](#) (prompt-to-prefix agreement  $9 \times 10^{-16}$  with RoPE and the offset, 1.3 without it and  $2 \times 10^{-15}$  with keys rotated by  $-m$ ; an affine head with  $W_o = 0$  although  $W_q^\top W_k \neq 0$ , and a non-affine one with  $W_q \neq 0$  and  $W_q^\top W_k = 0$ ) instantiate parts (e) and (f) as proved above: the criterion of (f) involves  $W_o$ ,  $W_q$  and  $W_k^\top b_q$ , not  $W_q^\top W_k$  ([Appendix E](#)).

## The catalogue

The atlas of Exposé VIII rests on a catalogue of 382 methods and 138 papers. This appendix says how the catalogue was assembled and fact-checked, how each method received its seven coordinates, its arrows and its obstructions, and how these were checked in turn. It ends with the complete tables: every method with its coordinates and its reference (Table 36), every recorded arrow with its map  $h$  (Table 37), and the arrows and obstructions counted by rule and by test (Table 35).

**Data files.** The data accompany this paper in machine-readable form, generated from the same source as the tables below and as the companion site.

**catalogue.json** Every method (382) and every paper (138), each with its verified metadata (title, full author list, year of the first arXiv version, venue, links to the paper and the code) and its BibTeX record. A method entry adds the formula, the trainable-parameter count, the locus, the modification kind, the merge verdict with its reason, the initialisation, the relations to other methods, the sources consulted, the seven coordinates with a justification for each, the categorical reading, and the method’s arrows and obstructions. The file also holds the definitions of the axis values, with the question that decides each axis, and the arrow rules.

**catalogue.csv** One row per method, with the main fields of `catalogue.json` and one column per coordinate.

**arrows.csv** Every recorded arrow  $M \rightarrow N$  with its rule, its map  $h$  and a note that names the instance of  $N$  it lands in (rank, pointing or regime).

**obstructions.csv** Every recorded obstruction  $M \not\rightarrow N$  with the test that rules the arrow out and the invariant that fails.

**propositions.json** Every numbered result (41 of them) with its statement, proof, review status, second-round verdict, public statement and referee notes.

**framework\_checks.py** The numerical checks of Appendix E.

### D.1 Families and sources

Every entry was written once and then attacked once, by a separate reader looking for what the first got wrong. The literature was split into the thirteen families of Table 34, then completed by four gap-filling batches, which added the methods a completeness review found missing, and by two addenda on 2 October 2026: methods the authors reported missing, and a sweep of the 2026 literature run in two passes (§D.3). The family boundaries owe much to earlier maps of the field: the unified view of He et al. (2021), the delta-tuning study of Ding et al. (2022), the guide of Lialin et al. (2023a), the survey of Han et al. (2024b) and the modular view of Pfeiffer et al. (2023).

**Merging.** The families overlap, so the 477 method records and 180 paper records were merged into 382 methods and 138 papers. Records were matched on a canonical identifier and a table of aliases, because matching by name would fold LoRA+ (Hayou et al., 2024b) into LoRA (Hu et al., 2021) and O-LoRA (Wang et al., 2023b) into OLoRA (Büyükyüz, 2024). Of the 382 entries, 358 are methods proper; the others are 17 training recipes, 4 software

**Table 34.** How the catalogue was assembled. Each family was written by one reader and attacked by another; the last column (Corr.) counts the logged corrections. The families overlap, so the records were merged into the catalogue on a canonical identifier.

| Family                        | What it covers                                         | Methods    | Papers     | Corr.      |
|-------------------------------|--------------------------------------------------------|------------|------------|------------|
| LoRA core                     | LoRA and its variants                                  | 20         | 12         | 21         |
| Initialisation and rank       | Base-point splits; rank allocation                     | 25         | 6          | 46         |
| Shape                         | Kronecker, Hadamard, tensor and sparse shapes          | 29         | 9          | 49         |
| Multiplicative                | Group and monoid actions on $W_0$                      | 32         | 9          | 54         |
| Sharing                       | Shared or random bases                                 | 28         | 4          | 46         |
| Selective                     | Subsets of existing weights                            | 26         | 7          | 37         |
| Adapters                      | Inserted modules and side networks                     | 31         | 10         | 30         |
| Prompts                       | Soft prompts, prefixes and learned states              | 27         | 13         | 39         |
| Quantisation and memory       | Quantised bases, gradient memory                       | 24         | 5          | 44         |
| Composition                   | Merging, routing, mixtures                             | 42         | 8          | 61         |
| HF PEFT                       | The library’s tuners and utilities                     | 74         | 10         | 53         |
| Frontier                      | 2025–2026: RL, reasoning, diffusion                    | 21         | 5          | 48         |
| Theory                        | Theory and foundations                                 | 0          | 50         | 23         |
| Gap-filling (4 batches)       | Methods a completeness review found missing            | 36         | 18         | 62         |
| Addendum (2 Oct 2026)         | Methods the authors reported missing                   | 1          | 0          | 2          |
| Addendum (2026 sweep, pass 1) | arXiv, January to September 2026; ICLR and ICML 2026   | 29         | 8          | 126        |
| Addendum (2026 sweep, pass 2) | Further candidates, mostly ICLR, ICML and NeurIPS 2026 | 32         | 6          | 164        |
| <b>All records</b>            | Before merging                                         | <b>477</b> | <b>180</b> | <b>905</b> |
| <b>Catalogue</b>              | After merging duplicates                               | <b>382</b> | <b>138</b> |            |

systems, 2 analyses and 1 baseline (linear probing). The first arXiv versions of the methods run from 2013 to 2026, and 47 methods were first posted in 2026.

**The bibliography.** The BibTeX record of each entry comes from the arXiv metadata of its identifier, with the full author list, for 493 identifiers, and was written by hand for the nine entries without one. These 502 records form the catalogue part of this paper’s bibliography, and the citation keys of Table 36 point into it.

## D.2 Fact-checking

Each field was checked against a primary source.

- Titles, authors and first-version years were checked against the arXiv metadata.
- Venues were checked against the official records: OpenReview, the ACL Anthology, PMLR and CVF open access, with Crossref, Semantic Scholar and DBLP for the remaining cases.
- Formulas and parameter counts were checked against the papers, and against the official code where an initialisation was in doubt.
- Every claim about Hugging Face PEFT (Mangrulkar et al., 2022) was checked against its source at commit 85be3f3 (85be3f3a1a58c4d153ffc4631c57196430af87a4) of 30 September 2026. The library’s code settled many questions of convention, such as the side of  $W$  on which a rotation acts, the default initialisation of each factor, and whether a scale such as  $\alpha/r$  is applied. The column “HF” of Table 36 marks the 82 methods available at that commit.



Every change was logged with its reason, 905 in all (Table 34). The merge verdict of each method is recorded beside its reason: 250 methods fold back into the frozen weights always, 41 under stated conditions and 91 not at all. This verdict concerns the deployed method; the merge coordinate of Exposé VIII refines it by the rewrite rules of Theorem VI.3.

### D.3 The 2026 sweep and the authors’ addendum

On 2 October 2026 the authors pointed out methods missing from the catalogue. LOFT (Zhao et al., 2026) was added at once as an addendum of its own, checked against arXiv and the paper (two corrections logged), and classified like every other method. Its coordinates are R, Orb, regular, none,  $\text{CO} \times \text{CO}$ , M1 and idempotent; it has arrows into OFT, LoRA-XS, PSOFT and LoRA, and for  $r \geq 4$  its first-order image rules out a smooth pointed arrow into HRA at HF’s paired initialisation (test O3). Two independent referees checked these coordinates, arrows and the obstruction, numerically and against the paper and the code. §II.B reads LOFT as a fixed leaf of HRA’s meet.

The other gaps were sought by a sweep of the 2026 literature, run in two passes on the same day. A pool of 265 candidates was drawn from six slices:

- three quarterly arXiv slices, for first versions posted from January to March, April to June and July to September 2026, with 60 candidates each;
- the accepted papers of ICLR 2026, 5,468 of them screened through their abstracts, with 48 candidates;
- the accepted papers of ICML 2026, 6,797 screened, with a second pass over ACL, EACL and CVPR 2026, together giving 31 candidates;
- the 2026 papers citing anchor methods (LoRA, DoRA, PiSSA, QLoRA, OFT, VeRA, HRA, GaLore, PSOFT and LoRA-GA), ranked by citations, recency and acceptance, with the 2026 release notes and pull requests of Hugging Face PEFT, giving 6 candidates.

Acceptance at ICLR and ICML 2026 was confirmed by an exact title match against the official lists of accepted papers or by the authors’ own arXiv comment. Candidates were matched against the catalogue by arXiv identifier and by name, so that near-homonyms stay apart: the entropy-guided FlexLoRA of 2026 (Liu et al., 2026c) is not the federated FlexLoRA of 2024 (arXiv:2402.11505), and the per-token MoLoRA of 2026 (arXiv:2603.15965) is not the MoLoRA of Zadouri et al. (2023). The first pass admitted 29 methods and 8 papers. The second re-screened the rest of the pool and admitted 32 methods and 6 papers, mostly from ICLR, ICML and NeurIPS 2026. Each admitted entry went through the same fact-checking as the families, with 126 and 164 logged corrections, and each of the 61 new methods received an independent check of its coordinates and arrows on the same day. The arXiv queries searched abstracts for LoRA, PEFT, adapter, parameter-efficient and prompt tuning, so a 2026 method paper whose abstract uses none of these terms lies outside the sweep’s reach.

### D.4 Coordinates, arrows and obstructions

**Assignment.** Each method was classified from its formula, in thirteen batches: ten of 32 methods for the catalogue as it stood on 1 October 2026, one for the authors’ addendum, and one for each pass of the 2026 sweep (29 and 32 methods). A first reader answered the seven questions of §VIII.A, which fix the kind, the shape of the image germ, the base point, the gauge, the covariance group, the merge type and the rebasing closure, following the decision procedures of §VIII.A and the conventions of §VIII.A. Each answer carries

a short justification, kept in `catalogue.json`. A null coordinate, printed “–”, means that the axis does not apply, for instance the shape of a method with no germ at  $\theta_0$ . Where an answer turns on the dimension of an image or a fibre, or on the first-order image at the initialisation, it was confirmed by a float64 Jacobian rank at a random point and at the base point (Background C.3).

Each arrow  $M \rightarrow N$  comes with an explicit pointed smooth map  $h$  satisfying  $\rho_N \circ h = \rho_M$ , obtained by one of the rules A2–A5 of §VIII.C, and with the instance of  $N$  it lands in: a rank, a pointing or a hyperparameter regime. Each obstruction  $M \nrightarrow N$  names the test O1–O3 of §VIII.C that rules the arrow out and the invariant that fails. The interval arrows  $\text{Frozen} \rightarrow M \rightarrow \text{Full FT}$  of rule A1 hold for every weight-space method and are not recorded. The 142 core methods are those named in the periodic table (Table 19) together with a curated set of others; the flag is not a coordinate.

**Independent check.** A second reader re-derived every coordinate from the formula and re-checked every arrow, by recomputing  $\rho_N \circ h$  and the pointing  $h(q_0^M)$ , and every obstruction, by recomputing the failing invariant. Each method carries a check note with the outcome. More than half of the notes record no change; the others record each change with its reason, and 47 record a numerical Jacobian computation (a rank, a fibre dimension or a deficiency).

A consolidation pass on 1 October 2026, over the 320 methods classified by then, unified the data and touched 135 methods, each change logged in its check note.

- (1) 137 coordinates used a value “n/a” that the axis definitions do not contain. 136 became null, matching the “–” of the tables, and the base point of GenKnowSub became “unpointed”, as for the other parameterless post-hoc operations. The merge axis keeps its legacy key `na`, read as null.
- (2) The row labels of the periodic table were unified across batches.
- (3) The covariance of LISA and BAdam changed from  $\Pi$  to  $\text{GL} \times \text{GL}$ , matching HiFT, surgical fine-tuning and BitFit: each stage frees whole boxes, and the layer partition breaks no neuron permutation.
- (4) Four obstruction reasons now say which instance of the target they concern (DSEE, GaLore, Spectral Adapter<sup>A</sup> and MoV, each against LoRA).

Where a formula disagreed with an earlier draft of the periodic table, the value the formula supports was kept and the table of Exposé VIII was revised to agree, in thirteen places. For example, LST, UniPELT and MAM are unpointed extensions, MPT and Compacter have torus gauges, the Houlby adapter’s GELU or swish leaves only the permutations of hidden units, and the gauges of DeLoRA and AdaLoRA are non-linear. The 62 later entries, LOFT and the 61 methods of the 2026 sweep, were classified and checked in the same way.

**Structural checks on the final data.** For this appendix the structural checks were re-run on the ancillary files. Every arrow and every obstruction joins two catalogued methods and names a valid rule or test; every arrow has a map  $h$  and every obstruction a reason; every coordinate is a value defined for its axis; and no arrow or obstruction is recorded twice. A check at the level of method identifiers flags four kinds of pattern, and each was resolved by the instance it concerns.

- *Pairs with both an arrow and an obstruction* (26). In every case the two concern different instances of the target: a different rank, pointing, regime or variant (§VIII.B). For example,

**Table 35.** The 403 recorded arrows by rule (top) and the 485 recorded obstructions by test (bottom). The rules and tests are those of §VIII.C; 161 of the arrows join two core methods. Every arrow is listed in Table 37, and every obstruction, with its failing invariant, in the ancillary file `obstructions.csv`.

| Rule | Count | How $h$ arises                                                           |
|------|-------|--------------------------------------------------------------------------|
| A2   | 61    | freezing a block of coordinates at its base value, $h(q') = (q', q_0'')$ |
| A3   | 329   | an explicit smooth pointed factorisation $\rho_M = \rho_N \circ h$       |
| A4   | 2     | a projection out of a fibre product                                      |
| A5   | 11    | a sum or an added stage, $h(q) = (q, q_0^N)$                             |

| Test | Count | What fails                                                                             |
|------|-------|----------------------------------------------------------------------------------------|
| O1   | 288   | an image invariant (Gram matrix, spectrum, row lines, zero pattern) or image inclusion |
| O2   | 129   | dimension or cut-rank                                                                  |
| O3   | 68    | first-order image or base-point stratum                                                |

ABBA is isomorphic to LoHa pointed at the factors of  $[W_0]_{r_1}$ , while the O3 obstruction concerns LoHa at HF’s random default pointing.

- *Arrows from a method to itself* (8). Each is an inclusion between instances of one family, such as  $\text{LoRA}_r \hookrightarrow \text{LoRA}_{r+1}$  or  $\text{ReLoRA}^K \hookrightarrow \text{ReLoRA}^{K+1}$ .
- *Cycles* (13 groups of mutually reachable methods). The cycles in these groups are of four kinds. Eight are isomorphisms: DoRA and BiDoRA, ReLoRA and COLA, AsyLoRA and LoRA-FA, AdaptFormer and the parallel adapter, S-LoRA and Punica, EFT and proxy tuning, LoRAMoE and MoLoRA, and AdaMerging and LoRA soups (CAT), which differ only in pointing. Three join methods with equal images that are not isomorphic: prompt tuning with MPT and with P-tuning, and QLoRA with IR-QLoRA. Five go up in rank or in the number of stages and are not isomorphisms: LoRA with ReLoRA, LoRA with PEANuT, LoRA-FA with Flora, surgical fine-tuning with HiFT, and KronA with LoKr. Initial-state tuning and prefix tuning simulate each other on S4 state-space layers only.
- *Obstructions  $M \not\rightarrow N$  with no direct arrow but a path of arrows from  $M$  to  $N$*  (20). In each, the obstruction concerns a fixed rank or pointing of the target and the path ends in another instance. FourierFT with  $n_c$  frequencies, for example, maps into  $\text{LoRA-FA}_{2n_c}$  and so into  $\text{LoRA}_{2n_c}$ , while the O2 obstruction concerns  $\text{LoRA}_r$  at a matched budget.

No arrow contradicts an obstruction once the stated rank or pointing is taken into account. The eight places where the data depart from the conventions of §VIII.A are listed in §VIII.B, and Table 23 counts the methods per coordinate value.

## D.5 The complete tables

Table 36 lists every method with its reference, the year of its first arXiv version and its seven coordinates. A method without an arXiv version carries the year it first became public. The year in a citation is that of the version cited, so the two can differ: LoRA-S, for instance, first appeared in 2025 and is cited as [Zheng and Wu \(2026\)](#), its ICLR 2026 version. The rows are grouped by kind, with the reparametrisations split as in the periodic table (additive, action, Hadamard and selective), and sorted by name within each group. The abbreviations are those of Table 23, with “unpt.” for unpointed, “nonlin.” for a non-linear gauge,  $\text{GL} \times \text{T}$  for GL times a Hadamard torus,  $\text{O} \times \text{O}$  for the spectral-frame value  $\text{CO} \times \text{CO}$ , “idemp.” for idempotent

and “sched.” for a scheme that is itself a rebasing schedule. The merge value na is the legacy key for backward methods, read as “-”.

Table 37 lists every recorded arrow, sorted by source, with its rule and its map  $h$ . Each  $h$  is stated in the notation of the target’s formula in `catalogue.json`, and the instance of the target (its rank, its pointing or its regime) is in the `note` column of `arrows.csv`.

**Table 36.** The complete catalogue: every method with its seven coordinates (§VIII). Core methods are marked •; HF marks methods available in Hugging Face PEFT at the pinned commit. “-” means the axis does not apply.

| Method              | Reference               | Year | Kind | Shape | Base pt. | Gauge   | Covar.  | Merge | Rebase | HF |
|---------------------|-------------------------|------|------|-------|----------|---------|---------|-------|--------|----|
| <b>R, additive</b>  |                         |      |      |       |          |         |         |       |        |    |
| ABBA                | Singhal et al., 2025    | 2025 | R    | Cone  | apex     | GL×T    | Mon×Mon | M1    | span   |    |
| • AdaLoRA           | Zhang et al., 2023g     | 2023 | R    | Cone  | apex     | nonlin. | GL×GL   | M1    | span   | ✓  |
| • AdaMSS            | Zheng et al., 2025      | 2025 | R    | Cone  | apex     | GL      | O×O     | M1    | span   | ✓  |
| AdaPaD              | Su et al., 2026         | 2026 | R    | Cone  | defect   | GL      | GL×GL   | M1    | span   |    |
| ALoRA               | Liu et al., 2024d       | 2024 | R    | Cone  | apex     | GL      | GL×GL   | M1    | span   |    |
| AltLoRA             | Yu et al., 2025         | 2025 | R    | Cone  | apex     | GL      | GL×GL   | M1    | span   |    |
| ApiQ                | Liao et al., 2024       | 2024 | R    | Cone* | defect   | GL      | Mon×Mon | Mq    | span   |    |
| AR1-ZO              | Chen et al., 2026b      | 2026 | R    | Cone  | apex     | GL      | GL×GL   | M1    | span   |    |
| • Astra             | Liu et al., 2026b       | 2026 | R    | Cone* | regular  | GL      | O×O     | M1    | span   | ✓  |
| AsyLoRA             | Zhu et al., 2024b       | 2024 | R    | Lin   | regular  | none    | frame   | M1    | idemp. |    |
| AutoLoRA            | Zhang et al., 2024b     | 2024 | R    | Cone  | apex     | GL      | GL×GL   | M1    | span   |    |
| • BD-LoRA           | Wang et al., 2025d      | 2025 | R    | Cone  | apex     | GL      | Π       | M1    | span   | ✓  |
| • C3A               | Chen et al., 2024a      | 2024 | R    | Lin   | regular  | none    | Π       | M1    | idemp. | ✓  |
| Concept Sliders     | Gandikota et al., 2023  | 2023 | R    | Cone  | apex     | GL      | GL×GL   | M1    | span   |    |
| • CorDA             | Yang et al., 2024c      | 2024 | R    | Cone* | regular  | GL      | O×O     | M1    | span   | ✓  |
| • DEFT              | Kumar et al., 2025      | 2025 | R    | Cone  | apex     | GL      | GL×GL   | M1    | span   | ✓  |
| • DeLoRA            | Bini et al., 2025       | 2025 | R    | Cone  | apex     | nonlin. | GL×GL   | M1    | span   | ✓  |
| DoRA (dynamic rank) | Mao et al., 2024a       | 2024 | R    | Cone  | apex     | nonlin. | GL×GL   | M1    | span   |    |
| DSEE                | Chen et al., 2021       | 2021 | R    | Cone  | apex     | GL      | Mon×Mon | M1    | span   |    |
| • DyLoRA            | Valipour et al., 2022   | 2022 | R    | Cone  | apex     | torus   | GL×GL   | M1    | span   |    |
| ElaLoRA             | Chang et al., 2025      | 2025 | R    | Cone  | apex     | nonlin. | GL×GL   | M1    | span   |    |
| • EVA               | Paischer et al., 2024   | 2024 | R    | Cone  | apex     | GL      | GL×GL   | M1    | span   | ✓  |
| • FacT              | Jie and Deng, 2022b     | 2022 | R    | Cone  | apex     | GL      | Π       | M1    | span   |    |
| FedRot-LoRA         | Zhang et al., 2026c     | 2026 | R    | Cone  | apex     | GL      | GL×GL   | M1    | span   |    |
| • FFA-LoRA          | Sun et al., 2024        | 2024 | R    | Lin   | regular  | none    | frame   | M1    | idemp. |    |
| Flat-LoRA           | Li et al., 2024c        | 2024 | R    | Cone  | apex     | GL      | GL×GL   | M1    | span   |    |
| FlexLoRA            | Liu et al., 2026c       | 2026 | R    | Cone  | apex     | nonlin. | GL×GL   | M1    | span   |    |
| FLoRA (Tucker core) | Si et al., 2024b        | 2024 | R    | Cone  | apex     | GL      | GL×GL   | M1    | span   |    |
| FLoRG               | Meng et al., 2026       | 2026 | R    | Cone  | defect   | GL      | frame   | M1    | span   |    |
| • FourierFT         | Gao et al., 2024        | 2024 | R    | Lin   | regular  | none    | Π       | M1    | idemp. | ✓  |
| FourTune            | Xue et al., 2026        | 2026 | R    | Cone  | defect   | GL      | Mon×Mon | Mq    | span   |    |
| FrameFT             | Adepu et al., 2026      | 2026 | R    | Lin   | regular  | none    | Π       | M1    | idemp. |    |
| • FRoD              | Wan et al., 2025        | 2025 | R    | Lin   | regular  | none    | O×O     | M1    | idemp. | ✓  |
| GeoRA               | Zhang et al., 2026d     | 2026 | R    | Cone* | regular  | GL      | Mon×Mon | M1    | span   |    |
| GiVA                | Gangwar et al., 2026    | 2026 | R    | Cone  | apex     | scalar  | O×O     | M1    | span   |    |
| GoRA                | He et al., 2025a        | 2025 | R    | Cone  | defect   | GL      | GL×GL   | M1    | span   |    |
| • GraLoRA           | Jung et al., 2025       | 2025 | R    | Cone  | apex     | GL      | Π       | M1    | span   | ✓  |
| HetLoRA             | Cho et al., 2024        | 2024 | R    | Cone  | apex     | GL      | GL×GL   | M1    | span   |    |
| Hybrid-LoRA         | Zhang et al., 2026a     | 2026 | R    | Cone  | apex     | GL      | GL×GL   | M1    | span   |    |
| IC-LoRA             | Huang et al., 2024      | 2024 | R    | Cone  | apex     | GL      | GL×GL   | M1    | span   |    |
| IncreLoRA           | Zhang et al., 2023a     | 2023 | R    | Cone  | apex     | nonlin. | GL×GL   | M1    | span   |    |
| • InfLoRA           | Liang and Li, 2024      | 2024 | R    | Lin   | regular  | none    | GL×CO   | M1    | sched. |    |
| Intrinsic SAID      | Aghajanyan et al., 2020 | 2020 | R    | Cone  | apex     | scalar  | Π       | M1    | span   |    |
| IR-QLoRA            | Qin et al., 2024a       | 2024 | R    | Cone  | defect   | GL      | Mon×Mon | Mq    | span   |    |
| KAdaptation         | He et al., 2022         | 2022 | R    | Cone  | apex     | GL      | Π       | M1    | span   |    |

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Table 36, continued

| Method                 | Reference                                | Year | Kind | Shape | Base pt. | Gauge   | Covar.  | Merge | Rebase | HF |
|------------------------|------------------------------------------|------|------|-------|----------|---------|---------|-------|--------|----|
| • KaSA                 | Wang et al., 2024a                       | 2024 | R    | Cone  | defect   | nonlin. | O×O     | M1    | span   | ✓  |
| KeepLoRA               | Luo et al., 2026                         | 2026 | R    | Lin   | regular  | none    | O×O     | M1    | sched. |    |
| KRAadapter             | Albert et al., 2025b                     | 2025 | R    | Cone  | apex     | torus   | $\Pi$   | M1    | span   |    |
| • KronA                | Edalati et al., 2022                     | 2022 | R    | Cone  | apex     | scalar  | $\Pi$   | M1    | span   |    |
| LaMDA                  | Azizi et al., 2024                       | 2024 | R    | Lin   | regular  | GL      | O×O     | M1    | idemp. |    |
| LoCA                   | Du et al., 2025                          | 2025 | R    | Lin   | regular  | none    | $\Pi$   | M1    | idemp. |    |
| LoDA                   | He et al., 2026b                         | 2026 | R    | Lin   | regular  | none    | GL×CO   | M1    | sched. |    |
| LoFT                   | Tastan et al., 2025                      | 2025 | R    | Cone  | apex     | GL      | GL×GL   | M1    | span   |    |
| • LoftQ                | Li et al., 2023b                         | 2023 | R    | Cone* | defect   | GL      | Mon×Mon | Mq    | span   | ✓  |
| • LoHa                 | Hyeon-Woo et al., 2021                   | 2021 | R    | Cone  | apex     | GL×T    | Mon×Mon | M1    | span   | ✓  |
| • LoKr                 | Yeh et al., 2023                         | 2023 | R    | Cone  | apex     | GL      | $\Pi$   | M1    | span   | ✓  |
| • LongLoRA             | Chen et al., 2023c                       | 2023 | R    | Cone  | apex     | GL      | Mon×GL  | M1    | span   |    |
| • LoRA                 | Hu et al., 2021                          | 2021 | R    | Cone  | apex     | GL      | GL×GL   | M1    | span   | ✓  |
| LoRA Dropout           | Lin et al., 2024b                        | 2024 | R    | Cone  | apex     | GL      | GL×GL   | M1    | span   |    |
| LoRA Without Regret    | Schulman and Thinking Machines Lab, 2025 | 2025 | R    | Cone  | apex     | GL      | GL×GL   | M1    | span   | ✓  |
| • LoRA+                | Hayou et al., 2024b                      | 2024 | R    | Cone  | apex     | GL      | GL×GL   | M1    | span   | ✓  |
| LoRA-DA                | Zhang et al., 2025b                      | 2025 | R    | Cone  | defect   | GL      | GL×GL   | M1    | span   |    |
| LoRA-Dash              | Si et al., 2024a                         | 2024 | R    | Cone  | apex     | GL      | O×O     | M1    | span   |    |
| LoRA-drop              | Zhou et al., 2024                        | 2024 | R    | Cone  | apex     | GL      | GL×GL   | M1    | span   |    |
| • LoRA-FA              | Zhang et al., 2023d                      | 2023 | R    | Lin   | regular  | none    | frame   | M1    | idemp. | ✓  |
| • LoRA-GA              | Wang et al., 2024e                       | 2024 | R    | Cone* | regular  | GL      | O×O     | M1    | span   | ✓  |
| LoRA-Mini              | Singh et al., 2024                       | 2024 | R    | Cone  | defect   | GL      | frame   | M1    | span   |    |
| LoRA-Null              | Tang et al., 2025                        | 2025 | R    | Cone* | regular  | GL      | GL×CO   | M1    | span   |    |
| LoRA-One               | Zhang et al., 2025d                      | 2025 | R    | Cone  | defect   | GL      | GL×GL   | M1    | span   |    |
| • LoRA-Pro             | Wang et al., 2024i                       | 2024 | R    | Cone  | apex     | GL      | GL×GL   | M1    | span   |    |
| • LoRA-RITE            | Yen et al., 2024                         | 2024 | R    | Cone  | apex     | GL      | GL×GL   | M1    | span   |    |
| LoRA-RLPO / LoRA-RLMO  | Zhang et al., 2026e                      | 2026 | R    | Cone  | apex     | GL      | GL×GL   | M1    | span   |    |
| LoRA-S (Adams, LRACS)  | Zheng and Wu, 2026                       | 2025 | R    | Cone  | apex     | GL      | GL×GL   | M1    | span   |    |
| • LoRA-SB              | Ponkshe et al., 2024                     | 2024 | R    | Lin   | defect   | none    | Mon×Mon | M1    | idemp. |    |
| • LoRA-XS              | Balazy et al., 2024                      | 2024 | R    | Lin   | regular  | none    | O×O     | M1    | idemp. |    |
| • LoRETTA              | Yang et al., 2024d                       | 2024 | R    | Cone* | regular  | GL      | $\Pi$   | M1    | span   |    |
| LoRI                   | Zhang et al., 2025a                      | 2025 | R    | Lin   | regular  | none    | frame   | M1    | idemp. |    |
| LoRTA                  | Hounie et al., 2024                      | 2024 | R    | Cone  | apex     | torus   | $\Pi$   | M1    | span   |    |
| LoTR                   | Bershatsky et al., 2024                  | 2024 | R    | Cone  | apex     | GL      | GL×GL   | M1    | span   |    |
| LQ-LoRA                | Guo et al., 2023                         | 2023 | R    | Cone* | defect   | GL      | Mon×Mon | Mq    | span   |    |
| MatryoshkaLoRA         | Modoranu et al., 2026                    | 2026 | R    | Cone  | apex     | GL      | GL×GL   | M1    | span   |    |
| MELoRA                 | Ren et al., 2024                         | 2024 | R    | Cone  | apex     | GL      | $\Pi$   | M1    | span   |    |
| MetaTT                 | Lopez-Piqueres et al., 2025              | 2025 | R    | Cone  | apex     | GL      | GL×GL   | M1    | span   |    |
| • MiCA                 | Rüdiger and Raschka, 2026                | 2026 | R    | Lin   | regular  | none    | O×O     | M1    | idemp. | ✓  |
| • MiLoRA               | Wang et al., 2024b                       | 2024 | R    | Cone* | regular  | GL      | O×O     | M1    | span   |    |
| • MiSS (formerly Bone) | Kang and Yin, 2024                       | 2024 | R    | Lin   | regular  | none    | $\Pi$   | M1    | idemp. | ✓  |
| ModuLoRA               | Yin et al., 2023                         | 2023 | R    | Cone  | defect   | GL      | Mon×Mon | Mq    | span   |    |
| MonteCLORA             | Sengupta et al., 2024                    | 2024 | R    | Cone  | apex     | GL      | GL×GL   | M1    | span   | ✓  |
| • MoRA                 | Jiang et al., 2024                       | 2024 | R    | Lin   | regular  | none    | $\Pi$   | M1    | idemp. |    |
| MoRe                   | Tan et al., 2024a                        | 2024 | R    | Cone  | apex     | torus   | $\Pi$   | M1    | span   |    |
| MoS                    | Wang et al., 2024f                       | 2024 | R    | Cone  | apex     | scalar  | $\Pi$   | M1    | span   |    |
| MoSA                   | Wang et al., 2026b                       | 2026 | R    | Lin   | regular  | none    | frame   | M1    | idemp. |    |
| MoSLoRA                | Wu et al., 2024b                         | 2024 | R    | Cone  | apex     | GL      | GL×GL   | M1    | span   |    |
| • NOLA                 | Koohpayegani et al., 2023                | 2023 | R    | Cone  | apex     | scalar  | frame   | M1    | span   |    |
| • O-LoRA               | Wang et al., 2023b                       | 2023 | R    | Cone  | apex     | GL      | GL×GL   | M1    | sched. |    |
| • OLoRA                | Büyükkakyüz, 2024                        | 2024 | R    | Cone* | regular  | GL      | $\Pi$   | M1    | span   | ✓  |

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Table 36, continued

| Method                                   | Reference                  | Year | Kind | Shape | Base pt. | Gauge   | Covar.  | Merge | Rebase | HF |
|------------------------------------------|----------------------------|------|------|-------|----------|---------|---------|-------|--------|----|
| Orthogonal Adaptation                    | Po et al., 2023            | 2023 | R    | Lin   | regular  | none    | frame   | M1    | idemp. |    |
| • OSF                                    | Nayak et al., 2025         | 2025 | R    | Lin   | regular  | nonlin. | O×O     | M1    | idemp. | ✓  |
| • PEANuT                                 | Zhong et al., 2024b        | 2024 | R    | Cone  | apex     | nonlin. | Mon×GL  | M1    | span   | ✓  |
| • PiSSA                                  | Meng et al., 2024a         | 2024 | R    | Cone* | regular  | GL      | O×O     | M1    | span   | ✓  |
| PLoP                                     | Hayou et al., 2025         | 2025 | R    | Cone  | apex     | GL      | GL×GL   | M1    | span   |    |
| PoLAR                                    | Lion et al., 2025          | 2025 | R    | Cone  | apex     | GL      | GL×GL   | M1    | span   |    |
| • PRISM                                  | Wang and Zhang, 2026       | 2026 | R    | Cone* | regular  | GL      | O×O     | M1    | span   |    |
| PRoLoRA                                  | Wang et al., 2024g         | 2024 | R    | Cone  | apex     | GL      | Π       | M1    | span   |    |
| • PVerA                                  | Fillioux et al., 2025      | 2025 | R    | Cone  | apex     | scalar  | frame   | M1    | span   | ✓  |
| • QA-LoRA                                | Xu et al., 2023b           | 2023 | R    | Cone  | defect   | GL      | Π       | M1    | span   | ✓  |
| QeRL                                     | Huang et al., 2025c        | 2025 | R    | Cone  | defect   | GL      | Mon×Mon | Mq    | span   |    |
| • QLoRA                                  | Dettmers et al., 2023      | 2023 | R    | Cone  | defect   | GL      | Mon×Mon | Mq    | span   | ✓  |
| QuanTA                                   | Chen et al., 2024f         | 2024 | R    | Cone* | regular  | GL      | Π       | M1    | span   |    |
| Quantum-PEFT                             | Koike-Akino et al., 2025   | 2025 | R    | Cone  | apex     | nonlin. | Π       | M1    | span   |    |
| • RandLoRA                               | Albert et al., 2025a       | 2025 | R    | Cone  | apex     | torus   | frame   | M1    | span   | ✓  |
| RaSA                                     | He et al., 2025b           | 2025 | R    | Cone  | apex     | GL      | GL×GL   | M1    | span   |    |
| • Riemannian LoRA                        | Zhang and Pilanci, 2024a   | 2024 | R    | Cone  | apex     | GL      | GL×GL   | M1    | span   | ✓  |
| Riemannion                               | Bogachev et al., 2025      | 2025 | R    | Cone* | regular  | none    | O×O     | M1    | span   |    |
| • RoSA                                   | Nikdan et al., 2024        | 2024 | R    | Cone  | apex     | GL      | Mon×Mon | M1    | span   |    |
| • rsLoRA                                 | Kalajdzievski, 2023        | 2023 | R    | Cone  | apex     | GL      | GL×GL   | M1    | span   | ✓  |
| SD-LoRA                                  | Wu et al., 2025c           | 2025 | R    | Cone  | defect   | GL      | GL×GL   | M1    | sched. |    |
| SEAL                                     | Zweiger et al., 2025       | 2025 | R    | Cone  | apex     | GL      | GL×GL   | M1    | span   |    |
| ShareLoRA                                | Song et al., 2024          | 2024 | R    | Cone  | apex     | GL      | GL×GL   | M1    | span   |    |
| Sine-LoRA                                | Ji et al., 2024            | 2024 | R    | Cone  | apex     | GL      | Mon×Mon | M1    | span   |    |
| SingLoRA                                 | Bensaïd et al., 2025       | 2025 | R    | Cone  | defect   | GL      | Π       | M1    | span   |    |
| SLAO (Merge before Forget)               | Qiao and Mahdavi, 2025     | 2025 | R    | Cone  | defect   | GL      | GL×GL   | M1    | span   |    |
| SLTrain                                  | Han et al., 2024a          | 2024 | R    | Cone  | defect   | GL      | Mon×Mon | M1    | span   |    |
| SoRA                                     | Ding et al., 2023          | 2023 | R    | Cone  | apex     | nonlin. | GL×GL   | M1    | span   |    |
| SparseLoRA                               | Khaki et al., 2025         | 2025 | R    | Cone  | apex     | GL      | GL×GL   | M1    | span   |    |
| Spectral Adapter <sup>A</sup>            | Zhang and Pilanci, 2024b   | 2024 | R    | Cone* | regular  | GL      | O×O     | M1    | span   |    |
| SRR (Structured Residual Reconstruction) | Cho et al., 2026           | 2026 | R    | Cone* | defect   | GL      | Mon×Mon | Mq    | span   |    |
| SSH                                      | Shen et al., 2025          | 2025 | R    | Lin   | regular  | none    | Π       | M1    | idemp. |    |
| Stable-LoRA                              | Wu et al., 2025d           | 2025 | R    | Cone  | apex     | GL      | GL×GL   | M1    | span   |    |
| StructLoRA                               | Xiao et al., 2026          | 2026 | R    | Lin   | defect   | nonlin. | GL×GL   | M1    | idemp. |    |
| • SVFT                                   | Lingam et al., 2024        | 2024 | R    | Lin   | regular  | none    | O×O     | M1    | idemp. |    |
| TeRA                                     | Gu et al., 2025            | 2025 | R    | Cone  | apex     | torus   | Π       | M1    | span   |    |
| Tied-LoRA                                | Renduchintala et al., 2023 | 2023 | R    | Cone  | apex     | torus   | Mon×GL  | M1    | span   |    |
| Tina                                     | Wang et al., 2025b         | 2025 | R    | Cone  | apex     | GL      | GL×GL   | M1    | span   | ✓  |
| • TinyLoRA                               | Morris et al., 2026        | 2026 | R    | Lin   | regular  | none    | O×O     | M1    | idemp. | ✓  |
| Trans-LoRA                               | Wang et al., 2024d         | 2024 | R    | Cone  | apex     | GL      | GL×GL   | M1    | span   |    |
| TT-LoRA                                  | Anjum et al., 2024         | 2024 | R    | Cone  | defect   | GL      | Π       | M1    | span   |    |
| • Uni-LoRA                               | Li et al., 2025a           | 2025 | R    | Cone  | defect   | none    | frame   | M1    | span   | ✓  |
| • VB-LoRA                                | Li et al., 2024d           | 2024 | R    | Cone  | defect   | none    | Π       | M1    | span   | ✓  |
| VeLoRA                                   | Miles et al., 2024         | 2024 | R    | Cone  | apex     | GL      | GL×GL   | M1    | span   | ✓  |
| • VeRA                                   | Kopiczko et al., 2023      | 2023 | R    | Cone  | apex     | scalar  | frame   | M1    | span   | ✓  |
| WASI                                     | Nguyen et al., 2025        | 2025 | R    | Cone* | defect   | GL      | O×O     | M1    | idemp. |    |
| • WaveFT                                 | Bilican et al., 2025       | 2025 | R    | Lin   | regular  | none    | Π       | M1    | idemp. | ✓  |
| <b>R, action</b>                         |                            |      |      |       |          |         |         |       |        |    |
| • (IA) <sup>3</sup>                      | Liu et al., 2022a          | 2022 | R    | Lin   | regular  | none    | Mon×GL  | M1→   | idemp. | ✓  |
| BiDoRA                                   | Qin et al., 2024b          | 2024 | R    | Sat   | apex     | GL      | Mon×GL  | M1    | span   |    |
| • BOFT                                   | Liu et al., 2023b          | 2023 | R    | Sat   | regular  | nonlin. | Π       | M1    | group  | ✓  |
| • DoRA                                   | Liu et al., 2024b          | 2024 | R    | Sat   | apex     | GL      | Mon×GL  | M1    | span   | ✓  |
| EfficientQAT                             | Chen et al., 2024b         | 2024 | R    | Lin   | defect   | none    | Π       | M1    | idemp. |    |
| ETHER                                    | Bini et al., 2024          | 2024 | R    | Orb   | unpt.    | torus   | GL×CO   | M1    | group  |    |

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Table 36, continued

| Method                          | Reference                | Year | Kind | Shape | Base pt. | Gauge   | Covar.  | Merge | Rebase | HF |
|---------------------------------|--------------------------|------|------|-------|----------|---------|---------|-------|--------|----|
| ETHER+                          | Bini et al., 2024        | 2024 | R    | Cone  | apex     | torus   | O×O     | M1    | group  |    |
| •GOFT                           | Ma et al., 2024          | 2024 | R    | Orb   | regular  | none    | Π       | M1    | group  |    |
| •HRA                            | Yuan et al., 2024        | 2024 | R    | Meet  | apex     | nonlin. | GL×CO   | M1    | group  | ✓  |
| L4Q                             | Jeon et al., 2024        | 2024 | R    | Lin   | defect   | nonlin. | Π       | M1    | span   |    |
| •LoCO                           | Nguyen et al., 2026      | 2026 | R    | Meet  | apex     | GL      | GL×CO   | M1    | group  |    |
| •LOFT                           | Zhao et al., 2026        | 2026 | R    | Orb   | regular  | none    | O×O     | M1    | idemp. |    |
| LoRA-X                          | Farhadzadeh et al., 2025 | 2025 | R    | Lin   | regular  | none    | O×O     | M1    | idemp. |    |
| LoRAPrune                       | Zhang et al., 2023f      | 2023 | R    | Sat   | apex     | GL      | Mon×GL  | M1    | span   |    |
| Masking                         | Zhao et al., 2020        | 2020 | R    | Orb   | defect   | torus   | Mon×Mon | M1    | idemp. |    |
| •OFT                            | Qiu et al., 2023         | 2023 | R    | Orb   | regular  | none    | GL×CO   | M1    | idemp. | ✓  |
| •OFTv2                          | Qiu et al., 2025         | 2025 | R    | Orb   | regular  | none    | GL×CO   | M1    | group  | ✓  |
| PaRa                            | Chen et al., 2024c       | 2024 | R    | Orb   | unpt.    | GL      | GL×CO   | M1    | group  | ✓  |
| •PEQA                           | Kim et al., 2023b        | 2023 | R    | Lin   | defect   | none    | Mon×Mon | M1    | idemp. |    |
| •Piggyback                      | Mallya et al., 2018      | 2018 | R    | Orb   | regular  | torus   | Mon×Mon | M1    | idemp. |    |
| •PSOFT                          | Wu et al., 2025a         | 2025 | R    | Sat   | apex     | scalar  | O×O     | M1    | group  | ✓  |
| •qGOFT                          | Ma et al., 2024          | 2024 | R    | Orb   | regular  | torus   | Π       | M1    | group  |    |
| QZO                             | Shang et al., 2025       | 2025 | R    | Lin   | defect   | none    | Π       | M1    | idemp. |    |
| •RED                            | Wu et al., 2024a         | 2024 | R    | Lin   | regular  | none    | Mon×GL  | M1→   | idemp. |    |
| RepAdapter                      | Luo et al., 2023a        | 2023 | R    | Cone  | apex     | GL      | Π       | M1→   | group  |    |
| Residual Adapters               | Rebuffi et al., 2017     | 2017 | R    | Lin   | regular  | torus   | GL×GL   | M1→   | idemp. |    |
| •RoAd                           | Liao and Monz, 2024      | 2024 | R    | Orb   | regular  | none    | Mon×GL  | M1    | idemp. | ✓  |
| Spectral Adapter <sup>R</sup>   | Zhang and Pilanci, 2024b | 2024 | R    | Orb   | regular  | none    | O×O     | M1    | idemp. |    |
| •SSF                            | Lian et al., 2022        | 2022 | R    | Lin   | regular  | none    | Mon×GL  | M1→   | idemp. |    |
| SVDiff                          | Han et al., 2023         | 2023 | R    | Lin   | regular  | none    | O×O     | M1    | idemp. |    |
| SVF (Transformer <sup>2</sup> ) | Sun et al., 2025         | 2025 | R    | Lin   | regular  | none    | O×O     | M1    | idemp. |    |
| <b>R, Hadamard</b>              |                          |      |      |       |          |         |         |       |        |    |
| FLoRA                           | Wen and Chaudhuri, 2023  | 2023 | R    | Cone  | apex     | GL      | Mon×Mon | M1    | group  |    |
| •GLoRA                          | Chavan et al., 2023      | 2023 | R    | Cone  | apex     | GL      | Mon×Mon | M1    | span   | ✓  |
| •HiRA                           | Huang et al., 2025b      | 2025 | R    | Cone  | apex     | GL      | Mon×Mon | M1    | span   | ✓  |
| LieRA                           | Si et al., 2025          | 2025 | R    | Cone  | apex     | GL      | Mon×Mon | M1    | span   |    |
| Low-Rank Ternary Adaptation     | Manolache et al., 2026   | 2026 | R    | Orb   | regular  | nonlin. | Π       | M1    | idemp. |    |
| SMoA                            | Liu et al., 2026e        | 2026 | R    | Cone  | apex     | GL      | Mon×Mon | M1    | span   |    |
| <b>R, selective</b>             |                          |      |      |       |          |         |         |       |        |    |
| •BEFT                           | Huang et al., 2025a      | 2025 | R    | Lin   | regular  | none    | GL×GL   | M1    | idemp. | ✓  |
| •BitFit                         | Ben-Zaken et al., 2021   | 2021 | R    | Lin   | regular  | none    | GL×GL   | M1    | idemp. |    |
| •Child-Tuning                   | Xu et al., 2021          | 2021 | R    | Lin   | regular  | none    | Mon×Mon | M1    | idemp. |    |
| DiaBlo                          | Gurses et al., 2025      | 2025 | R    | Lin   | regular  | none    | Π       | M1    | idemp. |    |
| •Diff Pruning                   | Guo et al., 2020         | 2020 | R    | Cone  | apex     | none    | Mon×Mon | M1    | span   |    |
| DiffFit                         | Xie et al., 2023         | 2023 | R    | Lin   | regular  | none    | Mon×GL  | M1→   | idemp. |    |
| ESFT                            | Wang et al., 2024j       | 2024 | R    | Lin   | regular  | none    | GL×GL   | M1    | idemp. |    |
| •FISH Mask                      | Sung et al., 2021        | 2021 | R    | Lin   | regular  | none    | Mon×Mon | M1    | idemp. |    |
| GPS                             | Zhang et al., 2023j      | 2023 | R    | Lin   | regular  | none    | Mon×Mon | M1    | idemp. |    |
| KappaTune                       | Ludwig, 2025             | 2025 | R    | Lin   | regular  | none    | O×O     | M1    | idemp. | ✓  |
| •Linear probing                 | Donahue et al., 2013     | 2013 | R    | Lin   | regular  | none    | GL×GL   | M1    | idemp. |    |
| •LN Tuning                      | Zhao et al., 2023        | 2023 | R    | Lin   | regular  | none    | Mon×GL  | M1→   | idemp. | ✓  |
| LoFit                           | Yin et al., 2024b        | 2024 | R    | Lin   | regular  | none    | GL×GL   | M1→   | idemp. |    |
| •LT-SFT                         | Ansell et al., 2021      | 2021 | R    | Lin   | regular  | none    | Mon×Mon | M1    | idemp. |    |
| MDPD                            | Zhang et al., 2026f      | 2026 | R    | Lin   | regular  | none    | Mon×GL  | M1→   | idemp. |    |
| PaFi                            | Liao et al., 2023a       | 2023 | R    | Lin   | regular  | none    | Mon×Mon | M1    | idemp. |    |
| SAM (sparse)                    | Fu et al., 2022          | 2022 | R    | Lin   | regular  | none    | Mon×Mon | M1    | idemp. |    |
| •SHIRA                          | Bhardwaj et al., 2024    | 2024 | R    | Lin   | regular  | none    | Mon×Mon | M1    | idemp. | ✓  |
| SIFT                            | Song et al., 2023        | 2023 | R    | Lin   | regular  | none    | Mon×Mon | M1    | idemp. |    |
| SMT                             | He et al., 2024a         | 2024 | R    | Lin   | regular  | none    | Π       | M1    | idemp. |    |
| Spectrum                        | Hartford et al., 2024    | 2024 | R    | Lin   | regular  | none    | O×O     | M1    | idemp. |    |
| SpIEL                           | Ansell et al., 2024      | 2024 | R    | Cone  | apex     | none    | Mon×Mon | M1    | span   |    |
| SPT                             | He et al., 2023          | 2023 | R    | Cone  | apex     | GL      | Mon×Mon | M1    | span   |    |

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Table 36, continued

| Method                           | Reference              | Year | Kind           | Shape | Base pt. | Gauge   | Covar.  | Merge | Rebase | HF |
|----------------------------------|------------------------|------|----------------|-------|----------|---------|---------|-------|--------|----|
| • Super-Tuning (Super / Supra)   | Ilin et al., 2026      | 2026 | R              | Lin   | regular  | none    | Mon×Mon | M1    | idemp. | ✓  |
| • Surgical FT                    | Lee et al., 2022       | 2022 | R              | Lin   | regular  | none    | GL×GL   | M1    | idemp. |    |
| S <sup>2</sup> FT                | Yang et al., 2024b     | 2024 | R              | Lin   | regular  | none    | Mon×GL  | M1    | idemp. |    |
| TinyTL                           | Cai et al., 2020       | 2020 | R              | Lin   | defect   | none    | Mon×Mon | M1    | idemp. |    |
| • Trainable Tokens               | team, 2025             | 2025 | R              | Lin   | regular  | none    | Mon×GL  | M1    | idemp. | ✓  |
| <b>E (pointed ext.)</b>          |                        |      |                |       |          |         |         |       |        |    |
| AdaMix                           | Wang et al., 2022a     | 2022 | E              | Fun   | neutral  | none    | GL×GL   | M∞    | –      |    |
| • Adapter (Houlsby)              | Houlsby et al., 2019   | 2019 | E              | Fun   | neutral  | none    | GL×GL   | M∞    | –      |    |
| Adapter+                         | Steitz and Roth, 2024  | 2024 | E              | Fun   | neutral  | torus   | GL×GL   | M∞    | –      |    |
| AdapterDrop                      | Rücklé et al., 2020    | 2020 | E              | Fun   | neutral  | torus   | GL×GL   | M∞    | –      |    |
| AdaptFormer                      | Chen et al., 2022a     | 2022 | E              | Fun   | neutral  | torus   | GL×GL   | M∞    | –      |    |
| AIM                              | Yang et al., 2023c     | 2023 | E              | Fun   | neutral  | none    | GL×GL   | M∞    | –      |    |
| AP (Activation Prompt)           | Zhang et al., 2023i    | 2023 | E              | Lin   | neutral  | none    | GL×GL   | M∞    | –      |    |
| CLIP-Adapter                     | Gao et al., 2021       | 2021 | E              | Fun   | defect   | torus   | GL×GL   | M∞    | –      |    |
| • Compacter                      | Mahabadi et al., 2021a | 2021 | E              | Fun   | neutral  | torus   | Π       | M∞    | –      |    |
| • ControlNet                     | Zhang et al., 2023e    | 2023 | E              | Fun   | neutral  | GL      | GL×GL   | M∞    | –      |    |
| Convpass                         | Jie and Deng, 2022a    | 2022 | E              | Fun   | neutral  | none    | GL×GL   | M∞    | –      |    |
| • Custom Diffusion               | Kumari et al., 2022    | 2022 | E              | Lin   | neutral  | none    | GL×GL   | M1    | idemp. |    |
| Initial State Tuning             | Galim et al., 2024     | 2024 | E              | Lin   | neutral  | none    | GL×GL   | M∞    | idemp. |    |
| IP-Adapter                       | Ye et al., 2023        | 2023 | E              | Fun   | defect   | torus   | GL×GL   | M∞    | –      |    |
| K-Adapter                        | Wang et al., 2020      | 2020 | E              | Fun   | neutral  | GL      | GL×GL   | M∞    | –      |    |
| • LLaMA-Adapter                  | Zhang et al., 2023h    | 2023 | E              | Fun   | neutral  | none    | GL×GL   | M∞    | –      | ✓  |
| LLaMA-Adapter V2                 | Gao et al., 2023       | 2023 | E              | Fun   | neutral  | torus   | Mon×GL  | M∞    | –      |    |
| • LoReFT                         | Wu et al., 2024d       | 2024 | E              | Fun   | defect   | GL      | GL×GL   | M∞    | –      |    |
| MAD-X                            | Pfeiffer et al., 2020c | 2020 | E              | Fun   | neutral  | torus   | Π       | M∞    | –      |    |
| Mona                             | Yin et al., 2024a      | 2024 | E              | Fun   | defect   | torus   | Mon×Mon | M∞    | –      |    |
| • Parallel Adapter               | He et al., 2021        | 2021 | E              | Fun   | neutral  | torus   | GL×GL   | M∞    | –      |    |
| • Pfeiffer Adapter               | Pfeiffer et al., 2020a | 2020 | E              | Fun   | neutral  | torus   | GL×GL   | M∞    | –      |    |
| PHM-Adapter                      | Mahabadi et al., 2021a | 2021 | E              | Fun   | neutral  | GL      | Π       | M∞    | –      |    |
| PrefixMemory-Tuning              | Wang et al., 2025a     | 2025 | E              | Fun   | neutral  | none    | GL×GL   | M∞    | –      |    |
| Res-Tuning                       | Jiang et al., 2023     | 2023 | E              | Fun   | neutral  | none    | GL×GL   | M∞    | –      |    |
| • ShadowPEFT                     | Li et al., 2026a       | 2026 | E              | Fun   | neutral  | GL      | Mon×Mon | M∞    | –      | ✓  |
| Side-Tuning                      | Zhang et al., 2019     | 2019 | E              | Fun   | neutral  | torus   | GL×GL   | M∞    | –      |    |
| ST-Adapter                       | Pan et al., 2022       | 2022 | E              | Fun   | neutral  | torus   | GL×GL   | M∞    | –      |    |
| State-offset Tuning              | Kang et al., 2025      | 2025 | E              | Lin   | neutral  | none    | GL×GL   | M∞    | idemp. |    |
| T2I-Adapter                      | Mou et al., 2023       | 2023 | E              | Fun   | defect   | torus   | GL×GL   | M∞    | –      |    |
| • Textual Inversion              | Gal et al., 2022       | 2022 | E              | Lin   | neutral  | none    | Mon×GL  | M1    | idemp. |    |
| Tip-Adapter                      | Zhang et al., 2022a    | 2022 | E              | Fun   | defect   | none    | O×O     | M∞    | –      |    |
| Visual Prompting                 | Bahng et al., 2022     | 2022 | E              | Lin   | defect   | none    | Π       | M1→   | idemp. |    |
| <b>E<sup>0</sup> (unpointed)</b> |                        |      |                |       |          |         |         |       |        |    |
| AutoPEFT                         | Zhou et al., 2023b     | 2023 | E <sup>0</sup> | Fun   | unpt.    | torus   | GL×GL   | M∞    | –      |    |
| • AutoPrompt                     | Shin et al., 2020      | 2020 | E <sup>0</sup> | Fun   | unpt.    | none    | GL×GL   | M∞    | –      |    |
| BBT                              | Sun et al., 2022       | 2022 | E <sup>0</sup> | Fun   | unpt.    | none    | frame   | M∞    | –      |    |
| • Cartridges                     | Eyuboglu et al., 2025  | 2025 | E <sup>0</sup> | Fun   | unpt.    | none    | GL×GL   | M∞    | –      | ✓  |
| • CoOp                           | Zhou et al., 2021      | 2021 | E <sup>0</sup> | Fun   | unpt.    | none    | GL×GL   | M1→   | –      |    |
| • CPT                            | Blau et al., 2024      | 2024 | E <sup>0</sup> | Fun   | unpt.    | none    | O×O     | M∞    | –      | ✓  |
| DePT                             | Shi and Lipani, 2023   | 2023 | E <sup>0</sup> | Fun   | unpt.    | GL      | GL×GL   | M∞    | –      |    |
| EasyControl                      | Zhang et al., 2025e    | 2025 | E <sup>0</sup> | Fun   | unpt.    | GL      | GL×GL   | M∞    | –      |    |
| LoRA layer replication (DUS)     | Kim et al., 2023a      | 2023 | E <sup>0</sup> | Fun   | unpt.    | GL      | GL×GL   | M1    | span   | ✓  |
| • LST                            | Sung et al., 2022      | 2022 | E <sup>0</sup> | Fun   | unpt.    | GL      | GL×GL   | M∞    | –      |    |
| • MAM Adapter                    | He et al., 2021        | 2021 | E <sup>0</sup> | Fun   | unpt.    | GL      | GL×GL   | M∞    | –      |    |
| MaPLe                            | Khattak et al., 2022   | 2022 | E <sup>0</sup> | Fun   | unpt.    | nonlin. | GL×GL   | M∞    | –      |    |
| MEFT                             | Liao et al., 2023b     | 2023 | E <sup>0</sup> | Fun   | unpt.    | none    | GL×GL   | M∞    | –      |    |

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Table 36, continued

| Method                    | Reference                  | Year | Kind           | Shape | Base pt. | Gauge   | Covar.  | Merge          | Rebase | HF |
|---------------------------|----------------------------|------|----------------|-------|----------|---------|---------|----------------|--------|----|
| • MPT                     | Wang et al., 2023d         | 2023 | E <sup>0</sup> | Fun   | unpt.    | torus   | GL×GL   | M <sub>∞</sub> | –      | ✓  |
| NOAH                      | Zhang et al., 2022b        | 2022 | E <sup>0</sup> | Fun   | unpt.    | GL      | GL×GL   | M <sub>∞</sub> | –      |    |
| OminiControl              | Tan et al., 2024b          | 2024 | E <sup>0</sup> | Fun   | unpt.    | GL      | GL×GL   | M <sub>∞</sub> | –      |    |
| • P-Tuning                | Liu et al., 2021d          | 2021 | E <sup>0</sup> | Fun   | unpt.    | nonlin. | GL×GL   | M <sub>∞</sub> | –      | ✓  |
| • P-Tuning v2             | Liu et al., 2021c          | 2021 | E <sup>0</sup> | Fun   | unpt.    | none    | GL×GL   | M <sub>∞</sub> | –      | ✓  |
| PPT                       | Gu et al., 2021            | 2021 | E <sup>0</sup> | Fun   | unpt.    | none    | GL×GL   | M <sub>∞</sub> | –      |    |
| • Prefix-Tuning           | Li and Liang, 2021         | 2021 | E <sup>0</sup> | Fun   | unpt.    | nonlin. | GL×GL   | M <sub>∞</sub> | –      | ✓  |
| • Prompt Tuning           | Lester et al., 2021        | 2021 | E <sup>0</sup> | Fun   | unpt.    | none    | GL×GL   | M <sub>∞</sub> | –      | ✓  |
| Residual Prompt Tuning    | Razdaibiedina et al., 2023 | 2023 | E <sup>0</sup> | Fun   | unpt.    | nonlin. | GL×GL   | M <sub>∞</sub> | –      |    |
| SPoT                      | Vu et al., 2021            | 2021 | E <sup>0</sup> | Fun   | unpt.    | none    | GL×GL   | M <sub>∞</sub> | –      |    |
| • UniPELT                 | Mao et al., 2021           | 2021 | E <sup>0</sup> | Fun   | unpt.    | GL      | GL×GL   | M <sub>∞</sub> | –      |    |
| • VPT                     | Jia et al., 2022           | 2022 | E <sup>0</sup> | Fun   | unpt.    | torus   | GL×GL   | M <sub>∞</sub> | –      |    |
| WARP                      | Hambardzumyan et al., 2021 | 2021 | E <sup>0</sup> | Fun   | unpt.    | none    | GL×GL   | M <sub>∞</sub> | –      |    |
| XPrompt                   | Ma et al., 2022            | 2022 | E <sup>0</sup> | Fun   | unpt.    | none    | II      | M <sub>∞</sub> | –      |    |
| <b>B (backward)</b>       |                            |      |                |       |          |         |         |                |        |    |
| AdaLomo                   | Lv et al., 2023a           | 2023 | B              | Lin   | regular  | none    | GL×GL   | na             | idemp. |    |
| Adam-mini                 | Zhang et al., 2024d        | 2024 | B              | Lin   | regular  | none    | GL×GL   | na             | idemp. |    |
| AGZO                      | Lin et al., 2026b          | 2026 | B              | Lin   | regular  | none    | frame   | –              | sched. |    |
| APOLLO                    | Zhu et al., 2024a          | 2024 | B              | Lin   | regular  | none    | GL×GL   | na             | idemp. |    |
| BAdam                     | Luo et al., 2024           | 2024 | B              | Lin   | regular  | none    | GL×GL   | na             | sched. |    |
| COLA                      | Xia et al., 2024           | 2024 | B              | Cone  | apex     | GL      | GL×GL   | M1             | sched. |    |
| Delta-LoRA                | Zi et al., 2023            | 2023 | B              | Cone  | apex     | GL      | GL×GL   | M1             | sched. |    |
| Fira                      | Chen et al., 2024e         | 2024 | B              | Lin   | regular  | none    | GL×GL   | na             | idemp. |    |
| • Flora                   | Hao et al., 2024           | 2024 | B              | Lin   | regular  | none    | frame   | na             | sched. |    |
| FLoRA (federated)         | Wang et al., 2024k         | 2024 | B              | Cone  | apex     | GL      | GL×GL   | M1             | sched. |    |
| FZOO                      | Dang et al., 2025          | 2025 | B              | Lin   | regular  | none    | Mon×Mon | –              | sched. |    |
| • GaLore                  | Zhao et al., 2024a         | 2024 | B              | Lin   | regular  | none    | O×O     | na             | sched. |    |
| GaLore 2                  | Su et al., 2025            | 2025 | B              | Lin   | regular  | none    | O×O     | na             | sched. |    |
| HiFT                      | Liu et al., 2024c          | 2024 | B              | Lin   | regular  | none    | GL×GL   | na             | sched. |    |
| LIFT                      | Liu et al., 2025           | 2025 | B              | Lin   | regular  | none    | Mon×Mon | na             | sched. |    |
| • LISA                    | Pan et al., 2024           | 2024 | B              | Lin   | regular  | none    | GL×GL   | na             | sched. |    |
| LOMO                      | Lv et al., 2023b           | 2023 | B              | Lin   | regular  | none    | GL×GL   | na             | idemp. |    |
| LoQT                      | Loeschcke et al., 2024     | 2024 | B              | Lin   | defect   | none    | Mon×Mon | Mq             | sched. |    |
| LoRA-Pre                  | Wang et al., 2026c         | 2026 | B              | Lin   | regular  | none    | GL×GL   | –              | –      |    |
| • LP-FT                   | Kumar et al., 2022         | 2022 | B              | Lin   | regular  | none    | GL×GL   | na             | sched. |    |
| • MeZO                    | Malladi et al., 2023       | 2023 | B              | Lin   | regular  | none    | frame   | na             | sched. |    |
| Online Subspace Descent   | Liang et al., 2024         | 2024 | B              | Lin   | regular  | none    | O×O     | na             | sched. |    |
| PeriodicLoRA              | Meng et al., 2024b         | 2024 | B              | Cone  | apex     | GL      | GL×GL   | M1             | sched. |    |
| POET-X                    | Qiu et al., 2026           | 2026 | B              | Orb   | regular  | none    | Mon×Mon | M1             | sched. |    |
| Q-GaLore                  | Zhang et al., 2024e        | 2024 | B              | Lin   | defect   | none    | Mon×Mon | na             | sched. |    |
| • ReLoRA                  | Lialin et al., 2023b       | 2023 | B              | Cone  | apex     | GL      | GL×GL   | M1             | sched. |    |
| ScaLoRA                   | Zhang et al., 2025c        | 2025 | B              | Cone* | regular  | GL      | GL×GL   | M1             | sched. |    |
| Sparse Memory Finetuning  | Lin et al., 2025           | 2025 | B              | Lin   | regular  | none    | Mon×GL  | na             | sched. |    |
| ZO-Muon                   | Lang et al., 2026          | 2026 | B              | Lin   | regular  | none    | frame   | –              | sched. |    |
| <b>P (post hoc)</b>       |                            |      |                |       |          |         |         |                |        |    |
| AdaMerging                | Yang et al., 2023b         | 2023 | P              | Lin   | defect   | none    | GL×GL   | M1             | idemp. |    |
| AdapterSoup               | Chronopoulou et al., 2023  | 2023 | P              | Fun   | unpt.    | –       | –       | M <sub>∞</sub> | –      |    |
| AdaRank                   | Lee et al., 2025           | 2025 | P              | Cone  | defect   | torus   | O×O     | M1             | span   |    |
| Compress-then-Merge (CtM) | He et al., 2026c           | 2026 | P              | Lin   | regular  | none    | O×O     | M1             | idemp. |    |
| Core Space                | Panariello et al., 2025    | 2025 | P              | Lin   | regular  | none    | O×O     | M1             | idemp. |    |
| • DARE                    | Yu et al., 2023            | 2023 | P              | Lin   | regular  | none    | Mon×Mon | M1             | idemp. | ✓  |
| DC-Merge                  | Zhang et al., 2026b        | 2026 | P              | Lin   | regular  | none    | O×O     | M1             | idemp. |    |
| EFT                       | Mitchell et al., 2023      | 2023 | P              | Fun   | neutral  | –       | GL×GL   | M <sub>∞</sub> | –      |    |

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Table 36, continued

| Method                           | Reference                 | Year | Kind  | Shape | Base pt. | Gauge   | Covar.  | Merge      | Rebase | HF |
|----------------------------------|---------------------------|------|-------|-------|----------|---------|---------|------------|--------|----|
| Fisher Merging                   | Matena and Raffel, 2021   | 2021 | P     | –     | unpt.    | –       | Mon×Mon | M1         | –      |    |
| GenKnowSub                       | Bagherifard et al., 2025  | 2025 | P     | Fun   | unpt.    | –       | O×O     | M $\infty$ | –      | ✓  |
| GradFix                          | Rinaldi et al., 2025      | 2025 | P     | Lin   | regular  | none    | Mon×Mon | M1         | idemp. |    |
| • Intruder-dimension reduction   | Shuttleworth et al., 2024 | 2024 | P     | Lin   | defect   | none    | O×O     | M1         | idemp. | ✓  |
| • KnOTS                          | Stoica et al., 2024       | 2024 | P     | Lin   | regular  | none    | O×O     | M1         | idemp. |    |
| • LCM-LoRA                       | Luo et al., 2023b         | 2023 | P     | Lin   | regular  | none    | GL×GL   | M1         | idemp. | ✓  |
| • LoRA Soups (CAT)               | Prabhakar et al., 2024    | 2024 | P     | Lin   | regular  | none    | GL×GL   | M1         | idemp. |    |
| LoRA Switch / LoRA Composite     | Zhong et al., 2024a       | 2024 | P     | Fun   | unpt.    | –       | –       | M $\infty$ | –      |    |
| LoRA-LEGO                        | Zhao et al., 2024c        | 2024 | P     | –     | unpt.    | –       | O×O     | M1         | –      |    |
| • LoraHub                        | Huang et al., 2023        | 2023 | P     | Cone  | apex     | none    | GL×GL   | M1         | –      |    |
| Mix-of-Show                      | Gu et al., 2023           | 2023 | P     | –     | unpt.    | –       | GL×GL   | M1         | –      |    |
| • Model Soups                    | Wortsman et al., 2022     | 2022 | P     | –     | unpt.    | –       | GL×GL   | M1         | –      |    |
| OrthoMerge                       | Yang et al., 2026a        | 2026 | P     | –     | unpt.    | –       | GL×CO   | M1         | –      |    |
| PEM Arithmetic                   | Zhang et al., 2023c       | 2023 | P     | –     | unpt.    | –       | GL×GL   | M1         | –      | ✓  |
| Pico                             | Tang and Yang, 2026       | 2026 | P     | –     | unpt.    | –       | O×O     | M1         | –      |    |
| • Proxy-Tuning                   | Liu et al., 2024a         | 2024 | P     | Fun   | neutral  | –       | GL×GL   | M $\infty$ | –      |    |
| RAIN-Merging                     | Huang et al., 2025d       | 2025 | P     | Lin   | regular  | none    | GL×CO   | M1         | idemp. |    |
| RegMean                          | Jin et al., 2022          | 2022 | P     | –     | unpt.    | –       | Mon×GL  | M1         | –      |    |
| • Task Arithmetic                | Ilharco et al., 2022      | 2022 | P     | Lin   | regular  | none    | GL×GL   | M1         | idemp. | ✓  |
| • TIES-Merging                   | Yadav et al., 2023        | 2023 | P     | Lin   | regular  | none    | Mon×Mon | M1         | idemp. | ✓  |
| TSV-Merge                        | Gargiulo et al., 2024     | 2024 | P     | Lin   | regular  | none    | O×O     | M1         | idemp. |    |
| • WiSE-FT                        | Wortsman et al., 2021     | 2021 | P     | Lin   | regular  | none    | GL×GL   | M1         | idemp. | ✓  |
| ZipLoRA                          | Shah et al., 2023         | 2023 | P     | Lin   | defect   | none    | Mon×GL  | M1         | idemp. |    |
| <b><math>F_T</math> (family)</b> |                           |      |       |       |          |         |         |            |        |    |
| • AdapterFusion                  | Pfeiffer et al., 2020a    | 2020 | $F_T$ | Fun   | defect   | GL      | GL×GL   | M $\infty$ | –      |    |
| • aLoRA                          | Greenewald et al., 2025   | 2025 | $F_T$ | Fun   | apex     | GL      | GL×GL   | M $\infty$ | –      | ✓  |
| • Arrow                          | Ostapenko et al., 2024    | 2024 | $F_T$ | Fun   | unpt.    | –       | –       | M $\infty$ | –      | ✓  |
| ATTEMPT                          | Asai et al., 2022         | 2022 | $F_T$ | Fun   | unpt.    | nonlin. | Mon×Mon | M $\infty$ | –      |    |
| Bayesian-LoRA                    | Lin et al., 2026a         | 2026 | $F_T$ | Cone  | defect   | nonlin. | GL×GL   | M $\infty$ | –      |    |
| BLoB                             | Wang et al., 2024h        | 2024 | $F_T$ | Cone  | apex     | torus   | GL×GL   | M $\infty$ | –      |    |
| CoA-LoRA                         | Ye et al., 2025           | 2025 | $F_T$ | Cone* | defect   | GL      | Mon×Mon | Mq         | –      |    |
| CoCoOp                           | Zhou et al., 2022         | 2022 | $F_T$ | Fun   | unpt.    | torus   | GL×GL   | M $\infty$ | –      |    |
| CODA-Prompt                      | Smith et al., 2022        | 2022 | $F_T$ | Fun   | unpt.    | torus   | Mon×GL  | M $\infty$ | –      |    |
| DnD                              | Liang et al., 2025        | 2025 | $F_T$ | Cone  | defect   | GL      | GL×GL   | M1         | –      |    |
| Doc-to-LoRA (D2L)                | Charakorn et al., 2026    | 2026 | $F_T$ | Cone  | defect   | GL      | GL×GL   | M1         | –      |    |
| DualPrompt                       | Wang et al., 2022b        | 2022 | $F_T$ | Fun   | unpt.    | torus   | O×O     | M $\infty$ | –      |    |
| FourierMoE                       | Jiang et al., 2026        | 2026 | $F_T$ | Fun   | neutral  | none    | $\Pi$   | M $\infty$ | –      |    |
| GenerativeAdapter                | Chen et al., 2024d        | 2024 | $F_T$ | Cone  | defect   | GL      | GL×GL   | M1         | –      |    |
| GOAT                             | Fan et al., 2025          | 2025 | $F_T$ | Fun   | defect   | GL      | –       | M $\infty$ | –      |    |
| • HydraLoRA                      | Tian et al., 2024         | 2024 | $F_T$ | Fun   | neutral  | GL      | –       | M $\infty$ | –      |    |
| • HyperDreamBooth                | Ruiz et al., 2023         | 2023 | $F_T$ | Cone  | defect   | GL      | frame   | M1         | sched. |    |
| • HyperFormer                    | Mahabadi et al., 2021b    | 2021 | $F_T$ | Fun   | defect   | GL      | GL×GL   | M $\infty$ | –      |    |
| ICEdit                           | Zhang et al., 2025f       | 2025 | $F_T$ | Fun   | neutral  | GL      | –       | M $\infty$ | –      |    |
| IDPG                             | Wu et al., 2022           | 2022 | $F_T$ | Fun   | unpt.    | GL      | $\Pi$   | M $\infty$ | –      |    |
| In-Place TTT                     | Feng et al., 2025         | 2025 | $F_T$ | Fun   | neutral  | torus   | GL×CO   | M $\infty$ | –      |    |
| K-LoRA                           | Ouyang et al., 2025       | 2025 | $F_T$ | –     | unpt.    | –       | Mon×Mon | M1         | –      |    |
| L2P                              | Wang et al., 2021         | 2021 | $F_T$ | Fun   | unpt.    | torus   | O×O     | M $\infty$ | –      |    |
| • Laplace-LoRA                   | Yang et al., 2023a        | 2023 | $F_T$ | Cone  | apex     | GL      | GL×GL   | M $\infty$ | –      |    |
| LD-MoLE                          | Zhuang et al., 2025       | 2025 | $F_T$ | Fun   | neutral  | GL      | GL×GL   | M $\infty$ | –      |    |
| • Lily                           | Zhong et al., 2024c       | 2024 | $F_T$ | Fun   | neutral  | GL      | –       | M $\infty$ | –      | ✓  |
| LiME                             | Kowsher et al., 2026      | 2026 | $F_T$ | Fun   | neutral  | GL×T    | $\Pi$   | M $\infty$ | –      |    |
| LoRA-Flow                        | Wang et al., 2024c        | 2024 | $F_T$ | Fun   | unpt.    | none    | –       | M $\infty$ | –      |    |
| LoRA-Mixer                       | Li et al., 2025b          | 2025 | $F_T$ | Fun   | unpt.    | GL      | GL×GL   | M $\infty$ | –      |    |
| • LoRAMoE                        | Dou et al., 2023          | 2023 | $F_T$ | Fun   | neutral  | GL      | –       | M $\infty$ | –      |    |

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Table 36, continued

| Method                 | Reference                  | Year | Kind  | Shape | Base pt. | Gauge   | Covar.         | Merge      | Rebase | HF |
|------------------------|----------------------------|------|-------|-------|----------|---------|----------------|------------|--------|----|
| LoraRetriever          | Zhao et al., 2024b         | 2024 | $F_T$ | Fun   | unpt.    | –       | –              | $M_\infty$ | –      |    |
| LPT                    | Liu et al., 2022b          | 2022 | $F_T$ | Fun   | unpt.    | torus   | $GL \times GL$ | $M_\infty$ | –      |    |
| LRAgent                | Jeon et al., 2026          | 2026 | $F_T$ | Cone  | apex     | GL      | $GL \times GL$ | $M_1$      | –      |    |
| MaLoRA / MaRA          | Dokme and Heck, 2026       | 2026 | $F_T$ | Fun   | neutral  | GL      | $GL \times GL$ | $M_\infty$ | –      |    |
| • MHR                  | Caccia et al., 2022        | 2022 | $F_T$ | Cone  | apex     | GL      | $\Pi$          | $M_1$      | span   | ✓  |
| MixLoRA                | Li et al., 2024b           | 2024 | $F_T$ | Fun   | neutral  | GL      | –              | $M_\infty$ | –      |    |
| MOELoRA                | Liu et al., 2023a          | 2023 | $F_T$ | Cone  | apex     | GL      | $GL \times GL$ | $M_1$      | span   |    |
| • MoLE                 | Wu et al., 2024c           | 2024 | $F_T$ | Fun   | unpt.    | scalar  | –              | $M_\infty$ | –      |    |
| MoV / MoLoRA           | Zadouri et al., 2023       | 2023 | $F_T$ | Fun   | neutral  | none    | –              | $M_\infty$ | –      |    |
| NaRA                   | Wang et al., 2026a         | 2026 | $F_T$ | Cone  | apex     | GL      | $GL \times GL$ | $M_1$      | –      |    |
| PERK                   | Chen et al., 2025          | 2025 | $F_T$ | Cone  | defect   | GL      | $GL \times GL$ | $M_1$      | –      |    |
| • PHATGOOSE            | Muqeeth et al., 2024       | 2024 | $F_T$ | Fun   | unpt.    | nonlin. | –              | $M_\infty$ | –      |    |
| • Polytropon (Poly)    | Ponti et al., 2022         | 2022 | $F_T$ | Cone  | apex     | GL      | $GL \times GL$ | $M_1$      | span   | ✓  |
| PreFT                  | Lanpouthakoun et al., 2026 | 2026 | $F_T$ | Fun   | apex     | GL      | $GL \times GL$ | $M_\infty$ | –      |    |
| Prompt2Effect          | Yang et al., 2026b         | 2026 | $F_T$ | Cone  | defect   | GL      | $GL \times GL$ | $M_1$      | –      |    |
| • Punica               | Chen et al., 2023b         | 2023 | $F_T$ | Cone  | –        | GL      | $GL \times GL$ | $M_1$      | –      |    |
| RotMoLE                | Sun et al., 2026           | 2026 | $F_T$ | Fun   | neutral  | GL      | $GL \times GL$ | $M_\infty$ | –      |    |
| • S-LoRA               | Sheng et al., 2023         | 2023 | $F_T$ | Cone  | –        | GL      | $GL \times GL$ | $M_1$      | –      |    |
| SHINE                  | Liu et al., 2026d          | 2026 | $F_T$ | Cone  | defect   | GL      | $GL \times GL$ | $M_1$      | –      |    |
| SmartFed (MoRE + EEQA) | Wu et al., 2025b           | 2025 | $F_T$ | Fun   | unpt.    | none    | $GL \times GL$ | $M_\infty$ | –      |    |
| SMEAR                  | Muqeeth et al., 2023       | 2023 | $F_T$ | Fun   | defect   | torus   | –              | $M_\infty$ | –      |    |
| • Text-to-LoRA (T2L)   | Charakorn et al., 2025     | 2025 | $F_T$ | Cone  | apex     | GL      | $GL \times GL$ | $M_1$      | –      |    |
| VAPT                   | Le et al., 2025            | 2025 | $F_T$ | Fun   | unpt.    | torus   | $GL \times GL$ | $M_\infty$ | –      |    |
| • X-LoRA               | Buehler and Buehler, 2024  | 2024 | $F_T$ | Fun   | unpt.    | none    | –              | $M_\infty$ | –      | ✓  |

Table 37. Every recorded simulation arrow  $M \rightarrow N$  (rules A2–A5 of §VIII), with its explicit map. Notes on the instances (rank, pointing) are in the ancillary file arrows.csv.

| Source $M$             | Target $N$        | Rule | Pointed map $h$ with $\rho_N \circ h = \rho_M$                                                                                                                     |
|------------------------|-------------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (IA) <sup>3</sup>      | DoRA              | A2   | $\ell \mapsto (B = 0, A = A_0, \mu = \ell \odot (\ w_{0,i}\ )_i)$                                                                                                  |
| (IA) <sup>3</sup>      | HiRA              | A3   | $\ell \mapsto (\ell - \mathbf{1}, \mathbf{1}^\top) (k, v); \quad \ell \mapsto (\mathbf{1}, (\ell - \mathbf{1})^\top) (W_{\text{down}})$                            |
| ABBA                   | LoHa              | A3   | $h(B_1, A_1, B_2, A_2) = (B_1, A_1, cB_2, A_2), \quad c = \frac{\alpha/\sqrt{r_1 r_2}}{\gamma_{\text{LoHa}}}$                                                      |
| ABBA                   | LoRA              | A3   | $h = (\frac{\alpha}{\sqrt{r_1 r_2}} B_1 \bullet B_2, (A_1^\top \bullet A_2^\top)^\top)$                                                                            |
| AdaLomo                | LOMO              | A3   | $h = \text{id}_\Theta$                                                                                                                                             |
| AdaLoRA                | LoRA              | A3   | $h(P, \Lambda, Q) = (P\Lambda, Q)$                                                                                                                                 |
| AdaLoRA                | PoLAR             | A3   | $h(P, \Lambda, Q) = (P(P^\top P)^{-1/2}, (P^\top P)^{1/2} \Lambda (QQ^\top)^{1/2}, Q^\top (QQ^\top)^{-1/2})$                                                       |
| Adam-mini              | LOMO              | A3   | $h = \text{id}_\Theta$                                                                                                                                             |
| AdaMerging             | LoRA Soups (CAT)  | A3   | $h = \text{id} : \lambda_i^l \mapsto \alpha_i^l$                                                                                                                   |
| AdaMix                 | Adapter (Houlsby) | A3   | $h((W_k^{\text{down}})_{k \leq M}, W^{\text{up}}) = (\frac{1}{M} \sum_k W_k^{\text{down}}, W^{\text{up}})$                                                         |
| AdaMSS                 | LoRA              | A3   | $h((C_k, B_k)_k) = ([E_{S_1} C_1 \cdots E_{S_K} C_K], [B_1 \tilde{A}_1; \dots; B_K \tilde{A}_K])$                                                                  |
| AdaPaD                 | LoRA              | A3   | $h = ([b_{i,1} \cdots b_{i,r_i}], [a_{i,1}^\top; \dots; a_{i,r_i}^\top])$                                                                                          |
| AdapterDrop            | Adapter (Houlsby) | A2   | $h(q_{>n}) = ((U_l=0, D_l=D_l^0)_{l \leq n}, q_{>n})$                                                                                                              |
| AdapterDrop            | Pfeiffer Adapter  | A2   | $h(q_{>n}) = ((U_l=0, D_l=D_l^0)_{l \leq n}, q_{>n})$                                                                                                              |
| AdaptFormer            | Parallel Adapter  | A3   | $h(W_{\text{down}}, b_{\text{down}}, W_{\text{up}}, b_{\text{up}}) = (W_{\text{down}}, b_{\text{down}}, \frac{0.1}{s} W_{\text{up}}, \frac{0.1}{s} b_{\text{up}})$ |
| AGZO                   | LoRA-FA           | A3   | $\lambda \mapsto \lambda R_t$                                                                                                                                      |
| ALoRA                  | LoRA              | A3   | $h(B, A) = (B \text{diag}(\alpha), A)$                                                                                                                             |
| AltLoRA                | LoRA              | A3   | $h = \text{id}_Q$                                                                                                                                                  |
| AltLoRA                | Riemannian LoRA   | A3   | $h = \text{id}_Q$                                                                                                                                                  |
| AP (Activation Prompt) | BitFit            | A3   | $\delta = \mathbf{1} \otimes v \mapsto \Delta b_{\text{down}}^{(l-1)} = v$                                                                                         |
| AR1-ZO                 | LoRA              | A3   | $h(B, A) = (\frac{\alpha}{s} B, A)$                                                                                                                                |

continued on the next page

Table 37, continued

| Source $M$                | Target $N$        | Rule | Pointed map $h$ with $\rho_N \circ h = \rho_M$                                                                                                                                      |
|---------------------------|-------------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Arrow                     | LoRA              | A3   | $h_x(*) = (\sum_i p_i(x) B_i, \sum_i p_i(x) A_i)$                                                                                                                                   |
| Arrow                     | LoRA Soups (CAT)  | A3   | $h_x(*) = (\alpha_i = p_i(x) s_i)_i$                                                                                                                                                |
| Astra                     | LoRA              | A3   | $h(B, A) = ([B, -B_0], [A; A_0])$                                                                                                                                                   |
| AsyLoRA                   | LoRA              | A2   | $h(B) = (B, Q)$                                                                                                                                                                     |
| AsyLoRA                   | LoRA-FA           | A3   | $h(B) = B Q A_0^+$                                                                                                                                                                  |
| AutoLoRA                  | LoRA              | A3   | $h(b, a, \beta) = ([\text{softmax}(\beta)_j b_j]_j, [a_j^\top]_j)$                                                                                                                  |
| AutoPrompt                | Prompt Tuning     | A3   | $h(w_1, \dots, w_\ell) = (E_{w_1}, \dots, E_{w_\ell})$                                                                                                                              |
| Bayesian-LoRA             | LoRA              | A3   | $h = (\tilde{B}, \tilde{A}), \tilde{B} = T_r^B \mathbb{E}_q[U^B] T_c^B, \tilde{A} = T_r^A \mathbb{E}_q[U^A] T_c^A$                                                                  |
| BBT                       | Prompt Tuning     | A3   | $h(z) = Az + p_0$                                                                                                                                                                   |
| BD-LoRA                   | GraLoRA           | A3   | $h(\{B_i\}, A) = (B_{ij} = B_i, A_{ij}^\top = A_{:,j}^{(i)})_{i,j \leq N}$                                                                                                          |
| BD-LoRA                   | LoRA              | A3   | $h(B_1, \dots, B_N, A) = (\text{diag}(B_1, \dots, B_N), A)$                                                                                                                         |
| BEFT                      | BitFit            | A2   | $h(\Delta b_v) = (\Delta b_v, 0_{\text{other biases}})$                                                                                                                             |
| BiDoRA                    | DoRA              | A3   | id                                                                                                                                                                                  |
| BitFit                    | SSF               | A2   | $h(\Delta b) = (\gamma = \mathbf{1}, \beta = \Delta b)$                                                                                                                             |
| BLoB                      | LoRA              | A3   | $h(M, G, B) = (B, M)$                                                                                                                                                               |
| BOFT                      | BOFT              | A2   | $(Q_1, \dots, Q_{m'}, s) \mapsto (Q_1, \dots, Q_{m'}, 0, s)$                                                                                                                        |
| Cartridges                | Prefix-Tuning     | A2   | $h(Z) = ([z_{\text{sink}}^{(l)}; Z_k^{(l)}], [z_{v,\text{sink}}^{(l)}; Z_v^{(l)}]_{l \leq L})$                                                                                      |
| Child-Tuning              | Diff Pruning      | A2   | $x \mapsto (\alpha_0, \iota_M x)$                                                                                                                                                   |
| Child-Tuning              | FISH Mask         | A3   | $\text{id}_{\mathbb{R}^M}$                                                                                                                                                          |
| CoA-LoRA                  | LQ-LoRA           | A3   | $h_C(\theta, L_1, L_2) = (L_1 (I_r + U_\theta(Q_C)), L_2)$                                                                                                                          |
| COLA                      | LoRA              | A3   | $h((B_i, A_i)_i) = ([B_1 \cdots B_M], [A_1; \dots; A_M])$                                                                                                                           |
| COLA                      | ReLoRA            | A3   | $h = \text{id}$ (stagewise)                                                                                                                                                         |
| Compacter                 | Adapter (Houlsby) | A3   | $h = (W_{\text{down}} = \sum_i A_i \otimes s_i^D t_i^{D^\top}, W_{\text{up}} = \sum_i A_i \otimes s_i^U t_i^{U^\top})$                                                              |
| Compacter                 | PHM-Adapter       | A3   | $h(\{A_i\}, \{s_i^{\bullet,\ell}, t_i^{\bullet,\ell}\}) = (A_i^{\bullet,\ell} = A_i, B_i^{\bullet,\ell} = s_i^{\bullet,\ell} t_i^{\bullet,\ell \top})_{\ell, \bullet \in \{D, U\}}$ |
| Compress-then-Merge (CtM) | LoRA              | A3   | $\lambda \mapsto (U \sum_t \lambda_t O^{(t)}, V^\top)$                                                                                                                              |
| Concept Sliders           | LoRA              | A3   | $h = \text{id}$                                                                                                                                                                     |
| Concept Sliders           | Task Arithmetic   | A3   | $h(\alpha_1, \dots, \alpha_K) = (\lambda_k = \alpha_k), \tau_k = B_k A_k$                                                                                                           |
| CoOp                      | CoCoOp            | A2   | $h(V) = (V, \theta \text{ with } W_2 = 0, b_2 = 0)$                                                                                                                                 |
| CoOp                      | Prompt Tuning     | A3   | $h = \text{id}_{\mathbb{R}^{\ell \times d}}$                                                                                                                                        |
| CorDA                     | LoRA              | A3   | $h(B, A) = ([B, -B_0], [A; A_0])$                                                                                                                                                   |
| Core Space                | Task Arithmetic   | A3   | $h = \text{id}_1$                                                                                                                                                                   |
| CPT                       | Prompt Tuning     | A2   | $h(E_{\text{ctx}}) = (E_{\text{ctx}}, E_{\text{label}}^0) \quad (\ E_{\text{ctx}} - E_{\text{ctx}}^0\  \leq \varepsilon)$                                                           |
| DC-Merge                  | LoRA              | A3   | $h(\alpha) = (\alpha \tilde{U} (\tilde{M} \odot M), \tilde{V}^\top)$                                                                                                                |
| DEFT                      | LoRA              | A3   | $(P, R) \mapsto (P, R - \text{relu}(P)^\top W_0) \quad (\text{QR: } (Q(P), R - Q(P)^\top W_0))$                                                                                     |
| DeLoRA                    | DoRA              | A3   | $(B, A, \lambda) \mapsto (B', A', \mu = (\ (W_0 + B' A')_{i,:}\ )_i, (B', A') \text{ as in the LoRA arrow})$                                                                        |
| DeLoRA                    | LoRA              | A3   | $(B, A, \lambda) \mapsto (\frac{\lambda}{r} B \text{diag}(1/\max(\ b_i\ , \varepsilon)), \text{diag}(1/\ a_i\ ) A D_w)$                                                             |
| Delta-LoRA                | LoRA              | A3   | $h(B, A) = ([ (1 + \lambda) B, -\lambda B_K ], [A; A_K])$                                                                                                                           |
| DiaBlo                    | Diff Pruning      | A2   | $D \mapsto (\alpha_0, \iota_{\text{blk}} D)$                                                                                                                                        |
| DiaBlo                    | FISH Mask         | A3   | $h = \text{id}, M = \text{supp blkdiag}(\mathbf{1}_{d_{\text{out}}/N \times d_{\text{in}}/N}, \dots)$                                                                               |
| DnD                       | LoRA              | A3   | $h_c(\phi) = \text{detok}(G_\phi(c))$                                                                                                                                               |
| Doc-to-LoRA (D2L)         | LoRA              | A3   | $h_c(\phi) = (\alpha_l [B_l^{(1)} \cdots B_l^{(K)}], [A_l^{(1)}; \dots; A_l^{(K)}])$                                                                                                |
| DoRA                      | BiDoRA            | A3   | id                                                                                                                                                                                  |
| DoRA                      | GLoRA             | A3   | $(B, A, \mu) \mapsto (B_A = c - \mathbf{1}, A_A = \mathbf{1}^\top; B_B = \text{diag}(c) s B, A_B = A), c = \mu \oslash (\ (W_0 + s B A)_{i,:}\ )_i$                                 |
| DoRA (dynamic rank)       | AdaLoRA           | A3   | $h = \text{id}$                                                                                                                                                                     |
| DoRA (dynamic rank)       | LoRA              | A3   | $h(B, c, A) = (B \text{diag}(c), A)$                                                                                                                                                |
| DoRA (dynamic rank)       | SoRA              | A3   | $h(B, c, A) = (B \text{diag}(c) \text{diag}(g_0)^{-1}, g_0, A)$                                                                                                                     |
| DSEE                      | LoRA              | A3   | $(U, V, S) \mapsto ([U   E_I \text{diag}(S_\Omega)], [V; E_J^\top]), E_I = [e_{i_1} \cdots e_{i_c}], E_J = [e_{j_1} \cdots e_{j_c}]$                                                |
| DyLoRA                    | DyLoRA            | A2   | $h(B_b, A_b) = ([\frac{r_{\max}}{b} B_b, 0], [A_b; A_{0,(b+1):}])$                                                                                                                  |

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Table 37, continued

| Source $M$          | Target $N$    | Rule | Pointed map $h$ with $\rho_N \circ h = \rho_M$                                                                                                                                                      |
|---------------------|---------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DyLoRA              | LoRA          | A3   | $h = \text{id}$                                                                                                                                                                                     |
| EasyControl         | Prefix-Tuning | A3   | $h_c(B, A) = (K_l(c; B, A), V_l(c; B, A))_{l \leq L}$                                                                                                                                               |
| EFT                 | Proxy-Tuning  | A3   | $h = \text{id}$                                                                                                                                                                                     |
| ElaLoRA             | AdaLoRA       | A3   | $h = \text{id}$                                                                                                                                                                                     |
| ElaLoRA             | LoRA          | A3   | $h(P, \Lambda, Q) = (P\Lambda, Q)$                                                                                                                                                                  |
| ETHER+              | LoRA          | A3   | $q \mapsto \left( [\hat{v}_j, -\hat{u}_j]_j, W_0[\hat{v}_j, -\hat{u}_j]_j, [\hat{v}_j, \hat{u}_j]_j^\top W_0 H^+; [\hat{v}_j, \hat{u}_j]_j^\top \right)$                                            |
| EVA                 | LoRA          | A3   | $h = \text{id} : (B, A) \mapsto (B, A)$                                                                                                                                                             |
| FacT                | FacT          | A3   | $h(U, P, C, V) = (U, (\sum_t P_{i,t} C_t)_i, V)$                                                                                                                                                    |
| FacT                | LoRA          | A3   | $h(U, \Sigma, V) = (sU\Sigma_i, V^\top)_i$                                                                                                                                                          |
| FedRot-LoRA         | LoRA          | A3   | $h = \text{id}$                                                                                                                                                                                     |
| FFA-LoRA            | LoRA          | A2   | $h(B) = (B, A_0)$                                                                                                                                                                                   |
| FFA-LoRA            | LoRA-FA       | A3   | $h = \text{id}$                                                                                                                                                                                     |
| FISH Mask           | Diff Pruning  | A2   | $x \mapsto (\alpha_0, \iota_M x)$                                                                                                                                                                   |
| FISH Mask           | RoSA          | A5   | $x \mapsto ((0, A_0), x)$                                                                                                                                                                           |
| Flat-LoRA           | LoRA          | A3   | $h = \text{id}_Q$                                                                                                                                                                                   |
| FlexLoRA            | AdaLoRA       | A3   | $h = \text{id}$                                                                                                                                                                                     |
| FlexLoRA            | ElaLoRA       | A3   | $h = \text{id}$                                                                                                                                                                                     |
| FlexLoRA            | LoRA          | A3   | $h(P, \Lambda, Q) = (P\Lambda, Q)$                                                                                                                                                                  |
| FLoRA               | HiRA          | A3   | $h(B, A) = ([\frac{1}{r} B \ -\mathbf{1}_m], [A; \mathbf{1}_n^\top])$                                                                                                                               |
| Flora               | LoRA-FA       | A3   | $C \mapsto s^{-1}C \quad (A_0 := A_t)$                                                                                                                                                              |
| Flora               | ReLoRA        | A2   | $C \mapsto (s^{-1}C, A_t)$                                                                                                                                                                          |
| FLoRA (federated)   | ReLoRA        | A3   | $h((B_k, A_k)_k) = ([p_1 B_1 \cdots p_K B_K], [A_1; \dots; A_K])$ per round                                                                                                                         |
| FLoRA (Tucker core) | LoRA          | A3   | $h(A, G, B) = (sAG, B^\top)$                                                                                                                                                                        |
| FLoRG               | LoRA          | A3   | $A \mapsto (sLA^\top, AR)$                                                                                                                                                                          |
| FLoRG               | SingLoRA      | A3   | $A \mapsto \sqrt{sr/\alpha} A^\top$                                                                                                                                                                 |
| FourierFT           | FourierFT     | A2   | $h(c_\Omega) = (c_\Omega, 0_{\Omega' \setminus \Omega})$                                                                                                                                            |
| FourierFT           | LoRA-FA       | A3   | $h(c) = \frac{\alpha}{mn} B_F \text{diag}(c, -c), \quad B_F = [\cos \theta_k \ \sin \theta_k]_k, \quad A_F = [\cos \varphi_k; \sin \varphi_k]_k$                                                    |
| FourierMoE          | LoRA-FA       | A3   | $h_x((c_i)_i, \Phi) = \eta \sum_{i \in S_\Phi(x)} g_{\Phi, i}(x) B(c_i), \quad B(c) = \frac{2}{d_{out} d_{in}} [a_k \cos \theta_k - b_k \sin \theta_k, -(a_k \sin \theta_k + b_k \cos \theta_k)]_k$ |
| FrameFT             | LoRA-FA       | A3   | $h(C_\Omega) = \frac{\alpha}{\sqrt{d_{out} d_{in}}} P_{out} C_{:,J}, \quad A_0 = (P_{in})_{:,J}^\top$                                                                                               |
| FRoD                | FRoD          | A2   | $h(\Sigma) = (\Sigma, S = 0)$                                                                                                                                                                       |
| GaLore              | LoRA          | A2   | $C \mapsto (P_k, s^{-1}C)$                                                                                                                                                                          |
| GaLore              | LoRA-FA       | A3   | $C \mapsto s^{-1}C \quad (A_0 := P_k^\top)$                                                                                                                                                         |
| GaLore 2            | GaLore        | A3   | $\text{id}$                                                                                                                                                                                         |
| GenerativeAdapter   | LoRA          | A3   | $h_c(A_1, A_2, B_1, B_2) = (A_1 U_r(c), V_r(c)^\top B_2), \quad U_r \Sigma_r V_r^\top = \text{SVD}_r(A_2 H_c^\top H_c B_1)$                                                                         |
| GeoRA               | LoRA          | A3   | $(B, A) \mapsto ([B, -B_0], [A; A_0])$                                                                                                                                                              |
| GiVA                | LoRA          | A3   | $(\gamma, \lambda) \mapsto (\Gamma B \Lambda, V_r^\top)$                                                                                                                                            |
| GiVA                | LoRA-FA       | A3   | $(\gamma, \lambda) \mapsto \Gamma B \Lambda$                                                                                                                                                        |
| GOAT                | LoRA          | A3   | $h_x(B, A, W_z) = ([s w^1(x) B^1 \cdots s w^E(x) B^E - \frac{s}{E} B_0^1 \cdots - \frac{s}{E} B_0^E], [A^1; \dots; A^E; A_0^1; \dots; A_0^E])$                                                      |
| GOFT                | BOFT          | A3   | $\theta \mapsto (t = \tan(\theta/2) \text{ on the kept } 2 \times 2 \text{ blocks, } 0 \text{ on the others, } s = 1)$                                                                              |
| GOFT                | qGOFT         | A3   | $\theta \mapsto \left( \tilde{G}_t = \begin{pmatrix} \cos \theta_t & -\sin \theta_t \\ \sin \theta_t & \cos \theta_t \end{pmatrix} \right)_t$                                                       |
| GoRA                | LoRA          | A3   | $h(B, A) = (cB, A), \quad c = \frac{\alpha/\sqrt{r\ell}}{s}$                                                                                                                                        |
| GPS                 | Diff Pruning  | A2   | $x \mapsto (\alpha_0, \iota_M x)$                                                                                                                                                                   |
| GradFix             | FISH Mask     | A3   | $h(\alpha) = \alpha (m \odot \tau_A)$                                                                                                                                                               |
| GraLoRA             | LoRA          | A3   | $h(\{U_{ij}, V_{ij}\}) = (\text{blockdiag}(L_1, \dots, L_k), S), \quad L_i = [U_{i1} \cdots U_{ik}], \quad S = \text{stack of the } V_{ij}^\top \text{ on column block } j$                         |
| HetLoRA             | DyLoRA        | A3   | $h = \text{id}$                                                                                                                                                                                     |
| HetLoRA             | LoRA          | A3   | $h = \text{id}$                                                                                                                                                                                     |
| HiFT                | Surgical FT   | A3   | $\text{id}_{\Theta_{G_j}}$                                                                                                                                                                          |
| HiRA                | GLoRA         | A2   | $h(B, A) = (\mathcal{A}\text{-path} = BA, \text{ all other GLoRA paths frozen at } 0)$                                                                                                              |

continued on the next page

Table 37, continued

| Source $M$                   | Target $N$       | Rule | Pointed map $h$ with $\rho_N \circ h = \rho_M$                                                                                                                                       |
|------------------------------|------------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HiRA                         | LoHa             | A2   | $h(B, A) = (U_0, V_0, B, A)$ , $U_0 V_0 = W_0 / \gamma$ a fixed rank-rk $W_0$ factorisation                                                                                          |
| HRA                          | LoRA             | A3   | $u \mapsto (W_0 B(u), A(u))$ , $B(u) = [-2H_{u_1} \cdots H_{u_{i-1}} \hat{u}_i]_{i=1}^r$ , $A(u) = [\hat{u}_1^\top; \dots; \hat{u}_r^\top]$                                          |
| HydraLoRA                    | LoRA             | A3   | $h_x(A, B_1, \dots, B_N, W_g) = (\sum_i \omega_i(x) B_i, A)$                                                                                                                         |
| HydraLoRA                    | LoRAMoE          | A3   | $h(A, (B_i)_i, W_g) = ((\frac{r}{\alpha} B_i)_i, (A, \dots, A), W_g)$                                                                                                                |
| HyperDreamBooth              | LoRA             | A3   | $h(B_{tr}, A_{tr}) = (P_{out} B_{tr}, A_{tr} P_{in})$                                                                                                                                |
| HyperDreamBooth              | NOLA             | A3   | $h(B_{tr}, A_{tr}) = (\alpha = \text{vec } A_{tr}, \beta = \text{vec } B_{tr})$ , NOLA bases $\{p_k e_j^\top\}, \{e_j q_l^\top\}$                                                    |
| IC-LoRA                      | LoRA             | A3   | $h = \text{id}$                                                                                                                                                                      |
| ICEdit                       | LoRA             | A3   | $h_x(B, A, g) = (\frac{\alpha}{r} [G_1(x) B_1 \cdots G_4(x) B_4], [A_1; \dots; A_4])$                                                                                                |
| In-Place TTT                 | LoRA             | A3   | $h_{c,i}(W_{target}, k) = (\eta [\hat{V}_{[1]}^\top \cdots \hat{V}_{[i-1]}^\top], [Z_{[1]}; \dots; Z_{[i-1]}])$                                                                      |
| IncreLoRA                    | AdaLoRA          | A3   | $h = \text{id} : (B, \Lambda, A) \mapsto (P, \Lambda, Q)$                                                                                                                            |
| IncreLoRA                    | IncreLoRA        | A5   | $h(q) = (q, (b_{\text{new}}, 0, a_{\text{new}}))$                                                                                                                                    |
| IncreLoRA                    | LoRA             | A3   | $h(B, \Lambda, A) = (B\Lambda, A)$                                                                                                                                                   |
| InfLoRA                      | LoRA             | A2   | $h(B) = (B, A_t)$                                                                                                                                                                    |
| InfLoRA                      | LoRA-FA          | A3   | $h = \text{id}$                                                                                                                                                                      |
| Initial State Tuning         | Prefix-Tuning    | A3   | $h(h') = (C_{\ell,l}^+ h'_l)_l$ , $C_{\ell,l} = [\bar{B}_l, \bar{A}_l \bar{B}_l, \dots, \bar{A}_l^{\ell-1} \bar{B}_l]$                                                               |
| Intruder-dimension reduction | SVDiff           | A3   | $h(\lambda) = \delta$ , $\delta_j = (\lambda - 1) \sigma_j$ ( $j \in I$ ), $\delta_j = 0$ otherwise                                                                                  |
| IR-QLoRA                     | QLoRA            | A3   | $(B, A, \beta_1, \beta_2) \mapsto (\frac{\alpha}{s} (B + \beta_2 E), A + \beta_1 P)$                                                                                                 |
| K-LoRA                       | LoRA Soups (CAT) | A3   | $h_t(*) = (\alpha_c^l, \alpha_s^l) = \begin{cases} (1, 0) & \text{layer } l \text{ selects content at } t \\ (0, 1) & \text{otherwise} \end{cases}$                                  |
| K-LoRA                       | ZipLoRA          | A3   | $h_t(*) = (m_c^l, m_s^l) \in \{(\mathbf{1}, 0), (0, \mathbf{1})\}$                                                                                                                   |
| KAdaptation                  | LoRA             | A3   | $h = ([A_1 \otimes u_1, \dots, A_n \otimes u_n], [I_n \otimes v_1^\top; \dots; I_n \otimes v_n^\top])$                                                                               |
| KaSA                         | LoRA             | A3   | $h(B, \Delta\sigma, A) = (\frac{\alpha}{r} B \text{diag}(\Delta\sigma), A)$                                                                                                          |
| KeepLoRA                     | LoRA             | A3   | $B \mapsto (B - B_t^{(0)}, A_t)$                                                                                                                                                     |
| KeepLoRA                     | LoRA-FA          | A3   | $B \mapsto B - B_t^{(0)}$                                                                                                                                                            |
| KnOTS                        | Task Arithmetic  | A3   | $h = \text{id}_\lambda$                                                                                                                                                              |
| KRApapter                    | LoHa             | A2   | $h(U, V) = (B_1 = S_1, A_1 = U, B_2 = S_2, A_2 = V)$ , $S_1, S_2$ the $\{0, 1\}$ row selectors of the Kronecker index                                                                |
| KronA                        | LoKr             | A3   | $h(A_k, B_k) = (C = A_k, \text{right factor} = B_k)$ (LoKr with neither factor decomposed)                                                                                           |
| LaMDA                        | LoRA             | A3   | $h(U, S) = (\alpha U S - U_r, \Sigma_r V_r^\top)$                                                                                                                                    |
| LaMDA                        | LoRA-FA          | A3   | $h(U, S) = \alpha U S - U_r$ into FA $(\Sigma_r V_r^\top)$                                                                                                                           |
| LaMDA                        | PiSSA            | A3   | $h(U, S) = (\alpha U S \Sigma_r^{1/2}, \Sigma_r^{1/2} V_r^\top)$                                                                                                                     |
| Laplace-LoRA                 | LoRA             | A3   | $h = \text{id}$ (the MAP object)                                                                                                                                                     |
| LCM-LoRA                     | LoRA             | A3   | $h(\lambda_1, \lambda_2) = ([\lambda_1 B' \lambda_2 B_{\text{LCM}}], [A'; A_{\text{LCM}}])$                                                                                          |
| LCM-LoRA                     | Task Arithmetic  | A3   | $h(\lambda_1, \lambda_2) = (\lambda_1, \lambda_2)$ , $\tau_1 = B' A'$ , $\tau_2 = B_{\text{LCM}} A_{\text{LCM}}$                                                                     |
| LD-MoLE                      | LoRA             | A3   | $h_x((B_i, A_i)_i, W_{gate}, f) = (\frac{\alpha}{rs} [p_{x,1} B_1 \cdots p_{x,E} B_E], [A_1; \dots; A_E])$                                                                           |
| LieRA                        | FLoRA            | A3   | $h(B, A) = ([\mathbf{I}_m B], [\mathbf{I}_n^\top; A])$                                                                                                                               |
| LieRA                        | HiRA             | A3   | $h(B, A) = (B, A)$                                                                                                                                                                   |
| LieRA                        | LoHa             | A3   | $h(B, A) = (B, A, U_0, V_0^\top)$ , $U_0 V_0^\top = W_0$                                                                                                                             |
| LIFT                         | Diff Pruning     | A2   | $x \mapsto (\alpha_0, \iota_{M_0} x)$                                                                                                                                                |
| LiME                         | LoRA             | A3   | $h_x(B, A, (p_i)_i, p_s, \gamma) = (\text{diag}(\mathcal{P}(x) + \gamma p_s) B, A)$                                                                                                  |
| Linear probing               | BitFit           | A2   | $h(V) = (\Delta b = 0, V)$                                                                                                                                                           |
| Linear probing               | LP-FT            | A5   | $V \mapsto (V, q_0^{(2)})$                                                                                                                                                           |
| Linear probing               | Surgical FT      | A2   | $h(V) = V$ (block = last layer)                                                                                                                                                      |
| LLaMA-Adapter                | LLaMA-Adapter V2 | A2   | $h(P, g) = (P, g, s = \mathbf{1}, b = 0, \gamma = \gamma_0, \text{visual branch at its zero gate})$                                                                                  |
| LN Tuning                    | SSF              | A3   | $(\gamma, \beta) \mapsto (\gamma' = \gamma \odot \gamma_0, \beta' = \beta - (\gamma \odot \gamma_0) \odot \beta_0)$ on SSF's post-norm wrappers; other wrappers at $(\mathbf{1}, 0)$ |
| LoCA                         | LoRA-FA          | A3   | $h(a) = [a_1 C_{l_1, \cdot}^\top \cdots a_B C_{l_B, \cdot}^\top]$ into FA $(A_0)$ , $A_0 = [D_{l_1^2, \cdot}; \dots; D_{l_B^2, \cdot}]$                                              |
| LoCO                         | LoRA             | A3   | $(U, V) \mapsto (2W_0 X (I_{2r} - Y^\top X)^{-1}, Y^\top)$ , $X = [U   -V]$ , $Y = [V   U]$                                                                                          |
| LoCO                         | OFT              | A3   | $(U, V) \mapsto Q = -(UV^\top - VU^\top)$                                                                                                                                            |

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Table 37, continued

| Source $M$                      | Target $N$       | Rule | Pointed map $h$ with $\rho_N \circ h = \rho_M$                                                                        |
|---------------------------------|------------------|------|-----------------------------------------------------------------------------------------------------------------------|
| LoDA                            | LoRA             | A3   | $(B_G, B_I) \mapsto ([w_G B_G, B_I], [A_G; A_I])$                                                                     |
| LoDA                            | LoRA-FA          | A3   | $(B_G, B_I) \mapsto [w_G B_G, B_I]$                                                                                   |
| LoFiT                           | BitFit           | A3   | $h(v) = (\Delta b_O^{(l)} = \sum_{i \in T_l} W_{(i)}^O v_i^l, \text{ other biases } 0)$                               |
| LoFT                            | LoRA             | A3   | $h = \text{id}_Q$                                                                                                     |
| LOFT                            | LoRA             | A3   | $E \mapsto (B = W_0 P_r^\top (C(E) - I_r), A = P_r)$                                                                  |
| LOFT                            | LoRA-XS          | A3   | $E \mapsto R = \Sigma_r (C(E) - I) \Sigma_r^{-1}$                                                                     |
| LOFT                            | OFT              | A3   | $E \mapsto Q = -P_r^\top E P_r \in \mathfrak{so}(n)$                                                                  |
| LOFT                            | PSOFT            | A3   | $E \mapsto (R = C(E)^\top, \alpha = \beta = \mathbf{1})$                                                              |
| LoftQ                           | LoRA             | A3   | $h(B, A) = ([B, -B_T], [A; A_T])$                                                                                     |
| LoHa                            | LoRA             | A3   | $h(B_1, A_1, B_2, A_2) = (\gamma B_1 \bullet B_2, (A_1^\top \bullet A_2^\top)^\top)$                                  |
| LoKr                            | KronA            | A3   | $h(C, B, A) = (C, BA)$                                                                                                |
| LoKr                            | LoRA             | A3   | $h(C, B, A) = (C \otimes B, I_{u_q} \otimes A)$                                                                       |
| LoQT                            | GaLore           | A3   | $B \mapsto sB$                                                                                                        |
| LoQT                            | LoRA             | A2   | $B \mapsto (P_k, \frac{s}{s'} B)$                                                                                     |
| LoRA                            | DoRA             | A3   | $h(B, A) = ((\ (W_0 + sBA)_{i,:}\ )_i, B, A)$                                                                         |
| LoRA                            | LoRA             | A3   | $h(B, A) = ([B \ 0], [A; a_0])$                                                                                       |
| LoRA                            | PEANuT           | A3   | $h(B, A) = ([M, -M], [B^\top; -B^\top]), \quad M = (W_0^+)^\top A^\top$                                               |
| LoRA                            | ReLoRA           | A5   | $q \mapsto (q, q_0^{(2)}, \dots, q_0^{(K)})$                                                                          |
| LoRA Dropout                    | LoRA             | A3   | $h = \text{id}_Q$                                                                                                     |
| LoRA Soups (CAT)                | AdaMerging       | A3   | $h = \text{id} : \alpha_i^l \mapsto \lambda_i^l$                                                                      |
| LoRA Soups (CAT)                | LoRA             | A3   | $h(\alpha) = ([\alpha_1 B_1 \cdots \alpha_N B_N], [A_1; \dots; A_N])$                                                 |
| LoRA Switch /<br>LoRA Composite | LoRA Soups (CAT) | A3   | $h_t(*) = w_{i(t)} e_{i(t)}$                                                                                          |
| LoRA Without<br>Regret          | LoRA             | A3   | $h = \text{id}_Q$                                                                                                     |
| LoRA+                           | LoRA             | A3   | $h = \text{id}_Q$                                                                                                     |
| LoRA-Dash                       | LoRA             | A3   | $h(B, A, \Delta\sigma) = ([B \ \bar{U} \text{diag}(\Delta\sigma)], [A; \bar{V}^\top])$                                |
| LoRA-drop                       | LoRA             | A3   | $h((B_i, A_i)_{\text{keep}}, (B_s, A_s)) = ((B_i, A_i)_{\text{keep}}, (B_s, A_s)_{j \in \text{share}})$               |
| LoRA-FA                         | AsyLoRA          | A3   | $h(B) = B A_0 Q^\top$                                                                                                 |
| LoRA-FA                         | Flora            | A5   | $B \mapsto (B, 0, \dots, 0)$                                                                                          |
| LoRA-FA                         | LoRA             | A2   | $h(B) = (B, A_0)$                                                                                                     |
| LoRA-Flow                       | LoRA Soups (CAT) | A3   | $h_x(W_{\text{gate}}, b) = (\alpha_i^l = w_i^l(x))_{i,l}$                                                             |
| LoRA-GA                         | LoRA             | A3   | $h(B, A) = ([B, -B_0], [A; A_0])$                                                                                     |
| LoRA-LEGO                       | LoRA Soups (CAT) | A3   | $h(*) = N^{-1/2} \mathbf{1}$                                                                                          |
| LoRA-Mini                       | LoRA             | A3   | $h(C, D) = (PC, DQ)$                                                                                                  |
| LoRA-Mixer                      | LoRA             | A3   | $h_x((B_e, A_e)_e, G) = ([\mathbb{1}_{e \in \text{TopK}(x)} p_e(x) B_e]_{e=1}^E, [A_1; \dots; A_E])$                  |
| LoRA-Null                       | LoRA             | A3   | $h(B, A) = ([B, -B_0], [A; A_0])$                                                                                     |
| LoRA-Pre                        | Adam-mini        | A3   | $h = \text{id}_Q$                                                                                                     |
| LoRA-Pro                        | LoRA             | A3   | $h = \text{id}_Q$                                                                                                     |
| LoRA-RITE                       | LoRA             | A3   | $h = \text{id}_Q$                                                                                                     |
| LoRA-RLPO /<br>LoRA-RLMO        | LoRA             | A3   | $h = \text{id} : (B, A) \mapsto (B, A)$                                                                               |
| LoRA-S (AdamS,<br>LRACS)        | LoRA             | A3   | $h = \text{id}_Q$                                                                                                     |
| LoRA-SB                         | LoRA-FA          | A4   | $h(R) = sU_r R$                                                                                                       |
| LoRA-X                          | LoRA             | A3   | $h(\Delta\Sigma) = (U_r, \Delta\Sigma V_r^\top)$                                                                      |
| LoRA-X                          | LoRA-XS          | A3   | $h(\Delta\Sigma) = \Delta\Sigma \Sigma_r^{-1}$                                                                        |
| LoRA-X                          | SVDiff           | A2   | $h(\delta\sigma) = (\delta\sigma_1, \dots, \delta\sigma_r, 0, \dots, 0)$                                              |
| LoRA-X                          | SVFT             | A2   | $h(\delta\sigma) = M, \quad M_{ii} = \delta\sigma_i \ (i \leq r), \ 0 \text{ otherwise}$                              |
| LoRA-XS                         | LoRA             | A3   | $h(R) = (U_r R, \Sigma_r V_r^\top)$                                                                                   |
| LoRA-XS                         | LoRA-FA          | A4   | $h(R) = U_r R \text{ into FA}(A_0), \ A_0 = \Sigma_r V_r^\top$                                                        |
| LoRA-XS                         | PiSSA            | A3   | $h(R) = (U_r \Sigma_r^{1/2} + U_r R \Sigma_r^{1/2}, \Sigma_r^{1/2} V_r^\top)$                                         |
| LoRA-XS                         | SVFT             | A3   | $h(R) = \begin{bmatrix} R \Sigma_r & 0 \\ 0 & 0 \end{bmatrix} \ (\Omega = \text{top-left } r \times r \text{ block})$ |
| LoraHub                         | LoRA             | A3   | $h(w) = (\sum_i w_i B_i, \sum_i w_i A_i)$                                                                             |

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Table 37, continued

| Source $M$           | Target $N$          | Rule | Pointed map $h$ with $\rho_N \circ h = \rho_M$                                                                              |
|----------------------|---------------------|------|-----------------------------------------------------------------------------------------------------------------------------|
| LoRAMoE              | LoRA                | A3   | $h_x(B, A, W_g) = (\frac{\alpha}{r} [\omega_1(x)B_1 \cdots \omega_N(x)B_N], [A_1; \dots; A_N])$                             |
| LoRAMoE              | MoV / MoLoRA        | A3   | $h(B_i, A_i, W_g) = (\frac{\alpha}{r} B_i, A_i, W_g^\top)$                                                                  |
| LoRAPrune            | DoRA                | A3   | $h(B, A, m) = (\mu = m \odot \ W_0 + sBA\ _{\text{row}}, B, A)$                                                             |
| LoraRetriever        | LoRA Soups (CAT)    | A3   | $h_x(*) = (\alpha_\phi = \mathbf{1}[\phi \in \Phi_x]/k)_\phi$                                                               |
| LoraRetriever        | LoraHub             | A3   | $h_x(*) = w = \mathbf{1}_{\Phi_x}/k$                                                                                        |
| LoReFT               | LoRA                | A3   | $h(R, W, b) = (R^\top, [W - R \mid b])$                                                                                     |
| LoRETTA              | LoRA                | A3   | $h(\{G_{up}^i\}, \{G_{down}^i\}) = (\text{TT}(\{G_{up}^i\}), \text{TT}(\{G_{down}^i\}))$                                    |
| LoRI                 | LoRA                | A2   | $h(X) = (\iota_M X, A_0)$                                                                                                   |
| LoRI                 | LoRA-FA             | A2   | $h(X) = \iota_M X$                                                                                                          |
| LoRTA                | FacT                | A3   | $h = (U = B, V = A, \Sigma_{(l,i)} = \text{Diag}(C_L[l, :] \odot C_M[i, :]))$                                               |
| LoRTA                | LoRA                | A3   | $h = (B \text{ Diag}(C_L[l, :] \odot C_M[i, :]), A^\top)_{(l,i)}, B = [B \text{ Diag}(C_H[H, :])$                           |
|                      |                     |      | $]; \dots; B \text{ Diag}(C_H[H, :])]$                                                                                      |
| LoRTA                | LoTR                | A3   | $h = (A_{\text{LoTR}} = A, B_{\text{LoTR}} = B, G_{(l,i)}^\top = \text{Diag}(C_L[l, :] \odot C_M[i, :]))$                   |
| LoRTA                | MetaTT              | A3   | $h = (G_1 = A, G_2[l] = \text{Diag}(C_L[l, :]), G_3[i] = \text{Diag}(C_M[i, :])$                                            |
|                      |                     |      | $], G_H[k] = \text{Diag}(C_H[k, :]), G_4 = B^\top)$                                                                         |
| LoTR                 | FacT                | A3   | $h(A, B, \{G_s\}) = (V = A, U = B, \Sigma_s = G_s^\top)$                                                                    |
| LoTR                 | FLoRA (Tucker core) | A3   | $h(A, B, \{G_s\}) = (A_F = B, G_F = G_s^\top, B_F = A)_s$                                                                   |
| LoTR                 | LoRA                | A3   | $h = (BG_s^\top, A^\top)_s$                                                                                                 |
| LoTR                 | ShareLoRA           | A3   | $h = (\{B_s = BG_s^\top\}_s, A^\top)$                                                                                       |
| LRAgent              | LoRA                | A3   | $h_i(A, B_1, \dots, B_N) = (B_i, A)$                                                                                        |
| LRAgent              | S-LoRA              | A3   | $(A, (B_i)_i) \mapsto ((B_i, A))_i$                                                                                         |
| LT-SFT               | Diff Pruning        | A2   | $x \mapsto (\alpha_0, \iota_M x)$                                                                                           |
| MaLoRA / MaRA        | LoRA                | A3   | $h_{x_{1:t}}(A, B, P, \text{Mamba}, w, b) = (\frac{\alpha}{r_s} \lambda(x_{1:t}) B, A)$                                     |
| MatryoshkaLoRA       | DyLoRA              | A3   | $h = \text{id}$                                                                                                             |
| MatryoshkaLoRA       | LoRA                | A3   | $(B, A) \mapsto (s_k B_{:,1:k}, A_{1:k,:})$                                                                                 |
| MatryoshkaLoRA       | SoRA                | A2   | $(B, A) \mapsto (B, p, A)$                                                                                                  |
| MDPD                 | LN Tuning           | A3   | $h = \text{id}$                                                                                                             |
| MELoRA               | GraLoRA             | A2   | $(B_i, A_i)_i \mapsto (B_{ii} = B_i, A_{ii} = A_i^\top; (B_{ij}, A_{ij}) = (0, A_{ij,0}) (i \neq j))$                       |
| MELoRA               | LoRA                | A2   | $(B_i, A_i)_i \mapsto (\text{blockdiag}(B_1, \dots, B_n), \text{blockdiag}(A_1, \dots, A_n))$                               |
| MetaTT               | LoRA                | A3   | $h = (G_4^\top G_3[m]^\top G_2[l]^\top, G_1^\top)_{(l,m)}$                                                                  |
| MetaTT               | LoTR                | A3   | $h(G_1, \{G_2[l]\}, \{G_3[m]\}, G_4) = (A = G_1, B = G_4^\top, G_{(l,m)} = G_2[l]G_3[m])$ per type $m$                      |
| MHR                  | LoRA                | A3   | $h_\tau(A, B, Z) = (B^\tau, A^\tau)$                                                                                        |
| MiCA                 | LoRA                | A2   | $h(A) = (U_{-r}, \frac{\alpha}{r} A)$                                                                                       |
| MiCA                 | MiLoRA              | A3   | $h(A) = (U_m \Sigma_m^{1/2}, \Sigma_m^{1/2} V_m^\top + \frac{\alpha}{r} \Sigma_m^{-1/2} A)$                                 |
| MiLoRA               | LoRA                | A3   | $h(B, A) = ([B, -B_0], [A; A_0])$                                                                                           |
| MiSS (formerly Bone) | LoRA                | A2   | $h(D) = (D, \mathbf{1}_{d_{\text{in}}/r}^\top \otimes I_r)$                                                                 |
| MiSS (formerly Bone) | LoRA-FA             | A3   | $h(D) = D, A_0 = \mathbf{1}_{d_{\text{in}}/r}^\top \otimes I_r$                                                             |
| Mix-of-Show          | RegMean             | A3   | $h(*) = (\alpha = 1)$                                                                                                       |
| Model Soups          | Task Arithmetic     | A3   | $h(*) = \mathbf{1}_S/ S $                                                                                                   |
| ModuLoRA             | QLoRA               | A3   | $h = \text{id}$                                                                                                             |
| MOELoRA              | LoRA                | A3   | $h_\tau(B, A, e, W_T) = ([\omega_{\tau 1} B_1 \cdots \omega_{\tau N} B_N], [A_1; \dots; A_N])$                              |
| MonteCLoRA           | LoRA                | A3   | $h(B, \mu, \xi) = (B, \mu)$                                                                                                 |
| MoRA                 | KronA               | A2   | $M \mapsto (A_k, B_k) = (I_{d/\hat{r}}, M/s)$                                                                               |
| MoRe                 | LoRA                | A2   | $(L, R) \mapsto (P_1 L P_2, R)$                                                                                             |
| MoS                  | LoRA                | A3   | $h(\text{pool}_B, \text{pool}_A) = (\text{Route}^c(\text{pool}_B, I_\ell^b), \text{Route}^r(\text{pool}_A, I_\ell^a))_\ell$ |
| MoSLoRA              | FLoRA (Tucker core) | A3   | $h = \text{id}$                                                                                                             |
| MoSLoRA              | LoRA                | A3   | $h(B, M, A) = (BM, A)$                                                                                                      |
| MoV / MoLoRA         | (IA) <sup>3</sup>   | A3   | $h_x(\ell_1, \dots, \ell_N, W_g) = \sum_i s_i(x) \ell_i$                                                                    |
| MoV / MoLoRA         | LoRAMoE             | A3   | $h(B_i, A_i, W_g) = (\frac{r}{\alpha} B_i, A_i, W_g^\top)$ (MoLoRA variant)                                                 |
| MPT                  | Prompt Tuning       | A3   | $h(P^*, u, v) = P^* \odot uv^\top$                                                                                          |
| NaRA                 | LoRA                | A3   | $h_\lambda(B, A, \phi) = (B C_\phi(\lambda), A)$                                                                            |

continued on the next page

Table 37, continued

| Source $M$              | Target $N$           | Rule | Pointed map $h$ with $\rho_N \circ h = \rho_M$                                                                                                            |
|-------------------------|----------------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| NaRA                    | MoSLoRA              | A3   | $h_\lambda(B, A, \phi) = (B, C_\phi(\lambda)/s, A)$                                                                                                       |
| NOLA                    | LoRA                 | A3   | $h(\alpha, \beta) = (\sum_j \beta_j B_j, \sum_i \alpha_i A_i)$                                                                                            |
| O-LoRA                  | LoRA                 | A5   | $h = ([B_1 \cdots B_T], [A_1; \dots; A_T])$                                                                                                               |
| O-LoRA                  | ReLoRA               | A3   | $h = \text{id}$ on each stage's $(B_t, A_t)$                                                                                                              |
| OFT                     | BOFT                 | A2   | $Q \mapsto (Q, s = \mathbf{1})$ in BOFT(1, $b$ )                                                                                                          |
| OFT                     | GOFT                 | A3   | $t \mapsto (\theta_k^{(1)} = 2 \arctan t_k, \theta^{(\ell)} = 0 \ (\ell \geq 2))$                                                                         |
| OFTv2                   | qGOFT                | A3   | $t \mapsto (\tilde{G}_k^{(1)} = (1 - t_k^2) \begin{pmatrix} 1 - t_k^2 & -2t_k \\ 2t_k & 1 - t_k^2 \end{pmatrix}, \tilde{G}^{(\ell)} = I \ (\ell \geq 2))$ |
| OLoRA                   | LoRA                 | A3   | $h(B, A) = ([B, -B_0], [A; A_0])$                                                                                                                         |
| Online Subspace Descent | LoRA                 | A2   | $h(X) = (P_t, X)$                                                                                                                                         |
| Orthogonal Adaptation   | LoRA                 | A2   | $h(B) = (B, A_i)$                                                                                                                                         |
| Orthogonal Adaptation   | LoRA-FA              | A3   | $h = \text{id}$                                                                                                                                           |
| OrthoMerge              | OFT                  | A3   | $* \mapsto Q_m = \frac{c}{N} \sum_i Q_i$                                                                                                                  |
| OSF                     | MiLoRA               | A3   | $(U, \Sigma, V) \mapsto (U \Sigma S_{lo}^{-1/2}, S_{lo}^{1/2} V^\top)$                                                                                    |
| P-Tuning                | Prompt Tuning        | A3   | $h(P, \psi) = f_\psi(P)$                                                                                                                                  |
| PaFi                    | Diff Pruning         | A2   | $x \mapsto (\alpha_0, \iota_M x)$                                                                                                                         |
| Parallel Adapter        | AdaptFormer          | A3   | $h(W_{down}, b_{down}, W_{up}, b_{up}) = (W_{down}, b_{down}, \frac{s}{0.1} W_{up}, \frac{s}{0.1} b_{up})$                                                |
| PEANuT                  | LoRA                 | A3   | $h(\Theta_1, \Theta_2) = (\Theta_2^\top, \sigma(\Theta_1^\top W_0))$                                                                                      |
| PEM Arithmetic          | Task Arithmetic      | A3   | $h(\lambda) = (\lambda, 1 - \lambda)$                                                                                                                     |
| PEQA                    | EfficientQAT         | A3   | $\Delta s \mapsto \Delta s \otimes \mathbf{1}_{n/g}^\top$                                                                                                 |
| PeriodicLoRA            | ReLoRA               | A3   | $h = \text{id}$ (stagewise)                                                                                                                               |
| PERK                    | LoRA                 | A3   | $h_c(\theta_{adapter}, \alpha, \psi) = \phi^{(N)}(\theta_{adapter}, \alpha, \psi; c)$                                                                     |
| PHATGOOSE               | LoRA                 | A3   | $h_x(v) = ([w_{x,1}(v)B_1 \cdots w_{x,N}(v)B_N], [A_1; \dots; A_N])$ , $w_{x,z} = 0$ off the top- $k$                                                     |
| PHM-Adapter             | Adapter (Houlsby)    | A3   | $h(\{A_i^D, B_i^D, A_i^U, B_i^U\}) = (W_{down} = \sum_i A_i^D \otimes B_i^D, W_{up} = \sum_i A_i^U \otimes B_i^U)$                                        |
| Pico                    | LoRA                 | A3   | $h(*) = ([\gamma S B_1 \cdots \gamma S B_T], [A_1; \dots; A_T])$                                                                                          |
| Piggyback               | Diff Pruning         | A3   | $M \mapsto (\alpha_0, -(\mathbf{1} - M) \odot W_0)$                                                                                                       |
| Piggyback               | HiRA                 | A3   | $M \mapsto (M - \mathbf{1}, I_n)$                                                                                                                         |
| PiSSA                   | LoRA                 | A3   | $h(B, A) = ([B, -B_0], [A; A_0])$                                                                                                                         |
| PLoP                    | LoRA                 | A2   | $h(q_T) = (q_T, q_0 _{T^c})$                                                                                                                              |
| PoLAR                   | LoRA                 | A3   | $h(X, \Theta, Y) = (X\Theta, Y^\top)$                                                                                                                     |
| Polytropon (Poly)       | LoRA                 | A3   | $h_\tau(A, B, Z) = (\sum_j \alpha_{\tau j} B_j, \sum_j \alpha_{\tau j} A_j)$                                                                              |
| Polytropon (Poly)       | MHR                  | A3   | $h(A, B, Z) = (A, B, Z \otimes \mathbf{1}_h)$                                                                                                             |
| PPT                     | Prompt Tuning        | A3   | $h = \text{id}_{\mathbb{R}^{\ell \times d}}$                                                                                                              |
| Prefix-Tuning           | Initial State Tuning | A3   | $h(x_{-\ell..-1}^{(l)}) = (\sum_{i=1}^\ell \bar{A}_i^{i-1} \bar{B}_i x_{-i}^{(l)})_l$                                                                     |
| Prefix-Tuning           | MAM Adapter          | A5   | $h(P', \theta) = (P', \theta, W_{up} = 0, W_{down}^0)$                                                                                                    |
| Prefix-Tuning           | P-Tuning v2          | A3   | $h(P', \theta) = (\text{MLP}_\theta(P'), W_{cls}^0)$                                                                                                      |
| PreFT                   | LoReFT               | A3   | $(W_1, W_2, b) \mapsto (R = (W_2 W_2^\top)^{-1/2} W_2, W = R + s_r (W_2 W_2^\top)^{1/2} W_1, b' = s_r (W_2 W_2^\top)^{1/2} b)$                            |
| PRISM                   | LoRA                 | A3   | $(B, A) \mapsto ([B, B_0], [A; -A_0])$                                                                                                                    |
| PRISM                   | PiSSA                | A3   | $(B, A) \mapsto (\sqrt{s} B, \sqrt{s} A)$                                                                                                                 |
| PRoLoRA                 | LoRA                 | A3   | $h(B_u, B_0, A_u, A_0) = (\mathcal{B}(B_u, B_0), \mathcal{A}(A_u, A_0))$                                                                                  |
| Prompt Tuning           | ATTEMPT              | A3   | $h(P) = (P_{\text{tgt}} = \frac{(t+1)P - \sum_{j \leq t} P_j}{t+2}, \gamma = 0, \beta = 0, W_{down}^0, W_{up}^0)$                                         |
| Prompt Tuning           | MPT                  | A3   | $h(P) = (P \odot u_0 v_0^\top, u_0, v_0)$                                                                                                                 |
| Prompt Tuning           | P-Tuning             | A3   | $h(P) = (P'_0, \psi_0 \text{ with } W_2 \mapsto W_2^0 + H_0^+(P - P_0))$                                                                                  |
| Prompt Tuning           | P-Tuning v2          | A3   | $h(p) = ((K_j(p), V_j(p))_{j \leq L}, W_{cls}^0)$                                                                                                         |
| Prompt2Effect           | LoRA                 | A3   | $h_c(\phi) = (\hat{B}_\phi(c, W_0), \hat{A}_\phi(c, W_0))$                                                                                                |
| Proxy-Tuning            | EFT                  | A3   | $h = \text{id}$                                                                                                                                           |
| PSOFT                   | LoRA-XS              | A3   | $(Q, \alpha, \beta) \mapsto \Sigma_r (D_\beta C(Q)^\top D_\alpha - I) \Sigma_r^{-1}$                                                                      |
| PSOFT                   | PiSSA                | A3   | $(Q, \alpha, \beta) \mapsto (U_r \Sigma_r D_\beta C(Q)^\top D_\alpha \Sigma_r^{-1/2}, \Sigma_r^{1/2} V_r^\top)$                                           |
| Punica                  | LoRA                 | A3   | $h_u(*) = (B_u, A_u)$                                                                                                                                     |

continued on the next page

Table 37, continued

| Source $M$                    | Target $N$        | Rule | Pointed map $h$ with $\rho_N \circ h = \rho_M$                                                                                                                     |
|-------------------------------|-------------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Punica                        | S-LoRA            | A3   | $h = \text{id}_{\prod_u \mathcal{Q}_{\text{LoRA}r_u}}$                                                                                                             |
| PVeRA                         | LoRA-FA           | A3   | $h(b, d_\mu, d_\sigma) = \alpha \Lambda_b B \Lambda_{d_\mu}$                                                                                                       |
| PVeRA                         | VeRA              | A3   | $h(b, d_\mu, d_\sigma) = (b, d_\mu)$                                                                                                                               |
| QA-LoRA                       | QLoRA             | A3   | $(B, A) \mapsto (\frac{s}{s'} B, AP)$                                                                                                                              |
| QLoRA                         | IR-QLoRA          | A2   | $(B, A) \mapsto (\frac{s}{\alpha} B, A, 0, 0)$                                                                                                                     |
| Quantum-PEFT                  | AdaLoRA           | A3   | $h(\theta_U, \theta_V, \lambda) = (P = QP(\theta_U)I_{d_{out} \times K}, \Lambda = \text{diag}(\lambda), Q = (QP(\theta_V)I_{d_{in} \times K})^\top)$              |
| Quantum-PEFT                  | LoRA              | A3   | $h = (U(\theta_U) \text{diag}(\lambda), V(\theta_V)^\top)$                                                                                                         |
| QZO                           | EfficientQAT      | A3   | $h = \text{id} : S \mapsto S$                                                                                                                                      |
| RandLoRA                      | LoRA              | A3   | $h(\Lambda, \Gamma) = ([B_1 \Lambda_1, \dots, B_n \Lambda_n], [\Lambda \Gamma_1; \dots; \Lambda \Gamma_n])$                                                        |
| RaSA                          | LoRA              | A3   | $h = ([\tilde{B}_i \ B_S] D_i, [\tilde{A}_i; A_S])_i$                                                                                                              |
| RED                           | SSF               | A2   | $(\ell_s, \ell_b) \mapsto ((\gamma, \beta)_{\text{FFN out}} = (\ell_s, \ell_b); \text{all other SSF wrappers at } (\mathbf{1}, 0))$                                |
| ReLoRA                        | COLA              | A3   | $h = \text{id}$ (stagewise)                                                                                                                                        |
| ReLoRA                        | LoRA              | A3   | $h((B_k, A_k)_k) = ([B_1 \cdots B_K], [A_1; \dots; A_K])$                                                                                                          |
| ReLoRA                        | ReLoRA            | A5   | $(q_1, \dots, q_K) \mapsto (q_1, \dots, q_K, q_0^{(K+1)})$                                                                                                         |
| RepAdapter                    | LoRA              | A3   | $h(W_u, W_d) = ((sW_0^{(x)} W_u, W_d))_{x \in \{q, k, v\}}$                                                                                                        |
| Residual Adapters             | LoRA-FA           | A3   | $h(\alpha, D_1, D_2) = D_2(I + \alpha D_1) - I$ as the trainable $B$ , with $A_0 = W_0, r = C$                                                                     |
| Residual Prompt Tuning        | Prompt Tuning     | A3   | $h(P, W_{\text{down}}, W_{\text{up}}, \text{LN}) = \text{LN}(\text{ReLU}(PW_{\text{down}})W_{\text{up}}) + P$                                                      |
| Riemannian LoRA               | LoRA              | A3   | $h = \text{id}_Q$                                                                                                                                                  |
| Riemannion                    | LoRA              | A3   | $X \mapsto ([X(B_0^+ X)^+, -B_0], [B_0^+ X; A_0]), \quad X_0 = B_0 A_0$                                                                                            |
| RotMoLE                       | LoRA              | A3   | $h_x((B_i, A_i, q_i)_i, W_g, W_\theta) = ([\mathbb{1}_{i \in \text{TopK}(x)} g_i(x) B_i R_i(x)]_{i=1}^n, [A_1; \dots; A_n])$                                       |
| rsLoRA                        | LoRA              | A3   | $h(B, A) = (\sqrt{r} B, A)$                                                                                                                                        |
| S-LoRA                        | LoRA Soups (CAT)  | A3   | $h_u(*) = e_u$                                                                                                                                                     |
| S-LoRA                        | Punica            | A3   | $h = \text{id}_{\prod_u \mathcal{Q}_{\text{LoRA}r_u}}$                                                                                                             |
| S <sup>2</sup> FT             | Diff Pruning      | A2   | $x \mapsto (\alpha_0, \iota_M x)$                                                                                                                                  |
| S <sup>2</sup> FT             | LoRA              | A2   | $V \mapsto (s^{-1}V, U_S^\top)$                                                                                                                                    |
| S <sup>2</sup> FT             | LoRA-FA           | A3   | $V \mapsto s^{-1}V \quad (A_0 := U_S^\top)$                                                                                                                        |
| SAM (sparse)                  | Diff Pruning      | A2   | $x \mapsto (\alpha_0, \iota_M x)$                                                                                                                                  |
| ScaLoRA                       | LoRA              | A3   | $h(B, A) = ([B, -\tilde{B}_t], [A; \tilde{A}_t])$                                                                                                                  |
| SD-LoRA                       | LoRA              | A3   | $h = ([\alpha_1 \hat{B}_1 \cdots \alpha_{t-1} \hat{B}_{t-1} \ \alpha_t B_t / \ B_t A_t\ _F], [\hat{A}_1; \dots; \hat{A}_{t-1}; A_t]), \ D_k = \hat{B}_k \hat{A}_k$ |
| SEAL                          | LoRA              | A3   | $h = \text{id}$                                                                                                                                                    |
| ShareLoRA                     | LoRA              | A3   | $h((B_\ell)_\ell, A) = (B_\ell, A)_\ell$                                                                                                                           |
| SHINE                         | LoRA              | A3   | $h_c(\Delta_{\text{meta}}, m, \psi) = \text{M2P}_\psi(\mathcal{M}(c))_I$                                                                                           |
| SHiRA                         | RoSA              | A5   | $h(\Delta) = (B = 0, A = A_0, S = M \odot \Delta)$                                                                                                                 |
| SIFT                          | Diff Pruning      | A2   | $x \mapsto (\alpha_0, \iota_M x)$                                                                                                                                  |
| SingLoRA                      | LoRA              | A3   | $h(A) = (\frac{\alpha}{r} u A_{[1:d_{out}]}, A_{[1:d_{in}]}^\top)$                                                                                                 |
| SLAO (Merge before Forget)    | LoRA              | A3   | $h = \text{id}_Q$                                                                                                                                                  |
| SLTrain                       | RoSA              | A3   | $h = \text{id}$                                                                                                                                                    |
| SmartFed (MoRE + EEQA)        | LoRA              | A3   | $h_x(W_{\text{router}}) = (\hat{B} \text{diag}(\tilde{g}_x(W_{\text{router}})), \hat{A})$                                                                          |
| SMEAR                         | Adapter (Houlsby) | A3   | $h_v(\theta_1, \dots, \theta_N, W_r) = \sum_i R(v)_i \theta_i$                                                                                                     |
| SMoA                          | HiRA              | A3   | $(B_k, A_k)_k \mapsto (P_{out}^\top \text{blkdiag}(B_1, \dots, B_K), \text{blkdiag}(A_1, \dots, A_K) P_{in})$                                                      |
| SMT                           | Diff Pruning      | A2   | $x \mapsto (\alpha_0, \iota_M x)$                                                                                                                                  |
| SoRA                          | LoRA              | A3   | $h(B, g, A) = (B \text{diag}(g), A)$                                                                                                                               |
| Sparse Memory Finetuning      | FISH Mask         | A3   | $h = \text{id}, \ M = \text{rows}(T_r)$                                                                                                                            |
| SparseLoRA                    | LoRA              | A3   | $h = \text{id}$                                                                                                                                                    |
| Spectral Adapter <sup>A</sup> | PiSSA             | A3   | $(A_U, A_V) \mapsto ((U_1 + A_U)S_1^{1/2}, S_1^{1/2}(V_1 + A_V)^\top)$                                                                                             |
| Spectral Adapter <sup>R</sup> | LoRA-XS           | A3   | $(Q_U, Q_V) \mapsto C(Q_U)S_1 C(Q_V)^\top S_1^{-1} - I$                                                                                                            |
| Spectral Adapter <sup>R</sup> | PiSSA             | A3   | $(Q_U, Q_V) \mapsto (U_1 C(Q_U)S_1 C(Q_V)^\top S_1^{-1/2}, S_1^{1/2}V_1^\top)$                                                                                     |

continued on the next page

Table 37, continued

| Source $M$                               | Target $N$                      | Rule | Pointed map $h$ with $\rho_N \circ h = \rho_M$                                                                                                                                  |
|------------------------------------------|---------------------------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SpIEL                                    | Diff Pruning                    | A2   | $(\eta, \phi) \mapsto (\alpha_0, \text{scatter}(\eta, \phi))$                                                                                                                   |
| SPoT                                     | Prompt Tuning                   | A3   | $h = \text{id}_{\mathbb{R}^{\ell \times d}}$                                                                                                                                    |
| SPT                                      | RoSA                            | A2   | $((B_s, A_s)_{s \in S}, (x_u)_{u \in U}) \mapsto ((B_s, A_s)_{s \in S}, (0, A_{0,u})_{u \in U}; \iota(x))$                                                                      |
| SRR (Structured Residual Reconstruction) | LoRA                            | A3   | $(B, A) \mapsto ([B, -B_0], [A; A_0])$                                                                                                                                          |
| SSF                                      | GLoRA                           | A3   | $(\gamma, \beta) \mapsto (B_A = \gamma - \mathbf{1}, A_A = \mathbf{1}^\top; D = \gamma - \mathbf{1}; E = \beta)$ (other paths at zero)                                          |
| Stable-LoRA                              | LoRA                            | A3   | $h = \text{id}_Q$                                                                                                                                                               |
| StructLoRA                               | LoRA                            | A3   | $(A, B, m) \mapsto (\frac{\alpha}{r} B \text{diag}(m), A)$                                                                                                                      |
| StructLoRA                               | SoRA                            | A3   | $(A, B, m) \mapsto (B, \frac{\alpha}{r} m, A)$                                                                                                                                  |
| Super-Tuning (Super / Supra)             | PaFi                            | A5   | $h(U) = (U, \gamma_0, \beta_0)$                                                                                                                                                 |
| Super-Tuning (Super / Supra)             | SHiRA                           | A3   | $h = \text{id}$                                                                                                                                                                 |
| Surgical FT                              | Diff Pruning                    | A2   | $x \mapsto (\alpha_0, \iota_M x)$                                                                                                                                               |
| Surgical FT                              | HiFT                            | A5   | $q \mapsto (\text{stage } j : q; \text{other stages} : 0)$                                                                                                                      |
| SVDiff                                   | SVF (Transformer <sup>2</sup> ) | A3   | $\delta \mapsto \text{ReLU}(\sigma + \delta) \oslash \sigma$                                                                                                                    |
| SVDiff                                   | SVFT                            | A3   | $\delta \mapsto \text{ReLU}(\sigma + \delta) - \sigma$                                                                                                                          |
| SVF (Transformer <sup>2</sup> )          | SVFT                            | A3   | $z \mapsto \sigma \odot (z - \mathbf{1})$                                                                                                                                       |
| SVFT                                     | LoRA-FA                         | A3   | $h(M) = U_{:,R} M_{R,C}, A_0 = V_{:,C}^\top$                                                                                                                                    |
| Task Arithmetic                          | AdaMerging                      | A3   | $h(\lambda) = (\lambda_i^l = \lambda_i)_{i,l}$                                                                                                                                  |
| Task Arithmetic                          | LoRA Soups (CAT)                | A3   | $h(\lambda) = (\alpha_i^l = \lambda_i)_{i,l}$                                                                                                                                   |
| Task Arithmetic                          | ZipLoRA                         | A3   | $h(\lambda_c, \lambda_s) = (\lambda_c \mathbf{1}, \lambda_s \mathbf{1})$                                                                                                        |
| TeRA                                     | VeRA                            | A3   | $h(d^{(1)}, d^{(2)}) = (b = d^{(1)}, d = d^{(2)})$                                                                                                                              |
| Text-to-LoRA (T2L)                       | LoRA                            | A3   | $h_z(\theta) = h_\theta(\text{concat}[f(z), E[m], E[l]])$                                                                                                                       |
| Textual Inversion                        | Custom Diffusion                | A2   | $h(v) = (\Delta W^k = 0, \Delta W^v = 0, V^* = v)$                                                                                                                              |
| Textual Inversion                        | Trainable Tokens                | A3   | $h = \text{id} : v \mapsto V_{\{s_v\}}$                                                                                                                                         |
| Tied-LoRA                                | LoRA                            | A3   | $h = (\Lambda_{v_\ell} B \Lambda_{u_\ell}, A)_\ell$                                                                                                                             |
| Tied-LoRA                                | ShareLoRA                       | A3   | $h(\{v_\ell, u_\ell\}_\ell, B, A) = (\{\Lambda_{v_\ell} B \Lambda_{u_\ell}\}_\ell, A)$                                                                                          |
| Tina                                     | LoRA                            | A3   | $h = \text{id}$                                                                                                                                                                 |
| TinyLoRA                                 | LoRA-XS                         | A3   | $h(v) = \Sigma_r^{1/2} \left( \sum_i v_i P_i \right) \Sigma_r^{-1/2}$                                                                                                           |
| TinyLoRA                                 | PiSSA                           | A3   | $h(v) = \left( U_r \Sigma_r^{1/2} (I + \sum_i v_i P_i), \Sigma_r^{1/2} V_r^\top \right)$                                                                                        |
| Trainable Tokens                         | LoRA                            | A2   | $h(X) = (S_T, X), X = V_T - E_{0,T}$                                                                                                                                            |
| Trainable Tokens                         | SHiRA                           | A3   | $h = \text{id}$                                                                                                                                                                 |
| Trans-LoRA                               | LoRA                            | A3   | $h = \text{id}$                                                                                                                                                                 |
| TT-LoRA                                  | LoRA                            | A3   | $h(C_1, \dots, C_D) = (C_1 \times^1 \dots \times^1 C_c, C_{c+1} \times^1 \dots \times^1 C_D)$ reshaped to $d_{out} \times r_c, r_c \times d_{in}$                               |
| Uni-LoRA                                 | LoRA                            | A3   | $h(\theta_d) = P \theta_d = (B^\ell(P \theta_d), A^\ell(P \theta_d))_\ell$                                                                                                      |
| VAPT                                     | VPT                             | A3   | $h_x(q) = ((P^{(l)}(\tilde{X}^{(l)}(x; q)))_{l \leq L}, \text{Head})$                                                                                                           |
| VB-LoRA                                  | LoRA                            | A3   | $h(\alpha, \sigma) = (B(\alpha, \sigma), A(\alpha, \sigma)), u = \sum_s \text{softmax}(\text{TopK}(\sigma, k))_s \alpha_s$                                                      |
| VeLoRA                                   | LoRA                            | A3   | $h = \text{id}_Q$                                                                                                                                                               |
| VeRA                                     | LoRA                            | A3   | $h(b, d) = (\Lambda_b B \Lambda_d, A)$                                                                                                                                          |
| VeRA                                     | LoRA-FA                         | A3   | $h(b, d) = \Lambda_b B \Lambda_d$                                                                                                                                               |
| VPT                                      | P-Tuning v2                     | A3   | $h(P, \text{Head}) = \left( (\text{LN}(P_{i-1}) W_k^{(i)} + \mathbf{1} b_k^{(i)\top}, \text{LN}(P_{i-1}) W_v^{(i)} + \mathbf{1} b_v^{(i)\top})_{i \leq L}, \text{Head} \right)$ |
| WaveFT                                   | LoRA-FA                         | A3   | $h(c) = \lambda B_W C_\Omega(c), A_W = [\psi_j^\top]_{j \in \text{cols}(\Omega)}$                                                                                               |
| WiSE-FT                                  | Concept Sliders                 | A3   | $h(\alpha) = (\alpha B_1, A_1)$                                                                                                                                                 |
| WiSE-FT                                  | Task Arithmetic                 | A3   | $h(\alpha) = (\lambda = \alpha)$                                                                                                                                                |
| X-LoRA                                   | LoRA Soups (CAT)                | A3   | $h_x(\phi) = (\alpha_i^l = \lambda_i^l(x; \phi) \alpha_i / r_i)$                                                                                                                |
| XPrompt                                  | Prompt Tuning                   | A2   | $h(P_S) = (P_S, 0_{S^c})$                                                                                                                                                       |
| ZipLoRA                                  | LoRA                            | A3   | $h(m_c, m_s) = ([B_c B_s], [A_c \text{diag}(m_c); A_s \text{diag}(m_s)])$                                                                                                       |
| ZO-Muon                                  | LoRA                            | A2   | $X \mapsto (P_k, X)$                                                                                                                                                            |

## Numerical checks and reproducibility

Claims marked [Num] in the main text are numerical checks and are not counted as proved. Most of them are re-run by one script, `framework_checks.py`, one of the data files that accompany this paper. This appendix says what the script computes and how to read its output (§E.1), lists every check with the result it supports and the value it prints (§E.2, Table 38), names the few numbers of the main text that come from elsewhere (§E.3), records the software (§E.4), describes how the companion site and the figures of this paper compute what they show (§E.5), and gives the settings and values behind each figure (§E.6).

### E.1 The script

The script runs with `python3 framework_checks.py` and needs PyTorch (Paszke et al., 2019), NumPy (Harris et al., 2020) and SciPy (Virtanen et al., 2020). It works in float64 throughout (`torch.set_default_dtype(torch.float64)`) and on the CPU. It seeds PyTorch’s generator at the start and again before several checks, so that a run prints the same numbers every time on a given machine. On a laptop (Apple M5) it finishes in about 50 seconds.

It computes three kinds of quantity.

- *Dimensions.* The expressive dimension  $d(M)$  of a method is the generic rank of its Jacobian (Background C.3). The script forms the Jacobian of  $\rho$  by automatic differentiation (`torch.autograd.functional.jacobian`) at a random point, or at the base point when it measures a first-order image  $S_1$ , and counts the singular values above  $10^{-9}$  times the largest. The points of lower rank have measure zero (Background C.21, Background C.22), so this computes the generic rank with probability one, up to the tolerance.
- *Optimizer runs.* Adam, decoupled weight decay and gradient descent are written out step by step, with the bias correction and the convention  $0/0 = 0$  at  $\varepsilon = 0$ , so that nothing is hidden in a library default. Gradient flows are integrated by explicit Euler steps or by the classical fourth-order Runge–Kutta method.
- *Identities and defects.* An identity or an equivariance is tested by computing both sides and printing the Frobenius norm of their difference.

Each line prints a value beside its prediction. Some lines are controls, cases in which the theory predicts that an identity fails; they show that the test would detect a failure.

**How sharp the rank decisions are.** The script takes 45 Jacobian ranks. In every one of them the smallest singular value it keeps is at least  $3.9 \times 10^{-3}$  times the largest, and the largest singular value it discards is at most  $4.3 \times 10^{-16}$  times the largest, which is rounding error. The threshold  $10^{-9}$  lies more than six orders of magnitude from both, so no rank in Table 38 depends on its choice. We measured these margins with a copy of the script whose rank function also records the two ratios; the copy prints the same output.

**What changes between machines.** The ranks and dimensions are integer decisions taken with the margins above, and the exact zero of check 3 comes from rescaling by a power of two, which is exact in IEEE arithmetic. These outputs are expected to be the same on any machine. Values of order  $10^{-15}$  are rounding errors, and their leading digits change with the processor and the linear-algebra library. Values of order one depend on the random draws, which follow from the seeds.



## E.2 The checks, section by section

The script has thirteen sections. Sections 1 to 11 are the numerical claims of the framework, section 12 re-runs the main numbers of the first referee round, and section 13 those of the second. Table 38 lists every check in the script’s order, with the result it supports, what it computes and what it prints in the run of 2 October 2026. Every printed value agrees with its prediction, and every control fails, by amounts between  $1.6 \times 10^{-2}$  and 52. Matrix shapes are written  $(m, n, r)$  for an  $m \times n$  weight and rank  $r$ ; a defect is the Frobenius norm of the difference between the two sides of the identity tested.

**Table 38.** The checks of `framework_checks.py`, in the script’s order, with the result each one supports, what it computes, and what it prints. Controls are cases in which the theory predicts a failure.

| Check                                                       | Supports               | What the script computes                                                                                                                                                                                                                                                          | What it prints                                                                                                                                                                                                                                                                                          |
|-------------------------------------------------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sections 1–11: the numerical claims of the framework</b> |                        |                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                         |
| 1                                                           | Thm II.7(b)            | $\text{HRA}_r$ without Gram–Schmidt, $u \mapsto W_0 H(u_1) \cdots H(u_r)$ , with an injective $W_0 \in \mathbb{R}^{(n+2) \times n}$ . Jacobian rank at a random point, and at HF’s paired initialisation ( $u_{2i-1} = u_{2i}$ , so $k = r/2$ distinct vectors).                  | At $(n, r) = (9, 2), (9, 4), (10, 2), (10, 4), (12, 6)$ : generic 15, 26, 17, 30, 51, equal to $rn - r(r+1)/2$ ; paired 8, 15, 9, 17, 30, equal to $kn - k(k+1)/2$ .                                                                                                                                    |
| 2                                                           | Thm III.12(b)          | The first Adam step from zero state with $\varepsilon = 0$ is sign descent. Its first-order weight update $dB A + B dA$ at $(B, A)$ and at $(Bg, g^{-1}A)$ , $(m, n, r) = (6, 5, 3)$ .                                                                                            | Defect $1.4 \times 10^{-15}$ for a signed permutation $g$ . Controls: 11.3 for $g = \text{diag}(2, 0.5, 3)$ and 6.6 for a random rotation. With per-channel rates $(\eta_B d_j, \eta_A/d_j)$ the diagonal $g = \text{diag}(d)$ gives $5.0 \times 10^{-16}$ .                                            |
| 3                                                           | Cor III.11, Thm III.10 | 200 steps of bias-corrected Adam with $\varepsilon = 0$ on a least-squares toy ( $W_0 \in \mathbb{R}^{7 \times 6}$ , $r = 2$ , $B_0 = 0$ ): LoRA+ with ratio $\lambda = 8$ and scale $s$ , against LoRA with scale $\lambda s$ and equal rates.                                   | Largest weight difference over the whole trajectory 0.0, exactly, since $\lambda$ is a power of two. Control without the rescaling: 0.93.                                                                                                                                                               |
| 4                                                           | Prop II.12             | GraLoRA with $k \times k$ blocks of rank $r/k$ : Jacobian rank, and the rank of $\Delta W$ at a random point.                                                                                                                                                                     | At $(m, n, r, k) = (8, 8, 4, 2), (12, 8, 4, 2), (12, 12, 6, 3)$ : dimension 48, 64, 108, equal to $r(m+n-r)$ ; rank of $\Delta W$ 8, 8, 12, against LoRA’s cap 4, 4, 6.                                                                                                                                 |
| 5                                                           | Prop II.12             | $\text{LoHa}_{r,r}, (B_1 A_1) \odot (B_2 A_2)$ : Jacobian rank.                                                                                                                                                                                                                   | At $(m, n, r) = (8, 7, 2), (12, 10, 2), (12, 10, 3)$ : 38, 59, 93, equal to the bound $2r(m+n-r) - (m+n-1)$ .                                                                                                                                                                                           |
| 6                                                           | Cor III.6(b)           | Euler integration ( $h = 10^{-4}$ , $T = 2$ ) of LoRA’s gradient flow with decay $\lambda = 0.5$ at $(m, n, r) = (5, 4, 2)$ ; the final charge $\Phi$ against its closed form.                                                                                                    | Coupled decay at $\eta_B/\eta_A = 4$ against $e^{-2\eta_A \lambda t} \Phi(0)$ : relative error 10.7 (control). Coupled decay at $\eta_A = \eta_B$ : $2.5 \times 10^{-4}$ . Per-time decay at $\eta_B/\eta_A = 4$ against $e^{-2\lambda t} \Phi(0)$ : $2.3 \times 10^{-4}$ . Both are integrator errors. |
| 7                                                           | Thm III.3, Prop III.4  | At a random point, $(m, n, r) = (7, 6, 2)$ , under a generic $g \in \text{GL}_2$ : the plain-GD cometric $GA^\top A + BB^\top G$ and the undamped scaled-GD cometric; and LoRA-Pro’s update $\Pi = P_U G + G P_V - P_U G P_V$ paired with a random tangent vector $T = XA + BY$ . | Gauge defect 52.3 for plain GD (control) and $1.1 \times 10^{-14}$ for scaled GD, which equals $P_U G + G P_V$ to $8.7 \times 10^{-16}$ . $\langle G - \Pi, T \rangle = 0.0$ , so $\Pi$ is the orthogonal projection onto the tangent space.                                                            |
| 8                                                           | Thm V.3                | The span of 200 random VeRA updates $\text{diag}(b) B \text{diag}(d) A$ with frozen $B$ and $A$ , $(m, n, r) = (6, 5, 2)$ .                                                                                                                                                       | Dimension 12 = $mr$ : the span is $\{XA\}$ , the reach of re-merged VeRA.                                                                                                                                                                                                                               |

| Check                                      | Supports               | What the script computes                                                                                                                                                                                                                | What it prints                                                                                                                                                                                                                                                                                 |
|--------------------------------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9                                          | Prop II.9              | RoAd <sub>1</sub> at $m = 8$ , one angle and one scale per output pair acting on $W_0$ by $2 \times 2$ blocks: Jacobian rank.                                                                                                           | $8 = m$ , the number of parameters: an injective map with an $m$ -dimensional image.                                                                                                                                                                                                           |
| 10                                         | Prop II.3              | The rank of $(dB, dA) \mapsto dB A + B dA$ with rank $A = r_A$ and rank $B = r_B$ , at $(m, n, r) = (7, 6, 3)$ .                                                                                                                        | $(r_A, r_B) = (3, 0), (3, 3), (2, 1)$ : 21, 30, 18, equal to $mr_A + nr_B - r_A r_B$ .                                                                                                                                                                                                         |
| 11                                         | Thm V.3                | The product of two rotations, each a product of two reflections in $\mathbb{R}^8$ , as in two cycles of HRA <sub>2</sub> .                                                                                                              | $\text{rank}(R_1 R_2 - I) = 4 \leq 2 + 2$ : subadditivity, attained.                                                                                                                                                                                                                           |
| <b>Section 12: the first referee round</b> |                        |                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                |
| 12a                                        | Prop II.8              | DoRA as in HF PEFT, $\text{diag}(\mu_i / \ V_i\ ) V$ with $V = W_0 + BA$ : Jacobian rank.                                                                                                                                               | $9 = mn$ at $(3, 3, 2)$ , where $r(m + n - r) + m = 11$ ; 36 at $(8, 8, 2)$ , matching the formula; 24 for a rank-one $W_0$ at $(6, 6, 2)$ , equal to $r(m + n - r) + m - r$ .                                                                                                                 |
| 12b                                        | Prop II.12             | The rank of GraLoRA's $\Delta W$ at a random point.                                                                                                                                                                                     | $8 = \min(kr, m, n)$ at $(16, 16, 4, 2)$ ; $12 = \min(m, n)$ at $(12, 12, 6, 3)$ .                                                                                                                                                                                                             |
| 12c                                        | Prop II.12             | LoHa at $(4, 4, 2)$ : Jacobian rank.                                                                                                                                                                                                    | $16 = mn$ , where the bound gives 17.                                                                                                                                                                                                                                                          |
| 12d                                        | Prop VI.4              | LoReFT on $d = 6$ with $r = 2$ , $(R, W, b) \mapsto (I + R^\top(W - R), R^\top b)$ with the rows of $R$ orthonormalised: Jacobian rank.                                                                                                 | $22 = r(2d + 1 - r)$ , against 26 for LoRA <sub>r</sub> with a free bias.                                                                                                                                                                                                                      |
| 12e                                        | Prop VI.4(b)           | The serial adapter $(I + UD)W_0$ at $(m, n, r) = (5, 3, 1)$ : Jacobian rank.                                                                                                                                                            | 7, equal to LoRA's $r(m + n - r)$ , since this $W_0$ is injective.                                                                                                                                                                                                                             |
| 12f                                        | Prop II.12, Prop II.13 | FourierFT atoms: the real parts of inverse two-dimensional DFTs of single frequencies on $8 \times 8$ .                                                                                                                                 | The frequencies $(1, 2)$ and $(7, 6)$ give the same atom; three frequencies containing one conjugate pair span 2 dimensions.                                                                                                                                                                   |
| 12g                                        | Cor II.6, Thm V.3      | HF's default Cayley–Neumann factor $R = I + 2Q + 2Q^2 + 2Q^3 + Q^4$ for a skew $Q$ of block size 8 with entries of size about 0.07; then 200 successive merges with entries of size about 0.05.                                         | $\ R^\top R - (I - Q^4)^2\  = 4.7 \times 10^{-16}$ and $\ R^\top R - I\  = 1.5 \times 10^{-2}$ . Relative drift of $K_{\text{out}}$ : 0.002, 0.010, 0.064, 0.224 after 1, 10, 50, 200 merges.                                                                                                  |
| 12h                                        | Cor II.6               | BOFT's $\text{diag}(s) W_0 R^\top$ with an exact Cayley $R$ and $s = (-0.5, 1, \dots, 1)$ .                                                                                                                                             | Largest change of a cosine between neurons 1.50; of an absolute cosine $5.6 \times 10^{-16}$ .                                                                                                                                                                                                 |
| 12i                                        | Prop V.4               | DARE with a fixed mask under an invertible diagonal $D$ with signs; TIES (top 20% kept, sign election by total mass) under scalings.                                                                                                    | $\text{DARE}(D\tau) - D \text{DARE}(\tau)$ : $1.8 \times 10^{-15}$ . TIES: $8.9 \times 10^{-16}$ for the factor 3.7, 0.0 for $-2$ , and 5.6 for a generic positive diagonal (control).                                                                                                         |
| 12j                                        | Prop II.16             | The whitened split $[W_0 L]_r L^{-1}$ , the paper-form CorDA split $[W_0 C]_r C^{-1}$ and PiSSA's $[W_0]_r$ at $(m, n, r) = (6, 5, 2)$ : the defect of $f(W_0 h^{-1}, h C h^\top)$ against $f(W_0, C) h^{-1}$ for an input change $h$ . | Under $h = 3O$ , $O \in O(n)$ : $2.2 \times 10^{-15}$ , $1.8 \times 10^{-15}$ , $8.1 \times 10^{-16}$ . Under a generic $h$ : $1.3 \times 10^{-15}$ , 0.63, 1.1. Whitened split with damping $C + 0.1I$ , generic $h$ : 0.14. The whitened split equals $\Pi_r W_0$ to $5.3 \times 10^{-15}$ . |
| 12k                                        | Thm III.12(b)          | The first Adam step with $\varepsilon = 10^{-8}$ and gradients of size $10^{-6}$ , under $g = \text{diag}(0.3, 1, 4)$ with per-channel rates.                                                                                           | Relative defect $1.6 \times 10^{-2}$ with $\varepsilon$ shared (control), $4.9 \times 10^{-17}$ with $\varepsilon$ rescaled per channel.                                                                                                                                                       |
| 12l                                        | Thm III.12             | LoRA-Pro's parameter step at $(m, n, r) = (8, 7, 3)$ , $s = 1.5$ , with the Sylvester-optimal free matrix $X$ and with $X = 0$ .                                                                                                        | Non-orthogonal $g$ : 1.2 with the Sylvester $X$ , $1.9 \times 10^{-15}$ with $X = 0$ . Orthogonal $g$ : $1.2 \times 10^{-15}$ and $1.4 \times 10^{-15}$ .                                                                                                                                      |

| Check                                       | Supports                                                 | What the script computes                                                                                                                                                                                                                        | What it prints                                                                                                                                                                                                                                         |
|---------------------------------------------|----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12m                                         | <a href="#">Thm II.2</a> ,<br><a href="#">Prop II.3</a>  | $\dim S_1$ at $(m, n, r) = (7, 5, 2)$ for the three strata and for HF's orthogonal initialisation.                                                                                                                                              | $T_A: 14 = mr$ ; $T_B: 10 = nr$ ; $S: 20 = r(m + n - r)$ . Orthogonal init: $ B_0 A_0  \leq 3.5 \times 10^{-18}$ and $\dim S_1 = 11 = m + n - 1$ , the rank formula at $\text{rank } A_0 = \text{rank } B_0 = 1$ .                                     |
| 12n                                         | <a href="#">Obs I.7</a> ,<br><a href="#">Prop I.8(a)</a> | $\dim S_1$ of a standard $\text{LoRA}_4$ and of $\text{LoRA}_2 \oplus \text{LoRA}_2$ with one $A_0$ used twice ( $m = 7, n = 6$ ); the dimension of the fibre product of $B \mapsto W_0 + B A_0$ and $A \mapsto W_0 + B_0 A$ at $(6, 8, 2)$ .   | 28 against 14; fibre product $4 = r^2$ .                                                                                                                                                                                                               |
| 12o                                         | <a href="#">Prop II.13</a>                               | Two HiRA cycles at $r = 1$ on a $6 \times 5$ weight with no zero entry; the rank of the relative update $(W_2 - W_0) \otimes W_0$ .                                                                                                             | 3, above $2r = 2$ and equal to $\min(2r + r^2, m, n)$ .                                                                                                                                                                                                |
| 12p                                         | <a href="#">Thm II.11</a>                                | The Kronecker rank of a generic $uv^\top$ for the $2 \times 4$ split of $\mathbb{R}^8$ ; the largest rank of a LoKr update $C \otimes D$ with $C \in \mathbb{R}^{4 \times 4}$ and a rank-one $D \in \mathbb{R}^{4 \times 4}$ over 200 draws.    | $4 = \min(m_1, m_2) \min(n_1, n_2)$ , the largest Kronecker rank for this split; 4.                                                                                                                                                                    |
| 12q                                         | <a href="#">Thm V.3(c)</a> ,<br>cor. 4                   | Connected components of the hypergraph of BOFT blocks of size $b = 4$ on $n = 64$ coordinates after $m'$ butterfly factors.                                                                                                                     | Component sizes 4, 8, 16, 32, 64 for $m' = 1, \dots, 5$ ; connected only at full depth.                                                                                                                                                                |
| 12r                                         | <a href="#">Cor III.6</a>                                | A long gradient flow ( $T = 200$ ) from $B_0 = 0$ at $\eta_B/\eta_A = 4$ , decay $\lambda = 0.1$ , $(m, n, r) = (12, 10, 3)$ : the final sizes of $\mu = B^\top B - A A^\top$ and of $\Phi$ , relative to $B^\top B$ and to $B^\top B/\eta_B$ . | Coupled decay: $\mu \ 1.0 \times 10^{-12}$ , $\Phi \ 3.0$ .<br>Per-time decay: $\mu \ 0.75$ , $\Phi \ 1.4 \times 10^{-12}$ .                                                                                                                           |
| 12s                                         | <a href="#">Cor III.13</a>                               | A $\text{GL}_r$ -equivariant example rule, $B \leftarrow B - \eta G_B(A A^\top)^{-1}$ and $A \leftarrow A - \eta A A^\top G_A$ , for 50 steps from $B_0 = 0$ , started at $A_0$ and at $g A_0$ .                                                | Difference of the final weights $1.8 \times 10^{-15}$ .                                                                                                                                                                                                |
| 12t                                         | <a href="#">Thm V.3</a> ,<br>cor. 4                      | The Lie algebra generated by the logarithms of 12 random HF Cayley–Neumann factors with $b = 4$ .                                                                                                                                               | Dimension 16 = $\dim \mathfrak{gl}(4)$ , against $\dim \mathfrak{so}(4) = 6$ . This identifies the generated group only; re-merging reaches a monoid (check 13n).                                                                                      |
| <b>Section 13: the second referee round</b> |                                                          |                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                        |
| 13a                                         | <a href="#">Obs I.7</a>                                  | $\dim S_1$ at $m = 7, n = 6$ for $\text{LoRA}_2 \oplus \text{LoRA}_2$ with independent $A_0$ 's, for $F(1) \oplus F(2)$ , and for a standard $F(3)$ .                                                                                           | 28; 21 and 21.                                                                                                                                                                                                                                         |
| 13b                                         | <a href="#">Thm II.2(e)</a>                              | HF's orthogonal initialisation against a $T_A$ point at $(m, n, r) = (4, 5, 2)$ .                                                                                                                                                               | $\dim S_1 = 8 = mr$ for both, but the two first-order images together span 14 dimensions, so they differ.                                                                                                                                              |
| 13c                                         | <a href="#">Cor II.6</a>                                 | BOFT with every $s_i < 0$ ; GOFT with its scalar $g = 1 \pm 0.05$ ; a conformal block $\sqrt{2} \text{Rot}(0.3)$ on two of eight inputs; the Cayley–Neumann factor at $\theta = 1$ with $b = 2$ .                                               | Cosine change $4.4 \times 10^{-16}$ . GOFT: cosine change 0.044, relative singular-value change 0.033, and $K_{\text{out}} = W_0 \text{diag}(g)^2 W_0^\top$ to $6.1 \times 10^{-15}$ . Conformal block: cosine change 0.19. $\ R\  = 0$ , so $R = 0$ . |
| 13d                                         | <a href="#">Thm II.7</a>                                 | The Hurwitz move $H_a H_b = H_{H_a b} H_a$ for reflections in $\mathbb{R}^6$ .                                                                                                                                                                  | Defect $4.4 \times 10^{-16}$ .                                                                                                                                                                                                                         |
| 13e                                         | <a href="#">Prop II.8</a>                                | DoRA at $(8, 8, 2)$ with one magnitude $\mu_i = 0$ : Jacobian rank.                                                                                                                                                                             | 34, against 36 when every $\mu_i \neq 0$ .                                                                                                                                                                                                             |
| 13f                                         | <a href="#">Thm II.11(e)</a>                             | The largest rank of a LoKr update $C \otimes B A$ over 50 draws.                                                                                                                                                                                | $(m_1, m_2, n_1, n_2) = (2, 8, 8, 2)$ with inner dimension 4: 4. $(4, 2, 2, 8)$ with inner dimension 5: 4. Predicted $\min(m_1, n_1) \min(r', m_2, n_2) = 4$ and 4.                                                                                    |

| Check | Supports        | What the script computes                                                                                                                                                                                                                            | What it prints                                                                                                                                                                                                                                                                                |
|-------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 13g   | Prop II.12      | LoHa <sub>r,r</sub> against LoRA <sub>2r</sub> ; the orbit of LoHa's exhibited gauge; HRA at $r \geq n - 1$ .                                                                                                                                       | (9, 9, 3): LoHa 73, LoRA <sub>6</sub> 72. (4, 4, 2): 16 and 16. LoHa at (4, 4, 2): orbit dimension 15, fibre dimension 16. HRA at $n = 5$ : 10 for $r = 4$ and for $r = 5$ , equal to $\dim \mathrm{SO}(5)$ .                                                                                 |
| 13h   | Prop II.13      | OFT transported by $T = 2I$ , through two merge-and-restart cycles of the rotation $-I$ from $\theta_0 = (1, 0)$ ; HiRA after two cycles at $r = 1, 2, 3$ on a $30 \times 30$ weight.                                                               | $\theta_2 = (9, 0)$ , of norm 9, off the rotation orbit of $\theta_0$ . HiRA's relative-update ranks 3, 8, 15, equal to $\min(2r + r^2, 30)$ .                                                                                                                                                |
| 13i   | Prop II.16      | Bottom- $r$ splits (CorDA's knowledge-preserving form on $W_0C$ ; MiLoRA on $W_0$ ) and whitened splits with absolute damping 0.1 and with trace-relative damping $0.1 \operatorname{tr} C/n$ , at $(m, n, r) = (7, 6, 2)$ .                        | Under $h = 3O$ : $3.1 \times 10^{-15}$ , $4.0 \times 10^{-15}$ , $2.3 \times 10^{-2}$ , $3.0 \times 10^{-15}$ . Under a generic $h$ : 4.9, 0.34, 4.6, 5.9. At $r = n$ , PiSSA under a generic $h$ : $3.1 \times 10^{-14}$ .                                                                   |
| 13j   | Thm III.5       | Per-channel rates $\eta_B = (1, 1, 1)$ and $\eta_A = (1, 1, 2)$ at $(m, n, r) = (5, 4, 3)$ on the non-quadratic loss $\sum_{ij} \log \cosh W_{ij} - \langle Y, W \rangle$ , by Runge–Kutta with $h = 10^{-3}$ for 1000 steps; one discrete GD step. | Drift of the charge of channels 1, 2 (equal products $\eta_{B,j} \eta_{A,j}$ ) $4.2 \times 10^{-12}$ ; of the form of channels 1, 3, 3.1. One GD step changes $\Phi$ by $\eta_B A G^T G A^T - \eta_A B^T G G^T B$ to $2.4 \times 10^{-15}$ .                                                  |
| 13k   | Cor III.6       | One Adam sign step: $\ \Delta\Phi\ $ divided by the learning rate.                                                                                                                                                                                  | 20.9, 21.1, 21.1 at learning rates $10^{-2}$ , $10^{-3}$ , $10^{-4}$ ; first order in the step size.                                                                                                                                                                                          |
| 13l   | Prop IV.3       | GaLore's field for $T = 1$ at $W = \operatorname{diag}(2, 1)$ with $L = \ W\ ^2/2$ ; the Lie bracket of $X_i(W) = u_1(W)e_i^T$ by automatic differentiation; the cometric of $B \mapsto BB^T$ at $B$ and at $BR$ , $R \in O(2)$ .                   | $[X_1, X_2] = -\frac{1}{3}e_2e_1^T$ to six digits, as predicted. Cometric difference $9.5 \times 10^{-16}$ .                                                                                                                                                                                  |
| 13m   | Prop V.2        | Blockwise absmax NF4 (blocks of 64) on $256 \times 256$ weights with $r = 8$ ; the defects of QLoRA, QPiSSA and one-step LoftQ; NF4 under addition.                                                                                                 | Gaussian $W$ : $\ e\  = 0.945$ , $\ e - \operatorname{SVD}_8(e)\  = 0.889$ , LoftQ with $s = 2$ : $0.945 = \ e\ $ , QPiSSA 0.887. Low rank plus noise: $\ e\  = 0.592$ , QPiSSA 0.227, LoftQ with $s = 1$ : 0.549. 96.7% of the entries of $\kappa(W) + \kappa(\Delta)$ lie off the NF4 grid. |
| 13n   | Thm V.3, cor. 4 | The operator norm of Cayley–Neumann factors for 2000 random skew $Q \in \mathbb{R}^{4 \times 4}$ with $\ Q\ _{\operatorname{op}} \leq 2^{1/4}$ ; 200 successive merges on $W \in \mathbb{R}^{6 \times 8}$ .                                         | Largest $\ R\ _{\operatorname{op}}$ : 1 to rounding. Largest eigenvalue of $K_{\operatorname{out}}(t+1) - K_{\operatorname{out}}(t)$ : $-5.5 \times 10^{-9}$ , so $K_{\operatorname{out}}$ decreases in the Loewner order at every merge.                                                     |
| 13o   | Prop V.4        | Factor-wise DARE on two adapters under a diagonal gauge of one adapter or of both; the LoraHub change formula with three modules; the fibre of Poly with $K = 3$ skills, $r = 2$ and $6 \times 6$ weights.                                          | Gauge on one adapter: 39.5 (control); on both: $5.6 \times 10^{-15}$ . LoraHub formula: $4.0 \times 10^{-15}$ . Poly fibre: 32 with 2 tasks, $4 = \dim \mathrm{GL}_2$ with 4 tasks.                                                                                                           |
| 13p   | Thm VI.3 (R5)   | Transposition under RoPE with $d_k = 8$ : $\langle R_m M q, R_n k \rangle$ against $\langle q, M^T R_{n-m} k \rangle$ , and against the slide onto the key side, $\langle R_m q, R_n M^T k \rangle$ .                                               | $2.2 \times 10^{-16}$ ; the key-side slide differs by 2.2 (control).                                                                                                                                                                                                                          |
| 13q   | Prop VI.4(b)    | The serial adapter at $(m, n, r) = (4, 8, 1)$ : Jacobian rank.                                                                                                                                                                                      | 7, against $d(\operatorname{LoRA}_1) = 11$ : with $m < n$ the update rows stay in row $W_0$ .                                                                                                                                                                                                 |

### E.3 Numbers from other sources

A few [Num] values in the main text are not printed by the script. They come from the replications that the referees ran in the two review rounds, recorded with the review notes of each result, and from the check scripts that accompanied the four proposals from which

**Table 39.** Software and hardware for the checks, the figures and the companion site.

| Component                                   | Version                             | Used for                                                                                      |
|---------------------------------------------|-------------------------------------|-----------------------------------------------------------------------------------------------|
| Python                                      | 3.9.6                               | the check script, the figure scripts, the generators of the tables and ancillary files        |
| PyTorch (Paszke et al., 2019)               | 2.8.0                               | float64 automatic differentiation, linear algebra, seeded random draws                        |
| NumPy (Harris et al., 2020)                 | 2.0.2                               | discrete Fourier transforms and ranks in the checks; the figure scripts                       |
| SciPy (Virtanen et al., 2020)               | 1.13.1                              | Sylvester equations (LoRA-Pro, check 12l) and matrix logarithms (check 12t)                   |
| Matplotlib (Hunter, 2007)                   | 3.9.4                               | the figures of this paper                                                                     |
| Hugging Face PEFT (Mangrulkar et al., 2022) | commit 85be3f3, 30 September 2026   | the implementation that every library claim was checked against; not needed to run the checks |
| TeX Live                                    | 2026 (pdfTeX 1.40.29, latexmk 4.88) | this document                                                                                 |
| D3 (Bostock et al., 2011)                   | 7.9.0                               | drawing the companion site's figures                                                          |
| MathJax (Cervone, 2012)                     | 3.2.2                               | typesetting the companion site's mathematics                                                  |
| Hardware                                    | Apple M5, macOS 26.7, CPU only      | the run reported in Table 38                                                                  |

the framework was synthesised. They are the rank formula of [Proposition II.3](#) in 400 random cases and its loss rate, the Adam steps after [Lemma II.4](#), the LOFT identities of §II.B, the RoAd<sub>2</sub> dimension ([Proposition II.9](#)), the 283 random networks of [Theorem II.10](#), VeRA's dimension  $m + r - 1$  and the LoHa dimension at (20, 16, 2) in the census of [Proposition II.12](#), the operator-Schmidt ranks after [Theorem II.11](#), the transported cometric and Adam runs of [Proposition II.13](#), the shared-frame and RoPE tests of [Observation II.15](#), the finite-difference cometrics of [Observation III.2](#) and [Theorem III.3](#), LoRA-FA under AdamW ([Proposition III.4](#)), the non-gradient signals and the clipping of [Theorem III.5](#), the Kempf–Ness minimum of [Proposition III.7](#), the closed form of [Theorem III.9](#), the warm-up,  $\varepsilon$ , decay and clipping runs of [Theorem III.10](#) and [Corollary III.11](#), LoRA-RITE, LoRA-Pro's second-order  $W$ -step defect and the zero- $B$  runs of [Theorem III.12](#) and [Corollary III.13](#), the GaLore runs of [Proposition IV.2](#), the general bracket formula of [Proposition IV.3](#), the NF4 example with a dominant low-rank part in [Proposition V.2](#) (check 13m runs an analogous one at another scale), the query-scaling and post-LN tests of [Theorem VI.3](#), and the decoder tests of [Theorem VI.5](#). Where a referee's replication and the script use different random draws, the main text may quote the referee's value; the conformal block of [Corollary II.6](#), for instance, changes some cosine by 0.17 in the referee's draw and by 0.19 in the script's (check 13c).

## E.4 Software

Table 39 lists the software behind the run of Table 38, the figures and this document. Only the first four rows are needed to re-run the checks.

## E.5 How the companion site and the figures compute what they show

The companion site is one static page. A build script assembles it from one HTML file per exposé, and a second script compiles its data file from the verified catalogue, the arXiv metadata, the classification batches, the axis definitions, the Rosetta stone and the numbered results. The data file is loaded as a script, so the page runs from a web host, from disk or in a



sandboxed viewer without fetching its data; it loads D3, MathJax and its fonts from public CDNs. The tables of Appendix D and the ancillary files are generated from the same data.

Each figure of the site is a small module in plain JavaScript that computes every number it shows in the reader’s browser, when it first comes into view and again at every interaction; nothing is typed in by hand. JavaScript numbers are IEEE doubles, so the arithmetic is float64, as in the script. A shared kernel provides dense matrix products, Kronecker and Hadamard products, a cyclic Jacobi eigensolver for symmetric matrices, singular values, numerical rank, random orthogonal matrices and a seeded generator (mulberry32, with Box–Muller normals), so every figure opens on the same default state. Figures that must see exact zeros, such as the operator-Schmidt spectra behind Figure 6 and the invariant lamps of the Gram lab, use a one-sided Jacobi SVD, which resolves singular values down to about  $10^{-16}$  of the norm, where an eigensolver applied to the Gram matrix stops near  $10^{-8}$ .

- The orbit explorer trains a seeded toy regression twice, by a hand-written bias-corrected Adam or by gradient descent. Its sliders move on a factor-of-two grid, so every torus element it can reach rescales doubles exactly, and at  $\varepsilon = 0$  without decay or clipping its meter  $\max_t \|W_t - W'_t\|_F$  reads exactly 0, as check 3 does.
- The gauge lab computes the update of SGD, Adam, Adam with per-channel rates, damped and undamped scaled GD and LoRA-Pro at  $(B, A)$  and at  $(Bg^{-1}, gA)$  and prints the relative defect, as checks 2, 7, 12k and 12l do. It recomputes task arithmetic and LoraHub from the factors at every gauge, as check 13o does.
- The Noether figure integrates scalar LoRA under gradient flow by fourth-order Runge–Kutta with  $h = 10^{-3}$ , under gradient descent by Euler steps and under Adam step by step, so that the charge is computed and never assumed. Its matrix panel uses the implicit midpoint rule, which keeps quadratic invariants to rounding, and compares the closed-form velocity of Theorem III.9 with the integrated one.
- The apex figure checks its counters by a Jacobian rank on mount, reproducing the [Num] lines of Theorem II.2 and Theorem II.7. The rank-versus-volume figure rebuilds each parametrisation, forms its Jacobian by central differences, which are exact here because each map is affine in every single coordinate, and takes its numerical rank by Householder QR with column pivoting; for a large layer it runs on a small copy with the same rank parameters. The plug workbench (§II.C) measures dimensions the same way.
- The merge game checks every rewrite of Theorem VI.3 on seeded random weights of a small pre-LN transformer block and prints the relative output difference, about  $10^{-16}$  for a legal rewrite, or the residual of a refused one. The prefix figure computes its attention weights from seeded synthetic tokens. The rebasing figure measures the rank of accumulated ReLoRA updates by Jacobi SVD against  $\min(Kr, m, n)$ . The lens figure runs two float64 checks on mount, of Proposition IV.2 and of the bracket of Proposition IV.3.
- The budget calculator evaluates parameter counts from each base model’s config.json and reproduces on screen the parameter count of the official checkpoint. The catalogue, the periodic table, the timeline, the category graph and the citation panel compute every count from the data file at load, and the graph’s layered layout is a deterministic Sugiyama procedure.

The static figures of this paper are drawn by Python scripts kept with the authors’ sources, most of them ports of the site’s modules, with NumPy and Matplotlib. They recompute every

plotted number in float64 from the site’s toys and seeds. Figure 8, for example, ports the orbit explorer and check 3, and Figure 9 ports the gauge lab and checks 2, 7, 12k, 12l and 13o.

## E.6 The numbers behind the figures

The captions of the figures describe what each panel shows. This subsection records the settings and the values behind them, as the figure scripts print them. Every value is recomputed in float64 when a script runs.

**Figure 4.** GraLoRA with  $k = 2, 4$  and 8 blocks reaches ranks 16, 32 and 64 at  $d = 65,472$ . HiRA<sub>8</sub>, the transport  $X \mapsto W_0 \odot X$  of LoRA<sub>8</sub> (Proposition II.13), reaches full rank for generic  $W_0$ . MoRA in its reshape form  $\Delta W = I_{16} \otimes M$  with  $M \in \mathbb{R}^{256 \times 256}$ , and C3A with  $256 \times 256$  circulant blocks, are injective, so  $d = |M|$ , and both reach full rank. FourierFT with  $n_c = 65,536$  frequencies from HF PEFT’s default draw (seed 777) has 120 conjugate pairs, so  $d = 65,416$ , and its maximum rank is known only through the bound  $\min(2n_c, m, n)$ . No KronA factor shapes have budget exactly 65,536. The closest,  $256 \times 256 \otimes 16 \times 16$ , cost 65,792 and give  $d = 65,791$  and rank  $\min(m_1, n_1) \min(m_2, n_2) = 4,096$  (Theorem II.11(b)). LoHa<sub>4,4</sub> is placed at the proved upper bound  $F = 57,313$ ; it reaches rank 16, as GraLoRA with  $k = 2$  does, with at least  $(m + n - 1) - 2r^2 = 8,159$  fewer dimensions than LoRA<sub>8</sub> (Proposition II.12(c)). Every dimension and maximum rank shown is confirmed by float64 Jacobian ranks and random-point ranks on a  $64 \times 64$  copy.

**Figure 6.** The vectors  $u, v$  and the factors  $C, D$  are seeded Gaussians, and the rank threshold is  $10^{-10}$ . The numerically zero values are at most  $1.3 \times 10^{-16}$ ; values below  $10^{-17}$ , exact zeros included, are drawn at  $10^{-17}$ , and every nonzero value is at least  $9.5 \times 10^{-4}$ . Each reshape permutes the entries, so the squared normalised singular values sum to 1 on every cut. A bond of dimension  $k$  costs  $k(4 \cdot 4 + 4 \cdot 4) = 32k$  parameters across any of the three cuts.

**Figure 7.** Panel (a) integrates gradient flow by fourth-order Runge–Kutta with step  $10^{-3}$ . The charge is integrated, not imposed, and stays at  $\Phi_0$  to within  $6 \times 10^{-11}$  on every run. From the zero- $B$  start  $(1.2, 0)$ ,  $\Phi_0 = -1.44$ , and Adam with  $\eta = 0.1$  ends at  $\Phi = -0.53$ . A GD step multiplies the charge by exactly  $1 - \eta^2 \eta_A \eta_B s^2 (w_k - w^*)^2$ ; the GD drift is  $1.2 \times 10^{-3}$  at  $\eta = 0.0025$  and 0.14 at  $\eta = 0.2$ , and  $5.8 \times 10^{-13}$  for the flow. The Adam drift stays between 0.55 and 1.5. In panel (b) the seeded target update has singular values 2.0 and 0.3, with  $s = 1$  and  $\eta_A = \eta_B = 1$ , so  $\mu = s\sqrt{\eta_A \eta_B} = 1$ . The flow is integrated by the implicit midpoint rule. The hollow circles are  $u_i^\top (BB^\top / \eta_B) u_i$  and  $v_i^\top (A^\top A / \eta_A) v_i$ ; the charge stays at  $-\kappa I$  to  $2 \times 10^{-14}$ , and the closed-form  $\dot{W}$  matches  $s(\dot{B}A + B\dot{A})$  to a relative  $1.3 \times 10^{-13}$ . The one-sided regime is where one gain exceeds ten times the other,  $\sigma < 0.70 \sigma^*$ , and the balanced regime where the two gains are within a factor two,  $\sigma > 2.83 \sigma^*$ .  $\sigma_1$  crosses  $\sigma^*$  at  $t \approx 0.49$  and settles at 2.0, where the  $A$ -update supplies 38% of its growth rate;  $\sigma_2$  settles at 0.3, where that share is 7%.

**Figure 8.** The toy has  $W_0 \in \mathbb{R}^{8 \times 6}$ , a rank-two target update and  $r = 2$ . Both runs take 300 steps of a hand-written, bias-corrected Adam in float64 and reach a loss of  $2.7 \times 10^{-7}$ ; the control ends at  $5.7 \times 10^{-2}$ . At  $\varepsilon = 0$  the runs are bit-identical at all 301 steps (check 3 of the script). The red curves use Adam’s  $\varepsilon = 10^{-8}$  or  $10^{-6}$  (the gap is proportional to  $\varepsilon$ ), PyTorch-style decoupled weight decay 0.01, and global-norm clipping at 1, which clips 8 steps of the LoRA+ run and 37 of the LoRA run. Their maximal gaps are  $4 \times 10^{-8}$ ,  $4 \times 10^{-6}$ ,  $7 \times 10^{-3}$  and 0.43, against 2.0 for the control. The per-block rescaling sets  $\varepsilon_B = 16 \varepsilon$ , decay  $16 \times 0.01$  on  $B$ , and  $B$ ’s clipping threshold multiplied by 16.

**Figure 9.** The state is  $B \in \mathbb{R}^{5 \times 2}$ ,  $A \in \mathbb{R}^{2 \times 4}$ . The rotation is by  $30^\circ$ , the stretch is  $\text{diag}(e^{0.5}, e^{-0.4})$ , and the signed permutation swaps the two rank channels and flips one. Adam is its first step from zero state, which at  $\varepsilon = 0$  is sign descent; the per-channel rescaling for  $g = \text{diag}(d)$  is  $\eta_{B,j} \mapsto \eta_{B,j}/d_j$ ,  $\eta_{A,j} \mapsto d_j \eta_{A,j}$ , and the damped scaled GD has  $\delta = 0.1$ . The first-order weight update is  $P_U G + G P_V$  for undamped scaled GD and  $P_U G + G P_V - P_U G P_V$  for LoRA-Pro. In panel (b) the weights are  $w = (0.6, 0.5, 0.4)$ . Task arithmetic changes by less than  $10^{-16}$  relative to its norm, the LoraHub merge by up to 3.4 times its norm at  $c = 1/8$ , and the closed form  $\sum_{l \neq 2} w_l w_2 [(c-1)B_2 A_l + (c^{-1}-1)B_l A_2]$  matches the recomputed change to  $10^{-15}$ .

**Figure 10.** In panel (a), ReLoRA reaches the full rank  $16 = \min(m, n)$  after 6 cycles for  $r = 3$  and after 4 for  $r = 5$ . LoRA-XS adds updates  $U_r R_k V_r^\top$  inside one  $r^2$ -dimensional subspace. In panel (b), BOFT's level  $\ell$  pairs coordinate  $c$  with  $c \oplus 2^\ell$ . Each light bar equals both the dimension of the Lie closure of the blocks' plane rotations under brackets and the measured dimension of the reachable set after 16 merge cycles of exact-Cayley factors (the rank of the right-trivialised Jacobian of the product map). Block OFT reaches 24 and 56 in one cycle and never more. BOFT goes from 16 to 24 and from 24 to 56 at depths  $m' = 2, 3$ , and only the full depth  $m' = 1 + \log_2(16/b) = 4$  connects the hypergraph: a single full-depth BOFT has dimension 32, and merging reaches 120 after 4 cycles. The 7 Givens planes leave components of sizes 4, 3, 2, 2 and five isolated coordinates, so they generate only 11 dimensions; the 15-plane tree reaches 120 after 9 cycles.

**Figure 12.** The query has  $q \in \mathbb{R}^4$  and  $\|q\| = 2.4$ ; the logits are  $\langle q, k \rangle/2$ , and the prefix keys are scaled by 1.2. In panel (a),  $\lambda_q = 0.44$ , so every content weight is multiplied by 0.56, and the six ratios agree to within  $10^{-15}$ . In panel (b),  $g = 0.5$  and the ratios are exactly 1. In panel (c), the two principal axes keep 88% of the variance of the values, and the LLaMA-Adapter output lies at distance 0.65 from the segment in  $\mathbb{R}^4$ . The script evaluates the joint softmax directly and recovers the identity of part (b) of [Theorem VI.5](#) to  $4 \times 10^{-16}$ .

## How this paper and its website were made

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This appendix records how the paper and its companion site were produced: who did what, how the material was found, how the results were generalised, unified and checked, and what it cost.

### F.1 In brief

- 1 October 2026, 17:06 AEST, to 4 October 2026, about 01:00 AEST, when the first full version of this appendix was written: about 56 hours of wall-clock time. At least one agent was running for about 23 of those hours. The rest was a 43-minute wait for an account usage reset, two interruptions, and long pauses between requests.
- The authors directed one orchestrating session of Claude (Anthropic’s Claude Opus 5.5) running in Claude Code, in short messages and a few answers to multiple-choice questions. That session wrote and launched 20 multi-agent workflows, which did most of the research, writing and checking, and two small ones that checked and redrafted this appendix. It did the planning, integration and publishing itself.
- Those workflows started 422 agent runs (420 distinct agent calls, two of which were started twice). 358 completed, 29 failed when the account reached a spend limit, and 35 were stopped or cut off.
- The agents processed about 117 million fresh tokens: prompts written to the cache plus generated text, of which 8.6 million were generated. They also re-read 4.1 billion tokens from the prompt cache. The orchestrating session added about 10 million fresh tokens and 0.2 billion cache reads.
- The outputs are a website (15 sections including a version of this appendix, 20 interactive figures plus an animated hero, and a checked catalogue of 382 methods and 138 papers) and a paper of about 400 pages with the same numbering.
- On 5 October a later session made a narrated film of the atlas, 16 minutes 44 seconds long, and a two-minute teaser. Its five agents processed about 5.0 million fresh tokens, and the session itself about 1.4 million.

### F.2 Who did what

The authors think that when the literature on parameter-efficient fine-tuning (PEFT) already holds several hundred methods, the field gains more from the approach Grothendieck pictured as a rising sea than from one more method. This atlas catalogues 382 of them.

In his memoir *Récoltes et Semailles*, written 1983–1986, Grothendieck compared a hard problem to a nut. One can strike the shell with hammer and chisel, at point after point, until it cracks. Or one can soak the nut in a softening liquid and let time pass, until the pressure of a hand is enough. On a larger scale the second way is the rising sea. It advances so slowly and quietly that nothing seems to change, until it surrounds the problem and the problem is dissolved by a theory that reaches well beyond it (Grothendieck, 2022; McLarty, 2007). In the authors’ analogy, a 383rd method would be one more strike of the chisel, and the sea is a theory in which the existing methods appear as instances of one notion.

The authors have packaged this way of working as a Claude Code skill. It asks for the natural home of a result, for objects defined by their relationships and for the relative point of view, so that hard results dissolve. Its binding rule is that every reframe must pay, by a shorter proof, a dropped assumption or a real unity, or be rejected.

This view shaped the question the authors chose, to re-found PEFT “as Grothendieck would” and to make the result an interactive site. They also made the main decisions of scope and presentation. Their instructions to the session were short messages: pointing out some missing methods in the literature, asking for a sweep of 2026 work, a mathematical appendix, more readable typefaces, a LaTeX paper and this appendix, supplying the author list and the title, and asking the session to resume after interruptions.

The orchestrating session did the following:

- planned the work and wrote the seed sketch of the theory;
- designed the visual identity, the data pipeline, the build and test tools and the LaTeX template;
- launched and resumed the workflows, read their reports and decided what to run next;
- applied some changes itself, such as the new typography rules, and rebuilt, tested and published the site, and built the paper.

The sub-agents researched the literature, proposed, judged and refereed the theory, classified the methods, wrote the sections, drew and tested the figures, and reviewed each other’s work. They also made the remaining decisions, including which 2026 candidates entered the catalogue and what was cut from the site in its final review.

A workflow is a script for Claude Code’s Workflow tool. It starts sub-agents in parallel or in sequence, hands each a prompt and files, and keeps a journal of their results. Every sub-agent ran the same model, `claude-opus-5-5`, whose training data ends in June 2026. All had web search and fetch and a shell on one Apple M5 laptop, with Python 3.9 (NumPy, PyTorch, SciPy, Matplotlib), Node.js, headless Chrome and TeX Live 2026. Below, an independent check means one made by a separate agent with its own context and instructions.

### F.3 Finding the material

The literature work was split into thirteen families:

- the Hugging Face PEFT library;
- LoRA and its training-dynamics variants;
- initialisation and rank allocation;
- sharing and extreme efficiency;
- structured shapes;
- multiplicative and orthogonal methods;
- adapters;
- prompts and states;
- selective fine-tuning;
- quantisation and memory;
- composition and merging;
- theory and foundations;
- the 2025–2026 frontier.



One agent swept each family. It worked from the arXiv API, Semantic Scholar, Open-Review, the ACL Anthology, the official conference listings and the Hugging Face PEFT source code. Each sweeper was told to read a paper’s method section before writing down its formula. As each family finished, an independent fact-checker re-verified every entry:

- title, first author and first-version year against the arXiv API;
- the venue against the proceedings;
- the formula against the method section;
- the parameter count, re-derived independently;
- mergeability and initialisation;
- Hugging Face PEFT support, against the repository at commit 85be3f3 (30 September 2026).

A completeness critic then compared the whole catalogue with the library and with its own knowledge of the field. It listed 34 missing methods, and 32 of them were researched and checked in four more batches. The checkers, including those of the four gap-filling batches, made 613 corrections before the first release. Merging duplicates on a canonical identifier and an alias table gave 320 methods and 124 papers. Matching by name would have folded LoRA+ into LoRA.

After the first build, the authors pointed out methods missing from the catalogue. They were added and refereed, and a dedicated sweep of 2026 work followed, in which six scouts covered:

- arXiv submissions from January to September 2026, one scout per quarter, by keyword queries over titles and abstracts;
- the accepted papers of ICLR 2026 and ICML 2026, with ACL and CVPR 2026;
- the citation graph of ten anchor PEFT papers and the 2026 Hugging Face PEFT releases.

A triage agent then chose which candidates entered the atlas. It admitted new 2026 methods with at least one signal (a 2026 venue, inclusion in Hugging Face PEFT, citations, a widely used codebase, or a structure the seven coordinates would treat differently), at most 36 per pass. The first triage saw only the first 105 of the 265 candidates not already in the atlas, the last of them cut off mid-entry, because the orchestrator had truncated the list it passed in. A second pass triaged the remaining 161 candidates and the scouts’ overflow lists. Together the passes added 61 methods and 14 theory papers. They were researched and fact-checked in the same way, giving 382 methods, 138 papers and 905 corrections in total.

The BibTeX of the 502 catalogue entries is built from arXiv metadata for the 493 that have an arXiv identifier, and the other nine are built from the catalogue’s own fields. Eight of those nine were corrected by hand for the paper. The paper’s 130 other references, mostly mathematics and related work, were written by the agents who cited them, who were told to check every field against the publisher, DOI or arXiv page.

#### F.4 From examples to a theory

The orchestrator wrote a seed sketch of eleven numbered points, about 1,000 words, which the architects could adopt, correct or reject. Its points were:

- methods as reparametrisations in Para and as objects of a pointed slice;
- frozen and full fine-tuning as initial and terminal;

- mergeability as a lifting problem;
- Klein’s Erlangen programme for group-type methods;
- the  $GL_r$  gauge of LoRA and its conserved charge;
- string diagrams for structured updates;
- a monoidal sum;
- base change;
- the lens duality between forward and backward methods;
- prompts and prefixes as learned input states;
- naturality, so that a method defined once per layer type extends to every network.

Four architect agents each developed a complete framework from this sketch, with definitions, propositions and proofs, from a different point of view:

- parametric lenses;
- Grothendieck’s relative point of view;
- the Erlangen programme with gauge symmetry;
- string diagrams.

Three judge agents scored all four proposals, each from an assigned point of view:

- a category theorist’s, for rigour;
- a PEFT researcher’s, for faithfulness to the methods;
- a science communicator’s, for clarity and visual potential.

All three recommended the relative point of view as the backbone. One synthesis agent then built the framework on it. As the judges advised, it took the lens and cometric from the parametric-lens proposal, the covariance and optimizer gauge from the Erlangen proposal and the diagrams from the string-diagram proposal, and it cut the constructions they judged decorative. The standing rule was that a categorical construction must explain, classify or predict something about PEFT, or be cut. The result was 41 numbered results, a seven-axis classification and a dictionary between the two vocabularies.

Generalisation usually followed one pattern:

1. Take an observation about a single method. Zero-initialised LoRA cannot move in some directions at the first step.
2. Find the invariant it depends on. Here it is the first-order image compared with the dimension of the image germ.
3. State the result for every method with that coordinate value. Any method whose base point is the vertex of a cone has the same deficit.
4. Look for consequences the source papers do not state. Householder reflection adaptation at its default start is also an apex.
5. Test the claim numerically and send it to referees.

The same pattern turned “LoRA+ helps” into an exact statement about Adam’s invariance to per-block rescaling. It turned “orthogonal methods preserve structure” into complete invariants of each group action. It turned “this adapter can be merged” into a local rewriting test. The test certifies exact merging under stated side conditions, names the weight that absorbs the update, and depends on the activation, the normalisation and the bias slots.

The seven coordinates are decidable from a method’s formula. That made it possible to classify the first 320 methods in ten batches of 32, and the 62 later additions in three more. In each batch a second agent re-derived every coordinate, checked every arrow’s factorisation and base point by hand, and spot-checked a few gauge dimensions and first-order deficits numerically. A consolidator then reconciled the first ten batches with each other and with the theory, and the final site review made the prose agree with the data. Every arrow between methods records its explicit map  $h$  with  $\rho_N \circ h = \rho_M$  (403 of them). Every recorded non-arrow names the test that rules it out (485).

## F.5 How category theory unifies the field here

The unifying move is to study each method through its map into weight space. A weight-space method is a pointed map from its small parameter space into weight space that sends its starting point to the pretrained weights. Comparing methods then means comparing maps:

- a method  $N$  simulates a method  $M$  when  $M$ ’s map factors through  $N$ ’s, that is  $\rho_N \circ h = \rho_M$  for a pointed map  $h$ ;
- doing nothing and full fine-tuning are the initial and terminal objects, so they bracket every weight-space method;
- three invariants of the map carry most of the content: its image germ, its fibres with the optimizer’s metric, and its behaviour under base change.

Extensions such as adapters and prefixes, which change the network itself, fit through a realisation map. Mergeability becomes the question of whether a lift exists.

Exposé XI lists what is analogy and what is theorem. The words Noether, Erlangen, gauge and Grothendieck are used in a precise, limited sense, and the paper says which.

## F.6 How the results were checked

Verification ran in layers:

- **Sources.** Every catalogue entry was checked by an agent other than the one that wrote it, against arXiv metadata, the paper and the library source.
- **Adversarial model referees, three rounds.** Each of the 41 results went to two referee agents with different lenses: mathematical correctness, and faithfulness to the papers and code. They were told to default to “refuted” when in doubt.
  - Round 1 produced 82 verdicts, 45 of them major objections (41 refutations). 27 results were repaired and none was false outright.
  - Round 2 re-examined 26 of the 27 repaired results in 48 reports. Six of its 54 runs were stopped by the spend limit, so Corollary III.13 received no second-round report and four results received one; in the final site review the section editors took the missing referees’ place. The reports raised 28 major objections against 18 results, and 19 results were narrowed again.
  - Each round-2 report ended with the strongest statement its referee would sign. These became one public statement per result in `propositions.json`. The writers and referees of the site and the paper were told to state nothing stronger, and checked for it.

- A third round, four mathematical referees during the writing of the paper, made 63 further fixes.
- Those fixes were then carried back into the site, the data and the theory files, 108 edits in all.
- **Numerical checks.** `framework_checks.py`, one of the paper’s data files, re-runs most claims marked [Num] in about a minute. The paper’s Appendix E lists the few that rest only on the referees’ own replications. Each interactive figure computes its numbers in the browser. The engineers of the theory-driven figures checked their key numbers against `framework_checks.py` or a separate NumPy computation, and in the final site review a second agent spot-checked the key numbers of every figure.
- **Engineering checks.** Every figure was rendered headlessly in light and dark themes at desktop and phone widths, and its builder looked at the images. A whole-site test drives Chrome to check console errors, unmounted figures, overflow, MathJax errors and broken in-page links. The paper is compiled from a clean copy of its sources with pdfLaTeX alone and checked for errors and undefined references.
- **This appendix.** Two agents checked every number and date in it against the workflow journals, the agents’ transcripts and the project files, and its text was corrected where they disagreed. Its opening was then redrafted by three agents writing independently, merged by a fourth and checked by two more, one for facts and one for style.

These checks do not amount to human peer review. Every proposer, writer and referee was the same model, so their errors can be correlated, and the referees of the paper also edited the files they reviewed. A numerical check is not a proof, and every check runs on small float64 examples. No fine-tuning run on a real model was made, so the eleven predictions of Exposé X are untested. As Exposé XI records, the dimension equalities for LoHa and, in general, for VeRA are checked numerically; one clause of corollary 4 of Theorem V.3 is labelled a proof sketch, although Appendix C gives an argument; and Conjectures A, C and D are open. The catalogue is a selection and can still miss methods. Its 2026 additions came from keyword queries on arXiv, four conference lists and one citation graph, filtered by a triage agent, and the completeness critic’s own knowledge of the field ends with the model’s training data. In the catalogue, 51 method entries and 7 paper entries carry a medium-confidence flag. Some methods were missing until the authors noticed them. The authors are responsible for what the site and the paper say.

## F.7 Building the site and the paper

The site is a static page. Its prose lives in fifteen HTML sections, counting a version of this appendix, and its data is generated from the verified catalogue and the theory files. Each of its 21 figure scripts was written by an engineer agent (four were finished by a second agent after the spend limit cut off the first), self-tested and later reviewed by another. The page runs from any static host, straight from disk and inside a sandboxed artifact. Its data is a script, so nothing is fetched at run time apart from D3, MathJax and the fonts, which come from public CDNs.

The typography was changed once, at the authors’ request. The session set one passage of the site in four typeface schemes, and the authors chose the one it recommended, with Source Serif 4 for text, IBM Plex Sans for labels and IBM Plex Mono for numbers. Seven engineer agents then raised the text inside all 21 figure modules to at least 12 pixels where a

reader must read it, and 11 pixels for tick labels. Later the authors found the display face of the title unattractive, and the title was reset in Source Serif 4 as well.

The paper uses a template designed for it: SGA-style exposés with Roman numerals, so that every result has the same number on the site and in the paper. Twenty-one writer agents converted the full 35,000-word theory into LaTeX, with complete proofs in Appendix C. Six figure engineers redrew the key figures as vector PDFs in the same typefaces. An assembler, four mathematical referees, a typesetting referee and a finalizer took it to a clean build of 393 pages, about 400 with this appendix.

## F.8 The film

On the evening of 5 October, at the authors’ request, a later session of Claude Code made a narrated film of the atlas, 16 minutes 44 seconds long, and then a two-minute teaser for X. The authors chose the narrating voice, Sulafat from Gemini 3.8 Flash TTS, and a quiet visual style from a short list of options.

The session wrote the script and checked each claim against the propositions and the paper, and a separate agent then fact-checked it adversarially. Every number on screen comes from a short computation in `video/sim/`. The opening, for example, trains LoRA+ and plain LoRA with a scaled alpha side by side, and the largest difference between their weights stays exactly zero. The session wrote a shared Manim library, a brief and one chapter, and four agents built the other seven chapters from the brief. The session then synthesised the narration beat by beat, timed each animation cue to the pauses detected in the audio, reviewed every chapter on contact sheets of its frames, and assembled the film with captions and chapter marks. The film plays on the companion site, and the title page of the paper links to it.

## F.9 Cost and interruptions

Table 40 breaks the agent runs and tokens down by phase.

**Table 40.** Agent runs and tokens by phase. Fresh tokens are prompts written to the cache plus generated text; re-reads from the prompt cache are not counted.

| Phase                                                     | Workflows | Agent runs | Fresh tokens   | Generated tokens |
|-----------------------------------------------------------|-----------|------------|----------------|------------------|
| Literature catalogue (first sweep, additions, 2026 sweep) | 4         | 75         | 24.6 M         | 0.43 M           |
| Theory and adversarial review (rounds 1–2)                | 2         | 145        | 16.3 M         | 2.03 M           |
| Classification, site sections and figures                 | 6         | 103        | 32.1 M         | 2.74 M           |
| Site review, appendix and typography                      | 4         | 49         | 22.1 M         | 1.68 M           |
| Paper (writing, referees, sync)                           | 4         | 50         | 22.0 M         | 1.75 M           |
| <b>All workflows</b>                                      | <b>20</b> | <b>422</b> | <b>117.1 M</b> | <b>8.63 M</b>    |
| Film (script check and chapters)                          | –         | 5          | 5.0 M          | 0.35 M           |
| <b>Total</b>                                              | <b>20</b> | <b>427</b> | <b>122.1 M</b> | <b>8.98 M</b>    |

The table counts every started agent run, one transcript each: the 422 runs of the workflows and the film’s five agents, a script fact-checker and four chapter builders. Cache re-reads are not counted (4.1 billion tokens for the workflows, 0.16 billion for the film’s agents), and neither are the sessions that directed the work: the orchestrating session of the workflows, and the film’s session, which added 1.4 million fresh tokens, 0.55 million of them generated.



Also not counted are the two later workflows that checked and redrafted this appendix (nine agents), and the film’s narration: about 15 minutes of speech from Gemini 3.8 Flash TTS, plus voice tests and a few retakes, which Google bills separately.

The work was interrupted three times:

- at 20:52 on 1 October an account spend limit made 29 agent runs fail, 12 of them mid-run and 17 within seconds of starting;
- at about 20:40 on 2 October the host machine crashed while the paper was being written;
- at about 23:10 on 2 October the session was cut off again, and it resumed at 02:04 the next morning.

Each time, the orchestrator inspected the workflow journals and launched fresh workflows containing only the unfinished items, telling each agent to read and complete its predecessor’s partial file. The six stopped second-round referee runs were not repeated; the final site review covered them instead. A first attempt to resume in place re-ran agents that had already finished. The workflow tool replays only the longest unchanged prefix of agent calls, so that attempt was stopped within two minutes. The other process lessons were:

- pass large inputs to agents through files, never as truncated prompt text;
- refereeing a repaired result again is worth it;
- corrections found while writing the paper must flow back to the site.

## F.10 Reproducibility

The catalogue and the checks accompany the paper as data files:

- `catalogue.csv` and `catalogue.json`, with `arrows.csv` and `obstructions.csv`;
- `propositions.json`, which holds each numbered result with its statement, proof, public statement and the notes of both review rounds;
- `framework_checks.py`, the numerical checks.

The LaTeX source carries the text and the generated tables, and the site’s figure code can be read in the page. The public repository, <https://github.com/xiaoyulics/peft-atlas>, holds the rest of the record:

- the raw and verified literature files and the classification batches, in `research/`;
- the full theory text and the round-2 reports, in `theory/`;
- the build and test scripts, in `site/build`, `site/test` and `paper/tools`, and the scripts of the paper’s figures, in `paper/figures/src`;
- the film’s script, scenes and computations, in `video/`.

The 20 workflow scripts, which contain every agent prompt, and the agents’ transcripts stay with the authors.